Linear Algebra

— ***EYNTKA*** **Series** —

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Contents

Introduction

Two linear equations in two unknowns describe two lines in the plane, and solving the system means finding where the lines meet. There are three possibilities and no others: the lines cross at a single point, or they are parallel and never meet, or the two equations describe the same line and every point on it is a solution. Three unknowns replace the lines by planes in space, and the count comes out the same: no solutions, or exactly one, or infinitely many. Nothing in that trichotomy depends on there being two or three unknowns, and neither does the procedure that decides which case holds. Scale an equation, add a multiple of one equation to another, repeat until the system is simple enough to read off. This is elimination, and it is the oldest thing in this book.

Elimination answers the question it was built for and leaves standing several it cannot even phrase. Once there are more equations than unknowns a system usually has no solution at all, and that is the ordinary situation whenever the numbers come from measurement rather than from exact arithmetic. Elimination reports the failure and halts. It has no way to ask for the best available answer, and no way to say what would make one candidate better than another. It cannot say what two systems with the same solutions have in common. It cannot separate the features of a system that are really about the system from the ones that are accidents of what the unknowns happened to be called. And it says nothing about what happens when the arithmetic is inexact, though every system solved by a machine is solved inexactly.

Those questions are all answered in this book, and the answers come from one place. Elimination uses nothing but addition, subtraction, multiplication and division, so it runs unchanged whether the coefficients are ordinary fractions or residues modulo a prime; any system of numbers closed under those four operations, a *field*, will serve. But "the best available answer" needs a way to compare two failures and call one of them smaller. Length and angle need a way to say that one vector is longer than another. "Inexact" needs a way to measure how far off an answer is. Each of the three needs the numbers to be *ordered*, and most fields are not. The complex numbers are not ordered either, and they carry instead an operation a general field lacks, conjugation, which sends $a + bi$ to $a - bi$ and restores a usable notion of length, since $z\bar{z}$ is the nonnegative real number $|z|^2$. An order and a conjugation are exactly what $\mathbb{R}$ and $\mathbb{C}$ add to a bare field, and this book is the account of what those two give.

The order in which these ideas were found is close to the reverse of the order in which they are now taught. Elimination is the oldest by a wide margin. The eighth chapter of the *Nine Chapters on the Mathematical Art*, compiled in China by the first century of the common era and titled *Rectangular Arrays*, solves systems of linear equations by setting the coefficients out as an array of counting rods and operating on it, by a procedure indistinguishable from the one taught now. The attachment of Gauss's name to that procedure dates only from the 1950s, by which time it was two thousand years old. Determinants came far later and in two places at once: Seki Takakazu in 1683 and Leibniz in a letter of 1693, with Cramer publishing in 1750 the rule that solves a system by ratios of determinants.

The nineteenth century gave the geometry and, eventually, the algebra. Cauchy proved in 1826, working on quadratic forms, that a real symmetric matrix has real eigenvalues and can be diagonalized. Gauss had computed the orbit of Ceres by least squares in 1801 and published the method in 1809, four years after Legendre published it independently. Sylvester named the matrix in 1850 and proved his law of inertia in 1852. Cayley's memoir of 1858 was the first treatment of matrices as an algebraic system in their own right, with a product, an inverse, and the assertion that a matrix satisfies its own characteristic equation, which Cayley checked for small matrices and left otherwise; the general proof is Frobenius's, twenty years later.

The axioms came last. Grassmann published an abstract theory of extended quantities in 1844 and was little read. Peano gave the modern axioms for a vector space in 1888, under the name *linear systems*. The axiomatic account came into general use only through the function spaces that Banach and Hilbert formalized around 1920. Two millennia of successful computation preceded the definition of the object being computed with, and this book is arranged to respect that: Part I does what was done first, and the abstraction enters once there is enough computation for it to organize.

Part I builds the concrete apparatus and presupposes no acquaintance with vector spaces. Elimination is the engine throughout: it solves systems, exposes the rank, records itself as a factorization of the coefficient matrix, and computes determinants, all before an abstract vector space is defined. The abstraction then enters as compression of what is already there rather than as a new subject, and the part closes with eigenvalues and with the obstruction to diagonalizing an operator, which is the question the rest of the book answers three separate times.

Part II introduces the inner product, and with it length, angle, orthogonality, projection, and the adjoint. Two payoffs follow immediately. Least squares answers what to do when a system has no solution, which is the generic situation whenever the coefficients are measured; the answer is to accept the inconsistency and minimize it, and the minimizer turns out to be an orthogonal projection. The spectral theorem settles when an operator can be diagonalized by a change of coordinates that preserves the geometry, and its proof by triangularization is the argument that *Functional Analysis* [16] later rebuilds in infinite dimensions.

Part III separates the two structures the scalars give and then recombines them. A quadratic form is a symmetric operator regarded as a function rather than as a map, and Sylvester's law of inertia says that a change of coordinates destroys everything about it except the signature, which is where the classification of critical points comes from. The conjugation settles the remainder: a real vector space carrying an operator that squares to -1 is a complex vector space in disguise, and from that observation come Hermitian forms, alternating forms, and the fact that a metric, a complex structure and a symplectic form, when they are compatible with one another, are linked tightly enough that any two of them determine the third. The singular value decomposition then draws the threads together. It exists for every matrix without exception, it displays the four fundamental subspaces, it solves the least-squares problem, and it produces the best approximation of any prescribed rank.

?? asks what becomes of all this when the arithmetic is inexact, a question that needs the order on $\mathbb{R}$ before it can even be posed. Conditioning measures how much a problem is capable of amplifying an error that was already present in its data, which is a property of the problem and not of the algorithm. Gershgorin and Weyl locate eigenvalues under perturbation. The power method computes an eigenvector by iterating rather than by factoring a characteristic polynomial, which is what is done in practice. The book closes with positivity, the last structure an ordered field gives: a matrix with positive entries has a positive eigenvalue strictly larger in absolute value than every other, and that single fact is enough to run a Markov chain.

A reader who finishes has the finite-dimensional theory in the form the rest of the series uses it. *Functional Analysis* [16] generalizes chapter 5 and chapter 7 to infinite dimensions, where closed bounded sets need not be compact and the spectral theorem has to be rebuilt from the ground up. *Ordinary Differential Equations* [8] solves $\dot{x} = Ax$ with the matrix exponential constructed in chapter 12. *Core Algebra* [9] develops the same subject over an arbitrary field and over a principal ideal domain, which is the account this book deliberately does not give: the two are companions rather than substitutes, and the boundary between them is exactly the order and the conjugation.

Part I assumes only *Preliminaries* [12] and does not presuppose that the reader has met a vector space. The later parts sit at the level of the rest of the series. What changes there is the quantity of background assumed and nothing else; in particular the standard of proof does not move, and every statement in Part I is proved where it is made.

0.0.1 Note: Readers Who Have Had a First Course A reader with a solid first course in linear algebra should begin at Part II. The two things Part I gives that such a reader may not already have are the four fundamental subspaces (section 2.3), which organize everything the singular value decomposition later says, and the treatment of the obstruction to diagonalizing in chapter 4. Both are short and can be read as reference when they are first cited.

Results Assumed from Other Volumes

Seven results are borrowed from elsewhere in the series and used without proof. The list is complete: nothing else in this book is asserted without an argument.

1. *Completeness of* $\mathbb{R}$. Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound. *Elementary Analysis* [6].
2. *Heine-Borel.* A subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded. *Analysis on Metric and Function Spaces* [3].
3. *Extreme value theorem.* A continuous real-valued function on a nonempty compact set attains a maximum and a minimum. *Analysis on Metric and Function Spaces* [3].
4. *Convergence in* $\mathbb{R}^n$. A sequence in $\mathbb{R}^n$ converges if and only if it converges in each coordinate, and $\mathbb{R}^n$ is complete. *Analysis on Metric and Function Spaces* [3].
5. *Taylor's theorem with second-order remainder.* A real-valued function with continuous second partial derivatives on an open set of $\mathbb{R}^n$ satisfies, near a point a of that set, $f(a+h) = f(a)+\nabla f(a)\cdot h+\frac{1}{2}h^\top Hh+r(h)$, where H is the matrix of second partials at a and $r(h)/\|h\|^2 \to 0$ as $h \to 0$. *Analysis on* $\mathbb{R}^n$. It is used once, in section 8.5.

6. *Fundamental theorem of algebra.* Every nonconstant polynomial with complex coefficients has a root in $\mathbb{C}$. *Core Algebra* [9, Fundamental Theorem of Algebra].
7. *The complex exponential.* There is a function $z \mapsto e^z$ on $\mathbb{C}$ with $e^0 = 1$ and $e^{z+w} = e^z e^w$, satisfying $|e^{a+bi}| = e^a$ for $a, b \in \mathbb{R}$. *Complex Analysis* [14, Exponential Function]. This is a named function, not a technique: the book uses these three properties and nothing else about it. It enters in section 12.5.

Item 6 enters in chapter 4, where it is what guarantees that a complex matrix has an eigenvalue at all. Items 1 through 4 are not used anywhere in Part I; they enter from Part II onward, wherever a quantity is produced by maximizing or minimizing something, or by iterating. Item 7 is confined to chapter 12, where the matrix exponential is built from the numbers e^λ, and it is the only analytic fact the book uses beyond those four. It is a named function rather than a technique, and the book never differentiates or integrates anything.

Notation and Conventions

- $M_{m\times n}(\mathbb{R})$ and $M_{m\times n}(\mathbb{C})$ denote the m by n matrices with real and complex entries; M_n abbreviates $M_{n\times n}$. Elements of $\mathbb{R}^n$ are columns, so that $\mathbb{R}^n$ is identified with $M_{n\times 1}(\mathbb{R})$ and the product Ax is defined whenever A has n columns.
- $e_1, \ldots, e_n$ is the standard basis of $\mathbb{R}^n$ or of $\mathbb{C}^n$.
- $A^\top$ is the transpose and A^* the conjugate transpose. Over $\mathbb{R}$ the two agree, and statements proved for A^* specialize to $A^\top$ without comment.
- $\langle -, - \rangle$ is linear in its first argument and conjugate-linear in its second. Over $\mathbb{R}$ conjugation is trivial, so the form is bilinear.
- k denotes $\mathbb{R}$ or $\mathbb{C}$ in a statement that holds over both, and the field is named explicitly whenever the distinction matters.

0.0.2 Reading the numbering A star marks *bonus material*: a chapter, section or subsection that belongs in the book but that nothing later depends on. Skipping it costs nothing downstream. The star sits on the number, as in §2.7⋆; in divisions written before this convention it sits at the front of the title instead, as in **Uniform Spaces*. The two mean the same thing.

Part I

Systems, Spaces, and Spectra

Elimination and its consequence are the focus throughout this part. The process on a system of linear equations exposes the rank, the operations it performs along the way assemble into a factorization of the coefficient matrix, the same bookkeeping computes determinants. The abstract vector space appears only after all of that has been extracted, and it appears as a name for what the computation has already produced.

The chapters follow that order. Chapter 1 sets up systems of linear equations and performs elimination on them. Chapter 2 classifies the objects elimination was manipulating and isolates the four subspaces attached to a matrix. Chapter 3 extracts the determinant from the same row operations and reads it as a signed volume. Chapter 4 asks which vectors a matrix only stretches, and finds that sometimes no basis of such vectors exists. That failure is the question the remaining three parts each answer in a different way.

1

Linear Systems and Elimination

A system of linear equations is the smallest object in this book, and a surprising amount of what follows is a statement about what can be done to one. This chapter develops two operations and the relationship between them: solving a system, and recording what the solving did.

Section 1.1 fixes the objects and settles the shape of a solution set before any algorithm is applied to it to makes the algorithm's output interpretable. Section 1.2 gives the algorithm and the normal form it produces. Section 1.3 reads the answer from that form: whether a solution exists at all, how many free parameters the solutions carry, and what the rank of a matrix counts. Section 1.4 turns the row operations themselves into matrices so that elimination becomes a product of matrices. Section 1.5 records that product, and the record is the LU factorization.

1.1 Linear Systems and Their Solution Sets

A single equation $2x + 3y = 7$ constrains the pair (x, y) without determining it: for each choice of x there is exactly one y making the equation true, while x itself is unconstrained. A second equation constrains the pair again, and the two together usually leave no freedom at all.

Example 1.1.1: Two Equations in Two Unknowns
Consider the pair of equations

$$\begin{aligned} 2x + 3y &= 7 \\ x - y &= 1 \end{aligned}$$

A pair (x, y) satisfying both equations satisfies every consequence of the two. Subtracting twice the second equation from the first removes x:

$$(2x + 3y) - 2(x - y) = 7 - 2 \cdot 1$$

The left side is $5y$ and the right side is 5, so $y = 1$. The second equation then reads $x - 1 = 1$, so $x = 2$.

Every step was forced, so $(2, 1)$ is the only candidate. It is indeed a solution: $2 \cdot 2 + 3 \cdot 1 = 7$ and $2 - 1 = 1$. The pair of equations is therefore satisfied by exactly one pair of numbers.

Each equation describes a line in the plane, and (x, y) satisfies both exactly when it lies on both lines. Above, the lines cross at one point. Only Two other things can happen. The equations $x - y = 1$ and $2x - 2y = 5$ describe parallel lines: doubling the first gives $2x - 2y = 2$, which contradicts the second, so no pair satisfies both. The equations $x - y = 1$ and $2x - 2y = 2$ describe the same line, and every point on it satisfies both.

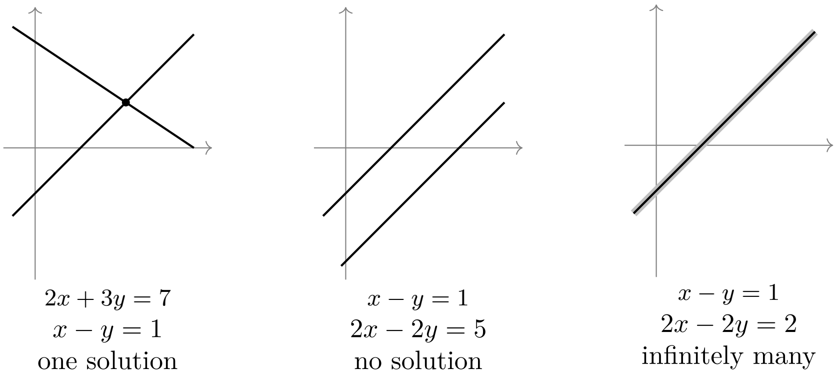


Figure 1.1: The three configurations available to two lines in the plane

Every equation above has the same shape: each unknown, in the above x and y, is multiplied by a constant, the results are added, the final result being a *linear equation*. Throughout, k denotes $\mathbb{R}$ or $\mathbb{C}$, and k^n denotes the set of ordered n-tuples $(s_1, \ldots, s_n)$ of elements of k.

Definition 1.1.2: Linear Equation

A *linear equation* in the unknowns $x_1, \ldots, x_n$ over k (recall k is either $\mathbb{R}$ or $\mathbb{C}$) is an equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

where the *coefficients* $a_1, \ldots, a_n$ and the *constant term* b are fixed elements of k. A *solution* of it is a tuple $s = (s_1, \ldots, s_n) \in k^n$ for which $a_1s_1 + a_2s_2 + \cdots + a_ns_n = b$.

Each unknown occurs in a linear equation exactly once, to the first power, and multiplied by a constant. Nothing else is permitted: $x^2 + y^2 = 1$, $xy = 1$ and $\sin x = y$ are all equations in two unknowns, and none of them is linear. These non-linear equations are much harder to solve, sometimes intractable, and are the subject of many further books in the EYNTKA series ([11], [17], [5], [2]) all of which need the theory of this book.

Definition 1.1.3: System of Linear Equations

A *system of m linear equations in n unknowns* over k is a list of m linear equations in the same unknowns $x_1, \dots, x_n$,

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

with every a_{ij} and every b_i in k. In the coefficient a_{ij} the first index i names the equation and runs from 1 to m, and the second index j names the unknown and runs from 1 to n.

A tuple satisfying the first equation need not satisfy the second, so it is the tuples meeting all m equations at once that get a name.

Definition 1.1.4: Solution Set

The *solution set* of the system in definition 1.1.3 is

$$S = \{\, s \in k^n \,\big|\, a_{i1}s_1 + a_{i2}s_2 + \cdots + a_{in}s_n = b_i \text{ for every } i = 1, \dots, m \,\}$$

the set of tuples solving all m equations at once. To *solve* the system means to describe S.

Solving is a question about a set, not a number, and the set can be empty. Each equation added to a system imposes a further requirement on the same n numbers, and requires can conflict.

Example 1.1.5: Systems With No Solution

1. The system
$$\begin{aligned} x - y &= 1 \\ 2x - 2y &= 5 \end{aligned}$$
has empty solution set. Doubling the first equation gives $2x - 2y = 2$, and any pair (x, y) satisfying both original equations satisfies this one as well, hence satisfies $2 = 5$, but $2 \neq 5$ so no such pair exists.

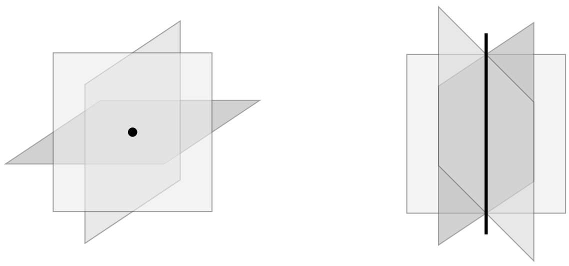
2. The system
$$\begin{aligned} x + y + z &= 1 \\ x + 2y + 3z &= 2 \\ 2x + 3y + 4z &= 5 \end{aligned}$$
has empty solution set. Adding the first equation to the second gives $2x + 3y + 4z = 3$, whose left side is the left side of the third equation. A triple (x, y, z) satisfying the first two equations would therefore satisfy both $2x + 3y + 4z = 3$ and $2x + 3y + 4z = 5$, hence $3 = 5$, so no triple can.

In neither case was a candidate solution produced and then rejected. The contradiction came from the equations themselves.

Definition 1.1.6: Consistent and Inconsistent Systems

A system of linear equations is *consistent* when its solution set is nonempty, and *inconsistent* when its solution set is empty.

The two systems of example 1.1.5 are inconsistent, and the system of example 1.1.1 is consistent. In three unknowns the geometry representing the algebra is the following. The triples satisfying one equation $a_1x_1 + a_2x_2 + a_3x_3 = b$, with the a_j not all zero, form a plane in space, and the solution set of a system of three such equations is the set of points lying on all three planes. Three planes can meet in a single point, in a whole line, in a whole plane when all three equations describe the same one, or in no point at all.



one common point a common line

Figure 1.2: Three planes meeting in a single point, and three planes meeting in a line

The unknowns did no work in either computation above. What was manipulated in example 1.1.1 were the coefficients and the constant terms; the symbols x and y were copied from line to line unchanged. Deleting them from the second system of example 1.1.5 leaves the rectangular array

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{pmatrix}$$

of coefficients, together with the separate list $1, 2, 5$ of constant terms.

Definition 1.1.7: Coefficient Matrix and Matrix Form

The *coefficient matrix* of the system in definition 1.1.3 is the array

$$A = (a_{ij}) \in M_{m\times n}(k)$$

with m rows and n columns, whose entry in row i and column j is a_{ij}. Elements of k^n are written as columns. Write x for the column of unknowns and b for the column of constant terms,

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \qquad b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}$$

and for $s \in k^n$ let As denote the element of k^m whose i-th entry is

$$(As)_i = a_{i1}s_1 + a_{i2}s_2 + \cdots + a_{in}s_n = \sum_{j=1}^{n} a_{ij}s_j$$

where $\sum_{j=1}^{n}$ abbreviates the sum of the n terms obtained by letting j run from 1 to n. The system is then the single equation

$$Ax = b$$

called its *matrix form*.

Definition 1.1.8: Augmented Matrix

The *augmented matrix* of the system $Ax = b$ is the array obtained by attaching b to A as an extra column on the right,

$$(A|b) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right)$$

an element of $M_{m\times(n+1)}(k)$. The vertical bar records which column holds the constant terms and has no other meaning.

For the second system of example 1.1.5 the matrix form and the augmented matrix are

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 3 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}, \qquad \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 2 \\ 2 & 3 & 4 & 5 \end{array}\right)$$

and the augmented matrix carries the whole system: no information was lost in dropping the unknowns, since their names can be restored from the column positions. A system of m equations in n unknowns has become a single array of $m(n+1)$ numbers.

Two elements of k^n are added coordinate by coordinate, and an element t of k multiplies each coordinate separately:

$$(u+v)_j = u_j + v_j, \qquad (tu)_j = tu_j$$

Write 0 for the tuple all of whose coordinates are 0, and $u - v$ for $u + (-1)v$. Both operations pass through the product of definition 1.1.7. For $u, v \in k^n$ and $t \in k$, the i-th entry of $A(u + v)$ is

$$\sum_{j=1}^{n} a_{ij}(u_j + v_j) = \sum_{j=1}^{n} a_{ij}u_j + \sum_{j=1}^{n} a_{ij}v_j$$

which is the i-th entry of $Au + Av$, and the i-th entry of $A(tu)$ is

$$\sum_{j=1}^{n} a_{ij}(tu_j) = t\sum_{j=1}^{n} a_{ij}u_j$$

which is the i-th entry of $t(Au)$. Since two elements of k^m agreeing in every entry are equal,

$$A(u + v) = Au + Av, \qquad A(tu) = t(Au)$$

for all $u, v \in k^n$ and all $t \in k$.

Only the coefficient matrix appears in these two identities; the constant terms take no part in either. Systems whose constant terms all vanish are singled out for that reason. Setting every constant term to zero in the first two equations of the second system of example 1.1.5 gives

$$\begin{aligned} x + y + z &= 0 \\ x + 2y + 3z &= 0 \end{aligned}$$

which is solved by $x = y = z = 0$, whatever the coefficients on the left had been.

Definition 1.1.9: Homogeneous System

A system $Ax = b$ is *homogeneous* when $b = 0$, that is, when every constant term is zero. A homogeneous system is consistent: every entry of $A0$ is a sum of terms $a_{ij} \cdot 0$ and so is zero, making $x = 0$ a solution, called the *trivial solution*. For an arbitrary system $Ax = b$, the homogeneous system $Ax = 0$ with the same coefficient matrix is the system *associated* to it.

Example 1.1.10: A Homogeneous System Whose Solutions Form a Line

Work over $\mathbb{R}$ and take the homogeneous system just displayed,

$$\begin{aligned} x + y + z &= 0 \\ x + 2y + 3z &= 0 \end{aligned}$$

Subtracting the first equation from the second removes x and leaves $y + 2z = 0$, so $y = -2z$. The first equation then gives $x = -y - z = 2z - z = z$. Every solution is therefore of the form $(z, -2z, z)$ for some $z \in \mathbb{R}$, and every such triple is a solution: $z - 2z + z = 0$ and $z - 4z + 3z = 0$. The solution set is

$$N = \{\, (t, -2t, t) \,|\, t \in \mathbb{R} \,\}$$

the line through the origin in the direction of $(1, -2, 1)$.

Now restore the constant terms 1 and 2, giving the first two equations of example 1.1.5(2):

$$\begin{aligned} x + y + z &= 1 \\ x + 2y + 3z &= 2 \end{aligned}$$

The same two steps apply. Subtracting the first equation from the second leaves $y + 2z = 1$, so $y = 1 - 2z$, and the first equation gives $x = 1 - y - z = 1 - (1 - 2z) - z = z$. The solution set is

$$\{\, (t, 1 - 2t, t) \,|\, t \in \mathbb{R} \,\} = \{\, (0, 1, 0) + (t, -2t, t) \,|\, t \in \mathbb{R} \,\}$$

which is N with the fixed triple $(0, 1, 0)$ added to each of its elements. The two lines are parallel, and the constant terms decided only which of the two the solutions lie on.

That the two solution sets differ by a fixed shift is not special to those numbers. It holds for every consistent system, and it reduces the description of any solution set to two separate pieces of information: one solution, and the solution set of the associated homogeneous system.

Theorem 1.1.11: The Solution Set Is a Translate of the Homogeneous Solution Set

Let $A \in M_{m\times n}(k)$ and $b \in k^m$, and let

$$N = \{\, h \in k^n \,|\, Ah = 0 \,\}$$

be the solution set of the associated homogeneous system. If $p \in k^n$ satisfies $Ap = b$, then the solution set of $Ax = b$ is

$$\{\, p + h \,|\, h \in N \,\}$$

***Proof*:**
Suppose $Ap = b$. If $h \in N$, then, since the product splits over sums,

$$A(p + h) = Ap + Ah = b + 0 = b$$

so $p + h$ solves $Ax = b$. Every element of $\{p + h \,|\, h \in N\}$ is therefore a solution.

Conversely, let $s \in k^n$ satisfy $As = b$ and put $h = s - p$. Applying the same identity to the sum $s = p + h$ gives $As = Ap + Ah$, so $b = b + Ah$ and hence $Ah = 0$. Thus $h \in N$, and $s = p + h$ belongs to the displayed set. The two inclusions give equality of the two sets, as we sought to show.

The theorem separates the two roles the data play. The set N depends on A alone, so all systems with the same coefficient matrix share it; the constant terms enter only through the single solution p, and only by translating N somewhere else. Changing b can destroy every solution, as in example 1.1.5, but if it does not, it moves the solution set rigidly without changing its shape. In example 1.1.10 the shape was a line, and the two lines produced were parallel.

The theorem also settles how many solutions a system can have, once one more observation is added: a nonzero element of N can be scaled, and scaling produces new solutions.

Theorem 1.1.12: Two Distinct Solutions Force Infinitely Many

Let $A \in M_{m\times n}(k)$ and $b \in k^m$. If $Ax = b$ has two distinct solutions, then it has infinitely many.

***Proof*:**
Let $u \neq v$ both solve $Ax = b$. By theorem 1.1.11 applied with $p = u$, the solution v has the form $v = u + h$ for some h with $Ah = 0$; comparing coordinates gives $h = v - u$, which is nonzero because $u \neq v$. For any $t \in k$, the identity $A(th) = t(Ah)$ gives $A(th) = 0$, so th solves the associated homogeneous system too, and theorem 1.1.11 makes $u + th$ a solution of $Ax = b$.

These solutions are distinct for distinct t. Choose an index j with $h_j \neq 0$, which exists because $h \neq 0$. If $u + th = u + t'h$ then the j-th coordinates give $u_j + th_j = u_j + t'h_j$, so $(t - t')h_j = 0$ and hence $t = t'$. Since k is $\mathbb{R}$ or $\mathbb{C}$ there are infinitely many available values of t, and they produce infinitely many distinct solutions $u + th$, completing the proof.

The proof produced the solutions explicitly: they are the points $u + t(v - u)$, which trace out the whole line through u and v. A linear system cannot contain two of its solutions without containing every point on the line joining them, and that single fact rules out every finite count above one.

Corollary 1.1.13: No Solution, One Solution, or Infinitely Many

A system of m linear equations in n unknowns over k has either no solution, or exactly one solution, or infinitely many solutions.

***Proof*:**
Let S be the solution set. If S is empty, the first case holds. If S has exactly one element, the second holds. Otherwise S contains two distinct elements, and theorem 1.1.12 makes S infinite, which is the third case. The three cases are mutually exclusive and exhaust the possibilities, as we sought to show.

A linear system with exactly two solutions does not exist, and neither does one with exactly seventeen. Linearity is what forces this, and it cannot be dropped: the single equation $x^2 = 1$ has exactly two solutions in $\mathbb{R}$, and x^2 is not a linear expression in x. Figure 1.1 is the corollary for two equations in two unknowns, and the two arrangements in fig. 1.2 are the second and third cases for three equations in three unknowns. What the corollary does not give is which of the three cases a given system falls into; that is decided by computation rather than by inspection.

Exercise 1.1.1

1. Solve the system
$$\begin{aligned} 3x + 2y &= 4 \\ x - y &= 3 \end{aligned}$$
over $\mathbb{R}$, and verify that the pair found satisfies both equations.

2. Decide whether the system
$$\begin{aligned} x + 2y - z &= 3 \\ 2x + 4y - 2z &= 6 \\ x + y + z &= 2 \end{aligned}$$
over $\mathbb{R}$ is consistent, and describe its solution set.

3. Show that the system
$$\begin{aligned} x + y + 2z &= 1 \\ 2x + 3y + z &= 3 \\ 3x + 4y + 3z &= 5 \end{aligned}$$
is inconsistent.

4. Find all real solutions of the homogeneous system
$$\begin{aligned} 2x - y + z &= 0 \\ x + y - z &= 0 \end{aligned}$$
and then all real solutions of
$$\begin{aligned} 2x - y + z &= 1 \\ x + y - z &= 2 \end{aligned}$$
Exhibit the second solution set as a translate of the first, as in theorem 1.1.11.

5. Write out the systems over $\mathbb{R}$ in the unknowns x_1, x_2, x_3 whose augmented matrices are
$$\left(\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 0 & 5 \end{array}\right), \qquad \left(\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right)$$
and decide for each whether it is consistent, describing the solution set when it is.

6. For which $c \in \mathbb{R}$ does the system
$$\begin{aligned} x + 2y &= 3 \\ 2x + cy &= 7 \end{aligned}$$
have exactly one solution, and for which does it have none? Can it have infinitely many?

7. A linear system in three unknowns over $\mathbb{R}$ has both $(1, 0, 1)$ and $(3, 2, -1)$ among its solutions. Write down three further solutions of it. Then explain why no linear system in the unknowns x, y can have solution set exactly $\{(0, 0), (1, 1)\}$, and give a single equation in one unknown whose solution set in $\mathbb{R}$ has exactly two elements.

8. Let $A \in M_{m\times n}(k)$ and $b \in k^m$. Show that if $Ax = b$ is consistent and the only solution of $Ax = 0$ is $x = 0$, then $Ax = b$ has exactly one solution. Show conversely that if $Ax = b$ has exactly one solution, then $Ax = 0$ has only the trivial solution.

1.2 Row Operations and the Echelon Forms

Section 1.1 settled what a solution set can look like without producing one. This section produces it. The method rewrites a system by three operations, each of which can be undone, and drives it into a shape from which the solutions can be read off.

The procedure can be carried out in full on a system of three equations in three unknowns, and nothing in it depends on that size.

Example 1.2.1: Elimination on Three Equations in Three Unknowns
Take the system

$$\begin{aligned} x + 2y + z &= 8 \\ 2x + 4y + 4z &= 22 \\ 3x + 7y + 4z &= 29 \end{aligned}$$

and record it as an augmented matrix (definition 1.1.8):

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 2 & 4 & 4 & 22 \\ 3 & 7 & 4 & 29 \end{array}\right)$$

Row i of the array is the ith equation, the first three columns carry the coefficients of x, y and z, and the column to the right of the bar carries the right-hand sides.

The first move removes x from the second equation. Subtracting twice the first equation from the second turns $2x + 4y + 4z = 22$ into $2z = 6$, and on the array this is the operation $R_2 \mapsto R_2 - 2R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 0 & 2 & 6 \\ 3 & 7 & 4 & 29 \end{array}\right)$$

Subtracting three times the first equation from the third, which is $R_3 \mapsto R_3 - 3R_1$, removes x from that equation as well:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 0 & 2 & 6 \\ 0 & 1 & 1 & 5 \end{array}\right)$$

The second row has no y in it either, so it cannot be used to remove y from the third. Exchanging the two rows, $R_2 \leftrightarrow R_3$, puts a usable coefficient of y in the second position:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 2 & 6 \end{array}\right)$$

The three rows now have their leftmost nonzero entries in columns 1, 2 and 3. Read from the bottom up, the third row says $2z = 6$, so $z = 3$; the second says $y + z = 5$, so $y = 2$; the first says $x + 2y + z = 8$, so $x = 8 - 4 - 3 = 1$.

The same three numbers can be extracted without any substitution, by continuing with row operations. Multiplying the last row by $\frac{1}{2}$, which is $R_3 \mapsto \frac{1}{2}R_3$, makes its leftmost nonzero entry equal to 1:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 8 \\ 0 & 1 & 1 & 5 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

The operations $R_2 \mapsto R_2 - R_3$ and $R_1 \mapsto R_1 - R_3$ then empty the third column above that entry:

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

and $R_1 \mapsto R_1 - 2R_2$ empties the second column above its leftmost nonzero entry:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right)$$

The last array is the system $x = 1$, $y = 2$, $z = 3$. Substituting these three numbers into the original equations gives $1 + 4 + 3 = 8$, $2 + 8 + 12 = 22$ and $3 + 14 + 12 = 29$, so the list $(1, 2, 3)$ is a solution of the system that was started with.

Each step above replaced one system by another and the answer was read off the last of them. That is legitimate only if none of the operations changes which lists of numbers satisfy the equations. Three operations were used, and they are the only ones that will be used.

Definition 1.2.2: Elementary Row Operations

Let $M \in M_{m\times n}(k)$ have rows $R_1, \ldots, R_m$. An *elementary row operation* on M is one of the following three changes.

1. Exchange two rows: $R_i \leftrightarrow R_j$, where $i \neq j$.
2. Multiply a row by a nonzero scalar: $R_i \mapsto cR_i$, where $c \in k$ and $c \neq 0$.
3. Add a scalar multiple of one row to a different row: $R_i \mapsto R_i + cR_j$, where $c \in k$ and $j \neq i$.

The three are called a *swap*, a *scaling* and a *replacement*.

Definition 1.2.3: Row Equivalence

Two matrices $M, N \in M_{m\times n}(k)$ are *row equivalent*, written $M \sim N$, if some finite sequence of elementary row operations carries M to N.

Every elementary row operation is undone by an elementary row operation of the same kind. A swap is undone by repeating it. The scaling $R_i \mapsto cR_i$ is undone by $R_i \mapsto c^{-1}R_i$, and this is the reason the definition requires $c \neq 0$. The replacement $R_i \mapsto R_i + cR_j$ is undone by $R_i \mapsto R_i - cR_j$, and this is the reason it insists on $j \neq i$: adding a multiple of a row to itself can destroy the row, as $R_1 \mapsto R_1 - R_1$ does. The relation $\sim$ is an equivalence relation. It is reflexive because the empty sequence of operations carries M to M; it is transitive because two sequences can be performed one after the other; and it is symmetric because reversing a sequence and undoing each operation in turn carries N back to M.

Lemma 1.2.4: Row Operations Preserve the Solution Set

Let S be a system of m linear equations in n unknowns over k (definition 1.1.3) with augmented matrix M, and let N be obtained from M by one elementary row operation. Then N is the augmented matrix of a system of m equations in n unknowns whose solution set equals that of S, and the same holds when N is obtained from M by any finite sequence of elementary row operations. In particular, if $P, Q \in M_{m\times n}(k)$ are row equivalent, then the homogeneous systems $Px = 0$ and $Qx = 0$ (definition 1.1.9) have the same solution set.

***Proof*:**
Write $E_1, \dots, E_m$ for the equations of S, so that E_i is $a_{i1}x_1 + \cdots + a_{in}x_n = b_i$ and row i of M is $(a_{i1}, \dots, a_{in} | b_i)$. A list $s = (s_1, \dots, s_n)$ of scalars is a solution of S exactly when substituting s_j for x_j makes each of $E_1, \dots, E_m$ a true equality (definition 1.1.4). Take the three operations in turn.
A swap produces the system whose equations are $E_1, \dots, E_m$ listed in a different order, and a list of scalars satisfies every equation of that system exactly when it satisfies every equation of S.
A scaling $R_i \mapsto cR_i$ produces the system with E_i replaced by

$$ca_{i1}x_1 + \cdots + ca_{in}x_n = cb_i$$

and every other equation retained. If s solves S, multiplying the equality E_i through by c gives the displayed equality, so s solves the new system. If s solves the new system, multiplying the displayed equality through by c^{-1}, which exists because $c \neq 0$, returns E_i, so s solves S.
A replacement $R_i \mapsto R_i + cR_j$ with $j \neq i$ produces the system with E_i replaced by

$$(a_{i1} + ca_{j1})x_1 + \cdots + (a_{in} + ca_{jn})x_n = b_i + cb_j$$

and every other equation retained, E_j among them. If s solves S then it satisfies E_i and E_j, and adding c times the second equality to the first gives the displayed equality. If s solves the new system then it satisfies the displayed equality and E_j, and subtracting c times E_j from the displayed equality returns E_i. Here $j \neq i$ is what guarantees that E_j survives into the new system.
A finite sequence of operations applies these three cases one after the other, and each leaves the solution set where it was, so the solution set at the end of the sequence is the solution set of S.
For the last statement, the augmented matrix of $Px = 0$ is P with a column of zeros attached. Each of the three operations sends a column of zeros to a column of zeros, so performing on that augmented matrix the sequence that carries P to Q produces Q with a column of zeros attached, which is the augmented matrix of $Qx = 0$. The first part of the lemma then gives the equality of the two solution sets, as we sought to show.

Row operations therefore move a system through a family of systems all posing the same question. Which member of the family is reached is a matter of choice, and elimination chooses one in which the answer can be read off the equations.

In two unknowns the choice can be drawn. The equations $x + y = 3$ and $2x + 3y = 8$ describe two lines in the plane, and the pair $(1, 2)$ lies on both. The replacement $R_2 \mapsto R_2 - 2R_1$ turns the second equation into $y = 2$, a different line through the same point, and the further replacement $R_1 \mapsto R_1 - R_2$ turns the first into $x = 1$. Each operation replaces a line by another line through the same point, and elimination continues until the lines are parallel to the axes.

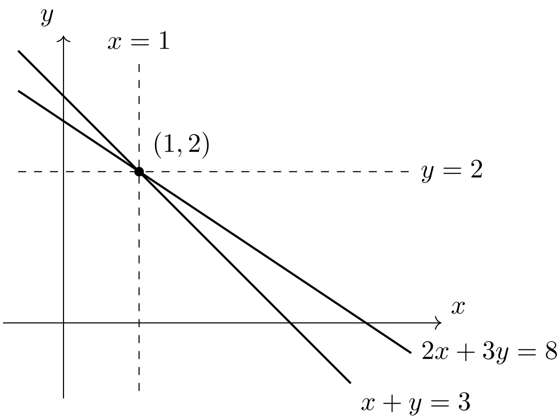


Figure 1.3: The solid lines are the two original equations and the dashed lines are what two row operations replace them by, each passing through the same point

What made the arrays in example 1.2.1 readable was the position of their leftmost nonzero entries: after the exchange, those entries sat in columns 1, 2 and 3, one step further right in each successive row. That shape has a name.

Definition 1.2.5: Echelon Form

The *leading entry* of a nonzero row is its leftmost nonzero entry. A matrix $M \in M_{m\times n}(k)$ is in *echelon form* if both of the following hold.

1. Every row all of whose entries are zero lies below every nonzero row.
2. For any two nonzero rows, the leading entry of the lower one lies in a column strictly to the right of the leading entry of the upper one.

Definition 1.2.6: Pivots and Pivot Columns

Let $M \in M_{m\times n}(k)$ be in echelon form. The leading entries of its nonzero rows are the *pivots* of M. The position of a pivot is a *pivot position*, a column containing a pivot is a *pivot column*, and a column containing no pivot is a *non-pivot column*.

A row has at most one leading entry, and condition (2) forces the pivots to descend strictly to the right, so no column holds two of them. Everything to the left of a pivot in its row is zero, and everything below a pivot in its column is zero.

$$\begin{array}{cccccc} \bullet & * & * & * & * & * \\ 0 & 0 & \bullet & * & * & * \\ 0 & 0 & 0 & 0 & \bullet & * \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}$$

Figure 1.4: A matrix in echelon form, where $\bullet$ marks a pivot and $*$ an entry of any value whatever, with every entry below the staircase equal to zero

The array reached after the exchange in example 1.2.1 is in echelon form, with pivots 1, 1 and 2 in positions $(1, 1)$, $(2, 2)$ and $(3, 3)$; the fourth column is a non-pivot column. The array reached at the end of that example satisfies two further conditions, and it is those that let the solution be read off without any arithmetic at all.

Definition 1.2.7: Reduced Echelon Form

A matrix $M \in M_{m\times n}(k)$ is in *reduced echelon form* if it is in echelon form, every pivot of M equals 1, and every pivot is the only nonzero entry in its column.

The final array of example 1.2.1 is in reduced echelon form: its three pivots are the entries 1 in positions $(1, 1)$, $(2, 2)$ and $(3, 3)$, and each column that contains a pivot contains nothing else. The corresponding equations are $x = 1$, $y = 2$ and $z = 3$, which is why nothing remained to be computed. That example reached both shapes, and it did so for reasons that had nothing to do with its entries.

Theorem 1.2.8: Existence of the Echelon Forms

Every matrix $M \in M_{m\times n}(k)$ is row equivalent to a matrix in echelon form. Moreover, a matrix U in echelon form is carried to a matrix in reduced echelon form by a further sequence of elementary row operations, none of which changes the columns in which the leading entries lie; consequently every $M \in M_{m\times n}(k)$ is row equivalent to a matrix in reduced echelon form.

***Proof*:**
For the first statement, argue by induction on the number m of rows. A matrix with one row is in echelon form already: it has at most one nonzero row, so condition (1) of definition 1.2.5 holds and condition (2) is vacuous.
Suppose $m > 1$ and that every matrix with $m - 1$ rows and any number of columns is row equivalent to a matrix in echelon form. If every entry of M is zero then M is in echelon form. Otherwise let j be the smallest index of a column of M containing a nonzero entry, so that columns 1 through $j - 1$ are entirely zero, and choose a row index i with $m_{ij} \neq 0$. The swap $R_1 \leftrightarrow R_i$, omitted when $i = 1$, puts a nonzero scalar a in position $(1, j)$ and moves no nonzero entry into the columns left of j, which are still zero. For each i' with $2 \leq i' \leq m$ write $c_{i'}$ for the entry standing in position (i', j) after that swap; the replacement $R_{i'} \mapsto R_{i'} - (c_{i'}/a)R_1$ makes the entry in position (i', j) equal to zero, and it alters no entry in columns 1 through $j - 1$, since row 1 is zero in those columns. After these $m - 1$ replacements every entry in columns 1 through j is zero except the entry a in position $(1, j)$.
Let M' be the matrix with $m - 1$ rows formed by rows $2, \ldots, m$ of the matrix just produced.

By the inductive hypothesis some finite sequence of elementary row operations carries M' to a matrix U' in echelon form. Every operation in that sequence involves only rows of M', so the same operation performed on rows $2, \ldots, m$ of the full matrix is an elementary row operation on the full matrix; performing the sequence leaves row 1 untouched and replaces rows $2, \ldots, m$ by the rows of U'. Every row of M' is zero in columns 1 through j, and adding multiples of such rows to one another, scaling them and exchanging them cannot produce a nonzero entry in those columns, so every nonzero row of U' has its leading entry in a column strictly to the right of j. The zero rows of U' lie below its nonzero rows, and row 1 of the full matrix is nonzero, so condition (1) holds for the full matrix; the leading entry of row 1 is in column j and those of the rows below it are strictly to the right of column j and increase strictly among themselves, so condition (2) holds as well. The full matrix is therefore in echelon form and is row equivalent to M, which completes the induction.
For the second statement, let U be in echelon form and let its nonzero rows be rows $1, \ldots, r$, with leading entries in columns $j_1 < j_2 < \cdots < j_r$. For each p the scaling $R_p \mapsto u_{pj_p}^{-1} R_p$, legitimate because $u_{pj_p} \neq 0$, makes the leading entry of row p equal to 1 and changes no entry from zero to nonzero, so the leading entries stay in columns $j_1, \ldots, j_r$. Now run through $p = r, r-1, \ldots, 2$, and for each p and each $q < p$ apply the replacement $R_q \mapsto R_q - u_{qj_p} R_p$, where u_{qj_p} is the entry of row q in column j_p at that moment. This makes the entry of row q in column j_p equal to zero. Row p is zero in every column to the left of j_p, so the replacement changes row q only in columns j_p and to the right; since $j_q < j_p$, the leading entry of row q is untouched and no nonzero entry appears to the left of it. Row p is also zero in the columns $j_{p+1}, \ldots, j_r$, because those columns were emptied above row p' for each $p' > p$ in the earlier passes, so the replacement does not disturb the zeros already created in row q. When the passes are finished each column j_p contains the entry 1 in row p and zeros elsewhere, the rows below row r are still zero, and the matrix is in reduced echelon form with its leading entries in the columns $j_1, \ldots, j_r$ where they started. Applying this to an echelon form of M, which exists by the first statement, produces a matrix in reduced echelon form row equivalent to M, completing the proof.

The proof is the procedure. Its first phase clears each column below a chosen nonzero entry and works downward, which is what was done in example 1.2.1 before the solution was read from the bottom row upward; its second phase scales each pivot to 1 and clears each pivot column upward, which is what was done after. All arithmetic in Part I is exact: every scalar is a real or a complex number and every division by a pivot is carried out exactly.

The next matrix returns in every chapter of this part. Its reduction is recorded by naming the operations rather than displaying every one of them, since the procedure has now been carried out in full once.

Example 1.2.9: The Running Matrix A

Let

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 1 & 2 & 5 \\ 3 & 4 & 11 \end{pmatrix}$$

so that A has $m = 4$ rows and $n = 3$ columns. The entry in position $(1, 1)$ is nonzero, so no swap is needed, and the three replacements $R_2 \mapsto R_2 - 2R_1$, $R_3 \mapsto R_3 - R_1$ and $R_4 \mapsto R_4 - 3R_1$ empty the first column below row 1:

$$\begin{pmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix}$$

The replacements $R_3 \mapsto R_3 - R_2$ and $R_4 \mapsto R_4 - R_2$ then empty the second column below row 2:

$$\begin{pmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

This matrix is in echelon form. Its pivots are the entries 1 in positions $(1, 1)$ and $(2, 2)$, so the first two columns are pivot columns and the third is a non-pivot column, and the last two rows are zero. Both pivots already equal 1, so the only step left is to empty the second column above its pivot, by $R_1 \mapsto R_1 - R_2$:

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

which is in reduced echelon form.

The third column of A is the first column plus twice the second, since $3 = 1 + 2 \cdot 1$, $8 = 2 + 2 \cdot 3$, $5 = 1 + 2 \cdot 2$ and $11 = 3 + 2 \cdot 4$. That relation is a statement about a homogeneous system in disguise. Writing $c_1, \ldots, c_n$ for the columns of A and grouping the defining formula $(Ax)_i = a_{i1}x_1 + \cdots + a_{in}x_n$ (definition 1.1.7) by columns instead of by rows gives

$$Ax = x_1c_1 + x_2c_2 + \cdots + x_nc_n$$

so a list x solves $Ax = 0$ exactly when that combination of the columns vanishes. Here $c_1 + 2c_2 - c_3 = 0$, so the list $x = (1, 2, -1)$ solves the homogeneous system $Ax = 0$ (definition 1.1.9). The same list solves the system read off the reduced form, whose two equations are $x_1 + x_3 = 0$ and $x_2 + 2x_3 = 0$. By lemma 1.2.4 the two homogeneous systems have identical solution sets, so a relation among the columns of A is visible in the reduced form and conversely. Elimination changes the columns of a matrix, but a list x with $Ax = 0$ is exactly a list with $Ex = 0$ for the reduced form E.

Both examples reached a reduced echelon form, but the operations used to get there were chosen and not forced. A row could have been scaled at any point, the replacements could have been performed in another order, and in example 1.2.1 the exchange could have been made earlier. Different choices can produce different echelon forms of the same matrix, since scaling a row of an echelon matrix leaves it in echelon form while changing its entries. The reduced form admits no such latitude.

Theorem 1.2.10: Uniqueness of the Reduced Echelon Form

Every matrix $M \in M_{m \times n}(k)$ is row equivalent to exactly one matrix in reduced echelon form. Moreover, every matrix in echelon form that is row equivalent to M has its leading entries in the same columns, namely in the pivot columns of that reduced echelon form.

***Proof*:**
Existence is theorem 1.2.8. For uniqueness, suppose B and C are in reduced echelon form and both are row equivalent to M. Since $\sim$ is an equivalence relation, $B \sim C$. The claim is that $B = C$, and it is proved by induction on the number n of columns.
Let $n = 1$. If B is the zero matrix then so is C, because every elementary row operation carries the zero matrix to itself, and the same reasoning applied to C shows that C is the zero matrix only if B is. So suppose neither is the zero matrix. Then B has a nonzero row, whose leading entry must lie in column 1; there is only one such row, since the leading entries of distinct nonzero rows lie in distinct columns, and it is row 1, since zero rows lie below nonzero ones. That leading entry is a pivot, so it equals 1 and is the only nonzero entry of column 1. Hence B has entry 1 in position $(1, 1)$ and zeros elsewhere, and the same argument applies to C, so $B = C$.
Let $n \geq 2$ and suppose the claim holds for matrices with $n - 1$ columns. Write B' and C' for the matrices obtained from B and C by deleting the last column. An elementary row operation acts on each column separately, so the sequence of operations carrying B to C carries B' to C', and $B' \sim C'$. Deleting the last column of a matrix in reduced echelon form leaves one in reduced echelon form: the only row that can become zero is a row whose leading entry lies in column n, and by condition (2) of definition 1.2.5 such a row is the lowest nonzero row, so after deletion the zero rows are still exactly the bottom rows, while the remaining leading entries keep their columns, keep the value 1, and remain alone in their columns. By the inductive hypothesis $B' = C'$, so B and C agree except possibly in column n.
Suppose $B \neq C$, so that $b_{in} \neq c_{in}$ for some row index i. By lemma 1.2.4 the systems $Bx = 0$ and $Cx = 0$ have the same solutions. Let $x = (x_1, \dots, x_n)$ be any solution of $Bx = 0$. It also solves $Cx = 0$, and subtracting the ith equation of the second system from the ith equation of the first gives

$$(b_{i1} - c_{i1})x_1 + \cdots + (b_{in} - c_{in})x_n = 0$$

All the coefficients but the last vanish because $B' = C'$, so $(b_{in} - c_{in})x_n = 0$ and therefore $x_n = 0$. Every solution of $Bx = 0$ thus has last entry zero.
This forces column n to be a pivot column of B. Suppose it were not. Let the nonzero rows of B be rows $1, \dots, r$ with pivots in columns $j_1 < \cdots < j_r$, all of them less than n, and set $x_n = 1$, set $x_{j_p} = -b_{pn}$ for each p with $1 \leq p \leq r$, and set $x_k = 0$ for every remaining index k. For a row p with $1 \leq p \leq r$, the corresponding equation of $Bx = 0$ evaluates on this list to $b_{pj_p}x_{j_p} + b_{pn}x_n$, because the coefficient in any other pivot column $j_{p'}$ with $p' \neq p$ is zero in reduced echelon form and the remaining entries of the list are zero; since $b_{pj_p} = 1$ this equals $-b_{pn} + b_{pn} = 0$. The rows below row r are zero and give the equation $0 = 0$. So the list is a solution of $Bx = 0$ with last entry 1, which contradicts the previous paragraph. Column n is therefore a pivot column of B, and the same argument applied to C shows it is a pivot column of C.
Let s be the number of pivot columns of B among the first $n-1$ columns. The leading entries of B' are exactly the leading entries of B lying in columns 1 through $n-1$, so s is also the number of pivot columns of C among the first $n-1$ columns, because $B' = C'$. In an echelon matrix the nonzero rows are the top rows and each carries exactly one pivot, so the pivot in the rightmost pivot column sits in the bottom nonzero row. Column n is the rightmost pivot column of B, and the remaining s pivots of B lie in the first $n - 1$ columns, so B has $s + 1$ nonzero rows and its pivot in column n sits in row $s + 1$. The same count applies to C. Since B and C are in reduced echelon form, column n of each consists of the entry 1 in row $s + 1$ and zeros elsewhere, so the two columns agree, contradicting $b_{in} \neq c_{in}$. Hence $B = C$.
For the last statement, let U be in echelon form with $U \sim M$. By theorem 1.2.8 a sequence of

row operations carries U to a matrix V in reduced echelon form whose leading entries lie in the same columns as those of U. Then $V \sim M$, so V is the unique matrix in reduced echelon form row equivalent to M, and the leading entries of U lie in its pivot columns, as we sought to show.

The reduced echelon form of a matrix is therefore an attribute of the matrix and not of the operations chosen to reach it, and it is the only shape reached by elimination with that property. Two consequences follow. First, the number of pivots and the columns they occupy depend only on M, since every echelon form of M has its leading entries in the pivot columns of the one reduced form. Second, two matrices are row equivalent exactly when they have the same reduced echelon form: if $M \sim N$ then the reduced echelon form of N is row equivalent to M and so equals that of M by the theorem, and conversely a common reduced echelon form links M to N through it.

Exercise 1.2.1

1. Bring the augmented matrix
$$\left(\begin{array}{ccc|c} 2 & 4 & -2 & 2 \\ 1 & 3 & 1 & 6 \\ 3 & 5 & -7 & -4 \end{array}\right)$$
to reduced echelon form, naming each operation used, and write down the solution of the system it records.

2. Find the reduced echelon form of
$$\begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 9 \\ -1 & -2 & 3 & 0 \end{pmatrix}$$
and list its pivot columns and its non-pivot columns.

3. Let B be the matrix obtained from the running matrix A of example 1.2.9 by changing the entry in position $(4,3)$ from 11 to 12. Find an echelon form of B and then its reduced echelon form, and compare the pivot columns with those of A.

4. For each of the four matrices
$$\begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 & 3 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 2 & 0 & 1 \\ 0 & 0 & 3 \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 4 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$
decide whether it is in echelon form and whether it is in reduced echelon form, and in each failing case name the condition of definition 1.2.5 or definition 1.2.7 that fails.

5. Find the reduced echelon form of
$$N = \begin{pmatrix} 1 & 2 & 5 \\ 0 & 1 & 3 \\ 2 & 3 & 7 \end{pmatrix}$$
then use it to produce a list $x \neq (0,0,0)$ with $Nx = 0$, verify that list against N itself, and say which result guarantees that the verification had to succeed.

6. Show that a matrix in $M_{2\times 2}(k)$ in reduced echelon form is one of
$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 1 & a \\ 0 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$
where $a \in k$ is arbitrary, and that no two matrices in this list are row equivalent unless they are equal.

7. Lemma 1.2.4 says that row equivalent augmented matrices record systems with the same solution set. Show that the converse fails, by exhibiting two systems of two equations in two unknowns that have the same solution set and whose augmented matrices are not row equivalent.

1.3 Rank, Consistency, and the Structure of the Solution Set

Elimination drives a system to echelon form, and the shape it lands in settles everything about the solutions. Two counts read that shape: the number of pivots the coefficients produce, and the number the augmented array produces. Together they decide whether a solution exists, and the first of them decides how many. When there are many, the same shape produces a formula listing all of them at once.

The two counts are easiest to see when the coefficients are held fixed and only the right-hand side moves.

Example 1.3.1: One Coefficient Matrix, Two Right-Hand Sides

1. Consider the system of $m = 3$ equations in $n = 4$ unknowns
$$\begin{array}{rcrcrcrcl} x_1 & + & 2x_2 & + & x_3 & + & x_4 & = & 1 \\ 2x_1 & + & 4x_2 & + & 3x_3 & + & 4x_4 & = & 4 \\ -x_1 & - & 2x_2 & & & + & x_4 & = & 1 \end{array}$$
The third equation has no x_3 term, so its coefficient of x_3 is 0. Write B for its coefficient matrix and c for its right-hand side. The augmented matrix (definition 1.1.8) is
$$(B|c) = \left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 1 \\ 2 & 4 & 3 & 4 & 4 \\ -1 & -2 & 0 & 1 & 1 \end{array}\right)$$
where the bar separates the coefficients from the right-hand side. The operations $R_2 \mapsto R_2 - 2R_1$ and $R_3 \mapsto R_3 + R_1$ clear the first column below its top entry:
$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 & 2 \end{array}\right)$$
and $R_3 \mapsto R_3 - R_2$ finishes the reduction:
$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

This matrix is in echelon form (definition 1.2.5), and its pivots (definition 1.2.6) sit in columns 1 and 3: two in all. The first four columns record what the same operations did to B on its own, and they are in echelon form too, with pivots in the same two columns. The coefficients contribute two pivots, and so does the whole augmented array.

2. Now change the last entry on the right from 1 to 0, leaving the coefficients unchanged, and call the new right-hand side c':

$$(B|c') = \left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 1 \\ 2 & 4 & 3 & 4 & 4 \\ -1 & -2 & 0 & 1 & 0 \end{array}\right)$$

The operations $R_2 \mapsto R_2 - 2R_1$ and $R_3 \mapsto R_3 + R_1$ act on the coefficients exactly as before, and only the last column comes out differently:

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 & 1 \end{array}\right)$$

Then $R_3 \mapsto R_3 - R_2$ gives

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & -1 \end{array}\right)$$

The last row records the equation $0 = -1$, which no choice of x_1, x_2, x_3, x_4 satisfies, so this system has no solution. In terms of pivots, the coefficients still contribute two, in columns 1 and 3, but the augmented array now has a third pivot, in its last column.

Both parts of the computation turned on a count of pivots, and the count deserves a name.

Definition 1.3.2: Rank

Let $A \in M_{m\times n}(k)$. The *rank* of A, written $\operatorname{rank} A$, is the number of pivots in the reduced echelon form of A.

The definition names the reduced echelon form because that form is unique (theorem 1.2.10), so the count belongs to A itself and not to the particular sequence of row operations that produced it. An echelon form that has not been reduced serves just as well, which is what makes the definition usable in practice. The second statement of theorem 1.2.10 says that every matrix in echelon form row equivalent to A carries its leading entries in the pivot columns of the reduced echelon form. All of them therefore have the same number of pivots, and $\operatorname{rank} A$ can be counted off whichever echelon form the elimination happened to produce. In example 1.3.1 this gives $\operatorname{rank} B = 2$, $\operatorname{rank}(B|c) = 2$ and $\operatorname{rank}(B|c') = 3$.

Two bounds come for free. Each row of an echelon form carries at most one pivot, and two different rows carry their pivots in two different columns, so $\operatorname{rank} A \le m$ and $\operatorname{rank} A \le n$. Reading a rank off costs nothing beyond a computation already performed: the echelon form reached in example 1.2.9 has two pivots and two zero rows below them, so the running matrix threaded through this part of the book has rank 2, inside both bounds since it has four rows and three columns.

In example 1.3.1 the second system failed for a visible reason. A row of zeros appeared among the coefficients while the entry to the right of the bar stayed nonzero, and the equation that row records cannot be satisfied. That is the only way elimination can report failure, and it is detected by comparing the two pivot counts.

Theorem 1.3.3: Consistency Criterion

Let $A \in M_{m\times n}(k)$ and let $b \in k^m$ be a column of m entries. The system $Ax = b$ is consistent (definition 1.1.6) if and only if

$$\operatorname{rank}(A|b) = \operatorname{rank}A$$

The alternative is $\operatorname{rank}(A|b) = \operatorname{rank}A + 1$, which happens exactly when an echelon form of the augmented matrix has a pivot in its last column.

***Proof*:**
Let E be the reduced echelon form of $(A|b)$, a matrix with m rows and $n+1$ columns. A row operation acts on whole rows and treats every column the same way, so the sequence of operations carrying $(A|b)$ to E carries A to the matrix E' consisting of the first n columns of E.

Suppose first that E has a pivot in column $n+1$. Going down an echelon form the pivots move strictly to the right, so a pivot in the rightmost column can only be the leading entry of the last nonzero row. That row is therefore $(0,\dots,0|1)$ and records the equation $0 = 1$. By lemma 1.2.4 the system with augmented matrix E has the same solutions as $Ax = b$, and no choice of $x_1,\dots,x_n$ satisfies $0 = 1$, so $Ax = b$ is inconsistent. Deleting the last column turns that row into a row of zeros, and since every row below it was already zero, E' is still in reduced echelon form; its pivots are those of E apart from the one in column $n+1$. So E' is the reduced echelon form of A, and $\operatorname{rank}A = \operatorname{rank}(A|b) - 1$.

Suppose instead that E has no pivot in column $n+1$. A row of E whose only nonzero entry lay in column $n+1$ would have its leading entry there, so no such row occurs, and E and E' have the same nonzero rows and the same pivots. Then E' is in reduced echelon form, hence is the reduced echelon form of A, and $\operatorname{rank}A = \operatorname{rank}(A|b)$. Write r for this common value, $j_1 < \cdots < j_r$ for the pivot columns, e_{ij} for the entry of E in row i and column j, and $d_1,\dots,d_m$ for the entries of the last column of E. The r pivots occupy the first r rows, one each, so rows $r+1$ through m of E are entirely zero and in particular $d_i = 0$ for $i > r$. Assign $x_j = 0$ for every j that is not a pivot column and $x_{j_i} = d_i$ for $1 \le i \le r$. Row i with $i \le r$ has the entry 1 in column j_i and zeros in the other pivot columns, so it records the equation $x_{j_i} + \sum_j e_{ij}x_j = d_i$ with j ranging over the non-pivot columns, and the assignment satisfies it; the remaining rows record $0 = 0$. So the assignment solves the system with augmented matrix E, and by lemma 1.2.4 it solves $Ax = b$.

The two cases are exhaustive. In the first the system is inconsistent and $\operatorname{rank}(A|b) = \operatorname{rank}A + 1$; in the second it is consistent and $\operatorname{rank}(A|b) = \operatorname{rank}A$. This establishes both asserted equivalences, as we sought to show.

The criterion converts a question about the existence of a solution into a comparison of two integers, each of which elimination computes on the way to anything else. It also says that the failure seen in example 1.3.1 is the only failure available: a system is unsolvable exactly when elimination manufactures a contradictory row, never for a subtler reason. Section 2.3 will restate the same condition as the requirement that b be built out of the columns of A.

Consistency settled, the remaining question is how many solutions there are, and the first system of example 1.3.1 is far enough along to answer it.

Example 1.3.4: A Consistent System with Two Free Unknowns
The first system of example 1.3.1 was carried to the echelon form

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 1 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

One more operation, $R_1 \mapsto R_1 - R_2$, clears the entry above the second pivot and produces the reduced echelon form (definition 1.2.7)

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & -1 & -1 \\ 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

The two nonzero rows record the equations

$$x_1 + 2x_2 - x_4 = -1, \qquad x_3 + 2x_4 = 2$$

and the third row records $0 = 0$, which constrains nothing. The pivots are in columns 1 and 3, and the two equations solve for x_1 and x_3 once x_2 and x_4 are known. Nothing constrains x_2 and x_4, so let $x_2 = s$ and $x_4 = t$ with s and t arbitrary real numbers. Then $x_1 = -1 - 2s + t$ and $x_3 = 2 - 2t$, so the solutions are exactly the columns

$$x = \begin{pmatrix} -1 - 2s + t \\ s \\ 2 - 2t \\ t \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 2 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ -2 \\ 1 \end{pmatrix}$$

as s and t range over $\mathbb{R}$.

Each piece can be checked against the original equations. Setting $s = t = 0$ gives $(-1, 0, 2, 0)^\top$, and the three left-hand sides evaluate to $-1+0+2+0 = 1$, $-2+0+6+0 = 4$ and $1-0+0+0 = 1$, matching the right-hand side c. Substituting $(-2, 1, 0, 0)^\top$ gives $-2+2+0+0 = 0$, $-4+4+0+0 = 0$ and $2-2+0+0 = 0$, and substituting $(1, 0, -2, 1)^\top$ gives $1+0-2+1 = 0$, $2+0-6+4 = 0$ and $-1-0+0+1 = 0$. The last two columns solve the homogeneous system $Bx = 0$ (definition 1.1.9).

The shape of the answer is the one theorem 1.1.11 predicted: one particular solution of the system, plus every solution of the corresponding homogeneous system. What the reduced echelon form adds is an explicit list of the homogeneous solutions rather than an abstract description of them, and the list has one entry for each unknown that the pivots left alone.

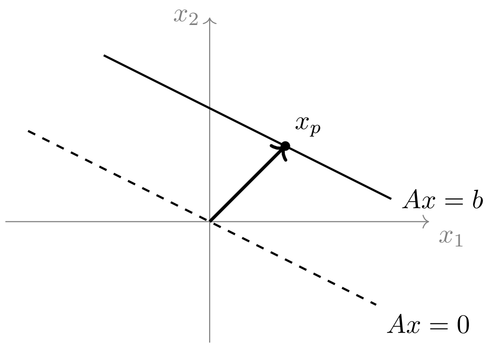


Figure 1.5: The solution set of a consistent system is the solution set of $Ax = 0$ shifted by any one particular solution x_p; here the single equation is $x_1 + 2x_2 = 3$ and its homogeneous version is $x_1 + 2x_2 = 0$

Definition 1.3.5: Pivot Variables and Free Variables

Let $A \in M_{m\times n}(k)$ and let R be the reduced echelon form of A, with pivots in columns $j_1 < \cdots < j_r$, so that $r = \operatorname{rank} A$. In a system $Ax = b$ with coefficient matrix A, the unknowns $x_{j_1}, \ldots, x_{j_r}$ are the *pivot variables* and the remaining $n - r$ unknowns are the *free variables*.

In example 1.3.4 the pivot variables are x_1 and x_3 and the free variables are x_2 and x_4, and the count $n - r = 4 - 2 = 2$ is the number of parameters the answer needed. The general picture is the same, because a reduced echelon form of a consistent augmented array always looks like

$$\left(\begin{array}{ccccc|c} 1 & * & 0 & * & 0 & d_1 \\ 0 & 0 & 1 & * & 0 & d_2 \\ 0 & 0 & 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

in this instance with $m = 4$, $n = 5$ and $r = 3$, where the entries marked $*$ may be anything and the pivots sit in columns 1, 3 and 5. The three nonzero rows give x_1, x_3 and x_5 in terms of x_2 and x_4, and the zero row says nothing. Choosing the two free variables therefore determines the three pivot variables, and choosing them differently gives a different solution, since x_2 and x_4 are themselves entries of the answer.

Theorem 1.3.6: Rank Determines the Number of Solutions

Let $A \in M_{m\times n}(k)$, let $b \in k^m$, and suppose $Ax = b$ is consistent. Put $r = \operatorname{rank} A$. Then:

1. Every assignment of values in k to the $n - r$ free variables determines the r pivot variables uniquely, and every solution of $Ax = b$ arises from exactly one such assignment.

2. If $r = n$, the system has exactly one solution.

3. If $r < n$, the system has infinitely many solutions.

***Proof*:**
Let E be the reduced echelon form of $(A|b)$ and E' the matrix of its first n columns; a row operation treats every column the same way, so the operations carrying $(A|b)$ to E carry A to E'. Since the system is consistent, theorem 1.3.3 says E has no pivot in column $n+1$. A row of E whose only nonzero entry lay in column $n+1$ would have its leading entry there, so no such row occurs, and E and E' have the same nonzero rows and the same pivots. Hence E' is in reduced echelon form, and being row equivalent to A it is the reduced echelon form of A. So E has exactly r pivots, in columns $j_1 < \cdots < j_r$ with $j_r \leq n$, one in each of its first r rows, and its rows $r+1$ through m are zero. Write F for the set of the $n-r$ columns among $1, \ldots, n$ that carry no pivot, write e_{ij} for the entry of E in row i and column j, and write $d_1, \ldots, d_m$ for its last column. Because E is reduced, row i with $i \leq r$ has the entry 1 in column j_i and 0 in the other pivot columns, so it records

$$x_{j_i} + \sum_{j \in F} e_{ij} x_j = d_i$$

and rows $r+1$ through m record $0 = 0$. By lemma 1.2.4 the solutions of $Ax = b$ are exactly the solutions of this system.

1. Fix values $x_j = c_j$ in k for every $j \in F$. For each $i \leq r$, the equation $x_{j_i} + \sum_{j \in F} e_{ij} x_j = d_i$ holds if and only if $x_{j_i} = d_i - \sum_{j \in F} e_{ij} c_j$, so the pivot variables are determined, and the column so built satisfies every row of E and hence solves $Ax = b$. Conversely, let x be any solution and let c_j be its entry in position j for $j \in F$. Then x satisfies each of those r equations, so its pivot entries are the ones the assignment $(c_j)_{j \in F}$ produces, and x is the column built from that assignment. No other assignment produces x, because the assignment is read off from the entries of x in the positions of F.
2. If $r = n$ then F is empty, so there is exactly one assignment, the empty one, and by part (1) exactly one solution.
3. If $r < n$ then F is nonempty. Two assignments differing in some position $j \in F$ produce solutions differing in entry j, so distinct assignments give distinct solutions. Both $\mathbb{R}$ and $\mathbb{C}$ are infinite, so there are infinitely many assignments and hence infinitely many solutions, completing the proof.

Corollary 1.1.13 said that a system has no solution, exactly one, or infinitely many. Two integers now decide which:

- $\operatorname{rank}(A|b) = \operatorname{rank} A + 1$: no solutions.
- $\operatorname{rank}(A|b) = \operatorname{rank} A = n$: exactly one solution.
- $\operatorname{rank}(A|b) = \operatorname{rank} A < n$: infinitely many solutions, parametrized by $n - \operatorname{rank} A$ free variables.

Both integers are produced by a single run of elimination, so the trichotomy is decided by the same computation that solves the system. The words *overdetermined* and *underdetermined* describe the shape of the array rather than the outcome: a system is overdetermined when it has more equations than unknowns and underdetermined when it has fewer. Neither word settles the trichotomy on its own. An overdetermined system is the one for which inconsistency is the ordinary case, but it may perfectly well be consistent; an underdetermined system may be inconsistent, though when it is consistent it has infinitely many solutions, since $n - \operatorname{rank} A \geq n - m > 0$ by theorem 1.3.6. Which case holds is decided by the two ranks and not by m and n.

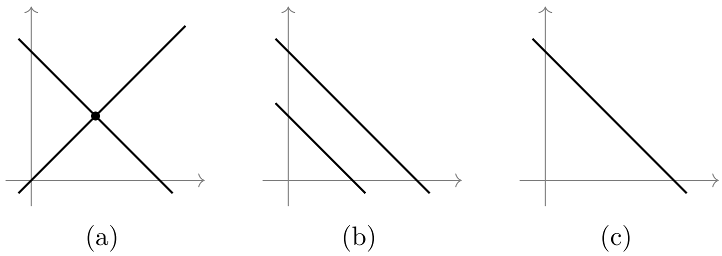


Figure 1.6: Two equations in two unknowns, one line for each: (a) $x_1 + x_2 = 2$ and $x_1 - x_2 = 0$, where $\operatorname{rank} A = \operatorname{rank}(A|b) = 2 = n$ and there is one solution; (b) $x_1 + x_2 = 2$ and $x_1 + x_2 = 1$, where $\operatorname{rank} A = 1$ and $\operatorname{rank}(A|b) = 2$ and there is none; (c) $x_1 + x_2 = 2$ and $2x_1 + 2x_2 = 4$, one line described twice, where $\operatorname{rank} A = \operatorname{rank}(A|b) = 1 < 2 = n$ and there are infinitely many

Two special cases are worth recording separately, since neither needs any computation beyond the counts. The first concerns a system with as many equations as unknowns.

Corollary 1.3.7: Square Systems of Full Rank

Let $A \in M_n(k)$. The system $Ax = b$ has exactly one solution for every $b \in k^n$ if and only if $\operatorname{rank} A = n$.

***Proof*:**
Suppose $\operatorname{rank} A = n$ and let R be the reduced echelon form of A. Then R has n pivots distributed among n columns and n rows, so every column and every row carries exactly one, and the pivot of row i is in column i. Each pivot equals 1 and the other entries of a pivot column are 0, so R has 1 in every diagonal position and 0 everywhere else. Given b, apply to $(A|b)$ the operations that carry A to R; they produce $(R|d)$ for some column d, which is in reduced echelon form with pivots in columns 1 through n and none in the last column. Hence $\operatorname{rank}(A|b) = n = \operatorname{rank} A$, so $Ax = b$ is consistent by theorem 1.3.3, and since $\operatorname{rank} A = n$ it has exactly one solution by theorem 1.3.6(2).

Suppose instead $\operatorname{rank} A = r < n$, and take $b = 0$. The system $Ax = 0$ is consistent, since the column of zeros solves it, so theorem 1.3.6(3) gives infinitely many solutions. There is therefore a b for which the solution is not unique, completing the proof.

Full rank for a square coefficient matrix is the condition under which elimination never stalls: it reaches a pivot in every column, leaves no free variables, and returns a single answer whatever the right-hand side. The second special case runs the other way and asks nothing of the coefficients at all.

Corollary 1.3.8: More Unknowns than Equations

Let $A \in M_{m\times n}(k)$ with $n > m$. Then the homogeneous system $Ax = 0$ has infinitely many solutions, and in particular it has a solution $x \neq 0$.

***Proof*:**
The column of zeros solves $Ax = 0$, so the system is consistent. Every echelon form of A has at most one pivot per row, so $\operatorname{rank} A \leq m < n$, and theorem 1.3.6(3) gives infinitely many solutions. A set with infinitely many elements has an element other than the column of zeros, as we sought to show.

The corollary settles a question about every system of its shape without looking at a single coefficient. The system of example 1.3.1 illustrates it: there $m = 3$ and $n = 4$, and the homogeneous system $Bx = 0$ has the solutions $s(-2, 1, 0, 0)^{\top} + t(1, 0, -2, 1)^{\top}$ computed in example 1.3.4, one for each pair (s, t) of real numbers.

Exercise 1.3.1

1. Compute the rank of each of

$$P = \begin{pmatrix} 1 & -1 & 2 \\ 2 & 1 & 1 \\ 3 & 0 & 3 \end{pmatrix}, \qquad Q = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 4 & 6 & 8 \end{pmatrix}$$

   by row reducing each to echelon form, and name the row operations used.

2. Decide whether the system

$$\begin{array}{rcrcrcrcl} x_1 & + & x_2 & - & x_3 & + & 2x_4 & = & 1 \\ 2x_1 & + & 3x_2 & + & x_3 & + & x_4 & = & 4 \\ x_1 & + & 2x_2 & + & 2x_3 & - & x_4 & = & 3 \end{array}$$

   is consistent. If it is, write down all of its solutions, and identify the pivot variables and the free variables.

3. For which real numbers t is the system

$$\begin{array}{rcrcrcl} x_1 & + & 2x_2 & + & 3x_3 & = & 1 \\ 2x_1 & + & 5x_2 & + & 4x_3 & = & 3 \\ x_1 & + & x_2 & + & 5x_3 & = & t \end{array}$$

   consistent? For each such t, write down all of the solutions.

4. Find all solutions of the homogeneous system whose coefficient matrix is

$$D = \begin{pmatrix} 1 & -2 & 1 & 3 \\ 2 & -4 & 3 & 1 \end{pmatrix}$$

   and check that the number of parameters in the answer is $n - \operatorname{rank} D$.

5. Let $A \in M_{m\times n}(k)$ and $b \in k^m$. Show that $\operatorname{rank}(A|b)$ is either $\operatorname{rank} A$ or $\operatorname{rank} A + 1$, and that no other value can occur.

6. A system of 4 equations in 3 unknowns has a coefficient matrix of rank 3. Which of the three cases of corollary 1.1.13 can occur, and what does the answer depend on?
7. Let $A \in M_{m\times n}(k)$ with $m < n$. Show that for no $b \in k^m$ does $Ax = b$ have exactly one solution.
8. Let $A \in M_{m\times n}(k)$ and suppose $Ax = b$ is consistent for every $b \in k^m$. Show that $\operatorname{rank} A = m$. (Use that every row operation can be undone by a row operation, so that row equivalence works in both directions.)

1.4 Matrix Algebra, Invertibility, and Elementary Matrices

Elimination has so far been a procedure performed on an array. This section makes it an algebraic operation. Matrices acquire a product, each of the three row operations becomes multiplication on the left by one fixed matrix, and a whole sequence of operations collapses into a single matrix. The question of whether a square system can be solved for every right hand side then becomes the question of whether one matrix has a multiplicative inverse.

Throughout, k stands for $\mathbb{R}$ or for $\mathbb{C}$, and every statement below holds over both.

Two changes of variable performed one after the other are what the product records. Suppose y_1 and y_2 are given in terms of x_1, x_2, x_3 by

$$y_1 = x_1 + 2x_2 - x_3, \qquad y_2 = 3x_1 - x_2 + 4x_3$$

and that z_1 and z_2 are given in terms of y_1 and y_2 by

$$z_1 = 2y_1 + y_2, \qquad z_2 = y_1 - y_2$$

Substituting the first pair into the second and collecting the coefficient of each x_j gives

$$\begin{aligned} z_1 &= 2(x_1 + 2x_2 - x_3) + (3x_1 - x_2 + 4x_3) = 5x_1 + 3x_2 + 2x_3 \\ z_2 &= (x_1 + 2x_2 - x_3) - (3x_1 - x_2 + 4x_3) = -2x_1 + 3x_2 - 5x_3 \end{aligned}$$

The three coefficient arrays involved are

$$C = \begin{pmatrix} 2 & 1 \\ 1 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 & -1 \\ 3 & -1 & 4 \end{pmatrix}, \qquad \begin{pmatrix} 5 & 3 & 2 \\ -2 & 3 & -5 \end{pmatrix}$$

where C carries the y variables to the z variables, B carries the x variables to the y variables, and the third array carries the x variables straight to the z variables. Each entry of the third array is obtained from a row of C and a column of B: the coefficient of x_2 in z_1 is $2 \cdot 2 + 1 \cdot (-1) = 3$, which runs along row 1 of C and down column 2 of B, multiplying entry by entry and adding. The same recipe produces every other entry, and it is taken as the definition.

Definition 1.4.1: Sum, Scalar Multiple, and Product of Matrices

Let $A = (a_{ij})$ and $A' = (a'_{ij})$ lie in $M_{m\times n}(k)$ and let $c \in k$. The *sum* $A + A'$ and the *scalar multiple* cA are the matrices in $M_{m\times n}(k)$ with entries

$$(A + A')_{ij} = a_{ij} + a'_{ij}, \qquad (cA)_{ij} = c\, a_{ij}$$

for $1 \le i \le m$ and $1 \le j \le n$. The *zero matrix* is the one all of whose entries are 0.
Let $A = (a_{ij}) \in M_{m\times n}(k)$ and $B = (b_{ij}) \in M_{n\times p}(k)$. Their *product* AB is the matrix in $M_{m\times p}(k)$ whose entry in row i and column j is

$$(AB)_{ij} = \sum_{l=1}^{n} a_{il} b_{lj}$$

for $1 \le i \le m$ and $1 \le j \le p$. The product AB is defined only when the number of columns of A equals the number of rows of B.

The size condition is not a technicality: the sum defining $(AB)_{ij}$ runs over the columns of A and simultaneously over the rows of B, so there has to be one index doing both jobs. The shape of the answer is recorded by $(m \times n)(n \times p) = (m \times p)$, in which the inner two numbers match and disappear.

Reading the definition one column at a time gives a second description. Let B_j denote the jth column of B, a column with n entries. Then $(AB_j)_i = \sum_l a_{il} b_{lj} = (AB)_{ij}$, so the jth column of AB is A times the jth column of B:

$$AB = \begin{pmatrix} AB_1 & AB_2 & \cdots & AB_p \end{pmatrix}$$

where each AB_j is the matrix-vector product of definition 1.1.7. Taking $p = 1$ recovers that product exactly, so the coefficient matrix acting on a column of unknowns is the case of the definition in which the second factor has a single column.

One instance of this is used repeatedly below. Write e_j for the column with 1 in position j and zeros elsewhere. Then

$$(Ae_j)_i = \sum_{l=1}^{n} a_{il}(e_j)_l = a_{ij}$$

since the only surviving term is $l = j$, so Ae_j is the jth column of A: multiplying a matrix by e_j selects a column.

Order matters, and it matters in two ways. If A is 2×3 and B is 3×2 then AB is 2×2 and BA is 3×3, so the two products are not even the same size. For square matrices both products have the same size and are still usually different. With

$$P = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

the entry of PQ in row 1, column 1 is $0 \cdot 0 + 1 \cdot 1 = 1$ and the remaining entries vanish, while the entry of QP in row 2, column 2 is $1 \cdot 1 + 0 \cdot 0 = 1$ and the rest vanish, so

$$PQ = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad QP = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

The same P shows that a product of two nonzero matrices can be zero, since every entry of PP is 0. No cancellation law is therefore available for free: from $XY = XZ$ one may not conclude $Y = Z$, and recovering such a law is exactly what the inverse will do.

One matrix does nothing at all. Multiplying the 2×3 matrix B above on the right by

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

returns B, because the columns of this array are e_1, e_2, e_3 and the jth column of the product is Be_j, the jth column of B.

Definition 1.4.2: Identity Matrix

The *identity matrix* $I_n \in M_n(k)$ is the matrix whose columns are $e_1, \ldots, e_n$; equivalently, its entry in row i and column j is 1 if $i = j$ and 0 otherwise.

Three properties of the product are needed before anything can be built on it. Associativity is the one that does real work later: a sequence of row operations will be recorded as a product of many matrices, and that record is worthless unless the factors can be regrouped at will.

Proposition 1.4.3: Associativity, Distributivity, and the Identity

1. If $A \in M_{m\times n}(k)$, $B \in M_{n\times p}(k)$ and $C \in M_{p\times q}(k)$, then $(AB)C = A(BC)$.
2. If $A \in M_{m\times n}(k)$ and $B, B' \in M_{n\times p}(k)$, then $A(B+B') = AB+AB'$; if $A, A' \in M_{m\times n}(k)$ and $B \in M_{n\times p}(k)$, then $(A + A')B = AB + A'B$; and $c(AB) = (cA)B = A(cB)$ for every $c \in k$.
3. If $A \in M_{m\times n}(k)$, then $I_m A = A = AI_n$.

***Proof*:**

1. Both sides are $m \times q$, so it suffices to compare the entry in row i and column j for $1 \le i \le m$ and $1 \le j \le q$. Expanding the outer product and then the inner one,
$$\big((AB)C\big)_{ij} = \sum_{l=1}^{p} (AB)_{il}\, c_{lj} = \sum_{l=1}^{p} \left(\sum_{r=1}^{n} a_{ir} b_{rl} \right) c_{lj} = \sum_{l=1}^{p} \sum_{r=1}^{n} a_{ir} b_{rl} c_{lj}$$
and in the other grouping
$$\big(A(BC)\big)_{ij} = \sum_{r=1}^{n} a_{ir} (BC)_{rj} = \sum_{r=1}^{n} a_{ir} \left(\sum_{l=1}^{p} b_{rl} c_{lj} \right) = \sum_{r=1}^{n} \sum_{l=1}^{p} a_{ir} b_{rl} c_{lj}$$
Each is the sum of the numbers $a_{ir} b_{rl} c_{lj}$ over all pairs (r, l) with $1 \le r \le n$ and $1 \le l \le p$, and a sum of finitely many numbers does not depend on the order in which they are added.
2. For the first identity,
$$\big(A(B + B')\big)_{ij} = \sum_{l=1}^{n} a_{il} (b_{lj} + b'_{lj}) = \sum_{l=1}^{n} a_{il} b_{lj} + \sum_{l=1}^{n} a_{il} b'_{lj} = (AB)_{ij} + (AB')_{ij}$$

The second is the same computation with the sum in the left factor, and the third follows from $\sum_l (ca_{il})b_{lj} = c\sum_l a_{il}b_{lj} = \sum_l a_{il}(cb_{lj})$.

3. The entry of I_m in row i and column l vanishes unless $l = i$, so in

$$(I_m A)_{ij} = \sum_{l=1}^{m} (I_m)_{il}\, a_{lj}$$

the only surviving term is $l = i$, which contributes a_{ij}. The entry of I_n in row l and column j vanishes unless $l = j$, so the same argument gives $(AI_n)_{ij} = a_{ij}$, as we sought to show.

Part (1) means that a product of several matrices may be written without parentheses: $E_s \cdots E_1 A$ is unambiguous, and every way of bracketing it gives the same matrix. Together with the failure of commutativity the situation is this: the order of the factors is rigid, the grouping is free.

Return now to systems. A square system $Ax = b$ has to be solved afresh for each new right hand side b, and elimination has to be rerun each time. Suppose instead that some matrix B satisfied both $BA = I_n$ and $AB = I_n$. Multiplying $Ax = b$ on the left by B would give

$$Bb = B(Ax) = (BA)x = I_n x = x$$

so a solution, if one exists, has to be Bb; and $A(Bb) = (AB)b = I_m b = b$, so Bb is a solution. The system would then have exactly one solution for every b, read off by a single multiplication rather than by a fresh elimination. Such a B sometimes exists. With

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$$

the four entries of AB are $2\cdot 1 + 1\cdot(-1) = 1$, $2\cdot(-1) + 1\cdot 2 = 0$, $1\cdot 1 + 1\cdot(-1) = 0$ and $1\cdot(-1) + 1\cdot 2 = 1$, so $AB = I_2$, and the same computation in the other order gives $BA = I_2$. The system $2x_1 + x_2 = b_1$, $x_1 + x_2 = b_2$ therefore has the solution $x = Bb$, that is $x_1 = b_1 - b_2$ and $x_2 = -b_1 + 2b_2$, whatever b_1 and b_2 are.

Definition 1.4.4: Invertible Matrix and Its Inverse

A matrix $A \in M_n(k)$ is *invertible* if there is a matrix $B \in M_n(k)$ with

$$AB = I_n = BA$$

and any such B is called an *inverse* of A. A square matrix with no inverse is called *singular*.

Not every square matrix is invertible. Take

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad x = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

so that Ax is the zero column. If B were an inverse of A, then $x = I_2 x = (BA)x = B(Ax)$ would be B times the zero column, which is the zero column; but x is not zero. So A is singular, and the obstruction is visible in the homogeneous system: a nonzero solution of $Ax = 0$ rules an inverse out.

The definition asks for a square matrix, and nothing is lost by that. Let $A \in M_{m\times n}(k)$ with $m \neq n$, and suppose $B \in M_{n\times m}(k)$ satisfied both $BA = I_n$ and $AB = I_m$. Whichever of A and B has more columns than rows is the coefficient matrix of a homogeneous system with more unknowns than equations, so by corollary 1.3.8 it sends some nonzero column x to zero. If $Ax = 0$ with $x \neq 0$ then $x = (BA)x = B(Ax) = 0$, a contradiction, and if $Bx = 0$ with $x \neq 0$ then $x = (AB)x = A(Bx) = 0$, likewise. So a two-sided inverse forces $m = n$.

Proposition 1.4.5: Uniqueness of the Inverse, and the Inverse of a Product

Let $A \in M_n(k)$.

1. If $B, C \in M_n(k)$ satisfy $BA = I_n$ and $AC = I_n$, then $B = C$. In particular an invertible matrix has exactly one inverse, written A^{-1}.
2. If A is invertible, so is A^{-1}, and $(A^{-1})^{-1} = A$.
3. If $A, B \in M_n(k)$ are invertible, so is AB, and $(AB)^{-1} = B^{-1}A^{-1}$.

***Proof*:**

1. Using proposition 1.4.3(3) and then (1),
$$B = BI_n = B(AC) = (BA)C = I_nC = C$$
If B and C are both inverses of A then in particular $BA = I_n$ and $AC = I_n$, so $B = C$.
2. The pair of identities $AA^{-1} = I_n = A^{-1}A$ required of A^{-1} is the pair required of A, with the roles of the two matrices exchanged. So A is an inverse of A^{-1}, and by (1) it is the only one.
3. Regrouping by proposition 1.4.3(1),
$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = AA^{-1} = I_n$$
and in the other order
$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_nB = B^{-1}B = I_n$$
so $B^{-1}A^{-1}$ is an inverse of AB, and by (1) it is the only one, completing the proof.

Part (3) extends to any number of factors by induction on s: if $A_1, \dots, A_s \in M_n(k)$ are invertible then so is $A_1 \cdots A_s$, with inverse $A_s^{-1} \cdots A_1^{-1}$, the factors reversed. Part (1) is stronger than uniqueness alone, since it assumes only that B is an inverse on the left and C an inverse on the right, and concludes that the two coincide.

The row operations can now be brought inside this algebra. Take

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{pmatrix}$$

and perform $R_2 \mapsto R_2 - 3R_1$ not on A but on I_3, which produces

$$E = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The entries of EA in row 2 are $(EA)_{2j} = -3a_{1j} + a_{2j}$, giving $3 - 3 = 0$, then $8 - 6 = 2$, then $1 - 3 = -2$; rows 1 and 3 of E agree with those of I_3, so rows 1 and 3 of EA agree with those of A. Hence

$$EA = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{pmatrix}$$

which is A after the operation $R_2 \mapsto R_2 - 3R_1$. Recording an operation by its effect on the identity matrix and then multiplying is the same as performing the operation.

Definition 1.4.6: Elementary Matrix

An *elementary matrix* of size m is a matrix in $M_m(k)$ obtained from I_m by performing a single row operation: one of $R_i \leftrightarrow R_j$, or $R_i \mapsto cR_i$ with $c \neq 0$, or $R_i \mapsto R_i + cR_j$ with $i \neq j$.

Theorem 1.4.7: Row Operations Are Left Multiplications

Let $A \in M_{m\times n}(k)$ and let $E \in M_m(k)$ be the elementary matrix obtained by performing a row operation ρ on I_m.

1. EA is the matrix obtained from A by performing ρ.
2. E is invertible and E^{-1} is again an elementary matrix, namely the one belonging to $R_i \leftrightarrow R_j$, to $R_i \mapsto c^{-1}R_i$, or to $R_i \mapsto R_i + (-c)R_j$, according as ρ is the swap $R_i \leftrightarrow R_j$, the scaling $R_i \mapsto cR_i$, or the replacement $R_i \mapsto R_i + cR_j$.
3. If B is obtained from A by performing the row operations $\rho_1, \ldots, \rho_s$ in that order, and $E_1, \ldots, E_s$ are their elementary matrices, then $B = E_s \cdots E_1 A$. In particular $B = MA$ for the invertible matrix $M = E_s \cdots E_1$.

***Proof*:**
Everything rests on one reading of the definition of the product. For each i and each j, $(EA)_{ij} = \sum_{l=1}^{m} e_{il}a_{lj}$, so row i of EA is the combination

$$\sum_{l=1}^{m} e_{il} \cdot (\text{row } l \text{ of } A)$$

whose coefficients are the entries of row i of E.

1. Take the three operations in turn. If ρ is the swap $R_i \leftrightarrow R_j$, then row i of E is row j of I_m, row j of E is row i of I_m, and every other row of E is the corresponding row of I_m. Row l of I_m has a single nonzero entry, a 1 in position l, so the combination above selects row l of A. Hence row i of EA is row j of A, row j of EA is row i of A, and all other rows are unchanged, which is A with rows i and j exchanged. If ρ is the scaling $R_i \mapsto cR_i$, then

row i of E has the single nonzero entry c in position i and the other rows are those of I_m, so row i of EA is c times row i of A and the rest are unchanged. If ρ is the replacement $R_i \mapsto R_i + cR_j$ with $i \neq j$, then row i of E has 1 in position i, c in position j and zeros elsewhere, so row i of EA is row i of A plus c times row j of A, and again the other rows are unchanged.

2. Let ρ' be the operation listed and E' its elementary matrix. By (1), $E'E$ is the result of performing ρ' on E, and E is the result of performing ρ on I_m. Performing ρ' after ρ returns I_m in each case: a swap performed twice restores the original two rows; scaling row i by c and then by c^{-1} restores row i, which is legitimate since $c \neq 0$; and adding c times row j to row i and then adding $-c$ times row j to row i restores row i, because $i \neq j$ leaves row j untouched by the first operation. So $E'E = I_m$. The reversing relation is symmetric: a swap reverses itself, $R_i \mapsto cR_i$ reverses $R_i \mapsto c^{-1}R_i$, and $R_i \mapsto R_i + cR_j$ reverses $R_i \mapsto R_i + (-c)R_j$. So the argument just given, run with ρ and ρ' interchanged, yields $EE' = I_m$. Hence E is invertible with $E^{-1} = E'$.

3. Induct on s. For $s = 1$ this is (1). Let $s \geq 2$ and let B' be the matrix obtained from A by performing $\rho_1, \ldots, \rho_{s-1}$, so that $B' = E_{s-1} \cdots E_1 A$ by the inductive hypothesis. Since B is obtained from B' by performing ρ_s, part (1) gives $B = E_s B' = E_s E_{s-1} \cdots E_1 A$. Each factor E_i is invertible by (2), so their product M is invertible by proposition 1.4.5(3) and induction, completing the proof.

Elimination is therefore a product. Whatever sequence of operations carries a matrix to an echelon form, the entire run is summarized by one invertible matrix M multiplying on the left, and the summary is available for any matrix at all, square or not. For the running matrix A of example 1.2.9, which has four rows and three columns, part (3) says that the elimination performed there is recorded by a single invertible $M \in M_4(\mathbb{R})$ with MA the echelon form reached; section 1.5 turns that record into a factorization of A itself.

Part (3) also converts the relation of definition 1.2.3 into an equation: B is row equivalent to A exactly when $B = MA$ for a matrix M that is a product of elementary matrices. One direction is part (3); the other is part (1) applied one factor at a time, reading the product MA from the innermost factor outward as the successive results of the corresponding operations.

For a square matrix the same machinery determines invertibility, using nothing beyond elimination. Three conditions that have appeared separately in this chapter, the absence of a nonzero solution of the homogeneous system, the shape of the reduced echelon form, and solvability for every right hand side, are the same condition, and each is equivalent to the existence of an inverse.

Theorem 1.4.8: Invertibility, Reduction to the Identity, and the Computation of the Inverse

Let $A \in M_n(k)$. The following are equivalent.

1. A is invertible.
2. The homogeneous system $Ax = 0$ has only the solution $x = 0$.
3. The reduced echelon form of A is I_n.
4. A is a product of elementary matrices.
5. For every $b \in k^n$ the system $Ax = b$ has a solution.

When these hold and $\rho_1, \ldots, \rho_s$ are row operations carrying A to I_n, performing $\rho_1, \ldots, \rho_s$ in the same order on I_n produces A^{-1}.

***Proof*:**
The five conditions are linked by the implications (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (1) together with (1) $\Rightarrow$ (5) $\Rightarrow$ (3), and the second chain rejoins the first at (3).

(1) implies (2). If A is invertible and $Ax = 0$, then $x = I_n x = (A^{-1}A)x = A^{-1}(Ax)$, which is A^{-1} times the zero column and hence zero.

(2) implies (3). Let R be the reduced echelon form of A, which exists by theorem 1.2.8 and is unique by theorem 1.2.10, and let $r = \operatorname{rank} A$ be its number of pivots (definition 1.3.2). By theorem 1.3.6 the solutions of $Ax = 0$ are described by $n - r$ free parameters, and hypothesis (2) leaves no free parameter, so $r = n$. An $n \times n$ matrix in reduced echelon form with n pivots is I_n: each pivot is the leading entry of a nonzero row, so n pivots occupy all n rows, one each, and their column indices $c_1 < c_2 < \cdots < c_n$ increase strictly; strict increase from $c_1 \geq 1$ forces $c_i \geq i$, while $c_n \leq n$ forces $c_i \leq i$, so $c_i = i$ for every i. In reduced echelon form each pivot equals 1 and is the only nonzero entry of its column, so column i of R is e_i and $R = I_n$.

(3) implies (4). By theorem 1.2.8 there are row operations carrying A to its reduced echelon form, which is I_n by hypothesis; let $E_1, \ldots, E_s$ be their elementary matrices, so that $E_s \cdots E_1 A = I_n$ by theorem 1.4.7(3). Each E_i is invertible with elementary inverse by theorem 1.4.7(2). Multiplying on the left by E_s^{-1}, then by E_{s-1}^{-1}, and so on down to E_1^{-1}, strips the factors one at a time and leaves

$$A = E_1^{-1} E_2^{-1} \cdots E_s^{-1}$$

which exhibits A as a product of elementary matrices.

(4) implies (1). Every elementary matrix is invertible by theorem 1.4.7(2), and a product of invertible matrices is invertible by proposition 1.4.5(3) and induction on the number of factors.

(1) implies (5). Given b, the column $x = A^{-1}b$ satisfies $Ax = A(A^{-1}b) = (AA^{-1})b = I_n b = b$.

(5) implies (3). Suppose instead that the reduced echelon form R of A is not I_n. An $n \times n$ matrix in reduced echelon form with n pivots is I_n, as shown in the proof that (2) implies (3), so R has fewer than n pivots and at least one of its n rows carries no pivot and is therefore zero; the zero rows of an echelon form sit at the bottom, so the last row of R is zero. Write $R = MA$ with M invertible (theorem 1.4.7(3)) and set $b = M^{-1}e_n$. Performing on the augmented matrix

$(A|b)$ the same operations that carried A to R produces $M(A|b)$, again by theorem 1.4.7(3), and each column of that product is M times the corresponding column of $(A|b)$, so

$$M(A|b) = (MA|Mb) = (R|e_n)$$

using $Mb = M(M^{-1}e_n) = e_n$. The last row of this augmented matrix reads $0x_1 + \cdots + 0x_n = 1$, an equation no x satisfies. By lemma 1.2.4 the system $Ax = b$ has the same solutions as this one, hence none, so (5) fails.

The computation of the inverse. Let $\rho_1, \ldots, \rho_s$ carry A to I_n, let $E_1, \ldots, E_s$ be their elementary matrices and put $M = E_s \cdots E_1$, which is invertible. Then $MA = I_n$ by theorem 1.4.7(3), so $A = M^{-1}(MA) = M^{-1}$, and therefore $A^{-1} = (M^{-1})^{-1} = M$ by proposition 1.4.5(2). Finally $M = MI_n$ is, by theorem 1.4.7(3) applied to I_n, exactly the matrix obtained from I_n by performing $\rho_1, \ldots, \rho_s$ in that order. So the operations that carry A to I_n carry I_n to A^{-1}, as we sought to show.

The last paragraph is an algorithm as it stands. Write the two blocks side by side as the augmented matrix $(A|I_n)$, run elimination on it, and stop when the left block is I_n; the right block is then A^{-1}, since every operation performed on the left block is performed on the right block at the same time. If instead a zero row appears in the left block, elimination cannot reach I_n, and A is singular. The equivalence of (2) and (5) says that for a square matrix solvability for every right hand side and uniqueness of solutions stand or fall together, so neither can fail alone (compare corollary 1.3.7).

Example 1.4.9: Inverting a 3×3 Matrix, and a Matrix That Has No Inverse

1. Let
$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 5 & 3 \\ 1 & 3 & 3 \end{pmatrix}$$
and start from $(A|I_3)$:
$$\left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 3 & 3 & 0 & 0 & 1 \end{array}\right)$$
Clear the first column with $R_2 \mapsto R_2 + (-2)R_1$ and $R_3 \mapsto R_3 + (-1)R_1$, which replace row 2 by $(0,1,1|-2,1,0)$ and row 3 by $(0,1,2|-1,0,1)$:
$$\left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 1 & 2 & -1 & 0 & 1 \end{array}\right)$$
Now $R_3 \mapsto R_3 + (-1)R_2$ replaces row 3 by $(0,0,1|1,-1,1)$:
$$\left(\begin{array}{ccc|ccc} 1 & 2 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array}\right)$$
The left block is in echelon form with three pivots, all equal to 1, so no scaling is needed and the reduction upward can begin. The operations $R_2 \mapsto R_2 + (-1)R_3$ and $R_1 \mapsto R_1 + (-1)R_3$

clear the third column, giving rows $(0,1,0\big|-3,2,-1)$ and $(1,2,0\big|0,1,-1)$:

$$\left(\begin{array}{ccc|ccc} 1 & 2 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & -3 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array}\right)$$

One operation remains. Applying $R_1 \mapsto R_1 + (-2)R_2$ replaces row 1 by $(1,0,0\big|6,-3,1)$:

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 6 & -3 & 1 \\ 0 & 1 & 0 & -3 & 2 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array}\right)$$

The left block is I_3, so A is invertible and

$$A^{-1} = \begin{pmatrix} 6 & -3 & 1 \\ -3 & 2 & -1 \\ 1 & -1 & 1 \end{pmatrix}$$

Checking AA^{-1} row by row: row 1 of A is $(1,2,1)$, and against the three columns of A^{-1} it gives $6-6+1=1$, then $-3+4-1=0$, then $1-2+1=0$. Row 2 of A gives $12-15+3=0$, then $-6+10-3=1$, then $2-5+3=0$; row 3 gives $6-9+3=0$, then $-3+6-3=0$, then $1-3+3=1$. So $AA^{-1}=I_3$.

2. Let

$$C = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 1 & 2 & 5 \end{pmatrix}$$

The operations $R_2 \mapsto R_2 + (-2)R_1$ and $R_3 \mapsto R_3 + (-1)R_1$ give rows $(0,0,1)$ and $(0,0,4)$, and then $R_3 \mapsto R_3 + (-4)R_2$ gives $(0,0,0)$:

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

This is an echelon form with two pivots, in columns 1 and 3, so the reduced echelon form of C is not I_3 and C is singular by theorem 1.4.8. A nonzero solution of $Cx=0$ is produced from the same computation: $R_1 \mapsto R_1 + (-1)R_2$ turns the first row into $(1,2,0)$, so the system reads $x_1 + 2x_2 = 0$ and $x_3 = 0$, and $x_2 = 1$ gives $x = (-2,1,0)$ written as a column. Indeed each row of C against that column gives $-2+2+0=0$, $-4+4+0=0$ and $-2+2+0=0$.

The criterion in theorem 1.4.8 refers to no formula for the entries of A^{-1} in terms of the entries of A. Such a formula exists and belongs to chapter 3; elimination reaches the inverse without it, and the computation above is the whole of what is required.

Exercise 1.4.1

1. Let

$$A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 4 & 1 \\ 0 & -2 \\ 2 & 3 \end{pmatrix}$$

Compute AB and BA. State the size of each before computing it.

2. Let
$$A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}$$
Compute AB and BA. Then show that neither A nor B is invertible, using only the two products just computed.

3. Compute the inverse of
$$B = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & 3 \\ 2 & 1 & 5 \end{pmatrix}$$
by row reducing $(B|I_3)$, naming each operation used, and verify the answer by computing BB^{-1}.

4. Decide whether
$$C = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 8 \\ 1 & 3 & 5 \end{pmatrix}$$
is invertible. If it is, compute C^{-1}; if it is not, exhibit a nonzero column x with $Cx = 0$.

5. Let $A = \begin{pmatrix} 2 & 4 \\ 3 & 7 \end{pmatrix}$. Find row operations carrying A to I_2, write down the elementary matrix of each, and use them to express A as a product of elementary matrices and to compute A^{-1}.

6. Let $A, B, C \in M_n(k)$ with A invertible. Show that $AB = AC$ forces $B = C$. Then give 2×2 matrices A, B, C with $AB = AC$ and $B \neq C$.

7. Let $A, B \in M_n(k)$ satisfy $AB = I_n$. Show that B is invertible, that $A = B^{-1}$, and hence that $BA = I_n$ as well. (A one-sided inverse of a square matrix is automatically two-sided.)

8. Let
$$E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$
Compute EA and AE, and describe in words what each product does to A.

1.5 The LU Factorization

Elimination discards everything except its answer. The echelon form of a matrix says which columns carry pivots and what the solutions are, and it keeps no trace of the multipliers that produced it. Those multipliers cost nothing to store, and storing them turns the procedure into an equation: the matrix elimination started from is a product of a matrix holding the multipliers with the echelon matrix it ended at.

Take
$$B = \begin{pmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{pmatrix}$$

and run elimination on it, writing down each multiplier as it is used. The pivot of the first stage is the entry 2 in the upper left corner. Row 2 begins with 4, which is 2 times the pivot, so $R_2 \mapsto R_2 - 2R_1$ clears it; row 3 begins with -2, which is -1 times the pivot, so $R_3 \mapsto R_3 + R_1$ clears it. The two operations give

$$\begin{pmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 8 & 3 \end{pmatrix}$$

The second stage has pivot -8 in row 2. The entry below it is 8, which is -1 times the pivot, so $R_3 \mapsto R_3 + R_2$ clears it:

$$\begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 8 & 3 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{pmatrix} = U$$

The matrix U is in echelon form (definition 1.2.5) and elimination has done its job. Three multipliers were used along the way: 2 to clear the entry in position $(2, 1)$, then -1 to clear $(3, 1)$, then -1 to clear $(3, 2)$. Collect them into a square array by putting each multiplier in the position it cleared, with ones on the diagonal and zeros above it:

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -1 & 1 \end{pmatrix}$$

The product LU is the matrix elimination started from. Its three rows are

$$\begin{aligned} 1 \cdot \begin{pmatrix} 2 & 1 & 1 \end{pmatrix} &= \begin{pmatrix} 2 & 1 & 1 \end{pmatrix} \\ 2 \begin{pmatrix} 2 & 1 & 1 \end{pmatrix} + 1 \cdot \begin{pmatrix} 0 & -8 & -2 \end{pmatrix} &= \begin{pmatrix} 4 & -6 & 0 \end{pmatrix} \\ (-1) \begin{pmatrix} 2 & 1 & 1 \end{pmatrix} + (-1) \begin{pmatrix} 0 & -8 & -2 \end{pmatrix} + 1 \cdot \begin{pmatrix} 0 & 0 & 1 \end{pmatrix} &= \begin{pmatrix} -2 & 7 & 2 \end{pmatrix} \end{aligned}$$

and these are the three rows of B.

The form of that check is the general rule for a matrix product. The (i, q) entry of LU is $\sum_t \ell_{it} u_{tq}$ (definition 1.4.1). Here ℓ_{it} is the entry of L in row i and column t, and u_{tq} is the entry of U in row t and column q. Reading this for every q at once, row i of LU is the combination of the rows of U whose coefficients are the entries of row i of L. So the identity $B = LU$ says that each row of B is a combination of the rows of U, with the multipliers as coefficients. That is elimination read backwards: row 3 of B was turned into row 3 of U by subtracting -1 times row 1 and then -1 times row 2, and adding those two multiples back is exactly what the third row of L prescribes.

Since B is square, the two factors also fit in one array. No entry of U sits below the diagonal, and every multiplier sits below it, so a multiplier can be kept in the very position it cleared without displacing an entry of U.

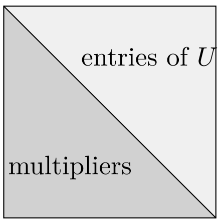


Figure 1.7: For a square matrix the two factors fit in one array: elimination leaves U on and above the diagonal, and the multiplier used to clear position (i, j) can be stored in that very position

Definition 1.5.1: LU Factorization

A matrix $U \in M_{m\times n}(k)$, where k is $\mathbb{R}$ or $\mathbb{C}$, is *upper triangular* if its entry u_{ij} in row i and column j is zero whenever $i > j$, that is, if every entry strictly below the diagonal is zero. A square matrix $L \in M_m(k)$ with entries ℓ_{ij} is *lower triangular* if $\ell_{ij} = 0$ whenever $j > i$, that is, if every entry strictly above the diagonal is zero, and *unit lower triangular* if in addition $\ell_{ii} = 1$ for every i: ones along the diagonal, zeros strictly above it, and no condition below it. An *LU factorization* of $A \in M_{m\times n}(k)$ is a pair (L, U) with $L \in M_m(k)$ unit lower triangular, $U \in M_{m\times n}(k)$ upper triangular, and $A = LU$.

Proposition 1.5.2: Arithmetic of Triangular Matrices

Let $L, L' \in M_n(k)$ be lower triangular with diagonal entries ℓ_{ii} and ℓ'_{ii}, and let $U, U' \in M_n(k)$ be upper triangular.

1. LL' is lower triangular and UU' is upper triangular, and in each case the ith diagonal entry of the product is the product of the ith diagonal entries of the factors.
2. If every $\ell_{ii} \neq 0$, then L is invertible and L^{-1} is lower triangular with ith diagonal entry ℓ_{ii}^{-1}. The same holds for U with "upper" throughout.
3. A matrix that is both lower triangular and upper triangular is diagonal.

***Proof*:**

1. By definition 1.4.1 the entry of LL' in row i and column j is $\sum_s \ell_{is}\ell'_{sj}$. A term is nonzero only if $s \le i$, since $\ell_{is} = 0$ for $s > i$, and only if $j \le s$, since $\ell'_{sj} = 0$ for $j > s$; so it requires $j \le s \le i$. When $j > i$ there is no such s and the entry vanishes, which is lower triangularity. When $j = i$ the only surviving term is $s = i$, giving $\ell_{ii}\ell'_{ii}$. For UU' the same count runs with the inequalities reversed: a nonzero term needs $i \le s \le j$, which is impossible for $j < i$ and leaves only $s = i$ when $j = i$.

2. Suppose every $\ell_{ii} \neq 0$. Solve $LX = I_n$ one column at a time. The jth column x of X must satisfy $Lx = e_j$, and by definition 1.4.1 row i of that system reads

$$\ell_{i1}x_1 + \cdots + \ell_{ii}x_i = (e_j)_i$$

because $\ell_{is} = 0$ for $s > i$. Reading these from the top, row 1 determines x_1 by a division by ℓ_{11}, and row i determines x_i from $x_1, \ldots, x_{i-1}$ by a division by ℓ_{ii}; so the system has exactly one solution. For $i < j$ the right-hand side is 0, and induction on i gives $x_i = 0$: each row expresses $\ell_{ii}x_i$ as a combination of earlier entries, all of which vanish. So the jth column of X is zero above row j, which is lower triangularity, and row j then reads $\ell_{jj}x_j = 1$, giving the diagonal entry ℓ_{jj}^{-1}.

   It remains to see that X is a two-sided inverse. By (1) the product XL is lower triangular with ith diagonal entry $\ell_{ii}^{-1}\ell_{ii} = 1$. Running the same column-by-column argument on X, whose diagonal entries are the nonzero numbers ℓ_{jj}^{-1}, produces a lower triangular Y with $XY = I_n$; then $L = LI_n = L(XY) = (LX)Y = Y$, so $XL = XY = I_n$. Hence L is invertible with $L^{-1} = X$ (definition 1.4.4, proposition 1.4.5). The upper triangular statement is the same argument reading the rows from the bottom.

3. Such a matrix has entry zero whenever $j > i$ and whenever $j < i$, so every off-diagonal entry vanishes.

The letters record the shapes: L is lower triangular and U is upper triangular. The sizes are forced by the product, since L has m columns and U has m rows, so a 4×3 matrix is factored by a 4×4 matrix L and a 4×3 matrix U.

An echelon matrix is upper triangular, so the output of elimination is always a candidate for U. Indeed, in an echelon matrix the nonzero rows come first and the pivot of each nonzero row lies strictly to the right of the pivot of the row above it, so the pivot of row i lies in a column of index at least i; every entry of row i to the left of its pivot is zero, and a zero row has no nonzero entry anywhere. Either way the entries of row i in columns $1, \dots, i-1$ vanish. The implication does not reverse: the 2×2 matrix with rows $(0, 1)$ and $(0, 1)$ is upper triangular and is not echelon, since its two pivots sit in the same column. Definition 1.5.1 asks only for upper triangularity, which is the condition the arguments below use.

Not every matrix has an LU factorization. The obstruction is visible in the one place elimination can stall: a zero in the position where a pivot is wanted.

Example 1.5.3: A Matrix With No LU Factorization

Elimination on

$$C = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 1 \\ 3 & 5 & 2 \end{pmatrix}$$

starts with the pivot 1 and clears the first column by $R_2 \mapsto R_2 - 2R_1$ and $R_3 \mapsto R_3 - 3R_1$, leaving

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & -5 \\ 0 & -1 & -7 \end{pmatrix}$$

The second stage searches rows 2 and 3 for the leftmost column carrying a nonzero entry. That column is the second, and the entry it has in row 2 is zero, so a swap is forced.

The matrix C has no LU factorization at all. Suppose $C = LU$ with L unit lower triangular and U upper triangular. Row 1 of LU is row 1 of U, so $u_{11} = 1$, $u_{12} = 2$ and $u_{13} = 3$. The entries of LU in positions $(2, 1)$ and $(3, 1)$ are $\ell_{21}u_{11}$ and $\ell_{31}u_{11}$, forcing $\ell_{21} = 2$ and $\ell_{31} = 3$. The entry in position $(2, 2)$ is $\ell_{21}u_{12} + u_{22} = 4 + u_{22}$, and it has to equal 4, so $u_{22} = 0$. But then the entry in position $(3, 2)$ is $\ell_{31}u_{12} + \ell_{32}u_{22} = 6 + 0 = 6$, while the corresponding entry of C is 5. No such L and U exist.

Performing the swap first repairs this. The operation $R_2 \leftrightarrow R_3$ is carried out by left multiplication by the elementary matrix of that operation (definition 1.4.6, theorem 1.4.7), called its *swap matrix*:

$$P = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad PC = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 5 & 2 \\ 2 & 4 & 1 \end{pmatrix}$$

Elimination on PC uses the multipliers 3 and 2 in its first stage, by $R_2 \mapsto R_2 - 3R_1$ and $R_3 \mapsto R_3 - 2R_1$, and finds the entry -1 waiting in the pivot position of its second stage, where the multiplier is 0 and no operation is performed. So

$$L = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}, \qquad U = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & -7 \\ 0 & 0 & -5 \end{pmatrix}, \qquad PC = LU$$

Three claims are now owed. Assembling the multipliers into L worked twice, and it has to be shown to work always. The factorization it produces should be the only one, at least when the matrix is square and elimination reaches a pivot in every row. And the failure just seen should always be repairable by a reordering of the rows decided in advance, as it was for C.

Stating the first claim precisely needs a name for the steps of the procedure. Elimination (theorem 1.2.8) proceeds in stages, one for each $j = 1, 2, \dots, m-1$. At the start of stage j the rows 1 through $j-1$ are settled and are never touched again. Among rows j through m the leftmost column carrying a nonzero entry is located; if the entry of that column in row j is zero, a swap brings a nonzero entry there. That entry is the pivot (definition 1.2.6), and a multiple of row j is subtracted from each row below it to clear the rest of the pivot's column. If rows j through m are all zero the procedure halts, and the matrix it has reached is in echelon form.

Theorem 1.5.4: Existence and Uniqueness of the LU Factorization

Let $A \in M_{m\times n}(k)$.

1. Suppose elimination drives A to an echelon matrix U without using a swap at any stage. For $1 \le j < i \le m$ let ℓ_{ij} be the multiple of row j that stage j subtracts from row i, taken to be 0 if nothing is subtracted, and let $L \in M_m(k)$ be the unit lower triangular matrix carrying these entries below its diagonal. Then $A = LU$.

2. Let $A \in M_n(k)$ be square and, in the situation of (1), suppose that elimination produces a pivot in every row. Then the pair (L, U) of (1) is the only LU factorization of A.

3. For every $A \in M_{m\times n}(k)$ there is a product P of swap matrices, that is, of elementary matrices performing the operations $R_i \leftrightarrow R_j$ (definition 1.4.6), such that PA has an LU factorization.

***Proof*:**

1. Write a_i for row i of A and u_i for row i of U. Let $A^{(0)} = A$ and let $A^{(t)}$ be the matrix left standing after stage t, so that $A^{(m-1)} = U$; if the procedure halts before stage $m-1$ then the remaining stages change nothing and their multipliers are 0, so the notation is still meaningful. Stage t leaves rows 1 through t unchanged and replaces row i, for each $i > t$, by that row minus ℓ_{it} times row t of $A^{(t-1)}$.

   Row t is altered by no stage from the t-th onwards, so row t of $A^{(t-1)}$ is u_t: the row subtracted at stage t is already in its final form. Row i, in turn, is altered at stages 1 through $i-1$ and at no other, so accumulating the alterations gives

   $$u_i = a_i - \sum_{t=1}^{i-1} \ell_{it} u_t$$

   Since $\ell_{ii} = 1$ and $\ell_{it} = 0$ for $t > i$, this rearranges to $a_i = \sum_{t=1}^{m} \ell_{it} u_t$. The q-th entry of the right-hand side is $\sum_t \ell_{it} u_{tq}$, which is the (i, q) entry of LU (definition 1.4.1). So row i of LU is a_i for every i, and $A = LU$.

2. The matrix U has n rows and n columns and a pivot in each row. Writing q_i for the column of the pivot in row i, the inequalities $1 \le q_1 < q_2 < \cdots < q_n \le n$ force $q_i = i$, so every diagonal entry u_{ii} is a pivot and is nonzero.

Let $A = L'U'$ be any LU factorization. For all i and q the (i,q) entry of $L'U'$ is $\sum_t \ell'_{it}u'_{tq}$, and $\ell'_{it} = 0$ for $t > i$ while $u'_{tq} = 0$ for $t > q$, so only the terms with $t \le \min(i,q)$ survive. Reading the resulting identity for $i \le q$, and using $\ell'_{ii} = 1$, gives

$$u'_{iq} = a_{iq} - \sum_{t<i} \ell'_{it}u'_{tq} \qquad (i \le q) \tag{1.1}$$

and reading it for $i > q$ gives

$$\ell'_{iq}u'_{qq} = a_{iq} - \sum_{t<q} \ell'_{it}u'_{tq} \qquad (i > q) \tag{1.2}$$

Both hold with (L,U) in place of (L',U'), since (L,U) is itself an LU factorization by (1).

The two factorizations now agree by induction on j, the claim at step j being that row j of U' is row j of U and column j of L' is column j of L. Assume it for all indices smaller than j, an empty assumption when $j = 1$. Taking $i = j$ in (1.1) expresses u'_{jq}, for each $q \ge j$, in terms of a_{jq}, of entries ℓ'_{jt} lying in columns $t < j$ of L', and of entries u'_{tq} lying in rows $t < j$ of U'. All of those agree with their unprimed counterparts, so $u'_{jq} = u_{jq}$ for $q \ge j$; for $q < j$ both entries are zero. In particular $u'_{jj} = u_{jj} \ne 0$. Taking $q = j$ in (1.2) now expresses $\ell'_{ij}u'_{jj}$, for each $i > j$, in terms of entries already known to agree, so $\ell'_{ij}u'_{jj} = \ell_{ij}u_{jj}$, and dividing by the nonzero u_{jj} gives $\ell'_{ij} = \ell_{ij}$. The remaining entries of column j in L' and in L are the diagonal 1 and zeros. The induction closes, so $L' = L$ and $U' = U$.

3. It suffices to find a product P of swap matrices for which elimination on PA uses no swap, since (1) then gives the factorization of PA. Such a P is produced by induction on the number of rows m.

   If $m = 1$ then elimination performs no operation and $P = I_1$ serves. Let $m \ge 2$ and assume the assertion for matrices with $m-1$ rows. If A is the zero matrix, $P = I_m$ serves. Otherwise let q be the leftmost column of A carrying a nonzero entry, say in row r, and let S be the swap matrix exchanging rows 1 and r, with $S = I_m$ when $r = 1$. The matrix SA has a nonzero entry in position $(1,q)$ and zeros everywhere to the left of column q, so the first stage of elimination on SA takes its pivot from position $(1,q)$ and needs no swap. That stage subtracts ℓ_i times row 1 from row i for $i = 2,\dots,m$, and produces a matrix B whose rows 2 through m are zero in columns 1 through q.

   Let B' be the $(m-1)\times n$ matrix formed by rows 2 through m of B. By the inductive hypothesis there is a product Q' of swap matrices for which elimination on $Q'B'$ uses no swap. Let Q be the product obtained from Q' by replacing each swap $R_a \leftrightarrow R_b$ occurring in it by the swap matrix of $R_{a+1} \leftrightarrow R_{b+1}$ on m-rowed matrices, in the same order. Every factor of Q fixes row 1 and exchanges two of the rows 2 through m, so for each $i \ge 2$ there is an index $\sigma(i) \ge 2$ such that row i of QM is row $\sigma(i)$ of M for every matrix M with m rows, and row 1 of QM is row 1 of M. The shift $i \mapsto i-1$ turns those swaps back into the swaps Q' performs, so rows 2 through m of QM form the matrix $Q'M'$, where M' is the matrix formed by rows 2 through m of M.

   The first stage of elimination applied to QSA produces QB. Row 1 of QSA is row 1 of SA, whose entry in column q is nonzero and whose entries to the left of column q vanish, so the pivot is again in position $(1,q)$ and no swap is used. Row i of QSA is row $\sigma(i)$ of SA, whose entry in column q is $\ell_{\sigma(i)}$ times the pivot, so the multiplier the stage computes for row i is $\ell_{\sigma(i)}$, and the row it leaves behind is row $\sigma(i)$ of B, which is row i of QB.

Every later stage of elimination on QB operates on rows 2 through m and depends on no other row, so those stages are the stages of elimination applied to the matrix formed by rows 2 through m of QB, which is $Q'B'$. By the choice of Q' none of them uses a swap. Elimination on QSA therefore uses no swap at any stage, and $P = QS$ is a product of swap matrices, completing the proof.

Part (1) says that elimination is a factorization algorithm, and that both factors are already present in the computation: U is what elimination returns and L is what it would otherwise throw away. The identity behind it, $a_i = \sum_{t\leq i} \ell_{it} u_t$, says that each row of A is a combination of the rows of U at or above its own index, with the multipliers as coefficients. Part (2) says that when the matrix is square with a pivot in every row, which is the case in which the system $Ax = b$ has exactly one solution for every b (corollary 1.3.7), there is nothing to choose: the pair elimination produces is the only pair there is. Part (3) says the only obstruction is the order in which the rows are presented, and that reordering them in advance removes it. A swap is forced there by a zero in a pivot position and by nothing else, since the arithmetic in this chapter is exact.

Example 1.5.5: The LU Factorization of the Running Matrix

The running matrix of example 1.2.9 is

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 1 & 2 & 5 \\ 3 & 4 & 11 \end{pmatrix}$$

and the elimination run there used no swap. Repeating it with the multipliers kept gives the factorization. The first stage has pivot 1 and multipliers 2, 1 and 3, so the operations $R_2 \mapsto R_2 - 2R_1$, $R_3 \mapsto R_3 - R_1$ and $R_4 \mapsto R_4 - 3R_1$ leave

$$\begin{pmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \\ 0 & 1 & 2 \end{pmatrix}$$

The second stage takes its pivot from the second column, where row 2 has the entry 1. That entry is nonzero, so no swap is needed. The multipliers are 1 for row 3 and 1 for row 4, and the operations $R_3 \mapsto R_3 - R_2$ and $R_4 \mapsto R_4 - R_2$ leave

$$U = \begin{pmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

Rows 3 and 4 are zero, so elimination halts and every remaining multiplier is 0. Putting each multiplier in the position it cleared,

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 3 & 1 & 0 & 1 \end{pmatrix}$$

and $A = LU$ by theorem 1.5.4(1). Multiplying out confirms it: row 3 of LU is $(1, 1, 3) + (0, 1, 2) = (1, 2, 5)$ and row 4 is $3(1, 1, 3) + (0, 1, 2) = (3, 4, 11)$.

The shapes are those definition 1.5.1 requires, L of size 4×4 and U of size 4×3. The third column of L is zero below the diagonal because elimination halted before a third stage had anything to clear, and U has two nonzero rows, so the rank of A is 2 (definition 1.3.2). None of this is special to A: a factorization records whatever elimination did, including its early stop.

The factorization pays for itself as soon as one coefficient matrix is used more than once. A single system $Ax = b$ costs one elimination whether or not the multipliers are kept. Several systems sharing the coefficient matrix and differing only in the right-hand side would cost one elimination each, and with $A = LU$ in hand they cost no further elimination: the factorization is computed once, and each right-hand side is then handled by two substitutions.

Theorem 1.5.6: Solving a System from an LU Factorization

Let $A \in M_{m\times n}(k)$ have an LU factorization $A = LU$, and let $b \in k^m$ be a column of m entries. Then the system $Ly = b$ has exactly one solution $y \in k^m$, whose entries are given in order by

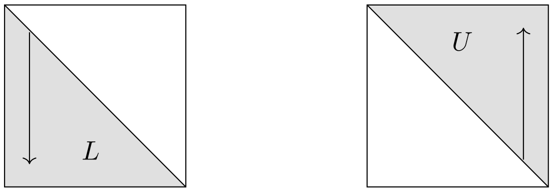
$$y_1 = b_1, \qquad y_i = b_i - \sum_{t=1}^{i-1} \ell_{it} y_t \quad (2 \leq i \leq m) \tag{1.3}$$

and a column $x \in k^n$ satisfies $Ax = b$ if and only if it satisfies $Ux = y$.

***Proof*:**
Equation i of the system $Ly = b$ reads $\sum_t \ell_{it} y_t = b_i$. Since $\ell_{it} = 0$ for $t > i$ and $\ell_{ii} = 1$, it reads $y_i + \sum_{t<i} \ell_{it} y_t = b_i$, which determines y_i from $y_1, \ldots, y_{i-1}$ and is what (1.3) records. Taking the equations in the order $i = 1, 2, \ldots, m$, each one has exactly one solution for its new unknown once the preceding ones are solved, so the system has exactly one solution, and it is the vector defined by (1.3).

Suppose $Ux = y$. Then $Ax = (LU)x = L(Ux) = Ly = b$, where the middle equality is associativity of the matrix product (proposition 1.4.3). Conversely suppose $Ax = b$, and put $z = Ux$. Then $Lz = L(Ux) = (LU)x = Ax = b$, so z solves $Ly = b$; by the uniqueness just proved, $z = y$, which is to say $Ux = y$, as we sought to show.

The two stages are named for the order in which they run. *Forward substitution* solves $Ly = b$ from the top down, each equation delivering one new entry of y. *Back substitution* solves $Ux = y$ from the bottom up. When U comes from elimination it is in echelon form, so the second stage is the reading-off already used on echelon systems in section 1.3, free variables and all. Nothing in theorem 1.5.6 assumes that stage has a unique solution, or any solution: $Ux = y$ is an echelon system like any other, and it may be inconsistent or carry free variables.



$Ly = b$ $Ux = y$

Figure 1.8: Forward substitution determines the entries of y from the top down; back substitution determines the entries of x from the bottom up

Take the matrix B factored at the start of this section and the right-hand side $b = (5, -2, 9)^\top$. Forward substitution through (1.3), with the entries of L read off row by row, gives

$$y_1 = 5, \qquad y_2 = -2 - 2(5) = -12, \qquad y_3 = 9 - (-1)(5) - (-1)(-12) = 2$$

Back substitution through $Ux = y$ then works upwards from the last equation $x_3 = 2$, through $-8x_2 - 2(2) = -12$, which gives $x_2 = 1$, to $2x_1 + 1 + 2 = 5$, which gives $x_1 = 1$. Substituting $x = (1, 1, 2)^\top$ into the original system confirms it: the three rows of B give $2+1+2 = 5$, $4-6+0 = -2$ and $-2+7+4 = 9$.

The work is now split cleanly between what depends on A and what depends on b. Computing y_i from (1.3) costs $i-1$ multiplications and $i-1$ subtractions, so forward substitution costs $n(n-1)/2$ of each on a square system, and back substitution costs the same again together with n divisions. Elimination is more expensive. On an $n \times n$ matrix with a pivot in every row, stage j divides once for each of the $n-j$ rows below the pivot. It then multiplies and subtracts once for each of the $n-j$ entries those rows carry to the right of the pivot's column. That is $(n-j)^2$ multiplications at stage j, and summing over $j = 1, \ldots, n-1$ gives $(n-1)n(2n-1)/6$. The ratio of the two counts is $(2n-1)/6$, so for a system of sixty equations the factorization costs about twenty times what a solve costs, and it is paid for once.

Exercise 1.5.1

1. Find the LU factorization of
$$D = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 5 \\ 4 & 6 & 8 \end{pmatrix}$$
by running elimination and recording the multipliers, and verify the factorization by multiplying L and U together.

2. Using the factorization of D found in the previous question, and running no further elimination, solve $Dx = b$ for $b = (1, 0, 2)^\top$ and then for $b = (2, 4, 6)^\top$.

3. Let A, L and U be as in example 1.5.5. Use theorem 1.5.6 to decide whether $Ax = b$ is consistent for $b = (1, 4, 2, 5)^\top$, and then for $b = (1, 4, 3, 5)^\top$; in the consistent case, describe every solution.

4. Show that
$$E = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 0 \end{pmatrix}$$

has no LU factorization, then find a swap matrix P for which PE has one and compute that factorization.

5. Let $A \in M_{m\times n}(k)$ have an LU factorization and suppose $a_{11} = 0$. Show that the whole first column of A is zero.

6. Show that
$$F = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$
has infinitely many LU factorizations, and say why theorem 1.5.4(2) does not apply to it. Show also that elimination on F uses a swap, so that the hypothesis of theorem 1.5.4(1) is sufficient for a factorization to exist but not necessary.

7. Let $A \in M_n(k)$ have an LU factorization $A = LU$ in which every diagonal entry of U is nonzero. Show that $Ax = b$ has exactly one solution for every $b \in k^n$.

2

Vector Spaces over $\mathbb{R}$ and $\mathbb{C}$

Elimination in chapter 1 kept producing sets that behaved alike. The solutions of a homogeneous system were closed under addition and under scaling; the solutions of an inhomogeneous one were a single solution shifted by that set; the columns of a matrix carried relations that elimination preserved. This chapter names the structure those observations share and shows that naming it needs nothing and settles several questions at once.

Section 2.1 isolates the closure conditions and the two operations built from them, span and linear independence. Section 2.2 produces a minimal generating set for any such object and proves that its size does not depend on how it was chosen, which is what makes dimension a number rather than an accident; the section closes by observing that every argument used only eight properties of arithmetic, and writing them down. Section 2.3 attaches four such objects to a single matrix and shows that their dimensions are determined by the rank alone. Section 2.4 shifts attention from matrices to the maps they represent, and section 2.5 asks what survives when the coordinates used to write a map down are changed.

2.1 Subspaces of $\mathbb{R}^n$ and $\mathbb{C}^n$; Span, Linear Independence, and Direct Sums

Arithmetic on columns is entrywise. For $u, v \in k^n$, where k is $\mathbb{R}$ or $\mathbb{C}$, the sum $u + v$ is the column whose ith entry is $u_i + v_i$, and for a scalar $c \in k$ the multiple cu is the column whose ith entry is cu_i. Every rule of arithmetic in k therefore holds entry by entry: $u+v = v+u$, $(u+v)+w = u+(v+w)$, $c(u+v) = cu+cv$, $(c+d)u = cu+du$, and $u+(-1)u = 0$, where 0 denotes the column of n zeros. The subsets of k^n this section picks out are those on which both operations can be performed without leaving the subset.

Four subsets of $\mathbb{R}^3$, each described by a condition on the entries, show what can happen.

Example 2.1.1: Four Subsets of $\mathbb{R}^3$

Write a column of $\mathbb{R}^3$ as $x = (x_1, x_2, x_3)^\top$.

1. Let $P = \{x \in \mathbb{R}^3 \big| x_1 + x_2 + x_3 = 0\}$. The zero column lies in P, since $0+0+0 = 0$. If u and v lie in P then the entries of $u+v$ sum to $(u_1+v_1)+(u_2+v_2)+(u_3+v_3) = 0+0 = 0$, so $u+v$ lies in P. If u lies in P and $c \in \mathbb{R}$ then the entries of cu sum to $cu_1 + cu_2 + cu_3 = c \cdot 0 = 0$, so cu lies in P. Concretely, $(1, -1, 0)^\top$ and $(0, 1, -1)^\top$ lie in P, and so does their sum $(1, 0, -1)^\top$.

2. Let $Q = \{x \in \mathbb{R}^3 \big| x_1 + x_2 + x_3 = 1\}$. The zero column does not lie in Q, since its entries sum to 0 and not to 1. Addition fails as well: $(1, 0, 0)^\top$ and $(0, 1, 0)^\top$ lie in Q, while their sum $(1, 1, 0)^\top$ has entry sum 2.

3. Let $C = \{x \in \mathbb{R}^3 \big| x_1 \geq 0,\ x_2 \geq 0,\ x_3 \geq 0\}$. The zero column lies in C, and a sum of two columns with nonnegative entries again has nonnegative entries, so C is closed under addition. Scalar multiplication fails: $(1, 0, 0)^\top$ lies in C and $(-1)(1, 0, 0)^\top = (-1, 0, 0)^\top$ does not.

4. Let $N = \{x \in \mathbb{R}^3 \big| x_1 x_2 = 0\}$, the columns with a zero in the first entry or in the second. The zero column lies in N, and if $x_1 x_2 = 0$ and $c \in \mathbb{R}$ then $(cx_1)(cx_2) = c^2 x_1 x_2 = 0$, so N is closed under scalar multiplication. Addition fails: $(1, 0, 0)^\top$ and $(0, 1, 0)^\top$ lie in N, while their sum $(1, 1, 0)^\top$ has $1 \cdot 1 = 1$.

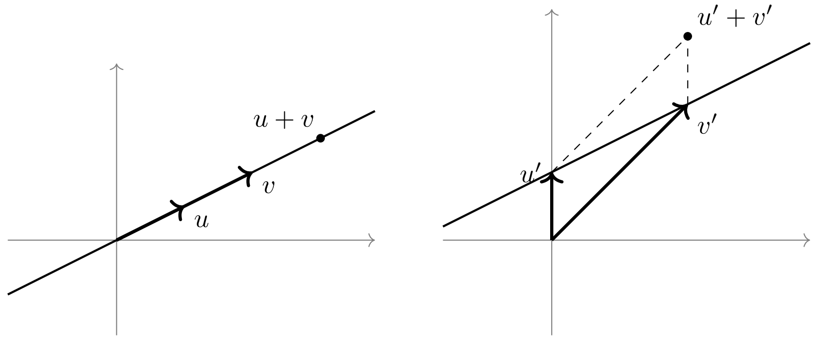
The last two subsets show that the two closure conditions are independent of one another: C has the first and not the second, and N has the second and not the first. Only P has both, and it is the shape that gets a name.

Definition 2.1.2: Subspace of k^n

A subset $W \subseteq k^n$ is a *subspace* of k^n if all three of the following hold.

1. The zero column lies in W.
2. If $u \in W$ and $v \in W$ then $u + v \in W$.
3. If $u \in W$ and $c \in k$ then $cu \in W$.

Conditions (2) and (3) are the two closures, and condition (1) rules out the empty set. Every k^n has two subspaces available with no computation at all: the whole of k^n, and the set $\{0\}$ containing the zero column alone. The set Q of example 2.1.1 misses (1) for a reason that recurs. It is the solution set of the single equation $x_1 + x_2 + x_3 = 1$, and theorem 1.1.11 describes such a set as a translate of the solutions of $x_1 + x_2 + x_3 = 0$, which is exactly P. Translating a subspace away from the origin destroys both closures at once.



(a) (b)

Figure 2.1: In (a) the line passes through the origin and the sum of two of its columns lies on it again, while in (b) the line misses the origin and the sum of two of its columns lies off it

Checking a subset against definition 2.1.2 means checking three conditions. Two of them can be merged, which shortens every verification that follows.

Proposition 2.1.3: Criterion for a Subspace

Let $W \subseteq k^n$.

1. W is a subspace of k^n if and only if W is nonempty and $u + cv \in W$ for all $u, v \in W$ and all $c \in k$.
2. If W is a subspace of k^n, if $p \geq 1$, if $w_1, \ldots, w_p \in W$ and if $a_1, \ldots, a_p \in k$, then

$$a_1 w_1 + a_2 w_2 + \cdots + a_p w_p \in W$$

***Proof*:**

1. Suppose W is a subspace. Then $0 \in W$, so W is nonempty. Given $u, v \in W$ and $c \in k$, condition (3) of definition 2.1.2 puts cv in W and condition (2) then puts $u + cv$ in W.

   Suppose instead that W is nonempty and closed under the operation $u + cv$. Choose $w \in W$, which is possible since W is nonempty. Taking $u = v = w$ and $c = -1$ gives $w + (-1)w = 0 \in W$, so condition (1) holds. Given $v \in W$ and $c \in k$, taking $u = 0$ gives $0 + cv = cv \in W$, so condition (3) holds. Given $u, v \in W$, taking $c = 1$ gives $u + 1 \cdot v = u + v \in W$, so condition (2) holds. All three conditions hold, so W is a subspace.

2. Argue by induction on p. For $p = 1$ the column $a_1 w_1$ lies in W by condition (3) of definition 2.1.2. Suppose $p \geq 2$ and that every linear combination of $p - 1$ columns of W lies in W. Then $a_1 w_1 + \cdots + a_{p-1} w_{p-1}$ lies in W by the inductive hypothesis and $a_p w_p$ lies in W by condition (3), so their sum lies in W by condition (2). That sum is $a_1 w_1 + \cdots + a_p w_p$, which completes the induction and the proof.

Part (2) extends the two closures from a single operation to every finite combination of operations:

a subspace absorbs not just one addition and one scaling but every linear combination of its own elements. The first subspaces come from chapter 1, and there are many.

Example 2.1.4: The Solution Set of a Homogeneous System

Let $A \in M_{m\times n}(k)$ and let $W = \{x \in k^n \big| Ax = 0\}$ be the solution set of the homogeneous system $Ax = 0$ (definition 1.1.9). Then W is a subspace of k^n.

The column of n zeros solves the system, so W is nonempty. Let $u, v \in W$ and $c \in k$. By definition 1.1.7 the ith entry of Ax is $a_{i1}x_1 + \cdots + a_{in}x_n$, so the ith entry of $A(u + cv)$ is

$$\sum_{j=1}^{n} a_{ij}(u_j + cv_j) = \sum_{j=1}^{n} a_{ij}u_j + c\sum_{j=1}^{n} a_{ij}v_j$$

which is the ith entry of Au plus c times the ith entry of Av. Both are zero, so every entry of $A(u + cv)$ is zero and $u + cv \in W$. By proposition 2.1.3(1) the set W is a subspace of k^n.

A concrete instance: take

$$G = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 4 \\ 1 & 2 & 1 & 3 \end{pmatrix}$$

with $m = 3$ rows and $n = 4$ columns. The replacements $R_2 \mapsto R_2 - 2R_1$ and $R_3 \mapsto R_3 - R_1$ both produce the row $(0, 0, 1, 2)$, and $R_3 \mapsto R_3 - R_2$ then empties the third row:

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Both pivots equal 1 and each is alone in its column, so this is the reduced echelon form of G (definition 1.2.7) and $\text{rank}G = 2$. Its two nonzero rows record $x_1 + 2x_2 + x_4 = 0$ and $x_3 + 2x_4 = 0$, with free variables x_2 and x_4 (definition 1.3.5). Setting $x_2 = s$ and $x_4 = t$ gives $x_1 = -2s - t$ and $x_3 = -2t$, so the members of W are exactly the columns

$$x = s\begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t\begin{pmatrix} -1 \\ 0 \\ -2 \\ 1 \end{pmatrix}$$

as s and t range over k. Substituting $(-2, 1, 0, 0)^\top$ into the three rows of G gives $-2+2+0+0 = 0$, $-4 + 4 + 0 + 0 = 0$ and $-2 + 2 + 0 + 0 = 0$; substituting $(-1, 0, -2, 1)^\top$ gives $-1 + 0 + 0 + 1 = 0$, $-2 + 0 - 2 + 4 = 0$ and $-1 + 0 - 2 + 3 = 0$.

Every homogeneous system therefore produces a subspace, and the trichotomy of corollary 1.1.13 reads on it as a statement about how large that subspace is: either it is $\{0\}$, or it is infinite. The solution set of $Ax = b$ with $b \neq 0$ is never a subspace, since a subspace contains the zero column while $A0 = 0 \neq b$.

The description of W in example 2.1.4 is of a different kind from the description of P in example 2.1.1. The set P was given by a condition its members must satisfy; the set W was given by a formula that manufactures its members, namely every combination $sw_1 + tw_2$ of two fixed columns. The second kind of description is available for any finite list of columns and deserves its own notation.

Definition 2.1.5: Linear Combination and Span

Let $v_1, \ldots, v_p \in k^n$ and let $a_1, \ldots, a_p \in k$. The column

$$a_1v_1 + a_2v_2 + \cdots + a_pv_p$$

is the *linear combination* of $v_1, \ldots, v_p$ with coefficients $a_1, \ldots, a_p$. The *span* of $v_1, \ldots, v_p$ is the set of all of them,

$$\operatorname{span}\{v_1, \ldots, v_p\} = \{a_1v_1 + \cdots + a_pv_p \big| a_1, \ldots, a_p \in k\}$$

The braces enclose the columns being combined; parentheses serve as well, and $\operatorname{span}(v_1, \ldots, v_p)$ denotes the same set. The span of the empty list is $\{0\}$. If $W = \operatorname{span}\{v_1, \ldots, v_p\}$, the list $v_1, \ldots, v_p$ is said to *span* W.

The notation records the list of columns and nothing more; the field is whichever of $\mathbb{R}$ and $\mathbb{C}$ is in force, and the choice matters, since a column can be a $\mathbb{C}$ multiple of another without being an $\mathbb{R}$ multiple of it. The coefficients range over all of k, so a span is never a finite set unless it is $\{0\}$. In this language example 2.1.4 says that the solution set of $Gx = 0$ is $\operatorname{span}\{(-2, 1, 0, 0)^\top, (-1, 0, -2, 1)^\top\}$. That set was shown to be a subspace by checking the criterion, and the same is true of every span.

Proposition 2.1.6: The Span Is the Smallest Subspace Containing a List

Let $v_1, \ldots, v_p \in k^n$ and write $S = \operatorname{span}\{v_1, \ldots, v_p\}$.

1. S is a subspace of k^n, and $v_i \in S$ for each i.
2. If W is a subspace of k^n with $v_1, \ldots, v_p \in W$, then $S \subseteq W$.

Proof:

1. Taking every coefficient equal to 0 gives $0v_1 + \cdots + 0v_p = 0$, so $0 \in S$ and S is nonempty. Let $u, w \in S$ and $c \in k$, say $u = a_1v_1 + \cdots + a_pv_p$ and $w = b_1v_1 + \cdots + b_pv_p$ with all $a_i, b_i \in k$. Collecting the multiples of each v_i gives

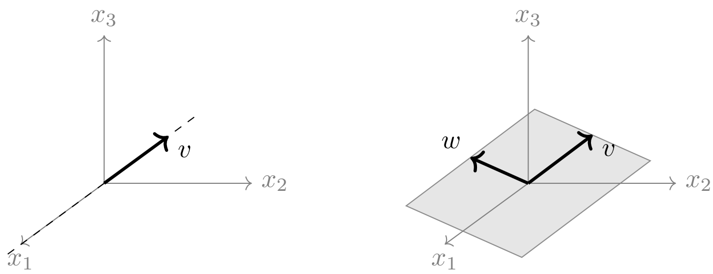
$$u + cw = (a_1 + cb_1)v_1 + \cdots + (a_p + cb_p)v_p$$

which is again a linear combination of $v_1, \ldots, v_p$, so $u + cw \in S$. By proposition 2.1.3(1) the set S is a subspace. Taking the coefficient of v_i equal to 1 and all others equal to 0 exhibits v_i as a linear combination, so $v_i \in S$. When $p = 0$ the list is empty and $S = \{0\}$, which is a subspace and has nothing to contain.

2. Let W be a subspace containing every v_i and let $x \in S$, say $x = a_1v_1 + \cdots + a_pv_p$. By proposition 2.1.3(2) this linear combination of elements of W lies in W, so $x \in W$ and $S \subseteq W$, as we sought to show.

Part (2) says that no subspace containing $v_1, \ldots, v_p$ can be smaller than their span, so $\operatorname{span}\{v_1, \ldots, v_p\}$ is the smallest one. Spans also settle the question chapter 1 began with. Writing $c_1, \ldots, c_n$ for the columns of $A \in M_{m\times n}(k)$, the column reading $Ax = x_1c_1 + \cdots + x_nc_n$ recorded in section 1.2

says that a solution of $Ax = b$ is precisely a list of coefficients expressing b as a linear combination of the columns. So $Ax = b$ is consistent (definition 1.1.6) if and only if $b \in \operatorname{span}\{c_1, \dots, c_n\}$, and theorem 1.3.3 is a test for membership in that span.



(a) (b)

Figure 2.2: In $\mathbb{R}^3$ the span of a single nonzero column is the line through the origin carrying it, and the span of two columns not on a common line through the origin is the shaded plane

The running matrix of example 1.2.9 shows that a list can carry a vector contributing nothing to its span.

Example 2.1.7: The Span of the Columns of the Running Matrix

The running matrix

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 1 & 2 & 5 \\ 3 & 4 & 11 \end{pmatrix}$$

of example 1.2.9 has columns $c_1 = (1, 2, 1, 3)^\top$, $c_2 = (1, 3, 2, 4)^\top$ and $c_3 = (3, 8, 5, 11)^\top$ in $\mathbb{R}^4$, and $c_3 = c_1 + 2c_2$, since $1 + 2 = 3$, $2 + 6 = 8$, $1 + 4 = 5$ and $3 + 8 = 11$. Substituting this relation into a general combination of the three columns gives

$$a_1c_1 + a_2c_2 + a_3c_3 = a_1c_1 + a_2c_2 + a_3(c_1 + 2c_2) = (a_1 + a_3)c_1 + (a_2 + 2a_3)c_2$$

so every combination of c_1, c_2, c_3 is already a combination of c_1 and c_2. The reverse inclusion needs only $a_3 = 0$, so

$$\operatorname{span}\{c_1, c_2, c_3\} = \operatorname{span}\{c_1, c_2\}$$

and the third column contributes nothing.

Membership in this subspace is decided by elimination. The column $b = (2, 5, 3, 7)^\top$ lies in it, since $b = c_1 + c_2$ entry by entry. The column $b' = (1, 0, 0, 0)^\top$ does not. Applying to $(A|b')$ the replacements $R_2 \mapsto R_2 - 2R_1$, $R_3 \mapsto R_3 - R_1$ and $R_4 \mapsto R_4 - 3R_1$ gives

$$\left(\begin{array}{ccc|c} 1 & 1 & 3 & 1 \\ 0 & 1 & 2 & -2 \\ 0 & 1 & 2 & -1 \\ 0 & 1 & 2 & -3 \end{array}\right)$$

and then $R_3 \mapsto R_3 - R_2$, $R_4 \mapsto R_4 - R_2$ and $R_4 \mapsto R_4 + R_3$ give

$$\left(\begin{array}{ccc|c} 1 & 1 & 3 & 1 \\ 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

This is in echelon form with a pivot in the last column, so $\operatorname{rank}(A|b') = 3$ while $\operatorname{rank} A = 2$, and the system $Ax = b'$ is inconsistent by theorem 1.3.3. Hence $b' \notin \operatorname{span}\{c_1, c_2\}$, and this span is a subspace of $\mathbb{R}^4$ smaller than $\mathbb{R}^4$ itself.

The relation $c_3 = c_1 + 2c_2$ is what made the third column droppable, and it is the same relation as the solution $(1, 2, -1)^\top$ of $Ax = 0$ found in example 1.2.9: the coefficients of a relation among the columns are the entries of a solution of the homogeneous system. Deciding whether a list carries such a relation is therefore an elimination.

Example 2.1.8: Testing Two Lists for a Relation

1. Take $u_1 = (1, 2, 1)^\top$, $u_2 = (2, 3, 0)^\top$ and $u_3 = (1, 1, 2)^\top$ in $\mathbb{R}^3$. A relation $a_1u_1 + a_2u_2 + a_3u_3 = 0$ is a solution of $Ex = 0$, where E is the matrix with these three columns. The replacements $R_2 \mapsto R_2 - 2R_1$ and $R_3 \mapsto R_3 - R_1$ give

$$E = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 1 & 0 & 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 1 \\ 0 & -1 & -1 \\ 0 & -2 & 1 \end{pmatrix}$$

   and $R_3 \mapsto R_3 - 2R_2$ gives third row $(0, 0, 3)$, so

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & 3 \end{pmatrix}$$

   is in echelon form with pivots in all three columns. So $\operatorname{rank} E = 3$, which is the number of columns of E, and theorem 1.3.6(2) gives exactly one solution of $Ex = 0$. That solution is the zero column, so no relation among u_1, u_2, u_3 has a nonzero coefficient.

2. Change the third column and take $w_1 = (1, 2, 1)^\top$, $w_2 = (2, 3, 0)^\top$ and $w_3 = (0, 1, 2)^\top$. With F the matrix of these columns, the replacements $R_2 \mapsto R_2 - 2R_1$ and $R_3 \mapsto R_3 - R_1$ give

$$F = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 3 & 1 \\ 1 & 0 & 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \\ 0 & -2 & 2 \end{pmatrix}$$

   and $R_3 \mapsto R_3 - 2R_2$ empties the third row. The scaling $R_2 \mapsto -R_2$ and the replacement $R_1 \mapsto R_1 - 2R_2$ then produce the reduced echelon form

$$\begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

   with $\operatorname{rank} F = 2$ and x_3 free. The two equations are $x_1 + 2x_3 = 0$ and $x_2 - x_3 = 0$, so $x_3 = 1$ gives $x = (-2, 1, 1)^\top$ and hence $-2w_1 + w_2 + w_3 = 0$. Entry by entry this reads $-2 + 2 + 0 = 0$, $-4 + 3 + 1 = 0$ and $-2 + 0 + 2 = 0$. Rearranged, $w_3 = 2w_1 - w_2$.

3. The same test runs over $\mathbb{C}$. Take $v_1 = (1, i)^\top$, $v_2 = (i, -1)^\top$ and $v_3 = (1, 1)^\top$ in $\mathbb{C}^2$. Then $iv_1 = (i, i^2)^\top = (i, -1)^\top = v_2$, so $iv_1 - v_2 = 0$ is a relation with nonzero coefficients. For v_1 and v_3 the single replacement $R_2 \mapsto R_2 - iR_1$ carries $\left(\begin{smallmatrix} 1 & 1 \\ i & 1 \end{smallmatrix}\right)$ to $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1-i \end{smallmatrix}\right)$, which has pivots in both columns, so the only relation between v_1 and v_3 has both coefficients zero.

Two of the four lists tested admit a relation with a nonzero coefficient and two do not. The two cases have names.

Definition 2.1.9: Linear Independence

A list $v_1, \ldots, v_p$ of columns in k^n is *linearly independent* if the only scalars $a_1, \ldots, a_p \in k$ with

$$a_1 v_1 + a_2 v_2 + \cdots + a_p v_p = 0$$

are $a_1 = a_2 = \cdots = a_p = 0$. Otherwise the list is *linearly dependent*, and a list of scalars $a_1, \ldots, a_p$, not all zero, satisfying the displayed equation is a *linear dependence relation* among $v_1, \ldots, v_p$. The empty list is linearly independent.

By example 2.1.8 the list u_1, u_2, u_3 is linearly independent and the list w_1, w_2, w_3 is linearly dependent, with dependence relation $(-2, 1, 1)$. The columns of the running matrix are linearly dependent, with dependence relation $(1, 2, -1)$. In each case the test was a single elimination: by the column reading, the list of columns of a matrix A is linearly independent exactly when $Ax = 0$ has only the zero solution, which by theorem 1.3.6 happens exactly when $\operatorname{rank} A$ equals the number of columns. Dependence is the statement that one member of the list is already available from the others.

Proposition 2.1.10: A Dependent List Has a Redundant Member

Let $v_1, \ldots, v_p \in k^n$ with $p \geq 1$. The list is linearly dependent if and only if there is an index i with

$$v_i \in \operatorname{span}\{v_j \,\big|\, j \neq i\}$$

the span on the right being taken over the list with v_i deleted. For any such i,

$$\operatorname{span}\{v_1, \ldots, v_p\} = \operatorname{span}\{v_j \,\big|\, j \neq i\}$$

***Proof*:**
Suppose the list is linearly dependent, and let $a_1, \ldots, a_p$ be a dependence relation, so that $a_1 v_1 + \cdots + a_p v_p = 0$ and $a_i \neq 0$ for some index i. Fix that i. Subtracting $a_i v_i$ from both sides and multiplying by $-a_i^{-1}$, which exists because $a_i \neq 0$, gives

$$v_i = \sum_{j \neq i} (-a_i^{-1} a_j) v_j$$

a linear combination of the list with v_i deleted, so $v_i \in \operatorname{span}\{v_j \,\big|\, j \neq i\}$. When $p = 1$ the sum on the right is empty and the equation reads $v_1 = 0$, which is consistent with the convention that the span of the empty list is $\{0\}$.

Suppose conversely that $v_i = \sum_{j \neq i} b_j v_j$ for some index i and scalars b_j. Setting $a_i = -1$ and $a_j = b_j$ for $j \neq i$ gives $a_1 v_1 + \cdots + a_p v_p = \sum_{j \neq i} b_j v_j - v_i = 0$, and the coefficient $a_i = -1$ is nonzero, so the list is linearly dependent.

For the last claim, fix an index i with $v_i = \sum_{j \neq i} b_j v_j$. The inclusion $\operatorname{span}\{v_j \big| j \neq i\} \subseteq \operatorname{span}\{v_1, \dots, v_p\}$ holds because a combination of the shorter list is a combination of the whole list with coefficient 0 on v_i. For the other inclusion, take $x = a_1 v_1 + \cdots + a_p v_p$ and substitute the expression for v_i:

$$x = \sum_{j \neq i} a_j v_j + a_i \sum_{j \neq i} b_j v_j = \sum_{j \neq i} (a_j + a_i b_j) v_j$$

which lies in $\operatorname{span}\{v_j \big| j \neq i\}$. The two inclusions give equality, completing the proof.

Deleting a redundant member shortens a list without changing what it spans. The shortened list can be tested again, and since each deletion removes one member the process stops after finitely many steps, at a list with no redundant member spanning what the original list spanned. By proposition 2.1.10 such a list is linearly independent. So every subspace of the form $\operatorname{span}\{v_1, \dots, v_p\}$ is spanned by a linearly independent list. In example 2.1.7 one deletion suffices: the columns of the running matrix span $\operatorname{span}\{c_1, c_2\}$, and c_1, c_2 are linearly independent, since a relation $a_1 c_1 + a_2 c_2 = 0$ has first entry $a_1 + a_2 = 0$ and second entry $2a_1 + 3a_2 = 0$, and subtracting twice the first from the second gives $a_2 = 0$ and then $a_1 = 0$.

Two subspaces of k^n can also be added together, and the case worth naming is the one in which the two summands of a column can be recovered from the column. Take $P = \{x \in \mathbb{R}^3 \big| x_1 + x_2 + x_3 = 0\}$ from example 2.1.1 and $L = \operatorname{span}\{(1,1,1)^\top\}$, a subspace by proposition 2.1.6(1). A column $t(1,1,1)^\top$ of L lies in P exactly when $3t = 0$, that is when $t = 0$, so the two subspaces meet only in the zero column. Every column of $\mathbb{R}^3$ is a sum of one from each: given x, put $t = (x_1 + x_2 + x_3)/3$ and $u = x - t(1,1,1)^\top$, whose entries sum to $(x_1 + x_2 + x_3) - 3t = 0$, so that $u \in P$ and $x = u + t(1,1,1)^\top$. For $x = (1,2,6)^\top$ this gives $t = 3$ and $u = (1,2,6)^\top - (3,3,3)^\top = (-2,-1,3)^\top$, whose entries sum to 0.

Definition 2.1.11: Sum and Direct Sum of Subspaces

Let U and W be subspaces of k^n. Their *sum* is the set

$$U + W = \{u + w \big| u \in U,\ w \in W\}$$

of all columns obtained by adding a member of U to a member of W. The sum is *direct* if $U \cap W = \{0\}$, and it is then written $U \oplus W$. A subspace V of k^n is the *direct sum* of U and W, written $V = U \oplus W$, if $V = U + W$ and $U \cap W = \{0\}$.

The computation above says $\mathbb{R}^3 = P \oplus L$. The condition $U \cap W = \{0\}$ looks like a restriction on how the two subspaces sit relative to one another, and its content is that the decomposition it permits is unique.

Proposition 2.1.12: Directness and Unique Decomposition

Let U and W be subspaces of k^n.

1. $U + W$ is a subspace of k^n containing both U and W, and it is contained in every subspace of k^n containing both.
2. Every element of $U + W$ has exactly one expression $u + w$ with $u \in U$ and $w \in W$ if and only if $U \cap W = \{0\}$.

***Proof*:**

1. The zero column is $0 + 0$ with $0 \in U$ and $0 \in W$, so $U + W$ is nonempty. Let $x = u + w$ and $x' = u' + w'$ lie in $U + W$, with $u, u' \in U$ and $w, w' \in W$, and let $c \in k$. Rearranging entrywise,
$$x + cx' = (u + cu') + (w + cw')$$
and $u + cu' \in U$ and $w + cw' \in W$ by proposition 2.1.3(1) applied to each subspace. So $x + cx' \in U + W$, and $U + W$ is a subspace by that criterion. It contains U, since $u = u + 0$, and W, since $w = 0 + w$. If Z is a subspace with $U \subseteq Z$ and $W \subseteq Z$, then for $u \in U$ and $w \in W$ both u and w lie in Z, so $u + w \in Z$ by condition (2) of definition 2.1.2, giving $U + W \subseteq Z$.
2. Suppose $U \cap W = \{0\}$ and let $u + w = u' + w'$ with $u, u' \in U$ and $w, w' \in W$. Then $u + (-1)u' = w' + (-1)w$. The left side lies in U and the right side lies in W, and they are equal, so this common column lies in $U \cap W$ and is therefore 0. Hence $u = u'$ and $w = w'$, and the expression is unique.

   Suppose instead that every element of $U + W$ has exactly one such expression, and let $x \in U \cap W$. Then $x = x + 0$ with $x \in U$ and $0 \in W$, and also $x = 0 + x$ with $0 \in U$ and $x \in W$. These are two expressions of the same element, so uniqueness forces $x = 0$. Thus $U \cap W = \{0\}$, completing the proof.

Definition 2.1.13: Direct Sum of Several Subspaces

Let $W_1, \dots, W_m$ be subspaces of k^n. Their *sum* is the set
$$W_1 + \cdots + W_m = \{w_1 + \cdots + w_m \big| w_i \in W_i \text{ for each } i\}$$
A subspace V of k^n is the *direct sum* of $W_1, \dots, W_m$, written $V = W_1 \oplus \cdots \oplus W_m$, if every $x \in V$ has exactly one expression $x = w_1 + \cdots + w_m$ with $w_i \in W_i$ for each i.

For $m = 2$ this agrees with definition 2.1.11 by proposition 2.1.12(2). For larger m the criterion is stated in terms of the zero vector rather than in terms of intersections.

Proposition 2.1.14: Directness of a Sum of Several Subspaces

Let $W_1, \dots, W_m$ be subspaces of k^n and let $V = W_1 + \cdots + W_m$. Then $V = W_1 \oplus \cdots \oplus W_m$ if and only if the only way to write $0 = w_1 + \cdots + w_m$ with $w_i \in W_i$ is with $w_1 = \cdots = w_m = 0$.

***Proof*:**
Suppose the sum is direct and $0 = w_1 + \cdots + w_m$ with $w_i \in W_i$. The zero vector also has the expression $0 = 0 + \cdots + 0$ with each summand in its W_i, and uniqueness forces the two expressions to agree, so every $w_i = 0$.

Conversely, suppose the only such expression of 0 is the trivial one, and let $x \in V$ have two expressions $x = w_1 + \cdots + w_m$ and $x = w_1' + \cdots + w_m'$ with $w_i, w_i' \in W_i$. Subtracting, $0 = (w_1 - w_1') + \cdots + (w_m - w_m')$, and $w_i - w_i' \in W_i$ by proposition 2.1.3(1). So every $w_i - w_i' = 0$, that is $w_i = w_i'$ for each i, and the expression is unique.

For $m \geq 3$ the pairwise conditions $W_i \cap W_{i'} = \{0\}$ do *not* suffice. The three lines in $\mathbb{R}^2$ spanned by $(1,0)^\top$, $(0,1)^\top$ and $(1,1)^\top$ meet one another only at the origin, yet $(1,0)^\top + (0,1)^\top - (1,1)^\top = 0$ is a nontrivial expression of the zero vector, so their sum is not direct.

A direct sum decomposition $V = U \oplus W$ therefore assigns to each $x \in V$ a pair, its U part and its W part, with no ambiguity. In the decomposition $\mathbb{R}^3 = P \oplus L$ the column $(1,2,6)^\top$ has P part $(-2,-1,3)^\top$ and L part $(3,3,3)^\top$, and no other pair from P and L sums to it. The condition $U \cap W = \{0\}$ cannot be dropped: taking $U = P$ and $W = \operatorname{span}\{(1,-1,0)^\top\}$, which lies inside P, gives $U + W = P$, and the column $(1,-1,0)^\top$ of P is both $(1,-1,0)^\top + 0$ and $0 + (1,-1,0)^\top$.

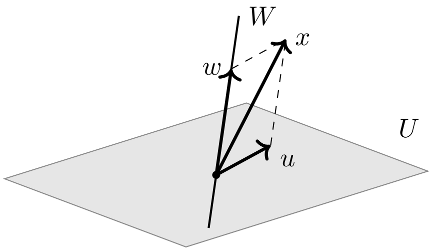


Figure 2.3: A plane U and a line W meeting only at the origin, with the unique decomposition $x = u + w$ of a column of $U \oplus W$

Exercise 2.1.1

1. Decide which of the following subsets of $\mathbb{R}^4$ are subspaces. For each that is, verify the conditions of definition 2.1.2; for each that is not, name a condition that fails and exhibit a specific failure.

$$W_1 = \{x \mid x_1 - 2x_2 + x_4 = 0\} \quad W_2 = \{x \mid x_1 + x_2 = x_3 + 1\}$$
$$W_3 = \{x \mid x_1 x_4 = x_2 x_3\} \quad W_4 = \{x \mid x_1 = x_2 = x_3 = x_4\}$$

2. Let $v_1 = (1,1,1)^\top$ and $v_2 = (1,2,3)^\top$ in $\mathbb{R}^3$. Decide whether $b = (3,4,5)^\top$ lies in $\operatorname{span}\{v_1, v_2\}$ and whether $b' = (1,0,2)^\top$ does, in each case by elimination, and write down the coefficients when they exist.

3. Decide whether the list $v_1 = (1,0,1)^\top$, $v_2 = (2,1,3)^\top$, $v_3 = (1,1,2)^\top$ in $\mathbb{R}^3$ is linearly independent. If it is dependent, write down a dependence relation, name an index i for which proposition 2.1.10 applies, and give a linearly independent list with the same span.

4. Let
$$M = \begin{pmatrix} 1 & 1 & 2 & 1 \\ 2 & 2 & 5 & 4 \\ 1 & 1 & 3 & 3 \end{pmatrix}$$
Write the solution set of $Mx = 0$ as the span of an explicit list of columns of $\mathbb{R}^4$, and show that the list you produce is linearly independent.

5. Let $U = \{x \in \mathbb{R}^3 \big| x_3 = 0\}$ and $W = \operatorname{span}\{(1,2,3)^\top\}$. Show that $\mathbb{R}^3 = U \oplus W$, and compute the U part and the W part of $x = (4,5,6)^\top$.

6. Let $v_1, \ldots, v_p \in k^n$ with $p > n$. Show that the list is linearly dependent. (Assemble the columns into a matrix and apply corollary 1.3.8.)

7. Let U and W be subspaces of k^n. Show that $U \cap W$ is a subspace of k^n, and that $U \cup W$ is a subspace of k^n if and only if $U \subseteq W$ or $W \subseteq U$.

8. Let $v_1, \ldots, v_p$ and $w_1, \ldots, w_q$ be lists of columns in k^n. Show that $\operatorname{span}\{v_1, \ldots, v_p\} \subseteq \operatorname{span}\{w_1, \ldots, w_q\}$ if and only if $v_i \in \operatorname{span}\{w_1, \ldots, w_q\}$ for every i. Then use this to show that
$$\operatorname{span}\{(1,1,0)^\top, (0,1,1)^\top\} = \operatorname{span}\{(1,0,-1)^\top, (1,2,1)^\top\}$$

2.2 Basis, Dimension, and Coordinates

Section 2.1 described a subspace as the span of a list of vectors. Nothing in that description forces the list to be short, and a list can carry vectors that contribute nothing to its span. This section cuts a spanning list down to one that is linearly independent, shows that every such list for a given subspace has the same length, and uses one of them to attach to each vector of the subspace a column of scalars that determines it.

Two facts about subspaces are used throughout, and both come from section 2.1. Recall that a subspace of k^n is closed under addition and under scalar multiplication (definition 2.1.2), so by proposition 2.1.3 it contains every linear combination of finitely many of its own elements; consequently, if each of $x_1, \ldots, x_s$ lies in a subspace U, then $\operatorname{span}\{x_1, \ldots, x_s\} \subseteq U$. Recall also that a linearly dependent list has a member lying in the span of the others, and that deleting such a member leaves the span unchanged (proposition 2.1.10).

Take the three vectors
$$u_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad u_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \qquad u_3 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$
in $\mathbb{R}^3$ and let $U = \operatorname{span}\{u_1, u_2, u_3\}$, the set of all combinations $au_1 + bu_2 + cu_3$ with $a, b, c \in \mathbb{R}$ (definition 2.1.5). Adding u_1 and u_2 entry by entry gives u_3, so u_3 lies in the span of the other two and deleting it leaves the span unchanged: $U = \operatorname{span}\{u_1, u_2\}$. The third vector was carried along at no benefit, and its presence had a visible cost. The vector u_3 is both $1u_1 + 1u_2 + 0u_3$ and $0u_1 + 0u_2 + 1u_3$, so a vector of U can be written as a combination of the three in more than one way.

The shortened list has neither defect. A combination of u_1 and u_2 is
$$au_1 + bu_2 = \begin{pmatrix} a \\ a+b \\ b \end{pmatrix}$$

whose first and third entries are a and b. Such a combination is the zero vector only when $a = b = 0$, so the list u_1, u_2 is linearly independent (definition 2.1.9); and two combinations $au_1 + bu_2$ and $a'u_1 + b'u_2$ that are equal have equal first entries and equal third entries, so $a = a'$ and $b = b'$ and the two combinations were written the same way. Every vector of U therefore has exactly one expression in terms of u_1 and u_2. The subspace U itself is the plane $x_1 - x_2 + x_3 = 0$: the displayed combination satisfies $a - (a + b) + b = 0$, and conversely a vector with $x_1 - x_2 + x_3 = 0$ is $x_1 u_1 + x_3 u_2$.

A list with these two properties gets a name.

Definition 2.2.1: Basis

Let W be a subspace of k^n. A *basis* of W is an ordered list $B = (v_1, \dots, v_p)$ of vectors of W that is linearly independent and satisfies

$$\operatorname{span}\{v_1, \dots, v_p\} = W$$

The list is ordered, so (v_1, v_2) and (v_2, v_1) count as different bases of the same subspace.

The empty list is admitted, with the conventions that a sum of no vectors is the zero vector, so that the span of the empty list is $\{0\}$, and that the empty list is linearly independent, since it admits no relation at all. The subspace $\{0\}$ therefore has the empty list as a basis.

Example 2.2.2: The Standard Basis and Two Small Bases

1. Let $e_1, \dots, e_n$ be the standard vectors of k^n, so that e_j has entry 1 in position j and 0 in every other position. For $x \in k^n$ the combination $x_1 e_1 + \cdots + x_n e_n$ has entry x_j in position j, hence equals x, so $\operatorname{span}\{e_1, \dots, e_n\} = k^n$. The same computation with $x = 0$ shows that $c_1 e_1 + \cdots + c_n e_n = 0$ forces $c_j = 0$ for every j, so the list is linearly independent. Thus $(e_1, \dots, e_n)$ is a basis of k^n, the *standard basis*.
2. The line $N = \{\, (t, -2t, t)^\top \,|\, t \in \mathbb{R} \,\}$ of example 1.1.10 is $\operatorname{span}\{(1, -2, 1)^\top\}$, since $(t, -2t, t)^\top = t(1, -2, 1)^\top$. A list consisting of a single nonzero vector v is linearly independent, because $cv = 0$ with $c \neq 0$ gives $v = c^{-1}(cv) = 0$. So $\big((1, -2, 1)^\top\big)$ is a basis of N, with one vector in it.
3. The subspace $\{0\}$ of k^n has the empty list as a basis, by the convention just fixed.

Of the two properties in definition 2.2.1, spanning says that every vector of W can be reached by a combination and independence says that no vector is reached twice. The computation with u_1 and u_2 above exhibited both, and the link between them is general.

Theorem 2.2.3: Independence Is Uniqueness of Expression

Let $v_1, \dots, v_p$ be vectors of k^n and put $W = \operatorname{span}\{v_1, \dots, v_p\}$. The list $(v_1, \dots, v_p)$ is a basis of W if and only if every $w \in W$ can be written as

$$w = c_1 v_1 + c_2 v_2 + \cdots + c_p v_p$$

for exactly one list $c_1, \dots, c_p$ of scalars in k.

***Proof*:**
Suppose first that the list is a basis of W. Let $w \in W$. Since the list spans W, at least one expression $w = c_1v_1 + \cdots + c_pv_p$ exists. Suppose $w = c'_1v_1 + \cdots + c'_pv_p$ as well. Subtracting the second expression from the first gives

$$(c_1 - c'_1)v_1 + \cdots + (c_p - c'_p)v_p = 0$$

and linear independence forces $c_j - c'_j = 0$ for every j, so $c_j = c'_j$ and the two expressions are the same one. Exactly one expression exists.
Suppose instead that every $w \in W$ has exactly one such expression. The list spans W by the definition of W. For independence, let $c_1, \ldots, c_p$ be scalars with $c_1v_1 + \cdots + c_pv_p = 0$. The vector 0 lies in W and has the expression $0v_1 + \cdots + 0v_p$, so by the uniqueness hypothesis applied to $w = 0$ these two expressions coincide, giving $c_j = 0$ for every j. The list is therefore linearly independent and is a basis of W, as we sought to show.

The scalars produced by the theorem are what make a basis useful: they turn a vector of W, which may be an unwieldy object, into a list of p numbers that determines it.

Definition 2.2.4: Coordinates and the Coordinate Column

Let W be a subspace of k^n and let $B = (v_1, \ldots, v_p)$ be a basis of W. For $w \in W$, the unique scalars $c_1, \ldots, c_p$ with

$$w = c_1v_1 + \cdots + c_pv_p$$

given by theorem 2.2.3 are the *coordinates of w with respect to B*, and the column

$$[w]_B = \begin{pmatrix} c_1 \\ \vdots \\ c_p \end{pmatrix} \in k^p$$

is the *coordinate column* of w with respect to B.

Taking coordinates does more than label the vectors of W: it carries the two operations of W across to k^p intact, and that is what makes it useful.

Proposition 2.2.5: The Coordinate Map Is a Bijection Respecting Both Operations

Let $B = (v_1, \ldots, v_p)$ be a basis of a subspace $W \subseteq k^n$. The assignment $w \mapsto [w]_B$ is a bijection from W to k^p, and for all $w, w' \in W$ and $t \in k$,

$$[w + w']_B = [w]_B + [w']_B, \qquad [tw]_B = t[w]_B$$

***Proof*:**
The assignment reaches every column, since any $c_1, \ldots, c_p \in k$ produce the vector $c_1v_1 + \cdots + c_pv_p$ of W; and it separates distinct vectors, since two vectors with the same coordinate column are the same combination of $v_1, \ldots, v_p$. So it is a bijection. Adding the expressions of w and w' term by term gives an expression for $w + w'$ as a combination of $v_1, \ldots, v_p$, and that expression is unique by theorem 2.2.3, so $[w + w']_B = [w]_B + [w']_B$. Multiplying the expression of w by t

gives in the same way $[tw]_B = t[w]_B$, completing the proof.

A choice of basis therefore replaces the subspace W by the concrete space k^p, in a way that carries sums to sums and scalar multiples to scalar multiples. The price is that the replacement depends on the basis chosen, and different bases assign different coordinate columns to the same vector.

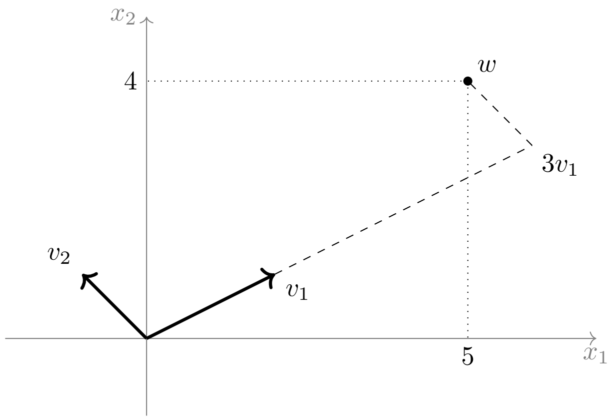


Figure 2.4: The vector w has coordinate column $(5,4)^\top$ with respect to the standard basis and $(3,1)^\top$ with respect to the basis $B = (v_1, v_2)$ with $v_1 = (2,1)^\top$ and $v_2 = (-1,1)^\top$

Example 2.2.6: Coordinates in Two Bases of $\mathbb{R}^3$

Let
$$v_1 = \begin{pmatrix}1\\0\\1\end{pmatrix}, \qquad v_2 = \begin{pmatrix}1\\1\\0\end{pmatrix}, \qquad v_3 = \begin{pmatrix}0\\1\\1\end{pmatrix}$$
and put $B = (v_1, v_2, v_3)$. A combination is
$$av_1 + bv_2 + cv_3 = \begin{pmatrix}a+b\\b+c\\a+c\end{pmatrix}$$
Setting this equal to 0 gives $a+b=0$, $b+c=0$ and $a+c=0$; adding the three equations gives $2(a+b+c)=0$, so $a+b+c=0$, and subtracting each of the three original equations from this in turn gives $c=0$, $a=0$ and $b=0$. The list is therefore linearly independent. Setting the combination equal to an arbitrary $x \in \mathbb{R}^3$ gives the system with augmented matrix
$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & x_1 \\ 0 & 1 & 1 & x_2 \\ 1 & 0 & 1 & x_3 \end{array}\right)$$
whose coefficient columns are v_1, v_2 and v_3. The operations $R_3 \mapsto R_3 - R_1$ and then $R_3 \mapsto R_3 + R_2$ carry the coefficient matrix to the echelon form with rows $(1,1,0)$, $(0,1,1)$ and $(0,0,2)$, which has three pivots, so its rank is 3 (definition 1.3.2). The system is consistent for every x by corollary 1.3.7, so the list spans $\mathbb{R}^3$ and B is a basis.

Take $w = (1, 2, 3)^\top$. The two operations just named turn the augmented matrix into

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 2 & 4 \end{array}\right)$$

since the last column becomes $(1, 2, 3-1)^\top = (1, 2, 2)^\top$ and then $(1, 2, 2+2)^\top = (1, 2, 4)^\top$. The last row gives $2c = 4$, so $c = 2$; the second gives $b + 2 = 2$, so $b = 0$; the first gives $a + 0 = 1$, so $a = 1$. Hence

$$[w]_B = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

and the check $1v_1 + 0v_2 + 2v_3 = (1, 2, 3)^\top$ confirms it. With respect to the standard basis $E = (e_1, e_2, e_3)$ the same vector has $[w]_E = (1, 2, 3)^\top$, its own list of entries. Reordering B to $B' = (v_3, v_1, v_2)$ permutes the coordinates the same way, giving $[w]_{B'} = (2, 1, 0)^\top$.

The subspace U of the opening computation had a basis with two vectors, and $\mathbb{R}^3$ has the standard basis with three. Nothing so far rules out U having a second basis with five vectors in it, and until that is excluded the number of vectors in a basis is a property of the basis rather than of the subspace. What excludes it is a comparison between spanning lists and independent lists, in which the spanning list is always the longer of the two. The comparison is proved by trading the vectors of the spanning list for the vectors of the independent list one at a time.

Lemma 2.2.7: An Independent List Is No Longer Than a Spanning List

Let $w_1, \dots, w_q$ be vectors of k^n and put $W = \operatorname{span}\{w_1, \dots, w_q\}$. Let $v_1, \dots, v_p$ be linearly independent vectors, all of them in W. Then $p \le q$, and after renumbering $w_1, \dots, w_q$,

$$W = \operatorname{span}\{v_1, \dots, v_p, w_{p+1}, \dots, w_q\}$$

***Proof*:**
For $0 \le i \le \min(p, q)$ write L_i for the list $v_1, \dots, v_i, w_{i+1}, \dots, w_q$, formed by taking the first i vectors of the independent list followed by the last $q - i$ vectors of the spanning list. The claim is that for each such i the vectors $w_1, \dots, w_q$ can be renumbered so that $\operatorname{span} L_i = W$, and the proof is by induction on i.
For $i = 0$ the list L_0 is $w_1, \dots, w_q$ itself and $\operatorname{span} L_0 = W$ by hypothesis, with no renumbering needed. Suppose $0 \le i < p$ and that the vectors have been renumbered so that $\operatorname{span} L_i = W$. The vector v_{i+1} lies in W, so there are scalars with

$$v_{i+1} = a_1 v_1 + \cdots + a_i v_i + b_{i+1} w_{i+1} + \cdots + b_q w_q$$

Suppose every b_j in this expression were zero, a case that includes $i = q$, where the second group of terms is absent. Then $v_{i+1} = a_1 v_1 + \cdots + a_i v_i$, so

$$a_1 v_1 + \cdots + a_i v_i + (-1) v_{i+1} = 0$$

which is a relation among $v_1, \dots, v_p$ whose coefficient on v_{i+1} is -1 and so is not the trivial one, contradicting linear independence. Hence $i < q$ and some b_j with $i + 1 \le j \le q$ is nonzero. Renumber $w_{i+1}, \dots, w_q$ so that $b_{i+1} \ne 0$; this permutes the last $q - i$ entries of L_i and so changes neither spanL_i nor W. Solving the displayed expression for w_{i+1}, which is legitimate because $b_{i+1} \neq 0$, gives

$$w_{i+1} = b_{i+1}^{-1}\big(v_{i+1} - a_1 v_1 - \cdots - a_i v_i - b_{i+2} w_{i+2} - \cdots - b_q w_q\big)$$

so w_{i+1} lies in spanL_{i+1}.
Every vector of L_i now lies in spanL_{i+1}: the vectors $v_1, \ldots, v_i$ and $w_{i+2}, \ldots, w_q$ belong to L_{i+1} outright, and w_{i+1} was just exhibited. A span is a subspace (proposition 2.1.6), so it contains every linear combination of vectors lying in it, and therefore $W = \text{span}L_i \subseteq \text{span}L_{i+1}$. In the other direction every vector of L_{i+1} lies in W, since the v's do by hypothesis and the w's do by the definition of W, so span$L_{i+1} \subseteq W$. The two inclusions give span$L_{i+1} = W$, completing the induction.
If $p = 0$ the inequality $p \leq q$ is immediate. Otherwise each step of the induction, performed for a value $i < p$, established $i < q$ along the way; taking $i = p - 1$ gives $p - 1 < q$, that is $p \leq q$. The induction run to $i = p$ gives the displayed equality, as we sought to show.

The lemma is a counting statement dressed as an exchange procedure. Its content is that independence and spanning pull in opposite directions: independence limits how long a list can be, spanning limits how short, and a list that does both is squeezed to a single length.

Theorem 2.2.8: Any Two Bases of a Subspace Have the Same Length

Let W be a subspace of k^n and let $(v_1, \ldots, v_p)$ and $(w_1, \ldots, w_q)$ be bases of W. Then $p = q$.

***Proof*:**
The vectors $v_1, \ldots, v_p$ are linearly independent and lie in $W = \text{span}\{w_1, \ldots, w_q\}$, so lemma 2.2.7 gives $p \leq q$. Exchanging the roles of the two bases, the vectors $w_1, \ldots, w_q$ are linearly independent and lie in $W = \text{span}\{v_1, \ldots, v_p\}$, so the same lemma gives $q \leq p$. Hence $p = q$, as we sought to show.

The theorem attaches an integer to a subspace, provided the subspace has a basis at all. That proviso is removed next, and the argument also shows how to build a basis: keep adding independent vectors until no more can be added, or keep discarding redundant ones until none is left.

Theorem 2.2.9: Existence, Extension, and Extraction of Bases

Let W be a subspace of k^n.

1. Every linearly independent list of vectors of W can be extended to a basis of W by appending finitely many further vectors of W. In particular W has a basis, and every basis of W has at most n vectors.

2. Every finite list of vectors spanning W has a sublist that is a basis of W.

***Proof*:**
First a bound. A linearly independent list of vectors of W lies in $W \subseteq k^n = \text{span}\{e_1, \ldots, e_n\}$ by example 2.2.2(1), so lemma 2.2.7 says it has at most n vectors. Every linearly independent list of vectors of W therefore has length at most n.

1. Let $u_1, \ldots, u_s$ be linearly independent vectors of W. Consider the linearly independent lists of vectors of W whose first s vectors are $u_1, \ldots, u_s$. There is at least one such list, namely $u_1, \ldots, u_s$ itself, and each has length at most n by the bound just proved, so among them there is one of greatest length. Write it as $u_1, \ldots, u_s, u_{s+1}, \ldots, u_p$ and let $w \in W$. The list $u_1, \ldots, u_p, w$ consists of vectors of W and begins with $u_1, \ldots, u_s$, and it is longer than the chosen list, so it is not linearly independent. There are therefore scalars $c_1, \ldots, c_p, c$, not all zero, with

$$c_1u_1 + \cdots + c_pu_p + cw = 0$$

   If $c = 0$ this is a nontrivial relation among $u_1, \ldots, u_p$, contradicting their independence, so $c \neq 0$ and

$$w = -c^{-1}(c_1u_1 + \cdots + c_pu_p)$$

   lies in $\operatorname{span}\{u_1, \ldots, u_p\}$. Hence $W \subseteq \operatorname{span}\{u_1, \ldots, u_p\}$, and the reverse inclusion holds because each u_j lies in the subspace W. So $\operatorname{span}\{u_1, \ldots, u_p\} = W$ and the list is a basis of W extending the one given. Applying this to the empty list, which is linearly independent, produces a basis of W; and every basis of W is a linearly independent list of vectors of W, so it has at most n vectors by the bound.

2. Let $w_1, \ldots, w_q$ span W. Among the sublists of $w_1, \ldots, w_q$ that still span W there is one of least length, since the lengths are integers between 0 and q and the full list is one of them. Renumber so that it is $w_1, \ldots, w_t$. If this sublist were linearly dependent, one of its vectors would lie in the span of the others and could be deleted without changing the span (proposition 2.1.10), producing a spanning sublist of length $t - 1$ and contradicting the choice of t. So $w_1, \ldots, w_t$ is linearly independent and spans W, which makes it a basis of W, completing the proof.

Every subspace of k^n therefore has a basis, and by theorem 2.2.8 all of its bases have the same length. That common length is the invariant the section was after.

Definition 2.2.10: Dimension

Let W be a subspace of k^n. The *dimension* of W, written $\dim W$, is the number of vectors in a basis of W. The number does not depend on which basis is used, by theorem 2.2.8, and a basis exists, by theorem 2.2.9(1).

Three values are immediate. The subspace $\{0\}$ has the empty list as a basis, so $\dim\{0\} = 0$. The whole space k^n has the standard basis, so $\dim k^n = n$: the count of independent directions in k^n was a description before and is a theorem now. Any subspace W of k^n has a basis of at most n vectors by theorem 2.2.9(1), so $0 \leq \dim W \leq n$. In $\mathbb{R}^3$ the subspaces of dimension 1 are the lines through the origin and those of dimension 2 are the planes through the origin, the plane U of the opening computation among them.

The dimension of a span is computed by the elimination of chapter 1. The bridge is the column reading $Mx = x_1c_1 + \cdots + x_pc_p$ of a matrix product, where $c_1, \ldots, c_p$ are the columns of M: a list of columns is linearly dependent exactly when the homogeneous system $Mx = 0$ has a solution other than $x = 0$, and the solutions of that system are exactly the relations among the columns.

Example 2.2.11: A Basis for a Span, Extracted by Elimination

1. Let
$$w_1 = \begin{pmatrix}1\\0\\1\\2\end{pmatrix}, \quad w_2 = \begin{pmatrix}2\\1\\3\\5\end{pmatrix}, \quad w_3 = \begin{pmatrix}0\\1\\1\\1\end{pmatrix}, \quad w_4 = \begin{pmatrix}1\\1\\2\\3\end{pmatrix}$$
in $\mathbb{R}^4$ and let $V = \operatorname{span}\{w_1, w_2, w_3, w_4\}$. Assemble the four vectors as the columns of
$$M = \begin{pmatrix}1 & 2 & 0 & 1\\0 & 1 & 1 & 1\\1 & 3 & 1 & 2\\2 & 5 & 1 & 3\end{pmatrix}$$
The replacements $R_3 \mapsto R_3 - R_1$ and $R_4 \mapsto R_4 - 2R_1$ turn rows 3 and 4 into $(0, 1, 1, 1)$, and then $R_3 \mapsto R_3 - R_2$ and $R_4 \mapsto R_4 - R_2$ empty them:
$$\begin{pmatrix}1 & 2 & 0 & 1\\0 & 1 & 1 & 1\\0 & 0 & 0 & 0\\0 & 0 & 0 & 0\end{pmatrix}$$
This is in echelon form with pivots in columns 1 and 2, so $\operatorname{rank} M = 2$, and $R_1 \mapsto R_1 - 2R_2$ produces the reduced echelon form with rows $(1, 0, -2, -1)$ and $(0, 1, 1, 1)$ and two zero rows. The homogeneous system it records is $x_1 - 2x_3 - x_4 = 0$ and $x_2 + x_3 + x_4 = 0$, with x_3 and x_4 free. Taking $x_3 = 1$ and $x_4 = 0$ gives $x = (2, -1, 1, 0)^\top$, that is $2w_1 - w_2 + w_3 = 0$, so $w_3 = w_2 - 2w_1$; taking $x_3 = 0$ and $x_4 = 1$ gives $x = (1, -1, 0, 1)^\top$, that is $w_1 - w_2 + w_4 = 0$, so $w_4 = w_2 - w_1$. Both relations check against the entries: $w_2 - 2w_1 = (0, 1, 1, 1)^\top = w_3$ and $w_2 - w_1 = (1, 1, 2, 3)^\top = w_4$.
Each of w_3 and w_4 lies in $\operatorname{span}\{w_1, w_2\}$, so deleting them in turn leaves the span unchanged (proposition 2.1.10) and $V = \operatorname{span}\{w_1, w_2\}$. The remaining two vectors are linearly independent: if $aw_1 + bw_2 = 0$, the second entry gives $b = 0$ and the first then gives $a = 0$. So (w_1, w_2) is a basis of V and $\dim V = 2$.

2. The running matrix
$$A = \begin{pmatrix}1 & 1 & 3\\2 & 3 & 8\\1 & 2 & 5\\3 & 4 & 11\end{pmatrix}$$
of example 1.2.9 has columns c_1, c_2, c_3 satisfying $c_3 = c_1 + 2c_2$, as the entries $3 = 1 + 2$, $8 = 2 + 6$, $5 = 1 + 4$ and $11 = 3 + 8$ record. Deleting c_3 leaves the span unchanged, so $\operatorname{span}\{c_1, c_2, c_3\} = \operatorname{span}\{c_1, c_2\}$. If $ac_1 + bc_2 = 0$ then the first two entries give $a + b = 0$ and $2a + 3b = 0$, and subtracting twice the first equation from the second gives $b = 0$ and then $a = 0$, so c_1 and c_2 are linearly independent. Hence (c_1, c_2) is a basis of $\operatorname{span}\{c_1, c_2, c_3\}$ and that subspace of $\mathbb{R}^4$ has dimension 2. The reduced echelon form computed in example 1.2.9 has two pivots, so $\operatorname{rank} A = 2$ as well (definition 1.3.2); that the two counts agree is accounted for in section 2.3, which explains it.

Dimension behaves as a measure of size should: a smaller subspace has a smaller dimension, and the only way for a subspace to have the dimension of a subspace containing it is to be that subspace.

Corollary 2.2.12: Monotonicity of Dimension

Let U and W be subspaces of k^n with $U \subseteq W$. Then $\dim U \leq \dim W$, and $\dim U = \dim W$ holds if and only if $U = W$.

***Proof*:**
Let $(u_1, \dots, u_p)$ be a basis of U and $(w_1, \dots, w_q)$ a basis of W, both available by theorem 2.2.9(1). The vectors $u_1, \dots, u_p$ are linearly independent and lie in $U \subseteq W = \operatorname{span}\{w_1, \dots, w_q\}$, so $p \leq q$ by lemma 2.2.7, that is $\dim U \leq \dim W$.
If $U = W$ the two dimensions are equal. Conversely suppose $p = q$. The list $u_1, \dots, u_p$ is linearly independent and consists of vectors of W, so by theorem 2.2.9(1) it extends to a basis of W, which has $q = p$ vectors by theorem 2.2.8. The extension therefore appended nothing, and $u_1, \dots, u_p$ is itself a basis of W. Hence $W = \operatorname{span}\{u_1, \dots, u_p\} = U$, as we sought to show.

Verifying that a list is a basis means checking two conditions, and the next corollary says that when the list has exactly the right length one of the two checks can be skipped. This is what makes dimension worth computing before anything else about a subspace.

Corollary 2.2.13: Independence and Spanning for a List of Length $\dim W$

Let W be a subspace of k^n with $\dim W = p$ and let $v_1, \dots, v_p$ be a list of exactly p vectors of W. The following are equivalent.

1. The list is linearly independent.
2. $\operatorname{span}\{v_1, \dots, v_p\} = W$.
3. The list is a basis of W.

***Proof*:**
Statement (3) is the conjunction of (1) and (2), so it implies each of them, and it suffices to show that (1) implies (3) and that (2) implies (3).
Suppose the list is linearly independent. By theorem 2.2.9(1) it extends to a basis of W, and that basis has $\dim W = p$ vectors by theorem 2.2.8 and definition 2.2.10. The list already has p vectors, so the extension appended nothing and the list is itself a basis of W.
Suppose instead that the list spans W. By theorem 2.2.9(2) some sublist of it is a basis of W, and that sublist has p vectors, again by theorem 2.2.8. A sublist of a list of p vectors having p vectors is the whole list, so the list is a basis of W, completing the proof.

For a list of three vectors in $\mathbb{R}^3$, the corollary says that independence and spanning are the same condition, and example 2.2.6 could have stopped after the independence check instead of solving the general system. The saving grows with n.

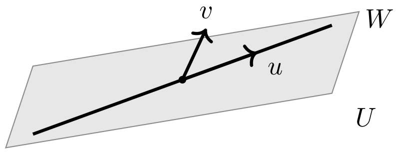


Figure 2.5: A line $U = \mathrm{span}\{u\}$ inside a plane $W = \mathrm{span}\{u, v\}$ in $\mathbb{R}^3$, with $\dim U = 1$ and $\dim W = 2$

Every proof in this section manipulated vectors through addition and scalar multiplication and through nothing else. The entries of a vector were used in the computed examples and in a single argument, the bound of n on the length of an independent list in k^n; no proof used that a vector is a list of numbers. What the proofs did use is the following list of eight properties, each of which holds in k^n because it holds entry by entry in k.

Definition 2.2.14: Vector Space over ℝ or ℂ

Let k be $\mathbb{R}$ or $\mathbb{C}$. A *vector space over* k is a set V equipped with an addition, assigning to each pair $u, v \in V$ an element $u + v \in V$, and a scalar multiplication, assigning to each $t \in k$ and $v \in V$ an element $tv \in V$, such that the following eight conditions hold for all $u, v, w \in V$ and all $s, t \in k$.

1. $u + v = v + u$.
2. $(u + v) + w = u + (v + w)$.
3. There is an element $0 \in V$ with $v + 0 = v$ for every $v \in V$.
4. For each $v \in V$ there is an element $-v \in V$ with $v + (-v) = 0$.
5. $s(u + v) = su + sv$.
6. $(s + t)v = sv + tv$.
7. $(st)v = s(tv)$.
8. $1v = v$.

The elements of V are called *vectors* and the elements of k are called *scalars*.

The space k^n is a vector space over k. Its addition and scalar multiplication are performed entry by entry, so each of the eight conditions reduces to the corresponding statement about the arithmetic of k: condition (1) holds because $u_j + v_j = v_j + u_j$ in k for each j, condition (3) holds with the vector all of whose entries are 0, condition (4) holds with the vector whose entries are $-v_j$, and the remaining five are checked the same way. Every subspace W of k^n is a vector space over k as well. The closure of W under the two operations is what makes the operations of k^n restrict to operations on W (definition 2.1.2(2) and (3)). Six of the eight conditions are then inherited from k^n: conditions (1), (2) and (5) through (8) are statements holding for all vectors of k^n, in particular for those of W. The other two assert that a vector of a certain kind belongs to the set, which inheritance does not deliver. Condition (3) holds because the zero column lies in W by definition 2.1.2(1), and condition (4) holds because the inverse of $v \in W$ is the vector $(-1)v$, which lies in W because W is closed under scalar multiplication.

The definitions of span, linear independence, basis and coordinates use nothing beyond the eight conditions, and neither do the proofs of theorem 2.2.3, lemma 2.2.7 and theorem 2.2.8, so each of them holds in any vector space over k word for word. The one argument that used more is theorem 2.2.9, which bounded the length of an independent list by exhibiting a spanning list of n vectors for the ambient space; a vector space with no finite spanning list is outside its reach. That is the only obstruction, and it deserves a name.

Definition 2.2.15: Finite-Dimensional Vector Space

A vector space V over k is *finite-dimensional* if some finite list of its vectors spans it. Its *dimension* $\dim V$ is then the length of any basis, which is well defined by theorem 2.2.8.

2.2.16 The Chapter Applies to Any Finite-Dimensional Space Every subspace of k^n is finite-dimensional, and for a finite-dimensional V the whole of this chapter applies verbatim. That includes the material on maps: definition 2.4.2, definition 2.4.4, definition 2.4.5, proposition 2.4.6, proposition 2.4.7, proposition 2.4.3, theorem 2.4.8, theorem 2.4.9, proposition 2.2.5, theorem 2.4.10 and the change-of-basis results of section 2.5 are stated above for subspaces of k^n only because that was the setting at hand. Their statements and proofs use nothing but the eight conditions and the existence of a basis. Part II works with finite-dimensional vector spaces that are not presented as subspaces of k^n, and appeals to the results of this chapter in that generality without further comment; so does chapter 12, where the space in question is a space of matrices.

Exercise 2.2.1

1. Decide whether

$$v_1 = \begin{pmatrix}1\\2\\1\end{pmatrix}, \qquad v_2 = \begin{pmatrix}2\\1\\3\end{pmatrix}, \qquad v_3 = \begin{pmatrix}1\\-1\\2\end{pmatrix}$$

   is a basis of $\mathbb{R}^3$. If it is not, find a basis of $\operatorname{span}\{v_1, v_2, v_3\}$ and give the dimension of that subspace.

2. Let $N \subseteq \mathbb{R}^4$ be the solution set of the homogeneous system

$$\begin{aligned} x_1 + 2x_2 - x_3 + x_4 &= 0 \\ 2x_1 + 4x_2 - x_3 + 3x_4 &= 0 \end{aligned}$$

   Find a basis of N and compute $\dim N$, checking each basis vector against both equations.

3. Let $B = (v_1, v_2, v_3)$ be the basis of $\mathbb{R}^3$ of example 2.2.6. Compute $[u]_B$ for $u = (2, 4, 4)^\top$, and find the vector $z \in \mathbb{R}^3$ whose coordinate column with respect to B is $(1, 2, 3)^\top$.

4. Let $v_1 = (1, 1, 0)^\top$ and $v_2 = (0, 0, 1)^\top$ in $\mathbb{R}^3$. For which $j \in \{1, 2, 3\}$ is (v_1, v_2, e_j) a basis of $\mathbb{R}^3$? Then show, without computing anything further, that at least one j had to work.

5. Let $W = \{ x \in \mathbb{R}^4 | x_1 + x_2 + x_3 + x_4 = 0 \}$. Compute $\dim W$, and then show that

$$f_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \\ 0 \end{pmatrix}, \qquad f_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \\ 0 \end{pmatrix}, \qquad f_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ -1 \end{pmatrix}$$

is a basis of W without checking that these three vectors span it.

6. Show that no list of four vectors of $\mathbb{R}^3$ is linearly independent, and that no list of two vectors spans $\mathbb{R}^3$. Then let $(v_1, \ldots, v_p)$ be a basis of a subspace W of k^n and let $t \in k$ be nonzero. Show that $B' = (tv_1, v_2, \ldots, v_p)$ is again a basis of W, and express $[w]_{B'}$ in terms of $[w]_B$.

2.3 The Four Fundamental Subspaces

Chapter 1 attached a single number to a matrix, its rank, and read off from that number whether a system was solvable and how many parameters its solutions carried. A matrix carries more than a number. Four subspaces are attached to it, two inside k^n and two inside k^m, and this section produces a basis and a dimension for each of the four. All four dimensions are determined by m, n and the rank.

Take the running matrix A of example 1.2.9 and write c_1, c_2, c_3 for its columns,

$$c_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 3 \end{pmatrix}, \qquad c_2 = \begin{pmatrix} 1 \\ 3 \\ 2 \\ 4 \end{pmatrix}, \qquad c_3 = \begin{pmatrix} 3 \\ 8 \\ 5 \\ 11 \end{pmatrix}$$

each a column with four entries. The column reading of the product, established after example 1.2.9, says that $Ax = x_1c_1 + x_2c_2 + x_3c_3$ for every $x \in \mathbb{R}^3$. The columns b for which $Ax = b$ can be solved are therefore exactly the combinations of c_1, c_2 and c_3, which is the set $\operatorname{span}\{c_1, c_2, c_3\}$ of definition 2.1.5. That set decides which right hand sides are reachable, and it is the first of the four subspaces.

Definition 2.3.1: Column Space

Let $A \in M_{m\times n}(k)$ have columns $c_1, \ldots, c_n$, each an element of k^m. The *column space* of A is

$$\operatorname{col} A = \operatorname{span}\{c_1, \ldots, c_n\}$$

a subspace of k^m by proposition 2.1.6.

The column space has a second description that avoids naming the columns. By the column reading, $Ax = x_1c_1 + \cdots + x_nc_n$, so the elements of k^m of the form Ax are exactly the linear combinations of $c_1, \ldots, c_n$; that is,

$$\operatorname{col} A = \{ Ax | x \in k^n \}$$

Consistency is now a membership question. A system $Ax = b$ is consistent (definition 1.1.6) exactly when some x satisfies $Ax = b$, and by the display that happens exactly when $b \in \operatorname{col} A$. This is theorem 1.3.3 stated without reference to elimination: solvability means that b can be built out of the columns of A. Deciding the membership still takes a computation, and the pivot count is that computation.

Example 2.3.2: Consistency Read from the Column Space

Keep the running matrix A and its columns c_1, c_2, c_3. By example 2.1.7, $\operatorname{span}\{c_1, c_2, c_3\} = \operatorname{span}\{c_1, c_2\}$, so $\operatorname{col} A = \operatorname{span}\{c_1, c_2\}$ and deciding whether a given $b \in \mathbb{R}^4$ lies in $\operatorname{col} A$ means deciding whether $\alpha c_1 + \beta c_2 = b$ has a solution (α, β).

Take $b = (2, 5, 3, 7)^\top$. The first two entries force $\alpha + \beta = 2$ and $2\alpha + 3\beta = 5$, so $\beta = 1$ and $\alpha = 1$. The remaining two entries hold with these values: $1 + 2 = 3$ and $3 + 4 = 7$. So $b = c_1 + c_2$ lies in $\operatorname{col} A$, the system $Ax = b$ is consistent, and $x = (1, 1, 0)^\top$ is one solution.

Take instead $b' = (1, 0, 0, 0)^\top$. The first two entries force $\alpha + \beta = 1$ and $2\alpha + 3\beta = 0$, so $\beta = -2$ and $\alpha = 3$. The third entry then requires $\alpha + 2\beta = 0$, and $3 - 4 = -1$ instead. No pair (α, β) works, so $b' \notin \operatorname{col} A$ and $Ax = b'$ is inconsistent.

Theorem 1.3.3 reaches the same two verdicts by counting pivots. For b', the operations $R_2 \mapsto R_2 - 2R_1$, $R_3 \mapsto R_3 - R_1$ and $R_4 \mapsto R_4 - 3R_1$, followed by $R_3 \mapsto R_3 - R_2$ and $R_4 \mapsto R_4 - R_2$ and then $R_4 \mapsto R_4 + R_3$, carry the augmented matrix to

$$\left(\begin{array}{ccc|c} 1 & 1 & 3 & 1 \\ 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

which is in echelon form with pivots in columns 1, 2 and 4. So $\operatorname{rank}(A \,|\, b') = 3$ while $\operatorname{rank} A = 2$, and the criterion reports inconsistency.

The second subspace was computed in chapter 1 and left unnamed. The reduced echelon form of the running matrix A is

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and by lemma 1.2.4 the lists x with $Ax = 0$ are exactly the lists satisfying $x_1 + x_3 = 0$ and $x_2 + 2x_3 = 0$. Setting $x_3 = t$ gives $x = t(-1, -2, 1)^\top$, so those lists form the line $\operatorname{span}\{(-1, -2, 1)^\top\}$ in $\mathbb{R}^3$. The same line is $\operatorname{span}\{(1, 2, -1)^\top\}$, obtained at $t = -1$, and $(1, 2, -1)^\top$ is the list of coefficients in the column relation $c_1 + 2c_2 - c_3 = 0$.

Definition 2.3.3: Null Space and Nullity

Let $A \in M_{m \times n}(k)$. The *null space* of A is

$$\operatorname{null} A = \{\, x \in k^n \,|\, Ax = 0 \,\}$$

the solution set of the homogeneous system $Ax = 0$ (definition 1.1.9), a subspace of k^n by example 2.1.4. Its dimension is the *nullity* of A, written $\operatorname{nullity} A$.

The column space records what A produces; the null space records what A sends to zero. The two live in different spaces: $\operatorname{col} A$ sits inside k^m, where the right hand sides live, and $\operatorname{null} A$ sits inside k^n, where the unknowns live. Theorem 1.1.11 joins them, since the solution set of a consistent system $Ax = b$ is a translate of null$A$, and the translate exists exactly when $b \in \operatorname{col} A$.

Elimination operates on rows, not on columns, so the rows of A deserve the same treatment. The rows of the running matrix are $(1, 1, 3)$, $(2, 3, 8)$, $(1, 2, 5)$ and $(3, 4, 11)$. Two relations hold among them: the third row is the second minus the first, and the fourth is the first plus the second. Every combination of the four rows is therefore a combination of the first two. Rows are lists of n numbers while the vectors of this book are columns, so a row is turned into a column before its span is taken. That is what the transpose does.

Definition 2.3.4: Transpose

Let $A = (a_{ij}) \in M_{m\times n}(k)$. The *transpose* of A is the matrix $A^\top \in M_{n\times m}(k)$ whose entry in row j and column i is a_{ij}.

The columns of $A^\top$ are the rows of A written as columns, and its rows are the columns of A written as rows. Applying the operation twice returns A, since the entry of $(A^\top)^\top$ in row i and column j is the entry of $A^\top$ in row j and column i, which is a_{ij}. Write $\rho_1, \dots, \rho_m \in k^n$ for the rows of A written as columns, so that ρ_i is the ith column of $A^\top$.

Proposition 2.3.5: Arithmetic of the Transpose

Let $A, A' \in M_{m\times n}(k)$, let $X \in M_{m\times n}(k)$ and $Y \in M_{n\times p}(k)$, and let $c \in k$.

1. $(A^\top)^\top = A$.
2. $(A + A')^\top = A^\top + (A')^\top$ and $(cA)^\top = cA^\top$.
3. $(XY)^\top = Y^\top X^\top$: transposition reverses the order of a product.
4. If $A \in M_n(k)$ is invertible, so is $A^\top$, and $(A^\top)^{-1} = (A^{-1})^\top$.

***Proof*:**

1. This is the computation made just above.
2. By definition 2.3.4 the entry of $(A + A')^\top$ in row j and column i is the entry of $A + A'$ in row i and column j, namely $a_{ij} + a'_{ij}$, which is the corresponding entry of $A^\top + (A')^\top$. The scalar statement is the same computation with ca_{ij} in place of a_{ij}.
3. Both sides lie in $M_{p\times m}(k)$. By definition 2.3.4 and definition 1.4.1, the entry of $(XY)^\top$ in row j and column i is
$$(XY)_{ij} = \sum_{s=1}^{n} x_{is} y_{sj}$$
while the entry of $Y^\top X^\top$ in the same position is
$$\sum_{s=1}^{n} (Y^\top)_{js}(X^\top)_{si} = \sum_{s=1}^{n} y_{sj} x_{is}$$
The two sums have the same terms, so the matrices agree entry by entry.

4. Transposing $A^{-1}A = I_n$ and $AA^{-1} = I_n$ with (3), and using $I_n^\top = I_n$, gives $A^\top(A^{-1})^\top = I_n$ and $(A^{-1})^\top A^\top = I_n$. Those are the two conditions of definition 1.4.4 for the matrix $(A^{-1})^\top$, so $A^\top$ is invertible with the stated inverse.

Over $\mathbb{C}$ the transpose is not the conjugate transpose of definition 5.5.12, and the two have different arithmetic: part (3) here has no conjugation in it, whereas $(XY)^* = Y^*X^*$ carries one on each factor. Statements about $\mathbb{R}$ may use either, since conjugation is the identity there.

Definition 2.3.6: Row Space

Let $A \in M_{m\times n}(k)$ and let $\rho_1, \ldots, \rho_m \in k^n$ be its rows written as columns. The *row space* of A is

$$\operatorname{row} A = \operatorname{span}\{\rho_1, \ldots, \rho_m\}$$

a subspace of k^n. Since $\rho_1, \ldots, \rho_m$ are the columns of $A^\top$, the row space of A is the column space of its transpose: $\operatorname{row} A = \operatorname{col} A^\top$.

For the running matrix, the two relations among the rows are $\rho_1 - \rho_2 + \rho_3 = 0$ and $-\rho_1 - \rho_2 + \rho_4 = 0$. Each is recorded by its list of coefficients, namely $(1, -1, 1, 0)$ and $(-1, -1, 0, 1)$, and each such list is a column with $m = 4$ entries. The lists recording relations among the rows form the fourth subspace.

Definition 2.3.7: Left Null Space

Let $A \in M_{m\times n}(k)$. The *left null space* of A is the null space of its transpose,

$$\operatorname{null} A^\top = \{\, y \in k^m \,|\, A^\top y = 0 \,\}$$

a subspace of k^m.

Three descriptions of that subspace agree. The columns of $A^\top$ are $\rho_1, \ldots, \rho_m$, so the column reading gives $A^\top y = y_1\rho_1 + \cdots + y_m\rho_m$, and y lies in $\operatorname{null} A^\top$ exactly when that combination of the rows of A vanishes. The name comes from the third description. The product $y^\top A$ is defined, since $y^\top$ has one row and m columns while A has m rows, and its entry in column j is $\sum_{i=1}^m y_i a_{ij}$, which is the jth entry of $y_1\rho_1 + \cdots + y_m\rho_m$. So $y^\top A = 0$ holds exactly when $A^\top y = 0$ does, and the elements of $\operatorname{null} A^\top$ are the columns that annihilate A from the left.

The four subspaces $\operatorname{col} A$, $\operatorname{null} A$, $\operatorname{row} A$ and $\operatorname{null} A^\top$ are called the *four fundamental subspaces* of A. Two of them are spans of explicit lists of vectors and two are solution sets of homogeneous systems, which is why a basis for each is within reach of elimination alone. The row space comes first, because it is the one elimination leaves untouched.

Theorem 2.3.8: The Nonzero Rows of an Echelon Form Are a Basis of the Row Space

Let $A \in M_{m\times n}(k)$ and let U be a matrix in echelon form (definition 1.2.5) that is row equivalent to A. Then $\operatorname{row} U = \operatorname{row} A$, and the nonzero rows of U, written as columns, form a basis of $\operatorname{row} A$. In particular

$$\dim \operatorname{row} A = \operatorname{rank} A$$

***Proof*:**
Let N be obtained from $M \in M_{m\times n}(k)$ by a single elementary row operation, and write $\mu_1, \dots, \mu_m$ and $\nu_1, \dots, \nu_m$ for the rows of M and of N written as columns. Each ν_i is a linear combination of $\mu_1, \dots, \mu_m$: a swap makes each ν_i equal to some $\mu_{i'}$, a scaling $R_i \mapsto cR_i$ makes $\nu_i = c\mu_i$ and leaves the other rows unchanged, and a replacement $R_i \mapsto R_i + cR_j$ makes $\nu_i = \mu_i + c\mu_j$ and leaves the other rows unchanged. Every combination of $\nu_1, \dots, \nu_m$ is therefore a combination of $\mu_1, \dots, \mu_m$, so $\operatorname{row} N \subseteq \operatorname{row} M$. Each of the three operations is undone by an operation of the same kind, as recorded after definition 1.2.3, and applying that reasoning to the inverse operation gives $\operatorname{row} M \subseteq \operatorname{row} N$. The two row spaces are equal. A finite sequence of operations repeats the argument once per operation, so $\operatorname{row} U = \operatorname{row} A$.

Let $u_1, \dots, u_r$ be the nonzero rows of U written as columns, with leading entries in columns $j_1 < j_2 < \cdots < j_r$; the inequalities are condition (2) of definition 1.2.5, and the zero rows lie below by condition (1). Those zero rows contribute nothing to a span, so $\operatorname{row} U = \operatorname{span}\{u_1, \dots, u_r\}$. Suppose $t_1u_1 + \cdots + t_ru_r = 0$ with $t_1, \dots, t_r \in k$, and fix an index p with $t_1 = \cdots = t_{p-1} = 0$, which holds vacuously for $p = 1$. For $q > p$ the leading entry of u_q lies in column $j_q > j_p$, so the entry of u_q in position j_p is zero. The entry of the combination in position j_p is therefore t_p times the entry of u_p in position j_p, which is the leading entry of u_p and is nonzero. Hence $t_p = 0$. Running p from 1 to r gives $t_1 = \cdots = t_r = 0$, so $u_1, \dots, u_r$ are linearly independent (definition 2.1.9) and form a basis of $\operatorname{row} A$ (definition 2.2.1).

Finally, each nonzero row of U carries exactly one leading entry, so r is the number of pivots of U (definition 1.2.6). By the second statement of theorem 1.2.10 the leading entries of U occupy the pivot columns of the reduced echelon form of A, one each, so $r = \operatorname{rank} A$ by definition 1.3.2. Therefore $\dim \operatorname{row} A = r = \operatorname{rank} A$, as we sought to show.

The row space is thus computed by running elimination and stopping: the nonzero rows of whatever echelon form appears are a basis. The column space admits no such shortcut, because elimination changes it. The column $c_1 = (1, 2, 1, 3)^\top$ lies in $\operatorname{col} A$ for the running matrix, while every column of the reduced echelon form of A has its last two entries zero, so c_1 lies outside the column space of that reduced form. The two column spaces are different. What elimination does preserve is which combinations of the columns vanish, and that is enough to locate a basis among the columns of A itself.

Theorem 2.3.9: The Pivot Columns Are a Basis of the Column Space

Let $A \in M_{m\times n}(k)$ have columns $c_1, \dots, c_n$, let R be the reduced echelon form of A, and let $j_1 < \cdots < j_r$ be the pivot columns of R, so that $r = \operatorname{rank} A$. Then $c_{j_1}, \dots, c_{j_r}$ form a basis of $\operatorname{col} A$. Moreover, if j is a non-pivot column then

$$c_j = \sum_{p=1}^{r} R_{pj}\, c_{j_p}$$

where R_{pj} is the entry of R in row p and column j. In particular $\dim \operatorname{col} A = \operatorname{rank} A$.

***Proof*:**
Write $\tilde{c}_1, \dots, \tilde{c}_n$ for the columns of R. For any $x \in k^n$ the column reading gives $Ax = x_1c_1 + \cdots + x_nc_n$ and $Rx = x_1\tilde{c}_1 + \cdots + x_n\tilde{c}_n$, and $Ax = 0$ holds if and only if $Rx = 0$ by lemma 1.2.4. So a list $(x_1, \dots, x_n)$ of scalars satisfies $\sum_j x_jc_j = 0$ if and only if it satisfies $\sum_j x_j\tilde{c}_j = 0$: the columns of A and the columns of R satisfy exactly the same linear relations.

The columns of R are easy to name. Since R is in reduced echelon form (definition 1.2.7) its r pivots equal 1, occupy rows $1, \dots, r$ in that order, and are alone in their columns, so $\tilde{c}_{j_p} = e_p$, the column of k^m with 1 in position p and zeros elsewhere. Rows $r+1$ through m of R are zero, so every column of R has its last $m - r$ entries zero, and consequently

$$\tilde{c}_j = \sum_{p=1}^{r} R_{pj}\, e_p = \sum_{p=1}^{r} R_{pj}\, \tilde{c}_{j_p}$$

for every j.

Take j a non-pivot column and let $x \in k^n$ have $x_j = 1$, $x_{j_p} = -R_{pj}$ for $1 \le p \le r$, and $x_{j'} = 0$ for every remaining index j'; the positions named are distinct, since j is not a pivot column. The displayed identity says $\sum_{j'} x_{j'}\tilde{c}_{j'} = 0$, so the same combination of the columns of A vanishes, which is the asserted formula $c_j = \sum_p R_{pj}c_{j_p}$.

For linear independence, suppose $t_1c_{j_1} + \cdots + t_rc_{j_r} = 0$ and let $x \in k^n$ have $x_{j_p} = t_p$ for $1 \le p \le r$ and $x_j = 0$ for every non-pivot column j. Then $\sum_j x_jc_j = 0$, so $\sum_j x_j\tilde{c}_j = 0$, which reads $t_1e_1 + \cdots + t_re_r = 0$. Comparing entries gives $t_1 = \cdots = t_r = 0$.

For spanning, put $W = \operatorname{span}\{c_{j_1}, \dots, c_{j_r}\}$, a subspace of k^m by proposition 2.1.6. Every column of A lies in W: a pivot column is one of the listed vectors, and a non-pivot column is the combination displayed above. A subspace containing $c_1, \dots, c_n$ contains every linear combination of them, so $\operatorname{col} A \subseteq W$, and the reverse inclusion holds because each c_{j_p} is a column of A. Hence $W = \operatorname{col} A$, and $c_{j_1}, \dots, c_{j_r}$ form a basis of it. A basis with r elements gives $\dim \operatorname{col} A = r = \operatorname{rank} A$, completing the proof.

Both bases are read off one elimination, and they are read off differently: the row space basis is taken from the echelon form, the column space basis from the original matrix at the positions the echelon form marks. Taking the columns of the echelon form instead is the standard error, and the running matrix shows what it costs, since those columns span a different subspace of $\mathbb{R}^4$.

Run both theorems once on a matrix that has not appeared before. Let

$$B = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 1 & 5 \end{pmatrix} \in M_{3\times 4}(\mathbb{R})$$

The replacements $R_2 \mapsto R_2 - 2R_1$ and $R_3 \mapsto R_3 - 3R_1$, followed by $R_3 \mapsto R_3 - R_2$, produce

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

which is already in reduced echelon form: both pivots equal 1, and each pivot is alone in its column. So $\operatorname{rank} B = 2$, with pivot columns 1 and 3. A basis of $\operatorname{row} B$ is $(1, 2, 0, 1)^\top$ and $(0, 0, 1, 2)^\top$, and a basis of $\operatorname{col} B$ is the first and third columns of B, namely $(1, 2, 3)^\top$ and $(0, 1, 1)^\top$. The two non-pivot columns of the reduced form are $(2,0,0)^\top = 2e_1$ and $(1,2,0)^\top = e_1 + 2e_2$, so theorem 2.3.9 predicts $c_2 = 2c_1$ and $c_4 = c_1 + 2c_3$ for the columns of B itself. Both hold: $2(1,2,3)^\top = (2,4,6)^\top$ and $(1,2,3)^\top + 2(0,1,1)^\top = (1,4,5)^\top$.

The two theorems produce bases with the same number of elements, namely $\operatorname{rank} A$. A count made along the rows and a count made along the columns therefore agree, although the subspaces being counted sit in different spaces and are usually unrelated as sets.

Theorem 2.3.10: Row Rank Equals Column Rank

Let $A \in M_{m\times n}(k)$. Then

$$\dim \operatorname{row} A = \dim \operatorname{col} A = \operatorname{rank} A$$

and consequently $\operatorname{rank} A^\top = \operatorname{rank} A$.

***Proof*:**
The two equalities are the last statements of theorem 2.3.8 and theorem 2.3.9. For the consequence, apply the second equality to the matrix $A^\top \in M_{n\times m}(k)$ in place of A:

$$\operatorname{rank} A^\top = \dim \operatorname{col} A^\top$$

and $\operatorname{col} A^\top = \operatorname{row} A$ by definition 2.3.6, whose dimension is $\operatorname{rank} A$ by the first equality. Hence $\operatorname{rank} A^\top = \operatorname{rank} A$, as we sought to show.

Rank was defined in definition 1.3.2 as a pivot count, which is an artifact of a procedure applied to rows. The theorem says the number so produced measures the columns just as well. Transposing a matrix interchanges the row space and the column space, and it therefore leaves the rank unchanged, which is not visible from the definition: elimination on A and elimination on $A^\top$ are unrelated computations that halt with the same pivot count.

The null space is next, and its dimension was counted in chapter 1 without being called a dimension. There the free variables were the unknowns left unconstrained by the pivots, and there were $n - \operatorname{rank} A$ of them. Each one now becomes a basis vector.

Theorem 2.3.11: Rank Plus Nullity Equals the Number of Columns

Let $A \in M_{m\times n}(k)$. Then

$$\operatorname{rank} A + \operatorname{nullity} A = n$$

that is, $\dim \operatorname{null} A = n - \operatorname{rank} A$, where n is the number of columns of A.

***Proof*:**
Put $r = \operatorname{rank} A$, let R be the reduced echelon form of A with pivot columns $j_1 < \cdots < j_r$, and let F be the set of the $n - r$ remaining column indices, so that the free variables of $Ax = 0$ are the x_j with $j \in F$ (definition 1.3.5). The system $Ax = 0$ is consistent, since the zero column solves it, so theorem 1.3.6(1) applies: assigning values in k to the free variables determines the pivot variables uniquely, and every element of $\operatorname{null} A$ arises from exactly one such assignment.

For each $j \in F$ let $w_j \in \operatorname{null} A$ be the solution produced by the assignment giving x_j the value 1 and every other free variable the value 0. The entry of w_j in position j', for $j' \in F$, is 1 if $j' = j$

and 0 otherwise, since those entries are the assigned values themselves.

These $n-r$ columns are linearly independent. Suppose $\sum_{j\in F} t_j w_j = 0$ with each $t_j \in k$, and fix $j' \in F$. The entry of the combination in position j' is $t_{j'}$, by the previous paragraph, so $t_{j'} = 0$. This holds for every $j' \in F$.

They also span $\mathrm{null} A$. Let $x \in \mathrm{null} A$ and let $s_j \in k$ be its entry in position j for each $j \in F$. The column $w = \sum_{j\in F} s_j w_j$ is a linear combination of elements of the subspace $\mathrm{null} A$, hence lies in $\mathrm{null} A$, and its entry in position $j' \in F$ is $s_{j'}$. So x and w both solve $Ax = 0$ and arise from the same assignment of values to the free variables, and theorem 1.3.6(1) gives $x = w$. Thus every element of $\mathrm{null} A$ is a combination of the w_j.

The columns w_j with $j \in F$ therefore form a basis of $\mathrm{null} A$, and they are indexed by the $n-r$ non-pivot columns, so $\mathrm{nullity} A = n - r$, completing the proof.

Each of the n columns of A is counted exactly once by the theorem. The r pivot columns give the basis of the column space of theorem 2.3.9, and the $n-r$ non-pivot columns give the basis of the null space just constructed. Rank and nullity are two ways of tallying the same list.

Applying the theorem to $A^\top$ as well as to A settles all four dimensions at once.

Corollary 2.3.12: Dimensions of the Four Fundamental Subspaces

Let $A \in M_{m\times n}(k)$ and put $r = \mathrm{rank} A$. Then:

1. $\mathrm{col}\, A$ is a subspace of k^m of dimension r;
2. $\mathrm{row}\, A$ is a subspace of k^n of dimension r;
3. $\mathrm{null} A$ is a subspace of k^n of dimension $n-r$;
4. $\mathrm{null} A^\top$ is a subspace of k^m of dimension $m-r$.

In particular $\dim \mathrm{row}\, A + \dim \mathrm{null} A = n$ and $\dim \mathrm{col}\, A + \dim \mathrm{null} A^\top = m$.

***Proof*:**
Parts (1) and (2) are theorem 2.3.10, together with the ambient spaces recorded in definition 2.3.1 and definition 2.3.6. Part (3) is theorem 2.3.11, and $\mathrm{null} A$ lies in k^n by definition 2.3.3.

For part (4), apply theorem 2.3.11 to $A^\top$, which lies in $M_{n\times m}(k)$ and so has m columns. It gives $\dim \mathrm{null} A^\top = m - \mathrm{rank} A^\top$, and $\mathrm{rank} A^\top = \mathrm{rank} A = r$ by theorem 2.3.10. Hence $\dim \mathrm{null} A^\top = m - r$, and $\mathrm{null} A^\top$ is a subspace of k^m by definition 2.3.7. The two final identities are (2) with (3) and (1) with (4), as we sought to show.

For a square matrix the same count settles invertibility, in the form that gets used whenever a change of basis is assembled column by column.

Proposition 2.3.13: A Square Matrix Is Invertible Exactly When Its Columns Are a Basis

Let $P \in M_n(k)$ have columns $p_1, \ldots, p_n$. The following are equivalent.

1. P is invertible.
2. $p_1, \ldots, p_n$ is linearly independent.
3. $p_1, \ldots, p_n$ is a basis of k^n.

***Proof*:**
For $x = (x_1, \ldots, x_n)^\top$ the ith entry of Px is $\sum_{j=1}^n p_{ij}x_j$, which is the ith entry of $x_1p_1 + \cdots + x_np_n$, so

$$Px = x_1p_1 + \cdots + x_np_n$$

Hence $Px = 0$ has only the solution $x = 0$ exactly when the only vanishing linear combination of the columns has all coefficients zero, which is definition 2.1.9.

(1) implies (2). If P is invertible and $Px = 0$, then $x = I_nx = (P^{-1}P)x = P^{-1}(Px) = 0$, so the columns are independent.

(2) implies (1). Suppose the columns are independent, so $\mathrm{null}P = \{0\}$ and $\mathrm{rank}P = n$ by theorem 2.3.11. By definition 1.3.2 the reduced echelon form R of P has n pivots. Since R is $n \times n$ and no row and no column holds two pivots, each of the n rows and each of the n columns holds exactly one. The pivot columns increase strictly with the row index (definition 1.2.5), and the only strictly increasing list of n indices in $\{1, \ldots, n\}$ is $1, 2, \ldots, n$, so the pivot of row i sits in column i. Each pivot equals 1 and is the only nonzero entry of its column (definition 1.2.7), so $R = I_n$ and P is invertible by theorem 1.4.8.

(2) is equivalent to (3). A list of n independent vectors in k^n is a basis by corollary 2.2.13, and a basis is independent by definition 2.2.1.

The two subspaces sitting inside k^n have dimensions summing to n, and so do the two sitting inside k^m. The sums say nothing about how the subspaces meet. Take $P = \begin{pmatrix} 1 & i \end{pmatrix} \in M_{1\times 2}(\mathbb{C})$, whose one row gives $\mathrm{row}\,P = \mathrm{span}\{(1, i)^\top\}$ and whose one equation $x_1 + ix_2 = 0$ gives $\mathrm{null}P = \mathrm{span}\{(-i, 1)^\top\}$. Both have dimension 1, as the corollary requires with $n = 2$ and $r = 1$, and the two are the same line, since $i(-i, 1)^\top = (1, i)^\top$. What the corollary does give is the size of all four subspaces from the single number r, computed by one run of elimination.

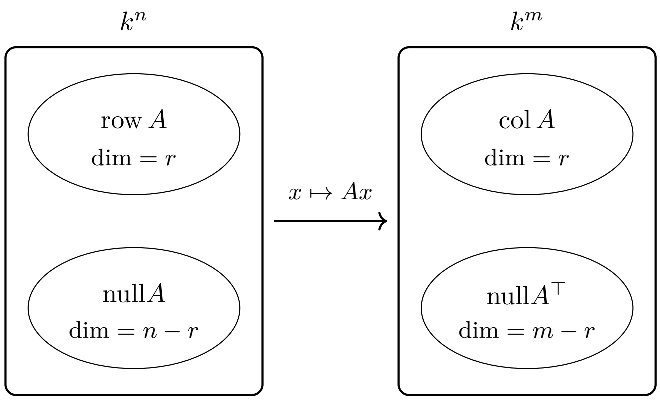


Figure 2.6: The four fundamental subspaces of $A \in M_{m\times n}(k)$, with $r = \text{rank}A$; the two on the left are subspaces of k^n and the two on the right are subspaces of k^m

Example 2.3.14: The Four Subspaces of the Running Matrix

Take the running matrix A of example 1.2.9, with $m = 4$, $n = 3$ and rank$A = 2$. Its reduced echelon form is

$$R = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

with pivots in columns 1 and 2. Corollary 2.3.12 predicts dimensions 2, 2, $3 - 2 = 1$ and $4 - 2 = 2$ for the column space, the row space, the null space and the left null space.

1. *Column space.* By theorem 2.3.9 the first two columns of A itself,

$$c_1 = (1, 2, 1, 3)^\top, \qquad c_2 = (1, 3, 2, 4)^\top$$

form a basis of col A, a subspace of $\mathbb{R}^4$ of dimension 2. The third column of R is $(1, 2, 0, 0)^\top$, so the theorem also gives $c_3 = c_1 + 2c_2$, which is the relation recorded in example 1.2.9.

2. *Row space.* By theorem 2.3.8 the nonzero rows of R, written as columns,

$$(1, 0, 1)^\top, \qquad (0, 1, 2)^\top$$

form a basis of row A, a subspace of $\mathbb{R}^3$ of dimension 2. The four rows of A are recovered from them: $(1, 1, 3) = (1, 0, 1) + (0, 1, 2)$, $(2, 3, 8) = 2(1, 0, 1) + 3(0, 1, 2)$, $(1, 2, 5) = (1, 0, 1) + 2(0, 1, 2)$ and $(3, 4, 11) = 3(1, 0, 1) + 4(0, 1, 2)$.

3. *Null space.* The two nonzero rows of R record $x_1 + x_3 = 0$ and $x_2 + 2x_3 = 0$, with x_3 the only free variable. Setting $x_3 = t$ gives $x = t(-1, -2, 1)^\top$, so

$$\text{null}A = \text{span}\{(-1, -2, 1)^\top\} = \text{span}\{(1, 2, -1)^\top\}$$

a subspace of $\mathbb{R}^3$ of dimension 1, as $n - r = 3 - 2$ requires. The second spanning vector is the coefficient list of the column relation $c_1 + 2c_2 - c_3 = 0$.

4. *Left null space.* Here
$$A^\top = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 2 & 4 \\ 3 & 8 & 5 & 11 \end{pmatrix} \in M_{3\times 4}(\mathbb{R})$$
The replacements $R_2 \mapsto R_2 - R_1$ and $R_3 \mapsto R_3 - 3R_1$ give second row $(0,1,1,1)$ and third row $(0,2,2,2)$; then $R_3 \mapsto R_3 - 2R_2$ empties the third row, and $R_1 \mapsto R_1 - 2R_2$ clears the second column above its pivot:
$$\begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$
So $\operatorname{rank} A^\top = 2$, matching theorem 2.3.10. The equations are $y_1 - y_3 + y_4 = 0$ and $y_2 + y_3 + y_4 = 0$, with y_3 and y_4 free. The assignments $(y_3, y_4) = (1,0)$ and $(y_3, y_4) = (0,1)$ give the basis
$$(1,-1,1,0)^\top, \qquad (-1,-1,0,1)^\top$$
of $\operatorname{null} A^\top$, a subspace of $\mathbb{R}^4$ of dimension 2, as $m - r = 4 - 2$ requires.

All four dimensions agree with corollary 2.3.12. The two basis vectors of the left null space are the coefficient lists of the two relations among the rows of A: the first says $\rho_1 - \rho_2 + \rho_3 = 0$ and the second says $-\rho_1 - \rho_2 + \rho_4 = 0$, both checked entry by entry from $(1,1,3)$, $(2,3,8)$, $(1,2,5)$ and $(3,4,11)$.

Each relation among the rows imposes a condition on the reachable right hand sides. If $b = Ax$ then $b_i = (Ax)_i$ is the ith entry of the column Ax, and combining the four entries with the coefficients $(1,-1,1,0)$ gives
$$b_1 - b_2 + b_3 = \sum_{j=1}^{3} (a_{1j} - a_{2j} + a_{3j}) x_j = 0$$
since $\rho_1 - \rho_2 + \rho_3$ vanishes coordinatewise. The second relation gives $-b_1 - b_2 + b_4 = 0$ in the same way. Both conditions hold for $b = (2,5,3,7)^\top$, since $2 - 5 + 3 = 0$ and $-2 - 5 + 7 = 0$, and the first fails for $b' = (1,0,0,0)^\top$, which is the inconsistency found in example 2.3.2.

Exercise 2.3.1

1. Let
$$M = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{pmatrix} \in M_3(\mathbb{R})$$
Compute a basis for each of $\operatorname{col} M$, $\operatorname{row} M$, $\operatorname{null} M$ and $\operatorname{null} M^\top$, naming the row operations used, and check each of the four dimensions against corollary 2.3.12.

2. Let A be the running matrix of example 1.2.9. Decide for each of
$$b = (0,1,1,1)^\top, \qquad b' = (1,1,1,1)^\top$$
whether it lies in $\operatorname{col} A$. For the one that does, write down every solution of $Ax = b$.

3. Produce a matrix $M \in M_{3\times 4}(\mathbb{R})$ of rank 2 whose null space contains both $(1,1,0,0)^\top$ and $(0,0,1,1)^\top$, and verify that these two columns are in fact a basis of null$M$. What are the dimensions of the other three subspaces of your matrix?
4. Let $A \in M_{m\times n}(k)$ and $b \in k^m$. Show that $\operatorname{col}(A|b) = \operatorname{col} A$ if and only if $Ax = b$ is consistent, where $(A|b)$ is the augmented matrix of definition 1.1.8.
5. Let $A \in M_{m\times n}(k)$ and let $E \in M_m(k)$ be invertible (definition 1.4.4). Show that $\operatorname{null}(EA) = \operatorname{null}A$ and that $\operatorname{rank}(EA) = \operatorname{rank}A$. Then give an A and an E for which $\operatorname{col}(EA) \neq \operatorname{col} A$.
6. Let $A \in M_{m\times n}(k)$. Show that $\operatorname{null}A = \{0\}$ if and only if $\operatorname{rank}A = n$, and deduce that $\operatorname{null}A = \{0\}$ forces $n \leq m$.
7. Let $A \in M_{m\times n}(k)$ and $C \in M_{n\times p}(k)$. Show that $\operatorname{col}(AC) \subseteq \operatorname{col} A$ and that $\operatorname{null}C \subseteq \operatorname{null}(AC)$, and deduce that
$$\operatorname{rank}(AC) \leq \operatorname{rank}A \qquad \text{and} \qquad \operatorname{rank}(AC) \leq \operatorname{rank}C$$
Then exhibit nonzero A and C in $M_2(\mathbb{R})$ with $AC = 0$, so that both inequalities can be strict.

2.4 Linear Maps, Their Matrix Representations, and Idempotent Projections

Section 2.3 read four subspaces off a matrix. Those subspaces are statements about a function: the matrix A sends each column x to the column Ax, the null space collects the columns sent to 0, and the column space collects the columns that are reached. This section makes the function itself the object. Which functions arise from a matrix in this way is settled below, as is how the matrix is recovered from the function once a basis has been chosen. The section closes with the maps that leave their own values unchanged, and with the splitting of a space that each such map produces.

Example 2.4.1: Three Rules on $\mathbb{R}^2$

Elements of $\mathbb{R}^2$ are columns, and each of the three rules below assigns a column to a column. Throughout, $u = \left(\begin{smallmatrix} u_1 \\ u_2 \end{smallmatrix}\right)$ and $v = \left(\begin{smallmatrix} v_1 \\ v_2 \end{smallmatrix}\right)$ are elements of $\mathbb{R}^2$ and $t \in \mathbb{R}$.

1. The quarter turn
$$R\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$
Adding first and then applying R gives
$$R(u+v) = \begin{pmatrix} -(u_2+v_2) \\ u_1+v_1 \end{pmatrix} = \begin{pmatrix} -u_2 \\ u_1 \end{pmatrix} + \begin{pmatrix} -v_2 \\ v_1 \end{pmatrix} = Ru + Rv$$
and scaling first gives
$$R(tu) = \begin{pmatrix} -tu_2 \\ tu_1 \end{pmatrix} = t\begin{pmatrix} -u_2 \\ u_1 \end{pmatrix} = tRu$$
so both operations may be performed before or after R with the same result. The rule is also a matrix product: by definition 1.1.7,
$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0\cdot x + (-1)y \\ 1\cdot x + 0\cdot y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$

2. The shear
$$S\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}x+2y\\y\end{pmatrix}$$
The same two checks succeed:
$$S(u+v)=\begin{pmatrix}(u_1+v_1)+2(u_2+v_2)\\u_2+v_2\end{pmatrix}=\begin{pmatrix}u_1+2u_2\\u_2\end{pmatrix}+\begin{pmatrix}v_1+2v_2\\v_2\end{pmatrix}=Su+Sv$$
and $S(tu)=\left(\begin{smallmatrix}tu_1+2tu_2\\tu_2\end{smallmatrix}\right)=tSu$. The matrix is $\left(\begin{smallmatrix}1&2\\0&1\end{smallmatrix}\right)$, since $1\cdot x+2y=x+2y$ and $0\cdot x+1\cdot y=y$.

3. The shift
$$F\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}x+1\\y\end{pmatrix}$$
Here the second check fails. Take $u=\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$ and $t=2$. Then $2u=\left(\begin{smallmatrix}2\\0\end{smallmatrix}\right)$ and
$$F(2u)=\begin{pmatrix}3\\0\end{pmatrix},\qquad 2F(u)=2\begin{pmatrix}2\\0\end{pmatrix}=\begin{pmatrix}4\\0\end{pmatrix}$$
which are different columns.

Those two checks are the whole test, and the rule $\left(\begin{smallmatrix}x\\y\end{smallmatrix}\right)\mapsto\begin{pmatrix}x^2\\y\end{pmatrix}$ fails the second one as well: at $u=\left(\begin{smallmatrix}1\\0\end{smallmatrix}\right)$ and $t=2$ it returns $\left(\begin{smallmatrix}4\\0\end{smallmatrix}\right)$ rather than $\left(\begin{smallmatrix}2\\0\end{smallmatrix}\right)$. The two conditions get a name. They are stated for maps between subspaces rather than only between $\mathbb{R}^n$ and $\mathbb{R}^m$, because a subspace is closed under both operations and so the conditions make sense there without change.

Definition 2.4.2: Linear Map

Let $V\subseteq k^n$ and $W\subseteq k^m$ be subspaces. A map $T\colon V\to W$ is *linear* if
$$T(u+v)=T(u)+T(v),\qquad T(tu)=tT(u)$$
for all $u,v\in V$ and all $t\in k$. The parentheses are dropped where no confusion results, so that Tv means $T(v)$. A linear map from V to itself is a *linear operator* on V.

A linear map sends 0 to 0: taking $t=0$ and $u=0$ in the second condition gives $T0=T(0\cdot 0)=0\cdot T0=0$. The two conditions also combine. For scalars s,t and vectors u,v in V,
$$T(su+tv)=T(su)+T(tv)=sTu+tTv$$
and repeating the same step $r-1$ times gives
$$T(t_1u_1+\cdots+t_ru_r)=t_1Tu_1+\cdots+t_rTu_r$$
for all $t_1,\dots,t_r\in k$ and $u_1,\dots,u_r\in V$. A linear map may therefore be pushed through any linear combination.

The example produced two linear maps out of matrices, and that is a general fact. Fix $A\in M_{m\times n}(k)$ and let $T_A\colon k^n\to k^m$ be the map $T_A(x)=Ax$. The identities $A(u+v)=Au+Av$ and $A(tu)=t(Au)$, verified entry by entry in section 1.1, say exactly that T_A is linear. The converse also holds: every linear map $k^n\to k^m$ is T_A for exactly one matrix A, and the route to that statement runs through bases.

The next result says that a linear map is fixed by very little data. Its values on a basis may be prescribed at will, and once prescribed they determine every other value.

Proposition 2.4.3: A Linear Map Is Determined by Its Values on a Basis

Let $V \subseteq k^n$ and $W \subseteq k^m$ be subspaces and let $v_1, \ldots, v_p$ be a basis of V.

1. If $S, T\colon V \to W$ are linear and $Sv_j = Tv_j$ for every j with $1 \le j \le p$, then $Sv = Tv$ for every $v \in V$.
2. Given any $w_1, \ldots, w_p \in W$, there is a linear map $T\colon V \to W$ with $Tv_j = w_j$ for every j with $1 \le j \le p$.

***Proof*:**

1. Let $v \in V$. By theorem 2.2.3 there are scalars $c_1, \ldots, c_p \in k$, uniquely determined by v, with $v = c_1v_1 + \cdots + c_pv_p$. Pushing S and T through this combination gives
$$Sv = c_1Sv_1 + \cdots + c_pSv_p = c_1Tv_1 + \cdots + c_pTv_p = Tv$$
where the middle equality is the hypothesis. Since v was arbitrary, the two maps agree everywhere.
2. Define T by
$$T(c_1v_1 + \cdots + c_pv_p) = c_1w_1 + \cdots + c_pw_p$$
Every $v \in V$ is $c_1v_1 + \cdots + c_pv_p$ for exactly one tuple $(c_1, \ldots, c_p)$, again by theorem 2.2.3, so this assigns to each v exactly one value. That value lies in W, since W is a subspace and so contains every linear combination of $w_1, \ldots, w_p$.
For linearity, let $v = c_1v_1 + \cdots + c_pv_p$ and $v' = c_1'v_1 + \cdots + c_p'v_p$. Then
$$v + v' = (c_1 + c_1')v_1 + \cdots + (c_p + c_p')v_p$$
and uniqueness makes $(c_1 + c_1', \ldots, c_p + c_p')$ the tuple belonging to $v + v'$. Hence
$$T(v + v') = (c_1 + c_1')w_1 + \cdots + (c_p + c_p')w_p = Tv + Tv'$$
The same argument applied to $tv = (tc_1)v_1 + \cdots + (tc_p)v_p$ gives $T(tv) = tTv$. Finally v_j itself has the tuple with 1 in position j and 0 elsewhere, so $Tv_j = w_j$, completing the proof.

Part (1) is a uniqueness statement and part (2) an existence statement, and together they say that linear maps $V \to W$ correspond to lists of p vectors of W, where p is the number of elements of a basis of V. For $V = \mathbb{R}^2$ and the basis e_1, e_2, the quarter turn of example 2.4.1 is the map sending $e_1 \mapsto \left(\begin{smallmatrix}0\\1\end{smallmatrix}\right)$ and $e_2 \mapsto \left(\begin{smallmatrix}-1\\0\end{smallmatrix}\right)$, and those two columns are the two columns of the matrix found there. Theorem 2.4.8 below shows that the columns of the matrix are always the images of the basis vectors.

Before the matrix comes the pair of subspaces a linear map carries with it. Consider the map

$T\colon \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$T\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x+y \\ 2x+2y \end{pmatrix}$$

which is linear, being the matrix rule for $\left(\begin{smallmatrix} 1 & 1 \\ 2 & 2 \end{smallmatrix}\right)$. A column is sent to 0 exactly when $x+y=0$ and $2x+2y=0$, and the second equation is twice the first, so the columns sent to 0 are those with $y=-x$, namely the multiples of $\left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$. A column is reached exactly when it is $\left(\begin{smallmatrix} s \\ 2s \end{smallmatrix}\right)$ for some $s \in \mathbb{R}$: every value of T has this shape with $s = x+y$, and every such column is the value of T at $\left(\begin{smallmatrix} s \\ 0 \end{smallmatrix}\right)$. So the columns reached are the multiples of $\left(\begin{smallmatrix} 1 \\ 2 \end{smallmatrix}\right)$. Both collections are subspaces of $\mathbb{R}^2$, and both are named.

Definition 2.4.4: Kernel

Let $V \subseteq k^n$ and $W \subseteq k^m$ be subspaces and let $T\colon V \to W$ be linear. The *kernel* of T is

$$\ker T = \{\, v \in V \,|\, Tv = 0 \,\}$$

the set of vectors of V that T sends to the zero vector of W.

Definition 2.4.5: Image

With T as above, the *image* of T is

$$\mathrm{im} T = \{\, Tv \,|\, v \in V \,\}$$

the set of vectors of W that T attains.

For the map $T_A(x) = Ax$ attached to a matrix $A \in M_{m\times n}(k)$ the two sets are already familiar. A column $x \in k^n$ lies in $\ker T_A$ exactly when $Ax = 0$, so $\ker T_A = \mathrm{null} A$ in the notation of definition 2.3.3. A column of k^m lies in $\mathrm{im} T_A$ exactly when it is Ax for some x, and by the column reading of the product this says it is a linear combination of the columns of A; so $\mathrm{im} T_A$ is the column space of A of definition 2.3.1. The kernel and the image are the null space and the column space read as statements about a map rather than about an array.

Proposition 2.4.6: The Kernel and the Image Are Subspaces

Let $V \subseteq k^n$ and $W \subseteq k^m$ be subspaces and let $T\colon V \to W$ be linear. Then $\ker T$ is a subspace of V and $\mathrm{im} T$ is a subspace of W.

***Proof*:**
Both are checked against proposition 2.1.3. The kernel contains 0, since $T0 = 0$. If $u, v \in \ker T$ and $t \in k$, then $T(u+v) = Tu + Tv = 0 + 0 = 0$ and $T(tu) = tTu = t \cdot 0 = 0$, so $u+v$ and tu lie in $\ker T$ as well.

The image contains 0, since $0 = T0$. If $w, w' \in \mathrm{im} T$, write $w = Tu$ and $w' = Tu'$ with $u, u' \in V$. Then $u + u' \in V$ and $tu \in V$, because V is a subspace, and

$$w + w' = Tu + Tu' = T(u+u'), \qquad tw = tTu = T(tu)$$

so $w + w'$ and tw are attained by T. Both sets therefore satisfy the criterion, as we sought to show.

The kernel measures how far T is from being injective, and it does so exactly.

Proposition 2.4.7: Injectivity Is Triviality of the Kernel

A linear map $T\colon V \to W$ is injective if and only if $\ker T = \{0\}$.

***Proof*:**
Suppose $\ker T = \{0\}$ and $Tu = Tv$. Then $T(u-v) = Tu - Tv = 0$, so $u - v \in \ker T$ and $u = v$. Conversely, suppose T is injective and $Tv = 0$. Since $T0 = 0$, this reads $Tv = T0$, and injectivity forces $v = 0$, as we sought to show.

A map is surjective, by definition, precisely when its image is all of W, so the two subspaces attached to T detect both halves of bijectivity.

A basis converts a vector of V into a column of scalars, and the next theorem converts a linear map into a matrix by the same device. The conversion needs one observation first. Fix a basis $v_1, \ldots, v_p$ of V, listed in a fixed order, and write $\mathcal{B} = (v_1, \ldots, v_p)$ for the list; recall from definition 2.2.4 that $[v]_{\mathcal{B}} \in k^p$ is the column whose j-th entry is the j-th coordinate of v in this basis. By proposition 2.2.5 the assignment $v \mapsto [v]_{\mathcal{B}}$ is a bijection $V \to k^p$ satisfying $[v+v']_{\mathcal{B}} = [v]_{\mathcal{B}} + [v']_{\mathcal{B}}$ and $[tv]_{\mathcal{B}} = t[v]_{\mathcal{B}}$, so it is itself a linear map in the sense just defined. Coordinates therefore match up V with k^p without losing or duplicating anything, and they do so compatibly with both operations.

Theorem 2.4.8: The Matrix of a Linear Map

Let $V \subseteq k^n$ and $W \subseteq k^m$ be nonzero subspaces, let $\mathcal{B} = (v_1, \ldots, v_p)$ be a basis of V and $\mathcal{C} = (w_1, \ldots, w_q)$ a basis of W, each listed in a fixed order. For each linear map $T\colon V \to W$ there is exactly one matrix $M \in M_{q\times p}(k)$, with q rows and p columns, satisfying

$$[Tv]_{\mathcal{C}} = M[v]_{\mathcal{B}} \quad \text{for every } v \in V$$

and its j-th column is $[Tv_j]_{\mathcal{C}}$. Conversely, every $M \in M_{q\times p}(k)$ arises in this way from exactly one linear map $T\colon V \to W$.

***Proof*:**
Let M be the matrix whose j-th column is $[Tv_j]_{\mathcal{C}} \in k^q$, for $1 \le j \le p$; it has q rows and p columns. Let $v \in V$ and write $[v]_{\mathcal{B}}$ for its coordinate column, with entries $c_1, \ldots, c_p$, so that $v = c_1v_1 + \cdots + c_pv_p$. Pushing T through the combination gives $Tv = c_1Tv_1 + \cdots + c_pTv_p$, and taking $\mathcal{C}$-coordinates, which is a linear operation by the paragraph preceding the theorem,

$$[Tv]_{\mathcal{C}} = c_1[Tv_1]_{\mathcal{C}} + \cdots + c_p[Tv_p]_{\mathcal{C}}$$

The right side is the combination of the columns of M with coefficients $c_1, \ldots, c_p$, which is $M[v]_{\mathcal{B}}$ by the column reading of the product. So M satisfies the displayed identity.

For uniqueness, suppose M' also satisfies it. The coordinate column of v_j is e_j, the column with 1 in position j and 0 elsewhere, so taking $v = v_j$ gives $M'e_j = [Tv_j]_{\mathcal{C}} = Me_j$. By the column reading, Me_j is the j-th column of M and $M'e_j$ is the j-th column of M'. The two matrices have the same columns and hence are equal.

For the converse, let $M \in M_{q\times p}(k)$ be given. Since coordinates match up W with k^q bijectively, there is for each j exactly one $u_j \in W$ whose coordinate column $[u_j]_{\mathcal{C}}$ is the j-th column of M. By proposition 2.4.3(2) there is a linear map $T\colon V \to W$ with $Tv_j = u_j$ for every j, and the matrix just constructed for this T has j-th column $[Tv_j]_{\mathcal{C}}$, which is the j-th column of M; so that matrix is M. If T' is a second linear map whose matrix is M, then $[T'v_j]_{\mathcal{C}}$ is the j-th column of M for every j, so $[T'v_j]_{\mathcal{C}} = [Tv_j]_{\mathcal{C}}$ and hence $T'v_j = Tv_j$; proposition 2.4.3(1) then gives $T' = T$, as we sought to show.

The matrix produced by the theorem is written $[T]_{\mathcal{CB}}$, with the basis of the target listed first and the basis of the source second, and when $V = W$ and $\mathcal{C} = \mathcal{B}$ it is abbreviated $[T]_{\mathcal{B}}$. The order of the two subscripts records the convention that the matrix multiplies coordinate columns *from the left*:

$$[Tv]_{\mathcal{C}} = [T]_{\mathcal{CB}}[v]_{\mathcal{B}}$$

This is the convention used throughout the book, and every later formula involving $[T]$ is written to agree with it.

Taking $V = k^n$ and $W = k^m$ with their standard bases makes the theorem concrete. The coordinate column of $x \in k^n$ in the standard basis is x itself, since $x = x_1e_1 + \cdots + x_ne_n$. The identity of the theorem then reads $Tx = Mx$, so every linear map $T\colon k^n \to k^m$ is the rule $x \mapsto Mx$ for exactly one matrix $M \in M_{m\times n}(k)$, whose j-th column is Te_j. Linear maps between the spaces k^n and matrices are therefore the same data in two notations. In example 2.4.1 the quarter turn sends e_1 to $\left(\begin{smallmatrix}0\\1\end{smallmatrix}\right)$ and e_2 to $\left(\begin{smallmatrix}-1\\0\end{smallmatrix}\right)$, and those are the columns of $\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$; the shear sends e_1 to e_1 and e_2 to $\left(\begin{smallmatrix}2\\1\end{smallmatrix}\right)$, and those are the columns of $\left(\begin{smallmatrix}1&2\\0&1\end{smallmatrix}\right)$.

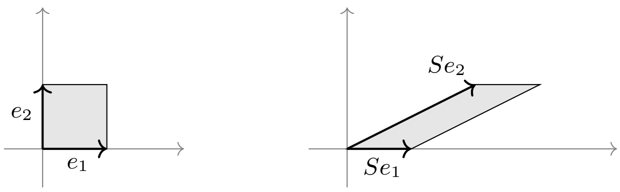


Figure 2.7: The shear S of example 2.4.1(2) on the unit square; the two arrows on the right are the columns of its matrix

Composing two linear maps multiplies their matrices, which is where the matrix product of definition 1.4.1 comes from.

Theorem 2.4.9: The Matrix of a Composite Is the Product of the Matrices

Let $V \subseteq k^n$, $W \subseteq k^m$ and $X \subseteq k^r$ be nonzero subspaces with bases $\mathcal{B}$, $\mathcal{C}$ and $\mathcal{D}$, and let $T\colon V \to W$ and $S\colon W \to X$ be linear. Then $S \circ T$ is linear and

$$[S \circ T]_{\mathcal{DB}} = [S]_{\mathcal{DC}}[T]_{\mathcal{CB}}$$

***Proof*:**
The composite is linear, since $S(T(u+v)) = S(Tu+Tv) = S(Tu) + S(Tv)$ and $S(T(tu)) = S(tTu) = tS(Tu)$. For every $v \in V$,

$$[(S\circ T)v]_{\mathcal{D}} = [S(Tv)]_{\mathcal{D}} = [S]_{\mathcal{DC}}[Tv]_{\mathcal{C}} = [S]_{\mathcal{DC}}\big([T]_{\mathcal{CB}}[v]_{\mathcal{B}}\big) = \big([S]_{\mathcal{DC}}[T]_{\mathcal{CB}}\big)[v]_{\mathcal{B}}$$

the last step by proposition 1.4.3. The uniqueness clause of theorem 2.4.8 then identifies the product as the matrix of the composite, completing the proof.

The rule for multiplying matrices, which was defined in section 1.4 by a formula, is the rule for composing the maps they represent.

The correspondence with matrices also transports the rank-nullity count of section 2.3 to maps. The statement there was about the columns of a matrix; here it is about the dimension of the source.

Theorem 2.4.10: Rank-Nullity for Linear Maps

Let $V \subseteq k^n$ and $W \subseteq k^m$ be subspaces and let $T\colon V \to W$ be linear. Then

$$\dim \ker T + \dim \mathrm{im} T = \dim V$$

***Proof*:**
Two degenerate cases are settled first. If $V = \{0\}$ then $\ker T$ and $\mathrm{im}T$ are both $\{0\}$ and the identity reads $0+0=0$. If $W = \{0\}$ then $Tv = 0$ for every v, so $\ker T = V$ and $\mathrm{im}T = \{0\}$, and the identity reads $\dim V + 0 = \dim V$. Assume from now on that neither V nor W is $\{0\}$, and choose bases $\mathcal{B} = (v_1, \dots, v_p)$ of V and $\mathcal{C} = (w_1, \dots, w_q)$ of W, which exist by theorem 2.2.9. Write $M = [T]_{\mathcal{CB}}$, a matrix with q rows and p columns, and write $\Phi(v) = [v]_{\mathcal{B}}$ and $\Psi(w) = [w]_{\mathcal{C}}$ for the two coordinate maps, each linear and bijective.

A linear bijection does not change dimensions. Suppose Λ is a linear map that is injective on a subspace U, and let $u_1, \dots, u_r$ be a basis of U. Every element of $\Lambda(U)$ is $\Lambda(c_1u_1 + \cdots + c_ru_r) = c_1\Lambda u_1 + \cdots + c_r\Lambda u_r$, so $\Lambda u_1, \dots, \Lambda u_r$ span $\Lambda(U)$; and if $c_1\Lambda u_1 + \cdots + c_r\Lambda u_r = 0$, then $\Lambda(c_1u_1 + \cdots + c_ru_r) = 0 = \Lambda 0$, so injectivity gives $c_1u_1 + \cdots + c_ru_r = 0$ and independence of the u_i gives $c_1 = \cdots = c_r = 0$. Hence $\Lambda u_1, \dots, \Lambda u_r$ is a basis of $\Lambda(U)$ and $\dim \Lambda(U) = \dim U$.

Apply this twice. First, $v \in \ker T$ holds exactly when $Tv = 0$, exactly when $[Tv]_{\mathcal{C}} = 0$ because Ψ is injective, exactly when $M[v]_{\mathcal{B}} = 0$ by theorem 2.4.8. So Φ carries $\ker T$ onto $\mathrm{null}M$, and $\dim \ker T = \dim \mathrm{null}M$. Second,

$$\Psi(\mathrm{im}T) = \{\, M[v]_{\mathcal{B}} \big| v \in V \,\} = \{\, Mx \big| x \in k^p \,\}$$

the second equality because Φ is onto k^p. By the column reading, the set on the right is the set of linear combinations of the columns of M, which is the column space of definition 2.3.1; its dimension is $\mathrm{rank}M$ by corollary 2.3.12. So $\dim \mathrm{im}T = \mathrm{rank}M$.

The matrix M has p columns, so theorem 2.3.11 gives $\dim \mathrm{null}M + \mathrm{rank}M = p$. Substituting the two dimensions computed above, and using $\dim V = p$, gives $\dim \ker T + \dim \mathrm{im}T = \dim V$, as we sought to show.

The theorem is a conservation statement. The source has $\dim V$ dimensions available; each dimension is either collapsed to 0 by T or survives into the image, and the two counts add up to $\dim V$. Nothing about the target W enters the identity, which is why enlarging W changes neither side.

The running matrix illustrates both counts at once. It is

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 1 & 2 & 5 \\ 3 & 4 & 11 \end{pmatrix} \in M_{4\times 3}(\mathbb{R})$$

of example 1.2.9, and $T_A\colon \mathbb{R}^3 \to \mathbb{R}^4$ is the map $x \mapsto Ax$. Example 2.3.14 found $\operatorname{rank} A = 2$, a basis $(1, 2, -1)^\top$ of the null space, and the first two columns of A as a basis of the column space. The null-space basis vector is verified directly: the four entries of $A(1, 2, -1)^\top$ are

$$1 + 2 - 3 = 0, \quad 2 + 6 - 8 = 0, \quad 1 + 4 - 5 = 0, \quad 3 + 8 - 11 = 0$$

So $\ker T_A = \operatorname{span}\{(1, 2, -1)^\top\}$ has dimension 1 and $\operatorname{im} T_A$ has dimension 2, and $1 + 2 = 3 = \dim \mathbb{R}^3$, in agreement with theorem 2.4.10. The map is not injective, since its kernel is larger than $\{0\}$, and not surjective, since a subspace of $\mathbb{R}^4$ of dimension 2 is not all of $\mathbb{R}^4$.

The last topic of the section is the operators satisfying $P \circ P = P$. One is computed in full before the name is attached.

Example 2.4.11: Projection onto a Line along Another Line

Work over $\mathbb{R}$ and let

$$P = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \text{so that} \qquad P\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ 0 \end{pmatrix}$$

Applying P twice changes nothing:

$$P^2 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 1\cdot 1 + 1\cdot 0 & 1\cdot 1 + 1\cdot 0 \\ 0\cdot 1 + 0\cdot 0 & 0\cdot 1 + 0\cdot 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = P$$

The values of P are the columns $\left(\begin{smallmatrix} s \\ 0 \end{smallmatrix}\right)$ with $s \in \mathbb{R}$: every value has this shape with $s = x + y$, and every such column is attained at $\left(\begin{smallmatrix} s \\ 0 \end{smallmatrix}\right)$. So $\operatorname{im} P = L$, where $L = \operatorname{span}\{e_1\}$ is the first coordinate axis. A column is sent to 0 exactly when $x + y = 0$, so

$$\ker P = \operatorname{span}\left\{\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right\} = L'$$

the line through the origin in the direction $\left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$.

The two lines fill $\mathbb{R}^2$ and meet only at 0. For the first,

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + y \\ 0 \end{pmatrix} + \begin{pmatrix} -y \\ y \end{pmatrix}$$

where the first summand lies in L and the second is $-y\left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right) \in L'$. For the second, a column in $L \cap L'$ is $\left(\begin{smallmatrix} s \\ 0 \end{smallmatrix}\right)$ and also $t\left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$ for some $s, t \in \mathbb{R}$; comparing second entries gives $-t = 0$, hence $t = 0$ and the column is 0. So $\mathbb{R}^2 = L \oplus L'$ in the sense of definition 2.1.11. On L the map P changes nothing, since $P\left(\begin{smallmatrix} s \\ 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} s \\ 0 \end{smallmatrix}\right)$, and on L' it is zero. Each column is moved parallel to L' until it lands on L: the displacement from $\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right)$ to its value is $\left(\begin{smallmatrix} x+y \\ 0 \end{smallmatrix}\right) - \left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = y\left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$.

The line L' is part of the data and not a consequence of L. The matrix $Q = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ also satisfies $Q^2 = Q$, and its values are again the columns $\left(\begin{smallmatrix} x \\ 0 \end{smallmatrix}\right)$, so $\operatorname{im} Q = L$ as well; but $Q\left(\begin{smallmatrix} x \\ y \end{smallmatrix}\right) = 0$ forces $x = 0$, so $\ker Q = \operatorname{span}\{e_2\}$, which is not L'. Two different maps share the image L and differ in the line along which they move columns onto it.

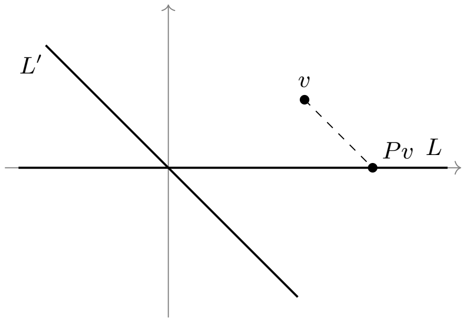


Figure 2.8: The map P of example 2.4.11 moves each column parallel to L' until it meets L

Definition 2.4.12: Projection

Let $V \subseteq k^n$ be a subspace. A linear operator $P\colon V \to V$ with $P \circ P = P$ is a *projection*, and is also called *idempotent*. A matrix $P \in M_n(k)$ with $P^2 = P$ is an *idempotent matrix*.

The two versions match under theorem 2.4.8: the matrix of $P \circ P$ relative to a single basis is the square of the matrix of P, so an operator on a nonzero subspace is a projection exactly when its matrix is idempotent. What example 2.4.11 exhibited, a splitting of the space into the image and the kernel with P leaving the first summand unchanged, holds for every projection, and every splitting arises this way.

Proposition 2.4.13: Idempotents and Direct Sum Decompositions

Let $V \subseteq k^n$ be a subspace.

1. If $P\colon V \to V$ is a projection, then $V = \operatorname{im} P \oplus \ker P$, and $Pu = u$ for every $u \in \operatorname{im} P$.
2. If $V = U \oplus Z$ for subspaces U and Z, then the map P sending $u + z$ to u, for $u \in U$ and $z \in Z$, is a projection with $\operatorname{im} P = U$ and $\ker P = Z$.

***Proof*:**

1. Let $v \in V$ and write $v = Pv + (v - Pv)$. The first summand lies in $\operatorname{im} P$. The second lies in $\ker P$, since
$$P(v - Pv) = Pv - P(Pv) = Pv - Pv = 0$$
using linearity of P and $P \circ P = P$. So every element of V is a sum of an element of $\operatorname{im} P$ and an element of $\ker P$.

   Next let $u \in \operatorname{im} P$, say $u = Pv$. Then $Pu = P(Pv) = Pv = u$, which is the second assertion. If moreover $u \in \ker P$, then $u = Pu = 0$. So $\operatorname{im} P \cap \ker P = \{0\}$, and with the previous paragraph this is the condition of definition 2.1.11 for $V = \operatorname{im} P \oplus \ker P$.
2. Since $V = U \oplus Z$, proposition 2.1.12 says that each $v \in V$ is $u + z$ with $u \in U$ and $z \in Z$ in exactly one way, so P assigns to each v exactly one value. For linearity, let $v = u + z$ and $v' = u' + z'$ with $u, u' \in U$ and $z, z' \in Z$. Then $v + v' = (u + u') + (z + z')$ with $u + u' \in U$

and $z + z' \in Z$, so uniqueness of the decomposition gives $P(v + v') = u + u' = Pv + Pv'$; likewise $tv = tu + tz$ gives $P(tv) = tu = tPv$.

For idempotence, $Pv = u$ decomposes as $u + 0$ with $u \in U$ and $0 \in Z$, so $P(Pv) = u = Pv$. Every value of P lies in U, and each $u \in U$ is the value of P at $u = u + 0$, so $\mathrm{im}P = U$. Finally $Pv = 0$ says $u = 0$, that is $v = z \in Z$, and conversely each $z \in Z$ decomposes as $0 + z$ and is sent to 0; so $\ker P = Z$, completing the proof.

Projections on V and decompositions of V into two complementary subspaces are therefore the same data. Part (1) turns a projection into a decomposition and part (2) turns a decomposition into a projection, and the two constructions undo one another: starting from P and forming the decomposition $V = \mathrm{im}P \oplus \ker P$ returns P, because the map of part (2) sends $v = Pv + (v - Pv)$ to Pv.

For a matrix the statement reads as follows. If $P \in M_n(k)$ satisfies $P^2 = P$, then k^n is the direct sum of the column space of P and the null space $\mathrm{null}P$, and $Px = x$ for every column x in the column space. Theorem 2.3.11 already gives $\mathrm{rank}P + \dim \mathrm{null}P = n$ for every matrix, idempotent or not. What the proposition adds for an idempotent is that these two particular subspaces meet only in 0 and together span k^n, which the dimension count on its own does not assert.

Nothing in this description singles out one complement of a subspace as preferable to another. In example 2.4.11 the lines L' and $\mathrm{span}\{e_2\}$ are two complements of L, and the two projections onto L that they determine are different maps; neither is the projection onto L. The apparatus that selects one complement, and with it one projection, is the inner product introduced in Part II.

Exercise 2.4.1

1. Decide which of the following maps $\mathbb{R}^2 \to \mathbb{R}^2$ are linear in the sense of definition 2.4.2, verifying both conditions for those that are and exhibiting a failure for those that are not. For the linear ones, write down the matrix relative to the standard bases.

$$T_1 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x - y \\ x + 3y \end{pmatrix}, \qquad T_2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ |y| \end{pmatrix}, \qquad T_3 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} xy \\ 0 \end{pmatrix}$$

2. Let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be the map $x \mapsto Mx$ with

$$M = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 2 & 3 \end{pmatrix}$$

Find a basis of $\ker T$ and a basis of $\mathrm{im}T$, and check the two dimensions against theorem 2.4.10.

3. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map whose matrix relative to the standard bases is $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, and let $\mathcal{B} = (b_1, b_2)$ with $b_1 = \left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)$ and $b_2 = \left(\begin{smallmatrix} 1 \\ -1 \end{smallmatrix}\right)$. Show that $\mathcal{B}$ is a basis of $\mathbb{R}^2$, compute $[T]_{\mathcal{B}}$, and verify the identity $[Tv]_{\mathcal{B}} = [T]_{\mathcal{B}}[v]_{\mathcal{B}}$ of theorem 2.4.8 for $v = \left(\begin{smallmatrix} 3 \\ 1 \end{smallmatrix}\right)$.

4. Let

$$P = \begin{pmatrix} 2 & -1 \\ 2 & -1 \end{pmatrix}$$

Verify that $P^2 = P$, find $\mathrm{im}P$ and $\ker P$, check that $\mathbb{R}^2 = \mathrm{im}P \oplus \ker P$ as proposition 2.4.13 predicts, and write $\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)$ explicitly as a sum of an element of the image and an element of the kernel.

5. Let $T\colon V \to W$ be linear, with $V \subseteq k^n$ and $W \subseteq k^m$ subspaces. Show that T is injective if and only if $\ker T = \{0\}$. Show further that if $\dim V = \dim W$, then T is injective if and only if T is surjective.

6. Let $P\colon V \to V$ be a projection in the sense of definition 2.4.12 and let id_V be the identity map of V. Show that $\mathrm{id}_V - P$, the map $v \mapsto v - Pv$, is also a projection, and that

$$\mathrm{im}(\mathrm{id}_V - P) = \ker P, \qquad \ker(\mathrm{id}_V - P) = \mathrm{im}P$$

Then compute the matrix of $\mathrm{id}_{\mathbb{R}^2} - P$ for the P of exercise 2.4.1 (4) and confirm the two identities for it.

7. Let $T\colon \mathbb{R}^3 \to \mathbb{R}^2$ be the unique linear map with

$$T\begin{pmatrix}1\\0\\0\end{pmatrix} = \begin{pmatrix}1\\2\end{pmatrix}, \qquad T\begin{pmatrix}1\\1\\0\end{pmatrix} = \begin{pmatrix}0\\1\end{pmatrix}, \qquad T\begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}3\\3\end{pmatrix}$$

whose existence and uniqueness are guaranteed by proposition 2.4.3. Find the matrix of T relative to the standard bases, then find a basis of $\ker T$ and decide whether T is surjective.

2.5 Change of Basis, Similarity, and the Trace

A linear map of k^n to itself acquires a matrix only after a basis has been fixed, and a different basis produces a different matrix for the same map. This section computes how the two matrices are related, and then extracts a number that every one of them shares.

Throughout, a basis is listed in a fixed order, since the coordinates of a vector are recorded in the order of the basis vectors. Write $\mathcal{B} = (b_1, \dots, b_n)$ for an ordered basis of k^n, and recall from definition 2.2.4 that $[v]_\mathcal{B} \in k^n$ denotes the column of coordinates of v in $\mathcal{B}$, so that $v = x_1 b_1 + \cdots + x_n b_n$ exactly when $[v]_\mathcal{B}$ is the column with entries $x_1, \dots, x_n$. Write $\mathcal{E} = (e_1, \dots, e_n)$ for the standard basis. For a linear map $T\colon k^n \to k^n$, write $[T]_\mathcal{B}$ for the matrix of T computed with $\mathcal{B}$ on both the input and the output side, so that $[T]_\mathcal{B}$ is the unique matrix with

$$[T(v)]_\mathcal{B} = [T]_\mathcal{B}\,[v]_\mathcal{B}$$

for every $v \in k^n$; this is theorem 2.4.8 with one basis used on both sides.

Example 2.5.1: Two Bases of $\mathbb{R}^2$ and the Matrix Between Them

Take the two columns

$$b_1 = \begin{pmatrix}1\\1\end{pmatrix}, \qquad b_2 = \begin{pmatrix}1\\2\end{pmatrix}$$

in $\mathbb{R}^2$. For an arbitrary column with entries s and t, the equation $xb_1 + yb_2 = (s,t)^\top$ is the system

$$\begin{aligned} x + y &= s \\ x + 2y &= t \end{aligned}$$

Subtracting the first equation from the second gives $y = t - s$, and the first then gives $x = s - y = 2s - t$. Every column is therefore a combination of b_1 and b_2 in exactly one way, so $\mathcal{B} = (b_1, b_2)$ is a basis of $\mathbb{R}^2$ (definition 2.2.1) and

$$\left[\begin{pmatrix} s \\ t \end{pmatrix}\right]_{\mathcal{B}} = \begin{pmatrix} 2s - t \\ t - s \end{pmatrix}$$

For $w = (5, 8)^\top$ this reads $2 \cdot 5 - 8 = 2$ and $8 - 5 = 3$, so $w = 2b_1 + 3b_2$.

Assemble b_1 and b_2 into the columns of a matrix,

$$P = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$$

By the column reading of a matrix product established in section 1.2, the product $P(2,3)^\top$ is $2b_1 + 3b_2 = w$: the matrix P converts $\mathcal{B}$-coordinates into standard coordinates. Elimination inverts it (theorem 1.4.8). Starting from P with the identity attached,

$$\left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array}\right) \xrightarrow{R_2 \mapsto R_2 - R_1} \left(\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & 1 & -1 & 1 \end{array}\right) \xrightarrow{R_1 \mapsto R_1 - R_2} \left(\begin{array}{cc|cc} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 1 \end{array}\right)$$

so

$$P^{-1} = \begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$$

which is the rule $(s, t)^\top \mapsto (2s - t, t - s)^\top$ computed above, written as a matrix.

Now let $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map

$$T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -x_1 + 3x_2 \\ -6x_1 + 8x_2 \end{pmatrix}$$

Its matrix in the standard basis is read off the coefficients,

$$[T]_{\mathcal{E}} = M = \begin{pmatrix} -1 & 3 \\ -6 & 8 \end{pmatrix}$$

Its matrix in $\mathcal{B}$ requires the images of b_1 and b_2 back in $\mathcal{B}$-coordinates. Those images are

$$T(b_1) = \begin{pmatrix} -1 + 3 \\ -6 + 8 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} = 2b_1, \qquad T(b_2) = \begin{pmatrix} -1 + 6 \\ -6 + 16 \end{pmatrix} = \begin{pmatrix} 5 \\ 10 \end{pmatrix} = 5b_2$$

so $[T(b_1)]_{\mathcal{B}} = (2, 0)^\top$ and $[T(b_2)]_{\mathcal{B}} = (0, 5)^\top$, and these are the columns of the matrix of T in $\mathcal{B}$:

$$[T]_{\mathcal{B}} = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}$$

The two matrices are related by P. Multiplying out,

$$P[T]_{\mathcal{B}}P^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 5 \\ 2 & 10 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 3 \\ -6 & 8 \end{pmatrix}$$

which is M.

The map T of the example does one thing to b_1 and one thing to b_2: it stretches the first by 2 and the second by 5. The matrix $\left(\begin{smallmatrix} 2 & 0 \\ 0 & 5 \end{smallmatrix}\right)$ says so and nothing else, while the matrix M says the same thing in coordinates that are not adapted to the map. Both matrices describe T completely, and the passage between them went through the single matrix P.

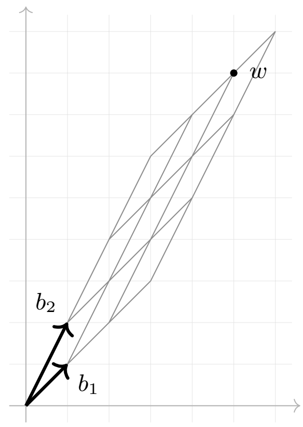


Figure 2.9: The lines through b_1 and b_2 rule the plane into a second grid, in which w sits two steps along b_1 and three along b_2

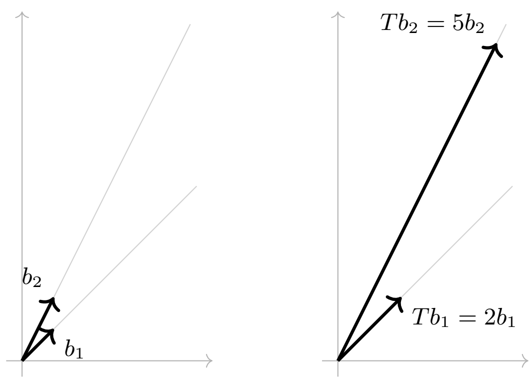


Figure 2.10: The map T of example 2.5.1 leaves both lines where they are and stretches b_1 by 2 and b_2 by 5, which is what the matrix $\left(\begin{smallmatrix} 2 & 0 \\ 0 & 5 \end{smallmatrix}\right)$ records

The matrix P of the example was built from one basis and the standard basis, but its construction used no property special to the standard basis. Two arbitrary bases of k^n produce such a matrix in the same way.

Definition 2.5.2: Change of Basis Matrix

Let $\mathcal{B} = (b_1, \dots, b_n)$ and $\mathcal{C} = (c_1, \dots, c_n)$ be ordered bases of k^n. The *change of basis matrix* from $\mathcal{B}$ to $\mathcal{C}$ is the matrix

$$P_{\mathcal{C}\mathcal{B}} \in M_n(k)$$

whose j-th column is $[b_j]_{\mathcal{C}}$, the coordinate column of the j-th vector of $\mathcal{B}$ written in the basis $\mathcal{C}$. The two subscripts are read from right to left: the matrix receives $\mathcal{B}$-coordinates and returns $\mathcal{C}$-coordinates.

The definition says what the columns of $P_{\mathcal{C}\mathcal{B}}$ are, not what the matrix does. What it does is convert coordinates, and it is the only matrix that converts them.

Proposition 2.5.3: Properties of the Change of Basis Matrix

Let $\mathcal{B}$, $\mathcal{C}$ and $\mathcal{D}$ be ordered bases of k^n.

1. For every $v \in k^n$, $[v]_{\mathcal{C}} = P_{\mathcal{C}\mathcal{B}}\,[v]_{\mathcal{B}}$, and $P_{\mathcal{C}\mathcal{B}}$ is the only matrix in $M_n(k)$ with this property.
2. $P_{\mathcal{B}\mathcal{B}} = I_n$ and $P_{\mathcal{D}\mathcal{C}}P_{\mathcal{C}\mathcal{B}} = P_{\mathcal{D}\mathcal{B}}$.
3. $P_{\mathcal{C}\mathcal{B}}$ is invertible, with inverse $P_{\mathcal{B}\mathcal{C}}$.

***Proof*:**

1. By proposition 2.2.5, taking coordinates in $\mathcal{C}$ satisfies $[u+w]_{\mathcal{C}} = [u]_{\mathcal{C}}+[w]_{\mathcal{C}}$ and $[\lambda u]_{\mathcal{C}} = \lambda[u]_{\mathcal{C}}$ for all $u, w \in k^n$ and $\lambda \in k$.

   Now write $x = [v]_{\mathcal{B}}$, so that $v = x_1b_1 + \cdots + x_nb_n$. Applying the two identities to this expression gives

   $$[v]_{\mathcal{C}} = x_1[b_1]_{\mathcal{C}} + \cdots + x_n[b_n]_{\mathcal{C}}$$

   The column $[b_j]_{\mathcal{C}}$ is the j-th column of $P_{\mathcal{C}\mathcal{B}}$, and the right-hand side is exactly the column reading of the product $P_{\mathcal{C}\mathcal{B}}x$ (section 1.2). Hence $[v]_{\mathcal{C}} = P_{\mathcal{C}\mathcal{B}}[v]_{\mathcal{B}}$.

   For uniqueness, suppose $Q \in M_n(k)$ satisfies $[v]_{\mathcal{C}} = Q[v]_{\mathcal{B}}$ for every $v \in k^n$. Take $v = b_j$. Its $\mathcal{B}$-coordinate column is e_j, since b_j is the combination of $b_1, \dots, b_n$ with coefficient 1 in position j and 0 elsewhere. So $Qe_j = [b_j]_{\mathcal{C}}$. The product Qe_j is the j-th column of Q, so Q and $P_{\mathcal{C}\mathcal{B}}$ agree in every column.

2. The j-th column of $P_{\mathcal{B}\mathcal{B}}$ is $[b_j]_{\mathcal{B}} = e_j$, which is the j-th column of I_n (definition 1.4.2). For the second identity, apply part (1) three times: for every $v \in k^n$,

   $$P_{\mathcal{D}\mathcal{C}}P_{\mathcal{C}\mathcal{B}}[v]_{\mathcal{B}} = P_{\mathcal{D}\mathcal{C}}[v]_{\mathcal{C}} = [v]_{\mathcal{D}}$$

   using associativity of the matrix product (proposition 1.4.3) in the first step. The matrix $P_{\mathcal{D}\mathcal{C}}P_{\mathcal{C}\mathcal{B}}$ therefore converts $\mathcal{B}$-coordinates into $\mathcal{D}$-coordinates, and the uniqueness in part (1) identifies it with $P_{\mathcal{D}\mathcal{B}}$.

3. Taking $\mathcal{D} = \mathcal{B}$ in part (2) gives $P_{\mathcal{B}\mathcal{C}}P_{\mathcal{C}\mathcal{B}} = P_{\mathcal{B}\mathcal{B}} = I_n$, and exchanging the roles of $\mathcal{B}$ and $\mathcal{C}$ gives $P_{\mathcal{C}\mathcal{B}}P_{\mathcal{B}\mathcal{C}} = I_n$. Both products are the identity, which is what definition 1.4.4 requires, completing the proof.

The case computed in practice is $\mathcal{C} = \mathcal{E}$. The coordinate column $[u]_{\mathcal{E}}$ is then u itself, because $u = u_1 e_1 + \cdots + u_n e_n$. The j-th column of $P_{\mathcal{E}\mathcal{B}}$ is then b_j itself: $P_{\mathcal{E}\mathcal{B}}$ is the matrix whose columns are the vectors of $\mathcal{B}$, which is how P was built in example 2.5.1. Two arbitrary bases are then handled by routing through $\mathcal{E}$, since part (2) of proposition 2.5.3 gives

$$P_{\mathcal{C}\mathcal{B}} = P_{\mathcal{C}\mathcal{E}} P_{\mathcal{E}\mathcal{B}} = \left(P_{\mathcal{E}\mathcal{C}}\right)^{-1} P_{\mathcal{E}\mathcal{B}}$$

and both factors on the right are written down by inspection, the second inverted by elimination.

With the conversion of coordinates settled, the matrix of a map in two different bases can be compared. The comparison is the computation of example 2.5.1, carried out in general.

Theorem 2.5.4: Change of Basis for the Matrix of a Linear Map

Let $T\colon k^n \to k^n$ be linear and let $\mathcal{B}$ and $\mathcal{C}$ be ordered bases of k^n. Put $P = P_{\mathcal{C}\mathcal{B}}$. Then

$$[T]_{\mathcal{C}} = P\,[T]_{\mathcal{B}}\,P^{-1}$$

***Proof*:**
Let $v \in k^n$. Converting to $\mathcal{B}$-coordinates, applying T there, and converting back gives

$$[T(v)]_{\mathcal{C}} = P\,[T(v)]_{\mathcal{B}} = P\,[T]_{\mathcal{B}}\,[v]_{\mathcal{B}} = P\,[T]_{\mathcal{B}}\,P^{-1}[v]_{\mathcal{C}}$$

where the first and third equalities are proposition 2.5.3(1), applied in the two directions and using $P^{-1} = P_{\mathcal{B}\mathcal{C}}$, and the second is the defining property of $[T]_{\mathcal{B}}$ from theorem 2.4.8.

The left-hand side is $[T]_{\mathcal{C}}[v]_{\mathcal{C}}$, again by theorem 2.4.8. So the two matrices $[T]_{\mathcal{C}}$ and $P[T]_{\mathcal{B}}P^{-1}$ send $[v]_{\mathcal{C}}$ to the same column, for every $v \in k^n$. Every column $x \in k^n$ arises as $[v]_{\mathcal{C}}$, namely for $v = x_1 c_1 + \cdots + x_n c_n$, so the two matrices agree on every $x \in k^n$. Taking $x = e_j$ compares their j-th columns, and the columns agree for every j, as we sought to show.

The matrix of a map therefore depends on the basis, but only to the extent of being multiplied by an invertible matrix on the left and by its inverse on the right. Anything about a matrix that is left unchanged by that operation is a statement about the map rather than about the coordinates used to write it down. The operation is common enough to be given a name of its own.

Definition 2.5.5: Similar Matrices

Two matrices $A, B \in M_n(k)$ are *similar* when there is an invertible $P \in M_n(k)$ with

$$B = PAP^{-1}$$

Theorem 2.5.4 says that the matrices of one linear map in two bases are similar. The converse holds as well, so nothing is lost by working with the definition instead of with bases. Suppose $B = PAP^{-1}$ with P invertible, let T be the linear map $T(x) = Ax$, and let b_j be the j-th column of P^{-1}. These n columns form a basis $\mathcal{B}$ of k^n: they span, because any w equals $P^{-1}(Pw)$, which the column reading exhibits as a combination of them; and they are independent, because a combination $x_1 b_1 + \cdots + x_n b_n$ equal to 0 is the product $P^{-1}x$, so $x = P(P^{-1}x) = 0$. Since $P_{\mathcal{E}\mathcal{B}}$ has the b_j as its columns, $P_{\mathcal{E}\mathcal{B}} = P^{-1}$ and hence $P_{\mathcal{B}\mathcal{E}} = P$. Now $[T]_{\mathcal{E}} = A$, and theorem 2.5.4 gives $[T]_{\mathcal{B}} = PAP^{-1} = B$. So A and B are the two matrices of the single map T, read in the standard basis and in $\mathcal{B}$.

Similarity is therefore exactly the relation of representing one map in two bases, and a relation used to sort $M_n(k)$ this way has to pass three tests.

Proposition 2.5.6: Similarity Is Reflexive, Symmetric, and Transitive

Let $A, B, C \in M_n(k)$. Then A is similar to A; if A is similar to B, then B is similar to A; and if A is similar to B and B is similar to C, then A is similar to C.

***Proof*:**
The identity matrix is invertible and $I_n A I_n^{-1} = A$, which gives the first claim. For the second, suppose $B = PAP^{-1}$ with P invertible. Multiplying on the left by P^{-1} and on the right by P gives $P^{-1}BP = A$. The matrix P^{-1} is invertible with inverse P (definition 1.4.4 is symmetric in the two matrices), so $A = P^{-1}B(P^{-1})^{-1}$ exhibits B as similar to A.

For the third, suppose $B = PAP^{-1}$ and $C = QBQ^{-1}$ with P and Q invertible. By proposition 1.4.5 the product QP is invertible with $(QP)^{-1} = P^{-1}Q^{-1}$. Hence, regrouping by proposition 1.4.3,

$$C = QBQ^{-1} = Q(PAP^{-1})Q^{-1} = (QP)A(P^{-1}Q^{-1}) = (QP)A(QP)^{-1}$$

so A is similar to C, completing the proof.

A relation with these three properties is an equivalence relation, and it sorts $M_n(k)$ into classes. One class holds all the matrices of one linear map, one for each basis of k^n. Every invariant attached to a map in chapter 4 has to be constant on these classes.

What is wanted is a number computed from the entries of a matrix and left unchanged by the passage from A to PAP^{-1}. Adding all the entries will not serve: the two matrices of example 2.5.1 have entry sums $-1 + 3 - 6 + 8 = 4$ and $2 + 0 + 0 + 5 = 7$. Adding only the entries on the diagonal gives $-1 + 8 = 7$ in the first case and $2 + 5 = 7$ in the second. That sum is the one that works.

Definition 2.5.7: Trace of a Square Matrix

Let $A = (a_{ij}) \in M_n(k)$. The *trace* of A is the sum of its diagonal entries,

$$\operatorname{tr} A = a_{11} + a_{22} + \cdots + a_{nn} = \sum_{i=1}^{n} a_{ii}$$

The trace is defined only for square matrices, since a rectangular matrix has no diagonal running from one corner to the other. It adds and scales entrywise: from the definition, $\operatorname{tr}(A + B) = \operatorname{tr} A + \operatorname{tr} B$ and $\operatorname{tr}(\lambda A) = \lambda \operatorname{tr} A$ for $A, B \in M_n(k)$ and $\lambda \in k$, because the (i, i) entry of $A + B$ is $a_{ii} + b_{ii}$ and that of λA is λa_{ii}.

The trace of a product is a different matter. The diagonal of AB is not determined by the diagonals of A and B, so there is no reason from the definition alone to expect $\operatorname{tr}(AB)$ and $\operatorname{tr}(BA)$ to agree, and indeed the two products need not even have the same size. They do have the same trace.

Proposition 2.5.8: The Trace of a Product Is Unchanged by Cyclic Rotation

1. Let $A \in M_{m\times n}(k)$ and $B \in M_{n\times m}(k)$. Then $AB \in M_m(k)$ and $BA \in M_n(k)$ are both square, of sizes m and n respectively, and
$$\operatorname{tr}(AB) = \operatorname{tr}(BA)$$
2. Let $A \in M_{m\times n}(k)$, $B \in M_{n\times p}(k)$ and $C \in M_{p\times m}(k)$. Then
$$\operatorname{tr}(ABC) = \operatorname{tr}(BCA) = \operatorname{tr}(CAB)$$

***Proof*:**

1. The product AB has m rows and m columns, and BA has n of each, so both are square and both have a trace. Write $A = (a_{ij})$ with $1 \le i \le m$ and $1 \le j \le n$, and $B = (b_{ji})$ with $1 \le j \le n$ and $1 \le i \le m$. By definition 1.4.1 the (i,i) entry of AB is $\sum_{j=1}^{n} a_{ij}b_{ji}$, so
$$\operatorname{tr}(AB) = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}b_{ji}$$
The (j,j) entry of BA is $\sum_{i=1}^{m} b_{ji}a_{ij}$, so
$$\operatorname{tr}(BA) = \sum_{j=1}^{n}\sum_{i=1}^{m} b_{ji}a_{ij}$$
The two displayed expressions add the same mn products $a_{ij}b_{ji}$, one grouping them by i and the other by j. Addition in k does not depend on the order of the summands, so the two sums are equal.
2. Apply part (1) to the pair A and BC, which have sizes $m \times n$ and $n \times m$: this gives $\operatorname{tr}(A(BC)) = \operatorname{tr}((BC)A)$, that is $\operatorname{tr}(ABC) = \operatorname{tr}(BCA)$. Apply it again to the pair B and CA, of sizes $n \times p$ and $p \times n$: this gives $\operatorname{tr}(BCA) = \operatorname{tr}(CAB)$. Associativity (proposition 1.4.3) allows the brackets to be dropped throughout, completing the proof.

Part (2) rotates the three factors without changing their cyclic order. It does not permute them arbitrarily, and no such statement is available: with
$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$
one computes $AB = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$ and then $ABC = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, of trace 1, while $BA = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and then BAC is the zero matrix, of trace 0. Exchanging two adjacent factors can change the trace.

Rotation is all that a product PAP^{-1} asks for: one rotation carries the leading P to the far right, next to P^{-1}.

Corollary 2.5.9: The Trace Is a Similarity Invariant

Similar matrices in $M_n(k)$ have the same trace. Consequently, if $T\colon k^n \to k^n$ is linear, the number $\mathrm{tr}[T]_{\mathcal{B}}$ is the same for every ordered basis $\mathcal{B}$ of k^n; it is called the *trace of T* and written $\mathrm{tr}T$.

***Proof*:**
Let $B = PAP^{-1}$ with $P \in M_n(k)$ invertible. Apply proposition 2.5.8(1) to the pair PA and P^{-1}, both in $M_n(k)$:

$$\mathrm{tr}B = \mathrm{tr}\left((PA)P^{-1}\right) = \mathrm{tr}\left(P^{-1}(PA)\right) = \mathrm{tr}\left((P^{-1}P)A\right) = \mathrm{tr}(I_nA) = \mathrm{tr}A$$

where the regrouping is associativity (proposition 1.4.3). For the second claim, let $\mathcal{B}$ and $\mathcal{C}$ be ordered bases of k^n. By theorem 2.5.4 the matrices $[T]_{\mathcal{B}}$ and $[T]_{\mathcal{C}}$ are similar, so the first claim gives $\mathrm{tr}[T]_{\mathcal{B}} = \mathrm{tr}[T]_{\mathcal{C}}$, as we sought to show.

The trace is therefore attached to the map and not to any choice of coordinates, and it can be computed from whichever matrix is most convenient. It does not distinguish similarity classes from one another. The matrices I_2 and $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ both have trace 2, and they are not similar: $PI_2P^{-1} = I_2$ for every invertible P, so the only matrix similar to I_2 is I_2 itself.

Example 2.5.10: Traces of a Similar Pair and of the Two Products $A^\top A$ and $AA^\top$

1. Take
$$N = \begin{pmatrix}1 & 2\\ 3 & 4\end{pmatrix}, \qquad P = \begin{pmatrix}3 & 1\\ 2 & 1\end{pmatrix}, \qquad Q = \begin{pmatrix}1 & -1\\ -2 & 3\end{pmatrix}$$
Then $PQ = \left(\begin{smallmatrix}3-2 & -3+3\\ 2-2 & -2+3\end{smallmatrix}\right) = I_2$, and the product in the other order is I_2 as well, so P is invertible with $P^{-1} = Q$. Forming PNP^{-1} gives
$$PN = \begin{pmatrix}3+3 & 6+4\\ 2+3 & 4+4\end{pmatrix} = \begin{pmatrix}6 & 10\\ 5 & 8\end{pmatrix}, \qquad PNP^{-1} = \begin{pmatrix}6 & 10\\ 5 & 8\end{pmatrix}\begin{pmatrix}1 & -1\\ -2 & 3\end{pmatrix} = \begin{pmatrix}-14 & 24\\ -11 & 19\end{pmatrix}$$
No entry of the result agrees with the corresponding entry of N, and the traces agree: $1+4=5$ and $-14+19=5$, as corollary 2.5.9 requires.

2. The running matrix of example 1.2.9 is
$$A = \begin{pmatrix}1 & 1 & 3\\ 2 & 3 & 8\\ 1 & 2 & 5\\ 3 & 4 & 11\end{pmatrix} \in M_{4\times 3}(\mathbb{R})$$
Its transpose $A^\top$ lies in $M_{3\times 4}(\mathbb{R})$, so $A^\top A$ is 3×3 and $AA^\top$ is 4×4. These are matrices of different sizes, and proposition 2.5.8(1) predicts that they have the same trace. The (k,l) entry of $A^\top A$ is $\sum_{i=1}^4 a_{ik}a_{il}$, an expression symmetric in k and l, so six computations settle all nine entries:
$$\begin{aligned} &(1,1)\colon\ 1+4+1+9=15 && (1,2)\colon\ 1+6+2+12=21\\ &(1,3)\colon\ 3+16+5+33=57 && (2,2)\colon\ 1+9+4+16=30\\ &(2,3)\colon\ 3+24+10+44=81 && (3,3)\colon\ 9+64+25+121=219 \end{aligned}$$

that is

$$A^\top A = \begin{pmatrix} 15 & 21 & 57 \\ 21 & 30 & 81 \\ 57 & 81 & 219 \end{pmatrix}$$

whose trace is $15 + 30 + 219 = 264$.

For $AA^\top$ only the diagonal is needed, and its (i,i) entry is $\sum_{j=1}^{3} a_{ij}^2$, the sum of the squares of the entries in row i of A. The four values are

$$1+1+9 = 11, \quad 4+9+64 = 77, \quad 1+4+25 = 30, \quad 9+16+121 = 146$$

and they add to 264. Both traces count the same 12 squares a_{ij}^2, the first grouping them by column and the second by row, which is the proof of proposition 2.5.8(1) in this instance.

Exercise 2.5.1

1. Let $b_1 = (2,1)^\top$ and $b_2 = (1,1)^\top$ in $\mathbb{R}^2$, and let $\mathcal{B} = (b_1, b_2)$. Show that $\mathcal{B}$ is a basis, compute $[v]_\mathcal{B}$ for $v = (5,3)^\top$, write down $P_{\mathcal{E}\mathcal{B}}$, and compute $P_{\mathcal{B}\mathcal{E}}$ by elimination.
2. Let $\mathcal{B} = (b_1, b_2)$ with $b_1 = (1,1)^\top$, $b_2 = (1,2)^\top$, and let $\mathcal{C} = (c_1, c_2)$ with $c_1 = (1,0)^\top$, $c_2 = (1,1)^\top$, both bases of $\mathbb{R}^2$. Compute $P_{\mathcal{C}\mathcal{B}}$ by routing through the standard basis, and then check the answer against definition 2.5.2 by expressing b_1 and b_2 directly in terms of c_1 and c_2.
3. Let $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map sending $(x_1, x_2)^\top$ to $(4x_1 - x_2,\ 2x_1 + x_2)^\top$, and let $\mathcal{B} = (b_1, b_2)$ with $b_1 = (1,1)^\top$ and $b_2 = (1,2)^\top$. Compute $[T]_\mathcal{E}$ and then $[T]_\mathcal{B}$, first directly from the images of b_1 and b_2 and then from theorem 2.5.4. Compare the traces.
4. Let
$$A = \begin{pmatrix} 1 & 0 & 2 \\ 3 & 1 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 \\ 0 & 4 \\ 1 & -1 \end{pmatrix}$$
Compute AB and BA in full and verify that their traces agree although the two matrices have different sizes. Then show that there are no matrices $X, Y \in M_n(k)$ with $XY - YX = I_n$.
5. Show that $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ are similar by exhibiting an invertible P with $P\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)P^{-1} = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$, and identify the basis of $\mathbb{R}^2$ that P corresponds to.
6. Show that similar matrices have the same rank.
7. Let $P \in M_n(k)$ satisfy $P^2 = P$. Show that $\operatorname{tr} P = \operatorname{rank} P$.

3

Determinants, Volume, and Orientation

Elimination has so far produced counts: a rank, a nullity, a number of free parameters. This chapter extracts from the same computation a single scalar rather than a count, and that scalar carries geometric information the counts cannot. It reports the volume of the box spanned by the columns of a matrix, and its sign reports whether the matrix preserves or reverses handedness.

Section 3.1 computes areas and volumes directly in two and three dimensions and reads off the three properties any such measurement must have. Section 3.2 shows those three properties determine one function completely, constructs it, and derives from it the criterion for invertibility that chapter 1 deliberately left in terms of elimination. Section 3.3 produces the classical formulas, at a computational cost the section states plainly. Section 3.4 settles what the sign means. Section 3.5 collects the determinant identities that later chapters and later books draw on.

3.1 Signed Area and Signed Volume

Two vectors in the plane span a parallelogram, and three vectors in space span a box. This section measures both by hand, and records three properties of the measurement.

A pair of vectors in $\mathbb{R}^2$ spans a parallelogram in the following way: drawn from the origin, the two vectors are two of its edges, and the corner opposite the origin is their sum. Take $u = (3, 1)^\top$ and $v = (1, 2)^\top$, where the superscript $\top$ is the transpose of definition 2.3.4, so that u and v are columns. The parallelogram they span has corners $(0, 0)$, $(3, 1)$, $(4, 3)$ and $(1, 2)$. It sits inside the rectangle with corners $(0, 0)$ and $(4, 3)$, and what remains of that rectangle once the parallelogram is removed falls into six pieces, four triangles and two rectangles, each of whose areas can be written down at sight.

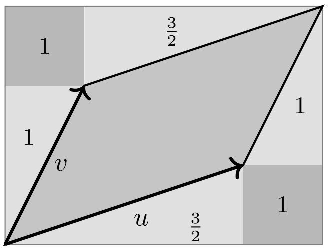


Figure 3.1: The parallelogram spanned by u and v inside its enclosing rectangle, with the six leftover pieces shaded and their areas marked

Example 3.1.1: The Area of a Parallelogram

Let $u = (3, 1)^\top$ and $v = (1, 2)^\top$ in $\mathbb{R}^2$. The enclosing rectangle of fig. 3.1 has width 4 and height 3, hence area 12. The six leftover pieces are the following.

1. The triangle with corners $(0, 0)$, $(3, 0)$, $(3, 1)$, whose legs have lengths 3 and 1, hence area $\frac{3}{2}$.
2. The rectangle with corners $(3, 0)$ and $(4, 1)$, of area 1.
3. The triangle with corners $(3, 1)$, $(4, 1)$, $(4, 3)$, whose legs have lengths 1 and 2, hence area 1.
4. The triangle with corners $(0, 0)$, $(1, 2)$, $(0, 2)$, whose legs have lengths 1 and 2, hence area 1.
5. The rectangle with corners $(0, 2)$ and $(1, 3)$, of area 1.
6. The triangle with corners $(1, 2)$, $(4, 3)$, $(1, 3)$, whose legs have lengths 3 and 1, hence area $\frac{3}{2}$.

These add to

$$\tfrac{3}{2} + 1 + 1 + 1 + 1 + \tfrac{3}{2} = 7$$

so the parallelogram has area $12 - 7 = 5$. The same 5 can be read off the array

$$\begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}$$

whose two columns are u and v: the product of the two entries on the main diagonal is $3 \cdot 2 = 6$, the product of the other two entries is $1 \cdot 1 = 1$, and the difference is $6 - 1 = 5$.

The same six pieces appear whatever the two vectors are, provided they are positioned as in fig. 3.1. Write $u = (a, c)^\top$ and $v = (b, d)^\top$, with all four numbers nonnegative and v steeper than u, so that the picture is the one drawn. The enclosing rectangle has width $a + b$ and height $c + d$. The two triangles cut off by the edges parallel to u have legs of lengths a and c, so together they contribute ac. The two cut off by the edges parallel to v have legs b and d and contribute bd. The two leftover rectangles each measure b by c and contribute $2bc$. Subtracting,

$$(a + b)(c + d) - ac - bd - 2bc = ac + ad + bc + bd - ac - bd - 2bc = ad - bc$$

so the area of the parallelogram is $ad - bc$. Setting $a = 3$, $c = 1$, $b = 1$ and $d = 2$ returns the 5 computed above.

Nothing so far distinguishes the pair (u, v) from the pair (v, u): they span the same parallelogram. The number $ad - bc$ does distinguish them. Interchanging u and v interchanges a with b and c with d, and turns $ad - bc$ into $bc - ad$. For the vectors of example 3.1.1 the second order gives $1 \cdot 1 - 3 \cdot 2 = -5$. The magnitude is the area and the sign is a record of the order in which the two edges were listed. The quantity gets a name.

Definition 3.1.2: Signed Area

Let $u = (a, c)^\top$ and $v = (b, d)^\top$ be elements of $\mathbb{R}^2$. The *signed area* of the ordered pair (u, v) is

$$D(u, v) = ad - bc$$

Its magnitude is the area of the parallelogram spanned by u and v, and its sign records the order in which the two edges were listed.

Which orderings receive a plus sign is settled in section 3.4, where the two classes of ordered basis are separated; the general definition, of which D is the two-dimensional case, is given in section 3.2. What matters here is that D is a number attached to an ordered pair of vectors, computed by one subtraction, and that it has three properties.

The first is that D is linear in each of its two arguments when the other is held fixed. Holding v fixed, replacing $u = (a, c)^\top$ by $tu = (ta, tc)^\top$ replaces $ad - bc$ by $tad - tbc = t(ad - bc)$, and replacing u by a sum $u + u' = (a + a', c + c')^\top$ gives

$$(a + a')d - b(c + c') = (ad - bc) + (a'd - bc')$$

which is $D(u, v) + D(u', v)$. The same two computations work in the second argument. Both statements have a geometric reading, and each needs a sign condition to be read literally. For $t > 0$, stretching one edge of a parallelogram by t while holding the other fixed stretches the region by the same factor; for $t < 0$ the edge reverses, so the area scales by $|t|$ and the sign of D flips, and for $t = 0$ the region collapses. Likewise two parallelograms built on a common edge v, from u and from u', combine into the one built on v and $u + u'$ when $D(u, v)$ and $D(u', v)$ have the same sign; when the signs differ the two areas subtract instead, which is what the additivity of D records and the unsigned area does not. As a check on the arithmetic, take $u = (3, 1)^\top$, $u' = (1, 0)^\top$ and $v = (1, 2)^\top$: then $D(u, v) = 5$ and $D(u', v) = 1 \cdot 2 - 1 \cdot 0 = 2$, while $u + u' = (4, 1)^\top$ has $D(u + u', v) = 4 \cdot 2 - 1 \cdot 1 = 7$.

The second property is that $D(u, u) = 0$ for every u. With $u = (a, c)^\top$ in both slots the formula reads $ac - ac$. Geometrically the parallelogram spanned by u and u is the segment from 0 to u traversed twice, a region with no area at all. More generally, if $v = tu$ is a multiple of u then the first property and then the second give

$$D(u, tu) = t\, D(u, u) = 0$$

so a pair in which one vector is a multiple of the other has signed area zero. Every dependent pair in the sense of definition 2.1.9 has that shape: if $\alpha u + \beta v = 0$ with α and β not both zero, then $v = -(\alpha/\beta)u$ when $\beta \neq 0$, and otherwise $u = 0 = 0 \cdot v$, so that u is a multiple of v and the same computation applies with the two arguments interchanged. A dependent pair therefore has signed area zero, and its two vectors lie on one line through the origin. The converse holds as well, and is the fourth exercise at the end of this section.

The third property is that the standard basis vectors $e_1 = (1, 0)^\top$ and $e_2 = (0, 1)^\top$, in that order, have $D(e_1, e_2) = 1 \cdot 1 - 0 \cdot 0 = 1$. They span the unit square, and the unit square is what area is measured against.

The first two properties combine into a statement about the third of the operations of definition 1.2.2, applied to columns rather than to rows. Adding a multiple of one vector to the other changes neither the signed area nor the parallelogram's area, since

$$D(u, v + tu) = D(u, v) + t\, D(u, u) = D(u, v)$$

using linearity in the second argument and then $D(u, u) = 0$. With $u = (3, 1)^\top$, $v = (1, 2)^\top$ and $t = 2$ this says that $(3, 1)^\top$ and $(7, 4)^\top$ span a parallelogram of area $3 \cdot 4 - 7 \cdot 1 = 5$, the same as before. The parallelogram has been sheared: one edge slid along the line through the other, which moves the corners without changing the base or the height.

That last sentence used something about area that has not been stated: that sliding an edge along the line through the other leaves the area unchanged. Two further such facts are used in this chapter, and none of the three can be proved here, because this book never defines area. They are recorded now, so that later appeals to them are appeals to something stated.

3.1.3 Convention: What Is Assumed About Area and Volume For the parallelograms and boxes of this chapter, three properties of area and volume are assumed without proof.

1. *Translation invariance.* A translate of a region has the same area or volume as the region.
2. *Slicing.* If a region is cut into slices parallel to a fixed line or plane, and each slice is then translated within its own line or plane, the area or volume is unchanged.
3. *Scaling one edge.* Stretching a parallelogram or box by a factor t along one edge, the others held fixed, multiplies the area or volume by $|t|$.

A construction of area and volume from which these three are theorems rather than assumptions belongs to measure theory; see *Measure Theory* [7].

The three properties of D do more than describe it. In the plane they already force it. Suppose a rule assigns a number $F(x, y)$ to every ordered pair of vectors in $\mathbb{R}^2$, is linear in each argument when the other is held fixed, vanishes whenever the two arguments are equal, and takes the value 1 at (e_1, e_2). Applying the second requirement to the pair $(e_1 + e_2, e_1 + e_2)$ and expanding by linearity in each slot gives

$$0 = F(e_1, e_1) + F(e_1, e_2) + F(e_2, e_1) + F(e_2, e_2) = 0 + 1 + F(e_2, e_1) + 0$$

so $F(e_2, e_1) = -1$. Now $u = (a, c)^\top$ is $ae_1 + ce_2$ and $v = (b, d)^\top$ is $be_1 + de_2$, and expanding by linearity in each argument gives

$$F(u, v) = ab\, F(e_1, e_1) + ad\, F(e_1, e_2) + cb\, F(e_2, e_1) + cd\, F(e_2, e_2) = ad - bc$$

so F is D, and no freedom was left anywhere. The corresponding statement for n vectors in $\mathbb{R}^n$ is theorem 3.2.3, and its proof runs on elimination rather than on a four-term expansion, but the mechanism is the one just seen: the three properties fix the value on the standard basis vectors and linearity carries it everywhere else.

Three vectors in $\mathbb{R}^3$ span a box in the same way that two vectors in $\mathbb{R}^2$ span a parallelogram: the three vectors drawn from the origin are edges, and the eight corners are the sums $su + tv + rw$ with each of s, t, r equal to 0 or 1. Its volume can be computed by hand in one convenient position, namely when two of the three edges lie in the xy-plane.

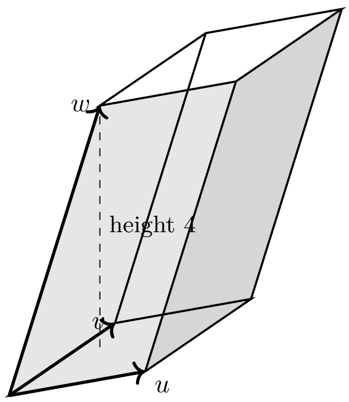


Figure 3.2: The box spanned by u, v and w, with its base in the xy-plane and the height of w above that plane marked

Example 3.1.4: The Volume of a Box

Let

$$u = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \qquad v = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}, \qquad w = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$$

in $\mathbb{R}^3$, and let B be the box they span, drawn in fig. 3.2. Its points are the combinations $su+tv+rw$ with s, t and r each running over the numbers from 0 to 1.

The third coordinate of $su + tv + rw$ is $4r$, since u and v have third coordinate 0. So the points of B at height z are exactly those with $r = z/4$, and they form the set of all $su + tv + (z/4)w$ with s and t running from 0 to 1. That set is the base parallelogram, the one spanned by u and v in the xy-plane, translated by the fixed vector $(z/4)w$. Every horizontal slice of B is therefore a copy of the base, moved but not reshaped, and the slices occupy the heights z from 0 to 4.

Since $w = (1, 2, 4)^\top$, the translation $(z/4)w$ moves the base sideways by $(z/4)(1, 2, 0)^\top$ and upward by z. Sliding each slice back by the sideways part $(z/4)(1, 2, 0)^\top$ leaves it at height z, stacks the slices directly above the base, and turns B into a vertical box of the same base and the same height, without altering any slice. Its volume is base times height. The base is spanned by $(2, 1)^\top$ and $(1, 3)^\top$ in the plane, of area $2 \cdot 3 - 1 \cdot 1 = 5$, and the height is 4, so

$$\text{volume of } B = 5 \cdot 4 = 20$$

which is the volume of the original box.

The computation used only that u and v lie in the xy-plane, so it applies to any box in that position, and the three properties found in the plane can be read off it. Doubling w to $(2, 4, 8)^\top$ doubles the height and leaves the base unchanged, giving volume 40; doubling u to $(4, 2, 0)^\top$ doubles the base area, again giving 40. Replacing w by $w + 3u = (7, 5, 4)^\top$ changes neither the base nor the height, so the volume stays 20, and the slices at each height are the same parallelograms moved sideways: the box has been sheared, exactly as the parallelogram was. Taking w equal to u flattens the box into the base parallelogram, of volume 0. Taking u, v and w to be e_1, e_2 and e_3 gives the unit cube, of volume 1.

Volume is a nonnegative quantity, and a sign has to be attached by hand, as it was in the plane. The choice made there was that $D(e_1, e_2) = 1$ rather than -1, and every other value followed. The same choice in $\mathbb{R}^3$ assigns the value 1 to the ordered triple (e_1, e_2, e_3), and it is the ordering that carries the information: the three properties know nothing about the box itself, only about the list of its edges. The geometric meaning of the resulting sign in $\mathbb{R}^n$, and the fact that there are exactly two possible answers rather than many, is the subject of section 3.4. That the magnitude really is the volume, for boxes in every position and not only the convenient one computed above, is theorem 3.4.5.

Exercise 3.1.1

1. Compute the signed area $D(u, v)$ for each of the following ordered pairs in $\mathbb{R}^2$, and in each case say whether the two vectors are dependent in the sense of definition 2.1.9.
   1. $u = (4, 1)^\top$, $v = (2, 3)^\top$.
   2. $u = (3, -6)^\top$, $v = (-1, 2)^\top$.
   3. $u = (0, 5)^\top$, $v = (2, 0)^\top$.
2. Take $u = (4, 1)^\top$ and $v = (2, 3)^\top$. Draw the parallelogram they span inside the rectangle with corners $(0, 0)$ and $(6, 4)$, list the six leftover pieces with their areas as in example 3.1.1, and check that the rectangle's area minus their total is $D(u, v)$.
3. Let $u = (a, c)^\top$, $v = (b, d)^\top$ and $v' = (b', d')^\top$ in $\mathbb{R}^2$, and let t be a real number. Verify directly from the formula $D(u, v) = ad - bc$ that $D(u, v + v') = D(u, v) + D(u, v')$ and that $D(u, tv) = t\,D(u, v)$. Verify also that $D(v, u) = -D(u, v)$, and then deduce $D(u + tv, v) = D(u, v)$ from linearity and antisymmetry alone, without using the formula again.
4. Let $u = (a, c)^\top$ and $v = (b, d)^\top$ in $\mathbb{R}^2$ with $ad - bc = 0$. Show that u and v are dependent. Together with the computation in the text, this says that the signed area vanishes exactly on the dependent pairs.
5. Let $u = (3, 1, 0)^\top$, $v = (1, 2, 0)^\top$ and $w = (2, 1, 5)^\top$ in $\mathbb{R}^3$. Compute the volume of the box they span by the method of example 3.1.4. Then compute the volumes of the boxes spanned by u, v and $w - 2u + v$, and by u, v and $3w$, and say which property of the measurement each answer illustrates.
6. Let $A \in M_2(\mathbb{R})$ have columns u and v, and write $D(A)$ for $D(u, v)$. For each of the three row operations of definition 1.2.2 applied to A, compute D of the resulting matrix in terms of $D(A)$, writing the scalar as t since c already names an entry of A. Compare the three answers with the effect of the same three operations applied to the columns of A instead of its rows.

3.2 The Determinant: Existence, Uniqueness, and Computation by Elimination

Section 3.1 measured boxes and read three properties off the measurement. This section takes those three properties as the definition of the determinant. Two questions come with such a definition: whether more than one function has the three properties, and whether any function has them at all. Neither answer is free, and the argument settling the first is an algorithm, since it computes the value of any such function by elimination.

Throughout, k stands for $\mathbb{R}$ or for $\mathbb{C}$, and every statement holds over both.

The measurement of section 3.1 attaches to a matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(k)$$

the number $ad - bc$. Write $r_1 = (a, b)$ and $r_2 = (c, d)$ for its two rows, each a list of two scalars. Three things are true of the number $ad - bc$, and they are the three the rest of the chapter is built on.

First, it is linear in each row separately. Holding r_2 fixed and replacing r_1 by $\lambda(a, b) + \mu(a', b') = (\lambda a + \mu a', \lambda b + \mu b')$ turns $ad - bc$ into

$$(\lambda a + \mu a')d - (\lambda b + \mu b')c = \lambda(ad - bc) + \mu(a'd - b'c)$$

and the same computation with the roles of the rows exchanged gives linearity in r_2. Second, it vanishes when the two rows are equal: if $(a, b) = (c, d)$ then $ad - bc = ab - ba = 0$. Third, its value at the identity matrix is $1 \cdot 1 - 0 \cdot 0 = 1$.

Nothing in the three statements refers to the size 2, so each can be asked of a function on $M_n(k)$ for any n. The three conditions are recorded for the rows rather than for the columns, because the three row operations of definition 1.2.2 are what elimination performs, and the conditions are about to be turned into a statement about elimination. Nothing is lost by the choice: corollary 3.2.12 recovers the column statements at the end of the section. For 2×2 matrices the two readings can be compared directly, since $ad - bc$ is linear in each of the columns (a, c) and (b, d) as well, and vanishes when those two columns agree.

Definition 3.2.1: Determinant Function

Let $n \geq 1$. A *row* of a matrix in $M_n(k)$ is a list of n scalars, and sums and scalar multiples of rows are formed entry by entry. For $A \in M_n(k)$ with rows $r_1, \ldots, r_n$, write $D(r_1, \ldots, r_n)$ for $D(A)$.
A function $D\colon M_n(k) \to k$ is a *determinant function* of size n if it satisfies the following three conditions.

1. *Linearity in each row.* For every index i, all rows $r_1, \ldots, r_n$ and u, v, and all scalars $\lambda, \mu \in k$,

$$D(r_1, \ldots, \lambda u + \mu v, \ldots, r_n) = \lambda\, D(r_1, \ldots, u, \ldots, r_n) + \mu\, D(r_1, \ldots, v, \ldots, r_n)$$

   where the displayed argument stands in position i and the remaining arguments are unchanged.

2. *Vanishing on a repeated row.* If $r_i = r_j$ for some pair of indices $i \neq j$, then $D(r_1, \ldots, r_n) = 0$.

3. *Normalization.* $D(I_n) = 1$.

Condition (1) is a statement about one row at a time, with the other $n - 1$ rows frozen; it says nothing about what happens when two rows are changed at once, and in particular it does not say that $D(A + B) = D(A) + D(B)$. Condition (2) is a condition on the position of the rows relative to one another rather than on their entries. Condition (3) fixes the scale: without it the function that is identically zero would satisfy the other two, and so would every constant multiple of a function that satisfies all three.

The function $D(A) = ad - bc$ on $M_2(k)$ is a determinant function of size 2, by the three computations performed above. For general n nothing yet says that a determinant function exists, and nothing yet says that there is only one. The second question is settled first, because settling it produces the computation.

The three row operations of definition 1.2.2 act on the rows of a matrix, and the conditions of definition 3.2.1 are conditions on the rows. So the conditions determine what each operation does to the value.

Lemma 3.2.2: The Row Operations and a Determinant Function

Let $D\colon M_n(k) \to k$ be a determinant function of size n and let $A \in M_n(k)$ have rows $r_1, \ldots, r_n$.

1. If B is obtained from A by multiplying row i by a scalar $c \in k$, then $D(B) = c\,D(A)$. For $c \neq 0$ this is the scaling $R_i \mapsto cR_i$ of definition 1.2.2.

2. If B is obtained from A by the swap $R_i \leftrightarrow R_j$ with $i \neq j$, then $D(B) = -D(A)$.

3. If B is obtained from A by the replacement $R_i \mapsto R_i + cR_j$ with $j \neq i$ and $c \in k$, then $D(B) = D(A)$.

4. If one row of A is a linear combination of the others, then $D(A) = 0$. In particular $D(A) = 0$ whenever some row of A is zero.

5. If A is upper triangular or lower triangular, that is if $a_{ij} = 0$ whenever $i > j$ or $a_{ij} = 0$ whenever $i < j$, then
$$D(A) = a_{11}a_{22}\cdots a_{nn}$$

***Proof*:**

1. Apply condition (1) of definition 3.2.1 in position i with $u = r_i$, $v = r_i$, $\lambda = c$ and $\mu = 0$. The left side is $D(B)$, since $cr_i + 0r_i = cr_i$, and the right side is $c\,D(A)$.

2. Let C be the matrix whose rows agree with those of A except that positions i and j both carry the row $r_i + r_j$. Its rows i and j are equal, so $D(C) = 0$ by condition (2). Expanding position i by condition (1), then position j in each of the two resulting terms, gives
$$0 = D(C) = D(\ldots r_i \ldots r_i \ldots) + D(\ldots r_i \ldots r_j \ldots) + D(\ldots r_j \ldots r_i \ldots) + D(\ldots r_j \ldots r_j \ldots)$$
where in each term the first displayed row occupies position i, the second occupies position j, and all other rows are those of A. The first and fourth terms have a repeated row and vanish by condition (2). The second term is $D(A)$ and the third is $D(B)$, since B is A with r_j in position i and r_i in position j. Hence $D(A) + D(B) = 0$.

3. Condition (1) applied in position i with $u = r_i$, $v = r_j$, $\lambda = 1$ and $\mu = c$ gives
$$D(B) = D(A) + c\,D(r_1, \ldots, r_j, \ldots, r_n)$$
where in the last term r_j stands in position i. That matrix carries r_j in position i and in position j, two distinct positions since $j \neq i$, so the last term vanishes by condition (2).

4. Suppose $r_i = \sum_{j \neq i} \lambda_j r_j$ with $\lambda_j \in k$. Applying condition (1) in position i repeatedly, once for each summand, gives

$$D(A) = \sum_{j \neq i} \lambda_j \, D(r_1, \ldots, r_j, \ldots, r_n)$$

where in the j-th term r_j stands in position i. Each such matrix carries r_j in the two distinct positions i and j, so each term vanishes by condition (2). If instead row i of A is zero, part (1) with $c = 0$ compares A with itself and gives $D(A) = 0 \cdot D(A) = 0$.

5. Take A upper triangular first, so that $a_{ij} = 0$ for $i > j$.

   Suppose every diagonal entry is nonzero. Run through $p = n, n-1, \ldots, 2$, and for each p and each $q < p$ apply the replacement $R_q \mapsto R_q - (a_{qp}/a_{pp})R_p$, where a_{qp} is the entry standing in position (q, p) at that moment. At the start of the pass belonging to p, row p has zeros in the columns $1, \ldots, p-1$ because A is upper triangular, and zeros in the columns $p+1, \ldots, n$ because those columns were emptied above row p' for each $p' > p$ in the earlier passes; so row p has the single nonzero entry a_{pp}, and the replacement alters row q only in column p, where it produces a zero. No diagonal entry is altered, since $q < p$. When the passes are finished every entry off the diagonal is zero, and by part (3) the value of D has not changed. The resulting matrix is obtained from I_n by scaling row i by a_{ii} for each i in turn, so n applications of part (1) and condition (3) give

$$D(A) = a_{11}a_{22} \cdots a_{nn} D(I_n) = a_{11}a_{22} \cdots a_{nn}$$

   Suppose instead that some diagonal entry vanishes, and let i be the largest index with $a_{ii} = 0$, so that $a_{pp} \neq 0$ for every $p > i$. Run through $p = i+1, i+2, \ldots, n$ in that order, applying $R_i \mapsto R_i - (a_{ip}/a_{pp})R_p$ with a_{ip} the entry standing in position (i, p) at that moment. Row p has zeros in the columns $1, \ldots, p-1$, so the replacement alters row i only in the columns $p, \ldots, n$; it produces a zero in column p and leaves untouched the zeros created in the columns $i+1, \ldots, p-1$ by the earlier steps. At the end row i has zeros in the columns $i+1, \ldots, n$, zeros in the columns $1, \ldots, i-1$ because A is upper triangular, and $a_{ii} = 0$ in column i, so it is a zero row. By part (3) the value of D is unchanged and by part (4) it is zero, which agrees with the product $a_{11} \cdots a_{nn}$, one of whose factors is $a_{ii} = 0$.

   For A lower triangular the same two arguments run with the direction of the passes reversed. If every diagonal entry is nonzero, run through $p = 1, 2, \ldots, n-1$ and for each $q > p$ apply $R_q \mapsto R_q - (a_{qp}/a_{pp})R_p$; at the start of the pass belonging to p, row p has the single nonzero entry a_{pp}, and the passes clear every entry below the diagonal, after which the scaling argument applies verbatim. If some diagonal entry vanishes, let i be the smallest index with $a_{ii} = 0$, so that $a_{pp} \neq 0$ for $p < i$, and run through $p = i-1, i-2, \ldots, 1$ applying $R_i \mapsto R_i - (a_{ip}/a_{pp})R_p$; row p has zeros in the columns $p+1, \ldots, n$, so each step clears column p of row i without disturbing the columns already cleared, and row i ends as a zero row, completing the proof.

Parts (1) to (3) say that a row operation multiplies the value of any determinant function by a scalar that depends on the operation and on nothing else: on c for a scaling, on -1 for a swap, on 1 for a replacement. A sequence of operations therefore multiplies the value by the product of those scalars, and that product is the same for every determinant function. Part (5) says that at the end of an elimination the value is forced outright. Those two observations are the whole proof of uniqueness, and they are also a procedure. Part (4) records what happens in the degenerate case, and by proposition 2.1.10 it may be read as saying that D vanishes exactly when the rows of A, written as columns, are linearly dependent.

Theorem 3.2.3: Uniqueness of a Determinant Function, and Its Computation by Elimination

Let $n \geq 1$, let $D\colon M_n(k) \to k$ be a determinant function of size n, let $A \in M_n(k)$, and let $\rho_1, \dots, \rho_s$ be elementary row operations carrying A to a matrix $U \in M_n(k)$ in echelon form. Attach to each operation the scalar

$$c \;\text{ for }\; R_i \mapsto cR_i, \qquad -1 \;\text{ for }\; R_i \leftrightarrow R_j, \qquad 1 \;\text{ for }\; R_i \mapsto R_i + cR_j$$

and let $\lambda \in k$ be the product of the s scalars so attached. Then U is upper triangular, $\lambda \neq 0$, and

$$D(A) = \lambda^{-1} u_{11} u_{22} \cdots u_{nn}$$

Moreover $u_{11}u_{22}\cdots u_{nn} \neq 0$ if and only if $\operatorname{rank} A = n$. Consequently at most one determinant function of size n exists.

***Proof*:**
Let the nonzero rows of U be rows $1, \dots, r$, with leading entries in columns $j_1 < j_2 < \cdots < j_r$; the ordering of the columns is condition (2) of definition 1.2.5 and the position of the zero rows is condition (1). Strict increase from $j_1 \geq 1$ forces $j_p \geq p$ for every p. Every entry of row p to the left of column j_p is zero, so $u_{pl} = 0$ whenever $l < j_p$, and in particular whenever $l < p$; the rows below row r are zero and satisfy the same condition. Hence U is upper triangular and lemma 3.2.2(5) gives $D(U) = u_{11}u_{22}\cdots u_{nn}$.

Each of $\rho_1, \dots, \rho_s$ multiplies the value of D by its attached scalar, by parts (1), (2) and (3) of lemma 3.2.2, so performing them one after the other gives $D(U) = \lambda D(A)$. A scaling is required by definition 1.2.2 to use a scalar $c \neq 0$, and -1 and 1 are nonzero, so λ is a product of nonzero scalars and is itself nonzero. Dividing by λ gives $D(A) = \lambda^{-1}u_{11}u_{22}\cdots u_{nn}$.

For the criterion, suppose first that $r < n$. Then row n of U is zero, so $u_{nn} = 0$ and the product vanishes. Suppose instead that $r = n$. Then $j_n \leq n$ together with the strict increase forces $j_p \leq p$, and with $j_p \geq p$ this gives $j_p = p$ for every p; so u_{pp} is the leading entry of row p and is nonzero, and the product of the diagonal entries is a product of n nonzero scalars. The number r of nonzero rows of U is the number of its leading entries, which by the second statement of theorem 1.2.10 is the number of pivot columns of the reduced echelon form of A, namely $\operatorname{rank} A$ (definition 1.3.2). So the product is nonzero exactly when $\operatorname{rank} A = n$.

Finally, let D and D' both be determinant functions of size n and let $A \in M_n(k)$. By theorem 1.2.8 there are elementary row operations carrying A to a matrix U in echelon form; fix one such sequence and let λ be its attached product. The displayed formula applies to D and to D' with the same U and the same λ, since neither depends on the function, so $D(A) = \lambda^{-1}u_{11}\cdots u_{nn} = D'(A)$. As A was arbitrary, $D = D'$, as we sought to show.

The theorem does two things at once. It says that the three conditions leave no freedom, so the phrase "the determinant" will be legitimate as soon as one determinant function is produced. It also says how to evaluate that function: run elimination, record a factor c for each scaling and a factor -1

for each swap, multiply the diagonal entries of the echelon form reached, and divide by the recorded product.

What the theorem does not do is produce a function. It says that a determinant function, should one exist, is evaluated by that recipe from any elimination whatever, so that the choice of elimination cannot matter. If no determinant function exists, the assertion is empty, and nothing proved so far forbids two eliminations of one matrix from returning different numbers. Constructing a function that satisfies the three conditions removes both doubts.

The construction is by induction on n. A 1×1 matrix has one entry, and that entry is its own measurement. For larger n the matrix is broken along its first column, each entry of that column being paired with the smaller matrix that survives when its row and the first column are deleted.

Theorem 3.2.4: Existence of a Determinant Function

For every $n \geq 1$ there is a determinant function of size n. It may be constructed as follows. For $n = 1$ set $D_1\big((a)\big) = a$. For $n \geq 2$ and $A = (a_{ij}) \in M_n(k)$, write $\widehat{A}_i \in M_{n-1}(k)$ for the matrix obtained from A by deleting row i and column 1, and set

$$D_n(A) = \sum_{i=1}^{n} (-1)^{i+1} a_{i1}\, D_{n-1}(\widehat{A}_i)$$

Then D_n is a determinant function of size n.

***Proof*:**
Induct on n. For $n = 1$ the three conditions read: D_1 is linear in the single entry, which it is; the condition on a repeated row is vacuous, since there is only one row; and $D_1(I_1) = D_1\big((1)\big) = 1$. Let $n \geq 2$ and suppose D_{n-1} is a determinant function of size $n-1$.

Step 1: normalization. For $A = I_n$ the entries of the first column are $a_{11} = 1$ and $a_{i1} = 0$ for $i \geq 2$, so only the term $i = 1$ survives. Deleting row 1 and column 1 of I_n leaves I_{n-1}, so

$$D_n(I_n) = (-1)^2 \cdot 1 \cdot D_{n-1}(I_{n-1}) = 1$$

by the inductive hypothesis.

Step 2: linearity in each row. Fix an index p and let A, A' and A'' be matrices in $M_n(k)$ agreeing in every row except row p, with row p of A equal to λ times row p of A' plus μ times row p of A''. Take the term $i = p$ first. Deleting row p removes the only row in which the three matrices differ, so $\widehat{A}_p = \widehat{A'}_p = \widehat{A''}_p$, while $a_{p1} = \lambda a'_{p1} + \mu a''_{p1}$ because the first entries of the three rows stand in that relation. So the term $i = p$ of $D_n(A)$ is λ times the term $i = p$ of $D_n(A')$ plus μ times the term $i = p$ of $D_n(A'')$. Now take a term with $i \neq p$. Here $a_{i1} = a'_{i1} = a''_{i1}$, since row i is common to the three matrices, while $\widehat{A}_i$, $\widehat{A'}_i$ and $\widehat{A''}_i$ agree in every row except the one inherited from row p, whose entries stand in the same relation with the same λ and μ. Condition (1) for D_{n-1} gives $D_{n-1}(\widehat{A}_i) = \lambda D_{n-1}(\widehat{A'}_i) + \mu D_{n-1}(\widehat{A''}_i)$, so the term i splits the same way. Adding the n terms gives $D_n(A) = \lambda D_n(A') + \mu D_n(A'')$.

Step 3: a repeated row. Suppose rows p and q of A are equal, with $p < q$. If $i \notin \{p, q\}$ then $\widehat{A}_i$ still carries both of those rows, so it has a repeated row and $D_{n-1}(\widehat{A}_i) = 0$ by the inductive hypothesis. Only the terms $i = p$ and $i = q$ survive, and $a_{p1} = a_{q1}$ because the two rows are equal.

Compare $\widehat{A}_p$ with $\widehat{A}_q$. Write $\hat{r}_i$ for row i of A with its first entry removed. The rows of $\widehat{A}_q$, in order, are

$$\hat{r}_1, \ldots, \hat{r}_p, \ldots, \hat{r}_{q-1}, \hat{r}_{q+1}, \ldots, \hat{r}_n$$

and the rows of $\widehat{A}_p$, in order, are

$$\hat{r}_1, \ldots, \hat{r}_{p-1}, \hat{r}_{p+1}, \ldots, \hat{r}_{q-1}, \hat{r}_q, \hat{r}_{q+1}, \ldots, \hat{r}_n$$

Since $\hat{r}_p = \hat{r}_q$, the second list is the first one with the entry $\hat{r}_p$ moved from position p to position $q-1$, past the $q-1-p$ rows $\hat{r}_{p+1}, \ldots, \hat{r}_{q-1}$. Each move past one row is an exchange of two rows in adjacent positions, and D_{n-1} is a determinant function, so lemma 3.2.2(2) applies once per exchange:

$$D_{n-1}(\widehat{A}_p) = (-1)^{q-1-p} D_{n-1}(\widehat{A}_q)$$

Substituting this into the two surviving terms gives

$$\begin{aligned} D_n(A) &= (-1)^{p+1} a_{p1} D_{n-1}(\widehat{A}_p) + (-1)^{q+1} a_{q1} D_{n-1}(\widehat{A}_q) \\ &= a_{p1} D_{n-1}(\widehat{A}_q) \Big[(-1)^{p+1}(-1)^{q-1-p} + (-1)^{q+1} \Big] \\ &= a_{p1} D_{n-1}(\widehat{A}_q) \Big[(-1)^{q} + (-1)^{q+1} \Big] \end{aligned}$$

and the bracket is zero. So $D_n(A) = 0$, which completes the verification of the three conditions and hence the induction, completing the proof.

The two theorems together license one name and one symbol.

Definition 3.2.5: The Determinant

Let $n \geq 1$. The unique determinant function of size n, which exists by theorem 3.2.4 and is unique by theorem 3.2.3, is written det, and $\det A$ is called the *determinant* of $A \in M_n(k)$. The notation $|A|$ for the same number is common elsewhere; this book writes $\det A$ throughout.

Two consequences of the definition are worth writing out at once. Since det satisfies the three conditions, everything proved about an arbitrary determinant function applies to it: lemma 3.2.2 describes what row operations do to det, and theorem 3.2.3 evaluates $\det A$ from any elimination of A. And since det is a determinant function, it agrees with the function D_n built in theorem 3.2.4, so the recursion

$$\det A = \sum_{i=1}^{n} (-1)^{i+1} a_{i1} \det \widehat{A}_i$$

holds, where $\widehat{A}_i$ is A with row i and column 1 deleted. This is the *expansion along the first column.* For $n = 2$ it reads

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

recovering the number of section 3.1, and for $n = 3$ it reads

$$\det\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{21}(a_{12}a_{33} - a_{13}a_{32}) \\ + a_{31}(a_{12}a_{23} - a_{13}a_{22})$$

which is six products with alternating signs once the brackets are opened.

Proposition 3.2.6: The Determinant of a Triangular Matrix

Let $A = (a_{ij}) \in M_n(k)$ be upper triangular or lower triangular. Then

$$\det A = a_{11}a_{22}\cdots a_{nn}$$

In particular $\det I_n = 1$, $\det \operatorname{diag}(d_1, \dots, d_n) = d_1 \cdots d_n$, and $\det(cI_n) = c^n$ for every $c \in k$.

***Proof*:**
By definition 3.2.5 the function det is a determinant function of size n, so lemma 3.2.2(5) applies and gives the displayed identity. A diagonal matrix is both upper and lower triangular, and I_n and cI_n are diagonal with diagonal entries $1, \dots, 1$ and $c, \dots, c$ respectively, which gives the three special cases, as we sought to show.

The determinant of a triangular matrix is therefore read off without any arithmetic beyond one product of n numbers, and elimination is exactly the procedure that replaces an arbitrary matrix by a triangular one. The next example runs the two together on a matrix that has not appeared before.

Example 3.2.7: A 4×4 Determinant Computed by Elimination
Let

$$B = \begin{pmatrix} 0 & 1 & 2 & 1 \\ 1 & 2 & 1 & 0 \\ 2 & 5 & 7 & 2 \\ 1 & 3 & 3 & 3 \end{pmatrix} \in M_4(\mathbb{R})$$

The entry in position $(1, 1)$ is zero, so elimination begins with the swap $R_1 \leftrightarrow R_2$, which contributes the scalar -1:

$$\begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 2 & 5 & 7 & 2 \\ 1 & 3 & 3 & 3 \end{pmatrix}$$

The replacements $R_3 \mapsto R_3 - 2R_1$ and $R_4 \mapsto R_4 - R_1$ empty the first column below row 1, replacing row 3 by $(2-2,\ 5-4,\ 7-2,\ 2-0) = (0, 1, 5, 2)$ and row 4 by $(1-1,\ 3-2,\ 3-1,\ 3-0) = (0, 1, 2, 3)$:

$$\begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 1 & 5 & 2 \\ 0 & 1 & 2 & 3 \end{pmatrix}$$

The replacements $R_3 \mapsto R_3 - R_2$ and $R_4 \mapsto R_4 - R_2$ empty the second column below row 2, replacing row 3 by $(0, 0, 3, 1)$ and row 4 by $(0, 0, 0, 2)$:

$$U = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 2 \end{pmatrix}$$

This is in echelon form, with four pivots, and it is upper triangular. Its diagonal entries multiply to $1 \cdot 1 \cdot 3 \cdot 2 = 6$. One swap and four replacements were used, so the attached scalars are -1 and four copies of 1, and $\lambda = -1$. By theorem 3.2.3,

$$\det B = \lambda^{-1} \cdot 6 = -6$$

The same number comes out of the expansion along the first column of B. The first entry of that column is zero, so three terms remain:

$$\det B = -1 \cdot \det \begin{pmatrix} 1 & 2 & 1 \\ 5 & 7 & 2 \\ 3 & 3 & 3 \end{pmatrix} + 2 \cdot \det \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 0 \\ 3 & 3 & 3 \end{pmatrix} - 1 \cdot \det \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 0 \\ 5 & 7 & 2 \end{pmatrix}$$

Each 3×3 determinant is expanded along its own first column. The first is $1(7 \cdot 3 - 2 \cdot 3) - 5(2 \cdot 3 - 1 \cdot 3) + 3(2 \cdot 2 - 1 \cdot 7) = 15 - 15 - 9 = -9$; the second is $1(1 \cdot 3 - 0 \cdot 3) - 2(2 \cdot 3 - 1 \cdot 3) + 3(2 \cdot 0 - 1 \cdot 1) = 3 - 6 - 3 = -6$; the third is $1(1 \cdot 2 - 0 \cdot 7) - 2(2 \cdot 2 - 1 \cdot 7) + 5(2 \cdot 0 - 1 \cdot 1) = 2 + 6 - 5 = 3$. Substituting,

$$\det B = -(-9) + 2(-6) - 3 = 9 - 12 - 3 = -6$$

in agreement with the elimination.

The two routes in the example cost different amounts. The elimination used five row operations; the expansion used three 3×3 determinants, each of which used three 2×2 determinants, and each of those used two products. The gap widens rapidly with n, and section 3.3 states it precisely.

Chapter 1 characterized invertibility by elimination alone, in theorem 1.4.8, and attached no number to the matrix. The criterion carried by theorem 3.2.3 gives one.

Corollary 3.2.8: A Matrix Is Invertible Exactly When Its Determinant Is Nonzero

Let $A \in M_n(k)$. Then A is invertible if and only if $\det A \neq 0$.

***Proof*:**
By theorem 1.2.8 there are elementary row operations carrying A to a matrix U in echelon form; fix one such sequence and let λ be the product of its attached scalars. By theorem 3.2.3, $\lambda \neq 0$ and $\det A = \lambda^{-1} u_{11} \cdots u_{nn}$, so $\det A \neq 0$ holds exactly when $u_{11} \cdots u_{nn} \neq 0$, which by the same theorem holds exactly when $\operatorname{rank} A = n$. It remains to identify that condition with invertibility. Suppose $\operatorname{rank} A = n$ and let R be the reduced echelon form of A, which has n pivots (definition 1.3.2), one in each of its n rows, in columns $j_1 < j_2 < \cdots < j_n$. Strict increase together with $j_1 \geq 1$ and $j_n \leq n$ forces $j_p = p$ for every p. In reduced echelon form each pivot equals 1 and is the only nonzero entry of its column (definition 1.2.7), so column p of R is e_p and $R = I_n$; hence A is invertible by theorem 1.4.8, condition (3). Conversely if A is invertible

then its reduced echelon form is I_n by the same condition, and I_n has n pivots, so $\operatorname{rank} A = n$, as we sought to show.

The corollary turns a yes-or-no question into the evaluation of a single scalar, and by theorem 3.2.3 that scalar is produced by the elimination one would have run anyway. It also extends theorem 1.4.8: the five equivalent conditions listed there acquire a sixth, namely $\det A \neq 0$. Combined with lemma 3.2.2(4), it says that a square matrix with one row a combination of the others is singular, which was visible before only after running elimination to the end.

Example 3.2.9: The Square Submatrices of the Running Matrix

The running matrix

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 1 & 2 & 5 \\ 3 & 4 & 11 \end{pmatrix}$$

of example 1.2.9 has four rows and three columns, so it has no determinant. Selecting three of its rows produces a matrix in $M_3(\mathbb{R})$ that does. Take the first three rows:

$$A' = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 1 & 2 & 5 \end{pmatrix}$$

Elimination needs three replacements and no swap. The operations $R_2 \mapsto R_2 - 2R_1$ and $R_3 \mapsto R_3 - R_1$ replace rows 2 and 3 by $(0, 1, 2)$ and $(0, 1, 2)$, and then $R_3 \mapsto R_3 - R_2$ replaces row 3 by $(0, 0, 0)$:

$$\begin{pmatrix} 1 & 1 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$$

No swap and no scaling was used, so $\lambda = 1$, and the diagonal entries multiply to $1 \cdot 1 \cdot 0 = 0$. Hence $\det A' = 0$. The expansion along the first column gives the same:

$$1(3 \cdot 5 - 8 \cdot 2) - 2(1 \cdot 5 - 3 \cdot 2) + 1(1 \cdot 8 - 3 \cdot 3) = -1 + 2 - 1 = 0$$

The answer is forced, and not only for these three rows. The third column of A is the first plus twice the second (section 1.2), so $x = (1, 2, -1)^\top$ satisfies $Ax = 0$. That relation holds in each coordinate separately, so it holds for the three coordinates surviving in any choice of three rows: the matrix A'' built from rows 1, 2 and 4,

$$A'' = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 3 & 4 & 11 \end{pmatrix}$$

also satisfies $A''x = 0$ with the same nonzero x. By theorem 1.4.8, condition (2), both A' and A'' are singular, so both determinants vanish by corollary 3.2.8. Every 3×3 matrix assembled from three rows of A has determinant zero, and the reason is the single column relation carried by A.

Two matrices can be multiplied, and the determinant respects that product. The statement is not forced by anything proved so far, since the entries of AB are sums of products of entries of A and B

and no manipulation of those sums is in sight. The proof does not manipulate them: it observes that one side of the identity, viewed as a function of A with B held fixed, satisfies the three conditions of definition 3.2.1, and then invokes uniqueness.

Theorem 3.2.10: The Determinant of a Product

Let $A, B \in M_n(k)$. Then
$$\det(AB) = \det A \det B$$

***Proof*:**
One reading of definition 1.4.1 is used throughout. For each i and each j, $(AB)_{ij} = \sum_{l=1}^{n} a_{il} b_{lj}$, so row i of AB is the combination

$$\sum_{l=1}^{n} a_{il} \cdot (\text{row } l \text{ of } B)$$

whose coefficients are the entries of row i of A. Row i of AB therefore depends on row i of A and on no other row of A, and depends on it linearly.

Case 1: $\det B \neq 0$. Define $F\colon M_n(k) \to k$ by

$$F(A) = \frac{\det(AB)}{\det B}$$

which makes sense because $\det B \neq 0$. Each of the three conditions of definition 3.2.1 holds for F. For condition (1), fix an index i and let A, A' and A'' agree in every row except row i, with row i of A equal to λ times row i of A' plus μ times row i of A''. By the reading above, AB, $A'B$ and $A''B$ agree in every row except row i, and row i of AB is λ times row i of $A'B$ plus μ times row i of $A''B$. Condition (1) for det gives $\det(AB) = \lambda \det(A'B) + \mu \det(A''B)$, and dividing by $\det B$ gives condition (1) for F. For condition (2), suppose rows i and j of A are equal with $i \neq j$. Then rows i and j of AB are the same combination of the rows of B, hence equal, so $\det(AB) = 0$ by condition (2) for det, and $F(A) = 0$. For condition (3), $I_n B = B$ by proposition 1.4.3(3), so $F(I_n) = \det B / \det B = 1$. Hence F is a determinant function of size n, and theorem 3.2.3 identifies it with det. So $\det(AB)/\det B = \det A$ for every A, which is the claim.

Case 2: $\det B = 0$. By corollary 3.2.8 the matrix B is singular, so by theorem 1.4.8, condition (2), there is a column $x \in k^n$ with $x \neq 0$ and $Bx = 0$. Regrouping by proposition 1.4.3(1),

$$(AB)x = A(Bx) = 0$$

so the homogeneous system $(AB)x = 0$ has a nonzero solution and AB is singular by the same condition. Hence $\det(AB) = 0$ by corollary 3.2.8, and $\det A \det B = \det A \cdot 0 = 0$ as well, completing the proof.

The identity converts a statement about a product of matrices into a statement about a product of scalars, and multiplication of scalars does not depend on the order of the factors: $\det(AB) = \det(BA)$ for all $A, B \in M_n(k)$, though AB and BA are usually different matrices. The determinants of the elementary matrices follow at once, by applying lemma 3.2.2 to I_n and using $\det I_n = 1$: the elementary matrix of a swap has determinant -1, that of the scaling $R_i \mapsto cR_i$ has determinant c, and that of a replacement has determinant 1. Since every invertible matrix is a product of elementary matrices (theorem 1.4.8, condition (4)), its determinant is the product of the scalars attached to the operations that build it.

Everything so far has been stated for the rows, while section 3.1 measured the box spanned by the columns. The transpose (definition 2.3.4) turns each reading into the other, and the next theorem says that the determinant does not notice the exchange.

Theorem 3.2.11: A Matrix and Its Transpose Have the Same Determinant

Let $A \in M_n(k)$. Then
$$\det(A^\top) = \det A$$

***Proof*:**
Three preliminary facts are used. The first is that transposing reverses a product: for $M \in M_{m\times n}(k)$ and $N \in M_{n\times p}(k)$,

$$\left((MN)^\top\right)_{ij} = (MN)_{ji} = \sum_{l=1}^{n} m_{jl}n_{li} = \sum_{l=1}^{n}(N^\top)_{il}(M^\top)_{lj} = (N^\top M^\top)_{ij}$$

for all i and j, using definition 2.3.4 in the first and third equalities and definition 1.4.1 in the second and fourth, so $(MN)^\top = N^\top M^\top$. The second is that A and $A^\top$ have the same rank: the columns of $A^\top$ are the rows of A (definition 2.3.4), so $\operatorname{col} A^\top = \operatorname{row} A$ by definition 2.3.6, and therefore
$$\operatorname{rank} A^\top = \dim \operatorname{col} A^\top = \dim \operatorname{row} A = \operatorname{rank} A$$
where the first equality is theorem 2.3.9 applied to $A^\top$ and the last is theorem 2.3.8. The third is that $\det M \neq 0$ holds exactly when $\operatorname{rank} M = n$, for every $M \in M_n(k)$: elimination carries M to an echelon form (theorem 1.2.8) and theorem 3.2.3 equates $\det M \neq 0$ with the nonvanishing of the diagonal product of that echelon form, which the same theorem equates with $\operatorname{rank} M = n$.

Case 1: $\operatorname{rank} A < n$. The third fact gives $\det A = 0$, and $\operatorname{rank} A^\top = \operatorname{rank} A < n$ by the second, so the third fact applied to $A^\top$ gives $\det(A^\top) = 0$. Hence $\det(A^\top) = 0 = \det A$.

Case 2: $\operatorname{rank} A = n$. The third fact gives $\det A \neq 0$, so A is invertible by corollary 3.2.8. By theorem 1.4.8, condition (4), there are elementary matrices $E_1, \dots, E_s$ with $A = E_1E_2\cdots E_s$, and applying the first preliminary fact repeatedly gives $A^\top = E_s^\top \cdots E_2^\top E_1^\top$. It suffices to show $\det(E^\top) = \det E$ for a single elementary matrix E, since theorem 3.2.10 then gives

$$\det(A^\top) = \det(E_s^\top)\cdots\det(E_1^\top) = \det(E_s)\cdots\det(E_1) = \det(E_1)\cdots\det(E_s) = \det A$$

the third equality holding because multiplication in k does not depend on the order of the factors. Take the three kinds of elementary matrix in turn (definition 1.4.6). The matrix of a swap $R_i \leftrightarrow R_j$ has entry 1 in the positions (l,l) for $l \notin \{i,j\}$ and in the positions (i,j) and (j,i), and zeros elsewhere; that list of positions is unchanged when the two indices in each pair are exchanged, so $E^\top = E$. The matrix of a scaling $R_i \mapsto cR_i$ is diagonal, so again $E^\top = E$. The matrix of a replacement $R_i \mapsto R_i + cR_j$ has 1 in every diagonal position, c in position (i,j), and zeros elsewhere, and $E^\top$ has 1 in every diagonal position, c in position (j,i), and zeros elsewhere; one of the two is upper triangular and the other lower triangular, and both have every diagonal entry equal to 1, so $\det E = 1 = \det(E^\top)$ by proposition 3.2.6, completing the proof.

Every statement proved about the rows now has a companion about the columns, and the companion needs no separate argument. Write $C_i \leftrightarrow C_j$, $C_j \mapsto cC_j$ with $c \neq 0$, and $C_i \mapsto C_i + cC_j$ with $i \neq j$ for the three operations of definition 1.2.2 performed on the columns of a matrix instead of on its rows.

Corollary 3.2.12: The Determinant in Terms of the Columns

Let $A \in M_n(k)$ have columns $v_1, \dots, v_n$.

1. det is linear in each column separately: for each index j, all columns $v_1, \dots, v_n$ and u, w, and all $\lambda, \mu \in k$, the determinant of the matrix whose j-th column is $\lambda u + \mu w$ and whose remaining columns are those of A equals λ times the determinant of the matrix with u in position j plus μ times the determinant of the matrix with w in position j.

2. If $v_i = v_j$ for some pair of indices $i \neq j$, or more generally if one column of A is a linear combination of the others, then $\det A = 0$.

3. The column operation $C_i \leftrightarrow C_j$ with $i \neq j$ changes $\det A$ to $-\det A$; the column operation $C_j \mapsto cC_j$ multiplies $\det A$ by c; and the column operation $C_i \mapsto C_i + cC_j$ with $i \neq j$ leaves $\det A$ unchanged.

***Proof*:**
By definition 2.3.4 the rows of $A^\top$ are the columns of A, and performing one of the three operations on the columns of A is the same as performing the corresponding operation on the rows of $A^\top$ and then transposing back. Each of (1), (2) and (3) is therefore the corresponding statement about the rows of $A^\top$, which is condition (1) of definition 3.2.1 for (1), condition (2) of definition 3.2.1 together with lemma 3.2.2(4) for (2), and parts (1), (2) and (3) of lemma 3.2.2 for (3). Transporting each back to A uses $\det(A^\top) = \det A$ from theorem 3.2.11, applied once to each of the matrices appearing in the statement, as we sought to show.

The three conditions of definition 3.2.1 could therefore have been imposed on the columns instead, with the same function resulting. That is what reconciles the algebra of this section with the geometry of section 3.1, where the box being measured was spanned by the columns. It also doubles the computational options: a determinant may be evaluated by column elimination as readily as by row elimination, and part (2) detects a singular matrix from a relation among its columns without any elimination at all, as in example 3.2.9.

One further consequence records how the determinant behaves under the change of basis of section 2.5.

Corollary 3.2.13: The Determinant of an Inverse and of a Similar Matrix

Let $A, B \in M_n(k)$.

1. If A is invertible, then $\det(A^{-1}) = (\det A)^{-1}$.

2. If A and B are similar (definition 2.5.5), then $\det A = \det B$.

***Proof*:**

1. By definition 1.4.4, $AA^{-1} = I_n$, so theorem 3.2.10 and $\det I_n = 1$ give $\det A \cdot \det(A^{-1}) = 1$. By corollary 3.2.8, $\det A \neq 0$, so dividing by it gives $\det(A^{-1}) = (\det A)^{-1}$.

2. Let $B = PAP^{-1}$ with P invertible. Applying theorem 3.2.10 twice and then part (1),

$$\det B = \det P \, \det A \, \det(P^{-1}) = \det A \, \det P \, (\det P)^{-1} = \det A$$

   where the middle equality uses that multiplication in k does not depend on the order of the factors, completing the proof.

By theorem 2.5.4 the matrices of one linear map $T\colon k^n \to k^n$ in two ordered bases are similar, so part (2) says that the number $\det[T]_{\mathcal{B}}$ is the same for every ordered basis $\mathcal{B}$ of k^n: it is a property of the map and not of the coordinates chosen to write it down. The trace was the first such number (corollary 2.5.9); the determinant is the second, and unlike the trace it decides invertibility.

Exercise 3.2.1

1. Compute

$$\det \begin{pmatrix} 2 & 1 & 3 \\ 4 & 3 & 8 \\ 2 & 4 & 5 \end{pmatrix}$$

   by elimination, naming each row operation used and stating the scalar it contributes, and then check the answer by expanding along the first column.

2. Compute

$$\det \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 \\ 2 & 5 & 3 & 2 \end{pmatrix}$$

   by elimination. State the echelon form reached and the value of λ.

3. For which $t \in \mathbb{R}$ is the matrix

$$M(t) = \begin{pmatrix} t & 1 & 0 \\ 1 & t & 1 \\ 0 & 1 & t \end{pmatrix}$$

   singular? Compute $\det M(t)$ as a polynomial in t and use corollary 3.2.8.

4. Let $A \in M_n(k)$ and $c \in k$. Show that $\det(cA) = c^n \det A$. Then let $A \in M_3(\mathbb{R})$ have $\det A = 5$ and compute $\det(2A)$, $\det(A^2)$, $\det(A^{-1})$ and $\det(A^\top A)$. Finally, give 2×2 matrices A and B with $\det(A + B) \neq \det A + \det B$.

5. Let $A \in M_n(\mathbb{R})$ satisfy $A^\top = -A$, with n odd. Show that $\det A = 0$. Then let $Q \in M_n(\mathbb{R})$ satisfy $Q^\top Q = I_n$ and show that $\det Q$ is 1 or -1.

6. Define $D\colon M_2(k) \to k$ by $D(A) = a_{11}a_{22} + a_{12}a_{21}$. Decide which of the three conditions of definition 3.2.1 the function D satisfies and which it fails, with proof in each case, and conclude that D is not det.

3.3 Cofactors, the Adjugate, and Cramer's Rule

The construction in section 3.2 computes an $n \times n$ determinant from n determinants of size $n-1$, obtained by deleting the first column and one row at a time. The first column earns its place in the induction, not in the answer. This section shows that every row and every column returns the same value, collects the n^2 smaller determinants into a single matrix, and reads off from that matrix a formula for the inverse and a formula for the solution of a square system.

Running the recipe in all six available directions on one 3×3 matrix costs nine small determinants and shows what to expect.

Example 3.3.1: The Nine Deleted Determinants of a 3×3 Matrix
Let

$$B = \begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \\ 4 & 1 & 5 \end{pmatrix}$$

For each position (i, j), delete row i and column j of B and take the determinant of the 2×2 matrix left behind. For a 2×2 matrix the construction of section 3.2 gives $\det \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) = ad - bc$, so each of the nine numbers is one subtraction of two products. Deleting the first row of B in turn with each of its three columns leaves

$$\det \begin{pmatrix} -1 & 2 \\ 1 & 5 \end{pmatrix} = -7, \qquad \det \begin{pmatrix} 0 & 2 \\ 4 & 5 \end{pmatrix} = -8, \qquad \det \begin{pmatrix} 0 & -1 \\ 4 & 1 \end{pmatrix} = 4$$

Deleting the second row leaves

$$\det \begin{pmatrix} 1 & 3 \\ 1 & 5 \end{pmatrix} = 2, \qquad \det \begin{pmatrix} 2 & 3 \\ 4 & 5 \end{pmatrix} = -2, \qquad \det \begin{pmatrix} 2 & 1 \\ 4 & 1 \end{pmatrix} = -2$$

and deleting the third row leaves

$$\det \begin{pmatrix} 1 & 3 \\ -1 & 2 \end{pmatrix} = 5, \qquad \det \begin{pmatrix} 2 & 3 \\ 0 & 2 \end{pmatrix} = 4, \qquad \det \begin{pmatrix} 2 & 1 \\ 0 & -1 \end{pmatrix} = -2$$

The expansion along the first column in section 3.2 attaches alternating signs to these numbers, starting with a plus in the top position. Attaching to position (i, j) the sign $+$ when $i + j$ is even and $-$ when $i + j$ is odd extends that pattern to the whole square,

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

and produces the table of signed numbers

$$\begin{pmatrix} -7 & 8 & 4 \\ -2 & -2 & 2 \\ 5 & -4 & -2 \end{pmatrix}$$

in which the entry in position (i, j) is the deleted determinant above with its sign attached.

Multiplying the entries of the first column of B by the entries of the first column of this table and adding gives

$$2(-7)+0(-2)+4(5)=-14+0+20=6$$

which is exactly the first-column expansion of section 3.2, so $\det B=6$. The same pairing along the second column gives

$$1(8)+(-1)(-2)+1(-4)=8+2-4=6$$

and along the third column

$$3(4)+2(2)+5(-2)=12+4-10=6$$

Pairing along rows instead of columns gives, for the three rows in order,

$$2(-7)+1(8)+3(4)=6, \qquad 0(-2)+(-1)(-2)+2(2)=6, \qquad 4(5)+1(-4)+5(-2)=6$$

All six directions return 6.

A second matrix, whose determinant can be read off without expanding anything, confirms the pattern. Let A' be the matrix formed by the first three rows of the running matrix A of example 1.2.9,

$$A'=\begin{pmatrix}1 & 1 & 3\\ 2 & 3 & 8\\ 1 & 2 & 5\end{pmatrix}$$

Write c_1, c_2, c_3 for its columns and $\det(u, v, w)$ for the determinant of the matrix with columns u, v and w. The columns satisfy $c_3 = c_1 + 2c_2$, the relation recorded in section 1.2, so linearity in the third column and the vanishing of the determinant on a matrix with two equal columns (corollary 3.2.12) give

$$\det A' = \det(c_1, c_2, c_1) + 2\det(c_1, c_2, c_2) = 0 + 0 = 0$$

The nine deleted determinants of A', with signs attached, are

$$\begin{pmatrix}-1 & -2 & 1\\ 1 & 2 & -1\\ -1 & -2 & 1\end{pmatrix}$$

and pairing along the first row, the third row, and the second column gives

$$1(-1)+1(-2)+3(1)=0, \qquad 1(-1)+2(-2)+5(1)=0, \qquad 1(-2)+3(2)+2(-2)=0$$

in agreement with $\det A'=0$.

The computation used one operation nine times, and that operation deserves a name.

Definition 3.3.2: Minor

Let $A \in M_n(k)$ with $n \geq 2$, and let $1 \leq i, j \leq n$. Write $A(i|j)$ for the $(n-1) \times (n-1)$ matrix obtained from A by deleting row i and column j and keeping all remaining rows and columns in their original order. The (i,j) *minor* of A is the number $\det A(i|j)$.

The nine numbers displayed first in example 3.3.1 are the minors of B. It was the signed version of them that the six expansions paired against the entries, and that version deserves a name of its own.

Definition 3.3.3: Cofactor

Let $A \in M_n(k)$ with $n \geq 2$. The (i,j) *cofactor* of A is the signed minor

$$C_{ij} = (-1)^{i+j} \det A(i|j)$$

The matrix $(C_{ij}) \in M_n(k)$ whose (i,j) entry is C_{ij} is the *cofactor matrix* of A.

The sign $(-1)^{i+j}$ is the checkerboard pattern of example 3.3.1: it is $+1$ on the diagonal and switches every time one moves one step along a row or a column. The two tables of signed numbers in that example are the cofactor matrices of B and of A', and each of the six sums computed there pairs one row or column of the matrix with the matching row or column of its cofactor matrix. In this language the first-column expansion that built the determinant in section 3.2 reads

$$\det A = \sum_{i=1}^{n} a_{i1} C_{i1}$$

and the example says that the first column has no monopoly on this.

Theorem 3.3.4: Cofactor Expansion Along Any Row or Column

Let $A = (a_{ij}) \in M_n(k)$ with $n \geq 2$, and let C_{ij} be its cofactors. For every row index i,

$$\det A = \sum_{j=1}^{n} a_{ij} C_{ij}$$

and for every column index j,

$$\det A = \sum_{i=1}^{n} a_{ij} C_{ij}$$

***Proof*:**
Swapping two columns of a matrix negates its determinant. This is the column form of the swap rule of lemma 3.2.2: interchanging columns p and q of A interchanges rows p and q of $A^\top$ (definition 2.3.4), and $\det A^\top = \det A$ by theorem 3.2.11, so the sign change recorded there for a row swap transfers to columns.

Fix a column index j and write $c_1, \ldots, c_n$ for the columns of A. Interchange column j with column $j-1$, then with column $j-2$, and so on until it reaches the front, there being nothing to do when $j = 1$. Each of these $j-1$ interchanges swaps two columns, and the columns other than c_j keep their relative order throughout, so the result is the matrix $\tilde{A}$ with columns

$$c_j,\ c_1,\ \ldots,\ c_{j-1},\ c_{j+1},\ \ldots,\ c_n$$

and $\det \tilde{A} = (-1)^{j-1} \det A$. The first column of $\tilde{A}$ has entries $a_{1j}, \ldots, a_{nj}$, and deleting that column together with row i leaves the columns $c_1, \ldots, c_{j-1}, c_{j+1}, \ldots, c_n$ with row i removed, which is $A(i|j)$. Expanding $\det \tilde{A}$ along its first column (theorem 3.2.4) therefore gives

$$\det \tilde{A} = \sum_{i=1}^{n} (-1)^{i+1} a_{ij} \det A(i|j)$$

Multiplying both sides by $(-1)^{j-1}$ and using $(-1)^{j-1}(-1)^{i+1} = (-1)^{i+j}$ yields

$$\det A = \sum_{i=1}^{n} (-1)^{i+j} a_{ij} \det A(i|j) = \sum_{i=1}^{n} a_{ij} C_{ij}$$

which is the expansion along column j.

The row statement follows by transposing. Fix a row index i and apply the column statement to $A^\top$ with column index i. The (p, i) entry of $A^\top$ is a_{ip}, and deleting row p and column i from $A^\top$ gives the transpose of the matrix obtained by deleting row i and column p from A, so $A^\top(p|i) = (A(i|p))^\top$ and these two have the same determinant by theorem 3.2.11. Hence

$$\det A = \det A^\top = \sum_{p=1}^{n} (-1)^{p+i} a_{ip} \det A(i|p) = \sum_{p=1}^{n} a_{ip} C_{ip}$$

which is the expansion along row i, as we sought to show.

The theorem turns the determinant into a computation with a choice in it. Any row or column may be taken, so one carrying zeros is worth taking: a zero entry kills its term, and with it an entire determinant of size $n - 1$. The expansion also settles what kind of function the determinant is.

Proposition 3.3.5: The Determinant Is a Polynomial in the Entries

For each $n \geq 1$ there is a polynomial in n^2 variables, with coefficients in k, whose value at the entries of any $A \in M_n(k)$ is $\det A$.

***Proof*:**
Induct on n. A 1×1 determinant is its single entry, which is a polynomial in that entry. For the inductive step, theorem 3.3.4 writes an $n \times n$ determinant as a sum of n products, each of an entry of A with $\pm$ a determinant of size $n - 1$. Each of those is a polynomial in the entries of the corresponding minor by the inductive hypothesis, and the entries of a minor are entries of A; a sum of products of polynomials is a polynomial, completing the proof.

Choice is not the same as speed. Let $\mu(n)$ count the multiplications used by a full expansion of an $n \times n$ determinant in which no entry is zero. Such an expansion computes n determinants of size

$n-1$ and then performs n further multiplications, so

$$\mu(n) = n\big(\mu(n-1)+1\big), \qquad \mu(1) = 0$$

Then $\mu(2) = 2$, and if $\mu(n-1) \geq (n-1)!$ then $\mu(n) \geq n\big((n-1)!+1\big) > n!$, so $\mu(n) \geq n!$ for every $n \geq 2$. Elimination is the alternative. Using only swaps and row replacements, the procedure of theorem 1.2.8 drives a square matrix to an upper triangular one; when the pivots land in the diagonal positions, clearing the entries below the k-th pivot costs $n-k$ divisions and $(n-k)^2$ multiplications, for a total of

$$\sum_{k=1}^{n-1}(n-k)^2 = \frac{(n-1)n(2n-1)}{6}$$

multiplications and $n(n-1)/2$ divisions, after which proposition 3.2.6 multiplies the n diagonal entries at a cost of $n-1$ further multiplications, and lemma 3.2.2 gives the sign from the swaps performed. For $n = 20$ the first route needs at least $20!$ multiplications, a number larger than $2 \cdot 10^{18}$, and the second needs $2470 + 19 = 2489$ multiplications and 190 divisions. Cofactor expansion is a tool for small matrices, for matrices with many zeros, and for proofs.

The expansion along row i pairs the entries of row i with the cofactors of row i. Pairing the entries of row i with the cofactors of a different row ℓ is just as easy to write down, and the answer is forced. Assembling all n^2 such pairings at once is a matrix product, provided the cofactors are stored transposed.

Definition 3.3.6: Adjugate

Let $A \in M_n(k)$ with $n \geq 2$ and let C_{ij} be its cofactors. The *adjugate* of A is the transpose (definition 2.3.4) of its cofactor matrix, that is, the matrix $\operatorname{adj}(A) \in M_n(k)$ whose (i,j) entry is C_{ji}.

For the matrix B of example 3.3.1, transposing the cofactor matrix computed there gives

$$\operatorname{adj}(B) = \begin{pmatrix} -7 & -2 & 5 \\ 8 & -2 & -4 \\ 4 & 2 & -2 \end{pmatrix}$$

and for the singular matrix A' of that example,

$$\operatorname{adj}(A') = \begin{pmatrix} -1 & 1 & -1 \\ -2 & 2 & -2 \\ 1 & -1 & 1 \end{pmatrix}$$

The transpose is what aligns the entries of a row of A with the cofactors of a row of A inside a matrix product, since the product pairs a row of the left factor with a column of the right one.

Theorem 3.3.7: The Adjugate Formula

Let $A \in M_n(k)$ with $n \geq 2$. Then

$$A\operatorname{adj}(A) = \operatorname{adj}(A)A = (\det A)I_n$$

***Proof*:**
Write $A = (a_{ij})$ and let C_{ij} be its cofactors, so the (j, ℓ) entry of $\operatorname{adj}(A)$ is $C_{\ell j}$ by definition 3.3.6. By the definition of the matrix product (definition 1.4.1), the (i, ℓ) entry of $A \operatorname{adj}(A)$ is

$$\sum_{j=1}^{n} a_{ij} C_{\ell j}$$

For $\ell = i$ this is the expansion of $\det A$ along row i, which is $\det A$ by theorem 3.3.4.

Suppose $\ell \neq i$, and let $\tilde{A}$ be the matrix obtained from A by replacing row ℓ with row i of A and leaving every other row unchanged. Rows i and ℓ of $\tilde{A}$ are then equal, so $\det \tilde{A} = 0$ by definition 3.2.1. The cofactors of $\tilde{A}$ along its row ℓ are the cofactors of A along row ℓ: deleting row ℓ deletes the only row in which the two matrices differ, so $\tilde{A}(\ell|j) = A(\ell|j)$ for every j, and the sign $(-1)^{\ell+j}$ does not depend on the matrix. The entries of row ℓ of $\tilde{A}$ are $a_{i1}, \ldots, a_{in}$, so expanding $\det \tilde{A}$ along that row gives

$$0 = \det \tilde{A} = \sum_{j=1}^{n} a_{ij} C_{\ell j}$$

The (i, ℓ) entry of $A \operatorname{adj}(A)$ is therefore $\det A$ when $i = \ell$ and 0 otherwise, which says $A \operatorname{adj}(A) = (\det A) I_n$.

The second identity is the same argument run on columns. The (i, ℓ) entry of $\operatorname{adj}(A)A$ is $\sum_j C_{ji} a_{j\ell}$, which for $i = \ell$ is the expansion of $\det A$ along column i. For $i \neq \ell$, let $\hat{A}$ be the matrix obtained from A by replacing column i with column ℓ of A. Columns i and ℓ of $\hat{A}$ are equal, so $\det \hat{A} = 0$, the determinant being zero on a matrix with two equal columns (corollary 3.2.12). Deleting column i deletes the only column in which A and $\hat{A}$ differ, so the cofactors of $\hat{A}$ along its column i are again $C_{1i}, \ldots, C_{ni}$, and the entries of that column are $a_{1\ell}, \ldots, a_{n\ell}$. Expanding $\det \hat{A}$ along column i gives $\sum_j a_{j\ell} C_{ji} = 0$, and so $\operatorname{adj}(A)A = (\det A) I_n$, completing the proof.

The off-diagonal entries are the whole content of the theorem: each is a cofactor expansion of a matrix with a repeated row or a repeated column, and vanishes for that reason alone. The diagonal entries are theorem 3.3.4 applied n times over. The two identities on the matrices of example 3.3.1 read

$$\begin{pmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \\ 4 & 1 & 5 \end{pmatrix} \begin{pmatrix} -7 & -2 & 5 \\ 8 & -2 & -4 \\ 4 & 2 & -2 \end{pmatrix} = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{pmatrix} = 6 I_3$$

and, for the singular A',

$$\begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 1 & 2 & 5 \end{pmatrix} \begin{pmatrix} -1 & 1 & -1 \\ -2 & 2 & -2 \\ 1 & -1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

The second display is the formula with $\det A' = 0$ on the right, and it shows that the adjugate carries no information about an inverse when there is none to carry. When there is one, dividing is all that remains.

Corollary 3.3.8: The Inverse as a Ratio of Determinants

Let $A \in M_n(k)$ with $n \geq 2$ be invertible. Then

$$A^{-1} = \frac{1}{\det A} \operatorname{adj}(A)$$

so the (i, j) entry of A^{-1} is $C_{ji}/\det A$, where the C_{ji} are the cofactors of A.

***Proof*:**
By corollary 3.2.8 an invertible matrix has $\det A \neq 0$, so the scalar $1/\det A$ is defined. Multiplying both identities of theorem 3.3.7 by it gives

$$A\left(\frac{1}{\det A} \operatorname{adj}(A)\right) = \left(\frac{1}{\det A} \operatorname{adj}(A)\right) A = I_n$$

so $\frac{1}{\det A} \operatorname{adj}(A)$ is an inverse of A, and the inverse is unique by proposition 1.4.5, as we sought to show.

For $n = 2$ the formula is one line. Each minor of $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is the single entry left by the two deletions, so the cofactor matrix is $\left(\begin{smallmatrix} d & -c \\ -b & a \end{smallmatrix}\right)$, its transpose is $\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$, and corollary 3.3.8 gives

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

which is the rule for inverting a 2×2 matrix. For the matrix B of example 3.3.1 it gives

$$B^{-1} = \frac{1}{6} \begin{pmatrix} -7 & -2 & 5 \\ 8 & -2 & -4 \\ 4 & 2 & -2 \end{pmatrix}$$

Each entry of A^{-1} is one determinant divided by another, both built from the entries of A, and both polynomial in those entries. One consequence follows at once: a matrix with integer entries and determinant ± 1 has an inverse with integer entries, since every cofactor is then an integer and the division is by ± 1. What the formula does not offer is a method: computing n^2 cofactors of size $n - 1$ costs far more than the elimination of theorem 1.4.8, and the gap widens with n.

Applied to a column b rather than to the identity, the same formula gives a closed expression for each unknown of a square system (definition 1.1.3) whose coefficient matrix is invertible.

Theorem 3.3.9: Cramer's Rule

Let $A \in M_n(k)$ with $n \geq 2$ and $\det A \neq 0$, and let $b \in k^n$. For each i, write $A_i(b)$ for the matrix obtained from A by replacing its i-th column with b. Then $Ax = b$ has exactly one solution, and its i-th entry is

$$x_i = \frac{\det A_i(b)}{\det A}$$

***Proof*:**
By corollary 3.2.8, $\det A \neq 0$ makes A invertible. If $Ax = b$ then multiplying on the left by A^{-1} gives $x = A^{-1}b$, and conversely $A(A^{-1}b) = b$, so $A^{-1}b$ is the one and only solution. By corollary 3.3.8 its i-th entry is

$$x_i = \frac{1}{\det A} \sum_{j=1}^{n} C_{ji} b_j$$

where the C_{ji} are the cofactors of A.

The sum is itself a cofactor expansion. The matrix $A_i(b)$ differs from A only in column i, so deleting column i from either leaves the same submatrix: for every j the cofactor of $A_i(b)$ in position (j, i) is C_{ji}. The entries of column i of $A_i(b)$ are $b_1, \dots, b_n$, so expanding $\det A_i(b)$ along that column (theorem 3.3.4) gives

$$\det A_i(b) = \sum_{j=1}^{n} b_j C_{ji}$$

Substituting this into the previous display gives $x_i = \det A_i(b)/\det A$, as we sought to show.

Example 3.3.10: A Three-Variable System Solved by Cramer's Rule
Take the matrix B of example 3.3.1 and the system $Bx = b$ with $b = (7, 5, 13)^\top$, that is,

$$\begin{aligned} 2x_1 + x_2 + 3x_3 &= 7 \\ -x_2 + 2x_3 &= 5 \\ 4x_1 + x_2 + 5x_3 &= 13 \end{aligned}$$

Since $\det B = 6 \neq 0$, Cramer's rule applies. Replacing the first column of B by b and expanding along the first row gives

$$\det B_1(b) = \det \begin{pmatrix} 7 & 1 & 3 \\ 5 & -1 & 2 \\ 13 & 1 & 5 \end{pmatrix} = 7(-7) - 1(25 - 26) + 3(5 + 13) = -49 + 1 + 54 = 6$$

Replacing the second column and expanding along the first column, which carries a zero, gives

$$\det B_2(b) = \det \begin{pmatrix} 2 & 7 & 3 \\ 0 & 5 & 2 \\ 4 & 13 & 5 \end{pmatrix} = 2(25 - 26) + 4(14 - 15) = -2 - 4 = -6$$

and replacing the third column and expanding along the first column gives

$$\det B_3(b) = \det \begin{pmatrix} 2 & 1 & 7 \\ 0 & -1 & 5 \\ 4 & 1 & 13 \end{pmatrix} = 2(-13 - 5) + 4(5 + 7) = -36 + 48 = 12$$

Dividing each by $\det B = 6$,

$$x_1 = \frac{6}{6} = 1, \qquad x_2 = \frac{-6}{6} = -1, \qquad x_3 = \frac{12}{6} = 2$$

The inverse computed from the adjugate returns the same answer:

$$B^{-1}b = \frac{1}{6}\begin{pmatrix} -7 & -2 & 5 \\ 8 & -2 & -4 \\ 4 & 2 & -2 \end{pmatrix}\begin{pmatrix} 7 \\ 5 \\ 13 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} -49-10+65 \\ 56-10-52 \\ 28+10-26 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 6 \\ -6 \\ 12 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

Substituting into the original system confirms it: $2-1+6=7$, $1+4=5$, and $4-1+10=13$.

Cramer's rule is a formula, not a procedure. Solving an $n \times n$ system by it requires $n+1$ determinants of size n, where elimination solves the system and delivers the determinant in one pass. What the formula gives instead is the dependence of the answer on the data: each unknown is displayed as a ratio of two polynomials in the entries of A and b, so the effect of changing one coefficient on one unknown can be read off rather than recomputed. For a 2×2 or 3×3 system with an invertible coefficient matrix it is also a serviceable hand method, which is how it is most often met.

Exercise 3.3.1

1. Let
$$P = \begin{pmatrix} 1 & 2 & 0 \\ 3 & -1 & 4 \\ 2 & 0 & 1 \end{pmatrix}$$
Compute all nine cofactors of P. Then evaluate $\det P$ by expanding along the second row, and again by expanding along the third column, and check that the two values agree.

2. Using the cofactors from the previous question, write down $\operatorname{adj}(P)$, verify by direct multiplication that $P \operatorname{adj}(P) = 9I_3$, and write down P^{-1}.

3. Solve $Px = b$ with $b = (0, 19, 7)^\top$ by Cramer's rule, and check the answer against $P^{-1}b$ using the inverse found in the previous question.

4. Let
$$Q = \begin{pmatrix} 2 & 0 & 0 & 1 \\ 5 & 3 & 0 & 0 \\ 1 & 4 & 2 & 0 \\ 0 & 1 & 0 & 3 \end{pmatrix}$$
Compute $\det Q$ by expanding at each stage along the row or column carrying the most zeros. Count the multiplications your computation used, and compare with $\mu(4) = 40$, the count for an expansion that skips nothing.

5. Let $A \in M_n(\mathbb{R})$ with $n \geq 2$ have integer entries and $\det A = \pm 1$. Show that A^{-1} has integer entries. (Begin by showing that a matrix with integer entries has an integer determinant.)

6. Let $A \in M_n(k)$ with $n \geq 2$ be invertible. Show first that $\det(cI_n) = c^n$ for every scalar c, and deduce that $\det(\operatorname{adj}(A)) = (\det A)^{n-1}$.

3.4 Orientation and the Sign of the Determinant

Two readings of the determinant are already in place. Section 3.1 read its absolute value as an area or a volume, and section 3.2 read its vanishing as the failure of invertibility. Neither reading uses the sign. This section says what the sign records: it sorts the ordered bases of $\mathbb{R}^n$ into two families, and it decides which invertible matrices carry each family into itself. A complex number has no sign, so all of this is a statement about $\mathbb{R}$, and every scalar below is real.

Example 3.4.1: Four Ordered Bases of $\mathbb{R}^2$ and the Signs of Their Determinants

Take the four lists

$$\mathcal{B} = (b_1, b_2), \qquad \mathcal{B}' = (b_2, b_1), \qquad \mathcal{C} = (c_1, c_2), \qquad \mathcal{D} = (e_1, -e_2)$$

where

$$b_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad b_2 = \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad c_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad c_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

and e_1, e_2 is the standard basis of $\mathbb{R}^2$. The list $\mathcal{B}'$ contains the same two vectors as $\mathcal{B}$, written in the other order.

Each list is an ordered basis. For $\mathcal{B}$, the equation $xb_1 + yb_2 = (s,t)^\top$ is the system

$$\begin{aligned} 2x + y &= s \\ x + 3y &= t \end{aligned}$$

Multiplying the first equation by 3 and subtracting the second gives $5x = 3s - t$, so $x = (3s - t)/5$, and the first equation then gives $y = s - 2x = (2t - s)/5$. Every column of $\mathbb{R}^2$ is thus a combination of b_1 and b_2, so the pair spans $\mathbb{R}^2$ (definition 2.1.5); and taking $s = t = 0$ forces $x = y = 0$, so the pair is linearly independent (definition 2.1.9). It is therefore a basis (definition 2.2.1). The list $\mathcal{B}'$ has the same two vectors, so it is a basis as well. For $\mathcal{C}$, the equation $xc_1 + yc_2 = (s,t)^\top$ reads $x + y = s$ and $-x + y = t$, with the unique solution $x = (s - t)/2$ and $y = (s + t)/2$; for $\mathcal{D}$, the combination $xe_1 + y(-e_2)$ is the column $(x, -y)^\top$, which equals $(s,t)^\top$ for $x = s$ and $y = -t$ and for no other pair. The same two observations, spanning from the existence of a solution and independence from the case $s = t = 0$, make both of these lists bases too.

Assemble each list into the columns of a matrix and take determinants. For a 2×2 matrix the determinant is $a_{11}a_{22} - a_{12}a_{21}$ (theorem 3.3.4), so

$$\det \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} = 6 - 1 = 5, \qquad \det \begin{pmatrix} 1 & 2 \\ 3 & 1 \end{pmatrix} = 1 - 6 = -5$$

for $\mathcal{B}$ and $\mathcal{B}'$, and

$$\det \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} = 1 + 1 = 2, \qquad \det \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = -1 - 0 = -1$$

for $\mathcal{C}$ and $\mathcal{D}$. Two of the four numbers are positive and two are negative.

Now compare the bases in pairs, using the change of basis matrices of definition 2.5.2. The matrix whose columns are the vectors of an ordered basis $\mathcal{B}$ is $P_{\mathcal{E}\mathcal{B}}$, since the coordinate column of b_j in the standard basis is b_j itself, and by proposition 2.5.3(2),

$$P_{\mathcal{C}\mathcal{B}} = P_{\mathcal{C}\mathcal{E}} P_{\mathcal{E}\mathcal{B}} = (P_{\mathcal{E}\mathcal{C}})^{-1} P_{\mathcal{E}\mathcal{B}}$$

The inverse is given by corollary 3.3.8, which for a 2×2 matrix of nonzero determinant reads

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

Hence
$$(P_{\mathcal{E}\mathcal{C}})^{-1} = \frac{1}{2}\begin{pmatrix}1 & -1\\ 1 & 1\end{pmatrix}, \qquad P_{\mathcal{C}\mathcal{B}} = \frac{1}{2}\begin{pmatrix}1 & -1\\ 1 & 1\end{pmatrix}\begin{pmatrix}2 & 1\\ 1 & 3\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1 & -2\\ 3 & 4\end{pmatrix}$$
whose determinant is $\frac{1}{4}(4+6) = \frac{5}{2}$. As a check on the two columns, $b_1 = \frac{1}{2}c_1 + \frac{3}{2}c_2$ and $b_2 = -c_1 + 2c_2$, which are the entries just computed.

The same procedure for the other two pairs uses $(P_{\mathcal{E}\mathcal{D}})^{-1} = P_{\mathcal{E}\mathcal{D}}$, since $\left(\begin{smallmatrix}1 & 0\\ 0 & -1\end{smallmatrix}\right)$ squares to I_2, and $(P_{\mathcal{E}\mathcal{B}'})^{-1} = -\frac{1}{5}\left(\begin{smallmatrix}1 & -2\\ -3 & 1\end{smallmatrix}\right)$. The products are
$$P_{\mathcal{D}\mathcal{B}} = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\begin{pmatrix}2 & 1\\ 1 & 3\end{pmatrix} = \begin{pmatrix}2 & 1\\ -1 & -3\end{pmatrix}, \qquad P_{\mathcal{B}'\mathcal{B}} = -\frac{1}{5}\begin{pmatrix}1 & -2\\ -3 & 1\end{pmatrix}\begin{pmatrix}2 & 1\\ 1 & 3\end{pmatrix} = \begin{pmatrix}0 & 1\\ 1 & 0\end{pmatrix}$$
with determinants $-6+1=-5$ and $0-1=-1$. The second answer is the expected one: $\mathcal{B}'$ is $\mathcal{B}$ with its two entries exchanged, so the matrix converting $\mathcal{B}$-coordinates into $\mathcal{B}'$-coordinates exchanges the two entries of a column.

The four bases have determinants 5, -5, 2 and -1, and the three change of basis matrices have determinants
$$\det P_{\mathcal{C}\mathcal{B}} = \tfrac{5}{2}, \qquad \det P_{\mathcal{D}\mathcal{B}} = -5, \qquad \det P_{\mathcal{B}'\mathcal{B}} = -1$$
The determinant of the change of basis matrix is positive for the one pair whose two bases have determinants of the same sign, and negative for the two pairs whose bases have determinants of opposite signs.

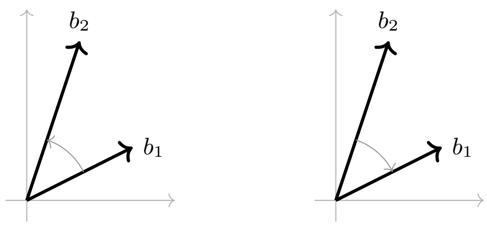


Figure 3.3: The two vectors of example 3.4.1 listed in the two possible orders; the turn from the first vector to the second runs counterclockwise on the left and clockwise on the right

Comparing two ordered bases is a statement about the change of basis matrix between them, and the sign of its determinant is the only feature of it that concerns this section.

Definition 3.4.2: Same Orientation; Positively and Negatively Oriented Bases

Let $\mathcal{B} = (b_1, \ldots, b_n)$ and $\mathcal{C} = (c_1, \ldots, c_n)$ be ordered bases of $\mathbb{R}^n$, and let $P_{\mathcal{CB}}$ be the change of basis matrix of definition 2.5.2. Then $\mathcal{B}$ and $\mathcal{C}$ *have the same orientation* when

$$\det P_{\mathcal{CB}} > 0$$

and *opposite orientations* when $\det P_{\mathcal{CB}} < 0$.
Write $\mathcal{E} = (e_1, \ldots, e_n)$ for the standard basis. An ordered basis $\mathcal{B}$ of $\mathbb{R}^n$ is *positively oriented* when it has the same orientation as $\mathcal{E}$, and *negatively oriented* when it has the opposite orientation to $\mathcal{E}$. The collection of all positively oriented ordered bases and the collection of all negatively oriented ordered bases are the two *orientations* of $\mathbb{R}^n$, and the first of them is the *standard orientation*.

The definition compares two bases through a matrix that must first be computed, as it was three times in example 3.4.1. The example also suggests that the computation is avoidable, since each answer was already recorded in the signs of the two determinants being compared. Write

$$[\mathcal{B}] \in M_n(\mathbb{R})$$

for the matrix whose j-th column is b_j. This is $P_{\mathcal{EB}}$: by definition 2.5.2 the j-th column of $P_{\mathcal{EB}}$ is the coordinate column of b_j in the standard basis, and that column is b_j itself.

Theorem 3.4.3: Orientation Sorts the Ordered Bases of $\mathbb{R}^n$ into Exactly Two Classes

Let $n \geq 1$.

1. A list $(v_1, \ldots, v_n)$ of n vectors in $\mathbb{R}^n$ is an ordered basis of $\mathbb{R}^n$ if and only if the matrix $V \in M_n(\mathbb{R})$ whose j-th column is v_j satisfies $\det V \neq 0$.
2. For ordered bases $\mathcal{B}$ and $\mathcal{C}$ of $\mathbb{R}^n$,

   $$\det P_{\mathcal{CB}} = \frac{\det[\mathcal{B}]}{\det[\mathcal{C}]}$$

   so $\mathcal{B}$ and $\mathcal{C}$ have the same orientation exactly when $\det[\mathcal{B}]$ and $\det[\mathcal{C}]$ have the same sign. In particular $\mathcal{B}$ is positively oriented exactly when $\det[\mathcal{B}] > 0$.
3. Having the same orientation is an equivalence relation on the set of ordered bases of $\mathbb{R}^n$, and it has exactly two classes, both nonempty: the positively oriented bases and the negatively oriented ones.

***Proof*:**

1. Suppose first that $\mathcal{V} = (v_1, \ldots, v_n)$ is an ordered basis. Then $V = P_{\mathcal{EV}}$, which is invertible by proposition 2.5.3(3), so $\det V \neq 0$ by corollary 3.2.8.

   Suppose instead that $\det V \neq 0$. Then V is invertible, again by corollary 3.2.8. If $x_1 v_1 + \cdots + x_n v_n = 0$ for scalars $x_1, \ldots, x_n$, that combination is the product Vx, where x is the column with entries $x_1, \ldots, x_n$, by the column reading of a matrix product established in section 1.2; multiplying by V^{-1} gives $x = V^{-1}Vx = 0$. So the list is linearly independent (definition 2.1.9), and a linearly independent list of n vectors in $\mathbb{R}^n$ is a basis by corollary 2.2.13.

2. By proposition 2.5.3(2), $P_{\mathcal{E}\mathcal{B}} = P_{\mathcal{E}\mathcal{C}}P_{\mathcal{C}\mathcal{B}}$, which in the present notation is $[\mathcal{B}] = [\mathcal{C}]\, P_{\mathcal{C}\mathcal{B}}$. Taking determinants and using theorem 3.2.10,

$$\det[\mathcal{B}] = \det[\mathcal{C}] \cdot \det P_{\mathcal{C}\mathcal{B}}$$

Both $\det[\mathcal{B}]$ and $\det[\mathcal{C}]$ are nonzero by part (1), so dividing by $\det[\mathcal{C}]$ gives $\det P_{\mathcal{C}\mathcal{B}} = \det[\mathcal{B}]/\det[\mathcal{C}]$. A quotient of two nonzero real numbers is positive exactly when the two numbers have the same sign and negative exactly when they do not, which is the second claim. For the third, take $\mathcal{C} = \mathcal{E}$: the matrix $[\mathcal{E}]$ is I_n, whose determinant is 1 by definition 3.2.1, so $\det[\mathcal{B}]$ and $\det[\mathcal{E}]$ have the same sign exactly when $\det[\mathcal{B}] > 0$.

3. Reflexivity holds because $P_{\mathcal{B}\mathcal{B}} = I_n$ by proposition 2.5.3(2), of determinant $1 > 0$. For symmetry, $P_{\mathcal{B}\mathcal{C}}$ is the inverse of $P_{\mathcal{C}\mathcal{B}}$ by proposition 2.5.3(3), and applying theorem 3.2.10 to $P_{\mathcal{B}\mathcal{C}}P_{\mathcal{C}\mathcal{B}} = I_n$ gives

$$\det P_{\mathcal{B}\mathcal{C}} \cdot \det P_{\mathcal{C}\mathcal{B}} = 1$$

so the two determinants are reciprocal and share a sign. For transitivity, let $\mathcal{D}$ be a third ordered basis with $\det P_{\mathcal{C}\mathcal{B}} > 0$ and $\det P_{\mathcal{D}\mathcal{C}} > 0$. Then $P_{\mathcal{D}\mathcal{B}} = P_{\mathcal{D}\mathcal{C}}P_{\mathcal{C}\mathcal{B}}$ by proposition 2.5.3(2), and its determinant is the product of two positive numbers, hence positive.

By part (2), two ordered bases lie in the same class exactly when $\det[\mathcal{B}]$ and $\det[\mathcal{C}]$ have the same sign, and a nonzero real number has one of two signs, so there are at most two classes and they are the two described. Both occur. The standard basis has $\det[\mathcal{E}] = \det I_n = 1 > 0$. For the other class, take the list $\mathcal{E}^- = (-e_1, e_2, \ldots, e_n)$, whose matrix of columns is the diagonal matrix $\mathrm{diag}(-1, 1, \ldots, 1)$, that is I_n with its first row scaled by -1. So $\det[\mathcal{E}^-] = -1 < 0$ by lemma 3.2.2, and part (1) confirms that $\mathcal{E}^-$ is an ordered basis, completing the proof.

Deciding whether two ordered bases have the same orientation is therefore a matter of computing two determinants and comparing their signs, and no change of basis matrix need be formed. The relation compares bases, but part (2) attaches to each basis on its own a single sign that decides every comparison it takes part in.

The names for the two classes come from the pictures. In $\mathbb{R}^2$ the positively oriented pairs are the ones drawn with the turn from the first vector to the second running counterclockwise, as in the left panel above. In $\mathbb{R}^3$ the positively oriented triples are the ones called right-handed: the thumb, index finger and middle finger of a right hand, held apart, point along v_1, v_2 and v_3 in that order, and the corresponding triples for a left hand are the negatively oriented ones.[1]

[1] These descriptions identify the two classes but do not prove anything about them. Turning “the turn runs counterclockwise” into a theorem would require moving one basis to another through a continuous family of bases, which is a statement about continuity rather than about determinants.

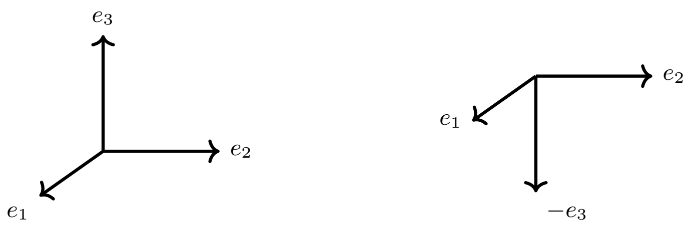


Figure 3.4: The triples (e_1, e_2, e_3) and $(e_1, e_2, -e_3)$ in $\mathbb{R}^3$, of determinant 1 and -1; the first is right-handed and the second is left-handed

The three operations of definition 1.2.2 can be applied to the entries of an ordered basis rather than to the rows of a matrix. Each of the three has a predictable effect on the orientation.

Corollary 3.4.4: Effect of the Three Operations on the Orientation of a Basis

Let $\mathcal{B} = (b_1, \ldots, b_n)$ be an ordered basis of $\mathbb{R}^n$, and let $\mathcal{B}'$ be the list obtained from $\mathcal{B}$ by one of the following changes.

1. Exchanging two entries b_i and b_j with $i \neq j$. Then $\mathcal{B}'$ is an ordered basis with the opposite orientation to $\mathcal{B}$.

2. Replacing b_i by cb_i for a scalar $c \neq 0$. Then $\mathcal{B}'$ is an ordered basis, with the same orientation as $\mathcal{B}$ when $c > 0$ and the opposite orientation when $c < 0$.

3. Replacing b_i by $b_i + cb_j$ for a scalar c and an index $j \neq i$. Then $\mathcal{B}'$ is an ordered basis with the same orientation as $\mathcal{B}$.

***Proof*:**
In each case $[\mathcal{B}']$ is obtained from $[\mathcal{B}]$ by performing the named operation on the columns rather than on the rows. A column operation on a matrix is the matching row operation on its transpose, and a matrix and its transpose have the same determinant (theorem 3.2.11), so the three rules of lemma 3.2.2 hold for columns as they do for rows (corollary 3.2.12). In each of the three cases the resulting determinant is nonzero, since $\det[\mathcal{B}] \neq 0$ by theorem 3.4.3(1) and $c \neq 0$ in case (2); that same part makes $\mathcal{B}'$ an ordered basis, and theorem 3.4.3(2) then settles the orientation by comparing the sign of $\det[\mathcal{B}']$ with the sign of $\det[\mathcal{B}]$.

1. Exchanging two columns gives $\det[\mathcal{B}'] = -\det[\mathcal{B}]$, of the opposite sign, so the two bases have opposite orientations.

2. Scaling a column by c gives $\det[\mathcal{B}'] = c\det[\mathcal{B}]$, which has the sign of $\det[\mathcal{B}]$ when $c > 0$ and the opposite sign when $c < 0$.

3. Adding a multiple of one column to a different column leaves the determinant unchanged, so the two signs agree and the two bases have the same orientation, completing the proof.

Part (3) says that sliding one basis vector along another changes the shape of the box the basis spans, sometimes drastically, and never changes its handedness. Part (2) says the same about lengths, since only the sign of c enters and not its size. Part (1) says that the order in which the vectors are listed is what the orientation is about. Two consequences of part (1) are worth naming. A list and the same list with two entries exchanged never have the same orientation, so of the two orderings of a basis of $\mathbb{R}^2$ exactly one is positively oriented; and for $n \geq 2$ any ordered basis of $\mathbb{R}^n$ can be made positively oriented by exchanging two of its entries, if it is not so already.

Orientation has so far been a property of bases. The determinant of a matrix governs it through the action of the matrix on a basis, and the same computation governs volume.

Theorem 3.4.5: The Absolute Value of $\det A$ Is the Factor by Which A Scales the Volume of a Box

Let $A \in M_n(\mathbb{R})$, let $v_1, \ldots, v_n \in \mathbb{R}^n$, and let $V \in M_n(\mathbb{R})$ be the matrix whose j-th column is v_j.

1. The matrix whose j-th column is Av_j is AV, and
$$\det(AV) = (\det A)(\det V)$$
2. Suppose A is invertible, let $\mathcal{B} = (b_1, \ldots, b_n)$ be an ordered basis of $\mathbb{R}^n$, and write $A\mathcal{B} = (Ab_1, \ldots, Ab_n)$. Then $A\mathcal{B}$ is an ordered basis of $\mathbb{R}^n$; it has the same orientation as $\mathcal{B}$ when $\det A > 0$ and the opposite orientation when $\det A < 0$. Which of the two occurs depends on A alone and not on $\mathcal{B}$.
3. Let $n = 2$. The parallelogram spanned by v_1 and v_2 has area $|\det V|$, whatever position the two vectors are in, and the parallelogram spanned by Av_1 and Av_2 therefore has area $|\det A|$ times that number. Let $n = 3$. The parallelepiped spanned by v_1, v_2 and v_3 has volume $|\det V|$, and the one spanned by Av_1, Av_2 and Av_3 has volume $|\det A|$ times that number.

***Proof*:**

1. The j-th column of a matrix $M \in M_n(\mathbb{R})$ is the product Me_j, by the column reading of section 1.2. Hence the j-th column of AV is
$$(AV)e_j = A(Ve_j) = Av_j$$
where the regrouping is associativity of the matrix product (proposition 1.4.3). So AV is the matrix described, and its determinant is $(\det A)(\det V)$ by theorem 3.2.10.
2. By part (1) the matrix $[A\mathcal{B}]$ is $A[\mathcal{B}]$, so
$$\det[A\mathcal{B}] = (\det A)\,(\det[\mathcal{B}])$$
Here $\det A \neq 0$ because A is invertible (corollary 3.2.8), and $\det[\mathcal{B}] \neq 0$ by theorem 3.4.3(1). Their product is nonzero, so the same part of that theorem makes $A\mathcal{B}$ an ordered basis. Multiplying a nonzero real number by $\det A$ preserves its sign when $\det A > 0$ and reverses it when $\det A < 0$, so theorem 3.4.3(2) compares $A\mathcal{B}$ with $\mathcal{B}$ exactly as claimed. The criterion mentions only the sign of $\det A$, so it is the same for every ordered basis $\mathcal{B}$.
3. Section 3.1 measured a parallelogram and a box by hand, but only in a convenient position: two vectors of the plane with every entry nonnegative and the second steeper than the first,

and three vectors of space two of which lie in the xy-plane. So the first task is to show that the area or volume of the region spanned by $v_1, \ldots, v_n$ is $|\det V|$ for every list, with $n = 2$ or $n = 3$. Granting that, the rest is immediate: the columns of AV are $Av_1, \ldots, Av_n$ by part (1), so the region spanned by those columns has area or volume

$$|\det(AV)| = |\det A|\,|\det V|$$

again by part (1), which is $|\det A|$ times the area or volume of the region spanned by $v_1, \ldots, v_n$.

Take the degenerate case first. If $\det V = 0$ then $(v_1, \ldots, v_n)$ is not an ordered basis by theorem 3.4.3(1), so it is not linearly independent, since an independent list of n vectors in $\mathbb{R}^n$ is a basis (corollary 2.2.13). By proposition 2.1.10 one of the vectors lies in the span of the others, so the whole region lies in the span of the remaining $n-1$ vectors, which sits inside a line when $n = 2$ and inside a plane when $n = 3$. A region inside a line has no area and a region inside a plane has no volume, so both sides are 0.

Suppose from now on that $\det V \neq 0$, and perform on the entries of the list the three operations of definition 1.2.2, as in corollary 3.4.4. Each of them multiplies the area or volume of the region and the number $|\det V|$ by one and the same factor. Exchanging two entries leaves both unchanged: the region is the set of combinations $s_1v_1 + \cdots + s_nv_n$ with each s_j between 0 and 1, which does not depend on the order of the list, and the determinant only changes sign (corollary 3.2.12(3)). Replacing v_i by cv_i with $c \neq 0$ multiplies both by $|c|$. For the determinant this is corollary 3.2.12(3) again. For the region and $c > 0$ it is the observation of section 3.1 that stretching one edge by a factor stretches the region by that factor, and for $c < 0$ the identity

$$s_icv_i = (1 - s_i)|c|v_i + cv_i$$

exhibits the region belonging to cv_i as the region belonging to $|c|v_i$ translated by cv_i, and a translate has the same area or volume by box 3.1.3(1); the change of length along that one edge scales it by $|c|$ by box 3.1.3(3).

The third operation replaces v_i by $v_i + tv_j$ with $j \neq i$, and it changes neither quantity. The determinant is unchanged by corollary 3.2.12(3). For the region, take as base the one spanned by the entries other than v_i, which the operation does not touch. The vector tv_j lies in the line or the plane of that base, so $v_i + tv_j$ stands at the same height above the base as v_i does, and each slice of the region parallel to the base is moved by a vector lying in the base's own line or plane. Every slice is therefore translated and none is reshaped, so the area or volume is unchanged by box 3.1.3(2). That is the shear of section 3.1, performed there with the base in the xy-plane, and the argument uses nothing about where the base lies.

It remains to reach a list whose region is known. Since $\det V \neq 0$ the matrix V is invertible (corollary 3.2.8), and so is $V^\top$, whose determinant is the same number (theorem 3.2.11). The reduced echelon form of $V^\top$ is therefore I_n (theorem 1.4.8, condition (3)), and theorem 1.2.8 gives row operations carrying $V^\top$ to it. A column operation on V is the matching row operation on $V^\top$, as in the proof of corollary 3.4.4, so the column operations matching that run carry V to $I_n^\top = I_n$, that is, carry the list $(v_1, \ldots, v_n)$ to the standard basis $\mathcal{E}$. The region spanned by $\mathcal{E}$ is the unit square when $n = 2$ and the unit cube when $n = 3$, of area or volume 1, and $|\det I_n| = 1$. Each operation performed along the way multiplied the area or volume of the region and the absolute value of the determinant by the same nonzero factor, and the two numbers agree at the end of the run, so they agreed at its start. Hence the region spanned by $v_1, \ldots, v_n$ has area or volume $|\det V|$, which is what the first paragraph left to prove, completing the proof.

Part (3) is stated for $n = 2$ and $n = 3$ because those are the dimensions in which an area and a volume were computed independently of any determinant. For larger n nothing here has constructed a notion of n-dimensional volume, and $\det V$ is what plays that role; part (1) is then the whole content of the phrase "A scales volume by $|\det A|$". Part (3) also puts a geometric reading on theorem 3.2.10: applying A and then B scales area by $|\det A|$ and then by $|\det B|$, and scaling factors multiply.

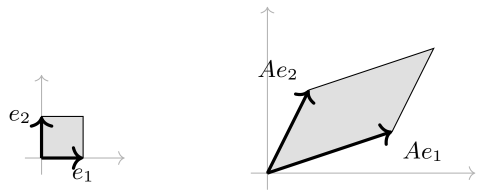


Figure 3.5: The unit square and its image under $A = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, of area 1 and area $5 = |\det A|$

Part (2) says that an invertible matrix treats every ordered basis the same way, so the two possibilities it describes are properties of the matrix.

Definition 3.4.6: Orientation-Preserving and Orientation-Reversing Matrices, and the Determinant of a Linear Map

Let $A \in M_n(\mathbb{R})$ be invertible. Then A is *orientation-preserving* when $\det A > 0$, and *orientation-reversing* when $\det A < 0$.
If $T\colon \mathbb{R}^n \to \mathbb{R}^n$ is linear, the number $\det[T]_{\mathcal{B}}$ is the same for every ordered basis $\mathcal{B}$ of $\mathbb{R}^n$, since the matrices of T in two bases are similar (theorem 2.5.4) and similar matrices have equal determinants (corollary 3.2.13). It is written $\det T$, and when $\det T \neq 0$ the map T is called orientation-preserving or orientation-reversing according to the sign of $\det T$.

The two words are justified by theorem 3.4.5(2): an orientation-preserving matrix carries every positively oriented basis to a positively oriented basis and every negatively oriented one to a negatively oriented one, while an orientation-reversing matrix exchanges the two classes. A matrix with $\det A = 0$ is neither: by theorem 3.4.5(1) the list $A\mathcal{B}$ then has $\det[A\mathcal{B}] = 0$ for every ordered basis $\mathcal{B}$, so it is never a basis and there is nothing to compare.

Example 3.4.7: Reflections of $\mathbb{R}^2$ and $\mathbb{R}^3$, and the Matrix $-I_n$

1. In $\mathbb{R}^2$, let
$$F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
which exchanges the two coordinates of a column. Its determinant is $0 \cdot 0 - 1 \cdot 1 = -1$, so F is orientation-reversing, and $|\det F| = 1$, so it leaves every area unchanged. Both facts are visible on the standard basis: $Fe_1 = e_2$ and $Fe_2 = e_1$, so $F\mathcal{E} = (e_2, e_1)$, which is the standard basis with its two entries exchanged and is therefore negatively oriented by corollary 3.4.4(1).

2. In $\mathbb{R}^3$, let $G = \operatorname{diag}(1, 1, -1)$, which reverses the sign of the third coordinate. It is triangular, so its determinant is the product of its diagonal entries (proposition 3.2.6), namely -1; the matrix is orientation-reversing and volume-preserving. On the standard basis it produces $(e_1, e_2, -e_3)$, the left-handed triple of fig. 3.4.

   Theorem 3.4.5(2) predicts the same answer for every other basis, and a second basis confirms it. Take
   $$\mathcal{B} = \left(\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right), \qquad [\mathcal{B}] = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$
   Expanding along the first row by theorem 3.3.4,
   $$\det[\mathcal{B}] = 1\,(1 \cdot 1 - 0 \cdot 1) - 0\,(1 \cdot 1 - 0 \cdot 0) + 1\,(1 \cdot 1 - 1 \cdot 0) = 1 + 1 = 2$$
   so $\mathcal{B}$ is positively oriented. Its image under G has matrix
   $$[G\mathcal{B}] = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & -1 & -1 \end{pmatrix}$$
   and expanding along the first row again gives
   $$1\,((1)(-1) - (0)(-1)) - 0\,((1)(-1) - (0)(0)) + 1\,((1)(-1) - (1)(0)) = -1 - 1 = -2$$
   which is $(\det G)(\det[\mathcal{B}]) = (-1)(2)$, and is negative, so $G\mathcal{B}$ is negatively oriented.

3. The matrix $-I_n$ sends every column x to $-x$. It is diagonal with every diagonal entry equal to -1, so $\det(-I_n) = (-1)^n$ by proposition 3.2.6. Hence $-I_n$ is orientation-preserving exactly when n is even. In $\mathbb{R}^2$ the map $x \mapsto -x$ carries (e_1, e_2) to $(-e_1, -e_2)$, whose matrix has determinant $(-1)(-1) = 1$; in $\mathbb{R}^3$ it carries the right-handed triple (e_1, e_2, e_3) to $(-e_1, -e_2, -e_3)$, of determinant -1. Reversing every coordinate at once is not the same operation in even and in odd dimensions.

Exercise 3.4.1

1. Decide whether each of the following ordered bases of $\mathbb{R}^3$ is positively or negatively oriented, computing the relevant determinant in full:
   $$\mathcal{B}_1 = \left(\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \right), \qquad \mathcal{B}_2 = \left(\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 5 \\ 6 \\ 0 \end{pmatrix} \right), \qquad \mathcal{B}_3 = (e_3, e_2, e_1)$$

2. Let $\mathcal{B} = (b_1, b_2)$ and $\mathcal{C} = (c_1, c_2)$ be the ordered bases of $\mathbb{R}^2$ with
   $$b_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad b_2 = \begin{pmatrix} 3 \\ 5 \end{pmatrix}, \quad c_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad c_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$
   Decide whether they have the same orientation, first by theorem 3.4.3(2) and then by computing $P_{\mathcal{C}\mathcal{B}}$ and its determinant directly. Check that the two answers agree.

3. Let
$$A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
Compute $\det A$, $\det B$ and $\det(AB)$, decide for each of A, B and AB whether it is orientation-preserving, and state by what factor each scales the area of a parallelogram. Then compute the image of the ordered basis $\mathcal{E}$ of $\mathbb{R}^2$ under AB and check the orientation directly.

4. Let $\mathcal{B} = (b_1, \dots, b_n)$ be an ordered basis of $\mathbb{R}^n$ and let $\mathcal{B}^{\text{rev}} = (b_n, b_{n-1}, \dots, b_1)$ be the same list reversed. Show that $\mathcal{B}^{\text{rev}}$ is an ordered basis and that it has the same orientation as $\mathcal{B}$ exactly when n leaves remainder 0 or 1 on division by 4.

5. Let $A \in M_n(\mathbb{R})$. Show that $\det(-A) = (-1)^n \det A$. Deduce that if A is invertible, then A and $-A$ are of the same type (definition 3.4.6) when n is even and of opposite types when n is odd, and illustrate both cases with a 2×2 and a 3×3 example.

6. Let $A, B \in M_n(\mathbb{R})$ be invertible. Show that the product of two orientation-preserving matrices is orientation-preserving, that the inverse of an orientation-preserving matrix is orientation-preserving, and that the product of two orientation-reversing matrices is orientation-preserving. Deduce that the orientation-reversing matrices are not closed under multiplication.

7. Suppose $A \in M_n(\mathbb{R})$ satisfies $A^2 = -I_n$. Show that n must be even and that $\det A = 1$ or $\det A = -1$. Exhibit such a matrix for $n = 2$ and determine which of A, $-A$ and A^{-1} are orientation-preserving for the matrix you chose.

3.5 Block Matrices, the Schur Complement, the Vandermonde Determinant, and Rank-One Updates

Cofactor expansion computes any determinant, and elimination computes it faster, but neither pays attention to how the matrix was assembled. Three kinds of structure make a determinant explicit: a matrix built out of smaller rectangular pieces, a matrix whose rows are the successive powers of n numbers, and a matrix obtained from one already understood by adding a single product $uv^\top$. The first of the three gives the tool that settles the last.

Here is a matrix with a corner of zeros:
$$M = \begin{pmatrix} 2 & 1 & 5 & 3 \\ 1 & 3 & 0 & 7 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 2 & 3 \end{pmatrix}$$
Only the first two entries of its first column are nonzero, so expanding along that column by theorem 3.3.4 gives
$$\det M = 2 \det \begin{pmatrix} 3 & 0 & 7 \\ 0 & 4 & 1 \\ 0 & 2 & 3 \end{pmatrix} - 1 \cdot \det \begin{pmatrix} 1 & 5 & 3 \\ 0 & 4 & 1 \\ 0 & 2 & 3 \end{pmatrix}$$
Each of the two 3×3 determinants has a first column with a single nonzero entry, and expanding each along that column gives
$$\det M = 2 \cdot 3 \cdot \det \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} - 1 \cdot 1 \cdot \det \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix} = (2 \cdot 3 - 1 \cdot 1) \cdot 10 = 5 \cdot 10 = 50$$

Determinants, Volume, and Orientation

The factor 10 is the determinant of the bottom right 2×2 corner of M, and the factor $5 = 2 \cdot 3 - 1 \cdot 1$ is the determinant of the top left 2×2 corner. The two corners never interacted: the zeros stopped the expansion from mixing them.

To state what happened, cut a square matrix into rectangular pieces. Fix $p, q \geq 1$ and let $M \in M_{p+q}(k)$. Breaking the rows after row p and the columns after column p produces four pieces, called *blocks*: the matrix $A \in M_p(k)$ in the first p rows and first p columns, the matrix $B \in M_{p \times q}(k)$ in the first p rows and last q columns, the matrix $C \in M_{q \times p}(k)$ in the last q rows and first p columns, and the matrix $D \in M_q(k)$ in the last q rows and last q columns. This is recorded as

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

where each letter now stands for a whole rectangular array rather than a single entry. In the matrix above, $p = q = 2$, and C is the zero matrix.

Proposition 3.5.1: Arithmetic of Block Matrices

Let M and N be square of size $p + q$, partitioned after row and column p into

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad N = \begin{pmatrix} A' & B' \\ C' & D' \end{pmatrix}$$

with $A, A' \in M_p(k)$ and $D, D' \in M_q(k)$, the off-diagonal blocks having the sizes those two force. Then the product and the transpose are computed blockwise:

$$MN = \begin{pmatrix} AA' + BC' & AB' + BD' \\ CA' + DC' & CB' + DD' \end{pmatrix}, \qquad M^\top = \begin{pmatrix} A^\top & C^\top \\ B^\top & D^\top \end{pmatrix}$$

The same formulas hold for a rectangular partition whenever the block sizes make each displayed product defined.

***Proof*:**
Index the rows and columns of M and N by $1, \ldots, p + q$ and write $I = \{1, \ldots, p\}$ and $J = \{p+1, \ldots, p+q\}$, so that the four blocks of M are its entries in the four index pairs $I \times I$, $I \times J$, $J \times I$ and $J \times J$. By definition 1.4.1 the entry of MN in row i and column j is $\sum_{s=1}^{p+q} m_{is} n_{sj}$, and splitting the range of s at p writes that sum as

$$\sum_{s \in I} m_{is} n_{sj} + \sum_{s \in J} m_{is} n_{sj}$$

Take $i \in I$ and $j \in I$. In the first sum both indices i and s lie in I, so m_{is} ranges over the entries of A, and $s, j \in I$ makes n_{sj} range over A'; that sum is therefore the corresponding entry of AA'. In the second, $i \in I$ and $s \in J$ makes m_{is} an entry of B, and $s \in J$, $j \in I$ makes n_{sj} an entry of C', giving the entry of BC'. So the $I \times I$ block of MN is $AA' + BC'$. The other three blocks are the same computation with i or j moved to J.

For the transpose, definition 2.3.4 puts the entry of $M^\top$ in row j and column i equal to m_{ij}. A pair $(j, i) \in I \times J$ therefore reads off an entry with $(i, j) \in J \times I$, that is an entry of C; so the $I \times J$ block of $M^\top$ is $C^\top$. The other three blocks follow in the same way.

Theorem 3.5.2: The Determinant of a Block Triangular Matrix

Let $p, q \geq 1$, let $A \in M_p(k)$ and let $D \in M_q(k)$.

1. For every $B \in M_{p\times q}(k)$,
$$\det \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det(A)\det(D)$$
where the zero block has size $q \times p$.
2. For every $C \in M_{q\times p}(k)$,
$$\det \begin{pmatrix} A & 0 \\ C & D \end{pmatrix} = \det(A)\det(D)$$
where the zero block has size $p \times q$.

***Proof*:**

1. Fix q and D, and induct on p. Write M for the block matrix and a_{ij} for the entries of A. Suppose first that $p = 1$. Then the first column of M is $(a_{11}, 0, \dots, 0)^\top$, and deleting its first row and first column leaves exactly D. Expanding along the first column by theorem 3.3.4 gives $\det M = a_{11} \det D$, and $a_{11} = \det A$ because A is the 1×1 matrix (a_{11}).
Now let $p \geq 2$ and assume the statement for $p-1$. The first column of M is $(a_{11}, \dots, a_{p1}, 0, \dots, 0)^\top$, so expanding along it by theorem 3.3.4 leaves at most p nonzero terms,
$$\det M = \sum_{i=1}^{p} (-1)^{i+1} a_{i1} \det M_i$$
where M_i is obtained from M by deleting row i and column 1. Deleting a row with index $i \leq p$ and the first column removes a row and a column from the top left block and a row from the top right block; the bottom left block loses a column but stays zero, and D is untouched. Hence
$$M_i = \begin{pmatrix} A_i & B_i \\ 0 & D \end{pmatrix}$$
where $A_i \in M_{p-1}(k)$ is A with row i and column 1 deleted and $B_i \in M_{(p-1)\times q}(k)$ is B with row i deleted. The inductive hypothesis gives $\det M_i = \det(A_i)\det(D)$, so
$$\det M = \left(\sum_{i=1}^{p} (-1)^{i+1} a_{i1} \det A_i \right) \det D$$
The bracketed sum is the expansion of $\det A$ along the first column of A, again by theorem 3.3.4, so $\det M = \det(A)\det(D)$.
2. Transposing interchanges the two shapes: by definition 2.3.4,
$$\begin{pmatrix} A & 0 \\ C & D \end{pmatrix}^\top = \begin{pmatrix} A^\top & C^\top \\ 0 & D^\top \end{pmatrix}$$

Applying theorem 3.2.11 twice and part (1) once,

$$\det\begin{pmatrix} A & 0 \\ C & D \end{pmatrix} = \det\begin{pmatrix} A^\top & C^\top \\ 0 & D^\top \end{pmatrix} = \det(A^\top)\det(D^\top) = \det(A)\det(D)$$

as we sought to show.

The hypothesis is on the shape of one block only. Nothing is asked of B in part (1) or of C in part (2), and nothing is asked of A or D beyond being square: a single zero corner is enough to split a $(p+q)\times(p+q)$ determinant into a product of two smaller ones. Taking $p = q = 1$ recovers the familiar statement that a 2×2 matrix with a zero in one off-diagonal position has determinant the product of its diagonal entries. Applying the theorem $r-1$ times in succession handles a matrix whose nonzero entries all sit in square diagonal blocks $A_1, \ldots, A_r$: its determinant is $\det(A_1)\cdots\det(A_r)$.

What is not true is the guess suggested by the 2×2 formula $ad - bc$, namely that a general block matrix should have determinant $\det(A)\det(D) - \det(B)\det(C)$. Take $p = q = 2$ with $A = D = 0$ and $B = C = I_2$, so that

$$M = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

This matrix is obtained from I_4 by the two swaps $R_1 \leftrightarrow R_3$ and $R_2 \leftrightarrow R_4$, so $\det M = (-1)^2 = 1$ by lemma 3.2.2, while $\det(A)\det(D) - \det(B)\det(C) = 0 - 1 = -1$. A correct formula in the four blocks needs a hypothesis, and invertibility of one block is enough.

Elimination is what gives it. Consider

$$M = \begin{pmatrix} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 3 & 1 \\ 0 & 1 & 1 & 3 \end{pmatrix}$$

and clear the first two columns below the diagonal, as in section 1.2. The operations $R_2 \mapsto R_2 - \frac{1}{2}R_1$ and $R_3 \mapsto R_3 - \frac{1}{2}R_1$ give

$$\begin{pmatrix} 2 & 1 & 1 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & 1 \\ 0 & \frac{1}{2} & \frac{5}{2} & 1 \\ 0 & 1 & 1 & 3 \end{pmatrix}$$

and then $R_3 \mapsto R_3 - R_2$ and $R_4 \mapsto R_4 - 2R_2$ give

$$\begin{pmatrix} 2 & 1 & 1 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 2 & 1 \end{pmatrix}$$

Every operation used was a replacement, so the determinant did not change by lemma 3.2.2, and the result is block triangular with corners of determinant $2\cdot\frac{1}{2} = 1$ and $3\cdot 1 - 0\cdot 2 = 3$. Hence $\det M = 3$ by theorem 3.5.2.

The block left in the bottom right corner, $\begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}$, can be written down without running the elimination. The four blocks of M are

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad D = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

and $\det A = 1$, so A is invertible by corollary 3.2.8, with $A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ as multiplying out confirms. Then

$$CA^{-1}B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}, \qquad D - CA^{-1}B = \begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}$$

which is the block the elimination produced. The combination $D - CA^{-1}B$ deserves a name.

Definition 3.5.3: The Schur Complement

Let $p, q \geq 1$ and let $M \in M_{p+q}(k)$ have block form

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \qquad A \in M_p(k),\ B \in M_{p\times q}(k),\ C \in M_{q\times p}(k),\ D \in M_q(k)$$

If A is invertible, the *Schur complement of A in M* is the $q \times q$ matrix

$$M/A = D - CA^{-1}B$$

If D is invertible, the *Schur complement of D in M* is the $p \times p$ matrix

$$M/D = A - BD^{-1}C$$

For the matrix just computed, $M/A = \begin{pmatrix} 3 & 0 \\ 2 & 1 \end{pmatrix}$ and $\det M = \det(A)\det(M/A) = 1 \cdot 3$. That identity is general.

Theorem 3.5.4: The Determinant of a Block Matrix via the Schur Complement

Let $p, q \geq 1$ and let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \in M_{p+q}(k)$ be as in definition 3.5.3.

1. If A is invertible, then

$$\det M = \det(A)\det(D - CA^{-1}B)$$

2. If D is invertible, then

$$\det M = \det(D)\det(A - BD^{-1}C)$$

***Proof*:**

1. Write $E = CA^{-1} \in M_{q\times p}(k)$ and let $e_{i\ell}$ be its entries, with $1 \le i \le q$ and $1 \le \ell \le p$. Let $r_1, \ldots, r_{p+q}$ be the rows of M. For each such pair (i, ℓ) apply the row operation $R_{p+i} \mapsto R_{p+i} - e_{i\ell}R_\ell$, which is a replacement in the sense of definition 1.2.2 and therefore leaves the determinant unchanged by lemma 3.2.2. None of these pq operations alters a row of index at most p, so each of them subtracts a multiple of an original row, and after all of them row $p+i$ has become

$$r_{p+i} - \sum_{\ell=1}^{p} e_{i\ell} r_\ell$$

Read this off in the two groups of columns. In columns 1 through p the row r_{p+i} contributes the ith row of C and the row r_ℓ contributes the ℓth row of A, so the new entry in column m is $c_{im} - \sum_\ell e_{i\ell}a_{\ell m}$, which is the (i, m) entry of $C - EA$. Since $EA = CA^{-1}A = C$, that block is zero. In columns $p+1$ through $p+q$ the same reading contributes the ith row of D and the ℓth row of B, so the new entry is the (i, m) entry of $D - EB = D - CA^{-1}B$.

The operations have therefore produced

$$\begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}$$

whose determinant is $\det(A)\det(D - CA^{-1}B)$ by theorem 3.5.2. No operation changed the determinant, so this is $\det M$.

2. Run the same argument upward. Write $F = BD^{-1} \in M_{p\times q}(k)$ with entries $f_{i\ell}$, and for $1 \le i \le p$ and $1 \le \ell \le q$ apply $R_i \mapsto R_i - f_{i\ell}R_{p+\ell}$. These are again replacements, none of which alters a row of index greater than p, and after all of them row i has become $r_i - \sum_\ell f_{i\ell}r_{p+\ell}$. In columns $p+1$ through $p+q$ this is the ith row of $B - FD = B - BD^{-1}D = 0$, and in columns 1 through p it is the ith row of $A - FC = A - BD^{-1}C$. The matrix produced is

$$\begin{pmatrix} A - BD^{-1}C & 0 \\ C & D \end{pmatrix}$$

and theorem 3.5.2 gives its determinant as $\det(A - BD^{-1}C)\det(D)$, completing the proof.

The Schur complement is what elimination leaves behind. Consider the system

$$M\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$$

where x collects the first p unknowns, y the last q, and $b_1 \in k^p$ and $b_2 \in k^q$ are the two halves of the right-hand side. The operations in the proof turn it into $Ax + By = b_1$ together with $(M/A)y = b_2 - CA^{-1}b_1$: the first p unknowns have been eliminated, and M/A is the matrix of the system that remains in y. The multiplier CA^{-1} plays the part that a single number c/a played in section 1.2, and the factorization $\det(A)\det(M/A)$ is the block version of reading a determinant off the pivots. With $p = q = 1$ the formula is $\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a(d - ca^{-1}b) = ad - bc$, valid whenever $a \neq 0$.

If A is invertible, then $\det M = \det(A)\det(M/A)$ is nonzero exactly when $\det(M/A)$ is nonzero, so by corollary 3.2.8 the matrix M is invertible if and only if its Schur complement M/A is. A question about a $(p+q)\times(p+q)$ matrix has become a question about a $q \times q$ one.

The second family of structured determinants comes from a question about polynomials. A polynomial $f(t) = c_0 + c_1 t + \cdots + c_{n-1}t^{n-1}$ is determined by its n coefficients, and prescribing its values at n points $x_1, \ldots, x_n$ imposes n linear equations on those coefficients, namely $c_0 + c_1 x_i + \cdots + c_{n-1}x_i^{n-1} = y_i$ for each i. The coefficient matrix of that system has (i,j) entry x_i^{j-1}, and whether the interpolation problem has exactly one solution is a question about its determinant.

Example 3.5.5: The Vandermonde Determinant on Three Nodes

Take $n = 3$ and the three numbers $x_1 = 1$, $x_2 = 2$, $x_3 = 3$. The matrix whose (i,j) entry is x_i^{j-1} is

$$V = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix}$$

Expanding along the first row by theorem 3.3.4,

$$\det V = \det\begin{pmatrix} 2 & 4 \\ 3 & 9 \end{pmatrix} - \det\begin{pmatrix} 1 & 4 \\ 1 & 9 \end{pmatrix} + \det\begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix} = (18-12) - (9-4) + (3-2) = 2$$

The value 2 is the product $(x_2 - x_1)(x_3 - x_1)(x_3 - x_2) = 1 \cdot 2 \cdot 1$, one factor for each pair of nodes, with the later node minus the earlier one.

The same number comes out of a sequence of row operations that will reappear in the general proof. Transposing changes nothing by theorem 3.2.11, and the transpose

$$W = V^\top = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}$$

has the powers running down the columns: row i of W is $(x_1^{i-1}, x_2^{i-1}, x_3^{i-1})$. Subtracting $x_1 = 1$ times the row above from each row, starting at the bottom, means $R_3 \mapsto R_3 - R_2$ and then $R_2 \mapsto R_2 - R_1$, and produces

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 2 & 6 \end{pmatrix}$$

Both are replacements, so the determinant is unchanged by lemma 3.2.2. Expanding along the first column leaves

$$\det W = \det\begin{pmatrix} 1 & 2 \\ 2 & 6 \end{pmatrix}$$

Its first column has the common factor $x_2 - x_1 = 1$ and its second column the common factor $x_3 - x_1 = 2$; pulling both out, which corollary 3.2.12 licenses because the determinant is linear in each column separately, gives

$$\det W = 1 \cdot 2 \cdot \det\begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} = 1 \cdot 2 \cdot (3-2) = 2$$

The last determinant is the two-node case built from x_2 and x_3, and its value $x_3 - x_2 = 1$ is the one factor that pair contributes.

Theorem 3.5.6: The Vandermonde Determinant

Let $n \geq 1$ and let $x_1, \ldots, x_n \in k$. Let $V(x_1, \ldots, x_n) \in M_n(k)$ be the *Vandermonde matrix* of the *nodes* $x_1, \ldots, x_n$, whose (i, j) entry is x_i^{j-1}, with the convention $x^0 = 1$ for every x:

$$V(x_1, \ldots, x_n) = \begin{pmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ 1 & x_2 & x_2^2 & \cdots & x_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^{n-1} \end{pmatrix}$$

Then

$$\det V(x_1, \ldots, x_n) = \prod_{1 \leq i < j \leq n} (x_j - x_i)$$

where a product over an empty set of pairs is read as 1.

***Proof*:**
Induct on n. For $n = 1$ the matrix is the 1×1 matrix (1), whose determinant is 1 by definition 3.2.1, and there are no pairs $i < j$, so the product is empty and also equals 1.

Let $n \geq 2$ and assume the formula for any $n-1$ nodes. Write $V = V(x_1, \ldots, x_n)$ and $W = V^\top$, so that the (i, j) entry of W is x_j^{i-1} and row i of W is $(x_1^{i-1}, x_2^{i-1}, \ldots, x_n^{i-1})$. By theorem 3.2.11 it suffices to compute $\det W$. Apply, for $i = n$, then $i = n-1$, and so on down to $i = 2$, the operation $R_i \mapsto R_i - x_1 R_{i-1}$. Because the indices decrease, each of these uses the row above it in its original form. Each is a replacement in the sense of definition 1.2.2, so by lemma 3.2.2 none of them changes the determinant.

Row 1 is untouched and remains $(1, 1, \ldots, 1)$. For $i \geq 2$ the new jth entry of row i is

$$x_j^{i-1} - x_1 x_j^{i-2} = x_j^{i-2}(x_j - x_1)$$

In particular the entries in column 1 vanish for $i \geq 2$, since $x_1 - x_1 = 0$, so the first column of the new matrix is $(1, 0, \ldots, 0)^\top$. Expanding along that column by theorem 3.3.4 leaves a single term with sign $(-1)^{1+1} = 1$, so the determinant equals $\det N$, where $N \in M_{n-1}(k)$ is what remains after deleting row 1 and column 1: its (a, b) entry is $x_{b+1}^{a-1}(x_{b+1} - x_1)$ for $1 \leq a, b \leq n-1$.

Column b of N is the scalar $x_{b+1} - x_1$ times the column with entries x_{b+1}^{a-1}. The determinant is linear in each column separately by corollary 3.2.12, so each of these scalars may be taken outside one column at a time, and

$$\det N = \left(\prod_{b=1}^{n-1} (x_{b+1} - x_1) \right) \det \left(V(x_2, \ldots, x_n)^\top \right)$$

because the matrix left behind has (a, b) entry x_{b+1}^{a-1}, which is the transpose of the Vandermonde matrix of the $n-1$ nodes $x_2, \ldots, x_n$. Using theorem 3.2.11 once more and then the inductive hypothesis,

$$\det V = \prod_{j=2}^{n} (x_j - x_1) \cdot \prod_{2 \leq i < j \leq n} (x_j - x_i)$$

The first product runs over exactly the pairs $i < j$ with $i = 1$ and the second over exactly the pairs with $i \geq 2$, so together they run over every pair once, giving $\det V = \prod_{1 \leq i < j \leq n} (x_j - x_i)$, as we sought to show.

The formula answers the interpolation question completely. A product of numbers is nonzero exactly when every factor is, so $\det V(x_1, \ldots, x_n) \neq 0$ if and only if $x_j \neq x_i$ for all $i < j$, that is, exactly when the n nodes are distinct. By corollary 3.2.8 the matrix V is then invertible, and the system for the coefficients $c_0, \ldots, c_{n-1}$ has exactly one solution for every list of prescribed values: through any n points with distinct first coordinates there passes exactly one polynomial of degree at most $n - 1$. If two nodes coincide the matrix has two equal rows and its determinant vanishes, as it must: no function takes two different values at a single point, so no matrix can solve the interpolation problem for every list of prescribed values.

The third structure is a change so small that a determinant already computed should not have to be computed again. Let $u, v \in k^n$ be columns. The product $uv^\top$ is an $n \times n$ matrix, its (i, j) entry being $u_i v_j$, and every one of its columns is a multiple of u, the jth being $v_j u$. The column space of $uv^\top$ therefore sits inside $\operatorname{span}\{u\}$, which has dimension at most 1, so it too has dimension at most 1 by corollary 2.2.12. Since theorem 2.3.9 makes the rank of a matrix equal to the dimension of its column space, the rank of $uv^\top$ is at most 1, and it equals 1 exactly when u and v are both nonzero, since otherwise the product is the zero matrix. Passing from A to $A + uv^\top$ is therefore called a *rank-one update* of A. Changing the single entry in position (i, j) of A by an amount t is the case $u = te_i$ and $v = e_j$, and adding t times the vector u to a single column of A is the case $v = te_j$.

Take A to be the unit upper triangular matrix on the left below, whose determinant is 1 by proposition 3.2.6, and change its $(3, 1)$ entry from 0 to 1, which is the update by $u = e_3$ and $v = e_1$:

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad A + e_3 e_1^\top = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$$

Expanding the second matrix along its first column gives $1 \cdot (1 - 0) + 1 \cdot (1 - 0) = 2$. The new determinant is twice the old one, and the factor 2 has a closed form.

Theorem 3.5.7: The Determinant of a Rank-One Update

Let $A \in M_n(k)$ be invertible and let $u, v \in k^n$. Then

$$\det(A + uv^\top) = (1 + v^\top A^{-1} u) \det(A)$$

where $v^\top A^{-1} u$ is a 1×1 matrix, read as a scalar.

***Proof*:**
Build the $(n + 1) \times (n + 1)$ matrix

$$M = \begin{pmatrix} A & -u \\ v^\top & 1 \end{pmatrix}$$

in the block form of definition 3.5.3 with $p = n$ and $q = 1$: the top left block is A, the top right block is the column $-u$, the bottom left block is the row $v^\top$, and the bottom right block is the 1×1 matrix (1). Both diagonal blocks are invertible, A by hypothesis and (1) because it is I_1, so both parts of theorem 3.5.4 apply to M.

Part (1) gives

$$\det M = \det(A)\det\left(1 - v^\top A^{-1}(-u)\right) = \det(A)\left(1 + v^\top A^{-1}u\right)$$

the last step because the determinant of a 1×1 matrix is its single entry, which is what definition 3.2.1 gives for $n = 1$. Part (2) gives

$$\det M = \det(I_1)\det\left(A - (-u)I_1 v^\top\right) = 1 \cdot \det(A + uv^\top)$$

Both expressions equal $\det M$, so they equal each other, completing the proof.

Once A^{-1} is known, the factor $1 + v^\top A^{-1}u$ costs one product $A^{-1}u$ and one further product with the row $v^\top$, which is less work than a fresh determinant. In the example above, A^{-1} is the matrix with rows $(1,-1,1)$, $(0,1,-1)$ and $(0,0,1)$, so $v^\top A^{-1}u = e_1^\top A^{-1}e_3$ is the $(1,3)$ entry of A^{-1}, which is 1; the factor is $1 + 1 = 2$, agreeing with the direct expansion. The same identity says when the update destroys invertibility: since $\det A \neq 0$, the matrix $A + uv^\top$ fails to be invertible exactly when $v^\top A^{-1}u = -1$.

Away from that one value, the inverse of the updated matrix is also a rank-one update, of A^{-1}.

Theorem 3.5.8: The Sherman-Morrison Formula

Let $A \in M_n(k)$ be invertible and let $u, v \in k^n$ satisfy $1 + v^\top A^{-1}u \neq 0$. Then $A + uv^\top$ is invertible and

$$(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1}uv^\top A^{-1}}{1 + v^\top A^{-1}u}$$

***Proof*:**
Write $\alpha = 1 + v^\top A^{-1}u$, a nonzero scalar by hypothesis. By theorem 3.5.7, $\det(A + uv^\top) = \alpha\det(A) \neq 0$, so $A + uv^\top$ is invertible by corollary 3.2.8. Let Y denote the matrix on the right of the claimed identity and compute the product $(A + uv^\top)Y$, expanding it into four terms:

$$(A + uv^\top)Y = AA^{-1} - \frac{AA^{-1}uv^\top A^{-1}}{\alpha} + uv^\top A^{-1} - \frac{uv^\top A^{-1}uv^\top A^{-1}}{\alpha}$$

The first term is I_n and the second is $\frac{1}{\alpha}uv^\top A^{-1}$. In the fourth term the middle factor $v^\top A^{-1}u$ is a scalar, equal to $\alpha - 1$, so that term is $\frac{\alpha-1}{\alpha}uv^\top A^{-1}$. Collecting the three terms that carry the factor $uv^\top A^{-1}$,

$$(A + uv^\top)Y = I_n + \left(-\frac{1}{\alpha} + 1 - \frac{\alpha - 1}{\alpha}\right)uv^\top A^{-1} = I_n + \frac{-1 + \alpha - \alpha + 1}{\alpha}uv^\top A^{-1} = I_n$$

Multiplying $(A + uv^\top)Y = I_n$ on the left by the inverse of $A + uv^\top$, which exists by the first line of the proof, gives $Y = (A + uv^\top)^{-1}$, as we sought to show.

Example 3.5.9: A Rank-One Update of a Triangular Matrix

Let

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad u = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \qquad v = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

The matrix A is upper triangular with all diagonal entries 1, so $\det A = 1$ by proposition 3.2.6, and its inverse is

$$A^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

as the product $AA^{-1} = I_3$ confirms. The update itself is

$$uv^\top = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}, \qquad A + uv^\top = \begin{pmatrix} 2 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

The two vectors needed by theorem 3.5.7 and theorem 3.5.8 are

$$A^{-1}u = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \qquad v^\top A^{-1} = \begin{pmatrix} 1 & 0 & 0 \end{pmatrix}$$

the first being the sum of columns 1 and 3 of A^{-1} and the second the sum of rows 1 and 2. Hence $v^\top A^{-1}u = (1, 0, 0)u = 1$ and $\alpha = 1 + 1 = 2$, so

$$\det(A + uv^\top) = 2 \cdot 1 = 2$$

Expanding the updated matrix along its first column checks this: the terms are $2(1 \cdot 1 - 1 \cdot 1) = 0$ from row 1, nothing from row 2, and $+1 \cdot (2 \cdot 1 - 0 \cdot 1) = 2$ from row 3.

For the inverse, $(A^{-1}u)(v^\top A^{-1})$ is the 3×3 matrix with first column $(2, -1, 1)^\top$ and the other two columns zero, so theorem 3.5.8 gives

$$(A + uv^\top)^{-1} = \begin{pmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 2 & 0 & 0 \\ -1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 & 1 \\ \frac{1}{2} & 1 & -1 \\ -\frac{1}{2} & 0 & 1 \end{pmatrix}$$

Multiplying out confirms it: the first row of $A + uv^\top$ against the three columns on the right gives $2 \cdot 0 + 2 \cdot \frac{1}{2} + 0 = 1$, then $-2 + 2 + 0 = 0$, then $2 - 2 + 0 = 0$, and the second and third rows give $(0, 1, 0)$ and $(0, 0, 1)$ in the same way.

Exercise 3.5.1

1. Compute the determinants of

$$M = \begin{pmatrix} 3 & 1 & 7 & 2 \\ 2 & 1 & 0 & 5 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 5 & 3 \end{pmatrix}, \qquad N = \begin{pmatrix} 3 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 9 & 8 & 4 & 1 \\ 7 & 6 & 5 & 3 \end{pmatrix}$$

naming the result used at each step.

2. Let
$$M = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 1 & 3 & 2 & 0 \\ 2 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \end{pmatrix}$$
Compute the Schur complement M/A of its top left 2×2 block and use it to compute $\det M$. Then eliminate the first two columns of M below the diagonal by row operations and compare the bottom right 2×2 block of the result with M/A.

3. Compute $\det V(-1, 0, 2, 3)$. Then find every real number t for which the nodes $-1, 0, 2, t$ give a Vandermonde matrix that is not invertible, by writing $\det V(-1, 0, 2, t)$ as a polynomial in t.

4. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, $u = (1, 1)^\top$ and $v = (0, 1)^\top$. Compute $\det(A + uv^\top)$ by theorem 3.5.7 and check the answer directly. Then compute $(A + uv^\top)^{-1}$ by theorem 3.5.8 and check that answer by multiplication.

5. Let $u, v \in k^n$. Show that $\det(I_n + uv^\top) = 1 + v^\top u$, and deduce that $I_n + uv^\top$ is invertible if and only if $v^\top u \neq -1$. Verify the formula by direct expansion when $n = 2$, $u = (1, 2)^\top$ and $v = (3, -1)^\top$.

6. Find the unique polynomial $f(t) = c_0 + c_1 t + c_2 t^2$ with $f(1) = 3$, $f(2) = 1$ and $f(3) = 5$, and say which result guarantees in advance that there is exactly one such polynomial.

7. Let $A \in M_n(k)$ be invertible, let $r \geq 1$, let $U, V \in M_{n\times r}(k)$, and suppose the $r \times r$ matrix $S = I_r + V^\top A^{-1} U$ is invertible. Show that
$$(A + UV^\top)^{-1} = A^{-1} - A^{-1} U S^{-1} V^\top A^{-1}$$
by verifying that the product of $A + UV^\top$ with the right-hand side is I_n. Which case of this is theorem 3.5.8?

4

Eigenvalues and Diagonalization

Every matrix so far has been studied through what it does to all of k^n at once. This chapter asks a narrower question: which vectors does a matrix *stretch*, leaving their direction unchanged? The vectors that answer it, when there are enough of them, produce a basis in which the matrix becomes diagonal.

There are not always enough of them, and the two ways they can run short are what the chapter is really about. Over $\mathbb{R}$ a matrix may have no such vector at all, as a quarter turn of the plane does not. Over $\mathbb{C}$ that failure cannot happen, but a second one can: the eigenvectors may exist and still be too few to span. Section 4.1 produces the polynomial whose roots are the stretching factors and settles the first failure. Section 4.2 shows what having enough eigenvectors gives and exhibits a matrix that does not. Section 4.3 measures the failure of diagonalizability by two counts attached to each eigenvalue. Section 4.4 finds the polynomial that detects that failure exactly, and proves along the way that a matrix satisfies its own characteristic equation. Section 4.5 applies the diagonal form to powers and on recurrences.

This chapter closes Part I. The obstruction it isolates is answered three separate times in what follows: by imposing a geometry that forces enough eigenvectors to exist, by enlarging $\mathbb{R}$ to $\mathbb{C}$ and back, and by settling for a canonical form short of diagonal.

4.1 Eigenvalues, Eigenvectors, and the Characteristic Polynomial

A matrix sends a column to another column, and the two need not lie on one line through the origin. For some columns they do. When the image of v is a multiple of v, the matrix acts on that column exactly as a single number does, and the number can be read off. The first task is to find those columns and those numbers.

Example 4.1.1: Two Invariant Lines of a 2×2 Matrix

Let

$$M = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \in M_2(\mathbb{R})$$

Multiplying M by the standard basis column $e_1 = (1, 0)^\top$ gives, by definition 1.4.1,

$$Me_1 = \begin{pmatrix} 1 \cdot 1 + 2 \cdot 0 \\ 2 \cdot 1 + 1 \cdot 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

No scalar c satisfies $(1, 2)^\top = c(1, 0)^\top$, since the second entry of the left side is 2 and the second entry of the right side is 0. The image of e_1 is therefore off the line through e_1.

Two other columns behave differently. Put $u = (1, 1)^\top$ and $w = (1, -1)^\top$. By definition 1.4.1,

$$Mu = \begin{pmatrix} 1 \cdot 1 + 2 \cdot 1 \\ 2 \cdot 1 + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3u, \qquad Mw = \begin{pmatrix} 1 \cdot 1 + 2 \cdot (-1) \\ 2 \cdot 1 + 1 \cdot (-1) \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -w$$

So M stretches u by the factor 3 and reverses w.

Every multiple of u behaves the same way. For $c \in \mathbb{R}$, proposition 1.4.3(2) gives $M(cu) = c(Mu) = c(3u) = 3(cu)$, and the same computation gives $M(cw) = -(cw)$. The whole line $\operatorname{span}\{u\}$ is carried into itself by M, and so is the whole line $\operatorname{span}\{w\}$.

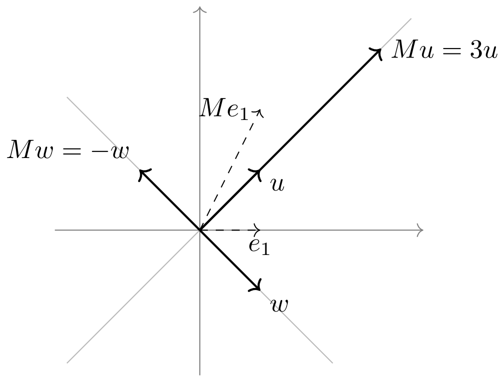


Figure 4.1: The two lines that M carries into itself, and one column that M moves off its line

The two numbers 3 and -1 and the two lines they belong to are what get names.

Definition 4.1.2: Eigenvalue

Let $A \in M_n(k)$. A scalar $\lambda \in k$ is an *eigenvalue* of A if there is a column $v \in k^n$ with

$$v \neq 0 \qquad \text{and} \qquad Av = \lambda v$$

Definition 4.1.3: Eigenvector

Let $A \in M_n(k)$ and let $\lambda \in k$. A column $v \in k^n$ is an *eigenvector of A belonging to λ* if $v \neq 0$ and $Av = \lambda v$.

So λ is an eigenvalue of A exactly when some eigenvector belongs to it. The requirement $v \neq 0$ is what makes the definition record something: the zero column satisfies $A0 = 0 = \lambda 0$ for every scalar λ at once, so admitting it would make every scalar an eigenvalue of every matrix. No such restriction is placed on λ itself. The value $\lambda = 0$ is allowed, and 0 is an eigenvalue of A exactly when $Av = 0$ for some nonzero v, which by theorem 1.4.8, conditions (1) and (2), says exactly that A is singular.

In example 4.1.1 the eigenvalues of M found so far are 3 and -1, and the columns belonging to them fill out two lines. Collecting all columns attached to one λ, including the zero column, gives a set already studied in chapter 2.

Definition 4.1.4: Eigenspace

Let $A \in M_n(k)$ and $\lambda \in k$. The *eigenspace* of A for λ is

$$E_\lambda(A) = \{\, v \in k^n \,\big|\, Av = \lambda v \,\}$$

the set of columns that A multiplies by λ.

The eigenspace is the null space of one matrix built from A and λ, and that identification is what makes it computable by elimination.

Proposition 4.1.5: The Eigenspace Is a Null Space

Let $A \in M_n(k)$ and $\lambda \in k$. Then

$$E_\lambda(A) = \operatorname{null}(A - \lambda I_n)$$

so $E_\lambda(A)$ is a subspace of k^n of dimension $n - \operatorname{rank}(A - \lambda I_n)$.

***Proof*:**
For $v \in k^n$ the identity $(\lambda I_n)v = \lambda(I_n v) = \lambda v$ holds by proposition 1.4.3(2) and (3), so proposition 1.4.3(2) applied to the sum $A + (-\lambda I_n)$ gives

$$(A - \lambda I_n)v = Av - \lambda v$$

Hence $Av = \lambda v$ holds exactly when $(A - \lambda I_n)v = 0$, which is the asserted equality of sets. A null space is a subspace (example 2.1.4), and its dimension is $n - \operatorname{rank}(A - \lambda I_n)$ by theorem 2.3.11, completing the proof.

The eigenspace contains the zero column whatever λ is, so it is never empty; the eigenvectors belonging to λ are its nonzero elements, and λ is an eigenvalue of A exactly when $E_\lambda(A) \neq \{0\}$. For the matrix M above, $E_3(M) = \operatorname{span}\{(1,1)^\top\}$ and $E_{-1}(M) = \operatorname{span}\{(1,-1)^\top\}$, both of dimension 1.

An eigenspace has a property worth naming on its own, because later chapters use it for subspaces that are not eigenspaces: A maps it into itself.

Definition 4.1.6: Invariant Subspace

Let $A \in M_n(k)$ and let W be a subspace of k^n. Then W is *invariant* under A, or A-*invariant*, if $Aw \in W$ for every $w \in W$. In that case the map $w \mapsto Aw$ from W to W is called the *restriction* of A to W, written $A|_W$.

Proposition 4.1.7: Every Eigenspace Is Invariant

Let $A \in M_n(k)$ and $\lambda \in k$. Then $E_\lambda(A)$ is invariant under A, and $A|_{E_\lambda(A)}$ is multiplication by λ.

***Proof*:**
Let $v \in E_\lambda(A)$, so $Av = \lambda v$ by definition 4.1.4. The eigenspace is a subspace by proposition 4.1.5, so it is closed under scalar multiplication and $\lambda v \in E_\lambda(A)$. Hence $Av \in E_\lambda(A)$, which is invariance, and the displayed equality $Av = \lambda v$ says exactly that the restriction is multiplication by λ.

The zero subspace and k^n itself are invariant under every A, and a one-dimensional subspace $\operatorname{span}\{v\}$ with $v \neq 0$ is invariant exactly when v is an eigenvector.

The two eigenspaces of M between them account for its whole action. Any $(x, y)^\top \in \mathbb{R}^2$ can be written as

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{x+y}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{x-y}{2}\begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

since the two coefficients add to x in the first entry and cancel to y in the second. Applying M and using proposition 1.4.3(2) on each term gives

$$M\begin{pmatrix} x \\ y \end{pmatrix} = 3\,\frac{x+y}{2}\begin{pmatrix} 1 \\ 1 \end{pmatrix} - \frac{x-y}{2}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} x+2y \\ 2x+y \end{pmatrix}$$

which is what definition 1.4.1 gives directly. Once a column is written in terms of u and w, the four entries of M are not needed: the numbers 3 and -1 do all the work. Section 4.2 asks which matrices admit such a description and which do not.

Finding the eigenvalues by guessing columns is not a procedure. The reformulation just made turns it into one. A scalar λ is an eigenvalue of A exactly when the homogeneous system $(A - \lambda I_n)x = 0$ has a nonzero solution, and by theorem 1.4.8, conditions (1) and (2), that happens exactly when $A - \lambda I_n$ is singular. Chapter 3 detects singularity by a single number: by corollary 3.2.8 the matrix $A - \lambda I_n$ is singular exactly when its determinant vanishes. The unknown λ therefore has to satisfy one scalar equation.

The order of the subtraction does not matter. Entry by entry, $\lambda I_n - A = (-1)(A - \lambda I_n)$, so proposition 1.4.3(2) gives $(\lambda I_n - A)v = -\big((A - \lambda I_n)v\big)$ for every $v \in k^n$; hence $A - \lambda I_n$ sends v to zero exactly when $\lambda I_n - A$ does, and by theorem 1.4.8, conditions (1) and (2), the two matrices are singular together. The order fixed below is $\lambda I_n - A$, for a reason that appears with proposition 4.1.8.

Run this on the matrix M of example 4.1.1, writing t for the unknown scalar. The entries are

$$tI_2 - M = \begin{pmatrix} t-1 & -2 \\ -2 & t-1 \end{pmatrix}$$

and the determinant of $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ is $ad - bc$, computed in section 3.2, so

$$\det(tI_2 - M) = (t-1)^2 - (-2)(-2) = t^2 - 2t + 1 - 4 = t^2 - 2t - 3$$

This is a quadratic in t, and $t^2 - 2t - 3 = (t-3)(t+1)$, so it vanishes exactly at $t = 3$ and $t = -1$. Those are the two eigenvalues found by inspection, and the computation shows there are no others.

The same construction works for any square matrix, and the object it produces is always a polynomial of a predictable shape. Two of its coefficients are quantities already attached to A in chapter 2 and chapter 3.

Proposition 4.1.8: The Determinant of $tI_n - A$ Is a Monic Polynomial of Degree n

Let $n \geq 1$ and let $A = (a_{ij}) \in M_n(k)$. There are unique scalars $c_0, c_1, \ldots, c_{n-1} \in k$ such that

$$\det(tI_n - A) = t^n + c_{n-1}t^{n-1} + \cdots + c_1 t + c_0$$

for every $t \in k$. Moreover

$$c_{n-1} = -\mathrm{tr}A \qquad \text{and} \qquad c_0 = (-1)^n \det A$$

***Proof*:**
Step 1: the coefficients are unique. Suppose $c_0, \ldots, c_{n-1}$ and $d_0, \ldots, d_{n-1}$ both satisfy the displayed identity for every $t \in k$. Subtracting the two identities gives

$$\sum_{l=0}^{n-1} (c_l - d_l)t^l = 0$$

for every $t \in k$, the two leading terms t^n having cancelled. Take the n scalars $s_1 = 0$, $s_2 = 1$, $\ldots$, $s_n = n-1$, which are distinct elements of k because k is $\mathbb{R}$ or $\mathbb{C}$. Let $x = (c_0 - d_0, \ldots, c_{n-1} - d_{n-1})^\top \in k^n$ and let $V = V(s_1, \ldots, s_n)$ be the Vandermonde matrix of theorem 3.5.6, whose entry in row i and column j is s_i^{j-1}. By definition 1.4.1, the ith entry of Vx is

$$\sum_{j=1}^{n} s_i^{j-1}(c_{j-1} - d_{j-1}) = \sum_{l=0}^{n-1} (c_l - d_l)s_i^l$$

which is zero by the display above. So $Vx = 0$. The nodes $s_1, \ldots, s_n$ are distinct, so every factor of the product $\prod_{i<j}(s_j - s_i)$ is nonzero and theorem 3.5.6 gives $\det V \neq 0$; hence V is invertible by corollary 3.2.8, and theorem 1.4.8, conditions (1) and (2), forces $x = 0$. So $c_l = d_l$ for every l.

Step 2: a degree bound. The following claim is used in Step 3. Let $m \geq 1$ and let $N(t) \in M_m(k)$ be a matrix depending on t whose entry in row i and column j is $\alpha_{ij}t + \beta_{ij}$ for fixed scalars $\alpha_{ij}, \beta_{ij} \in k$. Then there is a polynomial of degree at most m whose value at every $t \in k$ is $\det N(t)$. Induct on m. For $m = 1$ the determinant is the single entry $\alpha_{11}t + \beta_{11}$, of degree at most 1. For $m \geq 2$, expand along the first column by theorem 3.3.4:

$$\det N(t) = \sum_{i=1}^{m} (-1)^{i+1}(\alpha_{i1}t + \beta_{i1}) \det N(t)(i|1)$$

where $N(t)(i|1)$ is $N(t)$ with row i and column 1 deleted (definition 3.3.2). Every entry of $N(t)(i|1)$ is again of the form $\alpha t + \beta$, and the matrix has size $m-1$, so by the inductive hypothesis its determinant is given by a polynomial of degree at most $m-1$. Each of the m terms is therefore a product of a polynomial of degree at most 1 with a polynomial of degree at most $m-1$, hence has degree at most m, and a sum of m such terms has degree at most m.

Step 3: monic of degree n, with $c_{n-1} = -\mathrm{tr}A$. Write $B(t) = tI_n - A$, so that the entry of $B(t)$ in row i and column i is $t - a_{ii}$ and the entry in row i and column j with $i \neq j$ is $-a_{ij}$. The claim, proved by induction on n, is that $\det B(t)$ is given by a polynomial $t^n + c_{n-1}t^{n-1} + \cdots + c_0$ with $c_{n-1} = -\mathrm{tr}A$.

For $n = 1$, $\det B(t) = t - a_{11}$, which is monic of degree 1 with $c_0 = -a_{11} = -\mathrm{tr}A$ by definition 2.5.7.

Let $n \geq 2$ and assume the statement for matrices of size $n-1$. Expanding $\det B(t)$ along its first column by theorem 3.3.4,

$$\det B(t) = (t - a_{11}) \det B(t)(1|1) + \sum_{i=2}^{n} (-1)^{i+1}(-a_{i1}) \det B(t)(i|1)$$

Take the first term. Deleting row 1 and column 1 from $B(t)$ leaves the matrix whose entry in row p and column q, for $1 \leq p, q \leq n-1$, is the entry of $B(t)$ in row $p+1$ and column $q+1$: that entry is $t - a_{p+1,p+1}$ when $p = q$ and $-a_{p+1,q+1}$ otherwise. So $B(t)(1|1) = tI_{n-1} - A(1|1)$, and the inductive hypothesis applies to the matrix $A(1|1) \in M_{n-1}(k)$:

$$\det B(t)(1|1) = t^{n-1} - \big(\mathrm{tr}A(1|1)\big)t^{n-2} + g(t)$$

where g has degree at most $n-3$, read as $g = 0$ when $n = 2$. The diagonal entries of $A(1|1)$ are $a_{22}, \ldots, a_{nn}$, so $\mathrm{tr}A(1|1) = a_{22} + \cdots + a_{nn}$ by definition 2.5.7.

Now take a term with $i \geq 2$, and bound the degree of $\det B(t)(i|1)$. Row 1 of $B(t)$ is $(t - a_{11},\ -a_{12},\ \ldots,\ -a_{1n})$, and deleting column 1 removes the only entry of that row in which t occurs. Since $i \geq 2$, row 1 is not deleted, and it is the first row of $B(t)(i|1)$, with entries $-a_{12}, \ldots, -a_{1n}$, all independent of t. If $n = 2$ then $B(t)(2|1)$ is the 1×1 matrix $(-a_{12})$, whose determinant is the constant $-a_{12}$, of degree at most $n - 2 = 0$. If $n \geq 3$ then $B(t)(i|1)$ has size $n - 1 \geq 2$, and expanding its determinant along that first row by theorem 3.3.4 gives a sum of $n-1$ terms, each a constant times a determinant of size $n-2$ whose entries are entries of $B(t)$ and so of the form $\alpha t + \beta$. Step 2 bounds each of those determinants by degree $n-2$, so $\det B(t)(i|1)$ has degree at most $n-2$ in either case.

Write $h(t)$ for the sum over $i \geq 2$, a polynomial of degree at most $n-2$. Multiplying out the first term,

$$\det B(t) = t^n - \big(\mathrm{tr}A(1|1)\big)t^{n-1} - a_{11}t^{n-1} + t\,g(t) + a_{11}\big(\mathrm{tr}A(1|1)\big)t^{n-2} - a_{11}g(t) + h(t)$$

The four displayed corrections have degrees at most $n-2$, $n-2$, $n-3$ and $n-2$ respectively, so the coefficient of t^n is 1 and the coefficient of t^{n-1} is

$$-\big(\mathrm{tr}A(1|1)\big) - a_{11} = -(a_{22} + \cdots + a_{nn}) - a_{11} = -\mathrm{tr}A$$

again by definition 2.5.7. This completes the induction.

Step 4: the constant term. Setting $t = 0$ in the identity of the statement kills every term except c_0, so $c_0 = \det(0 \cdot I_n - A) = \det(-A)$. By definition 1.4.1 and definition 1.4.2 the entry of $(-I_n)A$ in row i and column j is $\sum_l (-I_n)_{il} a_{lj} = -a_{ij}$, so $(-I_n)A = -A$. The matrix $-I_n$ is diagonal with every diagonal entry equal to -1, so $\det(-I_n) = (-1)^n$ by proposition 3.2.6, and theorem 3.2.10 gives

$$c_0 = \det\big((-I_n)A\big) = \det(-I_n)\det A = (-1)^n \det A$$

as we sought to show.

The proposition licenses one name and one symbol.

Definition 4.1.9: The Characteristic Polynomial

Let $A \in M_n(k)$ with $n \geq 1$. The *characteristic polynomial* of A is the polynomial

$$\chi_A(t) = \det(tI_n - A) = t^n + c_{n-1}t^{n-1} + \cdots + c_1 t + c_0$$

whose coefficients are the unique scalars given by proposition 4.1.8. It is monic of degree n.

The order of the two matrices in $tI_n - A$ is a convention, and this book keeps it in every chapter. Some references write $\det(A - tI_n)$ instead. The two differ by a sign: for each $t \in k$ the matrix $A - tI_n$ is $(-I_n)(tI_n - A)$, so the computation of Step 4 above, run with $tI_n - A$ in place of A, gives

$$\det(A - tI_n) = (-1)^n \det(tI_n - A) = (-1)^n \chi_A(t)$$

Since $(-1)^n \neq 0$, the two vanish at the same scalars and nothing about eigenvalues depends on the choice; the coefficients, however, differ in sign when n is odd. Every statement below uses $\chi_A(t) = \det(tI_n - A)$, which is the choice that makes χ_A monic.

The computation that motivated the definition now becomes a theorem, and it brings a bound on how many eigenvalues a matrix can have.

Theorem 4.1.10: The Eigenvalues Are the Roots of the Characteristic Polynomial

Let $A \in M_n(k)$ with $n \geq 1$ and let $\lambda \in k$.

1. The following are equivalent: λ is an eigenvalue of A; the matrix $A - \lambda I_n$ is singular; $\chi_A(\lambda) = 0$.
2. A has at most n distinct eigenvalues in k.

***Proof*:**

1. By definition 4.1.2 the scalar λ is an eigenvalue of A exactly when there is a nonzero $v \in k^n$ with $Av = \lambda v$, and the identity $(A - \lambda I_n)v = Av - \lambda v$ established above turns this into the existence of a nonzero v with $(A - \lambda I_n)v = 0$. By theorem 1.4.8, conditions (1) and (2), such a v exists exactly when $A - \lambda I_n$ is not invertible, that is, exactly when it is singular (definition 1.4.4). By corollary 3.2.8 that happens exactly when $\det(A - \lambda I_n) = 0$. Finally $\lambda I_n - A = (-I_n)(A - \lambda I_n)$, since multiplying by $-I_n$ negates every entry, so theorem 3.2.10 together with $\det(-I_n) = (-1)^n$ from proposition 3.2.6 gives

$$\chi_A(\lambda) = \det(\lambda I_n - A) = (-1)^n \det(A - \lambda I_n)$$

the first equality being definition 4.1.9. Since $(-1)^n \neq 0$, the two determinants vanish together, which gives the third condition.

2. Write $\chi_A(t) = c_n t^n + c_{n-1}t^{n-1} + \cdots + c_0$ with $c_n = 1$, as proposition 4.1.8 permits. Suppose $\lambda_1, \ldots, \lambda_{n+1}$ were $n+1$ distinct eigenvalues of A. By part (1) each satisfies $\sum_{l=0}^{n} c_l \lambda_i^l = 0$. Let $V = V(\lambda_1, \ldots, \lambda_{n+1}) \in M_{n+1}(k)$ be the Vandermonde matrix of theorem 3.5.6 and put $c = (c_0, c_1, \ldots, c_n)^\top \in k^{n+1}$. The entry of V in row i and column j is λ_i^{j-1}, so by definition 1.4.1 the ith entry of Vc is $\sum_{j=1}^{n+1} \lambda_i^{j-1} c_{j-1} = \sum_{l=0}^{n} c_l \lambda_i^l = 0$, and $Vc = 0$. The nodes are distinct, so $\det V \neq 0$ by theorem 3.5.6, so V is invertible by corollary 3.2.8, and theorem 1.4.8, conditions (1) and (2), gives $c = 0$. This contradicts $c_n = 1$, completing the proof.

The theorem converts a search over all of k^n into two steps that elimination and chapter 3 both handle: compute one determinant to get χ_A, find its roots, and for each root run elimination on $A - \lambda I_n$ to get a basis of the eigenspace. Part (2) says the first step cannot produce more than n answers, however large n^2 the number of entries of A may be. The next example runs the whole procedure on a matrix that has not appeared before.

Example 4.1.11: The Characteristic Polynomial and Eigenspaces of a 3×3 Matrix
Let

$$C = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \\ 0 & 5 & -1 \end{pmatrix} \in M_3(\mathbb{R})$$

Subtracting C from tI_3 entry by entry gives

$$tI_3 - C = \begin{pmatrix} t-1 & -2 & -1 \\ 0 & t-3 & -1 \\ 0 & -5 & t+1 \end{pmatrix}$$

Its first column has two zero entries, so expanding along that column by theorem 3.3.4 leaves one term:

$$\chi_C(t) = (t-1)\det\begin{pmatrix} t-3 & -1 \\ -5 & t+1 \end{pmatrix} = (t-1)\big((t-3)(t+1) - (-1)(-5)\big)$$

Expanding the bracket, $(t-3)(t+1) = t^2 - 2t - 3$, and subtracting 5 gives $t^2 - 2t - 8$, which factors as $(t-4)(t+2)$ since $(-4) \cdot 2 = -8$ and $-4 + 2 = -2$. Hence

$$\chi_C(t) = (t-1)(t-4)(t+2)$$

a monic cubic, as proposition 4.1.8 requires. By theorem 4.1.10(1) the eigenvalues of C are 1, 4 and -2, and by part (2) there are no others.

Each eigenspace is the null space of one matrix (definition 4.1.4), found by elimination. For $\lambda = 1$,

$$C - I_3 = \begin{pmatrix} 0 & 2 & 1 \\ 0 & 2 & 1 \\ 0 & 5 & -2 \end{pmatrix}$$

The replacement $R_2 \mapsto R_2 - R_1$ empties the second row, and then $R_3 \mapsto 2R_3 - 5R_1$, which is the scaling $R_3 \mapsto 2R_3$ followed by the replacement $R_3 \mapsto R_3 - 5R_1$, replaces the third row by $(0,\ 10-10,\ -4-5) = (0, 0, -9)$:

$$\begin{pmatrix} 0 & 2 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & -9 \end{pmatrix}$$

Neither operation changes the solution set of the homogeneous system (lemma 1.2.4). A solution $(x,y,z)^\top$ therefore has $-9z=0$, hence $z=0$, and then $2y+z=0$ gives $y=0$, while x is unconstrained. So $E_1(C)=\operatorname{span}\{(1,0,0)^\top\}$, and indeed $C(1,0,0)^\top=(1,0,0)^\top$.

For $\lambda=4$,

$$C-4I_3=\begin{pmatrix}-3 & 2 & 1\\ 0 & -1 & 1\\ 0 & 5 & -5\end{pmatrix}$$

The replacement $R_3\mapsto R_3+5R_2$ empties the third row. The second row then reads $-y+z=0$, so $z=y$, and the first row reads $-3x+2y+z=0$, which with $z=y$ becomes $-3x+3y=0$ and gives $x=y$. Taking $y=1$ produces $(1,1,1)^\top$, so $E_4(C)=\operatorname{span}\{(1,1,1)^\top\}$. As a check, $C(1,1,1)^\top=(1+2+1,\,0+3+1,\,0+5-1)^\top=(4,4,4)^\top=4(1,1,1)^\top$.

For $\lambda=-2$,

$$C+2I_3=\begin{pmatrix}3 & 2 & 1\\ 0 & 5 & 1\\ 0 & 5 & 1\end{pmatrix}$$

The replacement $R_3\mapsto R_3-R_2$ empties the third row. The second row reads $5y+z=0$, so $z=-5y$, and the first row reads $3x+2y+z=0$, which with $z=-5y$ becomes $3x-3y=0$ and gives $x=y$. Taking $y=1$ produces $(1,1,-5)^\top$, so $E_{-2}(C)=\operatorname{span}\{(1,1,-5)^\top\}$. As a check, $C(1,1,-5)^\top=(1+2-5,\,0+3-5,\,0+5+5)^\top=(-2,-2,10)^\top=-2(1,1,-5)^\top$.

Each of the three eigenspaces above has dimension 1, and no two of the spanning columns $(1,0,0)^\top$, $(1,1,1)^\top$ and $(1,1,-5)^\top$ are multiples of one another: the first has second entry 0 and the other two have second entry 1, while the last two agree in their first two entries and differ in the third. Whether a matrix always gives enough such columns, and what happens when it does not, is the question section 4.2 takes up.

One class of matrices needs no computation at all, because the determinant of tI_n-A is already a product.

Proposition 4.1.12: The Eigenvalues of a Triangular Matrix Are Its Diagonal Entries

Let $A=(a_{ij})\in M_n(k)$ be upper triangular or lower triangular. Then

$$\chi_A(t)=(t-a_{11})(t-a_{22})\cdots(t-a_{nn})$$

and the eigenvalues of A are exactly the diagonal entries $a_{11},\dots,a_{nn}$.

***Proof*:**
Fix $t\in k$. The entry of tI_n-A in row i and column j with $i\neq j$ is $-a_{ij}$, so tI_n-A has a zero in every position where A has one, off the diagonal; hence tI_n-A is triangular of the same kind as A. Its diagonal entries are $t-a_{11},\dots,t-a_{nn}$, so proposition 3.2.6 gives

$$\det(tI_n-A)=(t-a_{11})\cdots(t-a_{nn})$$

Multiplying the n factors out produces a monic polynomial of degree n in t whose value at every $t\in k$ is $\det(tI_n-A)$, so by the uniqueness in proposition 4.1.8 it is χ_A.

By theorem 4.1.10(1) a scalar λ is an eigenvalue of A exactly when $(\lambda-a_{11})\cdots(\lambda-a_{nn})=0$.

A product of scalars in $\mathbb{R}$ or in $\mathbb{C}$ vanishes exactly when one of its factors does, so this holds exactly when $\lambda = a_{ii}$ for some i, as we sought to show.

A diagonal matrix is the smallest case: $\operatorname{diag}(d_1, \ldots, d_n)$ has eigenvalues $d_1, \ldots, d_n$, and $\operatorname{diag}(d_1, \ldots, d_n)e_j = d_j e_j$ exhibits e_j as an eigenvector belonging to d_j. The eigenvalues of a triangular matrix are read off the diagonal, and none of the entries above or below it enters the answer. Replacing a matrix by a similar triangular one therefore exhibits its eigenvalues, and chapter 7 carries that replacement out.

Nothing so far guarantees that any eigenvalue exists. Over $\mathbb{R}$ none need.

Example 4.1.13: The Quarter Turn of the Plane

Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$$

Then $Je_1 = (0,1)^\top$ and $Je_2 = (-1,0)^\top$, so J turns the plane by a quarter turn counterclockwise. Its characteristic polynomial is

$$\chi_J(t) = \det \begin{pmatrix} t & 1 \\ -1 & t \end{pmatrix} = t^2 - (1)(-1) = t^2 + 1$$

For every $t \in \mathbb{R}$ the value $t^2 + 1$ is at least 1, so χ_J has no real root, and by theorem 4.1.10(1) the matrix J has no eigenvalue in $\mathbb{R}$. Every nonzero column of $\mathbb{R}^2$ is moved off the line it spans.

Regarded instead as an element of $M_2(\mathbb{C})$, the same array has two eigenvalues, since $t^2 + 1 = (t - i)(t + i)$ vanishes at $t = i$ and $t = -i$. The eigenvectors are found as before. For $\lambda = i$,

$$J - iI_2 = \begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix}$$

The replacement $R_1 \mapsto R_1 + iR_2$ empties the first row, since $-i + i = 0$ and $-1 + i(-i) = -1 + 1 = 0$, so the system reduces to the single equation $x - iy = 0$ carried by the second row, that is $x = iy$. Taking $y = 1$ gives $E_i(J) = \operatorname{span}\{(i,1)^\top\}$. As a check, $J(i,1)^\top = (-1,i)^\top$ and $i(i,1)^\top = (i^2,i)^\top = (-1,i)^\top$. The same computation with i replaced by $-i$ gives $E_{-i}(J) = \operatorname{span}\{(-i,1)^\top\}$.

Nothing is special about the quarter turn. Rotation of the plane by an angle θ is given by

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad tI_2 - R_\theta = \begin{pmatrix} t - \cos\theta & \sin\theta \\ -\sin\theta & t - \cos\theta \end{pmatrix}$$

so that

$$\chi_{R_\theta}(t) = (t - \cos\theta)^2 - \sin\theta(-\sin\theta) = (t - \cos\theta)^2 + \sin^2\theta = t^2 - 2(\cos\theta)t + 1$$

the last step using $\cos^2\theta + \sin^2\theta = 1$. A real root requires $(t - \cos\theta)^2 = -\sin^2\theta$, and a square of a real number is never negative, so a real root exists only when $\sin\theta = 0$, that is only when R_θ is I_2 or $-I_2$.

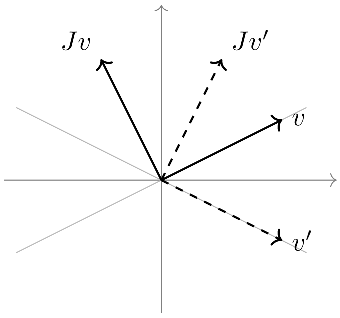


Figure 4.2: The quarter turn carries each nonzero column off the line it spans

Over $\mathbb{C}$ the obstruction disappears, and it disappears for one reason only: a nonconstant complex polynomial has a root. That statement is the fundamental theorem of algebra, item 6 of *Results Assumed from Other Volumes* in the front matter, proved in *Core Algebra* [9, Fundamental Theorem of Algebra]. This is its first use in the book, and it does more than produce a single eigenvalue: applied repeatedly it splits χ_A into linear factors.

Theorem 4.1.14: Existence of an Eigenvalue and the Splitting of χ_A over $\mathbb{C}$

Let $n \geq 1$ and $A \in M_n(\mathbb{C})$.

1. There are $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$, not necessarily distinct, with

$$\chi_A(t) = (t - \lambda_1)(t - \lambda_2) \cdots (t - \lambda_n)$$

   for every $t \in \mathbb{C}$.

2. A has at least one eigenvalue in $\mathbb{C}$, and the eigenvalues of A are exactly the scalars occurring in the list $\lambda_1, \ldots, \lambda_n$.

***Proof*:**

1. It suffices to prove the following statement for every $d \geq 1$: a polynomial $p(t) = t^d + b_{d-1}t^{d-1} + \cdots + b_0$ with complex coefficients satisfies $p(t) = (t - \mu_1) \cdots (t - \mu_d)$ for some $\mu_1, \ldots, \mu_d \in \mathbb{C}$. Applying it to χ_A, which is monic of degree $n \geq 1$ by proposition 4.1.8, gives the claim.

   Induct on d. For $d = 1$, $p(t) = t + b_0 = t - (-b_0)$.

   Let $d \geq 2$. The polynomial p is nonconstant, so the fundamental theorem of algebra gives $\mu \in \mathbb{C}$ with $p(\mu) = 0$. Write $b_d = 1$, so that $p(t) = \sum_{j=0}^{d} b_j t^j$. Since $p(\mu) = 0$,

$$p(t) = p(t) - p(\mu) = \sum_{j=1}^{d} b_j (t^j - \mu^j)$$

the terms with $j = 0$ cancelling. For each $j \geq 1$ the product

$$(t-\mu)\sum_{l=0}^{j-1} t^{j-1-l}\mu^l = \sum_{l=0}^{j-1} t^{j-l}\mu^l - \sum_{l=0}^{j-1} t^{j-1-l}\mu^{l+1} = \sum_{l=0}^{j-1} t^{j-l}\mu^l - \sum_{l=1}^{j} t^{j-l}\mu^l$$

has all terms with $1 \leq l \leq j-1$ cancelling in pairs, leaving $t^j - \mu^j$. Substituting this into the previous display,

$$p(t) = (t-\mu)\, q(t), \qquad q(t) = \sum_{j=1}^{d} b_j \sum_{l=0}^{j-1} t^{j-1-l}\mu^l$$

Every power of t occurring in q has exponent at most $d-1$, and the exponent $d-1$ occurs only for $j = d$ and $l = 0$, with coefficient $b_d = 1$. So q is monic of degree $d-1$, and the inductive hypothesis gives $q(t) = (t-\mu_1)\cdots(t-\mu_{d-1})$. Setting $\mu_d = \mu$ completes the induction.

2. Since $n \geq 1$ the list is nonempty, and $\chi_A(\lambda_1) = \prod_i(\lambda_1 - \lambda_i) = 0$ because the factor with $i = 1$ vanishes; so λ_1 is an eigenvalue by theorem 4.1.10(1). Conversely a scalar λ is an eigenvalue exactly when $\prod_i(\lambda - \lambda_i) = 0$, and a product of complex numbers vanishes exactly when one of its factors does, which happens exactly when $\lambda = \lambda_i$ for some i, completing the proof.

The theorem constrains the field rather than the matrix. One array of real numbers can have no eigenvalue when it is read as an element of $M_n(\mathbb{R})$, while its characteristic polynomial acquires n linear factors when the same array is read as an element of $M_n(\mathbb{C})$; the matrix J of example 4.1.13 does this with $n = 2$. Which of the two readings is in force is therefore part of the data of a question about eigenvalues, and this book names the field whenever the answer depends on it.

A factorization of χ_A into linear factors carries more than the list of roots. Two of its coefficients were computed in proposition 4.1.8 without any reference to eigenvalues, and comparing the two descriptions identifies them.

Proposition 4.1.15: The Trace Is the Sum and the Determinant the Product of the Eigenvalues

Let $n \geq 1$ and $A \in M_n(k)$, and suppose there are $\lambda_1, \ldots, \lambda_n \in k$ with

$$\chi_A(t) = (t-\lambda_1)(t-\lambda_2)\cdots(t-\lambda_n)$$

for every $t \in k$. Then

$$\mathrm{tr} A = \lambda_1 + \lambda_2 + \cdots + \lambda_n, \qquad \det A = \lambda_1\lambda_2\cdots\lambda_n$$

The hypothesis holds for every $A \in M_n(\mathbb{C})$ by theorem 4.1.14(1).

***Proof*:**
First expand the product. The claim is that

$$(t-\lambda_1)\cdots(t-\lambda_n) = t^n - (\lambda_1 + \cdots + \lambda_n)t^{n-1} + r(t)$$

for a polynomial r of degree at most $n-2$, read as $r=0$ when $n=1$. Induct on n. For $n=1$ the product is $t-\lambda_1$, which is the displayed form. For $n \geq 2$, the inductive hypothesis applied to the first $n-1$ factors gives

$$(t-\lambda_1)\cdots(t-\lambda_{n-1}) = t^{n-1} - \Big(\sum_{i=1}^{n-1} \lambda_i\Big) t^{n-2} + r_0(t)$$

with $\deg r_0 \leq n-3$. Multiplying by $t-\lambda_n$,

$$(t-\lambda_1)\cdots(t-\lambda_n) = t^n - \Big(\sum_{i=1}^{n-1} \lambda_i\Big) t^{n-1} + t\,r_0(t) - \lambda_n t^{n-1} + \lambda_n \Big(\sum_{i=1}^{n-1} \lambda_i\Big) t^{n-2} - \lambda_n r_0(t)$$

The three corrections $t\,r_0(t)$, $\lambda_n(\sum_{i<n} \lambda_i)t^{n-2}$ and $\lambda_n r_0(t)$ have degrees at most $n-2$, $n-2$ and $n-3$, so the coefficient of t^{n-1} is $-\sum_{i=1}^n \lambda_i$ and the polynomial is monic of degree n. This completes the induction.

The hypothesis now presents χ_A in the form $t^n + c_{n-1}t^{n-1} + \cdots + c_0$ with $c_{n-1} = -(\lambda_1 + \cdots + \lambda_n)$, and proposition 4.1.8 presents it in the same form with $c_{n-1} = -\mathrm{tr} A$. The coefficients in that form are unique by the same proposition, so $-\mathrm{tr} A = -(\lambda_1 + \cdots + \lambda_n)$, which is the first identity.

For the second, evaluate both descriptions at $t=0$. The product gives $\prod_{i=1}^n (0-\lambda_i) = (-1)^n \lambda_1 \cdots \lambda_n$, and proposition 4.1.8 gives $\chi_A(0) = c_0 = (-1)^n \det A$. Equating the two and dividing by $(-1)^n$, which is nonzero, gives $\det A = \lambda_1 \cdots \lambda_n$, as we sought to show.

Under the hypothesis of the proposition the list $\lambda_1, \ldots, \lambda_n$ consists of the eigenvalues of A and of nothing else: by theorem 4.1.10(1) a scalar λ is an eigenvalue exactly when $(\lambda-\lambda_1)\cdots(\lambda-\lambda_n) = 0$, and a product of scalars in $\mathbb{R}$ or in $\mathbb{C}$ vanishes exactly when one factor does. The list carries repetitions: a scalar λ occurs in it as many times as $t-\lambda$ occurs as a factor of χ_A, and section 4.3 gives that count a name. The repetition is part of the statement. For $n=2$ and $\lambda_1 = \lambda_2 = 2$ the sum is 4 and the product is 4, while the single value 2 would give the wrong answer for both.

For the matrix C of example 4.1.11 the two identities read $\mathrm{tr} C = 1 + 3 + (-1) = 3$ against $1 + 4 + (-2) = 3$, and $\det C = 1 \cdot 4 \cdot (-2) = -8$. Expanding $\det C$ along the first column, which has two zero entries, confirms the second: $\det C = 1 \cdot (3 \cdot (-1) - 1 \cdot 5) = -8$. Two numbers attached to A in earlier chapters by unrelated constructions, a sum of n diagonal entries and a value produced by elimination, are the two extreme coefficients of one polynomial.

Section 2.5 established that the matrices of one linear map in two ordered bases are similar (definition 2.5.5), and that a number attached to a matrix deserves attention when similar matrices share it. The characteristic polynomial is shared, and with it the n coefficients $c_0, \ldots, c_{n-1}$.

Corollary 4.1.16: Similar Matrices Have the Same Characteristic Polynomial

Let $A, B \in M_n(k)$ be similar. Then $\chi_B = \chi_A$, and A and B have the same eigenvalues. Consequently, if $T\colon k^n \to k^n$ is linear, the polynomial $\chi_{[T]_{\mathcal{B}}}$ is the same for every ordered basis $\mathcal{B}$ of k^n; it is called the *characteristic polynomial of* T and written χ_T, and its roots are called the *eigenvalues of* T.

***Proof*:**
By definition 2.5.5 there is an invertible $P \in M_n(k)$ with $B = PAP^{-1}$. Fix $t \in k$. Regrouping by proposition 1.4.3(1), moving the scalar by proposition 1.4.3(2), using proposition 1.4.3(3) for $PI_n = P$ and then definition 1.4.4 for $PP^{-1} = I_n$,

$$P(tI_n)P^{-1} = t\big((PI_n)P^{-1}\big) = t(PP^{-1}) = tI_n$$

and by proposition 1.4.3(2) again,

$$P(tI_n - A)P^{-1} = P(tI_n)P^{-1} - PAP^{-1} = tI_n - B$$

Applying theorem 3.2.10 twice and then corollary 3.2.13(1), which applies because P is invertible,

$$\det(tI_n - B) = \det P\, \det(tI_n - A)\, \det(P^{-1}) = \det(tI_n - A)$$

the last step using that multiplication in k does not depend on the order of the factors, so that $\det P \cdot (\det P)^{-1} = 1$ may be cancelled. The two polynomials χ_A and χ_B therefore take the same value at every $t \in k$, and both are monic of degree n (definition 4.1.9), so the uniqueness in proposition 4.1.8 makes their coefficients agree. Hence $\chi_B = \chi_A$, and theorem 4.1.10(1) turns the equality of the polynomials into the equality of the two sets of eigenvalues.

For the last claim, let $\mathcal{B}$ and $\mathcal{C}$ be ordered bases of k^n. By theorem 2.5.4 the matrices $[T]_{\mathcal{B}}$ and $[T]_{\mathcal{C}}$ are similar, so the first part gives $\chi_{[T]_{\mathcal{B}}} = \chi_{[T]_{\mathcal{C}}}$, as we sought to show.

The corollary is stronger than the two invariance statements already available. Corollary 2.5.9 and corollary 3.2.13(2) assert that similar matrices share one number each; the corollary asserts that they share the n coefficients $c_0, \dots, c_{n-1}$ of χ_A, among which proposition 4.1.8 identifies $c_{n-1} = -\mathrm{tr}A$ and $c_0 = (-1)^n \det A$. It also settles what an eigenvalue belongs to. A linear map $T\colon k^n \to k^n$ has many matrices, one for each ordered basis, and they can differ in every entry (example 2.5.10); the eigenvalues do not move. The condition $T(v) = \lambda v$ makes no reference to coordinates in the first place, which is what the corollary confirms at the level of matrices.

The converse fails. The matrices

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

are both upper triangular with diagonal entries $1, 1$, so proposition 4.1.12 gives both the characteristic polynomial $(t-1)^2$. They are not similar, since $PI_2P^{-1} = I_2$ for every invertible P, so the only matrix similar to I_2 is I_2 itself. Equal characteristic polynomials therefore do not force two matrices to represent one map. What separates these two is the dimension of E_1. For I_2 the matrix $I_2 - I_2$ is zero, so $E_1(I_2) = k^2$ and $\dim E_1(I_2) = 2$; writing N for the second matrix, $N - I_2 = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ has null space cut out by $y = 0$, so $E_1(N) = \mathrm{span}\{e_1\}$ and $\dim E_1(N) = 1$. Section 4.3 compares that dimension with the number of times $t - \lambda$ occurs as a factor of χ_A.

Exercise 4.1.1

1. Let
$$A = \begin{pmatrix} 5 & -2 \\ 6 & -2 \end{pmatrix} \in M_2(\mathbb{R})$$
Compute χ_A, factor it, list the eigenvalues, and compute a basis of each eigenspace by

elimination. Check each basis vector against definition 4.1.3, and check the two identities of proposition 4.1.15.

2. Let
$$B = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 2 \\ 0 & 0 & -1 \end{pmatrix} \in M_3(\mathbb{R})$$
Write down χ_B and the eigenvalues of B without expanding any determinant, naming the result used. Then compute a basis of $E_3(B)$ and of $E_{-1}(B)$, and give the dimension of each.

3. Let
$$D = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$
Compute χ_D. Regarding D as an element of $M_3(\mathbb{R})$, find every eigenvalue and a basis of every eigenspace. Then regard D as an element of $M_3(\mathbb{C})$ and do the same. Check proposition 4.1.15 in the complex case.

4. Let $A \in M_n(k)$. Show that $\chi_{A^\top} = \chi_A$, and deduce that A and $A^\top$ have the same eigenvalues. Then show that they need not have the same eigenvectors, by computing $E_1(A)$ and $E_1(A^\top)$ for $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 2 \end{smallmatrix}\right)$.

5. Let $A \in M_n(k)$ and let v be an eigenvector of A belonging to λ. Write A^m for the product of m copies of A.

   1. Show that λ^m is an eigenvalue of A^m for every integer $m \geq 1$, with v an eigenvector belonging to it.
   2. Show that if A is invertible then $\lambda \neq 0$, and that λ^{-1} is an eigenvalue of A^{-1} with the same eigenvector v.
   3. Decide whether every eigenvalue of A^2 has the form λ^2 for an eigenvalue λ of A, with a proof or a counterexample in $M_2(\mathbb{R})$.

6. Let $A \in M_n(k)$.

   1. Show that 0 is an eigenvalue of A if and only if $\det A = 0$, and that in that case $E_0(A) = \text{null}A$.
   2. Suppose every row of A has the same sum s, that is $\sum_{j=1}^n a_{ij} = s$ for every i. Show that s is an eigenvalue of A and exhibit an eigenvector belonging to it.

7. Let $A \in M_n(k)$ and $c \in k$.

   1. Show that $\chi_{A+cI_n}(t) = \chi_A(t-c)$ for every $t \in k$, that the eigenvalues of $A + cI_n$ are exactly the scalars $\lambda + c$ with λ an eigenvalue of A, and that $E_{\lambda+c}(A + cI_n) = E_\lambda(A)$.
   2. Let $N \in M_3(\mathbb{R})$ be the matrix all of whose nine entries equal 1. Compute χ_N, and compute a basis of each eigenspace of N, using theorem 2.3.11 to predict the dimension of $E_0(N)$ before computing it.
   3. Use (a) and (b) to write down the eigenvalues and the eigenspaces of
   $$\begin{pmatrix} 3 & 1 & 1 \\ 1 & 3 & 1 \\ 1 & 1 & 3 \end{pmatrix}$$

without computing a determinant.

4.2 Diagonalization and Its Obstruction

Section 4.1 attached to a square matrix its eigenvalues, and to each eigenvalue the vectors that the matrix scales. One eigenvector describes the matrix along one line. This section asks what happens when the eigenvectors are plentiful enough to describe all of k^n, and exhibits the two ways they can run short.

Take the matrix

$$A_0 = \begin{pmatrix} 3 & 1 \\ 2 & 2 \end{pmatrix} \in M_2(\mathbb{R})$$

and compute its characteristic polynomial $\chi_{A_0}(t) = \det(tI_2 - A_0)$ of definition 4.1.9, where I_2 is the 2×2 identity matrix of definition 1.4.2:

$$\chi_{A_0}(t) = \det \begin{pmatrix} t-3 & -1 \\ -2 & t-2 \end{pmatrix} = (t-3)(t-2) - (-1)(-2) = t^2 - 5t + 4 = (t-1)(t-4)$$

By theorem 4.1.10 the eigenvalues of A_0 are the roots of χ_{A_0}, namely 1 and 4.

Three consequences of definition 1.4.1 are used throughout the section. For $M, N \in M_n(k)$ and $v \in k^n$, the ith entry of $(M-N)v$ is $\sum_{j=1}^n (m_{ij} - n_{ij})v_j = \sum_{j=1}^n m_{ij}v_j - \sum_{j=1}^n n_{ij}v_j$, so $(M-N)v = Mv - Nv$. For $w_1, \ldots, w_s \in k^n$ and scalars $c_1, \ldots, c_s \in k$, the ith entry of $M(c_1w_1 + \cdots + c_sw_s)$ is

$$\sum_{j=1}^n m_{ij}\big(c_1(w_1)_j + \cdots + c_s(w_s)_j\big) = c_1 \sum_{j=1}^n m_{ij}(w_1)_j + \cdots + c_s \sum_{j=1}^n m_{ij}(w_s)_j$$

so $M(c_1w_1 + \cdots + c_sw_s) = c_1Mw_1 + \cdots + c_sMw_s$. Finally $(\lambda I_n)v = \lambda v$ for a scalar λ, because the ith row of λI_n carries λ in position i and zeros elsewhere.

An eigenvector of A_0 for the eigenvalue λ is a nonzero v with $A_0v = \lambda v$. Subtracting, $A_0v - \lambda v = A_0v - (\lambda I_2)v = (A_0 - \lambda I_2)v$, so the eigenvectors for λ are exactly the nonzero solutions of the homogeneous system $(A_0 - \lambda I_2)x = 0$ in the unknown column $x = (x_1, x_2)^\top$, which is the kind of system elimination solves (section 1.3). For $\lambda = 1$,

$$A_0 - I_2 = \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \xrightarrow{R_2 \mapsto R_2 - R_1} \begin{pmatrix} 2 & 1 \\ 0 & 0 \end{pmatrix} \xrightarrow{R_1 \mapsto \frac{1}{2}R_1} \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 0 \end{pmatrix}$$

so x_2 is free and $x_1 = -\frac{1}{2}x_2$. Taking $x_2 = -2$ clears the fraction and gives $p_1 = (1, -2)^\top$, and indeed $A_0p_1 = (3-2,\, 2-4)^\top = (1, -2)^\top = 1 \cdot p_1$. For $\lambda = 4$,

$$A_0 - 4I_2 = \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \xrightarrow{R_2 \mapsto R_2 + 2R_1} \begin{pmatrix} -1 & 1 \\ 0 & 0 \end{pmatrix} \xrightarrow{R_1 \mapsto -R_1} \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix}$$

so $x_1 = x_2$ with x_2 free, and $x_2 = 1$ gives $p_2 = (1, 1)^\top$ with $A_0p_2 = (3+1,\, 2+2)^\top = (4, 4)^\top = 4 \cdot p_2$.

Now place the two eigenvectors side by side as the columns of one matrix,

$$P = \begin{pmatrix} 1 & 1 \\ -2 & 1 \end{pmatrix}$$

The matrix $Q = \frac{1}{3}\begin{pmatrix} 1 & -1 \\ 2 & 1 \end{pmatrix}$ satisfies

$$PQ = \frac{1}{3}\begin{pmatrix} 1+2 & -1+1 \\ -2+2 & 2+1 \end{pmatrix} = I_2, \qquad QP = \frac{1}{3}\begin{pmatrix} 1+2 & 1-1 \\ 2-2 & 2+1 \end{pmatrix} = I_2$$

so P is invertible with $P^{-1} = Q$ (definition 1.4.4). Multiplying out A_0P first, and then $P^{-1}(A_0P)$, which by proposition 1.4.3 is $P^{-1}A_0P$,

$$A_0P = \begin{pmatrix} 3-2 & 3+1 \\ 2-4 & 2+2 \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ -2 & 4 \end{pmatrix}, \qquad P^{-1}A_0P = \frac{1}{3}\begin{pmatrix} 1+2 & 4-4 \\ 2-2 & 8+4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$$

The result has zeros off the diagonal, and its diagonal carries the two eigenvalues in the order in which their eigenvectors were listed.

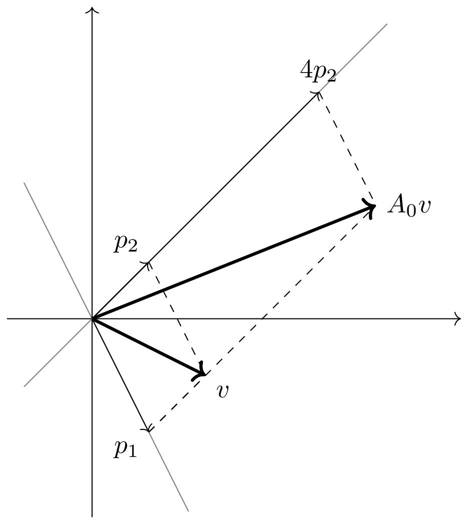


Figure 4.3: The two eigenlines of A_0, and the decomposition $v = p_1 + p_2$ carried to $A_0v = p_1 + 4p_2$

The picture behind the computation is that A_0 acts on each of the two eigenlines by a scalar, and every vector splits along them. The vector $v = (2,-1)^\top$ is $p_1 + p_2$, and multiplying by A_0 scales the first summand by 1 and the second by 4, giving $A_0v = p_1 + 4p_2 = (1,-2)^\top + (4,4)^\top = (5,2)^\top$, which agrees with the direct product $A_0v = (6-1,\, 4-2)^\top$. The matrix $\left(\begin{smallmatrix} 1 & 0 \\ 0 & 4 \end{smallmatrix}\right)$ records exactly this: two scalings with no interaction between them. The shape produced by the computation is what the following definition names.

Definition 4.2.1: Diagonalizable Matrix

Let k be $\mathbb{R}$ or $\mathbb{C}$. A matrix $D = (d_{ij}) \in M_n(k)$ is *diagonal* if $d_{ij} = 0$ whenever $i \neq j$; the notation $\operatorname{diag}(d_1, \ldots, d_n)$ denotes the diagonal matrix whose (j,j) entry is d_j.
A matrix $A \in M_n(k)$ is *diagonalizable over k* if there is an invertible $P \in M_n(k)$ such that $P^{-1}AP$ is diagonal. Such a P is said to *diagonalize* A.

The computation above shows that A_0 is diagonalizable over $\mathbb{R}$. Two features of the definition should be recorded at once. First, A and $P^{-1}AP$ are similar in the sense of definition 2.5.5, so diagonalizability is a property of the whole similarity class: if A is diagonalizable and B is similar to A, then B is diagonalizable, because similarity is transitive (proposition 2.5.6) and a matrix similar to a diagonal matrix is by definition diagonalizable. Second, the field is part of the statement. A real matrix may fail to be diagonalizable over $\mathbb{R}$ and succeed over $\mathbb{C}$, and example 4.2.8 exhibits this.

As it stands, the definition asks for a matrix P drawn from all invertible matrices, and that is not a condition anyone can check by searching. The computation on A_0 did not search: it built P out of eigenvectors, one per column. That construction is the entire content of the definition.

Theorem 4.2.2: Diagonalizable Exactly When an Eigenbasis Exists

Let k be $\mathbb{R}$ or $\mathbb{C}$ and let $A \in M_n(k)$. Then A is diagonalizable over k if and only if k^n has a basis $v_1, \ldots, v_n$ every member of which is an eigenvector of A. In that case, writing λ_j for the eigenvalue of v_j and P for the matrix whose jth column is v_j, the matrix P is invertible and

$$P^{-1}AP = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$$

***Proof*:**
Let $P \in M_n(k)$ have columns $p_1, \ldots, p_n$ and let $D = \operatorname{diag}(d_1, \ldots, d_n)$. By definition 1.4.1, the (i, j) entry of AP is $\sum_{l=1}^{n} a_{il}p_{lj}$, which is the ith entry of Ap_j; so the jth column of AP is Ap_j. The (i, j) entry of PD is $\sum_{l=1}^{n} p_{il}d_{lj}$, and $d_{lj} = 0$ for $l \neq j$ while $d_{jj} = d_j$, so this sum is $p_{ij}d_j$ and the jth column of PD is d_jp_j. Two matrices are equal exactly when their corresponding columns are equal, so $AP = PD$ holds if and only if $Ap_j = d_jp_j$ for every $j = 1, \ldots, n$.

Suppose P is invertible. Multiplying $AP = PD$ on the left by P^{-1} gives $P^{-1}AP = D$, and multiplying $P^{-1}AP = D$ on the left by P gives $AP = PD$; both steps use proposition 1.4.3 together with $PP^{-1} = P^{-1}P = I_n$ (definition 1.4.4) and $I_nM = MI_n = M$ (definition 1.4.2). So for invertible P the equation $P^{-1}AP = D$ says precisely that $Ap_j = d_jp_j$ for every j.

Next, P is invertible if and only if its columns form a basis of k^n, which is proposition 2.3.13.

Both directions now follow. If A is diagonalizable, choose an invertible P with $P^{-1}AP = D$ diagonal. The columns of P form a basis of k^n by the previous paragraph, and satisfy $Ap_j = d_jp_j$ by the first two. Each p_j is nonzero, since $p_j = 0$ would give the vanishing combination $1 \cdot p_j = 0$ with a nonzero coefficient, so each p_j is an eigenvector of A with eigenvalue d_j (definition 4.1.3), and A has a basis of eigenvectors. Conversely, let $v_1, \ldots, v_n$ be a basis of k^n with $Av_j = \lambda_jv_j$ for each j, and let P be the matrix with jth column v_j. Then P is invertible, and with $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ the relations $Av_j = \lambda_jv_j$ say exactly that $AP = PD$, whence $P^{-1}AP = D$ is diagonal and A is diagonalizable by the stated formula, as we sought to show.

The question has changed shape. “Is there an invertible P making $P^{-1}AP$ diagonal” was a search over a set of matrices; “does A have n linearly independent eigenvectors” is a question elimination answers eigenvalue by eigenvalue, since the eigenvectors for λ are the nonzero solutions of $(A - \lambda I_n)x = 0$. The remaining work is to decide how many independent eigenvectors are available. One count is free, and it comes from the eigenvalues alone.

Theorem 4.2.3: Eigenvectors for Distinct Eigenvalues Are Independent

Let $A \in M_n(k)$, let $\lambda_1, \ldots, \lambda_r \in k$ be pairwise distinct eigenvalues of A, and for each i let v_i be an eigenvector of A for λ_i. Then $v_1, \ldots, v_r$ are linearly independent.

***Proof*:**
The argument is an induction on r. For $r = 1$, an eigenvector is nonzero by definition 4.1.3; if $c_1v_1 = 0$ with $c_1 \neq 0$, then $v_1 = c_1^{-1}(c_1v_1) = 0$, a contradiction, so $c_1 = 0$ and the one-entry list is independent.

Let $r \geq 2$ and assume the statement for any $r-1$ eigenvectors attached to pairwise distinct eigenvalues. Suppose $c_1v_1 + \cdots + c_rv_r = 0$ with $c_1, \ldots, c_r \in k$. Multiplying on the left by A and using the identity $A(c_1v_1 + \cdots + c_rv_r) = c_1Av_1 + \cdots + c_rAv_r$ recorded at the start of the section, together with $Av_i = \lambda_iv_i$, gives

$$c_1\lambda_1v_1 + \cdots + c_r\lambda_rv_r = 0$$

Subtracting λ_r times the original relation from this one, the rth term becomes $c_r(\lambda_r - \lambda_r)v_r = 0$ and what remains is

$$c_1(\lambda_1 - \lambda_r)v_1 + \cdots + c_{r-1}(\lambda_{r-1} - \lambda_r)v_{r-1} = 0$$

The vectors $v_1, \ldots, v_{r-1}$ are eigenvectors for the pairwise distinct eigenvalues $\lambda_1, \ldots, \lambda_{r-1}$, so the inductive hypothesis makes them linearly independent, and therefore every coefficient in the last display vanishes: $c_i(\lambda_i - \lambda_r) = 0$ for $i \leq r-1$. Each scalar $\lambda_i - \lambda_r$ is nonzero because the eigenvalues are distinct, and a product of two real or complex numbers vanishes only if one of the factors does, so $c_i = 0$ for every $i \leq r-1$. The original relation now reads $c_rv_r = 0$ with $v_r \neq 0$, which forces $c_r = 0$ by the argument used for $r = 1$. All coefficients vanish, completing the induction and the proof.

Corollary 4.2.4: A Matrix with n Distinct Eigenvalues Is Diagonalizable

Let $A \in M_n(k)$ and suppose χ_A has n distinct roots $\lambda_1, \ldots, \lambda_n$ in k. Then A is diagonalizable over k: choosing an eigenvector v_j for each λ_j and letting P be the matrix with columns $v_1, \ldots, v_n$ gives $P^{-1}AP = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.

***Proof*:**
Each λ_j is a root of χ_A lying in k, hence an eigenvalue of A by theorem 4.1.10, so an eigenvector v_j for λ_j exists by definition 4.1.2. The eigenvalues $\lambda_1, \ldots, \lambda_n$ are pairwise distinct, so $v_1, \ldots, v_n$ are linearly independent by theorem 4.2.3, and n independent vectors in k^n form a basis by corollary 2.2.13. That basis consists of eigenvectors of A, so theorem 4.2.2 applies and yields both the invertibility of P and the displayed factorization, as we sought to show.

The hypothesis is sufficient and not necessary. The identity matrix I_n is already diagonal, so it is diagonalizable, while $tI_n - I_n$ is triangular with every diagonal entry $t-1$, so $\chi_{I_n}(t) = (t-1)^n$ by proposition 3.2.6 together with the uniqueness clause of proposition 4.1.8 and I_n has the single eigenvalue 1. Counting distinct eigenvalues therefore cannot settle the question in general. What is needed instead is a count of the independent eigenvectors carried by each eigenvalue separately, and that count is a dimension.

Fix $\lambda \in k$ and $A \in M_n(k)$. By proposition 4.1.5 the eigenspace E_λ is the null space $\operatorname{null}(A - \lambda I_n)$, a subspace of k^n of dimension $n - \operatorname{rank}(A - \lambda I_n)$, and elimination on $A - \lambda I_n$ produces a basis of it (section 2.3). These are the numbers the next result compares against n.

Proposition 4.2.5: Assembling P from Bases of the Eigenspaces

Let $A \in M_n(k)$ and let $\lambda_1, \dots, \lambda_r$ be the distinct eigenvalues of A, which by theorem 4.1.10 are the distinct roots of χ_A lying in k (the case $r = 0$, in which there are none, is allowed), and for each i let $v_{i,1}, \dots, v_{i,d_i}$ be a basis of E_{λ_i}, where $d_i = \dim E_{\lambda_i}$.

1. The concatenated list $v_{1,1}, \dots, v_{1,d_1}, v_{2,1}, \dots, v_{r,d_r}$ is linearly independent, and $d_1 + \cdots + d_r \leq n$.

2. A is diagonalizable over k if and only if $d_1 + \cdots + d_r = n$. In that case the matrix P whose columns are the vectors of that list, in that order, is invertible and $P^{-1}AP$ is the diagonal matrix whose jth diagonal entry is the eigenvalue of the jth column, so that λ_i occupies d_i consecutive diagonal positions.

***Proof*:**

1. Suppose $\sum_{i=1}^{r} \sum_{s=1}^{d_i} c_{i,s} v_{i,s} = 0$ for scalars $c_{i,s} \in k$, and set $w_i = \sum_{s=1}^{d_i} c_{i,s} v_{i,s}$ for each i. Each w_i lies in E_{λ_i}, which is a subspace, and $w_1 + \cdots + w_r = 0$. Let $S = \{\, i \,|\, w_i \neq 0 \,\}$. For $i \in S$ the vector w_i is a nonzero element of E_{λ_i}, hence an eigenvector of A for λ_i (definition 4.1.3), and the eigenvalues λ_i with $i \in S$ are pairwise distinct, so the vectors w_i with $i \in S$ are linearly independent by theorem 4.2.3. But $\sum_{i \in S} w_i = 0$ is a vanishing linear combination of them in which every coefficient equals 1, so S must be empty, and $w_i = 0$ for every i. Since $v_{i,1}, \dots, v_{i,d_i}$ is a basis of E_{λ_i} and hence linearly independent (definition 2.2.1), $w_i = 0$ forces $c_{i,1} = \cdots = c_{i,d_i} = 0$. Every coefficient vanishes, so the concatenated list is linearly independent. The standard basis $e_1, \dots, e_n$ spans k^n with n vectors, so by lemma 2.2.7 no linearly independent list in k^n has more than n entries, giving $d_1 + \cdots + d_r \leq n$.

2. Suppose first that $d_1 + \cdots + d_r = n$. By part (1) the concatenated list is independent and has n entries, so it is a basis of k^n by corollary 2.2.13. Each of its entries is a nonzero element of some E_{λ_i} and hence an eigenvector of A for λ_i, so theorem 4.2.2 applies: the matrix P built from the list is invertible and $P^{-1}AP$ is the diagonal matrix carrying the eigenvalue of the jth column in the jth diagonal position. Because the list was concatenated eigenspace by eigenspace, that diagonal is λ_1 repeated d_1 times, then λ_2 repeated d_2 times, and so on. In particular A is diagonalizable.

   Conversely, suppose A is diagonalizable. By theorem 4.2.2 there is a basis $u_1, \dots, u_n$ of k^n consisting of eigenvectors of A. Let μ_j be the eigenvalue of u_j; it is unique, since $\mu u_j = \mu' u_j$ with $u_j \neq 0$ gives $(\mu - \mu')u_j = 0$ and hence $\mu = \mu'$. Each μ_j is an eigenvalue of A lying in k, so $\mu_j = \lambda_{i(j)}$ for exactly one index $i(j)$, and $u_j \in E_{\lambda_{i(j)}}$. Write n_i for the number of indices j with $i(j) = i$; then $n_1 + \cdots + n_r = n$. Fix i. The vectors u_j with $i(j) = i$ form a sublist of the basis $u_1, \dots, u_n$ and are therefore linearly independent, because a vanishing combination of the sublist becomes a vanishing combination of the whole list once the remaining entries are given the coefficient 0. Those n_i independent vectors lie in E_{λ_i}, which is spanned by the d_i vectors $v_{i,1}, \dots, v_{i,d_i}$, so $n_i \leq d_i$ by lemma 2.2.7. Summing over i gives $n = n_1 + \cdots + n_r \leq d_1 + \cdots + d_r$, and part (1) gives the reverse inequality, so $d_1 + \cdots + d_r = n$, completing the proof.

Nothing in the assembly of P used the fact that the summands were eigenspaces beyond their being invariant, and recording the general statement now saves repeating the argument when the summands are larger. A direct sum of invariant subspaces turns A into a block diagonal matrix, one block per summand.

Proposition 4.2.6: An Invariant Direct Sum Gives a Block Diagonal Matrix

Let $A \in M_n(k)$ and suppose $k^n = W_1 \oplus \cdots \oplus W_m$ (definition 2.1.13) with each W_i invariant under A (definition 4.1.6) and $n_i = \dim W_i$. For each i choose a basis of W_i, and let $P \in M_n(k)$ be the matrix whose columns are those bases concatenated in the order $W_1, \ldots, W_m$. Then P is invertible and

$$P^{-1}AP = \begin{pmatrix} B_1 & 0 & \cdots & 0 \\ 0 & B_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & B_m \end{pmatrix}$$

where $B_i \in M_{n_i}(k)$ is the matrix of $A|_{W_i}$ in the chosen basis of W_i.

***Proof*:**
Write the chosen basis of W_i as $v_{i,1}, \ldots, v_{i,n_i}$, and let $u_1, \ldots, u_n$ denote the concatenated list, so that u_j is the jth column of P.

Step 1: the concatenated list is a basis of k^n. It spans, because every $x \in k^n$ is a sum $w_1 + \cdots + w_m$ with $w_i \in W_i$ (definition 2.1.13) and each w_i is a combination of the basis of W_i. For independence, suppose $\sum_{i=1}^{m} \sum_{s=1}^{n_i} c_{i,s} v_{i,s} = 0$ and put $w_i = \sum_{s=1}^{n_i} c_{i,s} v_{i,s} \in W_i$. Then $w_1 + \cdots + w_m = 0$, so every $w_i = 0$ by proposition 2.1.14, and since $v_{i,1}, \ldots, v_{i,n_i}$ is a basis of W_i it is independent, forcing $c_{i,1} = \cdots = c_{i,n_i} = 0$. Every coefficient vanishes, so the list is a basis. In particular it has n entries, so $n_1 + \cdots + n_m = n$ and P is invertible by proposition 2.3.13.

Step 2: the blocks.

Fix i and s, and consider the column $Av_{i,s}$. Since W_i is invariant, $Av_{i,s} \in W_i$, so it is a combination of the basis of W_i alone:

$$Av_{i,s} = \sum_{r=1}^{n_i} (B_i)_{rs}\, v_{i,r}$$

and the scalars $(B_i)_{rs}$ so determined are by definition the entries of the matrix B_i of $A|_{W_i}$ in that basis. The point is which coefficients are absent: no $v_{i',r}$ with $i' \neq i$ occurs, because $Av_{i,s}$ lies in W_i and the expression of a vector in the basis $u_1, \ldots, u_n$ is unique.

Now let j be the position of $v_{i,s}$ in the concatenated list, so $Pe_j = u_j = v_{i,s}$. Then $APe_j = Av_{i,s}$, and the displayed identity expresses $Av_{i,s}$ as a combination of exactly those u_r that belong to the ith block of positions, with coefficients $(B_i)_{rs}$. Since $P^{-1}u_r = e_r$, applying P^{-1} gives that $P^{-1}APe_j$ is the column whose entries are $(B_i)_{rs}$ in the ith block of positions and zero elsewhere. That column is the jth column of the asserted block diagonal matrix, and j was arbitrary, so the two matrices agree column by column.

Proposition 4.2.5 is the case in which every W_i is an eigenspace, where each B_i is $\lambda_i I_{n_i}$ by proposition 4.1.7 and the block diagonal matrix is diagonal. Chapter 12 uses the general form with summands on which A acts in a more complicated way than by a scalar.

The proposition is a procedure as well as a criterion: compute χ_A, find its roots in k, run elimination on $A - \lambda_i I_n$ for each root to get a basis of E_{λ_i}, add up the dimensions, and compare the total with n. The next example runs it in full on a 3×3 matrix, and then uses corollary 4.2.4 to reach the same verdict on a second matrix without computing a single eigenspace.

Example 4.2.7: Two Diagonalizable 3×3 Matrices

1. Let
$$C = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 2 & -4 & 2 \end{pmatrix} \in M_3(\mathbb{R})$$
The third column of $tI_3 - C$ has zeros in its first two entries, so the cofactor expansion of theorem 3.3.4 along that column has one surviving term, with sign $(-1)^{3+3} = 1$:
$$\chi_C(t) = \det \begin{pmatrix} t-1 & -2 & 0 \\ 0 & t-3 & 0 \\ -2 & 4 & t-2 \end{pmatrix} = (t-2)\det \begin{pmatrix} t-1 & -2 \\ 0 & t-3 \end{pmatrix} = (t-1)(t-2)(t-3)$$
The roots are 1, 2 and 3, all in $\mathbb{R}$, and these are the eigenvalues by theorem 4.1.10. Eliminating on $C - \lambda I_3$ for each in turn,
$$C - I_3 = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 2 & 0 \\ 2 & -4 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$
by $R_2 \mapsto R_2 - R_1$, then $R_1 \leftrightarrow R_3$ and $R_2 \leftrightarrow R_3$, then $R_1 \mapsto \frac{1}{2}R_1$ and $R_2 \mapsto \frac{1}{2}R_2$, then $R_1 \mapsto R_1 + 2R_2$. So x_3 is free, $x_2 = 0$ and $x_1 = -\frac{1}{2}x_3$; taking $x_3 = -2$ gives $E_1 = \operatorname{span}\{(1,0,-2)^\top\}$ and $d_1 = 1$. Next
$$C - 2I_3 = \begin{pmatrix} -1 & 2 & 0 \\ 0 & 1 & 0 \\ 2 & -4 & 0 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$
by $R_3 \mapsto R_3 + 2R_1$, then $R_1 \mapsto -R_1$, then $R_1 \mapsto R_1 + 2R_2$. So $x_1 = x_2 = 0$ with x_3 free, giving $E_2 = \operatorname{span}\{(0,0,1)^\top\}$ and $d_2 = 1$. Finally
$$C - 3I_3 = \begin{pmatrix} -2 & 2 & 0 \\ 0 & 0 & 0 \\ 2 & -4 & -1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 0 \end{pmatrix}$$
by $R_3 \mapsto R_3 + R_1$, then $R_2 \leftrightarrow R_3$, then $R_1 \mapsto -\frac{1}{2}R_1$ and $R_2 \mapsto -\frac{1}{2}R_2$, then $R_1 \mapsto R_1 + R_2$. So x_3 is free with $x_1 = x_2 = -\frac{1}{2}x_3$; taking $x_3 = -2$ gives $E_3 = \operatorname{span}\{(1,1,-2)^\top\}$ and $d_3 = 1$. The dimensions add to $1 + 1 + 1 = 3 = n$, so proposition 4.2.5(2) declares C diagonalizable over $\mathbb{R}$, with
$$P = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 1 \\ -2 & 1 & -2 \end{pmatrix}, \qquad P^{-1}CP = \operatorname{diag}(1,2,3)$$
The verification is the column identity from the proof of theorem 4.2.2: the columns of CP are $C(1,0,-2)^\top = (1,0,-2)^\top$, $C(0,0,1)^\top = (0,0,2)^\top$ and $C(1,1,-2)^\top = (3,3,-6)^\top$, which are 1, 2 and 3 times the corresponding columns of P, so $CP = P\operatorname{diag}(1,2,3)$.

2. Eigenvalues belong to square matrices: for the running matrix A of example 1.2.9, which is 4×3, the equation $Ax = \lambda x$ cannot hold at all, since Ax has four entries and λx has three. Its first three rows do form a square matrix,

$$B = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 1 & 2 & 5 \end{pmatrix} \in M_3(\mathbb{R})$$

and the column relation $c_3 = c_1 + 2c_2$ found in section 1.2 holds after the truncation:

$$B \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1+2-3 \\ 2+6-8 \\ 1+4-5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

so 0 is an eigenvalue of B with eigenvector $(1, 2, -1)^\top$. The first row of

$$tI_3 - B = \begin{pmatrix} t-1 & -1 & -3 \\ -2 & t-3 & -8 \\ -1 & -2 & t-5 \end{pmatrix}$$

is $(t-1, -1, -3)$, and the cofactor expansion of theorem 3.3.4 along it attaches the signs $+, -, +$ to the three minors, so

$$\begin{aligned} \chi_B(t) &= (t-1)\big[(t-3)(t-5) - (-8)(-2)\big] - (-1)\big[(-2)(t-5) - (-8)(-1)\big] \\ &\qquad + (-3)\big[(-2)(-2) - (t-3)(-1)\big] \\ &= (t-1)(t^2 - 8t - 1) + (-2t + 2) - 3(t+1) \\ &= t^3 - 9t^2 + 7t + 1 - 2t + 2 - 3t - 3 = t(t^2 - 9t + 2) \end{aligned}$$

The quadratic factor has discriminant $81 - 8 = 73 > 0$, so its roots $\frac{1}{2}(9 \pm \sqrt{73})$ are real and distinct, and neither is 0 because their product is 2. Thus χ_B has three distinct real roots, and corollary 4.2.4 shows that B is diagonalizable over $\mathbb{R}$. No eigenspace had to be computed to reach that conclusion.

Both matrices of example 4.2.7 turned out to be diagonalizable, and proposition 4.2.5(2) is an equivalence, so the matrices it rules out deserve to be seen as well. There are two ways for the dimension count to fall short of n: an eigenvalue can carry an eigenspace smaller than the number of times it occurs as a root of χ_A, or χ_A can fail to have enough roots in k at all. The next example realizes each.

Example 4.2.8: The Shear and the Quarter Turn

1. Let $N = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, which sends e_1 to e_1 and e_2 to $e_1 + e_2$. The matrix $tI_2 - N = \left(\begin{smallmatrix} t-1 & -1 \\ 0 & t-1 \end{smallmatrix}\right)$ is upper triangular, so $\chi_N(t) = (t-1)^2$ by proposition 3.2.6 together with the uniqueness clause of proposition 4.1.8, and the only root, over $\mathbb{R}$ and over $\mathbb{C}$ alike, is 1. Now $N - I_2 = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ is already in reduced echelon form, with its single pivot in column 2, so $(N - I_2)x = 0$ says $x_2 = 0$ with x_1 free and $E_1 = \operatorname{span}\{(1, 0)^\top\}$ has dimension 1. The eigenspace dimensions sum to 1, which is less than 2, so proposition 4.2.5(2) shows N is not diagonalizable over $\mathbb{R}$. Both computations are unchanged over $\mathbb{C}$: the characteristic polynomial is the same, its

only root is again 1, and the same reduced echelon form gives $\dim_{\mathbb{C}} E_1 = 1$. So N is not diagonalizable over $\mathbb{C}$ either.

2. Let $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, which sends e_1 to e_2 and e_2 to $-e_1$. Then

$$\chi_J(t) = \det \begin{pmatrix} t & 1 \\ -1 & t \end{pmatrix} = t^2 + 1$$

which has no real root, since $t^2 \geq 0$ for real t. By theorem 4.1.10, J has no eigenvalue in $\mathbb{R}$, hence no eigenvector in $\mathbb{R}^2$, so $\mathbb{R}^2$ has no basis of eigenvectors and theorem 4.2.2 shows J is not diagonalizable over $\mathbb{R}$. Over $\mathbb{C}$ the same polynomial factors as $t^2 + 1 = (t - i)(t + i)$, with the two distinct roots i and $-i$, so corollary 4.2.4 applies and J is diagonalizable over $\mathbb{C}$. Explicitly, the second row of $J - iI_2 = \left(\begin{smallmatrix} -i & -1 \\ 1 & -i \end{smallmatrix}\right)$ gives $x_1 = ix_2$, so $(i, 1)^\top$ is an eigenvector for i, and the second row of $J + iI_2 = \left(\begin{smallmatrix} i & -1 \\ 1 & i \end{smallmatrix}\right)$ gives $x_1 = -ix_2$, so $(-i, 1)^\top$ is an eigenvector for $-i$. The two products $J(i, 1)^\top = (-1, i)^\top = i \cdot (i, 1)^\top$ and $J(-i, 1)^\top = (-1, -i)^\top = -i \cdot (-i, 1)^\top$ confirm both, and with $P = \left(\begin{smallmatrix} i & -i \\ 1 & 1 \end{smallmatrix}\right)$ one gets $P^{-1}JP = \operatorname{diag}(i, -i)$.

The two obstructions behave differently under enlargement of the field. For J, passing from $\mathbb{R}$ to $\mathbb{C}$ gives the roots that were missing and the obstruction disappears. For N, passing to $\mathbb{C}$ changes neither the characteristic polynomial nor the reduced echelon form of $N - I_2$, so the eigenspace stays one-dimensional and no enlargement of the field repairs it: the eigenvalue is present and the eigenvectors are too few. The discrepancy is visible in the two numbers attached to the eigenvalue 1 of N, which occurs twice as a root of $\chi_N(t) = (t - 1)^2$ while $\dim E_1 = 1$, and it is that discrepancy which the dimension count of proposition 4.2.5 detects.

Exercise 4.2.1

1. Let $A = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix} \in M_2(\mathbb{R})$. Compute χ_A, find an eigenvector for each eigenvalue, and produce an invertible P and a diagonal D with $P^{-1}AP = D$. Verify the answer by computing AP and PD.

2. Let $M = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{pmatrix} \in M_3(\mathbb{R})$. Show that M is diagonalizable over $\mathbb{R}$ and exhibit P and D, even though χ_M has a repeated root.

3. Let $M = \begin{pmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ -1 & 1 & 2 \end{pmatrix} \in M_3(\mathbb{R})$. Show that M is diagonalizable neither over $\mathbb{R}$ nor over $\mathbb{C}$.

4. Let $A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$. Show that A is not diagonalizable over $\mathbb{R}$, that it is diagonalizable over $\mathbb{C}$, and exhibit a complex P with $P^{-1}AP$ diagonal.

5. Let $A \in M_n(k)$ and suppose $P^{-1}AP = D = \operatorname{diag}(d_1, \ldots, d_n)$ for some invertible P. Show that a scalar $\lambda \in k$ is an eigenvalue of A if and only if $\lambda = d_j$ for some j. Deduce that a diagonalizable matrix with exactly one eigenvalue λ equals λI_n, and use this to give a second proof that the matrix N of example 4.2.8(1) is not diagonalizable.

6. Let $A \in M_n(k)$ be diagonalizable and invertible, say $P^{-1}AP = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. Show that every λ_j is nonzero, that A^{-1} is diagonalizable, and that the same P works, with $P^{-1}A^{-1}P = \operatorname{diag}(\lambda_1^{-1}, \dots, \lambda_n^{-1})$.

7. Let $A \in M_n(k)$ satisfy $A^2 = A$, with $A \neq 0$ and $A \neq I_n$. Show that the eigenvalues of A are exactly 0 and 1, and that A is diagonalizable with $P^{-1}AP = \operatorname{diag}(0, \dots, 0, 1, \dots, 1)$ for a suitable P. (Begin from the identity $v = (v - Av) + Av$.)

4.3 Algebraic and Geometric Multiplicity

Section 4.2 decided diagonalizability by searching for a basis of eigenvectors. That criterion does not say how many independent eigenvectors a given eigenvalue can contribute, nor how many it would have to contribute for the search to succeed. Two integers attached to each eigenvalue answer both questions. One is the dimension of its eigenspace. The other is the number of times its linear factor occurs in the characteristic polynomial. This section computes both, proves that the first never exceeds the second, and turns the comparison into a test that runs off a factored characteristic polynomial and a rank.

The first of the two counts is a dimension, so it is computed by elimination once the eigenspace is recognized as the null space of a matrix built from A.

By proposition 4.1.5 the eigenspace is the null space $E_\lambda = \operatorname{null}(A - \lambda I_n)$, a subspace of k^n of dimension $n - \operatorname{rank}(A - \lambda I_n)$. That identification is what makes the geometric multiplicity computable.

Both quantities on the right are produced by elimination. Row reduce $A - \lambda I_n$ to echelon form, count the pivots to get the rank (definition 1.3.2), and subtract from n. The next computation does this for both eigenvalues of a 3×3 matrix, one of which is a repeated root of the characteristic polynomial.

Example 4.3.1: A Repeated Eigenvalue with a Plane of Eigenvectors

Let

$$B = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{pmatrix} \in M_3(\mathbb{R})$$

The first row of

$$tI_3 - B = \begin{pmatrix} t-3 & 0 & 0 \\ 0 & t-1 & -2 \\ 0 & -2 & t-1 \end{pmatrix}$$

has a single nonzero entry, so cofactor expansion along that row (theorem 3.3.4) leaves one term:

$$\chi_B(t) = (t-3)\big((t-1)^2 - 4\big) = (t-3)(t^2 - 2t - 3) = (t-3)^2(t+1)$$

The roots of χ_B are 3 and -1, so these are the eigenvalues of B (theorem 4.1.10).

For $\lambda = 3$,

$$B - 3I_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 2 & -2 \end{pmatrix}$$

and the operations $R_3 \mapsto R_3 + R_2$, then $R_2 \mapsto -\frac{1}{2}R_2$, then $R_1 \leftrightarrow R_2$ bring this to

$$\begin{pmatrix} 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

which is in reduced echelon form with a single pivot, in column 2. Hence $\operatorname{rank}(B - 3I_3) = 1$ and $\dim E_3 = 3 - 1 = 2$ by proposition 4.1.5. The surviving equation is $y = z$, with x and z free, so

$$E_3 = \operatorname{span}\{(1, 0, 0)^\top, (0, 1, 1)^\top\}$$

For $\lambda = -1$,

$$B + I_3 = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 2 \end{pmatrix}$$

and the operations $R_3 \mapsto R_3 - R_2$, then $R_1 \mapsto \frac{1}{4}R_1$, then $R_2 \mapsto \frac{1}{2}R_2$ bring this to

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

with pivots in columns 1 and 2. Hence $\operatorname{rank}(B + I_3) = 2$ and $\dim E_{-1} = 3 - 2 = 1$, with the equations $x = 0$ and $y = -z$ giving

$$E_{-1} = \operatorname{span}\{(0, 1, -1)^\top\}$$

The factor $t - 3$ occurs twice in χ_B and E_3 has dimension 2; the factor $t + 1$ occurs once and E_{-1} has dimension 1. The two counts agree at each eigenvalue, and $2 + 1 = 3$ independent directions of eigenvectors are available inside $\mathbb{R}^3$.

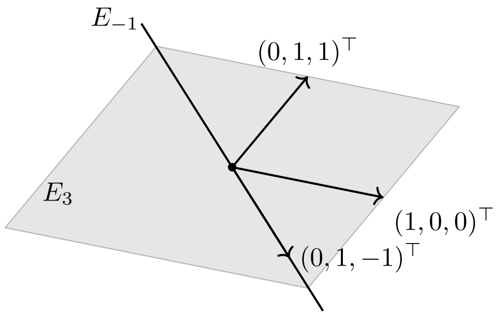


Figure 4.4: The eigenspaces of B: a plane of eigenvectors for $\lambda = 3$, and a line of eigenvectors for $\lambda = -1$

The computation produced two numbers at each eigenvalue, and they are of different kinds. One is the dimension of a subspace of k^n, read off from a rank. The other is an exponent in a factorization of χ_A, read off from a polynomial. Each gets a name.

Definition 4.3.2: Geometric Multiplicity

Let $A \in M_n(k)$ and let $\lambda \in k$ be an eigenvalue of A. The *geometric multiplicity* of λ is $\dim E_\lambda$, the dimension of its eigenspace.

Naming the exponent takes one preliminary fact about polynomials.

Lemma 4.3.3: The Degree of a Product

Let p and q be nonzero polynomials with coefficients in k. Then pq is not the zero polynomial, and

$$\deg(pq) = \deg p + \deg q$$

***Proof*:**
Write at^d for the leading term of p and bt^e for the leading term of q, so that $d = \deg p$ and $e = \deg q$ and the scalars a and b are nonzero. The coefficient of t^f in the product pq is the sum of the products $p_i q_j$ over all $i + j = f$, where p_i and q_j are the coefficients of t^i in p and of t^j in q. For $f = d + e$ the only term with both factors nonzero is ab, and for $f > d + e$ every term has $i > d$ or $j > e$ and so vanishes. Since a and b are nonzero scalars, ab is nonzero. Hence pq is not the zero polynomial, its leading term is abt^{d+e}, and $\deg(pq) = d + e$, completing the proof.

Now fix $\lambda \in k$ and consider the integers $m \geq 0$ for which $\chi_A(t) = (t - \lambda)^m q(t)$ holds for some polynomial q with coefficients in k. The value $m = 0$ qualifies, with $q = \chi_A$. If m qualifies with witness q, then q is nonzero, since χ_A is monic of degree n and hence not the zero polynomial (proposition 4.1.8); repeated use of lemma 4.3.3 then gives $n = m + \deg q$, so $m \leq n$. The qualifying integers form a nonempty set of integers between 0 and n, so they have a largest member.

Definition 4.3.4: Algebraic Multiplicity

Let $A \in M_n(k)$ and let $\lambda \in k$. The *algebraic multiplicity* of λ is the largest integer $m \geq 0$ for which

$$\chi_A(t) = (t - \lambda)^m q(t)$$

holds for some polynomial q with coefficients in k.

The definition attaches a number to every scalar, not only to every eigenvalue, and a scalar that is not an eigenvalue receives 0: an equation $\chi_A(t) = (t - \lambda)q(t)$ evaluated at $t = \lambda$ would give $\chi_A(\lambda) = 0$, which by theorem 4.1.10 makes λ an eigenvalue. That an eigenvalue receives at least 1 is part of theorem 4.3.7 below.

Asking for a largest exponent is not something a computation produces directly. In practice χ_A comes already factored, and the exponent can then be read off, provided the factor left over does not itself vanish at λ.

Proposition 4.3.5: Algebraic Multiplicity from a Factorization

Let $A \in M_n(k)$, let $\lambda \in k$, and suppose that

$$\chi_A(t) = (t-\lambda)^c q(t)$$

for an integer $c \geq 0$ and a polynomial q with coefficients in k satisfying $q(\lambda) \neq 0$. Then the algebraic multiplicity of λ is c.

***Proof*:**
The displayed factorization shows that c qualifies in definition 4.3.4, so the algebraic multiplicity m of λ satisfies $m \geq c$. Suppose $m \geq c+1$. Then $\chi_A(t) = (t-\lambda)^m h(t)$ for some polynomial h with coefficients in k, and putting $u(t) = q(t) - (t-\lambda)^{m-c}h(t)$ gives

$$(t-\lambda)^c u(t) = (t-\lambda)^c q(t) - (t-\lambda)^m h(t) = \chi_A(t) - \chi_A(t) = 0$$

If u were nonzero, then $(t-\lambda)^c u(t)$ would be a product of two nonzero polynomials, and lemma 4.3.3 makes such a product nonzero. So u is the zero polynomial, that is, $q(t) = (t-\lambda)^{m-c}h(t)$. Evaluating at $t = \lambda$ and using $m - c \geq 1$ gives $q(\lambda) = 0$, contradicting the hypothesis. Therefore $m = c$, as we sought to show.

Applied to example 4.3.1, where $\chi_B(t) = (t-3)^2(t+1)$, the proposition settles both exponents. Taking $q(t) = t+1$, which has $q(3) = 4 \neq 0$, gives algebraic multiplicity 2 for the eigenvalue 3; taking $q(t) = (t-3)^2$, which has $q(-1) = 16 \neq 0$, gives algebraic multiplicity 1 for the eigenvalue -1. The geometric multiplicities computed there were 2 and 1, so the two agree at both eigenvalues of B.

Nothing so far forces such an agreement. The characteristic polynomial is computed from A before any eigenspace is, and a different matrix with the same characteristic polynomial can have smaller eigenspaces.

Example 4.3.6: A Repeated Eigenvalue with a Line of Eigenvectors
Let

$$C = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -1 \end{pmatrix} \in M_3(\mathbb{R})$$

The matrix $tI_3 - C$ is upper triangular with diagonal entries $t-3$, $t-3$ and $t+1$, so its determinant is the product of those entries (proposition 3.2.6):

$$\chi_C(t) = (t-3)^2(t+1)$$

This is the characteristic polynomial of B in example 4.3.1, so the eigenvalues are again 3 and -1, with algebraic multiplicities 2 and 1 by proposition 4.3.5.

For $\lambda = 3$,

$$C - 3I_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -4 \end{pmatrix}$$

and the operations $R_2 \leftrightarrow R_3$, then $R_2 \mapsto -\frac{1}{4}R_2$, bring this to

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

with pivots in columns 2 and 3. Hence $\operatorname{rank}(C - 3I_3) = 2$ and $\dim E_3 = 3 - 2 = 1$, the equations being $y = 0$ and $z = 0$ with x free, so $E_3 = \operatorname{span}\{(1,0,0)^\top\}$.

For $\lambda = -1$,

$$C + I_3 = \begin{pmatrix} 4 & 1 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and the operations $R_1 \mapsto \frac{1}{4}R_1$, then $R_2 \mapsto \frac{1}{4}R_2$, then $R_1 \mapsto R_1 - \frac{1}{4}R_2$, bring this to

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

so $\operatorname{rank}(C + I_3) = 2$ and $\dim E_{-1} = 1$, with $E_{-1} = \operatorname{span}\{(0,0,1)^\top\}$.

The eigenvalue 3 has algebraic multiplicity 2 and geometric multiplicity 1. The matrix C is not diagonalizable, and this can be seen directly: every eigenvector of C lies in E_3 or in E_{-1}, hence is a nonzero multiple of $(1,0,0)^\top$ or of $(0,0,1)^\top$. Any three eigenvectors therefore include two, say au and bu with a, b and u nonzero, that are multiples of the same one of these two vectors; the relation $b(au) - a(bu) = 0$ makes those two dependent, and giving the third vector coefficient 0 extends it to a nontrivial relation among all three. So $\mathbb{R}^3$ has no basis of eigenvectors of C, and C is not diagonalizable by theorem 4.2.2.

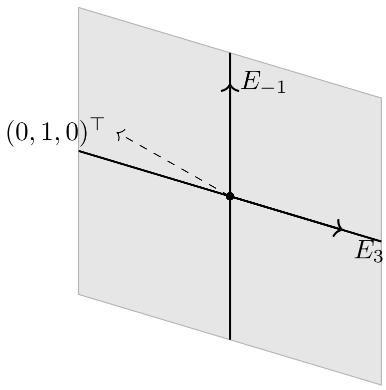


Figure 4.5: The eigenspaces of C: two lines, whose span is a plane that $(0,1,0)^\top$ lies outside

The two matrices B and C have the same characteristic polynomial and different eigenspace dimensions, so the characteristic polynomial alone does not determine the geometric multiplicities. What it does determine is a ceiling for them. The proof of that reads the ceiling off a change of basis: a basis of the eigenspace can be made the first block of a basis of k^n, and in the resulting coordinates the factor $t-\lambda$ appears g times by inspection.

Theorem 4.3.7: Geometric Multiplicity Is at Most Algebraic Multiplicity

Let $A \in M_n(k)$ and let $\lambda \in k$ be an eigenvalue of A with geometric multiplicity g. If $g = n$, then $A = \lambda I_n$ and $\chi_A(t) = (t-\lambda)^n$. If $g < n$, then there is a matrix $A' \in M_{n-g}(k)$ with

$$\chi_A(t) = (t-\lambda)^g \chi_{A'}(t)$$

In both cases the algebraic multiplicity of λ is at least g, and in particular every eigenvalue has algebraic multiplicity at least 1.

***Proof*:**
Case $g = n$. A basis of E_λ then consists of n linearly independent vectors of k^n, so it is a basis of k^n by corollary 2.2.13, and E_λ, containing the span of a basis of k^n, is all of k^n. Hence $Ae_j = \lambda e_j$ for every j. The vector Ae_j is the j-th column of A and λe_j is the j-th column of λI_n (definition 1.4.1), so $A = \lambda I_n$. Then $tI_n - A = (t-\lambda)I_n$ is diagonal, and proposition 3.2.6 gives $\chi_A(t) = (t-\lambda)^n$.

Case $g < n$. Choose a basis $v_1, \ldots, v_g$ of E_λ (theorem 2.2.9) and extend it to a basis $v_1, \ldots, v_n$ of k^n, which theorem 2.2.9(1) gives, applied to the subspace k^n: every linearly independent list of vectors of k^n extends to a basis of k^n, and every such basis has n vectors by theorem 2.2.8 (corollary 2.2.12), so the lengthening halts, and it halts only at a list that spans. Let P be the matrix whose j-th column is v_j. Its columns form a basis of k^n, so P is the change-of-basis matrix from that basis to the standard basis and is invertible (definition 2.5.2) and invertible by proposition 2.5.3, and $Pe_j = v_j$ for every j (definition 1.4.1).

For $j \leq g$ the vector v_j lies in E_λ, so

$$P^{-1}APe_j = P^{-1}Av_j = P^{-1}(\lambda v_j) = \lambda P^{-1}v_j = \lambda e_j$$

Thus the first g columns of $P^{-1}AP$ are $\lambda e_1, \ldots, \lambda e_g$. Writing the remaining $n-g$ columns as a $g \times (n-g)$ block Y stacked on an $(n-g) \times (n-g)$ block A',

$$P^{-1}AP = \begin{pmatrix} \lambda I_g & Y \\ 0 & A' \end{pmatrix}$$

Since A and $P^{-1}AP$ are similar (definition 2.5.5), they have the same characteristic polynomial (corollary 4.1.16). The matrix

$$tI_n - P^{-1}AP = \begin{pmatrix} (t-\lambda)I_g & -Y \\ 0 & tI_{n-g} - A' \end{pmatrix}$$

has entries that are polynomials in t, and its determinant is $\chi_{P^{-1}AP}(t)$. It is block upper triangular, so theorem 3.5.2 gives

$$\chi_A(t) = \det\big((t-\lambda)I_g\big) \cdot \det\big(tI_{n-g} - A'\big) = (t-\lambda)^g \chi_{A'}(t)$$

the first determinant being that of a diagonal matrix (proposition 3.2.6).

In both cases $\chi_A(t) = (t-\lambda)^g q(t)$ for a polynomial q with coefficients in k, namely $q = 1$ in the first case and $q = \chi_{A'}$ in the second. So g qualifies in definition 4.3.4 and the algebraic multiplicity of λ is at least g. An eigenvalue has an eigenvector, which is nonzero, so its eigenspace has dimension at least 1 and $g \geq 1$, completing the proof.

The inequality bounds each eigenspace by the exponent of its factor. An eigenvalue whose factor occurs m times in χ_A gives at most m independent eigenvectors, and example 4.3.6 shows it may give fewer. A repeated root is therefore where diagonalizability can fail, and it is the only place it can fail once the characteristic polynomial factors into linear factors.

Counting the eigenvectors available in total requires knowing that eigenvalues do not compete for the same directions. Eigenvectors for distinct eigenvalues are linearly independent (theorem 4.2.3), and the next proof upgrades that to the statement that bases of the separate eigenspaces concatenate to an independent list. The number of independent eigenvectors available is then exactly the sum of the geometric multiplicities, and diagonalizability is the statement that this sum reaches n.

Theorem 4.3.8: Diagonalizability and the Sum of the Geometric Multiplicities

Let $A \in M_n(k)$ and let $\lambda_1, \ldots, \lambda_r$ be its distinct eigenvalues in k, with geometric multiplicities $g_1, \ldots, g_r$ and algebraic multiplicities $m_1, \ldots, m_r$. Then $g_1 + \cdots + g_r \leq n$, and A is diagonalizable over k if and only if

$$g_1 + \cdots + g_r = n$$

When this holds, $m_i = g_i$ for every i and

$$\chi_A(t) = (t - \lambda_1)^{m_1} \cdots (t - \lambda_r)^{m_r}$$

***Proof*:**
Step 1: bases of the separate eigenspaces concatenate to an independent list. For each i choose a basis $v_{i,1}, \ldots, v_{i,g_i}$ of E_{λ_i} (theorem 2.2.9), and consider a relation

$$\sum_{i=1}^{r} \sum_{j=1}^{g_i} c_{ij} v_{i,j} = 0$$

among the $g_1 + \cdots + g_r$ vectors so listed. Put $w_i = \sum_{j=1}^{g_i} c_{ij} v_{i,j}$, so that $w_i \in E_{\lambda_i}$, the eigenspace being a subspace (proposition 4.1.5), and $w_1 + \cdots + w_r = 0$. Suppose some w_i is nonzero, and let S be the set of indices i with $w_i \neq 0$. For $i \in S$ the vector w_i is a nonzero member of E_{λ_i}, hence an eigenvector of A for λ_i, and the λ_i with $i \in S$ are distinct, so the vectors w_i with $i \in S$ are linearly independent (theorem 4.2.3). But $\sum_{i \in S} w_i = 0$ is a relation among them with every coefficient equal to 1 and S nonempty, which contradicts that independence. So $w_i = 0$ for every i, and the independence of $v_{i,1}, \ldots, v_{i,g_i}$ inside E_{λ_i} forces $c_{ij} = 0$ for all i and j. The concatenated list is therefore linearly independent, its span is a subspace of k^n of dimension $g_1 + \cdots + g_r$, and corollary 2.2.12 gives $g_1 + \cdots + g_r \leq n$.

Step 2: the criterion. Suppose $g_1 + \cdots + g_r = n$. Step 1 then produces n linearly independent vectors in k^n, each a nonzero member of some E_{λ_i} and hence an eigenvector of A. They form a basis of k^n by corollary 2.2.13, so A is diagonalizable by theorem 4.2.2.

Conversely, suppose A is diagonalizable, and let $u_1, \ldots, u_n$ be a basis of k^n consisting of eigenvectors of A (theorem 4.2.2). Each u_j satisfies $Au_j = \mu_j u_j$ for exactly one scalar μ_j: if $\mu u_j = \mu' u_j$ then $(\mu - \mu')u_j = 0$, and $u_j \neq 0$ forces $\mu = \mu'$. Each μ_j is an eigenvalue of A and so equals λ_i for exactly one i. Let c_i be the number of indices j with $\mu_j = \lambda_i$, so that $c_1 + \cdots + c_r = n$. The c_i vectors u_j with $\mu_j = \lambda_i$ lie in E_{λ_i} and are linearly independent, since a relation among them becomes a relation among $u_1, \ldots, u_n$ once the remaining vectors are given coefficient 0. Their span is a subspace of E_{λ_i} of dimension c_i, so $c_i \le g_i$ by corollary 2.2.12. Combining with Step 1,

$$n = c_1 + \cdots + c_r \le g_1 + \cdots + g_r \le n$$

so both inequalities are equalities. In particular $g_1 + \cdots + g_r = n$, and since $c_i \le g_i$ holds for each i with the two sums equal, $c_i = g_i$ for every i.

Step 3: the multiplicities agree. Assume $g_1 + \cdots + g_r = n$ and let $u_1, \ldots, u_n$ be the basis of eigenvectors assembled in Step 2 from the bases of the eigenspaces, in which exactly g_i of the vectors lie in E_{λ_i}. By proposition 4.2.5, $P^{-1}AP = D$, where P has columns $u_1, \ldots, u_n$ and $D = \operatorname{diag}(\mu_1, \ldots, \mu_n)$ with $Au_j = \mu_j u_j$; the value λ_i occurs exactly g_i times among $\mu_1, \ldots, \mu_n$. By corollary 4.1.16, $\chi_A = \chi_D$, and $tI_n - D$ is diagonal with entries $t - \mu_1, \ldots, t - \mu_n$, so proposition 3.2.6 gives

$$\chi_A(t) = \prod_{j=1}^{n} (t - \mu_j) = (t - \lambda_1)^{g_1} \cdots (t - \lambda_r)^{g_r}$$

Fix i and set $q(t) = \prod_{l \ne i} (t - \lambda_l)^{g_l}$. Then $q(\lambda_i) = \prod_{l \ne i} (\lambda_i - \lambda_l)^{g_l}$ is a product of nonzero scalars, the eigenvalues being distinct, so $q(\lambda_i) \ne 0$. Applying proposition 4.3.5 to $\chi_A(t) = (t - \lambda_i)^{g_i} q(t)$ gives $m_i = g_i$, and the displayed factorization is therefore $\chi_A(t) = (t - \lambda_1)^{m_1} \cdots (t - \lambda_r)^{m_r}$, completing the proof.

The criterion is now a finite computation. Factor χ_A over k, list the distinct roots, compute $n - \operatorname{rank}(A - \lambda_i I_n)$ at each, and add. The sum reaches n exactly when A is diagonalizable over k. For B of example 4.3.1 the sum is $2 + 1 = 3$, so B is diagonalizable; for C of example 4.3.6 it is $1 + 1 = 2$, so C is not, which is the conclusion reached there by hand.

Over $\mathbb{R}$ the sum can fall short of n for two separate reasons. The characteristic polynomial may fail to factor into linear factors at all, as for the rotation of example 4.1.13, or it may factor while some eigenspace is smaller than its exponent permits, as for C. Over $\mathbb{C}$ the first reason disappears, and the criterion collapses to a comparison of the two multiplicities at each eigenvalue.

Corollary 4.3.9: Multiplicities over the Complex Numbers

Let $A \in M_n(\mathbb{C})$ with $n \ge 1$, let $\lambda_1, \ldots, \lambda_r$ be its distinct eigenvalues, and let m_i and g_i be the algebraic and geometric multiplicities of λ_i. Then

1. $\chi_A(t) = (t - \lambda_1)^{m_1} \cdots (t - \lambda_r)^{m_r}$, and $m_1 + \cdots + m_r = n$;
2. A is diagonalizable over $\mathbb{C}$ if and only if $g_i = m_i$ for every i.

***Proof*:**

1. The first claim is that χ_A is a product of n linear factors, and this is proved by induction on n. For $n = 1$ the matrix is $A = (a)$ and $\chi_A(t) = t - a$. Let $n \ge 2$ and assume the claim for every complex square matrix of size between 1 and $n - 1$. By theorem 4.1.14 the matrix A has an eigenvalue $\mu \in \mathbb{C}$; let g be its geometric multiplicity. If $g = n$, then theorem 4.3.7 gives $\chi_A(t) = (t - \mu)^n$, a product of n linear factors. If $g < n$, the same theorem gives $A' \in M_{n-g}(\mathbb{C})$ with $\chi_A(t) = (t - \mu)^g \chi_{A'}(t)$, and $1 \le n - g \le n - 1$, so the inductive hypothesis writes $\chi_{A'}$ as a product of $n - g$ linear factors; multiplying by $(t - \mu)^g$

writes χ_A as a product of $g + (n - g) = n$ linear factors.

Write that factorization as $\chi_A(t) = (t - \mu_1) \cdots (t - \mu_n)$ and collect equal values among $\mu_1, \ldots, \mu_n$: this gives distinct scalars $\nu_1, \ldots, \nu_s \in \mathbb{C}$ and integers $c_1, \ldots, c_s \geq 1$ with $c_1 + \cdots + c_s = n$ and

$$\chi_A(t) = (t - \nu_1)^{c_1} \cdots (t - \nu_s)^{c_s}$$

Each ν_l satisfies $\chi_A(\nu_l) = 0$ and so is an eigenvalue of A (theorem 4.1.10). Conversely, if λ is an eigenvalue then $\chi_A(\lambda) = 0$, and a product of nonzero complex numbers is nonzero, so $\lambda - \nu_l = 0$ for some l. The scalars $\nu_1, \ldots, \nu_s$ are therefore exactly the eigenvalues $\lambda_1, \ldots, \lambda_r$; renumber so that $s = r$ and $\nu_i = \lambda_i$. Fixing i and setting $q(t) = \prod_{l \neq i}(t - \lambda_l)^{c_l}$, the value $q(\lambda_i)$ is a product of nonzero scalars and hence nonzero, so proposition 4.3.5 gives $m_i = c_i$. Thus $\chi_A(t) = (t - \lambda_1)^{m_1} \cdots (t - \lambda_r)^{m_r}$ and $m_1 + \cdots + m_r = c_1 + \cdots + c_s = n$.

2. By theorem 4.3.7, $g_i \leq m_i$ for every i. If $g_i = m_i$ for every i, then $g_1 + \cdots + g_r = m_1 + \cdots + m_r = n$ by part (1), so A is diagonalizable by theorem 4.3.8. Conversely, if A is diagonalizable, then $g_1 + \cdots + g_r = n$ by theorem 4.3.8, and $m_1 + \cdots + m_r = n$ by part (1), so the two sums are equal; since $g_i \leq m_i$ holds termwise, a strict inequality at a single index would make the first sum the smaller of the two, so $g_i = m_i$ for every i, as we sought to show.

Over $\mathbb{C}$ the algebraic multiplicities always sum to n, so the characteristic polynomial always offers exactly n eigenvector directions and the only way to fall short is for some eigenspace to be too small. The test is two computations per eigenvalue: the exponent of $t - \lambda$ in the factored χ_A, and the number $n - \operatorname{rank}(A - \lambda I_n)$. The matrix is diagonalizable over $\mathbb{C}$ exactly when these agree at every root.

Exercise 4.3.1

1. Let
$$M = \begin{pmatrix} 4 & 0 & 1 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} \in M_3(\mathbb{R})$$
Compute χ_M, find the eigenvalues of M, and compute the algebraic and geometric multiplicity of each. Decide whether M is diagonalizable over $\mathbb{R}$.

2. Let $A' \in M_3(\mathbb{R})$ be the matrix formed by the first three rows of the running matrix A of example 1.2.9, whose characteristic polynomial $\chi_{A'}(t) = t^3 - 9t^2 + 2t$ was computed in example 4.2.7(2). Compute the algebraic and geometric multiplicity of the eigenvalue 0, and decide whether A' is diagonalizable over $\mathbb{R}$.

3. Let
$$N = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \in M_3(\mathbb{R})$$
Compute both multiplicities at every eigenvalue of N, and decide whether N is diagonalizable over $\mathbb{R}$.

4. For which real numbers a is
$$\begin{pmatrix} 3 & a \\ 0 & 3 \end{pmatrix} \in M_2(\mathbb{R})$$

diagonalizable over $\mathbb{R}$?

5. Let $A \in M_n(k)$ and let $\lambda \in k$ be an eigenvalue of A. Show that the geometric multiplicity of λ equals n if and only if $A = \lambda I_n$.

6. Let $A, P \in M_n(k)$ with P invertible, and put $B = P^{-1}AP$. Write $E_\lambda(A)$ and $E_\lambda(B)$ for the eigenspaces of A and of B at a scalar λ. Show that A and B have the same eigenvalues, and that at each eigenvalue their algebraic multiplicities agree and their geometric multiplicities agree.

7. Let $A \in M_n(\mathbb{C})$ have exactly one eigenvalue λ. Show that A is diagonalizable over $\mathbb{C}$ if and only if $A = \lambda I_n$.

4.4 The Minimal Polynomial, Cayley-Hamilton, and the Diagonalizability Criterion

Section 4.3 decides diagonalizability by computing the dimension of an eigenspace, once for each eigenvalue. This section produces a second test. It attaches to a square matrix a single polynomial, built from the powers of that matrix by multiplication and addition alone, and diagonalizability is read off the way the polynomial factors.

Along the way the characteristic polynomial acquires a property its definition does not display. Evaluating it at the matrix itself, in place of the variable, gives the zero matrix. That is the Cayley-Hamilton theorem, and it is what keeps the new polynomial from having degree larger than n.

The relation between a matrix and its characteristic polynomial is already present in a 2×2 computation. Take

$$A_0 = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \in M_2(\mathbb{R})$$

and square it. By definition 1.4.1 the $(1,1)$ entry of A_0^2 is $2 \cdot 2 + 1 \cdot 1 = 5$, the $(1,2)$ entry is $2 \cdot 1 + 1 \cdot 2 = 4$, and the two remaining entries are $1 \cdot 2 + 2 \cdot 1 = 4$ and $1 \cdot 1 + 2 \cdot 2 = 5$, so

$$A_0^2 = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix}$$

The three matrices I_2, A_0 and A_0^2 satisfy a relation:

$$4A_0 - 3I_2 = \begin{pmatrix} 8 & 4 \\ 4 & 8 \end{pmatrix} - \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 5 & 4 \\ 4 & 5 \end{pmatrix} = A_0^2$$

so $A_0^2 - 4A_0 + 3I_2 = 0$. The coefficients $1, -4, 3$ are the coefficients of a polynomial, and the polynomial is one already attached to A_0: by definition 4.1.9,

$$\chi_{A_0}(t) = \det \begin{pmatrix} t-2 & -1 \\ -1 & t-2 \end{pmatrix} = (t-2)^2 - 1 = t^2 - 4t + 3$$

Whether every square matrix satisfies the relation that its own characteristic polynomial writes down is the question the Cayley-Hamilton theorem answers. Before it can be asked in general, the operation performed on A_0 needs a name.

Definition 4.4.1: Polynomial in a Matrix; Annihilating Polynomial

Let k be $\mathbb{R}$ or $\mathbb{C}$. A *polynomial over* k is a formal expression

$$p(t) = a_d t^d + a_{d-1} t^{d-1} + \cdots + a_1 t + a_0$$

with $a_0, \ldots, a_d \in k$; the a_i are its *coefficients*, and two polynomials are *equal* when their coefficients agree in every degree. For a number $x \in k$, the *value* $p(x) \in k$ is obtained by substituting x for t and computing in k, with $x^0 = 1$. For a matrix $A \in M_n(k)$, the matrix $p(A) \in M_n(k)$ is

$$p(A) = a_d A^d + a_{d-1} A^{d-1} + \cdots + a_1 A + a_0 I_n$$

where $A^0 = I_n$, so that the constant term contributes a multiple of the identity and not a multiple of a matrix of ones. The polynomial p *annihilates* A if $p(A) = 0$, the $n \times n$ matrix all of whose entries are zero; such a p is an *annihilating polynomial* for A.

Four conventions about polynomials are used throughout and are fixed here. If $a_d \neq 0$ then d is the *degree* $\deg p$ and a_d is the *leading coefficient*; a polynomial with leading coefficient 1 is *monic*; the zero polynomial, all of whose coefficients vanish, is assigned no degree. The *sum* of two polynomials adds coefficients in each degree, and the *product* of $f(t) = \sum_i a_i t^i$ and $g(t) = \sum_j b_j t^j$ is the polynomial whose coefficient in degree m is $\sum_{i+j=m} a_i b_j$. If f and g are both nonzero, so is fg, and $\deg(fg) = \deg f + \deg g$ by lemma 4.3.3. In particular a product of monic polynomials is monic. The product is commutative and associative: the formula $\sum_{i+j=m} a_i b_j$ is unchanged when f and g are interchanged, and for a third polynomial $h(t) = \sum_\ell c_\ell t^\ell$ the coefficient of t^m in each of $(fg)h$ and $f(gh)$ works out to $\sum_{i+j+\ell=m} a_i b_j c_\ell$. A product of several polynomials may therefore be written without brackets, and its factors may be reordered.

Polynomials are added and multiplied by rules involving only their coefficients, and evaluation at a fixed matrix respects both rules. Matrix multiplication is not commutative in general; two polynomials in the same matrix nonetheless commute. Both facts are used in every proof below.

Proposition 4.4.2: Evaluation Respects Sums and Products

Let f and g be polynomials over k, and let X be either a number in k or a matrix in $M_n(k)$. Then

$$(f+g)(X) = f(X) + g(X), \qquad (fg)(X) = f(X)g(X)$$

In particular $f(X)g(X) = g(X)f(X)$.

***Proof*:**
Write $f(t) = \sum_{i=0}^{d} a_i t^i$ and $g(t) = \sum_{j=0}^{e} b_j t^j$, and read X^0 as 1 when X is a number and as I_n when X is a matrix. The statement about sums is the definition of the sum of two polynomials read one degree at a time: the coefficient of t^m in $f+g$ is $a_m + b_m$, so $(f+g)(X) = \sum_m (a_m + b_m) X^m = \sum_m a_m X^m + \sum_m b_m X^m$.

For the product, expand and collect. Multiplication distributes over addition in both arguments, whether X is a number or a matrix, and a scalar factor may be moved out of either factor of a product, since by definition 1.4.1 each entry of a product is a sum of products of entries. Together with $X^i X^j = X^{i+j}$, which for matrices is proposition 1.4.3 applied $i+j-1$ times, this gives

$$f(X)g(X) = \Big(\sum_{i=0}^{d} a_i X^i\Big)\Big(\sum_{j=0}^{e} b_j X^j\Big) = \sum_{i=0}^{d}\sum_{j=0}^{e} a_i b_j X^{i+j} = \sum_{m=0}^{d+e}\Big(\sum_{i+j=m} a_i b_j\Big)X^m$$

the last step grouping the pairs (i, j) by the value of $i + j$. The final expression is $(fg)(X)$ by the definition of the product of two polynomials.

The product of polynomials is commutative, so $fg = gf$ and hence $f(X)g(X) = (fg)(X) = (gf)(X) = g(X)f(X)$, completing the proof.

Among the annihilating polynomials of A_0 there is one of smallest degree. A polynomial of degree 0 is a nonzero constant a_0, and $a_0 I_2 \neq 0$, so no constant annihilates. A monic polynomial of degree 1 is $t - c$, and $A_0 - cI_2 = 0$ would force the $(1, 2)$ entry of A_0 to be 0, which it is not. So $t^2 - 4t + 3$, already known to annihilate A_0, has the smallest degree available. That is the object the section is about.

Definition 4.4.3: Minimal Polynomial

Let $A \in M_n(k)$. A *minimal polynomial* of A is a monic polynomial μ over k such that $\mu(A) = 0$ and no monic polynomial of smaller degree annihilates A.

The definition describes a minimal polynomial without producing one, and it leaves open whether two different ones could exist. Both gaps close at once.

Theorem 4.4.4: Every Square Matrix Has Exactly One Minimal Polynomial

Let $A \in M_n(k)$. Then A has exactly one minimal polynomial, written μ_A, and $\deg \mu_A \geq 1$.

***Proof*:**
Step 1: some monic polynomial annihilates A. Consider the $n^2 + 1$ matrices

$$I_n,\ A,\ A^2,\ \ldots,\ A^{n^2}$$

of $M_n(k)$. Reading the n^2 entries of a matrix off in a fixed order, say row by row, produces from each $M \in M_n(k)$ a column $\sigma(M) \in k^{n^2}$. Addition of matrices and multiplication of a matrix by a scalar are performed entry by entry, so $\sigma(M + N) = \sigma(M) + \sigma(N)$ and $\sigma(cM) = c\,\sigma(M)$, and $\sigma(M) = 0$ holds exactly when $M = 0$. A relation $c_0 I_n + c_1 A + \cdots + c_{n^2} A^{n^2} = 0$ among these matrices is therefore the same thing as the relation $c_0\sigma(I_n) + \cdots + c_{n^2}\sigma(A^{n^2}) = 0$ among the corresponding $n^2 + 1$ columns of k^{n^2}.

Those columns are linearly dependent. If they were linearly independent, then, since they lie in $k^{n^2} = \operatorname{span}\{e_1, \ldots, e_{n^2}\}$ by example 2.2.2(1), the replacement lemma lemma 2.2.7 would give $n^2 + 1 \leq n^2$. So there are $c_0, \ldots, c_{n^2} \in k$, not all zero, with

$$c_0 I_n + c_1 A + c_2 A^2 + \cdots + c_{n^2} A^{n^2} = 0$$

Let d be the largest index with $c_d \neq 0$. Then $d \geq 1$: if d were 0 the relation would read $c_0 I_n = 0$ with $c_0 \neq 0$, forcing $I_n = 0$. Multiplying the relation by c_d^{-1} and dropping the terms above index

d, whose coefficients are zero, shows that the monic polynomial

$$p(t) = c_d^{-1}\big(c_0 + c_1 t + \cdots + c_{d-1}t^{d-1}\big) + t^d$$

satisfies $p(A) = 0$.

Step 2: a monic annihilating polynomial of least degree exists. The degrees of the monic polynomials annihilating A form a nonempty set of nonnegative integers, nonempty by Step 1, so it has a least element m; choose a monic μ of degree m with $\mu(A) = 0$. By definition 4.4.3 this μ is a minimal polynomial of A. The only monic polynomial of degree 0 is the constant 1, whose value at A is $I_n \neq 0$, so $m \geq 1$.

Step 3: uniqueness. Let μ and μ' both be minimal polynomials of A. Neither degree can be smaller than the other, by the defining property of each, so $\deg \mu = \deg \mu' = m$. Both are monic, so in the difference $h = \mu - \mu'$ the terms of degree m cancel and h is either 0 or of degree less than m. By proposition 4.4.2, $h(A) = \mu(A) - \mu'(A) = 0$. If h were nonzero, dividing it by its leading coefficient would produce a monic polynomial of degree less than m annihilating A, contradicting the minimality of m. Hence $h = 0$ and $\mu = \mu'$, as we sought to show.

Step 1 gives a crude bound as a by-product: the polynomial it produces has degree at most n^2, so $\deg \mu_A \leq n^2$. The true bound is n, and it comes from a different source. Reaching it requires comparing μ_A with other annihilating polynomials, and the comparison is made by dividing one polynomial by another with a remainder, exactly as one integer is divided by another. Such a division produces a quotient and a remainder whose degree is smaller than the divisor's, and it is the remainder that carries the comparison.

Lemma 4.4.5: Division with Remainder for Polynomials

Let f and g be polynomials over k with $g \neq 0$. Then there are polynomials q and r over k with

$$f = qg + r$$

and with $r = 0$ or $\deg r < \deg g$, and the pair (q, r) with these properties is unique.

***Proof*:**
Write $e = \deg g$ and let $b_e \neq 0$ be the leading coefficient of g.

Existence. The claim is that every f admits such a pair, and the proof is by strong induction on the degree of f. If $f = 0$, or if $\deg f < e$, take $q = 0$ and $r = f$; both requirements hold. Suppose instead that $f \neq 0$ with $\deg f = d \geq e$, let $a_d \neq 0$ be its leading coefficient, and suppose the claim holds for every polynomial that is zero or of degree less than d. Put

$$f_1 = f - \frac{a_d}{b_e} t^{d-e} g$$

The polynomial $\frac{a_d}{b_e} t^{d-e} g$ has degree $(d - e) + e = d$ and its coefficient in degree d is $\frac{a_d}{b_e} b_e = a_d$, which is the coefficient of f in degree d. Both polynomials being of degree d, the difference f_1 has zero coefficient in degree d and no coefficients above it, so $f_1 = 0$ or $\deg f_1 < d$. The inductive hypothesis gives q_1 and r with $f_1 = q_1 g + r$ and $r = 0$ or $\deg r < e$, and then

$$f = \frac{a_d}{b_e} t^{d-e} g + q_1 g + r = \Big(\frac{a_d}{b_e} t^{d-e} + q_1\Big) g + r$$

which is an expression of the required form.

Uniqueness. Suppose $qg + r = q'g + r'$ with each of r, r' either zero or of degree less than e. Rearranging gives $(q - q')g = r' - r$. If $q \neq q'$ then $q - q'$ is nonzero, so the left side is nonzero of degree $\deg(q - q') + e \geq e$. The right side is either zero or of degree less than e, since a difference of two polynomials each zero or of degree less than e is again zero or of degree less than e. These are incompatible, so $q = q'$, and then $r' - r = 0$, completing the proof.

Division by a polynomial of degree 1 leaves a constant remainder, and that single case carries three consequences used repeatedly below. The third of them is what licenses reading off coefficients from an identity between values, which is the step every appearance of a matrix with polynomial entries will need.

Corollary 4.4.6: Roots, Linear Factors, and Equality of Coefficients

Let p and q be polynomials over k.

1. If $\lambda \in k$ and $p(\lambda) = 0$, then $p(t) = (t - \lambda)h(t)$ for some polynomial h over k; and if $p \neq 0$ then $\deg h = \deg p - 1$.
2. If $p \neq 0$ and $\deg p = d$, then p has at most d roots in k, a root meaning a $\lambda \in k$ with $p(\lambda) = 0$.
3. If $p(x) = q(x)$ for every $x \in k$, then p and q have the same coefficients in every degree.

***Proof*:**

1. Apply lemma 4.4.5 with $g(t) = t - \lambda$, which is nonzero of degree 1: there are h and r with $p = h \cdot (t - \lambda) + r$ and $r = 0$ or $\deg r < 1$, so in either case r is a constant c. Taking values at λ and using proposition 4.4.2,
$$0 = p(\lambda) = h(\lambda)(\lambda - \lambda) + c = c$$
so $r = 0$ and $p = (t - \lambda)h$. If $p \neq 0$ then $h \neq 0$, and the degree of a product is the sum of the degrees, so $\deg p = \deg h + 1$.

2. Induct on d. If $d = 0$ then p is a nonzero constant and has no roots, and $0 \leq 0$. Let $d \geq 1$ and suppose the statement holds for nonzero polynomials of degree $d - 1$. If p has no root there is nothing to prove, since $0 \leq d$. Otherwise let λ be a root and write $p = (t - \lambda)h$ with h nonzero of degree $d - 1$, by part (1). If $\nu \in k$ is a root of p with $\nu \neq \lambda$, then $0 = p(\nu) = (\nu - \lambda)h(\nu)$ by proposition 4.4.2, and $\nu - \lambda$ is a nonzero number of k, so $h(\nu) = 0$. Every root of p other than λ is therefore a root of h, of which there are at most $d - 1$ by the inductive hypothesis. So p has at most $1 + (d - 1) = d$ roots.

3. Let $h = p - q$, the polynomial whose coefficients are the differences of those of p and q. By proposition 4.4.2, $h(x) = p(x) - q(x) = 0$ for every $x \in k$. Both $\mathbb{R}$ and $\mathbb{C}$ contain infinitely many numbers, so h has infinitely many roots, which by part (2) is impossible for a nonzero polynomial of any degree. Hence $h = 0$, which says that p and q have the same coefficients in every degree, as we sought to show.

Division with remainder now settles the relation between μ_A and the other annihilating polynomials of A: there is only one relation available, and it is divisibility. Say that a polynomial g *divides* a polynomial f, written $g \big| f$, when $f = qg$ for some polynomial q.

Theorem 4.4.7: The Minimal Polynomial Divides Every Annihilating Polynomial

Let $A \in M_n(k)$ and let p be a polynomial over k with $p(A) = 0$. Then $\mu_A \big| p$.

***Proof*:**
The minimal polynomial μ_A is monic, hence nonzero, so lemma 4.4.5 applies with $g = \mu_A$ and gives q and r with

$$p = q\mu_A + r$$

and $r = 0$ or $\deg r < \deg \mu_A$. Taking values at A and using both parts of proposition 4.4.2,

$$0 = p(A) = q(A)\mu_A(A) + r(A) = q(A)0 + r(A) = r(A)$$

Suppose $r \neq 0$, and let c be its leading coefficient. Then $c^{-1}r$ is monic of the same degree as r, and $(c^{-1}r)(A) = c^{-1}r(A) = 0$, so $c^{-1}r$ is a monic polynomial of degree less than $\deg \mu_A$ annihilating A. That contradicts definition 4.4.3. Hence $r = 0$ and $p = q\mu_A$, which is the assertion $\mu_A \big| p$, as we sought to show.

The theorem converts every annihilating polynomial that can be produced by any means into an upper bound on $\deg \mu_A$ and into a constraint on its factors. One annihilating polynomial is available for every matrix without any computation, and it is the one the opening computation on A_0 pointed at.

Its identification requires care about what is being substituted where. The characteristic polynomial was defined in definition 4.1.9 by $\chi_A(t) = \det(tI_n - A)$, and that identity relates two *numbers* for each choice of the number t. Replacing t by A in it produces on the left the matrix $\chi_A(A)$, in the sense of definition 4.4.1, and on the right the symbol $\det(AI_n - A)$, in which AI_n is a product of two $n \times n$ matrices rather than a scalar multiple of I_n. That symbol evaluates to $\det(0) = 0$, a number, while $\chi_A(A)$ is a matrix, and no equation between them has been asserted. The apparent one-line proof of the next theorem, which substitutes $t = A$ into $\chi_A(t) = \det(tI_n - A)$ and reads off $\det(A - A) = 0$, therefore proves nothing at all. A genuine proof keeps t a number throughout and compares coefficients only at the end.

Theorem 4.4.8: Cayley-Hamilton

Let $A \in M_n(k)$. Then $\chi_A(A) = 0$.

***Proof*:**
For $n = 1$ the matrix is $A = (a)$ with $a \in k$, and $tI_1 - A$ is the 1×1 matrix $(t - a)$, whose determinant is its single entry by proposition 3.2.6. So $\chi_A(t) = t - a$ and $\chi_A(A) = A - aI_1 = (a) - (a) = 0$. Assume from now on that $n \geq 2$, which is what definition 3.3.6 and theorem 3.3.7 require.

Step 1: the adjugate of $tI_n - A$ has entries polynomial in t. Write $A = (a_{i\ell})$. The (i, ℓ) entry of $tI_n - A$ is $t - a_{ii}$ when $i = \ell$ and $-a_{i\ell}$ otherwise, so each entry of $tI_n - A$ is given by a polynomial in t of degree at most 1. Fix i and j. By definition 3.3.6 and definition 3.3.3, the (i,j) entry of $\operatorname{adj}(tI_n - A)$ is

$$(-1)^{i+j}\det\big((tI_n - A)(j|i)\big)$$

the signed determinant of the $(n-1)\times(n-1)$ matrix obtained from $tI_n - A$ by deleting row j and column i (definition 3.3.2). By proposition 3.3.5 there is a polynomial F in $(n-1)^2$ variables, with coefficients in k, whose value at the entries of any $Y \in M_{n-1}(k)$ is $\det Y$. Substituting for those variables the entries of $(tI_n - A)(j|i)$, each a polynomial in t of degree at most 1, turns F into a sum of products of polynomials in t, which is again a polynomial in t. Hence each entry of $\operatorname{adj}(tI_n - A)$ is the value at t of a polynomial over k. Fix one such polynomial for each entry, let N be the largest degree occurring among the n^2 of them, and let $B_r \in M_n(k)$ be the matrix of their coefficients in degree r. Then

$$\operatorname{adj}(tI_n - A) = B_0 + tB_1 + \cdots + t^N B_N \tag{4.1}$$

holds for every $t \in k$.

Step 2: the adjugate identity, one number t at a time. For each fixed $t \in k$ the matrix $tI_n - A$ lies in $M_n(k)$, so theorem 3.3.7 gives

$$\operatorname{adj}(tI_n - A)\,(tI_n - A) = \det(tI_n - A)\,I_n = \chi_A(t)I_n$$

the last equality by definition 4.1.9. Substituting (4.1) and expanding, with the convention $B_{-1} = 0$ and $B_r = 0$ for $r > N$,

$$\sum_{r=0}^{N} t^{r+1}B_r - \sum_{r=0}^{N} t^r B_r A = \sum_{m=0}^{N+1} t^m\big(B_{m-1} - B_m A\big)$$

so that for every $t \in k$,

$$\sum_{m=0}^{N+1} t^m\big(B_{m-1} - B_m A\big) = \chi_A(t)I_n$$

Step 3: comparing coefficients. By proposition 4.1.8 the polynomial χ_A is monic of degree n; write $\chi_A(t) = c_0 + c_1t + \cdots + c_{n-1}t^{n-1} + t^n$, put $c_n = 1$, and put $c_m = 0$ for $m > n$. Let $M = \max(N+1, n)$. The previous display rearranges to

$$\sum_{m=0}^{M} t^m\big(B_{m-1} - B_m A - c_m I_n\big) = 0 \qquad \text{for every } t \in k$$

Reading this equation of matrices in position (i,j) gives a polynomial in t, with coefficients the (i,j) entries of the matrices $B_{m-1} - B_m A - c_m I_n$, whose value is 0 for every $t \in k$. By corollary 4.4.6(3) all of its coefficients vanish. Letting (i,j) range over all positions gives, for each m with $0 \le m \le M$,

$$B_{m-1} - B_m A = c_m I_n \tag{4.2}$$

Step 4: telescoping. Multiply (4.2) on the right by A^m and sum over $m = 0, 1, \ldots, M$:

$$\sum_{m=0}^{M}\big(B_{m-1}A^m - B_m A^{m+1}\big) = \sum_{m=0}^{M} c_m A^m$$

Write $u_m = B_{m-1}A^m$, so that the mth summand on the left is $u_m - u_{m+1}$ and the sum collapses to $u_0 - u_{M+1}$. Now $u_0 = B_{-1}I_n = 0$, and $M \geq N+1$ gives $B_M = 0$ and hence $u_{M+1} = B_M A^{M+1} = 0$. The left side is therefore 0. On the right, $c_m = 0$ for $m > n$ and $M \geq n$, so the sum is $c_0 I_n + c_1 A + \cdots + c_{n-1}A^{n-1} + A^n$, which is $\chi_A(A)$. Hence $\chi_A(A) = 0$, completing the proof.

The running matrix of example 1.2.9 has four rows and three columns, so it has no characteristic polynomial. Its top 3×3 block, already singled out in example 3.2.9, has one, and the theorem can be checked on it entry by entry.

Example 4.4.9: Cayley-Hamilton for the Top Block of the Running Matrix

Let
$$A' = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 1 & 2 & 5 \end{pmatrix}$$
the matrix of example 3.2.9. Its characteristic polynomial was computed in example 4.2.7(2) by a cofactor expansion of $\det(tI_3 - A')$ along the first row:
$$\chi_{A'}(t) = t(t^2 - 9t + 2) = t^3 - 9t^2 + 2t$$

Squaring A' entry by entry with definition 1.4.1, the first row of A'^2 is $(1+2+3,\ 1+3+6,\ 3+8+15) = (6, 10, 26)$, the second is $(2+6+8,\ 2+9+16,\ 6+24+40) = (16, 27, 70)$, and the third is $(1+4+5,\ 1+6+10,\ 3+16+25) = (10, 17, 44)$, so
$$A'^2 = \begin{pmatrix} 6 & 10 & 26 \\ 16 & 27 & 70 \\ 10 & 17 & 44 \end{pmatrix}, \qquad A'^3 = A'^2 A' = \begin{pmatrix} 52 & 88 & 228 \\ 140 & 237 & 614 \\ 88 & 149 & 386 \end{pmatrix}$$
where the entries of A'^3 are computed the same way; for instance its $(2,3)$ entry is $16 \cdot 3 + 27 \cdot 8 + 70 \cdot 5 = 48 + 216 + 350 = 614$.

Now assemble $\chi_{A'}(A') = A'^3 - 9A'^2 + 2A'$. Entry by entry,
$$A'^3 - 9A'^2 = \begin{pmatrix} 52-54 & 88-90 & 228-234 \\ 140-144 & 237-243 & 614-630 \\ 88-90 & 149-153 & 386-396 \end{pmatrix} = \begin{pmatrix} -2 & -2 & -6 \\ -4 & -6 & -16 \\ -2 & -4 & -10 \end{pmatrix}$$
which is exactly $-2A'$, so $A'^3 - 9A'^2 + 2A' = 0$ as theorem 4.4.8 requires.

Rewritten as $A'^3 = 9A'^2 - 2A'$, the identity expresses the third power in terms of the first two, and every higher power follows: multiplying by A' and substituting gives $A'^4 = 9A'^3 - 2A'^2 = 79A'^2 - 18A'$, then $A'^5 = 79A'^3 - 18A'^2 = 693A'^2 - 158A'$, and so on, so that A'^m is a combination of A'^2 and A' for every $m \geq 1$. The same reduction runs for every $A \in M_n(k)$, with n in place of 3: one identity of degree n controls all the powers.

The theorem has a second consequence of the same shape. Suppose $A \in M_n(k)$ is invertible and write $\chi_A(t) = t^n + c_{n-1}t^{n-1} + \cdots + c_1 t + c_0$. The constant term is $c_0 = \chi_A(0) = \det(-A)$ by definition 4.1.9, and $-A = (-I_n)A$, so $\det(-A) = \det(-I_n)\det A = (-1)^n \det A$ by theorem 3.2.10 and proposition 3.2.6. Since A is invertible, $\det A \neq 0$ by corollary 3.2.8, so $c_0 \neq 0$. Grouping the identity $\chi_A(A) = 0$ so that one factor of A comes out on the left,

$$A\bigl(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I_n\bigr) = -c_0 I_n$$

and dividing by $-c_0$ exhibits an inverse of A; by proposition 1.4.5 it is A^{-1}.

Corollary 4.4.10: The Inverse Is a Polynomial in the Matrix

Let $A \in M_n(k)$ be invertible and write $\chi_A(t) = t^n + c_{n-1}t^{n-1} + \cdots + c_1 t + c_0$. Then $c_0 \neq 0$ and

$$A^{-1} = -c_0^{-1}\bigl(A^{n-1} + c_{n-1}A^{n-2} + \cdots + c_1 I_n\bigr)$$

so the inverse of an invertible matrix is a polynomial in that matrix, with coefficients read off its characteristic polynomial.

Combining theorem 4.4.8 with theorem 4.4.7 gives the bound on $\deg \mu_A$ promised above.

Corollary 4.4.11: The Minimal Polynomial Divides the Characteristic Polynomial

Let $A \in M_n(k)$. Then $\mu_A \big| \chi_A$, and $1 \leq \deg \mu_A \leq n$.

***Proof*:**
By theorem 4.4.8 the polynomial χ_A annihilates A, so theorem 4.4.7 gives $\mu_A \big| \chi_A$, say $\chi_A = q\mu_A$. Both χ_A and μ_A are nonzero, so q is nonzero and the degree of a product is the sum of the degrees: $n = \deg \chi_A = \deg q + \deg \mu_A \geq \deg \mu_A$, the value $\deg \chi_A = n$ coming from proposition 4.1.8. The lower bound $\deg \mu_A \geq 1$ is theorem 4.4.4, completing the proof.

Two matrices can share a characteristic polynomial and differ in every other respect, and the minimal polynomial separates some such pairs.

Example 4.4.12: Three Matrices with the Characteristic Polynomial $(t-2)^3$
Take the three matrices

$$S = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \qquad N_1 = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \qquad N_2 = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$$

in $M_3(\mathbb{R})$. For each of the three, written B for the moment, the matrix $tI_3 - B$ is upper triangular with every diagonal entry $t-2$, so proposition 3.2.6 gives $\chi_S = \chi_{N_1} = \chi_{N_2} = (t-2)^3$. By proposition 4.4.2 the value of the polynomial $(t-2)^m$ at such a B is the power $(B - 2I_3)^m$, so the three minimal polynomials are found by taking powers of $B - 2I_3$. In each case theorem 4.4.4 gives $\deg \mu \geq 1$ together with the uniqueness that turns "monic, annihilating, of least degree" into an identification of μ.

1. $S = 2I_3$, so $S - 2I_3 = 0$ and $t-2$ annihilates S. It is monic of degree 1, and no smaller degree is available, so $\mu_S = t-2$.

2. No monic polynomial of degree 1 annihilates N_1: such a polynomial is $t-c$, and $N_1 - cI_3 = 0$

would force the $(1,2)$ entry 1 of N_1 to be 0. Meanwhile

$$N_1 - 2I_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad (N_1 - 2I_3)^2 = 0$$

the square being zero because the only nonzero entry of $N_1 - 2I_3$ sits in position $(1,2)$, and row 2 of that matrix is zero. So $(t-2)^2$ is monic of degree 2 and annihilates N_1, while degrees 0 and 1 have both been ruled out. Hence $\mu_{N_1} = (t-2)^2$.

3. For N_2, degree 1 fails for the same reason as for N_1. Degree 2 also fails: computing with definition 1.4.1,

$$N_2^2 = \begin{pmatrix} 4 & 4 & 1 \\ 0 & 4 & 4 \\ 0 & 0 & 4 \end{pmatrix}$$

and a monic polynomial of degree 2 is $t^2 + bt + c$, whose value at N_2 has $(1,3)$ entry $1 + b \cdot 0 + c \cdot 0 = 1$ for every choice of $b, c \in \mathbb{R}$, since the $(1,3)$ entries of N_2 and I_3 are both 0. So no monic polynomial of degree 2 annihilates N_2. On the other hand

$$N_2 - 2I_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad (N_2 - 2I_3)^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad (N_2 - 2I_3)^3 = 0$$

so $(t-2)^3$ is monic of degree 3 and annihilates N_2, while degrees 0, 1 and 2 have been ruled out. Hence $\mu_{N_2} = (t-2)^3$.

The three characteristic polynomials agree and the three minimal polynomials are $t-2$, $(t-2)^2$ and $(t-2)^3$.

The minimal polynomial, like the characteristic polynomial (corollary 4.1.16), depends only on the similarity class. Both statements come from one identity about polynomials and conjugation, worth recording on its own because later chapters use it in that form.

Proposition 4.4.13: Polynomials Commute with Conjugation, and Similar Matrices Share a Minimal Polynomial

Let $A \in M_n(k)$, let $P \in M_n(k)$ be invertible, and put $B = P^{-1}AP$.

1. $p(B) = P^{-1}p(A)P$ for every polynomial p over k.
2. $p(B) = 0$ if and only if $p(A) = 0$, so A and B have the same annihilating polynomials.
3. $\mu_B = \mu_A$. Similar matrices have the same minimal polynomial, and by theorem 4.4.15 are either both diagonalizable or both not.

***Proof*:**
For (1), first $B^m = P^{-1}A^mP$ for every $m \geq 0$, by induction: at $m = 0$ both sides are I_n, and if

$B^m = P^{-1}A^mP$ then

$$B^{m+1} = B^mB = (P^{-1}A^mP)(P^{-1}AP) = P^{-1}A^m(PP^{-1})AP = P^{-1}A^{m+1}P$$

For $p(t) = \sum_m a_m t^m$, taking the corresponding combination gives

$$p(B) = \sum_m a_m P^{-1}A^mP = P^{-1}\Big(\sum_m a_m A^m\Big)P = P^{-1}p(A)P$$

the middle step because a scalar multiple and a sum may be moved through a matrix product (definition 1.4.1).

For (2), if $p(A) = 0$ then $p(B) = P^{-1}0P = 0$, and if $p(B) = 0$ then $p(A) = Pp(B)P^{-1} = 0$.

For (3), by (2) the two matrices have the same annihilating polynomials, hence the same monic annihilating polynomials of least degree. So μ_A is a minimal polynomial of B, and the uniqueness clause of theorem 4.4.4 gives $\mu_B = \mu_A$. The last sentence is then theorem 4.4.15 applied to each.

The three matrices of the example do not differ in their eigenvalues: all three have $\chi(t) = (t-2)^3$, whose only root is 2, and each of the three minimal polynomials again has 2 as its only root. The two polynomials have the same roots for every square matrix.

Theorem 4.4.14: The Minimal and Characteristic Polynomials Have the Same Roots

Let $A \in M_n(k)$ and let $\lambda \in k$. The following are equivalent.

1. $\mu_A(\lambda) = 0$.
2. $\chi_A(\lambda) = 0$.
3. λ is an eigenvalue of A.

***Proof*:**
That (2) and (3) are equivalent is theorem 4.1.10.

Suppose (1) holds. By corollary 4.4.11 there is a polynomial q with $\chi_A = q\mu_A$, so $\chi_A(\lambda) = q(\lambda)\mu_A(\lambda) = 0$ by proposition 4.4.2, which is (2).

Suppose (3) holds, so that by definition 4.1.2 there is a column $v \in k^n$ with $v \neq 0$ and $Av = \lambda v$. Then $A^mv = \lambda^m v$ for every $m \geq 0$: this holds for $m = 0$ since $I_nv = v$, and if $A^mv = \lambda^mv$ then $A^{m+1}v = A(A^mv) = A(\lambda^mv) = \lambda^m(Av) = \lambda^{m+1}v$. Writing $\mu_A(t) = \sum_m a_mt^m$ and applying $\mu_A(A)$ to v,

$$0 = \mu_A(A)v = \sum_m a_mA^mv = \sum_m a_m\lambda^mv = \mu_A(\lambda)v$$

Since $v \neq 0$, some entry v_i of v is nonzero, and the ith entry of $\mu_A(\lambda)v$ is $\mu_A(\lambda)v_i$; this being 0 forces $\mu_A(\lambda) = 0$, which is (1), completing the proof.

The two polynomials therefore vanish at exactly the same numbers of k, and they are nonetheless different polynomials in general. In example 4.4.12 the characteristic polynomial is $(t-2)^3$ in all three cases, while the minimal polynomials are $t-2$, $(t-2)^2$ and $(t-2)^3$, so it is the minimal polynomial that separates the three matrices. The criterion below says that the separation it makes is exactly diagonalizability.

Theorem 4.4.15: Diagonalizable Exactly When the Minimal Polynomial Is a Product of Distinct Linear Factors

Let $A \in M_n(k)$. Then A is diagonalizable over k if and only if

$$\mu_A(t) = (t-\lambda_1)(t-\lambda_2)\cdots(t-\lambda_r)$$

for some $r \geq 1$ and some $\lambda_1, \ldots, \lambda_r \in k$ that are pairwise distinct.

***Proof*:**
Step 1: a diagonalizable matrix has such a minimal polynomial. Suppose A is diagonalizable, so by definition 4.2.1 there is an invertible $P \in M_n(k)$ with $P^{-1}AP = D$ diagonal, say with diagonal entries $d_1, \ldots, d_n$. Multiplying two diagonal matrices multiplies their diagonal entries in place (definition 1.4.1), so $D^m = \operatorname{diag}(d_1^m, \ldots, d_n^m)$ for every $m \geq 0$, and hence for every polynomial p over k,

$$p(D) = \operatorname{diag}\bigl(p(d_1), \ldots, p(d_n)\bigr)$$

Also $A = PDP^{-1}$, and $A^m = PD^mP^{-1}$ for every $m \geq 0$ by theorem 4.5.1(1). Taking a linear combination of these identities gives $p(A) = Pp(D)P^{-1}$ for every polynomial p.

Let $\lambda_1, \ldots, \lambda_r$ be the distinct numbers occurring among $d_1, \ldots, d_n$, so $r \geq 1$, and put $p(t) = (t-\lambda_1)\cdots(t-\lambda_r)$, which is monic of degree r. Each d_j equals some λ_i, so $p(d_j) = 0$ by proposition 4.4.2, whence $p(D) = 0$ and therefore $p(A) = P0P^{-1} = 0$. By theorem 4.4.7 there is a polynomial q with $p = q\mu_A$, and q is nonzero because p is, so the degrees add and $\deg \mu_A \leq \deg p = r$.

In the other direction, $\mu_A(A) = 0$ gives $\mu_A(D) = P^{-1}\mu_A(A)P = 0$, so $\mu_A(d_j) = 0$ for every j and in particular $\mu_A(\lambda_i) = 0$ for $i = 1, \ldots, r$. The polynomial μ_A is nonzero with r distinct roots, so $\deg \mu_A \geq r$ by corollary 4.4.6(2). Hence $\deg \mu_A = r$, and in $p = q\mu_A$ the factor q has degree $r - r = 0$ and leading coefficient 1, since p and μ_A are monic. So $q = 1$ and $\mu_A = p$, which is the stated form.

Step 2: the converse, when $r = 1$. Suppose $\mu_A(t) = t - \lambda_1$. Then $A - \lambda_1 I_n = \mu_A(A) = 0$, so $A = \lambda_1 I_n$ is already diagonal, and $I_n^{-1}AI_n = A$ exhibits A as diagonalizable in the sense of definition 4.2.1.

Step 3: the converse, when $r \geq 2$. Suppose $\mu_A(t) = (t-\lambda_1)\cdots(t-\lambda_r)$ with the λ_i pairwise distinct and $r \geq 2$. For each i put

$$c_i = \prod_{j \neq i}(\lambda_i - \lambda_j), \qquad p_i(t) = \frac{1}{c_i}\prod_{j \neq i}(t - \lambda_j)$$

Each factor $\lambda_i - \lambda_j$ with $j \neq i$ is a nonzero number of k, so $c_i \neq 0$ and p_i is a polynomial over k of degree $r-1$. By proposition 4.4.2 the value of a product of polynomials is the product of their values, so $p_i(\lambda_i) = c_i/c_i = 1$, while for $j \neq i$ the product computing $p_i(\lambda_j)$ contains the factor $\lambda_j - \lambda_j = 0$ and hence $p_i(\lambda_j) = 0$.

Let $h = p_1 + \cdots + p_r - 1$. Every p_i has degree $r-1$, so h is zero or of degree at most $r-1$. For each i,

$$h(\lambda_i) = p_i(\lambda_i) + \sum_{j \neq i} p_j(\lambda_i) - 1 = 1 + 0 - 1 = 0$$

so h has the r distinct roots $\lambda_1, \ldots, \lambda_r$. A nonzero polynomial of degree at most $r-1$ has at most $r-1$ roots by corollary 4.4.6(2), so $h = 0$, that is $p_1 + \cdots + p_r = 1$. Taking values at A and using proposition 4.4.2,

$$p_1(A) + p_2(A) + \cdots + p_r(A) = I_n \tag{4.3}$$

Each $p_i(A)$ sends every column of k^n into the null space of $A - \lambda_i I_n$. Multiplying the polynomial p_i by $t - \lambda_i$ restores the full product, so $(t - \lambda_i)p_i(t) = \frac{1}{c_i}\mu_A(t)$, and proposition 4.4.2 turns this into

$$(A - \lambda_i I_n)p_i(A) = \frac{1}{c_i}\mu_A(A) = 0$$

Hence for every $x \in k^n$ the column $p_i(A)x$ satisfies $(A - \lambda_i I_n)p_i(A)x = 0$.

Now build a basis of eigenvectors. Consider the rn columns

$$v_{ij} = p_i(A)e_j, \qquad 1 \leq i \leq r, \quad 1 \leq j \leq n$$

Let S be the span of the v_{ij}. By proposition 2.1.6(1) it is a subspace of k^n containing each v_{ij}, and it is therefore closed under addition, so applying (4.3) to e_j puts $e_j = v_{1j} + \cdots + v_{rj}$ in S for every j. Now apply proposition 2.1.6(2) with the list $e_1, \ldots, e_n$ and the subspace S: it gives $\operatorname{span}\{e_1, \ldots, e_n\} \subseteq S$, and the left side is k^n by example 2.2.2(1). So the list of the v_{ij} spans k^n, and by theorem 2.2.9(2) some sublist of it is a basis of k^n.

Every member of that sublist is an eigenvector of A. A basis is linearly independent (definition 2.2.1), and a list containing the zero column is not, since giving that column the coefficient 1 and all others the coefficient 0 produces a nontrivial relation (definition 2.1.9). So each member of the sublist is some v_{ij} with $v_{ij} \neq 0$. Taking $x = e_j$ in the identity $(A - \lambda_i I_n)p_i(A)x = 0$ established two paragraphs above gives $(A - \lambda_i I_n)v_{ij} = 0$, that is $Av_{ij} = \lambda_i v_{ij}$, which makes v_{ij} an eigenvector of A for the eigenvalue λ_i (definition 4.1.3). Thus k^n has a basis consisting of eigenvectors of A, and theorem 4.2.2 makes A diagonalizable, completing the proof.

The criterion decides diagonalizability from the factorization of μ_A alone, without the eigenspace dimensions that theorem 4.3.8 requires. In example 4.4.12 the matrix S has $\mu_S = t - 2$, a product of one distinct linear factor, so S is diagonalizable, as it visibly is. The matrix N_1 has $\mu_{N_1} = (t-2)^2$, which is not of the stated form: a product $(t - \lambda_1)\cdots(t - \lambda_r)$ with distinct λ_i has each λ_i as a root, and the only root of $(t-2)^2$ is 2, so r would be 1 and the degree 1 rather than 2. So N_1 is not diagonalizable, and the same argument rules out N_2.

For the top block A' of the running matrix the criterion answers a question example 4.4.9 did not. There $\chi_{A'}(t) = t^3 - 9t^2 + 2t = t(t^2 - 9t + 2)$, and the quadratic factor has roots $(9 \pm \sqrt{73})/2$, both real since $81 - 8 > 0$, distinct from each other, and both nonzero since their product is 2. So $\chi_{A'}$ has three distinct roots in $\mathbb{R}$; each is a root of $\mu_{A'}$ by theorem 4.4.14, so $\deg \mu_{A'} \geq 3$ by corollary 4.4.6(2), while $\deg \mu_{A'} \leq 3$ by corollary 4.4.11. Two monic polynomials of degree 3, one dividing the other, are equal, so $\mu_{A'} = \chi_{A'}$, which is the product of the three distinct linear factors t, $t - (9 + \sqrt{73})/2$ and $t - (9 - \sqrt{73})/2$. Hence A' is diagonalizable over $\mathbb{R}$, in agreement with corollary 4.2.4, which reaches the same conclusion from the three distinct eigenvalues.

Exercise 4.4.1

1. Let $M = \left(\begin{smallmatrix} 3 & 1 \\ 0 & 3 \end{smallmatrix}\right) \in M_2(\mathbb{R})$. Compute χ_M, verify $\chi_M(M) = 0$ by computing M^2, determine μ_M, and decide whether M is diagonalizable over $\mathbb{R}$.
2. Let $A = \left(\begin{smallmatrix} 1 & 2 \\ 3 & 2 \end{smallmatrix}\right) \in M_2(\mathbb{R})$. Compute χ_A and use theorem 4.4.8 to write both A^{-1} and A^4 as polynomials in A, then evaluate each as an explicit matrix. Check the inverse by multiplying it against A.
3. Let $P \in M_n(k)$ satisfy $P^2 = P$, with $P \neq 0$ and $P \neq I_n$. Show that $\mu_P(t) = t^2 - t$ and deduce that P is diagonalizable over k. Then carry this out for $P = \left(\begin{smallmatrix} 2 & -1 \\ 2 & -1 \end{smallmatrix}\right) \in M_2(\mathbb{R})$, computing χ_P as well.
4. Let $R = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Compute R^2 and χ_R, and determine μ_R. Show that R is not diagonalizable when regarded as an element of $M_2(\mathbb{R})$, and that it is diagonalizable when regarded as an element of $M_2(\mathbb{C})$.
5. Let A' be the matrix of example 4.4.9. Using only the identity $\chi_{A'}(A') = 0$ and the value of A'^2 computed there, express A'^4 as a combination of A'^2 and A', evaluate it as an explicit matrix, and check the answer by squaring A'^2.
6. Let $A \in M_n(k)$ and let $P \in M_n(k)$ be invertible. Show that $p(P^{-1}AP) = P^{-1}p(A)P$ for every polynomial p over k, and deduce that $\mu_{P^{-1}AP} = \mu_A$. Conclude that similar matrices (definition 2.5.5) are either both diagonalizable or both not.
7. Let $A \in M_2(k)$. Show that $\mu_A = \chi_A$ unless $A = cI_2$ for some $c \in k$, in which case $\mu_A = t - c$ while $\chi_A = (t - c)^2$.

4.5 Application: Powers, Iteration, and Linear Recurrences

Raising a diagonal matrix to a power raises each diagonal entry to that power and changes nothing else. A matrix similar to a diagonal one inherits this, and the inheritance produces one formula covering every power at once. Two uses of that formula are worked out here: iteration, and a closed expression for the m-th term of a sequence defined by a recurrence.

The second use reverses the direction of section 4.1. There a polynomial was attached to a given matrix; a recurrence comes with its polynomial already attached, so what is needed is a matrix whose characteristic polynomial is a prescribed one. The companion matrix is that matrix.

Powers of a square matrix arise whenever one step of a process is applied over and over. Suppose the state of the process at step m is a column $x_m \in k^n$, and one step replaces it by $x_{m+1} = Bx_m$ for a fixed $B \in M_n(k)$. Then $x_1 = Bx_0$, then $x_2 = B(Bx_0) = B^2x_0$, and after m steps $x_m = B^m x_0$. Here $B^0 = I_n$ and $B^{m+1} = B^m B$ for $m \geq 0$, which is a legitimate product because B is square: a rectangular matrix cannot be multiplied by itself. A question about the state after many steps is a question about B^m for large m.

Computing B^m one product at a time answers the question for one value of m at a time. For

$$B = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix}$$

two products give

$$B^2 = \begin{pmatrix} 14 & -10 \\ 5 & -1 \end{pmatrix}, \qquad B^3 = \begin{pmatrix} 46 & -38 \\ 19 & -11 \end{pmatrix}$$

and the top left entries $4, 14, 46$ give no rule for producing the next one.

A diagonal matrix behaves differently. Multiplying $\mathrm{diag}(2, 3)$ by itself,

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}$$

so the two diagonal entries are squared and the zeros stay. A matrix with a diagonal form is PDP^{-1}, and its powers are the powers of D with one product on each side.

Theorem 4.5.1: Powers of a Diagonalizable Matrix

Let $A \in M_n(k)$ be diagonalizable (definition 4.2.1): fix an invertible $P \in M_n(k)$ and scalars $\lambda_1, \ldots, \lambda_n \in k$ with

$$A = PDP^{-1}, \qquad D = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$$

1. For every integer $m \geq 0$,

$$A^m = P\mathrm{diag}(\lambda_1^m, \ldots, \lambda_n^m)P^{-1}$$

   where $\lambda^0 = 1$ for every $\lambda \in k$.

2. A is invertible if and only if $\lambda_i \neq 0$ for every i, and in that case the formula of (1) holds for every integer m, negative exponents included, where A^{-r} means $(A^{-1})^r$ for $r > 0$.

3. Let $v_1, \ldots, v_n$ be the columns of P; each v_i is an eigenvector of A for λ_i. Every $v \in k^n$ can be written as $v = c_1 v_1 + \cdots + c_n v_n$ for exactly one list $c_1, \ldots, c_n \in k$, and for every integer $m \geq 0$,

$$A^m v = c_1 \lambda_1^m v_1 + \cdots + c_n \lambda_n^m v_n$$

***Proof*:**

1. First, a product of two diagonal matrices is diagonal, with the diagonal entries multiplied in pairs. Let $F = \mathrm{diag}(f_1, \ldots, f_n)$ and $G = \mathrm{diag}(g_1, \ldots, g_n)$, so the (i, l) entry f_{il} of F is 0 unless $l = i$, and the (l, j) entry g_{lj} of G is 0 unless $j = l$. By definition 1.4.1 the (i, j) entry of FG is $\sum_{l=1}^n f_{il} g_{lj}$, and the only term that can be nonzero is the one with $l = i$, which is $f_i g_{ij}$. That vanishes unless $j = i$, and equals $f_i g_i$ when $j = i$. Hence

$$\mathrm{diag}(f_1, \ldots, f_n)\mathrm{diag}(g_1, \ldots, g_n) = \mathrm{diag}(f_1 g_1, \ldots, f_n g_n)$$

   Induction on m now gives $D^m = \mathrm{diag}(\lambda_1^m, \ldots, \lambda_n^m)$ for every $m \geq 0$: at $m = 0$ this reads $D^0 = I_n = \mathrm{diag}(1, \ldots, 1)$, which is the formula with $\lambda_i^0 = 1$, and if it holds at m then the displayed rule applied to D^m and D gives $D^{m+1} = D^m D = \mathrm{diag}(\lambda_1^m \lambda_1, \ldots, \lambda_n^m \lambda_n) = \mathrm{diag}(\lambda_1^{m+1}, \ldots, \lambda_n^{m+1})$.

   A second induction, again on m, gives the stated formula. For $m = 0$ the claim is $I_n = PI_nP^{-1}$, which holds because $PP^{-1} = I_n$ (definition 1.4.4). Assume $A^m = PD^mP^{-1}$. Then

$$A^{m+1} = A^m A = (PD^mP^{-1})(PDP^{-1}) = PD^m(P^{-1}P)DP^{-1} = PD^m I_n DP^{-1} = PD^{m+1}P^{-1}$$

where the brackets were moved by proposition 1.4.3 and $P^{-1}P = I_n$ by definition 1.4.4. Combining this with the formula for D^m gives the claim for every $m \geq 0$.

2. Suppose every $\lambda_i \neq 0$. Then $\mathrm{diag}(\lambda_1, \ldots, \lambda_n)\mathrm{diag}(\lambda_1^{-1}, \ldots, \lambda_n^{-1}) = \mathrm{diag}(1, \ldots, 1) = I_n$ by the product rule proved in (1), and the same computation in the other order gives I_n as well, so D is invertible with $D^{-1} = \mathrm{diag}(\lambda_1^{-1}, \ldots, \lambda_n^{-1})$. A product of invertible matrices is invertible (proposition 1.4.5), so $A = PDP^{-1}$ is invertible. Conversely, suppose some $\lambda_j = 0$. The j-th column of D is then zero, and definition 1.4.1 gives $(De_j)_i = \sum_l d_{il}(e_j)_l = d_{ij}$, the i-th entry of that column, so $De_j = 0$ for the j-th standard basis vector e_j. If D had an inverse E, then $e_j = I_n e_j = (ED)e_j = E(De_j) = E0 = 0$, and $e_j \neq 0$, so D has no inverse. Multiplying $A = PDP^{-1}$ by P^{-1} on the left and by P on the right gives $D = P^{-1}AP$, so an invertible A would make D a product of invertible matrices and hence invertible (proposition 1.4.5); so A is not invertible either.

   Assume now that every $\lambda_i \neq 0$, and let $m < 0$, say $m = -r$ with $r > 0$. Regrouping by proposition 1.4.3,

$$(PDP^{-1})(PD^{-1}P^{-1}) = PD(P^{-1}P)D^{-1}P^{-1} = PDD^{-1}P^{-1} = PP^{-1} = I_n$$

   and the same computation with the two factors interchanged also gives I_n, so $A^{-1} = PD^{-1}P^{-1}$ by uniqueness of the inverse (proposition 1.4.5). Part (1) applied to A^{-1}, which is diagonalized by the same P with diagonal entries λ_i^{-1}, gives

$$A^m = (A^{-1})^r = P\mathrm{diag}(\lambda_1^{-r}, \ldots, \lambda_n^{-r})P^{-1} = P\mathrm{diag}(\lambda_1^m, \ldots, \lambda_n^m)P^{-1}$$

   which is the formula of (1) at the negative exponent m.

3. Write p_{ij} for the entries of P, so that the i-th entry of v_j is p_{ij}. By definition 1.4.1 the i-th entry of Pb, for a column $b = (b_1, \ldots, b_n)^\top \in k^n$, is $\sum_{j=1}^n p_{ij}b_j$, which is the i-th entry of $b_1v_1 + \cdots + b_nv_n$; so Pb is that combination of the columns of P, and in particular $Pe_i = v_i$. Multiplying $A = PDP^{-1}$ on the right by P gives $AP = PD$, and applying both sides to e_i gives $Av_i = A(Pe_i) = P(De_i) = \lambda_i v_i$, where $De_i = \lambda_i e_i$ is the i-th column of D and the brackets were moved by proposition 1.4.3. Here $v_i \neq 0$, since $v_i = 0$ would force $e_i = P^{-1}(Pe_i) = 0$, so each v_i is an eigenvector of A for λ_i (definition 4.1.3).

   Now let $v \in k^n$ and set $c = P^{-1}v \in k^n$. Then $Pc = P(P^{-1}v) = (PP^{-1})v = I_nv = v$, regrouping by proposition 1.4.3, and Pc is the combination $c_1v_1 + \cdots + c_nv_n$ of the columns of P, so $v = c_1v_1 + \cdots + c_nv_n$. If $c_1', \ldots, c_n'$ is another such list, then $Pc' = v = Pc$, and multiplying by P^{-1} on the left gives $c' = c$, so the list is unique.

   Multiplying the identity of (1) on the right by P gives $A^mP = PD^m$. Hence

$$A^mv = A^m(Pc) = (A^mP)c = (PD^m)c = P(D^mc)$$

   again by proposition 1.4.3. The column D^mc has i-th entry $\lambda_i^m c_i$, by definition 1.4.1 applied to the diagonal matrix D^m computed in (1). Applying to $b = D^mc$ the reading of Pb above therefore gives

$$A^mv = \lambda_1^m c_1v_1 + \cdots + \lambda_n^m c_nv_n$$

   which is the stated formula, completing the proof.

Part (1) trades $m - 1$ matrix products for n scalar powers and two matrix products, and the two matrix products do not depend on m. Its right-hand side is also a formula in m, so it can be examined for every m at once instead of being evaluated one exponent at a time.

Part (3) restates this in the coordinates given by the eigenvectors. In the directions $v_1, \ldots, v_n$ the matrix does not mix: it scales the i-th component by λ_i and leaves the others unchanged, so repeating it m times scales that component by λ_i^m. Whether $A^m v$ grows, shrinks, or settles down is therefore decided component by component in these coordinates, by the sizes of the n numbers λ_i^m. The next example computes both parts on two matrices, one where the components grow and one where all but one of them decay.

Example 4.5.2: Powers of a 2×2 Matrix and a Migration Process

1. Take the matrix $B = \left(\begin{smallmatrix} 4 & -2 \\ 1 & 1 \end{smallmatrix}\right)$ of the computation above. Its characteristic polynomial (definition 4.1.9) is

$$\chi_B(t) = \det \begin{pmatrix} t-4 & 2 \\ -1 & t-1 \end{pmatrix} = (t-4)(t-1) + 2 = t^2 - 5t + 6 = (t-2)(t-3)$$

so the eigenvalues are 2 and 3 (theorem 4.1.10). For $\lambda = 2$, the eigenvectors are the nonzero columns $u = (u_1, u_2)^\top$ with $(2I_2 - B)u = 0$, and

$$2I_2 - B = \begin{pmatrix} -2 & 2 \\ -1 & 1 \end{pmatrix}$$

has both rows proportional to $(1, -1)$, so the condition is $u_1 = u_2$ and $v_1 = (1, 1)^\top$ is an eigenvector. For $\lambda = 3$,

$$3I_2 - B = \begin{pmatrix} -1 & 2 \\ -1 & 2 \end{pmatrix}$$

gives $u_1 = 2u_2$, and $v_2 = (2, 1)^\top$ is an eigenvector. The two eigenvalues are distinct, so B is diagonalizable (corollary 4.2.4), and proposition 4.2.5 takes

$$P = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}, \qquad D = \operatorname{diag}(2, 3)$$

with the columns of P in the order matching the diagonal of D. Since $\det P = 1 \cdot 1 - 2 \cdot 1 = -1$, corollary 3.3.8 gives

$$P^{-1} = \frac{1}{-1} \begin{pmatrix} 1 & -2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}$$

and multiplying out confirms $PP^{-1} = I_2$. Now theorem 4.5.1(1) gives, for every $m \geq 0$,

$$B^m = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2^m & 0 \\ 0 & 3^m \end{pmatrix} \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 2^m & 2 \cdot 3^m \\ 2^m & 3^m \end{pmatrix} \begin{pmatrix} -1 & 2 \\ 1 & -1 \end{pmatrix}$$

and one further product gives

$$B^m = \begin{pmatrix} 2 \cdot 3^m - 2^m & 2^{m+1} - 2 \cdot 3^m \\ 3^m - 2^m & 2^{m+1} - 3^m \end{pmatrix}$$

At $m = 0$ this reads $\left(\begin{smallmatrix} 2-1 & 2-2 \\ 1-1 & 2-1 \end{smallmatrix}\right) = I_2$, at $m = 1$ it reads $\left(\begin{smallmatrix} 6-2 & 4-6 \\ 3-2 & 4-3 \end{smallmatrix}\right) = B$, and at $m = 3$ it reads $\left(\begin{smallmatrix} 54-8 & 16-54 \\ 27-8 & 16-27 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 46 & -38 \\ 19 & -11 \end{smallmatrix}\right)$, which is the B^3 computed by hand above.

2. Two towns exchange residents once a year. Each year one tenth of the people in the first town move to the second, and two tenths of the people in the second move to the first. Writing $x_m \in \mathbb{R}^2$ for the two populations after m years, one year is $x_{m+1} = Mx_m$ with

$$M = \frac{1}{10}\begin{pmatrix} 9 & 2 \\ 1 & 8 \end{pmatrix}$$

since the first town keeps nine tenths of its own people and receives two tenths of the other town's. Each column of M has entries summing to 1, so for any column $y = (y_1, y_2)^\top$ the two entries of My add up to $\frac{9y_1+2y_2}{10} + \frac{y_1+8y_2}{10} = y_1 + y_2$: nobody is created or lost. The characteristic polynomial is

$$\chi_M(t) = \det\begin{pmatrix} t - \frac{9}{10} & -\frac{2}{10} \\ -\frac{1}{10} & t - \frac{8}{10} \end{pmatrix} = t^2 - \frac{17}{10}t + \frac{72}{100} - \frac{2}{100} = t^2 - \frac{17}{10}t + \frac{7}{10}$$

Multiplying out confirms $t^2 - \frac{17}{10}t + \frac{7}{10} = (t-1)\left(t - \frac{7}{10}\right)$, so the eigenvalues are 1 and $\frac{7}{10}$, and they are distinct. Writing $u = (u_1, u_2)^\top$ for a candidate eigenvector, the matrix $I_2 - M = \frac{1}{10}\left(\begin{smallmatrix} 1 & -2 \\ -1 & 2 \end{smallmatrix}\right)$ imposes $u_1 = 2u_2$ for $\lambda = 1$, with eigenvector $w_1 = (2,1)^\top$, and $\frac{7}{10}I_2 - M = \frac{1}{10}\left(\begin{smallmatrix} -2 & -2 \\ -1 & -1 \end{smallmatrix}\right)$ imposes $u_1 = -u_2$ for $\lambda = \frac{7}{10}$, with eigenvector $w_2 = (1,-1)^\top$.

Start with 100 people in the first town and 200 in the second, so $x_0 = (100, 200)^\top$. Solving $a_1(2,1)^\top + a_2(1,-1)^\top = (100,200)^\top$ means $2a_1 + a_2 = 100$ and $a_1 - a_2 = 200$; adding the two equations gives $3a_1 = 300$, so $a_1 = 100$ and $a_2 = -100$. By theorem 4.5.1(3),

$$x_m = M^m x_0 = 100 \cdot 1^m \begin{pmatrix} 2 \\ 1 \end{pmatrix} - 100\left(\tfrac{7}{10}\right)^m \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

At $m = 1$ this is $(200,100)^\top - 70(1,-1)^\top = (130,170)^\top$, which agrees with the direct computation $Mx_0 = \frac{1}{10}(900+400, 100+1600)^\top = (130,170)^\top$. At $m = 2$ it is $(200,100)^\top - 49(1,-1)^\top = (151,149)^\top$, agreeing with $Mx_1 = \frac{1}{10}(1170+340, 130+1360)^\top = (151,149)^\top$.

The first term does not move, because its eigenvalue is 1. The second is multiplied by $\frac{7}{10}$ every year, and $\left(\frac{7}{10}\right)^{10} = 0.0282\ldots$, so after ten years each town is within 3 people of the split $(200,100)$. The populations can be driven as close to that split as desired by waiting long enough, and the split itself is an eigenvector for the eigenvalue 1: it is the distribution the exchange fixes.

Part (2) shows what the eigenvalue 1 does in a process that conserves a total, and what an eigenvalue of size less than 1 does to everything else.

The Fibonacci sequence is defined by $F_0 = 0$, $F_1 = 1$, and

$$F_{m+2} = F_{m+1} + F_m \qquad (m \geq 0)$$

so it begins $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55$. The definition computes F_{100} in ninety-nine additions and says nothing about how large it is. No single number is carried from one step to the next, but a pair is: F_{m+2} and F_{m+1} are determined by F_{m+1} and F_m. Stacking the pair into a column,

$$\begin{pmatrix} F_{m+2} \\ F_{m+1} \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} F_{m+1} \\ F_m \end{pmatrix}$$

where the first entry of the product is $F_{m+1} + F_m$, which is F_{m+2}, and the second entry is F_{m+1}. The recurrence has become one matrix applied repeatedly, which is what theorem 4.5.1 handles.

The characteristic polynomial of this matrix is

$$\det\begin{pmatrix} t-1 & -1 \\ -1 & t \end{pmatrix} = t(t-1) - 1 = t^2 - t - 1$$

and the recurrence rearranges to $F_{m+2} - F_{m+1} - F_m = 0$, with the same coefficients $1, -1, -1$. The matrix was assembled from the recurrence, and its characteristic polynomial came back out. That is the pattern to record in general.

Definition 4.5.3: Companion Matrix

Let $n \geq 1$ and let

$$p(t) = t^n + c_{n-1}t^{n-1} + \cdots + c_1 t + c_0$$

be a monic polynomial with coefficients $c_0, \ldots, c_{n-1} \in k$, that is, a polynomial of degree n whose leading coefficient is 1. The *companion matrix* of p is the matrix $C_p \in M_n(k)$ whose first row is $(-c_{n-1}, -c_{n-2}, \ldots, -c_1, -c_0)$, whose $(i, i-1)$ entry is 1 for $2 \leq i \leq n$, and whose other entries are 0:

$$C_p = \begin{pmatrix} -c_{n-1} & -c_{n-2} & \cdots & -c_1 & -c_0 \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}$$

For $n = 1$ there are no entries below the first row and C_p is the 1×1 matrix $(-c_0)$.

For $p(t) = t^2 - t - 1$ the coefficients are $c_1 = -1$ and $c_0 = -1$, so the first row of C_p is $(1, 1)$ and the entry below the diagonal is 1: this is the Fibonacci matrix $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ above. For $p(t) = t^3 + 5t - 2$ the coefficients are $c_2 = 0$, $c_1 = 5$ and $c_0 = -2$, and

$$C_p = \begin{pmatrix} 0 & -5 & 2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

A companion matrix is therefore built by reading the coefficients of p backwards into the top row with their signs reversed, and filling in a line of ones below the diagonal.

The observation to prove is the one the Fibonacci case suggested: the polynomial that goes into the construction comes back out as the characteristic polynomial. It is useful twice over, since it gives the roots needed to diagonalize C_p, and it shows that every monic polynomial of degree n with coefficients in k is the characteristic polynomial of some matrix in $M_n(k)$.

Proposition 4.5.4: The Characteristic Polynomial of a Companion Matrix

Let $p(t) = t^n + c_{n-1}t^{n-1} + \cdots + c_1 t + c_0$ be monic of degree $n \geq 1$ with coefficients in k, and let $C_p \in M_n(k)$ be its companion matrix (definition 4.5.3). Then

$$\chi_{C_p}(t) = \det(tI_n - C_p) = p(t)$$

***Proof*:**
Induct on n. For $n = 1$ the matrix is $C_p = (-c_0)$, so $tI_1 - C_p = (t + c_0)$ and its determinant is $t + c_0$, which is $p(t)$.

Let $n \geq 2$ and assume the statement for monic polynomials of degree $n-1$. Write $N = tI_n - C_p$, whose entries are read off from definition 4.5.3: the first row is $(t + c_{n-1}, c_{n-2}, \dots, c_1, c_0)$, and for $2 \leq i \leq n$ the i-th row has -1 in column $i-1$, t in column i, and 0 elsewhere,

$$N = \begin{pmatrix} t + c_{n-1} & c_{n-2} & \cdots & c_1 & c_0 \\ -1 & t & \cdots & 0 & 0 \\ 0 & -1 & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & -1 & t \end{pmatrix}$$

Its last column has c_0 in row 1, t in row n, and 0 in every other row. Expanding the determinant along that column (theorem 3.3.4) leaves two terms,

$$\det N = (-1)^{1+n} c_0 \det N_{1n} + (-1)^{n+n} t \det N_{nn}$$

where N_{1n} and N_{nn} are obtained from N by deleting the indicated row and column.

Deleting row 1 and column n leaves the rows $2, \dots, n$ of N in the columns $1, \dots, n-1$. In this $(n-1) \times (n-1)$ matrix, the entry -1 of row i sits in position $(i-1, i-1)$ and the entry t of row i sits in position $(i-1, i)$, so N_{1n} is upper triangular with every diagonal entry equal to -1. By proposition 3.2.6, $\det N_{1n} = (-1)^{n-1}$, and the first term above is $(-1)^{1+n}(-1)^{n-1} c_0 = (-1)^{2n} c_0 = c_0$.

Deleting row n and column n leaves the rows $1, \dots, n-1$ of N in the columns $1, \dots, n-1$: the first row becomes $(t + c_{n-1}, c_{n-2}, \dots, c_1)$, and for $2 \leq i \leq n-1$ the i-th row has -1 in column $i-1$ and t in column i. Comparing this entry by entry with the display for N one size down, $N_{nn} = tI_{n-1} - C_q$, where C_q is the companion matrix of the monic polynomial of degree $n-1$

$$q(t) = t^{n-1} + c_{n-1} t^{n-2} + c_{n-2} t^{n-3} + \cdots + c_2 t + c_1$$

whose coefficients, read from the top row of C_q, are $c_{n-1}, \dots, c_1$. The inductive hypothesis gives $\det N_{nn} = q(t)$, so

$$\det N = c_0 + t\, q(t) = c_0 + t^n + c_{n-1} t^{n-1} + c_{n-2} t^{n-2} + \cdots + c_1 t$$

which is $p(t)$, as we sought to show.

The proof also shows why the first row of definition 4.5.3 carries minus signs: they are what the subtraction in $tI_n - C_p$ turns back into the coefficients of p. With the convention $\chi_A(t) = \det(tI_n - A)$ of section 4.1, the polynomial comes out monic and with no stray factor of $(-1)^n$; the opposite convention would produce $(-1)^n p(t)$ here.

A sequence obeying a recurrence can now be treated exactly as the Fibonacci sequence was. The state is the block of the last n terms, one step of the recurrence is multiplication by a companion matrix, and m steps is a power of it.

Theorem 4.5.5: Solutions of a Linear Recurrence with Distinct Roots

Let $n \geq 1$, let $c_0, \dots, c_{n-1} \in k$ with $c_0 \neq 0$, and consider the sequences $x_0, x_1, x_2, \dots$ of elements of k satisfying

$$x_{m+n} = -c_{n-1}x_{m+n-1} - \cdots - c_1 x_{m+1} - c_0 x_m \qquad \text{for every } m \geq 0$$

Let $p(t) = t^n + c_{n-1}t^{n-1} + \cdots + c_1 t + c_0$ and suppose p has n distinct roots $\lambda_1, \dots, \lambda_n$ in k.

1. For any $a_1, \dots, a_n \in k$, the sequence given by $x_m = a_1\lambda_1^m + \cdots + a_n\lambda_n^m$ satisfies the recurrence.
2. Every sequence satisfying the recurrence has this form for exactly one list $a_1, \dots, a_n \in k$, and that list is determined by the first n terms $x_0, \dots, x_{n-1}$.

***Proof*:**
Since $p(0) = c_0 \neq 0$, no root of p is 0, so every power λ_i^m appearing below has a nonzero base.

1. Fix $a_1, \dots, a_n \in k$ and set $x_m = a_1\lambda_1^m + \cdots + a_n\lambda_n^m$. For each index i and each $m \geq 0$, factoring λ_i^m out of the polynomial identity $p(\lambda_i) = 0$ gives

   $$\lambda_i^{m+n} + c_{n-1}\lambda_i^{m+n-1} + \cdots + c_1\lambda_i^{m+1} + c_0\lambda_i^m = \lambda_i^m\, p(\lambda_i) = 0$$

   Multiply this by a_i and add the n equations so obtained. For each exponent r between m and $m+n$ the terms carrying $\lambda_1^r, \dots, \lambda_n^r$ collect into $a_1\lambda_1^r + \cdots + a_n\lambda_n^r = x_r$, so the sum is

   $$x_{m+n} + c_{n-1}x_{m+n-1} + \cdots + c_1 x_{m+1} + c_0 x_m = 0$$

   which is the recurrence with every term moved to one side.

2. Let $x_0, x_1, \dots$ satisfy the recurrence, and collect its terms n at a time into the columns

   $$X_m = (x_{m+n-1}, x_{m+n-2}, \dots, x_m)^\top \in k^n \qquad (m \geq 0)$$

   so the j-th entry of X_m is x_{m+n-j}. Then $X_{m+1} = C_p X_m$, where C_p is the companion matrix of p. For the first entry, the first row of C_p is $(-c_{n-1}, \dots, -c_0)$, so by definition 1.4.1 the first entry of $C_p X_m$ is $-c_{n-1}x_{m+n-1} - \cdots - c_0 x_m$, which the recurrence says is x_{m+n}, the first entry of X_{m+1}. For $2 \leq j \leq n$, the j-th row of C_p has its only nonzero entry, a 1, in column $j-1$, so the j-th entry of $C_p X_m$ is the $(j-1)$-th entry of X_m, namely $x_{m+n-j+1}$; and the j-th entry of X_{m+1} is $x_{(m+1)+n-j}$, the same number. Iterating $X_{m+1} = C_p X_m$ from $m = 0$ gives $X_m = C_p^m X_0$ for every $m \geq 0$.

   Next, C_p is diagonalized by the roots. By proposition 4.5.4 the characteristic polynomial of C_p is p, so by theorem 4.1.10 the eigenvalues of C_p are exactly $\lambda_1, \dots, \lambda_n$. For each i put

   $$v_i = (\lambda_i^{n-1}, \lambda_i^{n-2}, \dots, \lambda_i, 1)^\top \in k^n$$

   whose j-th entry is λ_i^{n-j}. The first entry of $C_p v_i$ is $-c_{n-1}\lambda_i^{n-1} - \cdots - c_1\lambda_i - c_0$, which equals λ_i^n because $p(\lambda_i) = 0$ rearranges to $\lambda_i^n = -c_{n-1}\lambda_i^{n-1} - \cdots - c_0$; and $\lambda_i^n = \lambda_i \cdot \lambda_i^{n-1}$ is λ_i times the first entry of v_i. For $2 \leq j \leq n$ the j-th entry of $C_p v_i$ is the $(j-1)$-th entry of v_i, namely $\lambda_i^{n-j+1} = \lambda_i \cdot \lambda_i^{n-j}$, which is λ_i times the j-th entry of v_i. Hence $C_p v_i = \lambda_i v_i$, and $v_i \neq 0$ because its last entry is 1, so v_i is an eigenvector of C_p for λ_i (definition 4.1.3).

The λ_i are distinct, so $v_1, \ldots, v_n$ is linearly independent by theorem 4.2.3. It is a list of n vectors in k^n, and $\dim k^n = n$ because the standard basis of example 2.2.2(1) has n vectors (definition 2.2.10), so corollary 2.2.13 makes it a basis of k^n. It consists of eigenvectors of C_p, so C_p is diagonalizable (theorem 4.2.2), and proposition 4.2.5 takes P to be the matrix with columns $v_1, \ldots, v_n$ and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$, so that $C_p = PDP^{-1}$.

Now apply theorem 4.5.1(3) to C_p and the column X_0: there is exactly one list $a_1, \ldots, a_n \in k$ with $X_0 = a_1 v_1 + \cdots + a_n v_n$, and for every $m \geq 0$,

$$X_m = C_p^m X_0 = a_1 \lambda_1^m v_1 + \cdots + a_n \lambda_n^m v_n$$

Reading the last entry of each side, and using that the last entry of v_i is 1 while the last entry of X_m is x_m, gives $x_m = a_1 \lambda_1^m + \cdots + a_n \lambda_n^m$ for every $m \geq 0$. The list $a_1, \ldots, a_n$ is the unique one expressing X_0, and $X_0 = (x_{n-1}, \ldots, x_0)^\top$ is built from the first n terms of the sequence, so those terms determine the list.

Only one list can produce the sequence. If $a'_1, \ldots, a'_n$ also satisfies $x_m = a'_1 \lambda_1^m + \cdots + a'_n \lambda_n^m$ for every $m \geq 0$, then the column $a'_1 v_1 + \cdots + a'_n v_n$ has j-th entry $\sum_i a'_i \lambda_i^{n-j} = x_{n-j}$, so it equals X_0; and the expression of X_0 in the basis $v_1, \ldots, v_n$ is unique by theorem 2.2.3, so $a'_i = a_i$ for every i, completing the proof.

The theorem converts a rule for producing the next term into a formula for the m-th term, and the conversion is entirely mechanical: factor the polynomial, then solve one system of n equations in the n unknowns $a_1, \ldots, a_n$ using the first n terms. Part (2) also records that such a sequence is determined by its first n terms, since those fix the column X_0 of the proof, and X_0 fixes every later block X_m.

The hypothesis that the n roots are distinct cannot be dropped. Take $n = 2$ and $p(t) = t^2 - 2t + 1$, so $c_1 = -2$ and $c_0 = 1 \neq 0$, and the only root is 1, repeated. The recurrence is $x_{m+2} = 2x_{m+1} - x_m$, and the sequence $x_m = m$ solves it, because $2(m+1) - m = m + 2$. A combination $a_1 1^m + a_2 1^m$ is the constant sequence with value $a_1 + a_2$, and no constant sequence is $x_m = m$. What replaces the formula when a root repeats needs a canonical form for matrices that are not diagonalizable, and is settled in chapter 12.

The Fibonacci sequence has distinct roots, so the theorem applies to it.

Example 4.5.6: Binet's Formula for the Fibonacci Numbers

The Fibonacci recurrence $F_{m+2} = F_{m+1} + F_m$ rearranges to $F_{m+2} - F_{m+1} - F_m = 0$, which is the recurrence of theorem 4.5.5 with $n = 2$, $c_1 = -1$ and $c_0 = -1 \neq 0$, so

$$p(t) = t^2 - t - 1$$

By the quadratic formula its roots are $\frac{1 \pm \sqrt{1+4}}{2} = \frac{1 \pm \sqrt{5}}{2}$. Write

$$\varphi = \frac{1 + \sqrt{5}}{2}, \qquad \psi = \frac{1 - \sqrt{5}}{2}$$

These are two distinct real numbers, because $\sqrt{5} \neq 0$, so the theorem applies over $k = \mathbb{R}$ and every solution of the recurrence is $x_m = a_1 \varphi^m + a_2 \psi^m$.

The Fibonacci sequence is the solution with $F_0 = 0$ and $F_1 = 1$. Setting $m = 0$ and $m = 1$ in the formula gives

$$a_1 + a_2 = 0, \qquad a_1\varphi + a_2\psi = 1$$

The first equation gives $a_2 = -a_1$, and substituting into the second gives $a_1(\varphi - \psi) = 1$. Now $\varphi - \psi = \frac{(1+\sqrt{5})-(1-\sqrt{5})}{2} = \sqrt{5}$, so $a_1 = 1/\sqrt{5}$ and $a_2 = -1/\sqrt{5}$, and

$$F_m = \frac{1}{\sqrt{5}}\left(\left(\frac{1+\sqrt{5}}{2}\right)^m - \left(\frac{1-\sqrt{5}}{2}\right)^m\right)$$

which is *Binet's formula.*

The formula checks against the first three terms. At $m = 0$ the two powers are both 1 and it gives $0 = F_0$. At $m = 1$ it gives $(\varphi - \psi)/\sqrt{5} = \sqrt{5}/\sqrt{5} = 1 = F_1$. At $m = 2$, both φ and ψ satisfy $t^2 = t + 1$, so $\varphi^2 - \psi^2 = (\varphi + 1) - (\psi + 1) = \varphi - \psi = \sqrt{5}$, and the formula gives $1 = F_2$.

Numerically $\varphi = 1.6180\ldots$ and $\psi = -0.6180\ldots$. Since $|\psi| < 1$, the second term satisfies $|\psi^m/\sqrt{5}| \le 1/\sqrt{5} = 0.4472\cdots < \frac{1}{2}$ for every $m \ge 0$, so F_m is the integer nearest to $\varphi^m/\sqrt{5}$. At $m = 10$ this reads $\varphi^{10}/\sqrt{5} = 122.99\ldots/2.2360\cdots = 55.003\ldots$, and $F_{10} = 55$.

The formula answers what the recurrence could not: F_m grows like φ^m, multiplying by roughly 1.618 at each step, and the rule $F_{m+2} = F_{m+1} + F_m$ displays no such number anywhere. The right-hand side is built out of $\sqrt{5}$ and nonetheless takes an integer value at every m, since it equals F_m, and every F_m is an integer by the recurrence.

Exercise 4.5.1

1. Let $B = \begin{pmatrix} 5 & -6 \\ 3 & -4 \end{pmatrix}$. Find a formula for B^m valid for every integer $m \ge 0$, and check it at $m = 0$, $m = 1$ and $m = 2$ against direct computation.

2. Write down the companion matrix C_p of $p(t) = t^3 - 2t^2 - t + 2$, compute $\det(tI_3 - C_p)$ directly by a cofactor expansion, and confirm that it equals $p(t)$. Then factor p and list the eigenvalues of C_p.

3. Let $x_0 = 1$, $x_1 = 8$, and $x_{m+2} = x_{m+1} + 6x_m$ for $m \ge 0$. Find a closed formula for x_m and verify it at $m = 0, 1, 2$.

4. A quantity is redistributed between two containers by $x_{m+1} = Mx_m$, where $M = \frac{1}{5}\begin{pmatrix} 4 & 1 \\ 1 & 4 \end{pmatrix}$ and $x_0 = (10, 0)^\top$. Find a formula for x_m, identify the column the states approach, and find the smallest m for which both entries of x_m are within 0.1 of that column.

5. Let $C = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ be the Fibonacci matrix. Prove by induction that

$$C^m = \begin{pmatrix} F_{m+1} & F_m \\ F_m & F_{m-1} \end{pmatrix} \qquad (m \ge 1)$$

   and deduce from theorem 3.2.10 that $F_{m+1}F_{m-1} - F_m^2 = (-1)^m$ for every $m \ge 1$.

6. Consider $x_{m+2} = 4x_{m+1} - 4x_m$. Show that $x_m = m2^m$ is a solution, and that it is not of the form $a\lambda^m$ for any constants a and λ. Which hypothesis of theorem 4.5.5 fails, and why does

the theorem not contradict this?

Part II

Geometry

Part I treated a matrix as a bookkeeping device: it recorded a system, and elimination extracted what the system had to say. Nothing available there could speak of length, of angle, or of one candidate answer being better than another. This part gives that vocabulary, and gives it from the scalars rather than from any new assumption about matrices. An ordering makes it possible to call a quantity nonnegative; a conjugation makes it possible to force $\langle v, v\rangle$ to be real. Those two are what $\mathbb{R}$ and $\mathbb{C}$ have and a general field does not, and everything in this part follows from them.

Chapter 5 introduces the inner product and extracts what it carries: length, orthogonality, orthonormal bases, and the adjoint of a linear map. Chapter 6 spends that on the question Part I could not phrase, namely what to do with a system that has no solution, and finds the best available answer to be a projection. Chapter 7 asks which operators become diagonal in coordinates that preserve the geometry, and answers it completely for the operators that matter.

5

Inner Product Spaces

The dot product of two columns in $\mathbb{R}^n$ is an old acquaintance, and almost everything this chapter does is already contained in it: the length of a column is the square root of its dot product with itself, two columns are perpendicular when their dot product vanishes, and the projection of one column onto another is written with dot products alone. What the chapter does is isolate the properties that make those three statements work, keep them, and discard the formula.

Two of the properties are the ones Part II was built on. The dot product of a column with itself is a sum of squares, so it is real and never negative, and it vanishes only for the zero column; over $\mathbb{C}$ the naive sum $\sum z_i^2$ has neither property, and repairing it is what forces a conjugate into the definition.

Section 5.1 makes the repair and derives the Cauchy-Schwarz inequality from it. Section 5.2 extracts the length function and identifies which lengths come from an inner product at all. Section 5.3 produces bases in which every computation is a dot product, and gives the procedure that manufactures them. Section 5.4 records that procedure as a factorization of a matrix, three different ways. Section 5.5 turns the geometry back on linear maps: every linear functional is an inner product against a fixed vector, and every map acquires an adjoint. The last of these is what *Functional Analysis* [16] later rebuilds in infinite dimensions, where none of it is automatic.

5.1 Inner Products over $\mathbb{R}$ and $\mathbb{C}$; the Cauchy-Schwarz Inequality

The dot product on $\mathbb{R}^n$ is where length and angle come from, and the properties that produce them are few enough to isolate. This section names those properties, repairs the one of them that fails over $\mathbb{C}$, and proves that the product of two vectors never exceeds the product of their lengths.

Example 5.1.1: The Dot Product on $\mathbb{R}^n$

For columns $x, y \in \mathbb{R}^n$ set

$$x \cdot y = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

a real number obtained by multiplying the two lists entry by entry and adding. For $x = (2, 1, -2)^\top$ and $y = (1, 3, 1)^\top$ in $\mathbb{R}^3$,

$$x \cdot y = 2 \cdot 1 + 1 \cdot 3 + (-2) \cdot 1 = 3, \qquad x \cdot x = 4 + 1 + 4 = 9, \qquad y \cdot y = 1 + 9 + 1 = 11$$

The middle computation is the one that measures length. In $\mathbb{R}^2$ the Pythagorean theorem gives the length of $(x_1, x_2)^\top$ as $\sqrt{x_1^2 + x_2^2}$, and applying it twice in $\mathbb{R}^3$, once in the horizontal plane and once in the vertical triangle over the resulting segment, gives the length of $(x_1, x_2, x_3)^\top$ as $\sqrt{x_1^2 + x_2^2 + x_3^2}$. So $\sqrt{x \cdot x} = 3$ is the length of the x above.

Three properties can be read off the formula at once, and they are the only ones this section will use.

1. $x \cdot y = y \cdot x$, because $x_i y_i = y_i x_i$ for each i.
2. The product is linear in its first argument:

   $$(ax + a'x') \cdot y = \sum_{i=1}^{n} (ax_i + a'x'_i) y_i = a \sum_{i=1}^{n} x_i y_i + a' \sum_{i=1}^{n} x'_i y_i = a(x \cdot y) + a'(x' \cdot y)$$

   for all $x, x', y \in \mathbb{R}^n$ and all $a, a' \in \mathbb{R}$. By (1) it is linear in its second argument as well.
3. $x \cdot x = x_1^2 + \cdots + x_n^2 \geq 0$, since a square of a real number is nonnegative. A sum of nonnegative terms vanishes exactly when every term does, so $x \cdot x = 0$ if and only if $x_1 = \cdots = x_n = 0$, that is, if and only if $x = 0$.

The running matrix A of example 1.2.9 has columns $c_1 = (1, 2, 1, 3)^\top$, $c_2 = (1, 3, 2, 4)^\top$ and $c_3 = (3, 8, 5, 11)^\top$ in $\mathbb{R}^4$, and

$$c_1 \cdot c_1 = 1 + 4 + 1 + 9 = 15, \qquad c_1 \cdot c_2 = 1 + 6 + 2 + 12 = 21, \qquad c_2 \cdot c_2 = 1 + 9 + 4 + 16 = 30$$

The column relation $c_3 = c_1 + 2c_2$ recorded in example 1.2.9 can be checked against property (2): it predicts $c_1 \cdot c_3 = c_1 \cdot c_1 + 2(c_1 \cdot c_2) = 15 + 42 = 57$, and the direct computation gives $c_1 \cdot c_3 = 3 + 16 + 5 + 33 = 57$.

Over $\mathbb{C}$ the same formula loses property (3), and it loses the whole of it, not just the inequality. Take $z = (1, i)^\top \in \mathbb{C}^2$ and form $z_1 z_1 + z_2 z_2$:

$$1 \cdot 1 + i \cdot i = 1 + i^2 = 0$$

so a nonzero vector has been assigned the same number as the zero vector, and a length defined as the square root of that number would be 0 for z. The source of the failure is that w^2 is not a measure of the size of a complex number w, whereas $w\overline{w} = |w|^2$ is, where $\overline{w}$ is the complex conjugate of w and $|w|$ its modulus. Conjugating the second factor,

$$z_1 \overline{z_1} + z_2 \overline{z_2} = |z_1|^2 + |z_2|^2$$

is a nonnegative real number, equal to $1 + 1 = 2$ for the z above, and it vanishes exactly when every entry of z does.

Conjugating the second factor changes the other two properties. Property (1) becomes $\sum_i y_i\overline{x_i} = \overline{\sum_i x_i\overline{y_i}}$, so swapping the two arguments conjugates the answer rather than preserving it, and property (2) survives in the first argument but not in the second, where scalars now come out conjugated. The definition below records exactly this compromise, and over $\mathbb{R}$, where conjugation does nothing, it reduces to properties (1), (2) and (3) of example 5.1.1.

Definition 5.1.2: Inner Product and Inner Product Space

Let k be $\mathbb{R}$ or $\mathbb{C}$ and let V be a vector space over k (definition 2.2.14). An *inner product* on V is a function $\langle -,-\rangle\colon V\times V\to k$, the two dashes marking the slots into which a pair of vectors is placed, such that the following hold for all $u,u',v\in V$ and all $a\in k$.

1. (Linearity in the first argument) $\langle u+u',v\rangle = \langle u,v\rangle + \langle u',v\rangle$ and $\langle au,v\rangle = a\langle u,v\rangle$.
2. (Conjugate symmetry) $\langle v,u\rangle = \overline{\langle u,v\rangle}$, the bar denoting complex conjugation on k, which is the identity when $k=\mathbb{R}$.
3. (Positive definiteness) $\langle v,v\rangle > 0$ for every $v\neq 0$. The inequality is meaningful because (2) applied with $u=v$ gives $\langle v,v\rangle = \overline{\langle v,v\rangle}$, so $\langle v,v\rangle$ is a real number.

A vector space over k carrying an inner product is an *inner product space* over k.

Over $\mathbb{R}$ condition (2) is plain symmetry, and (1) together with (2) makes the product linear in each argument separately. Over $\mathbb{C}$ the second argument behaves differently, and how it behaves is the content of proposition 5.1.6(2) below. The convention fixed here, linear in the first slot and conjugate-linear in the second, is in force for the whole book.

Example 5.1.3: The Standard Inner Product on $\mathbb{C}^n$

On k^n, for k equal to $\mathbb{R}$ or $\mathbb{C}$, the *standard inner product* is

$$\langle x,y\rangle = x_1\overline{y_1} + x_2\overline{y_2} + \cdots + x_n\overline{y_n}$$

Each condition of definition 5.1.2 is a computation with the entries. For (1),

$$\langle x+x',y\rangle = \sum_{i=1}^{n}(x_i+x_i')\overline{y_i} = \sum_{i=1}^{n}x_i\overline{y_i} + \sum_{i=1}^{n}x_i'\overline{y_i} = \langle x,y\rangle + \langle x',y\rangle$$

and $\langle ax,y\rangle = \sum_i ax_i\overline{y_i} = a\langle x,y\rangle$. For (2), conjugation is additive and multiplicative and satisfies $\overline{\overline{w}} = w$, so

$$\overline{\langle x,y\rangle} = \sum_{i=1}^{n}\overline{x_i\overline{y_i}} = \sum_{i=1}^{n}\overline{x_i}\,y_i = \sum_{i=1}^{n}y_i\overline{x_i} = \langle y,x\rangle$$

For (3), $\langle x,x\rangle = \sum_i x_i\overline{x_i} = \sum_i |x_i|^2$, a sum of nonnegative real numbers, which is positive unless every x_i is zero.

Take $z=(1,i)^\top$ and $w=(2i,3)^\top$ in $\mathbb{C}^2$. Then

$$\langle z,w\rangle = 1\cdot\overline{2i} + i\cdot\overline{3} = -2i+3i = i, \qquad \langle w,z\rangle = 2i\cdot\overline{1} + 3\cdot\overline{i} = 2i-3i = -i$$

and the two are indeed conjugate. Their self-products are $\langle z,z\rangle = 1+1 = 2$ and $\langle w,w\rangle = 4+9 = 13$, both real and positive. When $k=\mathbb{R}$ every conjugation is the identity and the formula is the dot product of example 5.1.1, so that example is the case $k=\mathbb{R}$ of this one.

A vector space admits many different inner products, and changing the inner product changes every length measured with it. The next example is the simplest family exhibiting this.

Example 5.1.4: Weighted Inner Products on k^n

Fix real numbers $w_1, \ldots, w_n$ with $w_i > 0$ for each i, and define

$$\langle x, y\rangle_w = w_1 x_1 \overline{y_1} + \cdots + w_n x_n \overline{y_n}$$

for $x, y \in k^n$. Condition (1) of definition 5.1.2 is the same computation as in example 5.1.3 with the constant w_i carried along inside each term. Condition (2) uses that each w_i is real, so $\overline{w_i} = w_i$ and

$$\overline{\langle x, y\rangle_w} = \sum_{i=1}^{n} \overline{w_i}\,\overline{x_i}\, y_i = \sum_{i=1}^{n} w_i y_i \overline{x_i} = \langle y, x\rangle_w$$

Condition (3) uses that each w_i is positive: $\langle x, x\rangle_w = \sum_i w_i |x_i|^2$ is a sum of nonnegative terms, and it vanishes only when every $|x_i|^2$ does, since each $w_i \neq 0$.

On $\mathbb{R}^2$ with $w = (1, 4)$ the products of a vector with itself are $\langle x, x\rangle = x_1^2 + x_2^2$ and $\langle x, x\rangle_w = x_1^2 + 4x_2^2$. The vector $(0, 1)^\top$ has self-product 1 for the first and 4 for the second, so the second inner product measures the vertical direction as twice as long. The vectors with $\langle x, x\rangle_w = 1$ form the ellipse $x_1^2 + 4x_2^2 = 1$, which meets the horizontal axis at ± 1 and the vertical axis at $\pm\frac{1}{2}$.

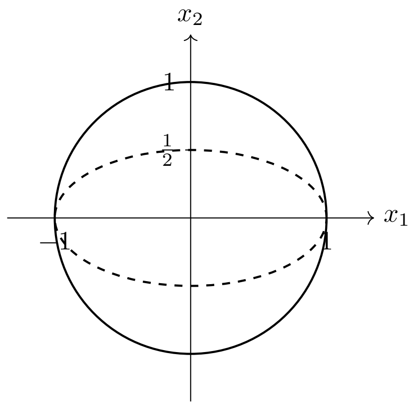


Figure 5.1: The vectors $x \in \mathbb{R}^2$ with $\langle x, x\rangle = 1$ for the standard inner product (solid) and with $\langle x, x\rangle_w = 1$ for the weighted inner product with $w = (1, 4)$ (dashed)

Each of the three products on $\mathbb{R}^2$ below satisfies some of the conditions of definition 5.1.2 and fails another, which shows what each condition excludes.

Example 5.1.5: Three Products That Are Not Inner Products

Each of the following is defined for $x, y \in \mathbb{R}^2$.

1. $\langle x, y\rangle_1 = x_1 y_1 - x_2 y_2$. This is linear in the first argument and symmetric, so conditions (1) and (2) of definition 5.1.2 hold. Condition (3) fails at $x = (0, 1)^\top$, where $\langle x, x\rangle_1 = -1 < 0$, and it fails again at $x = (1, 1)^\top$, where $\langle x, x\rangle_1 = 1 - 1 = 0$ with $x \neq 0$.

2. $\langle x, y\rangle_2 = x_1 y_1$. Conditions (1) and (2) hold, and $\langle x, x\rangle_2 = x_1^2 \geq 0$ for every x. Condition (3) still fails, because $x = (0,1)^\top$ is nonzero with $\langle x, x\rangle_2 = 0$, so nonnegativity of $\langle x, x\rangle_2$ does not deliver condition (3).

3. $\langle x, y\rangle_3 = x_1 y_1 + x_1 y_2 + x_2 y_2$. This is linear in the first argument, and completing the square gives
$$\langle x, x\rangle_3 = x_1^2 + x_1 x_2 + x_2^2 = \left(x_1 + \tfrac{1}{2}x_2\right)^2 + \tfrac{3}{4}x_2^2$$
which is positive unless $x_2 = 0$ and then $x_1 = 0$, so conditions (1) and (3) hold. Condition (2) fails: with e_1, e_2 the standard basis columns, $\langle e_1, e_2\rangle_3 = 0 + 1 + 0 = 1$ while $\langle e_2, e_1\rangle_3 = 0 + 0 + 0 = 0$.

Item (1) satisfies conditions (1) and (2) without (3), and item (3) satisfies conditions (1) and (3) without (2), so neither of the last two conditions is a consequence of the other two.

The rest of this chapter argues from definition 5.1.2 and not from any formula in coordinates, so the four identities below are stated once here and cited thereafter. The fourth expands the self-product of a sum, and section 5.2, section 5.3 and section 5.4 each use it.

Proposition 5.1.6: Basic Identities in an Inner Product Space

Let V be an inner product space over k, where k is $\mathbb{R}$ or $\mathbb{C}$. For all $u, v, w \in V$ and all $\lambda \in k$, the following hold.

1. $\langle v, v\rangle$ is a real number with $\langle v, v\rangle \geq 0$, and $\langle v, v\rangle = 0$ if and only if $v = 0$.
2. $\langle u, v\rangle = \overline{\langle v, u\rangle}$, $\langle u, v + w\rangle = \langle u, v\rangle + \langle u, w\rangle$, and $\langle u, \lambda v\rangle = \overline{\lambda}\langle u, v\rangle$.
3. $\langle 0, v\rangle = \langle v, 0\rangle = 0$.
4. $\langle u + v, u + v\rangle = \langle u, u\rangle + 2\operatorname{Re}\langle u, v\rangle + \langle v, v\rangle$, which over $\mathbb{R}$ reads $\langle u, u\rangle + 2\langle u, v\rangle + \langle v, v\rangle$.

***Proof*:**

1. That $\langle v, v\rangle$ is real is the observation recorded in definition 5.1.2(3): condition (2) of that definition applied with $u = v$ gives $\langle v, v\rangle = \overline{\langle v, v\rangle}$, and a scalar equal to its own conjugate is real. If $v \neq 0$ then $\langle v, v\rangle > 0$ by definition 5.1.2(3). If $v = 0$, use additivity in the first argument (definition 5.1.2(1)) together with $0 + 0 = 0$ in V (definition 2.2.14(3) applied to the vector 0) to get
$$\langle 0, v\rangle = \langle 0 + 0, v\rangle = \langle 0, v\rangle + \langle 0, v\rangle$$
for every $v \in V$, whence $\langle 0, v\rangle = 0$; taking $v = 0$ gives $\langle 0, 0\rangle = 0$. So $\langle v, v\rangle \geq 0$ in both cases, and it vanishes exactly when $v = 0$.

2. The first identity is definition 5.1.2(2) with the names of the two vectors exchanged. For the second, apply that condition, then additivity in the first argument, then that condition twice more, using that conjugation is additive:
$$\langle u, v + w\rangle = \overline{\langle v + w, u\rangle} = \overline{\langle v, u\rangle + \langle w, u\rangle} = \overline{\langle v, u\rangle} + \overline{\langle w, u\rangle} = \langle u, v\rangle + \langle u, w\rangle$$
For the third, apply the same condition, then homogeneity in the first argument, then

multiplicativity of conjugation:

$$\langle u, \lambda v\rangle = \overline{\langle \lambda v, u\rangle} = \overline{\lambda\langle v, u\rangle} = \overline{\lambda}\,\overline{\langle v, u\rangle} = \overline{\lambda}\langle u, v\rangle$$

3. The identity $\langle 0, v\rangle = 0$ is the display in the proof of (1). The second equality follows from it by definition 5.1.2(2): $\langle v, 0\rangle = \overline{\langle 0, v\rangle} = \overline{0} = 0$.

4. Expanding first in the left argument by definition 5.1.2(1) and then in the right argument by part (2),

$$\langle u+v, u+v\rangle = \langle u, u+v\rangle + \langle v, u+v\rangle = \langle u, u\rangle + \langle u, v\rangle + \langle v, u\rangle + \langle v, v\rangle$$

By part (2) again, $\langle v, u\rangle = \overline{\langle u, v\rangle}$, and any $z \in \mathbb{C}$ satisfies $z + \overline{z} = 2\operatorname{Re} z$, so the two middle terms combine into $2\operatorname{Re}\langle u, v\rangle$. When $k = \mathbb{R}$ every scalar is its own real part, giving the stated form, and completing the proof.

Part (1) is what makes a length available: for every v the number $\langle v, v\rangle$ is a nonnegative real, so it has a nonnegative square root, and the notation

$$\|v\| = \sqrt{\langle v, v\rangle}$$

is used from here on, with $\|v\|^2 = \langle v, v\rangle$ by construction and $\|v\| = 0$ exactly when $v = 0$. On $\mathbb{R}^n$ with the dot product this is the Euclidean length computed in example 5.1.1. The properties of $\|\cdot\|$ as a function on V, and the question of which such functions arise from an inner product at all, are taken up in section 5.2.

Part (2) records what the conjugate does to the second argument: a scalar pulled out of it comes conjugated. Over $\mathbb{R}$ this changes nothing and the product is linear in each argument; over $\mathbb{C}$ it is the reason $\langle \lambda v, \lambda v\rangle = |\lambda|^2\langle v, v\rangle$ rather than $\lambda^2\langle v, v\rangle$, which is what keeps self-products real and nonnegative.

A number is now attached to a pair of vectors, and a number is attached to each of them separately. Whether the first is bounded by the product of the other two is the question the next theorem settles, and the bound is what will let $\langle u, v\rangle$ be interpreted as an angle. The proof subtracts from v the multiple cu for which $\langle v - cu, u\rangle = 0$, which forces $c\langle u, u\rangle = \langle v, u\rangle$, and then uses that the remainder has nonnegative self-product.

Theorem 5.1.7: The Cauchy-Schwarz Inequality

Let V be an inner product space over k, where k is $\mathbb{R}$ or $\mathbb{C}$. For all $u, v \in V$,

$$|\langle u, v\rangle| \leq \sqrt{\langle u, u\rangle}\sqrt{\langle v, v\rangle}$$

that is, $|\langle u, v\rangle| \leq \|u\|\,\|v\|$.

***Proof*:**
Suppose first that $u = 0$. Then $\langle u, v\rangle = 0$ by proposition 5.1.6(3) and $\langle u, u\rangle = 0$ by proposition 5.1.6(1), so both sides of the asserted inequality are 0.

Suppose now that $u \neq 0$, so that $\langle u, u\rangle > 0$ by proposition 5.1.6(1) and the scalar

$$c = \frac{\langle v, u\rangle}{\langle u, u\rangle}$$

is defined. Expanding $\langle v - cu, v - cu\rangle$ in the first argument by definition 5.1.2(1) and then in the second by proposition 5.1.6(2) gives

$$\langle v - cu, v - cu\rangle = \langle v, v\rangle - \bar{c}\,\langle v, u\rangle - c\,\langle u, v\rangle + c\bar{c}\,\langle u, u\rangle$$

Since $\langle u, u\rangle$ is a positive real number it is its own conjugate, so $\bar{c} = \overline{\langle v, u\rangle}/\langle u, u\rangle = \langle u, v\rangle/\langle u, u\rangle$, the second equality by proposition 5.1.6(2). Write $q = \langle u, v\rangle$ for brevity, so that $\langle v, u\rangle = \bar{q}$ and $q\bar{q} = |q|^2$. The three terms after $\langle v, v\rangle$ are then

$$\bar{c}\,\langle v, u\rangle = \frac{q\bar{q}}{\langle u, u\rangle} = \frac{|q|^2}{\langle u, u\rangle}, \qquad c\,\langle u, v\rangle = \frac{\bar{q}q}{\langle u, u\rangle} = \frac{|q|^2}{\langle u, u\rangle}, \qquad c\bar{c}\,\langle u, u\rangle = \frac{\bar{q}q}{\langle u, u\rangle^2}\langle u, u\rangle = \frac{|q|^2}{\langle u, u\rangle}$$

Substituting the three and cancelling, the expansion becomes

$$\langle v - cu, v - cu\rangle = \langle v, v\rangle - \frac{|\langle u, v\rangle|^2}{\langle u, u\rangle} \tag{5.1}$$

The left side is nonnegative by proposition 5.1.6(1), so $|\langle u, v\rangle|^2 \leq \langle u, u\rangle\langle v, v\rangle$ after multiplying through by the positive number $\langle u, u\rangle$. Both sides are nonnegative reals, and the square root is an increasing function on $[0, \infty)$, so taking square roots preserves the inequality and gives $|\langle u, v\rangle| \leq \sqrt{\langle u, u\rangle}\sqrt{\langle v, v\rangle}$, as we sought to show.

The identity (5.1) measures by how much the inequality fails to be an equality, so it also settles when equality occurs.

Corollary 5.1.8: The Equality Case of Cauchy-Schwarz

Let V be an inner product space over k and let $u, v \in V$. Then $|\langle u, v\rangle| = \|u\|\,\|v\|$ if and only if one of u, v is a scalar multiple of the other, that is, if and only if the list u, v is linearly dependent in the sense of definition 2.1.9, whose statement uses only the vector space operations and so reads verbatim in V.

***Proof*:**
The two formulations of the condition agree. If $v = \lambda u$ then $\lambda u + (-1)v = 0$ is a dependence relation with a nonzero coefficient, and symmetrically if $u = \mu v$. Conversely, a dependence relation $au + bv = 0$ has $a \neq 0$ or $b \neq 0$; in the first case $u = (-b/a)v$ and in the second $v = (-a/b)u$.

Suppose $|\langle u, v\rangle| = \|u\|\,\|v\|$. If $u = 0$ then $u = 0 \cdot v$ and the list is dependent. If $u \neq 0$, squaring the assumed equality gives $|\langle u, v\rangle|^2 = \langle u, u\rangle\langle v, v\rangle$, so dividing by the positive number $\langle u, u\rangle$ turns the right side of (5.1) into $\langle v, v\rangle - \langle v, v\rangle = 0$. Hence $\langle v - cu, v - cu\rangle = 0$ with $c = \langle v, u\rangle/\langle u, u\rangle$, and proposition 5.1.6(1) forces $v - cu = 0$, so $v = cu$.

Conversely, suppose one of the two is a multiple of the other. Assume first that $v = \lambda u$. Then $\langle u, v\rangle = \langle u, \lambda u\rangle = \bar{\lambda}\langle u, u\rangle$ by proposition 5.1.6(2), so $|\langle u, v\rangle| = |\bar{\lambda}|\,\langle u, u\rangle = |\lambda|\langle u, u\rangle$, using that

$\langle u, u\rangle$ is a nonnegative real and that a scalar and its conjugate have the same modulus. Also

$$\langle v, v\rangle = \langle \lambda u, \lambda u\rangle = \lambda\overline{\lambda}\langle u, u\rangle = |\lambda|^2\langle u, u\rangle$$

by definition 5.1.2(1) and proposition 5.1.6(2), so $\|v\| = |\lambda|\,\|u\|$ and $\|u\|\,\|v\| = |\lambda|\,\|u\|^2 = |\lambda|\langle u, u\rangle$. The two quantities agree. If instead $u = \mu v$, the same computation with the two vectors exchanged gives $|\langle v, u\rangle| = \|v\|\,\|u\|$, and $|\langle u, v\rangle| = |\overline{\langle v, u\rangle}| = |\langle v, u\rangle|$ by proposition 5.1.6(2), which completes the proof.

The columns of the running matrix show both cases. With the values computed in example 5.1.1, $|c_1 \cdot c_2| = 21$ while $\|c_1\|\,\|c_2\| = \sqrt{15}\sqrt{30} = \sqrt{450}$, and $441 < 450$, so the inequality is strict. It has to be: c_2 is not a multiple of c_1, since matching first entries would force the multiplier to be 1 and the second entries are 3 and 2. Replacing c_2 by $2c_1$ produces the equality case, since $|\langle c_1, 2c_1\rangle| = 2 \cdot 15 = 30$ and $\|c_1\|\,\|2c_1\| = \sqrt{15} \cdot 2\sqrt{15} = 30$.

Over $\mathbb{R}$ the inequality says that for $u, v \neq 0$ the quotient

$$\frac{\langle u, v\rangle}{\|u\|\,\|v\|}$$

lies in the interval $[-1, 1]$, which is exactly the range of the cosine on $[0, \pi]$. In $\mathbb{R}^2$ with the dot product that quotient is the cosine of the angle between the two vectors, and the computation is short: a nonzero $u \in \mathbb{R}^2$ satisfies $(u_1/\|u\|)^2 + (u_2/\|u\|)^2 = 1$, so $u = \|u\|(\cos\alpha, \sin\alpha)^\top$ for some α, and likewise $v = \|v\|(\cos\beta, \sin\beta)^\top$, giving

$$u \cdot v = \|u\|\,\|v\|(\cos\alpha\cos\beta + \sin\alpha\sin\beta) = \|u\|\,\|v\|\cos(\alpha - \beta)$$

by the addition formula for the cosine. Theorem 5.1.7 says that the quotient stays in $[-1, 1]$ in every real inner product space, so an angle can be attached to a pair of nonzero vectors in any of them, and corollary 5.1.8 says the extreme values ± 1 occur exactly for dependent pairs, which is where the angle is 0 or π. Over $\mathbb{C}$ the quotient is a complex number and only its modulus is bounded by 1; the case $\langle u, v\rangle = 0$, which survives both fields intact, is the one section 5.3 builds on.

Exercise 5.1.1

1. In $\mathbb{R}^3$ with the standard inner product let $x = (1, -2, 2)^\top$ and $y = (3, 0, 4)^\top$. Compute $\langle x, y\rangle$, $\|x\|$ and $\|y\|$, verify theorem 5.1.7 for this pair by comparing the two numbers, and decide from corollary 5.1.8 whether the inequality is strict.

2. In $\mathbb{C}^2$ with the standard inner product of example 5.1.3 let $z = (1, i)^\top$ and $w = (i, 2)^\top$. Compute $\langle z, w\rangle$ and $\langle w, z\rangle$ and check that they are conjugate. Compute $\|z\|$ and $\|w\|$, and verify theorem 5.1.7 for this pair.

3. Let $w_1, \ldots, w_n \in \mathbb{C}$ and define $\langle x, y\rangle_w = \sum_{i=1}^n w_i x_i \overline{y_i}$ for $x, y \in \mathbb{C}^n$. Show that $\langle -, -\rangle_w$ is an inner product on $\mathbb{C}^n$ if and only if every w_i is real and positive. For the "only if" direction, test the conditions of definition 5.1.2 on the standard basis columns.

4. Let $a_1, \ldots, a_n$ be real numbers. Prove that

$$\left(\sum_{i=1}^n a_i\right)^2 \leq n\sum_{i=1}^n a_i^2$$

and that equality holds exactly when $a_1 = a_2 = \cdots = a_n$. (Apply theorem 5.1.7 and corollary 5.1.8 in $\mathbb{R}^n$ to a well-chosen pair of vectors.)

5. Let V be an inner product space over k.

   1. Show that if $v \in V$ satisfies $\langle v, u \rangle = 0$ for every $u \in V$, then $v = 0$.
   2. Deduce that if $S, T \colon V \to V$ are linear maps (definition 2.4.2) with $\langle Tv, u \rangle = \langle Sv, u \rangle$ for all $u, v \in V$, then $S = T$.

6. On $\mathbb{R}^2$ define $\langle x, y \rangle_B = x_1 y_1 + x_1 y_2 + x_2 y_1 + 2 x_2 y_2$.

   1. Show that $\langle -, - \rangle_B$ is an inner product on $\mathbb{R}^2$, proving positive definiteness by completing the square.
   2. Find nonzero $x, y \in \mathbb{R}^2$ with $\langle x, y \rangle_B = 0$ but $x \cdot y \neq 0$, so that the vanishing of an inner product is a statement about the inner product and not only about the two vectors.

5.2 Norms, the Parallelogram Law, and the Polarization Identity

An inner product attaches a scalar to a pair of vectors. Evaluated at the pair (v, v) it produces $\langle v, v \rangle$, which proposition 5.1.6(1) makes real and nonnegative. The square root of that number is a length for v, and the first task here is to establish the three properties that entitle it to the name.

Two identities then reverse the construction. The parallelogram law is a relation among lengths alone, satisfied by every length function that comes from an inner product, so a candidate violating it comes from none. The polarization identity goes further and writes $\langle u, v \rangle$ itself in terms of lengths, which makes the inner product recoverable from the lengths it produces.

In $\mathbb{R}^2$ with the dot product of example 5.1.1, the vector $v = (3, 4)^\top$ has $\langle v, v \rangle = 3^2 + 4^2 = 25$, and the nonnegative number whose square is 25 is 5, the length a right triangle with legs 3 and 4 assigns to its hypotenuse. Over $\mathbb{C}$ the recipe still returns a nonnegative real, which is what the conjugate in the second slot gives: in $\mathbb{C}^2$ with the inner product of example 5.1.3, the vector $z = (1, i)^\top$ has

$$\langle z, z \rangle = |1|^2 + |i|^2 = 2$$

and length $\sqrt{2}$.

The three properties recorded in the definition below are the ones this construction has. They are stated for an arbitrary function first, because the second half of the section asks which functions with these three properties arise from an inner product at all.

Definition 5.2.1: Norm and Induced Norm

Let V be a vector space over k, where $k = \mathbb{R}$ or $k = \mathbb{C}$. A *norm* on V is a function $\|\cdot\|: V \to \mathbb{R}$, written with the vector between the two bars, such that for all $u, v \in V$ and all $\lambda \in k$:

1. $\|v\| \geq 0$, and $\|v\| = 0$ if and only if $v = 0$;
2. $\|\lambda v\| = |\lambda|\,\|v\|$;
3. $\|u + v\| \leq \|u\| + \|v\|$.

If V carries an inner product $\langle -, - \rangle$ as in definition 5.1.2, the *induced norm* is

$$\|v\| := \sqrt{\langle v, v \rangle}$$

the nonnegative square root, which exists because $\langle v, v \rangle$ is a nonnegative real by proposition 5.1.6(1). A vector with $\|v\| = 1$ is a *unit vector*.

Nothing yet says the induced norm is a norm. Conditions (1) and (2) come directly from the inner product axioms, and condition (3) is where the work is; the next proposition disposes of the first two and records the two identities used repeatedly below.

Proposition 5.2.2: Basic Properties of the Induced Norm

Let V be a vector space over k carrying an inner product $\langle -, - \rangle$, and let $\|\cdot\|$ be the induced norm. For all $u, v \in V$ and all $\lambda \in k$:

1. $\|v\| \geq 0$, and $\|v\| = 0$ if and only if $v = 0$.
2. $\|\lambda v\| = |\lambda|\,\|v\|$.
3. $\|u + v\|^2 = \|u\|^2 + 2\,\mathrm{Re}\langle u, v \rangle + \|v\|^2$ and $\|u - v\|^2 = \|u\|^2 - 2\,\mathrm{Re}\langle u, v \rangle + \|v\|^2$.
4. $|\langle u, v \rangle| \leq \|u\|\,\|v\|$, with equality if and only if u and v are linearly dependent.

***Proof*:**

1. By proposition 5.1.6(1) the number $\langle v, v \rangle$ is real and nonnegative, so its nonnegative square root $\|v\|$ is defined and satisfies $\|v\| \geq 0$. That square root vanishes exactly when $\langle v, v \rangle = 0$, which by the second half of proposition 5.1.6(1) happens exactly when $v = 0$.
2. Expanding the defining expression,

$$\|\lambda v\|^2 = \langle \lambda v, \lambda v \rangle = \lambda \langle v, \lambda v \rangle = \lambda \overline{\lambda} \langle v, v \rangle = |\lambda|^2 \|v\|^2$$

   where the second equality is linearity in the first argument (definition 5.1.2), the third is proposition 5.1.6(2), and the last uses $\lambda\overline{\lambda} = |\lambda|^2$. The two numbers $\|\lambda v\|$ and $|\lambda|\,\|v\|$ are both nonnegative and have the same square, so they are equal.
3. The first identity is proposition 5.1.6(4) with $\langle u, u \rangle$ and $\langle v, v \rangle$ written as $\|u\|^2$ and $\|v\|^2$. For the second, apply the first identity to the pair u and $-v$. Part (2) with $\lambda = -1$ gives $\|-v\| = |-1|\,\|v\| = \|v\|$, and proposition 5.1.6(2) with $\lambda = -1$ gives $\langle u, -v \rangle =$

$\overline{(-1)}\langle u, v\rangle = -\langle u, v\rangle$, whose real part is $-\operatorname{Re}\langle u, v\rangle$. Substituting these two values into the first identity yields the second.

4. Theorem 5.1.7 gives $|\langle u, v\rangle|^2 \leq \langle u, u\rangle\langle v, v\rangle = \|u\|^2\|v\|^2$. Both $|\langle u, v\rangle|$ and $\|u\|\,\|v\|$ are nonnegative, so taking nonnegative square roots preserves the inequality and gives $|\langle u, v\rangle| \leq \|u\|\,\|v\|$. Equality of these two nonnegative numbers is equality of their squares, which by corollary 5.1.8 holds exactly when u and v are linearly dependent, as we sought to show.

Part (3) is the identity the rest of the section computes with. Over $\mathbb{R}$ the conjugation is trivial, the real part is the whole inner product, and the two identities read $\|u \pm v\|^2 = \|u\|^2 \pm 2\langle u, v\rangle + \|v\|^2$; over $\mathbb{C}$ only the real part of $\langle u, v\rangle$ survives into a statement about lengths. That asymmetry is why the polarization identity below needs four terms over $\mathbb{C}$ and only two over $\mathbb{R}$.

Condition (3) of definition 5.2.1 remains, and unlike (1) and (2) it is not a restatement of an inner product axiom. Cauchy-Schwarz is what gives it.

Theorem 5.2.3: The Triangle Inequality

Let V be a vector space over k carrying an inner product, and let $\|\cdot\|$ be the induced norm. For all $u, v \in V$,

$$\|u + v\| \leq \|u\| + \|v\|$$

Equality holds if and only if one of u and v is a nonnegative real multiple of the other.

***Proof*:**
Step 1: the inequality. By proposition 5.2.2(3),

$$\|u+v\|^2 = \|u\|^2+2\operatorname{Re}\langle u, v\rangle+\|v\|^2 \leq \|u\|^2+2|\langle u, v\rangle|+\|v\|^2 \leq \|u\|^2+2\|u\|\,\|v\|+\|v\|^2 = \big(\|u\|+\|v\|\big)^2$$

The first inequality holds because the real part of a complex number is at most its absolute value, and the second is proposition 5.2.2(4). Both $\|u + v\|$ and $\|u\| + \|v\|$ are nonnegative by proposition 5.2.2(1), so the inequality between their squares gives $\|u + v\| \leq \|u\| + \|v\|$.

Step 2: the equality case. Suppose first that $v = cu$ for a real $c \geq 0$. Then $u + v = (1 + c)u$, so proposition 5.2.2(2) gives $\|u+v\| = (1+c)\|u\| = \|u\|+c\|u\| = \|u\|+\|v\|$, using proposition 5.2.2(2) again for $\|v\| = c\|u\|$. The case $u = cv$ with $c \geq 0$ is the same computation with the roles exchanged.

Conversely, suppose $\|u+v\| = \|u\|+\|v\|$. Squaring, the two ends of the display in Step 1 are equal, so both inequalities in it are equalities. Equality in the second one reads $|\langle u, v\rangle| = \|u\|\,\|v\|$, so u and v are linearly dependent by proposition 5.2.2(4). Equality in the first reads $\operatorname{Re}\langle u, v\rangle = |\langle u, v\rangle|$; writing $\langle u, v\rangle = a + bi$ with $a, b \in \mathbb{R}$, this says $a = \sqrt{a^2 + b^2}$, which forces $b = 0$ and $a \geq 0$, so $\langle u, v\rangle$ is a nonnegative real number. If $u = 0$ then $u = 0 \cdot v$ and the claim holds. If $u \neq 0$, then linear dependence gives scalars α, β, not both zero, with $\alpha u + \beta v = 0$; here $\beta \neq 0$, since $\beta = 0$ would force $\alpha u = 0$ with $\alpha \neq 0$ and hence $u = 0$. Thus $v = cu$ with $c = -\alpha/\beta$, and

$$\langle u, v\rangle = \langle u, cu\rangle = \bar{c}\langle u, u\rangle = \bar{c}\,\|u\|^2$$

by proposition 5.1.6(2). Here $\|u\|^2$ is a positive real, since $u \neq 0$ and proposition 5.2.2(1) applies, and $\langle u, v\rangle$ is a nonnegative real as shown above; dividing, $\bar{c}$ is a nonnegative real, hence so is c. Therefore v is a nonnegative real multiple of u, completing the proof.

Proposition 5.2.2(1), proposition 5.2.2(2) and theorem 5.2.3 give conditions (1), (2) and (3) of definition 5.2.1 for the induced norm, so the induced norm of an inner product is a norm and the name is justified. The equality case says when the inequality is tight: if $v = cu$ with $c \geq 0$, the three points 0, u and $u + v = (1 + c)u$ are collinear with u between the other two, and the triangle whose sides they bound has degenerated to a segment.

Example 5.2.4: Lengths in $\mathbb{R}^3$, in $\mathbb{C}^3$, and along the Columns of A

1. In $\mathbb{R}^3$ with the dot product, the vector $x = (1, 2, 2)^\top$ has $\langle x, x\rangle = 1 + 4 + 4 = 9$, so $\|x\| = 3$. Scaling by $1/3$ produces a unit vector: by proposition 5.2.2(2),

$$\left\|\tfrac{1}{3}x\right\| = \left|\tfrac{1}{3}\right| \|x\| = 1$$

and the vector in question is $(1/3, 2/3, 2/3)^\top$.

2. In $\mathbb{C}^3$ with the inner product $\langle x, y\rangle = \sum_i x_i\overline{y_i}$, the vector $z = (1, i, 1 - i)^\top$ has

$$\langle z, z\rangle = |1|^2 + |i|^2 + |1 - i|^2 = 1 + 1 + 2 = 4$$

so $\|z\| = 2$. Multiplying z by i leaves the length unchanged, since $\|iz\| = |i| \|z\| = \|z\|$ by proposition 5.2.2(2), and the same holds for multiplication by any complex scalar of absolute value 1.

3. Take $\mathbb{R}^4$ with the dot product and let a_1, a_2, a_3 be the columns of the running matrix A of example 1.2.9, so that $a_1 = (1, 2, 1, 3)^\top$, $a_2 = (1, 3, 2, 4)^\top$, $a_3 = (3, 8, 5, 11)^\top$ and $a_3 = a_1 + 2a_2$. Directly,

$$\|a_1\|^2 = 1+4+1+9 = 15, \qquad \|a_2\|^2 = 1+9+4+16 = 30, \qquad \langle a_1, a_2\rangle = 1+6+2+12 = 21$$

The column relation now predicts $\|a_3\|$ without touching the entries of a_3: applying proposition 5.2.2(3) to the pair a_1 and $2a_2$, and using $\langle a_1, 2a_2\rangle = 2 \cdot 21 = 42$ together with $\|2a_2\|^2 = 4 \cdot 30 = 120$, gives

$$\|a_3\|^2 = \|a_1\|^2 + 2\,\mathrm{Re}\langle a_1, 2a_2\rangle + \|2a_2\|^2 = 15 + 84 + 120 = 219$$

The direct computation agrees: $9 + 64 + 25 + 121 = 219$.

4. The same three vectors show that theorem 5.2.3 and proposition 5.2.2(4) are both strict here. Neither a_1 nor a_2 is a scalar multiple of the other, since the first entries force the factor to be 1 while the second entries are 2 and 3; so a_1 and a_2 are linearly independent, as are a_1 and $2a_2$, and both equality cases fail. Numerically the two sides are close: $\|a_3\| = \sqrt{219} = 14.798\ldots$ against $\|a_1\| + \|2a_2\| = \sqrt{15} + 2\sqrt{30} = 14.827\ldots$, and $|\langle a_1, a_2\rangle| = 21$ against $\|a_1\| \|a_2\| = \sqrt{450} = 21.213\ldots$

Item (3) is the pattern: a linear relation among vectors becomes an arithmetic relation among lengths as soon as the cross term $\mathrm{Re}\langle u, v\rangle$ is known. The next result removes the cross term without evaluating it, by combining two instances of item (3) whose cross terms have opposite signs.

Theorem 5.2.5: The Parallelogram Law

Let V be a vector space over k carrying an inner product, and let $\|\cdot\|$ be the induced norm. For all $u, v \in V$,

$$\|u+v\|^2 + \|u-v\|^2 = 2\|u\|^2 + 2\|v\|^2$$

***Proof*:**
The two identities of proposition 5.2.2(3) are

$$\|u+v\|^2 = \|u\|^2 + 2\operatorname{Re}\langle u, v\rangle + \|v\|^2, \qquad \|u-v\|^2 = \|u\|^2 - 2\operatorname{Re}\langle u, v\rangle + \|v\|^2$$

Adding them, the terms $\pm 2\operatorname{Re}\langle u, v\rangle$ cancel and the remaining terms are $\|u\|^2 + \|v\|^2$ twice over, which is the claimed identity, as we sought to show.

The four vectors 0, u, $u+v$, v are the corners of a parallelogram whose sides have lengths $\|u\|$ and $\|v\|$, each occurring twice, and whose diagonals join 0 to $u+v$ and u to v. The second diagonal is the segment from u to v, of length $\|v-u\| = \|u-v\|$ by proposition 5.2.2(2). So the identity says that the two diagonals and the four sides of a parallelogram have the same total squared length.

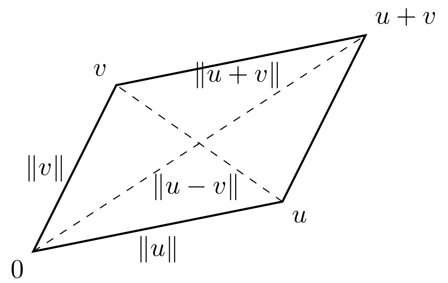


Figure 5.2: The two diagonals and the four sides of a parallelogram have equal total squared length

Every norm induced by an inner product satisfies this identity, so a norm failing it is induced by no inner product. That is a usable test, and the two standard norms on $\mathbb{R}^2$ below fail it.

Example 5.2.6: Two Norms on $\mathbb{R}^2$ That No Inner Product Induces
On $\mathbb{R}^2$ define

$$\|x\|_1 = |x_1| + |x_2|, \qquad \|x\|_\infty = \max\{|x_1|, |x_2|\}$$

Both are norms in the sense of definition 5.2.1. For condition (1), each is a sum or a maximum of absolute values, hence nonnegative, and each vanishes exactly when $|x_1| = |x_2| = 0$, which is $x = 0$. For condition (2), $|\lambda x_i| = |\lambda|\,|x_i|$ for each i, and the common factor $|\lambda|$ pulls out of a sum and out of a maximum of nonnegative numbers. For condition (3), the triangle inequality for absolute values in $\mathbb{R}$ gives $|x_i + y_i| \le |x_i| + |y_i|$ for $i = 1, 2$. Summing the two instances gives $\|x+y\|_1 \le \|x\|_1 + \|y\|_1$. For the other norm, use $|x_i| \le \|x\|_\infty$ and $|y_i| \le \|y\|_\infty$ to bound the right-hand side of each instance by $\|x\|_\infty + \|y\|_\infty$; both numbers $|x_i + y_i|$ are then at most $\|x\|_\infty + \|y\|_\infty$, and hence so is their maximum $\|x+y\|_\infty$.

Now take $u = (1,0)^\top$ and $v = (0,1)^\top$, so that $u+v = (1,1)^\top$ and $u - v = (1,-1)^\top$. For $\|\cdot\|_1$,

$$\|u+v\|_1^2 + \|u-v\|_1^2 = 2^2 + 2^2 = 8, \qquad 2\|u\|_1^2 + 2\|v\|_1^2 = 2 + 2 = 4$$

and for $\|\cdot\|_\infty$,

$$\|u+v\|_\infty^2 + \|u-v\|_\infty^2 = 1 + 1 = 2, \qquad 2\|u\|_\infty^2 + 2\|v\|_\infty^2 = 2 + 2 = 4$$

Neither pair of numbers agrees, so by theorem 5.2.5 neither norm is the norm induced by any inner product on $\mathbb{R}^2$.

The test is decisive in the other direction as well.

Theorem 5.2.7: Jordan-von Neumann: the Parallelogram Law Characterizes Inner Product Norms

Let $\|\cdot\|$ be a norm on a vector space V over k satisfying the identity of theorem 5.2.5. Then there is an inner product on V inducing it, and by theorem 5.2.9 exactly one.

The proof is omitted because its final step is analytic rather than algebraic and this chapter has no analysis in it; see Jordan and von Neumann [13].[1] What that converse needs is a way to produce an inner product out of lengths, and the identity giving it is the next result.

Theorem 5.2.8: The Polarization Identity

Let V be a vector space over k carrying an inner product, and let $\|\cdot\|$ be the induced norm. For all $u, v \in V$:

1. If $k = \mathbb{R}$, then
$$\langle u, v\rangle = \tfrac{1}{4}\left(\|u+v\|^2 - \|u-v\|^2\right)$$

2. If $k = \mathbb{C}$, then
$$\langle u, v\rangle = \frac{1}{4}\sum_{m=0}^{3} i^m \big\|u + i^m v\big\|^2 = \frac{1}{4}\Big(\|u+v\|^2 - \|u-v\|^2 + i\|u+iv\|^2 - i\|u-iv\|^2\Big)$$

***Proof*:**

1. Over $\mathbb{R}$ the inner product is real valued, so $\operatorname{Re}\langle u,v\rangle = \langle u,v\rangle$ and the two identities of proposition 5.2.2(3) read $\|u \pm v\|^2 = \|u\|^2 \pm 2\langle u,v\rangle + \|v\|^2$. Subtracting the second from the first cancels $\|u\|^2 + \|v\|^2$ and leaves $\|u+v\|^2 - \|u-v\|^2 = 4\langle u,v\rangle$. Dividing by 4 gives the stated formula.

2. Write $\langle u,v\rangle = a + bi$ with $a, b \in \mathbb{R}$, and set $S = \|u\|^2 + \|v\|^2$. Fix $m \in \{0,1,2,3\}$ and apply proposition 5.2.2(3) to the pair u and $i^m v$:
$$\|u + i^m v\|^2 = \|u\|^2 + 2\operatorname{Re}\langle u, i^m v\rangle + \|i^m v\|^2$$

[1] One shows first, using only the parallelogram law, that the expression of theorem 5.2.8 is additive in its first argument, which forces $\langle qu, v\rangle = q\langle u,v\rangle$ for every rational q; passing from rational to real scalars then needs a continuity argument, and this chapter uses none.

Here $\|i^m v\| = |i^m|\,\|v\| = \|v\|$ by proposition 5.2.2(2), since $|i^m| = 1$; so the two outer terms contribute S. For the middle term, proposition 5.1.6(2) gives $\langle u, i^m v\rangle = \overline{i^m}\langle u, v\rangle$, and the four values of $\overline{i^m}\,(a+bi)$ are

$$m=0\colon\ a+bi, \qquad m=1\colon\ -i(a+bi)=b-ai, \qquad m=2\colon\ -(a+bi), \qquad m=3\colon\ i(a+bi)=-b+ai$$

whose real parts are a, b, $-a$, $-b$ respectively. Consequently

$$\|u+v\|^2 = S+2a, \qquad \|u+iv\|^2 = S+2b, \qquad \|u-v\|^2 = S-2a, \qquad \|u-iv\|^2 = S-2b$$

using $i^2=-1$ and $i^3=-i$ to identify the last two. Summing with the coefficients i^m,

$$\frac{1}{4}\sum_{m=0}^{3} i^m\|u+i^m v\|^2 = \frac{1}{4}\Big((S+2a)+i(S+2b)-(S-2a)-i(S-2b)\Big) = \frac{1}{4}\big(4a+4bi\big) = a+bi$$

which is $\langle u,v\rangle$, as we sought to show.

The four-term form over $\mathbb{C}$ is forced by the observation made after proposition 5.2.2: the lengths of combinations of u and v determine only the real part of $\langle u,v\rangle$. The imaginary part b has to be reached some other way, and the $m=1$ line of the proof reaches it as $\mathrm{Re}\langle u, iv\rangle = b$, which is why iv enters the formula. Over $\mathbb{R}$ there is no imaginary part and two terms suffice.

Both right-hand sides involve lengths and nothing else, so a question about the inner product can always be converted into a question about the induced norm. The next theorem records the two conversions used later.

Theorem 5.2.9: An Inner Product Is Determined by Its Norm

Let V be a vector space over k.

1. If $\langle -,-\rangle_1$ and $\langle -,-\rangle_2$ are inner products on V with $\langle v,v\rangle_1 = \langle v,v\rangle_2$ for every $v\in V$, then $\langle u,v\rangle_1 = \langle u,v\rangle_2$ for all $u,v\in V$.
2. Suppose V carries an inner product with induced norm $\|\cdot\|$, and let $T\colon V\to V$ be a linear map with $\|Tv\| = \|v\|$ for every $v\in V$. Then $\langle Tu, Tv\rangle = \langle u,v\rangle$ for all $u,v\in V$.

***Proof*:**

1. Let $\|\cdot\|_1$ and $\|\cdot\|_2$ be the two induced norms. For each v the numbers $\|v\|_1$ and $\|v\|_2$ are the nonnegative square roots of $\langle v,v\rangle_1$ and $\langle v,v\rangle_2$, which are equal by hypothesis, so $\|v\|_1 = \|v\|_2$ for every $v\in V$. Now fix $u,v\in V$. If $k=\mathbb{R}$, theorem 5.2.8(1) applied to each inner product gives
$$\langle u,v\rangle_1 = \tfrac{1}{4}\big(\|u+v\|_1^2 - \|u-v\|_1^2\big) = \tfrac{1}{4}\big(\|u+v\|_2^2 - \|u-v\|_2^2\big) = \langle u,v\rangle_2$$
the middle equality because the two norms agree at the vectors $u+v$ and $u-v$. If $k=\mathbb{C}$, the same argument with theorem 5.2.8(2) uses the four vectors $u+i^m v$ instead.
2. Fix $u,v\in V$ and $m\in\{0,1,2,3\}$. Since T is linear (definition 2.4.2), $Tu + i^m Tv = T(u+i^m v)$, and the hypothesis applied to the vector $u+i^m v$ gives
$$\|Tu + i^m Tv\| = \|T(u+i^m v)\| = \|u+i^m v\|$$

Substituting this into theorem 5.2.8(2) for the pair Tu, Tv,

$$\langle Tu, Tv\rangle = \frac{1}{4}\sum_{m=0}^{3} i^m \|Tu + i^m Tv\|^2 = \frac{1}{4}\sum_{m=0}^{3} i^m \|u + i^m v\|^2 = \langle u, v\rangle$$

Over $\mathbb{R}$ the same substitution into theorem 5.2.8(1) uses only the two vectors $T(u+v)$ and $T(u-v)$ and gives the same conclusion, as we sought to show.

Part (1) says that two different inner products on the same space always produce two different length functions, so passing from an inner product to its induced norm discards nothing; the parallelogram law is then the condition deciding which length functions occur at all. Part (2) is the form in which the identity gets used later: a linear map assumed only to preserve lengths preserves the inner product as well, so a hypothesis about one vector at a time yields a conclusion about pairs. The linear maps satisfying that hypothesis are taken up in chapter 7.

Exercise 5.2.1

1. In $\mathbb{C}^3$ with the inner product $\langle x, y\rangle = \sum_i x_i \overline{y_i}$, let $u = (1, i, 0)^\top$ and $v = (0, 1, i)^\top$. Compute $\|u\|$, $\|v\|$, $\langle u, v\rangle$ and $\|u+v\|$, and check that these four values satisfy proposition 5.2.2(3).
2. In $\mathbb{C}^2$ with the same inner product, let $u = (1, 1)^\top$ and $v = (1, i)^\top$. Compute the four numbers $\|u+v\|^2$, $\|u-v\|^2$, $\|u+iv\|^2$ and $\|u-iv\|^2$, then verify both theorem 5.2.5 and theorem 5.2.8(2) for this pair.
3. Let a_1 and a_2 be the first two columns of the running matrix A of example 1.2.9, viewed in $\mathbb{R}^4$ with the dot product. Compute $\|a_1 + a_2\|^2$ and $\|a_1 - a_2\|^2$ directly from the entries, verify theorem 5.2.5 for the pair, and recover $\langle a_1, a_2\rangle$ from those two numbers using theorem 5.2.8(1).
4. Let $\|\cdot\|$ be the norm induced by an inner product on V. Prove that $\big|\, \|u\| - \|v\| \,\big| \le \|u - v\|$ for all $u, v \in V$.
5. Let $\|\cdot\|$ be the norm induced by an inner product on a vector space V over k. Show that $\|u+v\| = \|u-v\|$ if and only if $\operatorname{Re}\langle u, v\rangle = 0$, and that when $k = \mathbb{R}$ the condition on the right reads $\langle u, v\rangle = 0$.
6. Regard $\mathbb{R}$ as a one-dimensional vector space over itself and let $\|\cdot\|$ be any norm on it in the sense of definition 5.2.1. Show that there is exactly one inner product on $\mathbb{R}$ whose induced norm is $\|\cdot\|$, and write it down in terms of the number $\|1\|$.

5.3 Orthonormal Bases and the Gram-Schmidt Process

A basis writes each vector of V as a unique linear combination of its members (theorem 2.2.3), and recovering the coefficients from the vector means solving a linear system (chapter 1). When the basis vectors are pairwise orthogonal and of length one, the coefficients are inner products instead, one per basis vector, and the inner product of two vectors becomes the standard inner product of their coordinate vectors.

Such a basis exists in every finite-dimensional inner product space, and the Gram-Schmidt process produces one starting from any basis at all. The process repeats a single step, the subtraction of an orthogonal projection, so the projection is defined before the process is stated.

Throughout this section V is a finite-dimensional vector space over $k = \mathbb{R}$ or $k = \mathbb{C}$ carrying an inner product $\langle -, - \rangle$ in the sense of definition 5.1.2, linear in the first argument and conjugate-linear in the second, and $\|v\| = \sqrt{\langle v, v \rangle}$ is the associated norm (definition 5.2.1).

In $\mathbb{R}^3$ with the standard inner product (example 5.1.1), the vectors $u = (1,0,1)^\top$ and $v = (1,0,-1)^\top$ satisfy

$$\langle u, v \rangle = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot (-1) = 0$$

while u and $v' = (1,1,0)^\top$ satisfy $\langle u, v' \rangle = 1$.

Definition 5.3.1: Orthogonal Vectors

Let $u, v \in V$. Then u and v are *orthogonal*, written $u \perp v$, if $\langle u, v \rangle = 0$. A list $v_1, \ldots, v_m$ in V is *orthogonal* if $v_i \perp v_j$ whenever $i \neq j$.
For a subset $S \subseteq V$, write $u \perp S$ to mean that $u \perp s$ for every $s \in S$.

The relation is symmetric: by proposition 5.1.6(2) one has $\langle v, u \rangle = \overline{\langle u, v \rangle}$, and a complex number vanishes exactly when its conjugate does, so $u \perp v$ if and only if $v \perp u$. The zero vector is orthogonal to everything, since $\langle 0, v \rangle = 0$ for all v by proposition 5.1.6(3), and it is the only vector orthogonal to itself, since $\langle v, v \rangle = 0$ forces $v = 0$ by proposition 5.1.6(1).

Orthogonality is what makes the length of a sum computable from the lengths of its terms: the expansion in proposition 5.1.6(4) carries the extra term $2\,\mathrm{Re}\langle u, v \rangle$, and $\langle u, v \rangle = 0$ removes it.

Theorem 5.3.2: Pythagorean Theorem

Let $v_1, \ldots, v_m \in V$ be an orthogonal list. Then

$$\|v_1 + \cdots + v_m\|^2 = \|v_1\|^2 + \cdots + \|v_m\|^2$$

***Proof*:**
Put $v = v_1 + \cdots + v_m$. The inner product is additive in its first argument by definition 5.1.2 and additive in its second by proposition 5.1.6(2), which states that it is conjugate-linear there; expanding both arguments therefore gives

$$\langle v, v \rangle = \sum_{i=1}^{m} \sum_{j=1}^{m} \langle v_i, v_j \rangle$$

Each term with $i \neq j$ is zero, since the list is orthogonal (definition 5.3.1). Each term with $i = j$ is $\langle v_i, v_i \rangle = \|v_i\|^2$ by definition 5.2.1. The double sum therefore collapses to $\sum_{i=1}^{m} \|v_i\|^2$, which is $\|v\|^2 = \|v_1 + \cdots + v_m\|^2$, as we sought to show.

For $m = 2$ over $\mathbb{R}$ this is the classical Pythagorean relation between two sides u and v of a triangle and its third side $u + v$, which is what makes definition 5.3.1 the algebraic form of a right angle. Over $\mathbb{R}$ the converse also holds: proposition 5.1.6(4) reads $\|u + v\|^2 = \|u\|^2 + 2\langle u, v \rangle + \|v\|^2$ there, so the two sides of the Pythagorean identity agree exactly when $\langle u, v \rangle = 0$. Over $\mathbb{C}$ the same expansion carries $2\,\mathrm{Re}\langle u, v \rangle$, which can vanish while $\langle u, v \rangle$ does not, and the converse fails; exercise 5.3.1 (3) exhibits such a pair in $\mathbb{C}^2$.

The standard basis $e_1, \ldots, e_n$ of k^n is orthogonal for the standard inner product $\langle x, y\rangle = \sum_i x_i \overline{y_i}$, and each of its members has norm 1: the value of $\langle e_i, e_j\rangle$ is 1 when $i = j$ and 0 otherwise. A list with both properties makes the coordinate formulas of theorem 5.3.5 below free of constants.

Definition 5.3.3: Orthonormal List and Orthonormal Basis

A list $q_1, \ldots, q_m$ in V is *orthonormal* if

$$\langle q_i, q_j\rangle = \delta_{ij} \qquad \text{for all } 1 \leq i, j \leq m$$

where δ_{ij} denotes 1 when $i = j$ and 0 when $i \neq j$. Equivalently, the list is orthogonal and every member has norm 1. An orthonormal list that is a basis of V is an *orthonormal basis*.

The standard basis of k^n is an orthonormal basis for the standard inner product. A less trivial instance in $\mathbb{R}^3$: the vectors $q_1 = \frac{1}{\sqrt{2}}(1,0,1)^\top$ and $q_2 = \frac{1}{\sqrt{2}}(1,0,-1)^\top$ satisfy $\langle q_1, q_2\rangle = \frac{1}{2}(1+0-1) = 0$ and $\langle q_1, q_1\rangle = \langle q_2, q_2\rangle = \frac{1}{2}(1+0+1) = 1$, so q_1, q_2 is an orthonormal list. It is not a basis of $\mathbb{R}^3$, having only two members.

Proposition 5.3.4: Orthogonal Lists of Nonzero Vectors Are Independent

Let $v_1, \ldots, v_m$ be an orthogonal list in V with $v_i \neq 0$ for every i. Then $v_1, \ldots, v_m$ is linearly independent. In particular every orthonormal list is linearly independent.

***Proof*:**
Let $a_1, \ldots, a_m \in k$ satisfy $a_1 v_1 + \cdots + a_m v_m = 0$, and fix an index j. Pairing both sides with v_j and using proposition 5.1.6(3) on the right gives

$$\left\langle \sum_{i=1}^{m} a_i v_i,\, v_j \right\rangle = \langle 0, v_j\rangle = 0$$

The inner product is linear in its first argument (definition 5.1.2), so the left side equals $\sum_{i=1}^{m} a_i\langle v_i, v_j\rangle$, and every term with $i \neq j$ vanishes because the list is orthogonal. What survives is $a_j\langle v_j, v_j\rangle = a_j\|v_j\|^2 = 0$. Since $v_j \neq 0$, proposition 5.1.6(1) gives $\|v_j\|^2 > 0$, so $a_j = 0$. The index j was arbitrary, so every coefficient in the relation is zero, which is definition 2.1.9. An orthonormal list is orthogonal and has $\|q_i\| = 1 \neq 0$ for every i, so the first statement applies to it, completing the proof.

Independence of a list in k^n is ordinarily decided by elimination (theorem 1.2.8, definition 1.3.2). For an orthogonal list of nonzero vectors none is needed: pairing a relation with one member of the list exhibits that member's coefficient as zero. An orthonormal list of $n = \dim V$ vectors is therefore a basis of V by corollary 2.2.13.

The same pairing computes coordinates. Pairing a vector with one member of an orthonormal basis annihilates every term of its expansion but one, and what survives is the coefficient sought.

Theorem 5.3.5: Coordinates in an Orthonormal Basis Are Inner Products

Let V be as above.

1. Let $q_1, \dots, q_n$ be an orthonormal basis of V. Then every $v \in V$ satisfies
$$v = \sum_{i=1}^{n} \langle v, q_i \rangle \, q_i$$
so the coordinate vector of v in the ordered basis $q_1, \dots, q_n$ (definition 2.2.4) is $(\langle v, q_1 \rangle, \dots, \langle v, q_n \rangle)^\top$.
2. Let $q_1, \dots, q_m$ be an orthonormal list in V and put $W = \operatorname{span}\{q_1, \dots, q_m\}$. For $v \in V$ set $p = \sum_{i=1}^{m} \langle v, q_i \rangle \, q_i$. Then $p \in W$ and $v - p \perp W$, and p is the only vector of W with that property.

***Proof*:**

1. Since $q_1, \dots, q_n$ is a basis, theorem 2.2.3 gives unique scalars $a_1, \dots, a_n \in k$ with $v = \sum_{i=1}^{n} a_i q_i$. Fix j and pair with q_j. Using linearity in the first argument (definition 5.1.2) and then $\langle q_i, q_j \rangle = \delta_{ij}$ from definition 5.3.3,
$$\langle v, q_j \rangle = \sum_{i=1}^{n} a_i \langle q_i, q_j \rangle = a_j$$
So $a_j = \langle v, q_j \rangle$ for every j, which is the displayed expansion; the coordinate vector is the list of these scalars by definition 2.2.4.
2. The vector p is a linear combination of $q_1, \dots, q_m$ and so lies in $W = \operatorname{span}\{q_1, \dots, q_m\}$ by definition 2.1.5. For each $j \leq m$, linearity in the first argument and $\langle q_i, q_j \rangle = \delta_{ij}$ give
$$\langle v - p, q_j \rangle = \langle v, q_j \rangle - \sum_{i=1}^{m} \langle v, q_i \rangle \langle q_i, q_j \rangle = \langle v, q_j \rangle - \langle v, q_j \rangle = 0$$
A general $w \in W$ has the form $w = \sum_{j=1}^{m} c_j q_j$ with $c_j \in k$, and the inner product is conjugate-linear in its second argument by proposition 5.1.6(2), so $\langle v - p, w \rangle = \sum_{j=1}^{m} \overline{c_j} \langle v - p, q_j \rangle = 0$. Hence $v - p \perp W$.

   For uniqueness, suppose $p' \in W$ also satisfies $v - p' \perp W$. Then $p - p' = (v - p') - (v - p)$, so for every $w \in W$ additivity in the first argument gives $\langle p - p', w \rangle = \langle v - p', w \rangle - \langle v - p, w \rangle = 0$. Now W is a subspace by proposition 2.1.6, so $p - p' \in W$ and this applies with $w = p - p'$, giving $\langle p - p', p - p' \rangle = 0$. By proposition 5.1.6(1) this forces $p - p' = 0$, that is $p' = p$, completing the proof.

Part (1) replaces a linear system by n inner products: no elimination is performed, and the i-th coordinate of v is the single number $\langle v, q_i \rangle$. Part (2) applies the same formula to an orthonormal list that does not span V, and its content is the splitting it produces. The sum p lies in W, the remainder $v - p$ is orthogonal to every vector of W, and no other vector of W has that second property. Since the characterization mentions only W, the vector p is attached to the subspace and not to the list used to write it down.

Corollary 5.3.6: Parseval's Identity

Let $q_1, \ldots, q_n$ be an orthonormal basis of V and let $v, w \in V$. Then

$$\langle v, w\rangle = \sum_{i=1}^{n} \langle v, q_i\rangle \overline{\langle w, q_i\rangle}$$

and in particular $\|v\|^2 = \sum_{i=1}^{n} |\langle v, q_i\rangle|^2$.

***Proof*:**
Write $a_i = \langle v, q_i\rangle$ and $b_i = \langle w, q_i\rangle$. By theorem 5.3.5(1), $v = \sum_{i=1}^{n} a_i q_i$ and $w = \sum_{j=1}^{n} b_j q_j$. Expanding by linearity in the first argument (definition 5.1.2) and conjugate-linearity in the second (proposition 5.1.6(2)),

$$\langle v, w\rangle = \sum_{i=1}^{n}\sum_{j=1}^{n} a_i \overline{b_j} \langle q_i, q_j\rangle = \sum_{i=1}^{n} a_i \overline{b_i}$$

the second equality because $\langle q_i, q_j\rangle = \delta_{ij}$. Taking $w = v$ gives $\|v\|^2 = \sum_i a_i \overline{a_i} = \sum_i |a_i|^2$, using definition 5.2.1 and $z\overline{z} = |z|^2$ for $z \in \mathbb{C}$, as we sought to show.

The coordinate map of proposition 2.2.5 therefore does more in an orthonormal basis than transport the vector space structure: it carries $\langle -, -\rangle$ to the standard inner product on k^n. Every length and every angle computed in V can be computed instead from coordinate vectors, by the same formula that defines the standard inner product.

Example 5.3.7: Coordinates in a Rotated Basis of $\mathbb{R}^2$
Take $V = \mathbb{R}^2$ with the standard inner product, and set

$$q_1 = \tfrac{1}{\sqrt{2}}(1, 1)^\top, \qquad q_2 = \tfrac{1}{\sqrt{2}}(1, -1)^\top$$

Then $\langle q_1, q_2\rangle = \frac{1}{2}(1-1) = 0$ and $\langle q_1, q_1\rangle = \langle q_2, q_2\rangle = \frac{1}{2}(1+1) = 1$, so q_1, q_2 is an orthonormal list; it has two members and $\dim \mathbb{R}^2 = 2$, so it is an orthonormal basis by proposition 5.3.4 and corollary 2.2.13. Let $v = (3, 1)^\top$.

1. *Coordinates.* The two inner products are

$$\langle v, q_1\rangle = \tfrac{1}{\sqrt{2}}(3+1) = 2\sqrt{2}, \qquad \langle v, q_2\rangle = \tfrac{1}{\sqrt{2}}(3-1) = \sqrt{2}$$

   and theorem 5.3.5(1) predicts $v = 2\sqrt{2}\, q_1 + \sqrt{2}\, q_2$. Checking: $2\sqrt{2} \cdot \frac{1}{\sqrt{2}}(1, 1)^\top + \sqrt{2} \cdot \frac{1}{\sqrt{2}}(1, -1)^\top = 2(1, 1)^\top + (1, -1)^\top = (3, 1)^\top$.

2. *Parseval.* The squared coordinates sum to $(2\sqrt{2})^2 + (\sqrt{2})^2 = 8 + 2 = 10$, and $\|v\|^2 = 3^2 + 1^2 = 10$, as corollary 5.3.6 requires.

3. *A partial sum.* Apply theorem 5.3.5(2) to the shorter orthonormal list consisting of q_1 alone, so that $W = \mathrm{span}\{q_1\}$ is the line through $(1,1)^\top$. The prescription gives $p = \langle v, q_1\rangle q_1 = 2\sqrt{2} \cdot \frac{1}{\sqrt{2}}(1,1)^\top = (2,2)^\top$, and the remainder is $v - p = (1,-1)^\top$, which satisfies $\langle v - p, q_1\rangle = \frac{1}{\sqrt{2}}(1-1) = 0$. So $v - p \perp W$, since every element of W is a scalar multiple of q_1.

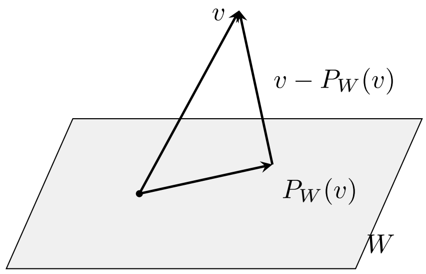
The construction in item (3) works for any subspace carrying an orthonormal basis.

Definition 5.3.8: Orthogonal Projection onto a Subspace

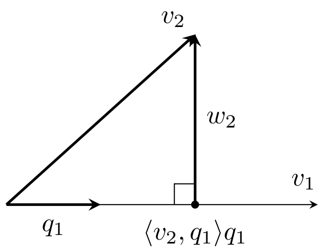
Let $W \subseteq V$ be a subspace possessing an orthonormal basis $q_1, \ldots, q_m$. The *orthogonal projection of $v \in V$ onto W* is

$$P_W(v) = \sum_{i=1}^{m} \langle v, q_i\rangle\, q_i$$

The vector $P_W(v)$ is determined by W and v alone, and does not depend on the orthonormal basis used to write it down. Indeed, if $q'_1, \ldots, q'_m$ is another orthonormal basis of W, then theorem 5.3.5(2) applied to each of the two lists identifies both $\sum_i \langle v, q_i\rangle q_i$ and $\sum_i \langle v, q'_i\rangle q'_i$ as *the* unique vector $p \in W$ with $v - p \perp W$; the characterization refers only to W, so the two sums agree. The same part of that theorem records the two properties used from here on: $P_W(v) \in W$ and $v - P_W(v) \perp W$. Corollary 5.3.11 below shows that every subspace of V has an orthonormal basis, so P_W is in fact defined for every subspace W.



Everything so far has been conditional on an orthonormal list being available. The Gram-Schmidt process produces one. It works on a list one vector at a time: subtract from each vector its projection onto the span of those already processed, then scale what remains to unit length. Each step changes only the vector it acts on, so the span of the first j vectors is never disturbed, and the theorem records that equality of spans alongside the orthonormality.



Theorem 5.3.9: The Gram-Schmidt Process

Let V be as above.

1. Let $v_1, \ldots, v_m$ be a linearly independent list in V. Define vectors recursively by

$$w_1 = v_1, \qquad w_j = v_j - \sum_{i=1}^{j-1} \langle v_j, q_i \rangle \, q_i \quad (2 \le j \le m), \qquad q_j = \frac{w_j}{\|w_j\|} \quad (1 \le j \le m)$$

   Then $w_j \neq 0$ for every j, so each q_j is defined; the list $q_1, \ldots, q_m$ is orthonormal; and

$$\operatorname{span}\{q_1, \ldots, q_j\} = \operatorname{span}\{v_1, \ldots, v_j\} \quad \text{for every } 1 \le j \le m$$

2. Let $v_1, \ldots, v_m$ be an arbitrary list in V, and let $i_1 < \cdots < i_r$ be exactly those indices j with $v_j \notin \operatorname{span}\{v_1, \ldots, v_{j-1}\}$, where the span of the empty list is $\{0\}$. Then $v_{i_1}, \ldots, v_{i_r}$ is linearly independent with $\operatorname{span}\{v_{i_1}, \ldots, v_{i_r}\} = \operatorname{span}\{v_1, \ldots, v_m\}$, so part (1) applied to it produces an orthonormal basis of $\operatorname{span}\{v_1, \ldots, v_m\}$.

***Proof*:**

1. The claim proved by induction on j is the conjunction of three statements: $w_j \neq 0$; the list $q_1, \ldots, q_j$ is orthonormal; and $\operatorname{span}\{q_1, \ldots, q_j\} = \operatorname{span}\{v_1, \ldots, v_j\}$.

   Base case $j = 1$. If $v_1 = 0$ then $1 \cdot v_1 + 0 \cdot v_2 + \cdots + 0 \cdot v_m = 0$ is a relation with a nonzero coefficient, contradicting definition 2.1.9; so $w_1 = v_1 \neq 0$ and $\|w_1\| > 0$ by proposition 5.1.6(1) and definition 5.2.1. Scaling multiplies the norm by the modulus of the scalar (proposition 5.2.2), so $\|q_1\| = \|w_1\|^{-1}\|w_1\| = 1$ and the one-member list q_1 is orthonormal. Each of q_1 and v_1 is a nonzero scalar multiple of the other, so $\operatorname{span}\{q_1\} = \operatorname{span}\{v_1\}$ by definition 2.1.5.

   Inductive step. Let $2 \le j \le m$ and assume the three statements for $j-1$. Put $W = \operatorname{span}\{q_1, \ldots, q_{j-1}\}$, which by hypothesis equals $\operatorname{span}\{v_1, \ldots, v_{j-1}\}$. The list $q_1, \ldots, q_{j-1}$ is orthonormal, and $w_j = v_j - p$ where $p = \sum_{i=1}^{j-1} \langle v_j, q_i \rangle q_i$ is the vector of theorem 5.3.5(2) for that list; so $p \in W$ and $w_j \perp W$.

   w_j is nonzero. If $w_j = 0$ then $v_j = p \in W = \operatorname{span}\{v_1, \ldots, v_{j-1}\}$, say $v_j = \sum_{i<j} c_i v_i$ with $c_i \in k$ (definition 2.1.5). Then $\sum_{i<j} c_i v_i - v_j = 0$ is a relation among $v_1, \ldots, v_m$ whose coefficient at v_j is -1, contradicting definition 2.1.9. Hence $w_j \neq 0$, and $\|w_j\| > 0$ as in the base case, so $q_j = w_j / \|w_j\|$ is defined and $\|q_j\| = 1$ by proposition 5.2.2.

   Orthonormality. For $l < j$ the vector q_l lies in W, and $w_j \perp W$, so $\langle w_j, q_l \rangle = 0$; linearity in the first argument (definition 5.1.2) gives $\langle q_j, q_l \rangle = \|w_j\|^{-1} \langle w_j, q_l \rangle = 0$. Together with $\|q_j\| = 1$ and the inductive hypothesis for the pairs with both indices below j, the list $q_1, \ldots, q_j$ satisfies $\langle q_a, q_b \rangle = \delta_{ab}$, which is definition 5.3.3.

   The span invariant. A span is the smallest subspace containing its generators (definition 2.1.5, proposition 2.1.6), so it suffices to check that each list lies in the span of the other. For one inclusion, $q_1, \ldots, q_{j-1}$ lie in $W = \operatorname{span}\{v_1, \ldots, v_{j-1}\}$ and $q_j = \|w_j\|^{-1}(v_j - p)$ lies in $\operatorname{span}\{v_1, \ldots, v_j\}$ because $p \in W$; hence $\operatorname{span}\{q_1, \ldots, q_j\} \subseteq \operatorname{span}\{v_1, \ldots, v_j\}$. For the other, $v_1, \ldots, v_{j-1}$ lie in $W = \operatorname{span}\{q_1, \ldots, q_{j-1}\}$ and $v_j = p + \|w_j\| q_j$ lies in $\operatorname{span}\{q_1, \ldots, q_j\}$;

hence $\operatorname{span}\{v_1,\dots,v_j\} \subseteq \operatorname{span}\{q_1,\dots,q_j\}$. The two spans are equal, which closes the induction.

2. *Independence of the sublist.* Suppose $\sum_{l=1}^{r} c_l v_{i_l} = 0$ with the $c_l \in k$ not all zero, and let L be the largest index with $c_L \neq 0$. Then $v_{i_L} = -c_L^{-1}\sum_{l<L} c_l v_{i_l}$, so $v_{i_L} \in \operatorname{span}\{v_{i_1},\dots,v_{i_{L-1}}\}$. Every i_l with $l < L$ satisfies $i_l < i_L$, so this span is contained in $\operatorname{span}\{v_1,\dots,v_{i_L-1}\}$, and $v_{i_L} \in \operatorname{span}\{v_1,\dots,v_{i_L-1}\}$ contradicts the defining property of the index i_L. Hence all c_l vanish and the sublist is independent (definition 2.1.9).

   The sublist spans. Write $U_j = \operatorname{span}\{v_{i_l} \,|\, i_l \leq j\}$, a subspace by proposition 2.1.6, and note that $U_{j-1} \subseteq U_j$ because the generating list of the first is a sublist of that of the second. Induct on j to show $v_j \in U_j$. If j is one of the i_l then $v_j \in U_j$ by definition 2.1.5. Otherwise $v_j \in \operatorname{span}\{v_1,\dots,v_{j-1}\}$ by the definition of the index set, so v_j is a linear combination of $v_1,\dots,v_{j-1}$, each of which lies in $U_{j-1} \subseteq U_j$ by the inductive hypothesis; a subspace contains every linear combination of its elements (proposition 2.1.3), so $v_j \in U_j$. For $j = 1$ this second case reads $v_1 \in \operatorname{span}\{\,\} = \{0\}$, and $v_1 = 0$ lies in the subspace U_1. Taking $j = m$, every v_j lies in $U_m = \operatorname{span}\{v_{i_1},\dots,v_{i_r}\}$, so $\operatorname{span}\{v_1,\dots,v_m\} \subseteq U_m$; the reverse inclusion holds because each v_{i_l} is one of the v_j. The two spans are equal.

   Conclusion. Part (1) applied to the independent list $v_{i_1},\dots,v_{i_r}$ produces an orthonormal list $q_1,\dots,q_r$ with $\operatorname{span}\{q_1,\dots,q_r\} = \operatorname{span}\{v_{i_1},\dots,v_{i_r}\} = \operatorname{span}\{v_1,\dots,v_m\}$. It is linearly independent by proposition 5.3.4 and spans that subspace, so it is a basis of it (definition 2.2.1), completing the proof.

Part (2) does not require the retained indices $i_1 < \cdots < i_r$ to be identified in advance; running the recursion of part (1) on the full list in order, and discarding any index at which w_j comes out zero, finds them. At index j the vectors produced so far are exactly those the recursion of part (1) produces from the retained entries among $v_1,\dots,v_{j-1}$: by induction on j, the two runs have produced the same vectors before each retained index, so they perform the same subtraction there, and a discarded index contributes nothing to either. By part (2) applied to $v_1,\dots,v_{j-1}$ those vectors form an orthonormal basis of $W = \operatorname{span}\{v_1,\dots,v_{j-1}\}$, so $w_j = v_j - P_W(v_j)$. Then theorem 5.3.5(2) makes $w_j = 0$ equivalent to $v_j \in W$: if $v_j \in W$ then v_j itself is a vector of W with $v_j - v_j = 0 \perp W$, so uniqueness forces $P_W(v_j) = v_j$; and conversely $w_j = 0$ puts $v_j = P_W(v_j) \in W$. The indices at which w_j vanishes are therefore exactly the indices omitted from the sublist.

Example 5.3.10: Gram-Schmidt on Three Vectors in $\mathbb{R}^3$

Take $V = \mathbb{R}^3$ with the standard inner product and the list

$$v_1 = (1,1,0)^\top, \qquad v_2 = (1,0,1)^\top, \qquad v_3 = (0,1,1)^\top$$

This list is linearly independent. Indeed, a relation $a_1v_1 + a_2v_2 + a_3v_3 = 0$ reads entrywise as $a_1 + a_2 = 0$, $a_1 + a_3 = 0$ and $a_2 + a_3 = 0$; adding the three equations gives $2(a_1 + a_2 + a_3) = 0$, and subtracting each equation in turn from $a_1 + a_2 + a_3 = 0$ leaves $a_3 = 0$, $a_2 = 0$ and $a_1 = 0$. So theorem 5.3.9(1) applies. This is the first run of the process, so every step is displayed.

Step 1. $w_1 = v_1 = (1,1,0)^\top$ and $\|w_1\|^2 = 1 + 1 + 0 = 2$, so

$$q_1 = \tfrac{1}{\sqrt{2}}(1,1,0)^\top$$

Step 2. The single coefficient is $\langle v_2, q_1\rangle = \frac{1}{\sqrt{2}}(1\cdot 1 + 0\cdot 1 + 1\cdot 0) = \frac{1}{\sqrt{2}}$, so

$$w_2 = (1,0,1)^\top - \tfrac{1}{\sqrt{2}}\cdot\tfrac{1}{\sqrt{2}}(1,1,0)^\top = (1,0,1)^\top - \tfrac{1}{2}(1,1,0)^\top = \left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right)^\top$$

with $\|w_2\|^2 = \frac{1}{4} + \frac{1}{4} + 1 = \frac{3}{2}$ and $\|w_2\| = \frac{\sqrt{6}}{2}$. Scaling,

$$q_2 = \frac{2}{\sqrt{6}}\left(\tfrac{1}{2}, -\tfrac{1}{2}, 1\right)^\top = \frac{1}{\sqrt{6}}(1, -1, 2)^\top$$

Step 3. The two coefficients are $\langle v_3, q_1\rangle = \frac{1}{\sqrt{2}}(0+1+0) = \frac{1}{\sqrt{2}}$ and $\langle v_3, q_2\rangle = \frac{1}{\sqrt{6}}(0-1+2) = \frac{1}{\sqrt{6}}$, so

$$w_3 = (0,1,1)^\top - \tfrac{1}{2}(1,1,0)^\top - \tfrac{1}{6}(1,-1,2)^\top = \left(-\tfrac{2}{3}, \tfrac{2}{3}, \tfrac{2}{3}\right)^\top$$

where the first entry is $0 - \frac{1}{2} - \frac{1}{6} = -\frac{2}{3}$, the second is $1 - \frac{1}{2} + \frac{1}{6} = \frac{2}{3}$, and the third is $1 - 0 - \frac{1}{3} = \frac{2}{3}$. Then $\|w_3\|^2 = 3 \cdot \frac{4}{9} = \frac{4}{3}$ and $\|w_3\| = \frac{2}{\sqrt{3}}$, giving

$$q_3 = \frac{\sqrt{3}}{2}\left(-\tfrac{2}{3}, \tfrac{2}{3}, \tfrac{2}{3}\right)^\top = \frac{1}{\sqrt{3}}(-1, 1, 1)^\top$$

Verification. The three inner products across pairs are $\langle q_1, q_2\rangle = \frac{1}{\sqrt{12}}(1-1+0) = 0$, $\langle q_1, q_3\rangle = \frac{1}{\sqrt{6}}(-1+1+0) = 0$ and $\langle q_2, q_3\rangle = \frac{1}{\sqrt{18}}(-1-1+2) = 0$, and each of $\frac{1}{2}(1+1+0)$, $\frac{1}{6}(1+1+4)$, $\frac{1}{3}(1+1+1)$ equals 1, so the list is orthonormal.

A projection. Let $W = \operatorname{span}\{q_1, q_2\}$ and $x = (1,2,3)^\top$. Then $\langle x, q_1\rangle = \frac{1}{\sqrt{2}}(1+2) = \frac{3}{\sqrt{2}}$ and $\langle x, q_2\rangle = \frac{1}{\sqrt{6}}(1-2+6) = \frac{5}{\sqrt{6}}$, so definition 5.3.8 gives

$$P_W(x) = \tfrac{3}{2}(1,1,0)^\top + \tfrac{5}{6}(1,-1,2)^\top = \tfrac{1}{3}(7,2,5)^\top$$

and $x - P_W(x) = \frac{1}{3}(3-7, 6-2, 9-5)^\top = \frac{4}{3}(-1,1,1)^\top$. This remainder pairs to zero against q_1 (its first two entries cancel) and against q_2 (since $-1-1+2=0$), as theorem 5.3.5(2) requires; and it equals $\frac{4}{\sqrt{3}}q_3$, which is the q_3 term of the expansion of x in the full basis.

The process needs an independent list to start from, and every finite-dimensional space gives one.

Corollary 5.3.11: Existence and Extension of Orthonormal Bases

Let V be as above.

1. Every subspace $W \subseteq V$ has an orthonormal basis. (For $W = \{0\}$ this is the empty list, under the convention that the empty list spans $\{0\}$.)
2. Every orthonormal list $q_1, \ldots, q_m$ in V extends to an orthonormal basis $q_1, \ldots, q_n$ of V, with the first m members unchanged.

***Proof*:**

1. Let $W \neq \{0\}$. The restriction of $\langle -, -\rangle$ to pairs of vectors of W satisfies every condition of definition 5.1.2, each being a condition on vectors of V that W inherits, so W is an inner product space in its own right. It is finite-dimensional with $\dim W \leq \dim V$ by corollary 2.2.12, so theorem 2.2.9 gives a basis $v_1, \ldots, v_r$ of W with $r = \dim W \geq 1$. A basis is linearly independent (definition 2.2.1), so theorem 5.3.9(1) applies and yields an orthonormal list $q_1, \ldots, q_r$ with $\operatorname{span}\{q_1, \ldots, q_r\} = \operatorname{span}\{v_1, \ldots, v_r\} = W$. It is linearly independent by proposition 5.3.4, hence a basis of W by definition 2.2.1.

2. An orthonormal list is linearly independent by proposition 5.3.4, so theorem 2.2.9(1) extends it to a basis $q_1, \dots, q_m, v_{m+1}, \dots, v_n$ of V. Apply theorem 5.3.9(1) to this basis. The first m outputs are the original vectors: by induction on $j \le m$, the recursion at step j has already returned $q_1, \dots, q_{j-1}$, so
$$w_j = q_j - \sum_{i=1}^{j-1} \langle q_j, q_i\rangle\, q_i = q_j$$
because $\langle q_j, q_i\rangle = 0$ for $i < j$ (definition 5.3.3), and $\|w_j\| = \|q_j\| = 1$ leaves q_j unscaled. The process therefore returns an orthonormal list of n vectors beginning with $q_1, \dots, q_m$, and it spans V because theorem 5.3.9(1) makes its span equal to that of the basis it started from. Being orthonormal it is independent (proposition 5.3.4), so it is an orthonormal basis of V, as we sought to show.

Part (1) removes the hypothesis attached to definition 5.3.8: P_W is defined for every subspace W of V, and by theorem 5.3.5(2) it is characterized without reference to any basis, as the unique vector of W whose difference from v is orthogonal to all of W. Part (2) allows a computation to begin from orthonormal vectors already produced, extending them instead of replacing them.

Exercise 5.3.1

1. Apply theorem 5.3.9(1) to the list $v_1 = (1,1,1)^\top$, $v_2 = (0,1,1)^\top$, $v_3 = (0,0,1)^\top$ in $\mathbb{R}^3$ with the standard inner product, displaying each w_j before it is scaled. Verify that the three vectors produced are pairwise orthogonal.
2. Let A be the matrix of example 1.2.9, whose columns are $c_1 = (1,2,1,3)^\top$, $c_2 = (1,3,2,4)^\top$ and $c_3 = (3,8,5,11)^\top$.
    1. Apply theorem 5.3.9(1) to c_1, c_2 and simplify the two vectors produced.
    2. The columns satisfy $c_3 = c_1 + 2c_2$. Use theorem 5.3.9(2) to say which orthonormal list the process produces from the full list c_1, c_2, c_3, and how its length compares with $\operatorname{rank} A$.
3. This question concerns the converse of theorem 5.3.2 over $\mathbb{C}$.
    1. Compute $\langle u, v\rangle$, $\|u\|$, $\|v\|$ and $\|u+v\|$ for $u = (1,0)^\top$ and $v = (i,0)^\top$ in $\mathbb{C}^2$ with the standard inner product, and conclude that the identity of theorem 5.3.2 can hold for a pair that is not orthogonal.
    2. Show that for u, v in an inner product space over $\mathbb{C}$ the identity $\|u+v\|^2 = \|u\|^2 + \|v\|^2$ holds if and only if $\operatorname{Re}\langle u, v\rangle = 0$.
4. Let $q_1, \dots, q_m$ be an orthonormal list in V, not assumed to be a basis, and let $v \in V$. Show that
$$\sum_{i=1}^{m} |\langle v, q_i\rangle|^2 \le \|v\|^2$$
with equality if and only if $v \in \operatorname{span}\{q_1, \dots, q_m\}$.
5. Let $q_1, \dots, q_n$ be an orthonormal basis of V and let $T\colon V \to V$ be linear. The matrix $[T]$ of T in this basis (theorem 2.4.8) has as its j-th column the coordinate vector of Tq_j. Show that the (i,j) entry of $[T]$ is $\langle Tq_j, q_i\rangle$.

6. On $\mathbb{R}^2$ define $\langle x, y\rangle = 2x_1y_1 + 3x_2y_2$.

   1. Verify that this satisfies the conditions of definition 5.1.2.
   2. Apply theorem 5.3.9(1) to $v_1 = (1, 0)^\top$, $v_2 = (1, 1)^\top$ with respect to it.
   3. Check that the list produced is not orthonormal for the standard inner product on $\mathbb{R}^2$, and identify which of the two defining conditions of definition 5.3.3 fails.

5.4 The QR Factorization: Gram-Schmidt, Householder, and Givens

Gram-Schmidt (theorem 5.3.9) replaces a list of vectors by an orthonormal list spanning the same subspace at every stage. The coefficients it computes on the way are not discarded: each vector of the original list is a combination of the orthonormal vectors produced up to that point, and collecting those combinations into a matrix turns the whole procedure into one identity between matrices, $A = QR$. This section proves that identity, shows that it is unique once one scalar per column is pinned down, and then produces the same factorization twice more by routes that never form Q from the columns of A at all.

Those two routes reverse the order of the work. Gram-Schmidt builds Q out of the columns of A and obtains R from the expansion coefficients; the other two apply matrices that preserve the inner product to A until what remains is triangular, and assemble Q from the matrices used. Householder's matrices are reflections, one of which clears an entire column below the diagonal; Givens's act on two coordinates at a time and clear one entry apiece.

Two pieces of notation recur throughout. The inner product on k^m is the standard one, $\langle x, y\rangle = \sum_i x_i\overline{y_i}$ (section 5.1), and A^* is the conjugate transpose of A, the matrix whose entry in row i and column j is $\overline{a_{ji}}$; over $\mathbb{R}$ conjugation does nothing and A^* is the transpose of definition 2.3.4. These two are related by a computation used repeatedly below. Let $U \in M_{m\times n}(k)$ have columns $u_1, \dots, u_n \in k^m$, so that u_{il} is the entry of U in row i and column l. By definition 1.4.1 the entry of U^*U in row j and column l is

$$(U^*U)_{jl} = \sum_{i=1}^{m} (U^*)_{ji}u_{il} = \sum_{i=1}^{m} \overline{u_{ij}}u_{il} = \langle u_l, u_j\rangle$$

so $U^*U = I_n$ says exactly that $\langle u_l, u_j\rangle$ is 1 when $j = l$ and 0 otherwise, which is orthonormality of the columns (definition 5.3.3). One further consequence of definition 1.4.1 is used without comment below: the entry of a product CB in row i and column l is $\sum_t c_{it}b_{tl}$, which depends on B only through its lth column, so column l of CB is C times column l of B.

Definition 5.4.1: QR Factorization

Let $A \in M_{m\times n}(k)$ with $m \geq n$. Call a matrix $R = (r_{il})$ *upper triangular* when $r_{il} = 0$ for every $i > l$.

1. A *reduced QR factorization* of A is a factorization $A = QR$ in which $Q \in M_{m\times n}(k)$ has orthonormal columns and $R \in M_n(k)$ is upper triangular.
2. A *full QR factorization* of A is a factorization $A = QR$ in which $Q \in M_m(k)$ has orthonormal columns and $R \in M_{m\times n}(k)$ is upper triangular.

The following is a factorization of the first kind, and both of its conditions can be checked by hand. For

$$A = \begin{pmatrix} 3 & 1 \\ 4 & 7 \end{pmatrix} \in M_2(\mathbb{R}), \qquad Q = \frac{1}{5}\begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}, \qquad R = \frac{1}{5}\begin{pmatrix} 25 & 31 \\ 0 & 17 \end{pmatrix}$$

the columns of Q are $(3,4)^\top/5$ and $(-4,3)^\top/5$, whose inner product is $(-12+12)/25 = 0$ and each of whose inner products with itself is $(9+16)/25 = 1$, so $Q^*Q = I_2$. The first column of QR is $5q_1 = (3,4)^\top$ and the second is

$$\tfrac{31}{5}q_1 + \tfrac{17}{5}q_2 = \tfrac{1}{25}\big(31\cdot(3,4)^\top + 17\cdot(-4,3)^\top\big) = \tfrac{1}{25}(93-68,\ 124+51)^\top = (1,7)^\top$$

so $A = QR$. Where the entries of Q and R came from is what theorem 5.4.2 answers.

The two shapes in definition 5.4.1 carry the same information. If $A = QR$ is a full factorization, let $\widehat{Q} \in M_{m\times n}(k)$ consist of the first n columns of Q and $\widehat{R} \in M_n(k)$ of the first n rows of R. For $l \leq n$ the entries r_{tl} with $t > l$ vanish, so only the indices $t \leq n$ contribute to $(QR)_{il} = \sum_{t=1}^m q_{it}r_{tl}$, and that truncated sum is $(\widehat{Q}\widehat{R})_{il}$. Hence $A = \widehat{Q}\widehat{R}$, a reduced factorization. The passage in the other direction requires enlarging an orthonormal list of n vectors to one of m vectors, which is why the full shape is the one the algorithms of this section produce directly.

The route from Gram-Schmidt to a factorization is short, because theorem 5.3.9 already delivers both halves of what definition 5.4.1 asks for: the orthonormal vectors are the columns of Q, and the requirement that R be upper triangular is precisely the statement that the jth column of A involves only the first j orthonormal vectors.

Theorem 5.4.2: Existence of the QR Factorization

Let $A \in M_{m\times n}(k)$ have linearly independent columns $a_1, \ldots, a_n$. Then A has a reduced QR factorization $A = QR$ in which every diagonal entry r_{jj} is real and positive. Its factors satisfy

$$\operatorname{span}\{q_1, \ldots, q_j\} = \operatorname{span}\{a_1, \ldots, a_j\} \qquad \text{and} \qquad r_{ij} = \langle a_j, q_i\rangle \quad (i \leq j)$$

for every j, where $q_1, \ldots, q_n$ are the columns of Q.

***Proof*:**
Step 1: orthonormalize the columns. The list $a_1, \ldots, a_n$ is linearly independent by hypothesis, so theorem 5.3.9 applies to it and produces an orthonormal list $q_1, \ldots, q_n$ in k^m with $\operatorname{span}\{q_1, \ldots, q_j\} = \operatorname{span}\{a_1, \ldots, a_j\}$ for every $j \leq n$.
Step 2: expand each column in the list. Fix j and write $W_j = \operatorname{span}\{q_1, \ldots, q_j\}$. The list $q_1, \ldots, q_j$ is orthonormal, hence linearly independent by proposition 5.3.4, and it spans W_j by definition, so

it is a basis of W_j (definition 2.2.1) and an orthonormal one. Now $a_j \in \operatorname{span}\{a_1, \dots, a_j\} = W_j$, so theorem 5.3.5 applies to a_j in W_j and gives

$$a_j = \sum_{i=1}^{j} \langle a_j, q_i \rangle q_i$$

Set $r_{ij} = \langle a_j, q_i \rangle$ for $i \le j$ and $r_{ij} = 0$ for $i > j$, let $R = (r_{ij}) \in M_n(k)$, which is upper triangular by construction, and let $Q \in M_{m \times n}(k)$ be the matrix with columns $q_1, \dots, q_n$. By definition 1.4.1 the entry of QR in row i and column l is $\sum_{t=1}^{n} q_{it} r_{tl}$, so column l of QR is $\sum_{t=1}^{n} r_{tl} q_t = \sum_{t \le l} \langle a_l, q_t \rangle q_t$, which is a_l by the displayed expansion. Hence $A = QR$.
Step 3: the diagonal entries are nonzero. Suppose $r_{jj} = 0$ for some j. Then the expansion of a_j reads $a_j = \sum_{i<j} r_{ij} q_i$, and since $\operatorname{span}\{q_1, \dots, q_{j-1}\} = \operatorname{span}\{a_1, \dots, a_{j-1}\}$ there are scalars $\lambda_1, \dots, \lambda_{j-1}$ with $a_j = \sum_{i<j} \lambda_i a_i$. Rearranged, this is the relation $\lambda_1 a_1 + \cdots + \lambda_{j-1} a_{j-1} + (-1) a_j = 0$ among $a_1, \dots, a_n$, whose coefficient on a_j is $-1 \neq 0$. That contradicts linear independence (definition 2.1.9), so $r_{jj} \neq 0$ for every j.
Step 4: scale the columns to make the diagonal positive. Put $\theta_j = r_{jj}/|r_{jj}|$, a scalar of absolute value 1, and replace q_j by $q_j' = \theta_j q_j$. The new list is again orthonormal: for $i \neq j$, $\langle \theta_i q_i, \theta_j q_j \rangle = \theta_i \overline{\theta_j} \langle q_i, q_j \rangle = 0$ by definition 5.1.2 and proposition 5.1.6(2), and $\langle \theta_j q_j, \theta_j q_j \rangle = \theta_j \overline{\theta_j} \cdot 1 = |\theta_j|^2 = 1$. Since each θ_j is nonzero, $\operatorname{span}\{q_1', \dots, q_j'\} = \operatorname{span}\{q_1, \dots, q_j\}$ for every j, so Step 2 applies verbatim to the new list and produces the upper triangular matrix R' with entries $r_{ij}' = \langle a_j, q_i' \rangle = \overline{\theta_i} \langle a_j, q_i \rangle$. Its diagonal entries are

$$r_{jj}' = \overline{\theta_j} r_{jj} = \frac{\overline{r_{jj}}\, r_{jj}}{|r_{jj}|} = |r_{jj}| > 0$$

Writing Q' for the matrix with columns $q_1', \dots, q_n'$, the pair (Q', R') satisfies every clause of the statement, as we sought to show.

The factorization repackages three facts about the columns of A in one equation. The columns of Q are an orthonormal basis of the column space $\operatorname{col} A$ (definition 2.3.1), since they are independent by proposition 5.3.4 and span it by the equality $\operatorname{span}\{q_1, \dots, q_n\} = \operatorname{span}\{a_1, \dots, a_n\}$. The triangularity of R records that the first j columns of A and the first j columns of Q span the same subspace, for every j at once. And the diagonal entry r_{jj} measures how much of a_j is new: the expansion of Step 2 gives $a_j - \sum_{i<j} r_{ij} q_i = r_{jj} q_j$, whose norm is $|r_{jj}|\,\|q_j\| = r_{jj}$ by proposition 5.2.2, so r_{jj} is the length of the part of a_j left after the earlier columns have been subtracted off.

Nothing in definition 5.4.1 forces the diagonal of R to be positive, and without that constraint a factorization is not unique: the single column $A = (1,0)^\top \in M_{2\times 1}(\mathbb{R})$ is both $(1,0)^\top \cdot (1)$ and $(-1,0)^\top \cdot (-1)$, and both factorizations meet the definition. The positivity constraint removes exactly this freedom.

Theorem 5.4.3: Uniqueness of the QR Factorization

Let $A \in M_{m\times n}(k)$ have linearly independent columns, and suppose

$$A = Q_1R_1 = Q_2R_2$$

are two reduced QR factorizations in which every diagonal entry of R_1 and of R_2 is real and positive. Then $Q_1 = Q_2$ and $R_1 = R_2$.

***Proof*:**
Write $a_1, \ldots, a_n$ for the columns of A, write $q_1, \ldots, q_n$ and $p_1, \ldots, p_n$ for the columns of Q_1 and Q_2, and write r_{il} and s_{il} for the entries of R_1 and R_2. As in Step 2 of the previous proof, definition 1.4.1 makes column l of a product QR equal to $\sum_t r_{tl}q_t$, so for every l

$$a_l = \sum_{i\le l} r_{il}q_i = \sum_{i\le l} s_{il}p_i$$

the sums stopping at $i = l$ because both R_1 and R_2 are upper triangular. The claim is proved by induction on l: for each l, the vectors q_l and p_l agree and the entries r_{il} and s_{il} agree for all i. Fix l and assume $q_i = p_i$ for all $i < l$, which is vacuous when $l = 1$. For $t < l$, take the inner product of the displayed expansions with q_t. Orthonormality of $q_1, \ldots, q_n$ gives $\langle a_l, q_t\rangle = \sum_{i\le l} r_{il}\langle q_i, q_t\rangle = r_{tl}$, and the same computation with p_t in place of q_t, using orthonormality of $p_1, \ldots, p_n$, gives $\langle a_l, p_t\rangle = s_{tl}$. Since $p_t = q_t$ for $t < l$ by the inductive hypothesis, the two inner products are the same number, so $r_{tl} = s_{tl}$ for every $t < l$.
Subtracting those common terms from both expansions, which is legitimate on the right because $s_{il} = r_{il}$ and $p_i = q_i$ for $i < l$, leaves

$$w := a_l - \sum_{i<l} r_{il}q_i = r_{ll}q_l = s_{ll}p_l$$

Taking norms and using proposition 5.2.2 together with $\|q_l\| = \|p_l\| = 1$ gives $\|w\| = |r_{ll}| = r_{ll}$ and likewise $\|w\| = s_{ll}$, the absolute values disappearing because both diagonal entries are positive by hypothesis. Hence $r_{ll} = s_{ll}$, and this common value is nonzero, so $w \ne 0$ and $q_l = w/r_{ll} = w/s_{ll} = p_l$. The induction is complete, and with it the equalities $Q_1 = Q_2$ and $R_1 = R_2$, completing the proof.

The proof also describes the whole of the non-uniqueness that positivity removes. Suppose $A = QR$ is a reduced factorization with every $r_{jj} \ne 0$, and let $D = \operatorname{diag}(\theta_1, \ldots, \theta_n)$ with $|\theta_j| = 1$ for each j. Then QD has columns $\theta_1q_1, \ldots, \theta_nq_n$, which are orthonormal by the computation in Step 4 above; the product of $\operatorname{diag}(\theta_1, \ldots, \theta_n)$ with $\operatorname{diag}(\theta_1^{-1}, \ldots, \theta_n^{-1})$ is I_n in either order, so $D^{-1} = \operatorname{diag}(\theta_1^{-1}, \ldots, \theta_n^{-1})$ and $D^{-1}R$ has entries $\theta_i^{-1}r_{il}$ and is again upper triangular; and $(QD)(D^{-1}R) = Q(DD^{-1})R = QR = A$ by proposition 1.4.3. Choosing $\theta_j = r_{jj}/|r_{jj}|$ makes the diagonal entries of $D^{-1}R$ equal to $|r_{jj}| > 0$, so any such factorization is carried to the one theorem 5.4.3 pins down by a single rescaling of the columns of Q.

For an invertible $A \in M_n(\mathbb{R})$ the two theorems together say that A factors in exactly one way as a square matrix with orthonormal columns times an upper triangular matrix with positive diagonal. Splitting $R = DN$ with $D = \operatorname{diag}(r_{11}, \ldots, r_{nn})$ and $N = D^{-1}R$, whose diagonal entries are all 1, refines this to a unique factorization of A into three matrices of prescribed types, which is the Iwasawa decomposition developed in *Lie Groups* [18].

Example 5.4.4: QR Factorization of the Running Matrix

The running matrix of example 1.2.9 is

$$A = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 1 & 2 & 5 \\ 3 & 4 & 11 \end{pmatrix} \in M_{4\times 3}(\mathbb{R})$$

with columns $a_1 = (1,2,1,3)^\top$, $a_2 = (1,3,2,4)^\top$ and $a_3 = (3,8,5,11)^\top$ satisfying $a_3 = a_1 + 2a_2$ (section 1.2). That relation is a nontrivial dependence among the three columns, so the hypothesis of theorem 5.4.2 fails for A and the theorem says nothing about it. It does apply to the matrix $A' = \left(\begin{smallmatrix} 1 & 1 \\ 2 & 3 \\ 1 & 2 \\ 3 & 4 \end{smallmatrix}\right)$ of the first two columns, which are independent because neither is a scalar multiple of the other.

Run the construction of theorem 5.4.2 on A'. First, $\langle a_1, a_1\rangle = 1+4+1+9 = 15$, so $r_{11} = \|a_1\| = \sqrt{15}$ and

$$q_1 = \frac{1}{\sqrt{15}}(1,2,1,3)^\top$$

Next, $\langle a_2, a_1\rangle = 1+6+2+12 = 21$, so $r_{12} = \langle a_2, q_1\rangle = 21/\sqrt{15} = 7\sqrt{15}/5$. Then $r_{12}q_1 = \frac{21}{15}a_1$, and subtracting it leaves

$$a_2 - \frac{21}{15}a_1 = \left(1-\frac{7}{5},\ 3-\frac{14}{5},\ 2-\frac{7}{5},\ 4-\frac{21}{5}\right)^\top = \frac{1}{5}(-2,1,3,-1)^\top$$

whose inner product with itself is $(4+1+9+1)/25 = 3/5$, so its norm is $\sqrt{15}/5$ and

$$q_2 = \frac{1}{\sqrt{15}}(-2,1,3,-1)^\top$$

The check $\langle q_1, q_2\rangle = (-2+2+3-3)/15 = 0$ confirms orthogonality. Finally $r_{22} = \langle a_2, q_2\rangle = (-2+3+6-4)/\sqrt{15} = 3/\sqrt{15} = \sqrt{15}/5$, so

$$A' = QR, \qquad Q = \frac{1}{\sqrt{15}}\begin{pmatrix} 1 & -2 \\ 2 & 1 \\ 1 & 3 \\ 3 & -1 \end{pmatrix}, \qquad R = \frac{\sqrt{15}}{5}\begin{pmatrix} 5 & 7 \\ 0 & 1 \end{pmatrix}$$

with both diagonal entries positive. Multiplying out confirms it: the first column of QR is $\sqrt{15}\,q_1 = a_1$, and the second is

$$\frac{7\sqrt{15}}{5}q_1 + \frac{\sqrt{15}}{5}q_2 = \frac{7}{5}(1,2,1,3)^\top + \frac{1}{5}(-2,1,3,-1)^\top = (1,3,2,4)^\top = a_2$$

The dependence relation now carries A itself. Append to R the column $\frac{\sqrt{15}}{5}(19,2)^\top$, which is $R(1,2)^\top$, and call the result $\widetilde{R}$:

$$\widetilde{R} = \frac{\sqrt{15}}{5}\begin{pmatrix} 5 & 7 & 19 \\ 0 & 1 & 2 \end{pmatrix}$$

The first two columns of $Q\widetilde{R}$ are the two computed above, namely a_1 and a_2, and the third is $\frac{19}{5}(1,2,1,3)^\top + \frac{2}{5}(-2,1,3,-1)^\top = (3,8,5,11)^\top = a_3$, so $A = Q\widetilde{R}$. This is neither of the two shapes in definition 5.4.1, since Q is 4×2 and $\widetilde{R}$ is 2×3; it factors A through an orthonormal basis of its column space, and it exists because that column space is two dimensional.

The dependence among the columns of A blocked theorem 5.4.2 but not the question, and the next construction answers it for every matrix. Instead of building the columns of Q from those of A, apply to A a sequence of matrices that preserve the inner product and drive it toward upper triangular form. One matrix per column suffices, and the matrices are reflections.

Definition 5.4.5: Householder Reflection

Let $v \in k^m$ be nonzero. The *Householder reflection* determined by v is the matrix

$$H_v = I_m - \frac{2}{\langle v, v\rangle} vv^* \in M_m(k)$$

where $vv^* \in M_m(k)$ is the product of the column v with the row v^*, and $\langle v, v\rangle$ is a positive real number by proposition 5.1.6(1).

The definition is a matrix, but every argument below uses its action on a vector. For $x \in k^m$ the product v^*x is the 1×1 matrix with entry $\sum_i \overline{v_i} x_i = \langle x, v\rangle$, so $(vv^*)x = v(v^*x) = \langle x, v\rangle v$ by proposition 1.4.3, and therefore

$$H_v x = x - \frac{2\langle x, v\rangle}{\langle v, v\rangle} v \tag{5.2}$$

The subtracted vector is twice the multiple of v appearing in the splitting $x = \frac{\langle x,v\rangle}{\langle v,v\rangle} v + w$, where $w = x - \frac{\langle x,v\rangle}{\langle v,v\rangle} v$ satisfies $\langle w, v\rangle = \langle x, v\rangle - \frac{\langle x,v\rangle}{\langle v,v\rangle}\langle v, v\rangle = 0$. So H_v keeps the part of x orthogonal to v and reverses the part along v: it is the reflection in the set of vectors orthogonal to v.

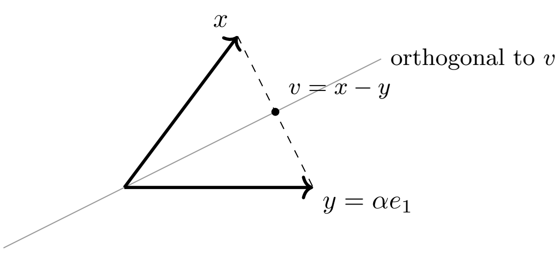


Figure 5.3: A Householder reflection carrying x to a multiple of e_1 of the same length

Proposition 5.4.6: Properties of a Householder Reflection

Let $v \in k^m$ be nonzero and let H_v be as in definition 5.4.5.

1. $H_v v = -v$, and $H_v x = x$ for every x with $\langle x, v\rangle = 0$.
2. $H_v^* = H_v$ and $H_v^2 = I_m$; in particular H_v is invertible and $H_v^{-1} = H_v$.
3. $\langle H_v x, H_v y\rangle = \langle x, y\rangle$ for all $x, y \in k^m$. Consequently $\|H_v x\| = \|x\|$ for every x, and the columns of H_v are orthonormal.
4. Let $x, y \in k^m$ satisfy $\|x\| = \|y\|$, $x \neq y$, and $\langle x, y\rangle \in \mathbb{R}$. Then $H_{x-y}\, x = y$.

***Proof*:**

1. Setting $x = v$ in (5.2) gives $H_v v = v - \frac{2\langle v,v\rangle}{\langle v,v\rangle} v = -v$, and setting $\langle x, v\rangle = 0$ gives $H_v x = x - 0 = x$.

2. For the first claim, compare entries. By definition 1.4.1 the entry of vv^* in row i and column j is $v_i\overline{v_j}$, and $\langle v, v\rangle$ is real by proposition 5.1.6(1), so the entry of H_v in row i and column j is $\delta_{ij} - 2v_i\overline{v_j}/\langle v, v\rangle$, where δ_{ij} is 1 for $i = j$ and 0 otherwise. Conjugating the entry in row j and column i gives
$$\overline{\delta_{ji} - \frac{2v_j\overline{v_i}}{\langle v,v\rangle}} = \delta_{ij} - \frac{2\overline{v_j}v_i}{\langle v,v\rangle}$$
which is the same number, so $H_v^* = H_v$. For the second claim, apply (5.2) twice. Writing $\lambda = 2\langle x, v\rangle/\langle v, v\rangle$, linearity of the inner product in its first argument (definition 5.1.2) gives
$$\langle H_v x, v\rangle = \langle x, v\rangle - \lambda\langle v, v\rangle = \langle x, v\rangle - 2\langle x, v\rangle = -\langle x, v\rangle$$
so applying (5.2) to $H_v x$ yields $H_v(H_v x) = (x - \lambda v) + \lambda v = x$. Two matrices carrying every vector to the same image have the same columns, since column j of B is Be_j by definition 1.4.1; hence $H_v^2 = I_m$. Since the product $H_v H_v$ is I_m read in either order, H_v is invertible with inverse H_v (definition 1.4.4).

3. Write $\lambda = 2\langle x, v\rangle/\langle v, v\rangle$ and $\mu = 2\langle y, v\rangle/\langle v, v\rangle$. Since $\langle v, v\rangle$ is real, $\overline{\mu} = 2\overline{\langle y, v\rangle}/\langle v, v\rangle = 2\langle v, y\rangle/\langle v, v\rangle$ by proposition 5.1.6(2). Expanding by linearity in the first argument and conjugate-linearity in the second,
$$\langle x - \lambda v,\ y - \mu v\rangle = \langle x, y\rangle - \overline{\mu}\langle x, v\rangle - \lambda\langle v, y\rangle + \lambda\overline{\mu}\langle v, v\rangle$$
The last three terms are $-\frac{2\langle v,y\rangle\langle x,v\rangle}{\langle v,v\rangle}$, $-\frac{2\langle x,v\rangle\langle v,y\rangle}{\langle v,v\rangle}$ and $+\frac{4\langle x,v\rangle\langle v,y\rangle}{\langle v,v\rangle}$, which cancel, leaving $\langle H_v x, H_v y\rangle = \langle x, y\rangle$. Taking $y = x$ and $\|z\| = \sqrt{\langle z, z\rangle}$ (definition 5.2.1) gives $\|H_v x\| = \|x\|$. Column j of H_v is $H_v e_j$, and $\langle H_v e_j, H_v e_l\rangle = \langle e_j, e_l\rangle$, which is 1 for $j = l$ and 0 otherwise; the columns are therefore orthonormal.

4. Put $v = x - y$, which is nonzero by hypothesis, so H_v is defined. By proposition 5.1.6(4), applied with $-y$ in the place of its second vector,
$$\langle v, v\rangle = \|x\|^2 + 2\operatorname{Re}\langle x, -y\rangle + \|y\|^2 = 2\|x\|^2 - 2\langle x, y\rangle$$
the second equality using $\|y\| = \|x\|$, then $\langle x, -y\rangle = -\langle x, y\rangle$ (proposition 5.1.6(2)), and finally the hypothesis that $\langle x, y\rangle$ is real, so that its real part is itself. Also $\langle x, v\rangle = \langle x, x\rangle - \langle x, y\rangle = \|x\|^2 - \langle x, y\rangle$. Hence $2\langle x, v\rangle = \langle v, v\rangle$, so the coefficient in (5.2) equals 1 and $H_v x = x - v = y$, as we sought to show.

Part (4) is what the triangularization below calls on: for any two vectors of equal length whose inner product is real, it produces a single reflection carrying the first to the second. Over $\mathbb{R}$ the condition $\langle x, y\rangle \in \mathbb{R}$ is automatic, and the construction is available for any target of the right length. Take

$x = (3,4)^\top \in \mathbb{R}^2$, of length 5, and $y = 5e_1$. Then $v = x - y = (-2,4)^\top$ has $\langle v, v\rangle = 20$, and

$$H_v = I_2 - \frac{2}{20}\begin{pmatrix} 4 & -8 \\ -8 & 16 \end{pmatrix} = \begin{pmatrix} 1-\frac{2}{5} & \frac{4}{5} \\ \frac{4}{5} & 1-\frac{8}{5} \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix}$$

which sends x to $\frac{1}{5}(9+16, 12-12)^\top = (5,0)^\top = y$, as part (4) predicts.

A reflection that carries a column to a multiple of e_1 empties that column of everything below its first entry in a single step. Applying such a reflection to the first column of A, then a second reflection that fixes the first coordinate and empties the second column below its second entry, and so on, drives A to upper triangular form in n steps.

Theorem 5.4.7: Triangularization by Householder Reflections

Let $A \in M_{m\times n}(k)$ with $m \ge n$. There are matrices $H_1, \dots, H_n \in M_m(k)$, each either I_m or a Householder reflection, such that

$$R := H_n \cdots H_1 A$$

is upper triangular. Setting $Q = H_1 H_2 \cdots H_n$ gives a full QR factorization $A = QR$. No hypothesis on the columns of A is needed.

***Proof*:**
Step 1: the inductive claim. For $0 \le j \le n$ let $P(j)$ be the statement that there are matrices $H_1, \dots, H_j \in M_m(k)$, with H_t for each $t \le j$ either I_m or a Householder reflection H_v whose vector v has $v_i = 0$ for all $i \le t-1$, such that the matrix $A^{(j)} = H_j \cdots H_1 A$ satisfies $(A^{(j)})_{il} = 0$ whenever $l \le j$ and $i > l$. The statement $P(0)$ holds with the empty list and $A^{(0)} = A$, since its condition is vacuous.
Step 2: the inductive step. Assume $P(j)$ with $j < n$ and let $B = A^{(j)}$. Define $x \in k^m$ by $x_i = b_{i,j+1}$ for $i \ge j+1$ and $x_i = 0$ for $i \le j$, so that x collects the part of column $j+1$ of B lying at or below row $j+1$. If $x_i = 0$ for every $i > j+1$, set $H_{j+1} = I_m$; then $A^{(j+1)} = B$ already satisfies the condition of $P(j+1)$, since column $j+1$ has zeros below row $j+1$ and the columns $l \le j$ are untouched. Otherwise put

$$\alpha = \begin{cases} \|x\| & \text{if } x_{j+1} = 0 \\ \dfrac{x_{j+1}}{|x_{j+1}|}\|x\| & \text{if } x_{j+1} \neq 0 \end{cases} \qquad \text{and} \qquad y = \alpha e_{j+1}$$

and check the three hypotheses of proposition 5.4.6(4). First, $\|y\| = |\alpha|\|e_{j+1}\| = \|x\|$ by proposition 5.2.2, since $\langle e_{j+1}, e_{j+1}\rangle = 1$ and $|\alpha| = \|x\|$ in both cases. Second, $x \neq y$: some x_i with $i > j+1$ is nonzero, while $y_i = 0$ for every $i \neq j+1$. Third, $\langle x, y\rangle = \overline{\alpha}\langle x, e_{j+1}\rangle = \overline{\alpha} x_{j+1}$, which is 0 in the first case and $\frac{\overline{x_{j+1}} x_{j+1}}{|x_{j+1}|}\|x\| = |x_{j+1}|\|x\|$ in the second, hence real in both. So $H_{j+1} := H_v$ with $v = x - y$ satisfies $H_{j+1}x = y$. Its vector v has $v_i = x_i - y_i = 0$ for $i \le j$, which is what $P(j+1)$ asks of its factor of index $t = j+1$; the earlier factors carry their own conditions over from $P(j)$ unchanged.
Step 3: the reflection preserves the zeros already made. Let $b \in k^m$ have $b_i = 0$ for every $i \ge j+1$. Then $\langle b, v\rangle = \sum_i b_i \overline{v_i} = 0$, because each term has either $i \le j$, where $v_i = 0$, or $i \ge j+1$, where $b_i = 0$; so $H_{j+1}b = b$ by proposition 5.4.6(1). Column l of B with $l \le j$ is such a vector, since $P(j)$ makes its entries vanish in all rows $i > l$ and $j+1 > l$. Those columns are therefore unchanged. For column $j+1$, write it as $u + x$ where $u_i = b_{i,j+1}$ for $i \le j$ and $u_i = 0$ for $i > j$. Then $\langle u, v\rangle = 0$ by the same argument, so by (5.2) and linearity of the inner product in its first

argument,

$$H_{j+1}(u+x) = (u+x) - \frac{2\langle x, v\rangle}{\langle v, v\rangle} v = u + H_{j+1}x = u + \alpha e_{j+1}$$

whose entries in rows below $j+1$ all vanish. Hence $A^{(j+1)} = H_{j+1}A^{(j)}$ satisfies $P(j+1)$, and the induction gives $P(n)$.

Step 4: assemble the factorization. Put $R = A^{(n)} = H_n \cdots H_1 A$, upper triangular by $P(n)$. Each H_t satisfies $H_t H_t = I_m$, by proposition 5.4.6(2) when it is a reflection and trivially when it is I_m. Multiplying $H_n \cdots H_1 A = R$ on the left by H_n, then by H_{n-1}, and so on, and using proposition 1.4.3 at each stage, gives $A = H_1 H_2 \cdots H_n R = QR$. Finally, each H_t satisfies $\langle H_t z, H_t z'\rangle = \langle z, z'\rangle$ by proposition 5.4.6(3), so applying this n times to $Qz = H_1(H_2(\cdots(H_n z)))$ gives $\langle Qz, Qz'\rangle = \langle z, z'\rangle$ for all $z, z' \in k^m$. Taking $z = e_p$ and $z' = e_q$ shows the columns $Qe_1, \ldots, Qe_m$ of Q to be orthonormal, so $A = QR$ is a full QR factorization, completing the proof.

Two features separate this from theorem 5.4.2. It needs no hypothesis on the columns of A, so it applies to the running matrix of example 5.4.4 directly, and it produces the full shape, with Q square. What it does not guarantee is a positive diagonal, since α was constrained only in absolute value and in the reality of $\overline{\alpha} x_{j+1}$. Suppose A has linearly independent columns and the reduced factorization extracted from $A = QR$ has no zero on the diagonal of $\widehat{R}$. Rescaling the columns of $\widehat{Q}$ by unit scalars as described after theorem 5.4.3 then makes that diagonal positive, and theorem 5.4.3 identifies the result with the factorization theorem 5.4.2 produces. Over $\mathbb{R}$ both signs $\alpha = \pm\|x\|$ are available at every step, and which of the two to take is a question about how errors grow in inexact arithmetic, which belongs to **??**.

A reflection clears a whole column at once, which is efficient when the column is full and wasteful when it is nearly empty. The remaining construction goes to the opposite extreme: each matrix changes two coordinates and creates one zero.

Definition 5.4.8: Givens Rotation

Let $1 \le i < j \le m$ and let $c, s \in k$ satisfy $|c|^2 + |s|^2 = 1$. The *Givens rotation* $G(i, j; c, s) \in M_m(k)$ is the matrix that agrees with I_m except in the four entries

$$g_{ii} = c, \qquad g_{ij} = s, \qquad g_{ji} = -\overline{s}, \qquad g_{jj} = \overline{c}$$

Its action on $x \in k^m$ is $(Gx)_i = cx_i + sx_j$, $(Gx)_j = -\overline{s}x_i + \overline{c}x_j$, and $(Gx)_t = x_t$ for $t \notin \{i, j\}$.

Over $\mathbb{R}$ the four entries form the block $\left(\begin{smallmatrix} c & s \\ -s & c \end{smallmatrix}\right)$ with $c^2 + s^2 = 1$, which is the rotation of the (i, j) coordinate plane through the angle whose cosine is c; the name comes from that case. The reason to isolate two coordinates is the following computation. Given $a, b \in k$ not both zero, set $r = \sqrt{|a|^2 + |b|^2} > 0$ and $c = \overline{a}/r$, $s = \overline{b}/r$. Then $|c|^2 + |s|^2 = (|a|^2 + |b|^2)/r^2 = 1$, so $G = G(i, j; c, s)$ is a Givens rotation, and for any x with $x_i = a$ and $x_j = b$,

$$(Gx)_i = \frac{\overline{a}a + \overline{b}b}{r} = r, \qquad (Gx)_j = \frac{-ba + ab}{r} = 0 \tag{5.3}$$

So one Givens rotation replaces the pair of coordinates (a, b) by $(r, 0)$, leaving every other coordinate unchanged. With $a = 3$, $b = 4$ and $k = \mathbb{R}$ this gives $r = 5$, $c = 3/5$, $s = 4/5$ and

$$G = \frac{1}{5}\begin{pmatrix} 3 & 4 \\ -4 & 3 \end{pmatrix}, \qquad G\begin{pmatrix} 3 \\ 4 \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 9 + 16 \\ -12 + 12 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \end{pmatrix}$$

which is the same transition as the Householder example above, carried out by a rotation rather than a reflection; the two matrices differ in the sign of their bottom row.

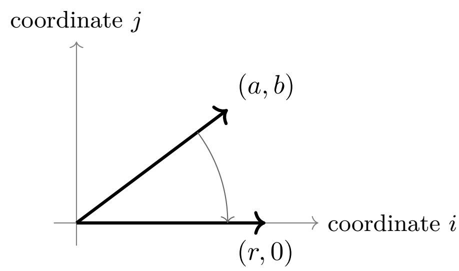


Figure 5.4: A Givens rotation in the (i, j) coordinate plane, sending (a, b) to $(r, 0)$ with $r^2 = |a|^2 + |b|^2$

Before the rotations are strung together, two facts about a single one are needed: it preserves the inner product, and its inverse is another rotation in the same pair of coordinates. Both are computations with the four entries. For the first, let $G = G(i, j; c, s)$ and let $x, y \in k^m$. The coordinates $t \notin \{i, j\}$ contribute $x_t\overline{y_t}$ to both $\langle Gx, Gy\rangle$ and $\langle x, y\rangle$, and the remaining contribution to $\langle Gx, Gy\rangle$ is

$$(cx_i + sx_j)\overline{(cy_i + sy_j)} + (-\overline{s}x_i + \overline{c}x_j)\overline{(-\overline{s}y_i + \overline{c}y_j)} = (|c|^2 + |s|^2)x_i\overline{y_i} + (|s|^2 + |c|^2)x_j\overline{y_j}$$

because the two cross terms in the first product, $c\overline{s}x_i\overline{y_j} + s\overline{c}x_j\overline{y_i}$, are cancelled by the two in the second, $-\overline{s}c\,x_i\overline{y_j} - \overline{c}s\,x_j\overline{y_i}$. Since $|c|^2 + |s|^2 = 1$, that contribution is $x_i\overline{y_i} + x_j\overline{y_j}$ and $\langle Gx, Gy\rangle = \langle x, y\rangle$. For the second, let $G' = G(i, j; \overline{c}, -s)$, which is a Givens rotation because $|\overline{c}|^2 + |-s|^2 = 1$, and whose action is $(G'z)_i = \overline{c}z_i - sz_j$ and $(G'z)_j = \overline{s}z_i + cz_j$. Then

$$(G'(Gx))_i = \overline{c}(cx_i + sx_j) - s(-\overline{s}x_i + \overline{c}x_j) = x_i, \qquad (G'(Gx))_j = \overline{s}(cx_i + sx_j) + c(-\overline{s}x_i + \overline{c}x_j) = x_j$$

and the other coordinates are untouched, so $G'G = I_m$; applying the same identity to G' in place of G, whose associated rotation is G itself, gives $GG' = I_m$. Hence $G^{-1} = G(i, j; \overline{c}, -s)$, again a Givens rotation in the coordinates i and j.

Theorem 5.4.9: Triangularization by Givens Rotations

Let $A \in M_{m\times n}(k)$ with $m \geq n$. There are Givens rotations $G_1, \dots, G_N \in M_m(k)$, with $N = \sum_{l=1}^{n}(m - l) = mn - \frac{1}{2}n(n + 1)$, such that

$$R := G_N \cdots G_1 A$$

is upper triangular, with R_{ll} real and nonnegative for every $l < m$. Setting $Q = G_1^{-1}G_2^{-1} \cdots G_N^{-1}$ gives a full QR factorization $A = QR$.

***Proof*:**
Step 1: the order of the entries. List the positions below the diagonal as (i, l) with $1 \leq l \leq n$ and $l < i \leq m$, ordered by column first and by row within a column: $(2, 1), (3, 1), \dots, (m, 1), (3, 2), \dots, (m, 2), \dots, (m, n)$. There are $m - l$ of them in column l, hence N in all. The rotation assigned to the position (i, l) will act on the coordinates l and i.
Step 2: one position. Suppose the positions preceding (i, l) in this order have been treated and

let B be the current matrix, so that $B_{i'l'} = 0$ for every position (i', l') already treated. Let $a = B_{ll}$ and $b = B_{il}$. If $a = b = 0$, take the rotation $G(l, i; 1, 0) = I_m$ and the entry at (i, l) is already 0. Otherwise take $G = G(l, i; c, s)$ with $c = \overline{a}/r$ and $s = \overline{b}/r$, where $r = \sqrt{|a|^2 + |b|^2}$. Column l' of GB is G times column l' of B, so (5.3) applied to column l, whose entries in the coordinates l and i are a and b, makes the new entry at (l, l) equal to r and the new entry at (i, l) equal to 0.
Step 3: nothing already zero is disturbed. The rotation G changes only rows l and i. Let $l' < l$. Both $l > l'$ and $i > l'$, so the positions (l, l') and (i, l') lie below the diagonal in column l'; every position in a column before l has already been treated, so the entries of B at those two positions are zero. The new entries in those two rows are combinations of the old two by definition 5.4.8, hence zero, so column l' keeps all its zeros. A treated position (i', l) in the current column has $l < i' < i$, so row i' is neither l nor i and is left alone. Hence after treating (i, l) every treated position carries a zero, and after all N positions the matrix R has $R_{il} = 0$ for all $i > l$, that is, R is upper triangular.
Step 4: the diagonal. Fix $l < m$, so that column l contains at least the position (m, l), and consider the last position treated in that column, which is (m, l). Step 2 leaves the entry at (l, l) equal to the nonnegative real number r produced there, or equal to 0 in the case $a = b = 0$. Every later rotation is assigned to a position (i', l') with $l' > l$ and therefore acts on the coordinates l' and i', both greater than l; row l is untouched by all of them. So R_{ll} is that nonnegative real number. The one diagonal entry this argument does not reach is R_{mm}, which occurs only when $m = n$, since column m then has no position below the diagonal.
Step 5: assemble the factorization. From $G_N \cdots G_1 A = R$, multiplying on the left successively by $G_N^{-1}, \dots, G_1^{-1}$ and using proposition 1.4.3 gives $A = G_1^{-1} \cdots G_N^{-1} R = QR$. Each G_t^{-1} is a Givens rotation, as computed before the theorem, so each preserves the inner product; applying this N times to $Qz = G_1^{-1}(\cdots(G_N^{-1} z))$ gives $\langle Qz, Qz' \rangle = \langle z, z' \rangle$ for all z, z'. Taking $z = e_p$ and $z' = e_q$ shows that the columns $Qe_1, \dots, Qe_m$ of Q are orthonormal, so $A = QR$ is a full QR factorization, as we sought to show.

The two counts are of different orders. Setting $m = n$ in $N = mn - \frac{1}{2}n(n+1)$ leaves $\frac{1}{2}n(n-1)$ rotations for a square matrix of size n, against the at most n reflections of theorem 5.4.7; the first count grows quadratically in n and the second linearly, so from $n = 4$ onward the rotations are the more numerous. What the rotations gain is locality. The rotation assigned to (i, l) changes only rows l and i, and when the two entries it reads are both zero it is the identity by Step 2, so on a matrix whose entries below the diagonal are already mostly zero most of the N rotations do nothing.

Example 5.4.10: Two Triangularizations of a 3×2 Matrix
Let
$$A = \begin{pmatrix} 1 & 6 \\ 2 & 1 \\ 2 & 2 \end{pmatrix} \in M_{3\times 2}(\mathbb{R})$$
with columns $a_1 = (1, 2, 2)^\top$ and $a_2 = (6, 1, 2)^\top$.

1. *By reflections.* The first column has $\|a_1\| = \sqrt{1 + 4 + 4} = 3$. Take $\alpha = 3$ and $y = 3e_1$, so $v = a_1 - 3e_1 = (-2, 2, 2)^\top$ with $\langle v, v \rangle = 12$, and
$$H_1 = I_3 - \frac{2}{12}\begin{pmatrix} 4 & -4 & -4 \\ -4 & 4 & 4 \\ -4 & 4 & 4 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ 2 & -2 & 1 \end{pmatrix}$$

Then $H_1 a_1 = \frac{1}{3}(1+4+4,\ 2+2-4,\ 2-4+2)^\top = (3,0,0)^\top$ and $H_1 a_2 = \frac{1}{3}(6+2+4,\ 12+1-4,\ 12-2+2)^\top = (4,3,4)^\top$, so

$$H_1 A = \begin{pmatrix} 3 & 4 \\ 0 & 3 \\ 0 & 4 \end{pmatrix}$$

For the second column, $x = (0,3,4)^\top$ has $\|x\| = 5$; take $\alpha = 5$ and $y = 5e_2$, so $v = (0,-2,4)^\top$ with $\langle v, v\rangle = 20$, and

$$H_2 = I_3 - \frac{2}{20}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 4 & -8 \\ 0 & -8 & 16 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{5} & \frac{4}{5} \\ 0 & \frac{4}{5} & -\frac{3}{5} \end{pmatrix}$$

Since H_2 fixes e_1 it leaves the first column of $H_1 A$ unchanged, and it sends $(4,3,4)^\top$ to $(4,\ \frac{9}{5}+\frac{16}{5},\ \frac{12}{5}-\frac{12}{5})^\top = (4,5,0)^\top$. Hence

$$R = H_2 H_1 A = \begin{pmatrix} 3 & 4 \\ 0 & 5 \\ 0 & 0 \end{pmatrix}, \qquad Q = H_1 H_2 = \frac{1}{15}\begin{pmatrix} 5 & 14 & 2 \\ 10 & -5 & 10 \\ 10 & -2 & -11 \end{pmatrix}$$

the product $H_1 H_2$ being computed entry by entry, for instance $(H_1H_2)_{12} = \frac{1}{3}\cdot 0 + \frac{2}{3}\cdot\frac{3}{5} + \frac{2}{3}\cdot\frac{4}{5} = \frac{14}{15}$. Each column of Q has inner product $225/225 = 1$ with itself, and the three pairwise inner products are $(70-50-20)/225$, $(10+100-110)/225$ and $(28-50+22)/225$, all zero.

2. *By rotations.* Three positions lie below the diagonal. For $(2,1)$, the pair is $(a,b) = (1,2)$ with $r = \sqrt{5}$, so $c = 1/\sqrt{5}$ and $s = 2/\sqrt{5}$; the new rows 1 and 2 are $\frac{1}{\sqrt{5}}\big((1,6) + 2(2,1)\big) = \frac{1}{\sqrt{5}}(5,8)$ and $\frac{1}{\sqrt{5}}\big(-(2,12) + (2,1)\big) = \frac{1}{\sqrt{5}}(0,-11)$, giving

$$G_1 A = \begin{pmatrix} \sqrt{5} & 8/\sqrt{5} \\ 0 & -11/\sqrt{5} \\ 2 & 2 \end{pmatrix}$$

For $(3,1)$, the pair is $(\sqrt{5}, 2)$ with $r = \sqrt{5+4} = 3$, so $c = \sqrt{5}/3$ and $s = 2/3$; the new rows 1 and 3 are $\frac{\sqrt{5}}{3}(\sqrt{5}, 8/\sqrt{5}) + \frac{2}{3}(2,2) = (3,4)$ and $-\frac{2}{3}(\sqrt{5}, 8/\sqrt{5}) + \frac{\sqrt{5}}{3}(2,2) = (0, -2/\sqrt{5})$, giving

$$G_2 G_1 A = \begin{pmatrix} 3 & 4 \\ 0 & -11/\sqrt{5} \\ 0 & -2/\sqrt{5} \end{pmatrix}$$

For $(3,2)$, the pair is $(-11/\sqrt{5}, -2/\sqrt{5})$ with $r = \sqrt{(121+4)/5} = 5$, so $c = -11/(5\sqrt{5})$ and $s = -2/(5\sqrt{5})$; the new entries in column 2 are $\frac{121}{25} + \frac{4}{25} = 5$ in row 2 and $\frac{2}{5\sqrt{5}}\cdot\frac{-11}{\sqrt{5}} + \frac{-11}{5\sqrt{5}}\cdot\frac{-2}{\sqrt{5}} = 0$ in row 3. The result is

$$G_3 G_2 G_1 A = \begin{pmatrix} 3 & 4 \\ 0 & 5 \\ 0 & 0 \end{pmatrix}$$

the same R as in (1), reached through intermediate entries that were not rational.

3. *The reduced factorization.* Discarding the third column of Q and the third row of R gives

$$\widehat{Q} = \frac{1}{15}\begin{pmatrix} 5 & 14 \\ 10 & -5 \\ 10 & -2 \end{pmatrix}, \qquad \widehat{R} = \begin{pmatrix} 3 & 4 \\ 0 & 5 \end{pmatrix}$$

with $A = \widehat{Q}\widehat{R}$ and positive diagonal. The columns of A are linearly independent, neither being a scalar multiple of the other, so theorem 5.4.3 says this is the only such factorization of A. Gram-Schmidt confirms it: $q_1 = a_1/3 = \frac{1}{15}(5, 10, 10)^\top$, then $\langle a_2, q_1\rangle = (6+2+4)/3 = 4$ and $a_2 - 4q_1 = \frac{1}{3}(14, -5, -2)^\top$, whose norm is $\frac{1}{3}\sqrt{196 + 25 + 4} = 5$, so $q_2 = \frac{1}{15}(14, -5, -2)^\top$.

The three constructions agree because theorem 5.4.3 leaves them no choice once the diagonal is positive, and they differ in what they need and what they deliver. Gram-Schmidt requires independent columns and returns the reduced shape with a positive diagonal. Householder and Givens require nothing of A and return the full shape, with the extra columns of Q completing an orthonormal basis of k^m; their diagonals need a rescaling to be made positive. In example 5.4.4 the difference is visible: the running matrix has dependent columns, so only the second and third constructions apply to it. Their output $A = QR$ has $Q \in M_4(\mathbb{R})$ with orthonormal columns and $R \in M_{4\times 3}(\mathbb{R})$ upper triangular, and the diagonal of R must contain a zero. Indeed $A(1, 2, -1)^\top = a_1 + 2a_2 - a_3 = 0$, and Q preserves norms, so $\|R(1, 2, -1)^\top\| = \|QR(1, 2, -1)^\top\| = 0$, whence $R(1, 2, -1)^\top = 0$ by proposition 5.1.6(1). Row 3 of R is $(0, 0, r_{33})$, so reading that relation in row 3 gives $-r_{33} = 0$.

Exercise 5.4.1

1. Compute the reduced QR factorization with positive diagonal of

$$B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 0 \end{pmatrix} \in M_{4\times 2}(\mathbb{R})$$

by the construction of theorem 5.4.2, and verify $Q^*Q = I_2$ and $QR = B$.

2. Let $x = (2, -1, 2)^\top \in \mathbb{R}^3$. There are exactly two vectors of the form αe_1 with $\|\alpha e_1\| = \|x\|$. Write down both, and compute for each the Householder reflection of proposition 5.4.6(4) carrying x to it. Check in both cases that the matrix is its own inverse.

3. Use one Givens rotation to triangularize $C = \left(\begin{smallmatrix} 3 & 5 \\ 4 & 0 \end{smallmatrix}\right) \in M_2(\mathbb{R})$, and write the resulting full QR factorization $C = QR$. The diagonal of R is not positive; produce from it the factorization with positive diagonal, as described after theorem 5.4.3.

4. Let $A \in M_n(\mathbb{R})$ have linearly independent columns and let $A = QR$ be its reduced QR factorization with positive diagonal, so that $Q \in M_n(\mathbb{R})$. Show that $\det Q = \pm 1$, and deduce that $|\det A| = r_{11}r_{22}\cdots r_{nn}$.

5. Let $A \in M_{m\times n}(k)$ have orthonormal columns. Show that its reduced QR factorization with positive diagonal is $Q = A$ and $R = I_n$.

6. Let $A \in M_{m\times n}(k)$ have columns $a_1, \ldots, a_n$ and let $A = QR$ be a reduced QR factorization, with columns $q_1, \ldots, q_n$ of Q, in which every diagonal entry of R is nonzero. Show that $\operatorname{span}\{a_1, \ldots, a_j\} = \operatorname{span}\{q_1, \ldots, q_j\}$ for every $j \le n$, without assuming that Q and R came from theorem 5.4.2.

5.5 Orthogonal Complements, Riesz Representation, and the Adjoint

Orthogonality has so far been a relation between two vectors. This section makes it a relation between a vector and a subspace, and then a relation between two subspaces. Three results come out of that. The first splits V into a subspace W and the set of vectors orthogonal to all of W. The second identifies every scalar-valued linear map on V with a single vector of V. The third attaches to each linear map T a second linear map T^* running the other way, which is the object chapter 6 and chapter 7 are stated in terms of.

One consequence of proposition 5.1.6(2) is used throughout. Conjugate symmetry converts a statement about the first argument of the inner product into a statement about the second: by proposition 5.1.6(2) the inner product is additive in its second argument and pulls a scalar out of that argument conjugated.

Take the plane $W = \operatorname{span}\{(1,2,0)^\top, (0,1,1)^\top\}$ in $\mathbb{R}^3$ with the standard inner product, and ask which vectors are orthogonal to every vector of W. A vector orthogonal to the two spanning vectors is orthogonal to every combination of them, by the display above, so the question is the pair of equations

$$\langle x, (1,2,0)^\top\rangle = x_1 + 2x_2 = 0, \qquad \langle x, (0,1,1)^\top\rangle = x_2 + x_3 = 0$$

Setting $x_2 = t$ gives $x_1 = -2t$ and $x_3 = -t$, so the vectors orthogonal to all of W are the multiples of $(-2,1,-1)^\top$: a line, meeting the plane W only at the origin, and of the dimension $3-2$ that leaves. The construction works for any subset of any inner product space.

Definition 5.5.1: Orthogonal Complement

Let V be an inner product space over k and let $S \subseteq V$ be a nonempty subset. The *orthogonal complement* of S is

$$S^\perp = \{\, v \in V \,\big|\, \langle v, s\rangle = 0 \text{ for every } s \in S \,\}$$

the set of vectors orthogonal to every element of S. The symbol is read "S perp".

The computation above found $W^\perp$ for a plane W in $\mathbb{R}^3$ by checking orthogonality against a spanning list rather than against every vector of W. That shortcut is the second part of the next proposition, and the first part says that $S^\perp$ is a subspace even when S is not.

Proposition 5.5.2: The Orthogonal Complement Is a Subspace

Let V be an inner product space over k and let $S \subseteq V$ be a nonempty subset.

1. $S^\perp$ is a subspace of V.
2. $S^\perp = (\operatorname{span} S)^\perp$. So a vector orthogonal to each of $w_1, \ldots, w_r$ is orthogonal to every vector of $\operatorname{span}\{w_1, \ldots, w_r\}$, and orthogonality to a subspace can be tested on a spanning list.
3. If W is a subspace of V, then $W \cap W^\perp = \{0\}$.

***Proof*:**

1. By proposition 5.1.6(3), $\langle 0, s\rangle = 0$ for every $s \in S$, so $0 \in S^{\perp}$. Let $u, v \in S^{\perp}$ and $\lambda \in k$. For every $s \in S$,

$$\langle u + v, s\rangle = \langle u, s\rangle + \langle v, s\rangle = 0, \qquad \langle \lambda u, s\rangle = \lambda\langle u, s\rangle = 0$$

both by linearity in the first argument (definition 5.1.2). So $S^{\perp}$ contains 0 and is closed under addition and under scalar multiplication, hence is a subspace by proposition 2.1.3.

2. Every element of S lies in span$S$ (definition 2.1.5), so a vector orthogonal to every element of span$S$ is orthogonal to every element of S; that is $(\text{span}S)^{\perp} \subseteq S^{\perp}$. For the reverse inclusion let $v \in S^{\perp}$ and let $w \in \text{span}S$, say $w = \lambda_1 s_1 + \cdots + \lambda_t s_t$ with $s_1, \ldots, s_t \in S$ and $\lambda_1, \ldots, \lambda_t \in k$. The opening display, applied repeatedly, gives

$$\langle v, w\rangle = \overline{\lambda_1}\langle v, s_1\rangle + \cdots + \overline{\lambda_t}\langle v, s_t\rangle = 0$$

since each $\langle v, s_i\rangle$ vanishes. So $v \in (\text{span}S)^{\perp}$, and the two sets are equal. The second sentence is this equality with $S = \{w_1, \ldots, w_r\}$.

3. Let $v \in W \cap W^{\perp}$. Then $v \in W$, and $v \in W^{\perp}$ means v is orthogonal to every element of W, so in particular $\langle v, v\rangle = 0$. By proposition 5.1.6(1) this forces $v = 0$. Conversely 0 lies in W and in $W^{\perp}$, both being subspaces, so the intersection is exactly $\{0\}$, as we sought to show.

In the $\mathbb{R}^3$ computation the plane and the line met only at the origin and their dimensions summed to 3, which is what it takes for the two of them to account for all of $\mathbb{R}^3$. The same holds for every subspace of every finite-dimensional inner product space, and the proof is the orthonormal basis of corollary 5.3.11 together with the coordinate formula of theorem 5.3.5.

Theorem 5.5.3: Orthogonal Decomposition

Let V be a finite-dimensional inner product space over k and let $W \subseteq V$ be a subspace. Then $V = W \oplus W^{\perp}$: every $v \in V$ can be written as $v = w + w'$ with $w \in W$ and $w' \in W^{\perp}$, and the pair (w, w') is determined by v.

***Proof*:**
Step 1: a decomposition exists. The conditions of definition 5.1.2 are conditions on pairs of vectors, and every pair of vectors of W is a pair of vectors of V, so the restriction of $\langle -, -\rangle$ to W makes W an inner product space; it is finite-dimensional, with $\dim W \leq \dim V$, by corollary 2.2.12. Corollary 5.3.11 therefore gives an orthonormal basis $q_1, \ldots, q_r$ of W, where $r = \dim W$. Given $v \in V$, put

$$w = \sum_{i=1}^{r} \langle v, q_i\rangle q_i, \qquad w' = v - w$$

with the convention that an empty sum is 0, so that $w = 0$ and $w' = v$ when $W = \{0\}$ and $r = 0$. By theorem 5.3.5(2) the vector w lies in W and $w' = v - w$ is orthogonal to every element of W, so $w' \in W^{\perp}$. Thus $v = w + w'$ with $w \in W$ and $w' \in W^{\perp}$.

Step 2: the two summands are determined by v. Suppose $v = w_1 + w_1' = w_2 + w_2'$ with $w_1, w_2 \in W$ and $w_1', w_2' \in W^{\perp}$. Rearranging gives $w_1 - w_2 = w_2' - w_1'$. The left side lies in W, a subspace, and the right side lies in $W^{\perp}$, a subspace by proposition 5.5.2(1); the two sides are the same vector, so that vector lies in $W \cap W^{\perp}$, which is $\{0\}$ by proposition 5.5.2(3). Hence $w_1 = w_2$ and $w_1' = w_2'$.

Every vector of V is therefore a sum of one vector of W and one of $W^{\perp}$, and $W \cap W^{\perp} = \{0\}$, which is what definition 2.1.11 asks of the notation $V = W \oplus W^{\perp}$; Step 2 is the uniqueness recorded in proposition 2.1.12, completing the proof.

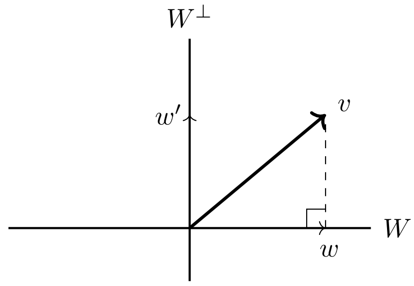


Figure 5.5: The decomposition $v = w + w'$ of theorem 5.5.3

The first summand is $\sum_i \langle v, q_i \rangle q_i$, which is the orthogonal projection of v onto W (definition 5.3.8). Step 1 computes it from one orthonormal basis of W and Step 2 shows that the answer does not depend on which orthonormal basis was used, since w is the only vector of W whose difference from v lies in $W^{\perp}$. Two corollaries follow, one about dimensions and one saying that taking the complement twice returns the subspace one started with.

Corollary 5.5.4: Dimension of an Orthogonal Complement

Let V be a finite-dimensional inner product space over k and let $W \subseteq V$ be a subspace. Then $\dim W + \dim W^{\perp} = \dim V$.

***Proof*:**
Both W and $W^{\perp}$ are subspaces of V, the second by proposition 5.5.2(1), so both are finite-dimensional by corollary 2.2.12 and both carry the restricted inner product. Corollary 5.3.11 therefore gives an orthonormal basis $u_1, \ldots, u_r$ of W and an orthonormal basis $z_1, \ldots, z_s$ of $W^{\perp}$, with $r = \dim W$ and $s = \dim W^{\perp}$ by definition 2.2.10; either list may be empty. The claim is that the concatenated list $u_1, \ldots, u_r, z_1, \ldots, z_s$ is a basis of V.

It spans V: given $v \in V$, theorem 5.5.3 writes $v = w + w'$ with $w \in W$ and $w' \in W^{\perp}$, and w is a combination of $u_1, \ldots, u_r$ while w' is a combination of $z_1, \ldots, z_s$, so v is a combination of the concatenated list.

It is independent: suppose $a_1 u_1 + \cdots + a_r u_r + b_1 z_1 + \cdots + b_s z_s = 0$ with all $a_i, b_j \in k$. Then

$$a_1 u_1 + \cdots + a_r u_r = -(b_1 z_1 + \cdots + b_s z_s)$$

The left side lies in W and the right side lies in $W^{\perp}$, so their common value lies in $W \cap W^{\perp} = \{0\}$ by proposition 5.5.2(3). Both sides therefore vanish, and each of the two lists is independent by proposition 5.3.4, so every a_i and every b_j is 0.

So V has a basis with $r+s$ entries, and $\dim V = r+s$ by theorem 2.2.8 and definition 2.2.10, as we sought to show.

Corollary 5.5.5: The Double Complement

Let V be a finite-dimensional inner product space over k and let $W \subseteq V$ be a subspace. Then $(W^\perp)^\perp = W$.

***Proof*:**
For the inclusion $W \subseteq (W^\perp)^\perp$, let $w \in W$ and let $v \in W^\perp$. Then $\langle v, w\rangle = 0$ by definition 5.5.1, so $\langle w, v\rangle = \overline{\langle v, w\rangle} = 0$ by proposition 5.1.6(2). As v was an arbitrary element of $W^\perp$, this says $w \in (W^\perp)^\perp$.

For the reverse inclusion, let $v \in (W^\perp)^\perp$ and use theorem 5.5.3 to write $v = w + w'$ with $w \in W$ and $w' \in W^\perp$. The first inclusion puts w in $(W^\perp)^\perp$, which is a subspace by proposition 5.5.2(1), so $w' = v - w$ lies in $(W^\perp)^\perp$ as well. Thus w' lies in $W^\perp \cap (W^\perp)^\perp$, which is $\{0\}$ by proposition 5.5.2(3) applied to the subspace $W^\perp$. Hence $w' = 0$ and $v = w \in W$, completing the proof.

For a nonempty subset S of a finite-dimensional inner product space the two complements recover the span rather than S itself: combining proposition 5.5.2(2) with corollary 5.5.5, applied to the subspace $\operatorname{span} S$, gives

$$(S^\perp)^\perp = \left((\operatorname{span} S)^\perp\right)^\perp = \operatorname{span} S$$

So a finite list of vectors and the orthogonality conditions it imposes determine each other, and each can be computed from the other by elimination.

The second construction of this section identifies scalar-valued linear maps with vectors. On $\mathbb{R}^3$ with the standard inner product, the map $\varphi(x) = 2x_1 - x_2 + 5x_3$ is $\varphi(x) = \langle x, a\rangle$ for $a = (2, -1, 5)^\top$, since $\langle x, a\rangle = 2x_1 - x_2 + 5x_3$ with real entries unaffected by conjugation. Over $\mathbb{C}$ the coefficients come conjugated: on $\mathbb{C}^2$ the map $\varphi(z) = 2z_1 + iz_2$ is $\varphi(z) = \langle z, a\rangle$ for $a = (2, -i)^\top$, because $\langle z, a\rangle = z_1\overline{2} + z_2\overline{(-i)} = 2z_1 + iz_2$. In both cases the vector is unique, and both cases are instances of one theorem.

Theorem 5.5.6: Riesz Representation

Let V be a finite-dimensional inner product space over k and let $\varphi\colon V \to k$ be a linear map, where k is regarded as a vector space over itself. Then there is exactly one $a \in V$ with

$$\varphi(v) = \langle v, a\rangle \qquad \text{for every } v \in V$$

***Proof*:**
Step 1: existence. Let $q_1, \ldots, q_n$ be an orthonormal basis of V, given by corollary 5.3.11, and put

$$a = \sum_{j=1}^{n} \overline{\varphi(q_j)}\, q_j$$

Let $v \in V$. By theorem 5.3.5, $v = \sum_{i=1}^{n} \langle v, q_i \rangle q_i$, so linearity of φ (definition 2.4.2) gives

$$\varphi(v) = \sum_{i=1}^{n} \langle v, q_i \rangle \varphi(q_i)$$

For the other side, the opening display of this section pulls each scalar out of the second argument conjugated:

$$\langle v, a \rangle = \left\langle v, \sum_{j=1}^{n} \overline{\varphi(q_j)}\, q_j \right\rangle = \sum_{j=1}^{n} \overline{\overline{\varphi(q_j)}} \langle v, q_j \rangle = \sum_{j=1}^{n} \varphi(q_j) \langle v, q_j \rangle$$

The two sums have the same terms, so $\varphi(v) = \langle v, a \rangle$.

Step 2: uniqueness. Suppose $\langle v, a \rangle = \langle v, b \rangle$ for every $v \in V$. The opening display, with $\lambda = -1$, gives $\langle v, a - b \rangle = \langle v, a \rangle - \langle v, b \rangle = 0$ for every v. Taking $v = a - b$ gives $\langle a - b, a - b \rangle = 0$, so $a - b = 0$ by proposition 5.1.6(1). Hence $a = b$, completing the proof.

The theorem says that $v \mapsto \langle v, a \rangle$ exhausts the scalar-valued linear maps on V as a runs over V, and that distinct a give distinct maps. Write φ_a for the map attached to a. The assignment $a \mapsto \varphi_a$ satisfies $\varphi_{a+b} = \varphi_a + \varphi_b$ and $\varphi_{\lambda a} = \overline{\lambda} \varphi_a$, both by the opening display read one term at a time; over $\mathbb{R}$ the conjugation is trivial and the assignment is a linear bijection between V and the scalar-valued linear maps on V, while over $\mathbb{C}$ it is a bijection that conjugates scalars. Finiteness of the dimension is what the proof used, through the existence of an orthonormal basis; nothing here needs a limit or a convergence argument.

Riesz representation is what makes the third construction possible. Start with a matrix. Take

$$B = \begin{pmatrix} 1 & i & 0 \\ 2 & 0 & 1 - i \end{pmatrix} \in M_{2\times 3}(\mathbb{C}), \qquad x = (1, 0, i)^{\top} \in \mathbb{C}^3, \qquad y = (1, 1)^{\top} \in \mathbb{C}^2$$

Then $Bx = (1,\ 2 + (1 - i)i)^{\top} = (1, 3 + i)^{\top}$, so $\langle Bx, y \rangle = 1 \cdot \overline{1} + (3 + i)\overline{1} = 4 + i$. Transposing B and conjugating its entries produces $B^* = \left(\begin{smallmatrix} 1 & 2 \\ -i & 0 \\ 0 & 1+i \end{smallmatrix}\right)$, and $B^* y = (3, -i, 1 + i)^{\top}$, so

$$\langle x, B^* y \rangle = 1 \cdot \overline{3} + 0 \cdot \overline{(-i)} + i \cdot \overline{(1 + i)} = 3 + i(1 - i) = 4 + i$$

The two agree. Moving B from the first argument of the inner product to the second replaced it by its conjugate transpose. The next theorem shows that the move is available for a linear map between any two finite-dimensional inner product spaces, with no matrix chosen.

Theorem 5.5.7: Existence and Uniqueness of the Adjoint

Let V and W be finite-dimensional inner product spaces over k, with inner products written $\langle -, - \rangle_V$ and $\langle -, - \rangle_W$, and let $T \colon V \to W$ be linear. There is exactly one map $T^* \colon W \to V$ satisfying

$$\langle Tv, w \rangle_W = \langle v, T^* w \rangle_V \qquad \text{for all } v \in V \text{ and } w \in W$$

and that map is linear.

***Proof*:**
Step 1: construction. Fix $w \in W$ and define $\varphi_w : V \to k$ by $\varphi_w(v) = \langle Tv, w\rangle_W$. It is linear: for $v_1, v_2 \in V$ and $\lambda \in k$,

$$\varphi_w(v_1 + \lambda v_2) = \langle Tv_1 + \lambda Tv_2, w\rangle_W = \langle Tv_1, w\rangle_W + \lambda\langle Tv_2, w\rangle_W = \varphi_w(v_1) + \lambda\varphi_w(v_2)$$

the first equality by linearity of T (definition 2.4.2) and the second by linearity of the inner product in its first argument (definition 5.1.2). Theorem 5.5.6 therefore gives exactly one $a_w \in V$ with $\varphi_w(v) = \langle v, a_w\rangle_V$ for every v. Setting $T^*w = a_w$ defines a map $W \to V$, and $\langle Tv, w\rangle_W = \langle v, T^*w\rangle_V$ holds by construction.

Step 2: linearity of T^*. Let $w_1, w_2 \in W$ and $\lambda \in k$. For every $v \in V$,

$$\langle v, T^*(w_1 + \lambda w_2)\rangle_V = \langle Tv, w_1 + \lambda w_2\rangle_W = \langle Tv, w_1\rangle_W + \overline{\lambda}\langle Tv, w_2\rangle_W = \langle v, T^*w_1\rangle_V + \overline{\lambda}\langle v, T^*w_2\rangle_V$$

where the second equality is the opening display of this section and the third is the identity of Step 1 applied to w_1 and to w_2. That same opening display, read from right to left, turns the last expression into $\langle v, T^*w_1 + \lambda T^*w_2\rangle_V$. So the two vectors $T^*(w_1 + \lambda w_2)$ and $T^*w_1 + \lambda T^*w_2$ represent the same scalar-valued linear map on V, and the uniqueness clause of theorem 5.5.6 makes them equal. Hence T^* is linear (definition 2.4.2).

Step 3: uniqueness of T^*. Suppose $S : W \to V$ also satisfies $\langle Tv, w\rangle_W = \langle v, Sw\rangle_V$ for all v and w. Fix w. Then $\langle v, T^*w\rangle_V = \langle v, Sw\rangle_V$ for every $v \in V$, so $T^*w = Sw$ by the uniqueness clause of theorem 5.5.6. Since w was arbitrary, $S = T^*$, completing the proof.

Definition 5.5.8: Adjoint

Let V and W be finite-dimensional inner product spaces over k and let $T : V \to W$ be linear. The *adjoint* of T is the unique linear map $T^* : W \to V$ of theorem 5.5.7, characterized by $\langle Tv, w\rangle_W = \langle v, T^*w\rangle_V$ for all $v \in V$ and $w \in W$.

The adjoint is defined by an identity rather than by a formula, and every property of it is proved the same way: compute $\langle v, ?w\rangle$ two ways and appeal to the uniqueness clause of theorem 5.5.6. The next proposition collects the properties later chapters use. Part (4) is the one that does geometric work, since it converts a statement about the kernel of T^* into a statement about the image of T.

Proposition 5.5.9: Properties of the Adjoint

Let U, V and W be finite-dimensional inner product spaces over k, let $S, T : V \to W$ and $R : W \to U$ be linear, and let $\lambda \in k$. Then:

1. $(S + T)^* = S^* + T^*$ and $(\lambda T)^* = \overline{\lambda}T^*$;
2. $(R \circ T)^* = T^* \circ R^*$;
3. $(T^*)^* = T$;
4. $\ker T^* = (\mathrm{im} T)^\perp$ and $\mathrm{im} T^* = (\ker T)^\perp$.

***Proof*:**

1. Let $w \in W$. For every $v \in V$,

$$\langle v, (S+T)^* w\rangle_V = \langle (S+T)v, w\rangle_W = \langle Sv, w\rangle_W + \langle Tv, w\rangle_W = \langle v, S^* w\rangle_V + \langle v, T^* w\rangle_V = \langle v, S^* w + T^* w\rangle_V$$

using definition 5.5.8 at the first equality, linearity in the first argument at the second, definition 5.5.8 again at the third, and the opening display at the fourth. Uniqueness in theorem 5.5.6 gives $(S+T)^* w = S^* w + T^* w$. Similarly

$$\langle v, (\lambda T)^* w\rangle_V = \langle \lambda T v, w\rangle_W = \lambda \langle Tv, w\rangle_W = \lambda \langle v, T^* w\rangle_V = \langle v, \overline{\lambda} T^* w\rangle_V$$

the last equality because pulling $\overline{\lambda}$ out of the second argument conjugates it back to λ. Uniqueness gives $(\lambda T)^* w = \overline{\lambda} T^* w$. As w was arbitrary, both identities hold as identities of maps.

2. Let $u \in U$. For every $v \in V$,

$$\langle v, (R \circ T)^* u\rangle_V = \langle R(Tv), u\rangle_U = \langle Tv, R^* u\rangle_W = \langle v, T^*(R^* u)\rangle_V$$

by definition 5.5.8 applied to $R \circ T$, then to R, then to T. Uniqueness gives $(R \circ T)^* u = T^*(R^* u) = (T^* \circ R^*)(u)$.

3. Let $v \in V$. For every $w \in W$,

$$\langle w, (T^*)^* v\rangle_W = \langle T^* w, v\rangle_V = \overline{\langle v, T^* w\rangle_V} = \overline{\langle Tv, w\rangle_W} = \langle w, Tv\rangle_W$$

using definition 5.5.8 for T^* at the first equality, proposition 5.1.6(2) at the second and fourth, and definition 5.5.8 for T at the third. Uniqueness gives $(T^*)^* v = Tv$.

4. Let $w \in \ker T^*$, so $T^* w = 0$. Every element of $\mathrm{im} T$ is Tv for some $v \in V$ (definition 2.4.5), and

$$\langle Tv, w\rangle_W = \langle v, T^* w\rangle_V = \langle v, 0\rangle_V = 0$$

by definition 5.5.8 and proposition 5.1.6(3). So $w \in (\mathrm{im} T)^\perp$. Conversely let $w \in (\mathrm{im} T)^\perp$. Then $\langle v, T^* w\rangle_V = \langle Tv, w\rangle_W = 0$ for every $v \in V$, and taking $v = T^* w$ gives $\langle T^* w, T^* w\rangle_V = 0$, so $T^* w = 0$ by proposition 5.1.6(1) and $w \in \ker T^*$. The two sets are therefore equal.

For the second identity, apply the first to the linear map $T^* \colon W \to V$ in place of T, which gives $\ker (T^*)^* = (\mathrm{im} T^*)^\perp$; by (3) the left side is $\ker T$. Both $\ker T$ and $\mathrm{im} T^*$ are subspaces of V (proposition 2.4.6), so corollary 5.5.5 applies to $\mathrm{im} T^*$ and gives

$$(\ker T)^\perp = \left((\mathrm{im} T^*)^\perp\right)^\perp = \mathrm{im} T^*$$

as we sought to show.

Coordinates turn the adjoint back into the conjugate transpose, provided the bases are orthonormal. The matrix operation deserves its own name and its own arithmetic, since later chapters manipulate it directly.

Definition 5.5.10: Entrywise Conjugate of a Matrix

Let $A = (a_{ij}) \in M_{m\times n}(k)$. The *conjugate* of A is the matrix $\overline{A} \in M_{m\times n}(k)$ whose entry in row i and column j is $\overline{a_{ij}}$. Over $\mathbb{R}$ the conjugation does nothing and $\overline{A} = A$. A matrix with $\overline{A} = A$ is said to have *real entries*, and for $A \in M_{m\times n}(\mathbb{C})$ this holds exactly when every a_{ij} is real.

Proposition 5.5.11: Arithmetic of the Entrywise Conjugate

Let $A, B \in M_{m\times n}(k)$, let $C \in M_{n\times p}(k)$ and let $\lambda \in k$.

1. $\overline{\overline{A}} = A$.
2. $\overline{A + B} = \overline{A} + \overline{B}$ and $\overline{\lambda A} = \overline{\lambda}\,\overline{A}$.
3. $\overline{AC} = \overline{A}\,\overline{C}$: unlike the transpose, entrywise conjugation does *not* reverse the order of a product.
4. If $A \in M_n(k)$ is invertible, so is $\overline{A}$, and $\overline{A}^{-1} = \overline{A^{-1}}$.

***Proof*:**

1. The entry in row i and column j is $\overline{\overline{a_{ij}}} = a_{ij}$, since conjugation of scalars is an involution.
2. Conjugation of scalars carries sums to sums and products to products, so the entry $\overline{a_{ij} + b_{ij}}$ equals $\overline{a_{ij}} + \overline{b_{ij}}$, and $\overline{\lambda a_{ij}} = \overline{\lambda}\,\overline{a_{ij}}$.
3. By definition 1.4.1 the entry of $\overline{AC}$ in row i and column j is
$$\overline{\sum_s a_{is} c_{sj}} = \sum_s \overline{a_{is}}\,\overline{c_{sj}}$$
again because conjugation carries sums to sums and products to products, and the right side is the corresponding entry of $\overline{A}\,\overline{C}$.
4. Conjugating $A^{-1}A = I_n$ and $AA^{-1} = I_n$ with (3), and using $\overline{I_n} = I_n$, gives $\overline{A^{-1}}\,\overline{A} = I_n$ and $\overline{A}\,\overline{A^{-1}} = I_n$; those are the two conditions of definition 1.4.4.

The conjugate transpose factors through this operation: $A^* = \overline{A}^\top$ by definition 2.3.4, so proposition 5.5.13 below could be read off from proposition 2.3.5 and the present proposition. The order-reversal in $(AC)^* = C^*A^*$ comes entirely from the transpose.

Definition 5.5.12: Conjugate Transpose

Let $A = (a_{ij}) \in M_{m\times n}(k)$. The *conjugate transpose* of A is the matrix $A^* \in M_{n\times m}(k)$ whose entry in row j and column i is $\overline{a_{ij}}$. Over $\mathbb{R}$ the conjugation does nothing and $A^* = A^\top$ in the sense of definition 2.3.4.

Proposition 5.5.13: Arithmetic of the Conjugate Transpose

Let $A, B \in M_{m\times n}(k)$, let $C \in M_{n\times p}(k)$ and let $\lambda \in k$. Then

1. $(A^*)^* = A$
2. $(A+B)^* = A^* + B^*$ and $(\lambda A)^* = \overline{\lambda}\, A^*$
3. $(AC)^* = C^* A^*$
4. A is invertible if and only if A^* is, and then $(A^*)^{-1} = (A^{-1})^*$

***Proof*:**
Throughout, write a_{ij}, b_{ij}, c_{ij} for the entries.

1. The entry of $(A^*)^*$ in row i and column j is the conjugate of the entry of A^* in row j and column i, which is $\overline{\overline{a_{ij}}} = a_{ij}$.
2. Both are entrywise: the entry of $(A+B)^*$ in row j, column i is $\overline{a_{ij} + b_{ij}} = \overline{a_{ij}} + \overline{b_{ij}}$, and that of $(\lambda A)^*$ is $\overline{\lambda a_{ij}} = \overline{\lambda}\,\overline{a_{ij}}$.
3. The entry of $(AC)^*$ in row j and column i is $\overline{(AC)_{ij}} = \overline{\sum_l a_{il} c_{lj}} = \sum_l \overline{c_{lj}}\,\overline{a_{il}}$ by definition 1.4.1, and the entry of C^*A^* in row j and column i is $\sum_l (C^*)_{jl}(A^*)_{li} = \sum_l \overline{c_{lj}}\,\overline{a_{il}}$. The two agree.
4. If $AB = BA = I_n$ then applying (3) gives $B^*A^* = A^*B^* = I_n^* = I_n$, so A^* is invertible with inverse $B^* = (A^{-1})^*$ by proposition 1.4.5. The converse follows by applying the same argument to A^* and using (1), completing the proof.

Theorem 5.5.14: The Matrix of the Adjoint in an Orthonormal Basis

Let V and W be finite-dimensional inner product spaces over k, let $\mathcal{B} = (q_1, \ldots, q_n)$ be an orthonormal basis of V, let $\mathcal{C} = (p_1, \ldots, p_m)$ be an orthonormal basis of W, and let $T\colon V \to W$ be linear.

1. $[T^*]_{\mathcal{BC}} = \big([T]_{\mathcal{CB}}\big)^*$. When $W = V$ and $\mathcal{C} = \mathcal{B}$ this reads $[T^*]_{\mathcal{B}} = \big([T]_{\mathcal{B}}\big)^*$.
2. Let $A \in M_{m\times n}(k)$ and give k^n and k^m the standard inner product. The map $T_A\colon k^n \to k^m$ sending $x \mapsto Ax$ is linear, and $(T_A)^* = T_{A^*}$; that is, $\langle Ax, y\rangle = \langle x, A^*y\rangle$ for all $x \in k^n$ and $y \in k^m$.

Both bases in (1) must be orthonormal. Example 5.5.15(2) exhibits an operator on $\mathbb{R}^2$ and a basis $\mathcal{B}$ of $\mathbb{R}^2$ for which $[T^*]_{\mathcal{B}}$ is not the transpose of $[T]_{\mathcal{B}}$.

***Proof*:**

1. Write $M = [T]_{\mathcal{CB}}$ and $N = [T^*]_{\mathcal{BC}}$. By theorem 2.4.8 the j-th column of M is $[Tq_j]_{\mathcal{C}}$, and by theorem 5.3.5 the i-th coordinate of a vector in the orthonormal basis $\mathcal{C}$ is its inner product with p_i. So $m_{ij} = \langle Tq_j, p_i\rangle_W$. The same two results applied to $T^*\colon W \to V$ and

the orthonormal basis $\mathcal{B}$ give $n_{ji} = \langle T^* p_i, q_j \rangle_V$. Now

$$n_{ji} = \langle T^* p_i, q_j \rangle_V = \overline{\langle q_j, T^* p_i \rangle_V} = \overline{\langle T q_j, p_i \rangle_W} = \overline{m_{ij}}$$

the second equality by proposition 5.1.6(2) and the third by the identity of definition 5.5.8 with $v = q_j$ and $w = p_i$. The entry of N in row j and column i is therefore $\overline{m_{ij}}$, which is exactly the entry of M^* in row j and column i. Hence $N = M^*$. The square case is this statement with $W = V$ and $\mathcal{C} = \mathcal{B}$, where the two subscripts coincide and the notation of theorem 2.4.8 abbreviates to a single one.

2. The i-th entry of $A(x + \lambda y)$ is $\sum_j a_{ij}(x_j + \lambda y_j) = \sum_j a_{ij} x_j + \lambda \sum_j a_{ij} y_j$ by definition 1.4.1, which is the i-th entry of $Ax + \lambda Ay$; so T_A is linear (definition 2.4.2). The standard basis $e_1, \ldots, e_n$ of k^n is orthonormal for the standard inner product, since $\langle e_i, e_j \rangle = \sum_l (e_i)_l \overline{(e_j)_l}$ has every term 0 unless $i = j = l$, giving 1 when $i = j$ and 0 otherwise; the same computation applies in k^m. The coordinate column of a vector of k^m in the standard basis is that vector itself (definition 2.2.4), and Ae_j is the j-th column of A by definition 1.4.1, so the matrix of T_A with respect to these two standard bases has j-th column equal to the j-th column of A; that is, the matrix of T_A is A. Part (1) then says that the matrix of $(T_A)^*$ with respect to the same bases is A^*, and by the uniqueness clause of theorem 2.4.8 a linear map is determined by its matrix, so $(T_A)^* = T_{A^*}$. Writing out definition 5.5.8 for this pair of maps gives $\langle Ax, y \rangle = \langle x, A^* y \rangle$, completing the proof.

Example 5.5.15: Adjoints Computed, With and Without an Orthonormal Basis

1. Let $M = \left(\begin{smallmatrix} 2 & 1-i \\ 0 & i \end{smallmatrix}\right) \in M_2(\mathbb{C})$ and give $\mathbb{C}^2$ the standard inner product. By theorem 5.5.14(2) the adjoint of $x \mapsto Mx$ is $y \mapsto M^* y$ with $M^* = \left(\begin{smallmatrix} 2 & 0 \\ 1+i & -i \end{smallmatrix}\right)$, obtained by transposing and conjugating each entry. Testing the identity on $x = (1, i)^\top$ and $y = (i, 1)^\top$:

$$Mx = \big(2 + (1-i)i,\ i \cdot i\big)^\top = (3+i,\ -1)^\top, \qquad \langle Mx, y \rangle = (3+i)\bar{i} + (-1)\bar{1} = -3i$$

since $(3+i)(-i) = -3i + 1$ and $1 - 1 = 0$ absorbs the second term. On the other side,

$$M^* y = \big(2i,\ (1+i)i - i\big)^\top = (2i,\ -1)^\top, \qquad \langle x, M^* y \rangle = 1 \cdot \overline{2i} + i \cdot \overline{(-1)} = -2i - i = -3i$$

The two agree, as definition 5.5.8 requires.

2. Let $V = \mathbb{R}^2$ with the standard inner product and let $\mathcal{B} = (b_1, b_2)$ with $b_1 = (1,0)^\top$ and $b_2 = (1,1)^\top$. This is a basis, since neither vector is a multiple of the other, but it is not orthonormal: $\langle b_1, b_2 \rangle = 1 \neq 0$. Define $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ by $Tb_1 = b_2$ and $Tb_2 = 0$, so that $[T]_{\mathcal{B}} = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$.

   To compute T^*, first find the matrix of T in the standard basis. Here $Te_1 = Tb_1 = (1,1)^\top$ and $Te_2 = T(b_2 - b_1) = -b_2 = (-1,-1)^\top$, so that matrix is $A = \left(\begin{smallmatrix} 1 & -1 \\ 1 & -1 \end{smallmatrix}\right)$. The standard basis is orthonormal, so theorem 5.5.14(2) gives $T^* y = A^\top y$ with $A^\top = \left(\begin{smallmatrix} 1 & 1 \\ -1 & -1 \end{smallmatrix}\right)$.

   Now express T^* in the basis $\mathcal{B}$. First $T^* b_1 = A^\top (1,0)^\top = (1,-1)^\top$, and $(1,-1)^\top = 2b_1 - b_2$ since $2(1,0)^\top - (1,1)^\top = (1,-1)^\top$; then $T^* b_2 = A^\top (1,1)^\top = (2,-2)^\top = 4b_1 - 2b_2$ by the same check. So

$$[T^*]_{\mathcal{B}} = \begin{pmatrix} 2 & 4 \\ -1 & -2 \end{pmatrix}, \qquad \big([T]_{\mathcal{B}}\big)^\top = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

The two matrices differ in every entry. So the identity $[T^*]_{\mathcal{B}} = ([T]_{\mathcal{B}})^*$ fails for this basis, and the orthonormality hypothesis in theorem 5.5.14(1) cannot be dropped. What survives is the defining identity itself, which mentions no basis: for instance with $v = b_1$ and $w = b_2$ both sides read $\langle Tb_1, b_2\rangle = \langle b_2, b_2\rangle = 2$ and $\langle b_1, T^*b_2\rangle = \langle (1,0)^\top, (2,-2)^\top\rangle = 2$.

The four subspaces attached to a matrix in section 2.3 were left with their dimensions settled and their relative position open. Corollary 2.3.12 gave $\dim \operatorname{row} A + \dim \operatorname{null} A = n$ and $\dim \operatorname{col} A + \dim \operatorname{null} A^\top = m$, and the discussion after it observed that a pair of subspaces can have complementary dimensions and still coincide. The adjoint settles the position.

Theorem 5.5.16: The Four Fundamental Subspaces Are Orthogonal in Pairs

Let $A \in M_{m\times n}(k)$ and give k^n and k^m the standard inner product. Then:

1. $\operatorname{null} A = (\operatorname{col} A^*)^\perp$ and $\operatorname{null} A^* = (\operatorname{col} A)^\perp$;
2. $k^n = \operatorname{col} A^* \oplus \operatorname{null} A$ and $k^m = \operatorname{col} A \oplus \operatorname{null} A^*$.

Over $\mathbb{R}$ the conjugation is trivial, so $A^* = A^\top$ and $\operatorname{col} A^\top = \operatorname{row} A$ by definition 2.3.6; part (1) then reads $\operatorname{null} A = (\operatorname{row} A)^\perp$ and $\operatorname{null} A^\top = (\operatorname{col} A)^\perp$.

***Proof*:**

1. Let $T_A : k^n \to k^m$ be the map $x \mapsto Ax$, linear by theorem 5.5.14(2). Its kernel is $\operatorname{null} A$: both are the set of $x \in k^n$ with $Ax = 0$, by definition 2.4.4 and definition 2.3.3. Its image is $\operatorname{col} A$: by definition 2.4.5 the image is $\{\, Ax | x \in k^n \,\}$, and $Ax = x_1c_1 + \cdots + x_nc_n$ for the columns $c_1, \ldots, c_n$ of A by definition 1.4.1, so the set of such Ax is $\operatorname{span}\{c_1, \ldots, c_n\}$ by definition 2.1.5, which is $\operatorname{col} A$ by definition 2.3.1.

   Theorem 5.5.14(2) also gives $(T_A)^* = T_{A^*}$, whose kernel is $\operatorname{null} A^*$ by the same identification. So proposition 5.5.9(4) applied to T_A reads $\operatorname{null} A^* = (\operatorname{col} A)^\perp$, which is the second identity. Applying that identity with A^* in place of A gives $\operatorname{null}(A^*)^* = (\operatorname{col} A^*)^\perp$, and $(A^*)^* = A$ because the entry of $(A^*)^*$ in row i and column j is $\overline{\overline{a_{ij}}} = a_{ij}$. So $\operatorname{null} A = (\operatorname{col} A^*)^\perp$.

2. The set $\operatorname{col} A^*$ is a subspace of k^n, being a span (definition 2.3.1 and proposition 2.1.6), so corollary 5.5.5 gives $\big((\operatorname{col} A^*)^\perp\big)^\perp = \operatorname{col} A^*$; combined with (1) this says $(\operatorname{null} A)^\perp = \operatorname{col} A^*$. Now $\operatorname{null} A$ is a subspace of k^n by definition 2.3.3, so theorem 5.5.3 applies to it and gives

$$k^n = \operatorname{null} A \oplus (\operatorname{null} A)^\perp = \operatorname{null} A \oplus \operatorname{col} A^*$$

   The second identity is this one applied to $A^* \in M_{n\times m}(k)$, whose null space lives in k^m, together with $(A^*)^* = A$ from part (1), completing the proof.

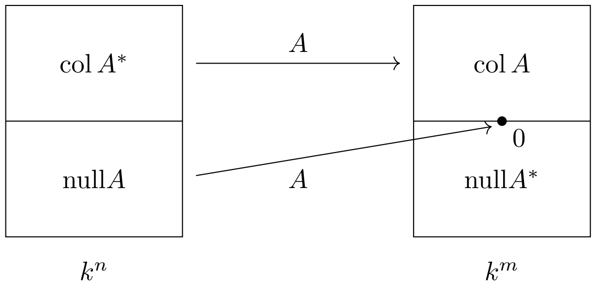


Figure 5.6: The two orthogonal decompositions of theorem 5.5.16, with null$A$ sent to 0

Over $\mathbb{C}$ the conjugate in $\operatorname{col} A^*$ cannot be dropped, and the example following corollary 2.3.12 shows why. There $P = \begin{pmatrix} 1 & i \end{pmatrix} \in M_{1\times 2}(\mathbb{C})$ had $\operatorname{row} P = \operatorname{span}\{(1,i)^\top\}$ and $\operatorname{null} P = \operatorname{span}\{(-i,1)^\top\}$, and those two lines are equal, since $i(-i,1)^\top = (1,i)^\top$. So over $\mathbb{C}$ the row space can fail to be a complement of the null space. The partner given by theorem 5.5.16 is instead $\operatorname{col} P^* = \operatorname{span}\{(1,-i)^\top\}$, the span of the conjugated row, and

$$\langle (-i,1)^\top, (1,-i)^\top \rangle = -i \cdot \overline{1} + 1 \cdot \overline{(-i)} = -i + i = 0$$

confirms the orthogonality that part (1) asserts, while part (2) gives $\mathbb{C}^2 = \operatorname{col} P^* \oplus \operatorname{null} P$. Over $\mathbb{R}$ the conjugation is trivial and no such adjustment arises: the row space is the orthogonal complement of the null space.

The running matrix illustrates the real case with both pairs at once. Example 2.3.14 computed, for $A = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 1 & 2 & 5 \\ 3 & 4 & 11 \end{pmatrix}$, the bases $(1,0,1)^\top, (0,1,2)^\top$ of $\operatorname{row} A$ and $(1,2,-1)^\top$ of $\operatorname{null} A$ inside $\mathbb{R}^3$, and the bases $c_1 = (1,2,1,3)^\top, c_2 = (1,3,2,4)^\top$ of $\operatorname{col} A$ and $(1,-1,1,0)^\top, (-1,-1,0,1)^\top$ of $\operatorname{null} A^\top$ inside $\mathbb{R}^4$. The first pair is orthogonal:

$$\langle (1,2,-1)^\top, (1,0,1)^\top \rangle = 1 + 0 - 1 = 0, \qquad \langle (1,2,-1)^\top, (0,1,2)^\top \rangle = 0 + 2 - 2 = 0$$

and so is the second, computed against the two spanning vectors of $\operatorname{col} A$ in turn:

$$\langle (1,-1,1,0)^\top, c_1 \rangle = 1 - 2 + 1 + 0 = 0, \qquad \langle (1,-1,1,0)^\top, c_2 \rangle = 1 - 3 + 2 + 0 = 0$$

$$\langle (-1,-1,0,1)^\top, c_1 \rangle = -1 - 2 + 0 + 3 = 0, \qquad \langle (-1,-1,0,1)^\top, c_2 \rangle = -1 - 3 + 0 + 4 = 0$$

By proposition 5.5.2(2) checking against a spanning list is enough, so these six numbers establish $\operatorname{null} A \subseteq (\operatorname{row} A)^\perp$ and $\operatorname{null} A^\top \subseteq (\operatorname{col} A)^\perp$, which theorem 5.5.16 strengthens to equality. The dimensions $1 + 2 = 3$ and $2 + 2 = 4$ are the ones corollary 2.3.12 predicted and corollary 5.5.4 now explains: each pair consists of a subspace and its orthogonal complement.

Exercise 5.5.1

1. In $\mathbb{R}^4$ with the standard inner product let $W = \operatorname{span}\{(1,1,0,0)^\top, (0,1,1,0)^\top\}$. Compute a basis of $W^\perp$ and check that $\dim W + \dim W^\perp = 4$, as corollary 5.5.4 requires.
2. In $\mathbb{C}^2$ with the standard inner product, let $\varphi(z) = (1+i)z_1 - 3iz_2$. Verify that φ is linear and find the unique $a \in \mathbb{C}^2$ with $\varphi(z) = \langle z, a \rangle$ for all z, as theorem 5.5.6 guarantees.

3. Let $B = \left(\begin{smallmatrix} 2 & -1 & 0 \\ 1 & 0 & 3 \end{smallmatrix}\right) \in M_{2\times 3}(\mathbb{R})$. Compute a basis of $\operatorname{null} B$ and a basis of $\operatorname{row} B$, verify by direct computation that each basis vector of the first is orthogonal to each basis vector of the second, and check that the two dimensions sum to 3.

4. Let $W_1 \subseteq W_2$ be subspaces of a finite-dimensional inner product space V. Show that $W_2^\perp \subseteq W_1^\perp$, and deduce that $W_1 = W_2$ if and only if $W_1^\perp = W_2^\perp$.

5. Let $A \in M_{m\times n}(k)$. Show that $\operatorname{null}(A^*A) = \operatorname{null} A$, and deduce that $\operatorname{rank}(A^*A) = \operatorname{rank} A$.

6. Let $C = \left(\begin{smallmatrix} 1 & i \\ 0 & 1+i \end{smallmatrix}\right) \in M_2(\mathbb{C})$ and give $\mathbb{C}^2$ the standard inner product. Compute C^* and verify $\langle Cx, y\rangle = \langle x, C^*y\rangle$ for $x = (1, i)^\top$ and $y = (i, 1)^\top$. Then compute $\langle x, C^\top y\rangle$ for the same pair and observe that the transpose alone does not represent the adjoint.

7. Let $S = \{(1, 1, 1)^\top, (1, 2, 3)^\top\} \subseteq \mathbb{R}^3$ with the standard inner product. Compute $S^\perp$, then compute $(S^\perp)^\perp$ and check directly that it equals $\operatorname{span} S$, as the discussion after corollary 5.5.5 predicts.

8. Let V be a finite-dimensional inner product space, let $T\colon V \to V$ be linear, and let $W \subseteq V$ be a subspace with $T(w) \in W$ for every $w \in W$. Show that $T^*(u) \in W^\perp$ for every $u \in W^\perp$.

6

Least Squares and the Pseudoinverse

Chapter 1 could decide whether a system had a solution and could produce one when it did. What it could not do was respond to a system with no solution at all, which is the ordinary situation once there are more equations than unknowns and the numbers come from measurement. Elimination reports the inconsistency and halts, and nothing in Part I could ask for the best available answer, because nothing there could compare two failures.

The inner product can. A candidate x produces a residual $b - Ax$, the residual has a length, and shorter is better. That turns an unsolvable equation into a minimization with a unique answer, and the answer is a projection.

Section 6.1 proves the fact the whole chapter rests on, that the orthogonal projection of a vector onto a subspace is the closest point of that subspace to it. Section 6.2 applies it to $\|Ax - b\|$ and extracts the equations characterizing the minimizers. Section 6.3 solves those equations through the factorization of section 5.4 instead, by back substitution. Section 6.4 removes the last hypothesis: when A has dependent columns the minimizer is no longer unique, and selecting the shortest one produces a matrix that inverts A as far as any matrix can.

6.1 Orthogonal Projection onto a Subspace

Section 5.3 defined the orthogonal projection $P_W(v)$ of a vector onto a subspace (definition 5.3.8) and characterized it by a single property: it is the vector of W whose difference from v is orthogonal to W. That definition never compared $P_W(v)$ with the other vectors of W, and the comparison was left for this chapter. This section makes it, and then records the projection as a matrix.

The distance between two vectors of an inner product space is the norm of their difference (definition 5.2.1), so the vector of W nearest to v is the one minimizing $\|v - w\|$ over $w \in W$. The set W is infinite as soon as it is nonzero, and nothing so far guarantees that a minimum over an infinite set is attained. The projection settles the question by exhibiting the minimizer: the candidate is written down first, and the comparison with an arbitrary competitor is then one application of theorem 5.3.2.

Theorem 6.1.1: The Orthogonal Projection Is the Closest Point of the Subspace

Let V be a finite-dimensional inner product space over k, let $W \subseteq V$ be a subspace, and let $v \in V$. Then

$$\|v - P_W(v)\| \le \|v - w\| \qquad \text{for every } w \in W$$

and equality holds exactly when $w = P_W(v)$.

***Proof*:**
Step 1: the difference $v - P_W(v)$ is orthogonal to W. By corollary 5.3.11(1) the subspace W has an orthonormal basis $q_1, \ldots, q_r$, so $P_W(v) = \sum_{i=1}^{r} \langle v, q_i \rangle q_i$ is defined (definition 5.3.8). That list is an orthonormal list in V whose span is W, which is the hypothesis of theorem 5.3.5(2); applied to v it says that $P_W(v)$ lies in W and that $v - P_W(v)$ is orthogonal to every vector of W.

Step 2: split $v - w$ into two orthogonal pieces. Fix $w \in W$ and write

$$v - w = \big(v - P_W(v)\big) + \big(P_W(v) - w\big)$$

The second summand lies in W, since $P_W(v)$ and w both do and a subspace is closed under differences (definition 2.1.2). By Step 1 the first summand is orthogonal to every vector of W, hence to the second, so the two summands form an orthogonal list (definition 5.3.1). Applying theorem 5.3.2 to that list gives

$$\|v - w\|^2 = \|v - P_W(v)\|^2 + \|P_W(v) - w\|^2$$

Step 3: compare the two sides. The term $\|P_W(v) - w\|^2$ is nonnegative by proposition 5.2.2(1), so the identity of Step 2 gives $\|v - w\|^2 \ge \|v - P_W(v)\|^2$. Both norms are nonnegative real numbers by the same part, and for real s, t with $0 \le s < t$ one has $s^2 \le st < t^2$; so $s^2 \ge t^2$ forces $s \ge t$, and therefore $\|v - w\| \ge \|v - P_W(v)\|$.

For the equality clause, the two nonnegative numbers $\|v - w\|$ and $\|v - P_W(v)\|$ are equal exactly when their squares are, and by the identity of Step 2 that happens exactly when $\|P_W(v) - w\|^2 = 0$. By proposition 5.2.2(1) this holds exactly when $P_W(v) - w = 0$, that is when $w = P_W(v)$, as we sought to show.

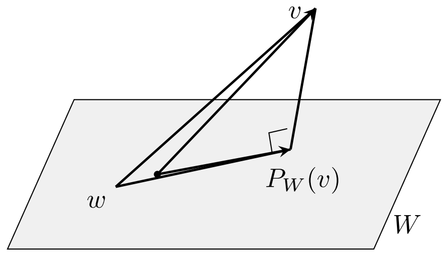


Figure 6.1: The splitting of theorem 6.1.1: the hypotenuse $v - w$ against the leg $v - P_W(v)$

The theorem turns a question about all of W into a formula. The distance from v to W, meaning the smallest of the numbers $\|v - w\|$ for $w \in W$, is attained, and it equals $\|v - P_W(v)\|$, which theorem 5.3.9 and definition 5.3.8 compute from any basis of W. No limit is taken anywhere in the argument: the minimizer is written down and then compared with an arbitrary competitor, and the comparison is theorem 5.3.2 applied to one splitting.

The equality clause also identifies the minimizer: any vector of W realizing the smallest distance equals $P_W(v)$.

Corollary 6.1.2: The Nearest Vector of a Subspace Is the Projection

Let V, W and v be as in theorem 6.1.1, and let $w_0 \in W$ satisfy $\|v - w_0\| \le \|v - w\|$ for every $w \in W$. Then $w_0 = P_W(v)$. In particular W contains exactly one vector nearest to v.

***Proof*:**
The vector $P_W(v)$ lies in W by theorem 5.3.5(2), so the hypothesis on w_0 applies to it and gives $\|v - w_0\| \le \|v - P_W(v)\|$. Theorem 6.1.1 applied to the vector w_0 of W gives the reverse inequality $\|v - P_W(v)\| \le \|v - w_0\|$. The two numbers are therefore equal, which is the equality case of theorem 6.1.1, and that case forces $w_0 = P_W(v)$. Existence is theorem 6.1.1 itself, since $P_W(v) \in W$ realizes the smallest distance, completing the proof.

Later arguments use one consequence of the corollary without further comment: $P_W(w) = w$ for every $w \in W$. Indeed $\|w - w\| = 0$ and every norm is nonnegative (proposition 5.2.2(1)), so w itself minimizes the distance from w to W, and corollary 6.1.2 identifies the minimizer as $P_W(w)$.

Chapter 5 characterized $P_W(v)$ by orthogonality of the difference; corollary 6.1.2 characterizes it by minimality. Both descriptions avoid mentioning a basis, and the two are used for different purposes: minimality is what makes the projection the answer to an approximation problem, and orthogonality is what turns that answer into a system of equations.

Example 6.1.3: Distance from a Point to a Plane in $\mathbb{R}^3$

Give $\mathbb{R}^3$ the standard inner product, and let

$$a_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad a_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \qquad W = \operatorname{span}\{a_1, a_2\}, \qquad v = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

The two vectors a_1, a_2 are not multiples of one another, so they are linearly independent and W

is a plane. Gram-Schmidt (theorem 5.3.9) turns them into an orthonormal basis of W. Since $\|a_1\|^2 = 1+1+0 = 2$,

$$q_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\\0\end{pmatrix}$$

and $\langle a_2, q_1\rangle = \frac{1}{\sqrt{2}}(0+1+0) = \frac{1}{\sqrt{2}}$, so the second vector loses its component along q_1:

$$w_2 = a_2 - \frac{1}{\sqrt{2}}\,q_1 = \begin{pmatrix}0\\1\\1\end{pmatrix} - \frac{1}{2}\begin{pmatrix}1\\1\\0\end{pmatrix} = \frac{1}{2}\begin{pmatrix}-1\\1\\2\end{pmatrix}$$

with $\|w_2\|^2 = \frac{1}{4}(1+1+4) = \frac{3}{2}$. Dividing by $\|w_2\| = \sqrt{3/2}$ gives

$$q_2 = \frac{1}{\sqrt{6}}\begin{pmatrix}-1\\1\\2\end{pmatrix}$$

The two coefficients of definition 5.3.8 are

$$\langle v, q_1\rangle = \frac{1}{\sqrt{2}}(1+2+0) = \frac{3}{\sqrt{2}}, \qquad \langle v, q_2\rangle = \frac{1}{\sqrt{6}}(-1+2+6) = \frac{7}{\sqrt{6}}$$

so the projection is

$$P_W(v) = \frac{3}{\sqrt{2}}\,q_1 + \frac{7}{\sqrt{6}}\,q_2 = \frac{3}{2}\begin{pmatrix}1\\1\\0\end{pmatrix} + \frac{7}{6}\begin{pmatrix}-1\\1\\2\end{pmatrix} = \frac{1}{3}\begin{pmatrix}1\\8\\7\end{pmatrix}$$

where the last equality collects entries: $\frac{3}{2} - \frac{7}{6} = \frac{1}{3}$, $\frac{3}{2} + \frac{7}{6} = \frac{8}{3}$ and $\frac{14}{6} = \frac{7}{3}$. The difference is

$$v - P_W(v) = \begin{pmatrix}1\\2\\3\end{pmatrix} - \frac{1}{3}\begin{pmatrix}1\\8\\7\end{pmatrix} = \frac{2}{3}\begin{pmatrix}1\\-1\\1\end{pmatrix}$$

and it is orthogonal to both spanning vectors, as theorem 5.3.5(2) predicts: $\langle v - P_W(v), a_1\rangle = \frac{2}{3}(1-1+0) = 0$ and $\langle v - P_W(v), a_2\rangle = \frac{2}{3}(0-1+1) = 0$. The distance from v to the plane is $\|v - P_W(v)\| = \frac{2}{3}\sqrt{1+1+1} = \frac{2}{\sqrt{3}}$.

Every other vector of W is farther from v. Take $w = a_1 + a_2 = (1,2,1)^\top$. Then $v - w = (0,0,2)^\top$ has $\|v-w\| = 2$, larger than $2/\sqrt{3}$. The Pythagorean identity in the proof of theorem 6.1.1 accounts for the excess exactly: $P_W(v) - w = \frac{1}{3}(1,8,7)^\top - (1,2,1)^\top = \frac{1}{3}(-2,2,4)^\top$ has $\|P_W(v) - w\|^2 = \frac{1}{9}(4+4+16) = \frac{8}{3}$, and

$$\|v - P_W(v)\|^2 + \|P_W(v) - w\|^2 = \frac{4}{3} + \frac{8}{3} = 4 = \|v - w\|^2$$

Fixing W and letting v vary makes P_W a map $V \to V$, and that map is linear. With $q_1, \ldots, q_r$ an orthonormal basis of W, the inner product is linear in its first argument (definition 5.1.2), so for

$u, v \in V$ and $\lambda \in k$

$$P_W(u + \lambda v) = \sum_{i=1}^{r} \langle u + \lambda v, q_i \rangle\, q_i = \sum_{i=1}^{r} \langle u, q_i \rangle\, q_i + \lambda \sum_{i=1}^{r} \langle v, q_i \rangle\, q_i = P_W(u) + \lambda P_W(v)$$

which is the condition of definition 2.4.2. When $V = k^m$ carries the standard inner product, a linear map is recorded by a matrix (theorem 2.4.8), and that matrix is what the rest of the chapter computes with.

Definition 6.1.4: Orthogonal Projection Matrix

Let $W \subseteq k^m$ be a subspace and give k^m the standard inner product. The *orthogonal projection matrix onto* W is the matrix of the linear map $P_W : k^m \to k^m$ relative to the standard basis: the matrix $P \in M_m(k)$ whose j-th column is $P_W(e_j)$, so that $Px = P_W(x)$ for every $x \in k^m$ (theorem 2.4.8). A matrix that arises this way from some subspace of k^m is called an *orthogonal projection matrix*.

Computing that matrix column by column from the definition would run Gram-Schmidt once and then evaluate the projection m times. One matrix product does the whole job, and its two factors have separate meanings: an orthonormal basis of W, assembled into the columns of a matrix Q, produces the coordinates of the projection when it appears as Q^* and reassembles the vector from those coordinates when it appears as Q.

Theorem 6.1.5: The Projection Matrix onto a Column Space

Let $W \subseteq k^m$ be a subspace with orthonormal basis $q_1, \ldots, q_r$, and let $Q \in M_{m\times r}(k)$ be the matrix whose columns are $q_1, \ldots, q_r$. Give k^m the standard inner product.

1. $Q^*Q = I_r$ and $\operatorname{col} Q = W$.
2. The orthogonal projection matrix onto W is QQ^*.
3. Let $A \in M_{m\times n}(k)$ have linearly independent columns and let $A = QR$ be its reduced QR factorization (theorem 5.4.2, unique by theorem 5.4.3), with $Q \in M_{m\times n}(k)$. The columns of this Q are an orthonormal basis of $\operatorname{col} A$, so parts (1) and (2) apply to it with $W = \operatorname{col} A$ and $r = n$: the orthogonal projection matrix onto $\operatorname{col} A$ is QQ^*.

***Proof*:**

1. Write $(q_i)_t$ for the t-th entry of q_i, so that the entry of Q in row t and column i is $(q_i)_t$ and the entry of Q^* in row i and column t is $\overline{(q_i)_t}$. By definition 1.4.1 the entry of Q^*Q in row i and column j is

$$\sum_{t=1}^{m} \overline{(q_i)_t}\,(q_j)_t = \sum_{t=1}^{m} (q_j)_t\,\overline{(q_i)_t} = \langle q_j, q_i \rangle$$

the last equality being the formula for the standard inner product on k^m. By definition 5.3.3 that number is 1 when $i = j$ and 0 otherwise, which says $Q^*Q = I_r$. For the second claim, $\operatorname{col} Q = \operatorname{span}\{q_1, \ldots, q_r\}$ by definition 2.3.1, and that span is W because $q_1, \ldots, q_r$ is a basis of W (definition 2.2.1).

2. Let $x \in k^m$. The same reading of definition 1.4.1 gives the i-th entry of Q^*x as

$$(Q^*x)_i = \sum_{t=1}^{m} \overline{(q_i)_t}\, x_t = \langle x, q_i \rangle$$

Multiplying a matrix by a column produces the combination of the matrix's columns with the entries of that column as coefficients (definition 1.4.1), so

$$Q(Q^*x) = \sum_{i=1}^{r} (Q^*x)_i\, q_i = \sum_{i=1}^{r} \langle x, q_i \rangle\, q_i = P_W(x)$$

the last equality being definition 5.3.8 for the orthonormal basis $q_1, \ldots, q_r$ of W. By proposition 1.4.3 the left side is $(QQ^*)x$, so $(QQ^*)x = P_W(x)$ for every x, which is what definition 6.1.4 asks of the projection matrix onto W.

3. Let $a_1, \ldots, a_n$ be the columns of A and $q_1, \ldots, q_n$ those of Q. The list $q_1, \ldots, q_n$ is orthonormal by definition 5.4.1, hence linearly independent by proposition 5.3.4. Taking $j = n$ in the span identity of theorem 5.4.2 gives $\operatorname{span}\{q_1, \ldots, q_n\} = \operatorname{span}\{a_1, \ldots, a_n\}$, and both spans are column spaces by definition 2.3.1, so $\operatorname{col} Q = \operatorname{col} A$. An independent spanning list is a basis (definition 2.2.1), so $q_1, \ldots, q_n$ is an orthonormal basis of $\operatorname{col} A$ and part (2) applies with $W = \operatorname{col} A$ and $r = n$, completing the proof.

The two products Q^*Q and QQ^* are different objects: the first is $r \times r$ and is always the identity, the second is $m \times m$ and is the identity only when W is all of k^m. Indeed $QQ^*x = P_W(x)$ lies in W for every x, so $QQ^* = I_m$ forces every $x \in k^m$ to lie in W; conversely if $W = k^m$ then $P_W(x) = x$ for every x, by the consequence of corollary 6.1.2 recorded above. Part (3) is the form used in practice: a subspace of k^m is usually presented as the column space of a matrix whose columns are neither orthogonal to one another nor of unit length, and the QR factorization (theorem 5.4.2) replaces those columns by an orthonormal basis of the same span.

Example 6.1.6: The Projection Matrix of a Plane in $\mathbb{R}^3$

Keep the plane $W = \operatorname{span}\{a_1, a_2\}$ of example 6.1.3 and its orthonormal basis

$$q_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\\0\end{pmatrix}, \qquad q_2 = \frac{1}{\sqrt{6}}\begin{pmatrix}-1\\1\\2\end{pmatrix}, \qquad Q = \begin{pmatrix}1/\sqrt{2} & -1/\sqrt{6}\\ 1/\sqrt{2} & 1/\sqrt{6}\\ 0 & 2/\sqrt{6}\end{pmatrix}$$

Over $\mathbb{R}$ the conjugation does nothing, so $Q^* = Q^\top$. The entry of $QQ^\top$ in row i and column j is $\sum_{t=1}^{2} (q_t)_i (q_t)_j$ by definition 1.4.1, which is the corresponding entry of $q_1 q_1^\top + q_2 q_2^\top$; computing the two summands,

$$q_1 q_1^\top = \frac{1}{2}\begin{pmatrix}1 & 1 & 0\\ 1 & 1 & 0\\ 0 & 0 & 0\end{pmatrix}, \qquad q_2 q_2^\top = \frac{1}{6}\begin{pmatrix}1 & -1 & -2\\ -1 & 1 & 2\\ -2 & 2 & 4\end{pmatrix}$$

and adding over a common denominator of 6 gives

$$P = QQ^\top = \frac{1}{6}\begin{pmatrix}4 & 2 & -2\\ 2 & 4 & 2\\ -2 & 2 & 4\end{pmatrix} = \frac{1}{3}\begin{pmatrix}2 & 1 & -1\\ 1 & 2 & 1\\ -1 & 1 & 2\end{pmatrix}$$

The matrix reproduces the projection computed by hand in example 6.1.3: for $v = (1, 2, 3)^\top$,

$$Pv = \frac{1}{3}\begin{pmatrix} 2+2-3 \\ 1+4+3 \\ -1+2+6 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 1 \\ 8 \\ 7 \end{pmatrix}$$

The matrix equals its own transpose, since its entry in row i and column j agrees with its entry in row j and column i for each of the three pairs $i \neq j$ in the display.

Write $M = 3P$, so that $m_{ij} = m_{ji}$ for all i, j. The entry of M^2 in row i and column j is $\sum_{t=1}^{3} m_{it}m_{tj}$ (definition 1.4.1); since $m_{it} = m_{ti}$ and $m_{tj} = m_{jt}$, that sum equals $\sum_{t=1}^{3} m_{jt}m_{ti}$, the entry of M^2 in row j and column i. So M^2 is symmetric and its six distinct entries, in positions $(1,1)$, $(1,2)$, $(1,3)$ on the first line and $(2,2)$, $(2,3)$, $(3,3)$ on the second, are

$$\begin{aligned}
&2\cdot 2 + 1\cdot 1 + (-1)(-1) = 6, \quad && 2\cdot 1 + 1\cdot 2 + (-1)\cdot 1 = 3, \quad && 2(-1) + 1\cdot 1 + (-1)\cdot 2 = -3, \\
&1\cdot 1 + 2\cdot 2 + 1\cdot 1 = 6, && 1(-1) + 2\cdot 1 + 1\cdot 2 = 3, && (-1)(-1) + 1\cdot 1 + 2\cdot 2 = 6
\end{aligned}$$

Hence

$$M^2 = \begin{pmatrix} 6 & 3 & -3 \\ 3 & 6 & 3 \\ -3 & 3 & 6 \end{pmatrix} = 3M$$

and dividing by 9 gives $P^2 = \frac{1}{9}M^2 = \frac{1}{3}M = P$.

Both properties the example exhibited, $P^2 = P$ and $P^* = P$, hold for every orthogonal projection matrix, and no other matrix has both. Chapter 2 already met the first on its own: a matrix with $P^2 = P$ is a projection in the sense of definition 2.4.12, and proposition 2.4.13 splits k^m into $\mathrm{im} P \oplus \ker P$ with P fixing the first summand. No inner product enters that statement, and the kernel is left free apart from meeting the image only at 0. The second condition is what pins the kernel down.

Theorem 6.1.7: Orthogonal Projection Matrices Are the Idempotent Self-Adjoint Matrices

Let $P \in M_m(k)$ and give k^m the standard inner product.

1. P is an orthogonal projection matrix (definition 6.1.4) if and only if $P^2 = P$ and $P^* = P$.
2. When these hold, P is the orthogonal projection matrix onto $W = \mathrm{col}\, P$, and $\mathrm{null} P = W^\perp$.

***Proof*:**
Step 1: a projection matrix satisfies both conditions. Let P be the orthogonal projection matrix onto a subspace $W \subseteq k^m$ and put $r = \dim W$. By corollary 5.3.11(1) the subspace W has an orthonormal basis $q_1, \ldots, q_r$; let Q have these as its columns. By theorem 6.1.5(2) the matrix QQ^* satisfies $QQ^*x = P_W(x)$ for every x, and P satisfies the same identity by definition 6.1.4; taking $x = e_j$ for each j shows that the two matrices have the same columns, since the j-th column of a matrix M is Me_j (definition 1.4.1), so $P = QQ^*$. Regrouping with proposition 1.4.3 and using $Q^*Q = I_r$ from theorem 6.1.5(1),

$$P^2 = (QQ^*)(QQ^*) = Q(Q^*Q)Q^* = QI_rQ^* = QQ^* = P$$

For the second condition, definition 1.4.1 gives the entry of $P = QQ^*$ in row i and column j as $p_{ij} = \sum_{t=1}^{r} (q_t)_i \overline{(q_t)_j}$. The entry of P^* in row i and column j is $\overline{p_{ji}}$, and

$$\overline{p_{ji}} = \overline{\sum_{t=1}^{r} (q_t)_j \overline{(q_t)_i}} = \sum_{t=1}^{r} \overline{(q_t)_j}\, (q_t)_i = p_{ij}$$

so $P^* = P$.

Step 2: the two conditions force P to be a projection matrix. Suppose $P^2 = P$ and $P^* = P$, and put $W = \operatorname{col} P$, a subspace of k^m by definition 2.3.1. Fix $x \in k^m$. By definition 1.4.1 the vector Px is the combination of the columns of P with the entries of x as coefficients, so Px lies in the span of those columns, which is W by definition 2.3.1; the same reading run backwards shows that every vector of W, being a combination of the columns of P, has the form Py for some $y \in k^m$.

By theorem 5.5.14(2) the identity $\langle Pu, y\rangle = \langle u, P^*y\rangle$ holds for all $u, y \in k^m$, and $P^* = P$ turns it into $\langle Pu, y\rangle = \langle u, Py\rangle$. Taking $u = x - Px$ and using $P(x - Px) = Px - P^2x = Px - Px = 0$, which is distributivity (proposition 1.4.3) together with the hypothesis $P^2 = P$, gives

$$\langle x - Px, Py\rangle = \langle P(x - Px), y\rangle = \langle 0, y\rangle = 0$$

for every y, the last equality by proposition 5.1.6(3). Since every vector of W is such a Py, the difference $x - Px$ is orthogonal to every vector of W.

The subspace W has an orthonormal basis by corollary 5.3.11(1), so theorem 5.3.5(2) applies to it and says that exactly one vector of W has its difference from x orthogonal to all of W, namely $P_W(x)$ as written in definition 5.3.8. The vector Px lies in W and has that property, so $Px = P_W(x)$. This holds for every $x \in k^m$, which is definition 6.1.4 for the subspace W.

Step 3: the column space and the null space. Let P satisfy the two conditions, so that by Step 2 it is the projection matrix onto some subspace W; that W is $\operatorname{col} P$ follows from $Px = P_W(x) \in W$ for every x, which gives $\operatorname{col} P \subseteq W$, together with $w = P_W(w) = Pw \in \operatorname{col} P$ for every $w \in W$, which gives the reverse inclusion. For the null space, $x \in \operatorname{null} P$ means $P_W(x) = 0$. If $P_W(x) = 0$ then $x = x - P_W(x)$ is orthogonal to every vector of W by theorem 5.3.5(2), so $x \in W^\perp$ by definition 5.5.1. Conversely if $x \in W^\perp$ then the vector 0 lies in W and $x - 0 = x$ is orthogonal to all of W, so the uniqueness in theorem 5.3.5(2) identifies 0 as $P_W(x)$ and $x \in \operatorname{null} P$. Hence $\operatorname{null} P = W^\perp$, completing the proof.

Among the matrices with $P^2 = P$, the second condition selects those whose null space is the orthogonal complement of their column space, which is theorem 6.1.7(2). An idempotent failing it still sends every vector into its column space, but not to the nearest vector of that space. Example 2.4.11 is the example: there $P = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ satisfies $P^2 = P$ with $\operatorname{im} P = L = \operatorname{span}\{e_1\}$ and $\ker P = \operatorname{span}\{(1, -1)^\top\}$, and $P^\top \neq P$. Its failure to minimize is visible at a single vector. Take $v = e_2 = (0, 1)^\top$, so that $Pv = (1, 0)^\top$ and $\|v - Pv\| = \|(-1, 1)^\top\| = \sqrt{2}$. The single vector e_1 is an orthonormal basis of L, so definition 5.3.8 gives $P_L(v) = \langle v, e_1\rangle e_1 = 0$, at distance $\|v\| = 1$ from v. The vector Pv lies in L, and it is not the vector of L closest to v.

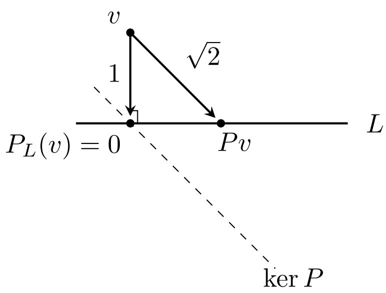


Figure 6.2: An idempotent that is not self-adjoint moves v parallel to $\ker P$ rather than perpendicularly to L

Exercise 6.1.1

1. Give $\mathbb{R}^3$ the standard inner product and let $W = \operatorname{span}\{(1,0,1)^\top, (0,1,1)^\top\}$ and $v = (2,1,0)^\top$. Produce an orthonormal basis of W with theorem 5.3.9, compute $P_W(v)$, check that $v - P_W(v)$ is orthogonal to both spanning vectors, and give the distance from v to W.

2. Let
$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix} \in M_{3\times 2}(\mathbb{R})$$
Compute the factor Q of a QR factorization of A, then use theorem 6.1.5(3) to write down the orthogonal projection matrix P onto $\operatorname{col} A$. Verify $P^2 = P$ and $P^\top = P$ by direct computation.

3. Let $u \in k^m$ be nonzero. Show that the orthogonal projection matrix onto $\operatorname{span}\{u\}$ is $\|u\|^{-2}\, uu^*$. Compute it for $u = (1,2,2)^\top \in \mathbb{R}^3$, and use it to find the distance from $(3,0,0)^\top$ to $\operatorname{span}\{u\}$.

4. Give $\mathbb{C}^2$ the standard inner product and let $W = \operatorname{span}\{(1,i)^\top\}$ and $v = (1,1)^\top$. Compute $P_W(v)$ and the distance from v to W, then compute the projection matrix onto W and check that it sends v to the same vector. (The inner product is conjugate-linear in its second argument, so the conjugation matters at every step.)

5. Let $W \subseteq k^m$ be a subspace with orthogonal projection matrix P, and let P' be the orthogonal projection matrix onto $W^\perp$. Show that $P + P' = I_m$.

6. Let V be a finite-dimensional inner product space over k and let $W \subseteq V$ be a subspace. Show that $\|P_W(v)\| \le \|v\|$ for every $v \in V$, with equality if and only if $v \in W$.

7. Let P be the orthogonal projection matrix onto a subspace $W \subseteq k^m$, and write $p_{11}, \ldots, p_{mm}$ for its diagonal entries.

   1. Show that $p_{ii} = \|Pe_i\|^2$, and deduce that $0 \le p_{ii} \le 1$ for every i.
   2. Show that $p_{11} + \cdots + p_{mm} = \dim W$. (Exercise 2.5.1 (7) identifies this sum already; only $\operatorname{rank} P$ remains to be computed.)

6.2 The Least-Squares Problem and the Normal Equations

Section 6.1 produced, for a subspace $W \subseteq k^m$ and a vector $b \in k^m$, the vector of W nearest to b. A system $Ax = b$ carries such a subspace with it: as x runs over k^n, the vector Ax runs over the column space of A (definition 2.3.1). The smallest value $\|b - Ax\|$ can take is therefore the distance from b to $\operatorname{col} A$, and theorem 6.1.1 already names the vector of that subspace realizing it. What is still missing is the vector x producing it. This section gives x, and the condition it satisfies turns out to be a square system, $A^*Ax = A^*b$, which has a solution for every A and every b whether or not $Ax = b$ does.

Throughout, $A \in M_{m\times n}(k)$ and $b \in k^m$, so the system has m equations and n unknowns, and k^m and k^n carry their standard inner products, conjugate-linear in the second argument. A system with more equations than unknowns has no solution unless its right-hand side happens to lie in the column space of the coefficient matrix, and that is the shape measured data produces: each measurement contributes one equation, the unknowns are the few parameters of a model, and the measurements disagree.

Example 6.2.1: An Inconsistent System of Four Equations in Two Unknowns
Four measurements of a quantity y, taken at the four settings $t = 0, 1, 2, 3$, record the pairs

$$(t, y) = (0, 1), \quad (1, 3), \quad (2, 4), \quad (3, 4)$$

A line $y = c + dt$ passing through all four points would have its two coefficients satisfy the four equations $c + dt_i = y_i$, which is the system $Ax = b$ with

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} \in M_{4\times 2}(\mathbb{R}), \qquad x = \begin{pmatrix} c \\ d \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 4 \end{pmatrix}$$

the first column of A recording the coefficient of c and the second the four settings t_i. Elimination on the augmented matrix settles whether such a line exists. The three replacements $R_2 \mapsto R_2 - R_1$, $R_3 \mapsto R_3 - R_1$ and $R_4 \mapsto R_4 - R_1$ clear the first column below the pivot, and then $R_3 \mapsto R_3 - 2R_2$ and $R_4 \mapsto R_4 - 3R_2$ clear the second:

$$\left(\begin{array}{cc|c} 1 & 0 & 1 \\ 1 & 1 & 3 \\ 1 & 2 & 4 \\ 1 & 3 & 4 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 2 & 3 \\ 0 & 3 & 3 \end{array}\right) \longrightarrow \left(\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & -1 \\ 0 & 0 & -3 \end{array}\right)$$

The third row of the last matrix is the equation $0c + 0d = -1$, which no pair (c, d) satisfies, so the system has no solution and no line passes through the four points.

The columns of A are linearly independent, and seeing this needs no elimination: if $Ax = 0$ then the first entry of Ax is c, so $c = 0$, and the second entry is $c + d$, so $d = 0$ as well.

Since no x makes $b - Ax$ vanish, the entries of $b - Ax$ are compared instead. Two trial lines:

$$x = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \;\text{ gives }\; Ax = \begin{pmatrix} 0 \\ 1 \\ 2 \\ 3 \end{pmatrix}, \quad b - Ax = \begin{pmatrix} 1 \\ 2 \\ 2 \\ 1 \end{pmatrix}, \quad \|b - Ax\|^2 = 1 + 4 + 4 + 1 = 10$$

$$x = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{gives} \quad Ax = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}, \quad b - Ax = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}, \quad \|b - Ax\|^2 = 0 + 1 + 1 + 0 = 2$$

The line $y = 1 + t$ misses the data by less than the line $y = t$ does. Whether any line misses it by less than $\sqrt{2}$, and which line misses it by least, are the questions this section answers.

The example replaces the demand that $b - Ax$ vanish by the demand that it be small, and the size of a vector is its norm (definition 5.2.1). Two names are attached to the objects involved.

Definition 6.2.2: Residual

Let $A \in M_{m \times n}(k)$ and $b \in k^m$. The *residual* of a vector $x \in k^n$ is

$$r(x) = b - Ax \in k^m$$

Its i-th entry is the amount by which x fails to satisfy the i-th equation of the system $Ax = b$, so x solves that system exactly when $r(x) = 0$.

Definition 6.2.3: Least-Squares Solution

Let $A \in M_{m \times n}(k)$ and $b \in k^m$, and give k^m its standard inner product. A *least-squares solution* of $Ax = b$ is a vector $\hat{x} \in k^n$ whose residual has the smallest norm:

$$\|b - A\hat{x}\| \leq \|b - Ax\| \qquad \text{for every } x \in k^n$$

The *least-squares problem* for A and b is the problem of producing the vectors $\hat{x}$ with this property.

The name records the quantity made least. Writing $r(x) = (r_1, \ldots, r_m)^\top$, the standard inner product gives $\|r(x)\|^2 = \sum_{i=1}^m |r_i|^2$, which over $k = \mathbb{R}$ is the sum of the squares of the m discrepancies; minimizing the norm and minimizing its square are the same demand, since $t \mapsto t^2$ is increasing on $t \geq 0$. When the system does have solutions the definition selects exactly those, since their residual has norm 0 and no norm is smaller (proposition 5.2.2).

The definition quantifies over all of k^n, which is infinite, and by itself gives no reason for a minimizer to exist and no way to compute one. Section 6.1 repairs both: the vectors Ax form a subspace, and the nearest vector of a subspace to b was exhibited there rather than shown to exist. Carrying that answer back from $\operatorname{col} A$ to k^n is what the next theorem does, and the condition it lands on is linear in x.

Theorem 6.2.4: Least-Squares Solutions Are the Solutions of the Normal Equations

Let $A \in M_{m\times n}(k)$ and $b \in k^m$, give k^m and k^n their standard inner products, and write $W = \operatorname{col} A$. For $\hat{x} \in k^n$ the following are equivalent.

1. $\hat{x}$ is a least-squares solution of $Ax = b$ (definition 6.2.3).
2. $A\hat{x} = P_W(b)$, where $P_W(b)$ denotes the orthogonal projection of b onto W (definition 5.3.8).
3. $A^*A\hat{x} = A^*b$.

The system in (3) is called the *normal equations* of $Ax = b$. It has at least one solution for every A and every b, so every system $Ax = b$ has at least one least-squares solution. No hypothesis on the shape of A or on its columns is used.

***Proof*:**
Step 1: the vectors Ax are exactly the vectors of W. Let $a_1, \dots, a_n$ be the columns of A. By definition 1.4.1 the product Ax is the combination $x_1a_1 + \cdots + x_na_n$, so the vectors of the form Ax with $x \in k^n$ are exactly the linear combinations of $a_1, \dots, a_n$, which form $\operatorname{span}\{a_1, \dots, a_n\}$ by definition 2.1.5 and are W by definition 2.3.1.

Step 2: (1) and (2) are equivalent. Suppose $\hat{x}$ is a least-squares solution. Every $w \in W$ is Ax for some x by Step 1, so definition 6.2.3 gives $\|b - A\hat{x}\| \le \|b - w\|$ for every $w \in W$, and $A\hat{x}$ lies in W, again by Step 1. Those are the hypotheses of corollary 6.1.2 on the vector $A\hat{x}$ of W, and its conclusion is $A\hat{x} = P_W(b)$. Conversely, suppose $A\hat{x} = P_W(b)$. By theorem 6.1.1 the inequality $\|b - P_W(b)\| \le \|b - w\|$ holds for every $w \in W$; applying it to $w = Ax$, which lies in W by Step 1, gives $\|b - A\hat{x}\| \le \|b - Ax\|$ for every $x \in k^n$, which is definition 6.2.3.

Step 3: a vector of k^m is orthogonal to W exactly when A^* annihilates it. Let $u \in k^m$. By theorem 5.5.14(2) the identity $\langle Ay, u\rangle = \langle y, A^*u\rangle$ holds for all $y \in k^n$. Suppose u is orthogonal to every vector of W. Then the left side vanishes for every y, since $Ay \in W$ by Step 1, so $\langle y, A^*u\rangle = 0$ for every $y \in k^n$; taking $y = A^*u$ gives $\|A^*u\|^2 = 0$, hence $A^*u = 0$ by proposition 5.2.2. Suppose instead that $A^*u = 0$. Then $\langle Ay, u\rangle = \langle y, 0\rangle = 0$ for every y by proposition 5.1.6, and the vectors Ay exhaust W by Step 1, so u is orthogonal to every vector of W.

Step 4: (2) and (3) are equivalent. By corollary 5.3.11(1) the subspace W has an orthonormal basis, so theorem 5.3.5(2) applies to W: exactly one vector of W has its difference from b orthogonal to every vector of W, and that vector is $P_W(b)$. Since $A\hat{x}$ lies in W by Step 1, the equality $A\hat{x} = P_W(b)$ therefore holds exactly when $b - A\hat{x}$ is orthogonal to every vector of W, which by Step 3 holds exactly when $A^*(b - A\hat{x}) = 0$. Distributing the matrix product (proposition 1.4.3), that last equation reads $A^*b - A^*A\hat{x} = 0$, which is (3).

Step 5: existence. The vector $P_W(b)$ lies in W, so by Step 1 there is some $\hat{x} \in k^n$ with $A\hat{x} = P_W(b)$. That $\hat{x}$ satisfies (2), hence (3) by Step 4 and (1) by Step 2, so the normal equations and the least-squares problem both have solutions, completing the proof.

The theorem trades an unbounded search for a square system. The matrix A^*A has n rows and n columns, since A^* has n rows and m columns and A has m rows and n columns (definition 1.4.1), so the normal equations are n equations in n unknowns however large m is, and the m measurements enter only through the two products A^*A and A^*b. The existence clause is worth separating from the rest: $Ax = b$ may be inconsistent, as in example 6.2.1, while $A^*Ax = A^*b$ is consistent always.

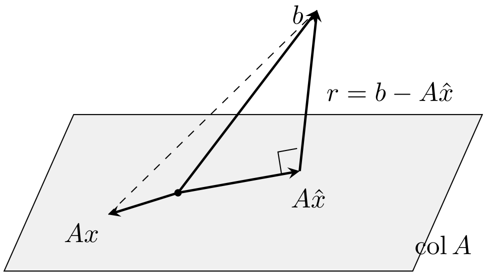


Figure 6.3: As x ranges over k^n the vector Ax ranges over col A, and the residual is shortest when it is perpendicular to that subspace

Whether the normal equations determine $\hat{x}$ is a separate question from whether they have a solution, and the answer depends on A alone: two vectors are both least-squares solutions exactly when A sends both to $P_W(b)$, so they differ by a vector of null$A$. The next two results turn this into a criterion on A^*A.

Lemma 6.2.5: The Null Space of A^*A

Let $A \in M_{m\times n}(k)$ and give k^m and k^n their standard inner products. Then $\text{null}(A^*A) = \text{null}A$.

***Proof*:**
Let $x \in \text{null}A$, so $Ax = 0$ by definition 2.3.3. Then $A^*Ax = A^*0 = 0$, since a matrix times the zero column is the zero column (definition 1.4.1), and therefore $x \in \text{null}(A^*A)$. This is the inclusion $\text{null}A \subseteq \text{null}(A^*A)$.

For the reverse inclusion, let $x \in \text{null}(A^*A)$, so $A^*Ax = 0$. By theorem 5.5.14(2) the identity $\langle Ay, u\rangle = \langle y, A^*u\rangle$ holds for all $y \in k^n$ and $u \in k^m$; taking $y = x$ and $u = Ax$ gives

$$\|Ax\|^2 = \langle Ax, Ax\rangle = \langle x, A^*Ax\rangle = \langle x, 0\rangle = 0$$

where the first equality is definition 5.2.1 and the last is proposition 5.1.6. A vector of norm 0 is the zero vector (proposition 5.2.2), so $Ax = 0$ and $x \in \text{null}A$. The two inclusions give the stated equality, as we sought to show.

The conjugation in A^* is what the second inclusion runs on, and the plain transpose does not serve. For $A = (1, i)^\top \in M_{2\times 1}(\mathbb{C})$ the product $A^\top A$ is the 1×1 matrix $1 + i^2 = 0$, whose null space is all of $\mathbb{C}$, while $Ax = (x, ix)^\top$ vanishes only for $x = 0$. The conjugate transpose gives instead $A^*A = 1 - i^2 = 2$, with null space $\{0\}$, matching the lemma.

Theorem 6.2.6: A^*A Is Invertible Exactly When the Columns of A Are Independent

Let $A \in M_{m\times n}(k)$ have columns $a_1, \dots, a_n$, and give k^m and k^n their standard inner products. The following are equivalent.

1. The list $a_1, \dots, a_n$ is linearly independent.
2. $\mathrm{null} A = \{0\}$.
3. $A^*A \in M_n(k)$ is invertible.

***Proof*:**
Step 1: (1) and (2) are equivalent. By definition 1.4.1 the product Ax equals $x_1a_1 + \cdots + x_na_n$, so the equation $Ax = 0$ says that the scalars $x_1, \dots, x_n$ are the coefficients of a vanishing combination of $a_1, \dots, a_n$. The condition $\mathrm{null} A = \{0\}$ (definition 2.3.3) therefore says that the only such coefficients are $x_1 = \cdots = x_n = 0$, which is linear independence of the list.

Step 2: (2) implies (3). Suppose $\mathrm{null} A = \{0\}$. By lemma 6.2.5 the null space of A^*A is $\{0\}$ as well, so the homogeneous system $(A^*A)x = 0$ has only the solution $x = 0$. The matrix A^*A is square, of size $n \times n$ (definition 1.4.1), so that is condition (2) of theorem 1.4.8 for A^*A, and condition (1) of that theorem makes A^*A invertible.

Step 3: (3) implies (2). Suppose A^*A is invertible and let $x \in \mathrm{null}(A^*A)$, so $A^*Ax = 0$. Multiplying on the left by $(A^*A)^{-1}$ and regrouping the product (proposition 1.4.3),

$$x = I_nx = \big((A^*A)^{-1}(A^*A)\big)x = (A^*A)^{-1}\big((A^*A)x\big) = (A^*A)^{-1}0 = 0$$

the second equality by definition 1.4.4. So $\mathrm{null}(A^*A) = \{0\}$, and lemma 6.2.5 gives $\mathrm{null} A = \{0\}$, completing the proof.

A matrix whose columns are linearly independent is said to have *full column rank*; by Step 1 and theorem 2.3.11 this is the condition $\mathrm{rank} A = n$. That is the shape a least-squares problem usually comes in, with many more measurements than unknowns, and it is the shape in which the answer is a formula.

Corollary 6.2.7: The Unique Least-Squares Solution and the Projection Matrix

Let $A \in M_{m\times n}(k)$ have linearly independent columns, let $b \in k^m$, and write $W = \mathrm{col}\, A$.

1. The system $Ax = b$ has exactly one least-squares solution, namely
$$\hat{x} = (A^*A)^{-1}A^*b$$
2. The orthogonal projection matrix onto W (definition 6.1.4) is $A(A^*A)^{-1}A^*$.

***Proof*:**

1. By theorem 6.2.6 the matrix A^*A is invertible. Multiplying the equation $A^*Ax = A^*b$ on the left by $(A^*A)^{-1}$, then regrouping with proposition 1.4.3 and using definition 1.4.4,

turns it into $x = (A^*A)^{-1}A^*b$; multiplying the latter equation on the left by A^*A and regrouping the same way returns the former. The two equations therefore have the same solutions, and the second has exactly one. By theorem 6.2.4 the least-squares solutions of $Ax = b$ are exactly the solutions of the first, so there is exactly one and it is the stated vector.

2. Let $b \in k^m$ be arbitrary and let $\hat{x}$ be the vector of part (1). By theorem 6.2.4 the equality $A\hat{x} = P_W(b)$ holds, and regrouping the product (proposition 1.4.3) writes its left side as

$$\big(A(A^*A)^{-1}A^*\big)b = P_W(b)$$

This holds for every $b \in k^m$, which is exactly what definition 6.1.4 requires of the orthogonal projection matrix onto W, so $A(A^*A)^{-1}A^*$ is that matrix, completing the proof.

Part (2) is the second formula for a projection matrix in this chapter, and the two run on different data. Theorem 6.1.5(2) takes an orthonormal basis of W, assembles it into the columns of Q, and returns QQ^*; the corollary takes any basis of W, assembles it into the columns of A, and pays for the loss of orthonormality with the inverse of A^*A. The two agree when the columns of A are already orthonormal, since then $A^*A = I_n$ by theorem 6.1.5(1).

When the columns of A are dependent the formula is unavailable, but theorem 6.2.4 still applies: least-squares solutions exist, and by lemma 6.2.5 the difference of any two solutions of $A^*Ax = A^*b$ lies in $\operatorname{null}(A^*A) = \operatorname{null} A$, which is now nonzero. Fixing one solution $\hat{x}$, the full set of them is $\hat{x} + \operatorname{null} A$, so there are infinitely many. Section 6.4 will single one out by a further condition.

What all least-squares solutions share is their residual, and its defining property is orthogonality.

Theorem 6.2.8: The Residual of a Least-Squares Solution

Let $A \in M_{m\times n}(k)$ and $b \in k^m$, write $W = \operatorname{col} A$, let $\hat{x}$ be a least-squares solution of $Ax = b$, and let $r = b - A\hat{x}$ be its residual (definition 6.2.2).

1. $A^*r = 0$, and r is orthogonal to every vector of W.
2. $r = b - P_W(b)$. In particular all least-squares solutions of $Ax = b$ have the same residual, and $\|r\|$ is the distance from b to W.
3. $\|b\|^2 = \|A\hat{x}\|^2 + \|r\|^2$.
4. $r = 0$ if and only if $Ax = b$ has a solution, and in that case the least-squares solutions of $Ax = b$ are exactly its solutions.

***Proof*:**

1. By theorem 6.2.4 the vector $\hat{x}$ satisfies $A^*A\hat{x} = A^*b$, so distributing the matrix product (proposition 1.4.3) gives $A^*r = A^*b - A^*A\hat{x} = 0$. Step 3 of the proof of theorem 6.2.4 shows that a vector annihilated by A^* is orthogonal to every vector of W, which is the second claim. Stated in terms of the four subspaces, r lies in $\operatorname{null}(A^*)$, and theorem 5.5.16 identifies that null space with $W^\perp$.
2. By theorem 6.2.4 the equality $A\hat{x} = P_W(b)$ holds, so $r = b - A\hat{x} = b - P_W(b)$. The

right side does not mention $\hat{x}$, so every least-squares solution has this residual. Its norm $\|b - P_W(b)\|$ is the smallest of the numbers $\|b - w\|$ for $w \in W$ by theorem 6.1.1, and that smallest number is the distance from b to W.

3. The vector $A\hat{x}$ lies in W, being a combination of the columns of A by definition 1.4.1 and definition 2.3.1, so part (1) makes r orthogonal to it and $A\hat{x}, r$ an orthogonal list (definition 5.3.1). Since $b = A\hat{x} + r$, applying theorem 5.3.2 to that list gives $\|b\|^2 = \|A\hat{x}\|^2 + \|r\|^2$.

4. If $r = 0$ then $b = A\hat{x}$, so $\hat{x}$ solves $Ax = b$. Conversely, suppose $Ax_0 = b$ for some $x_0 \in k^n$. Then b lies in W by definition 1.4.1 and definition 2.3.1, and the difference $b - b = 0$ is orthogonal to every vector of W (proposition 5.1.6); since W has an orthonormal basis (corollary 5.3.11(1)), the uniqueness clause of theorem 5.3.5(2) identifies b itself as $P_W(b)$, and part (2) gives $r = b - P_W(b) = 0$. For the last claim, assume $Ax = b$ is consistent. Any solution x_0 has $\|b - Ax_0\| = 0$, which no norm undercuts (proposition 5.2.2), so x_0 is a least-squares solution; conversely a least-squares solution has $r = 0$ by what was just proved, that is $A\hat{x} = b$, completing the proof.

Part (1) is what the word "normal" in the name refers to: the normal equations say that the residual is perpendicular to the column space, one equation for each column of A. Part (3) splits the squared length of b into the part the columns of A reproduce and the part they do not, and part (4) places ordinary solving inside the least-squares problem as the case where the second part is 0.

Example 6.2.9: The Line of Best Fit for Four Data Points

Return to the matrix A and the vector b of example 6.2.1. The entries are real, so the conjugate transpose is the transpose (definition 2.3.4) and $A^* = A^\top$. By definition 1.4.1 the entry of $A^\top A$ in row i and column j is the inner product of the i-th and j-th columns of A, so with $a_1 = (1, 1, 1, 1)^\top$ and $a_2 = (0, 1, 2, 3)^\top$,

$$A^\top A = \begin{pmatrix} 4 & 6 \\ 6 & 14 \end{pmatrix}, \qquad A^\top b = \begin{pmatrix} 12 \\ 23 \end{pmatrix}$$

the four entries of $A^\top A$ being $1+1+1+1 = 4$, then $0+1+2+3 = 6$ twice, then $0+1+4+9 = 14$, and the entries of $A^\top b$ being $1+3+4+4 = 12$ and $0+3+8+12 = 23$. The normal equations of theorem 6.2.4 are

$$4c + 6d = 12, \qquad 6c + 14d = 23$$

Three times the first is $12c + 18d = 36$ and twice the second is $12c + 28d = 46$; subtracting gives $10d = 10$, so $d = 1$, and then $4c = 12 - 6 = 6$ gives $c = 3/2$. The columns of A are linearly independent (example 6.2.1), so corollary 6.2.7 applies and

$$\hat{x} = \begin{pmatrix} 3/2 \\ 1 \end{pmatrix}$$

is the only least-squares solution. The line of best fit is $y = \frac{3}{2} + t$.

Its values at the four settings, and the residual, are

$$A\hat{x} = \begin{pmatrix} 3/2 \\ 5/2 \\ 7/2 \\ 9/2 \end{pmatrix}, \qquad r = b - A\hat{x} = \frac{1}{2}\begin{pmatrix} -1 \\ 1 \\ 1 \\ -1 \end{pmatrix}, \qquad \|r\|^2 = 4 \cdot \tfrac{1}{4} = 1$$

so no line misses the four measurements by less than 1, against the $\sqrt{2}$ and $\sqrt{10}$ of the two trial lines in example 6.2.1. Theorem 6.2.8(1) is visible in r: its inner products with the two columns of A are $\frac{1}{2}(-1+1+1-1) = 0$ and $\frac{1}{2}(0+1+2-3) = 0$. Theorem 6.2.8(3) checks against $\|b\|^2 = 1+9+16+16 = 42$ and $\|A\hat{x}\|^2 = \frac{1}{4}(9+25+49+81) = \frac{164}{4} = 41$, since $41+1 = 42$.

The projection matrix of corollary 6.2.7(2) is assembled from the same two products. Here $4 \cdot 14 - 6 \cdot 6 = 20$, and

$$(A^\top A)^{-1} = \frac{1}{20}\begin{pmatrix} 14 & -6 \\ -6 & 4 \end{pmatrix}$$

since multiplying this matrix by $A^\top A$ gives $\frac{1}{20}(14 \cdot 4 - 6 \cdot 6) = 1$ and $\frac{1}{20}(14 \cdot 6 - 6 \cdot 14) = 0$ in the first row and $\frac{1}{20}(-6 \cdot 4 + 4 \cdot 6) = 0$ and $\frac{1}{20}(-6 \cdot 6 + 4 \cdot 14) = 1$ in the second, which is I_2. Multiplying by $A^\top$ on the right and then by A on the left,

$$(A^\top A)^{-1}A^\top = \frac{1}{20}\begin{pmatrix} 14 & 8 & 2 & -4 \\ -6 & -2 & 2 & 6 \end{pmatrix}, \qquad A(A^\top A)^{-1}A^\top = \frac{1}{20}\begin{pmatrix} 14 & 8 & 2 & -4 \\ 8 & 6 & 4 & 2 \\ 2 & 4 & 6 & 8 \\ -4 & 2 & 8 & 14 \end{pmatrix}$$

where the four columns of the first product are the first column of $(A^\top A)^{-1}$ plus $0, 1, 2, 3$ times its second column, because the columns of $A^\top$ are $(1, t_i)^\top$; and the four rows of the second product are the first row of the first product plus $0, 1, 2, 3$ times its second row, because the rows of A are $(1, t_i)$. Writing P for the second product, P equals its transpose, as theorem 6.1.7 requires of a projection matrix over $\mathbb{R}$, and applying it to b returns the fitted values: the four entries of Pb are $\frac{1}{20}(14+24+8-16) = \frac{3}{2}$, $\frac{1}{20}(8+18+16+8) = \frac{5}{2}$, $\frac{1}{20}(2+12+24+32) = \frac{7}{2}$ and $\frac{1}{20}(-4+6+32+56) = \frac{9}{2}$, which are the entries of $A\hat{x}$.

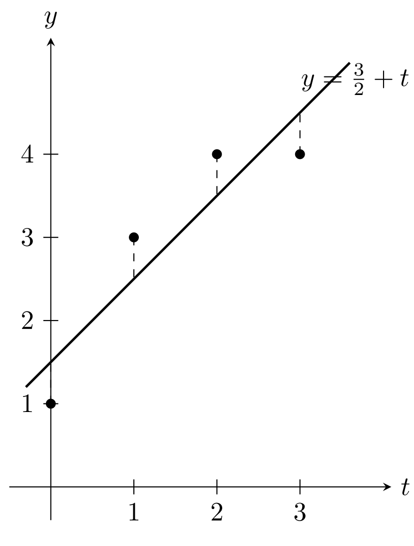


Figure 6.4: The four measurements of example 6.2.1 and the line of example 6.2.9, whose dashed vertical offsets are the entries of the residual

The dashed segments are the four entries of r, and the line minimizes the sum of their squares rather than the sum of their absolute values or the largest of them. Fitting data this way is what statistics calls linear regression. The choice of the norm is what makes the problem a projection: a different measure of size for the residual gives a different line, and no orthogonality relation characterizes it.

Exercise 6.2.1

1. Let
$$A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix} \in M_{3\times 2}(\mathbb{R}), \qquad b = \begin{pmatrix} 1 \\ 5 \\ 3 \end{pmatrix}$$
Form the normal equations, solve them, and give the residual r and the smallest value of $\|b - Ax\|$. Verify that $A^\top r = 0$ and that $\|b\|^2 = \|A\hat{x}\|^2 + \|r\|^2$.

2. Find the line $y = c + dt$ minimizing $\sum_{i=1}^{4}(y_i - c - dt_i)^2$ for the data
$$(t, y) = (-1, 1), \quad (0, 0), \quad (1, 1), \quad (2, 4)$$
and give the residual and its norm.

3. Let
$$A = \begin{pmatrix} 1 & 2 \\ 1 & 2 \\ 1 & 2 \end{pmatrix} \in M_{3\times 2}(\mathbb{R}), \qquad b = \begin{pmatrix} 1 \\ 2 \\ 6 \end{pmatrix}$$
Show that $Ax = b$ has no solution, write down the normal equations, and describe the set of all least-squares solutions. Compute $A\hat{x}$ and the residual for two different least-squares solutions and confirm that both come out the same, as theorem 6.2.8(2) requires.

4. Let $A \in M_{m\times n}(k)$. Using lemma 6.2.5 and theorem 2.3.11, show that $\operatorname{rank}(A^*A) = \operatorname{rank} A$.

5. Let $A \in M_{m\times n}(k)$ have linearly independent columns and put $P = A(A^*A)^{-1}A^*$. Using proposition 5.5.13 and proposition 1.4.5, verify directly that $P^* = P$ and $P^2 = P$, and then use theorem 6.1.7 to identify $\operatorname{col} P$ and $\operatorname{null} P$.

6. Give $\mathbb{C}^2$ its standard inner product and let
$$A = \begin{pmatrix} 1 \\ i \end{pmatrix} \in M_{2\times 1}(\mathbb{C}), \qquad b = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$
Compute A^*A and A^*b, solve the normal equations, and give the residual and its norm. Check that $A^*r = 0$.

7. Let $W \subseteq k^m$ be a subspace with basis $a_1, \ldots, a_n$, let A be the matrix whose columns are $a_1, \ldots, a_n$, and let $b \in k^m$. Show that $Ax = b$ has exactly one least-squares solution $\hat{x}$, and that its entries are the coordinates of $P_W(b)$ in that basis, that is $P_W(b) = \hat{x}_1 a_1 + \cdots + \hat{x}_n a_n$.

8. Let $\hat{x}$ be a least-squares solution of $Ax = b$. Show that
$$\|b - Ax\|^2 = \|b - A\hat{x}\|^2 + \|A(x - \hat{x})\|^2 \qquad \text{for every } x \in k^n$$
and deduce that the least-squares solutions of $Ax = b$ are exactly the vectors x with $A(x - \hat{x}) = 0$.

6.3 Solving Least Squares by QR

Section 6.2 characterized the least-squares solutions of $Ax = b$ as the solutions of $A^*Ax = A^*b$. Producing one still asks for the $n \times n$ matrix A^*A to be formed and a square system with it to be solved. The factorization $A = QR$ of section 5.4 reaches the same vector without either step: its factor Q carries an orthonormal basis of the column space, which is what an orthogonal projection is computed from, and its factor R expresses the columns of A in that basis. This section gives the recursion that solves a triangular system, shows that the least-squares problem for a matrix with independent columns is the triangular system $R\hat{x} = Q^*b$, and computes the length of the residual from b and Q^*b alone.

Throughout, k is $\mathbb{R}$ or $\mathbb{C}$, the matrix $A \in M_{m\times n}(k)$ has linearly independent columns $a_1, \dots, a_n$, the vector $b \in k^m$ is the right-hand side, and k^m carries the standard inner product $\langle u, v\rangle = \sum_{i=1}^{m} u_i\overline{v_i}$, conjugate-linear in its second argument. By theorem 5.4.2 there is a factorization $A = QR$ in which the columns $q_1, \dots, q_n$ of $Q \in M_{m\times n}(k)$ are orthonormal and $R = (r_{ij}) \in M_n(k)$ is upper triangular with positive diagonal entries, and by theorem 5.4.3 it is the only factorization of that shape.

Two consequences of the orthonormality of $q_1, \dots, q_n$ are used in every proof below. First, $Q^*Q = I_n$ (theorem 6.1.5(1)), so for every $z \in k^n$

$$\|Qz\|^2 = \langle Qz, Qz\rangle = \langle z, Q^*Qz\rangle = \langle z, z\rangle = \|z\|^2$$

the second equality being the defining property of the adjoint (definition 5.5.8), with Q^* the matrix of the adjoint of the map $z \mapsto Qz$ (theorem 5.5.14). Both $\|Qz\|$ and $\|z\|$ are nonnegative real numbers (proposition 5.2.2(1)), so $\|Qz\| = \|z\|$: multiplication by Q preserves lengths. Second, writing $(q_i)_t$ for the t-th entry of q_i, the entry of Q^* in row i and column t is $\overline{(q_i)_t}$, so definition 1.4.1 gives

$$(Q^*y)_i = \sum_{t=1}^{m} \overline{(q_i)_t}\, y_t = \langle y, q_i\rangle$$

for every $y \in k^m$: the entries of Q^*y are the inner products of y with the columns of Q.

The system the factorization leaves behind is triangular, and section 1.5 already solved one such system by hand: one pass through the equations in reverse order. The last equation involves one unknown and fixes it by a division. The equation above it involves that unknown and one more, so it fixes the next, and so on to the top. What was a procedure there is recorded here as a statement, because the argument to come needs two things a worked example cannot give: that the pass always terminates with the unique solution, and that its existence makes R invertible. The only hypothesis either claim needs is that no diagonal entry vanishes.

Proposition 6.3.1: Back Substitution Solves a Triangular System

Let $R = (r_{ij}) \in M_n(k)$ be upper triangular, so that $r_{ij} = 0$ whenever $i > j$, and suppose $r_{ii} \neq 0$ for every i.

1. For every $c \in k^n$ the system $Rx = c$ has exactly one solution, and its entries are produced from the last to the first by

$$x_n = \frac{c_n}{r_{nn}}, \qquad x_i = \frac{1}{r_{ii}}\Big(c_i - \sum_{j=i+1}^{n} r_{ij}x_j\Big) \quad \text{for } i = n-1, n-2, \ldots, 1$$

2. R is invertible.

***Proof*:**
By definition 1.4.1 the i-th entry of Rx is $\sum_{j=1}^{n} r_{ij}x_j$, and every r_{ij} with $j < i$ vanishes, so the system $Rx = c$ is the list of equations

$$r_{ii}x_i + \sum_{j=i+1}^{n} r_{ij}x_j = c_i, \qquad i = 1, \ldots, n$$

the sum being empty when $i = n$. Call this the row-i equation. Since $r_{ii} \neq 0$, dividing by it shows that the row-i equation holds if and only if

$$x_i = \frac{1}{r_{ii}}\Big(c_i - \sum_{j=i+1}^{n} r_{ij}x_j\Big)$$

which is the displayed recursion.

Step 1: at most one solution. Let x satisfy $Rx = c$. The row-n equation reads $r_{nn}x_n = c_n$ and involves no other entry, so $x_n = c_n/r_{nn}$ is determined by R and c. Fix $i < n$ and suppose the entries $x_{i+1}, \ldots, x_n$ of every solution are determined by R and c. The displayed equivalence expresses x_i in terms of those entries alone, so x_i is determined as well. Downward induction on i determines every entry, so two solutions have the same entries.

Step 2: at least one solution. Define numbers $\xi_n, \xi_{n-1}, \ldots, \xi_1$ in that order by the same recursion; the formula for ξ_i refers only to $\xi_{i+1}, \ldots, \xi_n$ and to entries of R and c, all available when it is used, so the definition is legitimate. By the displayed equivalence the vector $\xi = (\xi_1, \ldots, \xi_n)^\top$ satisfies the row-i equation for every i, hence $R\xi = c$. With Step 1 this gives (1).

Step 3: invertibility. Taking $c = 0$, the recursion returns $\xi_n = 0$ and then $\xi_i = 0$ for each i in turn, so the only solution of $Rx = 0$ is $x = 0$, that is $\operatorname{null} R = \{0\}$ (definition 2.3.3). By theorem 2.3.11 applied to R, which has n columns, $\operatorname{rank} R = n - \dim \operatorname{null} R = n$, and a square matrix whose rank equals its number of columns is invertible (theorem 1.4.8). This is (2), completing the proof.

The recursion is an algorithm, and one run of it shows what each step costs. For

$$R = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \\ 0 & 0 & 5 \end{pmatrix}, \qquad c = \begin{pmatrix} 10 \\ 10 \\ 10 \end{pmatrix}$$

the last row gives $x_3 = 10/5 = 2$; the middle row, $4x_2 + x_3 = 10$, gives $x_2 = (10-2)/4 = 2$; and the first row, $2x_1 - x_2 + 3x_3 = 10$, gives $x_1 = (10+2-6)/2 = 3$. The solution is $x = (3,2,2)^\top$, and substituting it back reproduces c, the three rows evaluating to $6-2+6 = 10$, $8+2 = 10$ and 10. No row operation was performed: the triangular shape is what elimination works to produce, and here it is given in advance.

What the factorization contributes is that the least-squares problem becomes a system of exactly this shape. The reason is a splitting of the residual, valid for every x and not only for the minimizer: the part of b that no vector of the column space reaches does not depend on x, and the part that does depend on x is measured in the orthonormal basis $q_1, \ldots, q_n$, where measuring it changes nothing because Q preserves lengths.

Theorem 6.3.2: Least Squares Through the QR Factorization

Let $A \in M_{m\times n}(k)$ have linearly independent columns, let $A = QR$ be its QR factorization (theorem 5.4.2), and let $b \in k^m$.

1. For every $x \in k^n$,
$$\|b - Ax\|^2 = \|b - QQ^*b\|^2 + \|Q^*b - Rx\|^2$$
2. R is invertible, and the least-squares problem for (A, b) (definition 6.2.3) has exactly one solution $\hat{x}$, namely the unique solution of the triangular system
$$R\hat{x} = Q^*b$$
so that $\hat{x} = R^{-1}Q^*b$.
3. $A\hat{x} = QQ^*b$, the orthogonal projection of b onto $\operatorname{col} A$.

***Proof*:**
Write $W = \operatorname{col} A$. By theorem 6.1.5(3) the columns of Q are an orthonormal basis of W, the column space of Q is W, and QQ^* is the orthogonal projection matrix onto W; the last statement says $QQ^*y = P_W(y)$ for every $y \in k^m$ (definition 6.1.4).

Step 1: the splitting. Fix $x \in k^n$. Adding and subtracting QQ^*b, using $A = QR$, and regrouping the products (proposition 1.4.3),

$$b - Ax = \big(b - QQ^*b\big) + \big(QQ^*b - QRx\big) = \big(b - QQ^*b\big) + Q\big(Q^*b - Rx\big)$$

The second summand is Q times a vector of k^n, hence a combination of the columns of Q (definition 1.4.1), so it lies in $\operatorname{col} Q = W$. The first summand is $b - P_W(b)$, which is orthogonal to every vector of W by theorem 5.3.5(2), applied to the orthonormal basis $q_1, \ldots, q_n$ of W. The two summands are therefore orthogonal (definition 5.3.1), and theorem 5.3.2 applied to that orthogonal list gives

$$\|b - Ax\|^2 = \|b - QQ^*b\|^2 + \|Q(Q^*b - Rx)\|^2$$

Multiplication by Q preserves lengths, as computed at the head of this section, so the last term is $\|Q^*b - Rx\|^2$. This is (1).

Step 2: the triangular system. The diagonal entries of R are positive, hence nonzero (theorem 5.4.2), so proposition 6.3.1 applies: R is invertible and the system $Rx = Q^*b$ has exactly one solution, $\hat{x} = R^{-1}Q^*b$. Evaluating (1) at $\hat{x}$, where the second term vanishes,

$$\|b - A\hat{x}\|^2 = \|b - QQ^*b\|^2$$

Step 3: comparison with an arbitrary x. Let $x \in k^n$. The second term in (1) is nonnegative (proposition 5.2.2(1)), so $\|b - Ax\|^2 \geq \|b - QQ^*b\|^2 = \|b - A\hat{x}\|^2$ by Step 2. For nonnegative real numbers s and t with $s < t$ one has $s^2 \leq st < t^2$, so $s^2 \geq t^2$ forces $s \geq t$; applied to $s = \|b - Ax\|$ and $t = \|b - A\hat{x}\|$ this gives $\|b - A\hat{x}\| \leq \|b - Ax\|$. So $\hat{x}$ is a least-squares solution (definition 6.2.3).

Conversely, let x be a least-squares solution, so that $\|b - Ax\| \leq \|b - A\hat{x}\|$; with the inequality just proved this gives $\|b - Ax\| = \|b - A\hat{x}\|$. Subtracting the identity of Step 2 from (1) at this x leaves $\|Q^*b - Rx\|^2 = 0$, hence $Rx = Q^*b$ by proposition 5.2.2(1), hence $x = \hat{x}$ by the uniqueness in proposition 6.3.1. This proves (2).

Finally $A\hat{x} = QR\hat{x} = Q(Q^*b) = QQ^*b$, regrouping with proposition 1.4.3 and using $R\hat{x} = Q^*b$, and $QQ^*b = P_W(b)$ as recorded at the start of the proof. This is (3), completing the proof.

The least-squares problem has become two operations on the data. The first forms Q^*b, the list of n inner products $\langle b, q_i\rangle$; the second runs the recursion of proposition 6.3.1 on $R\hat{x} = Q^*b$. Neither A^*A nor its inverse appears anywhere. Part (3) agrees with theorem 6.2.8(2), which identified $A\hat{x}$ as the orthogonal projection of b onto the column space for every least-squares solution of every matrix; here that projection is already factored, as QQ^*b.

The route through the normal equations produces the same vector, and the factorization shows why. Taking adjoints reverses a product (proposition 5.5.13(3)), so $A^* = (QR)^* = R^*Q^*$, and with $Q^*Q = I_n$,

$$A^*A = R^*Q^*QR = R^*R, \qquad A^*b = R^*Q^*b$$

so the normal equations $A^*Ax = A^*b$ of theorem 6.2.4 read $R^*Rx = R^*Q^*b$. The matrix R^* is invertible with inverse $(R^{-1})^*$: applying the adjoint to $R^{-1}R = I_n$ and to $RR^{-1} = I_n$ gives $R^*(R^{-1})^* = I_n$ and $(R^{-1})^*R^* = I_n$, which is what definition 1.4.4 asks. Multiplying $R^*Rx = R^*Q^*b$ on the left by $(R^*)^{-1}$ returns $Rx = Q^*b$, and multiplying that on the left by R^* returns the normal equations, so the two systems have the same solutions, and the formula $\hat{x} = (A^*A)^{-1}A^*b$ of corollary 6.2.7 is the vector $R^{-1}Q^*b$. Which of the two systems to solve when the arithmetic is inexact is a question about how each one amplifies error in its data, and it belongs to **??**.

Example 6.3.3: The Line of Best Fit Through Its QR Factorization

Take the four measurements of example 6.2.9, recording the value y at the times $t = 0, 1, 2, 3$, so that the line $y = c + dt$ through all four points would solve $Ax = b$ with $x = (c, d)^\top$ and

$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 4 \end{pmatrix}$$

The columns $a_1 = (1, 1, 1, 1)^\top$ and $a_2 = (0, 1, 2, 3)^\top$ were shown there to be linearly independent, so theorem 6.3.2 applies.

The factorization. Run the construction of theorem 5.3.9 on the two columns. Since $\|a_1\|^2 = 1+1+1+1 = 4$,

$$r_{11} = \|a_1\| = 2, \qquad q_1 = \frac{1}{2}\begin{pmatrix}1\\1\\1\\1\end{pmatrix}$$

Next $r_{12} = \langle a_2, q_1\rangle = \frac{1}{2}(0+1+2+3) = 3$, and subtracting from a_2 its component along q_1,

$$a_2 - 3q_1 = \begin{pmatrix}0\\1\\2\\3\end{pmatrix} - \frac{3}{2}\begin{pmatrix}1\\1\\1\\1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}-3\\-1\\1\\3\end{pmatrix}$$

whose squared norm is $\frac{1}{4}(9+1+1+9) = 5$. So $r_{22} = \sqrt{5}$ and $q_2 = \frac{1}{2\sqrt{5}}(-3,-1,1,3)^\top$, giving

$$Q = \frac{1}{2}\begin{pmatrix}1 & -3/\sqrt{5}\\ 1 & -1/\sqrt{5}\\ 1 & 1/\sqrt{5}\\ 1 & 3/\sqrt{5}\end{pmatrix}, \qquad R = \begin{pmatrix}2 & 3\\ 0 & \sqrt{5}\end{pmatrix}$$

Both diagonal entries of R are positive, as theorem 5.4.2 requires. The product QR is A column by column: its first column is $2q_1 = (1,1,1,1)^\top$, and its second is $3q_1 + \sqrt{5}\,q_2 = \frac{3}{2}(1,1,1,1)^\top + \frac{1}{2}(-3,-1,1,3)^\top = (0,1,2,3)^\top$.

The triangular system. Over $\mathbb{R}$ the conjugation does nothing, so $Q^* = Q^\top$ (definition 2.3.4), and the two entries of $Q^\top b$ are the inner products of b with the columns of Q:

$$\langle b, q_1\rangle = \tfrac{1}{2}(1+3+4+4) = 6, \qquad \langle b, q_2\rangle = \frac{-3-3+4+12}{2\sqrt{5}} = \frac{10}{2\sqrt{5}} = \sqrt{5}$$

Back substitution (proposition 6.3.1) on $R\hat{x} = (6, \sqrt{5})^\top$ takes the second row first: $\sqrt{5}\,\hat{x}_2 = \sqrt{5}$ gives $\hat{x}_2 = 1$, and then $2\hat{x}_1 + 3 = 6$ gives $\hat{x}_1 = \frac{3}{2}$. So

$$\hat{x} = \begin{pmatrix}3/2\\1\end{pmatrix}$$

which is the line $y = \frac{3}{2} + t$ that example 6.2.9 obtained from the normal equations. Neither $A^\top A$ nor its inverse was formed.

The residual. The fitted values are the entries of $A\hat{x} = (\frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \frac{9}{2})^\top$, so

$$r = b - A\hat{x} = \frac{1}{2}\begin{pmatrix}-1\\1\\1\\-1\end{pmatrix}, \qquad \|r\| = \tfrac{1}{2}\sqrt{1+1+1+1} = 1$$

which is the residual computed in example 6.2.9, entry by entry.

The example produced the length of the residual by first computing $\hat{x}$ and then subtracting. The splitting of theorem 6.3.2(1) makes that detour unnecessary: its first term is already the squared residual, and it is built from b and Q alone.

Theorem 6.3.4: The Norm of the Least-Squares Residual

Let $A \in M_{m\times n}(k)$ have linearly independent columns, let $A = QR$ be its QR factorization, let $b \in k^m$, let $\hat{x}$ be the least-squares solution of $Ax = b$, and let $r = b - A\hat{x}$ be its residual (definition 6.2.2).

1. $r = b - QQ^*b$, and
$$\|r\|^2 = \|b\|^2 - \|Q^*b\|^2$$

2. The subspace $(\operatorname{col} A)^\perp$ has dimension $m - n$. Let $q_{n+1}, \ldots, q_m$ be an orthonormal basis of it. Then $q_1, \ldots, q_m$ is an orthonormal basis of k^m, and
$$r = \sum_{j=n+1}^{m} \langle b, q_j \rangle \, q_j, \qquad \|r\|^2 = \sum_{j=n+1}^{m} |\langle b, q_j \rangle|^2$$
both sums being empty, and $r = 0$, when $m = n$.

***Proof*:**
Write $W = \operatorname{col} A$, so that $q_1, \ldots, q_n$ is an orthonormal basis of W and $QQ^*y = P_W(y)$ for every $y \in k^m$, both by theorem 6.1.5(3) with definition 6.1.4.

1. Subtracting theorem 6.3.2(3) from b gives $r = b - QQ^*b$. The vector $QQ^*b = P_W(b)$ lies in W, and $r = b - P_W(b)$ is orthogonal to every vector of W by theorem 5.3.5(2), hence to QQ^*b in particular. Since $b = QQ^*b + r$, theorem 5.3.2 applied to that orthogonal pair gives
$$\|b\|^2 = \|QQ^*b\|^2 + \|r\|^2$$
Multiplication by Q preserves lengths, as computed at the head of this section, so $\|QQ^*b\| = \|Q^*b\|$. Substituting and rearranging gives $\|r\|^2 = \|b\|^2 - \|Q^*b\|^2$.

2. The list $q_1, \ldots, q_n$ is a basis of W, so $\dim W = n$ (definition 2.2.10) and $\dim W^\perp = m - n$ by corollary 5.5.4. By corollary 5.3.11(1) the subspace $W^\perp$ has an orthonormal basis, and any basis of it has $m - n$ members; call one $q_{n+1}, \ldots, q_m$.

   The combined list $q_1, \ldots, q_m$ is orthonormal: each member has norm 1; two members drawn from the same half are orthogonal because each half is an orthonormal list (definition 5.3.3); and if $i \leq n < j$ then $q_j \in W^\perp$ and $q_i \in W$, so $\langle q_j, q_i \rangle = 0$ by definition 5.5.1 and therefore $\langle q_i, q_j \rangle = \overline{0} = 0$ by proposition 5.1.6(2). The list spans k^m: by theorem 5.5.3 every $v \in k^m$ is $w + w'$ with $w \in W$ and $w' \in W^\perp$, and w is a combination of $q_1, \ldots, q_n$ while w' is a combination of $q_{n+1}, \ldots, q_m$. An orthonormal list is linearly independent (proposition 5.3.4), so the combined list is an orthonormal basis of k^m (definition 2.2.1).

   By part (1) the vector r is orthogonal to every vector of W, so $r \in W^\perp$ (definition 5.5.1); and $b - r = QQ^*b$ lies in W, hence is orthogonal to every vector of $W^\perp$, again by definition 5.5.1 with the conjugate symmetry of proposition 5.1.6(2). Now theorem 5.3.5(2), applied to the subspace $W^\perp$ with its orthonormal basis $q_{n+1}, \ldots, q_m$ and to the vector

b, says that exactly one vector of $W^\perp$ has its difference from b orthogonal to all of $W^\perp$, namely $\sum_{j=n+1}^m \langle b, q_j\rangle q_j$ (definition 5.3.8). The vector r has both properties, so it is that vector.

The summands are pairwise orthogonal, since for $j \neq l$

$$\big\langle \langle b, q_j\rangle q_j,\, \langle b, q_l\rangle q_l \big\rangle = \langle b, q_j\rangle\, \overline{\langle b, q_l\rangle}\, \langle q_j, q_l\rangle = 0$$

by proposition 5.1.6(2) and definition 5.3.3, and each has norm $\|\langle b, q_j\rangle q_j\| = |\langle b, q_j\rangle|$ because $\|q_j\| = 1$ (proposition 5.2.2). So theorem 5.3.2 applied to that orthogonal list gives $\|r\|^2 = \sum_{j=n+1}^m |\langle b, q_j\rangle|^2$, completing the proof.

Part (1) computes the distance from b to $\operatorname{col} A$ from two lengths that are available before any system is solved: the length of b, and the length of the vector Q^*b that back substitution takes as its right-hand side. It also decides in one comparison whether the fit is exact. The identity makes $\|b\|^2 - \|Q^*b\|^2$ nonnegative, so $\|Q^*b\| \leq \|b\|$ for every b, with equality exactly when $r = 0$, which by theorem 6.3.2(3) happens exactly when $b = A\hat{x}$, that is when the system $Ax = b$ has an ordinary solution.

Part (2) names the coordinates responsible. In the basis $q_1, \ldots, q_m$ the vector b has coordinates $\langle b, q_i\rangle$, the first n of which are the entries of Q^*b. Those n the triangular system matches exactly, since R is invertible and Rx runs over all of k^n as x does; the remaining $m - n$ coordinates of b are the directions in which no vector of the column space has a component at all, so they survive in the residual whatever x is, and their squares add up to $\|r\|^2$.

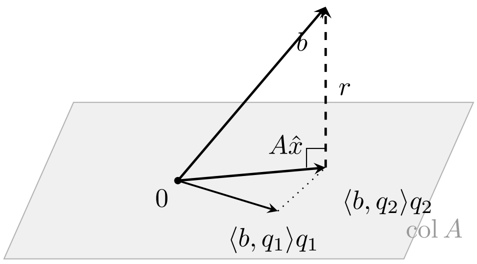


Figure 6.5: The coordinates of b in the basis of theorem 6.3.4(2): the first n assemble $A\hat{x}$, and the remaining $m - n$ assemble r

For the data of example 6.3.3 the first part reads $\|b\|^2 = 1+9+16+16 = 42$ and $\|Q^\top b\|^2 = 36+5 = 41$, so $\|r\|^2 = 1$, which is the residual computed there. For the second, the orthogonal complement of $\operatorname{col} A$ in $\mathbb{R}^4$ has dimension $4 - 2 = 2$, and

$$q_3 = \frac{1}{2}\begin{pmatrix} -1 \\ 1 \\ 1 \\ -1 \end{pmatrix}, \qquad q_4 = \frac{1}{2\sqrt{5}}\begin{pmatrix} 1 \\ -3 \\ 3 \\ -1 \end{pmatrix}$$

is an orthonormal basis of it. Their squared norms are $\frac{1}{4}(1+1+1+1) = 1$ and $\frac{1}{20}(1+9+9+1) = 1$; they are orthogonal to one another, since $(-1)(1) + (1)(-3) + (1)(3) + (-1)(-1) = 0$; and each is orthogonal to a_1 and to a_2, since $-1+1+1-1 = 0$ and $0+1+2-3 = 0$ for q_3, and $1-3+3-1 = 0$ and $0-3+6-3 = 0$ for q_4. A vector orthogonal to a_1 and a_2 is orthogonal to every combination of them, the inner product being conjugate-linear in its second argument (proposition 5.1.6(2)), so both lie in $(\operatorname{col} A)^\perp$. The two coordinates of b in these directions are

$$\langle b, q_3\rangle = \tfrac{1}{2}(-1+3+4-4) = 1, \qquad \langle b, q_4\rangle = \frac{1-9+12-4}{2\sqrt{5}} = 0$$

so $r = 1\cdot q_3 + 0\cdot q_4 = \frac{1}{2}(-1,1,1,-1)^\top$ and $\|r\|^2 = 1^2+0^2 = 1$, matching example 6.3.3 entry by entry. The measurements miss the fitted line in the direction q_3 and not at all in the direction q_4.

Exercise 6.3.1

1. Let
$$A = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$$
Compute the QR factorization of A with theorem 5.3.9, form $Q^\top b$, and solve $R\hat{x} = Q^\top b$ by back substitution. Compute the residual $r = b - A\hat{x}$ and check that $\|r\|^2 = \|b\|^2 - \|Q^\top b\|^2$, as theorem 6.3.4(1) predicts.

2. Solve $Rx = c$ by back substitution, where
$$R = \begin{pmatrix} 3 & -1 & 2 & 1 \\ 0 & 2 & 0 & -4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 2 \end{pmatrix}, \qquad c = \begin{pmatrix} 2 \\ -2 \\ 5 \\ 6 \end{pmatrix}$$
and check the answer in all four equations. Then replace the entry $r_{33} = 1$ by 0, leaving everything else unchanged, and show that the resulting system has no solution at all. Which hypothesis of proposition 6.3.1 fails?

3. Give $\mathbb{C}^2$ the standard inner product and let
$$A = \begin{pmatrix} 1 \\ i \end{pmatrix} \in M_{2\times 1}(\mathbb{C}), \qquad b = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$
Compute Q and R, then Q^*b, then the least-squares solution $\hat{x} \in \mathbb{C}$ and the residual r. Check that r is orthogonal to the single column of A and that $\|r\|^2 = \|b\|^2 - \|Q^*b\|^2$. (The inner product is conjugate-linear in its second argument, so the conjugation matters at every step.)

4. Let
$$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 1 \\ 1 & 0 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$$
Compute Q and R, and use theorem 6.3.4(1) to find the distance from b to $\operatorname{col} A$ before solving any system. Then solve $R\hat{x} = Q^\top b$, compute $r = b - A\hat{x}$ directly, and check that the two answers agree.

5. Let $A \in M_{m\times n}(k)$ have linearly independent columns, let $A = QR$, and let $\hat{x}$ and r be the least-squares solution and residual for a right-hand side $b \in k^m$. Show that $\|r\| \le \|b\|$, that equality holds if and only if $Q^*b = 0$, and that $Q^*b = 0$ holds if and only if b is orthogonal to every column of A. Show that in that case $\hat{x} = 0$.

6. Let $A \in M_{m\times n}(k)$ have linearly independent columns $a_1, \dots, a_n$ with $n \geq 2$, and let $A = QR$. Show that r_{nn} is the distance from a_n to $\operatorname{span}\{a_1, \dots, a_{n-1}\}$, in the sense of theorem 6.1.1. (Use the span identity of theorem 5.4.2 and the expression of a_n as a combination of the columns of Q.) Verify the statement on example 6.3.3.
7. Let $R \in M_n(k)$ be upper triangular with $r_{ii} \neq 0$ for every i. Running the recursion of proposition 6.3.1 on the n systems $Rx = e_j$, show that R^{-1} is upper triangular and that its diagonal entries are $1/r_{11}, \dots, 1/r_{nn}$.

6.4 The Moore-Penrose Pseudoinverse

Section 6.2 produced a least-squares solution for every pair (A, b), and made it unique exactly when the columns of A are linearly independent. When they are not, the minimizers form an infinite set and nothing said so far selects one of them. This section selects one for every A and every b at once, in a way that turns out to be linear in b, so that the choice is recorded by a single matrix.

The construction does not begin with the minimization problem. It begins with the two ways in which $x \mapsto Ax$ fails to be invertible, namely that it sends the nonzero vectors of $\operatorname{null} A$ to 0 and that its values fill only the subspace $\operatorname{col} A$ of k^m, and it removes both by shrinking the domain and the codomain until what is left is a bijection. Inverting that bijection and sending the rest of k^m to 0 produces a matrix A^+. The least-squares statement is then a theorem about A^+ rather than its definition.

Throughout, k is $\mathbb{R}$ or $\mathbb{C}$, the matrix is $A \in M_{m\times n}(k)$, and k^n and k^m carry their standard inner products. Two subspaces of k^n appear constantly: the null space $\operatorname{null} A$ (definition 2.3.3) and the column space of the conjugate transpose, $\operatorname{col}(A^*)$. Over $\mathbb{R}$ the columns of $A^* = A^\top$ are the rows of A, so $\operatorname{col}(A^*)$ is the row space of A (definition 2.3.6); over $\mathbb{C}$ the entries are conjugated as well. Theorem 5.5.16 records how these subspaces sit with respect to one another: $\operatorname{null} A$ and $\operatorname{col}(A^*)$ are orthogonal complements of each other inside k^n, and $\operatorname{null}(A^*)$ and $\operatorname{col} A$ are orthogonal complements of each other inside k^m.

Those two orthogonal splittings are what the construction uses. The first says that k^n is the direct sum of the subspace A kills and a subspace of the same dimension as $\operatorname{col} A$; the following lemma says that A carries the second of these onto $\operatorname{col} A$ bijectively.

Lemma 6.4.1: A Maps $\operatorname{col}(A^*)$ Bijectively onto $\operatorname{col} A$

Let $A \in M_{m\times n}(k)$ and write $r = \operatorname{rank} A$.

1. $\operatorname{col}(A^*) \cap \operatorname{null} A = \{0\}$, and $\dim \operatorname{col}(A^*) = r = \dim \operatorname{col} A$.
2. For every $y \in \operatorname{col} A$ there is exactly one $x \in \operatorname{col}(A^*)$ with $Ax = y$.

***Proof*:**

1. By theorem 5.5.16 the subspaces $\operatorname{null} A$ and $\operatorname{col}(A^*)$ are orthogonal complements of each other in k^n, so a vector x lying in both is orthogonal to itself, that is $\langle x, x\rangle = 0$. An inner product is positive definite (definition 5.1.2), so $x = 0$, and the intersection is $\{0\}$.

For the dimensions, $\mathrm{col}(A^*) = (\mathrm{null}A)^\perp$ by the same theorem, so corollary 5.5.4 gives $\dim\mathrm{col}(A^*) = n - \dim\mathrm{null}A$. Applying theorem 2.3.11 to A, whose domain k^n has dimension n, gives $\dim\mathrm{null}A = n - r$, whence $\dim\mathrm{col}(A^*) = r$. That $\dim\mathrm{col}\,A = r$ is corollary 2.3.12.

2. *Existence.* Let $y \in \mathrm{col}\,A$. By definition 2.3.1 there is $z \in k^n$ with $Az = y$. Apply theorem 5.5.3 to the subspace $W = \mathrm{col}(A^*)$ of k^n: it writes $z = x + u$ with $x \in W$ and $u \in W^\perp$. By theorem 5.5.16 the complement $W^\perp$ is $\mathrm{null}A$, so $Au = 0$ and

$$Ax = A(z-u) = Az - Au = Az = y$$

which exhibits the required x.

Uniqueness. Suppose $x_1, x_2 \in \mathrm{col}(A^*)$ satisfy $Ax_1 = Ax_2 = y$. Their difference lies in $\mathrm{col}(A^*)$, since a subspace is closed under differences (definition 2.1.2), and satisfies $A(x_1 - x_2) = 0$, so it lies in $\mathrm{null}A$ as well. By part (1) the intersection of the two subspaces is $\{0\}$, so $x_1 = x_2$, completing the proof.

The lemma says that the restriction of $x \mapsto Ax$ to $\mathrm{col}(A^*)$, with its values taken in $\mathrm{col}\,A$, is a bijection. On $\mathrm{col}(A^*)$ nothing is sent to 0 except 0 itself, and the values already fill $\mathrm{col}\,A$; the two subspaces have the same dimension r, and A matches them up one to one. What was discarded is exactly $\mathrm{null}A$ on the domain side and $(\mathrm{col}\,A)^\perp = \mathrm{null}(A^*)$ on the codomain side.

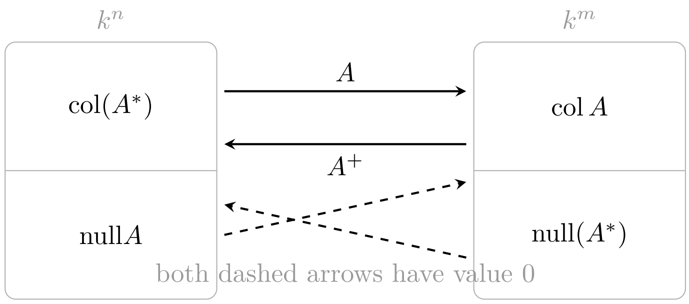


Figure 6.6: The pseudoinverse inverts the bijection of lemma 6.4.1 and sends the remaining summand to zero

Extending the inverse bijection to all of k^m now requires one decision, and the geometry of section 6.1 makes it: a vector $b \in k^m$ splits uniquely as a vector of $\mathrm{col}\,A$ plus a vector orthogonal to $\mathrm{col}\,A$, and only the first summand is in the range of A. The definition keeps that summand and discards the other.

Definition 6.4.2: Moore-Penrose Pseudoinverse

Let $A \in M_{m\times n}(k)$ and let $b \in k^m$. Write

$$b = y + z, \qquad y \in \operatorname{col} A, \qquad z \in (\operatorname{col} A)^{\perp}$$

for the orthogonal decomposition of b along $\operatorname{col} A$ (theorem 5.5.3), so that $y = P_{\operatorname{col} A}(b)$ in the notation of definition 5.3.8. The *Moore-Penrose pseudoinverse* of A is the map $A^+ : k^m \to k^n$ sending b to the unique $x \in \operatorname{col}(A^*)$ with $Ax = y$.

The assignment is unambiguous. The splitting of b is unique by theorem 5.5.3, and once y is fixed the vector x exists and is unique by lemma 6.4.1(2), since y lies in $\operatorname{col} A$. Both ingredients of the recipe are geometric: the projection discards the part of b that no vector Ax can reach, and the lemma inverts what is left.

Theorem 6.4.3: The Pseudoinverse Is Linear

Let $A \in M_{m\times n}(k)$. The map $A^+ : k^m \to k^n$ of definition 6.4.2 is linear, and therefore is recorded by a matrix in $M_{n\times m}(k)$, which is written A^+ as well. Moreover

1. $A^+(Ax) = x$ for every $x \in \operatorname{col}(A^*)$;
2. $\operatorname{col}(A^+) = \operatorname{col}(A^*)$, so $\operatorname{rank} A^+ = \operatorname{rank} A$;
3. $\operatorname{null}(A^+) = (\operatorname{col} A)^{\perp} = \operatorname{null}(A^*)$.

***Proof*:**
Step 1: linearity. Let $b_1, b_2 \in k^m$ and $\lambda \in k$, and put $x_i = A^+ b_i$, so that $x_i \in \operatorname{col}(A^*)$ and $Ax_i = P_{\operatorname{col} A}(b_i)$ by definition 6.4.2. The vector $x_1 + \lambda x_2$ lies in $\operatorname{col}(A^*)$, a subspace (definition 2.1.2), and

$$A(x_1 + \lambda x_2) = Ax_1 + \lambda Ax_2 = P_{\operatorname{col} A}(b_1) + \lambda P_{\operatorname{col} A}(b_2) = P_{\operatorname{col} A}(b_1 + \lambda b_2)$$

the first equality because $x \mapsto Ax$ is linear (theorem 2.4.8) and the last because $P_{\operatorname{col} A}$ is linear, which is what definition 6.1.4 records by exhibiting its matrix. So $x_1 + \lambda x_2$ is a vector of $\operatorname{col}(A^*)$ that A sends to the $\operatorname{col} A$-component of $b_1 + \lambda b_2$, and the uniqueness in definition 6.4.2 identifies it as $A^+(b_1 + \lambda b_2)$. This is the condition of definition 2.4.2. A linear map $k^m \to k^n$ is recorded by a matrix in $M_{n\times m}(k)$ (theorem 2.4.8), and that matrix is again written A^+.

Step 2: the three identities.

1. Let $x \in \operatorname{col}(A^*)$ and put $b = Ax$. Then $b \in \operatorname{col} A$, so its orthogonal decomposition along $\operatorname{col} A$ is $b = b + 0$ and $P_{\operatorname{col} A}(b) = b$. The vector x therefore lies in $\operatorname{col}(A^*)$ and satisfies $Ax = P_{\operatorname{col} A}(b)$, so $A^+ b = x$ by the uniqueness in definition 6.4.2.
2. Every value $A^+ b$ lies in $\operatorname{col}(A^*)$ by definition 6.4.2, so $\operatorname{col}(A^+) \subseteq \operatorname{col}(A^*)$; and every $x \in \operatorname{col}(A^*)$ is the value $A^+(Ax)$ by part (1), so the inclusion is an equality. Taking dimensions and using lemma 6.4.1(1), $\operatorname{rank} A^+ = \dim \operatorname{col}(A^*) = \operatorname{rank} A$.
3. By definition 6.4.2 the vector $A^+ b$ is the unique $x \in \operatorname{col}(A^*)$ with $Ax = P_{\operatorname{col} A}(b)$. If $A^+ b = 0$ then $P_{\operatorname{col} A}(b) = A0 = 0$, so the decomposition of b is $b = 0 + b$ and $b \in (\operatorname{col} A)^{\perp}$.

Conversely if $b \in (\operatorname{col} A)^\perp$ then that same decomposition gives $P_{\operatorname{col} A}(b) = 0$, and $x = 0$ is a vector of $\operatorname{col}(A^*)$ with $Ax = 0$, so $A^+ b = 0$. Finally $(\operatorname{col} A)^\perp = \operatorname{null}(A^*)$ by theorem 5.5.16, completing the proof.

Part (2) says that A^+ has the same rank as A, so nothing is gained or lost in passing to it, and part (3) names the vectors it sends to 0: precisely those the columns of A cannot help to represent. Part (1) is the sense in which A^+ inverts A: it does so on $\operatorname{col}(A^*)$ and nowhere else, because nowhere else can it.

Example 6.4.4: Two Pseudoinverses Computed from the Definition

1. Let
$$N = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \in M_2(\mathbb{R})$$
Both columns equal $(1,1)^\top$, so $\operatorname{col} N = \operatorname{span}\{(1,1)^\top\}$, and N is its own transpose, so $\operatorname{col}(N^\top)$ is the same line. A vector $(x_1, x_2)^\top$ satisfies $Nx = 0$ exactly when $x_1 + x_2 = 0$, so $\operatorname{null} N = \operatorname{span}\{(1,-1)^\top\}$. On the line $\operatorname{col}(N^\top)$ the matrix acts by $N(1,1)^\top = (2,2)^\top = 2\,(1,1)^\top$, so the bijection of lemma 6.4.1 is multiplication by 2 and its inverse is multiplication by $\frac{1}{2}$:
$$N^+ \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad N^+ \begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0$$
the second because $(1,-1)^\top$ is orthogonal to $\operatorname{col} N$, hence is sent to 0 by theorem 6.4.3(3). Writing $e_1 = \frac{1}{2}\left[(1,1)^\top + (1,-1)^\top\right]$ and $e_2 = \frac{1}{2}\left[(1,1)^\top - (1,-1)^\top\right]$ and using linearity,
$$N^+ e_1 = \frac{1}{4} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad N^+ e_2 = \frac{1}{4} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \text{so} \qquad N^+ = \frac{1}{4} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$
A check: $N^2 = 2N$, so $NN^+N = \frac{1}{4}N^3 = \frac{1}{4} \cdot 4N = N$.

2. Let A be the running matrix of example 1.2.9, with columns
$$c_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 3 \end{pmatrix}, \qquad c_2 = \begin{pmatrix} 1 \\ 3 \\ 2 \\ 4 \end{pmatrix}, \qquad c_3 = \begin{pmatrix} 3 \\ 8 \\ 5 \\ 11 \end{pmatrix} = c_1 + 2c_2$$
and let $b = (1,1,1,1)^\top$. The relation $c_3 = c_1 + 2c_2$ makes the columns linearly dependent, so by theorem 6.2.6 the matrix $A^\top A$ is not invertible and corollary 6.2.7 does not apply: this A is exactly the case the present section exists for.

   The first two columns are not multiples of one another, so $\operatorname{col} A = \operatorname{span}\{c_1, c_2\}$ is a plane and $B = (c_1\ c_2) \in M_{4\times 2}(\mathbb{R})$ has full column rank. Projecting b onto that plane is section 6.2 applied to B: from $c_1 \cdot c_1 = 1+4+1+9 = 15$, $c_1 \cdot c_2 = 1+6+2+12 = 21$, $c_2 \cdot c_2 = 1+9+4+16 = 30$, $c_1 \cdot b = 1+2+1+3 = 7$ and $c_2 \cdot b = 1+3+2+4 = 10$,
$$B^\top B = \begin{pmatrix} 15 & 21 \\ 21 & 30 \end{pmatrix}, \qquad B^\top b = \begin{pmatrix} 7 \\ 10 \end{pmatrix}$$

Multiplying the first equation $15\alpha + 21\beta = 7$ by 21 and the second $21\alpha + 30\beta = 10$ by 15 and subtracting gives $9\beta = 3$, so $\beta = \frac{1}{3}$ and then $15\alpha = 0$. Hence

$$y = P_{\operatorname{col} A}(b) = 0 \cdot c_1 + \frac{1}{3}c_2 = \frac{1}{3}\begin{pmatrix}1\\3\\2\\4\end{pmatrix}, \qquad b - y = \frac{1}{3}\begin{pmatrix}2\\0\\1\\-1\end{pmatrix}$$

and the residual is orthogonal to both columns, as theorem 6.2.8(1) predicts: $\frac{1}{3}(2+0+1-3) = 0$ and $\frac{1}{3}(2+0+2-4) = 0$.

Now solve $Ax = y$ inside $\operatorname{col}(A^\top)$. One solution is $x^{(0)} = (0, \frac{1}{3}, 0)^\top$, since $Ax^{(0)} = \frac{1}{3}c_2 = y$, and the null space is $\operatorname{null} A = \operatorname{span}\{v\}$ with $v = (1, 2, -1)^\top$, so every solution has the form $x^{(0)} + tv$ with $t \in \mathbb{R}$. Such a vector lies in $\operatorname{col}(A^\top) = (\operatorname{null} A)^\perp$ (theorem 5.5.16) exactly when it is orthogonal to v, and

$$\langle x^{(0)} + tv,\, v\rangle = \left(0 + \frac{2}{3} + 0\right) + t\,\|v\|^2 = \frac{2}{3} + 6t$$

vanishes at $t = -\frac{1}{9}$. Therefore

$$A^+ b = x^{(0)} - \frac{1}{9}v = \begin{pmatrix}0\\1/3\\0\end{pmatrix} - \frac{1}{9}\begin{pmatrix}1\\2\\-1\end{pmatrix} = \frac{1}{9}\begin{pmatrix}-1\\1\\1\end{pmatrix}$$

Two checks. First, $A(A^+b) = \frac{1}{9}(-c_1 + c_2 + c_3) = \frac{1}{9}(3, 9, 6, 12)^\top = y$, so the value does map to the projection. Second, $(-1, 1, 1)^\top$ is a combination of the rows $(1, 1, 3)^\top$ and $(2, 3, 8)^\top$ of A, namely $-5(1, 1, 3)^\top + 2(2, 3, 8)^\top = (-1, 1, 1)^\top$, so it does lie in $\operatorname{col}(A^\top)$.

The recipe of definition 6.4.2 is geometric, and checking that a proposed matrix is A^+ by following it means computing two subspaces first. Four matrix identities do the same job with no subspace in sight, and they hold because the two products AA^+ and A^+A are the projections onto the two subspaces the construction used.

Theorem 6.4.5: The Penrose Conditions

Let $A \in M_{m\times n}(k)$ and let $A^+ \in M_{n\times m}(k)$ be as in definition 6.4.2.

1. AA^+ is the orthogonal projection matrix onto $\operatorname{col} A$, and A^+A is the orthogonal projection matrix onto $\operatorname{col}(A^*)$.
2. The four identities

$$AA^+A = A, \qquad A^+AA^+ = A^+, \qquad (AA^+)^* = AA^+, \qquad (A^+A)^* = A^+A$$

hold.

3. A^+ is the only matrix in $M_{n\times m}(k)$ satisfying all four identities of part (2).

***Proof*:**
Step 1: the two products are projection matrices. Let $b \in k^m$. By definition 6.4.2 the vector A^+b satisfies $A(A^+b) = P_{\mathrm{col}\,A}(b)$, so $(AA^+)b = P_{\mathrm{col}\,A}(b)$ for every b, which is what definition 6.1.4 asks of the orthogonal projection matrix onto $\mathrm{col}\,A$.

For the other product, let $x \in k^n$ and write $x = u+w$ with $u \in \mathrm{col}(A^*)$ and $w \in \mathrm{col}(A^*)^\perp = \mathrm{null}A$, using theorem 5.5.3 and theorem 5.5.16. Then $Ax = Au + Aw = Au$, and theorem 6.4.3(1) applied to u gives $A^+(Ax) = A^+(Au) = u$. The uniqueness clause of theorem 5.5.3 identifies u as $P_{\mathrm{col}(A^*)}(x)$, so $(A^+A)x = P_{\mathrm{col}(A^*)}(x)$ for every x, which is definition 6.1.4 for the subspace $\mathrm{col}(A^*)$ of k^n.

Step 2: the four identities. An orthogonal projection matrix P satisfies $P^2 = P$ and $P^* = P$ by theorem 6.1.7. Applying this to the two matrices of Step 1 gives the third and fourth identities at once. For the first, let $x \in k^n$; then $Ax \in \mathrm{col}\,A$, so $P_{\mathrm{col}\,A}(Ax) = Ax$ and

$$(AA^+A)x = (AA^+)(Ax) = P_{\mathrm{col}\,A}(Ax) = Ax$$

holds for every x, whence $AA^+A = A$. For the second, let $b \in k^m$; then $A^+b \in \mathrm{col}(A^*)$ by definition 6.4.2, so $P_{\mathrm{col}(A^*)}(A^+b) = A^+b$ and

$$(A^+AA^+)b = (A^+A)(A^+b) = P_{\mathrm{col}(A^*)}(A^+b) = A^+b$$

holds for every b, whence $A^+AA^+ = A^+$.

Step 3: uniqueness. Let $X, Y \in M_{n\times m}(k)$ both satisfy the four identities, so that

$$AXA = A, \quad XAX = X, \quad (AX)^* = AX, \quad (XA)^* = XA$$

and the same four with Y in place of X. Throughout, the adjoint of a product reverses the order of the factors, $(MN)^* = N^*M^*$ (proposition 5.5.9). First,

$$AX = (AX)^* = X^*A^* = X^*(AYA)^* = X^*A^*Y^*A^* = (AX)^*(AY)^* = (AX)(AY)$$

where the third equality replaces A by AYA and the last uses the third identity for X and for Y. Regrouping the right-hand side with proposition 1.4.3 and using $AXA = A$,

$$AX = (AXA)Y = AY$$

The same argument on the other side gives

$$XA = (XA)^* = A^*X^* = (AYA)^*X^* = A^*Y^*A^*X^* = (YA)^*(XA)^* = (YA)(XA) = Y(AXA) = YA$$

Combining the two,

$$X = XAX = X(AX) = X(AY) = (XA)Y = (YA)Y = Y(AY) = YAY = Y$$

so any two matrices satisfying the four identities are equal. Part (2) exhibits A^+ as one such matrix, so it is the only one, completing the proof.

The four identities are a characterization: they mention no subspace, no projection and no basis, and a candidate matrix can be tested against them by multiplication alone. The first, $AA^+A = A$, says that A^+ returns each value Ax to a vector that A sends back to it; the second says the same of A^+

with the roles of the two matrices exchanged. The third and fourth are what force the two products of part (1) to be orthogonal projections rather than idempotent matrices only, and they are the sole place the inner product enters, through theorem 6.1.7. This is Penrose's 1955 characterization of an object introduced by E. H. Moore in 1920, and it is the source of both halves of the name. Chapter 11 constructs the same matrix a second way.

The definition selected one vector out of the solutions of $Ax = y$, and it did so before the least-squares problem was mentioned. The following theorem identifies what was selected.

Theorem 6.4.6: The Pseudoinverse Solves Least Squares with Smallest Norm

Let $A \in M_{m\times n}(k)$ and $b \in k^m$.

1. A^+b is a least-squares solution of $Ax = b$ (definition 6.2.3).
2. The set of least-squares solutions of $Ax = b$ is $A^+b + \text{null}A = \{A^+b + w \big| w \in \text{null}A\}$.
3. $\|A^+b\| \le \|\hat{x}\|$ for every least-squares solution $\hat{x}$, with equality only for $\hat{x} = A^+b$.

***Proof*:**
Write $W = \text{col}\, A$.

1. By definition 6.4.2 the vector A^+b satisfies $A(A^+b) = P_W(b)$. Let $x \in k^n$. Then $Ax \in W$ by definition 2.3.1, so theorem 6.1.1 applied to the subspace W and the vector b gives $\|b - P_W(b)\| \le \|b - Ax\|$, that is $\|b - A(A^+b)\| \le \|b - Ax\|$. Since x was arbitrary, this is the condition of definition 6.2.3.

2. Let $\hat{x}$ be a least-squares solution. By theorem 6.2.8(2) the vector $A\hat{x}$ is the orthogonal projection of b onto W, so $A\hat{x} = P_W(b)$, and by part (1) the same holds for A^+b. Subtracting, $A(\hat{x} - A^+b) = 0$, so $w = \hat{x} - A^+b$ lies in null$A$ and $\hat{x} = A^+b + w$. Conversely let $w \in \text{null}A$ and put $x = A^+b + w$. Then $Ax = A(A^+b) + Aw = A(A^+b)$, so x has the same residual as A^+b and part (1) makes it a least-squares solution as well.

3. Let $\hat{x}$ be a least-squares solution and write $\hat{x} = A^+b + w$ with $w \in \text{null}A$, as part (2) permits. By definition 6.4.2 the vector A^+b lies in $\text{col}(A^*)$, and null$A$ is the orthogonal complement of $\text{col}(A^*)$ in k^n (theorem 5.5.16), so A^+b and w are orthogonal (definition 5.3.1). Theorem 5.3.2 applied to this orthogonal pair gives

$$\|\hat{x}\|^2 = \|A^+b\|^2 + \|w\|^2$$

and $\|w\|^2 \ge 0$ by proposition 5.2.2(1), so $\|\hat{x}\|^2 \ge \|A^+b\|^2$. Both norms are nonnegative real numbers by the same part, and squaring is increasing on nonnegative reals, so $\|\hat{x}\| \ge \|A^+b\|$. Equality of the norms forces equality of their squares, hence $\|w\|^2 = 0$, hence $w = 0$ by proposition 5.2.2(1) and $\hat{x} = A^+b$, as we sought to show.

Part (2) describes the whole ambiguity in the least-squares problem: the approximation $A\hat{x}$ to b never varies, and the coordinate vectors producing it differ by elements of null$A$, so the solution set is a translate of null$A$. Part (3) is what distinguishes A^+ from every other matrix X with $AXA = A$: the translate meets $\text{col}(A^*)$ in exactly one point, that point is the one closest to the origin, and A^+b is it. When $\text{null}A = \{0\}$ the translate is a single point and part (3) says nothing new; when null$A \neq \{0\}$ it is the entire content of the definition.

Example 6.4.4(2) can now be read as a least-squares computation. The vector $\frac{1}{9}(-1,1,1)^\top$ is a least-squares solution of $Ax = (1,1,1,1)^\top$ for the running matrix, the full set of them is $\frac{1}{9}(-1,1,1)^\top +$ span$\{(1,2,-1)^\top\}$, and among these the one computed has the smallest norm: it has norm $\sqrt{3}/9$, while the competitor $(0,\frac{1}{3},0)^\top$ found on the way has the larger norm $\frac{1}{3} = 3/9$.

When the columns of A are independent the null space is trivial, and section 6.2 already produced a formula for the unique least-squares solution. The two answers must agree, and they do.

Corollary 6.4.7: The Pseudoinverse of a Matrix of Full Column Rank

Let $A \in M_{m\times n}(k)$ have linearly independent columns. Then A^*A is invertible and

$$A^+ = (A^*A)^{-1}A^*$$

In particular $A^+b = (A^*A)^{-1}A^*b$ is the unique least-squares solution of $Ax = b$.

***Proof*:**
Invertibility of A^*A is theorem 6.2.6. Put $X = (A^*A)^{-1}A^* \in M_{n\times m}(k)$; by theorem 6.4.5(3) it suffices to verify the four identities for X, since A^+ is the only matrix satisfying them.

Two preliminary facts. First, $(A^*)^* = A$, since conjugating and transposing twice returns every entry to its place (proposition 5.5.13(1)), so $(A^*A)^* = A^*(A^*)^* = A^*A$ by proposition 5.5.9. Second, taking adjoints in $(A^*A)(A^*A)^{-1} = I_n$ and using proposition 5.5.9 gives $\big((A^*A)^{-1}\big)^*(A^*A)^* = I_n$, and by the first fact this reads $\big((A^*A)^{-1}\big)^*(A^*A) = I_n$; the same computation on the other side gives $(A^*A)\big((A^*A)^{-1}\big)^* = I_n$, so $\big((A^*A)^{-1}\big)^* = (A^*A)^{-1}$ by proposition 1.4.5.

Now regroup products with proposition 1.4.3 throughout. From $(A^*A)^{-1}(A^*A) = I_n$,

$$XA = (A^*A)^{-1}(A^*A) = I_n$$

which is self-adjoint, giving the fourth identity, and gives the first two at once:

$$AXA = A(XA) = AI_n = A, \qquad XAX = (XA)X = I_nX = X$$

For the third, $AX = A(A^*A)^{-1}A^*$, so by proposition 5.5.9 and the two preliminary facts

$$(AX)^* = (A^*)^*\big((A^*A)^{-1}\big)^*A^* = A(A^*A)^{-1}A^* = AX$$

So X satisfies all four identities and equals A^+.

For the last sentence, null$A = \{0\}$, since a nonzero vector of null$A$ would record a nontrivial linear relation among the columns of A (definition 2.3.3), so the set of least-squares solutions given by theorem 6.4.6(2) is the single vector A^+b, completing the proof.

The formula is the one section 6.2 reached by a different route: theorem 6.2.4(3) makes the least-squares solutions the solutions of $A^*A\hat{x} = A^*b$, and when A^*A is invertible that system has the single solution $(A^*A)^{-1}A^*b$ (corollary 6.2.7). So the pseudoinverse extends the full-column-rank formula to every matrix, and agrees with it whenever that formula makes sense. The extension is not obtained by writing $(A^*A)^{-1}$ in a wider setting, which is impossible once the columns are dependent, but by the subspace construction, which never inverts anything that is not already a bijection.

The remaining case fixes the relationship between A^+ and the inverse matrix of definition 1.4.4.

Corollary 6.4.8: The Pseudoinverse of an Invertible Matrix

Let $A \in M_n(k)$ be invertible. Then $A^+ = A^{-1}$.

***Proof*:**
Every $b \in k^n$ satisfies $b = A(A^{-1}b)$, so $\operatorname{col} A = k^n$ by definition 2.3.1, and consequently $(\operatorname{col} A)^\perp = \{0\}$: a vector orthogonal to every vector of k^n is orthogonal to itself, hence is 0 by positive definiteness (definition 5.1.2). If $Ax = 0$ then $x = A^{-1}(Ax) = 0$, so $\operatorname{null} A = \{0\}$, and therefore $\operatorname{col}(A^*) = (\operatorname{null} A)^\perp = k^n$ by theorem 5.5.16, since every vector of k^n is orthogonal to 0 (proposition 5.1.6).

Let $b \in k^n$. Its orthogonal decomposition along $\operatorname{col} A = k^n$ is $b = b + 0$, so definition 6.4.2 makes A^+b the unique $x \in \operatorname{col}(A^*) = k^n$ with $Ax = b$. That vector is $A^{-1}b$, so $A^+b = A^{-1}b$ for every b and the two matrices are equal, as we sought to show.

So A^+ agrees with A^{-1} whenever the inverse exists, and is defined for every matrix of every shape, including the zero matrix, whose pseudoinverse is the zero matrix of the transposed shape. What it is not is an inverse: the products AA^+ and A^+A are the projections of theorem 6.4.5(1), and they equal the identity only when the corresponding subspace is everything, which is $\operatorname{col} A = k^m$ for the first and independence of the columns for the second.

Exercise 6.4.1

1. Let $v \in k^m$ be nonzero, regarded as a matrix in $M_{m\times 1}(k)$. Show that $v^+ = \|v\|^{-2} v^*$, and that vv^+ is the orthogonal projection matrix onto the line $\operatorname{span}\{v\}$.
2. Compute the pseudoinverse of
$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \in M_2(\mathbb{R})$$
directly from definition 6.4.2, by first determining $\operatorname{col} A$, $\operatorname{col}(A^\top)$ and $\operatorname{null} A$. Verify the four identities of theorem 6.4.5(2) for your answer.
3. Let A be the running matrix of example 1.2.9 and let $b = (5, 13, 8, 18)^\top$. Check that $Ax = b$ is consistent, describe all of its solutions, and compute A^+b.
4. Show that $A^+A = I_n$ if and only if the columns of A are linearly independent, and that $AA^+ = I_m$ if and only if $\operatorname{col} A = k^m$.
5. Show that $(A^+)^+ = A$ for every $A \in M_{m\times n}(k)$, using only theorem 6.4.5.
6. Show that $(A^*)^+ = (A^+)^*$ for every $A \in M_{m\times n}(k)$.
7. Find real matrices $A \in M_{1\times 2}(\mathbb{R})$ and $B \in M_{2\times 1}(\mathbb{R})$ with $(AB)^+ \neq B^+A^+$, and compute both sides. (Compare with the identity $(AB)^{-1} = B^{-1}A^{-1}$ for invertible square matrices.)

7

The Spectral Theorem

Chapter 4 asked when an operator becomes diagonal in some basis and answered with a condition on eigenvectors: enough of them, independent. The basis that condition produces is arbitrary in every other respect. Its vectors may sit at any angles to one another and may have any lengths, and the change-of-basis matrix that implements the diagonalization can distort the geometry Part II has spent two chapters building.

This chapter asks the question again with that geometry protected. The coordinates are now required to be orthonormal, so the change of basis is unitary, and the demand is far more stringent: the eigenbasis must itself be orthogonal. Remarkably, the operators meeting it can be named exactly. Over $\mathbb{C}$ they are the operators commuting with their own adjoint, and the condition $T^*T = TT^*$, which looks like a piece of algebraic bookkeeping, turns out to be equivalent to a complete geometric description. That equivalence is the spectral theorem, and it is the most useful theorem in this book.

Section 7.1 introduces the three classes of operator the adjoint distinguishes and establishes what each forces on eigenvalues and eigenvectors. Section 7.2 proves that every complex matrix is unitarily triangularizable, which is weaker than diagonal but available with no hypothesis at all; the two spectral theorems are then read off from it, over $\mathbb{C}$ in section 7.3 and for real symmetric matrices in section 7.4. The last three sections spend the result. Section 7.5 classifies the orthogonal operators, recovering the fact that a rotation of space turns about an axis. Section 7.6 diagonalizes many operators at once and finds commutativity to be exactly the obstruction. Section 7.7 exhibits an infinite family of matrices sharing a single eigenbasis, and identifies that basis as the discrete Fourier transform.

7.1 Self-Adjoint, Unitary, and Normal Operators

Throughout this section V is a finite-dimensional inner product space over k, where k is $\mathbb{R}$ or $\mathbb{C}$ (definition 5.1.2, definition 2.2.15), and an *operator* on V means a linear map $T\colon V \to V$

(definition 2.4.2). The space k^n always carries the standard inner product $\langle x, y\rangle = x_1\overline{y_1} + \cdots + x_n\overline{y_n}$ of example 5.1.3, and id_V denotes the identity operator on V, so that $\mathrm{id}_V v = v$ for every v. Every operator has an adjoint T^* (theorem 5.5.7, definition 5.5.8), again an operator on V, characterized by

$$\langle Tu, v\rangle = \langle u, T^*v\rangle \qquad \text{for all } u, v \in V$$

Three relations between T and T^* single out the operators this chapter diagonalizes: $T^* = T$, $T^*T = \mathrm{id}_V$, and $T^*T = TT^*$. This section defines the three classes, exhibits matrices separating them, and proves the constraints each class places on eigenvalues and eigenvectors.

Definition 7.1.1: Self-Adjoint Operator and Hermitian Matrix

An operator $T\colon V \to V$ is *self-adjoint* if $T^* = T$.
A matrix $A \in M_n(k)$ is *self-adjoint*, or *Hermitian*, if $A^* = A$, where A^* is the conjugate transpose of definition 5.5.12. Over $\mathbb{R}$ the conjugation does nothing and the condition reads $A^\top = A$ (definition 2.3.4); such a matrix is called *symmetric*.

The condition $T^* = T$ has an equivalent form that mentions no adjoint. If $T^* = T$ then the display above reads $\langle Tu, v\rangle = \langle u, Tv\rangle$ for all $u, v \in V$; conversely, an operator S satisfying $\langle Tu, v\rangle = \langle u, Sv\rangle$ for all u and v equals T^* by the uniqueness clause of theorem 5.5.7, so the identity $\langle Tu, v\rangle = \langle u, Tv\rangle$ forces $T^* = T$. A self-adjoint operator may therefore be moved from one argument of the inner product to the other without changing the value.

The two halves of definition 7.1.1 are the same condition read in two languages, and the translation is used without comment below. Let $A \in M_n(k)$ and let $T_A\colon k^n \to k^n$ be the operator $x \mapsto Ax$. Its adjoint is T_{A^*} by theorem 5.5.14(2). Composition of such operators corresponds to multiplication of matrices, since $T_A(T_B x) = A(Bx) = (AB)x$ by proposition 1.4.3, and two matrices defining the same operator are equal, since the j-th column of a matrix M is Me_j (definition 1.4.1). So each of the three conditions on T_A and $(T_A)^*$ holds exactly when the corresponding condition on A and A^* does, and every statement proved below for operators may be read as a statement about matrices.

Example 7.1.2: The Coordinate Swap of $\mathbb{R}^2$ and the Householder Reflections

Let

$$S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$$

so that $Sx = (x_2, x_1)^\top$ for $x = (x_1, x_2)^\top$. The matrix is its own transpose, and over $\mathbb{R}$ the conjugate transpose is the transpose, so $S^* = S$ and S is self-adjoint. The defining identity can also be checked directly: for $x, y \in \mathbb{R}^2$,

$$\langle Sx, y\rangle = x_2y_1 + x_1y_2 = \langle x, Sy\rangle$$

both sides being the same sum of two products.

Squaring S returns each coordinate to its place:

$$S^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2$$

by definition 1.4.1, so $S^*S = S^2 = I_2$ as well.

The map S fixes every point of the line $x_2 = x_1$ and sends $(1, -1)^\top$ to $-(1, -1)^\top$, so it is the reflection of the plane across that line. It is the Householder reflection H_v of definition 5.4.5 for

$v = (1,-1)^\top$: there $\langle v, v\rangle = 1+1 = 2$ and $vv^\top = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$, so

$$H_v = I_2 - \frac{2}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = S$$

Two eigenvectors are visible: $S(1,1)^\top = (1,1)^\top$ and $S(1,-1)^\top = -(1,-1)^\top$, so 1 and -1 are eigenvalues of S (definition 4.1.2). Both are real, and the two eigenvectors are orthogonal, since $\langle (1,1)^\top, (1,-1)^\top\rangle = 1-1 = 0$.

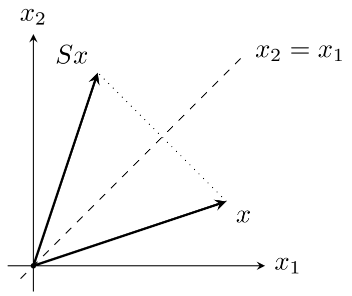


Figure 7.1: The operator S of example 7.1.2 exchanges the two coordinates, reflecting each vector across the line $x_2 = x_1$

The reflection just computed satisfies a second identity, $S^*S = I_2$, and it is that identity rather than $S^* = S$ that defines the next class.

Definition 7.1.3: Unitary Operator, Unitary Matrix, Orthogonal Matrix

An operator $T\colon V \to V$ is *unitary* if $T^*T = \mathrm{id}_V$. Over $\mathbb{R}$ such an operator is also called *orthogonal*.
A matrix $A \in M_n(k)$ is *unitary* if $A^*A = I_n$. Over $\mathbb{R}$ the condition reads $A^\top A = I_n$ and such a matrix is called *orthogonal*.

The matrix S of example 7.1.2 is orthogonal, since $S^*S = S^2 = I_2$ there. More generally every Householder reflection H_v is unitary and self-adjoint at once: Proposition 5.4.6(2) gives $H_v^* = H_v$ and $H_v^2 = I_m$, so $H_v^*H_v = H_v^2 = I_m$.

The definition is an identity between operators, and by itself says nothing about lengths or angles. The next theorem gives four equivalent forms of it, three of them geometric. The step from preserving lengths to preserving inner products is the one with content: it recovers the inner product from the norm through the polarization identity (theorem 5.2.8), whose shape differs over $\mathbb{R}$ and over $\mathbb{C}$.

Theorem 7.1.4: Characterizations of a Unitary Operator

Let V be a finite-dimensional inner product space over k with $n = \dim V$, and let $T\colon V \to V$ be linear. The following are equivalent.

1. $T^*T = \mathrm{id}_V$.
2. $\langle Tu, Tv\rangle = \langle u, v\rangle$ for all $u, v \in V$.
3. $\|Tv\| = \|v\|$ for all $v \in V$.
4. There is an orthonormal basis $q_1, \ldots, q_n$ of V for which $Tq_1, \ldots, Tq_n$ is again an orthonormal basis of V.
5. T is invertible and $T^{-1} = T^*$.

When these hold, $TT^* = \mathrm{id}_V$ as well, and T carries *every* orthonormal basis of V to an orthonormal basis of V.

***Proof*:**
Step 1: (1) implies (2). For $u, v \in V$, the defining identity of definition 5.5.8 applied to the pair u and Tv gives

$$\langle Tu, Tv\rangle = \langle u, T^*Tv\rangle = \langle u, \mathrm{id}_V v\rangle = \langle u, v\rangle$$

the middle equality being the hypothesis.

Step 2: (2) implies (3). Taking $u = v$ in (2) gives $\|Tv\|^2 = \langle Tv, Tv\rangle = \langle v, v\rangle = \|v\|^2$ by definition 5.2.1. Both $\|Tv\|$ and $\|v\|$ are nonnegative real numbers (proposition 5.2.2(1)), and two nonnegative reals with equal squares are equal, so $\|Tv\| = \|v\|$.

Step 3: (3) implies (2). This is exactly theorem 5.2.9(2), proved in chapter 5: a linear map preserving the norm preserves the inner product, because the polarization identity recovers the one from the other. The identity has a different shape over $\mathbb{R}$ and over $\mathbb{C}$, and both cases are carried out there.

Step 4: (2) implies (1). Let $v \in V$. For every $u \in V$, definition 5.5.8 and (2) give $\langle u, T^*Tv\rangle = \langle Tu, Tv\rangle = \langle u, v\rangle$, so $\langle u, T^*Tv - v\rangle = 0$ by additivity of the inner product in its second argument (proposition 5.1.6(2)). Taking $u = T^*Tv - v$ makes this $\langle w, w\rangle = 0$ for $w = T^*Tv - v$, so $w = 0$ by proposition 5.1.6(1). Hence $T^*Tv = v$ for every v, which is (1).

Step 5: (1) and (5) are equivalent, and (1) gives $TT^* = \mathrm{id}_V$. Assume (1). If $Tv = 0$ then $v = T^*Tv = T^*0 = 0$, the last equality because T^* is linear, so $\ker T = \{0\}$ and T is injective (proposition 2.4.7). By theorem 2.4.10 this gives $\dim \mathrm{im} T = \dim V - 0 = \dim V$, and $\mathrm{im} T$ is a subspace of V (proposition 2.4.6), so $\mathrm{im} T = V$ by corollary 2.2.12. Thus T is a bijection and has an inverse operator T^{-1}. Composing $T^*T = \mathrm{id}_V$ with T^{-1} on the right gives $T^* = T^{-1}$, which is (5); and then $TT^* = TT^{-1} = \mathrm{id}_V$. Conversely (5) gives $T^*T = T^{-1}T = \mathrm{id}_V$, which is (1).

Step 6: (1) implies (4), and the last clause. Let $q_1, \ldots, q_n$ be any orthonormal basis of V; at least one exists by corollary 5.3.11(1) applied to $W = V$. By Step 1 and definition 5.3.3,

$$\langle Tq_i, Tq_j\rangle = \langle q_i, q_j\rangle = \delta_{ij}$$

so $Tq_1, \ldots, Tq_n$ is an orthonormal list, hence linearly independent by proposition 5.3.4. It spans V: by Step 5 the operator T is invertible, so a given $w \in V$ is $T(T^{-1}w)$, and writing

$T^{-1}w = c_1q_1 + \cdots + c_nq_n$ in the basis $q_1, \ldots, q_n$ gives $w = c_1Tq_1 + \cdots + c_nTq_n$. An independent spanning list is a basis (definition 2.2.1), so $Tq_1, \ldots, Tq_n$ is an orthonormal basis. This proves (4), and since the basis $q_1, \ldots, q_n$ was arbitrary it proves the last clause of the theorem.

Step 7: (4) implies (1). Let $q_1, \ldots, q_n$ be as in (4) and fix j. For every i, definition 5.5.8 and the orthonormality of the two lists give

$$\langle q_i, T^*Tq_j\rangle = \langle Tq_i, Tq_j\rangle = \delta_{ij} = \langle q_i, q_j\rangle$$

so $\langle q_i, w\rangle = 0$ for $w = T^*Tq_j - q_j$ and every i, again by proposition 5.1.6(2). Conjugate symmetry (proposition 5.1.6(2)) turns this into $\langle w, q_i\rangle = 0$ for every i, and theorem 5.3.5(1) expands w in the orthonormal basis as $w = \sum_i \langle w, q_i\rangle q_i = 0$. So T^*T and id_V agree on the basis $q_1, \ldots, q_n$, hence are equal by proposition 2.4.3(1), which is (1).

Steps 1 to 4 give (1) $\Leftrightarrow$ (2) $\Leftrightarrow$ (3), Step 5 gives (1) $\Leftrightarrow$ (5) together with $TT^* = \mathrm{id}_V$, and Steps 6 and 7 give (1) $\Leftrightarrow$ (4) together with the last clause, completing the proof.

Condition (2) says that a unitary operator preserves every inner product, hence every length and every angle recorded by the inner product; condition (3) says that preserving lengths alone is already enough, so a candidate can be tested on norms without any inner product being computed. Condition (5) says that the inverse of a unitary operator is its adjoint, which for a matrix means that inversion is a conjugate transposition. Condition (4) is the one used when a construction has to produce such an operator: prescribing where one orthonormal basis goes, subject to the image being orthonormal, determines a unitary operator. In coordinates that last reading becomes a statement about the columns of a matrix.

Proposition 7.1.5: Unitary Matrices Are the Matrices with Orthonormal Columns

Let $A \in M_n(k)$ with columns $a_1, \ldots, a_n \in k^n$ and rows $r_1, \ldots, r_n$, each row read as a column of k^n. Give k^n the standard inner product. The following are equivalent.

1. A is unitary.
2. $a_1, \ldots, a_n$ is an orthonormal basis of k^n.
3. $r_1, \ldots, r_n$ is an orthonormal list in k^n.

When these hold, $A^*A = AA^* = I_n$, and A is invertible with $A^{-1} = A^*$.

***Proof*:**
Write $A = (a_{ij})$, so that the j-th column has entries $(a_j)_t = a_{tj}$ and the i-th row, read as a column, has entries $(r_i)_t = a_{it}$.

Step 1: the entries of A^*A and AA^* are inner products. By definition 5.5.12 the entry of A^* in row i and column t is $\overline{a_{ti}}$, so definition 1.4.1 gives the entry of A^*A in row i and column j as

$$\sum_{t=1}^{n} \overline{a_{ti}}\, a_{tj} = \sum_{t=1}^{n} (a_j)_t\, \overline{(a_i)_t} = \langle a_j, a_i\rangle$$

and the entry of AA^* in row i and column j as

$$\sum_{t=1}^{n} a_{it}\,\overline{a_{jt}} = \sum_{t=1}^{n} (r_i)_t\,\overline{(r_j)_t} = \langle r_i, r_j\rangle$$

Step 2: (1) is equivalent to (2). By Step 1 the identity $A^*A = I_n$ says exactly that $\langle a_j, a_i\rangle = \delta_{ij}$ for all i and j, which is the condition of definition 5.3.3 on the list $a_1, \ldots, a_n$. So (1) holds if and only if that list is orthonormal. It remains to see that an orthonormal list arising this way is a basis. The standard basis $e_1, \ldots, e_n$ of k^n is orthonormal, and $Ae_j = a_j$ by definition 1.4.1; when A is unitary the operator $x \mapsto Ax$ is unitary, so the last clause of theorem 7.1.4 carries $e_1, \ldots, e_n$ to an orthonormal basis, which is $a_1, \ldots, a_n$. Conversely (2) contains the orthonormality of the columns, which gives (1).

Step 3: (1) is equivalent to (3). If A is unitary then $AA^* = I_n$ by the corresponding clause of theorem 7.1.4, read through the correspondence between A and the operator $x \mapsto Ax$; by Step 1 this says $\langle r_i, r_j\rangle = \delta_{ij}$, which is (3). Conversely (3) gives $AA^* = I_n$ by Step 1, and $(A^*)^* = A$ by proposition 5.5.13(1), so $(A^*)^*A^* = I_n$ and A^* is unitary. The clause $TT^* = \mathrm{id}_V$ of theorem 7.1.4, applied to the operator $x \mapsto A^*x$, then gives $A^*(A^*)^* = I_n$, that is $A^*A = I_n$.

Step 4: the final clause. Suppose the three conditions hold. Then $A^*A = I_n$ by (1), and $AA^* = I_n$ was obtained in Step 3. Those two identities are the two conditions of definition 1.4.4 for the matrix A^*, so A is invertible with $A^{-1} = A^*$.

Example 7.1.6: The Rotations of $\mathbb{R}^2$

For $\theta \in \mathbb{R}$ let

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \in M_2(\mathbb{R})$$

Its two columns are $(\cos\theta, \sin\theta)^\top$ and $(-\sin\theta, \cos\theta)^\top$. Each has squared norm $\cos^2\theta + \sin^2\theta = 1$, and their inner product is $-\cos\theta\sin\theta + \sin\theta\cos\theta = 0$, so the columns are orthonormal and R_θ is orthogonal by proposition 7.1.5. By theorem 7.1.4(3) it changes no length, which is the geometric content of a rotation.

Its transpose is

$$R_\theta^\top = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} = R_{-\theta}$$

which equals R_θ exactly when $\sin\theta = -\sin\theta$, that is when $\sin\theta = 0$. So for every θ with $\sin\theta \neq 0$ the matrix R_θ is orthogonal and not symmetric.

Such an R_θ has no eigenvector in $\mathbb{R}^2$. Subtracting R_θ from tI_2 and expanding the resulting 2×2 determinant as $ad - bc$,

$$\chi_{R_\theta}(t) = \det\begin{pmatrix} t - \cos\theta & \sin\theta \\ -\sin\theta & t - \cos\theta \end{pmatrix} = (t - \cos\theta)^2 + \sin^2\theta$$

For $t \in \mathbb{R}$ the right side is a sum of two squares of which the second is positive, so it never vanishes: χ_{R_θ} has no real root, and by theorem 4.1.10 the matrix R_θ has no eigenvalue in $\mathbb{R}$, hence no eigenvector in $\mathbb{R}^2$ (definition 4.1.3). By theorem 4.2.2 it is therefore not diagonalizable over $\mathbb{R}$. Over $\mathbb{C}$ the same polynomial factors as $\big(t - (\cos\theta + i\sin\theta)\big)\big(t - (\cos\theta - i\sin\theta)\big)$, since the product of the two factors is $(t - \cos\theta)^2 - (i\sin\theta)^2 = (t - \cos\theta)^2 + \sin^2\theta$; the two eigenvalues of R_θ in $\mathbb{C}$ are $\cos\theta \pm i\sin\theta$.

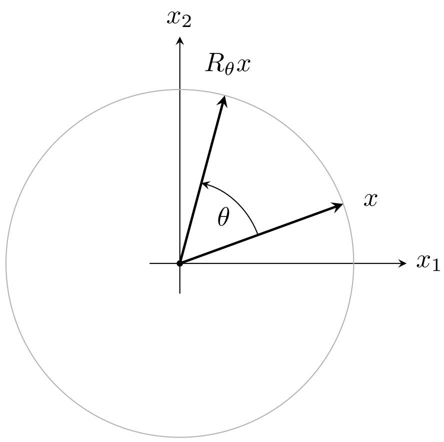


Figure 7.2: A rotation moves x along the circle through it, changing no length (example 7.1.6)

Neither of the two classes so far contains the other. The rotation R_θ with $\sin\theta \neq 0$ is orthogonal and not symmetric (example 7.1.6), while $2I_2$ is symmetric and satisfies $(2I_2)^*(2I_2) = 4I_2 \neq I_2$. What both classes do satisfy is one weaker condition: T commutes with T^*. For a self-adjoint T the two composites T^*T and TT^* are both T^2, and for a unitary T both are id_V. That condition is the one to name, since it admits operators lying in neither class.

Definition 7.1.7: Normal Operator and Normal Matrix

An operator $T\colon V \to V$ is *normal* if $T^*T = TT^*$. A matrix $A \in M_n(k)$ is *normal* if $A^*A = AA^*$.

Both earlier classes are normal. If $T^* = T$ then $T^*T = TT = TT^*$. If T is unitary then $T^*T = \mathrm{id}_V$ by definition 7.1.3 and $TT^* = \mathrm{id}_V$ by the last clause of theorem 7.1.4, so again the two products agree. In particular the rotation of example 7.1.6 is normal without being self-adjoint. The next example gives a matrix that is normal and in neither earlier class, and one that is not normal at all.

Example 7.1.8: A Non-Normal Shift and a Normal Diagonal Matrix

Let
$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in M_2(\mathbb{R})$$
the matrix carrying e_2 to e_1 and e_1 to 0. Here $N^* = N^\top = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, and definition 1.4.1 gives
$$N^*N = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad NN^* = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$
The two products differ, so N is not normal, and by the previous paragraph it is neither self-adjoint nor orthogonal.

Now let $D = \mathrm{diag}(d_1, \ldots, d_n) \in M_n(\mathbb{C})$ be any diagonal matrix. Its conjugate transpose is $\mathrm{diag}(\overline{d_1}, \ldots, \overline{d_n})$, and multiplying two diagonal matrices multiplies their diagonal entries in place, by definition 1.4.1, so
$$D^*D = \mathrm{diag}\big(|d_1|^2, \ldots, |d_n|^2\big) = DD^*$$

Every diagonal matrix is therefore normal. Taking $D = \operatorname{diag}(1, 2i) \in M_2(\mathbb{C})$ gives a normal matrix that is not self-adjoint, since $\overline{2i} = -2i \neq 2i$, and not unitary, since $D^*D = \operatorname{diag}(1, 4) \neq I_2$.

Normality is stated as an identity between two composites, and verifying it as written means computing both. The next two results replace it by a condition on lengths, which can be tested one vector at a time. The step that makes the replacement possible is a statement about self-adjoint operators alone: such an operator vanishes as soon as all the numbers $\langle Sv, v\rangle$ do.

Lemma 7.1.9: A Self-Adjoint Operator with Vanishing Quadratic Values Is Zero

Let V be a finite-dimensional inner product space over k and let $S\colon V \to V$ be self-adjoint. If $\langle Sv, v\rangle = 0$ for every $v \in V$, then $S = 0$.

***Proof*:**
Let $u, v \in V$. Linearity of S (definition 2.4.2), linearity of the inner product in its first argument (definition 5.1.2(1)) and additivity in its second (proposition 5.1.6(2)) expand the value at $u + v$ into four terms:

$$\langle S(u + v), u + v\rangle = \langle Su, u\rangle + \langle Su, v\rangle + \langle Sv, u\rangle + \langle Sv, v\rangle$$

The hypothesis makes the left side zero, and it makes the first and last terms on the right zero as well, leaving $\langle Su, v\rangle + \langle Sv, u\rangle = 0$. The second term is the conjugate of the first:

$$\langle Sv, u\rangle = \langle v, S^*u\rangle = \langle v, Su\rangle = \overline{\langle Su, v\rangle}$$

by definition 5.5.8, then $S^* = S$, then conjugate symmetry (proposition 5.1.6(2)). Adding a complex number to its conjugate doubles its real part, so

$$\operatorname{Re}\langle Su, v\rangle = 0 \qquad \text{for all } u, v \in V$$

Over $\mathbb{R}$ the inner product takes real values, so the last display already says $\langle Su, v\rangle = 0$ for all u and v. Over $\mathbb{C}$, apply it with iv in place of v: by proposition 5.1.6(2), $\langle Su, iv\rangle = \bar{i}\,\langle Su, v\rangle = -i\langle Su, v\rangle$, and writing $\langle Su, v\rangle = a + bi$ with $a, b \in \mathbb{R}$ gives $-i(a + bi) = b - ai$, whose real part is b. So $b = 0$, while $a = \operatorname{Re}\langle Su, v\rangle = 0$ by the display itself; hence $\langle Su, v\rangle = 0$ for all u and v over $\mathbb{C}$ as well.

Fix $u \in V$ and take $v = Su$. Then $\|Su\|^2 = \langle Su, Su\rangle = 0$, so $Su = 0$ by proposition 5.2.2(1). As u was arbitrary, $S = 0$, as we sought to show.

The hypothesis that S be self-adjoint cannot be dropped over $\mathbb{R}$. The rotation $J = R_{\pi/2} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ of example 7.1.6 satisfies $\langle Jx, x\rangle = -x_2x_1 + x_1x_2 = 0$ for every $x \in \mathbb{R}^2$, and $J \neq 0$.

Proposition 7.1.10: Normality Is the Agreement of $\|Tv\|$ with $\|T^*v\|$

Let V be a finite-dimensional inner product space over k and let $T\colon V \to V$ be linear. Then T is normal if and only if

$$\|Tv\| = \|T^*v\| \qquad \text{for every } v \in V$$

***Proof*:**
Put $S = T^*T - TT^*$, an operator on V.

Step 1: S is self-adjoint. By proposition 5.5.9(2) and (3),

$$(T^*T)^* = T^*(T^*)^* = T^*T, \qquad (TT^*)^* = (T^*)^*T^* = TT^*$$

and proposition 5.5.9(1), applied to the sum of T^*T and $(-1)TT^*$ with $\overline{-1} = -1$, gives $S^* = (T^*T)^* - (TT^*)^* = T^*T - TT^* = S$.

Step 2: the values of S measure the two norms. Let $v \in V$. The identity of definition 5.5.8 for the operator T^*, together with $(T^*)^* = T$ from proposition 5.5.9(3), gives $\langle T^*w, v\rangle = \langle w, Tv\rangle$ for all w; taking $w = Tv$ makes the left side $\langle T^*Tv, v\rangle$ and the right side $\langle Tv, Tv\rangle = \|Tv\|^2$. The same identity for T gives $\langle Tw, v\rangle = \langle w, T^*v\rangle$; taking $w = T^*v$ makes the left side $\langle TT^*v, v\rangle$ and the right side $\|T^*v\|^2$. Subtracting, and using linearity of the inner product in its first argument (definition 5.1.2(1)),

$$\langle Sv, v\rangle = \|Tv\|^2 - \|T^*v\|^2$$

Step 3: both directions. If T is normal then $S = 0$, so Step 2 gives $\|Tv\|^2 = \|T^*v\|^2$ for every v; both numbers are nonnegative reals (proposition 5.2.2(1)), so $\|Tv\| = \|T^*v\|$. Conversely, if the two norms agree for every v then Step 2 gives $\langle Sv, v\rangle = 0$ for every v, and S is self-adjoint by Step 1, so $S = 0$ by lemma 7.1.9. That is $T^*T = TT^*$, completing the proof.

The criterion is a single equality of numbers, checkable one vector at a time, and a single vector can refute normality: for the shift N of example 7.1.8 the image $Ne_1 = 0$ has norm 0, while $N^*e_1 = e_2$ has norm 1.

The eigenvalues of an operator are constrained by which of the three classes it belongs to, and the constraints are visible in the examples already computed: the reflection of example 7.1.2 has eigenvalues 1 and -1, both real, and the rotation of example 7.1.6 has eigenvalues $\cos\theta \pm i\sin\theta$, both of modulus $\sqrt{\cos^2\theta + \sin^2\theta} = 1$. Each is an instance of a general statement.

Proposition 7.1.11: The Eigenvalues of a Self-Adjoint Operator Are Real

Let V be a finite-dimensional inner product space over k, let $T\colon V \to V$ be self-adjoint, and suppose $Tv = \lambda v$ for some $\lambda \in k$ and some $v \in V$ with $v \neq 0$. Then $\lambda \in \mathbb{R}$.
In particular, if $A \in M_n(\mathbb{C})$ satisfies $A^* = A$, then every eigenvalue of A (definition 4.1.2) is real, and hence so is every root of χ_A.

***Proof*:**
Using linearity of the inner product in its first argument (definition 5.1.2(1)), then the hypothesis $Tv = \lambda v$, then definition 5.5.8 with $T^* = T$, then conjugate-linearity in the second argument (proposition 5.1.6(2)),

$$\lambda\|v\|^2 = \langle \lambda v, v\rangle = \langle Tv, v\rangle = \langle v, Tv\rangle = \langle v, \lambda v\rangle = \overline{\lambda}\,\|v\|^2$$

where $\|v\|^2 = \langle v, v\rangle$ by definition 5.2.1. Since $v \neq 0$, the number $\|v\|^2$ is a positive real (proposition 5.2.2(1)), so it may be cancelled, leaving $\lambda = \overline{\lambda}$. A scalar equal to its own conjugate is real.

For the matrix statement, let λ be an eigenvalue of A, so $Av = \lambda v$ for some $v \neq 0$ in $\mathbb{C}^n$ (definition 4.1.2). The operator $x \mapsto Ax$ on $\mathbb{C}^n$ is self-adjoint, by the correspondence between A and that operator recorded after definition 7.1.1, so the first part applies and gives $\lambda \in \mathbb{R}$. Over $\mathbb{C}$ the roots of χ_A are exactly the eigenvalues of A (theorem 4.1.10), completing the proof.

Proposition 7.1.12: The Eigenvalues of a Unitary Operator Have Modulus One

Let V be a finite-dimensional inner product space over k, let $T\colon V \to V$ be unitary, and suppose $Tv = \lambda v$ for some $\lambda \in k$ and some $v \in V$ with $v \neq 0$. Then $|\lambda| = 1$.
In particular, every eigenvalue of a unitary $A \in M_n(k)$ has modulus 1.

***Proof*:**
By theorem 7.1.4(3) and then proposition 5.2.2(2),

$$\|v\| = \|Tv\| = \|\lambda v\| = |\lambda|\,\|v\|$$

Since $v \neq 0$, the number $\|v\|$ is a positive real (proposition 5.2.2(1)) and may be cancelled, giving $|\lambda| = 1$. The matrix statement is this one applied to the unitary operator $x \mapsto Ax$ on k^n, whose eigenvectors are exactly those of A, as we sought to show.

For a normal operator neither restriction holds: the diagonal matrix $\operatorname{diag}(1, 2i)$ of example 7.1.8 is normal, and its eigenvalues are its diagonal entries 1 and $2i$ (proposition 4.1.12), the second being neither real nor of modulus 1. What survives is a statement relating T and T^* on the same vector, and the two propositions that close the section both rest on it.

Proposition 7.1.13: An Eigenvector of a Normal Operator Is an Eigenvector of Its Adjoint

Let V be a finite-dimensional inner product space over k, let $T\colon V \to V$ be normal, and let $v \in V$ satisfy $Tv = \lambda v$ for some $\lambda \in k$. Then

$$T^* v = \overline{\lambda}\, v$$

In particular, if $A \in M_n(k)$ is normal and $Av = \lambda v$, then $A^* v = \overline{\lambda} v$.

***Proof*:**
Step 1: the adjoint of $\lambda \mathrm{id}_V$. For all $u, w \in V$ one has $\langle \mathrm{id}_V u, w\rangle = \langle u, w\rangle = \langle u, \mathrm{id}_V w\rangle$, so $\mathrm{id}_V^* = \mathrm{id}_V$ by the uniqueness clause of theorem 5.5.7. Proposition 5.5.9(1) then gives $(\lambda\,\mathrm{id}_V)^* = \overline{\lambda}\,\mathrm{id}_V$.

Step 2: $S = T - \lambda\,\mathrm{id}_V$ *is normal.* By proposition 5.5.9(1) and Step 1, $S^* = T^* - \overline{\lambda}\,\mathrm{id}_V$. Composition of linear maps distributes over addition: for operators F, G, L on V and $v \in V$, $\big((F+G)L\big)v = F(Lv)+G(Lv)$ directly from the definition of a sum of maps, and $\big(F(G+L)\big)v = F(Gv + Lv) = F(Gv) + F(Lv)$ by linearity of F. Expanding both composites accordingly,

$$S^*S = T^*T - \lambda T^* - \overline{\lambda}T + |\lambda|^2\,\mathrm{id}_V, \qquad SS^* = TT^* - \overline{\lambda}T - \lambda T^* + |\lambda|^2\,\mathrm{id}_V$$

where $\lambda\overline{\lambda} = |\lambda|^2$. The two right-hand sides differ only in their first terms, which agree because T is normal, so $S^*S = SS^*$.

Step 3: conclusion. Since S is normal, proposition 7.1.10 gives $\|Sv\| = \|S^*v\|$. The hypothesis says $Sv = Tv - \lambda v = 0$, so $\|S^*v\| = \|Sv\| = 0$ and therefore $S^*v = 0$ by proposition 5.2.2(1). That is $T^*v = \overline{\lambda}v$. The matrix statement is this one applied to the normal operator $x \mapsto Ax$ on k^n, whose adjoint is $x \mapsto A^*x$ by theorem 5.5.14(2), completing the proof.

Proposition 7.1.14: Eigenvectors of a Normal Operator for Distinct Eigenvalues Are Orthogonal

Let V be a finite-dimensional inner product space over k, let $T\colon V \to V$ be normal, and let $u, v \in V$ satisfy $Tu = \lambda u$ and $Tv = \mu v$ with $\lambda \neq \mu$. Then $\langle u, v\rangle = 0$.
In particular, eigenvectors of a normal matrix $A \in M_n(k)$ belonging to distinct eigenvalues are orthogonal in k^n.

***Proof*:**
By linearity of the inner product in its first argument (definition 5.1.2(1)), then the hypothesis on u, then definition 5.5.8, then proposition 7.1.13 applied to v, then conjugate-linearity in the second argument (proposition 5.1.6(2)),

$$\lambda\langle u, v\rangle = \langle \lambda u, v\rangle = \langle Tu, v\rangle = \langle u, T^*v\rangle = \langle u, \overline{\mu}\,v\rangle = \mu\langle u, v\rangle$$

the last equality because the conjugate of $\overline{\mu}$ is μ. Hence $(\lambda - \mu)\langle u, v\rangle = 0$, and $\lambda \neq \mu$ forces $\langle u, v\rangle = 0$, which is orthogonality in the sense of definition 5.3.1. The matrix statement is this one applied to the normal operator $x \mapsto Ax$ on k^n, as we sought to show.

Both propositions use the hypothesis of normality, and both conclusions fail without it. For the shift N of example 7.1.8 the vector e_1 satisfies $Ne_1 = 0$, so it is an eigenvector for the eigenvalue 0; but $N^*e_1 = e_2 \neq 0$, so the conclusion of proposition 7.1.13 fails for N. For $B = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}$, which is not normal because

$$B^\top B = \begin{pmatrix} 1 & 1 \\ 1 & 5 \end{pmatrix} \qquad \text{and} \qquad BB^\top = \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}$$

the columns e_1 and $(1,1)^\top$ are eigenvectors for the distinct eigenvalues 1 and 2 (proposition 4.1.12), since $Be_1 = e_1$ and $B(1,1)^\top = (2,2)^\top$; their inner product is $\langle e_1, (1,1)^\top\rangle = 1 \neq 0$, so the conclusion of proposition 7.1.14 fails as well.

The rotation shows the two propositions at work over $\mathbb{C}$. Fix θ with $\sin\theta \neq 0$, write $R = R_\theta$ and $\lambda = \cos\theta + i\sin\theta$, and read R as an element of $M_2(\mathbb{C})$. Its entries are real, so its conjugate transpose is still $R^\top$. Example 7.1.6 found the columns of R to be orthonormal, so the closing clause of proposition 7.1.5 gives $R^*R = I_2 = RR^*$, and R is normal in $M_2(\mathbb{C})$. Solving $(R - \lambda I_2)v = 0$ uses the first row $-i\sin\theta\, v_1 - \sin\theta\, v_2 = 0$, which gives $v_2 = -iv_1$; taking $v_1 = 1$ and $v_2 = -i$, the second row evaluates to $\sin\theta \cdot 1 + (-i\sin\theta)(-i) = \sin\theta + i^2\sin\theta = 0$. So $v = (1, -i)^\top$ is an eigenvector for λ, and the same computation with $\overline{\lambda}$ gives the eigenvector $w = (1, i)^\top$. Three predictions can now be checked. The modulus of λ is 1, as proposition 7.1.12 requires. The adjoint acts on v by $\overline{\lambda}$, as proposition 7.1.13 requires:

$$R^*v = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} \cos\theta - i\sin\theta \\ -\sin\theta - i\cos\theta \end{pmatrix} = (\cos\theta - i\sin\theta) \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

the second entry because $(\cos\theta - i\sin\theta)(-i) = -i\cos\theta - \sin\theta$. And the two eigenvectors are orthogonal, as proposition 7.1.14 requires: $\langle v, w\rangle = 1 \cdot \overline{1} + (-i)\overline{i} = 1 + (-i)(-i) = 1 - 1 = 0$.

Exercise 7.1.1

1. Let
$$A = \begin{pmatrix} 2 & 1+i \\ 1-i & 3 \end{pmatrix} \in M_2(\mathbb{C})$$
Show that A is self-adjoint, hence normal, and that it is not unitary. Compute χ_A and its roots, confirming that both are real, and find an eigenvector for each; verify directly that the two eigenvectors are orthogonal in $\mathbb{C}^2$.

2. Let
$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \in M_2(\mathbb{C})$$
Show that U is unitary and not self-adjoint. Compute its two eigenvalues, check that each has modulus 1, and find an eigenvector for each; verify that the two are orthogonal.

3. Let $u = \frac{1}{3}(1, 2, 2)^\top \in \mathbb{R}^3$ and $H = I_3 - 2uu^\top$. Verify that $\|u\| = 1$, compute the nine entries of H, and check that its columns are orthonormal, so that H is both symmetric and orthogonal. Then show that $Hu = -u$ and that $Hx = x$ for every x orthogonal to u, and use this to name the eigenvalues of H and the dimensions of their eigenspaces.

4. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{R})$. Compute $A^\top A$ and $AA^\top$ entry by entry and deduce that A is normal if and only if either A is symmetric or $A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ for some $a, b \in \mathbb{R}$.

5. Let T be a normal operator on a finite-dimensional inner product space V. Show that $\ker T = \ker T^*$, and deduce from proposition 5.5.9(4) that $\mathrm{im}T = \mathrm{im}T^*$.

6. Let T be a unitary operator on V. Show that T is self-adjoint if and only if $T^2 = \mathrm{id}_V$. Then decide for which θ the rotation R_θ of example 7.1.6 satisfies the condition, and check that the swap S of example 7.1.2 does.

7. Let F and G be self-adjoint operators on V. Show that $F + G$ is self-adjoint, and that the composite FG is self-adjoint if and only if $FG = GF$. Show also that T^*T and $T + T^*$ are self-adjoint for every operator T on V.

8. Let V be a finite-dimensional inner product space over $\mathbb{C}$ and let T be an operator on V. Show that there are unique self-adjoint operators A and B on V with $T = A + iB$, and identify them in terms of T and T^*. Then show that T is normal if and only if $AB = BA$.

7.2 Schur Triangularization

Section 7.1 sorted operators by their relation to the adjoint and recorded what each class does to eigenvalues and eigenvectors. None of that sorting produced a basis. This section produces one, for every operator on a complex inner product space at once, at the price of asking for less than a diagonal matrix.

Chapter 4 identified the obstruction. An operator is diagonalizable exactly when its eigenvectors span the space (theorem 4.2.2), and an operator with too few eigenvectors has no diagonal matrix in any basis whatever. Upper triangular is the shape that is always available over $\mathbb{C}$, and the basis achieving it may be taken orthonormal, which is what makes the associated change of coordinates unitary.

The construction uses one eigenvector to cut the dimension by one. If u is an eigenvector of the adjoint T^*, then the orthogonal complement of $\operatorname{span}\{u\}$ is carried into itself by T, and the restriction of T to that complement is an operator on a space of dimension $n-1$, where induction takes over. The eigenvector of T^* rather than of T is what makes the complement invariant: the defining property $\langle Tw, u\rangle = \langle w, T^*u\rangle$ of the adjoint (definition 5.5.8) converts a condition on Tw into a condition on T^*u, and it is T^*u that is under control.

Theorem 7.2.1: Schur Triangularization

Let $n \geq 1$.

1. Let V be a complex inner product space with $\dim V = n$ and let $T\colon V \to V$ be linear. Then V has an orthonormal basis $u_1, \ldots, u_n$ in which the matrix of T is upper triangular, meaning that its entry in row i and column j is 0 whenever $i > j$.
2. Let $A \in M_n(\mathbb{C})$. Then there are a unitary $U \in M_n(\mathbb{C})$ and an upper triangular $R \in M_n(\mathbb{C})$ with
$$U^*AU = R \qquad \text{and} \qquad A = URU^*$$

***Proof*:**

1. The argument is an induction on n.

   Base case $n = 1$. By corollary 5.3.11(1) the space V has an orthonormal basis, which here is a single vector u_1. The matrix of T in it is a 1×1 matrix, and there is no pair $i > j$ with $1 \leq i, j \leq 1$, so the condition on the entries holds vacuously.

   Inductive step. Let $n \geq 2$ and assume the statement for every complex inner product space of dimension $n-1$. Let $\dim V = n$ and let $T\colon V \to V$ be linear.

   Step 1: the adjoint has a unit eigenvector. The adjoint $T^*\colon V \to V$ exists and is linear by theorem 5.5.7. Fix an orthonormal basis $b_1, \ldots, b_n$ of V (corollary 5.3.11(1)) and let $B \in M_n(\mathbb{C})$ be the matrix of T^* relative to it (theorem 2.4.8). Since B is a complex $n \times n$

matrix with $n \geq 1$, theorem 4.1.14 gives $\mu \in \mathbb{C}$ and a nonzero $z \in \mathbb{C}^n$ with $Bz = \mu z$ (definition 4.1.2, definition 4.1.3). Put $w = \sum_{i=1}^n z_i b_i$, the vector of V whose coordinate vector in the basis $b_1, \ldots, b_n$ is z. It is nonzero, since $z \neq 0$ and $b_1, \ldots, b_n$ is linearly independent (definition 2.2.1). By theorem 2.4.8 the coordinate vector of T^*w is $Bz = \mu z$, and μz is the coordinate vector of μw; a vector is determined by its coordinate vector in a basis (definition 2.2.1), so

$$T^*w = \mu w$$

Set $u = w/\|w\|$, which is defined because $\|w\| \neq 0$ (proposition 5.2.2(1)). Then $\|u\| = 1$ by the homogeneity of the norm (proposition 5.2.2), and $T^*u = \mu u$ because T^* is linear.

Step 2: the orthogonal complement of u is T-invariant. Let

$$W = \operatorname{span}\{u\}^\perp$$

be the orthogonal complement of the subspace $\operatorname{span}\{u\}$ (definition 5.5.1). A vector v lies in W exactly when $\langle v, u\rangle = 0$: one direction is the case $\alpha = 1$ of the defining condition, and conversely $\langle v, \alpha u\rangle = \bar{\alpha}\langle v, u\rangle$ for every $\alpha \in \mathbb{C}$, since the inner product is conjugate-linear in its second argument (definition 5.1.2). Now take $v \in W$. Then

$$\langle Tv, u\rangle = \langle v, T^*u\rangle = \langle v, \mu u\rangle = \bar{\mu}\langle v, u\rangle = 0$$

the first equality by definition 5.5.8, the second by Step 1, the third again by conjugate-linearity in the second argument, and the fourth because $v \in W$. So $Tv \in W$, and T restricts to a map $T|_W : W \to W$. That restriction is linear, since the identity $T(v + \lambda v') = Tv + \lambda Tv'$ of definition 2.4.2 holds in particular for $v, v' \in W$, and W with the inner product of V restricted to it satisfies every condition of definition 5.1.2, so W is again a complex inner product space.

Step 3: an orthonormal basis of W extends by u to one of V. By theorem 5.5.3, $V = \operatorname{span}\{u\} \oplus W$. Let $w_1, \ldots, w_m$ be an orthonormal basis of W, which exists by corollary 5.3.11(1). The list $w_1, \ldots, w_m, u$ is orthonormal: the first m vectors are orthonormal among themselves, $\|u\| = 1$, and $\langle w_i, u\rangle = 0$ for each i because $w_i \in W$. It spans V, since by the direct sum decomposition every $v \in V$ is $\alpha u + w'$ with $\alpha \in \mathbb{C}$ and $w' \in W$, and w' is a combination of $w_1, \ldots, w_m$. An orthonormal list is linearly independent (proposition 5.3.4), and an independent spanning list is a basis (definition 2.2.1), so $w_1, \ldots, w_m, u$ is a basis of V and therefore $m + 1 = n$ (definition 2.2.10). In particular $\dim W = n - 1$.

Step 4: induction and the shape of the matrix. Since $\dim W = n-1$, the inductive hypothesis applies to the operator $T|_W$ on W and produces an orthonormal basis $u_1, \ldots, u_{n-1}$ of W in which the matrix of $T|_W$ is upper triangular. Write r_{ij} for its entries, so that $r_{ij} = 0$ for $i > j$ and, by theorem 2.4.8,

$$Tu_j = (T|_W)u_j = \sum_{i=1}^{n-1} r_{ij} u_i = \sum_{i=1}^{j} r_{ij} u_i \qquad (1 \leq j \leq n-1)$$

Set $u_n = u$. By Step 3 the list $u_1, \ldots, u_{n-1}, u_n$ is an orthonormal basis of V, and Tu_n is some combination $\sum_{i=1}^n r_{in} u_i$ of it. The matrix of T in this basis has as its j-th column the coordinate vector of Tu_j (theorem 2.4.8). For $j \leq n-1$ that coordinate vector is $(r_{1j}, \ldots, r_{jj}, 0, \ldots, 0)$ by the display, so its entries in rows $i > j$ vanish; for $j = n$ there is no row below row n, so no entry of the last column is required to vanish. Every entry with $i > j$ is therefore 0, and the matrix is upper triangular.

2. Give $\mathbb{C}^n$ the standard inner product and let $T\colon \mathbb{C}^n \to \mathbb{C}^n$ be $T(x) = Ax$. This map is linear, since $\big(A(x+\lambda y)\big)_i = \sum_j a_{ij}(x_j + \lambda y_j) = (Ax)_i + \lambda(Ay)_i$ for every i (definition 1.4.1), which is the condition of definition 2.4.2. Part (1) gives an orthonormal basis $u_1, \dots, u_n$ of $\mathbb{C}^n$ and scalars r_{ij}, with $r_{ij} = 0$ for $i > j$, such that

$$Au_j = \sum_{i=1}^{n} r_{ij} u_i \qquad (1 \le j \le n) \tag{7.1}$$

Let $U \in M_n(\mathbb{C})$ be the matrix whose j-th column is u_j and let $R = (r_{ij}) \in M_n(\mathbb{C})$, which is upper triangular. Since the columns of U form an orthonormal basis of $\mathbb{C}^n$, the matrix U is unitary and $U^*U = UU^* = I_n$, by proposition 7.1.5 and its closing clause.

Now compare AU and UR column by column. The j-th column of AU is Au_j and the j-th column of UR is the combination of the columns of U with the entries of the j-th column of R as coefficients, namely $\sum_i r_{ij} u_i$ (definition 1.4.1); these agree by (7.1), so $AU = UR$. Multiplying on the left by U^* and using proposition 1.4.3 with $U^*U = I_n$ gives $U^*AU = R$; multiplying $AU = UR$ on the right by U^* and using $UU^* = I_n$ gives $A = URU^*$, completing the proof.

The two identities $U^*U = UU^* = I_n$ established in the proof say that U is invertible with $U^{-1} = U^*$ (definition 1.4.4), so the passage from A to $R = U^*AU$ is a similarity (definition 2.5.5) of a restricted kind: the conjugating matrix is unitary. Two features distinguish it from an arbitrary similarity. It preserves the inner product, since the columns of U are an orthonormal basis, so lengths and angles computed in the new coordinates are the ones computed in the old. And it is the numerically well-behaved kind of similarity, for reasons **??** makes precise.

Upper triangularity of the matrix of T is a statement about a chain of subspaces. Writing $V_j = \operatorname{span}\{u_1, \dots, u_j\}$, the displayed formula $Tu_j = \sum_{i \le j} r_{ij} u_i$ says that $Tu_j \in V_j$, and since V_j is spanned by $u_1, \dots, u_j$ this gives $T(V_j) \subseteq V_j$ for every j. So the theorem produces a chain $V_1 \subseteq V_2 \subseteq \cdots \subseteq V_n = V$ of subspaces of dimensions $1, 2, \dots, n$, each carried into itself by T, and each spanned by the first few members of one orthonormal basis. Diagonalizing asks for n independent one-dimensional invariant subspaces; triangularizing asks only for an increasing chain, and theorem 7.2.1 says that much is always available over $\mathbb{C}$. Which operators admit the stronger conclusion, a diagonal matrix in an orthonormal basis, is the question of section 7.3.

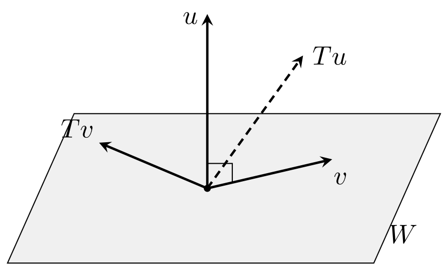


Figure 7.3: Step 2 of theorem 7.2.1: every v orthogonal to u has Tv orthogonal to u as well, while Tu need not lie in W

The theorem gives no information about the entries of R above the diagonal. The diagonal entries, however, are determined by A up to the order in which they are listed, because unitary similarity is similarity.

Corollary 7.2.2: The Diagonal of a Schur Form

Let $A \in M_n(\mathbb{C})$, let $U \in M_n(\mathbb{C})$ be unitary and let $R = U^*AU$ be upper triangular, with diagonal entries $r_{11}, \ldots, r_{nn}$. Then A and R have the same characteristic polynomial, and the eigenvalues of A are exactly the numbers $r_{11}, \ldots, r_{nn}$, each occurring on the diagonal of R as often as its multiplicity as a root of that polynomial, that is, as often as its algebraic multiplicity.

***Proof*:**
The columns of U form an orthonormal basis of $\mathbb{C}^n$, so U is unitary and invertible with $U^{-1} = U^*$ (proposition 7.1.5 and its closing clause), and $R = U^*AU = U^{-1}AU$ exhibits R as similar to A (definition 2.5.5). By corollary 4.1.16 similar matrices have the same characteristic polynomial, so the characteristic polynomial of A is that of R. The matrix R is upper triangular, so proposition 4.1.12 gives

$$\chi_A(t) = \chi_R(t) = (t - r_{11})(t - r_{22}) \cdots (t - r_{nn})$$

and identifies the eigenvalues of R, hence of A, as exactly the numbers $r_{11}, \ldots, r_{nn}$.

It remains to count. Let λ be one of them and let c be the number of indices i with $r_{ii} = \lambda$. Collecting those c factors in the display gives $\chi_A(t) = (t - \lambda)^c q(t)$, where $q(t)$ is the product of the remaining factors $t - r_{ii}$ over the indices with $r_{ii} \neq \lambda$. Evaluating, $q(\lambda) = \prod_{r_{ii} \neq \lambda}(\lambda - r_{ii})$ is a product of nonzero complex numbers, so $q(\lambda) \neq 0$. By proposition 4.3.5 the algebraic multiplicity of λ is c, which is the number of times λ appears on the diagonal of R, as we sought to show.

The trace and the determinant of A follow from the diagonal of R, and neither statement mentions U.

Corollary 7.2.3: Trace and Determinant from a Schur Form

Let $A \in M_n(\mathbb{C})$ and let $R = U^*AU$ be as in corollary 7.2.2, with diagonal entries $r_{11}, \ldots, r_{nn}$. Then

$$\operatorname{tr} A = \sum_{i=1}^{n} r_{ii} \qquad \text{and} \qquad \det A = \prod_{i=1}^{n} r_{ii}$$

***Proof*:**
The proof of corollary 7.2.2 established the factorization $\chi_A(t) = (t - r_{11}) \cdots (t - r_{nn})$, valid for every t. That is precisely the hypothesis of proposition 4.1.15, which then computes the trace of A as the sum of the numbers $r_{11}, \ldots, r_{nn}$ and the determinant of A as their product, completing the proof.

The left-hand sides depend only on A, so the sum and the product of the diagonal entries are the same for every unitary U triangularizing A, even though the entries themselves need not be: example 7.2.4(1) exhibits two Schur factorizations of one matrix whose entries above the diagonal differ.

Example 7.2.4: Triangularizing a 2×2 and a 3×3 Matrix

1. Let
$$A = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix} \in M_2(\mathbb{R}) \subseteq M_2(\mathbb{C})$$
Expanding the determinant of $\lambda I_2 - A$ along either row gives the characteristic polynomial
$$\det(\lambda I_2 - A) = (\lambda - 2)\lambda + 1 = (\lambda - 1)^2$$
so 1 is the only eigenvalue and its algebraic multiplicity is 2. The null space of $A - I_2 = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$ consists of the vectors with $x_1 + x_2 = 0$, so the eigenspace of 1 is spanned by $(1, -1)^\top$ and is one-dimensional (definition 4.1.4). Two independent eigenvectors would be needed for an eigenbasis of $\mathbb{C}^2$, so A is not diagonalizable (theorem 4.2.2).
Follow the proof. The conjugate transpose is the transpose here, since the entries are real (definition 5.5.12), and
$$A^* = \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}, \qquad A^* - I_2 = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix}$$
whose null space is spanned by $(1, 1)^\top$. Normalizing gives the unit eigenvector $u = \frac{1}{\sqrt{2}}(1, 1)^\top$ of A^*, with $\mu = 1$. The orthogonal complement $W = \operatorname{span}\{u\}^\perp$ is the set of $(x_1, x_2)^\top$ with $x_1 + x_2 = 0$, so $u_1 = \frac{1}{\sqrt{2}}(1, -1)^\top$ is an orthonormal basis of W, and u_1, u is an orthonormal basis of $\mathbb{C}^2$. The two products needed are
$$Au_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 2-1 \\ -1-0 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = u_1, \qquad Au = \frac{1}{\sqrt{2}}\begin{pmatrix} 2+1 \\ -1+0 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 3 \\ -1 \end{pmatrix}$$
The entry of U^*AU in row i and column j is $\langle Ac_j, c_i \rangle$, where $c_1 = u_1$ and $c_2 = u$ are the columns of U, since $(U^*AU)_{ij} = \sum_s \overline{(c_i)_s}(Ac_j)_s$ by definition 1.4.1. The four numbers are
$$\langle Au_1, u_1 \rangle = 1, \qquad \langle Au_1, u \rangle = \langle u_1, u \rangle = 0, \qquad \langle Au, u_1 \rangle = \tfrac{1}{2}(3+1) = 2, \qquad \langle Au, u \rangle = \tfrac{1}{2}(3-1) = 1$$
so, with
$$U = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \qquad R = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$
one has $U^*AU = R$. Multiplying out confirms it:
$$AU = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 3 \\ -1 & -1 \end{pmatrix}, \qquad U^*(AU) = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 3 \\ -1 & -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 & 4 \\ 0 & 2 \end{pmatrix} = R$$
The diagonal of R is $1, 1$, which is the eigenvalue 1 repeated twice, as corollary 7.2.2 requires; and $\operatorname{tr} A = 2 = 1 + 1$ and $\det A = 0 + 1 = 1 = 1 \cdot 1$, as corollary 7.2.3 requires. Replacing u by $-u$ leaves u_1 and the two diagonal entries unchanged and replaces $\langle Au, u_1 \rangle = 2$ by -2, so the second Schur factorization
$$\tilde{U} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}, \qquad \tilde{R} = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$$
also satisfies $\tilde{U}^*A\tilde{U} = \tilde{R}$. Neither U nor R is determined by A.

2. Let
$$A = \begin{pmatrix} 1 & 2 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \in M_3(\mathbb{R}) \subseteq M_3(\mathbb{C})$$
Expanding the determinant of $\lambda I_3 - A$ along its third row, whose entries are -1, 0 and $\lambda - 1$, gives
$$\det(\lambda I_3 - A) = -\big(2 + (\lambda-1)\big) + (\lambda-1)\big((\lambda-1)^2 - 2\big) = (\lambda-1)^3 - 3\lambda + 1 = \lambda^2(\lambda - 3)$$
so the eigenvalues are 0, with algebraic multiplicity 2, and 3. Solving $Ax = 0$ gives $x_1 + 2x_2 + x_3 = 0$, $x_1 + x_2 + x_3 = 0$ and $x_1 + x_3 = 0$; subtracting the second equation from the first gives $x_2 = 0$, and then $x_3 = -x_1$, so the eigenspace of 0 is spanned by $(1, 0, -1)^\top$ and is one-dimensional. Again A has no eigenbasis and is not diagonalizable.

Every column of A sums to 3, so $A^*(1,1,1)^\top = 3(1,1,1)^\top$ and $u = \frac{1}{\sqrt{3}}(1,1,1)^\top$ is a unit eigenvector of A^* with $\mu = 3$. Its orthogonal complement W is the plane $x_1 + x_2 + x_3 = 0$, with orthonormal basis
$$q_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}, \qquad q_2 = \frac{1}{\sqrt{6}}\begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$$
Multiplying A by these three vectors gives
$$Aq_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \qquad Aq_2 = \frac{1}{\sqrt{6}}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \qquad Au = \frac{1}{\sqrt{3}}\begin{pmatrix} 4 \\ 3 \\ 2 \end{pmatrix}$$
and $\langle Aq_1, u\rangle = \langle Aq_2, u\rangle = 0$, confirming the invariance established in Step 2 of the proof. The matrix of the restriction $T|_W$ in the basis q_1, q_2 has entries $B_{ij} = \langle Aq_j, q_i\rangle$, namely
$$B = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{6} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}, \qquad \mathrm{tr}B = 0, \qquad \det B = -\tfrac{1}{4} + \tfrac{3}{12} = 0$$
so the characteristic polynomial of B is λ^2 and its only eigenvalue is 0. Running the same procedure on B: the null space of $B^* = B^\top$ is spanned by $(-\sqrt{3}, 1)^\top$, giving the unit vector $\frac{1}{2}(-\sqrt{3}, 1)^\top$, whose orthogonal complement in $\mathbb{R}^2$ is spanned by $\frac{1}{2}(1, \sqrt{3})^\top$. These are coordinate vectors relative to q_1, q_2; the corresponding vectors of W are
$$v_1 = \tfrac{1}{2}q_1 + \tfrac{\sqrt{3}}{2}q_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \qquad v_2 = -\tfrac{\sqrt{3}}{2}q_1 + \tfrac{1}{2}q_2 = \frac{1}{\sqrt{6}}\begin{pmatrix} -1 \\ 2 \\ -1 \end{pmatrix}$$
and v_1, v_2, u is an orthonormal basis of $\mathbb{C}^3$, as the six products $\langle v_1, v_2\rangle = \langle v_1, u\rangle = \langle v_2, u\rangle = 0$ and $\|v_1\| = \|v_2\| = \|u\| = 1$ confirm. With Q the matrix whose columns are v_1, v_2, u, the entries $\langle Ac_j, c_i\rangle$ assemble into
$$Q^*AQ = R = \begin{pmatrix} 0 & \frac{2}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ 0 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

The identity $AQ = QR$ checks the three columns at once: $Av_1 = 0$ because v_1 spans the eigenspace of 0; $Av_2 = \frac{1}{\sqrt{6}}(2,0,-2)^\top = \frac{2}{\sqrt{3}}v_1$; and $Au = \frac{1}{\sqrt{3}}(4,3,2)^\top = \frac{2}{\sqrt{6}}v_1 + 3u$, the last because $\frac{2}{\sqrt{6}}v_1 = \frac{1}{\sqrt{3}}(1,0,-1)^\top$ and $3u = \frac{1}{\sqrt{3}}(3,3,3)^\top$. The diagonal of R is $0, 0, 3$, the eigenvalues of A with their algebraic multiplicities. The entries above the diagonal cannot all be cleared by any change of basis whatever, since a diagonal matrix in some basis is exactly what theorem 4.2.2 denies A.

Every number produced along the way is real. The eigenvalue $\mu = 3$ of A^* used at the first step and the eigenvalue 0 of B used at the second are both real, so the homogeneous systems solved for the corresponding eigenvectors had real coefficients and were solved in $\mathbb{R}^3$ and $\mathbb{R}^2$. Corollary 7.2.5 states the general form of this.

Whether the construction can be carried out inside $\mathbb{R}$ depends only on the eigenvalues it uses. For a real matrix and a real eigenvalue, theorem 4.1.10 applied over $\mathbb{R}$ produces an eigenvector in $\mathbb{R}^n$, and every subsequent step is a computation with real numbers. The hypothesis that the eigenvalues be real cannot be dropped: the rotation $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ of the plane (compare example 7.1.6) has characteristic polynomial $\lambda^2 + 1$, whose roots $\pm i$ are not real, while the characteristic polynomial of a real upper triangular matrix has its diagonal entries as its roots (proposition 4.1.12) and so has only real ones.

Corollary 7.2.5: Real Schur Triangularization

Let $n \geq 1$ and let $A \in M_n(\mathbb{R})$ be a matrix every root of whose characteristic polynomial in $\mathbb{C}$ is real. Then there are $Q \in M_n(\mathbb{R})$ with

$$Q^\top Q = QQ^\top = I_n$$

and an upper triangular $R \in M_n(\mathbb{R})$ with $Q^\top AQ = R$, hence $A = QRQ^\top$. The diagonal entries of R are the eigenvalues of A, each occurring as often as its algebraic multiplicity.

***Proof*:**
Over $\mathbb{R}$ the conjugate transpose is the transpose (definition 5.5.12), so a real matrix Q with $Q^\top Q = QQ^\top = I_n$ is unitary, and the final sentence follows from corollary 7.2.2 applied to $Q^*AQ = R$ once Q and R are produced. The construction is again an induction on n, and the route differs from that of theorem 7.2.1 in one respect: the hypothesis available here is about the roots of the characteristic polynomial of A itself, so the vector used to cut the dimension is an eigenvector of A and is placed first rather than last.

Base case $n = 1$. Take $Q = (1)$ and $R = A$; a 1×1 matrix is upper triangular and $Q^\top Q = QQ^\top = I_1$.

Inductive step. Let $n \geq 2$ and assume the statement for $n - 1$.

Step 1: a real unit eigenvector of A. The characteristic polynomial of A is computed from the entries of A (definition 4.1.9), so it is the same polynomial whether A is read in $M_n(\mathbb{R})$ or in $M_n(\mathbb{C})$. Read in $M_n(\mathbb{C})$, the matrix A has an eigenvalue by theorem 4.1.14, and that eigenvalue is a root of the characteristic polynomial by theorem 4.1.10; call it λ_1, which is real by hypothesis. Reading A in $M_n(\mathbb{R})$ again and applying theorem 4.1.10 over $\mathbb{R}$ to the real number λ_1, there is a nonzero $v \in \mathbb{R}^n$ with $Av = \lambda_1 v$. Rescaling v by $1/\|v\|$, which is a real number, leaves v real and an eigenvector, and makes $\|v\| = 1$.

Step 2: an orthonormal basis of $\mathbb{R}^n$ *starting at* v. Let $W = \operatorname{span}\{v\}^\perp \subseteq \mathbb{R}^n$ (definition 5.5.1). By theorem 5.5.3, $\mathbb{R}^n = \operatorname{span}\{v\} \oplus W$, and the argument of Step 3 of the proof of theorem 7.2.1, which used only that v is a unit vector and that the decomposition is a direct sum, shows that an orthonormal basis $q_1, \ldots, q_m$ of W (corollary 5.3.11(1)) together with v is an orthonormal basis of $\mathbb{R}^n$, so $m = n-1$. Let $Q_1 \in M_n(\mathbb{R})$ have columns $c_1 = v$ and $c_i = q_{i-1}$ for $2 \le i \le n$. Its columns form an orthonormal basis of $\mathbb{R}^n$, so proposition 7.1.5 and its closing clause give $Q_1^\top Q_1 = Q_1 Q_1^\top = I_n$ and $Q_1^\top = Q_1^{-1}$.

Step 3: the shape of $M = Q_1^\top A Q_1$. By definition 1.4.1, the entry of M in row i and column j is $\sum_s (c_i)_s (Ac_j)_s = \langle Ac_j, c_i\rangle$, the inner product being the standard one on $\mathbb{R}^n$. For $j=1$ this is $\langle Av, c_i\rangle = \lambda_1\langle v, c_i\rangle$, which is λ_1 for $i=1$ and 0 for $i \ge 2$, by orthonormality of the columns. So

$$M = \begin{pmatrix} \lambda_1 & b^\top \\ 0 & B \end{pmatrix}, \qquad B_{ij} = \langle Aq_j, q_i\rangle, \qquad b_j = \langle Aq_j, v\rangle$$

with $B \in M_{n-1}(\mathbb{R})$ and $b \in \mathbb{R}^{n-1}$, the symbol 0 standing for the zero column of height $n-1$.

Step 4: B inherits the hypothesis. Since $Q_1^\top = Q_1^{-1}$, the matrices M and A are similar (definition 2.5.5) and have the same characteristic polynomial (corollary 4.1.16). Let $\nu \in \mathbb{C}$ be a root of the characteristic polynomial of B. If $\nu = \lambda_1$ then ν is real and there is nothing to prove, so assume $\nu \ne \lambda_1$. Reading B in $M_{n-1}(\mathbb{C})$, theorem 4.1.10 gives a nonzero $z \in \mathbb{C}^{n-1}$ with $Bz = \nu z$. Put

$$s = \frac{\sum_{j=1}^{n-1} b_j z_j}{\nu - \lambda_1}, \qquad y = (s, z_1, \ldots, z_{n-1})^\top \in \mathbb{C}^n$$

which is nonzero because z is. Computing My entry by entry with definition 1.4.1 and the shape of M from Step 3,

$$(My)_1 = \lambda_1 s + \sum_{j=1}^{n-1} b_j z_j = \lambda_1 s + (\nu - \lambda_1)s = \nu s, \qquad (My)_{i+1} = \sum_{j=1}^{n-1} B_{ij} z_j = (Bz)_i = \nu z_i$$

for $1 \le i \le n-1$, so $My = \nu y$ and ν is an eigenvalue of M. By theorem 4.1.10 it is a root of the characteristic polynomial of M, which is that of A, so ν is real by hypothesis. Every root of the characteristic polynomial of B is therefore real.

Step 5: the inductive hypothesis and the assembly. By Step 4 the matrix $B \in M_{n-1}(\mathbb{R})$ satisfies the hypothesis of the corollary in dimension $n-1$, so there are $Q_2 \in M_{n-1}(\mathbb{R})$ with $Q_2^\top Q_2 = Q_2 Q_2^\top = I_{n-1}$ and an upper triangular $R_1 \in M_{n-1}(\mathbb{R})$ with $Q_2^\top B Q_2 = R_1$. Let $Q_3 \in M_n(\mathbb{R})$ be the matrix with $(Q_3)_{11} = 1$, with $(Q_3)_{1j} = 0$ and $(Q_3)_{i1} = 0$ for $i, j \ge 2$, and with $(Q_3)_{i+1,j+1} = (Q_2)_{ij}$ for $1 \le i, j \le n-1$. Then definition 1.4.1 gives

$$(Q_3^\top Q_3)_{11} = 1, \qquad (Q_3^\top Q_3)_{1,j+1} = \sum_{s=1}^{n} (Q_3)_{s1}(Q_3)_{s,j+1} = 0, \qquad (Q_3^\top Q_3)_{i+1,j+1} = (Q_2^\top Q_2)_{ij}$$

for $1 \le i, j \le n-1$, the middle entry vanishing because the only nonzero entry of the first column of Q_3 is in row 1, where the $(j+1)$-st column of Q_3 has a 0; so $Q_3^\top Q_3 = I_n$, and the same computation with the roles of the two factors exchanged gives $Q_3 Q_3^\top = I_n$. Writing $N = Q_3^\top M Q_3$ and using definition 1.4.1 twice,

$$N_{11} = \lambda_1, \qquad N_{i+1,1} = \sum_{s=1}^{n} (Q_3)_{s,i+1} M_{s1} = 0, \qquad N_{i+1,j+1} = \sum_{s,t=2}^{n} (Q_2)_{s-1,i} M_{st} (Q_2)_{t-1,j} = (R_1)_{ij}$$

for $1 \le i, j \le n-1$: the second because $(Q_3)_{1,i+1} = 0$ while $M_{s1} = 0$ for $s \ge 2$, and the third because $(Q_3)_{s,i+1}$ and $(Q_3)_{t,j+1}$ vanish unless $s, t \ge 2$, where $M_{st} = B_{s-1,t-1}$. Since R_1 is upper triangular, so is N. Finally put $Q = Q_1 Q_3$. Then $Q^\top Q = Q_3^\top Q_1^\top Q_1 Q_3 = Q_3^\top Q_3 = I_n$ and likewise $QQ^\top = I_n$, using proposition 5.5.13 for $(Q_1 Q_3)^\top = Q_3^\top Q_1^\top$ and proposition 1.4.3 to regroup, and

$$Q^\top A Q = Q_3^\top (Q_1^\top A Q_1) Q_3 = Q_3^\top M Q_3 = N$$

is upper triangular with real entries, completing the induction and the proof.

Corollary 7.2.5 carries a hypothesis that theorem 7.2.1 does not, since a real matrix can have non-real eigenvalues. What it returns in exchange is a factorization in which no complex number occurs, so every quantity computed from A, Q and R is real. Both matrices of example 7.2.4 satisfy its hypothesis, and both factorizations computed there are real.

Exercise 7.2.1

1. Let $A = \begin{pmatrix} 3 & 1 \\ -1 & 1 \end{pmatrix} \in M_2(\mathbb{R})$. Compute the characteristic polynomial of A and show that A is not diagonalizable. Then follow the proof of theorem 7.2.1: find a unit eigenvector of A^*, complete it to an orthonormal basis of $\mathbb{C}^2$ by taking the orthogonal complement, and exhibit a unitary U and an upper triangular R with $U^*AU = R$. Verify the factorization by multiplying out U^*AU, and check the diagonal of R against corollary 7.2.2.

2. Let $A = \begin{pmatrix} 0 & 0 \\ 1 & i \end{pmatrix} \in M_2(\mathbb{C})$. Carry out the same construction, taking care that the inner product is conjugate-linear in its second argument, and produce a unitary U and an upper triangular R with $U^*AU = R$. Verify that the diagonal entries of R are the eigenvalues of A and that $\operatorname{tr} A$ and $\det A$ agree with corollary 7.2.3.

3. Let $A \in M_n(\mathbb{C})$ and let $R = U^*AU$ be a Schur form of A as in theorem 7.2.1(2). Show that 0 occurs among the diagonal entries of R if and only if there is a nonzero $x \in \mathbb{C}^n$ with $Ax = 0$, and that the number of times it occurs does not depend on the choice of U.

4. Let V be a complex inner product space of dimension n, let $T\colon V \to V$ be linear, and suppose $V_1 \subseteq V_2 \subseteq \cdots \subseteq V_n = V$ are subspaces with $\dim V_j = j$ and $T(V_j) \subseteq V_j$ for every j. Produce an orthonormal basis of V in which the matrix of T is upper triangular, by choosing a basis $v_1, \ldots, v_n$ of V with $\operatorname{span}\{v_1, \ldots, v_j\} = V_j$ for each j and applying theorem 5.3.9 to it. This is the converse of the reading of upper triangularity given after theorem 7.2.1.

5. Let $A \in M_n(\mathbb{C})$ satisfy $A^k = 0$ for some integer $k \ge 1$. Show that every eigenvalue of A is 0, deduce from corollary 7.2.2 that every diagonal entry of every Schur form of A is 0, and conclude that $\operatorname{tr} A = 0$ and $\det A = 0$. Show also that $A^n = 0$.

6. Let $A \in M_n(\mathbb{R})$ and suppose there is a $Q \in M_n(\mathbb{R})$ with $Q^\top Q = QQ^\top = I_n$ such that $Q^\top A Q$ is upper triangular. Show that every root in $\mathbb{C}$ of the characteristic polynomial of A is real. Together with corollary 7.2.5 this makes the hypothesis of that corollary exactly the right one. Then exhibit a matrix in $M_2(\mathbb{R})$ for which no such Q exists.

7. Let $A \in M_n(\mathbb{C})$, let $U \in M_n(\mathbb{C})$ be unitary and let $R = U^*AU$ be upper triangular with diagonal entries $r_{11}, \ldots, r_{nn}$. Show that $U^*A^2U = R^2$, that R^2 is upper triangular, and that its diagonal entries are $r_{11}^2, \ldots, r_{nn}^2$. Conclude that the eigenvalues of A^2 are the squares of the eigenvalues of A, counted with algebraic multiplicity.

7.3 The Spectral Theorem for Normal Operators over $\mathbb{C}$

Section 7.2 produced, for every operator on a finite-dimensional complex inner product space, an orthonormal basis in which the matrix of that operator is upper triangular. Section 7.1 produced a class of operators, those commuting with their own adjoint. This section combines the two: the triangular matrix of a normal operator has no entries above the diagonal at all.

The comparison that forces this is between two products. For $A = (a_{ij}) \in M_n(k)$, the entry of A^*A in row j and column j is $\sum_t |a_{tj}|^2$, the sum of the squared moduli down column j; the entry of AA^* in the same position is $\sum_t |a_{jt}|^2$, the sum of the squared moduli across row j. Normality equates the two numbers for every j, and triangularity puts the zeros of A in the column sum and the row sum at different places. In the 2×2 case that is already decisive: for $A = \left(\begin{smallmatrix} a & b \\ 0 & c \end{smallmatrix}\right)$ the entry in position $(1,1)$ is $|a|^2$ in A^*A and $|a|^2 + |b|^2$ in AA^*, so $A^*A = AA^*$ forces $|b|^2 = 0$ and hence $b = 0$. The general case is this computation run down the diagonal.

Lemma 7.3.1: A Normal Triangular Matrix Is Diagonal

Let k be $\mathbb{R}$ or $\mathbb{C}$ and let $A = (a_{ij}) \in M_n(k)$ be upper triangular, so that $a_{ij} = 0$ whenever $i > j$. If A is normal, that is if $A^*A = AA^*$, then A is diagonal.

***Proof*:**
Step 1: the two diagonals. Fix j. By definition 5.5.12 the entry of A^* in row j and column t is $\overline{a_{tj}}$, so definition 1.4.1 gives

$$(A^*A)_{jj} = \sum_{t=1}^{n} (A^*)_{jt}\, a_{tj} = \sum_{t=1}^{n} \overline{a_{tj}}\, a_{tj} = \sum_{t=1}^{n} |a_{tj}|^2$$

and, reading the same two results with the factors exchanged,

$$(AA^*)_{jj} = \sum_{t=1}^{n} a_{jt}\, (A^*)_{tj} = \sum_{t=1}^{n} a_{jt}\, \overline{a_{jt}} = \sum_{t=1}^{n} |a_{jt}|^2$$

The hypothesis $A^*A = AA^*$ says that these two numbers agree, for each j separately.

Step 2: induction down the diagonal. The claim to be proved is that $a_{jt} = 0$ for every $t > j$, for each $j = 1, \ldots, n$. Fix j and suppose the claim holds for every index smaller than j: that is, suppose $a_{it} = 0$ whenever $i < j$ and $t > i$. For $j = 1$ this supposition is empty and the argument below is unaffected by it.

Split the column sum of Step 1 at the diagonal. A term with $t > j$ has $a_{tj} = 0$ because A is upper triangular. A term with $t < j$ has $a_{tj} = 0$ by the supposition, applied with $i = t$, since $t < j$ and $j > t$. Only the term $t = j$ survives:

$$(A^*A)_{jj} = |a_{jj}|^2$$

Split the row sum the same way. A term with $t < j$ has $a_{jt} = 0$ by upper triangularity, so

$$(AA^*)_{jj} = |a_{jj}|^2 + \sum_{t>j} |a_{jt}|^2$$

Equating the two displays gives $\sum_{t>j} |a_{jt}|^2 = 0$. Each summand is a nonnegative real number, so a vanishing sum forces every summand to vanish, and $|a_{jt}| = 0$ gives $a_{jt} = 0$ for every $t > j$. This is the claim for j, and the induction is complete.

So $a_{jt} = 0$ whenever $t > j$, and $a_{jt} = 0$ whenever $t < j$ by upper triangularity. Every off-diagonal entry of A vanishes, which is what definition 4.2.1 asks of a diagonal matrix, as we sought to show.

Theorem 7.2.1 applies to every operator and asks nothing of it; the lemma applies to a matrix that is already triangular. Chaining them converts the algebraic identity $T^*T = TT^*$ into a statement about a basis, and the conversion runs in both directions, since an operator whose matrix in some orthonormal basis is diagonal commutes with its adjoint for the same reason two diagonal matrices commute.

Theorem 7.3.2: The Spectral Theorem for Normal Operators

Let V be a nonzero finite-dimensional inner product space over $\mathbb{C}$ and let $T\colon V \to V$ be linear. The following are equivalent:

1. T is normal, that is $T^*T = TT^*$;
2. V has an orthonormal basis every member of which is an eigenvector of T.

***Proof*:**
Throughout, $\mathcal{B}$ denotes an orthonormal basis of V and $[S]_{\mathcal{B}}$ the matrix of an operator S relative to it (theorem 2.4.8). Two facts are used in both directions. First, $[T^*]_{\mathcal{B}} = \big([T]_{\mathcal{B}}\big)^*$ by theorem 5.5.14(1), whose hypothesis is exactly that the basis is orthonormal. Second, $[T^*T]_{\mathcal{B}} = [T^*]_{\mathcal{B}}[T]_{\mathcal{B}}$ and $[TT^*]_{\mathcal{B}} = [T]_{\mathcal{B}}[T^*]_{\mathcal{B}}$ by theorem 2.4.9. Since theorem 2.4.8 assigns to each linear map exactly one matrix and to each matrix exactly one linear map, two operators on V are equal exactly when their matrices relative to $\mathcal{B}$ are equal.

Step 1: (2) implies (1). Let $\mathcal{B} = (u_1, \ldots, u_n)$ be an orthonormal basis with $Tu_j = \lambda_j u_j$ for scalars $\lambda_1, \ldots, \lambda_n$, as (2) gives (definition 4.1.3). By theorem 2.4.8 the j-th column of $[T]_{\mathcal{B}}$ is the coordinate column $[Tu_j]_{\mathcal{B}} = [\lambda_j u_j]_{\mathcal{B}}$, which is $\lambda_j e_j$ by definition 2.2.4. So $[T]_{\mathcal{B}} = D$, where $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.

By definition 5.5.12 the matrix D^* is $\operatorname{diag}(\overline{\lambda_1}, \ldots, \overline{\lambda_n})$. For any two diagonal matrices the product is diagonal: by definition 1.4.1 the entry of D^*D in row s and column t is $\sum_r (D^*)_{sr} d_{rt}$, the factor $(D^*)_{sr}$ vanishes unless $r = s$ and the factor d_{rt} vanishes unless $r = t$, so the whole sum vanishes unless $s = t$, and for $s = t$ it is $\overline{\lambda_s}\lambda_s = |\lambda_s|^2$. Exchanging the two factors changes nothing in that computation beyond the order of the two complex numbers multiplied, so

$$D^*D = DD^* = \operatorname{diag}\big(|\lambda_1|^2, \ldots, |\lambda_n|^2\big)$$

The matrices of T^*T and of TT^* relative to $\mathcal{B}$ are therefore equal, and by the uniqueness recorded above the two operators are equal, which is (1).

Step 2: (1) implies (2). Let T be normal. By theorem 7.2.1 there is an orthonormal basis $\mathcal{B} = (u_1, \ldots, u_n)$ of V for which $A := [T]_{\mathcal{B}}$ is upper triangular. By the two facts above, the matrix of T^*T relative to $\mathcal{B}$ is A^*A and the matrix of TT^* is AA^*; the operators T^*T and TT^* are equal by hypothesis, so their matrices are equal, giving $A^*A = AA^*$.

Thus A is upper triangular and normal, so A is diagonal by lemma 7.3.1; write $A = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$.

Reading theorem 2.4.8 in the direction used in Step 1, the j-th column of A is $[Tu_j]_{\mathcal{B}}$, and that column is $\lambda_j e_j$, so $Tu_j = \lambda_j u_j$. Each u_j has $\|u_j\| = 1$ by definition 5.3.3 and is in particular nonzero, so each u_j is an eigenvector of T (definition 4.1.3). The basis $\mathcal{B}$ is therefore an orthonormal basis of eigenvectors, completing the proof.

Theorem 4.2.2 characterized diagonalizability by the existence of an eigenbasis. That criterion cannot be checked by looking at the matrix: deciding it means computing eigenvectors and counting them. Normality is decided by two matrix multiplications. The spectral theorem exchanges the one condition for the other, and the exchange is available only over $\mathbb{C}$ and only for the stricter demand, an eigenbasis that is orthonormal rather than only independent. The implication between the two conditions runs one way: an orthonormal eigenbasis is an eigenbasis, so every normal operator is diagonalizable, while a diagonalizable operator need not be normal.

Part of the orthogonality was already available. Proposition 7.1.14 showed that any two eigenvectors of a normal operator belonging to different eigenvalues are orthogonal. What the theorem adds is that the eigenspaces span V, and that inside a single eigenspace an orthonormal basis may be chosen, which theorem 5.3.9 gives once the eigenspace is known to be there.

In coordinates the theorem is a factorization, and the factorization is the form in which the result is used.

Corollary 7.3.3: Normal Matrices Are Exactly the Unitarily Diagonalizable Ones

Let $A \in M_n(\mathbb{C})$. Then A is normal if and only if there are a unitary $U \in M_n(\mathbb{C})$ and a diagonal $D \in M_n(\mathbb{C})$ with

$$A = UDU^*$$

When this holds, the columns $u_1, \dots, u_n$ of U are an orthonormal basis of $\mathbb{C}^n$ with $Au_j = d_{jj}u_j$ for each j, and

$$\chi_A(t) = (t - d_{11})(t - d_{22}) \cdots (t - d_{nn})$$

so the diagonal entries of D are the eigenvalues of A, each repeated as often as its algebraic multiplicity.

***Proof*:**
Give $\mathbb{C}^n$ the standard inner product (example 5.1.3) and let $T_A \colon \mathbb{C}^n \to \mathbb{C}^n$ be the map $x \mapsto Ax$, which is linear.

Step 1: a normal matrix gives a normal operator. By theorem 5.5.14(2) the adjoint of T_A is T_{A^*}. For any $M, N \in M_n(\mathbb{C})$ and any x, associativity of the matrix product (proposition 1.4.3) gives $M(Nx) = (MN)x$, so $T_M \circ T_N = T_{MN}$. Hence

$$(T_A)^* T_A = T_{A^*A}, \qquad T_A (T_A)^* = T_{AA^*}$$

If A is normal then $A^*A = AA^*$, so these two operators coincide and T_A is normal (definition 7.1.7).

Step 2: the factorization. By theorem 7.3.2 applied to T_A, there is an orthonormal basis $u_1, \dots, u_n$ of $\mathbb{C}^n$ with $Au_j = \lambda_j u_j$ for each j. Let U be the matrix whose j-th column is u_j and put $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. The list $u_1, \dots, u_n$ is a basis of $\mathbb{C}^n$ consisting of eigenvectors of A, so theorem 4.2.2 says that U is invertible and $U^{-1}AU = D$. Its columns are orthonormal, so U is unitary and $U^{-1} = U^*$ by proposition 7.1.5. Substituting, $A = UDU^{-1} = UDU^*$.

Step 3: the converse. Suppose $A = UDU^*$ with U unitary and D diagonal, so that $U^*U = UU^* = I_n$. By proposition 5.5.13(3) applied twice and (1) applied to U,

$$A^* = (UDU^*)^* = (U^*)^*D^*U^* = UD^*U^*$$

Regrouping with proposition 1.4.3 and cancelling $U^*U = I_n$,

$$A^*A = UD^*(U^*U)DU^* = UD^*DU^*, \qquad AA^* = UD(U^*U)D^*U^* = UDD^*U^*$$

The computation in Step 1 of the proof of theorem 7.3.2 gives $D^*D = DD^*$ for any diagonal D, so the two right-hand sides agree and A is normal.

Step 4: the columns and the characteristic polynomial. Assume $A = UDU^*$ as above. The entry of U^*U in row s and column t is $\sum_r \overline{(u_s)_r}(u_t)_r = \langle u_t, u_s\rangle$ by definition 1.4.1 and the formula for the standard inner product, so $U^*U = I_n$ says exactly that $u_1, \ldots, u_n$ is an orthonormal list (definition 5.3.3); it is independent by proposition 5.3.4 and has length $n = \dim \mathbb{C}^n$, hence is a basis of $\mathbb{C}^n$ by corollary 2.2.13. Multiplying $A = UDU^*$ on the right by U and using $U^*U = I_n$ gives $AU = UD$. The j-th column of AU is Au_j and the j-th column of UD is $d_{jj}u_j$, both by definition 1.4.1, so $Au_j = d_{jj}u_j$. Finally $A = UDU^{-1}$ exhibits A and D as similar (definition 2.5.5), so $\chi_A = \chi_D$ by corollary 4.1.16; a diagonal matrix is in particular upper triangular, so proposition 4.1.12 computes $\chi_D(t)$ as the product of the factors $t - d_{jj}$, completing the proof.

Normality restricts the eigenvalues not at all: any diagonal matrix is normal, by corollary 7.3.3 with $U = I_n$, and its diagonal entries may be any complex numbers. The restrictions proved in section 7.1 belong to the self-adjoint and the unitary case. The next example runs the corollary on two matrices, the second of which is normal with eigenvalues that are neither real nor of modulus 1.

Example 7.3.4: Two Normal Matrices Diagonalized Unitarily

1. Let J be the quarter turn of the plane (example 4.1.13), read as a complex matrix:

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{C})$$

Its entries are real, so $J^* = J^\top = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ by definition 5.5.12, and

$$J^*J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad JJ^* = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

so J is normal. Its characteristic polynomial is

$$\chi_J(t) = \det\begin{pmatrix} t & 1 \\ -1 & t \end{pmatrix} = t^2 + 1 = (t-i)(t+i)$$

with roots i and $-i$, which are the eigenvalues by theorem 4.1.10.

For $\lambda = i$ the eigenspace is $\operatorname{null}(J - iI_2)$ (proposition 4.1.5), and

$$(J - iI_2)\begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix}\begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} -i+i \\ 1-1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

using $(-1)(-i) = i$ and $(-i)(-i) = i^2 = -1$. Likewise $(J + iI_2)(1, i)^\top = (i - i,\, 1 - 1)^\top = 0$. The two vectors have squared norms $|1|^2 + |-i|^2 = 2$ and $|1|^2 + |i|^2 = 2$, so dividing by $\sqrt{2}$ gives unit vectors

$$u_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-i\end{pmatrix}, \qquad u_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\i\end{pmatrix}, \qquad \langle u_1, u_2\rangle = \tfrac{1}{2}\big(1\cdot\bar{1} + (-i)\bar{i}\big) = \tfrac{1}{2}(1-1) = 0$$

where $(-i)\bar{i} = (-i)(-i) = -1$. So u_1, u_2 is an orthonormal list of length 2 in $\mathbb{C}^2$, hence an orthonormal basis (proposition 5.3.4, corollary 2.2.13).

Assembling them into a matrix and reading the eigenvalues in the same order,

$$U = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\-i & i\end{pmatrix}, \qquad U^* = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & i\\1 & -i\end{pmatrix}, \qquad D = \operatorname{diag}(i, -i)$$

The matrix U is unitary, as

$$U^*U = \frac{1}{2}\begin{pmatrix}1 & i\\1 & -i\end{pmatrix}\begin{pmatrix}1 & 1\\-i & i\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1+1 & 1-1\\1-1 & 1+1\end{pmatrix} = I_2$$

using $i(-i) = 1$ and $i \cdot i = -1$ in the first row and $(-i)(-i) = -1$, $(-i)i = 1$ in the second. Multiplying out the factorization of corollary 7.3.3 in two steps,

$$UD = \frac{1}{\sqrt{2}}\begin{pmatrix}i & -i\\1 & 1\end{pmatrix}, \qquad (UD)U^* = \frac{1}{2}\begin{pmatrix}i-i & -1-1\\1+1 & i-i\end{pmatrix} = \begin{pmatrix}0 & -1\\1 & 0\end{pmatrix} = J$$

where the entries of UD use $(-i)i = 1$ and $i(-i) = 1$ in the second row. The entries of J are real while those of U are not; section 7.4 takes up which real matrices can be diagonalized by a real matrix with orthonormal columns.

2. Let $B = I_2 + J = \left(\begin{smallmatrix}1 & -1\\1 & 1\end{smallmatrix}\right)$. Then $B^* = \left(\begin{smallmatrix}1 & 1\\-1 & 1\end{smallmatrix}\right)$ and

$$B^*B = \begin{pmatrix}1+1 & -1+1\\-1+1 & 1+1\end{pmatrix} = 2I_2 = \begin{pmatrix}1+1 & 1-1\\1-1 & 1+1\end{pmatrix} = BB^*$$

so B is normal. It is not self-adjoint, since $B^* \neq B$, and it is not unitary, since $B^*B = 2I_2 \neq I_2$.

Each eigenvector of J is again an eigenvector of B: from $Ju_1 = iu_1$ and $Ju_2 = -iu_2$,

$$Bu_1 = u_1 + Ju_1 = (1+i)u_1, \qquad Bu_2 = u_2 + Ju_2 = (1-i)u_2$$

So the same unitary U diagonalizes B, with $D' = \operatorname{diag}(1+i, 1-i) = I_2 + D$. Multiplying out confirms it: using $UU^* = I_2$ and distributing with proposition 1.4.3,

$$UD'U^* = U(I_2 + D)U^* = UU^* + UDU^* = I_2 + J = B$$

The eigenvalues $1 \pm i$ are not real and have modulus $\sqrt{2}$, which is consistent with B being neither self-adjoint nor unitary.

The factorization $A = UDU^*$ records the eigenvectors one column at a time, and those columns are not determined by A: when an eigenvalue is repeated, any orthonormal basis of its eigenspace may be used, and even a single column may be rescaled by any complex number of modulus 1. Grouping the columns by eigenvalue removes the choice. The eigenspaces are determined by A, and so is the orthogonal projection onto each of them, so a form of the theorem written in terms of those projections has no arbitrary content left in it.

Theorem 7.3.5: The Spectral Resolution of a Normal Matrix

Let $A \in M_n(\mathbb{C})$ be normal, let $\lambda_1, \dots, \lambda_m$ be its distinct eigenvalues, and for each i let $P_i \in M_n(\mathbb{C})$ be the orthogonal projection matrix (definition 6.1.4) onto the eigenspace $E_{\lambda_i}(A) \subseteq \mathbb{C}^n$. Then:

1. $\mathbb{C}^n = E_{\lambda_1}(A) \oplus \cdots \oplus E_{\lambda_m}(A)$, and $\langle x, y\rangle = 0$ whenever $x \in E_{\lambda_i}(A)$ and $y \in E_{\lambda_{i'}}(A)$ with $i \neq i'$;
2. $P_1 + \cdots + P_m = I_n$, and $P_i P_{i'} = 0$ whenever $i \neq i'$;
3. $A = \lambda_1 P_1 + \cdots + \lambda_m P_m$.

***Proof*:**
Step 1: the eigenbasis, grouped by eigenvalue. By corollary 7.3.3 there are a unitary U with columns $u_1, \dots, u_n$ and a diagonal D with $A = UDU^*$, the u_j forming an orthonormal basis of $\mathbb{C}^n$ with $Au_j = \mu_j u_j$, where $\mu_j = d_{jj}$. Each u_j has norm 1 and is in particular nonzero, so each μ_j is an eigenvalue of A (definition 4.1.2) and hence is one of $\lambda_1, \dots, \lambda_m$. Conversely each λ_i is a root of χ_A (theorem 4.1.10), and $\chi_A(t) = \prod_j (t - \mu_j)$ by corollary 7.3.3; a product of complex numbers vanishes only if one of its factors does, so $\lambda_i = \mu_j$ for at least one j. Put

$$S_i = \{\, j \,|\, \mu_j = \lambda_i \,\}$$

The sets $S_1, \dots, S_m$ are nonempty, pairwise disjoint because the λ_i are distinct, and their union is $\{1, \dots, n\}$.

Step 2: each eigenspace is spanned by its block of columns. Fix i. Every u_j with $j \in S_i$ satisfies $Au_j = \mu_j u_j = \lambda_i u_j$, hence lies in $\mathrm{null}(A - \lambda_i I_n) = E_{\lambda_i}(A)$ (proposition 4.1.5); the eigenspace is a subspace, so it contains their span (proposition 2.1.6). For the reverse inclusion let $x \in E_{\lambda_i}(A)$ and write $x = \sum_{j=1}^n c_j u_j$ in the basis. Then $Ax = \sum_j c_j \mu_j u_j$ and $\lambda_i x = \sum_j c_j \lambda_i u_j$, and $Ax = \lambda_i x$, so subtracting gives

$$\sum_{j=1}^{n} c_j (\mu_j - \lambda_i)\, u_j = 0$$

The u_j are linearly independent (proposition 5.3.4), so every coefficient vanishes: $c_j(\mu_j - \lambda_i) = 0$ for each j. For $j \notin S_i$ the factor $\mu_j - \lambda_i$ is nonzero, forcing $c_j = 0$. Hence x is a combination of the u_j with $j \in S_i$, and

$$E_{\lambda_i}(A) = \mathrm{span}\{\, u_j \,|\, j \in S_i \,\}$$

Those vectors are orthonormal, hence independent (proposition 5.3.4), so they are a basis of $E_{\lambda_i}(A)$ (definition 2.2.1) and $\dim E_{\lambda_i}(A) = |S_i|$.

Step 3: part (1). For orthogonality, let $i \neq i'$, let $x \in E_{\lambda_i}(A)$ and $y \in E_{\lambda_{i'}}(A)$, and write $x = \sum_{j \in S_i} c_j u_j$ and $y = \sum_{l \in S_{i'}} b_l u_l$ by Step 2. The inner product is linear in its first argument and conjugate-linear in its second (definition 5.1.2), so

$$\langle x, y\rangle = \sum_{j\in S_i}\sum_{l\in S_{i'}} c_j\overline{b_l}\,\langle u_j, u_l\rangle$$

and every pair occurring has $j \neq l$, since S_i and $S_{i'}$ are disjoint; each such $\langle u_j, u_l\rangle$ is 0 by definition 5.3.3. So $\langle x, y\rangle = 0$.

For the decomposition, let $x \in \mathbb{C}^n$ and write $x = \sum_{j=1}^n c_j u_j$. Grouping the indices by the sets S_i, which partition $\{1,\dots,n\}$, expresses x as $x_1 + \cdots + x_m$ with $x_i = \sum_{j\in S_i} c_j u_j \in E_{\lambda_i}(A)$. If also $x = x'_1 + \cdots + x'_m$ with $x'_i \in E_{\lambda_i}(A)$, then $\sum_i (x_i - x'_i) = 0$ with $x_i - x'_i \in E_{\lambda_i}(A)$; expanding each difference in the u_j with $j \in S_i$ produces a combination of the whole basis $u_1,\dots,u_n$ equal to zero, so all its coefficients vanish and $x_i = x'_i$ for every i. Every vector of $\mathbb{C}^n$ therefore has exactly one expression as a sum of vectors from the eigenspaces, which is the direct sum decomposition claimed (definition 2.1.13).

Step 4: the projections as sums of rank-one blocks. Fix i and let Q_i be the matrix whose columns are the u_j with $j \in S_i$, in increasing order of j. By Step 2 those columns are an orthonormal basis of $E_{\lambda_i}(A)$, which is the hypothesis of theorem 6.1.5(2); that result identifies the orthogonal projection matrix onto $E_{\lambda_i}(A)$ as $Q_iQ_i^*$, so $P_i = Q_iQ_i^*$. By definition 1.4.1 the entry of $Q_iQ_i^*$ in row s and column t is $\sum_{j\in S_i}(u_j)_s\overline{(u_j)_t}$, which is the entry in the same position of $\sum_{j\in S_i} u_ju_j^*$, each summand $u_ju_j^*$ having $(u_j)_s\overline{(u_j)_t}$ there. Hence

$$P_i = \sum_{j\in S_i} u_ju_j^*$$

Step 5: parts (2) and (3). Summing the last display over i and using that the S_i partition the indices, $\sum_{i=1}^m P_i = \sum_{j=1}^n u_ju_j^*$, and the entry computation of Step 4 applied to U itself identifies this with UU^*, which is I_n because U is unitary. For the products, note first that $u_s^*u_t$ is the 1×1 matrix with entry $\sum_r \overline{(u_s)_r}(u_t)_r = \langle u_t, u_s\rangle$, which is 0 for $s \neq t$ (definition 5.3.3). So for $i \neq i'$, regrouping each term with proposition 1.4.3,

$$P_iP_{i'} = \sum_{s\in S_i}\sum_{t\in S_{i'}} u_s(u_s^*u_t)u_t^* = 0$$

since every pair has $s \neq t$. For part (3), the entry of UDU^* in row s and column t is $\sum_j u_{sj}(DU^*)_{jt} = \sum_j \mu_j(u_j)_s\overline{(u_j)_t}$ by definition 1.4.1 and the diagonality of D, so $A = UDU^* = \sum_{j=1}^n \mu_ju_ju_j^*$. Grouping the indices by the sets S_i and using $\mu_j = \lambda_i$ for $j \in S_i$ together with Step 4,

$$A = \sum_{i=1}^m \lambda_i \sum_{j\in S_i} u_ju_j^* = \sum_{i=1}^m \lambda_iP_i$$

as we sought to show.

Nothing in the statement of the theorem mentions a basis. The eigenvalues λ_i are determined by A as the roots of χ_A, and each P_i is determined by A as the projection onto an eigenspace, so the expression $A = \sum_i \lambda_iP_i$ is attached to A alone. The next result uses two further identities, both given by theorem 6.1.7(1): each P_i, being an orthogonal projection matrix, satisfies $P_i^2 = P_i$ and $P_i^* = P_i$.

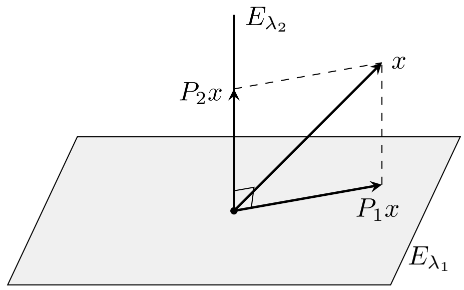


Figure 7.4: The resolution $x = P_1x + P_2x$ of theorem 7.3.5 for a matrix with two eigenvalues, one eigenspace drawn as a plane and the other as the line orthogonal to it

For the quarter turn J of example 7.3.4(1) the two eigenspaces are lines, spanned by u_1 and u_2, and Step 4 of the proof computes the projections as $P_1 = u_1u_1^*$ and $P_2 = u_2u_2^*$:

$$P_1 = \frac{1}{2}\begin{pmatrix}1\\-i\end{pmatrix}\begin{pmatrix}1 & i\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1 & i\\-i & 1\end{pmatrix}, \qquad P_2 = \frac{1}{2}\begin{pmatrix}1\\i\end{pmatrix}\begin{pmatrix}1 & -i\end{pmatrix} = \frac{1}{2}\begin{pmatrix}1 & -i\\i & 1\end{pmatrix}$$

where the entries in position $(2,2)$ are $(-i)\overline{(-i)} = (-i)(i) = 1$ and $i\bar{i} = i(-i) = 1$. Adding gives $P_1 + P_2 = I_2$; multiplying gives

$$P_1P_2 = \frac{1}{4}\begin{pmatrix}1-1 & -i+i\\-i+i & -1+1\end{pmatrix} = 0$$

and the resolution itself is

$$iP_1 + (-i)P_2 = \frac{i}{2}\begin{pmatrix}0 & 2i\\-2i & 0\end{pmatrix} = \begin{pmatrix}0 & -1\\1 & 0\end{pmatrix} = J$$

Both projections equal their own conjugate transpose, as theorem 6.1.7(1) requires.

In a product of two resolutions every cross term vanishes, by part (2) of the theorem, so the resolution computes powers of A without any matrix multiplication. The same computation handles an arbitrary polynomial in A, and it also runs backwards: each projection P_i is itself a polynomial in A.

Proposition 7.3.6: Polynomials in a Normal Matrix

Let $A \in M_n(\mathbb{C})$ be normal with distinct eigenvalues $\lambda_1, \dots, \lambda_m$ and projections $P_1, \dots, P_m$ as in theorem 7.3.5. Then:

1. $p(A) = p(\lambda_1)P_1 + \cdots + p(\lambda_m)P_m$ for every polynomial p with complex coefficients;
2. $P_i = p_i(A)$, where
$$p_i(t) = \prod_{l \neq i} \frac{t - \lambda_l}{\lambda_i - \lambda_l}$$
3. $A^* = q(A)$, where $q = \overline{\lambda_1}p_1 + \cdots + \overline{\lambda_m}p_m$.

***Proof*:**

1. First, $A^r = \sum_i \lambda_i^r P_i$ for every integer $r \geq 1$, by induction on r. The case $r = 1$ is theorem 7.3.5(3). Assuming the identity for r, expanding the product by distributivity (proposition 1.4.3) gives

$$A^{r+1} = A^r A = \Big(\sum_i \lambda_i^r P_i\Big)\Big(\sum_{i'} \lambda_{i'} P_{i'}\Big) = \sum_i \sum_{i'} \lambda_i^r \lambda_{i'} P_i P_{i'}$$

in which $P_i P_{i'} = 0$ for $i \neq i'$ by theorem 7.3.5(2) and $P_i P_i = P_i$ because P_i is an orthogonal projection matrix (theorem 6.1.7(1)). Only the diagonal terms survive, and they sum to $\sum_i \lambda_i^{r+1} P_i$.

Now write $p(t) = c_0 + c_1 t + \cdots + c_d t^d$, so that $p(A) = c_0 I_n + c_1 A + \cdots + c_d A^d$ (definition 4.4.1). Replacing I_n by $\sum_i P_i$, which is theorem 7.3.5(2), and each A^r by $\sum_i \lambda_i^r P_i$,

$$p(A) = \sum_i \Big(c_0 + c_1 \lambda_i + \cdots + c_d \lambda_i^d\Big) P_i = \sum_i p(\lambda_i) P_i$$

2. The eigenvalues $\lambda_1, \ldots, \lambda_m$ are distinct, so every denominator $\lambda_i - \lambda_l$ with $l \neq i$ is nonzero and p_i is a polynomial of degree $m-1$; when $m = 1$ the empty product is the constant polynomial 1. Evaluating, $p_i(\lambda_i)$ is a product of factors $(\lambda_i - \lambda_l)/(\lambda_i - \lambda_l) = 1$, so $p_i(\lambda_i) = 1$; and for $l' \neq i$ the factor indexed by $l = l'$ vanishes at $t = \lambda_{l'}$, so $p_i(\lambda_{l'}) = 0$. Part (1) applied to p_i therefore gives

$$p_i(A) = \sum_l p_i(\lambda_l) P_l = P_i$$

3. Taking conjugate transposes in theorem 7.3.5(3) and using proposition 5.5.13(2),

$$A^* = \Big(\sum_i \lambda_i P_i\Big)^* = \sum_i \overline{\lambda_i}\, P_i^* = \sum_i \overline{\lambda_i}\, P_i$$

the last equality because each P_i is self-adjoint (theorem 6.1.7(1)). On the other side, $q(\lambda_l) = \sum_i \overline{\lambda_i}\, p_i(\lambda_l) = \overline{\lambda_l}$ by the evaluations computed in (2), so part (1) applied to q gives $q(A) = \sum_l \overline{\lambda_l} P_l$. The two right-hand sides agree, so $A^* = q(A)$, completing the proof.

Part (3) says that the adjoint of a normal matrix is a polynomial in that matrix. The polynomial depends on the eigenvalues and so varies from one normal matrix to another. For the quarter turn, $\lambda_1 = i$ and $\lambda_2 = -i$ give $p_1(t) = (t+i)/(2i)$ and $p_2(t) = (t-i)/(-2i)$, whence

$$q(t) = \bar{i}\,\frac{t+i}{2i} + \overline{-i}\,\frac{t-i}{-2i} = -\frac{t+i}{2} - \frac{t-i}{2} = -t$$

and indeed $J^* = -J$. With the projections in hand, part (1) evaluates any polynomial in J with no further matrix multiplication: taking $p(t) = t^2$ gives $J^2 = i^2 P_1 + (-i)^2 P_2 = -(P_1 + P_2) = -I_2$, and squaring J directly gives the same matrix.

Exercise 7.3.1

1. Let $A = \left(\begin{smallmatrix} 2 & i \\ i & 2 \end{smallmatrix}\right) \in M_2(\mathbb{C})$. Show that A is normal but neither self-adjoint (that is, $A^* \neq A$) nor unitary, and find a unitary U and a diagonal D with $A = UDU^*$. Verify the factorization by multiplying it out.

2. Keep the matrix A of exercise 7.3.1 (1). Compute the two projections P_1 and P_2 of theorem 7.3.5 explicitly, check that $P_1 + P_2 = I_2$ and $P_1 P_2 = 0$, and check that $A = \lambda_1 P_1 + \lambda_2 P_2$. Then use proposition 7.3.6(3) to write A^* as a polynomial in A.

3. Let $A \in M_n(\mathbb{C})$ be normal. Using corollary 7.3.3, show that $A^* = A$ if and only if every eigenvalue of A is real, and that $A^*A = I_n$ if and only if every eigenvalue of A has modulus 1. (Section 7.1 proved one implication in each pair without assuming normality; the point here is the converse.)

4. Let $A \in M_n(\mathbb{C})$ be normal and let λ be an eigenvalue of A. Show that the geometric multiplicity of λ equals its algebraic multiplicity. (Use the characteristic polynomial computed in corollary 7.3.3 for one count and Step 2 of the proof of theorem 7.3.5 for the other.)

5. Exhibit a matrix in $M_2(\mathbb{C})$ that is diagonalizable (definition 4.2.1) but not normal, compute an eigenbasis for it, and say which requirement in theorem 7.3.2(2) that eigenbasis fails.

6. Let $A \in M_n(\mathbb{C})$ be normal and let p be a polynomial with complex coefficients. Show that $p(A)$ is normal, and that the eigenvalues of $p(A)$ are exactly the numbers $p(\lambda)$ with λ an eigenvalue of A.

7. Let $A = (a_{ij}) \in M_n(\mathbb{C})$ be normal and let $\lambda_1, \ldots, \lambda_n$ be its eigenvalues, each listed as often as its algebraic multiplicity. Show that
$$\sum_{i=1}^{n} \sum_{j=1}^{n} |a_{ij}|^2 = \sum_{j=1}^{n} |\lambda_j|^2$$
(Begin by identifying the left side as $\operatorname{tr}(A^*A)$, and use proposition 2.5.8.)

8. Let $A \in M_n(\mathbb{C})$ be normal and suppose $A^k = 0$ for some integer $k \geq 1$. Show that $A = 0$. Then exhibit a matrix $N \in M_2(\mathbb{C})$ with $N^2 = 0$ and $N \neq 0$, which by the first part cannot be normal.

7.4 The Spectral Theorem for Real Symmetric Matrices

Section 7.3 settled the question over $\mathbb{C}$: an operator on a finite-dimensional complex inner product space has an orthonormal basis of eigenvectors exactly when it is normal. Over $\mathbb{R}$ that criterion fails. A real matrix can satisfy $A^\top A = AA^\top$ and have no eigenvector anywhere in $\mathbb{R}^n$, and then no real change of coordinates, orthonormal or otherwise, makes it diagonal.

This section identifies the class of real matrices that does work and proves the theorem for it. The route is the real form of Schur triangularization (corollary 7.2.5) together with one observation about triangular matrices: a triangular matrix equal to its own transpose is diagonal.

A single matrix records the failure over $\mathbb{R}$, and it is one example 7.1.6 has already met as a normal operator that is not self-adjoint: the quarter turn of the plane, which rotates every nonzero vector off the line it spans and therefore fixes no direction at all.

Proposition 7.4.1: A Real Normal Matrix Need Not Have a Real Eigenvector

Let

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$$

Then $J^\top J = JJ^\top = I_2$, so J is normal. No nonzero vector of $\mathbb{R}^2$ is an eigenvector of J, and consequently there is no invertible $S \in M_2(\mathbb{R})$ for which $S^{-1}JS$ is diagonal.

***Proof*:**
The matrix J is the rotation R_θ of example 7.1.6 at $\theta = \pi/2$, where $\cos\theta = 0$ and $\sin\theta = 1$, and in particular $\sin\theta \neq 0$. That example establishes each of the three claims for such a θ: the columns of R_θ are orthonormal, so proposition 7.1.5 and its closing clause give $J^\top J = JJ^\top = I_2$ and J is normal; the characteristic polynomial $\chi_{R_\theta}(t) = (t-\cos\theta)^2 + \sin^2\theta$ has no real root, so J has no eigenvalue and hence no eigenvector in $\mathbb{R}^2$; and theorem 4.2.2 then rules out an invertible real S with $S^{-1}JS$ diagonal, since such an S would give a basis of $\mathbb{R}^2$ of eigenvectors. Its two eigenvalues in $\mathbb{C}$ are $\cos\theta \pm i\sin\theta = \pm i$.

Nothing about section 7.3 is at fault here. Read in $M_2(\mathbb{C})$ the same matrix does have an orthonormal eigenbasis: setting

$$u_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad u_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -i \end{pmatrix}$$

one computes $Ju_1 = \frac{1}{\sqrt{2}}(-i,1)^\top = -i\,u_1$ and $Ju_2 = \frac{1}{\sqrt{2}}(i,1)^\top = i\,u_2$, while $\langle u_1, u_2\rangle = \frac{1}{2}(1\cdot\overline{1} + i\cdot\overline{-i}) = \frac{1}{2}(1-1) = 0$ and $\|u_1\|^2 = \|u_2\|^2 = \frac{1}{2}(1+1) = 1$. This is the basis theorem 7.3.2 promises. What fails over $\mathbb{R}$ is that the eigenvalues $\pm i$ are not real, so the eigenvectors carrying them cannot be chosen in $\mathbb{R}^2$. Normality alone does not force real eigenvalues, and the real normal form that a matrix like J does admit is the subject of section 7.5.

The hypothesis over $\mathbb{R}$ therefore has to do two things: force the eigenvalues to be real, and keep the geometry. Self-adjointness does both. It forces real eigenvalues by proposition 7.1.11, and it is preserved by the transpose operation in the way the following proof exploits. In matrix terms a self-adjoint real matrix is a *symmetric* one: over $\mathbb{R}$ the conjugate transpose is the transpose (definition 5.5.12, definition 2.3.4), so the condition $A^* = A$ of definition 7.1.1 reads $A^\top = A$.

Theorem 7.4.2: The Spectral Theorem for Self-Adjoint Operators over $\mathbb{R}$

Let V be a finite-dimensional real inner product space with $\dim V = n \geq 1$, and let $T\colon V \to V$ be self-adjoint. Then there are an orthonormal basis $u_1, \dots, u_n$ of V and real numbers $\lambda_1, \dots, \lambda_n$ with $Tu_j = \lambda_j u_j$ for every j.

***Proof*:**
Step 1: record T as a symmetric matrix. By corollary 5.3.11 the space V has an orthonormal basis $e_1, \dots, e_n$. Let $A = (a_{ki}) \in M_n(\mathbb{R})$ be the matrix of T relative to this basis, so that $Te_i = \sum_{k=1}^n a_{ki}e_k$ (theorem 2.4.8). Because the basis is orthonormal, theorem 5.5.14 says that the matrix of T^* relative to it is the conjugate transpose A^*, which for a real matrix is $A^\top$ (definition 5.5.12, definition 2.3.4). A linear map is determined by its matrix relative to a fixed basis (theorem 2.4.8), so the hypothesis $T^* = T$ gives $A^\top = A$.

Step 2: every complex root of the characteristic polynomial of A is real. The polynomial $\det(tI_n - A)$ is computed from the entries of A by one formula, whichever of $M_n(\mathbb{R})$ and $M_n(\mathbb{C})$ the matrix is read in (definition 4.1.9), so the two readings have the same characteristic polynomial. Give $\mathbb{C}^n$ the standard inner product and let $S\colon \mathbb{C}^n \to \mathbb{C}^n$ be the map $S(x) = Ax$, whose matrix relative to the standard basis is A (theorem 2.4.8). The standard basis is orthonormal, so theorem 5.5.14 gives A^* as the matrix of S^*; the entries of A are real, so conjugating them changes nothing and $A^* = A^\top = A$ by Step 1. Hence S^* and S have the same matrix and are equal, which is definition 7.1.1 for S. Now let $\lambda \in \mathbb{C}$ be a root of the characteristic polynomial. By theorem 4.1.10 it is an eigenvalue of A over $\mathbb{C}$, hence an eigenvalue of S, and proposition 7.1.11 makes every eigenvalue of a self-adjoint operator real. So $\lambda \in \mathbb{R}$.

Step 3: triangularize, and check that the triangular factor is diagonal. Step 2 gives the hypothesis of corollary 7.2.5: A is a real matrix all of whose eigenvalues are real. That corollary produces a real orthogonal $Q \in M_n(\mathbb{R})$ and an upper triangular $R \in M_n(\mathbb{R})$ with $A = QRQ^\top$, equivalently $R = Q^\top AQ$, the two forms being interchangeable because $Q^\top Q = QQ^\top = I_n$; the columns $q_1, \ldots, q_n$ of Q form an orthonormal basis of $\mathbb{R}^n$ (proposition 7.1.5). Transposing $R = Q^\top AQ$ and using $(XY)^* = Y^*X^*$ (proposition 5.5.13), which over $\mathbb{R}$ is the same statement for the transpose, gives

$$R^\top = Q^\top A^\top (Q^\top)^\top = Q^\top AQ = R$$

since $A^\top = A$ by Step 1. So R is upper triangular and equal to its own transpose. Triangularity gives $r_{ij} = 0$ whenever $i > j$; and if $i < j$ then $j > i$, so $r_{ij} = r_{ji} = 0$ as well. Every off-diagonal entry vanishes, and R is the diagonal matrix $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ with $\lambda_j = r_{jj} \in \mathbb{R}$. Multiplying $A = QDQ^\top$ on the right by Q gives $AQ = QD$, and comparing j-th columns, $Aq_j = \lambda_j q_j$ for each j (definition 1.4.1, proposition 1.4.3).

Step 4: carry the basis back to V. Write q_{ij} for the entry of Q in row i and column j, and set

$$u_j = \sum_{i=1}^{n} q_{ij}\, e_i \in V \qquad (1 \le j \le n)$$

so that the coordinate column of u_j relative to $e_1, \ldots, e_n$ is exactly q_j. The inner product on V is linear in its first argument and, the scalars being real, in its second as well (definition 5.1.2, proposition 5.1.6), so

$$\langle u_j, u_k\rangle = \sum_{i=1}^{n}\sum_{l=1}^{n} q_{ij}q_{lk}\,\langle e_i, e_l\rangle = \sum_{i=1}^{n} q_{ij}q_{ik} = \langle q_j, q_k\rangle$$

the middle equality because $\langle e_i, e_l\rangle$ is 1 for $i = l$ and 0 otherwise (definition 5.3.3), and the last being the standard inner product of the two columns. By Step 3 that number is 1 for $j = k$ and 0 otherwise, so $u_1, \ldots, u_n$ is an orthonormal list and in particular a linearly independent one (definition 5.3.3, proposition 5.3.4). It spans V: given $v \in V$, write $v = \sum_i x_i e_i$ with $x \in \mathbb{R}^n$ (definition 2.2.1), expand $x = \sum_j c_j q_j$ in the basis $q_1, \ldots, q_n$ of $\mathbb{R}^n$, and substitute,

$$v = \sum_{i=1}^{n}\Big(\sum_{j=1}^{n} c_j q_{ij}\Big)e_i = \sum_{j=1}^{n} c_j \sum_{i=1}^{n} q_{ij}e_i = \sum_{j=1}^{n} c_j u_j$$

An independent spanning list is a basis (definition 2.2.1). Finally, each u_j is an eigenvector: using $Te_i = \sum_k a_{ki}e_k$ from Step 1 and $Aq_j = \lambda_j q_j$ from Step 3,

$$Tu_j = \sum_{i=1}^{n} q_{ij}\, Te_i = \sum_{k=1}^{n}\Big(\sum_{i=1}^{n} a_{ki}q_{ij}\Big)e_k = \sum_{k=1}^{n}(Aq_j)_k\, e_k = \lambda_j \sum_{k=1}^{n} q_{kj}e_k = \lambda_j u_j$$

So $u_1, \ldots, u_n$ is an orthonormal basis of V consisting of eigenvectors of T with real eigenvalues, as we sought to show.

The single hypothesis $T^* = T$ does two separate jobs, and it is worth separating them. Reality of the eigenvalues is the part that needs self-adjointness itself: it came from proposition 7.1.11, and normality alone would not have given it, since the matrix J of proposition 7.4.1 is normal with eigenvalues $\pm i$. Orthogonality across *distinct* eigenvalues is the part that needs less: a self-adjoint A is in particular normal, because $A^\top A = A^2 = AA^\top$, so proposition 7.1.14 already gives it. Inside a single eigenspace neither hypothesis gives anything, and the proof above conceals that by producing the whole basis at once; example 7.4.5 exhibits where the work goes.

Applied to $\mathbb{R}^n$ with its standard inner product the theorem becomes a factorization, which is the form later chapters cite.

Corollary 7.4.3: Orthogonal Diagonalization of a Real Symmetric Matrix

Let $A \in M_n(\mathbb{R})$. The following are equivalent.

1. $A^\top = A$.
2. $\mathbb{R}^n$ has an orthonormal basis consisting of eigenvectors of A.
3. $A = QDQ^\top$ for some real orthogonal $Q \in M_n(\mathbb{R})$ and some diagonal $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n) \in M_n(\mathbb{R})$.

When these hold, the j-th column of Q is an eigenvector of A for λ_j, and $\lambda_1, \ldots, \lambda_n$ are the roots of the characteristic polynomial of A, each listed as often as its multiplicity as a root.

***Proof*:**

(1) $\Rightarrow$ (2) Give $\mathbb{R}^n$ the standard inner product and let $T(x) = Ax$, a linear map whose matrix relative to the standard basis is A (theorem 2.4.8). That basis is orthonormal, so the matrix of T^* is $A^* = A^\top = A$ (theorem 5.5.14, definition 5.5.12), whence $T^* = T$ and T is self-adjoint (definition 7.1.1). Theorem 7.4.2 gives an orthonormal basis $q_1, \ldots, q_n$ of $\mathbb{R}^n$ with $Aq_j = \lambda_j q_j$ and $\lambda_j \in \mathbb{R}$.

(2) $\Rightarrow$ (3) Let $Q \in M_n(\mathbb{R})$ have columns $q_1, \ldots, q_n$ and let $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ carry the corresponding eigenvalues. The columns of Q are an orthonormal basis of $\mathbb{R}^n$, so proposition 7.1.5 and its closing clause give $Q^\top Q = QQ^\top = I_n$ and $Q^{-1} = Q^\top$. The j-th column of AQ is $Aq_j = \lambda_j q_j$, which is also the j-th column of QD (definition 1.4.1); so $AQ = QD$ and therefore $A = AQQ^\top = QDQ^\top$ (proposition 1.4.3).

(3) $\Rightarrow$ (1) A diagonal matrix equals its own transpose, so $(XY)^* = Y^*X^*$ over $\mathbb{R}$ (proposition 5.5.13) gives

$$A^\top = (QDQ^\top)^\top = (Q^\top)^\top D^\top Q^\top = QDQ^\top = A$$

For the last sentence of the statement, $AQ = QD$ was obtained above and says that the j-th column of Q is an eigenvector for λ_j. A diagonal matrix is in particular upper triangular, so $A = QDQ^\top$ is a Schur factorization of A and corollary 7.2.2 identifies its diagonal entries $\lambda_1, \ldots, \lambda_n$ with the roots of the characteristic polynomial of A, counted with multiplicity, completing the proof.

The factorization $A = QDQ^\top$ has a reading as a sum. Each $x \in \mathbb{R}^n$ satisfies $QDQ^\top x = \sum_j \lambda_j \langle x, q_j\rangle q_j$, by the two computations in the proof, and $\langle x, q_j\rangle q_j$ is $q_j q_j^\top x$; so

$$A = \sum_{j=1}^{n} \lambda_j \, q_j q_j^\top$$

where $q_j q_j^\top$ is the orthogonal projection matrix onto the line $\mathbb{R}q_j$, since $q_j q_j^\top x = \langle x, q_j\rangle q_j$ is what definition 5.3.8 computes for the orthonormal basis q_j of that line. Collecting the terms with equal λ_j turns this into the resolution of theorem 7.3.5 for the real symmetric case: the projections there are the sums of the $q_j q_j^\top$ over each eigenvalue's block of columns.

Example 7.4.4: Orthogonal Diagonalization of a 2×2 Symmetric Matrix

Let

$$A = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} \in M_2(\mathbb{R}), \qquad A^\top = A$$

Its characteristic polynomial is

$$\det(tI_2 - A) = (t-4)(t-1) - (-2)(-2) = t^2 - 5t + 4 - 4 = t(t-5)$$

so the eigenvalues are 5 and 0 (theorem 4.1.10), both real, as theorem 7.4.2 requires.

For $\lambda = 5$ the eigenspace is the null space of $A - 5I_2 = \begin{pmatrix} -1 & 2 \\ 2 & -4 \end{pmatrix}$ (proposition 4.1.5). The second row is -2 times the first, so the single condition is $-x_1 + 2x_2 = 0$, satisfied by $(2,1)^\top$ and its multiples. For $\lambda = 0$ the eigenspace is the null space of A itself: the second row of A is half the first, so the condition is $2x_1 + x_2 = 0$, satisfied by $(1,-2)^\top$ and its multiples. Both vectors have norm $\sqrt{4+1} = \sqrt{5}$, so

$$q_1 = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \qquad q_2 = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \qquad \langle q_1, q_2\rangle = \tfrac{1}{5}(2-2) = 0$$

The orthogonality was not arranged; it holds because the two eigenvalues are distinct (proposition 7.1.14).

Assembling the columns and the eigenvalues,

$$Q = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix}, \qquad D = \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix}$$

Here $Q^\top = Q$, and

$$Q^\top Q = \frac{1}{5} \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix} = I_2$$

so Q is orthogonal. Multiplying out,

$$QD = \frac{1}{\sqrt{5}} \begin{pmatrix} 10 & 0 \\ 5 & 0 \end{pmatrix}, \qquad (QD)Q^\top = \frac{1}{5} \begin{pmatrix} 10 & 0 \\ 5 & 0 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} = \frac{1}{5} \begin{pmatrix} 20 & 10 \\ 10 & 5 \end{pmatrix} = A$$

which is the factorization of corollary 7.4.3. Since $\lambda_2 = 0$, the sum $A = \sum_j \lambda_j q_j q_j^\top$ has a single surviving term, and it reproduces A:

$$5\, q_1 q_1^\top = 5 \cdot \frac{1}{5} \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} = A$$

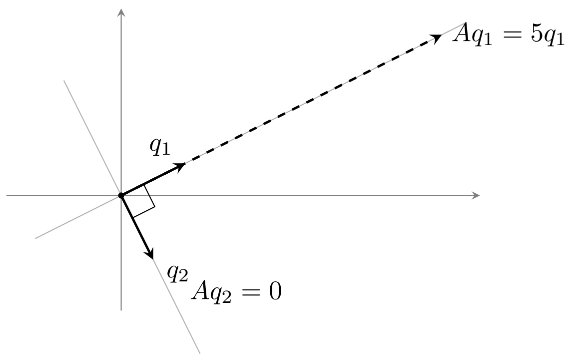


Figure 7.5: The eigendirections of example 7.4.4: A stretches the first by 5 and collapses the second

When an eigenvalue is repeated, its eigenspace has dimension larger than 1 and its vectors are not forced into any particular position relative to one another. An arbitrary basis of that eigenspace, such as the one row reduction returns, is usually not orthogonal, and the columns of Q must be produced from it by Gram-Schmidt (theorem 5.3.9). The next example is the smallest case where this happens.

Example 7.4.5: A Symmetric Matrix with a Repeated Eigenvalue

Let

$$A = \begin{pmatrix} 4 & 1 & 1 \\ 1 & 4 & 1 \\ 1 & 1 & 4 \end{pmatrix} \in M_3(\mathbb{R}), \qquad A^\top = A$$

Write $c = t - 4$, so that

$$tI_3 - A = \begin{pmatrix} c & -1 & -1 \\ -1 & c & -1 \\ -1 & -1 & c \end{pmatrix}$$

Expanding the determinant along the first row,

$$\det(tI_3 - A) = c(c^2 - 1) + \big((-1)c - (-1)(-1)\big) - \big((-1)(-1) - c(-1)\big) = c^3 - 3c - 2$$

and $c^3 - 3c - 2 = (c+1)^2(c-2)$, as multiplying out confirms: $(c^2 + 2c + 1)(c - 2) = c^3 - 3c - 2$. Substituting $c = t - 4$ back,

$$\det(tI_3 - A) = (t-3)^2(t-6)$$

so the eigenvalues are 6, simple, and 3, repeated twice (theorem 4.1.10).

The eigenspace for 6 is the null space of

$$A - 6I_3 = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix}$$

(proposition 4.1.5). Row reducing with $R_1 \leftrightarrow R_2$, then $R_2 \mapsto R_2 + 2R_1$ and $R_3 \mapsto R_3 - R_1$, then $R_3 \mapsto R_3 + R_2$, gives

$$\begin{pmatrix} 1 & -2 & 1 \\ 0 & -3 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

so $x_2 = x_3$ and $x_1 = 2x_2 - x_3 = x_2$: the eigenspace is spanned by $(1,1,1)^\top$, and $q_1 = \frac{1}{\sqrt{3}}(1,1,1)^\top$.

The eigenspace for 3 is the null space of $A - 3I_3$, all of whose entries are 1; the three rows are equal, so the single condition is $x_1 + x_2 + x_3 = 0$, and taking x_2, x_3 as the free variables gives the basis

$$v_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

These are not orthogonal: $\langle v_1, v_2 \rangle = 1$. Gram-Schmidt (theorem 5.3.9) repairs that inside the eigenspace. Since $\|v_1\|^2 = 2$, the first output is $q_2 = \frac{1}{\sqrt{2}}(-1,1,0)^\top$, and $\langle v_2, q_2 \rangle = \frac{1}{\sqrt{2}}(1+0+0) = \frac{1}{\sqrt{2}}$, so the second vector loses its component along q_2:

$$w = v_2 - \frac{1}{\sqrt{2}} q_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix}$$

with $\|w\|^2 = \frac{1}{4}(1+1+4) = \frac{3}{2}$. Dividing by $\|w\| = \sqrt{3/2}$ gives $q_3 = \frac{1}{\sqrt{6}}(-1,-1,2)^\top$. Both q_2 and q_3 still lie in the eigenspace for 3, since each is a combination of v_1 and v_2 and a null space is a subspace (definition 2.3.3); and both are orthogonal to q_1, as $\frac{1}{\sqrt{6}}(-1+1+0) = 0$ and $\frac{1}{\sqrt{18}}(-1-1+2) = 0$ record. That last orthogonality is proposition 7.1.14 again, since q_1 and the other two belong to distinct eigenvalues; the orthogonality of q_2 and q_3 to each other is the part Gram-Schmidt had to give.

Assembling,

$$Q = \begin{pmatrix} 1/\sqrt{3} & -1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \end{pmatrix}, \qquad D = \operatorname{diag}(6,3,3)$$

and the factorization is checked most quickly in the summed form $A = \sum_j \lambda_j q_j q_j^\top$. The three summands are

$$6q_1q_1^\top = \begin{pmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}, \quad 3q_2q_2^\top = \frac{3}{2}\begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad 3q_3q_3^\top = \frac{1}{2}\begin{pmatrix} 1 & 1 & -2 \\ 1 & 1 & -2 \\ -2 & -2 & 4 \end{pmatrix}$$

and adding them entrywise gives $2 + \frac{3}{2} + \frac{1}{2} = 4$ in positions $(1,1)$ and $(2,2)$, $2+0+2 = 4$ in position $(3,3)$, $2 - \frac{3}{2} + \frac{1}{2} = 1$ in positions $(1,2)$ and $(2,1)$, and $2+0-1 = 1$ in the four positions $(1,3)$, $(3,1)$, $(2,3)$, $(3,2)$. The sum is A.

The matrix Q is not the only one that works. Any orthonormal basis of the plane $x_1 + x_2 + x_3 = 0$ could have replaced q_2, q_3, and there are infinitely many. What corollary 7.4.3 pins down is D, whose entries are the roots of the characteristic polynomial with multiplicity, and not Q.

One consequence of the factorization is where the study of quadratic forms begins. The function $x \mapsto \langle Ax, x \rangle$ attached to a symmetric matrix becomes a sum of squares in the coordinates given by the eigenbasis, with the eigenvalues as coefficients, and the signs of those coefficients then decide which values the function takes.

Corollary 7.4.6: Principal Axes

Let $A \in M_n(\mathbb{R})$ be symmetric, and let $A = QDQ^\top$ be as in corollary 7.4.3, with columns $q_1, \dots, q_n$ of Q and $D = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. For $x \in \mathbb{R}^n$ write $y = Q^\top x$, so that $y_j = \langle x, q_j \rangle$. Then

$$\langle Ax, x \rangle = \sum_{j=1}^{n} \lambda_j y_j^2$$

***Proof*:**
For the standard inner product on $\mathbb{R}^n$ and any $u, v \in \mathbb{R}^n$ one has $\langle u, v \rangle = v^\top u$, a 1×1 matrix read as a scalar (definition 1.4.1). So

$$\langle Ax, x \rangle = x^\top Ax = x^\top QDQ^\top x = (Q^\top x)^\top D(Q^\top x) = y^\top Dy = \sum_{j=1}^{n} \lambda_j y_j^2$$

using $(Q^\top x)^\top = x^\top Q$ (proposition 5.5.13) and, in the last step, definition 1.4.1 for the diagonal matrix D. The identification $y_j = \langle x, q_j \rangle$ is the computation of $Q^\top x$ made in the proof of corollary 7.4.3.

The coordinates $y_1, \dots, y_n$ are those of x in the eigenbasis, so the corollary says that a symmetric matrix has no cross terms at all when the axes are chosen along its eigenvectors. Since each y_j^2 is nonnegative, the signs of $\lambda_1, \dots, \lambda_n$ then decide which values $x \mapsto \langle Ax, x \rangle$ takes; that criterion, and the determinant and factorization tests that apply it without computing eigenvalues, belong to chapter 8.

Exercise 7.4.1

1. Let $A = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} \in M_2(\mathbb{R})$. Find a real orthogonal Q and a real diagonal D with $A = QDQ^\top$, and verify the factorization by multiplying the three matrices out.
2. Let
$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{pmatrix} \in M_3(\mathbb{R})$$
Show that the null space of A is the plane $x_1 + 2x_2 + 3x_3 = 0$ and that $(1, 2, 3)^\top$ is an eigenvector, and use this to write down the eigenvalues of A with multiplicity. Then produce a real orthogonal Q and a diagonal D with $A = QDQ^\top$, applying theorem 5.3.9 inside the repeated eigenspace.
3. Let $A \in M_n(\mathbb{R})$ be symmetric with $A^2 = 0$. Show that $A = 0$. (The matrix $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ satisfies $A^2 = 0$ without being zero, so the symmetry hypothesis is doing work.)
4. Let $A = (a_{ij}) \in M_n(\mathbb{R})$ be symmetric with eigenvalues $\lambda_1, \dots, \lambda_n$ listed with multiplicity as in corollary 7.4.3. Show that $\sum_{i,j} a_{ij}^2 = \sum_j \lambda_j^2$. (First check that $\sum_{i,j} a_{ij}^2 = \operatorname{tr}(A^\top A)$.)

5. Let $A \in M_n(\mathbb{R})$ be symmetric with $A^2 = A$. Show that every eigenvalue of A is 0 or 1, and that $\mathrm{tr}A$ is the number of indices j with $\lambda_j = 1$. Conclude that $\mathrm{tr}A$ is a nonnegative integer.
6. Let $A \in M_n(\mathbb{R})$ be symmetric and suppose $\langle Ax, x\rangle = 0$ for every $x \in \mathbb{R}^n$. Use theorem 7.4.2 to show $A = 0$, and say which hypothesis of lemma 7.1.9 this argument is using.
7. Let $A \in M_n(\mathbb{R})$ satisfy $A^\top A = AA^\top$, and suppose every root of the characteristic polynomial of A is real. Show that $A^\top = A$. (Apply corollary 7.2.5, check that the resulting triangular matrix is itself normal, and use lemma 7.3.1.) Explain why proposition 7.4.1 does not contradict this.

7.5 The Real Normal Form for Orthogonal Operators; Euler's Theorem

Section 7.4 settled the real symmetric case: the eigenvalues are real and an orthonormal eigenbasis exists. The other class singled out in section 7.1 behaves in the opposite way over $\mathbb{R}$. Proposition 7.4.1 exhibits an orthogonal 2×2 matrix with no eigenvector in $\mathbb{R}^2$ at all, so for that matrix no real change of coordinates, orthonormal or otherwise, produces a diagonal matrix. This section proves what does happen instead. Every real orthogonal matrix becomes block diagonal in a suitable orthonormal basis of $\mathbb{R}^n$, with blocks of size at most two: the 2×2 blocks are the rotations R_θ of example 7.1.6, and the 1×1 blocks are 1 or -1.

The route is the one section 7.3 used, carried one step further. Read in $M_n(\mathbb{C})$ a real orthogonal matrix is unitary, hence normal, so it does have an orthonormal eigenbasis of $\mathbb{C}^n$; its eigenvalues have modulus 1, and the non-real ones occur in conjugate pairs. Each such pair carries a plane of $\mathbb{R}^n$ rather than a line, and the matrix turns that plane. Counting blocks in dimension three then forces a fixed line, which is Euler's theorem.

Read inside $M_n(\mathbb{C})$, a real orthogonal matrix is a unitary matrix, so section 7.1 already restricts its eigenvalues. Two of those restrictions, together with one fact peculiar to real entries, carry the whole construction.

Proposition 7.5.1: The Eigenvalues of a Real Orthogonal Matrix

Let $Q \in M_n(\mathbb{R})$ be orthogonal and read Q as an element of $M_n(\mathbb{C})$.

1. $Q^*Q = QQ^* = I_n$, so Q is unitary and normal. Every eigenvalue $\lambda \in \mathbb{C}$ of Q satisfies $|\lambda| = 1$, and an eigenvalue lying in $\mathbb{R}$ is therefore 1 or -1.
2. If $v \in \mathbb{C}^n$ is nonzero with $Qv = \lambda v$, then $Q\overline{v} = \overline{\lambda}\,\overline{v}$, where $\overline{v}$ is the vector of conjugated entries.
3. If in addition $\lambda \notin \mathbb{R}$, then $\langle v, \overline{v}\rangle = 0$. Writing $v = x + iy$ with $x, y \in \mathbb{R}^n$, this single condition says both
$$\|x\| = \|y\| \qquad \text{and} \qquad \langle x, y\rangle = 0$$
the inner products on the two sides being the standard ones of $\mathbb{C}^n$ and of $\mathbb{R}^n$.

***Proof*:**
Step 1: normality and the modulus. The entries of Q are real, so conjugating them changes nothing and $Q^* = Q^\top$ (definition 5.5.12, definition 2.3.4). The columns of Q are orthonormal in $\mathbb{R}^n$, since $Q^\top Q = I_n$ (definition 7.1.3, proposition 7.1.5), and the closing clause of proposition 7.1.5 gives $Q^\top Q = QQ^\top = I_n$. Hence $Q^*Q = QQ^* = I_n$, which is definition 7.1.3 for Q read in $M_n(\mathbb{C})$ and definition 7.1.7 at the same time. Proposition 7.1.12 now gives $|\lambda| = 1$ for every eigenvalue. A real number of modulus 1 is 1 or -1, which is (1).

Step 2: conjugating an eigenvector. Write $Q = (q_{ij})$ with $q_{ij} \in \mathbb{R}$. By definition 1.4.1 the i-th entry of Qv is $\sum_j q_{ij}v_j$. Conjugation on $\mathbb{C}$ preserves sums and products, so

$$\overline{(Qv)_i} = \sum_j \overline{q_{ij}}\,\overline{v_j} = \sum_j q_{ij}\,\overline{v_j} = (Q\overline{v})_i$$

the middle equality because each q_{ij} is real. The same conjugation applied to λv gives $\overline{(\lambda v)_i} = \overline{\lambda}\,\overline{v_i}$. So $Qv = \lambda v$ conjugates to $Q\overline{v} = \overline{\lambda}\,\overline{v}$, and $\overline{v} \neq 0$ because $v \neq 0$, which is (2).

Step 3: the two eigenvectors are orthogonal. Suppose $\lambda \notin \mathbb{R}$, so that $\overline{\lambda} \neq \lambda$. By Step 1 the matrix Q is normal in $M_n(\mathbb{C})$, by hypothesis v is an eigenvector for λ, and by Step 2 the vector $\overline{v}$ is an eigenvector for $\overline{\lambda}$. Proposition 7.1.14 therefore gives $\langle v, \overline{v}\rangle = 0$.

Step 4: what that condition says about x and y. The standard inner product of $\mathbb{C}^n$ is conjugate-linear in its second argument, so $\langle v, \overline{v}\rangle = \sum_j v_j\overline{\overline{v_j}} = \sum_j v_j^2$. Substituting $v_j = x_j + iy_j$ and expanding each square,

$$\sum_j v_j^2 = \sum_j \left(x_j^2 - y_j^2\right) + 2i\sum_j x_jy_j = \left(\|x\|^2 - \|y\|^2\right) + 2i\,\langle x, y\rangle$$

where the last equality is definition 5.2.1 and the formula for the standard inner product on $\mathbb{R}^n$. A complex number vanishes exactly when its real and imaginary parts both vanish, so $\|x\|^2 = \|y\|^2$ and $\langle x, y\rangle = 0$. Both $\|x\|$ and $\|y\|$ are nonnegative reals (proposition 5.2.2(1)) with equal squares, hence are equal, as we sought to show.

Part (3) is the reason a real orthogonal matrix has 2×2 blocks and not larger ones. A single scalar equation over $\mathbb{C}$, $\langle v, \overline{v}\rangle = 0$, delivers two real conditions at once: the real and imaginary parts of an eigenvector for a non-real eigenvalue have the same length and are perpendicular. After scaling they are an orthonormal pair in $\mathbb{R}^n$, and the plane they span is where the eigenvalue's geometry lives.

One property of the trigonometric functions is used from here on, and only to name an angle. A complex number λ with $|\lambda| = 1$ has $(\operatorname{Re}\lambda)^2 + (\operatorname{Im}\lambda)^2 = 1$, so the pair $(\operatorname{Re}\lambda, \operatorname{Im}\lambda)$ is a point of the unit circle in $\mathbb{R}^2$, and $\theta \mapsto (\cos\theta, \sin\theta)$ runs over that circle. Every eigenvalue of modulus 1 may therefore be written as $\cos\theta + i\sin\theta$ for some real θ. Every computation below uses only the two real numbers $\cos\theta$ and $\sin\theta$ and the identity $\cos^2\theta + \sin^2\theta = 1$.

Lemma 7.5.2: The Real Plane Attached to a Non-Real Eigenvalue

Let $Q \in M_n(\mathbb{R})$ be orthogonal, let $\lambda = \cos\theta + i\sin\theta$ with $\sin\theta \neq 0$ be an eigenvalue of Q in $\mathbb{C}$, and let $v \in \mathbb{C}^n$ satisfy $Qv = \lambda v$ and $\|v\| = 1$. Write $v = x + iy$ with $x, y \in \mathbb{R}^n$ and set

$$u_1 = \sqrt{2}\,x, \qquad u_2 = -\sqrt{2}\,y$$

Then:

1. u_1, u_2 is an orthonormal list in $\mathbb{R}^n$;
2. $Qu_1 = \cos\theta\, u_1 + \sin\theta\, u_2$ and $Qu_2 = -\sin\theta\, u_1 + \cos\theta\, u_2$;
3. $W = \operatorname{span}\{u_1, u_2\}$ has $\dim W = 2$ and $Q(W) = W$.

***Proof*:**
Step 1: the lengths. Since $\sin\theta \neq 0$ the eigenvalue λ is not real, so proposition 7.5.1(3) applies and gives $\|x\| = \|y\|$ and $\langle x, y\rangle = 0$. The hypothesis $\|v\| = 1$ reads, by definition 5.2.1 and the standard inner product of $\mathbb{C}^n$,

$$1 = \sum_j v_j \overline{v_j} = \sum_j |v_j|^2 = \sum_j \left(x_j^2 + y_j^2\right) = \|x\|^2 + \|y\|^2$$

Combined with $\|x\| = \|y\|$ this gives $2\|x\|^2 = 1$, so $\|x\| = \|y\| = 1/\sqrt{2}$. Then $\|u_1\| = \sqrt{2}\,\|x\| = 1$ and $\|u_2\| = |-\sqrt{2}|\,\|y\| = 1$ by proposition 5.2.2(2), while $\langle u_1, u_2\rangle = -2\langle x, y\rangle = 0$ by linearity of the inner product of $\mathbb{R}^n$ in each argument (definition 5.1.2, proposition 5.1.6). That is definition 5.3.3 for the list u_1, u_2, which is (1).

Step 2: separating $Qv = \lambda v$ into real and imaginary parts. Both Q and the vectors x, y are real, so Qx and Qy lie in $\mathbb{R}^n$. Expanding both sides of $Qv = \lambda v$,

$$Qx + i\,Qy = (\cos\theta + i\sin\theta)(x + iy) = \big(\cos\theta\, x - \sin\theta\, y\big) + i\big(\sin\theta\, x + \cos\theta\, y\big)$$

A vector of $\mathbb{C}^n$ determines its real and its imaginary part entry by entry, and all four of Qx, Qy, $\cos\theta\, x - \sin\theta\, y$ and $\sin\theta\, x + \cos\theta\, y$ lie in $\mathbb{R}^n$, so

$$Qx = \cos\theta\, x - \sin\theta\, y, \qquad Qy = \sin\theta\, x + \cos\theta\, y$$

Multiplying the first identity by $\sqrt{2}$ gives

$$Qu_1 = \sqrt{2}\,Qx = \cos\theta\left(\sqrt{2}\,x\right) + \sin\theta\left(-\sqrt{2}\,y\right) = \cos\theta\, u_1 + \sin\theta\, u_2$$

and multiplying the second by $-\sqrt{2}$ gives

$$Qu_2 = -\sqrt{2}\,Qy = -\sin\theta\left(\sqrt{2}\,x\right) + \cos\theta\left(-\sqrt{2}\,y\right) = -\sin\theta\, u_1 + \cos\theta\, u_2$$

which is (2). The sign in the definition of u_2 is what puts these two identities in the order that reproduces the rotation of example 7.1.6: the coordinate columns of Qu_1 and Qu_2 relative to u_1, u_2 are $(\cos\theta, \sin\theta)^\top$ and $(-\sin\theta, \cos\theta)^\top$, which are the columns of R_θ.

Step 3: the plane. The list u_1, u_2 is orthonormal by (1), hence linearly independent by proposition 5.3.4, hence a basis of its span (definition 2.2.1); so $\dim W = 2$ by definition 2.2.10. By (2) both Qu_1 and Qu_2 lie in W, and $Q(c_1u_1 + c_2u_2) = c_1Qu_1 + c_2Qu_2$ because $x \mapsto Qx$ is linear, so $Q(W) \subseteq W$. For the reverse inclusion put $z_1 = \cos\theta\, u_1 - \sin\theta\, u_2$ and $z_2 = \sin\theta\, u_1 + \cos\theta\, u_2$, both in W. Using (2) and $\cos^2\theta + \sin^2\theta = 1$,

$$Qz_1 = \cos\theta\, Qu_1 - \sin\theta\, Qu_2 = \big(\cos^2\theta + \sin^2\theta\big)u_1 = u_1$$

$$Qz_2 = \sin\theta\, Qu_1 + \cos\theta\, Qu_2 = \big(\sin^2\theta + \cos^2\theta\big)u_2 = u_2$$

the coefficients of the other basis vector cancelling in each line. So an arbitrary $c_1u_1 + c_2u_2$ of W equals $Q(c_1z_1 + c_2z_2)$ with $c_1z_1 + c_2z_2 \in W$, giving $W \subseteq Q(W)$ and hence $Q(W) = W$, completing the proof.

The lemma converts one complex eigenvector into a plane of $\mathbb{R}^n$ on which Q acts by the matrix R_θ. A real eigenvector, when there is one, gives a line on which Q acts by 1 or -1. The normal form is what happens when this is repeated: split off such a piece, pass to its orthogonal complement, and repeat. Two facts make the passage possible. The complement of a Q-invariant subspace is again Q-invariant, because Q preserves inner products; and the matrix recording Q on that complement is again orthogonal.

Theorem 7.5.3: The Real Normal Form for an Orthogonal Matrix

Let $n \geq 1$ and let $Q \in M_n(\mathbb{R})$ be orthogonal. Then there are integers $p, r, s \geq 0$ with $2p + r + s = n$, real numbers $\theta_1, \ldots, \theta_p$ with $\sin\theta_i \neq 0$, and an orthogonal $P \in M_n(\mathbb{R})$, such that

$$P^\top QP = \begin{pmatrix} I_r & 0 & 0 & & \\ 0 & -I_s & 0 & & \\ 0 & 0 & R_{\theta_1} & & \\ & & & \ddots & \\ & & & & R_{\theta_p} \end{pmatrix}$$

where I_r is the $r \times r$ identity matrix, R_θ is the rotation matrix of example 7.1.6, every entry outside the displayed blocks is zero, and a block whose index is 0 is absent. Moreover

$$\det Q = (-1)^s$$

***Proof*:**
Step 0: two standing facts. Give $\mathbb{R}^n$ the standard inner product and let $T_Q\colon \mathbb{R}^n \to \mathbb{R}^n$ be the map $x \mapsto Qx$, which is linear. By theorem 5.5.14(2) its adjoint is $x \mapsto Q^\top x$, and $Q^\top Q = I_n$ (definition 7.1.3), so $T_Q^* T_Q = \mathrm{id}_{\mathbb{R}^n}$ and T_Q is unitary. Theorem 7.1.4(2) then gives

$$\langle Qu, Qw\rangle = \langle u, w\rangle \qquad \text{for all } u, w \in \mathbb{R}^n \tag{7.2}$$

Second, for matrices split after row and column d, the product formula definition 1.4.1 gives

$$\begin{pmatrix} X & 0 \\ 0 & Y \end{pmatrix}\begin{pmatrix} X' & 0 \\ 0 & Y' \end{pmatrix} = \begin{pmatrix} XX' & 0 \\ 0 & YY' \end{pmatrix} \tag{7.3}$$

whenever $X, X' \in M_d(\mathbb{R})$ and $Y, Y' \in M_{n-d}(\mathbb{R})$: in the sum $\sum_t M_{it}N_{tj}$ computing the entry in row i and column j, a term with $t \leq d$ has $N_{tj} = 0$ unless $j \leq d$ and a term with $t > d$ has $M_{it} = 0$ unless $i > d$, so the sum vanishes when i and j lie on opposite sides of d, and otherwise reduces to the corresponding sum inside X and X', or inside Y and Y'. Transposition acts blockwise on such a matrix for the same reason.

Step 1: one block. Let $Q \in M_n(\mathbb{R})$ be orthogonal. Two cases.

Suppose first that Q has an eigenvalue ε in $\mathbb{R}$. By proposition 7.5.1(1) it is 1 or -1. Choose an eigenvector for it in $\mathbb{R}^n$ (definition 4.1.2) and divide it by its norm, which is a positive real by proposition 5.2.2(1); by proposition 5.2.2(2) the result u_1 is a unit vector, and it is still an eigenvector. Put $d = 1$, $W = \operatorname{span}\{u_1\}$ and $B = (\varepsilon)$, a 1×1 matrix. Then $Q(cu_1) = c\varepsilon u_1 \in W$, and $cu_1 = Q(c\varepsilon u_1)$ because $\varepsilon^2 = 1$, so $Q(W) = W$.

Suppose instead that Q has no eigenvalue in $\mathbb{R}$. Read as an element of $M_n(\mathbb{C})$ it has an eigenvalue $\lambda \in \mathbb{C}$ by theorem 4.1.14. That λ is not real: a real λ would be a real root of the characteristic polynomial of Q, hence an eigenvalue of Q over $\mathbb{R}$ by theorem 4.1.10. By proposition 7.5.1(1) it has modulus 1, so $\lambda = \cos\theta + i\sin\theta$ for some real θ, and $\sin\theta = \operatorname{Im}\lambda \neq 0$. Choose an eigenvector for λ in $\mathbb{C}^n$ and divide it by its norm to get a unit eigenvector v. Lemma 7.5.2 gives an orthonormal pair u_1, u_2 in $\mathbb{R}^n$; put $d = 2$, $W = \operatorname{span}\{u_1, u_2\}$ and $B = R_\theta$, so that $Q(W) = W$ and $Qu_j = \sum_{i \le 2} B_{ij} u_i$ for $j = 1, 2$ by part (2) of that lemma.

In both cases $u_1, \ldots, u_d$ is an orthonormal basis of W, the identity $Qu_j = \sum_{i \le d} B_{ij} u_i$ holds for $j \le d$, the matrix B is one of (1), (-1), R_θ with $\sin\theta \neq 0$, and $Q(W) = W$.

Step 2: splitting the block off. Keep W, d and B from Step 1. The orthogonal complement $W^\perp$ is a subspace of $\mathbb{R}^n$ (proposition 5.5.2(1)), finite-dimensional by corollary 2.2.12, and carries the restricted inner product, so corollary 5.3.11 gives an orthonormal basis $w_1, \ldots, w_m$ of $W^\perp$.

The concatenated list $p_1, \ldots, p_n$, where $p_j = u_j$ for $j \le d$ and $p_{d+k} = w_k$ for $1 \le k \le m$, is orthonormal: each of the two groups is orthonormal, and $\langle u_i, w_k \rangle = 0$ because $w_k \in W^\perp$ and $u_i \in W$ (definition 5.5.1, together with the symmetry of the inner product over $\mathbb{R}$ recorded in proposition 5.1.6(2)). It spans $\mathbb{R}^n$: by theorem 5.5.3 every $v \in \mathbb{R}^n$ is $w + w'$ with $w \in W$ and $w' \in W^\perp$, and w is a combination of the u_i while w' is a combination of the w_k. It is independent by proposition 5.3.4. So it is a basis of $\mathbb{R}^n$ (definition 2.2.1), and $d + m = n$ by definition 2.2.10.

The complement is Q-invariant. Let $w' \in W^\perp$ and let $u \in W$. As $Q(W) = W$ there is $u_0 \in W$ with $Qu_0 = u$, and then $\langle Qw', u \rangle = \langle Qw', Qu_0 \rangle = \langle w', u_0 \rangle = 0$ by (7.2) and the choice of w'. Since $u \in W$ was arbitrary, $Qw' \in W^\perp$ by definition 5.5.1.

Let $P_1 \in M_n(\mathbb{R})$ be the matrix with columns $p_1, \ldots, p_n$. Those columns are an orthonormal basis of $\mathbb{R}^n$, so P_1 is orthogonal and $P_1^{-1} = P_1^\top$, that is $P_1^\top P_1 = P_1 P_1^\top = I_n$ (proposition 7.1.5 and its closing clause). Put $D_1 = P_1^\top Q P_1$. Multiplying on the left by P_1 and using $P_1 P_1^\top = I_n$ gives $QP_1 = P_1 D_1$, and comparing j-th columns (definition 1.4.1),

$$Qp_j = \sum_{i=1}^{n} (D_1)_{ij}\, p_i$$

For $j \le d$ the vector $Qp_j = Qu_j$ equals $\sum_{i \le d} B_{ij} u_i$ by Step 1; subtracting the two expansions and using the independence of $p_1, \ldots, p_n$ gives $(D_1)_{ij} = B_{ij}$ for $i \le d$ and $(D_1)_{ij} = 0$ for $i > d$. For $j > d$ the vector $Qp_j = Qw_{j-d}$ lies in $W^\perp = \operatorname{span}\{w_1, \ldots, w_m\}$, so its expansion uses only the p_i with $i > d$, and $(D_1)_{ij} = 0$ for $i \le d$. Hence

$$D_1 = \begin{pmatrix} B & 0 \\ 0 & Q' \end{pmatrix} \qquad \text{for some } Q' \in M_m(\mathbb{R})$$

That Q' is orthogonal. Transposing $D_1 = P_1^\top Q P_1$ with proposition 5.5.13(1) and (3), which over $\mathbb{R}$ are statements about the transpose, gives $D_1^\top = P_1^\top Q^\top P_1$, and therefore

$$D_1^\top D_1 = P_1^\top Q^\top (P_1 P_1^\top) Q P_1 = P_1^\top (Q^\top Q) P_1 = P_1^\top P_1 = I_n$$

using proposition 1.4.3 to regroup. By (7.3) and the blockwise transposition recorded in Step 0, the left side is the block diagonal matrix with blocks $B^\top B$ and $(Q')^\top Q'$, so $(Q')^\top Q' = I_m$ and Q' is orthogonal (definition 7.1.3).

Step 3: the induction. The claim is that for every $n \geq 1$ and every orthogonal $Q \in M_n(\mathbb{R})$ there is an orthogonal $P \in M_n(\mathbb{R})$ for which $P^\top Q P$ is block diagonal with each diagonal block equal to (1), to (-1), or to R_θ for some θ with $\sin\theta \neq 0$. Argue by strong induction on n, the case $n = 1$ being covered by what follows since $d \geq 1$ forces $m = n - d < n$.

Let $n \geq 1$ and assume the claim for every size smaller than n. Steps 1 and 2 produce P_1, B of size d, and an orthogonal $Q' \in M_m(\mathbb{R})$ with $m = n - d < n$. If $m = 0$ then $D_1 = B$ and $P = P_1$ proves the claim. If $m \geq 1$, the inductive hypothesis gives an orthogonal $P' \in M_m(\mathbb{R})$ with $(P')^\top Q' P'$ block diagonal of the stated kind. Put

$$P_2 = \begin{pmatrix} I_d & 0 \\ 0 & P' \end{pmatrix}, \qquad P = P_1 P_2$$

By (7.3) and blockwise transposition, $P_2^\top P_2$ is block diagonal with blocks I_d and $(P')^\top P' = I_m$, so $P_2^\top P_2 = I_n$ and P_2 is orthogonal; then $P^\top P = P_2^\top (P_1^\top P_1) P_2 = P_2^\top P_2 = I_n$, so P is orthogonal. Regrouping with proposition 1.4.3 and applying (7.3) twice,

$$P^\top Q P = P_2^\top (P_1^\top Q P_1) P_2 = \begin{pmatrix} I_d & 0 \\ 0 & (P')^\top \end{pmatrix} \begin{pmatrix} B & 0 \\ 0 & Q' \end{pmatrix} \begin{pmatrix} I_d & 0 \\ 0 & P' \end{pmatrix} = \begin{pmatrix} B & 0 \\ 0 & (P')^\top Q' P' \end{pmatrix}$$

whose diagonal blocks are B followed by those of $(P')^\top Q' P'$, all of the stated kind. This proves the claim.

Step 4: grouping the blocks. Step 3 gives an orthogonal P for which $D = P^\top Q P$ is block diagonal with blocks $B_1, \ldots, B_t$ of the stated kinds, occupying consecutive index sets $J_1, \ldots, J_t$. Write $p_1, \ldots, p_n$ for the columns of P; as in Step 2 they are an orthonormal basis of $\mathbb{R}^n$ and $Qp_j = \sum_i D_{ij} p_i$, where $D_{ij} = 0$ unless i and j lie in the same J_k.

List the same n columns in a new order: first those whose block is (1), then those whose block is (-1), then those belonging to rotation blocks, two at a time and in their original order within each block. Write $p'_j = p_{\pi(j)}$ for the resulting list, π being the rearrangement of $\{1, \ldots, n\}$ just described, and let P' be the matrix with columns $p'_1, \ldots, p'_n$. The list is the same orthonormal basis in another order, so P' is orthogonal with $(P')^\top P' = P'(P')^\top = I_n$ (proposition 7.1.5). Re-indexing the expansion above,

$$Qp'_j = \sum_i D_{i\pi(j)}\, p_i = \sum_{i'} D_{\pi(i')\pi(j)}\, p'_{i'}$$

so $QP' = P'D'$ with $D'_{i'j} = D_{\pi(i')\pi(j)}$, and multiplying on the left by $(P')^\top$ gives $(P')^\top Q P' = D'$. Because π carries each consecutive group of the new listing onto one whole J_k preserving its internal order, D' is block diagonal with the same blocks in the new order: r blocks (1), then s blocks (-1), then p rotation blocks $R_{\theta_1}, \ldots, R_{\theta_p}$. Those r blocks assemble into I_r and those s blocks into $-I_s$, and $2p + r + s = n$ because the blocks partition n indices. This is the displayed matrix.

Step 5: the determinant. Write D' for the displayed matrix. By theorem 3.2.10, $\det D' = \det\big((P')^\top\big)\det Q \det P'$, and $\det\big((P')^\top\big)\det P' = \det\big((P')^\top P'\big) = \det I_n = 1$ by the same result together with proposition 3.2.6 applied to the diagonal matrix I_n. So $\det Q = \det D'$. Splitting the leading block from the rest and applying theorem 3.5.2(1) once for each of the $t-1$ splits,

$$\det D' = \det(I_r)\,\det(-I_s)\prod_{i=1}^{p}\det\left(R_{\theta_i}\right)$$

Here $\det(I_r) = 1$ and $\det(-I_s) = (-1)^s$ by proposition 3.2.6, both matrices being diagonal, and expanding the 2×2 determinant by theorem 3.3.4 gives $\det R_\theta = \cos^2\theta + \sin^2\theta = 1$. Hence $\det Q = (-1)^s$, completing the proof.

The theorem says that the only motions an orthogonal matrix performs are rotations of mutually perpendicular planes together with reflections in mutually perpendicular directions, and that these are performed independently of one another. The integers p, r, s are determined by Q. A vector fixed by the displayed matrix must vanish in every coordinate belonging to the $-I_s$ block and to each rotation block, since $-1 \neq 1$ and a rotation block with $\sin\theta_i \neq 0$ fixes only 0; so the eigenvalue 1 has eigenspace of dimension exactly r, and the same argument applied to $-Q$ gives dimension s for the eigenvalue -1. The remaining $2p$ complex eigenvalues are accounted for by the rotation blocks, each R_{θ_i} contributing the conjugate pair $\cos\theta_i \pm i\sin\theta_i$.

Write $\mathrm{O}(n)$ for the set of orthogonal matrices in $M_n(\mathbb{R})$ and $\mathrm{SO}(n)$ for those among them with determinant 1. The determinant clause says that $\det Q = \pm 1$ for every $Q \in \mathrm{O}(n)$, with the sign recording the parity of the number of reflected directions; the matrices of $\mathrm{SO}(n)$ are those with an even number of them.

Example 7.5.4: An Orthogonal 4×4 Matrix Brought to Normal Form

Let

$$Q = \frac{1}{2}\begin{pmatrix} 1 & -1 & -1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & -1 \\ 1 & -1 & 1 & 1 \end{pmatrix} \in M_4(\mathbb{R})$$

Each column has four entries of absolute value $\frac{1}{2}$, so each has squared norm $4\cdot\frac{1}{4} = 1$. For the six pairs of distinct columns, the four products of corresponding entries are $\pm\frac{1}{4}$ and cancel in pairs:

$$\tfrac{1}{4}(-1+1+1-1) = 0, \quad \tfrac{1}{4}(-1-1+1+1) = 0, \quad \tfrac{1}{4}(-1+1-1+1) = 0$$

for the pairs $(1,2)$, $(1,3)$, $(1,4)$, and

$$\tfrac{1}{4}(1-1+1-1) = 0, \quad \tfrac{1}{4}(1+1-1-1) = 0, \quad \tfrac{1}{4}(1-1-1+1) = 0$$

for the pairs $(2,3)$, $(2,4)$, $(3,4)$. So the columns are orthonormal and Q is orthogonal by proposition 7.1.5.

The diagonal entries of Q are all $\frac{1}{2}$ and $q_{ij} = -q_{ji}$ for $i \neq j$, so $Q + Q^\top = I_4$, that is $Q^\top = I_4 - Q$. Set $A = 2Q - I_4$. Then $A^\top = 2(I_4 - Q) - I_4 = -A$, and from $I_4 = Q^\top Q = (I_4 - Q)Q = Q - Q^2$ one gets $Q^2 = Q - I_4$ and hence

$$A^2 = 4Q^2 - 4Q + I_4 = 4(Q - I_4) - 4Q + I_4 = -3I_4$$

This identity produces an eigenvector without any elimination. For any $w \in \mathbb{C}^4$ the vector $z = i\sqrt{3}\,w + Aw$ satisfies $Az = i\sqrt{3}\,Aw + A^2w = i\sqrt{3}\,Aw - 3w = i\sqrt{3}\,z$, using $i\sqrt{3}\cdot i\sqrt{3} = -3$.

Taking $w = e_1$ and reading the first column of $A = 2Q - I_4$, which is $(1,1,1,1)^\top - e_1 = (0,1,1,1)^\top$,

$$v = i\sqrt{3}\,e_1 + (0,1,1,1)^\top = (i\sqrt{3},\ 1,\ 1,\ 1)^\top$$

Checking the four rows of Av against $i\sqrt{3}\,v$ directly: the first row of A is $(0,-1,-1,-1)$ and gives $-3 = i\sqrt{3}\cdot i\sqrt{3}$; the second row $(1,0,-1,1)$ gives $i\sqrt{3}-1+1 = i\sqrt{3}$; the third row $(1,1,0,-1)$ gives $i\sqrt{3}+1-1 = i\sqrt{3}$; the fourth row $(1,-1,1,0)$ gives $i\sqrt{3}-1+1 = i\sqrt{3}$. Since $Q = \frac{1}{2}(I_4 + A)$,

$$Qv = \tfrac{1}{2}\big(v + Av\big) = \frac{1+i\sqrt{3}}{2}\,v = \lambda v, \qquad \lambda = \cos\tfrac{\pi}{3} + i\sin\tfrac{\pi}{3}$$

Now run lemma 7.5.2. The squared norm of v is $|i\sqrt{3}|^2 + 1 + 1 + 1 = 6$, so $v/\sqrt{6}$ is a unit eigenvector, with real part $x = (0,1,1,1)^\top/\sqrt{6}$ and imaginary part $y = (\sqrt{3},0,0,0)^\top/\sqrt{6}$. The lemma's vectors are

$$u_1 = \sqrt{2}\,x = \frac{1}{\sqrt{3}}(0,1,1,1)^\top, \qquad u_2 = -\sqrt{2}\,y = -(1,0,0,0)^\top$$

using $\sqrt{2}/\sqrt{6} = 1/\sqrt{3}$ and $\sqrt{2}\sqrt{3}/\sqrt{6} = 1$. The lemma predicts $Qu_1 = \frac{1}{2}u_1 + \frac{\sqrt{3}}{2}u_2$, and indeed $Q(0,1,1,1)^\top$ is the sum of the last three columns of Q, namely $\frac{1}{2}(-3,1,1,1)^\top$, so

$$Qu_1 = \frac{1}{2\sqrt{3}}(-3,1,1,1)^\top = \frac{1}{2\sqrt{3}}(0,1,1,1)^\top - \frac{\sqrt{3}}{2}(1,0,0,0)^\top = \tfrac{1}{2}u_1 + \tfrac{\sqrt{3}}{2}u_2$$

Likewise $Qu_2 = -Qe_1 = -\frac{1}{2}(1,1,1,1)^\top$, which is $-\frac{\sqrt{3}}{2}u_1 + \frac{1}{2}u_2$ since $-\frac{\sqrt{3}}{2}\cdot\frac{1}{\sqrt{3}}(0,1,1,1)^\top - \frac{1}{2}(1,0,0,0)^\top = -\frac{1}{2}(1,1,1,1)^\top$.

The plane $W = \operatorname{span}\{u_1,u_2\}$ consists of the vectors whose last three coordinates are equal, so $W^\perp = \{\, z \in \mathbb{R}^4 \,|\, z_1 = 0,\ z_2 + z_3 + z_4 = 0 \,\}$, and

$$w_1 = \frac{1}{\sqrt{2}}(0,1,-1,0)^\top, \qquad w_2 = \frac{1}{\sqrt{6}}(0,1,1,-2)^\top$$

is an orthonormal basis of it: both are orthogonal to $(0,1,1,1)^\top$ and to e_1, their inner product is $(1-1+0)/\sqrt{12} = 0$, and their squared norms are $2/2$ and $6/6$. Computing on $W^\perp$, the second column of Q minus the third is $(0,1,0,-1)^\top$ and the second plus the third minus twice the fourth is $(0,-1,2,-1)^\top$, so

$$Qw_1 = \frac{1}{\sqrt{2}}(0,1,0,-1)^\top = \tfrac{1}{2}w_1 + \tfrac{\sqrt{3}}{2}w_2, \qquad Qw_2 = \frac{1}{\sqrt{6}}(0,-1,2,-1)^\top = -\tfrac{\sqrt{3}}{2}w_1 + \tfrac{1}{2}w_2$$

where the first identity follows from $\frac{1}{2\sqrt{2}}\big[(0,1,-1,0)^\top + (0,1,1,-2)^\top\big] = \frac{1}{\sqrt{2}}(0,1,0,-1)^\top$, using $\frac{\sqrt{3}}{2}\cdot\frac{1}{\sqrt{6}} = \frac{1}{2\sqrt{2}}$, and the second from $\frac{1}{2\sqrt{6}}\big[(0,-3,3,0)^\top + (0,1,1,-2)^\top\big] = \frac{1}{\sqrt{6}}(0,-1,2,-1)^\top$.

So with P the matrix whose columns are u_1, u_2, w_1, w_2,

$$P = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1/\sqrt{3} & 0 & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 0 & -2/\sqrt{6} \end{pmatrix}, \qquad P^\top Q P = \begin{pmatrix} R_{\pi/3} & 0 \\ 0 & R_{\pi/3} \end{pmatrix}$$

with $R_{\pi/3} = \left(\begin{smallmatrix} 1/2 & -\sqrt{3}/2 \\ \sqrt{3}/2 & 1/2 \end{smallmatrix}\right)$. Here $p = 2$ and $r = s = 0$, so $\det Q = 1$: the matrix turns two perpendicular planes of $\mathbb{R}^4$ through the same angle and fixes no direction.

In dimension two the theorem has only two outcomes, and each of them has a name. The determinant decides which occurs.

Corollary 7.5.5: Orthogonal 2×2 Matrices Are Rotations and Reflections

Let $Q \in M_2(\mathbb{R})$ be orthogonal.

1. If $\det Q = 1$, then $Q = R_\theta$ for some $\theta \in \mathbb{R}$.
2. If $\det Q = -1$, then there is an orthogonal $P \in M_2(\mathbb{R})$ with $P^\top Q P = \operatorname{diag}(1, -1)$. Its columns p_1, p_2 satisfy $Qp_1 = p_1$ and $Qp_2 = -p_2$, so Q fixes the line $\mathbb{R}p_1$ pointwise and negates the perpendicular line: it is the *reflection* of $\mathbb{R}^2$ in $\mathbb{R}p_1$. Moreover $Q^\top = Q$ and $Q^2 = I_2$.

***Proof*:**
Write $Q = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. A direct multiplication using definition 1.4.1 gives

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} ad-bc & 0 \\ 0 & ad-bc \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

both off-diagonal entries being $-ab + ba$ or $cd - dc$. By theorem 3.3.4 the scalar $ad - bc$ is $\det Q$.

For (1), assume $\det Q = 1$. The displayed identity then reads $Q\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right) = \left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)Q = I_2$, so $Q^{-1} = \left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$ by definition 1.4.4 and proposition 1.4.5. On the other hand $Q^{-1} = Q^\top = \left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right)$ by proposition 7.1.5. Comparing entries gives $d = a$ and $c = -b$, so $Q = \left(\begin{smallmatrix} a & b \\ -b & a \end{smallmatrix}\right)$ with $a^2 + b^2 = ad - bc = 1$. The pair $(a, -b)$ is therefore a point of the unit circle, so $a = \cos\theta$ and $-b = \sin\theta$ for some real θ, and $Q = R_\theta$.

For (2), assume $\det Q = -1$ and apply theorem 7.5.3 with $n = 2$. Its integers satisfy $2p + r + s = 2$ and $(-1)^s = \det Q = -1$, so s is odd; the only possibility is $s = 1$, and then $2p + r = 1$ forces $p = 0$ and $r = 1$. The normal form is therefore $\operatorname{diag}(1, -1)$, and with P the orthogonal matrix the theorem gives, $P^\top Q P = \operatorname{diag}(1, -1)$. Multiplying on the left by P and using $PP^\top = I_2$ (proposition 7.1.5) gives $QP = P\operatorname{diag}(1, -1)$, whose columns read $Qp_1 = p_1$ and $Qp_2 = -p_2$ (definition 1.4.1).

Finally $Q = P\operatorname{diag}(1, -1)P^\top$, again by multiplying on the right by $P^\top$. Transposing this with proposition 5.5.13(1) and (3), and using that $\operatorname{diag}(1, -1)$ is its own transpose, gives $Q^\top = P\operatorname{diag}(1, -1)P^\top = Q$; and regrouping with proposition 1.4.3,

$$Q^2 = P\operatorname{diag}(1, -1)\bigl(P^\top P\bigr)\operatorname{diag}(1, -1)P^\top = P\,\operatorname{diag}(1, 1)\,P^\top = I_2$$

as we sought to show.

Part (1) identifies the 2×2 blocks of theorem 7.5.3 as rotation matrices in the literal sense, each turning the plane it acts on through a definite angle. Part (2) exhausts the remaining case: an orthogonal 2×2 matrix of determinant -1 has both 1 and -1 as eigenvalues, so it is symmetric and section 7.4 already applies to it.

Dimension three is where the block count says something the reader can see. The sizes of the blocks add to 3, so at most one of them is a rotation block, and at least one is 1×1. The determinant then decides its sign.

Theorem 7.5.6: Euler's Rotation Theorem

Let $Q \in M_3(\mathbb{R})$ be orthogonal with $\det Q = 1$, that is $Q \in \mathrm{SO}(3)$. Then there are a unit vector $u \in \mathbb{R}^3$ with $Qu = u$, an orthonormal basis u, p_2, p_3 of $\mathbb{R}^3$, and a real number θ, such that the matrix P with columns u, p_2, p_3 is orthogonal and

$$P^\top QP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$

So Q fixes every vector of the line $\mathbb{R}u$ and carries the plane $\operatorname{span}\{p_2, p_3\} = (\mathbb{R}u)^\perp$ to itself, acting on it as the rotation R_θ in the basis p_2, p_3.

***Proof*:**
Step 1: the possible block counts. Apply theorem 7.5.3 to Q, obtaining $p, r, s \geq 0$ with $2p+r+s = 3$, angles θ_i with $\sin\theta_i \neq 0$, and an orthogonal P_0 with $P_0^\top QP_0$ in the displayed block form. The determinant clause gives $(-1)^s = \det Q = 1$, so s is even. From $2p + r + s = 3$ the integer p is 0 or 1, and the two cases are:

- $p = 1$: then $r + s = 1$ with s even, forcing $s = 0$ and $r = 1$, and $P_0^\top QP_0$ is $\operatorname{diag}(1)$ followed by the block R_{θ_1};
- $p = 0$: then $r + s = 3$ with s even, so $s = 0$ and $r = 3$, or $s = 2$ and $r = 1$; the normal form is I_3 in the first case and $\operatorname{diag}(1, -1, -1)$ in the second.

In every case $r \geq 1$, so the first diagonal block is (1).

Step 2: each case has the displayed shape. For $p = 1$ the normal form is already $\left(\begin{smallmatrix} 1 & 0 \\ 0 & R_{\theta_1} \end{smallmatrix}\right)$, which is the displayed matrix with $\theta = \theta_1$. For $p = 0$ and $r = 3$ the normal form is I_3, which is the displayed matrix with $\theta = 0$, since $\cos 0 = 1$ and $\sin 0 = 0$ make $R_0 = I_2$. For $p = 0$, $r = 1$ and $s = 2$ the normal form is $\operatorname{diag}(1, -1, -1)$, which is the displayed matrix with $\theta = \pi$, since $\cos\pi = -1$ and $\sin\pi = 0$ make $R_\pi = -I_2$. Take $P = P_0$ and let u, p_2, p_3 be its columns; they are an orthonormal basis of $\mathbb{R}^3$ by proposition 7.1.5.

Step 3: reading the columns. Multiplying $P^\top QP = D$ on the left by P and using $PP^\top = I_3$ gives $QP = PD$. Comparing first columns (definition 1.4.1), the first column of PD is $Pe_1 = u$, so $Qu = u$; comparing the second and third columns gives

$$Qp_2 = \cos\theta\, p_2 + \sin\theta\, p_3, \qquad Qp_3 = -\sin\theta\, p_2 + \cos\theta\, p_3$$

so Q carries $\operatorname{span}\{p_2, p_3\}$ into itself and its matrix there, relative to p_2, p_3, has columns $(\cos\theta, \sin\theta)^\top$ and $(-\sin\theta, \cos\theta)^\top$, which are the columns of R_θ.

Step 4: the plane is the perpendicular one. Both p_2 and p_3 are orthogonal to u, so $\operatorname{span}\{p_2, p_3\} \subseteq (\mathbb{R}u)^\perp$ by proposition 5.5.2(2), which lets orthogonality to a subspace be tested on a spanning list. The left side has dimension 2, and corollary 5.5.4 gives $\dim(\mathbb{R}u)^\perp = 3 - 1 = 2$ as well, so the two subspaces coincide by corollary 2.2.12. Finally $Q(cu) = cu$ for every scalar c, so every vector of $\mathbb{R}u$ is fixed, completing the proof.

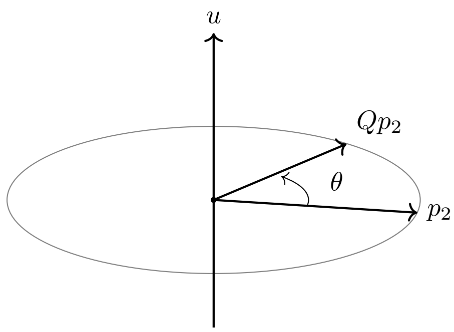


Figure 7.6: A matrix of SO(3) fixes the line $\mathbb{R}u$ and turns the plane $(\mathbb{R}u)^{\perp}$ through θ

Euler's theorem says that composing rotations of space about different lines through the origin produces a rotation about some third line, since the composite is again orthogonal with determinant 1. The fixed line is what makes the statement usable, and it deserves to be pinned down: except for the identity, it is exactly one line.

Corollary 7.5.7: The Axis of a Rotation of Space

Let $Q \in M_3(\mathbb{R})$ be orthogonal with $\det Q = 1$. The set of vectors fixed by Q is the eigenspace $E_1(Q) = \operatorname{null}(Q - I_3)$, and

$$\dim E_1(Q) = 1 \qquad \text{if and only if} \qquad Q \neq I_3$$

When $Q \neq I_3$ the line $E_1(Q)$ is called the *axis* of Q.

***Proof*:**
A vector v satisfies $Qv = v$ exactly when $(Q - I_3)v = 0$, so the fixed vectors are $\operatorname{null}(Q - I_3)$, which is $E_1(Q)$ by proposition 4.1.5 and definition 4.1.4.

If $Q = I_3$ then every vector is fixed and $\dim E_1(Q) = 3$, so the dimension is not 1. Suppose now $Q \neq I_3$, and take u, p_2, p_3 and θ as in theorem 7.5.6. Then $u \in E_1(Q)$, so $\dim E_1(Q) \geq 1$. Let $v = c_1u + c_2p_2 + c_3p_3$ be any fixed vector, the expansion being available because u, p_2, p_3 is a basis of $\mathbb{R}^3$. Using Step 3 of that proof,

$$Qv = c_1u + \big(c_2\cos\theta - c_3\sin\theta\big)p_2 + \big(c_2\sin\theta + c_3\cos\theta\big)p_3$$

Setting $Qv = v$ and comparing coefficients, which is legitimate because the three vectors are independent (proposition 5.3.4), gives

$$c_2 = c_2\cos\theta - c_3\sin\theta, \qquad c_3 = c_2\sin\theta + c_3\cos\theta$$

Multiply the first identity by c_2, the second by c_3, and add: the two terms in $c_2c_3\sin\theta$ cancel and

$$c_2^2 + c_3^2 = \big(c_2^2 + c_3^2\big)\cos\theta$$

Suppose $(c_2, c_3) \neq (0, 0)$. Then $c_2^2 + c_3^2$ is a nonzero real and may be cancelled, giving $\cos\theta = 1$ and hence $\sin^2\theta = 1 - \cos^2\theta = 0$. The two identities of Step 3 of theorem 7.5.6 then read

$Qp_2 = p_2$ and $Qp_3 = p_3$, which with $Qu = u$ gives $QP = P$ for the matrix P with columns u, p_2, p_3; multiplying on the right by $P^\top$ and using $PP^\top = I_3$ (proposition 7.1.5) gives $Q = I_3$, contrary to hypothesis. So $c_2 = c_3 = 0$ and $v \in \operatorname{span}\{u\}$. Hence $E_1(Q) = \operatorname{span}\{u\}$ and $\dim E_1(Q) = 1$, as we sought to show.

Example 7.5.8: A Rotation of $\mathbb{R}^3$ and Its Axis

Let

$$Q = \frac{1}{3}\begin{pmatrix} 2 & -1 & 2 \\ 2 & 2 & -1 \\ -1 & 2 & 2 \end{pmatrix} \in M_3(\mathbb{R}), \qquad M = 3Q$$

The three columns of Q are $(2,2,-1)^\top/3$, $(-1,2,2)^\top/3$ and $(2,-1,2)^\top/3$, each of squared norm $(4+4+1)/9 = 1$, with pairwise inner products

$$\tfrac{1}{9}(-2+4-2) = 0, \qquad \tfrac{1}{9}(4-2-2) = 0, \qquad \tfrac{1}{9}(-2-2+4) = 0$$

so Q is orthogonal by proposition 7.1.5.

Expanding $\det M$ along its first row by theorem 3.3.4,

$$\det M = 2\det\begin{pmatrix} 2 & -1 \\ 2 & 2 \end{pmatrix} - (-1)\det\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} + 2\det\begin{pmatrix} 2 & 2 \\ -1 & 2 \end{pmatrix} = 12 + 3 + 12 = 27$$

Since $M = 3I_3\,Q$, theorem 3.2.10 gives $\det M = \det(3I_3)\det Q$, and $\det(3I_3) = 27$ by proposition 3.2.6, so $\det Q = 1$ and $Q \in \mathrm{SO}(3)$. Theorem 7.5.6 therefore predicts a fixed line.

Finding it means solving $(M - 3I_3)v = 0$. Applying $R_2 \mapsto R_2 + 2R_1$ and then $R_3 \mapsto R_3 - R_1$ and $R_3 \mapsto R_3 + R_2$,

$$M - 3I_3 = \begin{pmatrix} -1 & -1 & 2 \\ 2 & -1 & -1 \\ -1 & 2 & -1 \end{pmatrix} \longrightarrow \begin{pmatrix} -1 & -1 & 2 \\ 0 & -3 & 3 \\ 0 & 0 & 0 \end{pmatrix}$$

The second row gives $v_2 = v_3$ and the first then gives $v_1 = -v_2 + 2v_3 = v_3$, so the fixed vectors are the multiples of $(1,1,1)^\top$ and the axis is $\mathbb{R}(1,1,1)^\top$, of dimension 1 as corollary 7.5.7 requires. Set $u = (1,1,1)^\top/\sqrt{3}$.

For the plane, take

$$p_2 = \frac{1}{\sqrt{2}}(1,-1,0)^\top, \qquad p_3 = \frac{1}{\sqrt{6}}(1,1,-2)^\top$$

Both are orthogonal to $(1,1,1)^\top$, since $1-1+0 = 0$ and $1+1-2 = 0$; their inner product is $(1-1+0)/\sqrt{12} = 0$; and their squared norms are $2/2$ and $6/6$. So u, p_2, p_3 is an orthonormal basis of $\mathbb{R}^3$. Computing $M(1,-1,0)^\top = (3,0,-3)^\top$ and $M(1,1,-2)^\top = (-3,6,-3)^\top$ entry by entry,

$$Qp_2 = \frac{1}{3\sqrt{2}}(3,0,-3)^\top = \frac{1}{\sqrt{2}}(1,0,-1)^\top, \qquad Qp_3 = \frac{1}{3\sqrt{6}}(-3,6,-3)^\top = \frac{1}{\sqrt{6}}(-1,2,-1)^\top$$

These are the combinations $\frac{1}{2}p_2 + \frac{\sqrt{3}}{2}p_3$ and $-\frac{\sqrt{3}}{2}p_2 + \frac{1}{2}p_3$. Indeed $\frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{6}} = \frac{1}{2\sqrt{2}}$, so

$$\tfrac{1}{2}p_2 + \tfrac{\sqrt{3}}{2}p_3 = \frac{1}{2\sqrt{2}}\left[(1,-1,0)^\top + (1,1,-2)^\top\right] = \frac{1}{\sqrt{2}}(1,0,-1)^\top$$

and $\frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}} = \frac{3}{2\sqrt{6}}$, so

$$-\tfrac{\sqrt{3}}{2}p_2 + \tfrac{1}{2}p_3 = \frac{1}{2\sqrt{6}}\big[-3(1,-1,0)^\top + (1,1,-2)^\top\big] = \frac{1}{\sqrt{6}}(-1,2,-1)^\top$$

So $\cos\theta = \frac{1}{2}$ and $\sin\theta = \frac{\sqrt{3}}{2}$, that is $\theta = \pi/3$, and with P the matrix with columns u, p_2, p_3,

$$P = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \end{pmatrix}, \qquad P^\top QP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/2 & -\sqrt{3}/2 \\ 0 & \sqrt{3}/2 & 1/2 \end{pmatrix}$$

The matrix Q turns space through $\pi/3$ about the line through $(1,1,1)^\top$.

Exercise 7.5.1

1. Let $Q = \frac{1}{5}\begin{pmatrix} 3 & 4 \\ 4 & -3 \end{pmatrix} \in M_2(\mathbb{R})$. Verify that Q is orthogonal and compute $\det Q$. Then find unit vectors p_1, p_2 with $Qp_1 = p_1$ and $Qp_2 = -p_2$, and check that the matrix P with these columns satisfies $P^\top QP = \operatorname{diag}(1,-1)$, as corollary 7.5.5(2) requires.

2. Let Q be the matrix of example 7.5.8 and put $Q' = -Q$. Show that Q' is orthogonal and that $\det Q' = -1$. Then use the matrix P computed in that example to write down $P^\top Q'P$, and read off the integers p, r, s and the values of $\cos\theta$ and $\sin\theta$ appearing in theorem 7.5.3 for Q'.

3. Let n be odd and let $Q \in M_n(\mathbb{R})$ be orthogonal. Show that if $\det Q = 1$ then $Qu = u$ for some unit $u \in \mathbb{R}^n$, and that if $\det Q = -1$ then $Qu = -u$ for some unit $u \in \mathbb{R}^n$. (Count the blocks in theorem 7.5.3 and use its determinant clause.)

4. Let $Q \in M_n(\mathbb{R})$ be orthogonal with no eigenvector in $\mathbb{R}^n$. Show that n is even and that $\det Q = 1$. Give such a Q for $n = 2$.

5. Let $Q \in M_n(\mathbb{R})$ be both orthogonal and symmetric. Show that every eigenvalue of Q is 1 or -1, that $Q^2 = I_n$, and that there is an orthogonal $P \in M_n(\mathbb{R})$ for which $P^\top QP$ is diagonal with every diagonal entry equal to 1 or -1. (Use corollary 7.4.3 and proposition 7.5.1(1).)

6. Let $Q \in M_3(\mathbb{R})$ be orthogonal with $\det Q = 1$, and let θ be as in theorem 7.5.6. Show that $\mathrm{tr}Q = 1 + 2\cos\theta$. Then compute the trace of the matrix of example 7.5.8 and recover $\cos\theta = \frac{1}{2}$ from it.

7. Let Q be the 4×4 matrix of example 7.5.4. Using the normal form computed there, show that $Q^3 = -I_4$ and hence $Q^6 = I_4$. Then obtain $Q^3 = -I_4$ a second way, from the matrix $A = 2Q - I_4$ and the identity $A^2 = -3I_4$ established in that example.

8. Let $Q \in M_n(\mathbb{R})$ be orthogonal and let $W \subseteq \mathbb{R}^n$ be a subspace with $Q(W) \subseteq W$. Show that $Q(W) = W$, and deduce that $Q(W^\perp) = W^\perp$. (For the first claim, use that Q is invertible together with theorem 2.3.11 and corollary 2.2.12; for the second, argue as in Step 2 of the proof of theorem 7.5.3.)

7.6 Commuting Families and Simultaneous Diagonalization

Theorem 7.3.2 produces an orthonormal basis of eigenvectors for one normal operator, and the basis it produces depends on that operator. Two normal matrices come with two such bases, and nothing established so far lets the two be taken to be the same list. This section decides when they can be.

One direction costs a computation with diagonal matrices: two matrices that are diagonal in a common orthonormal basis commute, because diagonal matrices do. The work is in the converse, and the mechanism is a single observation about eigenspaces, which is where two matrices first make contact.

Lemma 7.6.1: Commuting Operators Preserve Each Other's Eigenspaces

Let V be a finite-dimensional inner product space over k, let $S, T\colon V \to V$ be linear, and let $\lambda \in k$. Write

$$E_\lambda(T) = \{\, v \in V \big| Tv = \lambda v \,\}$$

extending to an operator the notation definition 4.1.4 uses for a matrix.

1. $E_\lambda(T)$ is a subspace of V.
2. If $ST = TS$, then $S\big(E_\lambda(T)\big) \subseteq E_\lambda(T)$.
3. If $ST = TS$ and T is normal, then $E_\lambda(T) = E_{\overline{\lambda}}(T^*)$ and $S^*\big(E_\lambda(T)\big) \subseteq E_\lambda(T)$.

For $A, B \in M_n(k)$ with $AB = BA$, part (2) says $B\big(E_\lambda(A)\big) \subseteq E_\lambda(A)$, the eigenspace being that of definition 4.1.4.

***Proof*:**

1. The zero vector satisfies $T0 = 0 = \lambda 0$, so it lies in $E_\lambda(T)$. If $Tv = \lambda v$ and $Tv' = \lambda v'$ then linearity of T gives $T(v + v') = Tv + Tv' = \lambda v + \lambda v' = \lambda(v + v')$, and $T(av) = a\,Tv = a\lambda v = \lambda(av)$ for $a \in k$. So $E_\lambda(T)$ contains 0 and is closed under addition and under scalar multiplication, which are the three conditions of definition 2.1.2 read in V.
2. Let $v \in E_\lambda(T)$. Then
$$T(Sv) = (TS)v = (ST)v = S(Tv) = S(\lambda v) = \lambda(Sv)$$
the second equality by the hypothesis $ST = TS$ and the last by linearity of S. So $Sv \in E_\lambda(T)$.
3. First, T^* is normal: proposition 5.5.9(3) gives $(T^*)^* = T$, so $(T^*)^*T^* = TT^* = T^*T = T^*(T^*)^*$, the middle equality because T is normal. Now let $v \in E_\lambda(T)$, so $Tv = \lambda v$; proposition 7.1.13, whose hypotheses are that V is a finite-dimensional inner product space and T is normal, gives $T^*v = \overline{\lambda}v$, so $v \in E_{\overline{\lambda}}(T^*)$. Conversely, if $T^*v = \overline{\lambda}v$, then the same proposition applied to the normal operator T^* gives $(T^*)^*v = \overline{\overline{\lambda}}\,v$, that is $Tv = \lambda v$. The two eigenspaces therefore coincide.

   Taking adjoints in $ST = TS$ with proposition 5.5.9(2) gives $T^*S^* = S^*T^*$, so part (2), applied to the commuting pair S^* and T^* and to the scalar $\overline{\lambda}$, gives $S^*\big(E_{\overline{\lambda}}(T^*)\big) \subseteq E_{\overline{\lambda}}(T^*)$. Rewriting both sides as $E_\lambda(T)$ is the claim.

For the matrix statement, give k^n the standard inner product and let $T_A(x) = Ax$ and $T_B(x) = Bx$, which are linear. Associativity of the matrix product (proposition 1.4.3) gives $T_A T_B = T_{AB} = T_{BA} = T_B T_A$, and $E_\lambda(T_A) = \{v \in k^n \,|\, Av = \lambda v\}$ is the eigenspace $E_\lambda(A)$ of definition 4.1.4. Part (2) applied to T_A and T_B is therefore the assertion made, completing the proof.

Part (2) is what makes a search for common eigenvectors finite. Rather than looking through all of k^n for a vector that both matrices scale, one may look inside a single eigenspace of A, knowing that B does not leave it. When B fails to commute with A that containment fails, and the failure can be read off a single eigenspace.

Example 7.6.2: Two Symmetric Matrices with No Common Eigenvector

Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

both in $M_2(\mathbb{R})$ and both equal to their own transposes, hence self-adjoint (definition 7.1.1) and in particular normal. Their two products are

$$AB = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad BA = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

computed entry by entry from definition 1.4.1: the $(1,2)$ entry of AB is $1 \cdot 1 + 0 \cdot 0 = 1$ while that of BA is $0 \cdot 0 + 1 \cdot (-1) = -1$. So $AB \neq BA$.

The matrix A scales e_1 by 1 and e_2 by -1, and for $v = (a,b)^\top$ the equation $Av = \lambda v$ reads $a = \lambda a$ and $-b = \lambda b$, which for $\lambda = 1$ forces $b = 0$ and for $\lambda = -1$ forces $a = 0$. Hence

$$E_1(A) = \operatorname{span}\{e_1\}, \qquad E_{-1}(A) = \operatorname{span}\{e_2\}$$

and every eigenvector of A is a nonzero multiple of e_1 or of e_2 (definition 4.1.3). Now $Be_1 = e_2$, which is not a multiple of e_1, and $Be_2 = e_1$, which is not a multiple of e_2. So B carries neither eigenspace of A into itself, and no nonzero vector of $\mathbb{R}^2$ is an eigenvector of both A and B.

Each of A and B here is orthogonally diagonalizable on its own, by corollary 7.4.3. What no basis achieves is diagonalizing both at once, and lemma 7.6.1 locates the reason in the failure of B to preserve $E_1(A)$.

To exploit the containment when it does hold, the spectral theorem has to be applied to B acting on the subspace $W = E_\lambda(A)$ rather than on all of k^n. That requires the restricted map to be a normal operator on W, and normality of B on k^n does not by itself deliver it: the adjoint of the restriction is computed inside W, and there is no reason for it to be the restriction of B^* unless W is carried into itself by B^* as well. Part (3) of the lemma gave exactly that second containment, so the following is the last piece needed.

Lemma 7.6.3: The Restriction of a Normal Operator to a Doubly Invariant Subspace Is Normal

Let V be a finite-dimensional inner product space over k, let $S\colon V \to V$ be linear, and let $W \subseteq V$ be a subspace with

$$S(W) \subseteq W \qquad \text{and} \qquad S^*(W) \subseteq W$$

Give W the inner product of V, restricted to pairs of vectors of W, and write $S|_W\colon W \to W$ for the map $w \mapsto Sw$. Then W is a finite-dimensional inner product space over k, the adjoint of $S|_W$ is

$$(S|_W)^* = (S^*)|_W$$

and if S is normal, then so is $S|_W$.

***Proof*:**
Step 1: W is a finite-dimensional inner product space. Each of the three conditions of definition 5.1.2 is a condition on vectors of V, so each holds in particular for all vectors of W, and the restricted function is an inner product on W. By corollary 5.3.11(1) the subspace W has an orthonormal basis, which is a finite list spanning W, so W is finite-dimensional (definition 2.2.15).

Step 2: the adjoint of the restriction. The hypothesis $S(W) \subseteq W$ makes $w \mapsto Sw$ a map from W to W, and it is linear because S is; the hypothesis $S^*(W) \subseteq W$ makes $(S^*)|_W$ a linear map from W to W in the same way. For $w, w' \in W$, the defining property of the adjoint in V (definition 5.5.8) gives

$$\langle S|_W\, w,\, w'\rangle = \langle Sw, w'\rangle = \langle w, S^*w'\rangle = \langle w,\, (S^*)|_W\, w'\rangle$$

every inner product here being that of V, which on vectors of W is the inner product of W. Theorem 5.5.7, applied to the linear map $S|_W$ on the finite-dimensional inner product space W of Step 1, says that exactly one map $W \to W$ has this property, so $(S|_W)^* = (S^*)|_W$.

Step 3: normality. Let S be normal and let $w \in W$. Then $Sw \in W$ and $S^*w \in W$, so both restrictions may be applied to them, and Step 2 gives

$$(S|_W)^*(S|_W)\, w = S^*(Sw) = (S^*S)w = (SS^*)w = S(S^*w) = (S|_W)(S|_W)^*\, w$$

the middle equality by normality of S. The two maps agree at every vector of W, so $(S|_W)^*(S|_W) = (S|_W)(S|_W)^*$ and $S|_W$ is normal (definition 7.1.7), as we sought to show.

The two lemmas now compose. Cutting $\mathbb{C}^n$ into the eigenspaces of A makes each piece invariant under B and under B^*, hence a space on which B acts normally, and the spectral theorem applies there.

Theorem 7.6.4: Commuting Normal Matrices Are Diagonalized by One Unitary

Let $A, B \in M_n(\mathbb{C})$ be normal. The following are equivalent.

1. $AB = BA$.
2. There is a unitary $U \in M_n(\mathbb{C})$ for which U^*AU and U^*BU are both diagonal.

When these hold, the columns of U are an orthonormal basis of $\mathbb{C}^n$ every member of which is an eigenvector of A and an eigenvector of B.

***Proof*:**
Step 1: (2) implies (1). Let U be unitary with $U^*AU = D$ and $U^*BU = E$ diagonal. By the closing clause of proposition 7.1.5, $U^*U = UU^* = I_n$. Multiplying $U^*AU = D$ on the left by U and on the right by U^*, and regrouping with proposition 1.4.3, gives $A = UDU^*$, and likewise $B = UEU^*$.

Two diagonal matrices commute. By definition 1.4.1 the entry of DE in row s and column t is $\sum_r d_{sr}e_{rt}$; the factor d_{sr} vanishes unless $r = s$ and the factor e_{rt} vanishes unless $r = t$, so the whole sum vanishes unless $s = t$, and for $s = t$ it is $d_{ss}e_{ss}$. Exchanging the two factors changes nothing in that computation beyond the order in which two complex numbers are multiplied, so $DE = ED$. Inserting $U^*U = I_n$ twice,

$$AB = UD(U^*U)EU^* = UDEU^* = UEDU^* = UE(U^*U)DU^* = BA$$

which is (1). The columns $u_1, \ldots, u_n$ of U are an orthonormal basis of $\mathbb{C}^n$ by proposition 7.1.5, and the closing clause of corollary 7.3.3, applied first to $A = UDU^*$ and then to $B = UEU^*$, gives $Au_j = d_{jj}u_j$ and $Bu_j = e_{jj}u_j$; each u_j has norm 1 and so is nonzero, hence is an eigenvector of both (definition 4.1.3). This proves the closing clause of the theorem in this direction.

Step 2: the eigenspaces of A and the restriction of B. Assume now that $AB = BA$. Let $\lambda_1, \ldots, \lambda_m$ be the distinct eigenvalues of A and put $W_i = E_{\lambda_i}(A)$, a subspace of $\mathbb{C}^n$ by proposition 4.1.5. Give $\mathbb{C}^n$ the standard inner product and let $T_A(x) = Ax$ and $T_B(x) = Bx$, both linear. By theorem 5.5.14(2) their adjoints are T_{A^*} and T_{B^*}, and associativity of the matrix product (proposition 1.4.3) gives $T_M T_N = T_{MN}$ for all $M, N \in M_n(\mathbb{C})$. Hence

$$(T_A)^*T_A = T_{A^*A} = T_{AA^*} = T_A(T_A)^*, \qquad T_AT_B = T_{AB} = T_{BA} = T_BT_A$$

the first line using normality of A and the second the hypothesis, so T_A is normal (definition 7.1.7) and T_A commutes with T_B; the same computation with B in place of A shows T_B is normal.

In the notation of lemma 7.6.1 the subspace W_i is $E_{\lambda_i}(T_A)$. Parts (2) and (3) of that lemma, whose hypotheses $T_BT_A = T_AT_B$ and T_A normal were just verified, give

$$T_B(W_i) \subseteq W_i \qquad \text{and} \qquad (T_B)^*(W_i) \subseteq W_i$$

so lemma 7.6.3, applied to the normal operator T_B and the subspace W_i, makes $T_B|_{W_i}$ a normal operator on the inner product space W_i.

Step 3: an orthonormal basis of each W_i made of common eigenvectors. Since λ_i is an eigenvalue of A, some nonzero vector lies in W_i (definition 4.1.2), so W_i is a nonzero finite-dimensional complex inner product space by Step 1 of the proof of lemma 7.6.3. Theorem 7.3.2, applied to the normal operator $T_B|_{W_i}$, gives an orthonormal basis $u_1^{(i)}, \ldots, u_{d_i}^{(i)}$ of W_i and scalars $\mu_r^{(i)}$ with $T_B|_{W_i} u_r^{(i)} = \mu_r^{(i)} u_r^{(i)}$, that is $B u_r^{(i)} = \mu_r^{(i)} u_r^{(i)}$. Each $u_r^{(i)}$ lies in $W_i = E_{\lambda_i}(A)$, so it also satisfies $A u_r^{(i)} = \lambda_i u_r^{(i)}$.

Step 4: assembling one basis of $\mathbb{C}^n$. By theorem 7.3.5(1), applied to the normal matrix A,

$$\mathbb{C}^n = W_1 \oplus \cdots \oplus W_m$$

and $\langle x, y \rangle = 0$ whenever $x \in W_i$ and $y \in W_{i'}$ with $i \neq i'$. Concatenate the lists of Step 3 into a single list $u_1, \ldots, u_N$. Every member has norm 1; two members from the same W_i are orthogonal because each list of Step 3 is orthonormal, and two members from different W_i are orthogonal by the clause just quoted. So the list is orthonormal, hence independent (proposition 5.3.4). Every $x \in \mathbb{C}^n$ can be written $x = w_1 + \cdots + w_m$ with $w_i \in W_i$ (definition 2.1.13), and each w_i is a linear combination of the i-th block, so the list spans $\mathbb{C}^n$ and is therefore a basis of $\mathbb{C}^n$ (definition 2.2.1); in particular $N = n$ (definition 2.2.10).

Step 5: the unitary. Let $U \in M_n(\mathbb{C})$ have columns $u_1, \ldots, u_n$. These are an orthonormal basis of $\mathbb{C}^n$, so U is unitary and $U^{-1} = U^*$ by proposition 7.1.5. Every column is an eigenvector of A by Step 3, so theorem 4.2.2 gives $U^{-1}AU = \operatorname{diag}(\lambda_{(1)}, \ldots, \lambda_{(n)})$, where $\lambda_{(j)}$ is the eigenvalue of A belonging to u_j; every column is an eigenvector of B, so the same result gives $U^{-1}BU = \operatorname{diag}(\mu_{(1)}, \ldots, \mu_{(n)})$. Both are diagonal and $U^{-1} = U^*$, which is (2), and the columns of U are common eigenvectors by construction, completing the proof.

Both hypotheses do separate work. Normality of A and of B separately is what gives each of them an orthonormal eigenbasis at all, and commutativity is what forces the two to be alignable into one. Neither can be dropped: example 7.1.8 exhibits a matrix with no orthonormal eigenbasis, and example 7.6.2 two normal matrices with no common eigenvector.

The theorem also explains what a second matrix contributes. When A has a repeated eigenvalue its orthonormal eigenbasis is not determined by A: any orthonormal basis of the eigenspace will serve. A commuting B selects one.

Example 7.6.5: A Common Orthonormal Eigenbasis for a Permutation and a Rank-Two Symmetric Matrix

Let

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

both in $M_3(\mathbb{R})$. Each equals its own transpose, so each is self-adjoint (definition 7.1.1) and therefore normal, since $A^*A = AA = AA^*$ and likewise for B.

They commute. Multiplying on the left by A exchanges rows 2 and 3, since the rows of A are $e_1^\top$, $e_3^\top$, $e_2^\top$; and multiplying on the right by A exchanges columns 2 and 3, since the columns of A are e_1, e_3, e_2 (definition 1.4.1). Both operations produce

$$AB = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} = BA$$

The eigenspaces of A are read off its action: $A(a,b,c)^\top = (a,c,b)^\top$, so $Av = \lambda v$ says $a = \lambda a$, $c = \lambda b$ and $b = \lambda c$. The last two give $b = \lambda^2 b$, so either $\lambda^2 = 1$ or $b = c = 0$; in the latter case

$a \neq 0$ is forced for $v \neq 0$ and then $\lambda = 1$. For $\lambda = 1$ the conditions reduce to $b = c$, and for $\lambda = -1$ to $a = 0$ and $c = -b$. Writing $f = (0,1,1)^\top$ and $g = (0,1,-1)^\top$,

$$E_1(A) = \operatorname{span}\{e_1, f\}, \qquad E_{-1}(A) = \operatorname{span}\{g\}$$

and $\langle e_1, f\rangle = 0$, so e_1, f is an orthogonal basis of the two-dimensional eigenspace $W = E_1(A)$.

Neither e_1 nor f is an eigenvector of B. Reading the columns of B,

$$Be_1 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = 2e_1 + f, \qquad Bf = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = 2e_1 + f$$

and $2e_1 + f$ is a multiple of neither e_1 nor f. Both images lie in W, as lemma 7.6.1(2) predicts.

Restricting B to W makes the two eigenvectors visible. A vector of W is $v = ae_1 + bf$, and linearity with the display above gives

$$Bv = a(2e_1 + f) + b(2e_1 + f) = (a+b)(2e_1 + f)$$

So $Bv = 0$ exactly when $a + b = 0$, and otherwise Bv is a nonzero multiple of $2e_1 + f$, which forces an eigenvector of B inside W with nonzero eigenvalue to be a multiple of $2e_1 + f$. Taking $a = 1$, $b = -1$ and then $a = 2$, $b = 1$,

$$B(e_1 - f) = 0, \qquad B(2e_1 + f) = 3(2e_1 + f)$$

where the second follows from $a+b = 3$. In coordinates $e_1 - f = (1,-1,-1)^\top$ and $2e_1 + f = (2,1,1)^\top$, and these are orthogonal, as proposition 7.1.14 requires of eigenvectors of the normal operator $B|_W$ for the distinct eigenvalues 0 and 3.

On the remaining eigenspace, Bg is the second column of B minus the third, namely $(1,1,0)^\top - (1,0,1)^\top = (0,1,-1)^\top = g$, so $Bg = g$. Normalizing the three vectors, whose squared norms are $4+1+1 = 6$, $1+1+1 = 3$ and $0+1+1 = 2$, gives the columns of

$$Q = \begin{pmatrix} 2/\sqrt{6} & 1/\sqrt{3} & 0 \\ 1/\sqrt{6} & -1/\sqrt{3} & 1/\sqrt{2} \\ 1/\sqrt{6} & -1/\sqrt{3} & -1/\sqrt{2} \end{pmatrix}$$

Its columns are pairwise orthogonal and of norm 1, so Q is unitary with $Q^{-1} = Q^* = Q^\top$ (proposition 7.1.5), and every column is an eigenvector of both matrices. By theorem 4.2.2,

$$Q^\top AQ = \operatorname{diag}(1,1,-1), \qquad Q^\top BQ = \operatorname{diag}(3,0,1)$$

Neither matrix alone determines this basis: A has the repeated eigenvalue 1, and B has none of the three vectors forced on it until A has cut $\mathbb{R}^3$ into W and $\operatorname{span}\{g\}$. The matrix Q came out real, which corollary 7.6.7 below shows is always available for commuting symmetric matrices.

Two matrices were treated only because the argument was stated for two. The eigenspaces of A are invariant under every matrix commuting with A, not just one, so the same cut can be made once and used by an entire collection. What changes is that a single cut no longer suffices: after restricting to $E_\lambda(A)$, the remaining members of the collection must be separated inside it, and that is an induction on dimension. The statement is also the one an application wants, since a collection of operators to be diagonalized together is rarely a pair.

Theorem 7.6.6: One Orthonormal Eigenbasis for a Commuting Family

Let k be $\mathbb{R}$ or $\mathbb{C}$, let V be a nonzero finite-dimensional inner product space over k, and let $\mathcal{F}$ be a set of linear maps $V \to V$, not assumed finite. Suppose that

1. every member of $\mathcal{F}$ is normal when $k = \mathbb{C}$, and self-adjoint when $k = \mathbb{R}$; and
2. $ST = TS$ for all $S, T \in \mathcal{F}$.

Then V has an orthonormal basis every member of which is an eigenvector of every member of $\mathcal{F}$.
Conversely, if V has an orthonormal basis every member of which is an eigenvector of every member of $\mathcal{F}$, then (1) and (2) hold.

***Proof*:**
Step 1: the converse. Let $u_1, \dots, u_n$ be an orthonormal basis of V with $Tu_j = \tau_j(T)\, u_j$ for every $T \in \mathcal{F}$ and every j, the scalars $\tau_j(T)$ lying in k. For $S, T \in \mathcal{F}$ and each j,

$$(ST)u_j = S\big(\tau_j(T)u_j\big) = \tau_j(T)\tau_j(S)\, u_j = T\big(\tau_j(S)u_j\big) = (TS)u_j$$

the middle equality because two scalars of k multiply in either order. An arbitrary $v \in V$ is $v = \sum_j a_j u_j$ for scalars a_j (definition 2.2.1), so linearity of ST and of TS gives $(ST)v = \sum_j a_j (ST)u_j = \sum_j a_j (TS)u_j = (TS)v$. Hence $ST = TS$, which is (2).

For (1) with $k = \mathbb{C}$: each $T \in \mathcal{F}$ has an orthonormal basis of eigenvectors of V, namely $u_1, \dots, u_n$, so T is normal by the implication (2) $\Rightarrow$ (1) of theorem 7.3.2, whose hypotheses are that V is a nonzero finite-dimensional complex inner product space. For (1) with $k = \mathbb{R}$: fix $T \in \mathcal{F}$ and write $\tau_j = \tau_j(T)$, a real number. Over $\mathbb{R}$ the inner product is linear in each argument separately (definition 5.1.2(1) and proposition 5.1.6(2), conjugation being the identity on $\mathbb{R}$), so for all s and t,

$$\langle Tu_s, u_t\rangle = \tau_s \langle u_s, u_t\rangle, \qquad \langle u_s, Tu_t\rangle = \tau_t \langle u_s, u_t\rangle$$

and both sides vanish when $s \neq t$ while both equal τ_s when $s = t$. Expanding arbitrary $v = \sum_s a_s u_s$ and $w = \sum_t b_t u_t$ by the same linearity gives $\langle Tv, w\rangle = \langle v, Tw\rangle$ for all $v, w \in V$. The map T therefore satisfies the property that theorem 5.5.7 attributes to exactly one map, so $T^* = T$ and T is self-adjoint (definition 7.1.1).

Step 2: every member of $\mathcal{F}$ is normal. Assume (1) and (2) from now on. Over $\mathbb{C}$ this is (1) itself; over $\mathbb{R}$, a self-adjoint T has $T^*T = TT = TT^*$, so T is normal (definition 7.1.7). The rest of the proof is an induction on $\dim V$.

Step 3: the base case $\dim V = 1$. By corollary 5.3.11(1) applied to $W = V$, the space V has an orthonormal basis, whose length is $\dim V = 1$ (definition 2.2.15); call its single member u. For $T \in \mathcal{F}$ the vector Tu lies in V, which u spans, so $Tu = \tau u$ for some $\tau \in k$; and $u \neq 0$ because $\|u\| = 1$. So u is an eigenvector of every member of $\mathcal{F}$.

Step 4: the inductive step, first case. Let $\dim V = d \geq 2$ and assume the conclusion for every nonzero finite-dimensional inner product space over k of dimension less than d, and every set of maps on it satisfying (1) and (2). Suppose first that every $T \in \mathcal{F}$ is of the form $\tau_T \operatorname{id}_V$ for some $\tau_T \in k$. Take any orthonormal basis of V (corollary 5.3.11(1)); each member u is nonzero and satisfies $Tu = \tau_T u$, so is an eigenvector of every member of $\mathcal{F}$, and the conclusion holds.

Step 5: the inductive step, second case. Otherwise fix $T_0 \in \mathcal{F}$ with $T_0 \neq \tau \operatorname{id}_V$ for every $\tau \in k$. Over $\mathbb{C}$ the operator T_0 is normal and theorem 7.3.2 applies; over $\mathbb{R}$ it is self-adjoint and theorem 7.4.2 applies. Either way V has an orthonormal basis $w_1, \dots, w_d$ with $T_0 w_j = \nu_j w_j$ for scalars $\nu_j \in k$. Let $\lambda_1, \dots, \lambda_m$ be the distinct values occurring among $\nu_1, \dots, \nu_d$, and put $W_i = E_{\lambda_i}(T_0)$ in the notation of lemma 7.6.1, a subspace of V by part (1) of that lemma.

Each W_i contains some w_j and so is nonzero. Each is also a proper subspace: the choice of T_0 gives a vector v with $T_0 v \neq \lambda_i v$, and that v is not in W_i. Consequently $\dim W_i < d$. Indeed, corollary 5.3.11(1) gives an orthonormal basis $q_1, \dots, q_{d_i}$ of W_i, which is an orthonormal list in V and so extends to an orthonormal basis of V with its first d_i members unchanged (corollary 5.3.11(2)); that basis has length d (definition 2.2.15), so $d_i \leq d$, and if $d_i = d$ the extension adds nothing, making $q_1, \dots, q_d$ a basis of V all of whose members lie in the subspace W_i. Then W_i would contain every linear combination of them, that is all of V, contradicting properness.

Every $v \in V$ is a linear combination of $w_1, \dots, w_d$ (definition 2.2.1), and grouping its terms according to which λ_i the corresponding ν_j equals writes v as a sum of vectors, one from each W_i. So $V = W_1 + \cdots + W_m$.

Step 6: the restricted family. Fix i and let $S \in \mathcal{F}$. Then $ST_0 = T_0 S$ by (2), and T_0 is normal by Step 2, so lemma 7.6.1, parts (2) and (3), gives $S(W_i) \subseteq W_i$ and $S^*(W_i) \subseteq W_i$. By lemma 7.6.3 the subspace W_i is a finite-dimensional inner product space over k, the restriction $S|_{W_i}$ is normal, and $(S|_{W_i})^* = (S^*)|_{W_i}$. Over $\mathbb{R}$ this last identity reads $(S|_{W_i})^* = S|_{W_i}$, since $S^* = S$ there, so $S|_{W_i}$ is self-adjoint. Hence $\mathcal{F}_i = \{\, S|_{W_i} \big| S \in \mathcal{F} \,\}$ satisfies (1) on W_i. It satisfies (2) as well: for $w \in W_i$ one has $Tw \in W_i$ and $Sw \in W_i$, so

$$(S|_{W_i})(T|_{W_i})\, w = S(Tw) = (ST)w = (TS)w = T(Sw) = (T|_{W_i})(S|_{W_i})\, w$$

Step 7: applying the induction hypothesis and assembling. Each W_i is a nonzero finite-dimensional inner product space over k with $\dim W_i = d_i < d$, and $\mathcal{F}_i$ satisfies (1) and (2), so the induction hypothesis gives an orthonormal basis $u_1^{(i)}, \dots, u_{d_i}^{(i)}$ of W_i every member of which is an eigenvector of every $S|_{W_i}$. Since $S|_{W_i} u = Su$ for $u \in W_i$, each of these vectors is an eigenvector of every $S \in \mathcal{F}$.

Concatenate the m lists. Every member has norm 1; two members of the same list are orthogonal by construction; and two members of different lists lie in W_i and $W_{i'}$ with $\lambda_i \neq \lambda_{i'}$, so they are eigenvectors of the normal operator T_0 for distinct eigenvalues and are orthogonal by proposition 7.1.14. The concatenation is therefore orthonormal, hence independent (proposition 5.3.4). It spans V, because $V = W_1 + \cdots + W_m$ by Step 5 and each W_i is spanned by its own list. So it is an orthonormal basis of V (definition 2.2.1) every member of which is an eigenvector of every member of $\mathcal{F}$. This is the conclusion for V, completing the induction and the proof.

The induction runs on $\dim V$ and never on the number of operators, which is why $\mathcal{F}$ may be infinite: each step handles the whole family at once, and each step strictly lowers the dimension, so the recursion terminates after at most $\dim V$ descents. The two fields enter at exactly one place, Step 5, where the operator T_0 needs an orthonormal eigenbasis of its own.

That is also where the real hypothesis has to be self-adjointness rather than normality. A real normal matrix need not have even one real eigenvector, by proposition 7.4.1, so Step 5 would have nothing to cut with. In matrix form over $\mathbb{R}$ the theorem reads as follows.

Corollary 7.6.7: Commuting Symmetric Matrices Are Simultaneously Orthogonally Diagonalizable

Let $n \geq 1$ and let $\mathcal{F} \subseteq M_n(\mathbb{R})$ be a set of symmetric matrices with $AB = BA$ for all $A, B \in \mathcal{F}$. Then there is a real orthogonal $Q \in M_n(\mathbb{R})$ such that $Q^\top AQ$ is diagonal for every $A \in \mathcal{F}$.

***Proof*:**
Give $\mathbb{R}^n$ the standard inner product and for $A \in \mathcal{F}$ let $T_A(x) = Ax$, a linear map $\mathbb{R}^n \to \mathbb{R}^n$. The standard basis is orthonormal, so theorem 5.5.14 gives the matrix of $(T_A)^*$ relative to it as A^*, which is $A^\top = A$ because the entries of A are real and A is symmetric (definition 5.5.12, definition 2.3.4). A linear map is determined by its matrix relative to a fixed basis (theorem 2.4.8), so $(T_A)^* = T_A$ and T_A is self-adjoint (definition 7.1.1). For $A, B \in \mathcal{F}$, associativity of the matrix product (proposition 1.4.3) gives $T_A T_B = T_{AB} = T_{BA} = T_B T_A$.

The family $\{\, T_A \big| A \in \mathcal{F} \,\}$ therefore satisfies hypotheses (1) and (2) of theorem 7.6.6 over $k = \mathbb{R}$, and $\mathbb{R}^n$ is a nonzero finite-dimensional real inner product space because $n \geq 1$. That theorem gives an orthonormal basis $q_1, \ldots, q_n$ of $\mathbb{R}^n$ with $Aq_j = \lambda_{A,j} q_j$ for every $A \in \mathcal{F}$ and every j, where $\lambda_{A,j} \in \mathbb{R}$.

Let $Q \in M_n(\mathbb{R})$ have columns $q_1, \ldots, q_n$. Its columns are an orthonormal basis of $\mathbb{R}^n$, so Q is unitary, that is real orthogonal, with $Q^{-1} = Q^* = Q^\top$ (proposition 7.1.5, definition 5.5.12). Fix $A \in \mathcal{F}$; its eigenvectors $q_1, \ldots, q_n$ form a basis of $\mathbb{R}^n$, so theorem 4.2.2 gives

$$Q^\top AQ = Q^{-1}AQ = \operatorname{diag}(\lambda_{A,1}, \ldots, \lambda_{A,n})$$

which is diagonal, and Q does not depend on A, as we sought to show.

The corollary is the form the geometry of quadratic forms uses: a family of symmetric matrices that commute can be put into diagonal form by one rotation of coordinates, so the coordinate system adapted to one of them is adapted to all. Two symmetric matrices that fail to commute admit no such coordinate system, and example 7.6.2 is the smallest instance.

Exercise 7.6.1

1. Let
$$A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$
in $M_2(\mathbb{R})$. Verify that $AB = BA$ by computing both products, then find a real orthogonal Q for which $Q^\top AQ$ and $Q^\top BQ$ are both diagonal, and give the two diagonal matrices.

2. Let
$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$
in $M_3(\mathbb{R})$. Show that $AB = BA$, and check that each of A and B has a repeated eigenvalue. Find a real orthogonal Q diagonalizing both, and give $Q^\top AQ$ and $Q^\top BQ$. Explain, using lemma 7.6.1, why the common eigenbasis is determined up to signs and reordering even though neither matrix determines a basis on its own.

3. Let $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$ be the matrix of proposition 7.4.1, read in $M_2(\mathbb{C})$, and let $C = 2I_2 + J$. Show that C is normal and that $JC = CJ$, then find a unitary $U \in M_2(\mathbb{C})$ for which U^*JU and U^*CU are both diagonal, and give the two diagonal matrices.
4. Let $A, B \in M_n(\mathbb{C})$ be normal with $AB = BA$. Show that $A + B$ and AB are normal. (Diagonalize both by one unitary and use corollary 7.3.3.)
5. Let
$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$
in $M_2(\mathbb{C})$. Verify that A and B are normal, that $AB \neq BA$, and that $A + B$ is not normal. Reconcile this with exercise 7.6.1 (4).
6. Let $A \in M_n(\mathbb{C})$ be normal with n distinct eigenvalues $\lambda_1, \dots, \lambda_n$. Show first that every eigenspace $E_{\lambda_j}(A)$ has dimension 1, and then that every $B \in M_n(\mathbb{C})$ with $AB = BA$ satisfies $Bu = \mu u$ for each eigenvector u of A and some scalar μ depending on u. Conclude that every such B is normal and is diagonalized by any unitary that diagonalizes A. Note that B is not assumed normal.
7. Let $A, B \in M_n(\mathbb{R})$ be symmetric. Show that AB is symmetric if and only if $AB = BA$, and that when this holds there is one real orthogonal Q putting A, B and AB simultaneously into diagonal form.

7.7 Application: Circulant Matrices and the Discrete Fourier Transform

Section 7.6 showed that a family of normal matrices is diagonalized in one orthonormal basis exactly when its members commute with one another. This section works out one such family completely: the basis is written down coordinate by coordinate, and the eigenvalues of every member of the family are given by an explicit formula in the entries of that member.

The family is the circulant matrices, whose rows are the successive cyclic shifts of the first row. All of them are polynomials in a single matrix, the shift that moves each coordinate of a vector one place, so the whole family is diagonalized as soon as that one matrix is. Two conventions are fixed first, since every formula below depends on them.

Throughout the section $n \geq 2$ is a fixed integer, and the coordinates of a vector in $\mathbb{C}^n$ are numbered $0, 1, \dots, n-1$ rather than $1, \dots, n$: a vector is $x = (x_0, x_1, \dots, x_{n-1})^\top$ and the standard basis is $e_0, \dots, e_{n-1}$. Rows and columns of a matrix in $M_n(\mathbb{C})$ are numbered the same way, so $A = (a_{lk})$ with $0 \leq l, k \leq n-1$. Any subscript on a coordinate, a row index or a column index is read modulo n, so that x_{-1} means x_{n-1} and e_n means e_0.

The second convention is a complex number. Put

$$\omega = \cos\frac{2\pi}{n} + i \sin\frac{2\pi}{n}$$

Five properties of ω are used below, and each is recorded here with its justification, since the arithmetic of the section is nothing but these five facts applied repeatedly.

1. $\omega^r = \cos\frac{2\pi r}{n} + i\sin\frac{2\pi r}{n}$ for every integer r. For $r \geq 0$ this is an induction on r, the case $r = 0$ reading $1 = \cos 0 + i \sin 0$ and the inductive step being the identity

   $$(\cos\alpha+i\sin\alpha)(\cos\beta+i\sin\beta) = (\cos\alpha\cos\beta-\sin\alpha\sin\beta)+i(\sin\alpha\cos\beta+\cos\alpha\sin\beta) = \cos(\alpha+\beta)+i\sin(\alpha+\beta)$$

   which is the product of two complex numbers followed by the addition formulas for cosine and sine. Since $\cos^2\theta + \sin^2\theta = 1$, every number of the form $\cos\theta + i\sin\theta$ has modulus 1; in particular $\omega\overline{\omega} = |\omega|^2 = 1$, so $\omega^{-1} = \overline{\omega} = \cos\frac{-2\pi}{n} + i\sin\frac{-2\pi}{n}$ because cosine is even and sine is odd, and the same induction applied to ω^{-1} covers $r < 0$.

2. $\omega^n = 1$, and ω^r depends only on the remainder of r modulo n. The first claim is (1) at $r = n$, where the angle is 2π; for the second, write $r = qn + s$ with q, s integers and $0 \leq s \leq n-1$, so that $\omega^r = (\omega^n)^q\omega^s = \omega^s$.

3. The numbers $\omega^0, \omega^1, \ldots, \omega^{n-1}$ are distinct. Suppose $\omega^r = \omega^{r'}$ with $0 \leq r < r' \leq n-1$. Each power of ω has modulus 1 by (1) and is therefore nonzero, so dividing gives $\omega^s = 1$ with $s = r' - r$ and $0 < s < n$. By (1) this says $\cos\frac{2\pi s}{n} = 1$ with $0 < \frac{2\pi s}{n} < 2\pi$, and the cosine takes the value 1 only at integer multiples of 2π.

4. $\overline{\omega^r} = \omega^{-r}$ for every integer r. By (1) the number ω^r has modulus 1, so $\omega^r\overline{\omega^r} = 1$, which exhibits $\overline{\omega^r}$ as the inverse of ω^r, and that inverse is ω^{-r}.

5. For every integer r,

   $$\sum_{k=0}^{n-1} \omega^{rk} = \begin{cases} n & \text{if } n \text{ divides } r \\ 0 & \text{otherwise} \end{cases}$$

   If n divides r then $\omega^r = 1$ by (2), so every summand $(\omega^r)^k$ equals 1 and the sum is n. If n does not divide r, put $z = \omega^r$; by (2) and (3), $z = \omega^s$ with $0 < s \leq n-1$ and $z \neq 1$. Expanding the product and cancelling in consecutive pairs gives $(z-1)\sum_{k=0}^{n-1} z^k = z^n - 1$, and $z^n = (\omega^n)^r = 1$, so $(z-1)\sum_{k=0}^{n-1} z^k = 0$; since $z \neq 1$, the sum vanishes.

The matrix that generates the whole section is the one that renumbers the coordinates of a vector cyclically.

Definition 7.7.1: The Cyclic Shift

The *cyclic shift* on $\mathbb{C}^n$ is the matrix $S \in M_n(\mathbb{C})$ determined by its action on the standard basis,

$$Se_k = e_{k-1} \qquad (k = 0, 1, \ldots, n-1)$$

the index $k-1$ read modulo n, so that $Se_0 = e_{n-1}$. Its entries are $S_{lk} = 1$ when $k \equiv l + 1 \pmod{n}$ and $S_{lk} = 0$ otherwise.

The two descriptions agree because the k-th column of a matrix A is Ae_k: by definition 1.4.1, $(Ae_k)_l = \sum_m A_{lm}(e_k)_m = A_{lk}$. So the k-th column of S is e_{k-1}, which has its single 1 in row $l = k-1$, that is at the position with $k \equiv l + 1$.

On coordinates S moves each entry one place towards the front:

$$Sx = \sum_k x_k Se_k = \sum_k x_k e_{k-1} = \sum_l x_{l+1} e_l, \qquad \text{so} \qquad (Sx)_l = x_{l+1}$$

with x_0 arriving in the last coordinate. For $n = 4$, with rows and columns numbered $0, 1, 2, 3$,

$$S = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad S \begin{pmatrix} x_0 \\ x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_0 \end{pmatrix}$$

A matrix whose entry in position (l, k) depends only on the difference $k - l$ is constant along each diagonal that wraps around, and its rows are the cyclic rearrangements of its first row.

Definition 7.7.2: Circulant Matrix

A matrix $C \in M_n(\mathbb{C})$ is a *circulant* if there are complex numbers $c_0, c_1, \ldots, c_{n-1}$ with

$$C_{lk} = c_{k-l} \qquad (0 \le l, k \le n - 1)$$

the index $k - l$ read modulo n. Taking $l = 0$ gives $C_{0k} = c_k$, so $c_0, \ldots, c_{n-1}$ are the entries of the first row of C and are determined by C; the vector $c = (c_0, \ldots, c_{n-1})^\top \in \mathbb{C}^n$ is called the first row of C throughout.

Row $l + 1$ of a circulant carries in column k the entry c_{k-l-1}, which is what row l carries in column $k - 1$: each row is the row above it moved one place to the right, with the entry pushed off the end reappearing at the front. For $n = 4$ this is

$$C = \begin{pmatrix} c_0 & c_1 & c_2 & c_3 \\ c_3 & c_0 & c_1 & c_2 \\ c_2 & c_3 & c_0 & c_1 \\ c_1 & c_2 & c_3 & c_0 \end{pmatrix}$$

The shift itself is the circulant with first row $(0, 1, 0, \ldots, 0)$, since $S_{lk} = 1$ exactly when $k - l \equiv 1$.

Before any circulant is analysed, the shift alone is diagonalized. Its columns are a rearrangement of the standard basis, which makes it unitary and so normal, and corollary 7.3.3 therefore guarantees an orthonormal eigenbasis without producing one. The point of the next result is that the eigenvectors can be written down: each is a list of powers of ω.

Proposition 7.7.3: The Eigenvectors of the Cyclic Shift

Let $S \in M_n(\mathbb{C})$ be the cyclic shift, and for $j = 0, 1, \ldots, n-1$ let $f_j \in \mathbb{C}^n$ be the vector with coordinates

$$(f_j)_k = \frac{1}{\sqrt{n}}\,\omega^{jk} \qquad (k = 0, 1, \ldots, n-1), \qquad \text{that is} \qquad f_j = \frac{1}{\sqrt{n}}\big(1, \omega^j, \omega^{2j}, \ldots, \omega^{(n-1)j}\big)^\top$$

Then:

1. $S^*S = SS^* = I_n$ and $S^{-1} = S^*$; in particular S is normal;
2. $S^r e_k = e_{k-r}$ for every integer $r \geq 0$, and $S^n = I_n$ and $S^* = S^{n-1}$;
3. $Sf_j = \omega^j f_j$ for each j;
4. $f_0, f_1, \ldots, f_{n-1}$ is an orthonormal basis of $\mathbb{C}^n$, and the scalars $\omega^0, \ldots, \omega^{n-1}$ are distinct, so the eigenspace $E_{\omega^j}(S)$ is the line $\operatorname{span}(f_j)$ for each j.

***Proof*:**
Throughout, $\mathbb{C}^n$ carries the standard inner product, for which $\langle x, y\rangle = \sum_k x_k \overline{y_k}$.

1. The columns of S are $Se_0, \ldots, Se_{n-1}$, that is $e_{n-1}, e_0, e_1, \ldots, e_{n-2}$: the standard basis vectors in a different order. For any two of them, $\langle e_a, e_b\rangle = \sum_k (e_a)_k \overline{(e_b)_k}$, and the k-th summand vanishes unless $k = a$ and $k = b$, so the inner product is 1 when $a = b$ and 0 otherwise. The columns of S are therefore an orthonormal list (definition 5.3.3), and proposition 7.1.5 gives $S^*S = SS^* = I_n$ and $S^{-1} = S^*$. The equality $S^*S = SS^*$ is normality.

2. Induct on r. For $r = 0$ the claim is $I_n e_k = e_k$, and if $S^r e_k = e_{k-r}$ then $S^{r+1} e_k = S(S^r e_k) = Se_{k-r} = e_{k-r-1}$ by definition 7.7.1, the index again read modulo n. At $r = n$ the index $k - n$ is congruent to k, so $S^n e_k = e_k$ for every k; two matrices whose columns agree are equal, so $S^n = I_n$. Consequently $S \cdot S^{n-1} = S^{n-1} \cdot S = S^n = I_n$, which makes S^{n-1} an inverse of S; inverses are unique (proposition 1.4.5), so $S^{n-1} = S^{-1} = S^*$ by (1).

3. The formula $(f_j)_k = \frac{1}{\sqrt{n}}\omega^{jk}$ is consistent with reading k modulo n, because $\omega^{j(k+n)} = \omega^{jk}$ by property (2) of ω. Using $(Sx)_l = x_{l+1}$,
$$(Sf_j)_l = (f_j)_{l+1} = \frac{1}{\sqrt{n}}\,\omega^{j(l+1)} = \omega^j \cdot \frac{1}{\sqrt{n}}\,\omega^{jl} = \omega^j (f_j)_l$$
for every l, which is the identity $Sf_j = \omega^j f_j$.

4. For any j and j', property (4) of ω gives $\overline{(f_{j'})_k} = \frac{1}{\sqrt{n}}\omega^{-j'k}$, so
$$\langle f_j, f_{j'}\rangle = \sum_{k=0}^{n-1} (f_j)_k \overline{(f_{j'})_k} = \frac{1}{n}\sum_{k=0}^{n-1} \omega^{(j-j')k}$$
which by property (5) is 1 if n divides $j - j'$ and 0 otherwise. For $0 \leq j, j' \leq n-1$ the difference $j - j'$ lies strictly between $-n$ and n, so n divides it exactly when $j = j'$. The list $f_0, \ldots, f_{n-1}$ is therefore orthonormal, and hence linearly independent by proposition 5.3.4.

It spans $\mathbb{C}^n$. Given $x \in \mathbb{C}^n$, put $a_j = \langle x, f_j \rangle = \frac{1}{\sqrt{n}} \sum_k x_k \omega^{-jk}$, again by property (4). Then for each coordinate l,

$$\sum_{j=0}^{n-1} a_j (f_j)_l = \frac{1}{n} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} x_k\, \omega^{-jk} \omega^{jl} = \frac{1}{n} \sum_{k=0}^{n-1} x_k \sum_{j=0}^{n-1} \omega^{j(l-k)} = \frac{1}{n} x_l \cdot n = x_l$$

where the inner sum was evaluated by property (5), which leaves only the term $k = l$. So $x = \sum_j a_j f_j$, and the list is a basis (definition 2.2.1).

The scalars $\omega^0, \ldots, \omega^{n-1}$ are distinct by property (3). If $Sx = \omega^j x$, expand $x = \sum_l a_l f_l$ in this basis; then $Sx = \sum_l a_l \omega^l f_l$ by (3), while $\omega^j x = \sum_l a_l \omega^j f_l$, and the coordinates of a vector in a basis are unique, so $a_l(\omega^l - \omega^j) = 0$ for every l. For $l \neq j$ the factor $\omega^l - \omega^j$ is nonzero, forcing $a_l = 0$. Hence x is a multiple of f_j, and since $Sf_j = \omega^j f_j$ the converse inclusion holds as well: $E_{\omega^j}(S) = \operatorname{span}(f_j)$ (definition 4.1.4), as we sought to show.

The n eigenvalues of S are the n numbers $\omega^0, \ldots, \omega^{n-1}$, which are the vertices of a regular n-gon inscribed in the unit circle of $\mathbb{C}$. Proposition 7.1.12 already forced them onto that circle, since S is unitary; what proposition 7.7.3 adds is that they are equally spaced around it and that each comes with a named eigenvector.

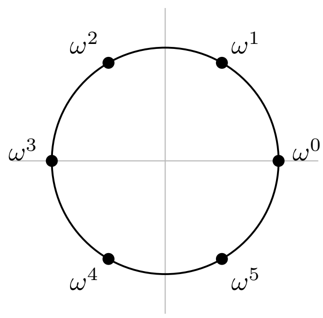


Figure 7.7: The eigenvalues of the cyclic shift, drawn in the complex plane for $n = 6$

Nothing so far connects S to a general circulant. The connection is that the powers of S are exactly the circulants whose first row is a standard basis vector, so a linear combination of powers of S is a circulant with an arbitrary first row.

Proposition 7.7.4: Circulants Are the Polynomials in the Shift

Let $C \in M_n(\mathbb{C})$ and let $c_0, \ldots, c_{n-1}$ be complex numbers. Then C is the circulant with first row $(c_0, \ldots, c_{n-1})$ if and only if

$$C = c_0 I_n + c_1 S + c_2 S^2 + \cdots + c_{n-1} S^{n-1}$$

In particular a matrix is a circulant if and only if it is a polynomial in S, and the coefficients $c_0, \ldots, c_{n-1}$ in the displayed expression are uniquely determined by C.

***Proof*:**
Fix r with $0 \le r \le n-1$. By proposition 7.7.3(2) the k-th column of S^r is $S^r e_k = e_{k-r}$, whose single nonzero entry is a 1 in row $k-r$. Reading this as a statement about entries,

$$(S^r)_{lk} = \begin{cases} 1 & \text{if } r \equiv k-l \pmod{n} \\ 0 & \text{otherwise} \end{cases}$$

Now fix l and k. Among the n values $r = 0, 1, \ldots, n-1$ exactly one is congruent to $k-l$ modulo n, namely the remainder of $k-l$ on division by n; call it r_0, so that $c_{r_0} = c_{k-l}$ under the convention that indices are read modulo n. Hence

$$\Big(\sum_{r=0}^{n-1} c_r S^r\Big)_{lk} = \sum_{r=0}^{n-1} c_r (S^r)_{lk} = c_{r_0} = c_{k-l}$$

So the matrix $\sum_r c_r S^r$ has entry c_{k-l} in position (l,k), which by definition 7.7.2 says exactly that it is the circulant with first row $(c_0, \ldots, c_{n-1})$. Both directions of the equivalence follow, since two matrices are equal precisely when all their entries agree.

For the last two claims: a polynomial in S is a linear combination of powers S^m with $m \ge 0$, and $S^m = S^r$ whenever $m \equiv r$ modulo n, because $S^n = I_n$ by proposition 7.7.3(2); collecting the exponents into their remainders puts any such polynomial in the displayed form, and every matrix of the displayed form is a polynomial in S. Uniqueness holds because the entry of $\sum_r c_r S^r$ in position $(0,k)$ is c_k, so the coefficients are the entries of the first row of C, completing the proof.

Every question about a circulant is now a question about a single matrix. Proposition 7.3.6(1) evaluates a polynomial p in a normal matrix on that matrix's spectral resolution, and returns the values of p at the eigenvalues; applied to S it predicts that the eigenvalues of the circulant with first row c are the values of $c_0 + c_1 t + \cdots + c_{n-1} t^{n-1}$ at $\omega^0, \ldots, \omega^{n-1}$. That prediction is confirmed below by direct computation, and the computation also produces the eigenvectors.

Membership in the family also has a description with no coefficients in it, which is the form in which circulants are recognized in practice.

Proposition 7.7.5: Circulants Are Exactly the Matrices Commuting with the Shift

Let $A \in M_n(\mathbb{C})$. Then A is a circulant if and only if $AS = SA$. Moreover any two circulants commute with each other, and the conjugate transpose of a circulant is a circulant, so every circulant is normal.

***Proof*:**
Step 1: the two products. For any $A \in M_n(\mathbb{C})$, definition 1.4.1 gives $(AS)_{lk} = \sum_r A_{lr} S_{rk}$. By definition 7.7.1 the factor S_{rk} vanishes unless $k \equiv r+1$, that is unless $r = k-1$, so $(AS)_{lk} = A_{l,k-1}$. In the other order, $(SA)_{lk} = \sum_r S_{lr} A_{rk}$, and S_{lr} vanishes unless $r = l+1$, so $(SA)_{lk} = A_{l+1,k}$.

Step 2: commuting with S forces the circulant pattern. Suppose $AS = SA$. Step 1 turns this into $A_{l,k-1} = A_{l+1,k}$ for all l and k; replacing k by $k+1$, which is legitimate because the indices run over all residues modulo n, gives

$$A_{l,k} = A_{l+1,k+1} \qquad (0 \le l, k \le n-1)$$

Applying this identity l times in the direction that decreases both indices gives $A_{lk} = A_{l-1,k-1} = \cdots = A_{0,k-l}$. Setting $c_r := A_{0r}$ for $r = 0, \ldots, n-1$, this reads $A_{lk} = c_{k-l}$, so A is the circulant with first row $(c_0, \ldots, c_{n-1})$ (definition 7.7.2).

Step 3: circulants commute with S, and with each other. Let A be a circulant. By proposition 7.7.4, $A = \sum_r c_r S^r$. Powers of the same matrix commute, since $S^r S^s = S^{r+s} = S^s S^r$, and multiplication distributes over addition (proposition 1.4.3), so

$$AS = \sum_r c_r S^r S = \sum_r c_r S^{r+1} = S \sum_r c_r S^r = SA$$

The same computation with S replaced by a second circulant $B = \sum_s d_s S^s$ gives $AB = \sum_{r,s} c_r d_s S^{r+s} = BA$, since the double sum is unchanged when the roles of the two factors are exchanged.

Step 4: the conjugate transpose, and normality. With $A = \sum_r c_r S^r$ as above, proposition 5.5.13(2) gives $A^* = \sum_r \overline{c_r}\,(S^r)^*$, and proposition 5.5.13(3), applied $r-1$ times to the product of r copies of S, gives $(S^r)^* = (S^*)^r$. By proposition 7.7.3(2), $S^* = S^{n-1}$, so $(S^*)^r = S^{(n-1)r}$, a power of S. Hence A^* is a polynomial in S and is therefore a circulant by proposition 7.7.4. Both A and A^* are circulants, so they commute by Step 3, which is the identity $A^*A = AA^*$ defining normality, completing the proof.

The set of circulants is thus closed under products, and consists of normal matrices that pairwise commute. Theorem 7.6.6 applies to the whole set at once and gives an orthonormal basis of $\mathbb{C}^n$ every member of which is an eigenvector of every circulant. That theorem produces the basis by an induction and does not say which basis it is. Proposition 7.7.3 has already exhibited one for S, and since every circulant is a polynomial in S, the same list of vectors serves for all of them. Both the list and the transform it computes get a name.

Definition 7.7.6: The Fourier Matrix and the Discrete Fourier Transform

The *Fourier matrix* is the matrix $F \in M_n(\mathbb{C})$ whose j-th column is the vector f_j of proposition 7.7.3, that is

$$F_{kj} = \frac{1}{\sqrt{n}}\,\omega^{jk} \qquad (0 \le j, k \le n-1)$$

The *discrete Fourier transform* of a vector $c = (c_0, \ldots, c_{n-1})^\top \in \mathbb{C}^n$ is the vector $\hat{c} \in \mathbb{C}^n$ with coordinates

$$\hat{c}_j = \sum_{k=0}^{n-1} c_k\,\omega^{jk} \qquad (0 \le j \le n-1)$$

The exponent jk is symmetric in its two indices, so $F_{kj} = F_{jk}$ and the two objects just defined are the same object twice: $(\sqrt{n}\,Fc)_j = \sqrt{n}\sum_k F_{jk}c_k = \sum_k c_k\omega^{jk} = \hat{c}_j$, so $\hat{c} = \sqrt{n}\,Fc$, and the transform is multiplication by F up to the scalar $\sqrt{n}$. The columns of F are orthonormal by proposition 7.7.3(4), so F is unitary with $F^*F = FF^* = I_n$ and $F^{-1} = F^*$ (proposition 7.1.5). Multiplying $\hat{c} = \sqrt{n}\,Fc$ by

$\frac{1}{\sqrt{n}}F^*$ therefore recovers c from $\hat{c}$: since $(F^*)_{mj} = \overline{F_{jm}} = \frac{1}{\sqrt{n}}\omega^{-jm}$ by property (4) of ω,

$$c_m = \frac{1}{n}\sum_{j=0}^{n-1} \hat{c}_j\,\omega^{-jm} \qquad (0 \leq m \leq n-1) \tag{7.4}$$

so no information is lost in passing from c to $\hat{c}$. For $n = 4$ the number ω is $\cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = i$, and

$$F = \frac{1}{2}\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}$$

whose entry in row k and column j is $\frac{1}{2}i^{jk}$, computed with $i^2 = -1$ and $i^4 = 1$.

With the basis fixed and assembled into a matrix, the diagonalization of the whole family is a single identity, and it identifies the diagonal entries as the transform of the first row.

Theorem 7.7.7: Circulants Are Exactly the Matrices Diagonalized by F

Let $A \in M_n(\mathbb{C})$. Then A is a circulant if and only if F^*AF is diagonal. When A is the circulant with first row c,

$$F^*AF = \operatorname{diag}(\hat{c}_0, \hat{c}_1, \ldots, \hat{c}_{n-1}), \qquad \text{equivalently} \qquad A = F\operatorname{diag}(\hat{c}_0, \ldots, \hat{c}_{n-1})F^*$$

***Proof*:**
Step 1: a circulant is diagonalized by F. Let A be the circulant with first row c, so $A = \sum_r c_r S^r$ by proposition 7.7.4. From $Sf_j = \omega^j f_j$ (proposition 7.7.3(3)) an induction on r gives $S^r f_j = \omega^{jr} f_j$: the case $r = 0$ is $I_n f_j = f_j$, and $S^{r+1}f_j = S(\omega^{jr}f_j) = \omega^{jr}\omega^j f_j$. Hence

$$Af_j = \sum_{r=0}^{n-1} c_r S^r f_j = \Big(\sum_{r=0}^{n-1} c_r \omega^{jr}\Big) f_j = \hat{c}_j f_j$$

by definition 7.7.6. The j-th column of AF is Af_j and the j-th column of $F\operatorname{diag}(\hat{c}_0, \ldots, \hat{c}_{n-1})$ is $\hat{c}_j f_j$, both by definition 1.4.1, so $AF = F\operatorname{diag}(\hat{c}_0, \ldots, \hat{c}_{n-1})$. Multiplying on the left by F^* and using $F^*F = I_n$ gives $F^*AF = \operatorname{diag}(\hat{c}_0, \ldots, \hat{c}_{n-1})$, and multiplying instead on the right by F^* and using $FF^* = I_n$ gives the second display.

Step 2: only circulants are diagonalized by F. Suppose $F^*AF = \Lambda$ is diagonal, so $A = F\Lambda F^*$. The shift is the circulant with first row $(0, 1, 0, \ldots, 0)$, whose transform has j-th coordinate ω^j, so Step 1 gives $S = F\Omega F^*$ with $\Omega = \operatorname{diag}(\omega^0, \ldots, \omega^{n-1})$. Two diagonal matrices commute: by definition 1.4.1 the entry of $\Lambda\Omega$ in position (j, j') is $\sum_r \Lambda_{jr}\Omega_{rj'}$, in which the first factor vanishes unless $r = j$ and the second unless $r = j'$, so the product is diagonal with j-th diagonal entry $\Lambda_{jj}\Omega_{jj}$, and exchanging the two matrices only exchanges the order of two complex numbers. Therefore, regrouping with proposition 1.4.3 and cancelling $F^*F = I_n$,

$$AS = F\Lambda F^* F\Omega F^* = F\Lambda\Omega F^* = F\Omega\Lambda F^* = F\Omega F^* F\Lambda F^* = SA$$

and A is a circulant by proposition 7.7.5, completing the proof.

The factorization $A = F\operatorname{diag}(\hat{c})F^*$ separates a circulant into a part depending on A and a part depending only on n. The matrix F is the same for every circulant of size n, so it is computed once; the only thing that varies from one circulant to the next is the diagonal, and that diagonal is the transform of the first row, which is n numbers rather than n^2. This is also why a product Cx with C circulant is computed in practice as $F(\operatorname{diag}(\hat{c})(F^*x))$, with a fast algorithm for multiplication by F standing in for the matrix product.[1]

Reading the factorization through corollary 7.3.3 converts it into statements about eigenvalues, determinant and trace.

Corollary 7.7.8: The Eigenvalues of a Circulant

Let $C \in M_n(\mathbb{C})$ be the circulant with first row $c = (c_0, \dots, c_{n-1})^\top$. Then:

1. $Cf_j = \hat{c}_j f_j$ for each j, and $\chi_C(t) = (t - \hat{c}_0)(t - \hat{c}_1)\cdots(t - \hat{c}_{n-1})$, so the eigenvalues of C are the distinct values among $\hat{c}_0, \dots, \hat{c}_{n-1}$, each occurring in the list as often as its algebraic multiplicity;
2. $\det C = \hat{c}_0\hat{c}_1\cdots\hat{c}_{n-1}$ and $\operatorname{tr} C = \hat{c}_0 + \cdots + \hat{c}_{n-1} = nc_0$.

***Proof*:**

1. By theorem 7.7.7, $C = FDF^*$ with $D = \operatorname{diag}(\hat{c}_0, \dots, \hat{c}_{n-1})$, and F is unitary. Corollary 7.3.3 applies to any such factorization: its closing clause states that the columns $f_0, \dots, f_{n-1}$ of F satisfy $Cf_j = d_{jj}f_j$, which is $Cf_j = \hat{c}_j f_j$, and that $\chi_C(t) = (t - d_{00})\cdots(t - d_{n-1,n-1})$, which is the displayed product. The eigenvalues of C are the roots of χ_C (theorem 4.1.10), and the roots of a product of linear factors are the numbers $\hat{c}_j$, each occurring with the multiplicity with which it appears in the list.

2. By theorem 3.2.10 applied twice, $\det C = \det F \cdot \det D \cdot \det F^*$, and the same result applied to $FF^* = I_n$ gives $\det F \cdot \det F^* = \det I_n = 1$. Both D and I_n are diagonal, hence upper triangular, so proposition 3.2.6 computes their determinants as the products of their diagonal entries; this gives $\det I_n = 1$ and $\det D = \hat{c}_0 \cdots \hat{c}_{n-1}$, whence $\det C = \hat{c}_0 \cdots \hat{c}_{n-1}$.

   For the trace, the diagonal entries of C are $C_{ll} = c_{l-l} = c_0$ for every l, so $\operatorname{tr} C = nc_0$. On the other side, exchanging the order of summation and applying property (5) of ω to the inner sum,
$$\sum_{j=0}^{n-1} \hat{c}_j = \sum_{j=0}^{n-1}\sum_{k=0}^{n-1} c_k\omega^{jk} = \sum_{k=0}^{n-1} c_k \sum_{j=0}^{n-1} \omega^{jk} = nc_0$$
   since the inner sum is n when $k = 0$ and 0 for $1 \le k \le n-1$. The two expressions for $\operatorname{tr} C$ agree, as we sought to show.

The determinant of an $n \times n$ matrix is a sum of $n!$ terms in general; for a circulant it is a product of n numbers, each of which is a sum of n terms. The same collapse happens for products: multiplying two circulants is multiplying two diagonal matrices once the transform has been taken, and the resulting operation on first rows is the one that reappears whenever periodic data is combined.

[1] The same factorization is read in *Unitary Representation Theory* [10] as the decomposition of the regular representation of $\mathbb{Z}/n\mathbb{Z}$ into one-dimensional pieces; the vectors f_j are what that reading produces.

Theorem 7.7.9: Products of Circulants and Cyclic Convolution

Let $B, C \in M_n(\mathbb{C})$ be the circulants with first rows b and c respectively. Then:

1. $BC = CB$, and BC is a circulant;
2. the first row a of BC is given by
$$a_m = \sum_{k=0}^{n-1} b_k\, c_{m-k} \qquad (0 \le m \le n-1)$$
the index $m-k$ read modulo n;
3. $\hat{a}_j = \hat{b}_j\, \hat{c}_j$ for every j;
4. C is invertible if and only if $\hat{c}_j \neq 0$ for every j, and in that case C^{-1} is the circulant whose first row d satisfies $\hat{d}_j = 1/\hat{c}_j$ for every j.

***Proof*:**
Write $B = \sum_r b_r S^r$ and $C = \sum_s c_s S^s$, as proposition 7.7.4 permits, both sums running over $0, \dots, n-1$.

1. This is Step 3 of the proof of proposition 7.7.5 for the commuting, and part (2) below, which exhibits BC as a linear combination of the powers $S^0, \dots, S^{n-1}$ and hence as a circulant by proposition 7.7.4.
2. Expanding the product by distributivity (proposition 1.4.3) gives $BC = \sum_r \sum_s b_r c_s S^{r+s}$. Since $S^n = I_n$ (proposition 7.7.3(2)), the matrix S^{r+s} equals S^m where m is the remainder of $r+s$ modulo n. Grouping the n^2 terms according to that remainder,
$$BC = \sum_{m=0}^{n-1} \Big(\sum_{\substack{0 \le r,s \le n-1 \\ r+s \equiv m}} b_r c_s \Big) S^m$$
For fixed r there is exactly one s in $\{0, \dots, n-1\}$ with $r+s \equiv m$, namely the remainder of $m-r$, so the inner sum is $\sum_r b_r c_{m-r}$ with the index read modulo n. By proposition 7.7.4 the first row of BC is the list of these coefficients, which is the stated formula after renaming r as k.
3. Using (2) and definition 7.7.6,
$$\hat{a}_j = \sum_{m=0}^{n-1} a_m \omega^{jm} = \sum_{m=0}^{n-1} \sum_{k=0}^{n-1} b_k c_{m-k}\, \omega^{jm}$$
Fix k and substitute s for the remainder of $m-k$ modulo n. As m runs over $0, \dots, n-1$ so does s, and $m \equiv k+s$, so $\omega^{jm} = \omega^{j(k+s)} = \omega^{jk}\omega^{js}$ by property (2) of ω. Hence
$$\hat{a}_j = \sum_{k=0}^{n-1} \sum_{s=0}^{n-1} b_k c_s\, \omega^{jk}\omega^{js} = \Big(\sum_k b_k \omega^{jk} \Big) \Big(\sum_s c_s \omega^{js} \Big) = \hat{b}_j \hat{c}_j$$

4. Put $D = \operatorname{diag}(\hat{c}_0, \dots, \hat{c}_{n-1})$, so $C = FDF^*$ by theorem 7.7.7. Suppose first that every $\hat{c}_j$ is nonzero and let $D' = \operatorname{diag}(1/\hat{c}_0, \dots, 1/\hat{c}_{n-1})$. The product of two diagonal matrices is diagonal with the products of the corresponding entries as its diagonal, by the computation in Step 2 of the proof of theorem 7.7.7, so $DD' = D'D = I_n$. Using $F^*F = FF^* = I_n$ and regrouping with proposition 1.4.3,

$$(FD'F^*)(FDF^*) = FD'DF^* = FF^* = I_n, \qquad (FDF^*)(FD'F^*) = FDD'F^* = I_n$$

so C is invertible with $C^{-1} = FD'F^*$ (definition 1.4.4, proposition 1.4.5). Then $F^*C^{-1}F = D'$ is diagonal, so C^{-1} is a circulant and, by theorem 7.7.7 again, the transform of its first row d is the diagonal of D', that is $\hat{d}_j = 1/\hat{c}_j$.

Conversely, suppose $\hat{c}_{j_0} = 0$ for some j_0. Then $Cf_{j_0} = \hat{c}_{j_0} f_{j_0} = 0$ by corollary 7.7.8(1), while $f_{j_0} \neq 0$ because it has norm 1. If C had an inverse C^{-1} then $f_{j_0} = C^{-1}(Cf_{j_0}) = C^{-1}0 = 0$, a contradiction, so C is not invertible, completing the proof.

Part (3) is the statement that the transform converts the operation in (2) into multiplication coordinate by coordinate. The operation in (2) is the cyclic convolution of b and c: each entry of the answer is a sum of products $b_k c_{m-k}$ over all ways of splitting the index m modulo n. Computing it directly costs n^2 products; computing it as $c \mapsto \hat{c}$, then a coordinatewise product, then the inversion (7.4), replaces those by n products and three transforms.

The example below runs the whole section on one 4×4 circulant, first computing its eigenvalues from the transform of its first row, then computing its characteristic polynomial independently and checking that the two agree.

Example 7.7.10: A Four by Four Circulant and a Product of Two

Take $n = 4$, so that $\omega = i$ and F is the matrix displayed after definition 7.7.6.

1. Let C be the circulant with first row $c = (1, 1, 0, 0)$. Its entries are $C_{lk} = c_{k-l}$, so

$$C = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix} = I_4 + S$$

the second equality by proposition 7.7.4. The transform of the first row is $\hat{c}_j = c_0 + c_1\omega^j = 1 + i^j$, that is

$$\hat{c} = (2,\ 1+i,\ 0,\ 1-i)$$

using $i^0 = 1$, $i^1 = i$, $i^2 = -1$ and $i^3 = -i$. By corollary 7.7.8 these four numbers are the eigenvalues of C, with eigenvectors f_0, f_1, f_2, f_3, and $\det C = 2(1+i) \cdot 0 \cdot (1-i) = 0$. As a check on the third of them, $f_2 = \frac{1}{2}(1, -1, 1, -1)^\top$ and

$$Cf_2 = \frac{1}{2} \begin{pmatrix} 1-1 \\ -1+1 \\ 1-1 \\ 1-1 \end{pmatrix} = 0 = \hat{c}_2 f_2$$

The characteristic polynomial can be computed without any of this. Writing $u = t - 1$, the matrix $tI_4 - C$ equals $uI_4 - S$, whose rows are

$$R_0 = (u, -1, 0, 0), \quad R_1 = (0, u, -1, 0), \quad R_2 = (0, 0, u, -1), \quad R_3 = (-1, 0, 0, u)$$

Adding a multiple of one row to another leaves the determinant unchanged and interchanging two rows reverses its sign (chapter 3). Perform $R_0 \mapsto R_0 + uR_3$, then $R_1 \mapsto R_1 + uR_0$, then $R_2 \mapsto R_2 + uR_1$, each using the row produced by the previous step:

$$(0, -1, 0, u^2), \quad (0, 0, -1, u^3), \quad (0, 0, 0, u^4 - 1), \quad (-1, 0, 0, u)$$

Now the three interchanges $R_0 \leftrightarrow R_3$, $R_1 \leftrightarrow R_3$, $R_2 \leftrightarrow R_3$ put these four rows in the order

$$\begin{pmatrix} -1 & 0 & 0 & u \\ 0 & -1 & 0 & u^2 \\ 0 & 0 & -1 & u^3 \\ 0 & 0 & 0 & u^4 - 1 \end{pmatrix}$$

which is upper triangular with determinant $(-1)^3(u^4 - 1)$ by proposition 3.2.6. Three interchanges contribute the sign $(-1)^3$, so

$$\chi_C(t) = \det(tI_4 - C) = u^4 - 1 = (t-1)^4 - 1$$

Its roots are the numbers t with $(t-1)^4 = 1$. The four distinct numbers $1, i, -1, -i$ all satisfy $z^4 = 1$ by property (2) of ω and are distinct by property (3), so they are all the roots of the degree-four polynomial $z^4 - 1$. Hence $t \in \{2, 1+i, 0, 1-i\}$, which is the list $\hat{c}$ computed above.

2. Let B be the circulant with first row $b = (1, 0, 1, 0)$, so that $B = I_4 + S^2$ and

$$B = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}, \qquad \hat{b}_j = 1 + i^{2j} = 1 + (-1)^j, \qquad \hat{b} = (2, 0, 2, 0)$$

By theorem 7.7.9(3) the transform of the first row a of BC is the coordinatewise product

$$\hat{a} = (2 \cdot 2,\ 0 \cdot (1+i),\ 2 \cdot 0,\ 0 \cdot (1-i)) = (4, 0, 0, 0)$$

and (7.4) recovers $a_m = \frac{1}{4} \cdot 4 \cdot i^0 = 1$ for every m, so $a = (1, 1, 1, 1)$. The convolution formula of theorem 7.7.9(2) gives the same list: with $b_1 = b_3 = 0$, only $k = 0$ and $k = 2$ contribute, and $a_m = c_m + c_{m-2}$, which is $c_0 + c_2 = 1$, $c_1 + c_3 = 1$, $c_2 + c_0 = 1$, $c_3 + c_1 = 1$.

Multiplying the two matrices confirms it. Row 0 of BC is row 0 plus row 2 of C, namely $(1, 1, 0, 0) + (0, 0, 1, 1) = (1, 1, 1, 1)$, and row 1 of BC is row 1 plus row 3 of C, namely $(0, 1, 1, 0) + (1, 0, 0, 1) = (1, 1, 1, 1)$; the remaining rows repeat these two, since B does. So BC is the matrix all of whose entries are 1, which is indeed the circulant with first row $(1, 1, 1, 1)$.

Exercise 7.7.1

1. Take $n = 4$, so that $\omega = i$, and let C be the circulant with first row $(0, 1, 0, 1)$. Write out C, compute $\hat{c}$, and give the eigenvalues of C together with $\chi_C(t)$ and $\det C$. Decide whether C is invertible, and exhibit an eigenvector for each distinct eigenvalue.
2. Take $n = 4$ and let B and C be the circulants with first rows $(1, 1, 1, 1)$ and $(1, -1, 1, -1)$.

Compute $\hat{b}$ and $\hat{c}$, then compute the first row of BC twice: once by theorem 7.7.9(2), and once by combining theorem 7.7.9(3) with (7.4). Conclude that $BC = 0$ although neither factor is 0, and explain why this does not contradict corollary 7.7.8(2).

3. Let C be the circulant with first row c. Show that C^* is the circulant with first row $(\overline{c_0}, \overline{c_{n-1}}, \overline{c_{n-2}}, \ldots, \overline{c_1})$, and that the transform of that row has j-th coordinate $\overline{\hat{c}_j}$. Deduce that C is self-adjoint if and only if every $\hat{c}_j$ is real, and say which half of that equivalence proposition 7.1.11 already gives.

4. Let C be the circulant with first row c. Show that $C^*C = I_n$ if and only if $|\hat{c}_j| = 1$ for every j. Then determine every $C \in M_4(\mathbb{C})$ that is a circulant with first row of the form $(c_0, c_1, 0, 0)$ and satisfies $C^*C = I_4$.

5. Let $J \in M_n(\mathbb{C})$ be the matrix all of whose entries are 1. Show that J is a circulant, compute the transform of its first row, and deduce that $J^2 = nJ$. Show also that $P = \frac{1}{n}J$ satisfies $P^2 = P$ and $P^* = P$, and that its column space is the line spanned by $(1, 1, \ldots, 1)^\top$.

6. Show that the set of circulants in $M_n(\mathbb{C})$ is a subspace of $M_n(\mathbb{C})$ (definition 2.1.2) and that $I_n, S, S^2, \ldots, S^{n-1}$ is a basis of it (definition 2.2.1), so that its dimension (definition 2.2.10) is n. For the independence, compute the first row of S^r.

7. Let C be the circulant with first row c and suppose $\hat{c}_j \neq 0$ for every j. Using theorem 7.7.9(4) and (7.4), show that C^{-1} is the circulant whose first row d is given by

$$d_m = \frac{1}{n} \sum_{j=0}^{n-1} \frac{\omega^{-jm}}{\hat{c}_j}$$

Carry this out for $n = 4$ and $c = (3, 1, 0, 1)$, and check the answer by computing the first row of the product of your matrix with C.

Part III

Forms, Structures, and Decompositions

Everything so far has acted *on* vectors. A matrix stood for a linear map, and the question was always what that map does: which subspaces it fixes, which directions it scales, which basis makes its action legible. The right notion of sameness followed from that, namely similarity, and every invariant so far has been a similarity invariant.

This part studies matrices that do something else. A bilinear form eats two vectors and returns a scalar, and a matrix records it just as faithfully; but changing coordinates now transforms that matrix by $P^\top BP$ rather than by $P^{-1}AP$. The two rules agree exactly when $P^\top = P^{-1}$, which is to say on the orthogonal matrices of chapter 7, and that coincidence is why the spectral theorem could be read as a statement about quadratic forms at all. Away from it the invariants part company: eigenvalues, which classify an operator, are not preserved by congruence, and what survives instead is a count of signs.

Chapter 8 carries this out and finds the classification to be complete: over $\mathbb{R}$ a symmetric form is determined up to congruence by three integers, and over $\mathbb{C}$ by one. Chapter 9 then asks what is lost and gained in passing between $\mathbb{R}$ and $\mathbb{C}$, treating the two fields as structures a space may carry rather than as a setting fixed once at the start. Chapter 10 puts a symmetric form, a skew form, and a complex structure on the same space and finds that any two of them determine the third. Chapter 11 extracts, from an arbitrary matrix, the one decomposition that needs no hypothesis whatever, and chapter 12 settles what can be achieved by similarity alone when diagonalization is unavailable.

8

Bilinear and Quadratic Forms; Sylvester's Law of Inertia

A symmetric matrix has appeared twice already in different roles. In chapter 7 it was the matrix of a self-adjoint operator, and the spectral theorem diagonalized it by an orthogonal change of basis whose diagonal entries were its eigenvalues. But the expression $x^\top Ax$, which corollary 7.4.6 turned into a sum of squares, never mentioned an operator at all: it fed two copies of a vector to a rule that is linear in each, and asked for a number.

That reading is the subject here, and it changes what counts as an invariant. The eigenvalues that classified the operator are attached to the wrong transformation rule for the form, and they move under a change of coordinates that any form-theoretic question would accept as harmless. Something coarser survives, and it turns out to be exactly a count of how many squares come out positive and how many negative.

Section 8.1 sets up the two transformation rules side by side and separates congruence from similarity. Section 8.2 diagonalizes a symmetric form by completing the square, a construction that needs no eigenvalue and no inner product. Section 8.3 proves that the resulting signs do not depend on how the diagonalization was reached, which is Sylvester's law of inertia and settles the classification. Section 8.4 specializes to the forms that are positive everywhere, giving the three tests for that condition which chapter 7 deferred, and section 8.5 spends the whole apparatus on the question of whether a critical point of a function of several variables is a maximum, a minimum, or neither.

8.1 Bilinear Forms; Congruence versus Similarity

Once a basis is fixed, a function of two vector arguments that is linear in each is recorded by a single square matrix. This section sets up that record, computes how it changes when the basis changes, and isolates the features of the record that do not change.

A linear map $T\colon V \to V$ takes one vector and returns another. The functions of this chapter take two vectors and return a scalar. The reader has met one of them at length already, namely the standard inner product on $\mathbb{R}^n$, which sends the pair (x, y) to $x_1y_1 + \cdots + x_ny_n$ (definition 5.1.2). That expression is linear in x with y held fixed and linear in y with x held fixed. Dropping the two conditions that distinguish an inner product, symmetry and the positivity of $\langle v, v\rangle$, leaves exactly the double linearity.

Definition 8.1.1: Bilinear Form

Let k be $\mathbb{R}$ or $\mathbb{C}$ and let V be a finite-dimensional vector space over k. A *bilinear form* on V is a function $\varphi\colon V \times V \to k$ that is linear in each of its two arguments when the other is held fixed. That is, for all $u, u', v, v' \in V$ and all $a \in k$:

1. $\varphi(u + u', v) = \varphi(u, v) + \varphi(u', v)$ and $\varphi(au, v) = a\,\varphi(u, v)$;
2. $\varphi(u, v + v') = \varphi(u, v) + \varphi(u, v')$ and $\varphi(u, av) = a\,\varphi(u, v)$.

Over $\mathbb{R}$ an inner product is a bilinear form, since conjugate symmetry (definition 5.1.2(2)) is plain symmetry there and carries the linearity of the first argument across to the second. Over $\mathbb{C}$ it is not: conjugate symmetry makes $\langle u, av\rangle = \overline{a}\langle u, v\rangle$, so the second argument is conjugate-linear rather than linear. A bilinear form over $\mathbb{C}$ is therefore a different object from an inner product over $\mathbb{C}$, and the two must be kept apart. The object that mixes the two behaviours, a form linear in one argument and conjugate-linear in the other, is Hermitian and belongs to chapter 10.

There are as many examples as there are matrices. Fix $B = (b_{ij}) \in M_n(k)$ and define

$$\varphi_B\colon k^n \times k^n \to k, \qquad \varphi_B(x, y) = x^\top By$$

where the right-hand side is a 1×1 matrix read as a scalar. The matrix product is distributive over addition and commutes with scalars in each factor separately (proposition 1.4.3(2)), so φ_B satisfies both conditions of definition 8.1.1. Written out with definition 1.4.1 twice, first for By and then for the product of the row $x^\top$ with that column,

$$\varphi_B(x, y) = \sum_{i=1}^{n}\sum_{j=1}^{n} b_{ij}\, x_iy_j$$

so φ_B is the general sum of terms x_iy_j with constant coefficients. For $n = 2$ and

$$B = \begin{pmatrix} 1 & 2 \\ -1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$$

the sum has three nonzero terms and reads $\varphi_B(x, y) = x_1y_1 + 2x_1y_2 - x_2y_1$. The four coefficients can be recovered from φ_B by feeding it the standard basis vectors: $\varphi_B(e_1, e_1) = 1$, $\varphi_B(e_1, e_2) = 2$, $\varphi_B(e_2, e_1) = -1$ and $\varphi_B(e_2, e_2) = 0$, which are the four entries of B in their positions. That recovery is what makes the correspondence between matrices and forms exact, and it works in any basis of any V.

Definition 8.1.2: The Matrix of a Bilinear Form

Let φ be a bilinear form on V and let $\mathcal{B} = (b_1, \dots, b_n)$ be an ordered basis of V (definition 2.2.1). The *matrix of φ relative to $\mathcal{B}$* is

$$[\varphi]_{\mathcal{B}} = \big(\varphi(b_i, b_j)\big) \in M_n(k)$$

the matrix whose entry in row i and column j is the value of φ on the i-th and j-th basis vectors, in that order.

The notation deliberately matches $[T]_{\mathcal{B}}$ for the matrix of a linear map (theorem 2.4.8) and $[v]_{\mathcal{B}}$ for the coordinate column of a vector, and the three are distinguished by what sits inside the brackets. The dictionary between forms and matrices is a bijection once the basis is fixed, and the value of the form is computed from the matrix by the same expression $x^\top By$ that produced the examples above.

Theorem 8.1.3: Forms and Matrices Correspond

Let φ be a bilinear form on V, let $\mathcal{B} = (b_1, \dots, b_n)$ be an ordered basis of V, and put $B = [\varphi]_{\mathcal{B}}$. Then:

1. for all $u, v \in V$,
$$\varphi(u, v) = [u]_{\mathcal{B}}^\top \, B \, [v]_{\mathcal{B}}$$
2. B is the only matrix in $M_n(k)$ satisfying the identity in (1);
3. the assignment $\varphi \mapsto [\varphi]_{\mathcal{B}}$ is a bijection from the set of bilinear forms on V onto $M_n(k)$.

***Proof*:**

1. Write $x = [u]_{\mathcal{B}}$ and $y = [v]_{\mathcal{B}}$, so that $u = x_1 b_1 + \cdots + x_n b_n$ and $v = y_1 b_1 + \cdots + y_n b_n$. Applying definition 8.1.1(1) to the first argument expands $\varphi(u, v)$ as $\sum_i x_i \, \varphi(b_i, v)$, and applying definition 8.1.1(2) to each $\varphi(b_i, v)$ expands that as $\sum_j y_j \, \varphi(b_i, b_j)$. Together,
$$\varphi(u, v) = \sum_{i=1}^{n} \sum_{j=1}^{n} x_i \, \varphi(b_i, b_j) \, y_j = \sum_{i=1}^{n} \sum_{j=1}^{n} x_i \, b_{ij} \, y_j$$
the second equality being definition 8.1.2. By definition 1.4.1 the i-th entry of By is $\sum_j b_{ij} y_j$, and the product of the row $x^\top$ with that column is $\sum_i x_i \sum_j b_{ij} y_j$, which is the double sum just computed.
2. Suppose $M \in M_n(k)$ also satisfies $\varphi(u, v) = [u]_{\mathcal{B}}^\top M [v]_{\mathcal{B}}$ for all u, v. Take $u = b_i$ and $v = b_j$. The coordinate column $[b_i]_{\mathcal{B}}$ is e_i, since b_i is the combination of $b_1, \dots, b_n$ with coefficient 1 in position i and 0 elsewhere. So $\varphi(b_i, b_j) = e_i^\top M e_j$, and definition 1.4.1 evaluates $e_i^\top M e_j$ as the entry of M in row i and column j. Hence that entry equals $\varphi(b_i, b_j) = b_{ij}$ for every i and j, so $M = B$.
3. Injectivity is (1): two forms with the same matrix take the same value on every pair (u, v), hence are equal. For surjectivity, let $B \in M_n(k)$ be given and define $\psi(u, v) = [u]_{\mathcal{B}}^\top B [v]_{\mathcal{B}}$. Taking coordinates respects both operations: if $u = \sum x_i b_i$ and $u' = \sum x_i' b_i$ then $u + u' =$

$\sum(x_i + x_i')b_i$ and $au = \sum(ax_i)b_i$, and since the coefficients expressing a vector in a basis are unique, these are the coordinate columns of $u+u'$ and of au. So $[u+u']_{\mathcal{B}} = [u]_{\mathcal{B}} + [u']_{\mathcal{B}}$ and $[au]_{\mathcal{B}} = a[u]_{\mathcal{B}}$. Feeding those two identities into $\psi(u,v) = [u]_{\mathcal{B}}^{\top} B[v]_{\mathcal{B}}$ and using the distributivity of the matrix product (proposition 1.4.3(2)) shows ψ bilinear. Its matrix is B: by the computation in (2) applied to ψ, the entry of $[\psi]_{\mathcal{B}}$ in position (i,j) is $e_i^{\top} B e_j = b_{ij}$, completing the proof.

The theorem says that a bilinear form on an n-dimensional space is, after a choice of basis, exactly a square matrix of size n, with no condition on the matrix whatsoever. This is the same situation as for linear maps (theorem 2.4.8), and the question to settle is likewise the same: the choice of basis is not part of the data of φ, so the matrix is not an invariant of the form, and what has to be determined is which matrices arise from one form as the basis varies.

One fact about transposition is needed before that question can be asked precisely: transposition reverses the order of a product, $(XY)^{\top} = Y^{\top}X^{\top}$, and carries the inverse of an invertible Q to $(Q^{\top})^{-1} = (Q^{-1})^{\top}$. Both are proposition 2.3.5(3) and (4). Over $\mathbb{C}$ this is not the conjugate transpose of definition 5.5.12 and not proposition 5.5.13(3): those carry a conjugation on each factor, and the rule for forms carries none.

The second computation is the change of basis itself, and it is where the theory of forms separates from the theory of maps.

Theorem 8.1.4: Change of Basis for the Matrix of a Form

Let φ be a bilinear form on V, let $\mathcal{B} = (b_1, \ldots, b_n)$ and $\mathcal{C} = (c_1, \ldots, c_n)$ be ordered bases of V, and let $Q = P_{\mathcal{B}\mathcal{C}}$ be the change of basis matrix from $\mathcal{C}$ to $\mathcal{B}$ (definition 2.5.2), whose j-th column is $[c_j]_{\mathcal{B}}$. Then

$$[\varphi]_{\mathcal{C}} = Q^{\top}\,[\varphi]_{\mathcal{B}}\,Q$$

***Proof*:**
Write $B = [\varphi]_{\mathcal{B}}$ and let q_{tj} denote the entry of Q in row t and column j, so that the j-th column of Q is $[c_j]_{\mathcal{B}}$ with entries $q_{1j}, \ldots, q_{nj}$.

Fix i and j. The entry of $[\varphi]_{\mathcal{C}}$ in row i and column j is $\varphi(c_i, c_j)$ by definition 8.1.2. Evaluating it by theorem 8.1.3(1) in the basis $\mathcal{B}$, whose hypothesis is only that $\mathcal{B}$ is an ordered basis of V,

$$\varphi(c_i, c_j) = [c_i]_{\mathcal{B}}^{\top}\, B\, [c_j]_{\mathcal{B}} = \sum_{s=1}^{n}\sum_{t=1}^{n} q_{si}\, b_{st}\, q_{tj}$$

where the second equality expands the two matrix products with definition 1.4.1, the entries of the row $[c_i]_{\mathcal{B}}^{\top}$ being $q_{1i}, \ldots, q_{ni}$.

The same double sum is the entry of $Q^{\top}BQ$ in row i and column j: by definition 1.4.1 that entry is $\sum_s \sum_t (Q^{\top})_{is}\, b_{st}\, q_{tj}$, and $(Q^{\top})_{is} = q_{si}$ by definition 2.3.4. The two matrices therefore agree in every entry, as we sought to show.

Combining the theorem with theorem 8.1.3(1) in the basis $\mathcal{C}$ gives the coordinate form of the statement, $\varphi(u,v) = [u]_{\mathcal{C}}^{\top}(Q^{\top}BQ)[v]_{\mathcal{C}}$: converting coordinates from $\mathcal{C}$ to $\mathcal{B}$ before evaluating is the same as replacing B by $Q^{\top}BQ$. The comparison with linear maps is now exact, and the two rules can be written with the same letter Q. The change of basis formula for a map (theorem 2.5.4), applied with $\mathcal{B}$ as the old basis and $\mathcal{C}$ as the new one, uses the matrix $P = P_{\mathcal{C}\mathcal{B}}$, which is Q^{-1} by proposition 2.5.3(3). Substituting $P = Q^{-1}$ into $[T]_{\mathcal{C}} = P[T]_{\mathcal{B}}P^{-1}$ turns it into $[T]_{\mathcal{C}} = Q^{-1}[T]_{\mathcal{B}}Q$. Side by side,

$$[T]_{\mathcal{C}} = Q^{-1}\,[T]_{\mathcal{B}}\,Q, \qquad\qquad [\varphi]_{\mathcal{C}} = Q^{\top}\,[\varphi]_{\mathcal{B}}\,Q$$

The only difference is Q^{-1} against $Q^{\top}$. Computing the inverse requires solving a system, computing the transpose requires nothing but relabelling entries, and the two relations these rules generate are different, as example 8.1.11 and example 8.1.12 below show. The operation on the right is common enough to be given a name of its own, in parallel with definition 2.5.5.

Definition 8.1.5: Congruent Matrices

Two matrices $A, B \in M_n(k)$ are *congruent* when there is an invertible $Q \in M_n(k)$ with

$$B = Q^{\top}AQ$$

Theorem 8.1.4 says that the matrices of one bilinear form in two bases are congruent. The converse holds as well, so nothing is lost by working with matrices instead of with bases: given $B = Q^{\top}AQ$ with Q invertible, let φ be the form on k^n with $[\varphi]_{\mathcal{E}} = A$ in the standard basis $\mathcal{E}$, which exists by theorem 8.1.3(3), and let c_j be the j-th column of Q. Those n columns are a basis $\mathcal{C}$ of k^n. They span: every $w \in k^n$ equals $Q(Q^{-1}w)$, and the column reading of the product exhibits Qx as the combination of the c_j with coefficients x_j. They are independent: a combination $x_1c_1 + \cdots + x_nc_n$ equal to 0 is the product Qx, whence $x = Q^{-1}(Qx) = 0$. Since $[c_j]_{\mathcal{E}} = c_j$ is the j-th column of Q, the matrix $P_{\mathcal{E}\mathcal{C}}$ of definition 2.5.2 equals Q, and theorem 8.1.4 gives $[\varphi]_{\mathcal{C}} = Q^{\top}AQ = B$. So A and B are the two matrices of a single form.

Congruence is therefore to bilinear forms what similarity is to linear maps, and the first thing to check is that it partitions $M_n(k)$ at all.

Proposition 8.1.6: Congruence Is an Equivalence Relation

Congruence is an equivalence relation on $M_n(k)$.

***Proof*:**

For reflexivity, take $Q = I_n$, which is invertible. Then $I_n^{\top} = I_n$ by definition 2.3.4, so $I_n^{\top}AI_n = A$ and every $A \in M_n(k)$ is congruent to itself.

For symmetry, suppose $B = Q^{\top}AQ$ with Q invertible. Then Q^{-1} is invertible, with inverse Q, by definition 1.4.4 read in the other direction. Substituting and regrouping with proposition 1.4.3,

$$(Q^{-1})^{\top}BQ^{-1} = (Q^{-1})^{\top}Q^{\top}A\,QQ^{-1} = \left(QQ^{-1}\right)^{\top}A\,I_n = A$$

where the middle step uses proposition 2.3.5(3) computed above, with $X = Q$ and $Y = Q^{-1}$. So A is congruent to B whenever B is congruent to A.

For transitivity, suppose $B = Q^{\top}AQ$ and $C = R^{\top}BR$ with Q and R invertible. The product QR is then invertible (proposition 1.4.5(3)). Substituting and applying proposition 2.3.5(3) to QR,

$$C = R^{\top}Q^{\top}AQR = (QR)^{\top}A(QR)$$

so A and C are congruent, completing the proof.

A congruence class is a bilinear form with its basis forgotten. Deciding which matrices lie in the same class requires quantities computable from a matrix and unchanged by congruence, and the first of them is the rank. The argument that rank survives is the same one that shows it survives multiplication by any invertible matrix, on either side.

Proposition 8.1.7: Congruence Preserves Rank

Let $A \in M_n(k)$ and let $Q \in M_n(k)$ be invertible. Then

$$\operatorname{rank}\bigl(Q^\top AQ\bigr) = \operatorname{rank} A$$

Consequently congruent matrices have equal rank, and the number $\operatorname{rank}[\varphi]_\mathcal{B}$ attached to a bilinear form φ does not depend on the basis $\mathcal{B}$.

***Proof*:**
Step 1: right multiplication by an invertible matrix. Let $C \in M_{m\times n}(k)$ and let $Q \in M_n(k)$ be invertible. The claim is that $\operatorname{rank}(CQ) = \operatorname{rank} C$. Let $(y_1, \ldots, y_s)$ be a basis of $\operatorname{null} C$ (definition 2.3.3), which exists by theorem 2.2.9(1), and put $x_r = Q^{-1}y_r$ for $1 \le r \le s$. Each x_r lies in $\operatorname{null}(CQ)$, since $CQx_r = CQQ^{-1}y_r = Cy_r = 0$, the regrouping by proposition 1.4.3. They are independent: if $a_1x_1 + \cdots + a_sx_s = 0$ then multiplying by Q and using $Qx_r = y_r$ gives $a_1y_1 + \cdots + a_sy_s = 0$, so every a_r vanishes. They span $\operatorname{null}(CQ)$: if $CQx = 0$ then $Qx \in \operatorname{null} C$, so $Qx = a_1y_1 + \cdots + a_sy_s$ for some scalars a_r, and multiplying by Q^{-1} gives $x = a_1x_1 + \cdots + a_sx_s$. Hence $(x_1, \ldots, x_s)$ is a basis of $\operatorname{null}(CQ)$ and $\operatorname{nullity}(CQ) = s = \operatorname{nullity} C$. Both C and CQ have n columns, so theorem 2.3.11 applied to each gives

$$\operatorname{rank}(CQ) = n - \operatorname{nullity}(CQ) = n - \operatorname{nullity} C = \operatorname{rank} C$$

Step 2: left multiplication, by transposing. Let A and Q be as in the statement. Applying Step 1 with $C = Q^\top A$ gives $\operatorname{rank}(Q^\top AQ) = \operatorname{rank}(Q^\top A)$. By theorem 2.3.10 a matrix and its transpose have the same rank, and proposition 2.3.5(3) with $(Q^\top)^\top = Q$ (definition 2.3.4) gives $(Q^\top A)^\top = A^\top Q$, so

$$\operatorname{rank}(Q^\top A) = \operatorname{rank}\bigl(A^\top Q\bigr) = \operatorname{rank} A^\top = \operatorname{rank} A$$

the middle equality by Step 1 applied to $C = A^\top$ and the last again by theorem 2.3.10. Chaining the two displays gives $\operatorname{rank}(Q^\top AQ) = \operatorname{rank} A$.

For the consequences, congruent matrices are of the form A and $Q^\top AQ$ with Q invertible (definition 8.1.5), and any two matrices of one form φ are congruent by theorem 8.1.4, completing the proof.

The rank of a form is therefore well defined, and it is the first invariant the classification uses. The determinant is not preserved, but the way it fails to be preserved is completely controlled, and over $\mathbb{R}$ one piece of it survives.

Proposition 8.1.8: The Determinant Under Congruence

Let $A \in M_n(k)$ and let $Q \in M_n(k)$ be invertible. Then

$$\det\left(Q^\top AQ\right) = (\det Q)^2 \det A$$

In particular:

1. whether $\det A$ vanishes is a congruence invariant over both $\mathbb{R}$ and $\mathbb{C}$;
2. for $k = \mathbb{R}$, the numbers $\det(Q^\top AQ)$ and $\det A$ are both zero, both positive, or both negative, so the *sign* of the determinant is a congruence invariant over $\mathbb{R}$;
3. for $k = \mathbb{C}$ and $\det A \neq 0$, every nonzero value of the determinant is attained on the congruence class of A, so nothing beyond (1) survives.

***Proof*:**
Applying theorem 3.2.10 twice, to the product $(Q^\top A)Q$ and then to $Q^\top A$,

$$\det\left(Q^\top AQ\right) = \det(Q^\top)\det A \det Q$$

and $\det(Q^\top) = \det Q$ by theorem 3.2.11, which gives the displayed formula.

Since Q is invertible, $\det Q \neq 0$ by corollary 3.2.8, so $(\det Q)^2 \neq 0$ and the two determinants vanish together, which is (1). For (2), a nonzero real number has positive square, so $(\det Q)^2 > 0$ and multiplication by it preserves the sign of $\det A$ and preserves the value 0.

For (3), let $c \in \mathbb{C}$ be nonzero and let $d \in \mathbb{C}$ satisfy $d^2 = c$, which exists because every complex number has a complex square root. Put $Q = \operatorname{diag}(d, 1, \ldots, 1)$. Then Q is invertible, since $\det Q = d \neq 0$ by proposition 3.2.6 and corollary 3.2.8, and the displayed formula gives $\det(Q^\top AQ) = c \det A$. As c ranges over the nonzero complex numbers so does $c \det A$, since $\det A \neq 0$, completing the proof.

Part (2) is the first sign of what section 8.3 proves in full. Over $\mathbb{R}$ a congruence class does not carry a determinant, but it carries one bit of the determinant, namely its sign, and that bit is a genuine feature of the form. Over $\mathbb{C}$ even that bit is gone: part (3) says the congruence class of an invertible matrix contains matrices of every nonzero determinant, so the only invariant of the kind produced so far is the rank. The two fields will end up with classifications of different sizes for exactly this reason.

Eigenvalues, by contrast, are not congruence invariants at all. A similarity $Q^{-1}AQ$ preserves the characteristic polynomial (corollary 4.1.16), and the congruence rule replaces Q^{-1} by $Q^\top$, for which the corresponding statement is false: example 8.1.11 below exhibits two congruent matrices whose eigenvalues differ. Before recording that example and its counterpart, one further piece of structure is needed, since the counterpart turns on it.

Definition 8.1.9: Symmetric and Skew-Symmetric Bilinear Forms

Let φ be a bilinear form on V. Then φ is *symmetric* if $\varphi(u, v) = \varphi(v, u)$ for all $u, v \in V$, and *skew-symmetric* if $\varphi(u, v) = -\varphi(v, u)$ for all $u, v \in V$. A matrix $A \in M_n(k)$ is *symmetric* if $A^\top = A$ and *skew-symmetric* if $A^\top = -A$.

For $k = \mathbb{R}$ the matrix condition $A^\top = A$ is the one already named in definition 7.1.1, so a real symmetric matrix in the present sense is a real symmetric matrix in the earlier sense. For $k = \mathbb{C}$ the two conditions differ: $A^\top = A$ is not the Hermitian condition $A^* = A$, and a complex symmetric matrix need not be Hermitian, the matrix $\operatorname{diag}(i, i)$ being an instance. The standard inner product on $\mathbb{R}^n$ has matrix I_n in the standard basis, since $\langle e_i, e_j \rangle$ is 1 for $i = j$ and 0 otherwise, and I_n is symmetric. The form φ_B of the running computation above, with $B = \left(\begin{smallmatrix} 1 & 2 \\ -1 & 0 \end{smallmatrix}\right)$, is neither symmetric nor skew-symmetric, since $\varphi_B(e_1, e_2) = 2$ while $\varphi_B(e_2, e_1) = -1$. It splits into a symmetric and a skew-symmetric piece, and every form does.

Proposition 8.1.10: The Symmetric and Skew-Symmetric Parts

Let φ be a bilinear form on V, let $\mathcal{B}$ be an ordered basis of V and put $A = [\varphi]_\mathcal{B}$.

1. φ is symmetric if and only if $A^\top = A$, and skew-symmetric if and only if $A^\top = -A$.
2. There are a unique symmetric form σ and a unique skew-symmetric form α on V with $\varphi = \sigma + \alpha$, namely
$$\sigma(u, v) = \tfrac{1}{2}\big(\varphi(u, v) + \varphi(v, u)\big), \qquad \alpha(u, v) = \tfrac{1}{2}\big(\varphi(u, v) - \varphi(v, u)\big)$$
and their matrices are $[\sigma]_\mathcal{B} = \frac{1}{2}(A + A^\top)$ and $[\alpha]_\mathcal{B} = \frac{1}{2}(A - A^\top)$.
3. If $A^\top = A$ then $(Q^\top A Q)^\top = Q^\top A Q$ for every $Q \in M_n(k)$, and if $A^\top = -A$ then $(Q^\top A Q)^\top = -Q^\top A Q$. So congruence preserves each of the two conditions, and a symmetric matrix is never congruent to a matrix that is not symmetric.

***Proof*:**

1. Write $\mathcal{B} = (b_1, \dots, b_n)$. If φ is symmetric then the entry of A in position (i, j) is $\varphi(b_i, b_j) = \varphi(b_j, b_i)$, which is the entry in position (j, i), so $A^\top = A$ by definition 2.3.4. Conversely, if $A^\top = A$ then for all u, v the scalar $\varphi(u, v) = [u]_\mathcal{B}^\top A [v]_\mathcal{B}$ of theorem 8.1.3(1) is a 1×1 matrix, hence equal to its own transpose, and proposition 2.3.5(3), applied twice, gives
$$[u]_\mathcal{B}^\top A [v]_\mathcal{B} = \big([u]_\mathcal{B}^\top A [v]_\mathcal{B}\big)^\top = [v]_\mathcal{B}^\top A^\top [u]_\mathcal{B} = [v]_\mathcal{B}^\top A [u]_\mathcal{B}$$
which is $\varphi(v, u)$. The skew-symmetric case is the same argument with the sign carried along. In the forward direction $a_{ij} = \varphi(b_i, b_j) = -\varphi(b_j, b_i) = -a_{ji}$, which is $A^\top = -A$, and in the converse direction substituting $A^\top = -A$ at the last step of the display produces $-\varphi(v, u)$.
2. The two displayed functions are bilinear, since each is a sum of two bilinear functions of (u, v) scaled by $\frac{1}{2}$, the second being $(u, v) \mapsto \pm\varphi(v, u)$, which satisfies definition 8.1.1 because φ does with the roles of the arguments exchanged. Exchanging u and v leaves σ unchanged and negates α, so σ is symmetric and α is skew-symmetric, and adding them gives φ. The scalar $\frac{1}{2}$ exists in k because k is $\mathbb{R}$ or $\mathbb{C}$; over a field in which $2 = 0$ the two displayed formulas are unavailable and the statement fails.

   For uniqueness, suppose $\varphi = \sigma' + \alpha'$ with σ' symmetric and α' skew-symmetric. Then
$$\varphi(u, v) + \varphi(v, u) = \big(\sigma'(u, v) + \sigma'(v, u)\big) + \big(\alpha'(u, v) + \alpha'(v, u)\big) = 2\sigma'(u, v)$$

because the first bracket is $2\sigma'(u,v)$ and the second is 0. Dividing by 2 identifies σ' with σ, and then $\alpha' = \varphi - \sigma' = \alpha$.

The matrices are computed entrywise: the entry of $[\sigma]_{\mathcal{B}}$ in position (i,j) is $\sigma(b_i, b_j) = \frac{1}{2}(a_{ij} + a_{ji})$, which is the entry of $\frac{1}{2}(A + A^\top)$ in that position by definition 2.3.4, and likewise for α.

3. Applying proposition 2.3.5(3) twice and using $(Q^\top)^\top = Q$ (definition 2.3.4),

$$(Q^\top A Q)^\top = Q^\top A^\top (Q^\top)^\top = Q^\top A^\top Q$$

Substituting $A^\top = A$ gives $Q^\top AQ$, and substituting $A^\top = -A$ gives $-Q^\top AQ$, since a scalar factor passes through a matrix product. For the last sentence, if A is symmetric and $B = Q^\top AQ$ then $B^\top = B$ by what has just been shown, so a matrix congruent to A is symmetric, completing the proof.

Part (3) separates congruence from similarity in one direction: congruence always preserves symmetry, and similarity does not, since example 8.1.12 below exhibits a symmetric matrix similar to a matrix that is not symmetric. Part (2) explains why the chapter can concentrate on the symmetric case. An arbitrary form splits as $\varphi = \sigma + \alpha$, and congruence acts on the two pieces separately: with $S = [\sigma]_{\mathcal{B}}$ and $N = [\alpha]_{\mathcal{B}}$, the product $Q^\top(S+N)Q$ equals $Q^\top SQ + Q^\top NQ$, whose two summands are again symmetric and skew-symmetric by part (3), and by theorem 8.1.4 they are the matrices of σ and α in the new basis. The skew-symmetric piece contributes nothing to the values $\varphi(v,v)$: skew-symmetry applied with $u = v$ gives $\alpha(v,v) = -\alpha(v,v)$, hence $2\alpha(v,v) = 0$ and $\alpha(v,v) = 0$, so $\varphi(v,v) = \sigma(v,v)$ for every v. The function $v \mapsto \varphi(v,v)$ is the subject of section 8.2, and it is determined by σ alone.

The two relations are now separated by explicit matrices, in both directions. The first pair is congruent without being similar.

Example 8.1.11: Congruent but Not Similar

Let

$$A = I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}, \qquad Q = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

all in $M_2(\mathbb{R})$. Then $Q^\top = Q$ by definition 2.3.4, and

$$Q^\top AQ = QI_2Q = Q^2 = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix} = B$$

so A and B are congruent, Q being invertible with $Q^{-1} = \operatorname{diag}(1, \frac{1}{2})$.

They are not similar. The trace of A is $1+1 = 2$ and the trace of B is $1+4 = 5$, and similar matrices have equal traces by corollary 2.5.9. Their eigenvalues differ as well: $Be_2 = 4e_2$ with $e_2 \neq 0$, so 4 is an eigenvalue of B (definition 4.1.2, definition 4.1.3), while $I_2 v = 4v$ gives $3v = 0$ and hence $v = 0$, so A has no eigenvector for the value 4 and 4 is not an eigenvalue of A.

Read as forms, A and B are one object. Let φ be the standard inner product on $\mathbb{R}^2$, so that $[\varphi]_{\mathcal{E}} = I_2 = A$ in the standard basis, and let $\mathcal{C} = (e_1, 2e_2)$. The change of basis matrix $P_{\mathcal{E}\mathcal{C}}$ has the vectors of $\mathcal{C}$ as its columns (definition 2.5.2), so it is Q, and theorem 8.1.4 recovers $[\varphi]_{\mathcal{C}} = B$ directly: $\varphi(2e_2, 2e_2) = 4$. Doubling a basis vector multiplies the corresponding diagonal entry by 4. The eigenvalues of the matrix of a form are thus an artifact of the basis and say nothing about the form, whereas the determinants 1 and 4 differ by the positive square $(\det Q)^2 = 4$, exactly as proposition 8.1.8 requires.

The second pair runs the other way, and the obstruction is the one just proved.

Example 8.1.12: Similar but Not Congruent
Let

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}, \qquad B = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

all in $M_2(\mathbb{R})$. The matrix P is invertible with $P^{-1} = \left(\begin{smallmatrix} 1 & -1 \\ 0 & 1 \end{smallmatrix}\right)$, as multiplying the two confirms, and

$$PA = \begin{pmatrix} 2 & 3 \\ 0 & 3 \end{pmatrix}, \qquad (PA)P^{-1} = \begin{pmatrix} 2 & -2+3 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix} = B$$

so A and B are similar (definition 2.5.5).

They are not congruent. The matrix A is symmetric, and B is not, since its entry in position $(1, 2)$ is 1 while the entry in position $(2, 1)$ is 0. By proposition 8.1.10(3) every matrix congruent to A is symmetric, so B is not among them.

The two matrices agree on every similarity invariant so far available, and they agree on them precisely because they are similar: equal characteristic polynomials, hence equal eigenvalues 2 and 3, by corollary 4.1.16 and theorem 4.1.10; equal trace 5 by corollary 2.5.9; equal determinant 6 by corollary 3.2.13. Spectral data therefore does not decide congruence.

The two relations therefore partition $M_n(k)$ differently, and neither partition refines the other: example 8.1.11 puts two matrices in one congruence class and two similarity classes, and example 8.1.12 does the reverse. The partitions do overlap. If Q is invertible with $Q^\top = Q^{-1}$, then $Q^\top AQ = Q^{-1}AQ$ and the two operations coincide. Over $\mathbb{R}$ such a Q is exactly an orthogonal matrix, the condition $Q^\top Q = I_n$ being the real case of unitarity (proposition 7.1.5) because the conjugate transpose is the transpose there, and this is why chapter 7 can be read twice. The factorization $A = QDQ^\top$ produced by corollary 7.4.3 for a real symmetric A exhibits D as similar to A, which is why its entries are the eigenvalues, and simultaneously exhibits D as congruent to A, which is why the sum of squares $\sum_j \lambda_j y_j^2$ of corollary 7.4.6 is a statement about the form and not about the coordinates chosen to write it. The two readings will be separated in section 8.2, where Q is only required to be invertible and the diagonal entries stop being eigenvalues while the count of their signs survives.

One property of a form remains to be named, since it is the hypothesis under which the classification is stated. A form can vanish on every pair whose first entry is some fixed nonzero vector.

Definition 8.1.13: Degenerate and Nondegenerate Forms

Let φ be a bilinear form on V. The *left radical* of φ is

$$\{\, v \in V \,|\, \varphi(v, w) = 0 \text{ for every } w \in V \,\}$$

and the *right radical* is the set of v with $\varphi(w, v) = 0$ for every $w \in V$. The form φ is *degenerate* if its left radical contains a nonzero vector, and *nondegenerate* otherwise.

The definition is stated with the left radical, and the next proposition shows that the choice is immaterial: the two radicals have the same dimension, and in particular one is zero exactly when the other is. Both are subspaces of V, by definition 2.1.2: each is closed under addition and scalar multiplication because φ is linear in the argument being varied.

Proposition 8.1.14: Nondegeneracy and Invertibility

Let φ be a bilinear form on V with $\dim V = n$, let $\mathcal{B}$ be an ordered basis of V and put $B = [\varphi]_{\mathcal{B}}$. Then the left radical of φ and the right radical of φ both have dimension $n - \mathrm{rank}B$, and the following are equivalent:

1. φ is nondegenerate;
2. B is invertible;
3. $\det B \neq 0$;
4. $\mathrm{rank}B = n$.

None of the four conditions depends on the choice of $\mathcal{B}$.

***Proof*:**
Step 1: the radicals in coordinates. Let $v \in V$ and put $x = [v]_{\mathcal{B}}$. By theorem 8.1.3(1), $\varphi(v, w) = x^\top B[w]_{\mathcal{B}}$ for every w, and proposition 2.3.5(3) computed above rewrites the row $x^\top B$ as $(B^\top x)^\top$. As w runs over V the column $[w]_{\mathcal{B}}$ runs over all of k^n, since $[w]_{\mathcal{B}} = y$ holds for $w = y_1 b_1 + \cdots + y_n b_n$. So v lies in the left radical exactly when $(B^\top x)^\top y = 0$ for every $y \in k^n$. Taking $y = e_j$ picks out the j-th entry of $B^\top x$, so that condition holds exactly when $B^\top x = 0$. That is, v lies in the left radical exactly when $[v]_{\mathcal{B}} \in \mathrm{null}B^\top$ (definition 2.3.3).

The two subspaces then have the same dimension. Let $(v_1, \ldots, v_s)$ be a basis of the left radical, available by theorem 2.2.9(1), and consider the columns $[v_1]_{\mathcal{B}}, \ldots, [v_s]_{\mathcal{B}}$, all of which lie in $\mathrm{null}B^\top$ by the previous paragraph. They are independent: taking coordinates respects sums and scalar multiples, as computed in the proof of theorem 8.1.3(3), so a vanishing combination $\sum_r a_r [v_r]_{\mathcal{B}}$ equals $[\sum_r a_r v_r]_{\mathcal{B}}$, and a vector with zero coordinate column is 0, so $\sum_r a_r v_r = 0$ and every a_r vanishes. They span: given $x \in \mathrm{null}B^\top$, the vector $v = x_1 b_1 + \cdots + x_n b_n$ has $[v]_{\mathcal{B}} = x$ and so lies in the left radical, hence $v = \sum_r a_r v_r$ for some scalars a_r, and taking coordinates gives $x = \sum_r a_r [v_r]_{\mathcal{B}}$. So $\dim \mathrm{null}B^\top = s$, and theorem 2.3.11 applied to $B^\top$ together with $\mathrm{rank}B^\top = \mathrm{rank}B$ (theorem 2.3.10) gives $s = n - \mathrm{rank}B$. The same computation with the two arguments of φ exchanged identifies the right radical with $\mathrm{null}B$, whose dimension is $n - \mathrm{rank}B$ by theorem 2.3.11.

Step 2: the equivalences. By Step 1 the left radical is $\{0\}$ exactly when $\mathrm{nullity}B^\top = 0$, that is when $\mathrm{null}B^\top = \{0\}$, which by theorem 1.4.8, equivalence of its conditions (1) and (2) applied to the square matrix $B^\top$, holds exactly when $B^\top$ is invertible. By corollary 3.2.8 that is $\det B^\top \neq 0$, and $\det B^\top = \det B$ by theorem 3.2.11, so it is $\det B \neq 0$, and by corollary 3.2.8 again that is the invertibility of B. So (1), (2) and (3) are equivalent. For (4), Step 1 gives $\mathrm{nullity}B^\top = n - \mathrm{rank}B$, so the left radical is $\{0\}$ exactly when $\mathrm{rank}B = n$.

Independence of the basis: condition (1) is stated without reference to any basis, so the equivalences just proved show that (2), (3) and (4) hold for the matrix of φ relative to one ordered basis exactly when they hold relative to every other. For (4) this is also immediate from proposition 8.1.7, since any two matrices of φ are congruent by theorem 8.1.4, completing the proof.

Nondegeneracy is thus decided by one determinant, and it is a property of the form rather than of any basis. Two cautions about what it does not say. It does not force $\varphi(v, v) \neq 0$: the skew-symmetric form on $\mathbb{R}^2$ with matrix $N = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ has $\det N = 1$, hence is nondegenerate by proposition 8.1.14, yet $\varphi(v, v) = v_1v_2 - v_2v_1 = 0$ for every v. Forms of that kind are the subject of chapter 10. And it does not force the values $\varphi(v, v)$ to have one sign: the form on $\mathbb{R}^2$ with matrix $\operatorname{diag}(1, -1)$ has determinant -1 and takes the value 1 at e_1 and -1 at e_2. Which sign patterns can occur, and how the count of signs is extracted from the matrix, is the content of section 8.3, and the case where all the signs agree is section 8.4.

Exercise 8.1.1

1. Let φ be the bilinear form on $\mathbb{R}^3$ given by

$$\varphi(x, y) = x_1y_1 + 3x_1y_3 - x_2y_2 + 2x_3y_1 + x_3y_3$$

   Write down $[\varphi]_{\mathcal{E}}$ for the standard basis $\mathcal{E}$. Then compute $[\varphi]_{\mathcal{C}}$ for $\mathcal{C} = (e_1 + e_2,\, e_2,\, e_3)$ twice: once directly from definition 8.1.2 by evaluating φ on the nine pairs of vectors of $\mathcal{C}$, and once as $Q^\top[\varphi]_{\mathcal{E}}Q$ for the change of basis matrix Q named in theorem 8.1.4. Check that the two answers agree.

2. With φ, $\mathcal{C}$ and Q as in the previous question, compute the symmetric and skew-symmetric parts of $[\varphi]_{\mathcal{E}}$ using proposition 8.1.10(2). Then verify by direct computation that the symmetric part of $[\varphi]_{\mathcal{C}}$ equals $Q^\top SQ$, where S is the symmetric part of $[\varphi]_{\mathcal{E}}$, and say which part of proposition 8.1.10 predicts this.

3. Consider the four matrices

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 2 & 0 \\ 0 & -3 \end{pmatrix}, \qquad I_2$$

   in $M_2(\mathbb{R})$. Show that the first three are pairwise congruent by exhibiting an explicit invertible Q for two of the pairs, two being enough by proposition 8.1.6, and show that I_2 is congruent to none of them. Then show that over $\mathbb{C}$ the matrices I_2 and $\operatorname{diag}(1, -1)$ *are* congruent, by exhibiting a $Q \in M_2(\mathbb{C})$, and say which part of proposition 8.1.8 explains why the real and complex answers differ.

4. Let φ be a bilinear form on V. Show that φ is skew-symmetric if and only if $\varphi(v, v) = 0$ for every $v \in V$. Which of the two directions uses the invertibility of 2 in k, and where?

5. Let $N = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) \in M_2(\mathbb{R})$. Show that $Q^\top NQ = (\det Q)\, N$ for every $Q \in M_2(\mathbb{R})$, by computing both sides for a general Q. Deduce that N and cN are congruent for every nonzero $c \in \mathbb{R}$, and that a nonzero skew-symmetric matrix in $M_2(\mathbb{R})$ is never congruent to 0.

6. Let φ be the bilinear form on $\mathbb{R}^2$ given by $\varphi(x, y) = x_1y_2$. Compute its left and right radicals (definition 8.1.13) directly from the definition, and check against the null spaces predicted by the proof of proposition 8.1.14. Conclude that the two radicals can have the same dimension without being equal.

7. Let $A = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \in M_2(\mathbb{R})$. Using corollary 7.4.3, find an orthogonal Q with $Q^\top AQ$ diagonal, and explain why for this Q the congruence is simultaneously a similarity. Then put $Q' = Q\operatorname{diag}(\frac{1}{2}, 1)$, compute $(Q')^\top AQ'$, and check that its diagonal entries are not the eigenvalues of A while its determinant agrees with the value proposition 8.1.8 predicts.

8.2 Diagonalization of a Quadratic Form

Section 8.1 attached a matrix to each bilinear form, one for each basis of the space, and replaced that matrix by $P^\top BP$ when the basis changed. This section looks for the simplest matrix a symmetric form has in some basis. The answer is a diagonal one, and a basis producing it can be computed from the entries of B by addition, multiplication and division alone.

A symmetric form is determined by its values on the pairs (x, x), so the computation can be run on a function of one variable, where it is the completion of a square. That determination needs 2 to be invertible in the scalars, which over $\mathbb{R}$ and $\mathbb{C}$ it is. The diagonalization reached this way is not the one chapter 7 produced: that route used an orthogonal P and placed the eigenvalues of B on the diagonal, while this one uses an invertible P and places numbers there that are neither the eigenvalues of B nor determined by B.

On $\mathbb{R}^2$ the function

$$q(x_1, x_2) = x_1^2 + 4x_1x_2 + x_2^2$$

is the restriction of a symmetric form to such pairs. The symmetric matrix producing it is $B = \left(\begin{smallmatrix} 1 & 2 \\ 2 & 1 \end{smallmatrix}\right)$, since theorem 8.1.3 evaluates $x^\top Bx = x_1^2 + 2x_1x_2 + 2x_2x_1 + x_2^2$: the coefficient 4 of the cross term is split evenly between the positions $(1, 2)$ and $(2, 1)$, which is what symmetry of B forces. Completing the square in x_1,

$$q(x) = (x_1 + 2x_2)^2 - 4x_2^2 + x_2^2 = (x_1 + 2x_2)^2 - 3x_2^2$$

so in the coordinates $y_1 = x_1 + 2x_2$ and $y_2 = x_2$ the function is $y_1^2 - 3y_2^2$, with no cross term left. Theorem 8.2.3 is this computation carried out in n variables.

Definition 8.2.1: Quadratic Form

Let k be $\mathbb{R}$ or $\mathbb{C}$ and let V be a finite-dimensional vector space over k. The *quadratic form associated with* a symmetric bilinear form φ on V (definition 8.1.9) is the function

$$q_\varphi : V \to k, \qquad q_\varphi(x) = \varphi(x, x)$$

A function $q : V \to k$ is a *quadratic form* on V if $q = q_\varphi$ for some symmetric bilinear form φ on V.

Take $V = k^n$ and let $B = (b_{ij}) \in M_n(k)$ be the matrix of φ relative to the standard basis (definition 8.1.2), so that $\varphi(x, y) = x^\top By$ by theorem 8.1.3 and $B^\top = B$. Write q_B for the associated quadratic form. Then $q_B(x) = x^\top Bx$, and expanding the product by definition 1.4.1 and pairing the term indexed (i, j) with the term indexed (j, i), which are equal because $b_{ij} = b_{ji}$,

$$q_B(x) = \sum_{i=1}^{n}\sum_{j=1}^{n} b_{ij}x_ix_j = \sum_{i=1}^{n} b_{ii}x_i^2 + 2\sum_{i<j} b_{ij}x_ix_j$$

A quadratic form on k^n is therefore a homogeneous polynomial of degree 2 in the coordinates, with the diagonal entries of B carrying the squares and twice the entries above the diagonal carrying the cross terms. The function $x_1^2 + 4x_1x_2 + x_2^2$ above is the case $n = 2$ with $b_{11} = b_{22} = 1$ and $b_{12} = 2$.

Nothing so far shows that φ can be reconstructed from q_φ. The values $\varphi(x, y)$ at pairs of distinct vectors are not among the values of q_φ, and the reconstruction has to produce them. Expanding $q_\varphi(x + y)$ produces them all at once.

Theorem 8.2.2: Recovery of a Symmetric Form from Its Quadratic Form

Let k be $\mathbb{R}$ or $\mathbb{C}$ and let V be a finite-dimensional vector space over k.

1. If φ is a symmetric bilinear form on V and $q = q_\varphi$, then for all $x, y \in V$
$$\varphi(x,y) = \tfrac{1}{2}\big(q(x+y) - q(x) - q(y)\big)$$
2. If φ and ψ are symmetric bilinear forms on V with $q_\varphi = q_\psi$, then $\varphi = \psi$. In particular, if $B, B' \in M_n(k)$ are symmetric and $x^\top Bx = x^\top B'x$ for every $x \in k^n$, then $B = B'$.

***Proof*:**

1. Bilinearity of φ (definition 8.1.1) expands the value at the pair $(x+y, x+y)$ into four terms,
$$q(x+y) = \varphi(x+y, x+y) = \varphi(x,x) + \varphi(x,y) + \varphi(y,x) + \varphi(y,y)$$
and symmetry gives $\varphi(y,x) = \varphi(x,y)$ (definition 8.1.9), so $q(x+y) = q(x) + 2\varphi(x,y) + q(y)$. Subtracting $q(x) + q(y)$ from both sides leaves $2\varphi(x,y)$ on the right. The scalar $2 = 1+1$ is a nonzero element of $\mathbb{R}$ or of $\mathbb{C}$ and is therefore invertible there, so dividing by it is permitted and gives the stated formula.
2. Applying (1) to φ and then to ψ, and using $q_\varphi = q_\psi$ in the middle,
$$\varphi(x,y) = \tfrac{1}{2}\big(q_\varphi(x+y) - q_\varphi(x) - q_\varphi(y)\big) = \tfrac{1}{2}\big(q_\psi(x+y) - q_\psi(x) - q_\psi(y)\big) = \psi(x,y)$$
for all $x, y \in V$, which is $\varphi = \psi$. For the matrix statement, let $\varphi(x,y) = x^\top By$ and $\psi(x,y) = x^\top B'y$ be the two bilinear forms on k^n with these matrices relative to the standard basis (theorem 8.1.3). Each is symmetric because its matrix is (definition 8.1.9). The hypothesis says exactly that $q_\varphi = q_\psi$, so the first part of (2) gives $\varphi = \psi$, and evaluating at pairs of standard basis vectors, $b_{ij} = e_i^\top Be_j = \varphi(e_i, e_j) = \psi(e_i, e_j) = b'_{ij}$ for all i and j (definition 1.4.1). Hence $B = B'$, completing the proof.

A symmetric bilinear form and its quadratic form therefore carry the same information, and either may be used to specify the other. Part (2) is why the phrase *the* symmetric matrix of a quadratic form is well defined: relative to a fixed basis there is exactly one symmetric B with $q(x) = x^\top Bx$, so reading off the coefficients of the polynomial and halving the cross terms cannot be done in two ways. The uniqueness holds only among symmetric matrices. A matrix that is not symmetric can produce the same function as a symmetric one, and exercise 8.2.1 (3) computes which symmetric matrix that is. Both parts of the theorem fail when 2 is not invertible in the scalars.[1]

The reduction of the opening example used no property of the coefficients beyond $b_{11} \neq 0$: having divided by it, the cross terms containing x_1 were absorbed into a square, and what remained involved x_2 alone. That is the inductive step, and the only obstacle is a matrix whose diagonal entries all vanish, since then there is no leading coefficient to divide by. One substitution repairs it.

[1] Over a field in which $1+1=0$ the form $\varphi(x,y) = x_1y_2 + x_2y_1$ on the two-dimensional space is symmetric and nonzero, while $q_\varphi(x) = 2x_1x_2 = 0$ for every x, so the quadratic form determines nothing. Quadratic forms in characteristic 2 are developed by a different route, and this book works over $\mathbb{R}$ and $\mathbb{C}$ only.

Theorem 8.2.3: Diagonalization of a Symmetric Matrix by Congruence

Let k be $\mathbb{R}$ or $\mathbb{C}$, let $n \geq 1$, and let $B \in M_n(k)$ satisfy $B^\top = B$. Then B is congruent (definition 8.1.5) to a diagonal matrix: there is an invertible $P \in M_n(k)$ for which $P^\top BP$ is diagonal.

***Proof*:**
Throughout, β denotes the bilinear form on k^n given by $\beta(u,v) = u^\top Bv$, which is the form whose matrix relative to the standard basis is B (theorem 8.1.3). It is symmetric because B is (definition 8.1.9). One identity is used at every step. If $Q \in M_n(k)$ has columns $q_1, \dots, q_n$, then row s of $Q^\top$ is the transpose of q_s (definition 2.3.4) and column t of BQ is Bq_t, so definition 1.4.1 evaluates the product $Q^\top(BQ)$ entry by entry as

$$(Q^\top BQ)_{st} = q_s^\top Bq_t = \beta(q_s, q_t) \tag{8.1}$$

Since β is symmetric, (8.1) shows that $Q^\top BQ$ is symmetric again, so each matrix produced below can be fed back into the argument.

The argument is an induction on n. For $n = 1$ every matrix in $M_1(k)$ is diagonal and $P = I_1$ serves. Let $n \geq 2$ and suppose every symmetric matrix in $M_{n-1}(k)$ is congruent to a diagonal matrix. Three cases cover the symmetric $B \in M_n(k)$: some diagonal entry of B is nonzero, or all of them vanish and $B = 0$, or all of them vanish and $B \neq 0$.

Case 1: $B = 0$. The matrix is already diagonal and $P = I_n$ serves.

Case 2: $b_{ii} \neq 0$ for some i. Move that entry into position $(1,1)$ first. Let S be I_n with its first and i-th columns exchanged, so that $Se_1 = e_i$, $Se_i = e_1$ and $Se_s = e_s$ for every other index ($S = I_n$ when $i = 1$). Column s of SS is $S(Se_s)$ (definition 1.4.1), which is e_s in each of the three cases, so $SS = I_n$ and S is invertible with $S^{-1} = S$ (definition 1.4.4). By (8.1) the matrix $S^\top BS$ is symmetric with

$$(S^\top BS)_{11} = \beta(Se_1, Se_1) = \beta(e_i, e_i) = b_{ii} \neq 0$$

Congruence is transitive (proposition 8.1.6), so a diagonal matrix congruent to $S^\top BS$ is congruent to B, and it is enough to treat $S^\top BS$. The rest of this case therefore assumes $b_{11} \neq 0$.

Set $w_1 = e_1$ and $w_j = e_j - (b_{1j}/b_{11})\,e_1$ for $2 \leq j \leq n$, and let $P_1 \in M_n(k)$ be the matrix with columns $w_1, \dots, w_n$: every diagonal entry is 1, the entry in row 1 and column j is $-b_{1j}/b_{11}$ for $j \geq 2$, and all other entries are 0. It is upper triangular, so $\det P_1 = 1$ by proposition 3.2.6 and P_1 is invertible by corollary 3.2.8. Bilinearity of β gives, for $j \geq 2$,

$$\beta(w_1, w_j) = \beta(e_1, e_j) - \frac{b_{1j}}{b_{11}}\,\beta(e_1, e_1) = b_{1j} - \frac{b_{1j}}{b_{11}}\,b_{11} = 0$$

while $\beta(w_1, w_1) = b_{11}$. By (8.1) the first row and the first column of $P_1^\top BP_1$ vanish away from position $(1,1)$, so

$$P_1^\top BP_1 = \begin{pmatrix} b_{11} & 0 \\ 0 & C \end{pmatrix}, \qquad C = \big(\beta(w_s, w_t)\big)_{2\leq s,t\leq n} \in M_{n-1}(k)$$

and C is symmetric because β is. The inductive hypothesis gives an invertible $R \in M_{n-1}(k)$ with $R^\top CR$ diagonal. Put

$$P_2 = \begin{pmatrix} 1 & 0 \\ 0 & R \end{pmatrix} \in M_n(k), \qquad P_2' = \begin{pmatrix} 1 & 0 \\ 0 & R^{-1} \end{pmatrix} \in M_n(k)$$

and note that $P_2^\top = \left(\begin{smallmatrix} 1 & 0 \\ 0 & R^\top \end{smallmatrix}\right)$ by definition 2.3.4. In a product of matrices of this shape the first row and the first column contribute only through position $(1,1)$, every other entry there being 0, while the entry in a position (s,t) with $s,t \geq 2$ is a sum over indices $u \geq 2$ or pairs $u,v \geq 2$ and is therefore the corresponding entry of the product of the lower blocks (definition 1.4.1). Hence $P_2P_2' = P_2'P_2 = I_n$, so P_2 is invertible (definition 1.4.4), and

$$P_2^\top \left(P_1^\top B P_1\right) P_2 = \begin{pmatrix} b_{11} & 0 \\ 0 & R^\top C R \end{pmatrix}$$

which is diagonal. So B is congruent to $P_1^\top B P_1$, which is congruent to this diagonal matrix, and transitivity (proposition 8.1.6) makes B congruent to it.

Case 3: every diagonal entry of B vanishes and $B \neq 0$. Some entry away from the diagonal is nonzero, say $b_{ij} \neq 0$ with $i \neq j$. Let T be I_n with its i-th column replaced by $e_i + e_j$ and its j-th column replaced by $e_i - e_j$, and let T' be I_n with its i-th column replaced by $\frac{1}{2}(e_i + e_j)$ and its j-th column replaced by $\frac{1}{2}(e_i - e_j)$. Those halves exist because k is $\mathbb{R}$ or $\mathbb{C}$, where 2 is invertible. Column s of TT' is $T(T'e_s)$ (definition 1.4.1), and

$$T\big(\tfrac{1}{2}(e_i + e_j)\big) = \tfrac{1}{2}\big((e_i+e_j) + (e_i - e_j)\big) = e_i, \qquad T\big(\tfrac{1}{2}(e_i - e_j)\big) = \tfrac{1}{2}\big((e_i+e_j) - (e_i - e_j)\big) = e_j$$

while $T'e_s = e_s = Te_s$ for $s \notin \{i,j\}$, so $TT' = I_n$. In the other order, $T'(Te_i) = T'(e_i + e_j) = \frac{1}{2}(e_i + e_j) + \frac{1}{2}(e_i - e_j) = e_i$ and $T'(Te_j) = T'(e_i - e_j) = \frac{1}{2}(e_i + e_j) - \frac{1}{2}(e_i - e_j) = e_j$, so $T'T = I_n$ as well and T is invertible with $T^{-1} = T'$ (definition 1.4.4).

By (8.1) and bilinearity, the entry of $T^\top B T$ in position (i,i) is

$$\beta(e_i + e_j, e_i + e_j) = \beta(e_i, e_i) + 2\beta(e_i, e_j) + \beta(e_j, e_j) = 0 + 2b_{ij} + 0$$

the middle term because $\beta(e_j, e_i) = \beta(e_i, e_j) = b_{ij}$ and the outer two because every diagonal entry of B vanishes in this case. Both 2 and b_{ij} are nonzero in k, so $2b_{ij} \neq 0$ and $T^\top B T$ is a symmetric matrix with a nonzero diagonal entry. The argument of Case 2 applies to it, and that argument invoked the inductive hypothesis only in size $n-1$, so using it here is not circular. It produces a diagonal matrix congruent to $T^\top BT$, and transitivity (proposition 8.1.6) makes B congruent to that same diagonal matrix.

The three cases are exhaustive, so the inductive step holds and every symmetric matrix in $M_n(k)$ is congruent to a diagonal one, as we sought to show.

The proof is a procedure. Each pass through Case 2 divides by one nonzero entry, forms $n-1$ combinations of the first coordinate with the others, and hands a symmetric matrix one size smaller to the next pass, so there are at most n passes, each preceded by at most one substitution of the kind in Case 3. Every operation is an addition, a multiplication or a division of entries already computed, and no root of a polynomial is extracted anywhere. This reduction is known as Lagrange's method.

In coordinates the passes are completions of squares. If P has columns $p_1, \ldots, p_n$ and $x = \sum_t y_t p_t$, then bilinearity and (8.1) give

$$q_B(x) = \beta(x,x) = \sum_{s=1}^{n}\sum_{t=1}^{n} y_s y_t\, \beta(p_s, p_t) = \sum_{s=1}^{n}\sum_{t=1}^{n} y_s y_t\, (P^\top B P)_{st} \tag{8.2}$$

For the P_1 of Case 2 this reads $q_B(x) = b_{11}y_1^2 + q_C(y_2, \ldots, y_n)$, where $y_1 = x_1 + \sum_{j\geq 2}(b_{1j}/b_{11})x_j$ and $y_s = x_s$ for $s \geq 2$, since $x = P_1 y$ recovers $x_1 = y_1 - \sum_{j \geq 2}(b_{1j}/b_{11})y_j$ and $x_s = y_s$. The square completed is the one containing x_1, and q_C is what is left over. The entries of that leftover form are available without recomputing anything: expanding $\beta(w_s, w_t)$ for $s, t \geq 2$ by bilinearity and using $\beta(e_1, e_s) = b_{1s}$ together with $b_{s1} = b_{1s}$,

$$C_{st} = b_{st} - \frac{b_{1t}}{b_{11}} b_{s1} - \frac{b_{1s}}{b_{11}} b_{1t} + \frac{b_{1s} b_{1t}}{b_{11}^2} b_{11} = b_{st} - \frac{b_{1s} b_{1t}}{b_{11}}$$

The statement about matrices carries a statement about bases, and that is the form in which the result is used.

Corollary 8.2.4: Every Symmetric Form Has an Orthogonal Basis

Let k be $\mathbb{R}$ or $\mathbb{C}$, let V be a vector space over k with $\dim V = n \geq 1$, and let φ be a symmetric bilinear form on V. Then V has a basis $w_1, \ldots, w_n$ with

$$\varphi(w_s, w_t) = 0 \qquad \text{whenever } s \neq t$$

Writing $d_s = \varphi(w_s, w_s)$, the associated quadratic form is

$$q_\varphi\Big(\sum_{s=1}^{n} a_s w_s\Big) = \sum_{s=1}^{n} d_s a_s^2 \qquad (a_1, \ldots, a_n \in k)$$

***Proof*:**
Choose any basis $v_1, \ldots, v_n$ of V (theorem 2.2.9) and let $B = (b_{ij})$ be the matrix of φ relative to it, so that $b_{ij} = \varphi(v_i, v_j)$ (definition 8.1.2). It is symmetric, since $b_{ij} = \varphi(v_i, v_j) = \varphi(v_j, v_i) = b_{ji}$ (definition 8.1.9). By theorem 8.2.3 there is an invertible $P = (p_{st}) \in M_n(k)$ with $D = P^\top B P$ diagonal.

Put $w_t = \sum_{s=1}^{n} p_{st} v_s$ for each t, the vector whose coordinate column relative to $v_1, \ldots, v_n$ is the t-th column of P. This list spans V: given $v = \sum_s x_s v_s$ with $x \in k^n$, set $c = P^{-1}x$, and then

$$\sum_{t=1}^{n} c_t w_t = \sum_{s=1}^{n} \Big(\sum_{t=1}^{n} p_{st} c_t\Big) v_s = \sum_{s=1}^{n} (Pc)_s v_s = \sum_{s=1}^{n} x_s v_s = v$$

using definition 1.4.1 for Pc and $Pc = P(P^{-1}x) = x$ (definition 1.4.4, proposition 1.4.3). It is independent: if $\sum_t c_t w_t = 0$ then the same computation gives $\sum_s (Pc)_s v_s = 0$, so $Pc = 0$ because $v_1, \ldots, v_n$ is independent, and then $c = P^{-1}(Pc) = 0$. An independent spanning list is a basis (definition 2.2.1). By construction the t-th column of P is the coordinate column of w_t relative to $v_1, \ldots, v_n$, so P is the change-of-basis matrix for these two bases (definition 2.5.2).

Theorem 8.1.4 therefore identifies the matrix of φ relative to $w_1, \ldots, w_n$ as $P^\top B P = D$, whose entry in position (s, t) is $\varphi(w_s, w_t)$ (definition 8.1.2). Since D is diagonal, $\varphi(w_s, w_t) = 0$ for $s \neq t$ and $\varphi(w_s, w_s) = d_{ss}$, which is the d_s of the statement. Finally, for $x = \sum_s a_s w_s$ bilinearity gives

$$q_\varphi(x) = \varphi(x, x) = \sum_{s=1}^{n} \sum_{t=1}^{n} a_s a_t\, \varphi(w_s, w_t) = \sum_{s=1}^{n} d_s a_s^2$$

the last equality because only the terms with $s = t$ survive, completing the proof.

A basis as in the corollary is called *orthogonal for* φ, by analogy with the orthogonal bases of chapter 5: the condition $\varphi(w_s, w_t) = 0$ for $s \neq t$ is the condition $\langle w_s, w_t \rangle = 0$ with the inner product replaced by φ. The analogy stops at the condition. An inner product satisfies $\langle x, x \rangle > 0$ for $x \neq 0$ (definition 5.1.2), so its orthogonal basis vectors can be scaled to unit length. A symmetric form carries no such condition, and the numbers $d_s = \varphi(w_s, w_s)$ may be positive, negative or zero.

How many of them are zero is not a matter of choice. The number of nonzero d_s equals $\operatorname{rank} B$ for the matrix B of φ in any basis: congruent matrices have equal rank (proposition 8.1.7), and for a diagonal matrix D the columns with $d_s \neq 0$ are the vectors $d_s e_s$ while the others are zero, so the column space of D is spanned by those e_s and has dimension the number of nonzero d_s (definition 2.3.1), which is $\operatorname{rank} D$ by theorem 2.3.10. In particular a diagonalization of a nondegenerate form (definition 8.1.13) has no zero on the diagonal at all, since nondegeneracy is invertibility of B (proposition 8.1.14) and hence $\operatorname{rank} B = n$.

Example 8.2.5: Completing the Square in Three Variables

Let $q = q_B$ on $\mathbb{R}^3$ for

$$B = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 3 & 3 \\ 1 & 3 & 1 \end{pmatrix}, \qquad q(x) = x_1^2 + 3x_2^2 + x_3^2 + 4x_1x_2 + 2x_1x_3 + 6x_2x_3$$

the coefficients of the cross terms being twice the entries above the diagonal. Since $b_{11} = 1 \neq 0$, Case 2 of theorem 8.2.3 applies with no preliminary exchange, and its coefficients are $b_{12}/b_{11} = 2$ and $b_{13}/b_{11} = 1$. The first substitution is therefore $y_1 = x_1 + 2x_2 + x_3$, with $y_2 = x_2$ and $y_3 = x_3$, and

$$y_1^2 = x_1^2 + 4x_2^2 + x_3^2 + 4x_1x_2 + 2x_1x_3 + 4x_2x_3$$

so that $q(x) - y_1^2 = -x_2^2 + 2x_2x_3$, a form in x_2 and x_3 alone with matrix $C = \left(\begin{smallmatrix} -1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. This is the C of the proof, and its entries agree with the formula $C_{st} = b_{st} - b_{1s}b_{1t}/b_{11}$ derived above: $3 - 4 = -1$, then $3 - 2 = 1$, then $1 - 1 = 0$.

Now run Case 2 on C. Its entry in position $(1, 1)$, which is the coefficient -1 of x_2^2, is nonzero, and $1/(-1) = -1$, so the second substitution is $y_2 = x_2 - x_3$. Then $-y_2^2 = -x_2^2 + 2x_2x_3 - x_3^2$, and

$$q(x) - y_1^2 + y_2^2 = x_3^2$$

Collecting the three steps,

$$q(x) = y_1^2 - y_2^2 + y_3^2, \qquad y_1 = x_1 + 2x_2 + x_3, \quad y_2 = x_2 - x_3, \quad y_3 = x_3$$

To exhibit the congruence, solve for x in terms of y: $x_3 = y_3$, then $x_2 = y_2 + y_3$, then $x_1 = y_1 - 2x_2 - x_3 = y_1 - 2y_2 - 3y_3$. The matrix P with $x = Py$ has these three expressions as its rows, so

$$P = \begin{pmatrix} 1 & -2 & -3 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det P = 1$$

by proposition 3.2.6, so P is invertible (corollary 3.2.8). Its columns are $p_1 = (1, 0, 0)^\top$, $p_2 = (-2, 1, 0)^\top$ and $p_3 = (-3, 1, 1)^\top$, and

$$Bp_1 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \qquad Bp_2 = \begin{pmatrix} -2+2 \\ -4+3 \\ -2+3 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \qquad Bp_3 = \begin{pmatrix} -3+2+1 \\ -6+3+3 \\ -3+3+1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

Taking the six products $p_s^\top B p_t$ with $s \le t$ as (8.1) prescribes gives 1, 0, 0 in the first row, then $(-2,1,0)\cdot(0,-1,1) = -1$ and $(-2,1,0)\cdot(0,0,1) = 0$, then $(-3,1,1)\cdot(0,0,1) = 1$, so

$$P^\top BP = \mathrm{diag}(1,-1,1)$$

Two checks. Expanding along the first row, $\det B = 1(3-9) - 2(2-3) + 1(6-3) = -6+2+3 = -1$ (theorem 3.3.4), which agrees with $\det \mathrm{diag}(1,-1,1) = -1$ as proposition 8.1.8 requires, the factor $(\det P)^2$ being 1 here. And $\det B \neq 0$ makes B invertible (corollary 3.2.8), so the form is nondegenerate (proposition 8.1.14), which is consistent with none of the three diagonal entries of $P^\top BP$ being zero.

A symmetric matrix can have every diagonal entry zero without being zero, and the smallest instance is the one a reader meets first.

Example 8.2.6: A Form with Zero Diagonal

Let $q = q_B$ on $\mathbb{R}^2$ for

$$B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad q(x) = 2x_1x_2$$

Here $q(e_1) = q(e_2) = 0$, so no exchange of coordinates produces a nonzero entry in position $(1,1)$ and there is no square to complete. The form is nevertheless nonzero, and it is even nondegenerate: $\det B = -1 \neq 0$ makes B invertible (corollary 3.2.8, proposition 8.1.14). Case 3 of theorem 8.2.3 applies with $i = 1$ and $j = 2$: the substitution is $x_1 = y_1 + y_2$ and $x_2 = y_1 - y_2$, under which

$$q(x) = 2(y_1+y_2)(y_1-y_2) = 2y_1^2 - 2y_2^2$$

In matrix terms $T = \left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ has columns $t_1 = (1,1)^\top$ and $t_2 = (1,-1)^\top$, with $Bt_1 = (1,1)^\top$ and $Bt_2 = (-1,1)^\top$, so (8.1) gives $t_1^\top Bt_1 = 2$, $t_1^\top Bt_2 = 0$ and $t_2^\top Bt_2 = -2$, that is

$$T^\top BT = \mathrm{diag}(2,-2)$$

Both diagonal entries are nonzero, matching $\mathrm{rank} B = 2$.

The vanishing of q is not confined to those two vectors: $q(x) = 0$ holds exactly on the union of the two coordinate axes, which is not a subspace of $\mathbb{R}^2$. What the vanishing of $q(e_1)$ and $q(e_2)$ does obstruct is the completion of a square, and the substitution above is what removes the obstruction.

The diagonal entries 2 and -2 produced here are not the eigenvalues of B, which are 1 and -1: the characteristic polynomial is $\det(tI_2 - B) = t^2 - 1$ (theorem 3.3.4, definition 4.1.9), with roots ± 1 (theorem 4.1.10). The signs of the two pairs agree, and the entries themselves do not. That is the general situation, and the next proposition says exactly how much freedom a diagonalizing congruence has.

Proposition 8.2.7: Rescaling a Diagonalization, and the Orthogonal Case

Let k be $\mathbb{R}$ or $\mathbb{C}$, let $B \in M_n(k)$ be symmetric, and let $P \in M_n(k)$ be invertible with $P^\top BP = \operatorname{diag}(d_1, \dots, d_n)$.

1. For any nonzero $t_1, \dots, t_n \in k$, the matrix $P \operatorname{diag}(t_1, \dots, t_n)$ is invertible and
$$\big(P\operatorname{diag}(t_1, \dots, t_n)\big)^\top B\big(P\operatorname{diag}(t_1, \dots, t_n)\big) = \operatorname{diag}\big(t_1^2 d_1, \dots, t_n^2 d_n\big)$$
2. Every symmetric $B \in M_n(\mathbb{C})$ is congruent to a diagonal matrix each of whose entries is 0 or 1. Every symmetric $B \in M_n(\mathbb{R})$ is congruent to a diagonal matrix each of whose entries is 0, 1 or -1.
3. If $k = \mathbb{R}$ and P is orthogonal, that is $P^\top P = I_n$, then $d_1, \dots, d_n$ are the roots of the characteristic polynomial of B, each listed as often as its multiplicity as a root.

***Proof*:**

1. Write $\Delta = \operatorname{diag}(t_1, \dots, t_n)$. By definition 1.4.1 the products of Δ with $\operatorname{diag}(t_1^{-1}, \dots, t_n^{-1})$ in either order are I_n, so Δ is invertible (definition 1.4.4). Then $P\Delta$ is invertible with inverse $\Delta^{-1}P^{-1}$, as proposition 1.4.3 gives $(P\Delta)(\Delta^{-1}P^{-1}) = P(\Delta\Delta^{-1})P^{-1} = I_n$ and the same computation in the other order. Column s of $P\Delta$ is $t_s p_s$, where p_s is column s of P (definition 1.4.1), and transposing multiplies no entry by anything, so $(t_s p_s)^\top = t_s p_s^\top$ (definition 2.3.4). Applying (8.1) to $P\Delta$ and then to P,
$$\big((P\Delta)^\top B(P\Delta)\big)_{st} = (t_s p_s)^\top B(t_t p_t) = t_s t_t\, p_s^\top B p_t = t_s t_t\, (P^\top BP)_{st}$$
The right-hand side is 0 for $s \neq t$ and $t_s^2 d_s$ for $s = t$, which is the stated diagonal matrix.
2. Theorem 8.2.3 gives an invertible P with $P^\top BP = \operatorname{diag}(d_1, \dots, d_n)$, so it is enough to choose the scalars t_s of part (1). Take $t_s = 1$ whenever $d_s = 0$, which leaves that entry at 0. Over $\mathbb{C}$, a nonzero d_s has $1/d_s \neq 0$, and every nonzero complex number is a square: writing $1/d_s = re^{i\theta}$ with $r > 0$, the number $t_s = \sqrt{r}\, e^{i\theta/2}$ satisfies $t_s^2 = 1/d_s$, whence $t_s^2 d_s = 1$. Over $\mathbb{R}$, a nonzero d_s has $|d_s| > 0$, so taking $t_s = |d_s|^{-1/2}$ gives $t_s^2 d_s = d_s/|d_s|$, which is 1 if $d_s > 0$ and -1 if $d_s < 0$. In both cases part (1) produces the required congruence.
3. Multiplying $P^\top P = I_n$ on the right by P^{-1} and regrouping with proposition 1.4.3 gives $P^\top = P^{-1}$, so
$$D := \operatorname{diag}(d_1, \dots, d_n) = P^\top BP = P^{-1}BP$$
and B and D are similar (definition 2.5.5). Similar matrices have the same characteristic polynomial (corollary 4.1.16), so $\det(tI_n - B) = \det(tI_n - D)$ (definition 4.1.9). The matrix $tI_n - D$ is diagonal, hence upper triangular, so proposition 3.2.6 computes its determinant as the product of its diagonal entries,
$$\det(tI_n - D) = (t - d_1)(t - d_2)\cdots(t - d_n)$$
Collecting equal factors and applying proposition 4.3.5 to each distinct value shows that the roots of this polynomial, listed with algebraic multiplicity, are $d_1, \dots, d_n$; and they are the roots of the characteristic polynomial of B, completing the proof.

Part (1) measures how far a Lagrange diagonalization is from being determined by B: any diagonal entry may be multiplied by the square of any nonzero scalar, and the resulting matrix is again congruent to B. Over $\mathbb{C}$ every nonzero scalar is a square, so no nonzero entry is distinguishable from any other and part (2) drives the diagonal down to 1s and 0s, the number of 1s being $\operatorname{rank} B$ by the rank count above. Over $\mathbb{R}$ the squares are exactly the positive numbers, so a rescaling can change the size of an entry but not its sign, and part (2) drives the diagonal down to 1s, -1s and 0s. What part (1) does not settle is whether a different diagonalization could produce a different number of 1s. Theorem 8.3.1 settles it, and the three counts it isolates are the subject of section 8.3. The criteria that decide the signs from B itself, without computing any diagonalization, are section 8.4's.

Part (3) locates the spectral theorem inside this picture. For a real symmetric B, corollary 7.4.3 produces an orthogonal Q with $Q^\top BQ$ diagonal, and for an orthogonal matrix the congruence $Q^\top BQ$ is also the similarity $Q^{-1}BQ$, so its diagonal carries the eigenvalues of B. The associated statement about the quadratic form is corollary 7.4.6, which is (8.2) for that particular Q. The two routes to a diagonal therefore differ in what they cost and in what they deliver. Lagrange's method is finitely many additions, multiplications and divisions, and the numbers it puts on the diagonal are whatever those operations return. The spectral route needs the roots of the characteristic polynomial, and it returns the eigenvalues, in coordinates that are orthonormal for the standard inner product. The extra cost is real: for $B = \left(\begin{smallmatrix}1&1\\1&2\end{smallmatrix}\right)$ the substitution $y_1 = x_1 + x_2$, $y_2 = x_2$ gives $q_B(x) = x_1^2 + 2x_1x_2 + 2x_2^2 = y_1^2 + y_2^2$ after a single division, with integer entries on the diagonal, while the eigenvalues are the roots $(3 \pm \sqrt{5})/2$ of $\det(tI_2 - B) = (t-1)(t-2) - 1 = t^2 - 3t + 1$.

Example 8.2.8: One Form Diagonalized Twice

Let $q = q_B$ on $\mathbb{R}^2$ for the matrix of the opening computation,

$$B = \begin{pmatrix}1 & 2\\ 2 & 1\end{pmatrix}, \qquad q(x) = x_1^2 + 4x_1x_2 + x_2^2$$

Lagrange's method. Case 2 of theorem 8.2.3 applies with $b_{11} = 1$ and $b_{12}/b_{11} = 2$, so the substitution is $y_1 = x_1 + 2x_2$ and $y_2 = x_2$, and $q(x) = y_1^2 - 3y_2^2$ as computed above. Solving $x_1 = y_1 - 2y_2$ and $x_2 = y_2$ gives

$$P = \begin{pmatrix}1 & -2\\ 0 & 1\end{pmatrix}, \qquad p_1 = \begin{pmatrix}1\\ 0\end{pmatrix}, \qquad p_2 = \begin{pmatrix}-2\\ 1\end{pmatrix}, \qquad \det P = 1$$

with $Bp_1 = (1,2)^\top$ and $Bp_2 = (-2+2, -4+1)^\top = (0,-3)^\top$, so (8.1) gives $p_1^\top Bp_1 = 1$, $p_1^\top Bp_2 = 0$ and $p_2^\top Bp_2 = -3$, that is $P^\top BP = \operatorname{diag}(1,-3)$.

The spectral route. The characteristic polynomial is

$$\det(tI_2 - B) = (t-1)^2 - 4 = t^2 - 2t - 3 = (t-3)(t+1)$$

so the eigenvalues are 3 and -1 (theorem 4.1.10), both real as proposition 7.1.11 requires of a symmetric matrix. For $\lambda = 3$ the equation $Bv = 3v$ reads $v_1 + 2v_2 = 3v_1$, that is $v_1 = v_2$; for $\lambda = -1$ it reads $v_1 + 2v_2 = -v_1$, that is $v_1 = -v_2$. Normalizing the two solutions $(1,1)^\top$ and $(1,-1)^\top$, each of norm $\sqrt{2}$,

$$Q = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}, \qquad Q^\top Q = \frac{1}{2}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} = I_2$$

so Q is orthogonal. Its columns q_1 and q_2 are scalar multiples of the two solutions found above, so $Bq_1 = 3q_1$ and $Bq_2 = -q_2$, and (8.1) together with $q_1^\top q_1 = q_2^\top q_2 = 1$ and $q_1^\top q_2 = 0$ gives

$Q^\top BQ = \operatorname{diag}(3, -1)$. This is the factorization of corollary 7.4.3, with the eigenvalues on the diagonal as proposition 8.2.7(3) predicts.

Comparison. The two diagonals are

$$\operatorname{diag}(1, -3) \qquad \text{and} \qquad \operatorname{diag}(3, -1)$$

Neither pair of entries is a permutation of the other, so the entries themselves are not an invariant of B. Three things do agree. The determinants: $\det B = 1 - 4 = -3$, and both diagonals have product -3, which proposition 8.1.8 forces since $\det P = 1$ and $\det Q = -1$ both have square 1. The number of positive entries and the number of negative entries: one each, on both sides. And the two are related by the rescaling of proposition 8.2.7(1) with $t_1 = \sqrt{3}$ and $t_2 = 1/\sqrt{3}$, since $t_1^2 \cdot 1 = 3$ and $t_2^2 \cdot (-3) = -1$. Rescaling instead by $t_1 = 1$, $t_2 = 1/\sqrt{3}$ in the first case and by $t_1 = 1/\sqrt{3}$, $t_2 = 1$ in the second sends both to $\operatorname{diag}(1, -1)$, which is proposition 8.2.7(2) run on this matrix.

The coordinates differ in one further respect. The columns of Q are orthonormal for the standard inner product of $\mathbb{R}^2$, so Q is orthogonal (proposition 7.1.5) and the map $x \mapsto Q^\top x$ preserves the inner product itself (theorem 7.1.4(2)); lengths and angles computed in the coordinates $z = Q^\top x$ therefore agree with those computed in x. The columns of P are not orthonormal, as $p_1^\top p_2 = -2 \neq 0$. The function q is the same in both, and only its expression changes.

Exercise 8.2.1

1. Let $q(x) = x_1^2 + 2x_2^2 + 3x_3^2 - 2x_1x_2 + 2x_1x_3 + 4x_2x_3$ on $\mathbb{R}^3$. Write down the symmetric matrix B with $q = q_B$, run the two substitutions of theorem 8.2.3 to write q as a combination of squares of new coordinates, and exhibit the invertible P with $P^\top BP$ diagonal. Verify the product using (8.1), and check your answer against proposition 8.1.8 by computing $\det B$.

2. Let $q(x) = 2x_1x_2 + 2x_1x_3 + 2x_2x_3$ on $\mathbb{R}^3$. Check that every diagonal entry of its symmetric matrix B vanishes, apply the substitution of Case 3 of theorem 8.2.3 to the indices $i = 1$, $j = 2$, and then complete the reduction to a diagonal matrix. Separately, verify that $(1, 1, 1)^\top$ satisfies $Bv = 2v$ and that every v with $v_1 + v_2 + v_3 = 0$ satisfies $Bv = -v$, so that the eigenvalues of B are $2, -1, -1$. Compare the signs of the eigenvalues with the signs of your diagonal entries.

3. Let $A \in M_n(k)$ be arbitrary, with k either $\mathbb{R}$ or $\mathbb{C}$, and put $q(x) = x^\top Ax$ for $x \in k^n$.

   1. Show that $q = q_S$ for the symmetric matrix $S = \frac{1}{2}(A + A^\top)$, and that S is the only symmetric matrix with this property. (For the first part, expand both $x^\top Ax$ and $x^\top A^\top x$ as double sums over the entries and exchange the two summation indices in one of them; for the second, use theorem 8.2.2(2).)
   2. Carry this out for $A = \begin{pmatrix} 1 & 5 \\ -1 & 2 \end{pmatrix}$, and then diagonalize the resulting form by theorem 8.2.3.

4. Let $B \in M_n(\mathbb{R})$ be symmetric with $\det B \neq 0$, and suppose $P^\top BP = \operatorname{diag}(d_1, \ldots, d_n)$ for some invertible $P \in M_n(\mathbb{R})$. Show that no d_s is zero and that the sign of $d_1 d_2 \cdots d_n$ equals the sign of $\det B$. Deduce that the number of negative d_s has the same parity for every diagonalization of B by congruence over $\mathbb{R}$.

5. Show that $\operatorname{diag}(1, 1)$ and $\operatorname{diag}(1, -1)$ are congruent as complex matrices but not as real ones. (For the first, exhibit an explicit diagonal $\Delta \in M_2(\mathbb{C})$ using proposition 8.2.7(1); for the

second, use exercise 8.2.1 (4).) Explain what this says about the diagonal forms available over the two fields.

6. Let φ be a symmetric bilinear form on a finite-dimensional space V over $\mathbb{R}$ and suppose $\varphi(x,x) = 0$ for every $x \in V$. Show that $\varphi = 0$, and say which hypothesis on the scalars the argument uses. Then exhibit a nondegenerate symmetric form on $\mathbb{R}^2$ (definition 8.1.13) whose matrix in the standard basis is *diagonal*, together with a nonzero $x \in \mathbb{R}^2$ at which its quadratic form vanishes, and explain why this is consistent with the first part.

7. Let $B = \left(\begin{smallmatrix} 5 & 2 \\ 2 & 2 \end{smallmatrix}\right) \in M_2(\mathbb{R})$. Diagonalize q_B twice: once by theorem 8.2.3, and once by producing an orthogonal Q as in corollary 7.4.3. Exhibit both diagonals, check both products against $\det B$ using proposition 8.1.8, and find the two nonzero scalars t_1, t_2 of proposition 8.2.7(1) that carry the first diagonal to the second.

8.3 Sylvester's Law of Inertia; Signature and Index

Corollary 8.2.4 produced, for every symmetric bilinear form on a finite-dimensional real vector space, a basis in which the matrix of the form is diagonal, and example 8.2.8 produced two such bases for one form with different diagonal entries. This section identifies what does not depend on the choice of basis: the number of positive entries, the number of negative ones, and the number of zeros. Over $\mathbb{R}$ those three integers determine the form up to congruence, and over $\mathbb{C}$ only the number of nonzero entries, the rank, does.

Two moves are available inside a diagonal representation, and comparing them isolates the candidates for an invariant. Proposition 8.2.7(1) records the first in matrix form; read through definition 8.1.2 it says that replacing a basis vector w_i by cw_i, for a nonzero scalar c, leaves the off-diagonal zeros where they are and replaces the diagonal entry d_i by $c^2 d_i$. Over $\mathbb{R}$ the factors c^2 available are exactly the positive reals, so a nonzero d_i can be moved to any real number of the same sign and to none of the opposite sign. Reordering the basis permutes the d_i and does nothing else. Normalizing as far as these moves allow is proposition 8.2.7(2): every diagonal representation can be brought to entries drawn from $\{1, -1, 0\}$ over $\mathbb{R}$, and from $\{1, 0\}$ over $\mathbb{C}$. The only data that could survive both moves is how many entries of each kind there are.

Theorem 8.3.1: Sylvester's Law of Inertia

Let V be a real vector space of dimension $n \geq 1$, let φ be a symmetric bilinear form on V, and let q be the associated quadratic form, $q(x) = \varphi(x, x)$ (definition 8.2.1). For a subspace $U \subseteq V$, the phrase $q > 0$ *on* $U \setminus \{0\}$ means that $q(u) > 0$ for every nonzero $u \in U$, and $q < 0$ *on* $U \setminus \{0\}$ is read the same way.

1. There are integers $p, m \geq 0$ with $p + m \leq n$ and a basis $v_1, \ldots, v_n$ of V in which the matrix of φ is
$$\operatorname{diag}(\underbrace{1, \ldots, 1}_{p}, \underbrace{-1, \ldots, -1}_{m}, \underbrace{0, \ldots, 0}_{n-p-m})$$
2. Let $u_1, \ldots, u_n$ be any basis of V in which the matrix of φ is diagonal, and let a and b be the number of positive and the number of negative entries on that diagonal. Then a is the largest dimension of a subspace $U \subseteq V$ with $q > 0$ on $U \setminus \{0\}$, and b is the largest dimension of a subspace $W \subseteq V$ with $q < 0$ on $W \setminus \{0\}$. In particular $a = p$ and $b = m$, so the two counts are the same for every basis that diagonalizes φ.

***Proof*:**

1. By corollary 8.2.4 there is a basis $w_1, \ldots, w_n$ of V with $\varphi(w_i, w_j) = 0$ for $i \neq j$. Put $d_i = \varphi(w_i, w_i)$, let p be the number of indices with $d_i > 0$ and let m be the number with $d_i < 0$. Since the two index sets are disjoint subsets of $\{1, \ldots, n\}$, $p + m \leq n$.

   Reordering the basis carries the diagonal entries with it. For a permutation σ of $\{1, \ldots, n\}$ the list $w_{\sigma(1)}, \ldots, w_{\sigma(n)}$ is again a basis of V, and the entry of the matrix of φ in that basis in row i and column j is $\varphi(w_{\sigma(i)}, w_{\sigma(j)})$ (definition 8.1.2), which vanishes for $i \neq j$ and is $d_{\sigma(i)}$ for $i = j$. Choose σ so that $d_{\sigma(i)} > 0$ for $i \leq p$, so that $d_{\sigma(i)} < 0$ for $p < i \leq p + m$, and so that $d_{\sigma(i)} = 0$ for $i > p + m$. After renaming, assume $d_1, \ldots, d_n$ is itself ordered this way.

   The rescaling now applied is that of proposition 8.2.7(2), carried out in basis language so that the resulting entries come in the chosen order, which the matrix statement there does not arrange. For $i \leq p + m$ the real number $|d_i|$ is positive, hence has a positive square root, and $v_i := w_i / \sqrt{|d_i|}$ is defined. For $i > p + m$ put $v_i := w_i$. Each v_i is $c_i w_i$ for a nonzero scalar c_i, and $\sum_i t_i v_i = \sum_i (t_i c_i) w_i$, so the list $v_1, \ldots, v_n$ spans V and is linearly independent because $w_1, \ldots, w_n$ is, hence is again a basis (definition 2.2.1). Bilinearity (definition 8.1.1) gives $\varphi(v_i, v_j) = 0$ for $i \neq j$, because the corresponding $\varphi(w_i, w_j)$ vanishes, and for $i \leq p + m$
$$\varphi(v_i, v_i) = \frac{1}{|d_i|} \varphi(w_i, w_i) = \frac{d_i}{|d_i|}$$
   which is 1 when $d_i > 0$ and -1 when $d_i < 0$, while $\varphi(v_i, v_i) = d_i = 0$ for $i > p + m$. The matrix of φ in the basis $v_1, \ldots, v_n$ is the one displayed in the statement.

2. Let the matrix of φ in the basis $u_1, \ldots, u_n$ be $\operatorname{diag}(c_1, \ldots, c_n)$ and put $S_+ = \{i | c_i > 0\}$ and $S_- = \{i | c_i < 0\}$, so that $a = |S_+|$ and $b = |S_-|$. Expanding by bilinearity and discarding

the terms carrying a factor $\varphi(u_i, u_j)$ with $i \neq j$, a vector $x = \sum_{i=1}^n t_i u_i$ has

$$q(x) = \varphi(x, x) = \sum_{i=1}^{n} c_i t_i^2 \tag{8.3}$$

Let N_+ be the largest dimension of a subspace $U \subseteq V$ with $q > 0$ on $U \setminus \{0\}$. There is at least one such subspace, namely $U = \{0\}$, for which the condition is vacuous, and every subspace of V has dimension at most n (corollary 2.2.12), so N_+ is a well-defined integer between 0 and n.

The subspace $U_0 = \operatorname{span}\{u_i \big| i \in S_+\}$ (definition 2.1.5) has dimension a, since a sublist of a basis is linearly independent and is therefore a basis of its span (definition 2.2.1, definition 2.2.10). For a nonzero $x = \sum_{i \in S_+} t_i u_i$ in U_0, equation (8.3) reads $q(x) = \sum_{i \in S_+} c_i t_i^2$, a sum of nonnegative terms at least one of which is positive, so $q(x) > 0$. Hence $N_+ \geq a$.

For the reverse inequality, let $W_0 = \operatorname{span}\{u_i \big| i \notin S_+\}$, which has dimension $n - a$ by the same argument, and note that (8.3) gives $q(y) = \sum_{i \notin S_+} c_i t_i^2 \leq 0$ for every $y = \sum_{i \notin S_+} t_i u_i$ in W_0, since $c_i \leq 0$ for $i \notin S_+$. Let U be a subspace with $q > 0$ on $U \setminus \{0\}$ and suppose $\dim U > a$. Choose a basis $x_1, \ldots, x_s$ of U and a basis $y_1, \ldots, y_t$ of W_0 (theorem 2.2.9). Then $s + t > a + (n - a) = n$. The combined list $x_1, \ldots, x_s, y_1, \ldots, y_t$ cannot be linearly independent: an independent list is a basis of its span (definition 2.2.1), so its length would be the dimension of a subspace of V, and that is at most n by corollary 2.2.12. So there are scalars $\alpha_1, \ldots, \alpha_s, \beta_1, \ldots, \beta_t$, not all zero, with $\sum_i \alpha_i x_i + \sum_j \beta_j y_j = 0$. Put

$$v = \sum_{i=1}^{s} \alpha_i x_i = -\sum_{j=1}^{t} \beta_j y_j$$

a vector lying in U and in W_0 at once. Were v zero, every α_i would vanish by independence of $x_1, \ldots, x_s$ and every β_j would vanish by independence of $y_1, \ldots, y_t$, against the choice of the scalars. So $v \neq 0$. Then $q(v) > 0$ because $v \in U \setminus \{0\}$, and $q(v) \leq 0$ because $v \in W_0$, which is impossible. Hence $\dim U \leq a$ for every such U, and with the previous paragraph $N_+ = a$.

The form $-\varphi$, defined by $(-\varphi)(x, y) = -\varphi(x, y)$, is again bilinear and symmetric, its associated quadratic form is $-q$, and its matrix in the basis $u_1, \ldots, u_n$ is $\operatorname{diag}(-c_1, \ldots, -c_n)$ (definition 8.1.2), whose positive entries are indexed by S_-. The previous two paragraphs applied to $-\varphi$ give $N_- = b$, where N_- is the largest dimension of a subspace on which $-q > 0$ away from 0, that is on which $q < 0$ away from 0.

The integers N_+ and N_- were defined from φ alone, with no basis entering their definition, so $a = N_+$ and $b = N_-$ hold for every basis in which the matrix of φ is diagonal. The basis produced in (1) is one of those, and its diagonal carries exactly p positive and m negative entries, so $N_+ = p$ and $N_- = m$, completing the proof.

The theorem separates the contribution of the basis from the contribution of the form.[2] The diagonal entries themselves are not determined by φ, as example 8.2.8 showed by producing two diagonals for one form. The three counts are determined, and clause (2) says why they can be. Each count has a

[2]The name is Sylvester's, from his 1852 paper on the reduction of a real quadratic form to a sum of positive and negative squares, where the invariance of the two counts is called a law of inertia.

description in which no basis appears, as the largest dimension of a subspace on which q keeps a strict sign. Two descriptions of one integer are what the proof reconciles, and the reconciliation is a dimension count: a subspace on which $q > 0$ and a subspace on which $q \leq 0$ meet only in 0, so their dimensions cannot exceed n together, and a diagonal representation with too few positive entries would violate that bound.

Definition 8.3.2: Inertia, Signature and Index

Let φ be a symmetric bilinear form on a real vector space V of dimension $n \geq 1$ and let p and m be the integers of theorem 8.3.1. The *inertia* of φ is the triple

$$(n_+, n_-, n_0) = (p, m, n - p - m)$$

of the number of positive, negative and zero entries in any diagonal representation of φ. The *signature* of φ is the integer $\sigma(\varphi) = n_+ - n_-$ and the *index* of φ is n_-. The same three integers are attached to a symmetric matrix $B \in M_n(\mathbb{R})$ through the form $\varphi_B(x, y) = x^\top By$ on $\mathbb{R}^n$, and are written $\big(n_+(B), n_-(B), n_0(B)\big)$ and $\sigma(B)$.

For $B = \operatorname{diag}(2, -3, 0)$ the matrix of φ_B in the standard basis is B itself (theorem 8.1.3), diagonal with one positive, one negative and one zero entry, so theorem 8.3.1(2) gives inertia $(1, 1, 1)$, signature 0 and index 1. Both conventions for the word *signature* are in use: many sources call the pair (n_+, n_-), or the whole triple, the signature and write it (p, q). This book always means the single integer $n_+ - n_-$, and always means n_- by the index. The index is the count that section 8.5 attaches to a critical point of a function.

Definition 8.3.2 reaches a matrix through a form, and the passage back and forth is congruence. Let k be $\mathbb{R}$ or $\mathbb{C}$, let $B \in M_n(k)$ be symmetric, and let $\varphi_B(x, y) = x^\top By$, whose matrix in the standard basis is B (theorem 8.1.3). If $v_1, \ldots, v_n$ is a basis of k^n and P is the matrix whose j-th column is v_j, then definition 1.4.1 gives

$$(P^\top BP)_{ij} = v_i^\top Bv_j = \varphi_B(v_i, v_j) \tag{8.4}$$

so $P^\top BP$ is the matrix of φ_B in the basis $v_1, \ldots, v_n$ (definition 8.1.2), and P is invertible, being the matrix relating that basis to the standard basis (definition 2.5.2, proposition 2.5.3). In the other direction, for invertible P the substitution $x = Py$ carries q_B to the quadratic form of $P^\top BP$, since

$$q_{P^\top BP}(y) = y^\top P^\top BPy = (Py)^\top B(Py) = q_B(Py) \tag{8.5}$$

where $(Py)^\top = y^\top P^\top$ holds because the i-th entry of Py is $\sum_j p_{ij}y_j$, which is also the i-th entry of the row $y^\top P^\top$ (definition 2.3.4, definition 1.4.1). The diagonal matrices congruent to B (definition 8.1.5) are therefore exactly the matrices of φ_B in the bases that diagonalize it. From here on $\operatorname{diag}(I_p, -I_m, 0)$ denotes the matrix of theorem 8.3.1(1), with an identity block of size p, the negative of an identity block of size m, and a zero block of size $n - p - m$. In this notation the two clauses of the theorem become a classification of symmetric matrices up to congruence.

Corollary 8.3.3: Real Symmetric Matrices Are Classified by Their Inertia

Let $n \geq 1$ and let $B, B' \in M_n(\mathbb{R})$ be symmetric. Then B and B' are congruent if and only if

$$\big(n_+(B), n_-(B), n_0(B)\big) = \big(n_+(B'), n_-(B'), n_0(B')\big)$$

Every symmetric $B \in M_n(\mathbb{R})$ is congruent to exactly one matrix of the form $\operatorname{diag}(I_p, -I_m, 0)$, namely the one with $p = n_+(B)$ and $m = n_-(B)$. Moreover $n_+(B) + n_-(B) = \operatorname{rank} B$, so $n_0(B) = n - \operatorname{rank} B$.

***Proof*:**
Step 1: congruent matrices have equal inertia. Suppose $B' = P^\top BP$ with P invertible, and let $U \subseteq \mathbb{R}^n$ be a subspace with $q_{B'} > 0$ on $U \setminus \{0\}$. The map $y \mapsto Py$ is linear and injective, since P is invertible, so the image $PU = \{Pu \big| u \in U\}$ is a subspace of $\mathbb{R}^n$ and a basis of U is carried to a linearly independent list spanning PU, whence $\dim PU = \dim U$. A nonzero element of PU is Pu for a nonzero $u \in U$, again by injectivity, and (8.5) gives $q_B(Pu) = q_{B'}(u) > 0$. So PU is a subspace of dimension $\dim U$ on which $q_B > 0$ away from 0, and theorem 8.3.1(2) applied to φ_B gives $\dim U \leq n_+(B)$. Taking U of dimension $n_+(B')$ yields $n_+(B') \leq n_+(B)$.

Multiplying $B' = P^\top BP$ on the left by $(P^{-1})^\top$ and on the right by P^{-1}, and using $(P^{-1})^\top P^\top = (PP^{-1})^\top = I_n$ (proposition 2.3.5(3)), gives $B = (P^{-1})^\top B' P^{-1}$, so the argument just given, with the roles of B and B' exchanged and P replaced by P^{-1}, yields $n_+(B) \leq n_+(B')$. Hence $n_+(B) = n_+(B')$. The same argument with the condition $q > 0$ on $U \setminus \{0\}$ replaced throughout by $q < 0$ on $U \setminus \{0\}$, which is the condition theorem 8.3.1(2) characterizes n_- by, gives $n_-(B) = n_-(B')$. The third counts agree as well, because all three sum to n.

Step 2: the normal form. Apply theorem 8.3.1(1) to φ_B: there is a basis $v_1, \dots, v_n$ of $\mathbb{R}^n$ in which the matrix of φ_B is $\operatorname{diag}(I_p, -I_m, 0)$ with $p = n_+(B)$ and $m = n_-(B)$. Let P be the matrix with columns $v_1, \dots, v_n$. By (8.4) and the invertibility recorded there, $P^\top BP = \operatorname{diag}(I_p, -I_m, 0)$, so B is congruent to that matrix. If B is congruent to $\operatorname{diag}(I_{p'}, -I_{m'}, 0)$ as well, then Step 1 says that this second diagonal matrix has the same inertia as B. The matrix of the form of a diagonal matrix in the standard basis is that matrix (theorem 8.1.3), so theorem 8.3.1(2) computes its inertia as $(p', m', n - p' - m')$, and therefore $p' = n_+(B)$ and $m' = n_-(B)$. This proves the uniqueness of the normal form, and it also proves the stated equivalence: if B and B' have equal inertia they are congruent to the same normal form, hence to each other by proposition 8.1.6, while the converse is Step 1.

Step 3: the rank. The nonzero columns of $D = \operatorname{diag}(I_p, -I_m, 0)$ are $e_1, \dots, e_p, -e_{p+1}, \dots, -e_{p+m}$, which are linearly independent and span the column space of D (definition 2.3.1), so that column space has dimension $p + m$ and $\operatorname{rank} D = p + m$ by theorem 2.3.10. Congruent matrices have equal rank (proposition 8.1.7), so $\operatorname{rank} B = p + m = n_+(B) + n_-(B)$ and $n_0(B) = n - \operatorname{rank} B$, as we sought to show.

Congruence classes of symmetric matrices in $M_n(\mathbb{R})$ are therefore in bijection with triples of non-negative integers summing to n, of which there are $\frac{1}{2}(n+1)(n+2)$. For $n = 2$ the six classes have representatives $\operatorname{diag}(1,1)$, $\operatorname{diag}(1,-1)$, $\operatorname{diag}(-1,-1)$, $\operatorname{diag}(1,0)$, $\operatorname{diag}(-1,0)$ and 0. The rank statement identifies $n_0(B)$ with $\operatorname{nullity} B$ through theorem 2.3.11, so $n_0(B) = 0$ exactly when B is invertible, which by proposition 8.1.14 is exactly when φ_B is nondegenerate. At the other end of the range, $n_+(B) = n$ says that $q_B(x) > 0$ for every nonzero x, and that case is the subject of section 8.4.

Over $\mathbb{R}$ one diagonal representation is always available from chapter 7. Let $B \in M_n(\mathbb{R})$ be symmetric and let $\lambda_1, \dots, \lambda_n$ be its eigenvalues, listed with algebraic multiplicity. They are real by proposition 7.1.11, and corollary 7.4.3 produces an orthogonal $Q \in M_n(\mathbb{R})$ with $Q^\top BQ = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$. Since $Q^\top = Q^{-1}$ this single equation exhibits B as both similar (definition 2.5.5) and congruent to that diagonal matrix, so Step 1 of the proof above and theorem 8.3.1(2) give

$$n_+(B) = \#\{i \big| \lambda_i > 0\}, \qquad n_-(B) = \#\{i \big| \lambda_i < 0\}, \qquad n_0(B) = \#\{i \big| \lambda_i = 0\} \tag{8.6}$$

where $\#$ counts indices. The identity behind this, q_B written in the coordinates of an orthonormal eigenbasis, is corollary 7.4.6. The inertia is thus what the spectrum of a symmetric matrix contributes to its congruence class: the signs of the eigenvalues are remembered and their sizes are not, which is why the congruent matrices of example 8.1.11 can fail to be similar. Two routes to the inertia are now available, one through the rational operations of theorem 8.2.3 and one through the roots of a characteristic polynomial, and they must agree.

Example 8.3.4: The Inertia of the Triangle Form

Let

$$B = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \in M_3(\mathbb{R}), \qquad q_B(x) = x^\top Bx = 2x_1x_2 + 2x_1x_3 + 2x_2x_3$$

the second display collecting the two equal off-diagonal contributions of each pair of indices.

Every diagonal entry of B vanishes, so no square carries a nonzero coefficient and there is nothing yet to complete. The substitution of example 8.2.6 comes first: put $x_1 = y_1 + y_2$, $x_2 = y_1 - y_2$ and $x_3 = y_3$. Then

$$2x_1x_2 = 2(y_1+y_2)(y_1-y_2) = 2y_1^2 - 2y_2^2, \qquad 2x_1x_3 + 2x_2x_3 = 2y_3\big((y_1+y_2)+(y_1-y_2)\big) = 4y_1y_3$$

so $q_B = 2y_1^2 - 2y_2^2 + 4y_1y_3$. The coefficient of y_1^2 is now nonzero, and completing the square in y_1 gives $2y_1^2 + 4y_1y_3 = 2(y_1+y_3)^2 - 2y_3^2$, so that in the coordinates $z_1 = y_1 + y_3$, $z_2 = y_2$, $z_3 = y_3$,

$$q_B = 2z_1^2 - 2z_2^2 - 2z_3^2$$

One positive and two negative coefficients, so theorem 8.3.1(2) gives inertia $(1, 2, 0)$, signature -1 and index 2.

The two substitutions are invertible, and composing them exhibits the congruence. From $x_1 = y_1 + y_2$, $x_2 = y_1 - y_2$, $x_3 = y_3$ and $y_1 = z_1 - z_3$, $y_2 = z_2$, $y_3 = z_3$,

$$x = P_1y, \quad y = P_2z, \qquad P_1 = \begin{pmatrix} 1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P_2 = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad P = P_1P_2 = \begin{pmatrix} 1 & 1 & -1 \\ 1 & -1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

Writing p_1, p_2, p_3 for the columns of P and using (8.4), the congruent matrix is computed from the nine numbers $p_i^\top Bp_j$. The three products Bp_j are

$$Bp_1 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \qquad Bp_2 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \qquad Bp_3 = \begin{pmatrix} 0 \\ 0 \\ -2 \end{pmatrix}$$

and pairing them against the columns gives $p_1^\top Bp_1 = 1 + 1 + 0 = 2$, $p_2^\top Bp_2 = -1 - 1 + 0 = -2$, $p_3^\top Bp_3 = 0 + 0 - 2 = -2$, while $p_2^\top Bp_1 = 1 - 1 + 0 = 0$, $p_3^\top Bp_1 = -1 - 1 + 2 = 0$ and $p_3^\top Bp_2 = 1 - 1 + 0 = 0$. So

$$P^\top BP = \operatorname{diag}(2, -2, -2)$$

which is the coefficient list of the completed square. Dividing each column of P by $\sqrt{2}$ divides each of these three numbers by 2, replacing the matrix by $\operatorname{diag}(1, -1, -1)$, which is the normal form of theorem 8.3.1(1) with $p = 1$ and $m = 2$.

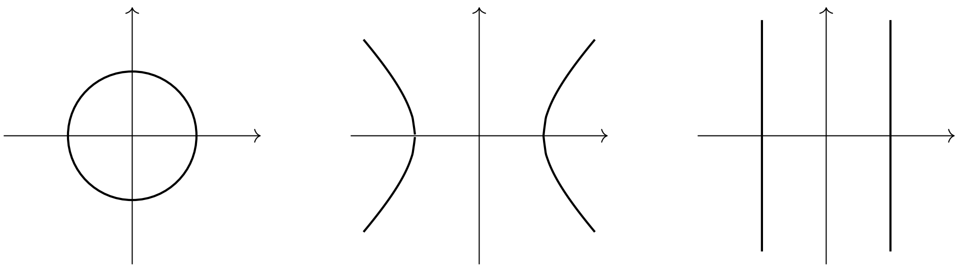
The second route asks for the eigenvalues. Expanding along the first row with theorem 3.3.4,

$$\chi_B(t) = \det \begin{pmatrix} t & -1 & -1 \\ -1 & t & -1 \\ -1 & -1 & t \end{pmatrix} = t(t^2-1) + \big((-1)t - (-1)(-1)\big) - \big((-1)(-1) - t(-1)\big) = t^3 - 3t - 2$$

and $t = 2$ is a root, since $8-6-2 = 0$, so dividing out gives $\chi_B(t) = (t-2)(t^2+2t+1) = (t-2)(t+1)^2$. The eigenvalues are therefore $2, -1, -1$ by theorem 4.1.10, all real as proposition 7.1.11 requires, and the sign count is one positive and two negative. The identity (8.6) therefore returns the inertia $(1, 2, 0)$ that completing the square gave, and the two diagonals reaching it are different: $(2, -2, -2)$ from the substitutions above, $(2, -1, -1)$ from the spectral theorem.

The determinants cross-check the congruence. Expanding $\det B$ along the first row gives $0(0-1) - 1(0-1) + 1(1-0) = 2$, and expanding $\det P$ along its last row gives $\det P = 1 \cdot (1 \cdot (-1) - 1 \cdot 1) = -2$. Then $(\det P)^2 \det B = 4 \cdot 2 = 8 = \det \operatorname{diag}(2, -2, -2)$, which is what proposition 8.1.8 predicts.

In two variables the classes with $n_+ \geq 1$ can be seen in the level set $q = 1$, drawn in fig. 8.1 in the coordinates of a normal-form basis: a circle when the inertia is $(2, 0, 0)$, a hyperbola when it is $(1, 1, 0)$, and a pair of parallel lines when it is $(1, 0, 1)$. For the three remaining classes on two variables $n_+ = 0$, so $q \leq 0$ everywhere and the set is empty.



$(n_+, n_-, n_0) = (2, 0, 0)$ $(n_+, n_-, n_0) = (1, 1, 0)$ $(n_+, n_-, n_0) = (1, 0, 1)$

Figure 8.1: The set where $q = 1$ in $\mathbb{R}^2$, for $q = x_1^2 + x_2^2$, for $q = x_1^2 - x_2^2$, and for $q = x_1^2$

Rescaling a basis vector by c multiplies the corresponding diagonal entry by c^2, and the argument above used the scalars being real in exactly one place, the inequality $c^2 > 0$. Over $\mathbb{C}$ every nonzero scalar is a square, so every nonzero diagonal entry can be sent to 1 and only the rank survives.

Corollary 8.3.5: Complex Symmetric Matrices Are Classified by Their Rank

Let $n \geq 1$ and let $B, B' \in M_n(\mathbb{C})$ be symmetric, meaning $B^\top = B$ with no conjugation. Then B is congruent to $\operatorname{diag}(I_r, 0)$ with $r = \operatorname{rank} B$, and B and B' are congruent if and only if $\operatorname{rank} B = \operatorname{rank} B'$.

***Proof*:**
Apply corollary 8.2.4 over $\mathbb{C}$ to the form φ_B on $\mathbb{C}^n$, whose matrix in the standard basis is B (theorem 8.1.3): there is a basis $w_1, \ldots, w_n$ of $\mathbb{C}^n$ with $\varphi_B(w_i, w_j) = 0$ for $i \neq j$. Put $d_i = \varphi_B(w_i, w_i)$ and, reordering the basis as in the proof of theorem 8.3.1(1), assume $d_i \neq 0$ for $i \leq r$ and $d_i = 0$ for $i > r$.

For $i \leq r$ the polynomial $t^2 - d_i$ has a root $c_i \in \mathbb{C}$ by the fundamental theorem of algebra, which the front matter imports, and $c_i \neq 0$ because $c_i^2 = d_i \neq 0$. Put $v_i = w_i/c_i$ for $i \leq r$ and $v_i = w_i$ for $i > r$. This list is again a basis, by the scaling argument in the proof of theorem 8.3.1(1), and bilinearity (definition 8.1.1) gives $\varphi_B(v_i, v_j) = 0$ for $i \neq j$ together with

$$\varphi_B(v_i, v_i) = \frac{1}{c_i^2}\varphi_B(w_i, w_i) = \frac{d_i}{d_i} = 1$$

for $i \leq r$, while $\varphi_B(v_i, v_i) = 0$ for $i > r$. So the matrix of φ_B in the basis $v_1, \ldots, v_n$ is $\operatorname{diag}(I_r, 0)$, and (8.4) turns that basis into an invertible P with $P^\top BP = \operatorname{diag}(I_r, 0)$.

The nonzero columns of $\operatorname{diag}(I_r, 0)$ are $e_1, \ldots, e_r$, which are linearly independent and span its column space (definition 2.3.1), so $\operatorname{rank}\operatorname{diag}(I_r, 0) = r$ by theorem 2.3.10. Congruent matrices have equal rank (proposition 8.1.7), so $r = \operatorname{rank} B$.

If $\operatorname{rank} B = \operatorname{rank} B'$, both matrices are congruent to the same $\operatorname{diag}(I_r, 0)$, hence to each other by proposition 8.1.6. Conversely congruent matrices have equal rank, again by proposition 8.1.7, completing the proof.

A symmetric bilinear form over $\mathbb{C}$ has one invariant, its rank, and the two further integers over $\mathbb{R}$ come from the order alone: the proof above used square roots of arbitrary nonzero scalars exactly where the real argument used square roots of positive ones. The congruence here is by $P^\top$ rather than by P^*, the conjugate-transpose analogue being the Hermitian theory of chapter 10.

The two classifications can be restated as the computation of a group. Fix $k = \mathbb{R}$ or $k = \mathbb{C}$ and consider the nondegenerate symmetric matrices over k, of all sizes $n \geq 1$ at once (definition 8.1.13, proposition 8.1.14). For $B \in M_n(k)$ and $B' \in M_{n'}(k)$ the *orthogonal sum* is

$$B \perp B' = \begin{pmatrix} B & 0 \\ 0 & B' \end{pmatrix} \in M_{n+n'}(k)$$

the matrix of the form acting as φ_B on the first n coordinates, as $\varphi_{B'}$ on the last n', and pairing the two groups of coordinates to 0. The *hyperbolic plane* is $H = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, and rH denotes the orthogonal sum of r copies of it, with $B \perp 0H$ read as B. Declare

$$B \approx B' \quad \text{if and only if} \quad B \perp rH \text{ is congruent to } B' \perp sH \text{ for some integers } r, s \geq 0$$

Write $W(k)$ for the set of $\approx$-classes of nondegenerate symmetric matrices over k, and carry $\perp$ over to it by $[B] \perp [B'] = [B \perp B']$. The corollary below makes $W(k)$ the *Witt group* of k.

Corollary 8.3.6: The Witt Groups of $\mathbb{R}$ and $\mathbb{C}$

1. The assignment $[B] \mapsto \sigma(B)$ is a well-defined bijection $W(\mathbb{R}) \to \mathbb{Z}$ carrying $\perp$ to addition. In particular $\perp$ is well defined on classes and $W(\mathbb{R}) \cong \mathbb{Z}$.

2. The assignment $[B] \mapsto n \bmod 2$, for $B \in M_n(\mathbb{C})$, is a well-defined bijection $W(\mathbb{C}) \to \mathbb{Z}/2$ carrying $\perp$ to addition, so $W(\mathbb{C}) \cong \mathbb{Z}/2$.

***Proof*:**
Block-diagonal matrices multiply blockwise: for $i, j \leq n$ the entry of $(B \perp B')(C \perp C')$ in row i and column j is $\sum_l (B \perp B')_{il}(C \perp C')_{lj}$ by definition 1.4.1, and the terms with $l > n$ vanish because $(B \perp B')_{il} = 0$ there, leaving $(BC)_{ij}$. The same count in the other three blocks gives $B'C'$ in the lower right and 0 in the two off-diagonal blocks. Transposing exchanges rows with columns, so it transposes each of the two diagonal blocks and exchanges the two zero blocks (definition 2.3.4), giving $(B \perp B')^\top = B^\top \perp B'^\top$. Together these two facts give

$$(P \perp P')^\top (B \perp B')(P \perp P') = (P^\top BP) \perp (P'^\top B'P') \tag{8.7}$$

and they also show that $P \perp P'$ is invertible with inverse $P^{-1} \perp P'^{-1}$ when P and P' are, and that $B \perp B'$ is invertible with inverse $B^{-1} \perp B'^{-1}$, hence nondegenerate by proposition 8.1.14 when B and B' are. So $\perp$ takes nondegenerate matrices to nondegenerate ones, and by (8.7) it takes congruent pairs to congruent pairs.

A nondegenerate $B \in M_n(k)$ is invertible (proposition 8.1.14), so $Bx = 0$ forces $x = B^{-1}0 = 0$ and $\text{null}B = \{0\}$, whence $\text{rank}B = n$ by theorem 2.3.11. Over $\mathbb{R}$ this gives $n_0(B) = n - \text{rank}B = 0$ by corollary 8.3.3, so a nondegenerate real symmetric matrix has inertia of the shape $(p, m, 0)$ with $p + m = n$.

The relation $\approx$ is an equivalence relation: it is reflexive with $r = s = 0$ and symmetric by proposition 8.1.6, and if $B \perp rH$ is congruent to $B' \perp sH$ and $B' \perp s'H$ is congruent to $B'' \perp tH$, then appending $s'H$ to the first congruence and sH to the second, which (8.7) permits, makes $B \perp (r + s')H$ congruent to $B' \perp (s + s')H$ and $B' \perp (s + s')H$ congruent to $B'' \perp (t + s)H$. Proposition 8.1.6 chains the two.

1. Let B and B' be nondegenerate symmetric matrices over $\mathbb{R}$ with inertia $(p, m, 0)$ and $(p', m', 0)$. Choose invertible P and P' with $P^\top BP = \text{diag}(I_p, -I_m)$ and $P'^\top B'P' = \text{diag}(I_{p'}, -I_{m'})$. By (8.7) the matrix $B \perp B'$ is congruent to a diagonal matrix with $p + p'$ positive and $m + m'$ negative entries, so theorem 8.3.1(2) and corollary 8.3.3 give

$$n_+(B \perp B') = n_+(B) + n_+(B'), \qquad n_-(B \perp B') = n_-(B) + n_-(B')$$

and therefore $\sigma(B \perp B') = \sigma(B) + \sigma(B')$.

The matrix H has inertia $(1, 1, 0)$ and signature 0, so $\sigma(B \perp rH) = \sigma(B)$ for every $r \geq 0$. If $B \approx B'$, say $B \perp rH$ congruent to $B' \perp sH$, then corollary 8.3.3 equates the signatures of those two matrices and hence $\sigma(B) = \sigma(B')$: the assignment $[B] \mapsto \sigma(B)$ is well defined on classes.

It is surjective: $\sigma(I_j) = j$ and $\sigma(-I_j) = -j$ for $j \geq 1$, and $\sigma(H) = 0$. It is injective. Suppose $\sigma(B) = \sigma(B')$, that is $p - m = p' - m'$. Then $B \perp m'H$ has inertia $(p + m', m + m', 0)$ and $B' \perp mH$ has inertia $(p' + m, m' + m, 0)$ by the additivity just proved. The second entries agree, and the first entries agree because $p + m' = p' + m$ is a rearrangement of $p - m = p' - m'$. Each of the two matrices has size equal to the sum of the entries of its inertia, so the sizes agree as well. Then they are congruent by corollary 8.3.3, giving $B \approx B'$.

A bijection onto $\mathbb{Z}$ that is additive on representatives makes $\perp$ well defined on classes: if $B \approx C$ and $B' \approx C'$, then $\sigma(B \perp B') = \sigma(B) + \sigma(B') = \sigma(C) + \sigma(C') = \sigma(C \perp C')$, and injectivity turns this into $B \perp B' \approx C \perp C'$. The operation on $W(\mathbb{R})$ therefore corresponds under the bijection to addition on $\mathbb{Z}$, which makes $W(\mathbb{R})$ a group isomorphic to $\mathbb{Z}$.

2. Let $B \in M_n(\mathbb{C})$ be nondegenerate symmetric. Then $\operatorname{rank} B = n$ by the preliminary computation above, so B is congruent to I_n by corollary 8.3.5. The matrix H is nondegenerate over $\mathbb{C}$, being invertible, so $B \perp rH$ is a nondegenerate symmetric matrix of size $n + 2r$ and is congruent to I_{n+2r} for the same reason.

   Congruent matrices have the same size, so for $B \in M_n(\mathbb{C})$ and $B' \in M_{n'}(\mathbb{C})$ the relation $B \approx B'$ holds precisely when $n + 2r = n' + 2s$ for some $r, s \geq 0$: if the sizes agree then both matrices are congruent to the identity of that size and hence to each other (proposition 8.1.6), and conversely a congruence forces the sizes to agree. Such r and s exist exactly when $n \equiv n' \pmod 2$, so $[B] \mapsto n \bmod 2$ is well defined and injective. It is surjective, as I_1 and I_2 show, and additive, since $B \perp B'$ has size $n + n'$. As in (1), a bijection onto $\mathbb{Z}/2$ additive on representatives makes $\perp$ well defined on classes and identifies $W(\mathbb{C})$ with $\mathbb{Z}/2$, completing the proof.

The two computations say that the signature is the only invariant of a nondegenerate real form once hyperbolic summands are discarded, and that over $\mathbb{C}$ only the parity of the size is left by the same discarding. The two Witt computations need no input beyond corollary 8.3.3 and corollary 8.3.5. The same construction makes sense over any field in which 2 is invertible, and this book does not take it up: the two cases computed here are the field carrying an order and the field in which every element is a square.

Exercise 8.3.1

1. Compute the inertia and the signature of the quadratic forms of

$$B_1 = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, \qquad B_2 = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

   by completing the square as in theorem 8.2.3, exhibiting the diagonalized form in each case. For B_1, confirm the answer by computing the eigenvalues.

2. Let $q(x) = 2x_1x_2 + 2x_3x_4$ on $\mathbb{R}^4$ and let B be its matrix. Find the inertia and signature of B, and exhibit an invertible $P \in M_4(\mathbb{R})$ with $P^\top BP$ equal to the normal form of theorem 8.3.1(1).

3. Let $B \in M_n(\mathbb{R})$ be symmetric with $n_0(B) = 0$. Show that $\det B \neq 0$ and that $\det B$ has the sign of $(-1)^{n_-(B)}$. Deduce that congruent nondegenerate real symmetric matrices have determinants of the same sign, and that the sign of the determinant alone does not determine the inertia when $n = 2$.

4. For $n \geq 2$ let $B_n \in M_n(\mathbb{R})$ have every diagonal entry 0 and every off-diagonal entry 1. Show that B_n has eigenvalues $n - 1$, occurring once, and -1, occurring $n - 1$ times. Conclude that

the inertia of B_n is $(1, n-1, 0)$ and its signature is $2-n$, and check the case $n=3$ against example 8.3.4.

5. Exhibit a matrix $P \in M_2(\mathbb{C})$ with $P^\top \operatorname{diag}(1,-1)P = \operatorname{diag}(1,1)$, and explain why no such P exists in $M_2(\mathbb{R})$. Then give two symmetric matrices in $M_2(\mathbb{R})$ with the same rank and the same determinant that are not congruent, and name the invariant that separates them.

6. Let φ be a nondegenerate symmetric bilinear form on $\mathbb{R}^{2m}$ with signature 0 and let q be its quadratic form. Show that there is a subspace $U \subseteq \mathbb{R}^{2m}$ of dimension m with $q(u) = 0$ for every $u \in U$, and that no subspace of dimension greater than m has that property.

7. For $t \in \mathbb{R}$ let $q_t(x) = x_1^2 + 2tx_1x_2 + x_2^2$ on $\mathbb{R}^2$. Determine the inertia of q_t for every value of t, and say at which values of t it changes.

8. Let $B \in M_n(\mathbb{R})$ be nondegenerate and symmetric. Show that B is congruent to rH for some $r \geq 1$ if and only if $\sigma(B) = 0$, and that then $n = 2r$. Deduce from corollary 8.3.6 that the identity element of $W(\mathbb{R})$ is the class of H, and that $[B]$ is that element exactly when B is congruent to an orthogonal sum of hyperbolic planes.

8.4 Positive Definiteness: Eigenvalue, Minor, and Cholesky Criteria

Section 8.3 classified real symmetric forms by their inertia and left the extreme case unexamined: the case $n_+ = n$, in which the quadratic form is positive at every nonzero vector. This section decides that case from the matrix itself, without producing a diagonalization first.

Three tests do it, and they use three different parts of the book. The first reads the signs of the eigenvalues, and it is the criterion chapter 7 deferred to this section. The second evaluates n determinants of submatrices and uses no spectral theory at all. The third attempts a factorization $A = LL^\top$ with L lower triangular, and succeeds exactly when the matrix is positive definite. The section closes with the square root of a positive semidefinite matrix, an object chapter 11 needs.

Definition 8.4.1: Definite, Semidefinite and Indefinite Forms

Let φ be a symmetric bilinear form (definition 8.1.9) on a real vector space V with $\dim V = n \geq 1$, and let $q(x) = \varphi(x,x)$ be its quadratic form (definition 8.2.1). Then φ, and also q, is

1. *positive definite* if $q(x) > 0$ for every $x \neq 0$;
2. *positive semidefinite* if $q(x) \geq 0$ for every $x \in V$;
3. *negative definite* if $q(x) < 0$ for every $x \neq 0$, and *negative semidefinite* if $q(x) \leq 0$ for every $x \in V$;
4. *indefinite* if $q(x) > 0$ for some $x \in V$ and $q(y) < 0$ for some $y \in V$.

A symmetric matrix $A \in M_n(\mathbb{R})$ takes the same four names through the form $\varphi_A(x,y) = x^\top Ay$ on $\mathbb{R}^n$, whose quadratic form is $q_A(x) = x^\top Ax$. Thus A is positive definite when $x^\top Ax > 0$ for every nonzero $x \in \mathbb{R}^n$, and positive semidefinite when $x^\top Ax \geq 0$ for every $x \in \mathbb{R}^n$.

On $\mathbb{R}^2$ the cases are separated by completing the square, which is the computation of theorem 8.2.3 run in two variables. For $A = \left(\begin{smallmatrix}2&1\\1&2\end{smallmatrix}\right)$,

$$q_A(x) = 2x_1^2 + 2x_1x_2 + 2x_2^2 = 2\Big(x_1 + \tfrac{x_2}{2}\Big)^2 + \tfrac{3}{2}x_2^2$$

a sum of two squares with positive coefficients. It vanishes only when $x_2 = 0$ and $x_1 + x_2/2 = 0$, hence only at $x = 0$, so A is positive definite. For $A = \left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$ the same expansion gives $q_A(x) = x_1^2 + 2x_1x_2 + x_2^2 = (x_1 + x_2)^2$, which is never negative and vanishes on the whole line $x_2 = -x_1$: positive semidefinite, and not positive definite. For $A = \operatorname{diag}(1, -1)$ the value $q_A(e_1) = 1$ is positive and $q_A(e_2) = -1$ is negative, so A is indefinite.

Each verdict came from a diagonal form of q_A, and what a diagonal form records is the inertia. Theorem 8.3.1 makes the three counts (n_+, n_-, n_0) of definition 8.3.2 independent of which diagonalization produced them, so every condition in definition 8.4.1 must be visible in that triple. Over $\mathbb{R}$ one diagonalization is always at hand from the spectral theorem, and it places the eigenvalues on the diagonal.

Theorem 8.4.2: Criteria for Positive Definiteness

Let $A \in M_n(\mathbb{R})$ be symmetric and let $\lambda_1, \ldots, \lambda_n$ be the roots of its characteristic polynomial, each listed as often as its multiplicity as a root. The following are equivalent.

1. A is positive definite.
2. $\lambda_j > 0$ for every j.
3. $n_+(A) = n$.
4. $A = C^\top C$ for some invertible $C \in M_n(\mathbb{R})$.

***Proof*:**
All λ_j are real, so the inequalities in (2) are meaningful: by corollary 7.4.3 there are a real orthogonal $Q \in M_n(\mathbb{R})$, with columns $q_1, \ldots, q_n$, and a real diagonal $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ with $A = QDQ^\top$, and the closing clause of that corollary identifies the diagonal entries of D as the roots of the characteristic polynomial with their multiplicities and each q_j as an eigenvector of A for λ_j. This Q and this D are fixed for the whole proof, which runs (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) $\Rightarrow$ (1). That the eigenvalues of a symmetric matrix are real is proposition 7.1.11.

(1) $\Rightarrow$ (2). The columns of Q are an orthonormal basis of $\mathbb{R}^n$ (proposition 7.1.5), so $q_j \neq 0$ and $q_j^\top q_j = \langle q_j, q_j \rangle = 1$. Since $Aq_j = \lambda_j q_j$,

$$q_A(q_j) = q_j^\top A q_j = q_j^\top(\lambda_j q_j) = \lambda_j\, q_j^\top q_j = \lambda_j$$

and positive definiteness applied to the nonzero vector q_j gives $\lambda_j > 0$.

(2) $\Rightarrow$ (3). The columns of Q form a basis of $\mathbb{R}^n$, and (8.4) identifies $Q^\top AQ$ as the matrix of φ_A in that basis. Since $Q^\top Q = QQ^\top = I_n$ (proposition 7.1.5),

$$Q^\top AQ = Q^\top QDQ^\top Q = D$$

so the matrix of φ_A in the basis $q_1, \ldots, q_n$ is diagonal with n positive entries. Theorem 8.3.1(2) computes the number of positive entries in any such basis as $n_+(A)$, whence $n_+(A) = n$.

(3) ⇒ (4). The three counts of definition 8.3.2 are $(p, m, n-p-m)$ with $p+m \le n$, so $n_+(A) = n$ forces $n_-(A) = 0$ and $n_0(A) = 0$. By corollary 8.3.3 the matrix A is then congruent to $\operatorname{diag}(I_n, -I_0, 0) = I_n$: there is an invertible $P \in M_n(\mathbb{R})$ with $P^\top AP = I_n$. Multiplying on the left by $(P^\top)^{-1}$ and on the right by P^{-1} gives $A = (P^\top)^{-1}P^{-1}$, and $(P^\top)^{-1} = (P^{-1})^\top$ by proposition 2.3.5(4). So $A = C^\top C$ for $C = P^{-1}$, which is invertible.

(4) ⇒ (1). Let $x \in \mathbb{R}^n$ be nonzero. Then $Cx \neq 0$, since $Cx = 0$ would give $x = C^{-1}(Cx) = 0$. Using $(Cx)^\top = x^\top C^\top$ (proposition 2.3.5(3)) and evaluating the product by definition 1.4.1,

$$q_A(x) = x^\top C^\top Cx = (Cx)^\top (Cx) = \sum_{i=1}^{n} \big((Cx)_i\big)^2$$

At least one entry $(Cx)_i$ is nonzero, so the sum is positive, and A is positive definite, completing the proof.

Condition (2) is the test chapter 7 deferred: a symmetric matrix is positive definite exactly when its spectrum lies in the positive reals. Condition (4) says the same thing as congruence to the identity, and reading it through (8.4) gives the geometric statement. If $A = C^\top C$ with C invertible and $p_1, \dots, p_n$ are the columns of $P = C^{-1}$, then $P^\top AP = (C^{-1})^\top C^\top CC^{-1} = I_n$, so $\varphi_A(p_i, p_j)$ is 1 when $i = j$ and 0 otherwise. A positive definite symmetric form on $\mathbb{R}^n$ is therefore an inner product (definition 5.1.2, whose three conditions over $\mathbb{R}$ are symmetry, bilinearity and positivity), and every inner product on $\mathbb{R}^n$ becomes the standard one in a suitable basis. The other cases of definition 8.4.1 follow by the same theorem applied to $-A$, whose eigenvalues are $-\lambda_1, \dots, -\lambda_n$: A is negative definite exactly when every $\lambda_j < 0$, that is exactly when $n_-(A) = n$, and A is indefinite exactly when $n_+(A) \ge 1$ and $n_-(A) \ge 1$.

One consequence of the factorization $A = QDQ^\top$ is used repeatedly below. Applying theorem 3.2.10 twice and theorem 3.2.11 once, $\det A = (\det Q)^2 \det D$, while $\det(Q^\top Q) = \det I_n = 1$ gives $(\det Q)^2 = 1$, and $\det D = \lambda_1 \cdots \lambda_n$ by proposition 3.2.6. Hence

$$\det A = \lambda_1 \lambda_2 \cdots \lambda_n \tag{8.8}$$

for every symmetric $A \in M_n(\mathbb{R})$.

Weakening the strict inequality in (1) weakens each of the other three conditions, and the proof is the same argument with $>$ relaxed to $\ge$. The one point needing care is (4), where the factor C is no longer invertible.

Corollary 8.4.3: Criteria for Positive Semidefiniteness

Let $A \in M_n(\mathbb{R})$ be symmetric with eigenvalues $\lambda_1, \dots, \lambda_n$ as in theorem 8.4.2. The following are equivalent.

1. A is positive semidefinite.
2. $\lambda_j \geq 0$ for every j.
3. $n_-(A) = 0$.
4. $A = C^\top C$ for some $C \in M_n(\mathbb{R})$.

Moreover A is positive semidefinite without being positive definite exactly when every $\lambda_j \geq 0$ and $\lambda_j = 0$ for at least one j, and then $\det A = 0$.

***Proof*:**
Keep Q, D and the columns $q_1, \dots, q_n$ of the previous proof.

(1) ⇒ (2). The computation $q_A(q_j) = \lambda_j$ made there is unchanged, and $q_A \geq 0$ gives $\lambda_j \geq 0$.

(2) ⇒ (3). The matrix of φ_A in the basis $q_1, \dots, q_n$ is D, as computed there. No diagonal entry is negative, so theorem 8.3.1(2) gives $n_-(A) = 0$.

(3) ⇒ (4). Put $p = n_+(A)$, so that $n_-(A) = 0$ and $n_0(A) = n - p$. By corollary 8.3.3 the matrix A is congruent to $\operatorname{diag}(I_p, -I_0, 0) = E$, the diagonal matrix with p entries equal to 1 and $n - p$ entries equal to 0: there is an invertible P with $P^\top AP = E$, and as in the previous proof $A = (P^{-1})^\top EP^{-1}$. The matrix E satisfies $E^\top = E$ and $E^2 = E$, both by definition 1.4.1 on a diagonal matrix whose entries are 0 and 1. Put $C = EP^{-1}$. Then proposition 2.3.5(3) gives $C^\top C = (P^{-1})^\top E^\top EP^{-1} = (P^{-1})^\top EP^{-1} = A$.

(4) ⇒ (1). For every $x \in \mathbb{R}^n$ the computation of the previous proof gives $q_A(x) = \sum_i ((Cx)_i)^2 \geq 0$.

For the final clause, condition (2) here says that A is positive semidefinite exactly when every $\lambda_j \geq 0$, and condition (2) of theorem 8.4.2 says that A is positive definite exactly when every $\lambda_j > 0$. The two conditions differ exactly when some λ_j vanishes, and then $\det A = 0$ by (8.8), as we sought to show.

The four conditions of theorem 8.4.2 and the four of corollary 8.4.3 express every case of definition 8.4.1 through the counts of section 8.3: positive definite is $n_+ = n$, positive semidefinite is $n_- = 0$, negative definite is $n_- = n$, and indefinite is $n_+ \geq 1$ together with $n_- \geq 1$. In particular definiteness depends only on the congruence class, since congruent matrices have equal inertia (corollary 8.3.3), and this is the form in which the observation is used below: if P is invertible then A and $P^\top AP$ are positive definite together.

What none of these criteria yet gives is a test that can be run on the entries of A. Condition (2) needs the roots of a degree n polynomial, and conditions (3) and (4) need a diagonalization to be produced. The next criterion needs neither.

Theorem 8.4.4: Sylvester's Criterion: the Leading Principal Minors

Let $A = (a_{ij}) \in M_n(\mathbb{R})$ be symmetric. For $1 \le k \le n$ let

$$A_k = (a_{ij})_{1 \le i,j \le k} \in M_k(\mathbb{R})$$

be the *leading principal submatrix* of order k, the block of A in the first k rows and the first k columns, and put $\Delta_k = \det A_k$, the *leading principal minor* of order k. Then A is positive definite if and only if

$$\Delta_k > 0 \qquad \text{for every } k = 1, \dots, n$$

***Proof*:**
Each A_k is symmetric, since $a_{ji} = a_{ij}$ holds in particular for $i, j \le k$. Write $U_k = \operatorname{span}\{e_1, \dots, e_k\}$. For $c \in \mathbb{R}^k$ and $u = c_1e_1 + \cdots + c_ke_k \in U_k$, expanding $q_A(u) = u^\top Au$ by definition 1.4.1 and discarding the coordinates of u beyond the k-th, which are zero,

$$q_A(u) = \sum_{i=1}^{k}\sum_{j=1}^{k} c_ic_j\, a_{ij} = q_{A_k}(c) \tag{8.9}$$

The entry of A_k in row i and column j is $a_{ij} = \varphi_A(e_i, e_j)$, so A_k is the matrix of the restriction of φ_A to U_k relative to the basis $e_1, \dots, e_k$ (definition 8.1.2), and (8.9) is what theorem 8.1.3(1) says about that restriction.

The criterion is necessary. Let A be positive definite and let $c \in \mathbb{R}^k$ be nonzero. Then $u = c_1e_1 + \cdots + c_ke_k$ is nonzero, the list $e_1, \dots, e_k$ being linearly independent, so $q_{A_k}(c) = q_A(u) > 0$ by (8.9). Hence A_k is positive definite, and theorem 8.4.2(4) writes $A_k = C^\top C$ with $C \in M_k(\mathbb{R})$ invertible. Applying proposition 8.1.8 to I_k and C gives $\Delta_k = \det(C^\top I_k C) = (\det C)^2$, and $\det C \neq 0$ by corollary 3.2.8, so $\Delta_k > 0$.

The criterion is sufficient. The argument is an induction on n. For $n = 1$ the hypothesis is $\Delta_1 = a_{11} > 0$, and $q_A(x) = a_{11}x_1^2 > 0$ for every $x_1 \neq 0$.

Let $n \ge 2$, assume the statement for symmetric matrices of size $n-1$, and let $A \in M_n(\mathbb{R})$ be symmetric with $\Delta_1, \dots, \Delta_n$ all positive. For $k \le n-1$ the leading principal submatrix of order k of A_{n-1} is A_k, so the leading principal minors of A_{n-1} are $\Delta_1, \dots, \Delta_{n-1}$, all positive, and the inductive hypothesis makes A_{n-1} positive definite. Write

$$A = \begin{pmatrix} A_{n-1} & b \\ b^\top & c \end{pmatrix}, \qquad b \in \mathbb{R}^{n-1}, \quad c = a_{nn}$$

the lower left block being $b^\top$ because A is symmetric. Since $\Delta_{n-1} \neq 0$, the matrix A_{n-1} is invertible (corollary 3.2.8), and A_{n-1}^{-1} is symmetric because $(A_{n-1}^{-1})^\top = ((A_{n-1})^\top)^{-1} = A_{n-1}^{-1}$ (proposition 2.3.5(4)).

Put $w = A_{n-1}^{-1}b \in \mathbb{R}^{n-1}$ and

$$P = \begin{pmatrix} I_{n-1} & -w \\ 0 & 1 \end{pmatrix} \in M_n(\mathbb{R})$$

which is upper triangular with every diagonal entry equal to 1, so $\det P = 1$ by proposition 3.2.6 and P is invertible by corollary 3.2.8. Multiplying blockwise (definition 1.4.1) and using

$A_{n-1}w = b$,

$$AP = \begin{pmatrix} A_{n-1} & -A_{n-1}w + b \\ b^\top & -b^\top w + c \end{pmatrix} = \begin{pmatrix} A_{n-1} & 0 \\ b^\top & s \end{pmatrix}, \qquad s = c - b^\top w$$

Now $P^\top = \left(\begin{smallmatrix} I_{n-1} & 0 \\ -w^\top & 1 \end{smallmatrix}\right)$ by definition 2.3.4, and $w^\top A_{n-1} = (A_{n-1}^\top w)^\top = (A_{n-1}w)^\top = b^\top$ by proposition 2.3.5(3) and the symmetry of A_{n-1}, so the lower left block of $P^\top(AP)$ vanishes and

$$P^\top AP = \begin{pmatrix} A_{n-1} & 0 \\ 0 & s \end{pmatrix}$$

The number s is positive. Indeed proposition 8.1.8 gives $\det(P^\top AP) = (\det P)^2 \det A = \Delta_n$, while the last row of $P^\top AP$ is $(0, \dots, 0, s)$, so expanding along it with theorem 3.3.4 gives $\det(P^\top AP) = s \det A_{n-1} = s\Delta_{n-1}$. Hence $s = \Delta_n/\Delta_{n-1} > 0$.

Write $M = P^\top AP$ and let $y \in \mathbb{R}^n$ be nonzero, with first $n-1$ entries $u \in \mathbb{R}^{n-1}$ and last entry $t \in \mathbb{R}$. Multiplying blockwise again, My has first $n-1$ entries $A_{n-1}u$ and last entry st, so $q_M(y) = u^\top A_{n-1}u + st^2$. If $u \neq 0$ the first term is positive because A_{n-1} is positive definite, and the second is not negative because $s > 0$. If $u = 0$ then $t \neq 0$ and $q_M(y) = st^2 > 0$. Either way $q_M(y) > 0$, so M is positive definite. By (8.5), $q_A(Py) = q_M(y)$ for every y, and every nonzero $x \in \mathbb{R}^n$ is Py for the nonzero vector $y = P^{-1}x$. Hence $q_A(x) > 0$ for every nonzero x, the inductive step holds, and the criterion is sufficient for every n, completing the proof.

The test is n determinants of matrices whose entries are the entries of A, with no root of a polynomial extracted anywhere. The proof also produced the quantity that makes the induction work, the number $s = a_{nn} - b^\top A_{n-1}^{-1} b$, together with the identity $\Delta_n = s\Delta_{n-1}$: the last minor is the previous one multiplied by what is left of the corner entry after the rest of the matrix has been subtracted off.

Both the statement and its proof used the strict inequality, and the semidefinite analogue obtained by relaxing $\Delta_k > 0$ to $\Delta_k \geq 0$ is false. Only one half of it remains true under the relaxation.

Proposition 8.4.5: The Minor Criterion and Semidefiniteness

Let $A = (a_{ij}) \in M_n(\mathbb{R})$ be symmetric. For a nonempty subset $S = \{i_1 < \cdots < i_k\}$ of $\{1, \dots, n\}$ write $A_S = (a_{i_r i_t}) \in M_k(\mathbb{R})$ for the submatrix of A in the rows and columns indexed by S, so that $\det A_S$ is a *principal minor* of A and $\det A_{\{1,\dots,k\}} = \Delta_k$.

1. If A is positive semidefinite then $\det A_S \geq 0$ for every such S. In particular $\Delta_k \geq 0$ for every k.
2. The converse fails for the leading principal minors alone. The matrix $A = \operatorname{diag}(0, -1)$ has $\Delta_1 = \Delta_2 = 0$, so both leading principal minors are nonnegative, and A is not positive semidefinite.
3. A is positive semidefinite if and only if $A + \varepsilon I_n$ is positive definite for every real $\varepsilon > 0$.

***Proof*:**

1. Let $S = \{i_1 < \cdots < i_k\}$ and let $c \in \mathbb{R}^k$. Put $u = c_1 e_{i_1} + \cdots + c_k e_{i_k} \in \mathbb{R}^n$. Expanding $q_A(u) = u^\top Au$ by definition 1.4.1 and discarding the coordinates of u outside S, which are

zero,

$$q_A(u) = \sum_{r=1}^{k}\sum_{t=1}^{k} c_r c_t \, a_{i_r i_t} = q_{A_S}(c)$$

The matrix A_S is symmetric, and $q_{A_S}(c) = q_A(u) \geq 0$ for every c, so A_S is positive semidefinite. By corollary 8.4.3(4) there is $C \in M_k(\mathbb{R})$ with $A_S = C^\top C$. That C need not be invertible, so proposition 8.1.8 does not apply; instead theorem 3.2.10 and theorem 3.2.11 give $\det A_S = \det(C^\top)\det C = (\det C)^2 \geq 0$. Taking $S = \{1, \ldots, k\}$ gives $\Delta_k \geq 0$.

2. For $A = \operatorname{diag}(0, -1)$ one has $A_1 = (0)$ and $A_2 = A$, so $\Delta_1 = 0$ and $\Delta_2 = 0 \cdot (-1) = 0$ (proposition 3.2.6). But $q_A(e_2) = -1 < 0$, so A is not positive semidefinite (definition 8.4.1). Its inertia is $(0, 1, 1)$, since A is already diagonal with one negative and one zero entry (theorem 8.3.1(2)).

3. Let $A = QDQ^\top$ be as in theorem 8.4.2, with Q orthogonal and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. For $\varepsilon > 0$ the matrix $A + \varepsilon I_n$ is symmetric, and $Q(\varepsilon I_n)Q^\top = \varepsilon QQ^\top = \varepsilon I_n$ (proposition 7.1.5) gives

$$A + \varepsilon I_n = Q\big(D + \varepsilon I_n\big)Q^\top = Q \operatorname{diag}(\lambda_1 + \varepsilon, \ldots, \lambda_n + \varepsilon)\, Q^\top$$

so the closing clause of corollary 7.4.3 identifies $\lambda_1 + \varepsilon, \ldots, \lambda_n + \varepsilon$ as the eigenvalues of $A + \varepsilon I_n$. If A is positive semidefinite then $\lambda_j \geq 0$ for every j by corollary 8.4.3(2), hence $\lambda_j + \varepsilon > 0$ for every $\varepsilon > 0$, and theorem 8.4.2(2) makes $A + \varepsilon I_n$ positive definite. If A is not positive semidefinite then $\lambda_i < 0$ for some i, again by corollary 8.4.3(2); taking $\varepsilon = -\lambda_i/2 > 0$ gives $\lambda_i + \varepsilon = \lambda_i/2 < 0$, so $A + \varepsilon I_n$ fails theorem 8.4.2(2) and is not positive definite, completing the proof.

Part (2) is the warning to carry away: the leading principal minors decide definiteness and do not decide semidefiniteness. The reason the induction of theorem 8.4.4 breaks is visible in $\operatorname{diag}(0, -1)$ at the first step, where $\Delta_1 = 0$ leaves no positive entry to divide by, and thereafter the vanishing of Δ_2 carries no information about the sign of a_{22}. Part (3) restores a usable test by adding a parameter: a symmetric A is positive semidefinite exactly when the n leading principal minors of $A + \varepsilon I_n$, each of them a polynomial in ε, are positive for every $\varepsilon > 0$.

Example 8.4.6: A Positive Semidefinite Matrix That Is Not Definite

Let

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \in M_3(\mathbb{R}), \qquad q_A(x) = 2x_1^2 + x_2^2 + x_3^2 + 2x_1x_2 + 2x_1x_3$$

the second display collecting the diagonal entries as coefficients of squares and twice each entry above the diagonal as the coefficient of a cross term. Grouping the two cross terms with the squares they came from,

$$q_A(x) = (x_1 + x_2)^2 + (x_1 + x_3)^2$$

which expands back to the displayed polynomial. So $q_A(x) \geq 0$ for every x, and A is positive semidefinite. It is not positive definite: the value is 0 exactly when $x_2 = -x_1$ and $x_3 = -x_1$, that is on the line spanned by $(1, -1, -1)^\top$, and that vector is not 0.

The three criteria of corollary 8.4.3 agree. Expanding along the first row with theorem 3.3.4,

$$\det(tI_3 - A) = (t-2)(t-1)^2 - (t-1) - (t-1) = (t-1)\big((t-2)(t-1) - 2\big) = t(t-1)(t-3)$$

so the eigenvalues are 0, 1 and 3, all nonnegative and one of them zero, which is the final clause of corollary 8.4.3. The inertia is therefore $(2, 0, 1)$ by (8.6), so $n_-(A) = 0$ and $\operatorname{rank} A = 2$. The two squares give a factorization $A = C^\top C$ directly: taking

$$C = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \qquad (Cx)_1 = x_1 + x_2, \quad (Cx)_2 = x_1 + x_3, \quad (Cx)_3 = 0$$

gives $q_A(x) = (Cx)_1^2 + (Cx)_2^2 + (Cx)_3^2$, and C is not invertible, as theorem 8.4.2(4) requires of a matrix that is not positive definite.

The leading principal minors are $\Delta_1 = 2$, $\Delta_2 = 2 - 1 = 1$ and $\Delta_3 = \det A = 0 \cdot 1 \cdot 3 = 0$ by (8.8). The first two are positive and only the last vanishes, and that pattern is not the inconclusive one: with $\Delta_1, \ldots, \Delta_{n-1}$ all positive, theorem 8.4.4 makes the leading A_{n-1} positive definite, the reduction in its proof gives $P^\top AP = \operatorname{diag}(A_{n-1}, \Delta_n/\Delta_{n-1})$, and $\Delta_n = 0$ leaves a positive semidefinite matrix. The failure recorded in proposition 8.4.5(2) needs a minor to vanish *earlier*, as in $\operatorname{diag}(0, -1)$, where Δ_1 already vanishes and the later entries are unconstrained.

The sum of two squares in the example was found by inspection. Producing such a decomposition in general is what Lagrange's method does (theorem 8.2.3), and when the matrix is positive definite that method has a normalized output: every coefficient can be absorbed into its square, and the change of coordinates that does it is triangular. Recording the coefficients gives a factorization of A itself.

Theorem 8.4.7: The Cholesky Factorization

Let $A \in M_n(\mathbb{R})$ be symmetric. Then A is positive definite if and only if

$$A = LL^\top$$

for some lower triangular $L \in M_n(\mathbb{R})$ whose diagonal entries are all positive. Such an L is unique.

***Proof*:**

Step 1: a factorization forces positive definiteness. Suppose $A = LL^\top$ with L lower triangular and $\ell_{11}, \ldots, \ell_{nn}$ all positive. Then L is invertible by proposition 1.5.2(2), hence so is $L^\top$, with $(L^\top)^{-1} = (L^{-1})^\top$ (proposition 2.3.5(4)). Put $C = L^\top$. Then $C^\top C = (L^\top)^\top L^\top = LL^\top = A$ by proposition 2.3.5(1), so theorem 8.4.2(4) makes A positive definite.

Step 2: existence, by induction on n. For $n = 1$ the entry $a_{11} = q_A(e_1)$ is positive, and $L = (\sqrt{a_{11}})$ serves.

Let $n \geq 2$, assume existence for positive definite symmetric matrices of size $n - 1$, and let $A \in M_n(\mathbb{R})$ be symmetric and positive definite. Again $a_{11} = q_A(e_1) > 0$. Write

$$A = \begin{pmatrix} a_{11} & b^\top \\ b & A' \end{pmatrix}, \qquad b \in \mathbb{R}^{n-1}, \quad A' \in M_{n-1}(\mathbb{R})$$

with A' symmetric and the upper right block equal to $b^\top$ by symmetry of A. Put $\ell = \sqrt{a_{11}} > 0$, $u = b/\ell \in \mathbb{R}^{n-1}$ and $S = A' - uu^\top \in M_{n-1}(\mathbb{R})$, which is symmetric because A' is and

$(uu^\top)^\top = uu^\top$ (proposition 2.3.5(1),(3)). Setting

$$R = \begin{pmatrix} \ell & u^\top \\ 0 & I_{n-1} \end{pmatrix}, \qquad R^\top = \begin{pmatrix} \ell & 0 \\ u & I_{n-1} \end{pmatrix}$$

and multiplying blockwise (definition 1.4.1),

$$R^\top \begin{pmatrix} 1 & 0 \\ 0 & S \end{pmatrix} R = R^\top \begin{pmatrix} \ell & u^\top \\ 0 & S \end{pmatrix} = \begin{pmatrix} \ell^2 & \ell u^\top \\ \ell u & uu^\top + S \end{pmatrix} = \begin{pmatrix} a_{11} & b^\top \\ b & A' \end{pmatrix} = A$$

using $\ell u = b$ and $uu^\top + S = A'$.

The matrix R is upper triangular with diagonal entries $\ell, 1, \ldots, 1$, all nonzero, so it is invertible (proposition 1.5.2(2)). Multiplying the identity just proved on the left by $(R^{-1})^\top = (R^\top)^{-1}$ (proposition 2.3.5(4)) and on the right by R^{-1} gives $\left(\begin{smallmatrix} 1 & 0 \\ 0 & S \end{smallmatrix}\right) = (R^{-1})^\top A R^{-1}$, so this matrix is congruent to A (definition 8.1.5) and is therefore positive definite by theorem 8.4.2(3) and corollary 8.3.3. Testing it on the vectors with first entry 0: for nonzero $v \in \mathbb{R}^{n-1}$ the vector $y \in \mathbb{R}^n$ with entries $(0, v)$ is nonzero and blockwise multiplication gives $y^\top \left(\begin{smallmatrix} 1 & 0 \\ 0 & S \end{smallmatrix}\right) y = v^\top S v$, which is therefore positive. So S is positive definite, and the inductive hypothesis gives a lower triangular $L' \in M_{n-1}(\mathbb{R})$ with positive diagonal entries and $S = L'(L')^\top$. Put

$$L = \begin{pmatrix} \ell & 0 \\ u & L' \end{pmatrix}, \qquad L^\top = \begin{pmatrix} \ell & u^\top \\ 0 & (L')^\top \end{pmatrix}$$

the second by definition 2.3.4. Then L is lower triangular with diagonal entries $\ell, \ell'_{11}, \ldots, \ell'_{n-1,n-1}$, all positive, and blockwise multiplication gives

$$LL^\top = \begin{pmatrix} \ell^2 & \ell u^\top \\ \ell u & uu^\top + L'(L')^\top \end{pmatrix} = \begin{pmatrix} a_{11} & b^\top \\ b & uu^\top + S \end{pmatrix} = A$$

which completes the induction.

Step 3: uniqueness. Let L_1 and L_2 both be lower triangular with positive diagonal entries and $L_1 L_1^\top = A = L_2 L_2^\top$, and write $\ell_{ii}^{(r)}$ for the i-th diagonal entry of L_r. Both matrices are invertible (proposition 1.5.2(2)), hence so are their transposes (proposition 2.3.5(4)). Multiplying $L_1 L_1^\top = L_2 L_2^\top$ on the left by L_2^{-1} and on the right by $(L_1^\top)^{-1}$ gives

$$L_2^{-1} L_1 = L_2^\top (L_1^\top)^{-1}$$

The left side is lower triangular: L_2^{-1} is lower triangular by proposition 1.5.2(2) and the product of two lower triangular matrices is lower triangular by proposition 1.5.2(1). The right side is upper triangular: $L_2^\top$ is upper triangular by definition 2.3.4, $(L_1^\top)^{-1} = (L_1^{-1})^\top$ (proposition 2.3.5(4)) is the transpose of a lower triangular matrix and hence upper triangular, and the product of two upper triangular matrices is upper triangular by proposition 1.5.2(1). A matrix that is both is diagonal (proposition 1.5.2(3)), so $M := L_2^{-1} L_1$ is diagonal, and by the diagonal-entry clauses of proposition 1.5.2(1) and (2) its i-th diagonal entry is $d_i = (\ell_{ii}^{(2)})^{-1} \ell_{ii}^{(1)} > 0$.

From $L_1 = L_2 M$ and $M^\top = M$ it follows that

$$L_2 L_2^\top = A = L_1 L_1^\top = L_2 M M^\top L_2^\top = L_2 M^2 L_2^\top$$

using proposition 2.3.5(3) for $(L_2 M)^\top = M^\top L_2^\top$. Multiplying on the left by L_2^{-1} and on the right by $(L_2^\top)^{-1}$ leaves $M^2 = I_n$, so $d_i^2 = 1$ for each i (definition 1.4.1), and $d_i > 0$ forces $d_i = 1$. Hence $M = I_n$ and $L_1 = L_2 M = L_2$, as we sought to show.

The existence proof is an algorithm, and it is the one the name refers to. Take the square root of the corner entry, divide the rest of the first column by it, subtract $uu^\top$ from the remaining block, and repeat on that block. Stage k takes one square root and divides a column of length $n-k$ by it, so a complete run costs n square roots and $n(n-1)/2$ divisions. Running it on a symmetric matrix is itself the test: the only extraction performed at each stage is the square root of the current corner entry, so the algorithm stops exactly when that entry fails to be positive, and Step 1 shows that a completed run certifies positive definiteness. This is the test used in practice, and the effect of inexact arithmetic on it belongs to **??**.

The Cholesky factor also computes the minors of theorem 8.4.4. For $i, j \le k$ the entry $(LL^\top)_{ij} = \sum_s \ell_{is}\ell_{js}$ has no term with $s > k$, since $\ell_{is} = 0$ for $s > i$, so the leading principal submatrix of order k of $A = LL^\top$ is $L_k L_k^\top$, where L_k is the leading principal submatrix of L. Then theorem 3.2.10, theorem 3.2.11 and proposition 3.2.6 give

$$\Delta_k = \det\left(L_k L_k^\top\right) = (\det L_k)^2 = \left(\ell_{11}\ell_{22}\cdots\ell_{kk}\right)^2 \tag{8.10}$$

so the positivity of the n leading principal minors and the existence of the n diagonal entries of L are the same statement, read forwards and backwards.

Example 8.4.8: One Matrix, Three Criteria

Let

$$A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \in M_3(\mathbb{R})$$

The eigenvalue criterion. Write $a = t-2$, so that $tI_3 - A$ is the matrix with a on the diagonal and -1 everywhere else. Expanding along its first row with theorem 3.3.4, the three minors are a^2-1, then $\det\left(\begin{smallmatrix} -1 & -1 \\ -1 & a \end{smallmatrix}\right) = -a-1$, then $\det\left(\begin{smallmatrix} -1 & a \\ -1 & -1 \end{smallmatrix}\right) = 1+a$, so

$$\det(tI_3 - A) = a(a^2-1) - (-1)(-a-1) + (-1)(1+a) = a^3 - 3a - 2$$

and $a^3 - 3a - 2 = (a+1)^2(a-2)$, as multiplying out confirms. In terms of t this is $(t-1)^2(t-4)$, so the eigenvalues are 1, 1 and 4. All are positive, so A is positive definite by theorem 8.4.2(2), and $\det A = 1 \cdot 1 \cdot 4 = 4$ by (8.8).

The minor criterion. The leading principal minors are $\Delta_1 = 2$, then $\Delta_2 = 2\cdot 2 - 1 = 3$, and $\Delta_3 = \det A = 4$, which theorem 3.3.4 confirms directly: $2(4-1) - 1(2-1) + 1(1-2) = 6 - 1 - 1 = 4$. All three are positive, so theorem 8.4.4 reaches the same verdict without any eigenvalue.

The Cholesky criterion. Run the algorithm of theorem 8.4.7. The corner entry is 2, so $\ell = \sqrt{2}$, and $b = (1,1)^\top$ gives $u = (1/\sqrt{2})(1,1)^\top$ and

$$S = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 3/2 & 1/2 \\ 1/2 & 3/2 \end{pmatrix}$$

Repeating on S: the corner entry is $3/2$, so the next diagonal entry is $\sqrt{3/2}$, the entry below it is $(1/2)/\sqrt{3/2} = 1/\sqrt{6}$, and what remains is $3/2 - 1/6 = 4/3$, whose square root is $2/\sqrt{3}$. Hence

$$L = \begin{pmatrix} \sqrt{2} & 0 & 0 \\ 1/\sqrt{2} & \sqrt{3/2} & 0 \\ 1/\sqrt{2} & 1/\sqrt{6} & 2/\sqrt{3} \end{pmatrix}$$

and multiplying out confirms $LL^\top = A$: the diagonal entries of $LL^\top$ are 2, $\frac{1}{2} + \frac{3}{2} = 2$ and $\frac{1}{2} + \frac{1}{6} + \frac{4}{3} = 2$, and the entries below the diagonal are $\sqrt{2}/\sqrt{2} = 1$, $\sqrt{2}/\sqrt{2} = 1$ and $\frac{1}{2} + \sqrt{3/2}/\sqrt{6} =$

$\frac{1}{2}+\frac{1}{2}=1$. The three criteria agree, and (8.10) ties two of them together: $\ell_{11}^2 = 2 = \Delta_1$, then $(\ell_{11}\ell_{22})^2 = 2\cdot\frac{3}{2} = 3 = \Delta_2$, then $(\ell_{11}\ell_{22}\ell_{33})^2 = 3\cdot\frac{4}{3} = 4 = \Delta_3$.

The Cholesky factor is a square root of A in the weak sense that it recovers A from L and $L^\top$, and it is triangular rather than symmetric. A different square root is available for every positive semidefinite matrix, and it is symmetric, positive semidefinite and unique. It is the object chapter 11 uses to define the singular values of an arbitrary matrix.

Theorem 8.4.9: The Positive Semidefinite Square Root

Let $A \in M_n(\mathbb{R})$ be symmetric and positive semidefinite. There is exactly one symmetric positive semidefinite $S \in M_n(\mathbb{R})$ with
$$S^2 = A$$
It is written $A^{1/2}$, and it is positive definite if and only if A is.

***Proof*:**
Existence. By corollary 7.4.3 there are a real orthogonal Q and $D = \operatorname{diag}(\lambda_1,\dots,\lambda_n)$ carrying the eigenvalues of A with $A = QDQ^\top$, and corollary 8.4.3(2) gives $\lambda_j \ge 0$ for every j, so the real numbers $\sqrt{\lambda_j}$ exist. Put $D^{1/2} = \operatorname{diag}(\sqrt{\lambda_1},\dots,\sqrt{\lambda_n})$ and $S = QD^{1/2}Q^\top$. It is symmetric, since $S^\top = (Q^\top)^\top(D^{1/2})^\top Q^\top = QD^{1/2}Q^\top$ by proposition 2.3.5(1),(3). Using $Q^\top Q = I_n$ (proposition 7.1.5) and $D^{1/2}D^{1/2} = D$ (definition 1.4.1 on diagonal matrices),
$$S^2 = QD^{1/2}Q^\top QD^{1/2}Q^\top = QD^{1/2}D^{1/2}Q^\top = QDQ^\top = A$$
The factorization $S = QD^{1/2}Q^\top$ has the shape of corollary 7.4.3(3), so its closing clause makes $\sqrt{\lambda_1},\dots,\sqrt{\lambda_n}$ the eigenvalues of S. They are all nonnegative, so S is positive semidefinite by corollary 8.4.3(2).

Uniqueness. Let $T \in M_n(\mathbb{R})$ be symmetric and positive semidefinite with $T^2 = A$. By corollary 7.4.3 there is an orthonormal basis $u_1,\dots,u_n$ of $\mathbb{R}^n$ with $Tu_i = \mu_i u_i$, and $\mu_i \ge 0$ for every i by corollary 8.4.3(2). Let $v \in \mathbb{R}^n$ satisfy $Av = \lambda v$ for a scalar λ, which is one of the eigenvalues of A when $v \ne 0$ and hence satisfies $\lambda \ge 0$. Write $v = c_1u_1 + \cdots + c_nu_n$. Applying T twice,
$$Av = T^2v = \sum_{i=1}^n c_i\mu_i^2 u_i, \qquad \lambda v = \sum_{i=1}^n \lambda c_i u_i$$
and the two are equal, so the coefficients agree in the basis $u_1,\dots,u_n$: for each i one has $c_i\mu_i^2 = \lambda c_i$. Whenever $c_i \ne 0$ this gives $\mu_i^2 = \lambda$, and since $\mu_i \ge 0$ and $\lambda \ge 0$ it gives $\mu_i = \sqrt{\lambda}$. Therefore
$$Tv = \sum_{i=1}^n c_i\mu_i u_i = \sqrt{\lambda}\sum_{i=1}^n c_i u_i = \sqrt{\lambda}\,v$$
the terms with $c_i = 0$ contributing nothing to either side.

The paragraph just finished used only that T is symmetric, positive semidefinite and squares to A, so it applies to S as well: for every eigenvector v of A with eigenvalue λ one has $Tv = \sqrt{\lambda}\,v = Sv$. Corollary 7.4.3 gives a basis $w_1,\dots,w_n$ of $\mathbb{R}^n$ consisting of eigenvectors of A, so $Tw_j = Sw_j$ for every j; writing an arbitrary $x \in \mathbb{R}^n$ as $x = \sum_j a_jw_j$ gives $Tx = Sx$, and taking $x = e_j$ for each j identifies the columns of T with those of S (definition 1.4.1). Hence $T = S$.

> For the last clause, the eigenvalues of S are the numbers $\sqrt{\lambda_j}$, and $\sqrt{\lambda_j} > 0$ holds for every j exactly when $\lambda_j > 0$ holds for every j. By theorem 8.4.2(2), applied once to S and once to A, the matrix S is positive definite exactly when A is, completing the proof.

The uniqueness clause is what makes the notation $A^{1/2}$ well defined, and positive semidefiniteness is the condition that selects one root from several. A symmetric matrix generally has many symmetric square roots: for $A = QDQ^\top$ the matrix $Q\operatorname{diag}(\pm\sqrt{\lambda_1}, \ldots, \pm\sqrt{\lambda_n})Q^\top$ squares to A for each of the 2^n sign patterns, and all of them are symmetric by the computation in the proof. Exactly one has all its eigenvalues nonnegative. Chapter 11 applies the theorem to $A = N^\top N$ for an arbitrary real $N \in M_{m\times n}(\mathbb{R})$, which is positive semidefinite because $x^\top N^\top N x = \sum_i \big((Nx)_i\big)^2 \geq 0$, and defines the singular values of N as the eigenvalues of $A^{1/2}$. The complex analogue of this section, with A^* in place of $A^\top$ and Hermitian forms in place of symmetric ones, belongs to chapter 10.

Exercise 8.4.1

1. Let
$$A = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 10 & 4 \\ 2 & 4 & 6 \end{pmatrix} \in M_3(\mathbb{R})$$
Compute the three leading principal minors of A and decide by theorem 8.4.4 whether A is positive definite. Then run the algorithm in the proof of theorem 8.4.7 to produce L with $A = LL^\top$, and check the result against (8.10).

2. For $t \in \mathbb{R}$ let
$$A_t = \begin{pmatrix} 1 & t & 0 \\ t & 1 & t \\ 0 & t & 1 \end{pmatrix} \in M_3(\mathbb{R})$$
Determine for which t the matrix A_t is positive definite. For the smallest positive t at which positive definiteness fails, write q_{A_t} as a sum of two squares, and exhibit a nonzero $x \in \mathbb{R}^3$ with $q_{A_t}(x) = 0$.

3. Let $A = (a_{ij}) \in M_n(\mathbb{R})$ be symmetric and positive semidefinite. Show that $a_{ii} \geq 0$ for every i, and that if $a_{ii} = 0$ then $a_{ij} = 0$ for every j. (For the second part, evaluate q_A at $te_i + e_j$ and choose t.) Then give a symmetric $A \in M_2(\mathbb{R})$ with both diagonal entries positive that is not positive semidefinite, and compute its inertia.

4. Let $A \in M_n(\mathbb{R})$ be symmetric and positive definite. Show that there are a unique lower triangular $L \in M_n(\mathbb{R})$ with every diagonal entry equal to 1 and a unique diagonal $D = \operatorname{diag}(d_1, \ldots, d_n)$ with every $d_i > 0$ such that $A = LDL^\top$, and that $d_k = \Delta_k/\Delta_{k-1}$ for $k = 1, \ldots, n$, where $\Delta_0 = 1$. (Use theorem 8.4.7 and (8.10).)

5. Compute $A^{1/2}$ for $A = \left(\begin{smallmatrix} 5 & 4 \\ 4 & 5 \end{smallmatrix}\right)$. Then exhibit a symmetric $T \in M_2(\mathbb{R})$ with $T^2 = A$ and $T \neq \pm A^{1/2}$, and say which hypothesis of theorem 8.4.9 it fails.

6. Let $A, B \in M_n(\mathbb{R})$ be symmetric and positive definite and let $c > 0$. Show that $A + B$ and cA are positive definite, that A is invertible, and that A^{-1} is symmetric and positive definite. Show also that $P^\top AP$ is positive definite for every invertible $P \in M_n(\mathbb{R})$, and give an example showing that AB need not be symmetric.

7. Let $v_1, \dots, v_n \in \mathbb{R}^m$ and let $G \in M_n(\mathbb{R})$ have entries $g_{ij} = \langle v_i, v_j \rangle$ for the standard inner product on $\mathbb{R}^m$. Show that G is symmetric and positive semidefinite, and that G is positive definite if and only if $v_1, \dots, v_n$ are linearly independent. Compute G and its inertia for $v_1 = (1,0)^\top$, $v_2 = (1,1)^\top$, $v_3 = (0,1)^\top$ in $\mathbb{R}^2$.

8. Let $A \in M_n(\mathbb{R})$ be symmetric and positive definite, written in the block form $A = \begin{pmatrix} A_{n-1} & b \\ b^\top & a_{nn} \end{pmatrix}$ of the proof of theorem 8.4.4, and put $s = a_{nn} - b^\top A_{n-1}^{-1} b$, so that $\det A = s \det A_{n-1}$. Using that A_{n-1}^{-1} is positive definite (exercise 8.4.1 (6)), show $s \le a_{nn}$, with equality if and only if $b = 0$. Deduce by induction on n that

$$\det A \le a_{11} a_{22} \cdots a_{nn}$$

with equality if and only if A is diagonal.

8.5 Application: the Second-Derivative Test and the Morse Index

A real-valued function of n variables has, at a point where its first derivatives vanish, a second-order term that is a quadratic form in the displacement. Section 8.4 decides when such a form takes only positive values, and section 8.3 attaches to it three integers that no change of coordinates disturbs. This section applies both. The definiteness of the second-order form separates a local minimum from a local maximum from a point that is neither, and the count n_- is the invariant of the critical point that survives every change of variables.

One analytic result is used, Taylor's theorem with second-order remainder, which the front matter declares among the results assumed from other volumes. It is quoted once, in (8.11) below, and everything after that is chapter 8 applied to the matrix it produces. In one variable the test is the familiar one: at a point where $f'(a) = 0$ the sign of $f''(a)$ decides between a minimum and a maximum, and $f''(a) = 0$ decides nothing. In n variables the single number $f''(a)$ becomes a symmetric matrix, so there is no one sign to consult, and what replaces the sign is the inertia of definition 8.3.2.

Throughout this section U is a nonempty open subset of $\mathbb{R}^n$, meaning that for each $a \in U$ there is $\delta > 0$ with $\{x \in \mathbb{R}^n \,\big|\, \|x - a\| < \delta\} \subseteq U$, and $f \colon U \to \mathbb{R}$ is a function whose second partial derivatives exist and are continuous on U. Norms are those induced by the standard inner product on $\mathbb{R}^n$ (definition 5.2.1), so $\|x\|^2 = x^\top x$.

Definition 8.5.1: Critical Points, Local Extrema and Saddle Points

Let $a \in U$ and write

$$\nabla f(a) = \Big(\frac{\partial f}{\partial x_1}(a), \dots, \frac{\partial f}{\partial x_n}(a)\Big)^\top \in \mathbb{R}^n$$

for the column of first partial derivatives of f at a. The point a is a *critical point* of f if $\nabla f(a) = 0$.
The function f has a *local minimum* at a if there is $\delta > 0$ such that $f(a+h) \geq f(a)$ for every $h \in \mathbb{R}^n$ with $\|h\| < \delta$, and a *strict local minimum* at a if there is $\delta > 0$ such that $f(a+h) > f(a)$ for every h with $0 < \|h\| < \delta$. Reversing both inequalities defines a *local maximum* and a *strict local maximum* at a. A critical point that is neither a local maximum nor a local minimum is a *saddle point* of f.

The condition $\nabla f(a) = 0$ says nothing about which of the three behaviours occurs, and the object that does say is the array of second derivatives.

Definition 8.5.2: Hessian Matrix

Let $a \in U$. The *Hessian matrix* of f at a is the matrix of second partial derivatives

$$H_f(a) = \Big(\frac{\partial^2 f}{\partial x_i \partial x_j}(a)\Big)_{1 \leq i,j \leq n} \in M_n(\mathbb{R})$$

It is symmetric,[a] so it is the matrix of a symmetric bilinear form on $\mathbb{R}^n$ (definition 8.1.9, theorem 8.1.3) and $h \mapsto h^\top H_f(a) h$ is the associated quadratic form (definition 8.2.1).

[a] Symmetry of $H_f(a)$ is the equality of the mixed second partial derivatives, which holds for a function whose second partials are continuous and is proved in analysis. Nothing below depends on importing it: the Hessian enters only through the quadratic form $h \mapsto h^\top H_f(a) h$, and $h^\top C h = h^\top C^\top h$ for every $C \in M_n(\mathbb{R})$ because a 1×1 matrix equals its own transpose, so a reader who prefers not to import the equality may read $H_f(a)$ as its symmetric part (proposition 8.1.10), which produces the same quadratic form and is the only symmetric matrix that does (theorem 8.2.2(2)).

For $f(x,y) = x^3 - 3xy + y^3$ on $U = \mathbb{R}^2$ the first partial derivatives are $\partial f/\partial x = 3x^2 - 3y$ and $\partial f/\partial y = -3x + 3y^2$. Differentiating each of these in each variable gives $\partial^2 f/\partial x^2 = 6x$, $\partial^2 f/\partial x\, \partial y = -3$, $\partial^2 f/\partial y\, \partial x = -3$ and $\partial^2 f/\partial y^2 = 6y$, so

$$\nabla f(x,y) = \begin{pmatrix} 3x^2 - 3y \\ -3x + 3y^2 \end{pmatrix}, \qquad H_f(x,y) = \begin{pmatrix} 6x & -3 \\ -3 & 6y \end{pmatrix}$$

The gradient vanishes at $(0,0)$ and at $(1,1)$, which are therefore critical points of f (definition 8.5.1), and example 8.5.5 shows they are the only ones. The Hessian varies from point to point. At the origin its quadratic form is $h \mapsto -6h_1h_2$, which takes the value -6 at $h = (1,1)^\top$ and the value 6 at $h = (1,-1)^\top$; at $(1,1)$ it is $h \mapsto 6h_1^2 - 6h_1h_2 + 6h_2^2$, which is positive away from 0. Those two sign patterns are what theorem 8.5.4 converts into statements about f itself.

The conversion runs through one expansion, which is the section's only import from analysis. Let $a \in U$, put $H = H_f(a)$, and let $\delta_0 > 0$ be such that $a + h \in U$ whenever $\|h\| < \delta_0$, which openness of U gives. For such h define

$$r(h) = f(a+h) - f(a) - \nabla f(a)^\top h - \tfrac{1}{2} h^\top H h \tag{8.11}$$

where $\nabla f(a)^\top h = \sum_i (\partial f/\partial x_i)(a)h_i$. Taylor's theorem with second-order remainder, declared in the front matter among the results assumed from other volumes, applies to f because its second partial derivatives are continuous on U, and it says that this r satisfies $r(h)/\|h\|^2 \to 0$ as $h \to 0$. Written out, for every $\varepsilon > 0$ there is $\delta \in (0, \delta_0]$ with

$$|r(h)| \le \varepsilon\|h\|^2 \qquad \text{whenever } 0 < \|h\| < \delta \tag{8.12}$$

Nothing else is taken from analysis, and every statement below is a statement about the matrix H.

The first use of (8.11) is to show that only critical points can be local extrema, which is why the rest of the section assumes $\nabla f(a) = 0$.

Proposition 8.5.3: A Local Extremum Is a Critical Point

Let f be twice continuously differentiable on an open set $U \subseteq \mathbb{R}^n$ and let $a \in U$. If $\nabla f(a) \ne 0$, then a is neither a local maximum nor a local minimum of f. Equivalently, every local extremum of f in U is a critical point (definition 8.5.1).

***Proof*:**
Suppose $u = \nabla f(a) \ne 0$ and apply (8.12) with $\varepsilon = 1$, giving a radius δ. For $0 < t < \delta/\|u\|$ the vectors $\pm tu$ have norm $t\|u\| < \delta$, and $u^\top(\pm tu) = \pm t\|u\|^2$ while $(\pm tu)^\top H(\pm tu) = t^2u^\top Hu$, so (8.11) gives

$$\left|f(a \pm tu) - f(a) \mp t\|u\|^2\right| \le \tfrac{t^2}{2}\left|u^\top Hu\right| + t^2\|u\|^2 = Ct^2, \qquad C = \tfrac{1}{2}\left|u^\top Hu\right| + \|u\|^2$$

Here $C > 0$ because $u \ne 0$. Restricting further to $t < \|u\|^2/(2C)$ makes $Ct^2 < \frac{1}{2}t\|u\|^2$, whence $f(a + tu) - f(a) > \frac{1}{2}t\|u\|^2 > 0$ and $f(a - tu) - f(a) < -\frac{1}{2}t\|u\|^2 < 0$. Both points $a \pm tu$ lie within $t\|u\|$ of a, and t may be taken as small as one likes, so every neighbourhood of a carries values of f above $f(a)$ and values below it. The point a is then neither a local maximum nor a local minimum, which is the contrapositive of the second statement.

At a critical point the linear term disappears and the quadratic term is what remains. It scales like t^2 along a ray $h = tw$, and by (8.12) so does a bound on the remainder, with a constant that can be made as small as required. The sign of $f(a + h) - f(a)$ is therefore the sign of $h^\top Hh$ as soon as that form is bounded away from 0 by a positive multiple of $\|h\|^2$, and positive definiteness gives exactly such a bound through the eigenvalues of H.

Theorem 8.5.4: The Second-Derivative Test

Let $a \in U$ be a critical point of f and let $H = H_f(a)$.

1. If H is positive definite (definition 8.4.1), then f has a strict local minimum at a.
2. If H is negative definite, then f has a strict local maximum at a.
3. If H is indefinite, then a is a saddle point of f.
4. Conversely, if f has a local minimum at a then H is positive semidefinite, and if f has a local maximum at a then H is negative semidefinite.

***Proof*:**
Since a is a critical point, $\nabla f(a)^\top h = 0$ for every h, so (8.11) reads

$$f(a+h) = f(a) + \tfrac{1}{2}h^\top Hh + r(h) \qquad (\|h\| < \delta_0) \tag{8.13}$$

with r subject to (8.12).

Two bounds on the quadratic form of H are used in the first two parts, and both come from the spectral theorem. The matrix H is real symmetric, so corollary 7.4.3 gives a real orthogonal $Q \in M_n(\mathbb{R})$ and a diagonal $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ with $H = QDQ^\top$, where $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of H listed with multiplicity and each is real (proposition 7.1.11). For $h \in \mathbb{R}^n$ put $y = Q^\top h$. Then corollary 7.4.6 gives $h^\top Hh = \sum_j \lambda_j y_j^2$, while $y^\top = h^\top Q$ (proposition 2.3.5) gives $\|y\|^2 = y^\top y = h^\top QQ^\top h = h^\top h = \|h\|^2$, the middle step because the columns of Q are an orthonormal basis of $\mathbb{R}^n$ (corollary 7.4.3), so that proposition 7.1.5 and its closing clause give $Q^\top Q = QQ^\top = I_n$. Write $m = \min_j \lambda_j$ and $M = \max_j \lambda_j$. Bounding every coefficient in the sum below by m and above by M, and using $\sum_j y_j^2 = \|y\|^2 = \|h\|^2$,

$$m\|h\|^2 \le h^\top Hh \le M\|h\|^2 \qquad (h \in \mathbb{R}^n) \tag{8.14}$$

1. Let H be positive definite. It is real symmetric, so theorem 8.4.2 applies and every eigenvalue of H is positive, giving $m > 0$. Apply (8.12) with $\varepsilon = m/4$ and let δ be the resulting radius. For $0 < \|h\| < \delta$, combining (8.13), (8.14) and (8.12),

   $$f(a+h) - f(a) = \tfrac{1}{2}h^\top Hh + r(h) \ge \tfrac{m}{2}\|h\|^2 - \tfrac{m}{4}\|h\|^2 = \tfrac{m}{4}\|h\|^2 > 0$$

   So $f(a+h) > f(a)$ whenever $0 < \|h\| < \delta$, which is a strict local minimum at a (definition 8.5.1).

2. Let H be negative definite, so $h^\top Hh < 0$ for every $h \neq 0$ (definition 8.4.1). Then $h^\top(-H)h = -h^\top Hh > 0$ for every $h \neq 0$, so $-H$ is positive definite, and it is real symmetric. Now $-H = Q(-D)Q^\top$ with $-D = \operatorname{diag}(-\lambda_1, \ldots, -\lambda_n)$, so the closing clause of corollary 7.4.3 identifies $-\lambda_1, \ldots, -\lambda_n$ as the eigenvalues of $-H$. By theorem 8.4.2 each of them is positive, so every λ_j is negative and $M < 0$. Apply (8.12) with $\varepsilon = -M/4$, a positive number, and let δ be the resulting radius. For $0 < \|h\| < \delta$ the upper bound in (8.14) gives

   $$f(a+h) - f(a) = \tfrac{1}{2}h^\top Hh + r(h) \le \tfrac{M}{2}\|h\|^2 - \tfrac{M}{4}\|h\|^2 = \tfrac{M}{4}\|h\|^2 < 0$$

   so f has a strict local maximum at a.

3. Let H be indefinite, so that its quadratic form takes both a positive and a negative value (definition 8.4.1): there are $u, v \in \mathbb{R}^n$ with $\alpha = u^\top Hu > 0$ and $\beta = v^\top Hv < 0$. Both are nonzero vectors, since $0^\top H0 = 0$. Apply (8.12) with $\varepsilon = \alpha/(4\|u\|^2)$ and let δ be the resulting radius. For $0 < t < \delta/\|u\|$ the vector $h = tu$ satisfies $0 < \|h\| = t\|u\| < \delta$, and $h^\top Hh = t^2\alpha$ while $\|h\|^2 = t^2\|u\|^2$, so (8.13) gives

   $$f(a+tu) - f(a) \ge \frac{t^2\alpha}{2} - \frac{\alpha}{4\|u\|^2}t^2\|u\|^2 = \frac{t^2\alpha}{4} > 0$$

   The same computation with v in place of u and $\varepsilon = -\beta/(4\|v\|^2)$ gives $f(a+tv) - f(a) \le t^2\beta/4 < 0$ for all small $t > 0$. Since $\|tu\| = t\|u\|$ and $\|tv\| = t\|v\|$ can be made as small as required, every neighbourhood of a contains a point where f exceeds $f(a)$ and a point where f is below $f(a)$. So a is neither a local maximum nor a local minimum, which makes it a saddle point (definition 8.5.1).

4. Let f have a local minimum at a, with $\delta_1 > 0$ such that $f(a+h) \geq f(a)$ whenever $\|h\| < \delta_1$. Fix $h \neq 0$ and let $\varepsilon > 0$. Choose δ as in (8.12) for this ε, shrink it so that $\delta \leq \delta_1$, and take any t with $0 < t < \delta/\|h\|$. Then $\|th\| < \delta$, so (8.13), the local-minimum inequality and (8.12) give

$$0 \leq f(a+th) - f(a) = \frac{t^2}{2} h^\top H h + r(th) \leq \frac{t^2}{2} h^\top H h + \varepsilon t^2 \|h\|^2$$

Dividing by $t^2 > 0$ and rearranging yields $h^\top H h \geq -2\varepsilon \|h\|^2$. The vector h is fixed and $\varepsilon > 0$ was arbitrary, so $h^\top H h \geq 0$. This holds for every $h \neq 0$, and $0^\top H 0 = 0$, so H is positive semidefinite (definition 8.4.1). If instead f has a local maximum at a, the same argument with both inequalities reversed gives $h^\top H h \leq 0$ for every h, which is negative semidefiniteness, completing the proof.

Parts (1) to (3) are implications and not equivalences, and part (4) locates the gap between them. A definite Hessian settles the question. A singular Hessian that is semidefinite without being definite is compatible with a strict minimum, with a minimum that is not strict, and with neither, as example 8.5.6 shows. In the language of definition 8.3.2 the three hypotheses of the test are $n_+ = n$, then $n_- = n$, then $n_+ > 0$ together with $n_- > 0$, and what they leave uncovered is $n_0 > 0$ with one of n_+, n_- equal to 0.

Example 8.5.5: The Critical Points of $x^3 - 3xy + y^3$

Let $f(x, y) = x^3 - 3xy + y^3$ on $U = \mathbb{R}^2$, whose gradient and Hessian were computed above. A point is critical when $3x^2 - 3y = 0$ and $-3x + 3y^2 = 0$, that is when $y = x^2$ and $x = y^2$. Substituting the first equation into the second gives $x = x^4$, so $x(x^3 - 1) = 0$ and $x \in \{0, 1\}$, since 1 is the only real cube root of 1. The critical points are therefore $(0, 0)$, with $y = 0^2 = 0$, and $(1, 1)$, with $y = 1^2 = 1$, and there are no others.

At the origin,

$$H_f(0,0) = \begin{pmatrix} 0 & -3 \\ -3 & 0 \end{pmatrix}, \qquad \det\big(\lambda I_2 - H_f(0,0)\big) = \lambda^2 - 9$$

so the eigenvalues are 3 and -3 (theorem 4.1.10). The quadratic form is $h \mapsto -6h_1h_2$, which takes the value -6 at $(1,1)^\top$ and the value 6 at $(1,-1)^\top$, so it is indefinite and theorem 8.5.4(3) makes $(0,0)$ a saddle point. Its eigenvalues are ± 3, so its inertia is $(n_+, n_-, n_0) = (1, 1, 0)$ by (8.6), and $\det H_f(0,0) = -9 \neq 0$.

At $(1, 1)$,

$$H_f(1,1) = \begin{pmatrix} 6 & -3 \\ -3 & 6 \end{pmatrix}, \qquad \det\big(\lambda I_2 - H_f(1,1)\big) = (\lambda - 6)^2 - 9 = (\lambda - 3)(\lambda - 9)$$

so the eigenvalues are 3 and 9, both positive, and theorem 8.4.2 makes the matrix positive definite. Its two leading principal minors are 6 and $36 - 9 = 27$, again both positive, which is the same verdict reached through theorem 8.4.4. By theorem 8.5.4(1) the point $(1, 1)$ is a strict local minimum, with value $f(1,1) = 1 - 3 + 1 = -1$; the matrix being positive definite, its inertia is $(2, 0, 0)$ by theorem 8.4.2(3).

The minimum is local and not global: $f(t,t) = 2t^3 - 3t^2$ tends to $-\infty$ as $t \to -\infty$, so f takes values far below -1. The test claims nothing else, since each of its hypotheses is a statement about H at the one point a.

The test is silent when H is singular without being definite, and the silence is forced rather than an artifact of the proof. Three functions can share a Hessian and behave differently at the point in question, so no criterion consulting that matrix alone could separate them.

Example 8.5.6: Three Functions with Hessian $\operatorname{diag}(0,2)$

On $U = \mathbb{R}^2$ consider

$$f_1(x,y) = x^4 + y^2, \qquad f_2(x,y) = -x^4 + y^2, \qquad f_3(x,y) = y^2$$

Their gradients are $(4x^3, 2y)^\top$, $(-4x^3, 2y)^\top$ and $(0, 2y)^\top$, each vanishing at the origin, so the origin is a critical point of all three (definition 8.5.1). Their Hessians are

$$\begin{pmatrix} 12x^2 & 0 \\ 0 & 2 \end{pmatrix}, \qquad \begin{pmatrix} -12x^2 & 0 \\ 0 & 2 \end{pmatrix}, \qquad \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}$$

and all three equal $A = \operatorname{diag}(0,2)$ at the origin. The associated quadratic form is $h \mapsto 2h_2^2$. It is never negative, so A is neither indefinite nor negative definite, and it vanishes at $h = (1,0)^\top \neq 0$, so A is not positive definite either (definition 8.4.1). No part of theorem 8.5.4 applies to any of the three functions. The matrix is diagonal, hence the matrix of its own form in the standard basis (theorem 8.1.3), so theorem 8.3.1(2) counts the entries on that diagonal and gives inertia $(n_+, n_-, n_0) = (1,0,1)$. The form is positive semidefinite and A is singular.

The behaviour at the origin differs in each case, and each case is settled by inspection.

1. $f_1(h) = h_1^4 + h_2^2 > 0 = f_1(0)$ for every $h \neq 0$, because a sum of two nonnegative numbers vanishes only when both do. The origin is a strict local minimum, and in fact a global one.
2. $f_2(t,0) = -t^4 < 0$ and $f_2(0,t) = t^2 > 0$ for every $t \neq 0$, so every neighbourhood of the origin carries values of f_2 on both sides of $f_2(0) = 0$. The origin is a saddle point.
3. $f_3(h) = h_2^2 \geq 0 = f_3(0)$ for every h, so the origin is a local minimum, and $f_3(t,0) = 0$ for every t shows that it is not a strict one.

Part (4) of the test is consistent with all three, since the common Hessian is positive semidefinite, as it must be at the two minima. Semidefiniteness by itself forbids nothing, which is what f_2 shows.

In each example the classification came down to the two counts n_+ and n_-. Positive definiteness is $n_+ = n$, negative definiteness is $n_- = n$, and an invertible indefinite Hessian is any of the cases strictly between. Once H is invertible the single count n_- therefore carries the classification, and at a critical point of a function it is given its own name.

Definition 8.5.7: Nondegenerate Critical Point and Morse Index

Let $a \in U$ be a critical point of f and let $H = H_f(a)$. The point a is a *nondegenerate* critical point of f if H is invertible, equivalently if the symmetric bilinear form $(u, v) \mapsto u^\top Hv$ is nondegenerate (definition 8.1.13, proposition 8.1.14). The *Morse index* of f at a is the index of H in the sense of definition 8.3.2,

$$\operatorname{ind}_a(f) = n_-(H)$$

the number of negative entries in any diagonal matrix congruent to H.

Nondegeneracy is also the condition $n_0(H) = 0$: section 8.3 recorded, just after corollary 8.3.3, that $n_0(B)$ is the nullity of B, so it vanishes exactly when B is invertible.

The Morse index has a description that mentions no diagonalization at all. Apply theorem 8.3.1(2) to the form $\varphi(u, v) = u^\top Hv$ on $\mathbb{R}^n$: the integer $n_-(H)$ is the largest dimension of a subspace $W \subseteq \mathbb{R}^n$ with $h^\top Hh < 0$ for every nonzero $h \in W$. Since $\frac{1}{2}h^\top Hh$ is the second-order term of f at a in the direction h, the index counts the independent directions in which f falls away from a to second order. Index 0 at a nondegenerate critical point is the case where no such direction exists, which theorem 8.5.4(1) turns into a strict local minimum; index n is the case where every direction is one, giving a strict local maximum; each value in between is a saddle carrying that many independent descending directions.

For the index to be an invariant of the critical point rather than of the coordinates in which f happens to be written, two things need checking. The Hessian must be determined by the values of f near a, rather than only by the recipe of differentiating twice in the given coordinates. And a change of coordinates must change H by a congruence, which theorem 8.3.1 then absorbs. Both hold, and the first is what makes the second available without differentiating anything again.

Proposition 8.5.8: The Second-Order Matrix Under a Change of Coordinates

Let $a \in U$ be a critical point of f and let $H = H_f(a)$.

1. H is the only symmetric matrix $S \in M_n(\mathbb{R})$ for which

$$f(a + h) = f(a) + \tfrac{1}{2}h^\top Sh + \rho(h) \qquad \text{with } \rho(h)/\|h\|^2 \to 0 \text{ as } h \to 0$$

2. Let $P \in M_n(\mathbb{R})$ be invertible and put $g(u) = f(a + Pu)$, defined for those $u \in \mathbb{R}^n$ with $a + Pu \in U$, a set containing every u with $\|u\| < \delta_1$ for some $\delta_1 > 0$. Then

$$g(u) = g(0) + \tfrac{1}{2}u^\top\big(P^\top HP\big)u + \rho(u) \qquad \text{with } \rho(u)/\|u\|^2 \to 0 \text{ as } u \to 0$$

and $P^\top HP$ is symmetric with the same inertia as H. In particular $n_-(P^\top HP) = n_-(H)$.

***Proof*:**
Throughout, r is the remainder (8.11) at the point a and δ_0 is the radius fixed there. Since $\nabla f(a) = 0$, that definition of r rearranges to

$$f(a + h) = f(a) + \tfrac{1}{2}h^\top Hh + r(h) \qquad (\|h\| < \delta_0) \tag{8.15}$$

and r satisfies (8.12).

1. The matrix H is such an S, by (8.15) with $\rho = r$. Let S be symmetric and satisfy the stated expansion with remainder ρ, and put $D = S - H$, symmetric because both S and H are. Subtracting the two expansions of $f(a+h)$ gives
$$\tfrac{1}{2}h^\top Dh = r(h) - \rho(h) \qquad (\|h\| < \delta_0)$$
Fix $h \neq 0$ and let $\varepsilon > 0$. Both r and ρ tend to 0 faster than $\|h\|^2$, so there is $\delta \in (0, \delta_0]$ with $|r(v)| \leq \varepsilon\|v\|^2$ and $|\rho(v)| \leq \varepsilon\|v\|^2$ for every v with $0 < \|v\| < \delta$. Take any t with $0 < t < \delta/\|h\|$ and evaluate the display at th, where $(th)^\top D(th) = t^2h^\top Dh$ and $\|th\|^2 = t^2\|h\|^2$:
$$\frac{t^2}{2}\left|h^\top Dh\right| = \left|r(th) - \rho(th)\right| \leq 2\varepsilon t^2\|h\|^2$$
Multiplying by $2/t^2$ gives $\left|h^\top Dh\right| \leq 4\varepsilon\|h\|^2$, and $\varepsilon > 0$ was arbitrary, so $h^\top Dh = 0$. This holds for every $h \neq 0$, and $0^\top D0 = 0$, so the symmetric matrices D and 0 have the same quadratic form. Theorem 8.2.2(2) makes them equal, so $S = H$.

2. The matrix $P^\top P$ is symmetric, since $(P^\top P)^\top = P^\top P$ by proposition 2.3.5. Let $P^\top P = Q'D'Q'^\top$ with Q' real orthogonal and $D' = \operatorname{diag}(\mu_1, \ldots, \mu_n)$ carrying the eigenvalues of $P^\top P$ (corollary 7.4.3), and put $\mu = \max_j \mu_j$. For $u \in \mathbb{R}^n$ set $w = Q'^\top u$. As in the proof of theorem 8.5.4, corollary 7.4.6 gives $u^\top P^\top Pu = \sum_j \mu_j w_j^2$ and proposition 7.1.5 gives $\|w\|^2 = \|u\|^2$, so replacing each μ_j by μ yields
$$\|Pu\|^2 = (Pu)^\top(Pu) = u^\top P^\top Pu \leq \mu\|u\|^2 \qquad (u \in \mathbb{R}^n)$$
where $(Pu)^\top = u^\top P^\top$ by proposition 2.3.5. Here $\mu > 0$: any $u \neq 0$ has $Pu \neq 0$ because P is invertible, so $0 < \|Pu\|^2 \leq \mu\|u\|^2$. Put $\delta_1 = \delta_0/\sqrt{\mu}$. For $\|u\| < \delta_1$ the bound gives $\|Pu\| \leq \sqrt{\mu}\|u\| < \delta_0$, so $a + Pu \in U$ and $g(u)$ is defined, which is the claim about the ball. For such u, substituting $h = Pu$ in (8.15) and using $(Pu)^\top H(Pu) = u^\top(P^\top HP)u$ (proposition 2.3.5, proposition 1.4.3) gives
$$g(u) = f(a+Pu) = f(a) + \tfrac{1}{2}u^\top\left(P^\top HP\right)u + r(Pu)$$
and $g(0) = f(a)$, so the stated expansion holds with $\rho(u) = r(Pu)$. For the remainder, let $\varepsilon > 0$ and choose $\delta \in (0, \delta_0]$ with $|r(v)| \leq (\varepsilon/\mu)\|v\|^2$ for $0 < \|v\| < \delta$, which (8.12) permits. If $0 < \|u\| < \delta/\sqrt{\mu}$ then $v = Pu$ is nonzero and has $\|v\| \leq \sqrt{\mu}\|u\| < \delta$, so
$$|\rho(u)| = |r(Pu)| \leq \frac{\varepsilon}{\mu}\|Pu\|^2 \leq \frac{\varepsilon}{\mu}\,\mu\|u\|^2 = \varepsilon\|u\|^2$$
which is $\rho(u)/\|u\|^2 \to 0$.

   Finally $(P^\top HP)^\top = P^\top H^\top P = P^\top HP$ (proposition 2.3.5), so $P^\top HP$ is symmetric, and it is congruent to H (definition 8.1.5) because P is invertible. Corollary 8.3.3 gives congruent real symmetric matrices the same inertia, so $\left(n_+(P^\top HP), n_-(P^\top HP), n_0(P^\top HP)\right)$ equals $\left(n_+(H), n_-(H), n_0(H)\right)$, as we sought to show.

Part (1) characterizes the Hessian by a property of f near a, so any procedure producing a symmetric matrix with that expansion produces H. Part (2) is the consequence: replacing the coordinates x

near a by coordinates u with $x = a + Pu$ replaces the matrix of the second-order term by $P^\top HP$ and leaves the inertia, the signature and the index unchanged. The two parts combine when g is itself twice continuously differentiable, as it is here, being f composed with an affine map: part (1) applied to g at 0 then identifies $H_g(0)$ with $P^\top HP$, so the Morse index computed from g agrees with the one computed from f. That identification is the only place the composition's smoothness is used. The proof obtains the second-order behaviour of g from that of f without differentiating g at all, which is what keeps the statement inside this section's single import from analysis. That the Morse index is well defined is therefore a consequence of theorem 8.3.1 and of nothing about calculus: the law of inertia is what makes the count a property of the critical point rather than of the coordinates.

A theorem of differential topology, the Morse lemma, extends the invariance from the second-order term to the function. It states that near a nondegenerate critical point a of Morse index k there are coordinates $u_1, \ldots, u_n$ centred at a, not linear in general, in which

$$f = f(a) - u_1^2 - \cdots - u_k^2 + u_{k+1}^2 + \cdots + u_n^2$$

holds exactly, with no remainder term. This is the normal form of corollary 8.3.3 promoted from the Hessian to f itself, and it says that the index is the only invariant a nondegenerate critical point has: two of them with the same index become the same function after a change of coordinates. The proof is not given here. It manufactures those coordinates with the inverse function theorem, an analytic tool this book does not develop, and it belongs with the differential topology built on it ([15, chapter 10.1]).

Exercise 8.5.1

1. Let $f(x, y) = x^4 + y^4 - 4xy$ on $\mathbb{R}^2$. Find every critical point of f, compute the Hessian at each one, classify each point by theorem 8.5.4, and give its Morse index.

2. Let $f(x, y) = x^3 - 3xy^2$ on $\mathbb{R}^2$. Show that the origin is the only critical point of f and that $H_f(0, 0) = 0$, so that no part of theorem 8.5.4 applies there. Then determine directly which kind of critical point the origin is, and give the inertia of $H_f(0, 0)$.

3. Let $f(x, y) = (x-y)^2 + x^4$ and $g(x, y) = (x-y)^2 - x^4$ on $\mathbb{R}^2$. Show that the origin is a critical point of both and that $H_f(0, 0) = H_g(0, 0)$, and compute the inertia of that matrix. Deduce that theorem 8.5.4 classifies neither point. Then show directly that the origin is a strict local minimum of f and a saddle point of g.

4. For $c \in \mathbb{R}$ let $f_c(x, y, z) = x^2 + y^2 + z^2 + 2c(xy + yz + zx)$ on $\mathbb{R}^3$. Show that the origin is a critical point of f_c for every c and compute $H_{f_c}(0)$. Using theorem 8.4.4, determine exactly which c make theorem 8.5.4(1) apply. For $c = 1$ compute the inertia of the Hessian and settle the behaviour of f_1 at the origin directly.

5. Let a be a nondegenerate critical point of $f\colon U \to \mathbb{R}$ with Morse index k, and use that the second partial derivatives of $-f$ are the negatives of those of f, so that $H_{-f}(a) = -H_f(a)$. Show that $n_-(-H) = n_+(H)$ for every symmetric $H \in M_n(\mathbb{R})$, deduce that a is a nondegenerate critical point of $-f$ of Morse index $n-k$, and conclude that f has a strict local maximum at a if and only if $k = n$.

6. A function f on an open subset of $\mathbb{R}^3$ has a critical point at a with

$$H_f(a) = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 1 & 2 \\ 0 & 2 & 1 \end{pmatrix}$$

Diagonalize the associated quadratic form by the method of theorem 8.2.3, that is by completing squares, and use the result to give the inertia of $H_f(a)$, the Morse index of f at a, and which kind of critical point a is. Check the answer against $\det H_f(a)$, using proposition 8.1.8.

7. Let $A \in M_n(\mathbb{R})$ be symmetric and let $q(x) = x^\top Ax$ on $\mathbb{R}^n$. Show that 0 is a critical point of q and that the symmetric matrix given by proposition 8.5.8(1) at 0 is $2A$. Then show that q has a strict local minimum at 0 if and only if A is positive definite, and that in that case $q(x) > q(0)$ for every $x \neq 0$.

9

Complexification, Real Forms, and Complex Structures

Every chapter so far has fixed the scalars at the start and worked over whichever of $\mathbb{R}$ and $\mathbb{C}$ the statement needed, occasionally noting that a result held over both. This chapter's focus is on what happens when changing scalars: to a real vector space when its scalars are changed to complex numbers, and conversely how complex space can be studied through real scalars. The chapter shall also study how to enlarge a real vector space to make it a complex vector space whose complex-action reduces to the original real action.

The question that organizes them is one of identifying the real space in the complex one. In particular, having passed to the complex side, one wants to know which objects there came from the real side. What identifies them is an extra piece of structure, a *conjugation*, and the objects defined over $\mathbb{R}$ are exactly those that are invariant to conjugation. The same criterion appears for subspaces, for operators, and for matrices, which is why it is worth isolating once.

Section 9.1 builds the complex space $V_{\mathbb{C}}$ from a real V and counts what the construction costs, finding that passing to $\mathbb{C}$ and back does not return the space one started with. Section 9.2 introduces the conjugation and shows that choosing one is the same as choosing a copy of a real space inside a complex one; there are many such copies, and none is distinguished. Section 9.3 turns the question around, asking what a real space needs in order to admit complex scalars directly, and finds a single operator J with $J^2 = -\mathrm{id}$; complexifying it splits $V_{\mathbb{C}}$ into two halves that the conjugation exchanges. That splitting is the piece the rest of the series uses. Section 9.4 does for operators what section 9.2 did for vectors, and settles when a complex operator is defined over $\mathbb{R}$.

9.1 Complexification and Realification

Over $\mathbb{C}$ every square matrix has an eigenvalue (theorem 4.1.14), and over $\mathbb{R}$ many have none: the characteristic polynomial (definition 4.1.9) of $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ is t^2+1, whose roots are $\pm i$. The standard response to a real matrix with no real eigenvalue is to read its entries as complex numbers and to let it act on $\mathbb{C}^n$ instead of on $\mathbb{R}^n$. That response is available because $\mathbb{R}^n$ sits inside $\mathbb{C}^n$ as the set of columns with real entries, and because every column of $\mathbb{C}^n$ is $x+iy$ for exactly one pair of real columns x and y.

An abstract real vector space has no natural ambient complex space to be embed into. This section builds one, by copying what $\mathbb{C}^n$ does to $\mathbb{R}^n$. In $\mathbb{C}^n$, write $z = x+iy$ with $x, y \in \mathbb{R}^n$ and let $a+bi$ be a scalar with $a, b \in \mathbb{R}$. Expanding and using $i^2 = -1$,

$$(a+bi)(x+iy) = ax + iay + ibx + i^2by = (ax - by) + i(bx + ay)$$

The two columns on the right, $ax - by$ and $bx + ay$, are assembled from x and y by real scalar multiplication and addition alone. So the pair (x, y) determines the pair $(ax - by, bx + ay)$ with no reference to multiplication by a complex number, and the same formula can therefore be imposed on a real vector space where no such multiplication is given:

Definition 9.1.1: Complexification of a Real Vector Space

Let V be a vector space over $\mathbb{R}$. The *complexification* of V is the set

$$V_{\mathbb{C}} = V \times V$$

of ordered pairs of vectors of V, whose element (u, v) is written $u + iv$, equipped with the addition

$$(u + iv) + (u' + iv') = (u + u') + i(v + v')$$

and with the multiplication by a scalar $a + bi$, where $a, b \in \mathbb{R}$, given by

$$(a + bi)(u + iv) = (au - bv) + i(bu + av) \tag{9.1}$$

The vectors u and v are the *components* of $u + iv$. Two elements of $V_{\mathbb{C}}$ are equal exactly when their first and second components agree, so each element of $V_{\mathbb{C}}$ is $u + iv$ for exactly one pair $u, v \in V$.

Three consequences of (9.1) fix the notation. First, taking $b = 0$ gives $a(u + iv) = au + i(av)$ for $a \in \mathbb{R}$, so a real scalar acts in each component separately. Taking $a = 0$ and $b = 1$ gives

$$i(v + i0) = (0v - 1 \cdot 0) + i(1v + 0 \cdot 0) = 0 + iv$$

And the rule $u \mapsto u + i0$ sends distinct vectors of V to distinct elements of $V_{\mathbb{C}}$, because the components of an element are unique. That rule is used to identify V with the subset $\{\, u + i0 \,|\, u \in V \,\}$ of $V_{\mathbb{C}}$ throughout the chapter, and under the identification the expression $u + iv$ is the sum of u with the product of the scalar i and the vector v, both computed in $V_{\mathbb{C}}$:

$$(u + i0) + i(v + i0) = (u + i0) + (0 + iv) = u + iv$$

So the symbol i in $u+iv$, introduced as a separator between two components, is the complex number i.[1]

Example 9.1.2: $\mathbb{C}^n$ as the Complexification of $\mathbb{R}^n$

Let $V=\mathbb{R}^n$. An element of $(\mathbb{R}^n)_{\mathbb{C}}$ is a pair of real columns, written $x+iy$, and it determines the column of $\mathbb{C}^n$ whose j-th entry is x_j+iy_j. Write $\Phi(x+iy)$ for that column, so that $\Phi\colon(\mathbb{R}^n)_{\mathbb{C}}\to\mathbb{C}^n$.

The map Φ is a bijection: each entry z_j of a column $z\in\mathbb{C}^n$ is x_j+iy_j for exactly one pair of real numbers x_j,y_j, so exactly one pair of real columns x,y satisfies $\Phi(x+iy)=z$.

It carries the addition of $(\mathbb{R}^n)_{\mathbb{C}}$ to the addition of $\mathbb{C}^n$: the j-th entry of $\Phi\big((x+iy)+(x'+iy')\big)$ is $(x_j+x'_j)+i(y_j+y'_j)$, which is the sum of the j-th entries of $\Phi(x+iy)$ and $\Phi(x'+iy')$. It carries the scalar action to the scalar action: for $\lambda=a+bi$ the j-th entry of $\Phi\big(\lambda(x+iy)\big)=\Phi\big((ax-by)+i(bx+ay)\big)$ is

$$(ax_j-by_j)+i(bx_j+ay_j)=(a+bi)(x_j+iy_j)$$

which is the j-th entry of $\lambda\,\Phi(x+iy)$. So $(\mathbb{R}^n)_{\mathbb{C}}$ is identified with $\mathbb{C}^n$, and under the identification the copy $\{x+i0\,|\,x\in\mathbb{R}^n\}$ of $\mathbb{R}^n$ is the set of columns with real entries, while the standard basis $e_1,\dots,e_n$ of $\mathbb{R}^n$ is the standard basis of $\mathbb{C}^n$.

For $n=2$ take $x=(1,-2)^\top$ and $y=(0,3)^\top$, so that $\Phi(x+iy)=(1,\ -2+3i)^\top$, and take $\lambda=2+i$, so that $a=2$ and $b=1$. Then (9.1) gives

$$(2+i)(x+iy)=(2x-y)+i(x+2y)=\begin{pmatrix}2\\-7\end{pmatrix}+i\begin{pmatrix}1\\4\end{pmatrix}$$

whose image under Φ is $(2+i,\ -7+4i)^\top$. Multiplying in $\mathbb{C}^2$ instead gives the same column, since $(2+i)\cdot 1=2+i$ and $(2+i)(-2+3i)=-4+6i-2i+3i^2=-7+4i$.

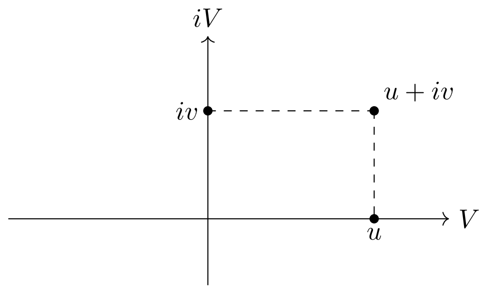


Figure 9.1: An element of $V_{\mathbb{C}}$ and its two components, with the copy of V drawn horizontally and the set iV of elements $0+iv$ drawn vertically

The set and the two operations are now fixed, and what remains is to check the eight conditions a vector space over $\mathbb{C}$ is required to meet:

[1] A reader who has met the tensor product will recognize $V_{\mathbb{C}}$ as $\mathbb{C}\otimes_{\mathbb{R}}V$, the extension of scalars of V along the inclusion of $\mathbb{R}$ in $\mathbb{C}$, with $u+iv$ corresponding to $1\otimes u+i\otimes v$; that construction is developed in *Core Algebra* [9, chapter 15.3], where this very case appears as an example of extension of scalars. Nothing below uses it, and every argument in this chapter computes with the pair.

Proposition 9.1.3: The Complexification Is a Complex Vector Space

Let V be a vector space over $\mathbb{R}$. Then $V_{\mathbb{C}}$, with the operations of definition 9.1.1, is a vector space over $\mathbb{C}$. Its zero vector is $0 + i0$ and the negative of $u + iv$ is $(-u) + i(-v)$.

***Proof*:**
The addition of $V_{\mathbb{C}}$ is performed in the two components separately, so each of the four additive conditions holds in $V_{\mathbb{C}}$ because it holds in V. Associativity and commutativity are the corresponding statements about V applied once in each component. The computation $(u + iv) + (0 + i0) = (u + 0) + i(v + 0) = u + iv$ exhibits the zero vector, and $\big((-u) + i(-v)\big) + (u + iv) = 0 + i0$ exhibits the negative.

For the four scalar conditions let $\lambda = a + bi$ and $\mu = c + di$ with $a, b, c, d \in \mathbb{R}$, and let $u + iv$ and $u' + iv'$ lie in $V_{\mathbb{C}}$.

Distributivity over a sum of vectors. Applying (9.1) to $(u + u') + i(v + v')$,

$$\lambda\big((u + iv) + (u' + iv')\big) = \big(a(u + u') - b(v + v')\big) + i\big(b(u + u') + a(v + v')\big)$$

while $\lambda(u + iv) + \lambda(u' + iv')$ is

$$\big((au - bv) + (au' - bv')\big) + i\big((bu + av) + (bu' + av')\big)$$

The two agree because a real scalar distributes over a sum in V, together with the commutativity and associativity of the addition of V.

Distributivity over a sum of scalars. Here $\lambda + \mu = (a + c) + (b + d)i$, so

$$(\lambda + \mu)(u + iv) = \big((a + c)u - (b + d)v\big) + i\big((b + d)u + (a + c)v\big)$$

while $\lambda(u + iv) + \mu(u + iv)$ is

$$\big((au - bv) + (cu - dv)\big) + i\big((bu + av) + (du + cv)\big)$$

The two agree because a sum of real scalars distributes over a vector of V, again with the commutativity and associativity of the addition.

Associativity of the scalar action. This is the condition the formula exists to make true. The product in $\mathbb{C}$ is $\lambda\mu = (ac - bd) + (ad + bc)i$, so

$$(\lambda\mu)(u + iv) = \big((ac - bd)u - (ad + bc)v\big) + i\big((ad + bc)u + (ac - bd)v\big)$$

Applying μ first gives $\mu(u + iv) = (cu - dv) + i(du + cv)$, and applying λ to that gives

$$\lambda\big(\mu(u + iv)\big) = \big(a(cu - dv) - b(du + cv)\big) + i\big(b(cu - dv) + a(du + cv)\big)$$

Expanding the first component gives $acu - adv - bdu - bcv$, which regroups as $(ac - bd)u - (ad + bc)v$; expanding the second gives $bcu - bdv + adu + acv$, which regroups as $(ad + bc)u + (ac - bd)v$. Both components agree with the display above, so $(\lambda\mu)w = \lambda(\mu w)$ for every $w \in V_{\mathbb{C}}$.

The unit scalar. Here $1 = 1 + 0i$, and (9.1) gives $(1 \cdot u - 0 \cdot v) + i(0 \cdot u + 1 \cdot v) = u + iv$, using $1u = u$ and $0u = 0$ in V, the second because $0u = (0 + 0)u = 0u + 0u$. All eight conditions hold, as we sought to show.

Multiplying in each component separately would already have made $V \times V$ a vector space over $\mathbb{R}$. The formula of definition 9.1.1 mixes the two components, the convention being that i carries $(u, 0)$ to $(0, u)$. Granted it, the rest is forced. Additivity makes the action of i on the pair (u, v) the sum of its actions on $(u, 0)$ and on $(0, v)$; the first is $(0, u)$ by the convention, and the second is $(-v, 0)$, since applying i twice must reverse the sign. So i carries (u, v) to $(-v, u)$, which is (9.1) with $a = 0$ and $b = 1$, and distributing $a + bi$ over its two summands recovers (9.1) in general.

Note that the convention is a choice that the eight conditions do not force. Let the scalar λ act on $V \times V$ as $\bar{\lambda}$ acts in (9.1), so that i carries (u, v) to $(v, -u)$. This second rule also mixes the two components, it also restricts on the real scalars to the componentwise action (since $\bar{a} = a$ for $a \in \mathbb{R}$), and it also meets all eight conditions, since conjugation preserves sums and products and fixes 1, its associativity over $\mathbb{C}$ satisfied since $\overline{\lambda\mu} = \bar{\lambda}\,\bar{\mu}$. What separates the two rules is which square root of -1 is chosen, and the $u + iv$ notation makes clear the chosen square-root is the one that sends (u, v) to $(-v, u)$. The choice will matter later: it is what tells the two summands of section 9.3 apart.

The construction so far doubles the underlying set, so what does it do to the dimension? A basis of V over $\mathbb{R}$ remains as a basis in the passage, leaving the dimension unchanged:

Theorem 9.1.4: A Real Basis Is a Complex Basis of the Complexification

Let V be a vector space over $\mathbb{R}$ and let $(v_1, \ldots, v_n)$ be a basis of V over $\mathbb{R}$. Then $(v_1, \ldots, v_n)$, read inside $V_{\mathbb{C}}$ through the identification $v_j = v_j + i0$, is a basis of $V_{\mathbb{C}}$ over $\mathbb{C}$.

***Proof*:**
Record first how a complex combination of the v_j splits. For $a_j, b_j \in \mathbb{R}$, (9.1) applied to $v_j + i0$ gives $(a_j + ib_j)v_j = a_j v_j + i(b_j v_j)$, and addition in $V_{\mathbb{C}}$ is performed in each component separately, so

$$\sum_{j=1}^{n}(a_j + ib_j)v_j = \Big(\sum_{j=1}^{n} a_j v_j\Big) + i\Big(\sum_{j=1}^{n} b_j v_j\Big) \tag{9.2}$$

Spanning. Let $w \in V_{\mathbb{C}}$ and write $w = u + iv$ with $u, v \in V$. The list $(v_1, \ldots, v_n)$ spans V over $\mathbb{R}$, so $u = \sum_j a_j v_j$ and $v = \sum_j b_j v_j$ for some real numbers $a_1, \ldots, a_n, b_1, \ldots, b_n$. Reading (9.2) from right to left, $w = \sum_j (a_j + ib_j)v_j$, a complex combination of the v_j.

Independence. Let $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$ satisfy $\sum_j \lambda_j v_j = 0$, the zero vector of $V_{\mathbb{C}}$ being $0 + i0$, and write $\lambda_j = a_j + ib_j$ with $a_j, b_j \in \mathbb{R}$. By (9.2) the left side has components $\sum_j a_j v_j$ and $\sum_j b_j v_j$, and the components of an element of $V_{\mathbb{C}}$ are unique (definition 9.1.1), so

$$\sum_{j=1}^{n} a_j v_j = 0 \qquad \text{and} \qquad \sum_{j=1}^{n} b_j v_j = 0$$

are two relations in V with real coefficients. The list is independent over $\mathbb{R}$, so $a_j = 0$ and $b_j = 0$ for every j, hence $\lambda_j = 0$ for every j.

The separation into components is the whole of the second half of the argument: one complex relation among the v_j is a pair of real relations, one per component, and independence over $\mathbb{R}$ disposes of them separately. Spanning and independence together make $(v_1, \ldots, v_n)$ a basis of $V_{\mathbb{C}}$ over $\mathbb{C}$ (definition 2.2.1), as we sought to show.

Corollary 9.1.5: The Dimension Is Unchanged

Let V be a finite-dimensional vector space over $\mathbb{R}$. Then $V_{\mathbb{C}}$ is finite-dimensional over $\mathbb{C}$ and

$$\dim_{\mathbb{C}} V_{\mathbb{C}} = \dim_{\mathbb{R}} V$$

***Proof*:**
Put $n = \dim_{\mathbb{R}} V$. If $n = 0$ then $V = \{0\}$, so $V_{\mathbb{C}}$ consists of the single element $0 + i0$ and both dimensions are 0. If $n \geq 1$ then theorem 2.2.9 gives a basis $(v_1, \dots, v_n)$ of V over $\mathbb{R}$, and theorem 9.1.4 makes the same list a basis of $V_{\mathbb{C}}$ over $\mathbb{C}$. So $V_{\mathbb{C}}$ has a finite basis, is finite-dimensional (definition 2.2.15), and has dimension the length of that basis (definition 2.2.10), which is n, completing the proof.

The list of vectors does not change in the passage from V to $V_{\mathbb{C}}$, only the scalars it may be combined with does. Thus, a computation carried out in a real basis is a computation in a basis after complexifying.

While complexification leaves the dimension unchanged over the new field, the underlying set has still doubled, so something must have doubled with it. Naming what did takes the reverse operation:

Definition 9.1.6: Realification of a Complex Vector Space

Let W be a vector space over $\mathbb{C}$. The *realification* of W, written $W_{\mathbb{R}}$, is the set W with its addition and with its scalar multiplication restricted to the real scalars, so that tw for $t \in \mathbb{R}$ and $w \in W$ is computed in W.

The realification is a vector space over $\mathbb{R}$. Its four additive conditions are those of W, which are unchanged, and each of its four scalar conditions is the corresponding condition for W read for those scalars that happen to be real. In consequence every subspace of W over $\mathbb{C}$ is a subspace of $W_{\mathbb{R}}$. The converse fails: the set $\mathbb{R}$ of real numbers is closed under addition and under real multiples, so it is a subspace of $\mathbb{C}_{\mathbb{R}}$ by proposition 2.1.3 (definition 2.1.2), and it is not closed under multiplication by i.

Example 9.1.7: The Realification of $\mathbb{C}$

Let $W = \mathbb{C}$, regarded as a vector space over $\mathbb{C}$. Its realification $\mathbb{C}_{\mathbb{R}}$ is the same set of complex numbers with only real scalars admitted. The list $(1, i)$ is a basis of $\mathbb{C}_{\mathbb{R}}$ over $\mathbb{R}$: every $z \in \mathbb{C}$ is $x \cdot 1 + y \cdot i$ for exactly one pair of real numbers x, y, and taking $z = 0$ in that statement gives independence. So $\dim_{\mathbb{R}} \mathbb{C}_{\mathbb{R}} = 2$ by definition 2.2.10.

The rule $x + iy \mapsto (x, y)^\top$ is a bijection $\mathbb{C}_{\mathbb{R}} \to \mathbb{R}^2$ Carrying addition to the addition, since real and imaginary parts add separately, and it carries $t(x + iy) = tx + i(ty)$ to $t(x, y)^\top$ for $t \in \mathbb{R}$. A complex vector space of dimension 1 is therefore the real plane, of dimension 2.

Multiplication by i is a map $m \colon \mathbb{C}_{\mathbb{R}} \to \mathbb{C}_{\mathbb{R}}$, and it is linear over $\mathbb{R}$: $i(z + z') = iz + iz'$ and $i(tz) = t(iz)$ for $t \in \mathbb{R}$ are the distributive and associative laws for complex multiplication. Since $m(1) = i$ and $m(i) = -1$, the matrix of m in the ordered basis $(1, i)$ has columns $(0, 1)^\top$ and $(-1, 0)^\top$ by theorem 2.4.8, so it is $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, the matrix with no real eigenvalue that opened this section.

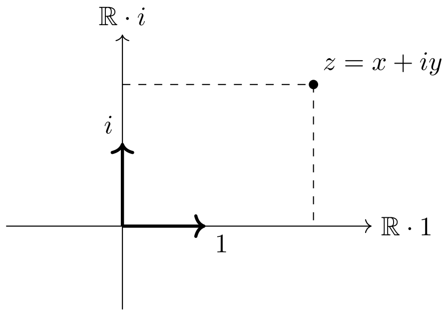


Figure 9.2: The realification of $\mathbb{C}$: the same numbers, with only real scalars admitted, and with $(1, i)$ a basis over $\mathbb{R}$

What happened to $\mathbb{C}$ happens in general:

Proposition 9.1.8: The Realification Has Twice the Dimension

Let W be a finite-dimensional vector space over $\mathbb{C}$ with $\dim_{\mathbb{C}} W = n$. Then $W_{\mathbb{R}}$ is finite-dimensional over $\mathbb{R}$ and

$$\dim_{\mathbb{R}} W_{\mathbb{R}} = 2n$$

Explicitly, if $(w_1, \ldots, w_n)$ is a basis of W over $\mathbb{C}$, then

$$(w_1,\ iw_1,\ w_2,\ iw_2,\ \ldots,\ w_n,\ iw_n)$$

is a basis of $W_{\mathbb{R}}$ over $\mathbb{R}$.

***Proof*:**
For $n = 0$ the space $W = \{0\}$ has the empty list as a basis, $W_{\mathbb{R}}$ is the same one-element set, and both counts are 0. Let $n \geq 1$ and fix a basis $(w_1, \ldots, w_n)$ of W over $\mathbb{C}$, which exists by theorem 2.2.9. Record next that for $\lambda = a + bi$ with $a, b \in \mathbb{R}$ and for $w \in W$, distributing λ over the sum $a + bi$ of scalars and then using associativity of the scalar action,

$$\lambda w = aw + (bi)w = aw + b(iw) \tag{9.3}$$

the last equality because $bi = ib$ in $\mathbb{C}$.

Spanning. Let $w \in W_{\mathbb{R}}$, which is to say $w \in W$. The list $(w_1, \ldots, w_n)$ spans W over $\mathbb{C}$, so $w = \sum_j \lambda_j w_j$ with $\lambda_j = a_j + ib_j$ and $a_j, b_j \in \mathbb{R}$. Applying (9.3) to each term,

$$w = \sum_{j=1}^{n} \big(a_j w_j + b_j (iw_j)\big)$$

which is a real combination of the $2n$ listed vectors.

Independence. Suppose $\sum_j \big(a_j w_j + b_j(iw_j)\big) = 0$ with all $a_j, b_j \in \mathbb{R}$. Reading (9.3) backwards turns the left side into $\sum_j (a_j + ib_j) w_j$, so the independence of $(w_1, \ldots, w_n)$ over $\mathbb{C}$ gives $a_j + ib_j = 0$ for every j. A complex number vanishes exactly when its real and imaginary parts both vanish, so $a_j = b_j = 0$ for every j.

The list is therefore a basis of $W_{\mathbb{R}}$ (definition 2.2.1) of length $2n$, so $W_{\mathbb{R}}$ is finite-dimensional (definition 2.2.15) with $\dim_{\mathbb{R}} W_{\mathbb{R}} = 2n$ (definition 2.2.10), completing the proof.

Complexification and Realification do not undo each other, and in each direction the dimension is out by a factor of two:

Proposition 9.1.9: Complexification and Realification Are Not Inverse

Let V be a finite-dimensional vector space over $\mathbb{R}$ with $\dim_{\mathbb{R}} V = n$ and let W be a finite-dimensional vector space over $\mathbb{C}$ with $\dim_{\mathbb{C}} W = m$.

1. Under the identification of definition 9.1.1 the space V is the subset $\{\, u + i0 \,|\, u \in V \,\}$ of $V_{\mathbb{C}}$; write iV for the subset $\{\, 0 + iv \,|\, v \in V \,\}$. Both are subspaces of $(V_{\mathbb{C}})_{\mathbb{R}}$ of real dimension n, and
$$(V_{\mathbb{C}})_{\mathbb{R}} = V \oplus iV$$
2. $\dim_{\mathbb{R}}(V_{\mathbb{C}})_{\mathbb{R}} = 2n$ and $\dim_{\mathbb{C}}(W_{\mathbb{R}})_{\mathbb{C}} = 2m$.
3. $(V_{\mathbb{C}})_{\mathbb{R}}$ has the same dimension as V only when $V = \{0\}$, and $(W_{\mathbb{R}})_{\mathbb{C}}$ has the same dimension as W only when $W = \{0\}$.

***Proof*:**

1. Both subsets contain $0 + i0$. For $u + i0$ and $u' + i0$ in the first and $t \in \mathbb{R}$, the operations of definition 9.1.1 give
$$(u + i0) + t(u' + i0) = (u + tu') + i0$$
since a real scalar acts in each component separately, so proposition 2.1.3, applied inside $(V_{\mathbb{C}})_{\mathbb{R}}$, makes the first subset a subspace; the same computation in the form $(0 + iv) + t(0 + iv') = 0 + i(v + tv')$ handles the second.

   Write $\alpha(u) = u + i0$ and $\beta(u) = 0 + iu$. Each is onto its subset by definition and injective because the components of an element are unique (definition 9.1.1), and the two displayed computations say exactly that $\alpha(u + tu') = \alpha(u) + t\alpha(u')$ and $\beta(u + tu') = \beta(u) + t\beta(u')$ for $t \in \mathbb{R}$.

   For the dimensions, take $n \geq 1$ and let $(v_1, \ldots, v_n)$ be a basis of V over $\mathbb{R}$, given by theorem 2.2.9; the case $n = 0$ has $V = \{0\}$, both subsets equal to $\{0 + i0\}$, and nothing to prove. Every element of the first subset is $\alpha(u)$ for some $u \in V$, and writing $u = \sum_j t_j v_j$ with $t_j \in \mathbb{R}$ gives $\alpha(u) = \sum_j t_j \alpha(v_j)$ by the additivity and real homogeneity just proved, so the list $(\alpha(v_1), \ldots, \alpha(v_n))$ spans. If $\sum_j t_j \alpha(v_j) = 0 + i0$ then $\alpha\big(\sum_j t_j v_j\big) = \alpha(0)$, so $\sum_j t_j v_j = 0$ because α is injective, and every $t_j = 0$ because $(v_1, \ldots, v_n)$ is independent. The list is therefore a basis of the first subset, which has dimension n by definition 2.2.10; the same argument with β gives dimension n for iV.

   Every element of $V_{\mathbb{C}}$ is $u + iv = (u + i0) + (0 + iv)$, so $(V_{\mathbb{C}})_{\mathbb{R}}$ is the sum of the two subspaces. If an element lies in both then $u + i0 = 0 + iv$, and equality of components gives $u = 0$ and $v = 0$, so the intersection is $\{0 + i0\}$ and the sum is direct (definition 2.1.11). The expression of an element as such a sum is then unique, by proposition 2.1.12.
2. By corollary 9.1.5 the space $V_{\mathbb{C}}$ has complex dimension n, and proposition 9.1.8 applied to

it gives $\dim_{\mathbb{R}}(V_{\mathbb{C}})_{\mathbb{R}} = 2n$. In the other order, proposition 9.1.8 gives $\dim_{\mathbb{R}} W_{\mathbb{R}} = 2m$, and corollary 9.1.5 applied to the real vector space $W_{\mathbb{R}}$ gives $\dim_{\mathbb{C}}(W_{\mathbb{R}})_{\mathbb{C}} = \dim_{\mathbb{R}} W_{\mathbb{R}} = 2m$.

3. The equality $2n = n$ holds exactly when $n = 0$, and likewise $2m = m$ exactly when $m = 0$. A space U with a nonzero vector u has $\dim U \geq 1$: the list (u) is independent, since $tu = 0$ with $t \neq 0$ gives $u = t^{-1}(tu) = 0$, so $\operatorname{span}\{u\}$ is a subspace of U of dimension 1, and corollary 2.2.12 gives $\dim U \geq 1$. Hence $n = 0$ says $V = \{0\}$ and $m = 0$ says $W = \{0\}$, completing the proof.

What $V_{\mathbb{C}}$ carries that makes it more than a complex space is the distinguished copy of V inside it (The first summand of proposition 9.1.9(1) is a distinguished subspace of $(V_{\mathbb{C}})_{\mathbb{R}}$). A complex vector space on its own has no such marking, and recovering a real space from inside it is a question with more than one answer, which is the subject of section 9.2.

Exercise 9.1.1

1. In $(\mathbb{R}^3)_{\mathbb{C}}$ let $u = (1, 0, 2)^\top$ and $v = (0, 3, -1)^\top$. Compute $(2 - i)(u + iv)$ from the scalar action of definition 9.1.1, then compute the same product in $\mathbb{C}^3$ under the identification of example 9.1.2, and check that the two agree entry by entry.

2. The set $M_2(\mathbb{R})$ is a vector space over $\mathbb{R}$ under entrywise addition and scalar multiplication. Exhibit a basis of it and compute $\dim_{\mathbb{R}} M_2(\mathbb{R})$; then use theorem 9.1.4 and corollary 9.1.5 to write down a basis of $\big(M_2(\mathbb{R})\big)_{\mathbb{C}}$ over $\mathbb{C}$ and its complex dimension. Finally, describe a bijection $\big(M_2(\mathbb{R})\big)_{\mathbb{C}} \to M_2(\mathbb{C})$ carrying the addition and the scalar action to those of $M_2(\mathbb{C})$, as Φ does in example 9.1.2, and compute the image of $(1 + 2i)(A + iB)$ for $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and $B = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$.

3. Let $W = \mathbb{C}^3$. Write down a basis of $W_{\mathbb{R}}$ over $\mathbb{R}$ using proposition 9.1.8 and state $\dim_{\mathbb{R}} W_{\mathbb{R}}$. Then compute $\dim_{\mathbb{C}}(W_{\mathbb{R}})_{\mathbb{C}}$ and $\dim_{\mathbb{R}} \big((W_{\mathbb{R}})_{\mathbb{C}}\big)_{\mathbb{R}}$.

4. Let $V = \mathbb{R}^2$ and work in $V_{\mathbb{C}}$. Put $w_1 = e_1 + ie_2$ and $w_2 = e_2 + i(-e_1)$. Compute iw_1 from the scalar action of definition 9.1.1 and deduce that the list (w_1, w_2) is linearly dependent over $\mathbb{C}$. Then show that the same list is linearly independent over $\mathbb{R}$ in the realification $(V_{\mathbb{C}})_{\mathbb{R}}$. (Independence depends on the field the coefficients are drawn from, which is what the second half of the proof of theorem 9.1.4 has to take care over.)

5. Let $V = \mathbb{R}$, of real dimension 1. Using example 9.1.2 with $n = 1$ and example 9.1.7, identify $V_{\mathbb{C}}$ with $\mathbb{C}$ and $(V_{\mathbb{C}})_{\mathbb{R}}$ with $\mathbb{R}^2$, and describe the two summands of proposition 9.1.9(1) as lines in $\mathbb{R}^2$. Then compute $\dim_{\mathbb{C}} \big((V_{\mathbb{C}})_{\mathbb{R}}\big)_{\mathbb{C}}$ and say which of V, $V_{\mathbb{C}}$ and $\big((V_{\mathbb{C}})_{\mathbb{R}}\big)_{\mathbb{C}}$ have equal dimension over their own fields.

6. Let W be a vector space over $\mathbb{C}$ and let $S\colon W \to W$ be a $\mathbb{C}$-linear map. Show that S is also a linear map $W_{\mathbb{R}} \to W_{\mathbb{R}}$ (definition 9.1.6, definition 2.4.2). Then show that the converse fails: check that the rule $f(x + iy) = x$, for $x, y \in \mathbb{R}$, is a linear map $\mathbb{C}_{\mathbb{R}} \to \mathbb{C}_{\mathbb{R}}$ that is not $\mathbb{C}$-linear, and compute the matrix of f as a map $\mathbb{R}^2 \to \mathbb{R}^2$ under the identification of example 9.1.7.

7. Let V be a vector space over $\mathbb{R}$ with basis $(v_1, \ldots, v_n)$. Show that iv_j is the element of $V_{\mathbb{C}}$ with components 0 and v_j, and that

$$(v_1,\, iv_1,\, \ldots,\, v_n,\, iv_n)$$

is a basis of $(V_{\mathbb{C}})_{\mathbb{R}}$ over $\mathbb{R}$. Explain how this is consistent with $\dim_{\mathbb{C}} V_{\mathbb{C}} = n$.

9.2 Conjugation and Real Forms

Section 9.1 manufactured a complex vector space from a real one, so that the complex space came with the real space already sitting inside it. This section runs the passage in the other direction. A complex vector space W is given and nothing else, and the question is which real subspaces of W could have produced it: for which subspaces V does every $w \in W$ have exactly one expression $u + iv$ with u and v in V.

Such subspaces exist, a complex space has many of them, and choosing one is the same as choosing a single map $W \to W$ that reverses multiplication by i.

Definition 9.2.1: Conjugate-Linear Maps and Conjugations

Let W and W' be vector spaces over $\mathbb{C}$. A map $f\colon W \to W'$ is *conjugate-linear* if

$$f(w + w') = f(w) + f(w') \qquad \text{and} \qquad f(\lambda w) = \overline{\lambda}\, f(w)$$

for all $w, w' \in W$ and all $\lambda \in \mathbb{C}$, where $\overline{\lambda}$ denotes the complex conjugate of the scalar λ. A *conjugation* on W is a conjugate-linear map $\sigma\colon W \to W$ satisfying $\sigma \circ \sigma = \mathrm{id}_W$.

A conjugate-linear map is additive, as a linear map is, and differs from a linear map (definition 2.4.2) only in pulling a scalar out with a bar on it. The reader has met that behaviour once already: over $\mathbb{C}$ the inner product is conjugate-linear in its second argument, which is the identity $\langle u, \lambda v\rangle = \overline{\lambda}\langle u, v\rangle$ of proposition 5.1.6(2).

The example the definition is built from is the complexification. Let V be a real vector space and let $V_{\mathbb{C}}$ be as in definition 9.1.1, with elements written $u + iv$ for $u, v \in V$. Define

$$\sigma(u + iv) = u - iv$$

which changes the second entry of the pair (u, v) to its negative and is therefore a well defined map $V_{\mathbb{C}} \to V_{\mathbb{C}}$. It is additive, because addition in $V_{\mathbb{C}}$ is performed in each entry separately:

$$\sigma\big((u + iv) + (u' + iv')\big) = \sigma\big((u + u') + i(v + v')\big) = (u + u') - i(v + v') = (u - iv) + (u' - iv')$$

It is conjugate-linear. Let $\lambda = a + bi$ with $a, b \in \mathbb{R}$. The scalar action of definition 9.1.1 gives $\lambda(u + iv) = (au - bv) + i(bu + av)$, whence

$$\sigma\big(\lambda(u + iv)\big) = (au - bv) - i(bu + av)$$

while applying the same scalar action to $\overline{\lambda} = a + (-b)i$ and to $\sigma(u + iv) = u + i(-v)$ gives

$$\overline{\lambda}\,\sigma(u + iv) = \big(au - (-b)(-v)\big) + i\big((-b)u + a(-v)\big) = (au - bv) - i(bu + av)$$

which is the same element. Finally $\sigma(\sigma(u + iv)) = \sigma(u + i(-v)) = u - i(-v) = u + iv$, so $\sigma \circ \sigma = \mathrm{id}$. Under the identification of $\mathbb{C}^n$ with $(\mathbb{R}^n)_{\mathbb{C}}$ from example 9.1.2 this σ is the map $z \mapsto \overline{z}$ that conjugates every entry of a column.

The two conditions in definition 9.2.1 act in opposite directions. Conjugate-linearity prevents σ from being a map of complex vector spaces, while restricting the scalars to $\mathbb{R}$ turns it into a linear map of the underlying real space, and the involution condition then makes that map its own inverse.

Proposition 9.2.2: Properties of a Conjugation

Let σ be a conjugation on a finite-dimensional complex vector space W, and write $W_{\mathbb{R}}$ for the realification of W (definition 9.1.6), that is, the set W with its addition and with scalar multiplication restricted to real scalars.

1. $\sigma(0) = 0$, and $\sigma(-w) = -\sigma(w)$ for every $w \in W$.
2. σ is a bijection, and $\sigma^{-1} = \sigma$.
3. $\sigma(tw) = t\,\sigma(w)$ for every $t \in \mathbb{R}$ and every $w \in W$. Consequently σ is a linear map $W_{\mathbb{R}} \to W_{\mathbb{R}}$.
4. σ is a linear map of W over $\mathbb{C}$ if and only if $W = \{0\}$.

***Proof*:**

1. Additivity applied to $w = w' = 0$ gives $\sigma(0) = \sigma(0) + \sigma(0)$, and adding $-\sigma(0)$ to both sides gives $0 = \sigma(0)$. Additivity applied to w and $-w$ then gives $\sigma(w) + \sigma(-w) = \sigma(0) = 0$, so $\sigma(-w) = -\sigma(w)$.
2. The condition $\sigma \circ \sigma = \mathrm{id}_W$ exhibits σ as a two-sided inverse of itself, so σ is a bijection with $\sigma^{-1} = \sigma$.
3. A real scalar satisfies $\bar{t} = t$, so conjugate-linearity reads $\sigma(tw) = \bar{t}\,\sigma(w) = t\,\sigma(w)$. Together with additivity this is exactly the pair of conditions in definition 2.4.2 for the space $W_{\mathbb{R}}$, whose scalars are the real numbers.
4. Conjugate-linearity gives $\sigma(iw) = \bar{i}\,\sigma(w) = -i\,\sigma(w)$ for every w. Suppose σ is $\mathbb{C}$-linear. Then $\sigma(iw) = i\,\sigma(w)$ as well, so $i\,\sigma(w) = -i\,\sigma(w)$ and hence $2i\,\sigma(w) = 0$. Multiplying by the scalar $1/(2i)$ gives $\sigma(w) = 0$ for every $w \in W$. By (2) the map σ is surjective, so $W = \{0\}$. Conversely, if $W = \{0\}$ the only map $W \to W$ is the zero map, which is $\mathbb{C}$-linear, completing the proof.

Proposition 9.2.3: A Conjugation Carries a Subspace to One of the Same Dimension

Let σ be a conjugation on a finite-dimensional complex vector space W and let $U \subseteq W$ be a complex subspace. Then $\sigma(U)$ is a complex subspace of W; if $w_1, \ldots, w_r$ is a basis of U over $\mathbb{C}$, then $\sigma(w_1), \ldots, \sigma(w_r)$ is a basis of $\sigma(U)$ over $\mathbb{C}$; and consequently

$$\dim_{\mathbb{C}} \sigma(U) = \dim_{\mathbb{C}} U$$

***Proof*:**
Step 1: $\sigma(U)$ is a complex subspace. It contains $\sigma(0) = 0$. For $z, z' \in \sigma(U)$ write $z = \sigma(u)$ and $z' = \sigma(u')$ with $u, u' \in U$; additivity of σ (definition 9.2.1) gives $z + z' = \sigma(u + u') \in \sigma(U)$, since $u + u' \in U$. For $\lambda \in \mathbb{C}$, conjugate-linearity gives $\lambda z = \lambda\sigma(u) = \sigma(\bar{\lambda}u)$, and $\bar{\lambda}u \in U$ because U is a complex subspace; so $\lambda z \in \sigma(U)$. That is proposition 2.1.3.

Step 2: the images of a basis span. Put $z_j = \sigma(w_j)$. Let $z \in \sigma(U)$, say $z = \sigma(u)$ with $u \in U$, and write $u = \lambda_1 w_1 + \cdots + \lambda_r w_r$. Applying σ and using additivity and conjugate-linearity,

$$z = \sigma(u) = \overline{\lambda_1}\, z_1 + \cdots + \overline{\lambda_r}\, z_r$$

so the z_j span $\sigma(U)$.

Step 3: the images of a basis are independent. Suppose $\mu_1 z_1 + \cdots + \mu_r z_r = 0$ with $\mu_j \in \mathbb{C}$. Applying σ to both sides and using $\sigma \circ \sigma = \mathrm{id}$ (proposition 9.2.2(2)) together with the same two properties,

$$\overline{\mu_1}\, w_1 + \cdots + \overline{\mu_r}\, w_r = \sigma(0) = 0$$

The independence of the w_j forces every $\overline{\mu_j} = 0$, hence every $\mu_j = 0$. So $z_1, \ldots, z_r$ is a basis of $\sigma(U)$ (definition 2.2.1) and the dimensions agree (definition 2.2.10).

Part (4) removes the option of treating σ as an operator on the complex space W, so no eigenvalue, determinant or characteristic polynomial of σ is available. What (3) gives instead is an operator on $W_{\mathbb{R}}$ whose square is the identity, and the subspace on which it acts as the identity is the object the section is after.

Definition 9.2.4: Real Form of a Complex Vector Space

Let W be a finite-dimensional vector space over $\mathbb{C}$. A *real form* of W is a subspace V of the realification $W_{\mathbb{R}}$ such that every $w \in W$ has exactly one expression

$$w = u + iv, \qquad u, v \in V$$

Write $iV = \{iv \,|\, v \in V\}$. This is again a subspace of $W_{\mathbb{R}}$: it contains $0 = i0$, and for $v, v' \in V$ and $t \in \mathbb{R}$ one has $iv + t(iv') = i(v + tv')$ with $v + tv' \in V$, so proposition 2.1.3(1) applies. The condition in definition 9.2.4 says that $W_{\mathbb{R}} = V + iV$ and that the expression of each element as a sum is unique, which by proposition 2.1.12(2) is the condition $V \cap iV = \{0\}$. So a real form is a subspace V of $W_{\mathbb{R}}$ with

$$W_{\mathbb{R}} = V \oplus iV$$

in the sense of definition 2.1.11. A real form is therefore never a subspace of W over $\mathbb{C}$ unless $W = \{0\}$: closure of V under multiplication by i would give $iV \subseteq V$ and hence $iV = V \cap iV = \{0\}$, so every $v \in V$ would satisfy $iv = 0$ and therefore $v = (-i)(iv) = 0$, leaving $W_{\mathbb{R}} = V \oplus iV = \{0\}$.

Example 9.2.5: Two Real Forms of $\mathbb{C}^2$

Take $W = \mathbb{C}^2$ and put

$$V_1 = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \,\middle|\, x_1, x_2 \in \mathbb{R} \right\} = \mathbb{R}^2, \qquad V_2 = \left\{ \begin{pmatrix} x_1 \\ ix_2 \end{pmatrix} \,\middle|\, x_1, x_2 \in \mathbb{R} \right\}$$

Both are subspaces of $(\mathbb{C}^2)_{\mathbb{R}}$. For V_1 this is proposition 2.1.3(1) applied entrywise: $\mathbb{R}^2$ contains the zero column, and a real multiple of a real column added to a real column is real. For V_2 the same criterion applies, since

$$\begin{pmatrix} x_1 \\ ix_2 \end{pmatrix} + t \begin{pmatrix} x_1' \\ ix_2' \end{pmatrix} = \begin{pmatrix} x_1 + tx_1' \\ i(x_2 + tx_2') \end{pmatrix}$$

for $t \in \mathbb{R}$, and V_2 contains the zero column.

V_1 is a real form. Given $z = (z_1, z_2)^\top \in \mathbb{C}^2$, each entry has exactly one expression $z_j = x_j + iy_j$ with $x_j, y_j \in \mathbb{R}$, so $z = x + iy$ with $x = (x_1, x_2)^\top$ and $y = (y_1, y_2)^\top$ in $\mathbb{R}^2$, and no other pair from $\mathbb{R}^2$ works.

V_2 is a real form. Let $u = (a_1, ia_2)^\top$ and $v = (b_1, ib_2)^\top$ lie in V_2, with $a_j, b_j \in \mathbb{R}$. Then

$$u + iv = \begin{pmatrix} a_1 + ib_1 \\ ia_2 + i(ib_2) \end{pmatrix} = \begin{pmatrix} a_1 + ib_1 \\ -b_2 + ia_2 \end{pmatrix}$$

Matching this against $z = (z_1, z_2)^\top$ determines the four real numbers and determines them uniquely: $a_1 = \operatorname{Re} z_1$ and $b_1 = \operatorname{Im} z_1$ from the first entry, and $b_2 = -\operatorname{Re} z_2$ and $a_2 = \operatorname{Im} z_2$ from the second.

The two are different subspaces of $(\mathbb{C}^2)_\mathbb{R}$: the column $(0, i)^\top$ lies in V_2, taking $x_1 = 0$ and $x_2 = 1$, and it does not lie in V_1 because i is not real. For a numerical instance take $z = (2 - 3i,\ 5 + 4i)^\top$. Its decomposition along V_1 is

$$z = \begin{pmatrix} 2 \\ 5 \end{pmatrix} + i \begin{pmatrix} -3 \\ 4 \end{pmatrix}$$

and its decomposition along V_2 is

$$z = \begin{pmatrix} 2 \\ 4i \end{pmatrix} + i \begin{pmatrix} -3 \\ -5i \end{pmatrix} = \begin{pmatrix} 2 - 3i \\ 4i + 5 \end{pmatrix}$$

the second entry of the last column being $4i - 5i^2 = 5 + 4i$.

The same column has two different decompositions, one along each form, and nothing in $\mathbb{C}^2$ selects either form over the other. The theorem below says that this is the general situation, and that each real form is recorded by a single map.

Theorem 9.2.6: Real Forms and Conjugations Correspond

Let W be a finite-dimensional vector space over $\mathbb{C}$.

1. If σ is a conjugation on W, then the fixed set
$$W^\sigma = \{\, w \in W \,|\, \sigma(w) = w \,\}$$
is a real form of W.
2. If V is a real form of W, then the map $\sigma_V : W \to W$ defined by $\sigma_V(u + iv) = u - iv$ for $u, v \in V$ is a conjugation on W, and $W^{\sigma_V} = V$.
3. The assignments $\sigma \mapsto W^\sigma$ and $V \mapsto \sigma_V$ are mutually inverse bijections between the set of conjugations on W and the set of real forms of W.

***Proof*:**

1. Three things are checked: that W^σ is a subspace of $W_\mathbb{R}$, that every $w \in W$ decomposes, and that the decomposition is unique.

   $\sigma(0) = 0$ by proposition 9.2.2(1), so $0 \in W^\sigma$ and the set is nonempty. Let $u, v \in W^\sigma$ and

$t \in \mathbb{R}$. Additivity and proposition 9.2.2(3) give

$$\sigma(u+tv) = \sigma(u) + \sigma(tv) = \sigma(u) + t\,\sigma(v) = u + tv$$

so $u + tv \in W^\sigma$, and proposition 2.1.3(1), applied in $W_\mathbb{R}$, makes W^σ a subspace of $W_\mathbb{R}$.

For the decomposition, let $w \in W$ and put

$$u = \tfrac{1}{2}\big(w + \sigma(w)\big), \qquad v = \tfrac{1}{2i}\big(w - \sigma(w)\big)$$

Then $iv = \frac{i}{2i}(w - \sigma(w)) = \frac{1}{2}(w - \sigma(w))$, so

$$u + iv = \tfrac{1}{2}\big(w + \sigma(w)\big) + \tfrac{1}{2}\big(w - \sigma(w)\big) = w$$

Both u and v are fixed by σ. For u, additivity together with proposition 9.2.2(3) applied to the real scalar $\frac{1}{2}$, and then $\sigma(\sigma(w)) = w$, give

$$\sigma(u) = \tfrac{1}{2}\,\sigma(w) + \tfrac{1}{2}\,\sigma(\sigma(w)) = \tfrac{1}{2}\,\sigma(w) + \tfrac{1}{2}w = u$$

For v, first record that $\frac{1}{2i} = -\frac{i}{2}$, since $(-\frac{i}{2})(2i) = -i^2 = 1$, so $v = -\frac{i}{2}(w - \sigma(w))$. Additivity and proposition 9.2.2(1) give $\sigma(w - \sigma(w)) = \sigma(w) - \sigma(\sigma(w)) = \sigma(w) - w$, and conjugate-linearity applied to the scalar $-\frac{i}{2}$, whose conjugate is $\frac{i}{2}$, gives

$$\sigma(v) = \tfrac{i}{2}\big(\sigma(w) - w\big) = -\tfrac{i}{2}\big(w - \sigma(w)\big) = v$$

Dividing by $2i$ rather than by 2 is what makes v fixed: the conjugate of $-i/2$ is $i/2$, and that change of sign cancels the one that σ introduces on $w - \sigma(w)$.

For uniqueness, suppose $u + iv = u' + iv'$ with $u, v, u', v' \in W^\sigma$. Applying σ and using additivity, conjugate-linearity and $\bar{i} = -i$,

$$\sigma(u + iv) = \sigma(u) + \bar{i}\,\sigma(v) = u - iv$$

and in the same way $\sigma(u' + iv') = u' - iv'$. So $u - iv = u' - iv'$. Adding this to $u + iv = u' + iv'$ gives $2u = 2u'$, and multiplying by $\frac{1}{2}$ gives $u = u'$; subtracting gives $2iv = 2iv'$, and multiplying by $\frac{1}{2i}$ gives $v = v'$. Hence W^σ is a real form.

2. Each $w \in W$ has exactly one expression $u+iv$ with $u, v \in V$, so the recipe $\sigma_V(u+iv) = u-iv$ defines a map on all of W.

   It is additive. If $w = u + iv$ and $w' = u' + iv'$ with all four vectors in V, then $w + w' = (u + u') + i(v + v')$ and both $u + u'$ and $v + v'$ lie in V by proposition 2.1.3(1) with $t = 1$. By uniqueness that is the expression of $w + w'$, so

   $$\sigma_V(w + w') = (u + u') - i(v + v') = (u - iv) + (u' - iv') = \sigma_V(w) + \sigma_V(w')$$

   It is conjugate-linear. Let $\lambda = a + bi$ with $a, b \in \mathbb{R}$. Expanding in W with $i^2 = -1$,

   $$\lambda w = (a + bi)(u + iv) = (au - bv) + i(bu + av)$$

   and $au - bv$ and $bu + av$ lie in V, again by proposition 2.1.3, since a and b are real. So $\sigma_V(\lambda w) = (au - bv) - i(bu + av)$. Expanding the other product the same way,

   $$\bar{\lambda}\,\sigma_V(w) = (a - bi)(u - iv) = au - iav - ibu + bi^2 v = (au - bv) - i(av + bu)$$

and the two agree.

It is an involution. The expression of $\sigma_V(w) = u + i(-v)$ has both entries u and $-v$ in V, so $\sigma_V(\sigma_V(w)) = u - i(-v) = u + iv = w$.

Its fixed set is V. If $u \in V$ then $u = u + i0$ is its expression and $\sigma_V(u) = u - i0 = u$, so $V \subseteq W^{\sigma_V}$. Conversely, if $\sigma_V(w) = w$ with $w = u + iv$, then $u - iv = u + iv$. Subtracting u from both sides and adding iv gives $2iv = 0$, and multiplying by $\frac{1}{2i}$ gives $v = 0$, so $w = u \in V$.

3. By (1) the first assignment sends conjugations to real forms and by (2) the second sends real forms to conjugations. The identity $W^{\sigma_V} = V$ of (2) says that one composite is the identity on real forms.

   For the other composite, let σ be a conjugation and put $V = W^\sigma$, a real form by (1). Let $w \in W$. The proof of (1) produced the expression $w = u + iv$ with $u = \frac{1}{2}(w + \sigma(w))$ and $v = \frac{1}{2i}(w - \sigma(w))$ in V, and showed it is the only one, so σ_V is computed on w from these two vectors:
$$\sigma_V(w) = u - iv = \tfrac{1}{2}\big(w + \sigma(w)\big) - \tfrac{1}{2}\big(w - \sigma(w)\big) = \sigma(w)$$
   using $iv = \frac{1}{2}(w - \sigma(w))$ as before. Hence $\sigma_{W^\sigma} = \sigma$, the two composites are identities, and the assignments are mutually inverse bijections, as we sought to show.

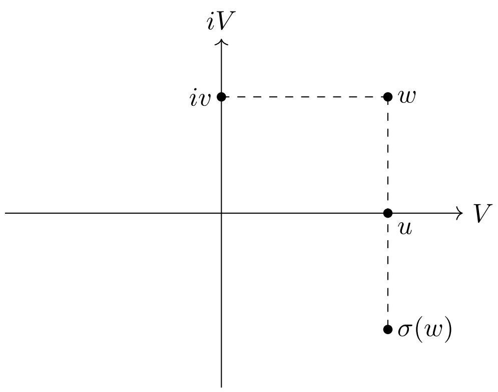


Figure 9.3: The decomposition $w = u + iv$ of theorem 9.2.6, in which u is the midpoint of w and $\sigma(w)$

The theorem replaces a subspace by a map: to fix a real structure on a complex space it is enough to say where one conjugation sends each vector, and the subspace is recovered as its fixed set. It also isolates what a complexification carries that a bare complex space does not. For $V_\mathbb{C}$ the conjugation $\sigma(u + iv) = u - iv$ constructed after definition 9.2.1 has fixed set $\{\, u + i0 \,|\, u \in V \,\}$: equality in $V_\mathbb{C}$ is equality of both entries of the pair, and $-v = v$ gives $2v = 0$ and hence $v = 0$. So $V_\mathbb{C}$ comes with a real form already selected, namely V itself, whereas $\mathbb{C}^2$ on its own offers V_1 and V_2 of example 9.2.5 with nothing to choose between them.

The size of a real form is not itself a choice. It is fixed by $\dim_\mathbb{C} W$, at exactly half the real dimension of the realification, and every basis of W over $\mathbb{C}$ produces a real form by taking real spans. Write $\operatorname{span}_\mathbb{R}\{w_1, \ldots, w_n\}$ for the set of all combinations $t_1 w_1 + \cdots + t_n w_n$ with $t_1, \ldots, t_n \in \mathbb{R}$, so that the subscript records which field the coefficients are drawn from; the subscripts on $\dim_\mathbb{R}$ and $\dim_\mathbb{C}$ do the same for dimensions.

Corollary 9.2.7: Dimension and Existence of Real Forms

Let W be a finite-dimensional vector space over $\mathbb{C}$, with $\dim_{\mathbb{C}} W = n$.

1. If V is a real form of W, then every basis of the real vector space V is a basis of W over $\mathbb{C}$. In particular $\dim_{\mathbb{R}} V = n$, which is half of $\dim_{\mathbb{R}} W_{\mathbb{R}}$.
2. If $w_1, \ldots, w_n$ is a basis of W over $\mathbb{C}$, then $\operatorname{span}_{\mathbb{R}}\{w_1, \ldots, w_n\}$ is a real form of W. In particular every finite-dimensional complex vector space has a real form.

***Proof*:**

1. By proposition 9.1.8 the space $W_{\mathbb{R}}$ is finite-dimensional with $\dim_{\mathbb{R}} W_{\mathbb{R}} = 2n$, so the subspace V satisfies $\dim_{\mathbb{R}} V \le 2n$ by corollary 2.2.12 and has a basis $v_1, \ldots, v_m$ over $\mathbb{R}$ by theorem 2.2.9.

   The list spans W over $\mathbb{C}$. Given $w \in W$, definition 9.2.4 gives $u, v \in V$ with $w = u + iv$. Writing $u = \sum_j a_j v_j$ and $v = \sum_j b_j v_j$ with all $a_j, b_j \in \mathbb{R}$ gives

   $$w = \sum_{j=1}^{m} a_j v_j + i \sum_{j=1}^{m} b_j v_j = \sum_{j=1}^{m} (a_j + i b_j) v_j$$

   The list is linearly independent over $\mathbb{C}$. Let $\lambda_1, \ldots, \lambda_m \in \mathbb{C}$ satisfy $\sum_j \lambda_j v_j = 0$, and write $\lambda_j = a_j + i b_j$ with $a_j, b_j \in \mathbb{R}$. Reading the display above backwards, $\sum_j \lambda_j v_j = u + iv$ with $u = \sum_j a_j v_j$ and $v = \sum_j b_j v_j$, both in V. The zero vector also has the expression $0 + i0$ with $0 \in V$, so the uniqueness clause of definition 9.2.4 forces $u = 0$ and $v = 0$. Independence of $v_1, \ldots, v_m$ over $\mathbb{R}$ then gives $a_j = b_j = 0$ for every j, hence $\lambda_j = 0$ for every j.

   So $v_1, \ldots, v_m$ is a basis of W over $\mathbb{C}$ (definition 2.2.1), and $m = n$ because the dimension of W over $\mathbb{C}$ does not depend on the basis used to compute it (definition 2.2.10).

2. Every $w \in W$ is $\sum_j \lambda_j w_j$ for exactly one list $\lambda_1, \ldots, \lambda_n$ of complex scalars: such a list exists because $w_1, \ldots, w_n$ spans W, and if $\sum_j \lambda_j w_j = \sum_j \mu_j w_j$ then $\sum_j (\lambda_j - \mu_j) w_j = 0$ and linear independence gives $\lambda_j = \mu_j$ for every j (definition 2.2.1). So

   $$\sigma\Big(\sum_{j=1}^{n} \lambda_j w_j\Big) = \sum_{j=1}^{n} \overline{\lambda_j}\, w_j$$

   is a well defined map $\sigma\colon W \to W$.

   It is a conjugation. Adding two vectors adds their coordinates, and $\overline{\lambda_j + \mu_j} = \overline{\lambda_j} + \overline{\mu_j}$, so σ is additive. Multiplying by $\mu \in \mathbb{C}$ multiplies each coordinate by μ, and $\overline{\mu \lambda_j} = \overline{\mu}\, \overline{\lambda_j}$, so $\sigma(\mu w) = \overline{\mu}\, \sigma(w)$. Applying σ twice conjugates each coordinate twice and returns it, so $\sigma \circ \sigma = \mathrm{id}_W$.

   Its fixed set is the real span. Since coordinates are unique, $\sigma(w) = w$ holds if and only if $\overline{\lambda_j} = \lambda_j$ for every j, that is if and only if every λ_j is real, that is if and only if $w \in \operatorname{span}_{\mathbb{R}}\{w_1, \ldots, w_n\}$. By theorem 9.2.6(1) that fixed set is a real form. A basis of W over $\mathbb{C}$ exists by theorem 2.2.9, so a real form exists, completing the proof.

Part (1) contains theorem 9.1.4 as the case $W = V_{\mathbb{C}}$ with V its own real form: there the real basis of V was shown to be a complex basis of $V_{\mathbb{C}}$, and here the same conclusion is reached for an arbitrary real form of an arbitrary complex space. Part (2) makes the real forms as numerous as the complex bases, and the two forms of example 9.2.5 are instances of it: V_1 is the real span of e_1, e_2 and V_2 is the real span of e_1, ie_2, and both lists are bases of $\mathbb{C}^2$ over $\mathbb{C}$.

For $W = \mathbb{C}^n$ the real form that every computation in the book has already used is $\mathbb{R}^n$, and the conjugation attached to it is entrywise conjugation of columns. Recording this puts the entrywise conjugate of definition 5.5.10 into the language of this section, and it characterizes the matrices with real entries by a condition that never mentions an entry.

Proposition 9.2.8: Real Columns and Real Matrices as Fixed Points

Let $\sigma_0 : \mathbb{C}^n \to \mathbb{C}^n$ be entrywise conjugation, $\sigma_0(z) = \overline{z}$ in the notation of definition 5.5.10.

1. σ_0 is a conjugation on $\mathbb{C}^n$ whose fixed set is $\mathbb{R}^n$, so $\mathbb{R}^n$ is a real form of $\mathbb{C}^n$ and $(\mathbb{C}^n)_{\mathbb{R}} = \mathbb{R}^n \oplus i\mathbb{R}^n$.
2. For every $A \in M_{m\times n}(\mathbb{C})$ and every $z \in \mathbb{C}^n$,
$$\overline{Az} = \overline{A}\,\overline{z}$$
Consequently $\overline{Az} = A\overline{z}$ holds for every $z \in \mathbb{C}^n$ if and only if $\overline{A} = A$.
3. $Ax \in \mathbb{R}^m$ for every $x \in \mathbb{R}^n$ if and only if $\overline{A} = A$.

***Proof*:**

1. A column of $\mathbb{C}^n$ is a matrix in $M_{n\times 1}(\mathbb{C})$, so proposition 5.5.11 applies to it. Part (2) of that proposition gives $\overline{z + z'} = \overline{z} + \overline{z'}$ and $\overline{\lambda z} = \overline{\lambda}\,\overline{z}$, which is conjugate-linearity, and part (1) gives $\overline{\overline{z}} = z$, which is $\sigma_0 \circ \sigma_0 = \mathrm{id}$. So σ_0 is a conjugation (definition 9.2.1). Its fixed set is $\{z \mid \overline{z} = z\}$, and by the closing sentence of definition 5.5.10 that condition holds exactly when every entry of z is real, so the fixed set is $\mathbb{R}^n$. By theorem 9.2.6(1) it is a real form, and the direct sum decomposition is the reformulation of definition 9.2.4 recorded after that definition.
2. The displayed identity is proposition 5.5.11(3) applied to the product of $A \in M_{m\times n}(\mathbb{C})$ with $z \in M_{n\times 1}(\mathbb{C})$. If $\overline{A} = A$, then substituting into it gives $\overline{Az} = A\overline{z}$ for every z. Conversely, suppose $\overline{Az} = A\overline{z}$ for every $z \in \mathbb{C}^n$. Combining with the identity gives $\overline{A}\,\overline{z} = A\overline{z}$ for every z. Taking $z = e_j$, which satisfies $\overline{e_j} = e_j$, gives $\overline{A}e_j = Ae_j$, and by definition 1.4.1 the column Be_j is the j-th column of B for any B. So $\overline{A}$ and A have the same j-th column for every j, hence $\overline{A} = A$.
3. Suppose $\overline{A} = A$ and let $x \in \mathbb{R}^n$, so $\overline{x} = x$. Then part (2) gives $\overline{Ax} = \overline{A}\,\overline{x} = Ax$, so Ax is fixed by entrywise conjugation and therefore has real entries (definition 5.5.10), that is $Ax \in \mathbb{R}^m$. Conversely, suppose $Ax \in \mathbb{R}^m$ for every $x \in \mathbb{R}^n$. Each standard basis column e_j lies in $\mathbb{R}^n$, so $Ae_j \in \mathbb{R}^m$. By definition 1.4.1 this column is the j-th column of A, so every entry of A is real and $\overline{A} = A$, as we sought to show.

Part (3) states realness of a matrix without looking at its entries: A has real entries exactly when the linear map it defines carries the real form $\mathbb{R}^n$ of its source into the real form $\mathbb{R}^m$ of its target. That condition still makes sense on a complex vector space carrying a chosen conjugation, where there are no entries to inspect.

Exercise 9.2.1

1. In $\mathbb{C}^3$ let σ_0 be entrywise conjugation (proposition 9.2.8) and let
$$w = \begin{pmatrix} 1+2i \\ 3 \\ -i \end{pmatrix}$$
Compute $u = \frac{1}{2}(w + \sigma_0(w))$ and $v = \frac{1}{2i}(w - \sigma_0(w))$, check that both lie in $\mathbb{R}^3$, and verify $w = u + iv$ entry by entry.

2. Let $V = \{(z, \overline{z})^\top \,|\, z \in \mathbb{C}\} \subseteq \mathbb{C}^2$. Show that V is a real form of $\mathbb{C}^2$ by exhibiting a basis of $\mathbb{C}^2$ over $\mathbb{C}$ whose real span is V, and then find the conjugation σ_V explicitly as a map $\mathbb{C}^2 \to \mathbb{C}^2$, justifying the answer by theorem 9.2.6(3).

3. For $A \in M_n(\mathbb{C})$ define $\sigma_A \colon \mathbb{C}^n \to \mathbb{C}^n$ by $\sigma_A(z) = \overline{Az}$. Show that σ_A is conjugate-linear for every A, and that σ_A is a conjugation if and only if $\overline{A}A = I_n$. Then take $n = 2$ and $A = \operatorname{diag}(1, i)$: verify the condition and describe the fixed set of σ_A as a real span of two columns.

4. Let σ be a conjugation on a complex vector space W and put $V = W^\sigma$. Show that
$$iV = \{\, w \in W \,|\, \sigma(w) = -w \,\}$$
and that $V \cap iV = \{0\}$. (The first equality says that the two summands of $W_\mathbb{R} = V \oplus iV$ are the two sets on which the map σ of proposition 9.2.2(3) acts as $+1$ and as -1.)

5. The set $M_n(\mathbb{C})$ is a vector space over $\mathbb{C}$ under entrywise addition and scalar multiplication. Show that $\tau(A) = \overline{A}$ is a conjugation on it and identify its fixed set. Check that the n^2 matrix units span $M_n(\mathbb{C})$ over $\mathbb{C}$, so that corollary 9.2.7 applies, and then use part (1) of that corollary with the real form you have identified to compute $\dim_\mathbb{C} M_n(\mathbb{C})$.

6. Let σ be a conjugation on a complex vector space $W \neq \{0\}$ and let $\lambda \in \mathbb{C}$. Show that $w \mapsto \lambda\,\sigma(w)$ is a conjugation on W if and only if $|\lambda| = 1$. Then take $W = \mathbb{C}$ and $\sigma(z) = \overline{z}$, and compute the fixed set of $z \mapsto i\overline{z}$. Finally, show that the real forms of the complex vector space $\mathbb{C}$ are exactly the sets $\mathbb{R}z_0$ with $z_0 \neq 0$.

7. Let σ be a conjugation on a complex vector space W and let $U \subseteq W$ be a subspace over $\mathbb{C}$ with $\sigma(U) = U$. Show that $U \cap W^\sigma$ is a real form of U. Then show that the hypothesis $\sigma(U) = U$ cannot be dropped: for $W = \mathbb{C}^2$ with entrywise conjugation and $U = \operatorname{span}_\mathbb{C}\{(1, i)^\top\}$, verify that $\sigma_0(U) \neq U$, that $U \cap \mathbb{R}^2 = \{0\}$, and that $\{0\}$ is not a real form of U.

9.3 Complex Structures on a Real Vector Space; the $(1,0)/(0,1)$ Decomposition

Section 9.1 enlarged a real space until it admitted complex scalars, and section 9.2 recovered the real space from inside the enlargement. This section asks when the enlargement is unnecessary: when a real vector space already carries multiplication by i, in the form of an operator on the space itself.

Such an operator is a single piece of data, and it does two things at once. It turns the real space into a complex vector space of half the real dimension, and it splits the complexification into two subspaces exchanged by the conjugation of section 9.2. The operator is also one of the three structures that chapter 10 fits together on a single space.

On $\mathbb{C}$ itself, multiplication by i is a function of one complex variable, and it is also an $\mathbb{R}$-linear map of the real plane. Writing $z = x + iy$ with $x, y \in \mathbb{R}$ and identifying z with the column $(x, y)^\top \in \mathbb{R}^2$,

$$iz = i(x + iy) = -y + ix$$

so multiplication by i carries $(x, y)^\top$ to $(-y, x)^\top$, which is the quarter turn of the plane. Performing it twice gives $i^2 z = -z$. Nothing else about the complex scalars needs recording: the scalar $a + bi$ acts as a times the identity plus b times this one operator. The condition that the operator squares to minus the identity is therefore the whole of the extra structure, and it is taken as the definition.

Definition 9.3.1: Complex Structure

Let V be a finite-dimensional real vector space. A *complex structure* on V is an $\mathbb{R}$-linear map $J \colon V \to V$ with

$$J \circ J = -\mathrm{id}_V$$

that is, with $J(Jv) = -v$ for every $v \in V$.

Example 9.3.2: The Standard Complex Structure on $\mathbb{R}^2$

Let $J_0 \colon \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map whose matrix in the standard basis is

$$M_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

so that $J_0(x_1, x_2)^\top = (-x_2, x_1)^\top$, the quarter turn computed above. Multiplying the matrix by itself (definition 1.4.1), the entry in row 1 and column 1 of M_0^2 is $0 \cdot 0 + (-1) \cdot 1 = -1$, the entry in row 1 and column 2 is $0 \cdot (-1) + (-1) \cdot 0 = 0$, the entry in row 2 and column 1 is $1 \cdot 0 + 0 \cdot 1 = 0$, and the entry in row 2 and column 2 is $1 \cdot (-1) + 0 \cdot 0 = -1$, so

$$M_0^2 = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} = -I_2$$

By theorem 2.4.9 the matrix of $J_0 \circ J_0$ is M_0^2, and the map $-\mathrm{id}_{\mathbb{R}^2}$ sends e_j to $-e_j$, so its matrix has columns $-e_1$ and $-e_2$ and equals $-I_2$. Two linear maps with the same matrix in the same basis are equal (theorem 2.4.8), so $J_0 \circ J_0 = -\mathrm{id}_{\mathbb{R}^2}$ and J_0 is a complex structure on $\mathbb{R}^2$.

Over $\mathbb{R}$ the map J_0 has no eigenvector at all. If $J_0 x = cx$ with $c \in \mathbb{R}$ and $x \neq 0$, then $-x = J_0(J_0 x) = c^2 x$, so $(c^2 + 1)x = 0$ and $c^2 = -1$, which no real number satisfies. No line through the origin is carried into itself.

Regarded as an element of $M_2(\mathbb{C})$, however, M_0 has eigenvalues. Multiplying out (definition 1.4.1),

$$M_0 \begin{pmatrix} 1 \\ -i \end{pmatrix} = \begin{pmatrix} i \\ 1 \end{pmatrix} = i \begin{pmatrix} 1 \\ -i \end{pmatrix}, \qquad M_0 \begin{pmatrix} 1 \\ i \end{pmatrix} = \begin{pmatrix} -i \\ 1 \end{pmatrix} = -i \begin{pmatrix} 1 \\ i \end{pmatrix}$$

so i and $-i$ are eigenvalues of M_0 with the displayed eigenvectors (definition 4.1.2, definition 4.1.3). There are no others: the computation of the previous paragraph, run over $\mathbb{C}$, turns $M_0 z = \lambda z$

with $z \neq 0$ into $\lambda^2 = -1$, whose only complex solutions are i and $-i$. By proposition 4.1.5 the eigenspace for i (definition 4.1.4) is $\operatorname{null}(M_0 - iI_2)$, and

$$M_0 - iI_2 = \begin{pmatrix} -i & -1 \\ 1 & -i \end{pmatrix}$$

has both rows equal to $(1, -i)$ after the row operation $R_1 \mapsto iR_1$, since $i(-i) = 1$ and $i(-1) = -i$. The surviving equation is $x_1 = ix_2$, so

$$\operatorname{null}(M_0 - iI_2) = \operatorname{span}\{(1, -i)^\top\}, \qquad \operatorname{null}(M_0 + iI_2) = \operatorname{span}\{(1, i)^\top\}$$

the second eigenspace being computed the same way from $M_0 + iI_2$, whose rows both become $(1, i)$ after $R_1 \mapsto -iR_1$.

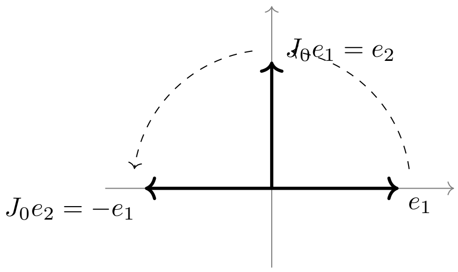


Figure 9.4: The standard complex structure on $\mathbb{R}^2$ is the quarter turn, and applying it twice reverses every vector

A line does not admit a complex structure. An $\mathbb{R}$-linear map of a one-dimensional space is multiplication by a scalar $c \in \mathbb{R}$, and $c^2 = -1$ has no real solution. The plane does admit one, and the obstruction separating the two cases is a parity condition on the dimension, which the determinant detects.

Proposition 9.3.3: A Complex Structure Exists Exactly in Even Dimension

Let V be a finite-dimensional real vector space. Then V carries a complex structure if and only if $\dim_{\mathbb{R}} V$ is even.

***Proof*:**
A complex structure forces the dimension to be even. Let J be a complex structure on V and put $n = \dim_{\mathbb{R}} V$. If $n = 0$ then n is even and there is nothing to prove, so assume $n \geq 1$ and fix a basis $\mathcal{B} = (v_1, \dots, v_n)$ of V, which exists by theorem 2.2.9. Let $A \in M_n(\mathbb{R})$ be the matrix of J in that basis (theorem 2.4.8). By theorem 2.4.9 the matrix of $J \circ J$ in the same basis is A^2. The map $-\mathrm{id}_V$ sends v_j to $-v_j$, whose coordinate column is $-e_j$, so the matrix of $-\mathrm{id}_V$ has columns $-e_1, \dots, -e_n$ and equals $-I_n$. Since $J \circ J = -\mathrm{id}_V$ and a linear map has exactly one matrix in a fixed basis (theorem 2.4.8),

$$A^2 = -I_n$$

Taking determinants and applying theorem 3.2.10, $(\det A)^2 = \det(-I_n)$. The matrix $-I_n$ is diagonal, hence triangular, so proposition 3.2.6 evaluates its determinant as the product of its diagonal entries, giving $\det(-I_n) = (-1)^n$. Now $\det A$ is a real number, so $(\det A)^2 \geq 0$, while $(-1)^n$ is 1 for even n and -1 for odd n. Odd n would give $(\det A)^2 = -1 < 0$, so n is even. The same identity now reads $(\det A)^2 = 1$, so $\det A \neq 0$ and A is invertible by corollary 3.2.8. Equivalently J is a bijection, and its inverse is $-J$, since $J(-Jv) = -J(Jv) = v$ and $(-J)(Jv) = v$ for every $v \in V$.

Even dimension gives one. Let $\dim_{\mathbb{R}} V = n = 2m$. If $m = 0$ then $V = \{0\}$, the only map $V \to V$ is the zero map, and it equals both id_V and $-\mathrm{id}_V$, so it is a complex structure. Assume $m \geq 1$, fix a basis $\mathcal{B}$ of V, and let $A \in M_n(\mathbb{R})$ be the block diagonal matrix with m diagonal blocks equal to the matrix M_0 of example 9.3.2 and all other entries 0. Multiplying blockwise (definition 1.4.1), the product A^2 is block diagonal with diagonal blocks $M_0^2 = -I_2$, so $A^2 = -I_n$. By the converse half of theorem 2.4.8 there is a linear map $J\colon V \to V$ whose matrix in the basis $\mathcal{B}$ is A, and then theorem 2.4.9 gives $A^2 = -I_n$ as the matrix of $J \circ J$, which is the matrix of $-\mathrm{id}_V$ computed above. Two maps with the same matrix in the same basis are equal, so $J \circ J = -\mathrm{id}_V$ and J is a complex structure on V, completing the proof.

The definition of a complex structure was extracted from multiplication by i on $\mathbb{C}$, and nothing was lost in the extraction. A single operator J with $J \circ J = -\mathrm{id}_V$ carries a complete complex scalar multiplication on V, since the scalar $a + bi$ can act only as a times the identity plus b times J. What has to be checked is that the rule so obtained satisfies the vector space axioms over $\mathbb{C}$, and that every complex scalar multiplication on V extending its real one arises from exactly one such J.

Theorem 9.3.4: Complex Structures Are Complex Vector Space Structures

Let V be a finite-dimensional real vector space, with its addition and its multiplication by real scalars fixed.

1. Let J be a complex structure on V. Then the same addition together with the rule

$$(a + bi) \cdot v = av + bJv \qquad (a, b \in \mathbb{R},\ v \in V) \tag{9.4}$$

   makes V a complex vector space, written V^J, and the restriction of (9.4) to real scalars is the given real scalar multiplication.

2. Conversely, let a complex scalar multiplication on V be given which makes V a complex vector space and which agrees with the given real scalar multiplication on real scalars. Then $Jv = i \cdot v$ defines a complex structure on V.

3. The two passages are mutually inverse. Complex structures on V therefore correspond one to one with complex vector space structures on V extending its real one.

4. Let J be a complex structure on V and let $(w_1, \ldots, w_m)$ be a basis of V^J over $\mathbb{C}$. Then $(w_1, Jw_1, w_2, Jw_2, \ldots, w_m, Jw_m)$ is a basis of V over $\mathbb{R}$. In particular $\dim_{\mathbb{C}} V^J = \frac{1}{2} \dim_{\mathbb{R}} V$.

***Proof*:**

1. The addition of V^J is that of V, so the vector space axioms mentioning only the addition hold in V^J because they hold in V. What remains are the four axioms in which scalars

appear. Let $\lambda = a + bi$ and $\mu = c + di$ with $a, b, c, d \in \mathbb{R}$, and let $u, v \in V$. Distributivity over vector addition follows from the additivity of J and the corresponding identity for real scalars:

$$\lambda \cdot (u + v) = a(u + v) + bJ(u + v) = (au + bJu) + (av + bJv) = \lambda \cdot u + \lambda \cdot v$$

For distributivity over scalar addition, $\lambda + \mu = (a + c) + (b + d)i$, and the corresponding real identity gives

$$(\lambda + \mu) \cdot v = (a + c)v + (b + d)Jv = (av + bJv) + (cv + dJv) = \lambda \cdot v + \mu \cdot v$$

Associativity of scalars is the axiom (9.4) exists to make true, and it is where $J \circ J = -\mathrm{id}_V$ enters. The product in $\mathbb{C}$ is $\lambda\mu = (ac - bd) + (ad + bc)i$, so

$$(\lambda\mu) \cdot v = (ac - bd)v + (ad + bc)Jv$$

while expanding $\lambda \cdot (\mu \cdot v)$, using the $\mathbb{R}$-linearity of J and then $J(Jv) = -v$,

$$\lambda\cdot(\mu\cdot v) = a(cv+dJv)+bJ(cv+dJv) = acv+adJv+bcJv+bdJ(Jv) = (ac-bd)v+(ad+bc)Jv$$

The two agree. For the unit axiom, $1 = 1 + 0i$ gives $1 \cdot v = 1v + 0Jv = v$. Finally, taking $b = 0$ in (9.4) gives $a \cdot v = av$, so the restriction to real scalars is the original scalar multiplication.

2. Write $\cdot$ for the given complex scalar multiplication. The map $Jv = i \cdot v$ is additive because $\cdot$ distributes over vector addition, and for $a \in \mathbb{R}$ the associativity of scalars applied twice, together with $ia = ai$ in $\mathbb{C}$ and the agreement of $\cdot$ with the real scalar multiplication, gives

$$J(av) = i \cdot (a \cdot v) = (ia) \cdot v = (ai) \cdot v = a \cdot (i \cdot v) = aJv$$

so J is $\mathbb{R}$-linear. Applying associativity of scalars once more, $J(Jv) = i \cdot (i \cdot v) = (i \cdot i) \cdot v = (-1) \cdot v = (-1)v$, the last step again by the agreement on real scalars, so that the final expression is computed in V. It remains to identify $(-1)v$ with $-v$, the additive inverse of v. The unit axiom and distributivity over scalar addition give $v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v$, and the same distributivity gives $0v = (0 + 0)v = 0v + 0v$, so adding the additive inverse of $0v$ to both sides yields $0v = 0$. Hence $v + (-1)v = 0$. Additive inverses are unique, since $v + x = 0$ and $v + y = 0$ give $x = x + (v + y) = (x + v) + y = y$ by associativity of the addition and the property of the zero vector. So $(-1)v = -v$ and $J(Jv) = -v$ for every v.

3. Start from a complex structure J and form V^J. Its scalar $i = 0 + 1i$ acts by $i \cdot v = 0v + 1Jv = Jv$, so the passage of (2) applied to V^J returns J. Start instead from a complex scalar multiplication $\cdot$ agreeing with the real one on $\mathbb{R}$, and let $Jv = i \cdot v$ be the complex structure of (2). The rule (9.4) built from this J sends $(a + bi, v)$ to

$$av + bJv = a \cdot v + b \cdot (i \cdot v) = a \cdot v + (bi) \cdot v = (a + bi) \cdot v$$

using the agreement on real scalars, then associativity of scalars, then distributivity over scalar addition. So the rule reproduces $\cdot$, and the two passages are mutually inverse.

4. First, V^J is finite-dimensional over $\mathbb{C}$: a basis $(v_1, \dots, v_n)$ of V over $\mathbb{R}$ spans V^J over $\mathbb{C}$, because every real linear combination of the v_j is in particular a complex one, so V^J has a finite spanning list (definition 2.2.15) and hence a basis (theorem 2.2.9). Let $(w_1, \dots, w_m)$ be one. Given $v \in V$, write $v = \lambda_1 \cdot w_1 + \cdots + \lambda_m \cdot w_m$ with $\lambda_j = a_j + b_j i$ and expand each term by (9.4):

$$v = (a_1 w_1 + b_1 J w_1) + \cdots + (a_m w_m + b_m J w_m)$$

which exhibits v as a real linear combination of $w_1, Jw_1, \dots, w_m, Jw_m$, so that list spans V over $\mathbb{R}$. If instead $a_1 w_1 + b_1 J w_1 + \cdots + a_m w_m + b_m J w_m = 0$ with all $a_j, b_j \in \mathbb{R}$, then reading the same computation backwards gives $(a_1 + b_1 i) \cdot w_1 + \cdots + (a_m + b_m i) \cdot w_m = 0$, and the independence of $(w_1, \dots, w_m)$ over $\mathbb{C}$ forces $a_j + b_j i = 0$, hence $a_j = b_j = 0$, for every j. So the list is independent over $\mathbb{R}$ and is a basis of V. It has $2m$ members, so $\dim_{\mathbb{R}} V = 2m = 2 \dim_{\mathbb{C}} V^J$, as we sought to show.

Part (4) recovers proposition 9.3.3 and adds the value of the halved dimension: a complex structure on a real space of dimension $2m$ presents it as a complex space of dimension m. Part (3) says that J and the scalar action determine each other, so a property established for one is a property of the other. The correspondence is between structures on the same underlying set: V^J and V have the same vectors and the same addition, and only the scalars differ.

Example 9.3.5: The Standard Complex Structure on $\mathbb{R}^4$

Let $V = \mathbb{R}^4$ and let J be the linear map whose matrix in the standard basis is

$$A = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

the block diagonal matrix with two copies of M_0, so that $Je_1 = e_2$, $Je_2 = -e_1$, $Je_3 = e_4$ and $Je_4 = -e_3$. The computation in the proof of proposition 9.3.3 gives $A^2 = -I_4$, so J is a complex structure.

The complex space V^J of theorem 9.3.4(1) has (e_1, e_3) as a basis over $\mathbb{C}$. Indeed $(a + bi) \cdot e_1 = ae_1 + bJe_1 = ae_1 + be_2$ and $(c + di) \cdot e_3 = ce_3 + de_4$, so

$$(x_1 + x_2 i) \cdot e_1 + (x_3 + x_4 i) \cdot e_3 = x_1 e_1 + x_2 e_2 + x_3 e_3 + x_4 e_4$$

which is an arbitrary vector of $\mathbb{R}^4$, giving spanning. The right side vanishes only when $x_1 = x_2 = x_3 = x_4 = 0$, since (e_1, e_2, e_3, e_4) is independent over $\mathbb{R}$, which gives independence over $\mathbb{C}$. So $\dim_{\mathbb{C}} V^J = 2 = \frac{1}{2} \dim_{\mathbb{R}} V$, and the display exhibits $x \mapsto (x_1 + ix_2,\ x_3 + ix_4)$ as the coordinate map of V^J for the basis (e_1, e_3), which identifies V^J with $\mathbb{C}^2$.

This J is not the only complex structure on $\mathbb{R}^4$. Let J' be the linear map with $J'e_1 = e_3$, $J'e_2 = e_4$, $J'e_3 = -e_1$ and $J'e_4 = -e_2$. Then $J'(J'e_j) = -e_j$ for each of the four basis vectors, so the matrix of $J' \circ J'$ has j-th column $-e_j$ and equals $-I_4$. That is also the matrix of $-\mathrm{id}_{\mathbb{R}^4}$, so the two maps are equal (theorem 2.4.8) and J' is a complex structure. It is a different one, since $J'e_1 = e_3 \neq e_2 = Je_1$. Complex structures are a choice, as real forms were in section 9.2.

A complex structure makes V itself complex, and that is one of the two complex spaces attached to (V, J). The other is the complexification $V_{\mathbb{C}}$ of definition 9.1.1, which has $\dim_{\mathbb{C}} V_{\mathbb{C}} = \dim_{\mathbb{R}} V$ by corollary 9.1.5 and so is twice as large over $\mathbb{C}$ as V^J is. The two are not the same space. What connects them is that J acts on $V_{\mathbb{C}}$ as well, and that on $V_{\mathbb{C}}$ it has eigenvectors, while on V it has none.

Let J be a complex structure on V and let $u + iv$, with $u, v \in V$, denote the element of $V_{\mathbb{C}}$ with components u and v (definition 9.1.1). Define

$$J_{\mathbb{C}} \colon V_{\mathbb{C}} \to V_{\mathbb{C}}, \qquad J_{\mathbb{C}}(u + iv) = Ju + iJv \tag{9.5}$$

which is a well-defined map because each element of $V_{\mathbb{C}}$ has exactly one such pair of components. It is $\mathbb{C}$-linear. Additivity holds because addition in $V_{\mathbb{C}}$ is componentwise and J is additive, and for $\lambda = a + bi$ the scalar action of definition 9.1.1 gives $\lambda(u + iv) = (au - bv) + i(bu + av)$, so

$$J_{\mathbb{C}}\big(\lambda(u + iv)\big) = J(au - bv) + iJ(bu + av) = (aJu - bJv) + i(bJu + aJv) = \lambda\big(Ju + iJv\big)$$

the middle equality by the $\mathbb{R}$-linearity of J and the last by the same scalar action applied to $Ju + iJv$. Applying (9.5) twice,

$$J_{\mathbb{C}}\big(J_{\mathbb{C}}(u + iv)\big) = J_{\mathbb{C}}(Ju + iJv) = J(Ju) + iJ(Jv) = (-u) + i(-v) = -(u + iv)$$

so $J_{\mathbb{C}} \circ J_{\mathbb{C}} = -\mathrm{id}_{V_{\mathbb{C}}}$. The same construction attaches a $\mathbb{C}$-linear operator on $V_{\mathbb{C}}$ to an arbitrary $\mathbb{R}$-linear operator on V, and section 9.4 carries it out in that generality.

Definition 9.3.6: The $(1, 0)$ and $(0, 1)$ Subspaces

Let V be a finite-dimensional real vector space with a complex structure J, and let $J_{\mathbb{C}}$ be as in (9.5). Put

$$V^{1,0} = \{\, w \in V_{\mathbb{C}} \,|\, J_{\mathbb{C}} w = iw \,\}, \qquad V^{0,1} = \{\, w \in V_{\mathbb{C}} \,|\, J_{\mathbb{C}} w = -iw \,\}$$

called respectively the *holomorphic* and *antiholomorphic* subspaces of $V_{\mathbb{C}}$, or the $(1, 0)$ and $(0, 1)$ subspaces.

For $V = \mathbb{R}^2$ with the quarter turn J_0 of example 9.3.2, identify $(\mathbb{R}^2)_{\mathbb{C}}$ with $\mathbb{C}^2$ as in example 9.1.2. Under that identification $(J_0)_{\mathbb{C}}$ is multiplication by M_0: for $z = u + iv$ with $u, v \in \mathbb{R}^2$, matrix multiplication gives $M_0 z = M_0 u + iM_0 v = J_0 u + iJ_0 v$, which is $(J_0)_{\mathbb{C}} z$ by (9.5). So the two subspaces are the two eigenspaces already computed in example 9.3.2,

$$V^{1,0} = \operatorname{span}\{(1, -i)^\top\} = \operatorname{span}\{e_1 - ie_2\}, \qquad V^{0,1} = \operatorname{span}\{(1, i)^\top\} = \operatorname{span}\{e_1 + ie_2\}$$

each a complex line in $\mathbb{C}^2$. The two together span $\mathbb{C}^2$: for $z = (z_1, z_2)^\top \in \mathbb{C}^2$,

$$z = \frac{z_1 + iz_2}{2} \begin{pmatrix} 1 \\ -i \end{pmatrix} + \frac{z_1 - iz_2}{2} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

as multiplying out and adding the two columns confirms. Both features hold for every complex structure.

Theorem 9.3.7: The $(1,0)/(0,1)$ Decomposition

Let V be a finite-dimensional real vector space with a complex structure J.

1. $V^{1,0}$ and $V^{0,1}$ are complex subspaces of $V_{\mathbb{C}}$.
2. $V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}$, and the unique expression of $w \in V_{\mathbb{C}}$ as a member of $V^{1,0}$ plus a member of $V^{0,1}$ is $w = w^{1,0} + w^{0,1}$ with
$$w^{1,0} = \tfrac{1}{2}\big(w - iJ_{\mathbb{C}}w\big), \qquad w^{0,1} = \tfrac{1}{2}\big(w + iJ_{\mathbb{C}}w\big)$$
3. $V^{1,0} = \{\, u - iJu \big| u \in V \,\}$ and $V^{0,1} = \{\, u + iJu \big| u \in V \,\}$. The map $\iota\colon V \to V_{\mathbb{C}}$ given by $\iota(u) = u - iJu$ is injective and $\mathbb{R}$-linear, and its image is $V^{1,0}$.
4. If $w \in V_{\mathbb{C}}$ is nonzero and $J_{\mathbb{C}}w = \lambda w$ for some $\lambda \in \mathbb{C}$, then $\lambda = i$ or $\lambda = -i$. If $V \neq \{0\}$, both values occur.

***Proof*:**

1. The zero vector satisfies $J_{\mathbb{C}}0 = 0 = i0$, so $V^{1,0}$ is nonempty. Let $w, w' \in V^{1,0}$ and $\lambda \in \mathbb{C}$. The $\mathbb{C}$-linearity of $J_{\mathbb{C}}$ established above gives
$$J_{\mathbb{C}}(w + \lambda w') = J_{\mathbb{C}}w + \lambda J_{\mathbb{C}}w' = iw + \lambda(iw') = i(w + \lambda w')$$
so $w + \lambda w' \in V^{1,0}$, and $V^{1,0}$ is a subspace by proposition 2.1.3. The same computation with $-i$ in place of i handles $V^{0,1}$.
2. Let $w \in V_{\mathbb{C}}$. Then $w^{1,0} + w^{0,1} = w$, since the two terms $-\frac{i}{2}J_{\mathbb{C}}w$ and $\frac{i}{2}J_{\mathbb{C}}w$ cancel. Using the $\mathbb{C}$-linearity of $J_{\mathbb{C}}$ and then $J_{\mathbb{C}}(J_{\mathbb{C}}w) = -w$,
$$J_{\mathbb{C}}w^{1,0} = \tfrac{1}{2}\big(J_{\mathbb{C}}w - iJ_{\mathbb{C}}(J_{\mathbb{C}}w)\big) = \tfrac{1}{2}\big(J_{\mathbb{C}}w + iw\big) = i \cdot \tfrac{1}{2}\big(w - iJ_{\mathbb{C}}w\big) = iw^{1,0}$$
the third equality because $i(w - iJ_{\mathbb{C}}w) = iw + J_{\mathbb{C}}w$. So $w^{1,0} \in V^{1,0}$, and the same computation with the signs reversed gives $J_{\mathbb{C}}w^{0,1} = -iw^{0,1}$, so $w^{0,1} \in V^{0,1}$. Hence $V_{\mathbb{C}} = V^{1,0} + V^{0,1}$. If $w \in V^{1,0} \cap V^{0,1}$ then $iw = J_{\mathbb{C}}w = -iw$, so $2iw = 0$ and multiplying by the scalar $1/(2i)$ gives $w = 0$. The sum is therefore direct (definition 2.1.11), and by proposition 2.1.12 each w has exactly one expression as a member of $V^{1,0}$ plus a member of $V^{0,1}$. The expression exhibited above is that one.
3. An element $u \in V$ has components u and 0 as an element of $V_{\mathbb{C}}$, so $J_{\mathbb{C}}u = Ju$ by (9.5). Applying part (2) to the vector $2u$ and using $\frac{1}{2}(2u - iJ_{\mathbb{C}}(2u)) = u - iJu$ places $u - iJu$ in $V^{1,0}$ for every $u \in V$. Conversely let $w = u + iv$ lie in $V^{1,0}$, with $u, v \in V$. Then
$$J_{\mathbb{C}}w = Ju + iJv, \qquad iw = i(u + iv) = -v + iu$$
and these are equal, so comparing components, which are unique, gives $Ju = -v$ and $Jv = u$. Hence $v = -Ju$ and $w = u + i(-Ju) = u - iJu$, which proves $V^{1,0} = \{u - iJu \big| u \in V\}$. Addition in $V_{\mathbb{C}}$ is componentwise and J is additive, so ι is additive, and for $a \in \mathbb{R}$ the scalar action gives $\iota(au) = au - iJ(au) = au - i(aJu) = a(u - iJu) = a\iota(u)$, so ι is $\mathbb{R}$-linear. If $\iota(u) = 0$ then both components vanish, so $u = 0$ and ι is injective. Its image is $V^{1,0}$ by the description just proved. The same argument applied to $w = u + iv \in V^{0,1}$ gives $Ju + iJv = -i(u + iv) = v - iu$, hence $Ju = v$ and $w = u + iJu$.

4. Let $w \neq 0$ with $J_{\mathbb{C}} w = \lambda w$. Applying $J_{\mathbb{C}}$ again and using its $\mathbb{C}$-linearity,

$$-w = J_{\mathbb{C}}(J_{\mathbb{C}} w) = J_{\mathbb{C}}(\lambda w) = \lambda J_{\mathbb{C}} w = \lambda^2 w$$

so $(\lambda^2+1)w = 0$. If λ^2+1 were nonzero, then multiplying by its inverse in $\mathbb{C}$ would give $w = 0$, so $\lambda^2 + 1 = 0$. Since $\lambda^2+1 = (\lambda - i)(\lambda + i)$ and $\mathbb{C}$ has no zero divisors, $\lambda = i$ or $\lambda = -i$. If $V \neq \{0\}$, pick $u \neq 0$ in V. By part (3) the vector $u - iJu$ lies in $V^{1,0}$ and is nonzero, since ι is injective, so the value i occurs. The vector $u + iJu$ lies in $V^{0,1}$ and is nonzero for the same reason, so the value $-i$ occurs, completing the proof.

Part (4) is the reason the decomposition exists. On V the operator J has no eigenvector at all, by the argument run for J_0 in example 9.3.2: $Jv = cv$ with $c \in \mathbb{R}$ and $v \neq 0$ gives $-v = J(Jv) = c^2 v$, hence $c^2 = -1$, which no real number satisfies. After complexifying the eigenvectors appear, and part (2) writes every vector of $V_{\mathbb{C}}$ as a member of $V^{1,0}$ plus a member of $V^{0,1}$. Fixing a basis of $V_{\mathbb{C}}$ and letting M be the matrix of $J_{\mathbb{C}}$ in it (theorem 2.4.8), a nonzero w with $J_{\mathbb{C}} w = \lambda w$ is an eigenvector of M belonging to λ (definition 4.1.3), so for $V \neq \{0\}$ the eigenvalues of M are i and $-i$ and no others, and by theorem 4.1.10 these are the roots in $\mathbb{C}$ of the characteristic polynomial of M (definition 4.1.9). In coordinates $V^{1,0}$ and $V^{0,1}$ are the eigenspaces of M for i and for $-i$ (definition 4.1.4), which proposition 4.1.5 computes as null spaces.

The spanning vectors recorded above for $\mathbb{R}^2$ illustrate part (3): $J_0 e_1 = e_2$, so $e_1 - iJ_0 e_1 = e_1 - ie_2$, which is the vector $(1, -i)^\top$ spanning $V^{1,0}$. Conjugating its entries produces $(1, i)^\top$, which spans $V^{0,1}$, and the same exchange happens for every complex structure.

Proposition 9.3.8: The Conjugation Interchanges the Two Summands

Let V be a finite-dimensional real vector space with a complex structure J, and let $\sigma\colon V_{\mathbb{C}} \to V_{\mathbb{C}}$, $\sigma(u+iv) = u - iv$, be the conjugation of $V_{\mathbb{C}}$ relative to V (definition 9.2.1). Then $\sigma \circ J_{\mathbb{C}} = J_{\mathbb{C}} \circ \sigma$ and

$$\sigma(V^{1,0}) = V^{0,1}, \qquad \sigma(V^{0,1}) = V^{1,0}$$

Moreover the restriction of σ is a bijection of $V^{1,0}$ onto $V^{0,1}$ which is additive and satisfies $\sigma(\lambda w) = \overline{\lambda}\,\sigma(w)$ for all $\lambda \in \mathbb{C}$ and $w \in V^{1,0}$.

***Proof*:**
For $u, v \in V$, both (9.5) and the formula for σ act on components, and

$$\sigma\big(J_{\mathbb{C}}(u+iv)\big) = \sigma(Ju + iJv) = Ju - iJv, \qquad J_{\mathbb{C}}\big(\sigma(u+iv)\big) = J_{\mathbb{C}}(u - iv) = Ju - iJv$$

the second because the components of $u - iv$ are u and $-v$, and $J(-v) = -Jv$. So the two composites agree on every element of $V_{\mathbb{C}}$.

Let $w \in V^{1,0}$. By definition 9.2.1 the map σ is additive, satisfies $\sigma(\lambda z) = \overline{\lambda}\sigma(z)$ for every $\lambda \in \mathbb{C}$ and $z \in V_{\mathbb{C}}$, and satisfies $\sigma \circ \sigma = \mathrm{id}_{V_{\mathbb{C}}}$. Using the commutation just proved and then this conjugate-linearity with $\lambda = i$,

$$J_{\mathbb{C}}(\sigma w) = \sigma(J_{\mathbb{C}} w) = \sigma(iw) = \overline{i}\,\sigma(w) = -i\,\sigma(w)$$

so $\sigma(w) \in V^{0,1}$, giving $\sigma(V^{1,0}) \subseteq V^{0,1}$. The same computation applied to $w \in V^{0,1}$ gives $J_{\mathbb{C}}(\sigma w) = \sigma(-iw) = \overline{-i}\,\sigma(w) = i\sigma(w)$, so $\sigma(V^{0,1}) \subseteq V^{1,0}$. Applying σ to the second inclusion and using $\sigma \circ \sigma = \mathrm{id}$,

$$V^{0,1} = \sigma\big(\sigma(V^{0,1})\big) \subseteq \sigma(V^{1,0})$$

so the first inclusion is an equality, and the same argument with the two subspaces exchanged gives the second. The restriction of σ to $V^{1,0}$ therefore maps onto $V^{0,1}$, and it is injective because $\sigma(w) = 0$ gives $w = \sigma(\sigma(w)) = \sigma(0) = 0$. Additivity and the conjugate-linearity identity are part of definition 9.2.1, completing the proof.

Concretely, theorem 9.3.7(3) writes a member of $V^{1,0}$ as $u - iJu$, and $\sigma(u - iJu) = u + iJu$ is the corresponding member of $V^{0,1}$. Because of this the antiholomorphic subspace is often written $\overline{V^{1,0}}$. Either summand therefore determines the other. What both are attached to is the real space V, which sits inside $V_{\mathbb{C}}$ as the fixed set of σ: the equation $\sigma(u+iv) = u+iv$ reads $u - iv = u + iv$, which forces $v = -v$ and hence $v = 0$, so the vectors fixed by σ are exactly those with second component 0. This is the case of theorem 9.2.6 in which the real form is V itself. It shows that V meets each summand only in 0, since a vector w fixed by σ and lying in $V^{1,0}$ satisfies $w = \sigma(w) \in V^{0,1}$ and therefore lies in $V^{1,0} \cap V^{0,1} = \{0\}$.

Corollary 9.3.9: The Two Summands Have Half the Dimension

Let V be a finite-dimensional real vector space with a complex structure J, and put $n = \dim_{\mathbb{R}} V$. Then
$$\dim_{\mathbb{C}} V^{1,0} = \dim_{\mathbb{C}} V^{0,1} = \frac{n}{2}$$

***Proof*:**
The number n is even by proposition 9.3.3, so $n/2$ is an integer. The space $V^{1,0}$ is a subspace of the finite-dimensional complex space $V_{\mathbb{C}}$, so it has a basis $(w_1, \ldots, w_r)$ over $\mathbb{C}$ by theorem 2.2.9, and $r = \dim_{\mathbb{C}} V^{1,0}$ by definition 2.2.10.

Step 1: the two summands have the same complex dimension. By proposition 9.3.8 the conjugation σ carries $V^{1,0}$ onto $V^{0,1}$, and by proposition 9.2.3 a conjugation carries a complex subspace to one of the same complex dimension. So $\dim_{\mathbb{C}} V^{0,1} = \dim_{\mathbb{C}} V^{1,0} = r$.

Step 2: the concatenation is a basis of $V_{\mathbb{C}}$. Let $w \in V_{\mathbb{C}}$. By theorem 9.3.7(2) it is $w' + w''$ with $w' \in V^{1,0}$ and $w'' \in V^{0,1}$, and expanding each in its own basis writes w as a complex linear combination of $w_1, \ldots, w_r, z_1, \ldots, z_r$. For independence, suppose
$$a_1 w_1 + \cdots + a_r w_r + b_1 z_1 + \cdots + b_r z_r = 0$$
with $a_j, b_j \in \mathbb{C}$. The first group lies in $V^{1,0}$ and the second in $V^{0,1}$, since both are subspaces by theorem 9.3.7(1) and a subspace contains every complex linear combination of its members (definition 2.1.2). The zero vector also decomposes as $0+0$, so the uniqueness in theorem 9.3.7(2) forces both groups to vanish separately, and then the independence of each basis forces every a_j and every b_j to be 0. So the concatenated list of $2r$ vectors is a basis of $V_{\mathbb{C}}$, and $\dim_{\mathbb{C}} V_{\mathbb{C}} = 2r$.

By corollary 9.1.5, $\dim_{\mathbb{C}} V_{\mathbb{C}} = \dim_{\mathbb{R}} V = n$. Hence $2r = n$ and $\dim_{\mathbb{C}} V^{1,0} = \dim_{\mathbb{C}} V^{0,1} = r = n/2$, as we sought to show.

Three dimensions are now in play. The real space V has real dimension n. The complex space V^J, built on the same set, has complex dimension $n/2$. The complexification $V_{\mathbb{C}}$ has complex dimension n and splits into two subspaces of complex dimension $n/2$ each, one carried onto the other by the conjugation. So V^J and $V^{1,0}$ have the same complex dimension, and theorem 9.3.7(3) does more than match their sizes: the map $\iota(u) = u - iJu$ carries V bijectively onto $V^{1,0}$, and it carries the action of

J on V to multiplication by i on $V^{1,0}$, since $\iota(Ju) = Ju - iJ(Ju) = Ju + iu = i(u - iJu) = i\iota(u)$. For the standard structure on $\mathbb{R}^4$ of example 9.3.5, the two summands are

$$V^{1,0} = \operatorname{span}\{e_1 - ie_2,\ e_3 - ie_4\}, \qquad V^{0,1} = \operatorname{span}\{e_1 + ie_2,\ e_3 + ie_4\}$$

each of complex dimension $2 = \frac{1}{2} \cdot 4$. To see the first, every member of $V^{1,0}$ is $u - iJu$ for a real $u = x_1e_1 + x_2e_2 + x_3e_3 + x_4e_4$, and the $\mathbb{R}$-linearity of ι expands it as $x_1\iota(e_1) + x_2\iota(e_2) + x_3\iota(e_3) + x_4\iota(e_4)$. Here $\iota(e_1) = e_1 - ie_2$ and $\iota(e_2) = e_2 + ie_1 = i(e_1 - ie_2)$, and likewise $\iota(e_3) = e_3 - ie_4$ and $\iota(e_4) = i(e_3 - ie_4)$, so every member is a complex combination of $e_1 - ie_2$ and $e_3 - ie_4$. The second summand is the image of the first under σ.

Exercise 9.3.1

1. Let $V = \mathbb{R}^4$ and let J be the linear map with

$$Je_1 = e_2, \qquad Je_2 = -e_1, \qquad Je_3 = \tfrac{1}{2}e_4, \qquad Je_4 = -2e_3$$

   Verify that J is a complex structure on $\mathbb{R}^4$ by checking $J(Je_j) = -e_j$ for each j, and then use theorem 9.3.7(3) to write down a basis of $V^{1,0}$ and a basis of $V^{0,1}$ over $\mathbb{C}$.

2. Keep the standard structure J on $\mathbb{R}^4$ of example 9.3.5. In the complex space V^J of theorem 9.3.4(1), compute $(2+3i)\cdot e_1$ and $(1-i)\cdot e_3$ as vectors of $\mathbb{R}^4$, and express e_2 and e_4 as complex multiples of members of the basis (e_1, e_3). Then compute the vector $(1+i)\cdot e_1 + (2-i)\cdot e_3$.

3. Let J be a complex structure on a real vector space V and let $P\colon V \to V$ be an invertible linear map. Show that $P \circ J \circ P^{-1}$ is again a complex structure on V. Taking $V = \mathbb{R}^2$, the map J_0 of example 9.3.2 and $P = \operatorname{diag}(1, 2)$, compute the matrix of $P \circ J_0 \circ P^{-1}$, verify directly that its square is $-I_2$, and compute a spanning vector of the corresponding $V^{1,0} \subseteq \mathbb{C}^2$.

4. Let J be a complex structure on a real vector space V with $\dim_{\mathbb{R}} V = 2m$ and $m \geq 1$. Using theorem 9.3.4(4), show that V has a basis in which the matrix of J is block diagonal with m blocks equal to $M_0 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Deduce that the matrix of J in *any* basis of V has determinant 1.

5. Let J be a complex structure on a real vector space $V \neq \{0\}$ and let $c \in \mathbb{R}$. Show that cJ is a complex structure on V if and only if $c = 1$ or $c = -1$. For $c = -1$, show that the $(1, 0)$ subspace attached to $-J$ is the $(0, 1)$ subspace attached to J, and conversely.

6. For a complex structure J on V define $\pi^{1,0}(w) = \frac{1}{2}(w - iJ_{\mathbb{C}}w)$ and $\pi^{0,1}(w) = \frac{1}{2}(w + iJ_{\mathbb{C}}w)$ for $w \in V_{\mathbb{C}}$. Show that both maps are $\mathbb{C}$-linear, that $\pi^{1,0} \circ \pi^{1,0} = \pi^{1,0}$ and $\pi^{0,1} \circ \pi^{0,1} = \pi^{0,1}$, that $\pi^{1,0} \circ \pi^{0,1} = 0$, and that $\sigma \circ \pi^{1,0} \circ \sigma = \pi^{0,1}$, where σ is the conjugation of definition 9.2.1.

7. Let J be a complex structure on V and let $W \subseteq V$ be a real subspace with $J(W) \subseteq W$. Show that $J(W) = W$, that the restriction of J to W is a complex structure on W, and that $\dim_{\mathbb{R}} W$ is therefore even. Then exhibit a subspace of $\mathbb{R}^2$, with the structure J_0 of example 9.3.2, that is not carried into itself by J_0.

9.4 Complexification of Operators; When a Complex Operator Is Defined over $\mathbb{R}$

The three preceding sections moved spaces between $\mathbb{R}$ and $\mathbb{C}$ and left the maps alone. Section 9.3 already needed one exception: the operator $J_{\mathbb{C}}$ on $V_{\mathbb{C}}$ was built from a complex structure J on V by acting on each component separately, and the construction was announced there as an instance of a general one. This section carries it out for an arbitrary $\mathbb{R}$-linear operator, and then answers the question the construction raises. Complexifying produces $\mathbb{C}$-linear operators on $V_{\mathbb{C}}$, but not all of them, and the ones it misses are exactly those that fail to commute with the conjugation.

Definition 9.4.1: Complexification of an Operator

Let V be a vector space over $\mathbb{R}$ and let $T\colon V \to V$ be linear. The *complexification* of T is the map

$$T_{\mathbb{C}}\colon V_{\mathbb{C}} \to V_{\mathbb{C}}, \qquad T_{\mathbb{C}}(u+iv) = Tu + iTv$$

where $u+iv$ denotes the element of $V_{\mathbb{C}}$ with components u and v (definition 9.1.1). Each element of $V_{\mathbb{C}}$ has exactly one pair of components, so the formula assigns exactly one value to each element and $T_{\mathbb{C}}$ is a well defined map.

Taking $v = 0$ gives $T_{\mathbb{C}}u = Tu$ for $u \in V$, so $T_{\mathbb{C}}$ extends T under the identification of V with $\{u + i0 \,|\, u \in V\}$ fixed in definition 9.1.1. Taking $T = J$ for a complex structure recovers the operator $J_{\mathbb{C}}$ of section 9.3.

A concrete instance is worth having before the general properties. Let $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ be the map $x \mapsto Ax$ with $A = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 3 \end{smallmatrix}\right)$, and take the element $e_1 + ie_2$ of $(\mathbb{R}^2)_{\mathbb{C}}$, whose components are the two standard basis columns. Then $Te_1 = (1,0)^\top$ and $Te_2 = (2,3)^\top$, so

$$T_{\mathbb{C}}(e_1 + ie_2) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + i\begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

Under the identification Φ of $(\mathbb{R}^2)_{\mathbb{C}}$ with $\mathbb{C}^2$ from example 9.1.2, the element $e_1 + ie_2$ is the column $(1,i)^\top$ and the value just computed is the column $(1+2i,\ 3i)^\top$. Multiplying in $\mathbb{C}^2$ instead,

$$\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} 1 \\ i \end{pmatrix} = \begin{pmatrix} 1+2i \\ 3i \end{pmatrix}$$

which is the same column. Complexifying an operator given by a real matrix therefore changes nothing about the matrix, only the columns it is allowed to act on. The next proposition proves that, together with the three algebraic properties the construction needs.

Proposition 9.4.2: The Complexification Is Linear and Keeps the Matrix

Let V be a vector space over $\mathbb{R}$ and let $S, T\colon V \to V$ be linear.

1. $T_{\mathbb{C}}$ is $\mathbb{C}$-linear, and it is the only $\mathbb{C}$-linear map $V_{\mathbb{C}} \to V_{\mathbb{C}}$ whose restriction to V is T.
2. $(S+T)_{\mathbb{C}} = S_{\mathbb{C}} + T_{\mathbb{C}}$ and $(tT)_{\mathbb{C}} = t\,T_{\mathbb{C}}$ for every $t \in \mathbb{R}$, while $(S \circ T)_{\mathbb{C}} = S_{\mathbb{C}} \circ T_{\mathbb{C}}$ and $(\mathrm{id}_V)_{\mathbb{C}} = \mathrm{id}_{V_{\mathbb{C}}}$.
3. Suppose $\dim_{\mathbb{R}} V = n \geq 1$, let $\mathcal{B} = (v_1, \ldots, v_n)$ be an ordered basis of V over $\mathbb{R}$, and let $A = (a_{ij}) \in M_n(\mathbb{R})$ be the matrix of T in $\mathcal{B}$, so that $Tv_j = \sum_i a_{ij} v_i$ for each j. Then $\mathcal{B}$ is an ordered basis of $V_{\mathbb{C}}$ over $\mathbb{C}$, and
$$[T_{\mathbb{C}} w]_{\mathcal{B}} = A\,[w]_{\mathcal{B}} \qquad \text{for every } w \in V_{\mathbb{C}}$$
so the same A is the matrix of $T_{\mathbb{C}}$ in $\mathcal{B}$.
4. Let $A \in M_n(\mathbb{R})$ and let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ be $x \mapsto Ax$. Under the identification $\Phi\colon (\mathbb{R}^n)_{\mathbb{C}} \to \mathbb{C}^n$ of example 9.1.2, the operator $T_{\mathbb{C}}$ is the map $z \mapsto Az$ on $\mathbb{C}^n$.

***Proof*:**

1. Addition in $V_{\mathbb{C}}$ is performed in each component separately (definition 9.1.1) and T is additive, so
$$T_{\mathbb{C}}\big((u+iv)+(u'+iv')\big) = T(u+u') + iT(v+v') = (Tu+iTv)+(Tu'+iTv')$$
which is $T_{\mathbb{C}}(u+iv) + T_{\mathbb{C}}(u'+iv')$. For a scalar $\lambda = a+bi$ with $a, b \in \mathbb{R}$, the scalar action of definition 9.1.1 gives $\lambda(u+iv) = (au-bv)+i(bu+av)$, so
$$T_{\mathbb{C}}\big(\lambda(u+iv)\big) = T(au-bv)+iT(bu+av) = (aTu-bTv)+i(bTu+aTv)$$
the second equality by the $\mathbb{R}$-linearity of T. Applying the same scalar action to λ and to $Tu+iTv$ produces that same element, so $T_{\mathbb{C}}(\lambda w) = \lambda\, T_{\mathbb{C}}(w)$, and with additivity this is definition 2.4.2 for $V_{\mathbb{C}}$ over $\mathbb{C}$.

   For uniqueness, let $S\colon V_{\mathbb{C}} \to V_{\mathbb{C}}$ be $\mathbb{C}$-linear with $Su = Tu$ for every $u \in V$. The closing computation of definition 9.1.1 writes $u+iv$ as $(u+i0)+i(v+i0)$, so $\mathbb{C}$-linearity gives $S(u+iv) = Su+iSv = Tu+iTv = T_{\mathbb{C}}(u+iv)$.

2. Each identity is checked at an arbitrary $u+iv$. For the first two, $(S+T)_{\mathbb{C}}(u+iv) = (S+T)u+i(S+T)v = (Su+iSv)+(Tu+iTv)$, and $(tT)_{\mathbb{C}}(u+iv) = tTu+itTv = t(Tu+iTv)$, the last equality because a real scalar acts in each component separately (definition 9.1.1). For the third,
$$S_{\mathbb{C}}\big(T_{\mathbb{C}}(u+iv)\big) = S_{\mathbb{C}}(Tu+iTv) = S(Tu)+iS(Tv) = (S\circ T)_{\mathbb{C}}(u+iv)$$
and the fourth is $(\mathrm{id}_V)_{\mathbb{C}}(u+iv) = u+iv$.

3. That $\mathcal{B}$ is a basis of $V_{\mathbb{C}}$ over $\mathbb{C}$ is theorem 9.1.4. Let $w \in V_{\mathbb{C}}$ and write $w = u+iv$ with $u, v \in V$. Expand $u = \sum_j c_j v_j$ and $v = \sum_j d_j v_j$ with $c_j, d_j \in \mathbb{R}$. The scalar action of

(9.2), established in the proof of theorem 9.1.4, gives

$$\sum_{j=1}^{n}(c_j + id_j)v_j = \Big(\sum_{j=1}^{n} c_j v_j\Big) + i\Big(\sum_{j=1}^{n} d_j v_j\Big)$$

whose right side is w. Hence the j-th entry of $[w]_{\mathcal{B}}$ is $c_j + id_j$ (definition 2.2.4). Applying T to each component of w and using $Tv_j = \sum_i a_{ij} v_i$,

$$T_{\mathbb{C}}w = Tu + iTv = \Big(\sum_i \Big(\sum_j a_{ij}c_j\Big)v_i\Big) + i\Big(\sum_i \Big(\sum_j a_{ij}d_j\Big)v_i\Big)$$

Reading (9.2) from right to left, with the two inner sums in place of c_i and d_i, the i-th entry of $[T_{\mathbb{C}}w]_{\mathcal{B}}$ is

$$\sum_j a_{ij}c_j + i\sum_j a_{ij}d_j = \sum_j a_{ij}(c_j + id_j)$$

the equality because each a_{ij} is real. By definition 1.4.1 the right side is the i-th entry of $A[w]_{\mathcal{B}}$. Taking $w = v_j$, whose coordinate column is e_j, identifies the j-th column of A as $[T_{\mathbb{C}}v_j]_{\mathcal{B}}$, which is the description of the matrix of a map in theorem 2.4.8.

4. Let $x, y \in \mathbb{R}^n$ and put $z = \Phi(x + iy)$, the column whose j-th entry is $x_j + iy_j$. Then $T_{\mathbb{C}}(x+iy) = Ax+iAy$, whose image under Φ is the column whose j-th entry is $(Ax)_j+i(Ay)_j$. By definition 1.4.1,

$$(Ax)_j + i(Ay)_j = \sum_k a_{jk}x_k + i\sum_k a_{jk}y_k = \sum_k a_{jk}(x_k + iy_k)$$

again because each a_{jk} is real, and the right side is the j-th entry of Az by definition 1.4.1. So $\Phi\big(T_{\mathbb{C}}(x + iy)\big) = Az$, completing the proof.

Part (3) is the computational content of the whole construction. A real matrix carries two meanings at once, one over $\mathbb{R}$ and one over $\mathbb{C}$, and complexifying an operator is the passage from the first to the second with the array of entries untouched. Nothing is recomputed, and every quantity built from the entries alone is unchanged in the passage. The characteristic polynomial is such a quantity, and it is what makes eigenvalues available on the complex side.

Proposition 9.4.3: The Characteristic Polynomial Is Unchanged

Let V be a vector space over $\mathbb{R}$ with $\dim_{\mathbb{R}} V = n \geq 1$, let $T\colon V \to V$ be linear, let $\mathcal{B}$ be an ordered basis of V over $\mathbb{R}$ and let $A \in M_n(\mathbb{R})$ be the matrix of T in $\mathcal{B}$.

1. The characteristic polynomial χ_A of definition 4.1.9 is one and the same polynomial whether A is read in $M_n(\mathbb{R})$ or in $M_n(\mathbb{C})$; it has degree n and all of its coefficients are real.

2. Over $\mathbb{C}$ it factors as $\chi_A(t) = (t-\lambda_1)\cdots(t-\lambda_n)$ for some $\lambda_1, \ldots, \lambda_n \in \mathbb{C}$, and for each j there is a nonzero $w \in V_{\mathbb{C}}$ with $T_{\mathbb{C}} w = \lambda_j w$. In particular such a scalar and such a vector exist.

3. For $\lambda \in \mathbb{R}$ the following three conditions are equivalent: $Tu = \lambda u$ for some nonzero $u \in V$; $T_{\mathbb{C}} w = \lambda w$ for some nonzero $w \in V_{\mathbb{C}}$; and $\chi_A(\lambda) = 0$.

***Proof*:**

1. By definition 4.1.9 the polynomial is $\chi_A(t) = \det(tI_n - A)$, monic of degree n, and every entry of $tI_n - A$ is $t - a_{ii}$ or $-a_{ij}$. Expanding that determinant along the first column by theorem 3.3.4, and iterating the expansion on the smaller matrices it produces until matrices of size 1 are reached, writes $\det(tI_n - A)$ as a signed sum of products of those entries. Multiplying the products out gives a polynomial in t each of whose coefficients is obtained from the real numbers a_{ij} by additions and multiplications, and is therefore real. The same expression is obtained whichever field the entries are read in, since $\mathbb{R}$ is a subfield of $\mathbb{C}$ and the two perform additions and multiplications of real numbers identically. So a single polynomial with real coefficients computes $\det(tI_n - A)$ at every $t \in \mathbb{C}$, and the uniqueness of the coefficients in definition 4.1.9 identifies it with χ_A in either reading.

2. Read A in $M_n(\mathbb{C})$. By theorem 4.1.14 the polynomial χ_A factors into n linear factors over $\mathbb{C}$, say $\chi_A(t) = (t-\lambda_1)\cdots(t-\lambda_n)$, and each λ_j is then a root of χ_A, hence an eigenvalue of A over $\mathbb{C}$ by theorem 4.1.10(1). So there is a nonzero column $c \in \mathbb{C}^n$ with $Ac = \lambda_j c$ (definition 4.1.2). Since $\mathcal{B}$ is a basis of $V_{\mathbb{C}}$ over $\mathbb{C}$ (theorem 9.1.4), there is a $w \in V_{\mathbb{C}}$ with $[w]_{\mathcal{B}} = c$, and $w \neq 0$ because the coordinate column of the zero vector is the zero column. By proposition 9.4.2(3),
$$[T_{\mathbb{C}} w]_{\mathcal{B}} = A[w]_{\mathcal{B}} = \lambda_j c = [\lambda_j w]_{\mathcal{B}}$$
the last equality because taking coordinates carries a scalar multiple of a vector to that scalar multiple of its coordinate column (definition 2.2.4). Coordinates relative to a basis are unique, so $T_{\mathbb{C}} w = \lambda_j w$. The list of λ_j is nonempty because $n \geq 1$.

3. Let $\lambda \in \mathbb{R}$. If $Tu = \lambda u$ with $u \in V$ nonzero, then u read in $V_{\mathbb{C}}$ is nonzero and $T_{\mathbb{C}} u = Tu = \lambda u$. Conversely let $T_{\mathbb{C}} w = \lambda w$ with $w = x + iy$ nonzero, where $x, y \in V$. Since λ is real, the scalar action of definition 9.1.1 gives $\lambda w = \lambda x + i\lambda y$, while $T_{\mathbb{C}} w = Tx + iTy$. Components are unique, so $Tx = \lambda x$ and $Ty = \lambda y$, and $w \neq 0$ forces $x \neq 0$ or $y \neq 0$; whichever of the two is nonzero is a vector of V that T scales by λ.

   Finally, expanding $u = \sum_j c_j v_j$ with $c_j \in \mathbb{R}$ and using $Tv_j = \sum_i a_{ij} v_i$ gives $Tu = \sum_i (\sum_j a_{ij} c_j) v_i$, so $[Tu]_{\mathcal{B}} = A[u]_{\mathcal{B}}$ by definition 1.4.1; this is the computation of proposition 9.4.2(3) with every imaginary component set to zero. Coordinates relative to $\mathcal{B}$ are a bijection $V \to \mathbb{R}^n$ carrying scalar multiples to scalar multiples and sending only the zero vector to the zero column (definition 2.2.4, definition 2.2.1). So $Tu = \lambda u$ for some nonzero $u \in V$ holds exactly when $Ac = \lambda c$ for some nonzero $c \in \mathbb{R}^n$, which is the condition of definition 4.1.2 for A over $\mathbb{R}$, and theorem 4.1.10(1) turns that into $\chi_A(\lambda) = 0$, completing the proof.

Part (3) is the sharp form of what is gained. A real operator on a real space may have no eigenvector at all, and complexifying gives one without creating any new real eigenvalue: the real spectrum is untouched, and what appears are the non-real roots that $\mathbb{R}$ could not accommodate. The smallest instance is

$$A_0 = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \qquad \chi_{A_0}(t) = (t-1)^2 + 1$$

whose value at every real t is at least 1, so χ_{A_0} has no real root and the operator $x \mapsto A_0 x$ on $\mathbb{R}^2$ has no eigenvector (theorem 4.1.10(1)). Over $\mathbb{C}$ the polynomial factors as $(t-(1+i))(t-(1-i))$ and the complexified operator has eigenvectors for both roots. The two roots are exchanged by conjugation, and that exchange is the mark left on the spectrum by the fact that A_0 has real entries. Locating that mark exactly is what the rest of the section does.

The condition on an operator of $V_{\mathbb{C}}$ that detects a real origin is stated in terms of the conjugation, and the conjugation is available because V is a real form of $V_{\mathbb{C}}$. Definition 9.2.4 asks for two things, and section 9.1 gives both: the copy $\{u + i0\}$ of V is a real subspace of the realification $(V_{\mathbb{C}})_{\mathbb{R}}$ (proposition 9.1.9(1)), and every element of $V_{\mathbb{C}}$ is $u+iv$ for exactly one pair $u, v \in V$ (definition 9.1.1). So theorem 9.2.6(2) gives the conjugation

$$\sigma\colon V_{\mathbb{C}} \to V_{\mathbb{C}}, \qquad \sigma(u+iv) = u - iv$$

whose fixed set is V. This is the σ used for the rest of the section.

Theorem 9.4.4: An Operator Comes from $\mathbb{R}$ Exactly When It Commutes with the Conjugation

Let V be a vector space over $\mathbb{R}$, let σ be the conjugation of $V_{\mathbb{C}}$ with fixed set V, and let $S\colon V_{\mathbb{C}} \to V_{\mathbb{C}}$ be $\mathbb{C}$-linear. The following are equivalent.

1. $S = T_{\mathbb{C}}$ for some linear $T\colon V \to V$.
2. $S \circ \sigma = \sigma \circ S$.
3. $S(V) \subseteq V$.

When these hold, the operator T in (1) is unique and is the restriction of S to V.

***Proof*:**
The implications are proved in the cycle (1) ⇒ (2) ⇒ (3) ⇒ (1).

(1) ⇒ (2). Let $S = T_{\mathbb{C}}$ and let $u, v \in V$. The element $u + iv$ has components u and v, and $\sigma(u+iv) = u - iv$ has components u and $-v$, so definition 9.4.1 gives

$$T_{\mathbb{C}}\bigl(\sigma(u+iv)\bigr) = Tu + iT(-v) = Tu - iTv, \qquad \sigma\bigl(T_{\mathbb{C}}(u+iv)\bigr) = \sigma(Tu + iTv) = Tu - iTv$$

the first using $T(-v) = -Tv$, which holds because T is linear. The two composites agree at every element of $V_{\mathbb{C}}$.

(2) ⇒ (3). Let $u \in V$. Then $\sigma(u) = u$, since V is the fixed set of σ, and therefore

$$\sigma(Su) = S\big(\sigma(u)\big) = Su$$

So Su is fixed by σ, and the fixed set of σ is V, whence $Su \in V$.

(3) ⇒ (1). By (3) the restriction $T = S|_V$ is a map $V \to V$. It is $\mathbb{R}$-linear: for $u, u' \in V$ and $t \in \mathbb{R}$ the $\mathbb{C}$-linearity of S gives $S(u + tu') = S(u) + tS(u')$, and the addition and the real scalar multiplication of $V_{\mathbb{C}}$ restrict to those of V, since $(u + i0) + (u' + i0) = (u + u') + i0$ and $t(u + i0) = tu + i0$ by definition 9.1.1. For $u, v \in V$ the closing computation of definition 9.1.1 writes $u + iv$ as $(u + i0) + i(v + i0)$, so

$$S(u + iv) = S(u) + iS(v) = Tu + iTv = T_{\mathbb{C}}(u + iv)$$

the last equality by definition 9.4.1. Hence $S = T_{\mathbb{C}}$.

For the final clause, suppose $T_{\mathbb{C}} = T'_{\mathbb{C}}$ for linear $T, T' \colon V \to V$. Evaluating both at $u + i0$ gives $Tu = T'u$ for every $u \in V$, so $T = T'$; and the operator produced in (3) ⇒ (1) is the restriction of S to V, as we sought to show.

Condition (3) is the geometric reading: an operator of $V_{\mathbb{C}}$ comes from V exactly when it carries the real form into itself. Condition (2) is the one that computes, because it is an identity between two maps rather than a statement about a subset, and it is the form the criterion takes in every later use. For $V = \mathbb{R}^n$ the conjugation is entrywise conjugation of columns, and the criterion becomes a condition on entries.

Corollary 9.4.5: The Matrix Criterion

Let $S \in M_n(\mathbb{C})$ and let $\Sigma \colon \mathbb{C}^n \to \mathbb{C}^n$ be the operator $z \mapsto Sz$. Identify $\mathbb{C}^n$ with $(\mathbb{R}^n)_{\mathbb{C}}$ as in example 9.1.2, so that entrywise conjugation $\sigma_0(z) = \overline{z}$ is the conjugation with fixed set $\mathbb{R}^n$ (proposition 9.2.8(1)). The following are equivalent.

1. $\Sigma = T_{\mathbb{C}}$ for some linear $T \colon \mathbb{R}^n \to \mathbb{R}^n$.
2. $\overline{Sz} = S\overline{z}$ for every $z \in \mathbb{C}^n$.
3. $\overline{S} = S$, that is, every entry of S is real.

When these hold, T is the map $x \mapsto Sx$ on $\mathbb{R}^n$.

***Proof*:**
The composite $\Sigma \circ \sigma_0$ sends z to $S\overline{z}$ and $\sigma_0 \circ \Sigma$ sends z to $\overline{Sz}$, so condition (2) says exactly that $\Sigma \circ \sigma_0 = \sigma_0 \circ \Sigma$. The operator Σ is $\mathbb{C}$-linear, since $S(z + \lambda z') = Sz + \lambda Sz'$ by definition 1.4.1, so theorem 9.4.4 applies and gives the equivalence of (1) and (2). The equivalence of (2) and (3) is the closing clause of proposition 9.2.8(2). When the three hold, $S \in M_n(\mathbb{R})$ by (3), and proposition 9.4.2(4) identifies the complexification of $x \mapsto Sx$ with $z \mapsto Sz$, which is Σ. The uniqueness clause of theorem 9.4.4 then names $x \mapsto Sx$ as the operator T, completing the proof.

The criterion can now be applied to a matrix by inspection: a complex matrix is the complexification of a real operator exactly when it already has real entries, and nothing has to be computed to decide it. What the theorem adds beyond this restatement is that the same criterion works on a space with no distinguished basis, where there are no entries to inspect and only the conjugation is available.

The criterion also constrains the spectrum. An operator commuting with σ has each of its eigenspaces carried by σ to an eigenspace, and since σ is conjugate-linear rather than linear, the eigenvalue moves to its conjugate.

Proposition 9.4.6: Non-Real Eigenvalues Come in Conjugate Pairs

Let V be a vector space over $\mathbb{R}$ with $\dim_{\mathbb{R}} V = n \geq 1$, let $T\colon V \to V$ be linear, let σ be the conjugation of $V_{\mathbb{C}}$ with fixed set V, and for $\lambda \in \mathbb{C}$ put

$$E_\lambda = \{\, w \in V_{\mathbb{C}} \,|\, T_{\mathbb{C}} w = \lambda w \,\}$$

1. E_λ is a complex subspace of $V_{\mathbb{C}}$ and $\sigma(E_\lambda) = E_{\overline{\lambda}}$. Consequently $E_\lambda \neq \{0\}$ if and only if $E_{\overline{\lambda}} \neq \{0\}$, and $\dim_{\mathbb{C}} E_\lambda = \dim_{\mathbb{C}} E_{\overline{\lambda}}$.
2. Let $\lambda = a + bi$ with $a, b \in \mathbb{R}$ and $b \neq 0$, and let $w = x + iy$ be a nonzero element of E_λ, with $x, y \in V$. Then
$$Tx = ax - by \qquad \text{and} \qquad Ty = bx + ay$$
3. In the situation of (2) the list x, y is linearly independent over $\mathbb{R}$, the subspace $U = \operatorname{span}_{\mathbb{R}}\{x, y\}$ of V has $\dim_{\mathbb{R}} U = 2$ and satisfies $T(U) \subseteq U$, and the matrix of the restriction of T to U in the ordered basis (x, y) is
$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

***Proof*:**

1. The zero vector satisfies $T_{\mathbb{C}} 0 = 0 = \lambda 0$, so E_λ is nonempty, and for $w, w' \in E_\lambda$ and $\mu \in \mathbb{C}$ the $\mathbb{C}$-linearity of $T_{\mathbb{C}}$ (proposition 9.4.2(1)) gives
$$T_{\mathbb{C}}(w + \mu w') = T_{\mathbb{C}} w + \mu T_{\mathbb{C}} w' = \lambda w + \mu \lambda w' = \lambda(w + \mu w')$$
so $w + \mu w' \in E_\lambda$, and proposition 2.1.3(1) makes E_λ a subspace of $V_{\mathbb{C}}$ over $\mathbb{C}$.
Let $w \in E_\lambda$. Implication (1) $\Rightarrow$ (2) of theorem 9.4.4, applied to $S = T_{\mathbb{C}}$, makes $T_{\mathbb{C}}$ and σ commute, and σ is conjugate-linear (definition 9.2.1), so
$$T_{\mathbb{C}}\big(\sigma(w)\big) = \sigma\big(T_{\mathbb{C}} w\big) = \sigma(\lambda w) = \overline{\lambda}\,\sigma(w)$$
Hence $\sigma(E_\lambda) \subseteq E_{\overline{\lambda}}$. The same inclusion with $\overline{\lambda}$ in place of λ reads $\sigma(E_{\overline{\lambda}}) \subseteq E_\lambda$, and applying σ to it and using $\sigma \circ \sigma = \mathrm{id}$ (definition 9.2.1) gives $E_{\overline{\lambda}} \subseteq \sigma(E_\lambda)$. The two inclusions together are the asserted equality. Since σ is a bijection fixing 0 (proposition 9.2.2(1) and (2)), one of the two subspaces is $\{0\}$ exactly when the other is.
For the dimensions, the equality just proved says that σ carries E_λ onto $E_{\overline{\lambda}}$, and proposition 9.2.3 then gives $\dim_{\mathbb{C}} E_{\overline{\lambda}} = \dim_{\mathbb{C}} E_\lambda$.
2. The element $w = x + iy$ has components x and y, so $T_{\mathbb{C}} w = Tx + iTy$ by definition 9.4.1, while the scalar action of definition 9.1.1 gives
$$\lambda w = (a + bi)(x + iy) = (ax - by) + i(bx + ay)$$

The two are equal by hypothesis and components are unique, so $Tx = ax - by$ and $Ty = bx + ay$.

3. Suppose the list x, y were linearly dependent over $\mathbb{R}$. Then $\operatorname{span}_{\mathbb{R}}\{x, y\}$ is spanned by a list of length at most 1. It is not $\{0\}$, since $x = y = 0$ would give $w = 0$, so there are a nonzero $z \in V$ and real numbers p, q, not both zero, with $x = pz$ and $y = qz$. Applying (9.2) to the one-element list (z) gives $w = (p + qi)z$. Put $\mu = p + qi$, which is nonzero. Using the $\mathbb{C}$-linearity of $T_{\mathbb{C}}$ and $T_{\mathbb{C}} z = Tz$,

$$\mu Tz = T_{\mathbb{C}}(\mu z) = T_{\mathbb{C}} w = \lambda w = \lambda \mu z$$

and multiplying by the scalar μ^{-1} gives $Tz = \lambda z$ in $V_{\mathbb{C}}$. The left side has components Tz and 0, while $\lambda z = az + i(bz)$ has components az and bz. Components are unique, so $bz = 0$, and multiplying by the real scalar $1/b$, which exists because $b \neq 0$, gives $z = 0$. That contradicts $z \neq 0$. So x, y are linearly independent over $\mathbb{R}$, and being a basis of U they give $\dim_{\mathbb{R}} U = 2$ (definition 2.2.10).

Part (2) writes Tx and Ty as real combinations of x and y, so for $s, t \in \mathbb{R}$ one has $T(sx + ty) = sTx + tTy \in U$, whence $T(U) \subseteq U$. The matrix of the restriction in the ordered basis (x, y) has as its j-th column the coordinate column of the image of the j-th basis vector (theorem 2.4.8), and part (2) gives those two coordinate columns as $(a, -b)^{\top}$ and $(b, a)^{\top}$, which are the columns of the displayed matrix, completing the proof.

The plane U carries no eigenvector of T. The matrix of the restriction has characteristic polynomial $t^2 - 2at + (a^2 + b^2)$, whose value at a real t is $(t - a)^2 + b^2 \geq b^2 > 0$, so it has no real root and theorem 4.1.10(1) leaves the restriction without an eigenvalue in $\mathbb{R}$. One non-real eigenvalue of $T_{\mathbb{C}}$ therefore accounts for a two-dimensional piece of V on which T has no invariant line, and the conjugate eigenvalue produces the same piece: by (1) the vector $\sigma(w) = x - iy$ lies in $E_{\overline{\lambda}}$, and its components are x and $-y$, whose real span is again U. The pair $\{\lambda, \overline{\lambda}\}$ corresponds to the plane, not either eigenvalue separately.

This is the mechanism behind section 7.5, stated in the present language. There $Q \in M_n(\mathbb{R})$ was orthogonal, and proposition 7.5.1(2) produced from an eigenvector v of Q for λ the eigenvector $\overline{v}$ for $\overline{\lambda}$. That passage $v \mapsto \overline{v}$ is the conjugation σ_0 of corollary 9.4.5, and the assertion is $\sigma_0(E_\lambda) = E_{\overline{\lambda}}$, which is part (1) above for the operator $x \mapsto Qx$. Lemma 7.5.2 then went further than part (3) does, and it went further by using the inner product: writing $\lambda = \cos\theta + i \sin\theta$ and normalizing v to $\|v\| = 1$, the rescaled vectors $u_1 = \sqrt{2}\, x$ and $u_2 = -\sqrt{2}\, y$ are orthonormal, which is lemma 7.5.2(1); the plane they span is then an orthogonal summand. The matrix $\left(\begin{smallmatrix} a & b \\ -b & a \end{smallmatrix}\right)$ of part (3) is taken in the basis (x, y), and the passage to (u_1, u_2) reverses the sign of the second basis vector, hence the sign of both off-diagonal entries: in the orthonormal basis the matrix is the transpose $\left(\begin{smallmatrix} a & -b \\ b & a \end{smallmatrix}\right)$, which for $a = \cos\theta$ and $b = \sin\theta$ is the rotation matrix of lemma 7.5.2(2). That refinement belongs to lemma 7.5.2 and is not reproved here, and theorem 7.5.3 assembles the resulting planes into the real normal form. What the present section gives is the part of the argument that never used the metric: for an arbitrary real operator, a non-real eigenvalue of the complexification is attached to a real invariant plane.

Example 9.4.7: A Real Matrix with a Conjugate Pair of Eigenvalues

Let

$$A = \begin{pmatrix} 1 & -1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \in M_3(\mathbb{R})$$

and let $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ be $x \mapsto Ax$. By proposition 9.4.2(4) the complexification $T_{\mathbb{C}}$ is the map $z \mapsto Az$ on $\mathbb{C}^3$.

The characteristic polynomial. The last row of $tI_3 - A$ is $(0, 0, t-2)$, so expanding along it with theorem 3.3.4 leaves one term, and evaluating the remaining 2×2 determinant as $ad - bc$,

$$\chi_A(t) = (t-2) \det \begin{pmatrix} t-1 & 1 \\ -1 & t-1 \end{pmatrix} = (t-2)\big((t-1)^2 + 1\big)$$

so the roots of χ_A in $\mathbb{C}$ are 2, $1+i$ and $1-i$. Only 2 is real, so by proposition 9.4.3(3) the operator T has exactly one eigenvalue in $\mathbb{R}$, namely 2, and $Ae_3 = 2e_3$ exhibits an eigenvector for it.

An eigenvector for $\lambda = 1+i$. Solving $(A - \lambda I_3)v = 0$ means solving

$$\begin{pmatrix} -i & -1 & 0 \\ 1 & -i & 0 \\ 0 & 0 & 1-i \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

The third equation is $(1-i)v_3 = 0$ with $1 - i \neq 0$, so $v_3 = 0$. The first is $-iv_1 - v_2 = 0$, so $v_2 = -iv_1$. Taking $v_1 = 1$ gives $v = (1, -i, 0)^\top$, and the second equation holds as well, since $1 + (-i)(-i) = 1 + i^2 = 0$. So v is a nonzero element of E_{1+i}, and conjugating its entries gives $\overline{v} = (1, i, 0)^\top$, which proposition 9.4.6(1) places in E_{1-i}. Directly,

$$A\overline{v} = \begin{pmatrix} 1-i \\ 1+i \\ 0 \end{pmatrix} = (1-i) \begin{pmatrix} 1 \\ i \\ 0 \end{pmatrix}$$

since $(1-i)i = i - i^2 = 1 + i$.

The real plane. The components of v are $x = (1,0,0)^\top = e_1$ and $y = (0,-1,0)^\top = -e_2$, and with $a = b = 1$ part (3) of the same proposition predicts that $U = \operatorname{span}_{\mathbb{R}}\{x, y\}$, which is the coordinate plane $\operatorname{span}_{\mathbb{R}}\{e_1, e_2\}$, is carried into itself by T, with matrix $\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ in the ordered basis (x, y). Both assertions are checked directly:

$$Tx = Ae_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = x - y, \qquad Ty = -Ae_2 = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} = x + y$$

whose coordinate columns in the basis (x, y) are $(1, -1)^\top$ and $(1, 1)^\top$, the two columns of that matrix.

So T carries both U and the line $\operatorname{span}_{\mathbb{R}}\{e_3\}$ into themselves, the line consists of eigenvectors for the eigenvalue 2, and the plane contains no eigenvector at all. Over $\mathbb{C}$ the situation is different: the three vectors v, $\overline{v}$ and e_3 are eigenvectors of $T_{\mathbb{C}}$ for the three distinct eigenvalues $1+i$, $1-i$ and 2, hence are linearly independent (theorem 4.2.3), and three independent vectors in the three-dimensional space $\mathbb{C}^3$ form a basis. So $T_{\mathbb{C}}$ is diagonalizable (theorem 4.2.2).

T itself is not, and the two facts above give the reason once they are put together. Since $e_3 \notin U$, the plane U and the line $\operatorname{span}_{\mathbb{R}}\{e_3\}$ meet only at 0 and together span $\mathbb{R}^3$, so $\mathbb{R}^3 = U \oplus \operatorname{span}_{\mathbb{R}}\{e_3\}$ (definition 2.1.11). Let $Tw = \lambda w$ with $w \neq 0$ real, and write $w = u + ce_3$ with $u \in U$ and $c \in \mathbb{R}$. Both summands are carried into their own subspace by T, so $Tw = Tu + 2ce_3$ with $Tu \in U$, while $\lambda w = \lambda u + \lambda ce_3$; the uniqueness of the decomposition (proposition 2.1.12(2)) forces $Tu = \lambda u$ and $2c = \lambda c$. If $u \neq 0$ then u is an eigenvector of T lying in U, which the previous paragraph excludes; so $u = 0$ and w is a multiple of e_3. Every real eigenvector of T therefore lies on a single line, and $\mathbb{R}^3$ has no basis of them, so T is not diagonalizable over $\mathbb{R}$ (theorem 4.2.2).

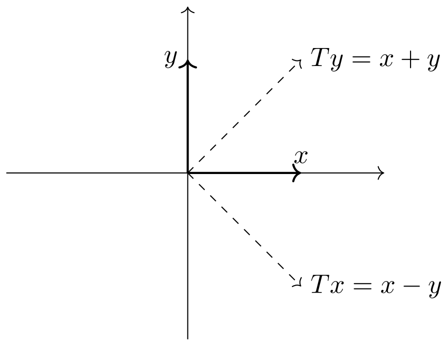


Figure 9.5: The invariant plane U of example 9.4.7, drawn in the basis (x, y), with the images of the two basis vectors; no line through the origin is carried into itself

Exercise 9.4.1

1. Let $A = \left(\begin{smallmatrix} 3 & -2 \\ 4 & -1 \end{smallmatrix}\right) \in M_2(\mathbb{R})$ and let $T\colon \mathbb{R}^2 \to \mathbb{R}^2$ be $x \mapsto Ax$. Compute χ_A and its roots in $\mathbb{C}$. For the root $\lambda = a + bi$ with $b > 0$, find a nonzero $v \in \mathbb{C}^2$ with $Av = \lambda v$, write $v = x + iy$ with $x, y \in \mathbb{R}^2$, and verify by direct computation that the matrix of T in the ordered basis (x, y) of $\mathbb{R}^2$ is $\left(\begin{smallmatrix} a & b \\ -b & a \end{smallmatrix}\right)$, as proposition 9.4.6(3) asserts.

2. For each of the following matrices, decide whether the operator $z \mapsto Sz$ on $\mathbb{C}^2$ is the complexification of a linear operator on $\mathbb{R}^2$, and when it is, name that operator:
$$S_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad S_2 = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad S_3 = \begin{pmatrix} 1 & 1+i \\ 1-i & 1 \end{pmatrix}, \quad S_4 = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$
For each one that is not, exhibit a single column $z \in \mathbb{C}^2$ with $\overline{Sz} \neq S\overline{z}$.

3. Let V be a vector space over $\mathbb{R}$ with $\dim_{\mathbb{R}} V = n \geq 1$ and let $T\colon V \to V$ be linear. Show that T is a bijection if and only if $T_{\mathbb{C}}$ is, and that in that case $(T^{-1})_{\mathbb{C}} = (T_{\mathbb{C}})^{-1}$.

4. Let V be a vector space over $\mathbb{R}$, let $T\colon V \to V$ be linear, and let U be a subspace of V. Write $U_{\mathbb{C}} = \{\, u + iv \,|\, u, v \in U \,\}$, a subset of $V_{\mathbb{C}}$. Show that $U_{\mathbb{C}}$ is a complex subspace of $V_{\mathbb{C}}$, that $U_{\mathbb{C}} \cap V = U$, and that $T(U) \subseteq U$ holds if and only if $T_{\mathbb{C}}(U_{\mathbb{C}}) \subseteq U_{\mathbb{C}}$.

5. Let V be a vector space over $\mathbb{R}$ with $\dim_{\mathbb{R}} V = n \geq 1$, let $\mathcal{B}$ be an ordered basis of V over $\mathbb{R}$, read as an ordered basis of $V_{\mathbb{C}}$ over $\mathbb{C}$, and let $S\colon V_{\mathbb{C}} \to V_{\mathbb{C}}$ be $\mathbb{C}$-linear with matrix $[S]_{\mathcal{B}} \in M_n(\mathbb{C})$. Show that S commutes with the conjugation σ of $V_{\mathbb{C}}$ if and only if every entry of $[S]_{\mathcal{B}}$ is real.

6. Let V be a finite-dimensional vector space over $\mathbb{R}$ carrying a complex structure J (definition 9.3.1), and let $T\colon V \to V$ be linear. Show that $T \circ J = J \circ T$ if and only if $T_{\mathbb{C}}(V^{1,0}) \subseteq V^{1,0}$, where $V^{1,0}$ is the holomorphic subspace of definition 9.3.6.

7. Let $\theta \in \mathbb{R}$ with $\sin\theta \neq 0$ and let $Q = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right) \in M_2(\mathbb{R})$. Compute χ_Q and show that its roots in $\mathbb{C}$ are $\cos\theta \pm i\sin\theta$. Find a $v \in \mathbb{C}^2$ with $Qv = (\cos\theta + i\sin\theta)v$ and $\|v\| = 1$, and check that its components x and y satisfy $\|x\| = \|y\|$ and $\langle x, y\rangle = 0$, as proposition 7.5.1(3) requires. Then compute the vectors $u_1 = \sqrt{2}\,x$ and $u_2 = -\sqrt{2}\,y$ of lemma 7.5.2 and identify the matrix of $x \mapsto Qx$ in the ordered basis (u_1, u_2).

10

Hermitian and Symplectic Forms; the Compatibility Triple

Chapter 8 classified the symmetric bilinear forms and chapter 9 gave the complex structures. Two things were left out. A form on a complex space that is linear in one argument and conjugate-linear in the other was named in chapter 8 and deferred; and a skew form, which the classification there treated only as the half of a bilinear form that the symmetric part discards, was never studied on its own. This chapter takes both, and then puts them on one space together with a complex structure.

What emerges is that the three are not independent. A Hermitian form on a complex space, viewed through real eyes, is a symmetric form and a skew form that determine each other through multiplication by i; and conversely a symmetric form, a skew form and a complex structure that fit together in the right way assemble into a single Hermitian form. Three structures turn out to be two pieces of information, and which two is a matter of convenience.

Section 10.1 classifies Hermitian forms, finding the same three integers that governed the real symmetric case, and extracts the real and imaginary parts that the rest of the chapter runs on. Section 10.2 studies a nondegenerate skew form by itself and finds it has no invariants at all beyond the dimension: every such form looks like every other in a suitable basis. Section 10.3 locates what a symplectic space does have, namely its subspaces on which the form vanishes identically, which have no analogue for an inner product. Section 10.4 assembles the three structures, singles out the positive ones, and identifies the parameter space they sweep out.

10.1 Hermitian Forms and Their Classification; the Real and Imaginary Parts

Chapter 8 classified real symmetric bilinear forms by three integers, and the same classification collapses over $\mathbb{C}$. By corollary 8.3.5 a complex symmetric matrix is determined up to congruence by its rank alone, because the substitution $v \mapsto iv$ multiplies a diagonal entry by $i^2 = -1$ and so turns a -1 into a $+1$. Nothing survives that a sign could be attached to.

The conjugation restores what is lost. A form that is conjugate-symmetric rather than symmetric takes real values on the diagonal, real numbers have signs, and the counting argument behind Sylvester's law runs again. This section defines those forms, proves the classification, and then splits one into a real and an imaginary part, a decomposition section 10.4 takes as its starting data.

Definition 10.1.1: Hermitian Forms, Their Matrices, and $*$-Congruence

Let V be a finite-dimensional vector space over $\mathbb{C}$. A *Hermitian form* on V is a function $h\colon V \times V \to \mathbb{C}$ such that, for all $u, u', v \in V$ and all $\lambda \in \mathbb{C}$,

1. (linearity in the first argument) $h(u+u', v) = h(u,v) + h(u', v)$ and $h(\lambda u, v) = \lambda\, h(u,v)$;
2. (conjugate symmetry) $h(v,u) = \overline{h(u,v)}$.

Write $q_h(v) = h(v,v)$ for the function recording the values of h on a repeated argument. Let $\mathcal{B} = (b_1, \ldots, b_n)$ be an ordered basis of V. The *matrix of h relative to $\mathcal{B}$* is

$$[h]_{\mathcal{B}} = \big(h(b_j, b_i)\big) \in M_n(\mathbb{C})$$

the matrix whose entry in row i and column j is $h(b_j, b_i)$, the two arguments taken in the order opposite to that of definition 8.1.2.
Two matrices $A, A' \in M_n(\mathbb{C})$ are *$*$-congruent* when there is an invertible $P \in M_n(\mathbb{C})$ with $A' = P^*AP$, where P^* is the conjugate transpose of definition 5.5.12.

The reversed index order is the one under which the evaluation formula and the change of basis rule of theorem 10.1.3 both come out with a conjugate transpose on the left and the matrix itself on the right. Running the computation in the proof of that theorem with the order of definition 8.1.2 instead produces $Q^\top [h]_{\mathcal{B}} \overline{Q}$, where the conjugation sits on one factor and not the other. Nothing else distinguishes the two conventions, and for a symmetric bilinear form, where no conjugate appears, they agree.

The standard inner product of definition 5.1.2 on $\mathbb{C}^n$ is the first example. Its two conditions (1) and (2) there are exactly (1) and (2) here, so $h_0(x,y) = x_1\overline{y_1} + \cdots + x_n\overline{y_n}$ is a Hermitian form on $\mathbb{C}^n$. A Hermitian form is an inner product precisely when it satisfies the third condition of definition 5.1.2 as well. In matrix language, $h_0(x,y) = y^*x$ by definition 1.4.1 and definition 5.5.12, and $h_0(e_j, e_i) = e_i^* e_j$ is 1 for $i = j$ and 0 otherwise, so $[h_0]_{\mathcal{E}} = I_n$ in the standard basis $\mathcal{E}$.

A second example carries a matrix that is not the identity. Put

$$H = \begin{pmatrix} 2 & 1-i \\ 1+i & 3 \end{pmatrix} \in M_2(\mathbb{C}), \qquad h(x,y) = y^*Hx$$

The conjugate of H has entries $2, 1+i, 1-i, 3$ in the same positions (definition 5.5.10), and transposing exchanges the two off-diagonal entries and returns H, so $H^* = H$ and h is a Hermitian form by theorem 10.1.3(2) below. Expanding the product with definition 1.4.1,

$$q_h(x) = x^*Hx = 2|x_1|^2 + 3|x_2|^2 + (1-i)\overline{x_1}x_2 + (1+i)\overline{x_2}x_1$$

and the last two terms are conjugates of one another, so their sum is $2\mathrm{Re}\big((1-i)\overline{x_1}x_2\big)$ and $q_h(x)$ is real. At $x = (1,1)^\top$ the value is $2 + 3 + 2\mathrm{Re}(1-i) = 7$. The matrix $\mathrm{diag}(i,i)$ is symmetric in the sense of definition 8.1.9 and is not Hermitian, so the two conditions on a complex matrix are different conditions.

Both examples took real values on the diagonal, and this is forced.

Proposition 10.1.2: Conjugate-Linearity in the Second Argument, and the Reality of $h(v,v)$

Let h be a Hermitian form on a complex vector space V. For all $u, v, v' \in V$ and all $\lambda \in \mathbb{C}$:

1. $h(u, v+v') = h(u,v) + h(u,v')$ and $h(u, \lambda v) = \overline{\lambda}\, h(u,v)$;
2. $q_h(v) = h(v,v)$ is a real number;
3. $q_h(\lambda v) = |\lambda|^2 q_h(v)$.

***Proof*:**

1. Conjugate symmetry (definition 10.1.1(2)) and additivity in the first argument (definition 10.1.1(1)) give

$$h(u, v+v') = \overline{h(v+v', u)} = \overline{h(v,u) + h(v',u)} = \overline{h(v,u)} + \overline{h(v',u)} = h(u,v) + h(u,v')$$

the third equality because conjugation of complex numbers carries sums to sums. The same two facts, with homogeneity in place of additivity, give $h(u,\lambda v) = \overline{h(\lambda v, u)} = \overline{\lambda\, h(v,u)} = \overline{\lambda}\,\overline{h(v,u)} = \overline{\lambda}\, h(u,v)$.

2. Conjugate symmetry applied with $u = v$ reads $h(v,v) = \overline{h(v,v)}$, and a complex number equal to its own conjugate has zero imaginary part.

3. Using definition 10.1.1(1) on the first argument and (1) above on the second, $h(\lambda v, \lambda v) = \lambda\, h(v, \lambda v) = \lambda\overline{\lambda}\, h(v,v) = |\lambda|^2 q_h(v)$, as we sought to show.

Clause (1) is the reason a Hermitian form is not a bilinear form and the apparatus of chapter 8 cannot be applied to it unchanged: the second argument absorbs a scalar with a conjugate attached. Clause (2) is what makes the classification below possible. A real symmetric form takes real values at every pair, so its quadratic form does too. A Hermitian form takes complex values off the diagonal, and clause (2) says that on the diagonal it does not, which is what allows an inequality $q_h(v) > 0$ to be written at all.

The correspondence between forms and matrices comes next, and it converts a change of basis into an operation on matrices, exactly as theorem 8.1.4 did for bilinear forms.

Theorem 10.1.3: Hermitian Forms and Hermitian Matrices Correspond; Change of Basis Is $*$-Congruence

Let V be a complex vector space with $\dim_{\mathbb{C}} V = n \geq 1$ and let $\mathcal{B} = (b_1, \ldots, b_n)$ be an ordered basis of V. Write $[u]_{\mathcal{B}} \in \mathbb{C}^n$ for the coordinate column of u relative to $\mathcal{B}$.

1. Let h be a Hermitian form on V and $H = [h]_{\mathcal{B}}$. Then $H^* = H$, and for all $u, v \in V$

$$h(u, v) = [v]_{\mathcal{B}}^* \, H \, [u]_{\mathcal{B}}$$

the right side being a 1×1 matrix, identified with its single entry. In particular $q_h(u) = [u]_{\mathcal{B}}^* H [u]_{\mathcal{B}}$.

2. The assignment $h \mapsto [h]_{\mathcal{B}}$ is a bijection from the set of Hermitian forms on V onto the set of Hermitian matrices in $M_n(\mathbb{C})$, that is, onto $\{H \in M_n(\mathbb{C}) \middle| H^* = H\}$ (definition 7.1.1). For $V = \mathbb{C}^n$ with the standard basis $\mathcal{E}$ the inverse assignment sends H to $h_H(x, y) = y^* H x$.

3. Let $\mathcal{C} = (c_1, \ldots, c_n)$ be a second ordered basis of V and let Q be the matrix whose j-th column is $[c_j]_{\mathcal{B}}$, the change of basis matrix of definition 2.5.2. Then Q is invertible and

$$[h]_{\mathcal{C}} = Q^* \, [h]_{\mathcal{B}} \, Q$$

4. Let $H' \in M_n(\mathbb{C})$ be Hermitian and let $S \in M_n(\mathbb{C})$ be invertible with columns $w_1, \ldots, w_n$. Then $\mathcal{W} = (w_1, \ldots, w_n)$ is an ordered basis of $\mathbb{C}^n$ and $[h_{H'}]_{\mathcal{W}} = S^* H' S$. Consequently two Hermitian matrices in $M_n(\mathbb{C})$ are $*$-congruent exactly when they are the matrices of one Hermitian form on $\mathbb{C}^n$ in two bases.

***Proof*:**

1. By definition 5.5.12 the entry of H^* in row i and column j is $\overline{H_{ji}}$, which is $\overline{h(b_i, b_j)}$ by definition 10.1.1, and conjugate symmetry turns that into $h(b_j, b_i) = H_{ij}$. So $H^* = H$.

   For the evaluation, write $x = [u]_{\mathcal{B}}$ and $y = [v]_{\mathcal{B}}$, so that $u = \sum_j x_j b_j$ and $v = \sum_i y_i b_i$. Expanding the first argument by definition 10.1.1(1) and then the second by proposition 10.1.2(1),

$$h(u, v) = \sum_{j=1}^{n} x_j \, h(b_j, v) = \sum_{i=1}^{n} \sum_{j=1}^{n} \overline{y_i} \, h(b_j, b_i) \, x_j = \sum_{i=1}^{n} \sum_{j=1}^{n} \overline{y_i} \, H_{ij} \, x_j$$

   By definition 1.4.1 the i-th entry of Hx is $\sum_j H_{ij} x_j$, and the entries of the row y^* are $\overline{y_1}, \ldots, \overline{y_n}$ (definition 5.5.12), so the product $y^*(Hx)$ is the double sum just computed.

2. Injectivity is immediate from (1): two Hermitian forms with the same matrix take the same value at every pair (u, v).

   For surjectivity, let $H \in M_n(\mathbb{C})$ satisfy $H^* = H$ and define $h(u, v) = [v]_{\mathcal{B}}^* H [u]_{\mathcal{B}}$. Coordinates are additive and homogeneous, since expanding u and u' in $\mathcal{B}$ and adding produces an expansion of $u + u'$, and the expansion of a vector in a basis is unique (definition 2.2.1). So $[u + u']_{\mathcal{B}} = [u]_{\mathcal{B}} + [u']_{\mathcal{B}}$ and $[\lambda u]_{\mathcal{B}} = \lambda [u]_{\mathcal{B}}$. Matrix multiplication is additive and homogeneous in the right factor by definition 1.4.1, which gives definition 10.1.1(1). For conjugate symmetry, the number $h(u, v)$ is a 1×1 matrix, and the conjugate transpose of a 1×1 matrix is its entrywise conjugate (definition 5.5.12), so

$$\overline{h(u,v)} = \left([v]_{\mathcal{B}}^* H [u]_{\mathcal{B}}\right)^* = [u]_{\mathcal{B}}^* H^* [v]_{\mathcal{B}} = [u]_{\mathcal{B}}^* H [v]_{\mathcal{B}} = h(v,u)$$

using proposition 5.5.13(3) twice, then proposition 5.5.13(1) on the factor $([v]_{\mathcal{B}}^*)^*$, and then $H^* = H$. Finally $[b_j]_{\mathcal{B}} = e_j$, so the entry of $[h]_{\mathcal{B}}$ in row i and column j is $h(b_j, b_i) = e_i^* H e_j = H_{ij}$ by definition 1.4.1. On $\mathbb{C}^n$ with $\mathcal{B} = \mathcal{E}$ the coordinate column of x is x itself, so the form just constructed is $h_H(x, y) = y^* H x$.

3. Let R be the matrix whose j-th column is $[b_j]_{\mathcal{C}}$. For any $w \in V$ with $[w]_{\mathcal{C}} = z$ one has $w = \sum_j z_j c_j$, so $[w]_{\mathcal{B}} = \sum_j z_j [c_j]_{\mathcal{B}}$ by additivity and homogeneity of coordinates, and the column reading of a matrix product (definition 1.4.1) identifies that combination as Qz. Applying this to $w = b_j$ gives $Q[b_j]_{\mathcal{C}} = e_j$, so $QR = I_n$. The same argument with the two bases exchanged gives $RQ = I_n$, and Q is invertible with $Q^{-1} = R$ (definition 1.4.4).

   Write $H = [h]_{\mathcal{B}}$ and let q_{si} be the entry of Q in row s and column i, so that $[c_i]_{\mathcal{B}}$ has entries $q_{1i}, \ldots, q_{ni}$. By definition 1.4.1 and definition 5.5.12 the entry of $Q^* H Q$ in row i and column j is

$$\sum_{s=1}^{n} \sum_{t=1}^{n} \overline{q_{si}}\, H_{st}\, q_{tj} = [c_i]_{\mathcal{B}}^*\, H\, [c_j]_{\mathcal{B}}$$

   and (1) evaluates the right side as $h(c_j, c_i)$, which is the entry of $[h]_{\mathcal{C}}$ in row i and column j by definition 10.1.1.

4. The columns of an invertible S span $\mathbb{C}^n$, since every $z \in \mathbb{C}^n$ equals $S(S^{-1}z)$ and the column reading of the product exhibits Sx as the combination of the w_j with coefficients x_j. They are linearly independent, since $x_1 w_1 + \cdots + x_n w_n = Sx = 0$ gives $x = S^{-1}(Sx) = 0$. So $\mathcal{W}$ is a basis. Its j-th vector has $[w_j]_{\mathcal{E}} = w_j$, the j-th column of S, so the change of basis matrix from $\mathcal{W}$ to $\mathcal{E}$ is S, and (3) applied to $h_{H'}$, whose matrix in $\mathcal{E}$ is H' by (2), gives $[h_{H'}]_{\mathcal{W}} = S^* H' S$.

   For the last sentence, suppose H and H' are Hermitian with $H' = P^* H P$ and P invertible. Then H and H' are the matrices of h_H in the bases $\mathcal{E}$ and $\mathcal{P}$, the latter being the columns of P, by the previous paragraph applied with $S = P$. Conversely, if H and H' are the matrices of one Hermitian form in two bases then (3) exhibits an invertible Q with $H' = Q^* H Q$, completing the proof.

Read against theorem 8.1.4, the theorem says that the operation a change of basis performs on the matrix of a Hermitian form is $H \mapsto P^* H P$ and not $H \mapsto P^\top H P$. Over $\mathbb{R}$ the two coincide (definition 5.5.12), and over $\mathbb{C}$ they do not, so chapter 8's classification of complex symmetric matrices under $P^\top A P$ says nothing about the present question. That classification is by rank alone (corollary 8.3.5), while the classification proved below is by three integers.

Two consequences of the definition of $*$-congruence are used without further comment. It is symmetric, since $H' = P^* H P$ gives $H = (P^{-1})^* H' P^{-1}$ by proposition 5.5.13(4) and proposition 1.4.3, and it is transitive, since $R^*(P^* H P)R = (PR)^* H (PR)$ by proposition 5.5.13(3), with PR invertible.

Before the classification, one identity is needed: a Hermitian form is determined by the single-variable function q_h. The corresponding statement for real symmetric forms is theorem 8.2.2, and there three evaluations of q sufficed. Over $\mathbb{C}$ the two cross terms in the expansion of $q_h(u + v)$ are conjugates of each other rather than equal, so they recover only the real part of $h(u,v)$, and a second family of evaluations is needed for the imaginary part.

Theorem 10.1.4: Recovery of a Hermitian Form from Its Diagonal Values

Let h be a Hermitian form on a complex vector space V and write $q = q_h$.

1. For all $u, v \in V$ and all $\lambda \in \mathbb{C}$,

$$q(u + \lambda v) = q(u) + \overline{\lambda}\, h(u,v) + \lambda\, \overline{h(u,v)} + |\lambda|^2 q(v) \tag{10.1}$$

2. For all $u, v \in V$,

$$h(u,v) = \tfrac{1}{4}\Big(q(u+v) - q(u-v) + i\,q(u+iv) - i\,q(u-iv)\Big)$$

3. If h and h' are Hermitian forms on V with $q_h = q_{h'}$ then $h = h'$. In particular, if $H, H' \in M_n(\mathbb{C})$ are Hermitian and $x^*Hx = x^*H'x$ for every $x \in \mathbb{C}^n$, then $H = H'$.

***Proof*:**

1. Expanding the first argument by definition 10.1.1(1) and then each of the two resulting terms in its second argument by proposition 10.1.2(1),

$$h(u + \lambda v, u + \lambda v) = h(u,u) + \overline{\lambda}\, h(u,v) + \lambda\, h(v,u) + \lambda\overline{\lambda}\, h(v,v)$$

and $h(v,u) = \overline{h(u,v)}$ by definition 10.1.1(2), while $\lambda\overline{\lambda} = |\lambda|^2$.

2. Substitute $\lambda = i^k$ in (10.1), multiply by i^k, and sum over $k = 0,1,2,3$. Since $|i^k| = 1$ and $\overline{i^k} = i^{-k}$, the sum splits as

$$\sum_{k=0}^{3} i^k\, q(u + i^k v) = \big(q(u) + q(v)\big)\sum_{k=0}^{3} i^k + h(u,v)\sum_{k=0}^{3} i^k i^{-k} + \overline{h(u,v)}\sum_{k=0}^{3} i^{2k}$$

Now $\sum_{k=0}^{3} i^k = 1 + i - 1 - i = 0$, the second sum is $1+1+1+1 = 4$, and $\sum_{k=0}^{3} i^{2k} = 1 - 1 + 1 - 1 = 0$, so the right side is $4h(u,v)$. On the left, $i^0 = 1$ and $i^0 v = v$ give the term $q(u+v)$; $k = 1$ gives $i\,q(u+iv)$; $i^2 = -1$ gives $-q(u-v)$; and $i^3 = -i$ gives $-i\,q(u-iv)$. Dividing by 4 gives the stated formula.

3. Every term on the right of the formula in (2) is a value of q, so $q_h = q_{h'}$ forces $h(u,v) = h'(u,v)$ for all u and v. For the matrix statement, let $h = h_H$ and $h' = h_{H'}$ be the Hermitian forms on $\mathbb{C}^n$ of theorem 10.1.3(2). Their diagonal values are $q_h(x) = x^*Hx$ and $q_{h'}(x) = x^*H'x$ by theorem 10.1.3(1), so the hypothesis says $q_h = q_{h'}$, whence $h = h'$, and theorem 10.1.3(2) is a bijection, so $H = H'$, completing the proof.

The two families of evaluations in (2) do different jobs. Setting $\lambda = 1$ in (10.1) isolates $h(u,v) + \overline{h(u,v)}$, which is twice the real part of $h(u,v)$, and setting $\lambda = i$ isolates the imaginary part. The weighted sum in the proof combines the two into a single formula for h. Over $\mathbb{R}$ the conjugation is the identity, the two cross terms coincide, and theorem 8.2.2 needs only the first evaluation.

The classification can now be proved. Its statement is that of theorem 8.3.1 with $\mathbb{C}$ in place of $\mathbb{R}$, and the two integers it produces are again counts of positive and negative diagonal entries. What has to be checked at each step is that the conjugate does not disturb the argument, and proposition 10.1.2(2) is what guarantees it: the diagonal entries being compared are real numbers.

Theorem 10.1.5: Sylvester's Law of Inertia for Hermitian Forms

Let V be a complex vector space with $\dim_{\mathbb{C}} V = n \geq 1$, let h be a Hermitian form on V and let $q = q_h$, a real-valued function by proposition 10.1.2(2). For a complex subspace $U \subseteq V$, the phrase $q > 0$ *on* $U \setminus \{0\}$ means that $q(u) > 0$ for every nonzero $u \in U$, and $q < 0$ *on* $U \setminus \{0\}$ is read the same way.

1. There are integers $p, m \geq 0$ with $p + m \leq n$ and a basis $v_1, \ldots, v_n$ of V in which the matrix of h is
$$\operatorname{diag}(\underbrace{1, \ldots, 1}_{p}, \underbrace{-1, \ldots, -1}_{m}, \underbrace{0, \ldots, 0}_{n-p-m})$$
2. Let $u_1, \ldots, u_n$ be any basis of V in which the matrix of h is diagonal, and let a and b be the number of positive and the number of negative entries on that diagonal. Then a is the largest dimension of a complex subspace $U \subseteq V$ with $q > 0$ on $U \setminus \{0\}$, and b is the largest dimension of a complex subspace $W \subseteq V$ with $q < 0$ on $W \setminus \{0\}$. In particular $a = p$ and $b = m$, so the two counts are the same for every basis in which the matrix of h is diagonal.

The diagonal entries in (2) are the numbers $h(u_i, u_i)$ by definition 10.1.1, hence real by proposition 10.1.2(2), so counting the positive and the negative ones is meaningful. Call a basis $v_1, \ldots, v_n$ with $h(v_i, v_j) = 0$ for $i \neq j$ an *h-orthogonal* basis. By definition 10.1.1 these are exactly the bases in which the matrix of h is diagonal.

***Proof*:**

1. An h-orthogonal basis exists, by induction on n. For $n = 1$ any basis v_1 works, the condition being vacuous.

   Let $n \geq 2$ and assume the statement for complex vector spaces of dimension $n - 1$. If $q(v) = 0$ for every $v \in V$, then theorem 10.1.4(2) writes every value of h as a combination of values of q and gives $h = 0$. Any basis of V, one of which exists by theorem 2.2.9, is then h-orthogonal. Otherwise fix $v_1 \in V$ with $d_1 := q(v_1) \neq 0$, a nonzero real number by proposition 10.1.2(2).

   The function $f\colon V \to \mathbb{C}$ defined by $f(x) = h(x, v_1)$ is linear (definition 2.4.2) by definition 10.1.1(1). Its image is a subspace of $\mathbb{C}$ (definition 2.4.5), so $\dim \operatorname{im} f \leq 1$ by corollary 2.2.12, while $f(v_1) = d_1 \neq 0$ makes $\operatorname{im} f$ nonzero and so $\dim \operatorname{im} f \geq 1$. Hence $\dim \operatorname{im} f = 1$, and theorem 2.4.10 gives $\dim W = n - 1$ for $W = \ker f$ (definition 2.4.4). Conjugate symmetry turns the defining condition of W into $h(v_1, w) = \overline{h(w, v_1)} = 0$, so h pairs v_1 with a vector of W to 0 in either order.

   The restriction of h to $W \times W$ satisfies the two conditions of definition 10.1.1, being their restriction, so it is a Hermitian form on W. By the inductive hypothesis W has an h-orthogonal basis $v_2, \ldots, v_n$. The list $v_1, v_2, \ldots, v_n$ is linearly independent: applying f to a relation $\lambda_1 v_1 + \cdots + \lambda_n v_n = 0$ and using $f(v_j) = 0$ for $j \geq 2$ leaves $\lambda_1 d_1 = 0$, hence $\lambda_1 = 0$, and then $\lambda_2 = \cdots = \lambda_n = 0$ by the independence of $v_2, \ldots, v_n$. Its span is therefore a subspace of V of dimension n (definition 2.2.1, definition 2.2.10), so it is V by corollary 2.2.12, and the list is a basis of V. It is h-orthogonal: $h(v_1, v_j) = 0$ and $h(v_j, v_1) = 0$ for $j \geq 2$ because $v_j \in W$, and $h(v_i, v_j) = 0$ for $2 \leq i \neq j \leq n$ by the choice of the basis of W.

Now let $v_1, \ldots, v_n$ be h-orthogonal and put $d_i = h(v_i, v_i) \in \mathbb{R}$. Let p be the number of indices with $d_i > 0$ and m the number with $d_i < 0$. The two index sets are disjoint subsets of $\{1, \ldots, n\}$, so $p + m \leq n$. For a permutation σ of $\{1, \ldots, n\}$ the list $v_{\sigma(1)}, \ldots, v_{\sigma(n)}$ is again a basis, and the entry of the matrix of h in it in row i and column j is $h(v_{\sigma(j)}, v_{\sigma(i)})$, which vanishes for $i \neq j$ and is $d_{\sigma(i)}$ for $i = j$. Choose σ so that the positive d's come first, then the negative ones, then the zeros, and rename so that $d_1, \ldots, d_n$ is itself ordered this way.

For $i \leq p + m$ the real number $|d_i|$ is positive and has a positive square root. Put $w_i = |d_i|^{-1/2} v_i$ for those i and $w_i = v_i$ for $i > p + m$. Each w_i is a nonzero scalar multiple $c_i v_i$, so $\sum_i t_i w_i = \sum_i (t_i c_i) v_i$ shows that $w_1, \ldots, w_n$ spans V and is independent, hence is a basis (definition 2.2.1). Orthogonality survives, since $h(w_i, w_j) = c_i \overline{c_j}\, h(v_i, v_j) = 0$ for $i \neq j$ by definition 10.1.1(1) and proposition 10.1.2(1). For $i \leq p + m$, proposition 10.1.2(3) with the real scalar $\lambda = |d_i|^{-1/2}$ gives

$$h(w_i, w_i) = \frac{1}{|d_i|}\, d_i$$

which is 1 when $d_i > 0$ and -1 when $d_i < 0$, while $h(w_i, w_i) = d_i = 0$ for $i > p + m$. The matrix of h in the basis $w_1, \ldots, w_n$ is the one displayed in the statement.

2. Let $u_1, \ldots, u_n$ be a basis in which the matrix of h is $\operatorname{diag}(c_1, \ldots, c_n)$, put $S_+ = \{i \big| c_i > 0\}$ and $S_- = \{i \big| c_i < 0\}$, so that $a = |S_+|$ and $b = |S_-|$. Expanding the first argument by definition 10.1.1(1) and the second by proposition 10.1.2(1), and discarding the terms with $i \neq j$, a vector $x = \sum_{i=1}^n t_i u_i$ has

$$q(x) = \sum_{i=1}^{n} \sum_{j=1}^{n} t_i \overline{t_j}\, h(u_i, u_j) = \sum_{i=1}^{n} |t_i|^2 c_i \tag{10.2}$$

a real number, as proposition 10.1.2(2) requires. Let N_+ be the largest dimension of a complex subspace $U \subseteq V$ with $q > 0$ on $U \setminus \{0\}$. At least one such subspace exists, namely $U = \{0\}$, for which the condition is vacuous, and every subspace of V has dimension at most n by corollary 2.2.12, so N_+ is a well-defined integer between 0 and n.

The subspace $U_0 = \operatorname{span}\{u_i \big| i \in S_+\}$ has dimension a, a sublist of a basis being independent and hence a basis of its span (definition 2.2.1, definition 2.2.10). For nonzero $x = \sum_{i \in S_+} t_i u_i$ in U_0, equation (10.2) reads $q(x) = \sum_{i \in S_+} |t_i|^2 c_i$, a sum of nonnegative terms at least one of which is positive, so $q(x) > 0$. Hence $N_+ \geq a$.

For the reverse inequality put $W_0 = \operatorname{span}\{u_i \big| i \notin S_+\}$, of dimension $n - a$ by the same argument, on which (10.2) gives $q(y) = \sum_{i \notin S_+} |t_i|^2 c_i \leq 0$, since $c_i \leq 0$ for $i \notin S_+$. Let U be a subspace with $q > 0$ on $U \setminus \{0\}$ and suppose $\dim U > a$. Choose a basis $x_1, \ldots, x_s$ of U and a basis $y_1, \ldots, y_t$ of W_0 (theorem 2.2.9), and note that $s + t > a + (n - a) = n$. The combined list cannot be linearly independent, since an independent list is a basis of its span and that span is a subspace of V, of dimension at most n by corollary 2.2.12. So there are complex scalars $\alpha_1, \dots, \alpha_s, \beta_1, \dots, \beta_t$, not all zero, with $\sum_i \alpha_i x_i + \sum_j \beta_j y_j = 0$, and the vector

$$z = \sum_{i=1}^{s} \alpha_i x_i = -\sum_{j=1}^{t} \beta_j y_j$$

lies in U and in W_0 at once. Were z zero, every α_i would vanish by independence of $x_1, \dots, x_s$ and every β_j by independence of $y_1, \dots, y_t$, against the choice of the scalars. So $z \neq 0$, whence $q(z) > 0$ because $z \in U \setminus \{0\}$ and $q(z) \leq 0$ because $z \in W_0$, which is impossible. Hence $\dim U \leq a$ for every such U, and with the previous paragraph $N_+ = a$.

The function $-h$, defined by $(-h)(x, y) = -h(x, y)$, satisfies the two conditions of definition 10.1.1 and is therefore Hermitian. Its diagonal function is $-q$ and its matrix in the basis $u_1, \dots, u_n$ is $\operatorname{diag}(-c_1, \dots, -c_n)$, whose positive entries are indexed by S_-. The previous two paragraphs applied to $-h$ give $N_- = b$, where N_- is the largest dimension of a subspace on which $-q > 0$ away from 0, that is on which $q < 0$ away from 0.

Neither N_+ nor N_- mentions a basis, so $a = N_+$ and $b = N_-$ hold for every basis in which the matrix of h is diagonal. The basis produced in (1) is one of them and its diagonal carries p positive and m negative entries, so $N_+ = p$ and $N_- = m$, completing the proof.

The argument is that of theorem 8.3.1 with two changes, and they are the only places where the conjugate could have broken it. In (10.2) the coefficient of c_i is $|t_i|^2$ rather than t_i^2, which is what keeps the coefficients nonnegative once the scalars are complex. The real argument would have failed here, since t_i^2 is negative for t_i imaginary. In the rescaling the factor $|d_i|^{-1/2}$ is real, so proposition 10.1.2(3) contributes $|\lambda|^2 = \lambda^2 > 0$ and cannot change a sign. Everything else is a dimension count, and a dimension count does not see the scalars.

Definition 10.1.6: Inertia and Signature of a Hermitian Form

Let h be a Hermitian form on a complex vector space V with $\dim_{\mathbb{C}} V = n \geq 1$ and let p and m be the integers of theorem 10.1.5. The *inertia* of h is the triple

$$(n_+, n_-, n_0) = (p,\ m,\ n - p - m)$$

of the number of positive, negative and zero entries in any diagonal representation of h, and the *signature* of h is the integer $\sigma(h) = n_+ - n_-$. The same three integers are attached to a Hermitian matrix $H \in M_n(\mathbb{C})$ through the form $h_H(x, y) = y^* H x$ on $\mathbb{C}^n$ of theorem 10.1.3(2), and are written $\big(n_+(H), n_-(H), n_0(H)\big)$ and $\sigma(H)$.

The words *inertia* and *signature* were already attached to a real symmetric form in definition 8.3.2, and the two usages are separate: a real symmetric matrix has an inertia in the sense of definition 8.3.2 and a Hermitian matrix has one in the sense above, and a real symmetric matrix, being Hermitian, has both. That they agree is seen by starting on the real side. Let $B \in M_n(\mathbb{R})$ be symmetric. Theorem 8.3.1(1) gives a basis of $\mathbb{R}^n$ diagonalizing φ_B, with real diagonal entries $c_1, \dots, c_n$. Those same n vectors are a basis of $\mathbb{C}^n$ over $\mathbb{C}$ (theorem 9.1.4), and since $B^* = B^\top = B$ over $\mathbb{R}$ the value $h_B(v_i, v_j) = v_j^* B v_i$ agrees with $\varphi_B(v_i, v_j)$ on real vectors; so the same basis is h_B-orthogonal with the same real diagonal entries. Theorem 10.1.5(2) computes the Hermitian inertia from those entries, and it is the inertia that theorem 8.3.1 attaches to the same list. The argument runs in this direction and not the other, because theorem 10.1.5(1) produces a basis of $\mathbb{C}^n$ that need not be real.

Corollary 10.1.7: Hermitian Matrices Are Classified by Their Inertia

Let $n \geq 1$ and let $H, H' \in M_n(\mathbb{C})$ be Hermitian. Then H and H' are $*$-congruent if and only if

$$\big(n_+(H), n_-(H), n_0(H)\big) = \big(n_+(H'), n_-(H'), n_0(H')\big)$$

Every Hermitian $H \in M_n(\mathbb{C})$ is $*$-congruent to exactly one matrix $\operatorname{diag}(I_p, -I_m, 0)$, namely the one with $p = n_+(H)$ and $m = n_-(H)$. Moreover $n_+(H) + n_-(H) = \operatorname{rank} H$, so $n_0(H) = n - \operatorname{rank} H$.

***Proof*:**

Step 1: every Hermitian matrix is $*$-congruent to a normal form. Let H be Hermitian with inertia (p, m, r). Applying theorem 10.1.5(1) to h_H produces a basis $\mathcal{V}$ of $\mathbb{C}^n$ with $[h_H]_{\mathcal{V}} = D := \operatorname{diag}(I_p, -I_m, 0)$. Since $[h_H]_{\mathcal{E}} = H$ in the standard basis by theorem 10.1.3(2), theorem 10.1.3(3) turns this into $D = R^* H R$ with R the change of basis matrix, which that clause states is invertible. So H is $*$-congruent to D.

Step 2: $*$-congruent matrices have equal inertia. Suppose $H' = P^* H P$ with P invertible, and let $D = R^* H R$ be as in Step 1. Symmetry of $*$-congruence gives $H = (P^{-1})^* H' P^{-1}$, so

$$D = R^*(P^{-1})^* H' P^{-1} R = (P^{-1}R)^* H' (P^{-1}R)$$

by proposition 5.5.13(3). Put $S = P^{-1}R$, which is invertible. By theorem 10.1.3(4) the columns of S are a basis $\mathcal{W}$ of $\mathbb{C}^n$ with $[h_{H'}]_{\mathcal{W}} = S^* H' S = D$. That matrix is diagonal with p positive and m negative entries, so theorem 10.1.5(2) applied to $h_{H'}$ gives $n_+(H') = p = n_+(H)$ and $n_-(H') = m = n_-(H)$. The third counts agree because all three sum to n.

Step 3: equal inertia forces $*$-congruence. If H and H' have the same inertia (p, m, r) then Step 1 gives invertible R and R' with $R^* H R = D = (R')^* H' R'$. Then $H' = ((R')^{-1})^* D (R')^{-1}$ by proposition 5.5.13(4), and substituting $D = R^* H R$ and regrouping with proposition 5.5.13(3) gives $H' = P^* H P$ for $P = R(R')^{-1}$, which is invertible. Together with Step 2 this proves the stated equivalence, and it makes the normal form unique: two matrices $\operatorname{diag}(I_p, -I_m, 0)$ and $\operatorname{diag}(I_{p'}, -I_{m'}, 0)$ are $*$-congruent only when their inertias agree, and theorem 10.1.5(2) reads those inertias off their diagonals as $(p, m, n-p-m)$ and $(p', m', n-p'-m')$.

Step 4: the rank. By definition 1.4.1 the i-th entry of Hx is $e_i^* H x = h_H(x, e_i)$, so $\operatorname{null} H$ (definition 2.3.3) consists of the $x \in \mathbb{C}^n$ with $h_H(x, e_i) = 0$ for $i = 1, \ldots, n$, and by proposition 10.1.2(1) that is the same as $h_H(x, y) = 0$ for every $y \in \mathbb{C}^n$. Let $u_1, \ldots, u_n$ be a basis of $\mathbb{C}^n$ in which the matrix of h_H is diagonal with entries $c_1, \ldots, c_n$, of which p are positive, m negative and $r = n - p - m$ zero. For $x = \sum_i t_i u_i$, expanding the first argument by definition 10.1.1(1) and using $h_H(u_i, u_j) = 0$ for $i \neq j$ gives $h_H(x, u_j) = t_j c_j$. So x lies in $\operatorname{null} H$ exactly when $t_j c_j = 0$ for every j, that is exactly when $t_j = 0$ for every j with $c_j \neq 0$. Hence $\operatorname{null} H = \operatorname{span}\{u_j \,|\, c_j = 0\}$, a subspace of dimension r, and theorem 2.3.11 gives $\operatorname{rank} H = n - r = p + m$, as we sought to show.

The corollary is corollary 8.3.3 with $P^\top$ replaced by P^*, and the count of classes is the same in both: the classes are indexed by pairs (p, m) with $p + m \leq n$. What changed is the field. For a symmetric bilinear form the rescaling $v \mapsto \lambda v$ multiplies a diagonal entry by λ^2, which no real λ can make negative and which $\lambda = i$ does make negative, and that is the collapse corollary 8.3.5 records. For a Hermitian form the same rescaling multiplies the entry by $|\lambda|^2$ (proposition 10.1.2(3)), which is positive for every nonzero $\lambda \in \mathbb{C}$, so the sign is preserved and the invariant survives.

Example 10.1.8: An Indefinite Hermitian Form on $\mathbb{C}^2$

Let

$$H = \begin{pmatrix} 1 & 1+i \\ 1-i & 1 \end{pmatrix} \in M_2(\mathbb{C}), \qquad h(x,y) = y^* H x$$

1. *H is Hermitian.* Its entrywise conjugate (definition 5.5.10) has $1, 1-i, 1+i, 1$ in the four positions, and transposing exchanges the off-diagonal entries and returns H, so $H^* = H$ and h is a Hermitian form by theorem 10.1.3(2).

2. *Two values of opposite sign.* Expanding $x^* H x$ with definition 1.4.1 gives $q_h(e_1) = H_{11} = 1$. For $x = (1, -1+i)^\top$ the product Hx has first entry $1 + (1+i)(-1+i) = 1 + (-1-1) = -1$ and second entry $(1-i) + (-1+i) = 0$, and $x^* = (1, -1-i)$, so $q_h(x) = 1 \cdot (-1) + (-1-i) \cdot 0 = -1$.

3. *The leading principal minors.* Writing Δ_k for the leading principal minors, as in theorem 8.4.4, one has $\Delta_1 = 1$ and $\Delta_2 = \det H = 1 - (1+i)(1-i) = 1 - 2 = -1$ by theorem 3.3.4. The criterion of that theorem is stated for real symmetric matrices and is not applied here; the two numbers are recorded for comparison with (4).

4. *An explicit $*$-congruence.* Run the induction in part (1) of the proof of theorem 10.1.5 with $v_1 = e_1$, for which $d_1 = 1$. The condition $h(w, e_1) = e_1^* H w = 0$ says $w_1 + (1+i) w_2 = 0$, so $v_2 = (-(1+i), 1)^\top$ spans the subspace W of that proof. Then Hv_2 has first entry $-(1+i) + (1+i) = 0$ and second entry $-(1-i)(1+i) + 1 = -2 + 1 = -1$, and $v_2^* = (-(1-i),\ 1)$, so $h(v_2, v_2) = v_2^* H v_2 = -1$. Taking P to be the matrix with columns v_1 and v_2,

$$P = \begin{pmatrix} 1 & -(1+i) \\ 0 & 1 \end{pmatrix}, \qquad P^* H P = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

as theorem 10.1.3(4) predicts and as the product confirms: HP has columns $He_1 = (1, 1-i)^\top$ and $Hv_2 = (0, -1)^\top$, and multiplying on the left by $P^* = \left(\begin{smallmatrix} 1 & 0 \\ -(1-i) & 1 \end{smallmatrix}\right)$ gives first row $(1, 0)$ and second row $\big(-(1-i) + (1-i),\ -1\big) = (0, -1)$. The inertia of H is $(1, 1, 0)$ and $\sigma(H) = 0$.

5. *The eigenvalues.* The operator $x \mapsto Hx$ on $\mathbb{C}^2$ is self-adjoint (definition 7.1.1), so its eigenvalues are real (proposition 7.1.11). They are the λ for which $H - \lambda I_2$ is singular (theorem 4.1.10(1)), hence by corollary 3.2.8 the solutions of $\det(H - \lambda I_2) = (1-\lambda)^2 - (1+i)(1-i) = 0$, that is of $(1-\lambda)^2 = 2$: they are $1 + \sqrt{2}$ and $1 - \sqrt{2}$. Their product is $1 - 2 = -1 = \det H$ and one is positive while the other is negative, matching the inertia found in (4).

Three of the five computations agree on the inertia by different routes. The two values of opposite sign in (2) rule out $n_+ = 2$ and $n_- = 2$ directly, the $*$-congruence in (4) produces the normal form, and the value $\Delta_2 = \det H = -1$ in (3) is what the normal form predicts. The matrix P of (4) is upper triangular with diagonal entries 1, and the entry of P^* in row i and column j is $\overline{P_{ji}}$ (definition 5.5.12), so P^* is lower triangular with diagonal entries $\overline{1} = 1$. By proposition 3.2.6 both determinants are 1, and theorem 3.2.10 then gives $\det H = \det(P^* H P) = \det \operatorname{diag}(1, -1) = -1$. The eigenvalue computation in (5) is the one that does not generalize from what is available here, since nothing proved so far attaches the signs of the roots of a characteristic polynomial to the inertia of a Hermitian matrix. In this instance the two were computed separately and compared.

Hermitian, the Symplectic Form, the Compatibility Triple

What remains is to record what a Hermitian form becomes when the complex scalars are forgotten. A complex vector space V carries a real vector space $V_{\mathbb{R}}$, its realification (definition 9.1.6), obtained by restricting the scalars to $\mathbb{R}$, of dimension $2\dim_{\mathbb{C}} V$ by proposition 9.1.8. Multiplication by i becomes an $\mathbb{R}$-linear operator on $V_{\mathbb{R}}$, and that operator is a complex structure in the sense of definition 9.3.1 by theorem 9.3.4(2), applied to the complex scalar multiplication V already carries. A Hermitian form, being complex-valued, splits into two real-valued forms on $V_{\mathbb{R}}$. The next theorem says which pairs of real forms arise this way and how each of the two determines h. Section 10.2 takes up the second of them on its own, and section 10.4 treats all three structures as independent data on one real space.

Theorem 10.1.9: The Real and Imaginary Parts of a Hermitian Form

Let V be a complex vector space with $\dim_{\mathbb{C}} V = n \geq 1$, let $V_{\mathbb{R}}$ be its realification and let $J\colon V_{\mathbb{R}} \to V_{\mathbb{R}}$ be the complex structure $Jv = iv$. Let h be a Hermitian form on V and define $g, \omega\colon V_{\mathbb{R}} \times V_{\mathbb{R}} \to \mathbb{R}$ by

$$g(v,w) = \mathrm{Re}h(v,w), \qquad \omega(v,w) = \mathrm{Im}h(v,w)$$

so that $h = g + i\omega$.

1. g and ω are bilinear forms on $V_{\mathbb{R}}$ (definition 8.1.1), g is symmetric and ω is skew-symmetric (definition 8.1.9), and both are J-invariant:

$$g(Jv, Jw) = g(v,w), \qquad \omega(Jv,Jw) = \omega(v,w)$$

2. Each of g and ω determines the other, by

$$g(v,w) = \omega(Jv,w), \qquad \omega(v,w) = g(v,Jw) = -g(Jv,w)$$

and each determines h, by

$$h(v,w) = \omega(Jv,w) + i\,\omega(v,w) = g(v,w) - i\,g(Jv,w)$$

3. Conversely, let g_0 be a symmetric bilinear form on $V_{\mathbb{R}}$ with $g_0(Jv,Jw) = g_0(v,w)$ for all v, w. Then $h_0(v,w) = g_0(v,w) - i\,g_0(Jv,w)$ is the unique Hermitian form on V with $\mathrm{Re}h_0 = g_0$. Let ω_0 be a skew-symmetric bilinear form on $V_{\mathbb{R}}$ with $\omega_0(Jv,Jw) = \omega_0(v,w)$ for all v, w. Then $h_0(v,w) = \omega_0(Jv,w) + i\,\omega_0(v,w)$ is the unique Hermitian form on V with $\mathrm{Im}h_0 = \omega_0$.

4. $\omega(v,v) = 0$ and $q_h(v) = g(v,v)$ for every v. If $w_1, \ldots, w_n$ is an h-orthogonal basis of V and $c_j = h(w_j, w_j)$, then $(w_1, Jw_1, w_2, Jw_2, \ldots, w_n, Jw_n)$ is a basis of $V_{\mathbb{R}}$ in which the matrix of g is $\mathrm{diag}(c_1, c_1, c_2, c_2, \ldots, c_n, c_n)$. Consequently the inertia of g in the sense of definition 8.3.2 is twice, entry by entry, the inertia of h in the sense of definition 10.1.6.

***Proof*:**

1. The form h is additive in each argument, by definition 10.1.1(1) and proposition 10.1.2(1), and for a real scalar t the same two clauses give $h(tv,w) = t\,h(v,w)$ and $h(v,tw) = \bar{t}\,h(v,w) = t\,h(v,w)$. So h is bilinear as a function on $V_{\mathbb{R}} \times V_{\mathbb{R}}$ with values in $\mathbb{C}$. Taking

real and imaginary parts preserves sums and commutes with multiplication by a real scalar, so g and ω are bilinear forms on $V_{\mathbb{R}}$ (definition 8.1.1).

Conjugate symmetry reads $g(w,v)+i\,\omega(w,v)=\overline{g(v,w)+i\,\omega(v,w)}=g(v,w)-i\,\omega(v,w)$, and two complex numbers are equal exactly when their real and imaginary parts agree, so $g(w,v)=g(v,w)$ and $\omega(w,v)=-\omega(v,w)$.

The identity from which everything else in the theorem follows is obtained by moving J into the first argument. Since $Jv=iv$, definition 10.1.1(1) gives

$$h(Jv,w)=i\,h(v,w)=i\big(g(v,w)+i\,\omega(v,w)\big)=-\omega(v,w)+i\,g(v,w) \tag{10.3}$$

and comparing real and imaginary parts,

$$g(Jv,w)=-\omega(v,w), \qquad \omega(Jv,w)=g(v,w)$$

Applying the second of these with w in the first slot and v in the second, and then the skew-symmetry of ω and the symmetry of g, gives $\omega(v,Jw)=-\omega(Jw,v)=-g(w,v)=-g(v,w)$. Substituting Jw for w in the first identity now gives $g(Jv,Jw)=-\omega(v,Jw)=g(v,w)$, and substituting Jw for w in the second gives $\omega(Jv,Jw)=g(v,Jw)$, while $g(v,Jw)=g(Jv,J(Jw))=g(Jv,-w)=-g(Jv,w)=\omega(v,w)$ by the J-invariance just proved, the identity $J(Jw)=-w$ (definition 9.3.1) and the first identity again. So $\omega(Jv,Jw)=\omega(v,w)$.

2. The three identities $g(v,w)=\omega(Jv,w)$, $\omega(v,w)=g(v,Jw)$ and $\omega(v,w)=-g(Jv,w)$ were all established in the proof of (1). Substituting the first into $h=g+i\omega$ gives $h(v,w)=\omega(Jv,w)+i\,\omega(v,w)$, and substituting the third gives $h(v,w)=g(v,w)-i\,g(Jv,w)$.

3. Let g_0 be symmetric, bilinear on $V_{\mathbb{R}}$ and J-invariant, and put $h_0(v,w)=g_0(v,w)-i\,g_0(Jv,w)$. It is additive in each argument and $\mathbb{R}$-homogeneous in each, being built from a bilinear form and the $\mathbb{R}$-linear map J. Replacing v by Jv and using $J(Jv)=-v$,

$$h_0(Jv,w)=g_0(Jv,w)-i\,g_0(J(Jv),w)=g_0(Jv,w)+i\,g_0(v,w)=i\,h_0(v,w)$$

For $\lambda=s+ti$ with $s,t\in\mathbb{R}$ the scalar action on V is $\lambda v=sv+tJv$, which is (9.4) of theorem 9.3.4, so $h_0(\lambda v,w)=s\,h_0(v,w)+t\,h_0(Jv,w)=(s+ti)h_0(v,w)$, which is definition 10.1.1(1). For conjugate symmetry, J-invariance gives $g_0(Jv,w)=g_0(J(Jv),Jw)=-g_0(v,Jw)$, so with the symmetry of g_0,

$$\overline{h_0(w,v)}=g_0(w,v)+i\,g_0(Jw,v)=g_0(v,w)+i\,g_0(v,Jw)=g_0(v,w)-i\,g_0(Jv,w)=h_0(v,w)$$

Since g_0 is real-valued, $\mathrm{Re}h_0=g_0$. Uniqueness: any Hermitian form h_1 with $\mathrm{Re}h_1=g_0$ satisfies $h_1(v,w)=g_0(v,w)-i\,g_0(Jv,w)$ by (2), so $h_1=h_0$.

Now let ω_0 be skew-symmetric, bilinear and J-invariant, and put $g_0(v,w)=\omega_0(Jv,w)$. Then g_0 is bilinear, and it is symmetric: J-invariance and $J(Jw)=-w$ give $\omega_0(v,Jw)=\omega_0(Jv,J(Jw))=-\omega_0(Jv,w)$, so $g_0(w,v)=\omega_0(Jw,v)=-\omega_0(v,Jw)=\omega_0(Jv,w)=g_0(v,w)$. It is J-invariant: $g_0(Jv,Jw)=\omega_0(J(Jv),Jw)=-\omega_0(v,Jw)=\omega_0(Jv,w)=g_0(v,w)$. Also $-g_0(Jv,w)=-\omega_0(J(Jv),w)=\omega_0(v,w)$, so the form attached to g_0 by the previous paragraph is $g_0(v,w)-i\,g_0(Jv,w)=\omega_0(Jv,w)+i\,\omega_0(v,w)$, which is the stated h_0. It is Hermitian by the previous paragraph, its imaginary part is ω_0, and any Hermitian form with imaginary part ω_0 equals it by (2).

4. Skew-symmetry gives $\omega(v,v) = -\omega(v,v)$, hence $\omega(v,v) = 0$, and therefore $q_h(v) = h(v,v) = g(v,v)$. For the basis, the complex vector space attached to $V_{\mathbb{R}}$ and J by theorem 9.3.4(1) is V itself, by clause (3) of that theorem, so clause (4) of it makes $(w_1, Jw_1, \ldots, w_n, Jw_n)$ a basis of $V_{\mathbb{R}}$ over $\mathbb{R}$.

   Its matrix for g is computed entry by entry from $h(w_j, w_k) = 0$ for $j \neq k$ and $h(w_j, w_j) = c_j$, a real number by proposition 10.1.2(2). For $j \neq k$ all four of $g(w_j, w_k)$, $g(w_j, Jw_k)$, $g(Jw_j, w_k)$ and $g(Jw_j, Jw_k)$ vanish: the first because g and ω are the parts of h, the second because $g(w_j, Jw_k) = \omega(w_j, w_k) = 0$ by (2), the third because $g(Jw_j, w_k) = -\omega(w_j, w_k) = 0$ by (2), and the fourth by the J-invariance in (1). For $j = k$ one has $g(w_j, w_j) = \mathrm{Re}\, c_j = c_j$, then $g(Jw_j, Jw_j) = c_j$ by J-invariance, and $g(w_j, Jw_j) = \omega(w_j, w_j) = 0$ by (2) and the first sentence of this clause, with $g(Jw_j, w_j) = 0$ by symmetry. So the matrix of g in the listed basis (definition 8.1.2) is $\operatorname{diag}(c_1, c_1, \ldots, c_n, c_n)$.

   That matrix is diagonal, and its positive, negative and zero entries occur in pairs, one pair for each c_j. So it has $2n_+$ positive, $2n_-$ negative and $2n_0$ zero entries, where (n_+, n_-, n_0) is the inertia of h. Theorem 8.3.1(2) identifies those three counts as the inertia of g, completing the proof.

The theorem gives three descriptions of one object. A Hermitian form on an n-dimensional complex space, a J-invariant symmetric form on the underlying $2n$-dimensional real space, and a J-invariant skew-symmetric form on that same real space are interchangeable data, and (2) is what converts each into the others. No information is lost in either direction, and clause (4) records what happens to the invariant: each positive direction for h becomes the two real directions w_j and Jw_j, on both of which g is positive, so all three counts double. In particular $q_h(v) > 0$ for every nonzero v exactly when g is positive definite in the sense of definition 8.4.1, since $q_h(v) = g(v,v)$ and the nonzero vectors of V are the nonzero vectors of $V_{\mathbb{R}}$.

Exercise 10.1.1

1. Let $H = \left(\begin{smallmatrix} 1 & i \\ -i & 2 \end{smallmatrix}\right)$ and $h(x,y) = y^* H x$ on $\mathbb{C}^2$. Verify that H is Hermitian, compute the leading principal minors Δ_1 and Δ_2, and run the induction in part (1) of the proof of theorem 10.1.5 from $v_1 = e_1$ to produce an invertible $P \in M_2(\mathbb{C})$ with $P^* H P = I_2$. State the inertia and the signature of H, and check your answer against the two solutions of $\det(H - \lambda I_2) = 0$.

2. Show that $\det \overline{A} = \overline{\det A}$ for every $A \in M_n(\mathbb{C})$, by induction on n using theorem 3.3.4, and deduce that $\det(A^*) = \overline{\det A}$. Conclude that the determinant of a Hermitian matrix is real, that $\det(P^* H P) = |\det P|^2 \det H$ for every invertible P, and that a Hermitian $H \in M_n(\mathbb{C})$ with $n_0(H) = 0$ has $\det H$ of the sign of $(-1)^{n_-(H)}$.

3. Show that every Hermitian $H \in M_2(\mathbb{C})$ has the form $\left(\begin{smallmatrix} a & b \\ \bar{b} & d \end{smallmatrix}\right)$ with $a, d \in \mathbb{R}$ and $b \in \mathbb{C}$, and that $\det H = ad - |b|^2$. Show that $q_H(x) = x^* H x > 0$ for every nonzero $x \in \mathbb{C}^2$ if and only if $a > 0$ and $\det H > 0$, and that H is of inertia $(1,1,0)$ if and only if $\det H < 0$. (For the second statement treat the cases $a \neq 0$ and $a = 0$ separately.)

4. Let h be a Hermitian form on a complex vector space V with $q_h(v) \geq 0$ for every $v \in V$. Using (10.1), show that
$$|h(u,v)|^2 \leq q_h(u)\, q_h(v)$$
for all $u, v \in V$. (Treat $q_h(v) = 0$ and $q_h(v) > 0$ separately; in the first case take $\lambda = -s\, h(u,v)$ with $s > 0$ real and let s grow.) Deduce that $h(u,v) = 0$ whenever $q_h(v) = 0$.

5. Take $V = \mathbb{C}$ with $h(z, w) = z\overline{w}$, and identify $V_{\mathbb{R}}$ with $\mathbb{R}^2$ by $z = a_1 + a_2 i \mapsto (a_1, a_2)^\top$. Compute $g = \mathrm{Re}h$ and $\omega = \mathrm{Im}h$ as bilinear forms on $\mathbb{R}^2$, showing that g is the standard inner product and that $\omega\big((a_1, a_2), (c_1, c_2)\big) = a_2 c_1 - a_1 c_2$. Write the matrices of g and ω in the standard basis, identify $\omega(u, v)$ as a determinant up to sign, and verify $\omega(Ju, v) = g(u, v)$ directly for $J(a_1, a_2)^\top = (-a_2, a_1)^\top$.

6. Show that the number of $*$-congruence classes of Hermitian matrices in $M_n(\mathbb{C})$ is $(n+1)(n+2)/2$, while the number of congruence classes of complex symmetric matrices in $M_n(\mathbb{C})$ under $A \mapsto P^\top AP$ is $n + 1$. Evaluate both for $n = 3$.

7. Let h be a Hermitian form on V with $\dim_{\mathbb{C}} V = n$ and inertia (p, m, r). Show that the largest dimension of a complex subspace $U \subseteq V$ with $h(u, u') = 0$ for all $u, u' \in U$ is $\min(p, m) + r$. (For the upper bound intersect U with the spans of the positive and of the negative part of a normal-form basis; for the lower bound use the vectors $v_j + v_{p+j}$.)

8. Let $H = \mathrm{diag}(1, -1)$ and $h(x, y) = y^* Hx$ on $\mathbb{C}^2$. Compute the matrices of $g = \mathrm{Re}h$ and $\omega = \mathrm{Im}h$ in the real basis (e_1, Je_1, e_2, Je_2) of $(\mathbb{C}^2)_{\mathbb{R}}$, where $Jv = iv$. Confirm that the inertia of g is twice the inertia of h, as theorem 10.1.9(4) requires, and that the matrix of ω is skew-symmetric with the two diagonal blocks $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

10.2 Symplectic Vector Spaces and the Standard Symplectic Basis

Section 10.1 settled the conjugate-symmetric case and chapter 8 the symmetric one. One condition named in definition 8.1.9 is still unexamined: $\omega(u, v) = -\omega(v, u)$. Its classification has a different shape from the other two. No signature appears, and the invariant that survives is the dimension alone, which is forced to be even.

The normal form is a basis paired off two vectors at a time, $\omega(e_i, f_i) = 1$ with every other pairing among the basis vectors equal to zero, and the theorem producing it is the linear Darboux theorem. This section proves that theorem over $\mathbb{R}$ and over $\mathbb{C}$ by an induction that strips off one such pair at each step, and derives from it the congruence classification of invertible skew-symmetric matrices.

The sign condition has a consequence on the diagonal, and over $\mathbb{R}$ and $\mathbb{C}$ that consequence is equivalent to it.

Proposition 10.2.1: Skew-Symmetry and Vanishing on the Diagonal

Let k be $\mathbb{R}$ or $\mathbb{C}$, let V be a finite-dimensional vector space over k, and let φ be a bilinear form on V (definition 8.1.1). Then φ is skew-symmetric (definition 8.1.9) if and only if $\varphi(v, v) = 0$ for every $v \in V$.

Section 8.1 set this as exercise 8.1.1 (4). It is stated and proved here because the rest of the section leans on it in argument position, where an exercise is not available to cite.

***Proof*:**
Suppose φ is skew-symmetric and let $v \in V$. Taking $u = v$ in $\varphi(u, v) = -\varphi(v, u)$ gives $\varphi(v, v) = -\varphi(v, v)$, hence $2\,\varphi(v, v) = 0$. The scalar 2 is nonzero in $\mathbb{R}$ and in $\mathbb{C}$, so it may be

divided out, and $\varphi(v, v) = 0$.

Conversely, suppose φ vanishes on the diagonal and let $u, v \in V$. Expanding $\varphi(u + v, u + v)$ by definition 8.1.1(1) in the first argument and then by definition 8.1.1(2) in the second,

$$0 = \varphi(u + v, u + v) = \varphi(u, u) + \varphi(u, v) + \varphi(v, u) + \varphi(v, v) = \varphi(u, v) + \varphi(v, u)$$

the last equality because the two diagonal terms vanish by hypothesis. Hence $\varphi(u, v) = -\varphi(v, u)$ for all u and v, as we sought to show.

Only the division by 2 in the first half is sensitive to the scalars, and it is why the equivalence is stated for $\mathbb{R}$ and $\mathbb{C}$ rather than in general. In a field where $2 = 0$ one has $-1 = 1$, so the sign condition reads $\varphi(u, v) = \varphi(v, u)$ and says only that φ is symmetric, while the vanishing condition remains strictly stronger.[1]

The vanishing on the diagonal is what separates this case from the two already settled. A symmetric bilinear form is recovered from the one-variable function $q(v) = \varphi(v, v)$ by theorem 8.2.2, and it is the signs taken by q that theorem 8.3.1 counts. For a skew-symmetric form that function is identically zero, so it records nothing about φ and no count of signs can be extracted from it. What replaces the signature is a normal form for the pairing of one basis vector against another, and producing it needs the nondegeneracy hypothesis of chapter 8.

Definition 10.2.2: Symplectic Form and Symplectic Vector Space

Let k be $\mathbb{R}$ or $\mathbb{C}$ and let V be a finite-dimensional vector space over k. A *symplectic form* on V is a bilinear form ω on V that is skew-symmetric (definition 8.1.9) and nondegenerate (definition 8.1.13). The pair (V, ω) is then a *symplectic vector space*. Two vectors $u, v \in V$ are *ω-orthogonal* when $\omega(u, v) = 0$. The relation is symmetric in u and v, since $\omega(v, u) = -\omega(u, v)$.

The smallest instance is $V = k^2$ with

$$\omega(x, y) = x_1 y_2 - x_2 y_1$$

Evaluating on the standard basis gives $\omega(e_1, e_1) = \omega(e_2, e_2) = 0$, $\omega(e_1, e_2) = 1$ and $\omega(e_2, e_1) = -1$, so the matrix of ω in that basis (definition 8.1.2) is $B = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, which satisfies $B^\top = -B$ and has $\det B = 0 \cdot 0 - 1 \cdot (-1) = 1 \neq 0$. So ω is skew-symmetric and, by proposition 8.1.14, nondegenerate: (k^2, ω) is a symplectic vector space. The number $\omega(x, y)$ is the determinant of the matrix whose columns are x and y, expanded along its first row by theorem 3.3.4. Over $\mathbb{R}$ that determinant is the signed area of the parallelogram spanned by x and y (chapter 3).

[1] The general-field theory takes the vanishing condition as the definition and calls such a form *alternating*. See *Core Algebra* [9, Alternating Bilinear Form]. Over $\mathbb{R}$ and $\mathbb{C}$ proposition 10.2.1 makes the two names interchangeable, and this book uses the sign condition of definition 8.1.9 throughout.

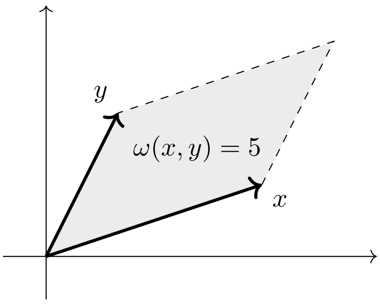


Figure 10.1: The value of the symplectic form on $\mathbb{R}^2$ at $x = (3, 1)^\top$ and $y = (1, 2)^\top$ is the signed area $3 \cdot 2 - 1 \cdot 1$ of the parallelogram they span

More generally, for any $B \in M_n(k)$ the form $\varphi_B(x, y) = x^\top By$ on k^n has matrix B in the standard basis $\mathcal{E}$, since the entry of $[\varphi_B]_\mathcal{E}$ in row i and column j is $\varphi_B(e_i, e_j) = e_i^\top Be_j = b_{ij}$ by definition 8.1.2 and definition 1.4.1. So (k^n, φ_B) is a symplectic vector space exactly when $B^\top = -B$ and B is invertible, by proposition 8.1.10(1) and proposition 8.1.14.

Skew-symmetry makes every vector ω-orthogonal to itself, and with it every pair x and cx: in k^2 the nonzero vectors e_1 and $2e_1$ satisfy $\omega(e_1, 2e_1) = 2\,\omega(e_1, e_1) = 0$. Nondegeneracy rules out only the extreme case, a nonzero vector ω-orthogonal to all of V. Over $\mathbb{R}$ an inner product allows neither, since $\langle x, cx\rangle = c\langle x, x\rangle$ is nonzero whenever c and x are, so a symplectic form is never an inner product. The two conditions can hold together only in even dimension, and the obstruction is visible in the determinant of the matrix of the form.

Proposition 10.2.3: A Symplectic Space Has Even Dimension

Let k be $\mathbb{R}$ or $\mathbb{C}$ and let (V, ω) be a symplectic vector space over k. Then $\dim V$ is even. Equivalently, a skew-symmetric matrix in $M_n(k)$ with n odd is not invertible.

The matrix half was set over $\mathbb{R}$ as exercise 3.2.1 (5) in chapter 3, with a worked answer. The proof below runs the same determinant chain, needed here over $\mathbb{C}$ as well and in the vector-space form that the induction of theorem 10.2.6 uses.

***Proof*:**
Put $n = \dim V$. If $n = 0$ there is nothing to prove, so assume $n \geq 1$, fix an ordered basis $\mathcal{B}$ of V, available by theorem 2.2.9(1), and put $B = [\omega]_\mathcal{B}$. Since ω is skew-symmetric, $B^\top = -B$ by proposition 8.1.10(1), and since ω is nondegenerate, B is invertible and $\det B \neq 0$ by proposition 8.1.14.

The matrix $-B$ is the product $(-I_n)B$: by definition 1.4.1 the entry of $(-I_n)B$ in row i and column j is $\sum_s (-I_n)_{is} b_{sj} = -b_{ij}$, the only nonzero term being $s = i$. The matrix $-I_n$ is diagonal, hence triangular, so $\det(-I_n) = (-1)^n$ by proposition 3.2.6. Combining theorem 3.2.11 with theorem 3.2.10,

$$\det B = \det\left(B^\top\right) = \det(-B) = \det(-I_n) \det B = (-1)^n \det B$$

Subtracting and factoring gives $\left(1 - (-1)^n\right) \det B = 0$, and $\det B \neq 0$ forces $(-1)^n = 1$, so n is even.

For the second statement, let $B \in M_n(k)$ be skew-symmetric and invertible. The form $\varphi_B(x, y) =$

$x^\top By$ on k^n has matrix B in the standard basis (definition 8.1.2 and definition 1.4.1), so it is skew-symmetric by proposition 8.1.10(1) and nondegenerate by proposition 8.1.14. Thus (k^n, φ_B) is symplectic and the first statement makes n even. Hence no skew-symmetric matrix of odd size is invertible, completing the proof.

A nondegenerate *symmetric* form exists on a space of every dimension, the standard inner product being one, so the parity restriction comes from the skew condition alone. In odd dimension every skew-symmetric form is degenerate, and its radical therefore contains a nonzero vector. The even-dimensional case is the whole of the theory, and it has a single model.

Definition 10.2.4: The Standard Symplectic Form and Symplectic Bases

Let k be $\mathbb{R}$ or $\mathbb{C}$, let $m \geq 1$, and let

$$J_0 = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix} \in M_{2m}(k)$$

be the matrix carrying the identity I_m in its upper right block, $-I_m$ in its lower left block, and zero blocks on the diagonal. The *standard symplectic form* on k^{2m} is

$$\omega_0(x, y) = x^\top J_0 y$$

Let (V, ω) be a symplectic vector space over k with $\dim V = 2m$. An ordered basis $\mathcal{B} = (e_1, \dots, e_m, f_1, \dots, f_m)$ of V is a *symplectic basis* when

$$\omega(e_i, e_j) = 0, \qquad \omega(f_i, f_j) = 0, \qquad \omega(e_i, f_j) = \delta_{ij}$$

for all $1 \leq i, j \leq m$, where δ_{ij} is 1 when $i = j$ and 0 otherwise. Equivalently, $[\omega]_{\mathcal{B}} = J_0$.

Three properties of J_0 are used throughout, and each is a blockwise computation. Transposing blockwise (proposition 3.5.1),

$$J_0^\top = \begin{pmatrix} 0^\top & (-I_m)^\top \\ I_m^\top & 0^\top \end{pmatrix} = \begin{pmatrix} 0 & -I_m \\ I_m & 0 \end{pmatrix} = -J_0$$

so J_0 is skew-symmetric. Multiplying blockwise (proposition 3.5.1),

$$J_0^2 = \begin{pmatrix} 0 \cdot 0 + I_m(-I_m) & 0 \cdot I_m + I_m \cdot 0 \\ (-I_m) \cdot 0 + 0 \cdot (-I_m) & (-I_m)I_m + 0 \cdot 0 \end{pmatrix} = \begin{pmatrix} -I_m & 0 \\ 0 & -I_m \end{pmatrix} = -I_{2m}$$

so $J_0(-J_0) = (-J_0)J_0 = I_{2m}$, and J_0 is invertible with $J_0^{-1} = -J_0$ (definition 1.4.4). Finally the entry of J_0 in row i and column j is $e_i^\top J_0 e_j$ by definition 1.4.1, which is $\omega_0(e_i, e_j)$, so J_0 is the matrix of ω_0 in the standard basis (definition 8.1.2). Consequently ω_0 is skew-symmetric (proposition 8.1.10(1)) and nondegenerate (proposition 8.1.14), and (k^{2m}, ω_0) is a symplectic vector space.

The four blocks of J_0, read in their positions, make the standard basis of k^{2m} a symplectic basis when it is taken in the order $e_1, \dots, e_m$ followed by $e_{m+1}, \dots, e_{2m}$: for $i, j \leq m$ the values $\omega_0(e_i, e_j) = (J_0)_{ij}$ and $\omega_0(e_{m+i}, e_{m+j}) = (J_0)_{m+i,m+j}$ are 0, coming from the two zero blocks, while $\omega_0(e_i, e_{m+j}) = (I_m)_{ij} = \delta_{ij}$. The equivalence at the end of definition 10.2.4 is the same computation in an arbitrary basis. If $\mathcal{B} = (e_1, \dots, e_m, f_1, \dots, f_m)$ satisfies the three pairing conditions, then cutting $[\omega]_{\mathcal{B}}$ after row and column m produces four blocks whose entries are $\omega(e_i, e_j) = 0$, $\omega(e_i, f_j) = \delta_{ij}$,

$\omega(f_i, e_j) = -\omega(e_j, f_i) = -\delta_{ij}$ and $\omega(f_i, f_j) = 0$, so $[\omega]_{\mathcal{B}} = J_0$. In the other direction, if $[\omega]_{\mathcal{B}} = J_0$ then those same four blocks return the three conditions.

Example 10.2.5: The Standard Symplectic Form on $\mathbb{R}^4$

Take $m = 2$ and $k = \mathbb{R}$. Then

$$J_0 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

and expanding $x^\top J_0 y$ by definition 1.4.1 keeps only the four nonzero entries of J_0:

$$\omega_0(x, y) = x_1 y_3 + x_2 y_4 - x_3 y_1 - x_4 y_2 \tag{10.4}$$

For $x = (1, 2, 3, 4)^\top$ and $y = (0, 1, -1, 2)^\top$ this gives

$$\omega_0(x, y) = 1 \cdot (-1) + 2 \cdot 2 - 3 \cdot 0 - 4 \cdot 1 = -1$$

while $\omega_0(y, x) = 0 \cdot 3 + 1 \cdot 4 - (-1) \cdot 1 - 2 \cdot 2 = 1$, confirming the sign change, and $\omega_0(x, x) = 1 \cdot 3 + 2 \cdot 4 - 3 \cdot 1 - 4 \cdot 2 = 0$, confirming proposition 10.2.1. The symplectic basis given by the standard basis is e_1, e_2, e_3, e_4 in that order, with $f_1 = e_3$ and $f_2 = e_4$: substituting into (10.4) gives $\omega_0(e_1, e_3) = \omega_0(e_2, e_4) = 1$ and $\omega_0(e_1, e_2) = \omega_0(e_1, e_4) = \omega_0(e_2, e_3) = \omega_0(e_3, e_4) = 0$.

Independent vectors can be ω_0-orthogonal, as e_1 and e_2 are, and an inner product allows that too. What has no analogue for an inner product is $\omega_0(x, x) = 0$: over $\mathbb{R}$ an inner product has $\langle x, x \rangle > 0$ for every nonzero x.

Nothing so far shows that a symplectic basis exists outside k^{2m}. One does, in every symplectic vector space over $\mathbb{R}$ or $\mathbb{C}$, and the proof constructs it two vectors at a time: pick any nonzero e_1, use nondegeneracy to find a partner f_1 pairing with it to 1, and then show that the rest of V splits off as a symplectic space of dimension two smaller.

Theorem 10.2.6: The Linear Darboux Theorem

Let k be $\mathbb{R}$ or $\mathbb{C}$ and let (V, ω) be a symplectic vector space over k with $\dim V = 2m$ and $m \geq 1$. Then V has a symplectic basis $(e_1, \ldots, e_m, f_1, \ldots, f_m)$ in the sense of definition 10.2.4, and the matrix of ω in that basis is J_0.

***Proof*:**

The argument is an induction on m. Steps 1 to 4 are the construction performed once, for an arbitrary $m \geq 1$, and Step 5 assembles them into the induction.

Step 1: a pair with $\omega(e_1, f_1) = 1$. Since $\dim V = 2m \geq 2$, there is a nonzero $e_1 \in V$. Nondegeneracy says that e_1 does not lie in the left radical of ω (definition 8.1.13), so there is $w \in V$ with $c = \omega(e_1, w) \neq 0$. Put $f_1 = c^{-1} w$. Linearity in the second argument (definition 8.1.1(2)) gives

$$\omega(e_1, f_1) = c^{-1} \omega(e_1, w) = c^{-1} c = 1$$

Step 2: $U = \operatorname{span}\{e_1, f_1\}$ has dimension 2. Suppose $a e_1 + b f_1 = 0$ with $a, b \in k$. Applying

$\omega(e_1, -)$ and using definition 8.1.1(2),

$$0 = \omega(e_1, ae_1 + bf_1) = a\,\omega(e_1, e_1) + b\,\omega(e_1, f_1) = a \cdot 0 + b \cdot 1 = b$$

where $\omega(e_1, e_1) = 0$ is proposition 10.2.1. With $b = 0$ the relation reads $ae_1 = 0$, and $e_1 \neq 0$ forces $a = 0$. So (e_1, f_1) is independent and is a basis of U, whence $\dim U = 2$.

Step 3: $V = U \oplus U^\omega$. Write

$$U^\omega = \{\, v \in V \big| \omega(u, v) = 0 \text{ for every } u \in U \,\}$$

This is a subspace of V (definition 2.1.2): it contains 0, and it is closed under addition and scalar multiplication because $\omega(u, -)$ is linear for each fixed u by definition 8.1.1(2). Since e_1 and f_1 span U, the same linearity shows that $v \in U^\omega$ if and only if $\omega(e_1, v) = 0$ and $\omega(f_1, v) = 0$.

First, $U \cap U^\omega = \{0\}$. Let $u = ae_1 + bf_1$ lie in U^ω. Then $0 = \omega(e_1, u) = a \cdot 0 + b \cdot 1 = b$, and $0 = \omega(f_1, u) = a\,\omega(f_1, e_1) + b\,\omega(f_1, f_1) = -a$, using $\omega(f_1, e_1) = -\omega(e_1, f_1) = -1$ and $\omega(f_1, f_1) = 0$. So $a = b = 0$ and $u = 0$.

Second, $V = U + U^\omega$. Given $v \in V$, put

$$u = \omega(v, f_1)\, e_1 + \omega(e_1, v)\, f_1 \in U$$

and test $v - u$ against e_1 and f_1. Expanding $\omega(e_1, u)$ and $\omega(f_1, u)$ by definition 8.1.1(2) and inserting the four values $\omega(e_1, e_1) = \omega(f_1, f_1) = 0$, $\omega(e_1, f_1) = 1$ and $\omega(f_1, e_1) = -1$,

$$\begin{aligned}
\omega(e_1, v - u) &= \omega(e_1, v) - \omega(v, f_1)\,\omega(e_1, e_1) - \omega(e_1, v)\,\omega(e_1, f_1) \\
&= \omega(e_1, v) - 0 - \omega(e_1, v) = 0 \\
\omega(f_1, v - u) &= \omega(f_1, v) - \omega(v, f_1)\,\omega(f_1, e_1) - \omega(e_1, v)\,\omega(f_1, f_1) \\
&= \omega(f_1, v) + \omega(v, f_1) - 0 = 0
\end{aligned}$$

the last equality because $\omega(v, f_1) = -\omega(f_1, v)$. So $v = u + (v - u)$ with $u \in U$ and $v - u \in U^\omega$, and $V = U \oplus U^\omega$ in the sense of definition 2.1.11.

Step 4: U^ω has dimension $2m - 2$ and ω restricts to a symplectic form on it. Let $(w_1, \ldots, w_d)$ be a basis of U^ω, available by theorem 2.2.9(1). The list $(e_1, f_1, w_1, \ldots, w_d)$ is a basis of V. It spans, because every $v \in V$ is $u + v'$ with $u \in U$ and $v' \in U^\omega$ by Step 3, and u is a combination of e_1, f_1 while v' is a combination of $w_1, \ldots, w_d$. It is independent: a relation $ae_1 + bf_1 + c_1 w_1 + \cdots + c_d w_d = 0$ puts $ae_1 + bf_1 = -(c_1 w_1 + \cdots + c_d w_d)$, a vector lying in U and in U^ω, hence equal to 0 by Step 3. Then $a = b = 0$ by Step 2, and every $c_r = 0$ because $(w_1, \ldots, w_d)$ is independent. Since every basis of V has $\dim V = 2m$ members (definition 2.2.10), $2 + d = 2m$ and $d = 2m - 2$.

Write ω' for the restriction of ω to $U^\omega \times U^\omega$. It is bilinear and skew-symmetric, both conditions being inherited from ω by restricting the quantifiers in definition 8.1.1 and definition 8.1.9. It is nondegenerate: let $v \in U^\omega$ satisfy $\omega(v, w) = 0$ for every $w \in U^\omega$, and let $z \in V$ be arbitrary. Writing $z = u + w$ with $u \in U$ and $w \in U^\omega$ by Step 3,

$$\omega(v, z) = \omega(v, u) + \omega(v, w) = -\omega(u, v) + 0 = 0$$

since $v \in U^\omega$ makes $\omega(u, v) = 0$. So v lies in the left radical of ω, which is $\{0\}$ by nondegeneracy, and $v = 0$. Hence (U^ω, ω') is a symplectic vector space of dimension $2m - 2$.

Step 5: the induction. For $m = 1$, Step 4 gives $\dim U^\omega = 0$, so $U^\omega = \{0\}$ and $V = U$ by Step 3. The pair (e_1, f_1) is then a basis of V satisfying $\omega(e_1, e_1) = \omega(f_1, f_1) = 0$ and $\omega(e_1, f_1) = 1$, which are the three conditions of definition 10.2.4 for $m = 1$.

Let $m \geq 2$ and assume the theorem for symplectic spaces of dimension $2m-2$. By Step 4, (U^ω, ω') is such a space, so it has a symplectic basis, which may be indexed as $(e_2, \dots, e_m, f_2, \dots, f_m)$. By the inductive hypothesis $\omega(e_i, e_j) = \omega(f_i, f_j) = 0$ and $\omega(e_i, f_j) = \delta_{ij}$ for all $2 \leq i, j \leq m$. By Step 4 the list $(e_1, f_1, e_2, \dots, e_m, f_2, \dots, f_m)$ is a basis of V, and reordering it as $(e_1, \dots, e_m, f_1, \dots, f_m)$ changes neither its span nor its independence, so the reordered list is a basis too.

It remains to check the pairings involving index 1. Each of $e_2, \dots, e_m, f_2, \dots, f_m$ lies in U^ω, so $\omega(e_1, e_j) = \omega(e_1, f_j) = \omega(f_1, e_j) = \omega(f_1, f_j) = 0$ for every $j \geq 2$ by the definition of U^ω, and skew-symmetry turns these into $\omega(e_j, e_1) = \omega(e_j, f_1) = 0$ as well. Together with $\omega(e_1, e_1) = \omega(f_1, f_1) = 0$ and $\omega(e_1, f_1) = 1$, the three conditions of definition 10.2.4 hold for all $1 \leq i, j \leq m$, so $(e_1, \dots, e_m, f_1, \dots, f_m)$ is a symplectic basis. The matrix of ω in it is J_0 by the equivalence in definition 10.2.4, completing the proof.

The construction is entirely free in its first move: any nonzero e_1 starts it, and the partner f_1 is only constrained by $\omega(e_1, f_1) = 1$. A symplectic space therefore carries many symplectic bases, and the theorem says that all of them present ω by the same matrix. Turning that into a statement about matrices classifies the skew-symmetric case of chapter 8 outright.

Corollary 10.2.7: Congruence Classification of Nondegenerate Skew Forms

Let k be $\mathbb{R}$ or $\mathbb{C}$.

1. Let $B \in M_n(k)$ be skew-symmetric and invertible. Then $n = 2m$ is even, and there is an invertible $P \in M_n(k)$ with $P^\top B P = J_0$.

2. Any two invertible skew-symmetric matrices in $M_n(k)$ are congruent (definition 8.1.5).

3. Let (V, ω) and (V', ω') be symplectic vector spaces over k with $\dim V = \dim V'$. Then there is an invertible linear map $T \colon V \to V'$ with $\omega'(Tu, Tv) = \omega(u, v)$ for all $u, v \in V$.

***Proof*:**

1. The form $\varphi_B(x, y) = x^\top B y$ on k^n has matrix B in the standard basis (definition 8.1.2 and definition 1.4.1), so it is skew-symmetric by proposition 8.1.10(1) and nondegenerate by proposition 8.1.14. Hence (k^n, φ_B) is a symplectic vector space, and $n = 2m$ is even by proposition 10.2.3. Let $\mathcal{C} = (c_1, \dots, c_n)$ be a symplectic basis of it, given by theorem 10.2.6, and let $P \in M_n(k)$ be the matrix whose j-th column is c_j. By definition 1.4.1 the entry of $P^\top B P$ in row i and column j is the product of the i-th row of $P^\top$, which is $c_i^\top$, with the column Bc_j, that is $c_i^\top B c_j = \varphi_B(c_i, c_j)$. Those numbers are the entries of $[\varphi_B]_\mathcal{C}$ (definition 8.1.2), which is J_0 by definition 10.2.4. Hence $P^\top B P = J_0$.

   The matrix P is invertible. Applying theorem 3.2.10 twice and theorem 3.2.11 once to that identity gives $(\det P)^2 \det B = \det J_0$, and $\det J_0 \neq 0$ because J_0 is invertible (corollary 3.2.8), so $\det P \neq 0$ and P is invertible by corollary 3.2.8 again.

2. Let B and B' be invertible and skew-symmetric in $M_n(k)$. By (1) each is congruent to J_0, and congruence is an equivalence relation by proposition 8.1.6, so B is congruent to B'.

3. Put $2m = \dim V = \dim V'$, even by proposition 10.2.3. If $m = 0$ then $V = V' = \{0\}$, the only map $T \colon V \to V'$ is the zero map, which is the identity of $\{0\}$ and hence invertible, and both sides of the required identity are 0; so assume $m \geq 1$, which is what theorem 10.2.6 needs. Choose symplectic bases $\mathcal{B} = (b_1, \ldots, b_{2m})$ of V and $\mathcal{C} = (c_1, \ldots, c_{2m})$ of V' by theorem 10.2.6. Define

$$T(x_1 b_1 + \cdots + x_{2m} b_{2m}) = x_1 c_1 + \cdots + x_{2m} c_{2m}$$

This is well defined because the coefficients expressing a vector in a basis are unique: the difference of two such expressions is a vanishing combination of an independent list, so its coefficients all vanish (definition 2.2.1). It is linear because coordinates in a basis add and scale entrywise: if $[u]_{\mathcal{B}} = x$ and $[v]_{\mathcal{B}} = y$ then $[u + av]_{\mathcal{B}} = x + ay$, so $T(u + av) = Tu + aTv$. The same recipe read from $\mathcal{C}$ to $\mathcal{B}$ produces a map inverse to T on both sides, so T is invertible. By construction $[Tv]_{\mathcal{C}} = [v]_{\mathcal{B}}$ for every v.

Both bases are symplectic, so $[\omega]_{\mathcal{B}} = [\omega']_{\mathcal{C}} = J_0$ by definition 10.2.4. Applying theorem 8.1.3(1) in V' and then in V,

$$\omega'(Tu, Tv) = [Tu]_{\mathcal{C}}^\top J_0 [Tv]_{\mathcal{C}} = [u]_{\mathcal{B}}^\top J_0 [v]_{\mathcal{B}} = \omega(u, v)$$

for all $u, v \in V$, as we sought to show.

The symmetric case counts differently. Over $\mathbb{R}$, the nondegenerate symmetric forms on an n-dimensional space fall into $n + 1$ congruence classes, one for each value of n_+ between 0 and n (corollary 8.3.3). Over $\mathbb{C}$ the signature collapses, and there is exactly one class in each dimension (corollary 8.3.5). For skew-symmetric forms the count is the same over both fields: one class in each even dimension, and none at all in odd dimension. A symplectic vector space carries no invariant beyond its dimension, so any statement proved about (k^{2m}, ω_0) and preserved by the maps of corollary 10.2.7(3) holds in every symplectic space of dimension $2m$.

The proof of theorem 10.2.6 is an algorithm as well as an argument, and running it on $(\mathbb{R}^4, \omega_0)$ from a starting vector other than e_1 produces a symplectic basis other than the standard one.

Example 10.2.8: A Non-Standard Symplectic Basis of $\mathbb{R}^4$

Work in $(\mathbb{R}^4, \omega_0)$ with ω_0 given by (10.4), and start the construction at $e_1 = (1, 1, 0, 0)^\top$.

Step 1. For $y \in \mathbb{R}^4$, formula (10.4) gives $\omega_0(e_1, y) = y_3 + y_4$. The vector $y = (0, 0, 1, 0)^\top$ makes this equal to 1, so $f_1 = (0, 0, 1, 0)^\top$ is an admissible partner and no rescaling is needed.

Step 3. With $U = \operatorname{span}\{e_1, f_1\}$, a vector v lies in U^ω exactly when $\omega_0(e_1, v) = 0$ and $\omega_0(f_1, v) = 0$. The first condition is $v_3 + v_4 = 0$. For the second, (10.4) with $x = f_1 = (0, 0, 1, 0)^\top$ reads $\omega_0(f_1, v) = -v_1$, so the second condition is $v_1 = 0$. Hence

$$U^\omega = \{\, v \in \mathbb{R}^4 \,|\, v_1 = 0,\ v_3 = -v_4 \,\}$$

a subspace of dimension 2, as Step 4 of the proof of theorem 10.2.6 predicts, with basis $(0, 1, 0, 0)^\top$ and $(0, 0, -1, 1)^\top$.

Step 5. Applying Step 1 inside U^ω: for $e_2 = (0, 1, 0, 0)^\top$ and $f_2 = (0, 0, -1, 1)^\top$,

$$\omega_0(e_2, f_2) = 0 \cdot (-1) + 1 \cdot 1 - 0 \cdot 0 - 0 \cdot 0 = 1$$

so no rescaling is needed here either, and $\mathcal{C} = (e_1, e_2, f_1, f_2)$ is the symplectic basis the construction returns. The six pairings among the four vectors confirm it:

$$\omega_0(e_1,e_2)=0,\quad \omega_0(e_1,f_1)=1,\quad \omega_0(e_1,f_2)=-1+1=0$$

$$\omega_0(e_2,f_1)=0,\quad \omega_0(e_2,f_2)=1,\quad \omega_0(f_1,f_2)=0$$

The matrix P whose columns are e_1,e_2,f_1,f_2 in that order is

$$P=\begin{pmatrix}1&0&0&0\\1&1&0&0\\0&0&1&-1\\0&0&0&1\end{pmatrix}$$

This is the change of basis matrix from $\mathcal{C}$ to the standard basis (definition 2.5.2), so theorem 8.1.4 gives $P^\top J_0P=[\omega_0]_{\mathcal{C}}=J_0$. Direct multiplication confirms it. Each row of J_0 has a single nonzero entry, so J_0P is obtained by moving rows of P:

$$J_0P=\begin{pmatrix}0&0&1&-1\\0&0&0&1\\-1&0&0&0\\-1&-1&0&0\end{pmatrix}$$

its first two rows being rows 3 and 4 of P and its last two rows the negatives of rows 1 and 2 of P. Multiplying on the left by $P^\top$, whose rows are $(1,1,0,0)$, $(0,1,0,0)$, $(0,0,1,0)$ and $(0,0,-1,1)$, forms four combinations of the rows of J_0P: the sum of rows 1 and 2, row 2, row 3, and row 4 minus row 3. These are $(0,0,1,0)$, $(0,0,0,1)$, $(-1,0,0,0)$ and $(0,-1,0,0)$, which stack to J_0.

Exercise 10.2.1

1. In $(\mathbb{R}^4,\omega_0)$ let $x=(1,0,2,-1)^\top$ and $y=(0,3,1,1)^\top$. Compute $\omega_0(x,y)$ and $\omega_0(y,x)$. Then describe the set of all $v\in\mathbb{R}^4$ with $\omega_0(x,v)=0$ by a single linear equation in the coordinates of v, give its dimension, and verify that x itself belongs to it.

2. Let ω be the bilinear form on $\mathbb{R}^4$ whose matrix in the standard basis is

$$B=\begin{pmatrix}0&1&0&0\\-1&0&1&0\\0&-1&0&1\\0&0&-1&0\end{pmatrix}$$

   Show that ω is symplectic. Then run the construction in the proof of theorem 10.2.6, starting at $e_1=(1,0,0,0)^\top$, to produce a symplectic basis of $(\mathbb{R}^4,\omega)$, and write down the matrix P of corollary 10.2.7(1) with $P^\top BP=J_0$.

3. Let φ be a skew-symmetric bilinear form on a finite-dimensional vector space V over k, and let N be its left radical (definition 8.1.13).

    1. Show that N is also the right radical of φ.
    2. Let W be any subspace with $V=N\oplus W$. Show that the restriction of φ to W is a symplectic form on W.

3. Deduce that $\operatorname{rank}[\varphi]_{\mathcal{B}}$ is even for every basis $\mathcal{B}$ of V, and that every skew-symmetric $B \in M_n(k)$ is congruent to a block diagonal matrix $\operatorname{diag}(J_0, 0)$ whose first block is the standard $2r \times 2r$ matrix of definition 10.2.4 and whose second block is the zero matrix of size $n - 2r$.

4. Let ω_0 be the standard symplectic form on k^2. Show that $\omega_0(x, y)$ is the determinant of the matrix with columns x and y, and deduce that $\omega_0(x, y) = 0$ if and only if x and y are linearly dependent. Then show that the corresponding statement on k^4 is false, by exhibiting an independent pair x, y with $\omega_0(x, y) = 0$.

5. Show that a pair (e, f) of vectors in $\mathbb{R}^2$ is a symplectic basis of $(\mathbb{R}^2, \omega_0)$ if and only if the matrix P with columns e and f satisfies $\det P = 1$. Conclude that $(\mathbb{R}^2, \omega_0)$ has infinitely many symplectic bases, and exhibit one in which neither vector is a multiple of e_1 or of e_2.

6. Let $A \in M_{2m}(\mathbb{R})$ satisfy $A^\top J_0 A = J_0$. Show that $\det A = \pm 1$ and that $A^{-1} = -J_0 A^\top J_0$. Show also that the set of such A is closed under multiplication and under taking inverses, and that J_0 itself belongs to it.

7. Over $k = \mathbb{C}$, let $\omega(x, y) = x^\top B y$ on $\mathbb{C}^2$ with $B = \left(\begin{smallmatrix} 0 & i \\ -i & 0 \end{smallmatrix}\right)$. Verify that B is skew-symmetric and invertible, so that ω is symplectic, and find a symplectic basis (e, f) of $(\mathbb{C}^2, \omega)$ together with the matrix P satisfying $P^\top B P = J_0$. Then decide whether B is skew-symmetric when regarded as a matrix over $\mathbb{C}$ in the sense of $B^* = -B$, where B^* is the conjugate transpose (definition 5.5.12).

10.3 Isotropic and Lagrangian Subspaces

Theorem 10.2.6 produces a symplectic basis in every symplectic space of dimension $2m$, so no invariant of such a space survives beyond that dimension. Its subspaces are another matter. A subspace W of an inner product space inherits an inner product, and theorem 5.5.3 writes $V = W \oplus W^\perp$. A subspace of a symplectic space inherits a skew-symmetric form that may be degenerate, and that may be zero.

This section defines the complement attached to ω, computes its dimension, and identifies the subspaces on which ω vanishes identically. The largest of those have dimension m. They are exactly the subspaces equal to their own complement, and each of them is spanned by the first half of a symplectic basis. Throughout, k is $\mathbb{R}$ or $\mathbb{C}$ and (V, ω) is a symplectic vector space over k (definition 10.2.2), whose dimension is even (proposition 10.2.3) and is written $\dim V = 2m$ with $m \geq 1$.

Definition 10.3.1: The Symplectic Complement of a Subspace

Let (V, ω) be a symplectic vector space over k and let $W \subseteq V$ be a subspace. The *symplectic complement* of W is

$$W^\omega = \{\, v \in V \,|\, \omega(w, v) = 0 \text{ for every } w \in W \,\}$$

the set of vectors v that are *ω-orthogonal* to every vector of W.

The set W^ω is a subspace of V (definition 2.1.2): the vector 0 lies in it because $\omega(w, 0) = 0$ for every w, by linearity of ω in its second argument (definition 8.1.1), and if $u, v \in W^\omega$ and $c \in k$ then $\omega(w, u + cv) = \omega(w, u) + c\,\omega(w, v) = 0$ for every $w \in W$, so $u + cv \in W^\omega$. The order in which the two arguments are written does not matter: $\omega(v, w) = -\omega(w, v)$, so one of them vanishes exactly when the other does. That is why the one-sided definition suffices here, where the two radicals of definition 8.1.13 are the same set, and not just two sets of the same dimension.

The definition copies definition 5.5.1 with the inner product replaced by ω, and the two constructions behave differently from the start. If $\langle -, - \rangle$ is an inner product then $W \cap W^\perp = \{0\}$, because a vector in the intersection satisfies $\langle w, w \rangle = 0$ and is therefore zero. For a symplectic form $\omega(v, v) = 0$ holds for every v (proposition 10.2.1), so every line $\operatorname{span}\{v\}$ is contained in its own symplectic complement. The complement of W can meet W, and can contain it. This construction already appeared inside the proof of theorem 10.2.6, which passed from the plane spanned by a pair e_1, f_1 to the vectors ω-orthogonal to it.

Complements are computed from a matrix of ω in the same way orthogonal complements are computed from a matrix. Fix an ordered basis $\mathcal{B} = (b_1, \ldots, b_{2m})$ of V and put $B = [\omega]_{\mathcal{B}}$ (definition 8.1.2), which is invertible because ω is nondegenerate (proposition 8.1.14). Let $w_1, \ldots, w_j$ be a basis of a subspace W of dimension $j \geq 1$ and let $S \in M_{2m\times j}(k)$ be the matrix whose columns are the coordinate columns $[w_1]_{\mathcal{B}}, \ldots, [w_j]_{\mathcal{B}}$. A vector v lies in W^ω exactly when $\omega(w_i, v) = 0$ for $i = 1, \ldots, j$: every $w \in W$ is $c_1 w_1 + \cdots + c_j w_j$ for some $c_i \in k$, and $\omega(w, v) = c_1\omega(w_1, v) + \cdots + c_j\omega(w_j, v)$ by linearity in the first argument. Each of those j conditions reads $[w_i]_{\mathcal{B}}^\top B\,[v]_{\mathcal{B}} = 0$ by theorem 8.1.3(1), and the j of them stacked into one matrix equation say that

$$v \in W^\omega \quad \text{if and only if} \quad S^\top B\,[v]_{\mathcal{B}} = 0$$

so in the coordinates of $\mathcal{B}$ the symplectic complement of W is the null space $\operatorname{null}(S^\top B)$ of definition 2.3.3.

Take $V = \mathbb{R}^4$ with the standard symplectic form ω_0 of definition 10.2.4 and let $\mathcal{B}$ be the standard symplectic basis e_1, e_2, f_1, f_2, so that $B = J_0$ with

$$J_0 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix}$$

Write $y = (y_1, y_2, y_3, y_4)^\top$ for the coordinate column of a vector relative to $\mathcal{B}$. For $W = \operatorname{span}\{e_1\}$ the matrix $S^\top J_0$ is the first row $(0, 0, 1, 0)$ of J_0, so W^ω is cut out by $y_3 = 0$ and equals $\operatorname{span}\{e_1, e_2, f_2\}$, a subspace of dimension 3 that contains W. For $W = \operatorname{span}\{e_1, f_1\}$ the matrix $S^\top J_0$ has rows $(0, 0, 1, 0)$ and $(-1, 0, 0, 0)$, the first and third rows of J_0, so W^ω is cut out by $y_3 = 0$ and $y_1 = 0$ and equals $\operatorname{span}\{e_2, f_2\}$, which meets W only in 0. For $W = \operatorname{span}\{e_1, e_2\}$ the matrix $S^\top J_0$ has rows $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, so W^ω is cut out by $y_3 = y_4 = 0$ and equals $\operatorname{span}\{e_1, e_2\} = W$. The three subspaces exhibit three of the relations that can hold between W and W^ω: containment $W \subseteq W^\omega$, intersection in 0 alone, and equality. In all three cases the two dimensions add to 4. The three do not exhaust the possibilities: for $W = \operatorname{span}\{e_1, e_2, f_1\}$ the rows of $S^\top J_0$ are the first three rows of J_0, which cut out $y_1 = y_3 = y_4 = 0$, so $W^\omega = \operatorname{span}\{e_2\}$ is a line lying inside W.

Proposition 10.3.2: Dimension and the Double Complement

Let (V, ω) be a symplectic vector space over k with $\dim V = 2m$, and let W, W_1, W_2 be subspaces of V.

1. $\dim W + \dim W^\omega = 2m$.
2. $(W^\omega)^\omega = W$.
3. If $W_1 \subseteq W_2$ then $W_2^\omega \subseteq W_1^\omega$.

***Proof*:**

1. If $W = \{0\}$ the defining condition of definition 10.3.1 is empty, so $W^\omega = V$ and the two dimensions are 0 and $2m$. Assume $j = \dim W \geq 1$, fix a basis $w_1, \dots, w_j$ of W (theorem 2.2.9), and keep the basis $\mathcal{B}$, the matrix $B = [\omega]_\mathcal{B}$ and the matrix $S \in M_{2m \times j}(k)$ of the coordinate description established before the proposition. Define

   $$T \colon V \longrightarrow k^j, \qquad T(v) = \big(\omega(w_1, v), \dots, \omega(w_j, v)\big)^\top$$

   which is linear (definition 2.4.2) because ω is linear in its second argument. That description says both that $T(v) = 0$ holds exactly when $v \in W^\omega$, so that $\ker T = W^\omega$ (definition 2.4.4), and that $T(v) = S^\top B\,[v]_\mathcal{B}$.

   The image of T (definition 2.4.5) is therefore the set of columns $S^\top B y$ with $y \in k^{2m}$, since every $y \in k^{2m}$ is the coordinate column of the vector $y_1 b_1 + \cdots + y_{2m} b_{2m}$. The matrix B is invertible, so every $u \in k^{2m}$ is By for $y = B^{-1}u$, and the two sets $\{S^\top B y \big| y \in k^{2m}\}$ and $\{S^\top u \big| u \in k^{2m}\}$ coincide. The second is the column space of $S^\top$ (definition 2.3.1), whose dimension is $\operatorname{rank} S^\top = \operatorname{rank} S$ by theorem 2.3.10.

   The rank of S is j. Indeed, writing $w_i = \sum_t s_{ti} b_t$ so that s_{ti} is the entry of S in row t and column i, a column $c \in k^j$ satisfies $Sc = 0$ exactly when $\sum_t \big(\sum_i c_i s_{ti}\big) b_t = 0$, which by the uniqueness of the coefficients of a vector in a basis (definition 2.2.1) holds exactly when $c_1 w_1 + \cdots + c_j w_j = 0$, hence only for $c = 0$ because the w_i are linearly independent. So $\operatorname{nullity} S = 0$ and theorem 2.3.11 gives $\operatorname{rank} S = j - 0 = j$.

   Combining the last two paragraphs, $\dim \operatorname{im} T = j$. Applying theorem 2.4.10 to T,

   $$\dim W^\omega + j = \dim \ker T + \dim \operatorname{im} T = \dim V = 2m$$

   which is the claim.

2. Let $w \in W$. Every $u \in W^\omega$ satisfies $\omega(w, u) = 0$ by definition 10.3.1, hence also $\omega(u, w) = -\omega(w, u) = 0$, so $w \in (W^\omega)^\omega$. This proves $W \subseteq (W^\omega)^\omega$. Applying (1) twice,

   $$\dim (W^\omega)^\omega = 2m - \dim W^\omega = 2m - (2m - \dim W) = \dim W$$

   and two subspaces one of which contains the other, with equal dimensions, are equal by corollary 2.2.12.

3. Let $v \in W_2^\omega$. Then $\omega(w, v) = 0$ for every $w \in W_2$, and in particular for every $w \in W_1$, since $W_1 \subseteq W_2$. Hence $v \in W_1^\omega$, as we sought to show.

Part (1) is the same count as in the orthogonal case, where theorem 5.5.3 gives $\dim W + \dim W^{\perp} = \dim V$. What does not carry over is the direct sum. The subspace $W = \operatorname{span}\{e_1\}$ of $\mathbb{R}^4$ computed above has $W^{\omega} = \operatorname{span}\{e_1, e_2, f_2\}$, so $W + W^{\omega}$ is that same three-dimensional subspace and is not all of $\mathbb{R}^4$, even though the dimensions add correctly. Part (2) says that no information is lost in passing to the complement, and part (3) says that the passage reverses inclusions, so a large subspace has a small complement. The three parts together make $W \mapsto W^{\omega}$ a self-inverse, inclusion-reversing correspondence on the subspaces of V.

The first of the $\mathbb{R}^4$ computations above, in which W was contained in W^{ω}, is the case with no counterpart in the orthogonal setting, and it is the one the rest of the section is about.

Definition 10.3.3: Isotropic Subspace

Let (V, ω) be a symplectic vector space over k. A subspace $W \subseteq V$ is *isotropic* if $\omega(u, v) = 0$ for all $u, v \in W$, that is, if the restriction of ω to W is the zero form.

Unwinding definition 10.3.1, a subspace W is isotropic exactly when $W \subseteq W^{\omega}$, and that reformulation is the one used below.[2] Every line $\operatorname{span}\{v\}$ with $v \neq 0$ is isotropic, since $\omega(av, bv) = ab\,\omega(v, v) = 0$ for all $a, b \in k$ by bilinearity and proposition 10.2.1. The zero subspace is isotropic for the same reason.

Isotropy is checked on a basis. If $u_1, \ldots, u_p$ span W and $u = a_1u_1 + \cdots + a_pu_p$ and $v = b_1u_1 + \cdots + b_pu_p$ are two vectors of W, then expanding in each argument in turn gives

$$\omega(u, v) = \sum_{i=1}^{p} \sum_{j=1}^{p} a_i b_j\, \omega(u_i, u_j) \tag{10.5}$$

so W is isotropic as soon as $\omega(u_i, u_j) = 0$ for all i and j. In $(\mathbb{R}^4, \omega_0)$ the subspace $\operatorname{span}\{e_1, e_2\}$ is therefore isotropic, because $\omega_0(e_i, e_j) = 0$ for $i, j \in \{1, 2\}$, and so is $\operatorname{span}\{e_1, f_2\}$, since $\omega_0(e_1, f_2) = \delta_{12} = 0$ and $\omega_0(f_2, e_1) = 0$, where δ_{ij} is 1 when $i = j$ and 0 otherwise. Neither $\operatorname{span}\{e_1, f_1\}$ nor $\mathbb{R}^4$ itself is isotropic, since $\omega_0(e_1, f_1) = 1$.

An isotropic subspace cannot be too large, and the bound follows from the dimension count alone.

Proposition 10.3.4: The Dimension Bound for Isotropic Subspaces

Let (V, ω) be a symplectic vector space over k with $\dim V = 2m$ and let $W \subseteq V$ be isotropic. Then $\dim W \leq m$.

***Proof*:**
Being isotropic, W is contained in W^{ω}, so $\dim W \leq \dim W^{\omega}$ by corollary 2.2.12. By proposition 10.3.2(1) the right-hand side is $2m - \dim W$, so $\dim W \leq 2m - \dim W$, that is $2 \dim W \leq 2m$. Dividing by 2 gives $\dim W \leq m$, as we sought to show.

Half the dimension is therefore the ceiling, and the subspaces attaining it carry a name.

[2] The same two definitions are made over an arbitrary field, where the alternating condition $\omega(v, v) = 0$ replaces skew-symmetry; see *Core Algebra* [9, Alternating Bilinear Form].

Definition 10.3.5: Lagrangian Subspace

Let (V, ω) be a symplectic vector space over k with $\dim V = 2m$. A subspace $L \subseteq V$ is *Lagrangian* if L is isotropic and $\dim L = m$.

Example 10.3.6: Lagrangian Subspaces of $\mathbb{R}^4$

Work in $(\mathbb{R}^4, \omega_0)$ with the symplectic basis e_1, e_2, f_1, f_2, so that $m = 2$ and a Lagrangian subspace is an isotropic plane. Four of them, each verified against definition 10.3.5:

1. $E = \operatorname{span}\{e_1, e_2\}$. It is isotropic because $\omega_0(e_1, e_2) = 0$, and its dimension is 2. The computation preceding proposition 10.3.2 found $E^\omega = E$.

2. $F = \operatorname{span}\{f_1, f_2\}$, isotropic because $\omega_0(f_1, f_2) = 0$.

3. $\operatorname{span}\{e_1, f_2\}$, isotropic because $\omega_0(e_1, f_2) = \delta_{12} = 0$. This plane meets both E and F in a line.

4. $\operatorname{span}\{e_1 + f_2,\ e_2 + f_1\}$. Expanding by bilinearity,
$$\omega_0(e_1 + f_2,\ e_2 + f_1) = \omega_0(e_1, e_2) + \omega_0(e_1, f_1) + \omega_0(f_2, e_2) + \omega_0(f_2, f_1) = 0 + 1 - 1 + 0 = 0$$
using $\omega_0(f_2, e_2) = -\omega_0(e_2, f_2) = -1$. With (10.5) and $\omega_0(v, v) = 0$ this makes the plane isotropic. The two vectors are independent, so this is a Lagrangian plane, and it meets neither E nor F except in 0.

The fourth belongs to a family. For $S \in M_2(\mathbb{R})$ let L_S be the span of the two columns of $\left(\begin{smallmatrix} I_2 \\ S \end{smallmatrix}\right) \in M_{4\times 2}(\mathbb{R})$, that is, the set of vectors with coordinate column $\left(\begin{smallmatrix} x \\ Sx \end{smallmatrix}\right)$ for $x \in \mathbb{R}^2$. Those two columns are independent, since their first two entries are the columns of I_2, so $\dim L_S = 2$. For $x, y \in \mathbb{R}^2$,
$$\omega_0\left(\begin{pmatrix} x \\ Sx \end{pmatrix}, \begin{pmatrix} y \\ Sy \end{pmatrix}\right) = \begin{pmatrix} x \\ Sx \end{pmatrix}^\top \begin{pmatrix} Sy \\ -y \end{pmatrix} = x^\top S y - (Sx)^\top y = x^\top (S - S^\top) y$$
where the first equality evaluates $J_0\left(\begin{smallmatrix} y \\ Sy \end{smallmatrix}\right)$ and the last uses $(Sx)^\top = x^\top S^\top$ (proposition 2.3.5(3)). Letting x and y run over the two standard basis columns of $\mathbb{R}^2$ picks out the individual entries of $S - S^\top$, so the displayed value vanishes for all x and y exactly when $S = S^\top$. So L_S is Lagrangian exactly when S is symmetric, and item (4) is L_S for $S = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, while item (1) is L_S for $S = 0$. Item (3) is not any L_S, since it meets F in the line $\operatorname{span}\{f_2\}$ while $L_S \cap F = \{0\}$ for every S.

For $m = 1$ the list is complete and short. A Lagrangian subspace of $(\mathbb{R}^2, \omega_0)$ is an isotropic line, and every line is isotropic, so the Lagrangian subspaces of $\mathbb{R}^2$ are exactly the lines through the origin. Here
$$\omega_0(x, y) = x_1 y_2 - x_2 y_1 = \det \begin{pmatrix} x_1 & y_1 \\ x_2 & y_2 \end{pmatrix}$$
is the signed area of chapter 3, and it vanishes on a pair exactly when the pair is linearly dependent. The graph family of example 10.3.6 becomes the family L_s of lines of slope s, which covers every line except the one spanned by f_1.

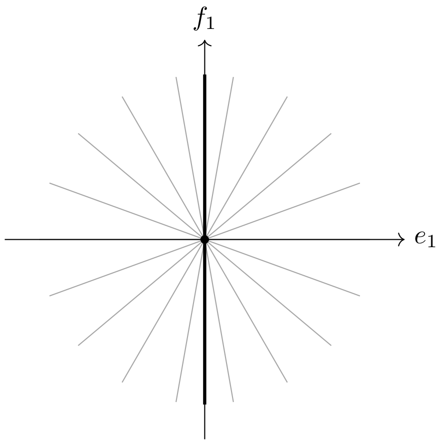


Figure 10.2: Every line through the origin in $\mathbb{R}^2$ is a Lagrangian subspace of $(\mathbb{R}^2, \omega_0)$, and the graphs L_s account for all of them except the thickened line $\operatorname{span}\{f_1\}$

Two descriptions of a Lagrangian subspace have appeared so far. One is the definition, half the dimension of V. The other showed up in the computation $E^\omega = E$ above, equality with its own symplectic complement. A third is maximality: being contained in no larger isotropic subspace. The three agree.

Theorem 10.3.7: Three Characterizations of a Lagrangian Subspace

Let (V, ω) be a symplectic vector space over k with $\dim V = 2m$ and let $W \subseteq V$ be a subspace. The following are equivalent.

1. W is Lagrangian.
2. $W = W^\omega$.
3. W is isotropic, and no isotropic subspace of V contains W properly.

***Proof*:**
The three implications $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (1)$ are proved in turn.

$(1) \Rightarrow (2)$. Let W be Lagrangian. Being isotropic it satisfies $W \subseteq W^\omega$, and proposition 10.3.2(1) with $\dim W = m$ gives $\dim W^\omega = 2m - m = m = \dim W$. Two subspaces one of which contains the other, with equal dimensions, are equal by corollary 2.2.12, so $W = W^\omega$.

$(2) \Rightarrow (3)$. Suppose $W = W^\omega$. Then $W \subseteq W^\omega$, so W is isotropic. Let U be an isotropic subspace with $W \subseteq U$. Applying proposition 10.3.2(3) to that inclusion gives $U^\omega \subseteq W^\omega = W$, and $U \subseteq U^\omega$ because U is isotropic, so $U \subseteq W$. Together with $W \subseteq U$ this forces $U = W$, so no isotropic subspace contains W properly.

$(3) \Rightarrow (1)$. Let W be isotropic and maximal in the stated sense, and put $j = \dim W$, so that $j \leq m$ by proposition 10.3.4. Suppose $j < m$. Then proposition 10.3.2(1) gives $\dim W^\omega = 2m - j > j = \dim W$, and since $W \subseteq W^\omega$ the two subspaces are unequal, again by corollary 2.2.12. Choose $v \in W^\omega$ with $v \notin W$, let $w_1, \ldots, w_j$ be a basis of W when $j \geq 1$, and put $U = \operatorname{span}\{w_1, \ldots, w_j, v\}$, with $U = \operatorname{span}\{v\}$ when $j = 0$. The list $w_1, \ldots, w_j, v$ is linearly independent, because a dependence with a nonzero coefficient on v would place v in $\operatorname{span}\{w_1, \ldots, w_j\} = W$ and a dependence with coefficient 0 on v is a dependence among the w_i, so $\dim U = j + 1$.

The subspace U is isotropic. A pair of its vectors is $u = w + cv$ and $u' = w' + c'v$ with $w, w' \in W$ and $c, c' \in k$, and bilinearity gives

$$\omega(u, u') = \omega(w, w') + c'\,\omega(w, v) + c\,\omega(v, w') + cc'\,\omega(v, v)$$

The first term is 0 because W is isotropic. The second is 0 because $v \in W^\omega$. The third is $-c\,\omega(w', v) = 0$ for the same reason. The fourth is 0 by proposition 10.2.1. So U is an isotropic subspace containing W properly, contradicting (3). Hence $j = m$, and W is Lagrangian, completing the proof.

Condition (2) is the equality $W = W^\omega$, which no subspace other than a Lagrangian one satisfies. Condition (3) identifies the Lagrangian subspaces as the maximal isotropic ones, so the bound $\dim W \leq m$ of proposition 10.3.4 is attained by every maximal isotropic subspace and by no other. The enlargement performed in the proof of (3) $\Rightarrow$ (1) also gives a procedure: starting from any isotropic subspace it can be repeated until the dimension reaches m, so every isotropic subspace lies inside a Lagrangian one. The exercises below ask for that argument.

A Lagrangian subspace L equals its own symplectic complement, so $L + L^\omega = L$, which is not V, and the symplectic complement cannot split the space the way an orthogonal complement does. Another Lagrangian subspace can, and the proof below produces one together with a basis adapted to it.

Proposition 10.3.8: Every Lagrangian Subspace Is Half of a Symplectic Basis

Let (V, ω) be a symplectic vector space over k with $\dim V = 2m$, $m \geq 1$, and let $L \subseteq V$ be Lagrangian. Then V has a symplectic basis $e_1, \ldots, e_m, f_1, \ldots, f_m$, so one with

$$\omega(e_i, e_j) = 0, \qquad \omega(f_i, f_j) = 0, \qquad \omega(e_i, f_j) = \delta_{ij}$$

for all i and j, such that $L = \operatorname{span}\{e_1, \ldots, e_m\}$. Consequently $L' = \operatorname{span}\{f_1, \ldots, f_m\}$ is a Lagrangian subspace with $V = L \oplus L'$.

***Proof*:**
Since L is Lagrangian its dimension is m, and a basis $e_1, \ldots, e_m$ of L is available by theorem 2.2.9. The three steps below produce the second half of the basis from it.

Step 1: vectors pairing with the e_i. Define

$$T\colon V \longrightarrow k^m, \qquad T(v) = \big(\omega(e_1, v), \ldots, \omega(e_m, v)\big)^\top$$

which is linear because ω is linear in its second argument. Since $e_1, \ldots, e_m$ span L and ω is linear in its first argument, $T(v) = 0$ holds exactly when $\omega(w, v) = 0$ for every $w \in L$, so $\ker T = L^\omega$. By theorem 10.3.7, applied to the Lagrangian subspace L, this kernel is L itself, of dimension m. Theorem 2.4.10 then gives $\dim \operatorname{im} T = 2m - m = m$, and $\operatorname{im} T$ is a subspace of k^m of dimension m, hence equal to k^m by corollary 2.2.12. So T takes every value, and there are $g_1, \ldots, g_m \in V$ with $T(g_j)$ equal to the j-th standard basis column of k^m, that is with

$$\omega(e_i, g_j) = \delta_{ij}$$

for all i and j.

Step 2: correcting the g_j so that they pair trivially with each other. Put $c_{ij} = \omega(g_i, g_j)$, so that $c_{ji} = -c_{ij}$ by skew-symmetry, and define

$$f_i = g_i + \tfrac{1}{2} \sum_{t=1}^{m} c_{ti}\, e_t$$

for $i = 1, \ldots, m$, which is available because 2 is invertible in k. Each f_i differs from g_i by a vector of L. Since L is isotropic, $\omega(e_i, e_t) = 0$ for all i and t, so bilinearity gives

$$\omega(e_i, f_j) = \omega(e_i, g_j) + \tfrac{1}{2} \sum_{t} c_{tj}\, \omega(e_i, e_t) = \delta_{ij}$$

and the pairing established in Step 1 is unchanged by the correction. For the pairs among the f's, expand by bilinearity and use $\omega(g_i, e_t) = -\omega(e_t, g_i) = -\delta_{ti}$ together with $\omega(e_t, g_j) = \delta_{tj}$:

$$\omega(f_i, f_j) = c_{ij} + \tfrac{1}{2} \sum_{t} c_{tj}\, \omega(g_i, e_t) + \tfrac{1}{2} \sum_{t} c_{ti}\, \omega(e_t, g_j) + \tfrac{1}{4} \sum_{t,s} c_{ti} c_{sj}\, \omega(e_t, e_s)$$

The last sum vanishes term by term. The first sum contributes $-\frac{1}{2} c_{ij}$, since only $t = i$ survives, and the second contributes $\frac{1}{2} c_{ji} = -\frac{1}{2} c_{ij}$, since only $t = j$ survives. Hence $\omega(f_i, f_j) = c_{ij} - \frac{1}{2} c_{ij} - \frac{1}{2} c_{ij} = 0$.

Step 3: the list is a basis, and the consequence. The list $e_1, \ldots, e_m, f_1, \ldots, f_m$ is linearly independent. Suppose $\sum_i a_i e_i + \sum_j b_j f_j = 0$ with $a_i, b_j \in k$. Applying $\omega(e_l, -)$ and using linearity in the second argument,

$$0 = \sum_i a_i\, \omega(e_l, e_i) + \sum_j b_j\, \omega(e_l, f_j) = b_l$$

because the first sum vanishes (L is isotropic) and the second is b_l by Step 2. So every b_l is 0, whence $\sum_i a_i e_i = 0$ and every a_i is 0 as well. The list is a basis of its span (definition 2.2.1), so that span has dimension $2m$ and is therefore all of V by corollary 2.2.12. Its three families of values are those displayed in the statement, so the list is a symplectic basis, and its first half spans L by construction.

For the consequence, $L' = \operatorname{span}\{f_1, \ldots, f_m\}$ has dimension m, and it is isotropic by (10.5), since $\omega(f_i, f_j) = 0$ for all i and j. So L' is Lagrangian. Every vector of V is a combination of the basis, hence lies in $L + L'$, and a vector $v \in L \cap L'$ can be written both as $\sum_i a_i e_i$ and as $\sum_j b_j f_j$, so that $\sum_i a_i e_i - \sum_j b_j f_j = 0$ forces all coefficients to vanish and $v = 0$. Therefore $V = L \oplus L'$ (definition 2.1.11), completing the proof.

A Lagrangian subspace therefore carries no invariant of its own. A symplectic basis can always be chosen so that L is the span of its first half, so a symplectic space of dimension $2m$ together with a Lagrangian subspace of it is (k^{2m}, ω_0) together with $\operatorname{span}\{e_1, \ldots, e_m\}$, read in a suitable basis, which extends corollary 10.2.7 from the space to the pair. The complement L' the proof produces is not determined by L. In $(\mathbb{R}^4, \omega_0)$ with $L = \operatorname{span}\{e_1, e_2\}$, the subspaces $\operatorname{span}\{f_1, f_2\}$ and $\operatorname{span}\{e_1 + f_2,\ e_2 + f_1\}$ of example 10.3.6 are both Lagrangian complements of L, and so is every L_S with S symmetric and invertible, since a vector of L_S with coordinate column $\left(\begin{smallmatrix} x \\ Sx \end{smallmatrix}\right)$ lies in L exactly when $Sx = 0$, hence only when $x = 0$. The orthogonal complement differs on both counts:

$W^\perp$ is determined by W and satisfies $V = W \oplus W^\perp$ (theorem 5.5.3), whereas L^ω is L again and each Lagrangian complement of L is a choice.

Exercise 10.3.1

1. Work in $(\mathbb{R}^6, \omega_0)$ with symplectic basis $e_1, e_2, e_3, f_1, f_2, f_3$ and let $W = \operatorname{span}\{e_1 + f_2,\ e_3\}$. Compute W^ω, giving a basis for it, and check the count of proposition 10.3.2(1). Decide whether W is isotropic and whether it is Lagrangian.

2. With $W = \operatorname{span}\{e_1 + f_2,\ e_3\}$ in $(\mathbb{R}^6, \omega_0)$ as in the previous question, exhibit a Lagrangian subspace L of $\mathbb{R}^6$ containing W, and then exhibit a Lagrangian subspace L' with $\mathbb{R}^6 = L \oplus L'$. Verify the isotropy conditions for both.

3. Let (V, ω) be symplectic and let W_1, W_2 be subspaces, and write $W_1 + W_2 = \{\, u + v \,|\, u \in W_1,\ v \in W_2 \,\}$. Show directly from definition 10.3.1 that $(W_1 + W_2)^\omega = W_1^\omega \cap W_2^\omega$, and deduce from proposition 10.3.2(2) that $(W_1 \cap W_2)^\omega = W_1^\omega + W_2^\omega$.

4. Let W be a subspace of a symplectic space (V, ω) of dimension $2m$ and let $\omega|_W$ be the restriction of ω to W, a bilinear form on W. Show that the left radical of $\omega|_W$ in the sense of definition 8.1.13 is $W \cap W^\omega$, so that $\omega|_W$ is nondegenerate exactly when $W \cap W^\omega = \{0\}$. Show that in that case $V = W \oplus W^\omega$ and $\omega|_{W^\omega}$ is nondegenerate as well.

5. Let L_1 and L_2 be Lagrangian subspaces of a symplectic space (V, ω) of dimension $2m$. Using the identity $(W_1 + W_2)^\omega = W_1^\omega \cap W_2^\omega$ established in the previous question, show that $L_1 \cap L_2 = \{0\}$ holds if and only if $L_1 + L_2 = V$. Then exhibit Lagrangian subspaces L_1, L_2 of $(\mathbb{R}^4, \omega_0)$ with $\dim(L_1 \cap L_2) = 1$.

6. In $(\mathbb{R}^{2m}, \omega_0)$ write $F = \operatorname{span}\{f_1, \ldots, f_m\}$, and for $S \in M_m(\mathbb{R})$ let L_S be the span of the columns of $\left(\begin{smallmatrix} I_m \\ S \end{smallmatrix}\right) \in M_{2m \times m}(\mathbb{R})$, generalizing example 10.3.6. Show that L_S is Lagrangian if and only if $S^\top = S$. Show further that a subspace $L \subseteq \mathbb{R}^{2m}$ is a Lagrangian subspace with $L \cap F = \{0\}$ if and only if $L = L_S$ for exactly one symmetric $S \in M_m(\mathbb{R})$.

7. Let (V, ω) be symplectic of dimension $2m$ and let $W \subseteq V$ be isotropic. Show that some Lagrangian subspace of V contains W. Deduce that V has a Lagrangian subspace, and that for every j with $0 \le j \le m$ it has an isotropic subspace of dimension j.

10.4 The Compatibility Triple (g, J, ω) and Positivity

Section 10.1 worked with a Hermitian form on a complex vector space, and section 10.2 with a symplectic form on a real one. A complex structure J (definition 9.3.1) is what converts between the two settings, so a real vector space carrying J can carry all three objects at once: a symmetric form g, a skew-symmetric form ω, and J itself. This section fixes the identity that ties the three together and proves that any two of them determine the third.

The identity is theorem 10.1.9 in the reverse direction: g and ω are the real and imaginary parts of one Hermitian form h, and J is multiplication by i. What the reverse direction adds is a positivity condition. Requiring g to be positive definite, and requiring nothing else, makes h positive definite and ω nondegenerate, and that implication is the linear-algebra content of the Riemann relations. The section closes by parametrizing the positive triples that share the standard symplectic form on $\mathbb{R}^{2m}$: they are indexed by the points of the Siegel upper half-space.

Two of the three conditions below say that J preserves a form. The third is the link between the two forms, and its exact shape is forced by the convention of definition 10.1.1 together with proposition 10.1.2(1): a Hermitian form is linear in its first variable by the definition, and conjugate-linear in its second by that proposition. With $\omega = \mathrm{Im} h$ the recovery identity of theorem 10.1.9 reads $h(v, w) = \omega(iv, w) + i\,\omega(v, w)$, so the real part of h is the form $(v, w) \mapsto \omega(iv, w)$. That is the third condition, with J in place of multiplication by i.

Definition 10.4.1: Compatible Triple

Let V be a finite-dimensional real vector space. A *compatible triple* on V consists of

1. a symmetric bilinear form g on V (definition 8.1.9);
2. a complex structure J on V (definition 9.3.1), that is, a linear map $J\colon V \to V$ with $J^2 = -\mathrm{id}_V$;
3. a skew-symmetric bilinear form ω on V (definition 8.1.9),

subject to the three conditions

$$g(Ju, Jv) = g(u, v) \tag{10.6}$$
$$\omega(Ju, Jv) = \omega(u, v) \tag{10.7}$$
$$g(u, v) = \omega(Ju, v) \tag{10.8}$$

for all $u, v \in V$. The triple is written (g, J, ω).

Conditions (10.6) and (10.7) are the statements that J preserves g and that J preserves ω. Condition (10.8) is the only one mentioning all three objects, and every computation in this section runs through it. Two rewritings of it are used constantly. Replacing u by Ju in (10.8) and using $J^2 = -\mathrm{id}_V$ gives

$$g(Ju, v) = \omega(J^2u, v) = -\omega(u, v) \tag{10.9}$$

and combining (10.9) with (10.6) gives $\omega(u, v) = -g(Ju, v) = -g(J^2u, Jv) = g(u, Jv)$.

The smallest instance is the plane, where all three objects are already familiar under other names.

Example 10.4.2: The Standard Triple on the Plane

On $V = \mathbb{R}^2$ take the standard inner product, the standard symplectic form, and the clockwise quarter turn:

$$g_0(u, v) = u_1v_1 + u_2v_2, \qquad \omega_0(u, v) = u_1v_2 - u_2v_1, \qquad J_0\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} y \\ -x \end{pmatrix}$$

In the standard basis J_0 has matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, which is the matrix J_0 of definition 10.2.4 at $m = 1$, and the letter is reused for that reason. It is the negative of the counterclockwise quarter turn also written J_0 in example 9.3.2, and the sign here is forced rather than chosen: (10.8) demands $\omega_0(J_0u, v) = g_0(u, v)$, whereas the counterclockwise turn $(x, y) \mapsto (-y, x)$ returns $-g_0(u, v)$. The identification of $\mathbb{R}^2$ with $\mathbb{C}$ used below, $(x, y) \mapsto y + ix$ rather than $x + iy$, is forced with it.

The form g_0 is symmetric and ω_0 is skew-symmetric directly from the formulas, $\omega_0(u, v)$ is the signed area of the parallelogram spanned by u and v (chapter 3), and $J_0^2(x, y) = J_0(y, -x) = (-x, -y)$,

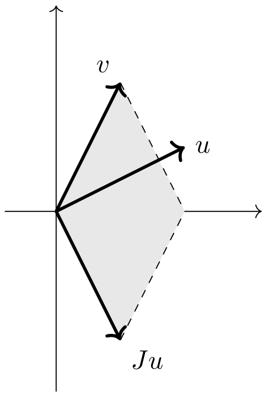


Figure 10.3: Condition (10.8) on $\mathbb{R}^2$ for $u = (2, 1)$ and $v = (1, 2)$: the shaded parallelogram spanned by Ju and v has signed area 4, which is the value of $g(u, v)$

so $J_0^2 = -\mathrm{id}$. The three conditions of definition 10.4.1 are three computations. For (10.6),

$$g_0(J_0 u, J_0 v) = u_2 v_2 + (-u_1)(-v_1) = u_1 v_1 + u_2 v_2 = g_0(u, v)$$

For (10.7),

$$\omega_0(J_0 u, J_0 v) = u_2(-v_1) - (-u_1)v_2 = u_1 v_2 - u_2 v_1 = \omega_0(u, v)$$

For (10.8),

$$\omega_0(J_0 u, v) = u_2 v_2 - (-u_1)v_1 = u_1 v_1 + u_2 v_2 = g_0(u, v)$$

so (g_0, J_0, ω_0) is a compatible triple.

Identify $\mathbb{R}^2$ with $\mathbb{C}$ by $\Phi(x, y) = y + ix$, which satisfies $\Phi(J_0(x, y)) = \Phi(y, -x) = -x + iy = i\Phi(x, y)$, so Φ carries J_0 to multiplication by i. Under Φ the two forms assemble into $h(z, w) = z\bar{w}$: writing $z = \Phi(u) = y + ix$ and $w = \Phi(v) = y' + ix'$,

$$z\bar{w} = (y + ix)(y' - ix') = (xx' + yy') + i(xy' - yx')$$

whose real part is $g_0(u, v)$ and whose imaginary part is $\omega_0(u, v)$.

Take $z = 1 + 2i$ and $w = 3 - i$, so that $h(z, w) = (1 + 2i)(3 + i) = 1 + 7i$ and therefore $g_0 = 1$ and $\omega_0 = 7$ on the corresponding pair of vectors. The recovery identity of theorem 10.1.9 is checked numerically by computing $iz = -2 + i$ and

$$\omega(iz, w) = \mathrm{Im}\big((-2 + i)(3 + i)\big) = \mathrm{Im}(-7 + i) = 1$$

which is g_0, so $\omega(iz, w) + i\,\omega(z, w) = 1 + 7i = h(z, w)$.

Each of the three objects in example 10.4.2 was written down independently, and the three conditions then held. That is not how the general situation behaves: two of the three objects, chosen freely, already pin the third down.

Theorem 10.4.3: Any Two of (g, J, ω) Determine the Third

Let V be a real vector space with $\dim V = n \geq 1$.

1. Let J be a complex structure on V and let g be a symmetric bilinear form with $g(Ju, Jv) = g(u, v)$ for all u, v. Then $\omega(u, v) := g(u, Jv)$ is the unique skew-symmetric bilinear form making (g, J, ω) a compatible triple, and ω is nondegenerate exactly when g is.

2. Let J be a complex structure on V and let ω be a skew-symmetric bilinear form with $\omega(Ju, Jv) = \omega(u, v)$ for all u, v. Then $g(u, v) := \omega(Ju, v)$ is the unique symmetric bilinear form making (g, J, ω) a compatible triple, and g is nondegenerate exactly when ω is.

3. Let g be a symmetric bilinear form and ω a nondegenerate skew-symmetric bilinear form on V, with matrices G and B respectively in a basis of V (definition 8.1.2). At most one complex structure J makes (g, J, ω) a compatible triple: the only candidate is the linear map whose matrix in that basis is $-B^{-1}G$, and it is one if and only if

$$(B^{-1}G)^2 = -I_n$$

a condition that does not depend on the chosen basis. When it holds, g is nondegenerate and n is even.

***Proof*:**

1. Bilinearity of $\omega(u, v) = g(u, Jv)$ is bilinearity of g together with linearity of J. It is skew-symmetric: using the symmetry of g, then J-invariance, then $J^2 = -\mathrm{id}_V$,

$$\omega(v, u) = g(v, Ju) = g(Jv, J^2u) = -g(Jv, u) = -g(u, Jv) = -\omega(u, v)$$

Condition (10.8) holds because $\omega(Ju, v) = g(Ju, Jv) = g(u, v)$, the hypothesis on g being applied to the pair (u, v). Condition (10.6) is the hypothesis. Condition (10.7) follows:

$$\omega(Ju, Jv) = g(Ju, J^2v) = -g(Ju, v) \qquad \text{and} \qquad \omega(u, v) = g(u, Jv) = g(Ju, J^2v) = -g(Ju, v)$$

the second chain using J-invariance of g on the pair (u, Jv).

For uniqueness, let ω' be any skew-symmetric form making (g, J, ω') compatible. Then ω' satisfies (10.8), so by the computation (10.9), which used only (10.8) and $J^2 = -\mathrm{id}_V$, one has $\omega'(u, v) = -g(Ju, v)$; and $-g(Ju, v) = -g(J^2u, Jv) = g(u, Jv) = \omega(u, v)$ by J-invariance of g. Hence $\omega' = \omega$.

Finally, J is a bijection of V, since $J(-Jv) = -J^2v = v$ exhibits an inverse. So for a fixed u the condition $\omega(u, v) = 0$ for all v says $g(u, Jv) = 0$ for all v, which as v ranges over V says $g(u, w) = 0$ for all w. The two forms therefore have the same degenerate vectors, and by definition 8.1.13 one is nondegenerate exactly when the other is.

2. Bilinearity of $g(u, v) = \omega(Ju, v)$ is again inherited. Applying the hypothesis $\omega(Jx, Jy) = \omega(x, y)$ to the pair $(x, y) = (u, Jv)$ gives $\omega(Ju, J^2v) = \omega(u, Jv)$, that is,

$$\omega(u, Jv) = -\omega(Ju, v) \tag{10.10}$$

Symmetry of g follows: $g(v,u) = \omega(Jv,u) = -\omega(u,Jv)$ by skew-symmetry, and $-\omega(u,Jv) = \omega(Ju,v) = g(u,v)$ by (10.10). Condition (10.8) holds by the definition of g, and (10.7) is the hypothesis. For (10.6),

$$g(Ju,Jv) = \omega(J^2u,Jv) = -\omega(u,Jv) = \omega(Ju,v) = g(u,v)$$

using (10.10) at the third equality. Uniqueness is immediate from (10.8), which expresses g in terms of ω and J alone. For the last clause, J is a bijection, so $g(u,v) = 0$ for all v says $\omega(Ju,v) = 0$ for all v, and $u \mapsto Ju$ carries the degenerate vectors of g onto those of ω.

3. Fix the basis and write M for the matrix of a linear map $J\colon V \to V$ in it (theorem 2.4.8). By definition 8.1.2 the entries of G and B are the values of g and ω on pairs of basis vectors, so a bilinear form is determined by its matrix, and $G^\top = G$ while $B^\top = -B$. By theorem 8.1.3 the three conditions of definition 10.4.1 read

$$M^\top GM = G, \qquad M^\top BM = B, \qquad G = M^\top B$$

the last because $\omega(Ju,v)$ has coordinate expression $(M[u])^\top B[v] = [u]^\top(M^\top B)[v]$, and $J^2 = -\mathrm{id}_V$ reads $M^2 = -I_n$. Since ω is nondegenerate, B is invertible (proposition 8.1.14), so $G = M^\top B$ forces $M^\top = GB^{-1}$ and hence

$$M = (GB^{-1})^\top = (B^{-1})^\top G^\top = -B^{-1}G$$

using $(B^{-1})^\top = (B^\top)^{-1} = -B^{-1}$ (proposition 2.3.5). This proves that at most one J can occur and identifies its matrix.

Let $M = -B^{-1}G$, so that $M^\top = GB^{-1}$ by the same computation read backwards, and $G = M^\top B$ holds. Then $M^2 = (-B^{-1}G)^2 = (B^{-1}G)^2$, so $M^2 = -I_n$ is exactly the stated condition, and it is necessary. It is also sufficient: assume $(B^{-1}G)^2 = B^{-1}GB^{-1}G = -I_n$. Then

$$M^\top GM = (GB^{-1})G(-B^{-1}G) = -G\big(B^{-1}GB^{-1}G\big) = G$$

and

$$M^\top BM = (GB^{-1})B(-B^{-1}G) = -GB^{-1}G = -B\big(B^{-1}GB^{-1}G\big) = B$$

so all three matrix conditions hold and the map with matrix M is a complex structure completing the triple. From $M^2 = -I_n$ the matrix M is invertible, so $G = M^\top B$ is a product of invertible matrices and g is nondegenerate by proposition 8.1.14; and n is even by proposition 9.3.3.

For basis independence, let P be the change of basis matrix to a second basis. By theorem 8.1.4 the new matrices of the two forms are $G' = P^\top GP$ and $B' = P^\top BP$, whence

$$(B')^{-1}G' = P^{-1}B^{-1}(P^\top)^{-1}P^\top GP = P^{-1}(B^{-1}G)P$$

Squaring, $((B')^{-1}G')^2 = P^{-1}(B^{-1}G)^2P$, which equals $-I_n$ exactly when $(B^{-1}G)^2$ does. The same identity shows $-(B')^{-1}G' = P^{-1}(-B^{-1}G)P$, which is how the matrix of one linear map transforms (theorem 2.5.4), so every basis produces the same candidate map, completing the proof.

Parts (1) and (2) also settle a redundancy in definition 10.4.1. Given (10.8), each of the two invariance conditions implies the other: part (1) derived (10.7) from (10.6), part (2) derived (10.6) from (10.7), and part (3) derived both from $M^2 = -I_n$. The definition lists all three because each is the natural hypothesis on a different pair.

The three pairs do not behave alike. A complex structure together with either form always completes, by an explicit formula. A pair of forms need not complete at all: on $\mathbb{R}^2$ with g of matrix $G = \mathrm{diag}(1, 2)$ and ω_0 of matrix $B = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, one has $B^{-1} = -B$ and

$$B^{-1}G = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}, \qquad (B^{-1}G)^2 = \begin{pmatrix} -2 & 0 \\ 0 & -2 \end{pmatrix}$$

which is $-2I_2$, so no complex structure is compatible with this pair even though both forms are nondegenerate and g is positive definite. The condition constrains the relative scale of the two forms: replacing ω_0 by $\sqrt{2}\,\omega_0$ multiplies $B^{-1}G$ by $1/\sqrt{2}$ and its square by $1/2$, turning $-2I_2$ into $-I_2$.

The next result explains why the three conditions of definition 10.4.1 are the ones to impose. Recall from theorem 9.3.4 that a complex structure J on V makes the same underlying set into a complex vector space, with scalar multiplication $(a + bi)v := av + bJv$ for $a, b \in \mathbb{R}$; write V_J for that complex vector space. For a complex-valued function F of two variables write $\mathrm{Re}F$ and $\mathrm{Im}F$ for the real-valued functions $(u, v) \mapsto \mathrm{Re}\big(F(u, v)\big)$ and $(u, v) \mapsto \mathrm{Im}\big(F(u, v)\big)$.

Theorem 10.4.4: Compatible Triples Are Hermitian Forms

Let V be a finite-dimensional real vector space and J a complex structure on V.

1. If (g, J, ω) is a compatible triple, then
$$h(u, v) := g(u, v) + i\,\omega(u, v)$$
is a Hermitian form on V_J (definition 10.1.1), with $\mathrm{Re}h = g$ and $\mathrm{Im}h = \omega$.
2. If h is a Hermitian form on V_J, then $g := \mathrm{Re}h$ and $\omega := \mathrm{Im}h$ make (g, J, ω) a compatible triple.

The two constructions are mutually inverse, so compatible triples on V with complex structure J correspond to Hermitian forms on V_J.

***Proof*:**
Both directions are theorem 10.1.9, read through theorem 9.3.4, which identifies V_J with V carrying the scalar action $(a + bi)v = av + bJv$; under that identification the operator J is multiplication by i, which is the hypothesis of the §1 theorem.

1. Let (g, J, ω) be a compatible triple. Its conditions (10.6) and (10.8) say that g is symmetric, J-invariant, and linked to ω by $g(v, w) = \omega(Jv, w)$, which are exactly the hypotheses of theorem 10.1.9(3). That part gives a unique Hermitian form on V_J with real part g, namely $h(v, w) = g(v, w) - i\,g(Jv, w)$, and theorem 10.1.9(2) rewrites $-g(Jv, w)$ as $\omega(v, w)$; so $h = g + i\omega$ and $\mathrm{Im}h = \omega$.
2. Let h be Hermitian on V_J. Theorem 10.1.9(1) gives that $g = \mathrm{Re}h$ and $\omega = \mathrm{Im}h$ are bilinear on $V_{\mathbb{R}}$, that g is symmetric and ω skew, and that both are J-invariant — conditions (1), (2) and (10.6), (10.7) of definition 10.4.1. Part (2) of the same theorem gives $g(v, w) = \omega(Jv, w)$, which is (10.8). So (g, J, ω) is a compatible triple.

The two constructions are inverse to each other, since a complex number is its real part plus i times its imaginary part, as we sought to show.

A compatible triple is therefore one object viewed three ways, and the third condition (10.8) is what makes the three views agree. In example 10.4.2 the Hermitian form produced by theorem 10.4.4 is $h(z, w) = z\bar{w}$ on $\mathbb{C}$, and the compatibility conditions were the statement that the real and imaginary parts of that product are the standard inner product and the standard area form.

Nothing so far requires either form to be nondegenerate, and theorem 10.1.5 allows a Hermitian form any inertia. The condition that singles out the geometrically usable triples is positivity of g, and it is a strong condition: it forces the inertia of h and the nondegeneracy of ω simultaneously.

Definition 10.4.5: Positive Triples and Positive Definite Hermitian Forms

A compatible triple (g, J, ω) on a real vector space V is *positive* if g is positive definite (definition 8.4.1), that is, if $g(v, v) > 0$ for every $v \neq 0$.
A Hermitian form h on a complex vector space W is *positive definite* if $h(w, w) > 0$ for every $w \neq 0$. The inequality is meaningful because $h(w, w)$ is real (proposition 10.1.2).

The triple of example 10.4.2 is positive, since g_0 is the standard inner product on $\mathbb{R}^2$ and $g_0(v, v) = v_1^2 + v_2^2$.

Theorem 10.4.6: Positivity of a Compatible Triple

Let (g, J, ω) be a compatible triple on a real vector space V with $\dim_{\mathbb{R}} V = 2m$, $m \geq 1$, and let $h = g + i\,\omega$ be the Hermitian form on V_J of theorem 10.4.4. The following are equivalent.

1. The triple is positive.
2. h is positive definite.
3. $\omega(Jv, v) > 0$ for every $v \neq 0$.

When they hold, ω is nondegenerate, so (V, ω) is a symplectic vector space (definition 10.2.2); g is an inner product on V for which $g(Jv, Jv) = g(v, v)$; and the inertia of h (definition 10.1.6) is $(m, 0, 0)$, so its signature is m.
Moreover, every complex structure J on V occurs in a positive compatible triple: if g_0 is any inner product on V (definition 5.1.2) then

$$g(u, v) := \tfrac{1}{2}\big(g_0(u, v) + g_0(Ju, Jv)\big)$$

is a J-invariant inner product, and (g, J, ω) with $\omega(u, v) = g(u, Jv)$ is a positive compatible triple.

***Proof*:**
The three conditions are equivalent. A skew-symmetric form on a real vector space vanishes on the diagonal (proposition 10.2.1), so

$$h(v, v) = g(v, v) + i\,\omega(v, v) = g(v, v)$$

for every v. Thus $h(v, v) > 0$ for all $v \neq 0$ says exactly that $g(v, v) > 0$ for all $v \neq 0$, which is (1)

$\Leftrightarrow$ (2). Setting $u = v$ in (10.8) gives $g(v, v) = \omega(Jv, v)$, which is (1) $\Leftrightarrow$ (3).

The consequences. Assume the three conditions. If $g(u, v) = 0$ for every v then in particular $g(u, u) = 0$, so $u = 0$; hence g is nondegenerate, and ω is nondegenerate by theorem 10.4.3(1), whose hypotheses hold because g is symmetric and satisfies (10.6). A nondegenerate skew-symmetric form is what definition 10.2.2 asks for. A symmetric positive definite bilinear form on a real vector space is an inner product (definition 5.1.2), and $g(Jv, Jv) = g(v, v)$ is (10.6) on the pair (v, v).

For the inertia, first record that $\dim_{\mathbb{C}} V_J = m$. A $\mathbb{C}$-basis $w_1, \ldots, w_k$ of V_J gives the list $w_1, Jw_1, \ldots, w_k, Jw_k$ in V, which spans V over $\mathbb{R}$ because $(a + bi)w = aw + bJw$, and which is $\mathbb{R}$-independent because a relation $\sum_j (a_j w_j + b_j Jw_j) = 0$ is the $\mathbb{C}$-relation $\sum_j (a_j + ib_j)w_j = 0$, forcing every $a_j + ib_j = 0$. So $2k = \dim_{\mathbb{R}} V = 2m$ and $k = m$. Now theorem 10.1.5 produces a basis $u_1, \ldots, u_m$ of V_J in which the matrix of h is $\operatorname{diag}(I_p, -I_q, 0)$ with $p + q + r = m$, where r counts the zero entries. If $q \geq 1$ or $r \geq 1$ then the corresponding basis vector u_j is nonzero with $h(u_j, u_j) \leq 0$, contradicting (2). Hence $q = r = 0$ and $p = m$, so the inertia of h is $(m, 0, 0)$ and $\sigma(h) = m$.

Existence for a given J. Let g_0 be an inner product on V and let g be the average displayed in the statement. It is bilinear and symmetric, being a sum of two symmetric bilinear forms, the second of which is $(u, v) \mapsto g_0(Ju, Jv)$. For $v \neq 0$ the vector Jv is nonzero, since J is a bijection with inverse $-J$, so $g(v, v) = \frac{1}{2}(g_0(v, v) + g_0(Jv, Jv)) > 0$ and g is an inner product. It is J-invariant: using $J^2 = -\mathrm{id}_V$ and bilinearity,

$$g(Ju, Jv) = \tfrac{1}{2}\big(g_0(Ju, Jv) + g_0(J^2u, J^2v)\big) = \tfrac{1}{2}\big(g_0(Ju, Jv) + g_0(u, v)\big) = g(u, v)$$

So theorem 10.4.3(1) applies and completes (g, J) to a compatible triple with $\omega(u, v) = g(u, Jv)$, which is positive because g is, completing the proof.

The equivalence of (1) and (2) is what makes the word "positive" unambiguous: the real form g and the Hermitian form h are positive definite together, because they take the same values on the diagonal. Condition (3) is the same statement written with ω and J alone, and it is the form in which the condition is usually imposed when ω is the given object and J is being chosen. Note the sign: (10.8) gives $\omega(v, Jv) = -\omega(Jv, v) = -g(v, v)$, so a positive triple has $\omega(v, Jv) < 0$ for every nonzero v. The opposite inequality appears in sources that take a Hermitian form conjugate-linear in its first variable, which is not the convention of definition 10.1.1.

A complex torus V/Λ carries the same three structures together with a lattice Λ, and the classical Riemann relations are the conditions that ω be J-invariant, take integer values on Λ, and pair with J to a positive definite g. Theorem 10.4.6 is the part of that statement which involves no lattice, and the lattice-level theorem belongs to *Abelian Varieties* [1].

The remainder of the section fixes $V = \mathbb{R}^{2m}$ and the standard symplectic form on it, and asks which complex structures complete it to a positive triple. Group the standard basis as $e_1, \ldots, e_m, f_1, \ldots, f_m$ with $f_j = e_{m+j}$, and write a vector as

$$u = \begin{pmatrix} x \\ y \end{pmatrix}, \qquad x, y \in \mathbb{R}^m$$

abbreviated $u = (x, y)$, so that x collects the first m coordinates and y the last m. In this grouping the standard symplectic form (definition 10.2.4) is

$$\omega_0(u, v) = x^\top y' - y^\top x' \qquad \text{for } u = (x, y),\ v = (x', y')$$

with matrix
$$J_0 = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix}, \qquad J_0^2 = -I_{2m}, \qquad J_0^{-1} = -J_0$$
the two identities being blockwise products (proposition 3.5.1). Taking g to be the standard inner product, with matrix I_{2m}, theorem 10.4.3(3) produces the candidate $-J_0^{-1} I_{2m} = J_0$, whose square is $-I_{2m}$: on $\mathbb{R}^{2m}$ the standard inner product and the standard symplectic form are compatible, and the complex structure they determine has J_0 for its matrix as well.

Positivity is what will cut the parameter set down. The next definition names that set; the proposition after it identifies what the set parametrizes.

Definition 10.4.7: The Siegel Upper Half-Space

Let $m \geq 1$. The *Siegel upper half-space of degree m* is
$$\mathcal{H}_m = \{\, \tau \in M_m(\mathbb{C}) \big| \tau^\top = \tau \text{ and } \mathrm{Im}\tau \text{ is positive definite} \,\}$$
where $\mathrm{Im}\tau \in M_m(\mathbb{R})$ is the entrywise imaginary part of τ. The second condition is positive definiteness of the real symmetric matrix $\mathrm{Im}\tau$ in the sense of definition 8.4.1, not an entrywise inequality: it says $x^\top(\mathrm{Im}\tau)x > 0$ for every nonzero $x \in \mathbb{R}^m$.

For $m = 1$ a matrix is its single entry, the symmetry condition is vacuous, and a 1×1 real matrix (q) is positive definite exactly when $q > 0$. So $\mathcal{H}_1 = \{\, p + iq \big| p, q \in \mathbb{R},\ q > 0 \,\}$ is the upper half plane: $1 + 2i$ lies in it and $1 - 2i$ does not. For $m = 2$ the matrix $\tau = \left(\begin{smallmatrix} 2i & 1 \\ 1 & i \end{smallmatrix}\right)$ is symmetric and has $\mathrm{Im}\tau = \mathrm{diag}(2, 1)$, which is positive definite, so $\tau \in \mathcal{H}_2$.

Proposition 10.4.8: Positive Triples and the Siegel Upper Half-Space

Let $m \geq 1$ and let ω_0 be the standard symplectic form on $\mathbb{R}^{2m}$. For $\tau \in M_m(\mathbb{C})$ write $\tau = P + iQ$ with $P, Q \in M_m(\mathbb{R})$, and let $\Phi_\tau \colon \mathbb{R}^{2m} \to \mathbb{C}^m$ be the $\mathbb{R}$-linear map $\Phi_\tau(x, y) = \tau x + y$.

1. Φ_τ is bijective if and only if Q is invertible. In that case the rule $J_\tau(u) := \Phi_\tau^{-1}\big(i\,\Phi_\tau(u)\big)$ defines a complex structure on $\mathbb{R}^{2m}$, whose matrix in the standard basis is
$$\begin{pmatrix} Q^{-1}P & Q^{-1} \\ -Q - PQ^{-1}P & -PQ^{-1} \end{pmatrix}$$

2. Let Q be invertible. There is a positive compatible triple (g, J_τ, ω_0) on $\mathbb{R}^{2m}$ if and only if $\tau \in \mathcal{H}_m$, and in that case g is unique and is given by
$$g(u, v) = \mathrm{Re}\big(\Phi_\tau(u)^\top Q^{-1} \overline{\Phi_\tau(v)}\big)$$

3. Every positive compatible triple on $\mathbb{R}^{2m}$ whose skew form is ω_0 equals (g, J_τ, ω_0) for exactly one $\tau \in \mathcal{H}_m$.

***Proof*:**

1. Regard $\mathbb{C}^m$ as a real vector space, with basis $\varepsilon_1, \ldots, \varepsilon_m, i\varepsilon_1, \ldots, i\varepsilon_m$, so that $\dim_{\mathbb{R}} \mathbb{C}^m = 2m = \dim_{\mathbb{R}} \mathbb{R}^{2m}$. For $x, y \in \mathbb{R}^m$ the vector $\tau x + y$ has imaginary part Qx and real part

$Px + y$. So $\Phi_\tau(x, y) = 0$ forces $Qx = 0$ and then $y = -Px$, and conversely every such pair is in the kernel; the map $x \mapsto (x, -Px)$ is an injective linear map from $\operatorname{null}(Q)$ onto $\ker \Phi_\tau$, so the two have equal dimension. By theorem 2.4.10 a linear map between real vector spaces of the same finite dimension is bijective exactly when its kernel is zero, so Φ_τ is bijective exactly when $\operatorname{null}(Q) = \{0\}$.

That condition is invertibility of Q. If Q is invertible then $Qx = 0$ gives $x = Q^{-1}Qx = 0$. Conversely, if $\operatorname{null}(Q) = \{0\}$ then $\operatorname{rank} Q = m$ by theorem 2.3.11, so the column space of Q is a subspace of $\mathbb{R}^m$ of dimension m and therefore all of $\mathbb{R}^m$ (corollary 2.2.12); the map $x \mapsto Qx$ is then a linear bijection of $\mathbb{R}^m$, and the matrix of its inverse is a two-sided inverse of Q (definition 1.4.4).

Let Q be invertible. Then J_τ is $\mathbb{R}$-linear as a composite of $\mathbb{R}$-linear maps, and $\Phi_\tau(J_\tau^2 u) = i\Phi_\tau(J_\tau u) = i^2\Phi_\tau(u)$, so $J_\tau^2 u = -u$ and J_τ is a complex structure. To compute its matrix, write $J_\tau(x, y) = (x', y')$, which by definition means $\tau x' + y' = i(\tau x + y)$. Comparing imaginary parts gives $Qx' = Px + y$, hence $x' = Q^{-1}Px + Q^{-1}y$; comparing real parts gives $Px' + y' = -Qx$, hence

$$y' = -Qx - P\big(Q^{-1}Px + Q^{-1}y\big) = \big(-Q - PQ^{-1}P\big)x - PQ^{-1}y$$

These are the two block rows of the displayed matrix acting on (x, y) (proposition 3.5.1).

2. By construction Φ_τ satisfies $\Phi_\tau(J_\tau u) = i\,\Phi_\tau(u)$, so for $a, b \in \mathbb{R}$,

$$\Phi_\tau\big((a + bi)u\big) = \Phi_\tau(au + bJ_\tau u) = a\Phi_\tau(u) + bi\Phi_\tau(u) = (a + bi)\Phi_\tau(u)$$

that is, Φ_τ is a $\mathbb{C}$-linear bijection from $(\mathbb{R}^{2m})_{J_\tau}$ to $\mathbb{C}^m$, and its inverse is $\mathbb{C}$-linear as well. The claim of part (2) is now equivalent to the following one: there is a positive definite Hermitian form H on $\mathbb{C}^m$ with

$$\operatorname{Im} H\big(\Phi_\tau(u), \Phi_\tau(v)\big) = \omega_0(u, v) \qquad \text{for all } u, v \in \mathbb{R}^{2m} \tag{10.11}$$

if and only if $\tau \in \mathcal{H}_m$. Indeed, given a positive compatible triple (g, J_τ, ω_0), the form $h = g + i\,\omega_0$ is Hermitian on $(\mathbb{R}^{2m})_{J_\tau}$ by theorem 10.4.4(1) and positive definite by theorem 10.4.6, and $H(z, z') := h(\Phi_\tau^{-1}z, \Phi_\tau^{-1}z')$ then satisfies every condition of definition 10.1.1, because Φ_τ^{-1} is $\mathbb{C}$-linear, and is positive definite because Φ_τ^{-1} is a bijection sending only 0 to 0; its imaginary part is (10.11). Conversely, given such an H, the form $h(u, v) := H(\Phi_\tau u, \Phi_\tau v)$ is Hermitian on $(\mathbb{R}^{2m})_{J_\tau}$ and positive definite for the same two reasons, so theorem 10.4.4(2) makes $(\operatorname{Re} h, J_\tau, \operatorname{Im} h)$ a compatible triple, $\operatorname{Im} h = \omega_0$ by (10.11), and the triple is positive by theorem 10.4.6 applied to h. Uniqueness of g is theorem 10.4.3(2), since ω_0 and J_τ determine it.

Necessity. Let H be such a form and let $S \in M_m(\mathbb{C})$ have entries $S_{jk} = H(\varepsilon_j, \varepsilon_k)$ for the standard basis $\varepsilon_1, \ldots, \varepsilon_m$ of $\mathbb{C}^m$, so that $H(z, z') = z^\top S\overline{z'}$ by expanding both arguments, and $S^* = S$ because $H(z', z) = \overline{H(z, z')}$. Evaluate (10.11) on three families of vectors. For $u = (0, y)$ and $v = (0, y')$ one has $\Phi_\tau(u) = y$ and $\Phi_\tau(v) = y'$, both real, and $\omega_0(u, v) = 0$, so $y^\top(\operatorname{Im} S)y' = 0$ for all $y, y' \in \mathbb{R}^m$ and therefore $\operatorname{Im} S = 0$. Thus S is real, and $S^* = S$ makes it real symmetric. For $u = (x, 0)$ and $v = (0, y')$ one has $\Phi_\tau(u) = \tau x$, $\Phi_\tau(v) = y'$ and $\omega_0(u, v) = x^\top y'$, while $H(\tau x, y') = x^\top\tau^\top Sy'$ has imaginary part $x^\top Q^\top Sy'$ because S is real. Hence $Q^\top S = I_m$, so $S = (Q^\top)^{-1}$; and S symmetric forces $Q^\top$ symmetric, so $Q^\top = Q$ and $S = Q^{-1}$. For $u = (x, 0)$ and $v = (x', 0)$ one has $\omega_0(u, v) = 0$ and

$$\tau^\top Q^{-1}\overline{\tau} = (P^\top + iQ)Q^{-1}(P - iQ) = P^\top Q^{-1}P + Q + i(P - P^\top)$$

using $Q^\top = Q$, so the imaginary part of $H(\tau x, \tau x') = x^\top \tau^\top Q^{-1} \overline{\tau} x'$ is $x^\top (P - P^\top) x'$, and this vanishes for all x, x' exactly when $P^\top = P$. Hence $\tau^\top = \tau$. Finally, for a nonzero real y the vector $\Phi_\tau(0, y) = y$ is nonzero and $H(y, y) = y^\top Q^{-1} y > 0$, so Q^{-1} is positive definite; and then $x^\top Q x = (Qx)^\top Q^{-1}(Qx) > 0$ for every nonzero x, using the symmetry of Q^{-1}, so $Q = \mathrm{Im}\tau$ is positive definite and $\tau \in \mathcal{H}_m$.

Sufficiency. Let $\tau \in \mathcal{H}_m$ and put $H(z, z') = z^\top Q^{-1} \overline{z'}$. The matrix Q^{-1} is real and symmetric, so H is linear in its first variable, conjugate-linear in its second, and satisfies $H(z', z) = \overline{H(z, z')}$; it is therefore Hermitian. Writing $z = a + ib$ with $a, b \in \mathbb{R}^m$ and using the symmetry of Q^{-1},

$$H(z, z) = (a + ib)^\top Q^{-1}(a - ib) = a^\top Q^{-1} a + b^\top Q^{-1} b$$

the cross terms cancelling, and Q^{-1} is positive definite by the computation $x^\top Q^{-1} x = (Q^{-1}x)^\top Q (Q^{-1} x) > 0$ for $x \neq 0$. So H is positive definite. It remains to verify (10.11). With $z = \tau x + y$ and $z' = \tau x' + y'$,

$$z^\top Q^{-1} \overline{z'} = x^\top \tau^\top Q^{-1} \overline{\tau} x' + x^\top \tau^\top Q^{-1} y' + y^\top Q^{-1} \overline{\tau} x' + y^\top Q^{-1} y'$$

The fourth term is real. The first term is real because $\tau^\top Q^{-1} \overline{\tau} = P Q^{-1} P + Q$ by the display above with $P^\top = P$. The second term is $x^\top (P + iQ) Q^{-1} y' = x^\top P Q^{-1} y' + i\, x^\top y'$, with imaginary part $x^\top y'$, and the third is $y^\top Q^{-1}(P - iQ)x' = y^\top Q^{-1} P x' - i\, y^\top x'$, with imaginary part $-y^\top x'$. Adding, $\mathrm{Im} H(z, z') = x^\top y' - y^\top x' = \omega_0(u, v)$, which is (10.11). The displayed formula for g is the real part of the same transport.

3. Let (g, J, ω_0) be a positive compatible triple and put $L = \mathrm{span}\{f_1, \ldots, f_m\}$, the set of vectors $(0, y)$. Then $\omega_0((0, y), (0, y')) = 0$ for all y, y', so L is isotropic (definition 10.3.3) of dimension m, hence Lagrangian (definition 10.3.5). If $w \in L$ and $Jw \in L$ then

$$g(w, w) = \omega_0(Jw, w) = 0$$

by (10.8) and isotropy, so $w = 0$ by positivity. Hence $L \cap JL = \{0\}$, since an element of the intersection is Jw for some $w \in L$ with $Jw \in L$.

Consequently the list $f_1, \ldots, f_m, Jf_1, \ldots, Jf_m$ is $\mathbb{R}$-independent. A relation $\sum_j a_j f_j + \sum_j b_j J f_j = 0$ with real coefficients puts $\sum_j a_j f_j = -J(\sum_j b_j f_j)$, and the left side lies in L while the right lies in JL, so both lie in $L \cap JL = \{0\}$. Then $a_j = 0$ for every j because $f_1, \ldots, f_m$ is independent, and $J(\sum_j b_j f_j) = 0$ gives $\sum_j b_j f_j = 0$ because J is injective, so $b_j = 0$ for every j as well. The span of these $2m$ vectors is therefore a subspace of $\mathbb{R}^{2m}$ of dimension $2m$, hence all of $\mathbb{R}^{2m}$ (corollary 2.2.12).

In $(\mathbb{R}^{2m})_J$ this says that $f_1, \ldots, f_m$ is a $\mathbb{C}$-basis: every v is $\sum_j a_j f_j + \sum_j b_j J f_j = \sum_j (a_j + i b_j) f_j$ for real a_j, b_j, and a $\mathbb{C}$-relation $\sum_j (a_j + i b_j) f_j = 0$ is the $\mathbb{R}$-relation just shown to force $a_j = b_j = 0$.

Let $\tau \in M_m(\mathbb{C})$ be the matrix whose k-th column holds the $\mathbb{C}$-coordinates of e_k in this basis, so that $e_k = \sum_j \tau_{jk} f_j$ in $(\mathbb{R}^{2m})_J$, and let $\Psi \colon \mathbb{R}^{2m} \to \mathbb{C}^m$ send a vector to its $\mathbb{C}$-coordinate vector. Then Ψ is a $\mathbb{C}$-linear bijection with $\Psi(f_j) = \varepsilon_j$ and $\Psi(e_k)$ the k-th column of τ, so for real x, y,

$$\Psi(x, y) = \sum_k x_k \Psi(e_k) + \sum_j y_j \varepsilon_j = \tau x + y = \Phi_\tau(x, y)$$

Thus $\Phi_\tau = \Psi$ is bijective, so $\mathrm{Im}\tau$ is invertible by part (1); and $\mathbb{C}$-linearity of Ψ says $\Phi_\tau(Ju) = i\Phi_\tau(u)$, so $J = J_\tau$. Part (2) now gives $\tau \in \mathcal{H}_m$ and identifies g.

For uniqueness, suppose $J_\sigma = J_\tau$ for some $\sigma \in \mathcal{H}_m$. The map Φ_σ is $\mathbb{C}$-linear from $(\mathbb{R}^{2m})_{J_\sigma}$ to $\mathbb{C}^m$ by the computation in part (2), it is bijective, and $\Phi_\sigma(f_j) = \varepsilon_j$, so $f_1, \dots, f_m$ is the $\mathbb{C}$-basis of $(\mathbb{R}^{2m})_{J_\sigma}$ whose coordinate map is Φ_σ. Its value on e_k is the k-th column of σ, so σ is the coordinate matrix constructed above from $J_\sigma = J_\tau$, and $\sigma = \tau$, completing the proof.

The parametrization is a statement about one fixed symplectic form: it holds ω_0 still and varies J, and theorem 10.4.3(2) then carries g along. What $\mathcal{H}_m$ records is the freedom in the complex structure, and the two conditions defining it are exactly the two halves of compatibility with positivity, as the proof of part (2) showed: symmetry of τ is J_τ-invariance of ω_0, and positive definiteness of $\mathrm{Im}\tau$ is positivity of g.

Example 10.4.9: The Upper Half-Plane and the Triples on $\mathbb{R}^2$

Take $m = 1$ and $\tau = 1 + 2i$, so $P = (1)$ and $Q = (2)$. The matrix of J_τ from proposition 10.4.8(1) is

$$M = \begin{pmatrix} Q^{-1}P & Q^{-1} \\ -Q - PQ^{-1}P & -PQ^{-1} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ -\frac{5}{2} & -\frac{1}{2} \end{pmatrix}$$

and squaring it entry by entry gives

$$M^2 = \begin{pmatrix} \frac{1}{4} - \frac{5}{4} & \frac{1}{4} - \frac{1}{4} \\ -\frac{5}{4} + \frac{5}{4} & -\frac{5}{4} + \frac{1}{4} \end{pmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

as proposition 10.4.8(1) requires. The matrix of g is $G = M^\top J_0$, since $g(u, v) = \omega_0(J_\tau u, v)$ has coordinate expression $(Mu)^\top J_0 v$:

$$G = \begin{pmatrix} \frac{1}{2} & -\frac{5}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} \frac{5}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

which is symmetric with leading principal minors $\frac{5}{2} > 0$ and $\frac{5}{4} - \frac{1}{4} = 1 > 0$, hence positive definite by theorem 8.4.4. The triple (g, J_τ, ω_0) is therefore positive.

The same numbers come out of the formula in proposition 10.4.8(2), where $H(z, z') = \frac{1}{2} z \overline{z'}$ because $Q^{-1} = \frac{1}{2}$. Here $\Phi_\tau(e_1) = \tau$ and $\Phi_\tau(e_2) = 1$, so

$$H(\tau, \tau) = \tfrac{1}{2}|1 + 2i|^2 = \tfrac{5}{2}, \qquad H(\tau, 1) = \tfrac{1}{2}(1 + 2i), \qquad H(1, 1) = \tfrac{1}{2}$$

whose real parts $\frac{5}{2}, \frac{1}{2}, \frac{1}{2}$ are the entries of G, and whose imaginary parts $0, 1, 0$ are the values $\omega_0(e_1, e_1), \omega_0(e_1, e_2), \omega_0(e_2, e_2)$.

At $\tau = i$ the matrix M becomes $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) = J_0$ and G becomes I_2, which is the triple of example 10.4.2. Varying τ over the upper half plane varies the complex structure on $\mathbb{R}^2$ compatible with the fixed area form ω_0, and proposition 10.4.8(3) says that every one of them arises exactly once.

Exercise 10.4.1

1. On $\mathbb{R}^2$ let g be the symmetric bilinear form with matrix $G = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and let ω_0 be the standard symplectic form, with matrix $B = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Verify that $(B^{-1}G)^2 = -I_2$, write down the matrix of the unique complex structure J compatible with the pair, decide whether the triple (g, J, ω_0) is positive, and find the $\tau \in \mathcal{H}_1$ with $J = J_\tau$.

2. On $\mathbb{R}^2$ let g_c have matrix $\operatorname{diag}(1, c)$ with $c \in \mathbb{R}$ nonzero, and let $\omega_\lambda = \lambda\,\omega_0$ with $\lambda \in \mathbb{R}$ nonzero. Determine exactly which pairs (c, λ) admit a complex structure J making (g_c, J, ω_λ) a compatible triple, and show that every such triple is positive.

3. Let (g, J, ω) be a compatible triple on a real vector space V. Show that $\omega(u, Ju) = -g(u, u)$ for every u, and that $(g, -J, -\omega)$ is again a compatible triple, positive exactly when (g, J, ω) is.

4. For $\tau \in \mathcal{H}_1$ let $G_\tau \in M_2(\mathbb{R})$ be the matrix of the form g in the positive triple (g, J_τ, ω_0) on $\mathbb{R}^2$ given by proposition 10.4.8. Compute the matrix of J_{it} and the matrix G_{it} for $t > 0$, and check that G_{it} is positive definite. Then show that $\det G_\tau = 1$ for every $\tau \in \mathcal{H}_1$, not only for the purely imaginary ones.

5. Let (g, J, ω) be a positive compatible triple on V with $\dim_{\mathbb{R}} V = 2m$ and let $W \subseteq V$ be a Lagrangian subspace. Show that $W \cap JW = \{0\}$, that $V = W \oplus JW$, and that JW is again Lagrangian.

6. Theorem 10.1.9(4) gives the inertia of $g = \mathrm{Re}h$ as twice, entry by entry, the inertia of h. Use it, without reproving it, to show that the signature of g in the sense of definition 8.3.2 is always an even integer. Deduce that a real symmetric bilinear form of odd signature is the real part of no Hermitian form, for any complex structure J whatever. Exhibit such a form on $\mathbb{R}^4$, and say why $\mathbb{R}^4$ rather than $\mathbb{R}^3$ makes the point: on $\mathbb{R}^3$ the obstruction is already the dimension (proposition 9.3.3), so nothing is learned from the signature there.

7. Let $\tau = \left(\begin{smallmatrix} 2i & 1 \\ 1 & i \end{smallmatrix}\right)$. Check that $\tau \in \mathcal{H}_2$, compute the four blocks of the matrix of J_τ from proposition 10.4.8(1), and verify blockwise that its square is $-I_4$.

11

The Singular Value Decomposition

Every factorization in this book so far has asked something of its matrix. The spectral theorem wanted a normal operator, Cholesky a positive definite one, and even the LU factorization wanted the pivots to fall in the right places. The factorization of this chapter asks nothing. Any matrix at all, of any shape, square or not, real or complex, invertible or badly degenerate, is a unitary matrix times a diagonal one times another unitary matrix.

What makes that possible is a change in what the two unitary factors are allowed to be. Diagonalizing an operator uses one basis at both ends, because the operator maps a space to itself and the two ends are the same space. A rectangular matrix has no such constraint: its input and output live in different spaces, and choosing an orthonormal basis for each independently is enough to make it diagonal. The price is that the diagonal entries are no longer eigenvalues. What they are instead is lengths, and they are called the singular values.

Section 11.1 defines them and proves the factorization exists. Section 11.2 reads the four fundamental subspaces off it and, as a consequence, produces the pseudoinverse of chapter 6 a second time from a different direction. Section 11.3 regroups the three factors into two and recovers the matrix analogue of writing a complex number as a modulus times a phase. Section 11.4 measures matrices, finds that two natural ways of doing so are both computed from the singular values, and proves that discarding the small ones is the best approximation of its rank available. Section 11.5 spends that last result on a table of data.

11.1 Singular Values and the Existence of the SVD

The spectral theorem of chapter 7 diagonalizes a self-adjoint operator, and it does so with a single orthonormal basis serving as the basis of the source and of the target at once. An arbitrary $A \in M_{m\times n}(k)$ admits no such basis: when $m \neq n$ the source and the target are not the same space. What replaces it is a pair of orthonormal bases, one of k^n and one of k^m, in which A becomes a rectangular matrix carrying nonnegative numbers on its main diagonal and zeros everywhere else. This section constructs that pair and identifies those numbers. The construction applies the spectral theorem to A^*A, which is square and self-adjoint whatever the shape of A.

Passing from A to A^*A replaces an $m \times n$ matrix by an $n \times n$ one, and three features of the replacement are what the construction runs on: A^*A is self-adjoint, the roots of its characteristic polynomial are nonnegative reals, and it is therefore diagonalized by an orthonormal basis of k^n with a nonnegative diagonal. All three come from a single identity, which converts a statement about A^*A into a statement about the length of Ax.

Proposition 11.1.1: The Matrix A^*A

Let k be $\mathbb{R}$ or $\mathbb{C}$, let $A \in M_{m\times n}(k)$, and give k^n and k^m their standard inner products (definition 5.1.2).

1. $(A^*A)^* = A^*A$, and $\langle A^*Ax, x\rangle = \|Ax\|^2$ for every $x \in k^n$.
2. $A^*Ax = 0$ if and only if $Ax = 0$.
3. Every root of the characteristic polynomial χ_{A^*A} is a nonnegative real number.
4. There are a unitary $V \in M_n(k)$, with columns $v_1, \dots, v_n$, and real numbers $d_1 \geq d_2 \geq \cdots \geq d_n \geq 0$ with
$$A^*A = V \operatorname{diag}(d_1, \dots, d_n) V^*$$
and $A^*Av_j = d_jv_j$ for every j. The list $d_1, \dots, d_n$ consists of the roots of χ_{A^*A}, each listed as often as its multiplicity as a root.
5. If $k = \mathbb{R}$, then $A^\top A$ is symmetric and positive semidefinite in the sense of definition 8.4.1.

***Proof*:**

1. Applying proposition 5.5.13(3) to the product A^*A and then (1) of the same proposition to A,
$$(A^*A)^* = A^*(A^*)^* = A^*A$$
For the identity, theorem 5.5.14(2) applied to A with the vectors $x \in k^n$ and $Ax \in k^m$ reads $\langle Ax, Ax\rangle = \langle x, A^*Ax\rangle$, and conjugate symmetry (definition 5.1.2(2)) turns this into
$$\langle A^*Ax, x\rangle = \overline{\langle x, A^*Ax\rangle} = \overline{\langle Ax, Ax\rangle} = \overline{\|Ax\|^2} = \|Ax\|^2$$
the last equality because $\|Ax\|^2$ is real (proposition 5.1.6(1)).
2. If $Ax = 0$ then $A^*Ax = A^*(Ax) = 0$, every entry of the product being a sum of terms with a factor 0 (definition 1.4.1). Conversely, if $A^*Ax = 0$ then (1) gives $\|Ax\|^2 = \langle 0, x\rangle = 0$ by proposition 5.1.6(3), and $\|Ax\| = 0$ forces $Ax = 0$ by proposition 5.2.2(1).
3. Read A^*A as an element of $M_n(\mathbb{C})$, which changes neither the matrix nor its characteristic polynomial when $k = \mathbb{R}$; the entries are then real, so the conjugate transpose taken in $M_n(\mathbb{C})$ is the transpose (definition 5.5.12) and (1) still reads $(A^*A)^* = A^*A$. So A^*A is self-adjoint (definition 7.1.1), and the closing clause of proposition 7.1.11 makes every root of χ_{A^*A} real. Let λ be such a root. Then $\lambda \in \mathbb{R} \subseteq k$, so theorem 4.1.10(1) gives a

nonzero $x \in k^n$ with $A^*Ax = \lambda x$, and (1) together with linearity in the first argument (definition 5.1.2(1)) gives

$$\|Ax\|^2 = \langle A^*Ax, x\rangle = \langle \lambda x, x\rangle = \lambda\|x\|^2$$

Here $\|x\|^2 > 0$ because $x \neq 0$ (proposition 5.2.2(1)), so $\lambda = \|Ax\|^2/\|x\|^2 \geq 0$.

4. Suppose first that $k = \mathbb{C}$. By (1) the matrix A^*A is self-adjoint, hence normal, since $(A^*A)^*(A^*A) = (A^*A)(A^*A) = (A^*A)(A^*A)^*$ (definition 7.1.7). Corollary 7.3.3 therefore produces a unitary $W \in M_n(\mathbb{C})$ and a diagonal D_0 with $A^*A = WD_0W^*$, and its closing clause says that the columns $w_1, \ldots, w_n$ of W satisfy $A^*Aw_j = (D_0)_{jj}w_j$ and that the diagonal entries of D_0 are the roots of χ_{A^*A}, each listed as often as its multiplicity. When $k = \mathbb{R}$ the matrix $A^\top A$ is symmetric by (1), so corollary 7.4.3 produces the same data with W real orthogonal, which is unitary over $\mathbb{R}$ (definition 7.1.3), and with the same closing clause. In both cases every diagonal entry of D_0 is a nonnegative real by (3).

   Choose a permutation π of $\{1, \ldots, n\}$ arranging those entries in decreasing order, put $d_j = (D_0)_{\pi(j)\pi(j)}$ and let V have columns $v_j = w_{\pi(j)}$. The columns of V are the columns of W in another order, hence again an orthonormal basis of k^n, so V is unitary with $VV^* = I_n$ (proposition 7.1.5). The j-th column of A^*AV is $A^*Av_j = d_jv_j$, which is also the j-th column of $V\operatorname{diag}(d_1, \ldots, d_n)$ (definition 1.4.1); so $A^*AV = V\operatorname{diag}(d_1, \ldots, d_n)$, and multiplying on the right by V^* gives the factorization. Reordering does not change which numbers occur on the diagonal or how often, so $d_1, \ldots, d_n$ are still the roots of χ_{A^*A} listed with multiplicity.

5. Over $\mathbb{R}$ the conjugate transpose is the transpose (definition 5.5.12), so (1) reads $(A^\top A)^\top = A^\top A$ and $A^\top A$ is symmetric. The standard inner product on $\mathbb{R}^n$ is $\langle y, x\rangle = \sum_i y_ix_i = x^\top y$, so taking $y = A^\top Ax$ gives $\langle A^\top Ax, x\rangle = x^\top A^\top Ax$, which is the value $q_{A^\top A}(x)$ of the quadratic form of $A^\top A$ (definition 8.4.1). By (1) that value is $\|Ax\|^2 \geq 0$ for every $x \in \mathbb{R}^n$, which is condition (2) of definition 8.4.1, completing the proof.

The identity in (1) carries the whole proposition: it evaluates a quadratic expression in A^*A as a squared length, and a squared length is a nonnegative real. Part (2) records that A^*A discards nothing about which vectors A kills, which is why replacing A by A^*A is not a loss of information about A. Part (5) is exactly the hypothesis of theorem 8.4.9, so over $\mathbb{R}$ the matrix $A^\top A$ has a unique symmetric positive semidefinite square root $(A^\top A)^{1/2}$; this is the object section 8.4 built and set aside.

The numbers $d_1, \ldots, d_n$ of part (4) are attached to A and not to any choice made in the proof, since part (4) identifies them as the roots of χ_{A^*A} listed with multiplicity. Their square roots are the numbers this chapter is about.

Definition 11.1.2: Singular Values and Singular Vectors

Let $A \in M_{m\times n}(k)$ and let $d_1 \geq d_2 \geq \cdots \geq d_n \geq 0$ be the roots of χ_{A^*A}, each listed as often as its multiplicity as a root and arranged in decreasing order; they are nonnegative reals by proposition 11.1.1(3). The *singular values* of A are their nonnegative square roots

$$\sigma_j = \sqrt{d_j}, \qquad j = 1, \dots, n$$

so that $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$. They are written $\sigma_j(A)$ when the matrix needs naming. Let σ be a singular value of A. A pair of unit vectors $u \in k^m$ and $v \in k^n$ with

$$Av = \sigma u \qquad \text{and} \qquad A^*u = \sigma v$$

is a pair of *singular vectors* of A for σ; the vector u is a *left singular vector* and v a *right singular vector*.

A matrix that is already rectangular diagonal shows what the ordering does. For

$$C = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 3 \end{pmatrix} \in M_{2\times 3}(\mathbb{R})$$

the entry of $C^\top C$ in row i and column j is the inner product of the i-th and j-th columns of C (definition 2.3.4, definition 1.4.1), so $C^\top C = \operatorname{diag}(4, 0, 9)$. The matrix $tI_3 - C^\top C$ is diagonal, so $\chi_{C^\top C}(t) = t(t-4)(t-9)$ by proposition 3.2.6, and its roots in decreasing order are $9, 4, 0$. The singular values of C are therefore $\sigma_1 = 3$, $\sigma_2 = 2$ and $\sigma_3 = 0$: the entries of C reappear, but sorted, and the sorting has exchanged them.

Over $\mathbb{R}$ the singular values are the eigenvalues of the square root recorded above. Write $S = (A^\top A)^{1/2}$, so that S is symmetric, positive semidefinite and $S^2 = A^\top A$ (theorem 8.4.9). By corollary 7.4.3 there are a real orthogonal Q and a diagonal $D' = \operatorname{diag}(\mu_1, \dots, \mu_n)$ carrying the roots of χ_S with multiplicity, with $S = QD'Q^\top$, and every $\mu_j \geq 0$ by corollary 8.4.3(2). Using $Q^\top Q = I_n$ (proposition 7.1.5),

$$A^\top A = S^2 = QD'Q^\top QD'Q^\top = Q(D')^2Q^\top$$

and $(D')^2 = \operatorname{diag}(\mu_1^2, \dots, \mu_n^2)$ by definition 1.4.1 on diagonal matrices. This factorization has the shape of corollary 7.4.3(3), so its closing clause identifies $\mu_1^2, \dots, \mu_n^2$ as the roots of $\chi_{A^\top A}$ listed with multiplicity. Every μ_j is nonnegative, so arranging the μ_j in decreasing order arranges the μ_j^2 in decreasing order as well, and definition 11.1.2 then reads the reordered list $\mu_1, \dots, \mu_n$ as $\sigma_1, \dots, \sigma_n$.

Definition 11.1.2 produced n numbers from A but no bases, and the pairs of singular vectors it names have not been shown to exist. Both come from one construction. The eigenvectors of A^*A are the right singular vectors, and their images under A, once scaled to unit length, are the left ones; what makes the construction work is that those images are already orthogonal to each other, which is proposition 11.1.1(1) applied one pair at a time.

Theorem 11.1.3: Existence of the Singular Value Decomposition

Let k be $\mathbb{R}$ or $\mathbb{C}$, let $A \in M_{m\times n}(k)$ with singular values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$ (definition 11.1.2), and let r be the number of indices j with $\sigma_j > 0$. Let $\Sigma \in M_{m\times n}(k)$ be the matrix whose entry in position (j, j) is σ_j for $1 \leq j \leq \min(m, n)$ and whose every other entry is 0. Then $r \leq \min(m, n)$, and there are unitary matrices $U \in M_m(k)$ and $V \in M_n(k)$ with

$$A = U\Sigma V^* \tag{11.1}$$

Writing $u_1, \dots, u_m$ for the columns of U and $v_1, \dots, v_n$ for the columns of V, the pair may be chosen so that in addition

1. $v_1, \dots, v_n$ is an orthonormal basis of k^n with $A^*Av_j = \sigma_j^2 v_j$ for every j;
2. $Av_j = \sigma_j u_j$ for $1 \leq j \leq \min(m, n)$, and $Av_j = 0$ for every $j > r$;
3. $A^*u_j = \sigma_j v_j$ for $1 \leq j \leq \min(m, n)$, and $A^*u_j = 0$ for every $j > r$.

***Proof*:**
Step 1: an orthonormal eigenbasis for A^*A. Proposition 11.1.1(4) gives a unitary $V \in M_n(k)$ whose columns $v_1, \dots, v_n$ satisfy $A^*Av_j = d_j v_j$, where $d_1 \geq \cdots \geq d_n \geq 0$ are the roots of χ_{A^*A} listed with multiplicity in decreasing order. By definition 11.1.2 these are $d_j = \sigma_j^2$. The columns of a unitary matrix form an orthonormal basis of k^n (proposition 7.1.5), so this is (1).

Step 2: the vectors $Av_1, \dots, Av_n$ are orthogonal, with lengths $\sigma_1, \dots, \sigma_n$. Fix i and j. By theorem 5.5.14(2), applied to A with the vectors $v_i \in k^n$ and $Av_j \in k^m$, and then by Step 1,

$$\langle Av_i, Av_j\rangle = \langle v_i, A^*Av_j\rangle = \langle v_i, \sigma_j^2 v_j\rangle = \sigma_j^2\langle v_i, v_j\rangle$$

the last equality by conjugate-linearity in the second argument (proposition 5.1.6(2)) together with $\overline{\sigma_j^2} = \sigma_j^2$, the number σ_j^2 being real. Since $v_1, \dots, v_n$ is orthonormal, $\langle v_i, v_j\rangle$ is 1 when $i = j$ and 0 otherwise (definition 5.3.3), so

$$\|Av_j\|^2 = \sigma_j^2, \qquad \langle Av_i, Av_j\rangle = 0 \text{ for } i \neq j \tag{11.2}$$

In particular $\|Av_j\| = \sigma_j$, both numbers being nonnegative reals.

Step 3: the first r columns of U. For $j > r$ the definition of r gives $\sigma_j = 0$, so $\|Av_j\| = 0$ by (11.2) and hence $Av_j = 0$ by proposition 5.2.2(1); this is the second half of (2). For $1 \leq j \leq r$ put

$$u_j = \frac{1}{\sigma_j}Av_j$$

which is defined because $\sigma_j > 0$. For $i, j \leq r$, linearity in the first argument (definition 5.1.2(1)) and conjugate-linearity in the second (proposition 5.1.6(2)), the scalars σ_i^{-1} and σ_j^{-1} being real, give

$$\langle u_i, u_j\rangle = \frac{1}{\sigma_i\sigma_j}\langle Av_i, Av_j\rangle$$

which by (11.2) is $\sigma_j^2/\sigma_j^2 = 1$ when $i = j$ and 0 otherwise. So $u_1, \dots, u_r$ is an orthonormal list in k^m (definition 5.3.3). It is linearly independent (proposition 5.3.4), so theorem 2.2.9(1) extends it to a basis of k^m by appending further vectors, and the same clause bounds every basis of k^m by m vectors; hence $r \leq m$. Also $r \leq n$, the list $\sigma_1, \dots, \sigma_n$ having n members. Hence $r \leq \min(m, n)$.

Step 4: completing U. By corollary 5.3.11(2) the orthonormal list $u_1, \dots, u_r$ extends to an orthonormal basis $u_1, \dots, u_m$ of k^m whose first r members are unchanged. Let $U \in M_m(k)$ have these columns; it is unitary by proposition 7.1.5.

Step 5: the factorization. The j-th column of AV is Av_j and the j-th column of $U\Sigma$ is $U(\Sigma e_j)$, both by definition 1.4.1. For $j \leq \min(m, n)$ the j-th column of Σ is $\sigma_j e_j \in k^m$, so $U\Sigma e_j = \sigma_j U e_j = \sigma_j u_j$; for $j > \min(m, n)$ that column is 0, so $U\Sigma e_j = 0$. Comparing the two matrices column by column: for $j \leq r$ the definition of u_j in Step 3 gives $Av_j = \sigma_j u_j$; for $r < j \leq \min(m, n)$ both columns vanish, since $Av_j = 0$ by Step 3 and $\sigma_j = 0$; and for $j > \min(m, n)$ both vanish for the same two reasons, such a j exceeding r by Step 3. Hence $AV = U\Sigma$, and multiplying on the right by V^*, with $VV^* = I_n$ (proposition 7.1.5), gives (11.1). The first half of (2) was established in this comparison.

Step 6: the adjoint relations. Taking conjugate transposes in (11.1) and applying proposition 5.5.13(3) twice and (1) once,

$$A^* = (U\Sigma V^*)^* = (V^*)^* \Sigma^* U^* = V\Sigma^* U^*$$

The entries of Σ are real, so $\Sigma^* \in M_{n\times m}(k)$ has entry σ_j in position (j, j) for $j \leq \min(m, n)$ and 0 elsewhere (definition 5.5.12). Since $u_j = Ue_j$ and $U^*U = I_m$ (proposition 7.1.5), one has $U^*u_j = e_j$, whence $A^*u_j = V\Sigma^* e_j$. That vector is $\sigma_j V e_j = \sigma_j v_j$ when $j \leq \min(m, n)$, and 0 when $j > \min(m, n)$; since $\sigma_j = 0$ for $j > r$, this is (3), completing the proof.

The theorem asks nothing of A beyond its shape. Clause (2) is where the content sits: on the orthonormal basis $v_1, \dots, v_n$ the matrix A sends each v_j to a nonnegative multiple of an orthonormal vector, the multiple being σ_j, and sends $v_{r+1}, \dots, v_n$ to 0. The matrix of $x \mapsto Ax$ relative to the basis $v_1, \dots, v_n$ of the source and $u_1, \dots, u_m$ of the target has j-th column the coordinate vector of Av_j (theorem 2.4.8), which by clause (2) is σ_j in position j and zero elsewhere; that matrix is Σ. Clause (3) says that A^* has the same description with the two bases exchanged, so the pairs (u_j, v_j) with $j \leq \min(m, n)$ are pairs of singular vectors in the sense of definition 11.1.2.

For $m = n = 2$ and $r = 2$ the description can be drawn. Let $x \in \mathbb{R}^2$ satisfy $\|x\| = 1$ and write $x = c_1 v_1 + c_2 v_2$ with $c_i = \langle x, v_i \rangle$ (theorem 5.3.5(1)). The vectors $c_1 v_1$ and $c_2 v_2$ are orthogonal and have lengths $|c_1|$ and $|c_2|$ (proposition 5.2.2(2)), so $c_1^2 + c_2^2 = \|x\|^2 = 1$ by theorem 5.3.2. Then $Ax = c_1\sigma_1 u_1 + c_2\sigma_2 u_2$, whose coordinates $(y_1, y_2) = (c_1\sigma_1, c_2\sigma_2)$ in the orthonormal basis u_1, u_2 satisfy

$$\left(\frac{y_1}{\sigma_1}\right)^2 + \left(\frac{y_2}{\sigma_2}\right)^2 = 1$$

The image of the unit circle is the ellipse with semi-axes $\sigma_1 u_1$ and $\sigma_2 u_2$, and every point of that ellipse is attained, since $(c_1, c_2) = (y_1/\sigma_1, y_2/\sigma_2)$ recovers a unit vector x with Ax the given point.

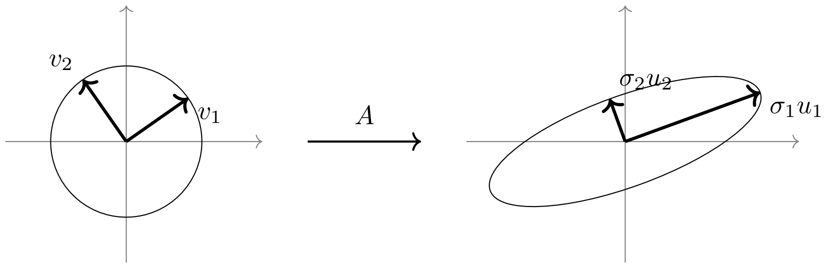


Figure 11.1: The unit circle of $\mathbb{R}^2$ and its image under a matrix $A \in M_2(\mathbb{R})$ with $\sigma_2 > 0$

The construction in the proof is also the recipe for computing a decomposition by hand: form A^*A, diagonalize it, take square roots, and divide.

Example 11.1.4: A Singular Value Decomposition of a 3×2 Matrix

Let

$$B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \in M_{3\times 2}(\mathbb{R})$$

so that $m = 3$ and $n = 2$. The entry of $B^\top B$ in row i and column j is the inner product of the i-th and j-th columns of B (definition 2.3.4, definition 1.4.1), and the two columns have squared lengths 2 and 2 and inner product 1, so

$$B^\top B = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \chi_{B^\top B}(t) = (t-2)^2 - 1 = (t-1)(t-3)$$

the characteristic polynomial computed from $\det\left(\begin{smallmatrix} t-2 & -1 \\ -1 & t-2 \end{smallmatrix}\right)$. Its roots in decreasing order are 3 and 1, so $\sigma_1 = \sqrt{3}$ and $\sigma_2 = 1$, and $r = 2$.

The right singular vectors. The null space of $B^\top B - 3I_2 = \left(\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$ is spanned by $(1,1)^\top$ and that of $B^\top B - I_2 = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ by $(1,-1)^\top$, so

$$v_1 = \tfrac{1}{\sqrt{2}}(1,1)^\top, \qquad v_2 = \tfrac{1}{\sqrt{2}}(1,-1)^\top$$

are unit eigenvectors, and they are orthogonal since $\langle v_1, v_2\rangle = (1-1)/2 = 0$.

The left singular vectors. Computing Bv_1 and Bv_2 and dividing by σ_1 and σ_2,

$$u_1 = \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{6}} \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \qquad u_2 = \frac{1}{1} \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$$

with squared lengths $6/6 = 1$ and $2/2 = 1$ and inner product $(0+1-1)/\sqrt{12} = 0$. Here $r = 2 < 3 = m$, so Step 4 of the proof has something to do: a unit vector u_3 orthogonal to both satisfies $2x_1 + x_2 + x_3 = 0$ and $x_2 - x_3 = 0$, whence $x_2 = x_3 = -x_1$, and normalizing $(-1,1,1)^\top$ gives $u_3 = \frac{1}{\sqrt{3}}(-1,1,1)^\top$.

The factorization. Assembling the three matrices,

$$U = \begin{pmatrix} 2/\sqrt{6} & 0 & -1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \end{pmatrix}, \qquad \Sigma = \begin{pmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

with U of size 3×3, Σ of size 3×2 and V of size 2×2. Multiplying out, the columns of $U\Sigma$ are $\sqrt{3}\,u_1$ and u_2, so

$$U\Sigma V^\top = \frac{1}{\sqrt{2}}\begin{pmatrix}2 & 0\\ 1 & 1\\ 1 & -1\end{pmatrix}\cdot\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} = \frac{1}{2}\begin{pmatrix}2 & 2\\ 2 & 0\\ 0 & 2\end{pmatrix} = B$$

Finally $B^\top u_3 = \frac{1}{\sqrt{3}}(-1+1+0,\,-1+0+1)^\top = 0$, as theorem 11.1.3(3) requires of an index $j=3>r$.

The example makes visible which parts of U and V do work. The column u_3 was produced by an extension step and is annihilated by $B^\top$; the corresponding row of Σ is zero. In general the columns $u_{r+1},\dots,u_m$ meet only zero entries of Σ, and so do the columns $v_{r+1},\dots,v_n$, so a smaller factorization carries the same information.

Corollary 11.1.5: The Reduced Singular Value Decomposition

Keep the notation of theorem 11.1.3 and assume $r\ge 1$. Let $U_r\in M_{m\times r}(k)$ have columns $u_1,\dots,u_r$, let $V_r\in M_{n\times r}(k)$ have columns $v_1,\dots,v_r$, and put $\Sigma_r=\operatorname{diag}(\sigma_1,\dots,\sigma_r)\in M_r(k)$. Then $U_r^*U_r = V_r^*V_r = I_r$, the matrix Σ_r is invertible, and

$$A = U_r\Sigma_r V_r^* = \sum_{j=1}^{r}\sigma_j\, u_j v_j^*$$

where each $u_jv_j^*$ is an $m\times n$ matrix whose t-th column is $\overline{(v_j)_t}\,u_j$.

***Proof*:**

By definition 5.5.12 and definition 1.4.1 the entry of $U_r^*U_r$ in row i and column j is $\sum_t \overline{(u_i)_t}(u_j)_t = \langle u_j,u_i\rangle$, which is 1 for $i=j$ and 0 otherwise, the list $u_1,\dots,u_r$ being orthonormal; so $U_r^*U_r=I_r$, and the same computation gives $V_r^*V_r=I_r$. The matrix Σ_r is diagonal with $\det\Sigma_r=\sigma_1\cdots\sigma_r$ (proposition 3.2.6), a product of positive numbers, so Σ_r is invertible by corollary 3.2.8.

The two matrices agree on the basis $v_1,\dots,v_n$. The j-th entry of $V_r^*v_i$ is $\langle v_i,v_j\rangle$ by the computation just made, so $V_r^*v_i=e_i\in k^r$ when $i\le r$ and $V_r^*v_i=0$ when $i>r$. Hence for $i\le r$

$$U_r\Sigma_rV_r^*v_i = U_r\Sigma_r e_i = \sigma_i U_r e_i = \sigma_i u_i$$

using definition 1.4.1 twice, and $U_r\Sigma_rV_r^*v_i=0$ for $i>r$. Theorem 11.1.3(2) gives Av_i the same two values. So the matrix $M=A-U_r\Sigma_rV_r^*$ satisfies $Mv_i=0$ for every i; since every $x\in k^n$ is a linear combination of $v_1,\dots,v_n$ and $x\mapsto Mx$ is linear, $Mx=0$ for every x, and taking $x=e_t$ shows every column of M is zero (definition 1.4.1). Thus $A=U_r\Sigma_rV_r^*$.

The sum of outer products. For $B\in M_{m\times r}(k)$ and $C\in M_{r\times n}(k)$, definition 1.4.1 gives $(BC)_{it}=\sum_{j=1}^r b_{ij}c_{jt}$, which is the entry in position (i,t) of $\sum_{j=1}^r b^{(j)}c^{(j)}$, where $b^{(j)}\in M_{m\times 1}(k)$ is the j-th column of B and $c^{(j)}\in M_{1\times n}(k)$ the j-th row of C. Apply this with $B=U_r\Sigma_r$, whose j-th column is $U_r\Sigma_re_j=\sigma_ju_j$, and with $C=V_r^*$, whose j-th row has t-th entry $\overline{(v_j)_t}$ and is therefore v_j^* (definition 5.5.12). This gives $A=\sum_{j=1}^r\sigma_ju_jv_j^*$. The t-th column of $u_jv_j^*$ is $u_jv_j^*e_t=\overline{(v_j)_t}\,u_j$, as we sought to show.

Each summand $\sigma_j u_j v_j^*$ has all its columns on the line spanned by u_j, so the corollary writes an arbitrary matrix as a sum of r matrices each of whose columns lie on a single line, the j-th summand carrying the scalar factor σ_j. The reduced form $U_r \Sigma_r V_r^*$ stores $r(m+n+1)$ numbers in place of mn, and its middle factor is invertible, which the full Σ is not unless A is square and $r = n$.

Theorem 11.1.3 produced one factorization and said nothing about how many there are. The middle factor is determined by A; the outer two are not, and both halves of that statement are used later.

Proposition 11.1.6: The Diagonal Factor Is Determined by A

Let $A \in M_{m\times n}(k)$. Suppose $W \in M_m(k)$ and $Z \in M_n(k)$ are unitary and $T \in M_{m\times n}(k)$ has entry s_j in position (j, j) for $1 \le j \le \min(m, n)$ and 0 in every other position, with $s_1 \ge s_2 \ge \cdots \ge s_{\min(m,n)} \ge 0$. If $A = WTZ^*$, then

$$s_j = \sigma_j(A) \qquad \text{for } 1 \le j \le \min(m, n)$$

***Proof*:**
By proposition 5.5.13(3), applied twice, and (1), $A^* = (WTZ^*)^* = ZT^*W^*$, so

$$A^*A = ZT^*W^*WTZ^* = Z(T^*T)Z^*$$

using $W^*W = I_m$ (proposition 7.1.5). The entry of T^*T in row j and column l is $\sum_t \overline{T_{tj}} T_{tl}$ (definition 5.5.12, definition 1.4.1), and $T_{tj} \ne 0$ only when $t = j \le \min(m, n)$, so $T^*T = D'$ where $D' \in M_n(k)$ is diagonal with $D'_{jj} = s_j^2$ for $j \le \min(m, n)$ and $D'_{jj} = 0$ otherwise. The numbers s_j are nonnegative and decreasing, so $D'_{11} \ge D'_{22} \ge \cdots \ge D'_{nn}$.

Since Z is unitary, $Z^{-1} = Z^*$ (proposition 7.1.5), so $D' = Z^*(A^*A)(Z^*)^{-1}$ and D' is similar to A^*A (definition 2.5.5). By corollary 4.1.16 the two have the same characteristic polynomial, and $tI_n - D'$ is diagonal, so

$$\chi_{A^*A}(t) = \chi_{D'}(t) = (t - D'_{11})(t - D'_{22}) \cdots (t - D'_{nn})$$

by proposition 3.2.6. The roots of χ_{A^*A}, listed with multiplicity and arranged in decreasing order, are therefore $D'_{11}, \ldots, D'_{nn}$, and definition 11.1.2 gives $\sigma_j(A)^2 = D'_{jj}$ for every j. For $j \le \min(m, n)$ this reads $\sigma_j(A)^2 = s_j^2$, and both numbers are nonnegative, so $\sigma_j(A) = s_j$, as we sought to show.

The proposition says that any factorization of the shape (11.1) recovers the same diagonal, so the singular values may be computed from any one of them. The unitary factors are genuinely not determined. For $A = I_2$ one has $A^*A = I_2$, whose characteristic polynomial is $(t-1)^2$ by proposition 3.2.6, so $\sigma_1 = \sigma_2 = 1$ and $\Sigma = I_2$; and for every unitary $W \in M_2(k)$ the identity $WI_2W^* = WW^* = I_2$ (proposition 7.1.5) is a factorization of I_2 of that shape with $U = V = W$. Taking $W = I_2$ and $W = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, whose columns are orthonormal, gives two different ones.

A repeated singular value is not what causes this. Fix any A, any decomposition as in theorem 11.1.3, any index $j \le r$ and any scalar $c \in k$ with $|c| = 1$, and replace u_j by cu_j and v_j by cv_j. The new lists are still orthonormal, since scaling one member of an orthonormal list by a scalar of modulus one preserves every inner product among them (definition 5.1.2(1), proposition 5.1.6(2)), so the new matrices U' and V' are again unitary (proposition 7.1.5). The j-th summand of corollary 11.1.5 is unchanged, because $(cv_j)^* = \bar{c} v_j^*$ (proposition 5.5.13(2)) and $c\bar{c} = |c|^2 = 1$, so $U'\Sigma(V')^* = A$ as well.

Over $\mathbb{R}$ the free scalar is a sign; over $\mathbb{C}$ it is a full circle of choices.

Nothing so far compares the singular values of a square matrix with its eigenvalues, and for a general square matrix the two lists are computed from different matrices: the eigenvalues put no upper bound on σ_1, as example 11.1.8(1) will show. When A commutes with A^* they agree up to absolute value, and the reason is that the spectral theorem then diagonalizes A and A^*A in the same basis.

Proposition 11.1.7: Singular Values of a Normal Matrix

Let $A \in M_n(\mathbb{C})$ be normal and let $\lambda_1, \ldots, \lambda_n$ be the roots of χ_A, each listed as often as its multiplicity as a root and indexed so that $|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_n|$. Then

$$\sigma_j(A) = |\lambda_j| \qquad \text{for } 1 \leq j \leq n$$

If $A \in M_n(\mathbb{R})$ is symmetric and positive semidefinite, then A is normal and its singular values are its eigenvalues: $\sigma_j(A) = \lambda_j$ with $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$.

***Proof*:**
By corollary 7.3.3 there are a unitary $U \in M_n(\mathbb{C})$ and a diagonal D with $A = UDU^*$, the columns of U being eigenvectors of A for the corresponding diagonal entries, and those entries being the roots of χ_A listed with multiplicity. The reordering carried out in the proof of proposition 11.1.1(4), applied here to A in place of A^*A, permutes the columns of U and the diagonal entries of D by one permutation without disturbing either the unitarity of U or the factorization, so assume $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$ with the stated indexing.

By proposition 5.5.13(3) and (1), $A^* = (UDU^*)^* = UD^*U^*$, so, using $U^*U = I_n$ (proposition 7.1.5),

$$A^*A = UD^*U^*UDU^* = U(D^*D)U^*$$

and $D^*D = \operatorname{diag}(\overline{\lambda_1}\lambda_1, \ldots, \overline{\lambda_n}\lambda_n) = \operatorname{diag}(|\lambda_1|^2, \ldots, |\lambda_n|^2)$ by definition 5.5.12 and definition 1.4.1 on diagonal matrices. Since $U^{-1} = U^*$, the matrices A^*A and D^*D are similar (definition 2.5.5), so they have the same characteristic polynomial (corollary 4.1.16), which proposition 3.2.6 evaluates as $\prod_{j=1}^n (t - |\lambda_j|^2)$. The roots of χ_{A^*A} listed with multiplicity and in decreasing order are therefore $|\lambda_1|^2 \geq \cdots \geq |\lambda_n|^2$, and definition 11.1.2 gives $\sigma_j(A)^2 = |\lambda_j|^2$, hence $\sigma_j(A) = |\lambda_j|$, both numbers being nonnegative.

For the last statement let $A \in M_n(\mathbb{R})$ be symmetric and positive semidefinite. Then $A^* = A^\top = A$ (definition 5.5.12), so $A^*A = A^2 = AA^*$ and A is normal (definition 7.1.7); reading A as an element of $M_n(\mathbb{C})$ changes neither χ_A nor χ_{A^*A} nor therefore the singular values. By corollary 8.4.3(2) every λ_j is a nonnegative real, so $|\lambda_j| = \lambda_j$ and ordering by $|\lambda_j|$ is ordering by λ_j. The first statement then gives $\sigma_j(A) = \lambda_j$, completing the proof.

Example 11.1.8: Singular Values Against Eigenvalues
All three matrices below lie in $M_2(\mathbb{R})$, and each is read in $M_2(\mathbb{C})$ when its complex eigenvalues are wanted.

1. Let $c \in \mathbb{R}$ and $N_c = \left(\begin{smallmatrix} 0 & c \\ 0 & 0 \end{smallmatrix}\right)$. Then $\chi_{N_c}(t) = \det\left(\begin{smallmatrix} t & -c \\ 0 & t \end{smallmatrix}\right) = t^2$ by proposition 3.2.6, so both roots are 0. But $N_c^\top N_c = \left(\begin{smallmatrix} 0 & 0 \\ 0 & c^2 \end{smallmatrix}\right)$, whose characteristic polynomial is $t(t - c^2)$, so $\sigma_1(N_c) = |c|$ and $\sigma_2(N_c) = 0$. Letting $|c|$ grow makes the largest singular value arbitrarily large while every

eigenvalue stays 0: no inequality bounds σ_1 from above in terms of the eigenvalues.

2. Let $S = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$. Both roots of $\chi_S(t) = (t-1)^2$ are 1, so every eigenvalue has absolute value 1. Here $S^\top S = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ with $\chi_{S^\top S}(t) = (t-1)(t-2) - 1 = t^2 - 3t + 1$, whose roots are $(3 \pm \sqrt{5})/2$. Since $\left(\frac{1+\sqrt{5}}{2}\right)^2 = \frac{6+2\sqrt{5}}{4} = \frac{3+\sqrt{5}}{2}$ and $\left(\frac{\sqrt{5}-1}{2}\right)^2 = \frac{3-\sqrt{5}}{2}$,
$$\sigma_1(S) = \frac{1+\sqrt{5}}{2} \approx 1.618, \qquad \sigma_2(S) = \frac{\sqrt{5}-1}{2} \approx 0.618$$
Proposition 11.1.7 does not apply, and indeed $S^\top S \neq SS^\top = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, so S is not normal.

3. Let $R = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$. Then $R^\top R = I_2 = RR^\top$, the diagonal entries being $\cos^2\theta + \sin^2\theta = 1$ and the off-diagonal ones $-\cos\theta\sin\theta + \sin\theta\cos\theta = 0$, so R is normal and $\sigma_1(R) = \sigma_2(R) = 1$. Its characteristic polynomial is $t^2 - 2t\cos\theta + 1$, whose roots $\cos\theta \pm i\sin\theta$ have absolute value 1. This is proposition 11.1.7 in the case where all the singular values coincide.

The contrast is sharpest in part (1): the eigenvalues of N_c record that N_c has no invariant direction on which it acts by a nonzero scalar, while $\sigma_1(N_c) = |c|$ records the length $\|N_c e_2\| = |c|$. The two lists answer different questions. Proposition 11.1.7 gives one direction of the comparison: when A is normal the answers agree. The converse also holds — if $\sigma_j(A) = |\lambda_j|$ for every j then A is normal — but it is a separate theorem, resting on the equality case of an inequality between the two lists that this book does not prove, and it is not used below.

Exercise 11.1.1

1. Let $C = \left(\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right) \in M_2(\mathbb{R})$. Compute $C^\top C$ and its characteristic polynomial, and determine $\sigma_1(C)$ and $\sigma_2(C)$. Then follow the proof of theorem 11.1.3 to produce orthogonal $U, V \in M_2(\mathbb{R})$ with $C = U\Sigma V^\top$, and verify the factorization by multiplying out.

2. Let $E = \left(\begin{smallmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{smallmatrix}\right) \in M_{2\times 3}(\mathbb{R})$. Compute the three singular values of E and a full singular value decomposition $E = U\Sigma V^\top$, with U of size 2×2, Σ of size 2×3 and V of size 3×3. Identify the value of r and check that the third column of V satisfies the second clause of theorem 11.1.3(2).

3. Let A be the running matrix of example 1.2.9, that is
$$A = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 3 & 8 \\ 1 & 2 & 5 \\ 3 & 4 & 11 \end{pmatrix} \in M_{4\times 3}(\mathbb{R})$$
Compute $A^\top A$ and verify that $A^\top A\,(1, 2, -1)^\top = 0$, so that $\sigma_3(A) = 0$. Then, using proposition 11.1.1(4) together with corollary 2.5.9, show that $\sigma_1(A)^2 + \sigma_2(A)^2 + \sigma_3(A)^2 = \operatorname{tr}(A^\top A)$, evaluate the right-hand side, and deduce that $\sigma_1(A) \geq 2\sqrt{33}$.

4. Let $A \in M_n(k)$. Show that A is invertible if and only if $\sigma_n(A) > 0$. Show further that when A is invertible the singular values of A^{-1} are $\sigma_n(A)^{-1} \geq \sigma_{n-1}(A)^{-1} \geq \cdots \geq \sigma_1(A)^{-1}$, by exhibiting a factorization of A^{-1} of the shape required by proposition 11.1.6. Check the answer against exercise 11.1.1 (1).

5. Let $A \in M_{m\times n}(k)$. Show that $\sigma_j(A^*) = \sigma_j(A)$ for every $j \leq \min(m, n)$, and that the remaining singular values of A^*, if any, are 0. Deduce that A and A^* have the same number

of nonzero singular values.

6. Let $A \in M_{m\times n}(k)$, let $W \in M_m(k)$ and $Z \in M_n(k)$ be unitary, and put $A' = WAZ$. Show that A' and A have the same singular values. (First check that a product of two unitary matrices of the same size is unitary.)

7. Let $x \in k^m$ and $y \in k^n$ be nonzero and put $A = xy^* \in M_{m\times n}(k)$. Show that $\sigma_1(A) = \|x\|\,\|y\|$ and $\sigma_j(A) = 0$ for $j \geq 2$, and exhibit a pair of singular vectors of A for $\sigma_1(A)$ in the sense of definition 11.1.2.

8. Let $A \in M_n(k)$. Show that A is unitary if and only if $\sigma_1(A) = \cdots = \sigma_n(A) = 1$. Deduce from proposition 11.1.7 that a normal $A \in M_n(\mathbb{C})$ all of whose eigenvalues have absolute value 1 is unitary, and give a non-normal $A \in M_2(\mathbb{R})$ with both eigenvalues of absolute value 1 that is not unitary.

11.2 The SVD and the Four Fundamental Subspaces

Section 11.1 produced, for every $A \in M_{m\times n}(k)$, a factorization $A = U\Sigma V^*$ with U and V unitary and Σ diagonal with nonnegative entries. The columns of V are an orthonormal basis of k^n and the columns of U an orthonormal basis of k^m, and the singular values divide each of those bases into two lists: the columns carrying a nonzero singular value, and the rest. This section identifies the four resulting lists as orthonormal bases of the four subspaces of section 2.3.

Two things follow from that identification. The rank of A is the number of nonzero singular values, so the invariant definition 1.3.2 obtained by counting pivots becomes a count of positive numbers attached to A itself. And the pseudoinverse of section 6.4, defined there by projecting onto $\operatorname{col} A$ and inverting a bijection, is the matrix $V\Sigma^+U^*$, where Σ^+ is obtained from Σ by inverting its nonzero entries and transposing.

Throughout the section $A \in M_{m\times n}(k)$ with $k = \mathbb{R}$ or $\mathbb{C}$, and

$$A = U\Sigma V^* \tag{11.3}$$

is a singular value decomposition of A in the sense of theorem 11.1.3: $U \in M_m(k)$ and $V \in M_n(k)$ are unitary, with columns $u_1, \ldots, u_m$ and $v_1, \ldots, v_n$ respectively, and $\Sigma \in M_{m\times n}(\mathbb{R})$ has entry σ_i in row i and column i for $1 \leq i \leq p = \min(m, n)$ and every other entry 0, where $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$ are the singular values of A (definition 11.1.2), of which Σ carries the first p on its diagonal. Write r for the number of indices i with $\sigma_i > 0$, so that

$$\sigma_1 \geq \cdots \geq \sigma_r > 0 = \sigma_{r+1} = \cdots = \sigma_n$$

with $r \leq p$ by theorem 11.1.3. Both k^n and k^m carry the standard inner product, conjugate linear in the second argument. One reading of the matrix product is used constantly below: for $w, x \in k^m$, the conjugate transpose w^* is the row vector with entries $\overline{w_t}$ (definition 5.5.12), so evaluating the product by definition 1.4.1 gives

$$w^*x = \sum_{t=1}^{m} \overline{w_t}\, x_t = \langle x, w\rangle \tag{11.4}$$

the 1×1 matrix on the left being identified with its single entry.

Multiplying (11.3) on the right by V and using $V^*V = I_n$ (proposition 7.1.5) gives $AV = U\Sigma$. Column i of a product MN is M times column i of N (definition 1.4.1), so column i of AV is Av_i, while column i of Σ is $\sigma_i e_i$ for $i \leq p$ and 0 for $i > p$, whence column i of $U\Sigma$ is $\sigma_i u_i$ for $i \leq p$ and 0 for $i > p$. Taking conjugate transposes in (11.3) by proposition 5.5.13(1) and (3), and using $\Sigma^* = \Sigma^\top$ because every entry of Σ is real, gives

$$A^* = V\Sigma^\top U^* \tag{11.5}$$

which has the same shape with the roles of U and V exchanged: $\Sigma^\top \in M_{n\times m}(\mathbb{R})$ carries the same numbers $\sigma_1 \geq \cdots \geq \sigma_p \geq 0$ on its diagonal and zeros elsewhere. The computation just made, applied to (11.5), gives $A^*U = V\Sigma^\top$. Reading off the columns of both products,

$$Av_i = \sigma_i u_i \quad \text{and} \quad A^* u_i = \sigma_i v_i \qquad (1 \leq i \leq r) \tag{11.6}$$

while $Av_j = 0$ for every $j > r$ and $A^* u_j = 0$ for every $j > r$, since the singular value attached to such an index is zero and the columns beyond the p-th vanish outright.

Everything in this section comes from (11.6). The first statement is the one the section is named for.

Theorem 11.2.1: Singular Vectors Are Bases of the Four Subspaces

Let $A \in M_{m\times n}(k)$ with singular value decomposition (11.3) and notation as above. Then:

1. $u_1, \ldots, u_r$ is an orthonormal basis of $\operatorname{col} A$;
2. $u_{r+1}, \ldots, u_m$ is an orthonormal basis of $\operatorname{null}(A^*)$;
3. $v_1, \ldots, v_r$ is an orthonormal basis of $\operatorname{col}(A^*)$;
4. $v_{r+1}, \ldots, v_n$ is an orthonormal basis of $\operatorname{null} A$.

Over $\mathbb{R}$ the conjugation is trivial, so $A^* = A^\top$ and $\operatorname{col}(A^*) = \operatorname{row} A$ by definition 2.3.6; part (3) then produces an orthonormal basis of the row space.

***Proof*:**
Step 1: each of the four lists is orthonormal. The columns $u_1, \ldots, u_m$ of the unitary matrix U form an orthonormal basis of k^m, and the columns $v_1, \ldots, v_n$ of V an orthonormal basis of k^n, by proposition 7.1.5. Any sublist of an orthonormal list is orthonormal, and an orthonormal list is linearly independent by proposition 5.3.4, its vectors being nonzero and pairwise orthogonal. So each of the four lists is a linearly independent orthonormal list, hence an orthonormal basis of its own span, and it remains to identify the four spans.

Step 2: the column space. By definition 1.4.1 the product Ax is $x_1c_1 + \cdots + x_nc_n$, where $c_1, \ldots, c_n$ are the columns of A, so

$$\{\, Ax \,|\, x \in k^n \,\} = \operatorname{span}\{c_1, \ldots, c_n\} = \operatorname{col} A$$

by definition 2.1.5 and definition 2.3.1. Every $x \in k^n$ satisfies $x = \sum_{i=1}^n \langle x, v_i\rangle v_i$ by theorem 5.3.5(1), and applying A and using (11.6),

$$Ax = \sum_{i=1}^n \langle x, v_i\rangle\, Av_i = \sum_{i=1}^r \sigma_i \langle x, v_i\rangle\, u_i \tag{11.7}$$

Every such vector lies in $\operatorname{span}\{u_1, \ldots, u_r\}$. Conversely $u_i = A(v_i/\sigma_i)$ for $i \leq r$ by (11.6), so each u_i with $i \leq r$ is of the form Ax, and hence so is every linear combination of $u_1, \ldots, u_r$, by (11.7) applied to the corresponding combination of the v_i. Therefore $\operatorname{col} A = \operatorname{span}\{u_1, \ldots, u_r\}$, which with Step 1 proves (1).

Step 3: the null space. Let $x \in k^n$. By (11.7) the vector Ax is the combination of the linearly independent list $u_1, \ldots, u_r$ with coefficients $\sigma_i\langle x, v_i\rangle$, so $Ax = 0$ holds exactly when $\sigma_i\langle x, v_i\rangle = 0$ for every $i \le r$, and since $\sigma_i > 0$ there, exactly when $\langle x, v_i\rangle = 0$ for every $i \le r$. The expansion $x = \sum_{i=1}^{n}\langle x, v_i\rangle v_i$ turns that condition into $x = \sum_{i>r}\langle x, v_i\rangle v_i$, which says exactly that $x \in \operatorname{span}\{v_{r+1}, \ldots, v_n\}$: one direction is the display, and for the other, a vector in that span has $\langle x, v_i\rangle = 0$ for $i \le r$ because the list $v_1, \ldots, v_n$ is orthonormal. Hence $\operatorname{null} A = \operatorname{span}\{v_{r+1}, \ldots, v_n\}$ by definition 2.3.3, and with Step 1 this proves (4).

Step 4: the remaining two subspaces. The factorization (11.5) presents $A^* \in M_{n\times m}(k)$ in exactly the shape (11.3): its left unitary factor is V, with columns $v_1, \ldots, v_n$; its right unitary factor is U, with columns $u_1, \ldots, u_m$; and its diagonal factor $\Sigma^\top$ carries the numbers $\sigma_1 \ge \cdots \ge \sigma_p \ge 0$, of which exactly r are nonzero. Steps 2 and 3 used nothing about A beyond that shape and the identities (11.6) it produces, so both apply to A^* with the roles of the u_i and the v_i interchanged. Step 2 then gives $\operatorname{col}(A^*) = \operatorname{span}\{v_1, \ldots, v_r\}$ and Step 3 gives $\operatorname{null}(A^*) = \operatorname{span}\{u_{r+1}, \ldots, u_m\}$, which with Step 1 proves (3) and (2). The final clause is definition 2.3.6, which identifies $\operatorname{col}(A^\top)$ with $\operatorname{row} A$, completing the proof.

Theorem 5.5.16 placed the four subspaces relative to one another, as two orthogonal splittings $k^n = \operatorname{col}(A^*) \oplus \operatorname{null} A$ and $k^m = \operatorname{col} A \oplus \operatorname{null}(A^*)$. The theorem just proved gives a basis adapted to both splittings at once, and (11.6) says how A moves one into the other: $v_i \mapsto \sigma_i u_i$ for $i \le r$, and $v_j \mapsto 0$ for $j > r$. Lemma 6.4.1 established that A carries $\operatorname{col}(A^*)$ bijectively onto $\operatorname{col} A$; in the bases $v_1, \ldots, v_r$ and $u_1, \ldots, u_r$ that bijection is multiplication of the i-th coordinate by σ_i, and nothing else.

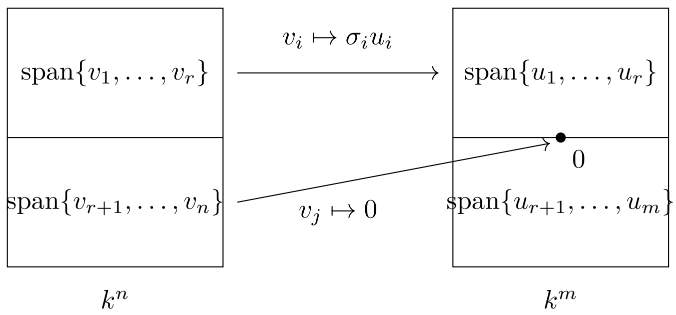


Figure 11.2: The two splittings of theorem 11.2.1, with the top halves $\operatorname{col}(A^*)$ and $\operatorname{col} A$ and the bottom halves $\operatorname{null} A$ and $\operatorname{null}(A^*)$

Counting the vectors in each of the four bases settles four dimensions, and the first of them is the rank.

Corollary 11.2.2: Rank Is the Number of Nonzero Singular Values

With the notation above:

1. $\operatorname{rank} A = r$;
2. $\operatorname{nullity} A = n - r$ and $\dim \operatorname{null}(A^*) = m - r$;
3. the columns of A are linearly independent if and only if $r = n$, and the system $Ax = b$ is consistent for every $b \in k^m$ if and only if $r = m$.

***Proof*:**

1. By theorem 11.2.1(1) the subspace $\operatorname{col} A$ has a basis with r elements, so $\dim \operatorname{col} A = r$ by definition 2.2.10. By corollary 2.3.12(1) that dimension is $\operatorname{rank} A$.
2. By theorem 11.2.1(4) the subspace $\operatorname{null} A$ has a basis with $n - r$ elements, and $\operatorname{nullity} A = \dim \operatorname{null} A$ by definition 2.3.3. By theorem 11.2.1(2) the subspace $\operatorname{null}(A^*)$ has a basis with $m - r$ elements.
3. Let $c_1, \ldots, c_n$ be the columns of A. By definition 1.4.1 the vector Ax is $x_1c_1 + \cdots + x_nc_n$, so the columns are linearly independent exactly when $Ax = 0$ forces $x = 0$, that is exactly when $\operatorname{null} A = \{0\}$ (definition 2.3.3). Suppose $r = n$ and $Ax = 0$. Then (11.7) gives $\sum_{i=1}^n \sigma_i \langle x, v_i \rangle u_i = 0$, so $\langle x, v_i \rangle = 0$ for every $i \leq n$ because the u_i are linearly independent and each $\sigma_i > 0$, and then $x = \sum_{i=1}^n \langle x, v_i \rangle v_i = 0$ by theorem 5.3.5(1). If instead $r < n$ then v_n lies in $\operatorname{null} A$ by theorem 11.2.1(4) and $\|v_n\| = 1$, so $\operatorname{null} A \neq \{0\}$.

   For the second equivalence, $Ax = b$ is consistent exactly when $b \in \operatorname{col} A$, since $\operatorname{col} A$ is the set of all Ax as computed in Step 2 of the previous proof. If $r = m$ then $\operatorname{col} A = \operatorname{span}\{u_1, \ldots, u_m\} = k^m$ by theorem 11.2.1(1) and proposition 7.1.5. If $r < m$ then $u_m \notin \operatorname{col} A$: writing $u_m = \sum_{i \leq r} c_i u_i$ and pairing with u_m would give $1 = \langle u_m, u_m \rangle = \sum_{i \leq r} c_i \langle u_i, u_m \rangle = 0$, since $u_1, \ldots, u_m$ is orthonormal. So the system fails to be consistent for $b = u_m$, as we sought to show.

Part (1) recovers, from the factorization alone, the number corollary 2.3.12 obtained by counting pivots, and part (2) then reproduces all four dimensions of that corollary. The gain is that r is now attached to a list of numbers belonging to A rather than to the outcome of an elimination: $\operatorname{rank} A$ counts how many of the σ_i are nonzero.

Example 11.2.3: The Four Subspaces of a Singular 3×3 Matrix

Let

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & 1 & 0 \end{pmatrix} \in M_3(\mathbb{R})$$

whose first two columns are equal, so it is not invertible. Its columns are $c_1 = c_2 = (1, 1, 1)^\top$ and $c_3 = (1, -1, 0)^\top$, and the entries of $A^\top A$ are the inner products $c_i^\top c_j$ by definition 1.4.1:

$c_1^\top c_1 = c_1^\top c_2 = c_2^\top c_2 = 3$, $c_1^\top c_3 = c_2^\top c_3 = 1 - 1 + 0 = 0$ and $c_3^\top c_3 = 2$. So

$$A^\top A = \begin{pmatrix} 3 & 3 & 0 \\ 3 & 3 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Three eigenvectors can be read off this matrix. The vector e_3 satisfies $A^\top A e_3 = 2e_3$; the vector $(1,1,0)^\top$ satisfies $A^\top A(1,1,0)^\top = (6,6,0)^\top = 6\,(1,1,0)^\top$; and $A^\top A(1,-1,0)^\top = 0$. The three are pairwise orthogonal and nonzero, hence linearly independent (proposition 5.3.4) and so a basis of $\mathbb{R}^3$; normalizing each and ordering the eigenvalues $6 > 2 > 0$ gives the orthonormal list

$$v_1 = \tfrac{1}{\sqrt{2}}(1,1,0)^\top, \qquad v_2 = (0,0,1)^\top, \qquad v_3 = \tfrac{1}{\sqrt{2}}(1,-1,0)^\top$$

with $\sigma_1 = \sqrt{6}$, $\sigma_2 = \sqrt{2}$ and $\sigma_3 = 0$, so $r = 2$. Multiplying,

$$Av_1 = \tfrac{1}{\sqrt{2}}(2,2,2)^\top, \qquad Av_2 = (1,-1,0)^\top, \qquad Av_3 = \tfrac{1}{\sqrt{2}}(0,0,0)^\top = 0$$

and dividing the first two by σ_1 and σ_2,

$$u_1 = \tfrac{1}{\sqrt{3}}(1,1,1)^\top, \qquad u_2 = \tfrac{1}{\sqrt{2}}(1,-1,0)^\top$$

both of norm 1. A unit vector orthogonal to these two completes the list: solving $u_1^\top y = 0$ and $u_2^\top y = 0$, that is $y_1 + y_2 + y_3 = 0$ and $y_1 = y_2$, gives the line spanned by $(1,1,-2)^\top$, and $u_3 = \frac{1}{\sqrt{6}}(1,1,-2)^\top$. Assembling,

$$U = \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \end{pmatrix}, \qquad \Sigma = \operatorname{diag}(\sqrt{6}, \sqrt{2}, 0), \qquad V = \begin{pmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 0 & 1 & 0 \end{pmatrix}$$

Both matrices have orthonormal columns, hence are orthogonal (proposition 7.1.5), and the three displayed products say exactly that $AV = U\Sigma$, since the columns of $U\Sigma$ are $\sqrt{6}\,u_1$, $\sqrt{2}\,u_2$ and 0. Multiplying on the right by $V^\top$ and using $VV^\top = I_3$ gives $A = U\Sigma V^\top$, so this is a singular value decomposition of A.

Theorem 11.2.1 now reads off the four subspaces without further computation:

$$\operatorname{col} A = \operatorname{span}\{u_1, u_2\}, \qquad \operatorname{null}(A^\top) = \operatorname{span}\{u_3\}, \qquad \operatorname{row} A = \operatorname{span}\{v_1, v_2\}, \qquad \operatorname{null} A = \operatorname{span}\{v_3\}$$

and corollary 11.2.2 gives $\operatorname{rank} A = 2$. Each answer is checkable against the matrix: the rows of A are $(1,1,1)$, $(1,1,-1)$ and $(1,1,0)$, all of which lie in $\operatorname{span}\{(1,1,0)^\top, (0,0,1)^\top\} = \operatorname{span}\{v_1, v_2\}$; the vector v_3 records the column relation $c_1 - c_2 = 0$; and $A^\top(1,1,-2)^\top = (1+1-2,\ 1+1-2,\ 1-1+0)^\top = 0$ puts u_3 in the left null space.

Section 6.4 attached to every A a matrix A^+ by a two-step geometric recipe: project b onto $\operatorname{col} A$, then invert the bijection of lemma 6.4.1 to reach the unique preimage lying in $\operatorname{col}(A^*)$. Both steps were just described in the singular bases. The projection keeps the components along $u_1, \ldots, u_r$, and the bijection divides the i-th component by σ_i. Inverting the nonzero entries of Σ performs both.

Definition 11.2.4: The Pseudoinverse of a Diagonal Factor

Let $\Sigma \in M_{m\times n}(\mathbb{R})$ be as in (11.3), with diagonal entries $\sigma_1 \geq \cdots \geq \sigma_r > 0 = \sigma_{r+1} = \cdots = \sigma_p$ and all other entries 0. Define $\Sigma^+ \in M_{n\times m}(\mathbb{R})$ to be the matrix with entry $1/\sigma_i$ in row i and column i for $1 \leq i \leq r$, and every other entry 0.

For the matrix of example 11.2.3, where $\Sigma = \operatorname{diag}(\sqrt{6}, \sqrt{2}, 0)$, the definition gives $\Sigma^+ = \operatorname{diag}(1/\sqrt{6}, 1/\sqrt{2}, 0)$: the zero singular value is left at zero rather than inverted, which is the only choice available and is what makes the next theorem true. When $m = n = r$ the matrix Σ is invertible and $\Sigma^+ = \Sigma^{-1}$.

Theorem 11.2.5: The SVD Computes the Moore-Penrose Pseudoinverse

Let $A \in M_{m\times n}(k)$ have singular value decomposition (11.3). Then

$$V\Sigma^+U^* = A^+$$

where A^+ is the Moore-Penrose pseudoinverse of definition 6.4.2. In particular the matrix $V\Sigma^+U^*$ does not depend on which singular value decomposition of A is used, and

$$A^+ = \sum_{i=1}^{r} \frac{1}{\sigma_i} v_i u_i^*$$

***Proof*:**
Step 1: the two products of Σ *with* Σ^+. Write $J_m \in M_m(\mathbb{R})$ for the diagonal matrix whose first r diagonal entries are 1 and whose other entries are 0, and $J_n \in M_n(\mathbb{R})$ for the analogous matrix of size n. By definition 1.4.1 the entry of $\Sigma\Sigma^+$ in row s and column t is $\sum_l \Sigma_{sl}(\Sigma^+)_{lt}$. The factor Σ_{sl} vanishes unless $l = s \leq p$, when it is σ_s, and $(\Sigma^+)_{lt}$ vanishes unless $t = l \leq r$, when it is $1/\sigma_l$. So the sum is $\sigma_s/\sigma_s = 1$ when $s = t \leq r$ and 0 otherwise, giving $\Sigma\Sigma^+ = J_m$. The same computation with the roles of the two matrices exchanged gives $\Sigma^+\Sigma = J_n$.

Multiplying by J_m on the left annihilates rows $r+1, \ldots, m$ and fixes the others (definition 1.4.1); the rows of Σ beyond the r-th are already zero, because $\sigma_i = 0$ for $r < i \leq p$ and Σ has no other nonzero entries. Hence $J_m\Sigma = \Sigma$, that is

$$\Sigma\Sigma^+\Sigma = \Sigma \tag{11.8}$$

and the same argument, with J_n and the rows of Σ^+ beyond the r-th, gives $\Sigma^+\Sigma\Sigma^+ = \Sigma^+$.

Step 2: the four Penrose identities. Put $X = V\Sigma^+U^*$, a matrix in $M_{n\times m}(k)$ since $V \in M_n(k)$, $\Sigma^+ \in M_{n\times m}(\mathbb{R})$ and $U^* \in M_m(k)$. Using $V^*V = I_n$ and $U^*U = I_m$ (proposition 7.1.5),

$$AX = U\Sigma V^*V\Sigma^+U^* = U\Sigma\Sigma^+U^* = UJ_mU^*, \qquad XA = V\Sigma^+U^*U\Sigma V^* = V\Sigma^+\Sigma V^* = VJ_nV^*$$

Therefore, using (11.8),

$$AXA = UJ_mU^*U\Sigma V^* = UJ_m\Sigma V^* = U\Sigma V^* = A$$

and likewise $XAX = VJ_nV^*V\Sigma^+U^* = VJ_n\Sigma^+U^* = V\Sigma^+U^* = X$. For the two self-adjointness identities, proposition 5.5.13(3) applied twice and (1) once give $(UJ_mU^*)^* = (U^*)^*J_m^*U^* = UJ_m^*U^*$, and $J_m^* = J_m$ because J_m is diagonal with real entries. Hence $(AX)^* = AX$, and the same computation gives $(XA)^* = XA$.

Step 3: identification. Theorem 6.4.5(3) states that A^+ is the only matrix in $M_{n\times m}(k)$ satisfying the four identities $AXA = A$, $XAX = X$, $(AX)^* = AX$ and $(XA)^* = XA$. Step 2 verified all four for $X = V\Sigma^+U^*$, which lies in $M_{n\times m}(k)$, so $X = A^+$. Since A^+ was defined in definition 6.4.2 without reference to any factorization, the value of $V\Sigma^+U^*$ is the same for every singular value decomposition of A.

Step 4: the sum of rank-one terms. By definition 1.4.1 the entry of $V\Sigma^+U^*$ in row s and column t is

$$\sum_{i,l} V_{si}(\Sigma^+)_{il}(U^*)_{lt} = \sum_{i=1}^{r} \frac{1}{\sigma_i} V_{si}\,\overline{U_{ti}}$$

the only surviving terms being those with $i = l \leq r$, and the entry of U^* in row i and column t being $\overline{U_{ti}}$ by definition 5.5.12. Since $V_{si} = (v_i)_s$ and $\overline{U_{ti}} = \overline{(u_i)_t}$, the displayed number is the entry in row s and column t of $\sum_{i=1}^{r} \sigma_i^{-1} v_i u_i^*$, again by definition 1.4.1. The two matrices agree entrywise, completing the proof.

This is the second construction of A^+ promised in section 6.4, and the uniqueness clause of theorem 6.4.5 is what makes it the same matrix rather than a competitor. The two constructions divide the labour in the expected way: definition 6.4.2 characterizes A^+ through subspaces and with no basis chosen, and theorem 11.2.5 writes it down once a singular value decomposition is available. When A is invertible both agree with the inverse. Then $m = n = r$, so $\Sigma^+ = \Sigma^{-1}$, and using $V^*V = UU^* = U^*U = VV^* = I_n$ (proposition 7.1.5),

$$A(V\Sigma^{-1}U^*) = U\Sigma V^*V\Sigma^{-1}U^* = UU^* = I_n, \qquad (V\Sigma^{-1}U^*)A = V\Sigma^{-1}U^*U\Sigma V^* = VV^* = I_n$$

so $V\Sigma^{-1}U^*$ is a two-sided inverse of A and $A^+ = A^{-1}$ by proposition 1.4.5.

The two products AA^+ and A^+A appeared in Step 2 as UJ_mU^* and VJ_nV^*, and those are projections written in the singular bases.

Corollary 11.2.6: The Four Projections from the SVD

With the notation above,

$$AA^+ = \sum_{i=1}^{r} u_i u_i^*, \qquad A^+A = \sum_{i=1}^{r} v_i v_i^*$$

and these are the orthogonal projection matrices onto $\operatorname{col} A$ and onto $\operatorname{col}(A^*)$. Moreover

$$I_m - AA^+ = \sum_{i=r+1}^{m} u_i u_i^*, \qquad I_n - A^+A = \sum_{i=r+1}^{n} v_i v_i^*$$

are the orthogonal projection matrices onto $\operatorname{null}(A^*)$ and onto $\operatorname{null} A$.

***Proof*:**
Step 2 of the previous proof gave $AA^+ = UJ_mU^*$, and the entry computation of Step 4, with J_m in place of Σ^+, identifies UJ_mU^* with $\sum_{i=1}^{r} u_i u_i^*$: the surviving terms are those with $i \leq r$, where the diagonal entry of J_m is 1. The same computation gives $A^+A = VJ_nV^* = \sum_{i=1}^{r} v_i v_i^*$. That these two products are the orthogonal projection matrices onto $\operatorname{col} A$ and $\operatorname{col}(A^*)$ is theorem 6.4.5(1).

Since U is unitary, $UU^* = I_m$ (proposition 7.1.5), and the same entry computation with I_m in place of J_m gives $I_m = \sum_{i=1}^m u_i u_i^*$. Subtracting the first display leaves $I_m - AA^+ = \sum_{i>r} u_i u_i^*$. For $x \in k^m$ the identity (11.4) gives $u_i^* x = \langle x, u_i \rangle$, so

$$\Big(\sum_{i>r} u_i u_i^*\Big) x = \sum_{i>r} u_i \,(u_i^* x) = \sum_{i>r} \langle x, u_i \rangle\, u_i$$

using proposition 1.4.3 for the first equality. By theorem 11.2.1(2) the list $u_{r+1}, \ldots, u_m$ is an orthonormal basis of $\operatorname{null}(A^*)$, so the right-hand side is $P_{\operatorname{null}(A^*)}(x)$ by definition 5.3.8. As this holds for every x, the matrix $I_m - AA^+$ is the orthogonal projection matrix onto $\operatorname{null}(A^*)$ by definition 6.1.4. The argument for $I_n - A^+A$ is the same with V, I_n and theorem 11.2.1(4) in place of U, I_m and theorem 11.2.1(2), as we sought to show.

Return to the matrix of example 11.2.3, where $r = 2$, $\sigma_1 = \sqrt{6}$ and $\sigma_2 = \sqrt{2}$. By theorem 11.2.5,

$$A^+ = \frac{1}{\sqrt{6}} v_1 u_1^\top + \frac{1}{\sqrt{2}} v_2 u_2^\top = \frac{1}{6}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & -1 & 0 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 3 & -3 & 0 \end{pmatrix}$$

the first product being $(\sqrt{6}\sqrt{2}\sqrt{3})^{-1}(1,1,0)^\top(1,1,1) = \frac{1}{6}(1,1,0)^\top(1,1,1)$ and the second $(\sqrt{2}\sqrt{2})^{-1}(0,0,1)^\top(1,-1,0)$. Multiplying out,

$$AA^+ = \frac{1}{6}\begin{pmatrix} 5 & -1 & 2 \\ -1 & 5 & 2 \\ 2 & 2 & 2 \end{pmatrix}, \qquad I_3 - AA^+ = \frac{1}{6}\begin{pmatrix} 1 & 1 & -2 \\ 1 & 1 & -2 \\ -2 & -2 & 4 \end{pmatrix} = u_3 u_3^\top$$

which is corollary 11.2.6 for this matrix: the second is the projection onto the line $\operatorname{null}(A^\top) = \operatorname{span}\{u_3\}$, and the first is the projection onto the plane $\operatorname{col} A$.

A matrix with independent columns gives a second check, because corollary 6.4.7 computed A^+ in that case by a closed formula. Let

$$B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \end{pmatrix} \in M_{3\times 2}(\mathbb{R}), \qquad B^\top B = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$$

The eigenvectors $(1,1)^\top$ and $(1,-1)^\top$ of $B^\top B$ have eigenvalues 4 and 2, so $\sigma_1 = 2$ and $\sigma_2 = \sqrt{2}$, and $r = 2$ matches the two independent columns as corollary 11.2.2(3) requires. Normalizing gives $v_1 = \frac{1}{\sqrt{2}}(1,1)^\top$ and $v_2 = \frac{1}{\sqrt{2}}(1,-1)^\top$, and then $Bv_1 = \frac{1}{\sqrt{2}}(2,0,2)^\top$ and $Bv_2 = \frac{1}{\sqrt{2}}(0,2,0)^\top$ give

$$u_1 = \frac{Bv_1}{\sigma_1} = \tfrac{1}{\sqrt{2}}(1,0,1)^\top, \qquad u_2 = \frac{Bv_2}{\sigma_2} = (0,1,0)^\top$$

so that

$$B^+ = \tfrac{1}{2} v_1 u_1^\top + \tfrac{1}{\sqrt{2}} v_2 u_2^\top = \frac{1}{4}\begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} + \frac{1}{2}\begin{pmatrix} 0 & 1 & 0 \\ 0 & -1 & 0 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 1 & 2 & 1 \\ 1 & -2 & 1 \end{pmatrix}$$

The closed formula gives the same matrix. Since $\det(B^\top B) = 8$, the inverse is $(B^\top B)^{-1} = \frac{1}{8}\left(\begin{smallmatrix} 3 & -1 \\ -1 & 3 \end{smallmatrix}\right)$, and

$$(B^\top B)^{-1} B^\top = \frac{1}{8}\begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}\begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \end{pmatrix} = \frac{1}{8}\begin{pmatrix} 2 & 4 & 2 \\ 2 & -4 & 2 \end{pmatrix} = \frac{1}{4}\begin{pmatrix} 1 & 2 & 1 \\ 1 & -2 & 1 \end{pmatrix}$$

in agreement with corollary 6.4.7. The formula of that corollary needs independent columns and inverts an $n \times n$ matrix; theorem 11.2.5 needs no hypothesis at all and inverts r numbers.

Exercise 11.2.1

1. Let
$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 2 & 4 \end{pmatrix} \in M_{3\times 2}(\mathbb{R})$$
Compute $A^\top A$, find a singular value decomposition of A, and use theorem 11.2.1 to give an orthonormal basis of each of the four subspaces $\operatorname{col} A$, $\operatorname{null}(A^\top)$, $\operatorname{row} A$ and $\operatorname{null} A$. Then compute A^+ by theorem 11.2.5 and verify $AA^+A = A$.

2. Let
$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix} \in M_3(\mathbb{R})$$
Find a singular value decomposition of A, give orthonormal bases for the four subspaces of theorem 11.2.1, and compute A^+. Verify directly that AA^+ is the orthogonal projection matrix onto $\operatorname{col} A$, as corollary 11.2.6 asserts.

3. Keep the matrix A of example 11.2.3, together with the vector $u_3 = \frac{1}{\sqrt{6}}(1, 1, -2)^\top$ computed there, and let $b = (1, 0, 0)^\top$. Compute A^+b and check that it lies in $\operatorname{row} A$. Then compute AA^+b and check that it equals $b - \langle b, u_3\rangle u_3$, which is the orthogonal projection of b onto $\operatorname{col} A$.

4. Let $A \in M_{m\times n}(k)$ have singular value decomposition $A = U\Sigma V^*$ with r nonzero singular values. Show that $A^*A = V(\Sigma^\top\Sigma)V^*$ is a singular value decomposition of A^*A whose nonzero singular values are $\sigma_1^2, \dots, \sigma_r^2$, and deduce from corollary 11.2.2 that
$$\operatorname{rank} A = \operatorname{rank}(A^*) = \operatorname{rank}(A^*A) = \operatorname{rank}(AA^*)$$
Deduce also that $\operatorname{col}(A^*A) = \operatorname{col}(A^*)$. (The companion identity $\operatorname{null}(A^*A) = \operatorname{null} A$ is already proposition 11.1.1(2); do not reprove it, but say which of the two routes — that proposition, or the decomposition above — is the shorter.)

5. Let $A = U\Sigma V^*$ be a singular value decomposition with $\sigma_1 = \sigma_2 > 0$. For $\theta \in \mathbb{R}$ put
$$u_1' = \cos\theta\, u_1 + \sin\theta\, u_2, \qquad u_2' = -\sin\theta\, u_1 + \cos\theta\, u_2$$
and define v_1', v_2' from v_1, v_2 by the same two formulas. Show that replacing the first two columns of U by u_1', u_2' and the first two columns of V by v_1', v_2' again gives a singular value decomposition of A, and that the four subspaces of theorem 11.2.1 are the same for both. Conclude that the unitary factors of a singular value decomposition are not determined by A, while the four subspaces are.

6. Let $W \subseteq k^m$ be a subspace of dimension d with $1 \le d \le m$, and let $P \in M_m(k)$ be the orthogonal projection matrix onto W (definition 6.1.4). Show that the singular values of P are 1, occurring d times, and 0, occurring $m - d$ times, and that $P^+ = P$.

7. Let $A \in M_{m\times n}(k)$. Show that $(A^+)^+ = A$, using the uniqueness clause theorem 6.4.5(3) rather than a computation with the singular value decomposition. Show also that $(A^*)^+ = (A^+)^*$, this time from theorem 11.2.5.

8. Let $A \in M_{m\times n}(k)$ be nonzero. Show that $A^+ = A^*$ if and only if every nonzero singular value of A equals 1, and that in that case $\|Ax\| = \|x\|$ for every $x \in \operatorname{col}(A^*)$. Give a 2×2 real example that is not unitary and satisfies these conditions.

11.3 The Polar Decomposition

Theorem 11.1.3 writes $A = U\Sigma V^*$ with two unitary matrices, and nothing so far ties one to the other. Throughout this section the letter U is needed for the unitary factor of the polar decomposition, which is what the notation $A = UP$ has meant since Gauss; so the left factor of a singular value decomposition is written W here, and U always denotes the polar factor. The other sections of this chapter keep the usage of theorem 11.1.3. Regrouping its three factors writes a square matrix as a product of exactly two: a unitary matrix, and a matrix whose quadratic form takes no negative value. This section proves that the regrouping always succeeds, decides which of the two factors is determined by A, and shows that the resulting factorization and the singular value decomposition each produce the other.

The case $n = 1$ is the model, and it is where the name comes from. A complex number z can be written $z = wr$ with $r = |z| \geq 0$ and $|w| = 1$, and w is determined by z as soon as $z \neq 0$. Read z, w and r as 1×1 matrices. The condition $|w| = 1$ is $\overline{w}w = 1$, which is the condition $W^*W = I_1$ of definition 7.1.3; and $r \geq 0$ says that the number $\overline{x}rx = r|x|^2$ is never negative. Both conditions read verbatim for $n \times n$ matrices.

Throughout the section k is $\mathbb{R}$ or $\mathbb{C}$ and k^n carries the standard inner product. Written with the conjugate transpose of definition 5.5.12, that inner product is

$$\langle a, b\rangle = \sum_{i=1}^{n} a_i\overline{b_i} = b^*a \tag{11.9}$$

for $a, b \in k^n$, the right-hand side being a 1×1 matrix read as its single entry (definition 1.4.1). In particular $\|a\|^2 = \langle a, a\rangle = a^*a$ by definition 5.2.1. A unitary $U \in M_n(k)$ changes no length, since

$$\|Ux\|^2 = (Ux)^*(Ux) = x^*U^*Ux = x^*x = \|x\|^2 \tag{11.10}$$

for every $x \in k^n$, using $(Ux)^* = x^*U^*$ from proposition 5.5.13(3) and $U^*U = I_n$ from definition 7.1.3. This is condition (3) of theorem 7.1.4 for the map $x \mapsto Ux$.

Definition 11.3.1: Positive Semidefinite Matrix

Let k be $\mathbb{R}$ or $\mathbb{C}$. A matrix $P \in M_n(k)$ is *positive semidefinite* if $P^* = P$ and

$$\langle Px, x\rangle \geq 0 \qquad \text{for every } x \in k^n$$

The inequality is meaningful, because $\langle Px, x\rangle = x^*Px$ is real for every x once $P^* = P$ holds: the conjugate transpose of a 1×1 matrix is its complex conjugate, and proposition 5.5.13(1),(3) give $\overline{x^*Px} = (x^*Px)^* = x^*P^*x = x^*Px$. Over $\mathbb{R}$ the condition reads $P^\top = P$ and $x^\top Px \geq 0$, which is positive semidefiniteness in the sense of definition 8.4.1.

The strict version costs one word and is needed later, so it is recorded here alongside its companion.

Definition 11.3.2: Positive Definite Matrix

Let k be $\mathbb{R}$ or $\mathbb{C}$. A matrix $P \in M_n(k)$ is *positive definite* if $P^* = P$ and

$$\langle Px, x\rangle > 0 \qquad \text{for every nonzero } x \in k^n$$

The quantity is real for the reason given in definition 11.3.1, and over $\mathbb{R}$ the condition is positive definiteness in the sense of definition 8.4.1. A positive definite matrix is positive semidefinite, since $\langle P0, 0\rangle = 0$.

Theorem 8.4.2 characterized the real case by the sign of the eigenvalues. The same characterization holds over $\mathbb{C}$, and the spectral theorem proves both at once.

Proposition 11.3.3: Definiteness Is the Sign of the Spectrum

Let k be $\mathbb{R}$ or $\mathbb{C}$ and let $P \in M_n(k)$ satisfy $P^* = P$, which over $\mathbb{R}$ reads $P^\top = P$. Let $\lambda_1, \ldots, \lambda_n$ be the roots of the characteristic polynomial of P, each listed as often as its multiplicity; they are real, by proposition 7.1.11 when $k = \mathbb{C}$ and by corollary 7.4.3 when $k = \mathbb{R}$.

1. P is positive semidefinite if and only if $\lambda_i \geq 0$ for every i.
2. P is positive definite if and only if $\lambda_i > 0$ for every i.

***Proof*:**
There is a unitary $U \in M_n(k)$ with $P = UDU^*$ and $D = \operatorname{diag}(\lambda_1, \ldots, \lambda_n)$. For $k = \mathbb{C}$ this is corollary 7.3.3, since a matrix with $P^* = P$ is normal (definition 7.1.7). For $k = \mathbb{R}$ it is corollary 7.4.3(3), which states that $P = QDQ^\top$ for a real orthogonal Q. Such a Q satisfies $Q^* = Q^\top = Q^{-1}$ and so is unitary in the sense of definition 7.1.3, and $Q^\top = Q^*$ makes the two factorizations the same equation. For $x \in k^n$ put $y = U^*x$, so that $x = Uy$ by theorem 7.1.4, and x is nonzero exactly when y is, because U^* is invertible. Then

$$\langle Px, x\rangle = \langle UDU^*x, x\rangle = \langle DU^*x, U^*x\rangle = \langle Dy, y\rangle = \sum_{i=1}^{n} \lambda_i |y_i|^2$$

the second equality by definition 5.5.8 and the last by definition 1.4.1, since the ith entry of Dy is $\lambda_i y_i$.

Suppose every $\lambda_i \geq 0$. Every term of the sum is then nonnegative, so $\langle Px, x\rangle \geq 0$ and P is positive semidefinite. Conversely, suppose P is positive semidefinite and fix j. Taking $y = e_j$, that is $x = Ue_j$, makes the sum λ_j, so $\lambda_j = \langle Px, x\rangle \geq 0$. This proves (1).

For (2), suppose every $\lambda_i > 0$ and let $x \neq 0$. Then $y \neq 0$, so some $|y_i|^2 > 0$, and every term of the sum is nonnegative, which makes $\langle Px, x\rangle > 0$. Conversely, if P is positive definite then $x = Ue_j$ is nonzero and gives $\lambda_j > 0$ as before.

Definition 11.3.4: Polar Decomposition

A *polar decomposition* of a matrix $A \in M_n(k)$ is a factorization

$$A = UP$$

in which $U \in M_n(k)$ is unitary (definition 7.1.3) and $P \in M_n(k)$ is positive semidefinite.

Here is one, produced by inspection. Take

$$A = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}, \qquad U = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad P = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

Then $U^\top U = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) = I_2$, so U is orthogonal; P is symmetric with $x^\top P x = x_1^2 + 2x_2^2 \geq 0$, so P is positive semidefinite; and

$$UP = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix} = A$$

The two factors act on $\mathbb{R}^2$ in different ways. The matrix P scales the first coordinate axis by 1 and the second by 2, so it carries the unit circle to an ellipse with semi-axes 1 and 2, while U changes no length by (11.10) and turns the plane a quarter turn.

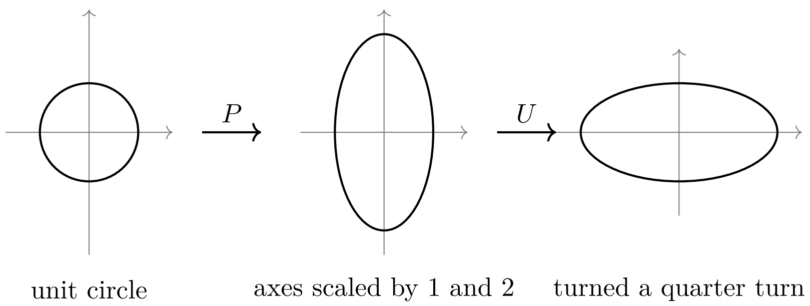


Figure 11.3: The image of the unit circle under the two factors of $A = UP$

That A was chosen so the factors could be written down. The singular value decomposition gives them for every square matrix at once, and the only work is to see which factors to group together.

Theorem 11.3.5: Existence of the Polar Decomposition

Let $A \in M_n(k)$. Then A has a polar decomposition. One is obtained from any singular value decomposition of theorem 11.1.3, written $A = W\Sigma V^*$ in the notation fixed at the head of this section, by setting

$$U = WV^*, \qquad P = V\Sigma V^*$$

Moreover every polar decomposition $A = UP$ of A satisfies $P^2 = A^*A$.

***Proof*:**
Step 1: the two candidate factors. Applying theorem 11.1.3 to the square matrix A gives unitary matrices $W, V \in M_n(k)$ and a diagonal matrix $\Sigma = \operatorname{diag}(\sigma_1, \ldots, \sigma_n)$ carrying the singular values of A (definition 11.1.2), so that $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$ and $A = W\Sigma V^*$. Put $U = WV^*$ and $P = V\Sigma V^*$. Being unitary, W and V satisfy $W^*W = WW^* = I_n$ and $V^*V = VV^* = I_n$, by the closing clause of proposition 7.1.5.

Step 2: U is unitary. By proposition 5.5.13(3) applied twice and then (1),

$$U^*U = (WV^*)^*(WV^*) = (V^*)^*W^*WV^* = VW^*WV^* = VV^* = I_n$$

which is the condition of definition 7.1.3.

Step 3: P is positive semidefinite. The matrix Σ is diagonal with real entries, so $\Sigma^* = \Sigma$ by definition 5.5.12. Hence, by proposition 5.5.13(3) and (1),

$$P^* = (V\Sigma V^*)^* = (V^*)^*(V\Sigma)^* = V\Sigma^*V^* = V\Sigma V^* = P$$

For positivity, fix $x \in k^n$ and put $y = V^*x \in k^n$, with entries $y_1, \ldots, y_n$. The same two parts of proposition 5.5.13 give $y^* = (V^*x)^* = x^*(V^*)^* = x^*V$, so by (11.9) and definition 1.4.1,

$$\langle Px, x\rangle = x^*V\Sigma V^*x = y^*\Sigma y = \sum_{j=1}^{n} \sigma_j \, \overline{y_j} \, y_j = \sum_{j=1}^{n} \sigma_j \, |y_j|^2$$

Every σ_j is nonnegative and every $|y_j|^2$ is nonnegative, so the sum is nonnegative.

Step 4: the product is A. Using $V^*V = I_n$ and associativity of the matrix product (proposition 1.4.3),

$$UP = (WV^*)(V\Sigma V^*) = W(V^*V)\Sigma V^* = W\Sigma V^* = A$$

so $A = UP$ is a polar decomposition.

Step 5: the square of the positive semidefinite factor. Let $A = UP$ be any polar decomposition of A, so $U^*U = I_n$ and $P^* = P$. Then proposition 5.5.13(3) gives

$$A^*A = (UP)^*(UP) = P^*U^*UP = P^*P = P^2$$

as we sought to show.

Every square matrix over $\mathbb{R}$ or $\mathbb{C}$ therefore acts in two stages: a scaling along n orthogonal directions, followed by a map that changes no length. The directions are the columns of V, since $Pv_j = V\Sigma V^*v_j = V\Sigma e_j = \sigma_j v_j$, using $V^*v_j = V^{-1}v_j = e_j$ from proposition 7.1.5; the scaling factors are the singular values; and the second stage is U, which changes no length by (11.10). The figure above is this statement for the matrix $\left(\begin{smallmatrix} 0 & -2 \\ 1 & 0 \end{smallmatrix}\right)$, whose positive semidefinite factor $\operatorname{diag}(1, 2)$ scales the two coordinate axes by 1 and by 2.

The last clause of theorem 11.3.5 pins P down as far as its square: whatever polar decomposition is used, P^2 is the matrix A^*A. Theorem 8.4.9 settles the real symmetric case of the question this raises, saying that a real symmetric positive semidefinite matrix has exactly one real symmetric positive semidefinite square root. Over $\mathbb{C}$ the factor P is Hermitian rather than symmetric, and the columns of V are already an orthonormal basis of eigenvectors of A^*A, which is what the argument needs. The unitary factor is determined only when A is invertible, and the next statement records both facts.

Proposition 11.3.6: What a Polar Decomposition Determines

Let $A \in M_n(k)$ have singular values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$.

1. There is exactly one positive semidefinite $P \in M_n(k)$ with $P^2 = A^*A$ — over $\mathbb{R}$ this is theorem 8.4.9, and the proof below extends it to $\mathbb{C}$ — and it is the positive semidefinite factor of every polar decomposition of A. It is written $(A^*A)^{1/2}$, and its eigenvalues, each listed as often as its multiplicity as a root of the characteristic polynomial, are $\sigma_1, \ldots, \sigma_n$.

2. If A is invertible, then $(A^*A)^{1/2}$ is invertible and $U = A\big((A^*A)^{1/2}\big)^{-1}$ is the only unitary matrix with $A = U(A^*A)^{1/2}$.

3. If A is not invertible, then A has more than one polar decomposition. Concretely, $(A^*A)^{1/2}z = 0$ for some $z \in k^n$ with $\|z\| = 1$, and for any such z and any unitary factor U of A the matrix $U(I_n - 2zz^*)$ is a second unitary factor, different from U.

***Proof*:**
Fix a singular value decomposition $A = W\Sigma V^*$ as in theorem 11.1.3, with $\Sigma = \operatorname{diag}(\sigma_1, \ldots, \sigma_n)$, and let $v_1, \ldots, v_n$ be the columns of V. They are an orthonormal basis of k^n (proposition 7.1.5), and $V^*v_j = V^{-1}v_j = e_j$. Write $P = V\Sigma V^*$, which theorem 11.3.5 showed to be a positive semidefinite factor of a polar decomposition of A. Two computations are used repeatedly: $Pv_j = V\Sigma e_j = \sigma_j v_j$, and

$$A^*A = P^2 = V\Sigma V^* V\Sigma V^* = V\Sigma^2 V^*, \qquad \text{so} \qquad A^*Av_j = V\Sigma^2 e_j = \sigma_j^2 v_j$$

the first equality being theorem 11.3.5. Note also that two matrices $M, N \in M_n(k)$ with $Mv_j = Nv_j$ for every j are equal: every $x \in k^n$ is a linear combination of $v_1, \ldots, v_n$, so $Mx = Nx$ for every x, and taking $x = e_i$ identifies the i-th columns (definition 1.4.1).

1. Let $Q \in M_n(k)$ be positive semidefinite with $Q^2 = A^*A$. Since $Q^* = Q$, the matrix Q satisfies $Q^*Q = Q^2 = QQ^*$ and so is normal (definition 7.1.7). Over $\mathbb{C}$, corollary 7.3.3 gives an orthonormal basis $w_1, \ldots, w_n$ of $\mathbb{C}^n$ with $Qw_j = \mu_j w_j$, and the μ_j are real because the operator $x \mapsto Qx$ is self-adjoint (theorem 5.5.14, definition 7.1.1) and proposition 7.1.11 makes the eigenvalues of a self-adjoint operator real. Over $\mathbb{R}$ the matrix Q is symmetric and corollary 7.4.3 gives the same data with $\mu_j \in \mathbb{R}$. In both cases

$$\mu_j = \mu_j\, w_j^* w_j = w_j^*(\mu_j w_j) = w_j^* Q w_j = \langle Qw_j, w_j \rangle \geq 0$$

by $\|w_j\| = 1$, (11.9) and positive semidefiniteness, so every $\mu_j \geq 0$. Applying Q twice gives $A^*Aw_j = Q^2 w_j = \mu_j^2 w_j$.
Now fix j and expand $v_j = c_1 w_1 + \cdots + c_n w_n$ in the basis $w_1, \ldots, w_n$. Then

$$A^*Av_j = \sum_{i=1}^{n} c_i \mu_i^2 w_i, \qquad \sigma_j^2 v_j = \sum_{i=1}^{n} \sigma_j^2 c_i w_i$$

and these two vectors are equal, so their coefficients in the basis $w_1, \ldots, w_n$ agree: $c_i\mu_i^2 = \sigma_j^2 c_i$ for every i. Whenever $c_i \neq 0$ this forces $\mu_i^2 = \sigma_j^2$, and since $\mu_i \geq 0$ and $\sigma_j \geq 0$ it forces $\mu_i = \sigma_j$. The terms with $c_i = 0$ contribute nothing to either sum below, so

$$Qv_j = \sum_{i=1}^{n} c_i \mu_i w_i = \sigma_j \sum_{i=1}^{n} c_i w_i = \sigma_j v_j = Pv_j$$

This holds for every j, so $Q = P$ by the remark above. Since P is a positive semidefinite matrix with $P^2 = A^*A$, the matrix P is the only one, and by the last clause of theorem 11.3.5 the positive semidefinite factor of any polar decomposition of A is such a matrix, hence is P. Finally $P = V\Sigma V^*$ with V unitary and Σ diagonal, which is the shape of corollary 7.3.3 over $\mathbb{C}$ and of corollary 7.4.3(3) over $\mathbb{R}$. The closing clause of either identifies the diagonal entries $\sigma_1, \ldots, \sigma_n$ of Σ as the roots of the characteristic polynomial of P, each listed as often as its multiplicity.

2. By part (1) the matrix P is $(A^*A)^{1/2}$. Since $VV^* = I_n$, theorem 3.2.10 gives $\det V \cdot \det V^* = \det I_n = 1$, and Σ is triangular, so $\det \Sigma = \sigma_1 \cdots \sigma_n$ by proposition 3.2.6. Applying theorem 3.2.10 twice more,

$$\det P = \det V \cdot \det \Sigma \cdot \det V^* = \sigma_1 \sigma_2 \cdots \sigma_n \tag{11.11}$$

The same theorem applied to the factorization $A = (WV^*)P$ of theorem 11.3.5 gives $\det A = \det(WV^*) \cdot \det P$, and $\det(WV^*) \neq 0$ because WV^* is unitary, hence invertible (proposition 7.1.5, closing clause), and corollary 3.2.8. So $\det A \neq 0$ if and only if $\det P \neq 0$, and corollary 3.2.8 turns this into: A is invertible if and only if P is. Assume now that A is invertible, so that P^{-1} exists. Theorem 11.3.5 produces a polar decomposition $A = U_0 P_0$, and $P_0 = P$ by (1), so $U_0 P = A$ and multiplying on the right by P^{-1} gives $U_0 = AP^{-1}$: the matrix AP^{-1} is a unitary factor. If U is any unitary matrix with $A = UP$, the same multiplication gives $U = AP^{-1}$, so there is no other.

3. Assume A is not invertible. By the previous paragraph $\det P = 0$, so (11.11) gives $\sigma_1 \cdots \sigma_n = 0$, and since the σ_j are ordered downward this forces $\sigma_n = 0$. The last column v_n of V then has norm 1 and satisfies $Pv_n = \sigma_n v_n = 0$, so a vector of the kind claimed exists. Let z be any vector with $\|z\| = 1$ and $Pz = 0$, and put $H = I_n - 2zz^*$, a matrix in $M_n(k)$ since z is $n \times 1$ and z^* is $1 \times n$. The scalar z^*z equals $\|z\|^2 = 1$, so proposition 1.4.3 gives $(zz^*)(zz^*) = z(z^*z)z^* = zz^*$, whence

$$H^* = I_n - 2(zz^*)^* = I_n - 2zz^* = H, \qquad H^2 = I_n - 4zz^* + 4zz^* = I_n$$

using proposition 5.5.13(1),(3) for $(zz^*)^* = (z^*)^*z^* = zz^*$. Hence $H^*H = H^2 = I_n$ and H is unitary. Next, $P^* = P$ and $Pz = 0$ give $z^*P = (P^*z)^* = (Pz)^* = 0$, so

$$HP = P - 2z(z^*P) = P$$

Therefore $(UH)P = U(HP) = UP = A$. The matrix UH is unitary, since proposition 5.5.13(3) gives $(UH)^*(UH) = H^*U^*UH = H^*H = I_n$. So $A = (UH)P$ is a second polar decomposition. It is a different one: if $UH = U$ then multiplying on the left by $U^{-1} = U^*$ gives $H = I_n$, hence $2zz^* = 0$, hence $0 = 2z(z^*z) = 2z$, contradicting $\|z\| = 1$. This completes the proof.

The two factors are therefore determined to different degrees. The positive semidefinite one is a function of A alone and can be computed without ever choosing a decomposition, since $(A^*A)^{1/2}$

depends only on the matrix A^*A. The unitary one is pinned down only on the vectors Px, and the null space of P is where the freedom sits: part (3) alters U by a matrix that fixes every vector of the form Px and negates one vector killed by P, which leaves the product unchanged. The extreme case is $A = 0$, where $P = 0$ is forced and $U \cdot 0 = 0$ holds for every unitary U, so every unitary matrix is a unitary factor of the zero matrix.

Example 11.3.7: The Polar Decomposition of a Singular Matrix

Let

$$A = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} \in M_2(\mathbb{R})$$

Then $A^\top = \left(\begin{smallmatrix} 1 & -1 \\ 1 & -1 \end{smallmatrix}\right)$ and

$$A^\top A = \begin{pmatrix} 1 & -1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$$

whose characteristic polynomial is $t^2 - 4t$, so its eigenvalues are 4 and 0 and the singular values of A are $\sigma_1 = 2$ and $\sigma_2 = 0$. Orthonormal eigenvectors are $v_1 = \frac{1}{\sqrt{2}}(1,1)^\top$ for 4 and $v_2 = \frac{1}{\sqrt{2}}(1,-1)^\top$ for 0, as the products $\left(\begin{smallmatrix} 2 & 2 \\ 2 & 2 \end{smallmatrix}\right)(1,1)^\top = (4,4)^\top$ and $\left(\begin{smallmatrix} 2 & 2 \\ 2 & 2 \end{smallmatrix}\right)(1,-1)^\top = (0,0)^\top$ confirm.

Assembling V from these columns and $\Sigma = \operatorname{diag}(2,0)$,

$$V = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad P = V\Sigma V^\top = 2v_1 v_1^\top = 2 \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

the middle equality because the second diagonal entry of Σ is 0, so only the first column of V contributes. Squaring confirms $P^2 = \left(\begin{smallmatrix} 2 & 2 \\ 2 & 2 \end{smallmatrix}\right) = A^\top A$, and P is symmetric with $x^\top P x = (x_1 + x_2)^2 \geq 0$, so $P = (A^\top A)^{1/2}$ by proposition 11.3.6(1).

For the unitary factor, a singular value decomposition $A = W\Sigma V^\top$ is the same as the identity $AV = W\Sigma$, since $V^\top = V^{-1}$; and comparing j-th columns turns that identity into $Av_j = \sigma_j u_j$, where u_1, u_2 are the columns of W. So $u_1 = Av_1/2$, and

$$Av_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1+1 \\ -1-1 \end{pmatrix} = \frac{2}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad \text{so} \qquad u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

while $\sigma_2 = 0$ leaves u_2 unconstrained apart from orthonormality, consistently with $Av_2 = \frac{1}{\sqrt{2}}(1 - 1, -1+1)^\top = 0$. A unit vector orthogonal to u_1 in $\mathbb{R}^2$ is $u_2 = \pm\frac{1}{\sqrt{2}}(1,1)^\top$, and both signs give an orthonormal basis. The two choices give the two matrices $W_+ = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right)$ and $W_- = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & -1 \\ -1 & -1 \end{smallmatrix}\right)$, and since $V^\top = V$ the recipe $U = WV^\top$ of theorem 11.3.5 gives

$$U_+ = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad U_- = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

Both are orthogonal, and multiplying out,

$$U_+ P = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} = A, \qquad U_- P = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} = A$$

so A has at least two polar decompositions, as proposition 11.3.6(3) requires of a singular matrix. Here $U_- = U_+(I_2 - 2v_2 v_2^\top)$, since $I_2 - 2v_2 v_2^\top = I_2 - \left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, which is the modification the proposition constructs from the unit vector v_2 in the null space of P.

The factorization $A = UP$ applies P first. Nothing forces that order, and conjugating P by U reverses it. The resulting factorization has the same unitary matrix and a positive semidefinite matrix built from AA^* instead of A^*A.

Corollary 11.3.8: The Left Polar Decomposition

Let $A \in M_n(k)$ and let $A = UP$ be a polar decomposition. Put $P' = UPU^*$. Then P' is positive semidefinite,

$$A = P'U, \qquad (P')^2 = AA^*$$

and $P' = (AA^*)^{1/2}$, so P' does not depend on the choice of U. Moreover P and P' have the same characteristic polynomial, so their eigenvalues with multiplicity are the same: the singular values of A.

***Proof*:**
By proposition 5.5.13(3) and (1), and $P^* = P$,

$$(P')^* = (UPU^*)^* = (U^*)^*P^*U^* = UPU^* = P'$$

For positivity, let $x \in k^n$ and put $y = U^*x$, so that $y^* = x^*U$ by proposition 5.5.13(1),(3). Then (11.9) gives

$$\langle P'x, x\rangle = x^*UPU^*x = y^*Py = \langle Py, y\rangle \geq 0$$

because P is positive semidefinite. So P' is positive semidefinite.

Since $U^*U = I_n$, the product $P'U = UPU^*U = UP = A$. For the square, proposition 5.5.13(3) and $P^* = P$ give

$$AA^* = (UP)(UP)^* = UPP^*U^* = UP^2U^* = (UPU^*)(UPU^*) = (P')^2$$

the fourth equality again using $U^*U = I_n$. Now $(A^*)^*A^* = AA^*$ by proposition 5.5.13(1), so P' is a positive semidefinite matrix whose square is $(A^*)^*A^*$; applying proposition 11.3.6(1) to the matrix A^* in place of A identifies it as $P' = (AA^*)^{1/2}$, a matrix built from A alone.

Finally $U^{-1} = U^*$ by proposition 7.1.5, so $P' = UPU^{-1}$ and P' is similar to P (definition 2.5.5). By corollary 4.1.16 the two have the same characteristic polynomial, and proposition 11.3.6(1) identifies its roots, with multiplicity, as the singular values of A, completing the proof.

For the matrix $\left(\begin{smallmatrix} 0 & -2 \\ 1 & 0 \end{smallmatrix}\right)$ above, $P' = UPU^\top = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) = \operatorname{diag}(2, 1)$, and indeed $\operatorname{diag}(2, 1)\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & -2 \\ 1 & 0 \end{smallmatrix}\right)$. The same quarter turn appears in both factorizations. What changes is whether the axes scaled by 1 and 2 are the coordinate axes or their images under the turn.

The polar decomposition was extracted from the singular value decomposition, so it carries no more information than that factorization does. The statement below is that it carries no less.

Proposition 11.3.9: The Two Factorizations Produce Each Other

Let $A \in M_n(k)$.

1. If $A = W\Sigma V^*$ is a singular value decomposition of A, then $(A^*A)^{1/2} = V\Sigma V^*$, and WV^* is a unitary factor of A, the only one when A is invertible.

2. Conversely, let $A = UP$ be a polar decomposition and let $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$ be the singular values of A, which by proposition 11.3.6(1) are the eigenvalues of P listed with multiplicity. Let $v_1, \ldots, v_n$ be an orthonormal basis of k^n with $Pv_j = \sigma_j v_j$, let $V \in M_n(k)$ have columns $v_1, \ldots, v_n$, and put $\Sigma = \operatorname{diag}(\sigma_1, \ldots, \sigma_n)$. Then

$$A = (UV)\Sigma V^*$$

is a singular value decomposition of A.

***Proof*:**

1. Theorem 11.3.5 exhibits $V\Sigma V^*$ as the positive semidefinite factor of the polar decomposition $A = (WV^*)(V\Sigma V^*)$, and proposition 11.3.6(1) identifies that factor as $(A^*A)^{1/2}$. The same theorem makes WV^* a unitary factor, and proposition 11.3.6(2) makes it the only one when A is invertible.

2. A basis of the stated kind exists. Over $\mathbb{C}$ the matrix P is Hermitian, hence normal (definition 7.1.7), and corollary 7.3.3 gives an orthonormal basis of $\mathbb{C}^n$ consisting of eigenvectors of P. Over $\mathbb{R}$ the matrix P is symmetric and corollary 7.4.3(2) gives the same. The eigenvalue attached to a unit eigenvector w of P is $\langle Pw, w\rangle$, since (11.9) gives

$$\langle Pw, w\rangle = w^*Pw = w^*(\mu w) = \mu\, w^*w = \mu$$

when $Pw = \mu w$; so every such eigenvalue is real and nonnegative (definition 11.3.4). By proposition 11.3.6(1) the eigenvalues of P with multiplicity are $\sigma_1, \ldots, \sigma_n$, so relabelling the basis arranges $Pv_j = \sigma_j v_j$ with the σ_j in nonincreasing order.

The columns of V are an orthonormal basis of k^n, so V is unitary (proposition 7.1.5) and $V^*v_j = V^{-1}v_j = e_j$. Hence $V\Sigma V^*v_j = V\Sigma e_j = \sigma_j Ve_j = \sigma_j v_j = Pv_j$ for every j, and two matrices agreeing on the basis $v_1, \ldots, v_n$ agree on every e_i and so have the same columns (definition 1.4.1). Therefore $P = V\Sigma V^*$. The matrix UV is unitary, since $(UV)^*(UV) = V^*U^*UV = V^*V = I_n$ by proposition 5.5.13(3) and proposition 7.1.5, and

$$(UV)\Sigma V^* = U(V\Sigma V^*) = UP = A$$

This exhibits A as a unitary matrix, times a diagonal matrix whose entries are nonnegative and nonincreasing, times the conjugate transpose of a unitary matrix, which is the shape of theorem 11.1.3; and the diagonal entries are the singular values of A by proposition 11.3.6(1). So it is a singular value decomposition of A, as we sought to show.

Each factorization is therefore computable from the other, and the choice between them is a choice of what to make visible. The singular value decomposition displays two orthonormal bases and the numbers relating them, which is what section 11.2 uses to produce bases for the four fundamental subspaces. The polar decomposition displays instead one matrix that changes no length and one matrix that scales n orthogonal directions, which is the form in which the two effects of A on lengths are separated.

Exercise 11.3.1

1. Let
$$A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix} \in M_2(\mathbb{R})$$
Compute $A^\top A$, its eigenvalues and an orthonormal basis of eigenvectors, and use them to compute $P = (A^\top A)^{1/2}$. Then compute the unique orthogonal U with $A = UP$, verify $U^\top U = I_2$ and $UP = A$, and compute the left polar factor $P' = UPU^\top$ of corollary 11.3.8, checking that $(P')^2 = AA^\top$.

2. Let $A = \left(\begin{smallmatrix} 2 & 2 \\ 1 & 1 \end{smallmatrix}\right) \in M_2(\mathbb{R})$. Compute $P = (A^\top A)^{1/2}$, and exhibit two orthogonal matrices U with $A = UP$, one with $\det U = 1$ and one with $\det U = -1$. Explain which hypothesis of proposition 11.3.6(2) fails, and identify the two matrices you found as U and $U(I_2 - 2zz^\top)$ for a unit vector z with $Pz = 0$.

3. Let $A \in M_n(k)$ and let $A = UP$ be a polar decomposition. Show that $P = I_n$ if and only if A is unitary, and that $P = A$ if and only if A is positive semidefinite, in which case I_n is one admissible unitary factor.

4. Let $A \in M_n(k)$ have singular values $\sigma_1 \geq \cdots \geq \sigma_n$ and let $P = (A^*A)^{1/2}$. The identity $\det P = \sigma_1\sigma_2\cdots\sigma_n$ was established inside the proof of proposition 11.3.6; take it as given and show that A is invertible if and only if $\sigma_n > 0$. Show also that when $k = \mathbb{R}$ every unitary factor satisfies $\det U = \pm 1$, and hence that $|\det A| = \sigma_1\sigma_2\cdots\sigma_n$.

5. Let $A \in M_{m\times n}(k)$ with $m \geq n$. Using theorem 11.1.3 and the first n columns of the $m \times m$ unitary factor it produces, show that there are a matrix $Q \in M_{m\times n}(k)$ with $Q^*Q = I_n$ and a positive semidefinite $P \in M_n(k)$ with
$$A = QP, \qquad P^2 = A^*A$$
Then show that $QQ^* = I_m$ holds if and only if $m = n$, by comparing the column space (definition 2.3.1) of QQ^* with that of I_m.

6. Let $A \in M_n(k)$ and let $A = UP$ be a polar decomposition. Show that A and P have the same null space (definition 2.3.3), and that the column space (definition 2.3.1) of A consists exactly of the vectors Uy with y in the column space of P. Deduce that $\operatorname{rank}A = \operatorname{rank}P$ and, using proposition 11.3.6(1), that $\operatorname{rank}A$ is the number of nonzero singular values of A.

7. Let $A \in M_n(k)$ and let $A = UP$ be a polar decomposition, with $P' = UPU^*$ as in corollary 11.3.8. Show that $A^* = U^*P'$ is a polar decomposition of A^*. Then show that A is normal (definition 7.1.7) if and only if $P = P'$, and give a 2×2 real example in which $P \neq P'$.

11.4 Operator and Frobenius Norms; Eckart-Young Low-Rank Approximation

Two functions on $M_{m\times n}(k)$ measure the size of a matrix, and the singular values compute both. The first is the largest factor by which A lengthens a vector; the second is the square root of the sum of the squares of the entries. This section defines them, evaluates each on the list $\sigma_1 \geq \cdots \geq \sigma_n$, and then settles the approximation question those two functions make precise: among the matrices of rank at most s, which one lies closest to A? The answer is the same for both norms, and it is obtained by keeping the s largest singular values in a decomposition of A and discarding the rest.

Throughout, $k = \mathbb{R}$ or $k = \mathbb{C}$, the spaces k^n and k^m carry the standard inner product $\langle x, y\rangle = \sum_t x_t\overline{y_t}$ with its induced norm (definition 5.2.1), and $A \in M_{m\times n}(k)$ has singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n \geq 0$ (definition 11.1.2). The symbol p denotes the smaller of the two dimensions, $p = \min(m, n)$, and r denotes $\operatorname{rank} A$; the letter s is reserved for the rank of an approximating matrix.

Definition 11.4.1: The Operator Norm

Let $A \in M_{m\times n}(k)$ have columns $a_1, \ldots, a_n \in k^m$. The set of real numbers $\|Ax\|$, as x ranges over the vectors of k^n with $\|x\| \leq 1$, is nonempty and bounded above. Indeed definition 1.4.1 gives $Ax = x_1a_1 + \cdots + x_na_n$ for $x = (x_1, \ldots, x_n)^\top$, so parts (2) and (3) of definition 5.2.1 give

$$\|Ax\| \leq |x_1|\,\|a_1\| + \cdots + |x_n|\,\|a_n\|$$

while $\langle x, e_j\rangle = x_j$ and $\|e_j\| = 1$ turn proposition 5.2.2(4) into $|x_j| \leq \|x\| \leq 1$; hence $\|Ax\| \leq \|a_1\| + \cdots + \|a_n\|$ for every such x. The *operator norm* of A is

$$\|A\|_{\mathrm{op}} = \sup\{\, \|Ax\| \big| x \in k^n,\ \|x\| \leq 1 \,\}$$

which exists because a nonempty set of real numbers bounded above has a least upper bound.

For $A = \operatorname{diag}(3, -2) \in M_2(\mathbb{R})$ the vector Ax is $(3x_1, -2x_2)^\top$, so $\|Ax\|^2 = 9x_1^2 + 4x_2^2 \leq 9(x_1^2 + x_2^2) = 9\|x\|^2$, with equality at $x = e_1$. Every member of the set is therefore at most 3, and 3 is a member, so $\|A\|_{\mathrm{op}} = 3$: the larger of the two absolute values on the diagonal. A diagonal matrix stretches each coordinate axis by its own factor and the operator norm records the largest of those factors. The general case reduces to exactly this once a singular value decomposition is available, since the decomposition presents A as a stretching of one orthonormal basis onto another.

Two consequences of the definition are worth isolating before the singular values arrive, because they need nothing but the supremum and are what makes the operator norm useful for estimating products and powers.

Proposition 11.4.2: The Operator Norm Bounds the Stretching

Let $A \in M_{m\times n}(k)$. Then

$$\|Ax\| \leq \|A\|_{\mathrm{op}}\,\|x\| \qquad \text{for every } x \in k^n$$

and $\|A\|_{\mathrm{op}}$ is the least constant with that property.

***Proof*:**
For $x = 0$ both sides are 0, since $A0 = 0$. For $x \neq 0$ put $u = x/\|x\|$, which satisfies $\|u\| = 1$ by definition 5.2.1(2), so u is one of the vectors in the set defining the supremum and $\|Au\| \leq \|A\|_{\text{op}}$. By definition 5.2.1(2) again, $\|Ax\| = \| \, \|x\| \, Au \, \| = \|x\| \, \|Au\| \leq \|A\|_{\text{op}} \|x\|$.

Now let C be any constant with $\|Ax\| \leq C\|x\|$ for all x. Every x with $\|x\| \leq 1$ then has $\|Ax\| \leq C\|x\| \leq C$, so C is an upper bound for the set whose supremum is $\|A\|_{\text{op}}$, giving $\|A\|_{\text{op}} \leq C$.

Proposition 11.4.3: The Operator Norm Is Submultiplicative

Let $A \in M_{m\times n}(k)$ and $B \in M_{n\times p}(k)$. Then

$$\|AB\|_{\text{op}} \leq \|A\|_{\text{op}} \, \|B\|_{\text{op}}$$

In particular $\|A^j\|_{\text{op}} \leq \|A\|_{\text{op}}^j$ for every integer $j \geq 1$ when A is square.

***Proof*:**
For any $x \in k^p$, two applications of proposition 11.4.2 give

$$\|(AB)x\| = \|A(Bx)\| \leq \|A\|_{\text{op}} \, \|Bx\| \leq \|A\|_{\text{op}} \, \|B\|_{\text{op}} \, \|x\|$$

using proposition 1.4.3(1) for the first equality. So $\|A\|_{\text{op}}\|B\|_{\text{op}}$ is one of the constants C of proposition 11.4.2, and that proposition names $\|AB\|_{\text{op}}$ the least of them, giving the inequality. The statement about powers follows by induction on j: it is an equality for $j = 1$, and if $\|A^{j-1}\|_{\text{op}} \leq \|A\|_{\text{op}}^{j-1}$ then $\|A^j\|_{\text{op}} = \|A^{j-1}A\|_{\text{op}} \leq \|A^{j-1}\|_{\text{op}}\|A\|_{\text{op}} \leq \|A\|_{\text{op}}^j$.

One identity from chapter 5 is used throughout this section. Corollary 5.3.6 states that for an orthonormal basis $q_1, \ldots, q_n$ of k^n and any x,

$$\|x\|^2 = \sum_{i=1}^{n} |\langle x, q_i \rangle|^2 \tag{11.12}$$

so the coordinates of a vector in an orthonormal basis carry its length.

Theorem 11.4.4: The Operator Norm Is the Largest Singular Value

Let $A \in M_{m\times n}(k)$ have rank r and singular values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$, and let $A = U\Sigma V^*$ be a singular value decomposition (theorem 11.1.3), with $u_1, \ldots, u_m$ the columns of U and $v_1, \ldots, v_n$ the columns of V.

1. $r \leq p$, and $\sigma_i = 0$ for every $i > r$, hence for every $i > p$. Moreover
$$Av_i = \sigma_i u_i \quad (1 \leq i \leq p), \qquad Av_i = 0 \quad (p < i \leq n)$$
2. For every $x \in k^n$,
$$Ax = \sum_{i=1}^{p} \sigma_i \langle x, v_i\rangle\, u_i \tag{11.13}$$
3. For every $x \in k^n$,
$$\|Ax\|^2 = \sum_{i=1}^{n} \sigma_i^2\, |\langle x, v_i\rangle|^2 \tag{11.14}$$
4. $\|Ax\| \leq \sigma_1\|x\|$ for every $x \in k^n$, and $\|A\|_{\text{op}} = \sigma_1$, the supremum of definition 11.4.1 being attained at $x = v_1$.

***Proof*:**
By proposition 7.1.5 the list $u_1, \ldots, u_m$ is an orthonormal basis of k^m, the list $v_1, \ldots, v_n$ is an orthonormal basis of k^n, and $V^*V = I_n$.

Step 1: proof of (1). By corollary 2.3.12 the column space $\operatorname{col} A$ is a subspace of k^m of dimension r and null$A$ is a subspace of k^n of dimension $n - r$. The first gives $r \leq m$ by corollary 2.2.12, and the second gives $r \leq n$ because $n - r \geq 0$; hence $r \leq p$. Then corollary 11.2.2 makes $\sigma_i = 0$ for every $i > r$, and every $i > p$ is such an i.

For the displayed identity, multiply $A = U\Sigma V^*$ on the right by V to get $AV = U\Sigma V^*V = U\Sigma$. The i-th column of AV is Av_i, and the i-th column of $U\Sigma$ is U applied to the i-th column of Σ (definition 1.4.1). That column of Σ is $\sigma_i e_i \in k^m$ for $i \leq p$ and is zero for $i > p$, because Σ has (i, i) entry σ_i for $i \leq p$ and no other nonzero entry. Applying U gives $\sigma_i u_i$ and 0 respectively.

Step 2: proof of (2) and (3). Fix $x \in k^n$. Applying theorem 5.3.5(1) to the orthonormal basis $v_1, \ldots, v_n$ and then linearity of $x \mapsto Ax$,
$$Ax = A\left(\sum_{i=1}^{n}\langle x, v_i\rangle v_i\right) = \sum_{i=1}^{n}\langle x, v_i\rangle\, Av_i = \sum_{i=1}^{p}\sigma_i\langle x, v_i\rangle\, u_i$$
the last equality by part (1), which kills the terms with $i > p$. This is (11.13). Its summands are pairwise orthogonal because $u_1, \ldots, u_p$ is orthonormal, so theorem 5.3.2 and proposition 5.2.2(2) give $\|Ax\|^2 = \sum_{i=1}^{p}\sigma_i^2|\langle x, v_i\rangle|^2$. The terms with $i > p$ carry the factor $\sigma_i^2 = 0$ by part (1), so appending them changes nothing and (11.14) holds.

Step 3: proof of (4). Every σ_i satisfies $\sigma_i \leq \sigma_1$, and the numbers $|\langle x, v_i\rangle|^2$ are nonnegative, so

(11.14) and then (11.12) applied to the orthonormal basis $v_1, \ldots, v_n$ give

$$\|Ax\|^2 \leq \sigma_1^2 \sum_{i=1}^{n} |\langle x, v_i \rangle|^2 = \sigma_1^2 \|x\|^2$$

and taking nonnegative square roots gives $\|Ax\| \leq \sigma_1 \|x\|$. If $\|x\| \leq 1$ this reads $\|Ax\| \leq \sigma_1$, so σ_1 is an upper bound for the set whose supremum defines $\|A\|_{\text{op}}$.

At $x = v_1$ orthonormality gives $\langle v_1, v_i \rangle = 0$ for $i \neq 1$ and $\langle v_1, v_1 \rangle = 1$, so (11.14) collapses to $\|Av_1\|^2 = \sigma_1^2$, whence $\|Av_1\| = \sigma_1$; and $\|v_1\| = 1 \leq 1$, so σ_1 is itself a member of that set. A member that is also an upper bound is the least upper bound, so $\|A\|_{\text{op}} = \sigma_1$ and the supremum is attained at v_1, completing the proof.

Three consequences follow. First, the supremum is a maximum: a unit vector achieving the largest stretch is exhibited rather than shown to exist, and the attainment came from the decomposition rather than from a compactness argument about the unit sphere of k^n. Second, part (4) makes $\|A\|_{\text{op}}$ the least constant C with $\|Ax\| \leq C\|x\|$ for all x, since any such C bounds the set of definition 11.4.1 above and hence bounds its supremum. Third, $\|\cdot\|_{\text{op}}$ satisfies the three conditions of definition 5.2.1 on the vector space $M_{m\times n}(k)$: it is nonnegative by construction and vanishes only for $A = 0$, since $\|A\|_{\text{op}} = 0$ forces $Ae_j = 0$ for every j and so kills every column; $\|\lambda A\|_{\text{op}} = |\lambda|\, \|A\|_{\text{op}}$ because $\|\lambda Ax\| = |\lambda|\, \|Ax\|$ for each admissible x; and $\|(A+B)x\| \leq \|Ax\| + \|Bx\| \leq \|A\|_{\text{op}} + \|B\|_{\text{op}}$ for $\|x\| \leq 1$ makes the right side an upper bound for the set defining $\|A+B\|_{\text{op}}$. For a square invertible matrix the ratio σ_1/σ_n of the largest singular value to the smallest is the condition number, developed in **??**.

The operator norm depends on σ_1 alone. A second measurement uses every singular value, and it is read directly off the entries of A.

Definition 11.4.5: The Frobenius Norm

Let $A = (a_{ij}) \in M_{m\times n}(k)$. The *Frobenius norm* of A is

$$\|A\|_F = \left(\sum_{i=1}^{m} \sum_{j=1}^{n} |a_{ij}|^2 \right)^{1/2}$$

the nonnegative square root of the sum of the squared absolute values of all mn entries.

Listing the entries of A as a single vector of k^{mn} makes $\|A\|_F$ the norm of that vector for the standard inner product on k^{mn}, so the three conditions of definition 5.2.1 hold without further checking. For $A = \operatorname{diag}(3, -2)$ the value is $\|A\|_F = \sqrt{9+0+0+4} = \sqrt{13}$, larger than $\|A\|_{\text{op}} = 3$ because the second diagonal entry now contributes. Writing $a_1, \ldots, a_n$ for the columns of A, the inner sum over i is $\|a_j\|^2$, so

$$\|A\|_F^2 = \|a_1\|^2 + \cdots + \|a_n\|^2 \tag{11.15}$$

which is the form in which the norm is computed below.

Proposition 11.4.6: The Frobenius Norm from the Trace and from the Singular Values

Let $A \in M_{m\times n}(k)$ have singular values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$. Then

$$\|A\|_F^2 = \operatorname{tr}(A^*A) = \sigma_1^2 + \sigma_2^2 + \cdots + \sigma_n^2$$

***Proof*:**
Write $A = (a_{ij})$. By definition 5.5.12 the entry of A^* in row j and column i is $\overline{a_{ij}}$, so definition 1.4.1 gives the (j,j) entry of $A^*A \in M_n(k)$ as $\sum_{i=1}^m \overline{a_{ij}}\, a_{ij} = \sum_{i=1}^m |a_{ij}|^2$. Summing over j is definition 2.5.7, and the result is the double sum of definition 11.4.5, so $\operatorname{tr}(A^*A) = \|A\|_F^2$.

Fix a singular value decomposition $A = U\Sigma V^*$ (theorem 11.1.3). By proposition 5.5.13 the conjugate transpose of the product is $A^* = V\Sigma^*U^*$, and $U^*U = I_m$ by proposition 7.1.5, so

$$A^*A = V\Sigma^*U^*U\Sigma V^* = V\left(\Sigma^*\Sigma\right)V^*$$

Applying proposition 2.5.8(1) to the two square factors V and $(\Sigma^*\Sigma)V^*$, and then $V^*V = I_n$, gives

$$\operatorname{tr}(A^*A) = \operatorname{tr}\big(V\cdot(\Sigma^*\Sigma)V^*\big) = \operatorname{tr}\big((\Sigma^*\Sigma)V^*\cdot V\big) = \operatorname{tr}(\Sigma^*\Sigma)$$

The matrix Σ has (i,i) entry σ_i for $i \leq p$ and no other nonzero entry, so the computation of the first paragraph applied to Σ identifies the (j,j) entry of $\Sigma^*\Sigma$ as σ_j^2 for $j \leq p$ and as 0 for $j > p$. Since $\sigma_j = 0$ for $j > p$ by theorem 11.4.4(1), the trace is $\sum_{j=1}^n \sigma_j^2$, as we sought to show.

The two norms therefore read one list in two ways: the operator norm takes its largest member and the Frobenius norm takes the length of the whole list. Both are consequently unchanged by the two unitary factors of a decomposition.

Proposition 11.4.7: Unitary Invariance of Both Norms

Let $A \in M_{m\times n}(k)$, let $W \in M_m(k)$ and $Z \in M_n(k)$ be unitary, and let $w_1, \ldots, w_n$ be any orthonormal basis of k^n.

1. $\|WAZ\|_{\mathrm{op}} = \|A\|_{\mathrm{op}}$.
2. $\|WAZ\|_F = \|A\|_F$.
3. $\|A\|_F^2 = \sum_{j=1}^n \|Aw_j\|^2$.

***Proof*:**

1. A unitary matrix preserves lengths: clause (3) of theorem 7.1.4 gives $\|Ry\| = \|y\|$ for every unitary R and every y, read through theorem 5.5.14(2) for the passage between the matrix and the operator it defines. Consequently $\|WAZx\| = \|A(Zx)\|$ for every $x \in k^n$, and $\|Zx\| = \|x\|$. So $x \mapsto Zx$ carries $\{x \,|\, \|x\| \leq 1\}$ into itself, and $y \mapsto Z^*y$ is an inverse for it doing the same, Z^* being unitary by proposition 7.1.5. The map is therefore a bijection of

that set onto itself, and

$$\{\, \|WAZx\| \big| \|x\| \leq 1 \,\} = \{\, \|Ay\| \big| \|y\| \leq 1 \,\}$$

Two equal sets of real numbers have equal suprema, which is the claim.

2. By proposition 5.5.13 and $W^*W = I_m$ (proposition 7.1.5),

$$(WAZ)^*(WAZ) = Z^*A^*W^*WAZ = Z^*\left(A^*A\right)Z$$

and proposition 2.5.8(1) applied to the square factors Z^* and $(A^*A)Z$, followed by $ZZ^* = I_n$, gives $\operatorname{tr}\big(Z^*(A^*A)Z\big) = \operatorname{tr}\big((A^*A)QQ^*\big) = \operatorname{tr}(A^*A)$. Now proposition 11.4.6 turns both traces into Frobenius norms, so $\|WAZ\|_F^2 = \|A\|_F^2$, and both norms are nonnegative.

3. Let $W \in M_n(k)$ be the matrix with columns $w_1, \ldots, w_n$, which is unitary by proposition 7.1.5. The j-th column of AW is Aw_j (definition 1.4.1), so (11.15) applied to AW gives $\|AW\|_F^2 = \sum_{j=1}^n \|Aw_j\|^2$. Part (2) with $W = I_m$ and $Z = W$ gives $\|AW\|_F = \|A\|_F$, completing the proof.

Part (3) says the Frobenius norm may be computed by testing A on any orthonormal basis, not only on the standard one, and that is what the second half of the approximation theorem needs: the basis can be chosen to suit the competing matrix rather than A. Part (2) also accounts for the shape of proposition 11.4.6. Neither norm can see the factors U and V^*, so both are functions of Σ alone, and the only data Σ carries is the list of singular values.

The approximation question can now be posed. A matrix of rank at most s is determined by far fewer numbers than a general $m \times n$ matrix, its column space having dimension at most s. Asking for the rank-s matrix closest to A therefore asks how well A is approximated by data of that reduced size, and the decomposition gives a candidate at once: discard the smallest singular values and keep the rest.

Definition 11.4.8: The Rank-s Truncation

Let $A \in M_{m\times n}(k)$, let $A = U\Sigma V^*$ be a singular value decomposition (theorem 11.1.3) with columns $u_1, \ldots, u_m$ of U and $v_1, \ldots, v_n$ of V, and let s be an integer with $1 \leq s \leq p$. The *rank-s truncation* of A relative to this decomposition is

$$A_s = \sum_{i=1}^{s} \sigma_i\, u_i v_i^* \in M_{m\times n}(k)$$

each summand being the product of a column $u_i \in M_{m\times 1}(k)$ with a row $v_i^* \in M_{1\times n}(k)$. The range of s stops at p because the decomposition gives exactly p such pairs: for $i > p$ one of u_i, v_i is absent, U having m columns and V having n, so no summand $\sigma_i u_i v_i^*$ with $i > p$ can be written down at all.

The truncation depends on the chosen decomposition, and a matrix generally has more than one; section 11.1 exhibits two for the identity matrix. The singular values, by contrast, are the same in every such factorization (proposition 11.1.6), and the error committed by the truncation depends on them alone. For $x \in k^n$ the scalar $v_i^* x$ is $\langle x, v_i \rangle$, both being $\sum_t x_t \overline{(v_i)_t}$, so

$$A_s x = \sum_{i=1}^{s} \sigma_i \langle x, v_i \rangle \, u_i \tag{11.16}$$

Comparison with (11.13) gives $A_p x = Ax$ for every x, hence $A_p = A$: the truncations run from A_1 up to A itself. Stopping the range at p loses nothing, since the singular values past the p-th are zero (theorem 11.4.4(1)) and A_p already reproduces A. The symbol A_s accordingly carries the range $1 \le s \le p$ wherever it appears.

Example 11.4.9: The Two Norms and the Rank-One Truncation of a 2×2 Matrix
Let

$$A = \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix} \in M_2(\mathbb{R})$$

Since A is lower triangular, its characteristic polynomial is $(3-\lambda)(5-\lambda)$ by proposition 3.2.6, so its eigenvalues are 3 and 5. The singular values come from $A^\top A$ instead.

The decomposition. Multiplying out,

$$A^\top A = \begin{pmatrix} 3 & 4 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 4 & 5 \end{pmatrix} = \begin{pmatrix} 25 & 20 \\ 20 & 25 \end{pmatrix}$$

whose characteristic polynomial is $(25-\lambda)^2 - 400$, with roots $\lambda = 25 \pm 20$, that is 45 and 5. Hence $\sigma_1 = \sqrt{45} = 3\sqrt{5}$ and $\sigma_2 = \sqrt{5}$. For $\lambda = 45$ the first row of the eigenvector equation reads $(25-45)x_1 + 20x_2 = 0$, that is $x_1 = x_2$; for $\lambda = 5$ it reads $20x_1 + 20x_2 = 0$, that is $x_2 = -x_1$. So

$$v_1 = \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad v_2 = \tfrac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

are orthonormal eigenvectors of $A^\top A$. Then $Av_1 = \frac{1}{\sqrt{2}}(3,9)^\top$ and $Av_2 = \frac{1}{\sqrt{2}}(3,-1)^\top$, so

$$u_1 = \frac{Av_1}{\sigma_1} = \tfrac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \qquad u_2 = \frac{Av_2}{\sigma_2} = \tfrac{1}{\sqrt{10}} \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

Both are unit vectors and $\langle u_1, u_2 \rangle = (3-3)/10 = 0$. With U and V the matrices carrying these columns and $\Sigma = \operatorname{diag}(3\sqrt{5}, \sqrt{5})$, multiplying back confirms the decomposition:

$$U\Sigma V^\top = \frac{1}{\sqrt{10}\sqrt{2}} \begin{pmatrix} 3\sqrt{5} & 3\sqrt{5} \\ 9\sqrt{5} & -\sqrt{5} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2\sqrt{5}} \begin{pmatrix} 6\sqrt{5} & 0 \\ 8\sqrt{5} & 10\sqrt{5} \end{pmatrix} = A$$

The two norms. By theorem 11.4.4(4), $\|A\|_{\mathrm{op}} = \sigma_1 = 3\sqrt{5}$. Directly from definition 11.4.5, $\|A\|_F = \sqrt{9+0+16+25} = \sqrt{50}$, and proposition 11.4.6 agrees: $\sigma_1^2 + \sigma_2^2 = 45 + 5 = 50$. Neither eigenvalue of A appears among these numbers, although $\sigma_1 \sigma_2 = 15$ is the absolute value of $\det A = 3 \cdot 5 - 0 \cdot 4 = 15$.

The rank-one truncation. By definition 11.4.8,

$$A_1 = \sigma_1 u_1 v_1^\top = \frac{3\sqrt{5}}{\sqrt{10}\sqrt{2}} \begin{pmatrix} 1 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} = \frac{3}{2} \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix}$$

whose two columns are equal and nonzero, so rank$A_1 = 1$. The error is

$$A - A_1 = \begin{pmatrix} 3 - \frac{3}{2} & 0 - \frac{3}{2} \\ 4 - \frac{9}{2} & 5 - \frac{9}{2} \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 3 & -3 \\ -1 & 1 \end{pmatrix}$$

with $\|A - A_1\|_F = \frac{1}{2}\sqrt{9+9+1+1} = \frac{1}{2}\sqrt{20} = \sqrt{5} = \sigma_2$. The same matrix is

$$\sigma_2 u_2 v_2^\top = \frac{\sqrt{5}}{\sqrt{10}\sqrt{2}}\begin{pmatrix} 3 \\ -1 \end{pmatrix}\begin{pmatrix} 1 & -1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 3 & -3 \\ -1 & 1 \end{pmatrix}$$

so $(A - A_1)x = \sigma_2(v_2^\top x)\, u_2 = \sigma_2\langle x, v_2\rangle u_2$, giving $\|(A - A_1)x\| = \sigma_2|\langle x, v_2\rangle|$ by proposition 5.2.2(2) and $\|u_2\| = 1$. This is at most $\sigma_2\|x\|\,\|v_2\| = \sigma_2\|x\|$ by proposition 5.2.2(4), with equality at $x = v_2$, so $\|A - A_1\|_{\mathrm{op}} = \sqrt{5} = \sigma_2$ as well. Both errors equal the discarded singular value.

No competitor was tested there, and the theorem now says none needs to be: no matrix of rank one lies closer to that A in either norm. The proof of the operator-norm half rests on a dimension count. A matrix B of rank at most s annihilates a subspace of dimension at least $n - s$, while the first $s + 1$ right singular vectors span a subspace of dimension $s + 1$ on which A stretches by at least σ_{s+1}; those two dimensions add to more than n, so the subspaces share a nonzero vector, and that vector gives the lower bound. The intersection argument is the one used for theorem 8.3.1(2), and what licenses it is corollary 2.2.12: a linearly independent list in k^n is a basis of its span, so it has at most n members.

Theorem 11.4.10: Eckart-Young: Best Low-Rank Approximation in the Operator Norm

Let $A \in M_{m\times n}(k)$ have rank r and singular values $\sigma_1 \geq \cdots \geq \sigma_n \geq 0$, fix a singular value decomposition $A = U\Sigma V^*$, and let s be an integer with $1 \leq s < r$. Let A_s be the rank-s truncation of definition 11.4.8. Then rank$A_s = s$, and

$$\|A - B\|_{\mathrm{op}} \;\geq\; \sigma_{s+1} \;=\; \|A - A_s\|_{\mathrm{op}}$$

for every $B \in M_{m\times n}(k)$ with rank$B \leq s$. Thus A_s minimizes $\|A - B\|_{\mathrm{op}}$ over the matrices of rank at most s, and the minimum value is σ_{s+1}. For $s \geq r$ the minimum is 0, attained at $B = A$.

***Proof*:**
Keep the notation of theorem 11.4.4: the columns $u_1, \ldots, u_m$ of U form an orthonormal basis of k^m and the columns $v_1, \ldots, v_n$ of V an orthonormal basis of k^n (proposition 7.1.5). Part (1) of that theorem gives $r \leq p$, so $s+1 \leq r \leq p$ and both A_s and σ_{s+1} are defined; and corollary 11.2.2 gives $\sigma_i > 0$ for every $i \leq r$, in particular for every $i \leq s + 1$.

Step 1: rank$A_s = s$. By (11.16) each column $A_s e_j$ of A_s lies in span$\{u_1, \ldots, u_s\}$, so col $A_s \subseteq$ span$\{u_1, \ldots, u_s\}$ (definition 2.3.1). Conversely, expanding v_j in the standard basis and using definition 1.4.1, the vector $A_s v_j$ is a linear combination of the columns of A_s and so lies in col A_s; and (11.16) with orthonormality of the v_i gives $A_s v_j = \sigma_j u_j$ for $j \leq s$, where $\sigma_j > 0$. Hence $u_j \in$ col A_s for $j \leq s$ and col $A_s =$ span$\{u_1, \ldots, u_s\}$. The list $u_1, \ldots, u_s$ is orthonormal, hence linearly independent (proposition 5.3.4), hence a basis of its span (definition 2.2.1), so that span has dimension s (definition 2.2.10) and rank$A_s = s$ by corollary 2.3.12(1).

Step 2: $\|A - A_s\|_{\mathrm{op}} = \sigma_{s+1}$. Subtracting (11.16) from (11.13) gives

$$(A - A_s)x = \sum_{i=s+1}^{p} \sigma_i \langle x, v_i\rangle\, u_i \tag{11.17}$$

for every $x \in k^n$. The summands are pairwise orthogonal, so theorem 5.3.2 and proposition 5.2.2(2) give $\|(A - A_s)x\|^2 = \sum_{i=s+1}^{p} \sigma_i^2 |\langle x, v_i\rangle|^2$. Since $\sigma_i \le \sigma_{s+1}$ for $i \ge s+1$, this is at most $\sigma_{s+1}^2 \sum_{i=1}^{n} |\langle x, v_i\rangle|^2 = \sigma_{s+1}^2 \|x\|^2$ by (11.12), so $\|x\| \le 1$ forces $\|(A - A_s)x\| \le \sigma_{s+1}$. At $x = v_{s+1}$ orthonormality collapses the sum to σ_{s+1}^2, and $\|v_{s+1}\| = 1$. So σ_{s+1} is both an upper bound for and a member of the set of definition 11.4.1 for $A - A_s$, hence is its supremum.

Step 3: no matrix of rank at most s does better. Let $B \in M_{m\times n}(k)$ with $\operatorname{rank} B \le s$. By corollary 2.3.12(3) the subspace $\operatorname{null} B \subseteq k^n$ has dimension $n - \operatorname{rank} B \ge n - s$. Put $S = \operatorname{span}\{v_1, \dots, v_{s+1}\}$; the list $v_1, \dots, v_{s+1}$ is orthonormal, hence linearly independent (proposition 5.3.4) and a basis of S (definition 2.2.1), so $\dim S = s+1$.

Choose a basis $y_1, \dots, y_q$ of $\operatorname{null} B$ (theorem 2.2.9(1)), where $q = \dim \operatorname{null} B \ge n - s$. The combined list $v_1, \dots, v_{s+1}, y_1, \dots, y_q$ has $s + 1 + q \ge n + 1$ members, all in k^n. It cannot be linearly independent: an independent list is a basis of its span (definition 2.2.1), so its length would be the dimension of a subspace of k^n, and that is at most n by corollary 2.2.12. So there are scalars $\alpha_1, \dots, \alpha_{s+1}, \beta_1, \dots, \beta_q$, not all zero, with $\sum_i \alpha_i v_i + \sum_j \beta_j y_j = 0$. Put

$$z = \sum_{i=1}^{s+1} \alpha_i v_i = -\sum_{j=1}^{q} \beta_j y_j$$

a vector lying in S and in $\operatorname{null} B$ at once. Were z zero, every α_i would vanish by independence of $v_1, \dots, v_{s+1}$ and every β_j by independence of $y_1, \dots, y_q$, against the choice of the scalars. So $z \ne 0$, and replacing z by $z/\|z\|$, which lies in the same two subspaces, arranges $\|z\| = 1$.

Since $z \in \operatorname{null} B$, the vector $(A - B)z$ equals Az. Since $z = \sum_{i \le s+1} \alpha_i v_i$, orthonormality gives $\langle z, v_j\rangle = 0$ for $j > s+1$, so (11.14) and (11.12) give

$$\|Az\|^2 = \sum_{i=1}^{s+1} \sigma_i^2 |\langle z, v_i\rangle|^2 \;\ge\; \sigma_{s+1}^2 \sum_{i=1}^{s+1} |\langle z, v_i\rangle|^2 = \sigma_{s+1}^2 \|z\|^2 = \sigma_{s+1}^2$$

using $\sigma_i \ge \sigma_{s+1}$ for $i \le s+1$. As $\|z\| = 1$, the number $\|(A - B)z\| = \|Az\| \ge \sigma_{s+1}$ belongs to the set defining $\|A - B\|_{\mathrm{op}}$, so $\|A - B\|_{\mathrm{op}} \ge \sigma_{s+1}$.

Steps 2 and 3 together make σ_{s+1} a lower bound for $\|A - B\|_{\mathrm{op}}$ over the matrices of rank at most s, attained at A_s, which has rank s by Step 1. Finally, if $s \ge r$ then $B = A$ itself has rank at most s and gives $\|A - B\|_{\mathrm{op}} = 0$, the least value a norm takes, completing the proof.

The theorem solves an optimization problem over a set that is neither a subspace nor cut out by equations, namely the matrices of rank at most s, and it solves it without a search: the minimizer is written down from a decomposition of A, and the minimum value is the first discarded singular value. In example 11.4.9 the number $\|A - A_1\|_{\mathrm{op}} = \sqrt{5}$ is therefore not one competitor's score but the exact distance from that A to the matrices of rank one. The minimizer need not be unique: only σ_{s+1} is determined by A, and a matrix B of rank at most s with $\|A - B\|_{\mathrm{op}} = \sigma_{s+1}$ can be produced without truncating anything (exercise 11.4.1 (4)).

The same truncation is optimal for the Frobenius norm. The proof changes, because the Frobenius norm registers every singular value and the error must therefore be accounted for in all n directions at once. Part (3) of proposition 11.4.7 is what makes that accounting possible.

Corollary 11.4.11: Eckart-Young in the Frobenius Norm

In the situation of theorem 11.4.10,

$$\|A - B\|_F^2 \;\geq\; \sum_{i=s+1}^{n} \sigma_i^2 \;=\; \|A - A_s\|_F^2$$

for every $B \in M_{m\times n}(k)$ with rank$B \leq s$. Thus the same matrix A_s minimizes $\|A-B\|_F$ over the matrices of rank at most s, and the minimum value is $(\sigma_{s+1}^2 + \cdots + \sigma_n^2)^{1/2}$.

***Proof*:**
Step 1: the value at A_s. Evaluate (11.17) at $x = v_j$. Orthonormality of $v_1, \ldots, v_n$ gives $(A - A_s)v_j = 0$ for $j \leq s$, then $(A - A_s)v_j = \sigma_j u_j$ for $s < j \leq p$, and $(A - A_s)v_j = 0$ for $j > p$. Applying proposition 11.4.7(3) to $A - A_s$ with the orthonormal basis $v_1, \ldots, v_n$, and using $\|u_j\| = 1$ together with proposition 5.2.2(2) and $\sigma_j = 0$ for $j > p$ (theorem 11.4.4(1)),

$$\|A - A_s\|_F^2 = \sum_{j=1}^{n} \|(A - A_s)v_j\|^2 = \sum_{j=s+1}^{p} \sigma_j^2 = \sum_{j=s+1}^{n} \sigma_j^2$$

Step 2: a basis adapted to B. Let rank$B \leq s$. By corollary 2.3.12(3) the subspace null$B$ has dimension $n - \text{rank}B \geq n - s$, and $n - s \geq 1$ because $s < r \leq n$. Take a basis of null$B$ (theorem 2.2.9(1)) and let W be the span of its first $n - s$ members; that sublist is linearly independent and is therefore a basis of W (definition 2.2.1), so $\dim W = n - s$ and $W \subseteq \text{null}B$. By corollary 5.3.11(1) the subspace W has an orthonormal basis $w_1, \ldots, w_{n-s}$, and by corollary 5.3.11(2) that list extends to an orthonormal basis $w_1, \ldots, w_n$ of k^n whose first $n - s$ members are unchanged. Since $Bw_j = 0$ for $j \leq n - s$, applying proposition 11.4.7(3) to $A - B$ gives

$$\|A - B\|_F^2 = \sum_{j=1}^{n} \|(A - B)w_j\|^2 \;\geq\; \sum_{j=1}^{n-s} \|(A - B)w_j\|^2 = \sum_{j=1}^{n-s} \|Aw_j\|^2$$

the inequality holding because the discarded terms are nonnegative.

Step 3: the weights. By (11.14) and an exchange of two finite sums,

$$\sum_{j=1}^{n-s} \|Aw_j\|^2 = \sum_{j=1}^{n-s}\sum_{i=1}^{n} \sigma_i^2 |\langle w_j, v_i\rangle|^2 = \sum_{i=1}^{n} \sigma_i^2\, t_i, \qquad t_i := \sum_{j=1}^{n-s} |\langle w_j, v_i\rangle|^2$$

Each t_i satisfies $0 \leq t_i \leq 1$: it is a sum of nonnegative terms, and $|\langle w_j, v_i\rangle| = |\langle v_i, w_j\rangle|$ by definition 5.1.2(2), so $t_i \leq \sum_{j=1}^{n} |\langle v_i, w_j\rangle|^2 = \|v_i\|^2 = 1$ by (11.12) applied to the orthonormal basis $w_1, \ldots, w_n$. Summing over i instead, and using (11.12) for the orthonormal basis $v_1, \ldots, v_n$,

$$\sum_{i=1}^{n} t_i = \sum_{j=1}^{n-s}\sum_{i=1}^{n} |\langle w_j, v_i\rangle|^2 = \sum_{j=1}^{n-s} \|w_j\|^2 = n - s$$

Step 4: the weighted sum is least when the weight is carried by the smallest singular values. Write $c_i = \sigma_i^2$, so that $c_1 \geq c_2 \geq \cdots \geq c_n \geq 0$, and put $\delta = \sum_{i=1}^{s} t_i$. Splitting the range of summation at s,

$$\sum_{i=1}^{n} c_i t_i - \sum_{i=s+1}^{n} c_i = \sum_{i=1}^{s} c_i t_i - \sum_{i=s+1}^{n} c_i(1-t_i)$$

The range $s+1, \ldots, n$ has $n-s$ members, so Step 3 gives $\sum_{i=s+1}^{n}(1-t_i) = (n-s) - \sum_{i=s+1}^{n} t_i = (n-s) - \big((n-s)-\delta\big) = \delta$. Each c_i with $i \leq s$ is at least c_s and each t_i is nonnegative, so $\sum_{i \leq s} c_i t_i \geq c_s \delta$; each c_i with $i > s$ is at most c_{s+1} and each $1-t_i$ is nonnegative, so $\sum_{i>s} c_i(1-t_i) \leq c_{s+1}\delta$. The difference above is therefore at least $(c_s - c_{s+1})\delta \geq 0$, which says $\sum_{i=1}^{n} \sigma_i^2 t_i \geq \sum_{i=s+1}^{n} \sigma_i^2$.

Steps 2, 3 and 4 give $\|A-B\|_F^2 \geq \sum_{i=s+1}^{n} \sigma_i^2$, which Step 1 identifies with $\|A-A_s\|_F^2$; and both norms are nonnegative, so the inequality holds without the squares, as we sought to show.

One matrix, A_s, is therefore optimal both for a norm that penalizes the largest error direction and for a norm that penalizes all of them at once. Two different norms on the same set usually have different minimizers, and exercise 11.4.1 (4) exhibits a matrix optimal for the operator norm and not for the Frobenius norm. What the two proofs above have in common is that each norm is computed from the singular values alone (theorem 11.4.4, proposition 11.4.6) and that truncation removes the smallest of them; the agreement of the minimizers is a conclusion of those two proofs rather than a general principle about norms. Step 1 also records a splitting of the total. Since $\|A\|_F^2 = \sum_i \sigma_i^2$ by proposition 11.4.6, the squared Frobenius norm of A is $\|A_s\|_F^2 + \|A-A_s\|_F^2$ (exercise 11.4.1 (8)), so the fraction $(\sigma_1^2 + \cdots + \sigma_s^2)/(\sigma_1^2 + \cdots + \sigma_n^2)$ is the proportion of $\|A\|_F^2$ that the rank-s truncation retains.

Exercise 11.4.1

1. Let $A = \left(\begin{smallmatrix} 0 & 2 \\ 3 & 0 \end{smallmatrix}\right) \in M_2(\mathbb{R})$. Compute $A^\top A$, the singular values of A, and both $\|A\|_{\mathrm{op}}$ and $\|A\|_F$. Then compute the eigenvalues of A and check that their absolute values are not the singular values, while the product of the singular values does equal $|\det A|$.

2. Let $A = \left(\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right) \in M_2(\mathbb{R})$. Compute a singular value decomposition of A, the rank-one truncation A_1 of definition 11.4.8, and the two numbers $\|A-A_1\|_{\mathrm{op}}$ and $\|A-A_1\|_F$. Explain why those two numbers agree for this A by computing the rank of $A-A_1$.

3. Let $u \in k^m$ and $v \in k^n$ be nonzero and put $C = uv^* \in M_{m\times n}(k)$. Show that $Cx = \langle x, v\rangle u$ for every $x \in k^n$, and deduce that $\|C\|_{\mathrm{op}} = \|C\|_F = \|u\|\,\|v\|$. Conclude that C has exactly one nonzero singular value, namely $\|u\|\,\|v\|$.

4. Let $A = \mathrm{diag}(5,3,3) \in M_3(\mathbb{R})$ and take $s=1$. Compute the minimum of $\|A-B\|_{\mathrm{op}}$ and the minimum of $\|A-B\|_F$ over the matrices $B \in M_3(\mathbb{R})$ of rank at most 1, naming a matrix that attains each. Then show that $B = \mathrm{diag}(4,0,0)$ attains the first minimum and not the second, so that the operator-norm minimizer is not unique.

5. Let $A \in M_{m\times n}(k)$. Show that $\|Ax\| \leq \|A\|_{\mathrm{op}}\|x\|$ for every $x \in k^n$, and that $\|A\|_{\mathrm{op}}$ is the smallest real number C for which $\|Ax\| \leq C\|x\|$ holds for every x. Deduce that $\|AB\|_{\mathrm{op}} \leq \|A\|_{\mathrm{op}}\|B\|_{\mathrm{op}}$ for every $B \in M_{n\times q}(k)$.

6. Let $A \in M_{m\times n}(k)$ be nonzero and put $r = \mathrm{rank} A$. Show that $\|A\|_{\mathrm{op}} \leq \|A\|_F \leq \sqrt{r}\,\|A\|_{\mathrm{op}}$, and that $\|A\|_F = \|A\|_{\mathrm{op}}$ holds if and only if $r=1$. Then, for each r with $1 \leq r \leq \min(m,n)$,

give a matrix of rank r attaining the right-hand equality.

7. Let $A = \left(\begin{smallmatrix} 3 & 0 \\ 4 & 5 \end{smallmatrix}\right)$ be the matrix of example 11.4.9 and let $Q = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 1 & -1 \\ 1 & 1 \end{smallmatrix}\right)$. Check that Q is orthogonal, compute the four entries of QA, and verify $\|QA\|_F = \|A\|_F$ from those entries. Note that the entries of QA are not a rearrangement of the entries of A, so proposition 11.4.7(2) is not an entrywise statement.

8. Let $A \in M_{m\times n}(k)$ have rank r, fix a singular value decomposition, and let $1 \le s < r$. Using proposition 11.4.7(3) with the orthonormal basis $v_1, \dots, v_n$, show that $\|A_s\|_F^2 = \sigma_1^2 + \cdots + \sigma_s^2$, and deduce $\|A\|_F^2 = \|A_s\|_F^2 + \|A - A_s\|_F^2$. Show also that $\|A_s\|_{\text{op}} = \|A\|_{\text{op}}$, so that truncation leaves the largest singular value unchanged.

11.5 Application: Principal Component Analysis

A table of measurements is a matrix that no linear system produced: m observations, each recording the same n real quantities, stacked one row per observation. The question asked of such a table is whether the n recorded quantities carry n separate pieces of information, or whether the observations in fact sit close to a subspace of $\mathbb{R}^n$ of dimension smaller than n, and if so which subspace and how close.

The singular value decomposition answers this, applied to the table after the average observation has been subtracted from every row. Its right singular vectors are the directions in $\mathbb{R}^n$ along which the observations spread out, its squared singular values measure how far they spread along each, and the statement that the resulting summary is the best one of its size is corollary 11.4.11 read in the Frobenius norm. Data are real numbers and a variance is a sum of squares, so this section works over $\mathbb{R}$ throughout. There $A^* = A^\top$, the inner product of definition 5.1.2 is $\langle x, y\rangle = x^\top y$, and the norm of definition 5.2.1 satisfies $\|y\|^2 = y^\top y$.

Definition 11.5.1: Data Matrix, Sample Mean, and Centering

Let $m \ge 2$ and $n \ge 1$. A *data matrix* is a matrix $D \in M_{m\times n}(\mathbb{R})$ read as m observations of n real quantities: the entry D_{ij} is the value of the j-th quantity on the i-th observation, so that the i-th row of D is $d_i^\top$, where $d_i \in \mathbb{R}^n$ collects the n quantities recorded on observation i. The *sample mean* of the data is the vector

$$\bar{d} = \frac{1}{m}\sum_{i=1}^{m} d_i \in \mathbb{R}^n$$

whose j-th entry is the average of the j-th column of D. Write $\mathbf{1} \in \mathbb{R}^m$ for the column vector with every entry equal to 1. The *centered data matrix* of D is

$$X = D - \mathbf{1}\,\bar{d}^\top \in M_{m\times n}(\mathbb{R})$$

whose i-th row is $(d_i - \bar{d})^\top$. A matrix $Y \in M_{m\times n}(\mathbb{R})$ is called *centered* when $\mathbf{1}^\top Y = 0$, that is, when the entries of each of its columns sum to zero.

The centered data matrix is centered, which is the one identity about X used below. Evaluating the products by definition 1.4.1, the row vector $\mathbf{1}^\top D$ has j-th entry $\sum_{i=1}^m D_{ij} = m\,\bar{d}_j$, so $\mathbf{1}^\top D = m\,\bar{d}^\top$, while $\mathbf{1}^\top \mathbf{1} = m$ gives $\mathbf{1}^\top(\mathbf{1}\bar{d}^\top) = m\,\bar{d}^\top$ as well. Hence

$$\mathbf{1}^\top X = m\,\bar{d}^\top - m\,\bar{d}^\top = 0 \tag{11.18}$$

Subtracting $\bar{d}$ from every observation moves the whole table by a single vector and leaves all differences $d_i - d_{i'}$ unchanged, so nothing about how the observations sit relative to one another is lost.

Four observations of two quantities make the arithmetic visible. Take $d_1 = (5,3)^\top$, $d_2 = (4,4)^\top$, $d_3 = (2,0)^\top$ and $d_4 = (1,1)^\top$, so that $m = 4$ and $n = 2$. Then

$$D = \begin{pmatrix} 5 & 3 \\ 4 & 4 \\ 2 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \bar{d} = \frac{1}{4}\begin{pmatrix} 5+4+2+1 \\ 3+4+0+1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \qquad X = \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ -1 & -2 \\ -2 & -1 \end{pmatrix}$$

and each column of X sums to zero, as (11.18) requires.

What the table records about a single quantity, once centered, is how far its values lie from their average; what it records about a pair of quantities is whether their deviations move together. Both are sums of products of entries of X, and all of them at once are the entries of a single matrix.

Definition 11.5.2: Sample Covariance Matrix

Let $D \in M_{m\times n}(\mathbb{R})$ be a data matrix with $m \geq 2$ and let X be its centered data matrix (definition 11.5.1). The *sample covariance matrix* of the data is

$$S = \frac{1}{m-1} X^\top X \in M_n(\mathbb{R})$$

Its entry in row j and column k is

$$S_{jk} = \frac{1}{m-1}\sum_{i=1}^{m}(D_{ij} - \bar{d}_j)(D_{ik} - \bar{d}_k)$$

called the *sample covariance* of the j-th and k-th quantities; the diagonal entry S_{jj} is the *sample variance* of the j-th quantity. For a list of m real numbers $t_1, \dots, t_m$ with average $\bar{t} = \frac{1}{m}\sum_i t_i$, the case $n = 1$ of the same expression,

$$\frac{1}{m-1}\sum_{i=1}^{m}(t_i - \bar{t})^2$$

is the *sample variance* of the list.[a]

[a] The divisor $m-1$ rather than m is one of two conventions in use. The two matrices differ by the factor $m/(m-1)$, which is positive and depends on nothing but m, so every statement in this section holds under the other convention with $m-1$ replaced by m throughout. The divisor $m-1$ is fixed for the rest of the section.

The displayed formula for S_{jk} is definition 1.4.1 applied to $X^\top X$, whose (j,k) entry is $\sum_{i=1}^m X_{ij}X_{ik}$, together with $X_{ij} = D_{ij} - \bar{d}_j$. Two properties of S follow at once. It is symmetric, since $(X^\top X)^\top = X^\top X$ by proposition 2.3.5, and it is positive semidefinite: the matrix $X^\top X$ is positive semidefinite by proposition 11.1.1, so $x^\top(X^\top X)x \geq 0$ for every $x \in \mathbb{R}^n$, and multiplying by the positive number $1/(m-1)$ preserves the inequality, so $x^\top Sx \geq 0$ as well. Symmetry is the hypothesis corollary 7.4.3 needs.

For the four observations above,

$$X^\top X = \begin{pmatrix} 4+1+1+4 & 2+2+2+2 \\ 2+2+2+2 & 1+4+4+1 \end{pmatrix} = \begin{pmatrix} 10 & 8 \\ 8 & 10 \end{pmatrix}, \qquad S = \frac{1}{3}\begin{pmatrix} 10 & 8 \\ 8 & 10 \end{pmatrix}$$

The two quantities have equal sample variance $10/3$, and their sample covariance $8/3$ is not zero.

Since S is symmetric, corollary 7.4.3 gives an orthonormal basis of $\mathbb{R}^n$ made of eigenvectors of S. Producing that basis looks like a second computation, performed on S after S has been formed. It is not: the basis is already present in any singular value decomposition of X, and so are the eigenvalues.

Theorem 11.5.3: Principal Directions Are the Right Singular Vectors

Let $D \in M_{m\times n}(\mathbb{R})$ be a data matrix with $m \geq 2$, let X be its centered data matrix, let $S = \frac{1}{m-1}X^\top X$ be its sample covariance matrix, and let

$$X = U\Sigma V^\top$$

be a singular value decomposition of X (theorem 11.1.3), so that $U \in M_m(\mathbb{R})$ and $V \in M_n(\mathbb{R})$ are unitary, which over $\mathbb{R}$ means $U^\top U = I_m$ and $V^\top V = I_n$, with columns $u_1, \dots, u_m$ and $v_1, \dots, v_n$ respectively, and $\Sigma \in M_{m\times n}(\mathbb{R})$ has (i,i) entry σ_i for $1 \leq i \leq p = \min(m,n)$ and all other entries zero, with $\sigma_1 \geq \cdots \geq \sigma_p \geq 0$. Put $\sigma_j = 0$ for $p < j \leq n$. Then:

1. $Sv_j = \dfrac{\sigma_j^2}{m-1}v_j$ for $1 \leq j \leq n$. Thus $v_1, \dots, v_n$ is an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of S, and
$$S = V\operatorname{diag}\Big(\frac{\sigma_1^2}{m-1}, \dots, \frac{\sigma_n^2}{m-1}\Big)V^\top$$
2. $Xv_j = \sigma_j u_j$ for $1 \leq j \leq p$, and $Xv_j = 0$ for $p < j \leq n$.
3. $(Xv_j)^\top(Xv_k) = 0$ whenever $j \neq k$, and $\|Xv_j\|^2 = \sigma_j^2$ for every j.

The vectors $v_1, \dots, v_n$ are the *principal directions* of the data. The vector $Xv_j \in \mathbb{R}^m$, whose i-th entry is the coordinate $(d_i - \bar{d})^\top v_j$ of the i-th centered observation along v_j, is the j-th *principal component* of the data.

***Proof*:**
Since $V^\top V = I_n$, the columns $v_1, \dots, v_n$ form an orthonormal basis of $\mathbb{R}^n$ by proposition 7.1.5, and the i-th entry of $V^\top v_j$ is $v_i^\top v_j$, which is 1 for $i = j$ and 0 otherwise; that is, $V^\top v_j = e_j$, the j-th standard basis vector of $\mathbb{R}^n$. Taking transposes with proposition 2.3.5 and using $U^\top U = I_m$,

$$X^\top X = (U\Sigma V^\top)^\top(U\Sigma V^\top) = V\Sigma^\top U^\top U\Sigma V^\top = V\Sigma^\top\Sigma V^\top$$

The matrix $\Sigma^\top\Sigma$ lies in $M_n(\mathbb{R})$ and its (j,k) entry is $\sum_{i=1}^m \Sigma_{ij}\Sigma_{ik}$ by definition 1.4.1. The entry Σ_{ij} is σ_j when $i = j \leq p$ and is zero otherwise, so the sum is σ_j^2 when $j = k \leq p$ and is zero in every other case. With the convention $\sigma_j = 0$ for $j > p$ this says

$$\Sigma^\top\Sigma = \operatorname{diag}(\sigma_1^2, \dots, \sigma_n^2) \tag{11.19}$$

1. Dividing the display above by $m-1$ and substituting (11.19),
$$S = \frac{1}{m-1} V \operatorname{diag}(\sigma_1^2, \dots, \sigma_n^2) V^\top = V \operatorname{diag}\Big(\frac{\sigma_1^2}{m-1}, \dots, \frac{\sigma_n^2}{m-1}\Big) V^\top$$
the second equality because a diagonal matrix scaled by $1/(m-1)$ is the diagonal matrix of the scaled entries. Applying this to v_j and using $V^\top v_j = e_j$, then $\operatorname{diag}(\dots) e_j = \frac{\sigma_j^2}{m-1} e_j$, then $V e_j = v_j$, gives $S v_j = \frac{\sigma_j^2}{m-1} v_j$.

2. $X v_j = U \Sigma V^\top v_j = U \Sigma e_j$, and Σe_j is the j-th column of Σ. That column is σ_j times the j-th standard basis vector of $\mathbb{R}^m$ when $j \le p$, and is zero when $j > p$. In the first case $U \Sigma e_j = \sigma_j U e_j = \sigma_j u_j$, and in the second $U \Sigma e_j = 0$.

3. Since $X^\top X = (m-1)S$, part (1) reads $X^\top X v_k = \sigma_k^2 v_k$. Hence, transposing the first factor with proposition 2.3.5,
$$(X v_j)^\top (X v_k) = v_j^\top X^\top X v_k = \sigma_k^2 \, v_j^\top v_k$$
which is 0 for $j \neq k$ and is σ_j^2 for $j = k$, the latter being $\|X v_j\|^2$ since $\|y\|^2 = y^\top y$.

This establishes all three parts, as we sought to show.

Part (1) says that a singular value decomposition of X *is* an orthogonal diagonalization of S: the matrix V is one of the orthogonal matrices corollary 7.4.3 promises for the symmetric matrix S, and the eigenvalues it exhibits are the numbers $\sigma_j^2/(m-1)$, listed in decreasing order because the singular values are. Part (2) says that the j-th principal component is σ_j times the j-th column of U. Part (3) says that different principal components are orthogonal in $\mathbb{R}^m$, and this has a reading in the language of definition 11.5.2: the list of coordinates $X v_j$ has entries summing to $\mathbf{1}^\top X v_j = 0$ by (11.18), so the sample covariance of the lists $X v_j$ and $X v_k$ is $\frac{1}{m-1}(X v_j)^\top (X v_k)$, which vanishes for $j \neq k$. The principal directions are thus a set of n new quantities, each a fixed linear combination of the recorded ones, whose sample covariances all vanish.

Example 11.5.4: Principal Components of Four Points in the Plane

Keep the four observations $d_1 = (5,3)^\top$, $d_2 = (4,4)^\top$, $d_3 = (2,0)^\top$, $d_4 = (1,1)^\top$, with
$$X = \begin{pmatrix} 2 & 1 \\ 1 & 2 \\ -1 & -2 \\ -2 & -1 \end{pmatrix}, \qquad X^\top X = \begin{pmatrix} 10 & 8 \\ 8 & 10 \end{pmatrix}$$
computed above. Here $m = 4$, $n = 2$ and $p = 2$. By theorem 11.5.3(1) the matrix $X^\top X$ equals $V \operatorname{diag}(\sigma_1^2, \sigma_2^2) V^\top$ with $V^{-1} = V^\top$, so $X^\top X$ and $\operatorname{diag}(\sigma_1^2, \sigma_2^2)$ are similar (definition 2.5.5) and have the same characteristic polynomial by corollary 4.1.16. That polynomial is
$$\det \begin{pmatrix} 10-\lambda & 8 \\ 8 & 10-\lambda \end{pmatrix} = (10-\lambda)^2 - 64$$
whose roots are $\lambda = 18$ and $\lambda = 2$. Hence $\{\sigma_1^2, \sigma_2^2\} = \{18, 2\}$, and since $\sigma_1 \ge \sigma_2 \ge 0$,
$$\sigma_1 = 3\sqrt{2}, \qquad \sigma_2 = \sqrt{2}$$

Solving $(X^\top X - 18I_2)y = 0$ for $y = (y_1, y_2)^\top$ gives $-8y_1 + 8y_2 = 0$ from both rows, so the eigenvectors for 18 are the nonzero multiples of $(1,1)^\top$; solving $(X^\top X - 2I_2)y = 0$ gives $8y_1 + 8y_2 = 0$, so the eigenvectors for 2 are the nonzero multiples of $(1,-1)^\top$. Normalizing,

$$v_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}, \qquad v_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$$

which are orthonormal, since $v_1^\top v_2 = (1-1)/2 = 0$ and each has length 1. These are the principal directions. Multiplying X by them row by row gives the two principal components

$$Xv_1 = \frac{1}{\sqrt{2}}\begin{pmatrix}3\\3\\-3\\-3\end{pmatrix}, \qquad Xv_2 = \frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\\1\\-1\end{pmatrix}$$

with $\|Xv_1\| = \sqrt{36/2} = 3\sqrt{2} = \sigma_1$ and $\|Xv_2\| = \sqrt{4/2} = \sqrt{2} = \sigma_2$, and with $(Xv_1)^\top(Xv_2) = (3-3-3+3)/2 = 0$, exactly as theorem 11.5.3(3) states. Dividing each by its length, as part (2) of that theorem prescribes,

$$u_1 = \frac{Xv_1}{\sigma_1} = \frac{1}{2}\begin{pmatrix}1\\1\\-1\\-1\end{pmatrix}, \qquad u_2 = \frac{Xv_2}{\sigma_2} = \frac{1}{2}\begin{pmatrix}1\\-1\\1\\-1\end{pmatrix}$$

and $u_1^\top u_2 = (1-1-1+1)/4 = 0$ with $\|u_1\| = \|u_2\| = 1$. Multiplying out the two rank-one matrices,

$$\sigma_1 u_1 v_1^\top + \sigma_2 u_2 v_2^\top = \frac{3}{2}\begin{pmatrix}1 & 1\\1 & 1\\-1 & -1\\-1 & -1\end{pmatrix} + \frac{1}{2}\begin{pmatrix}1 & -1\\-1 & 1\\1 & -1\\-1 & 1\end{pmatrix} = \begin{pmatrix}2 & 1\\1 & 2\\-1 & -2\\-2 & -1\end{pmatrix}$$

which is X, so the two rank-one terms are a reduced decomposition in the sense of corollary 11.1.5.

The eigenvalues of S carry the same information as the singular values of X, and the next result says what that information is: the eigenvalue attached to a principal direction is the sample variance of the coordinates the observations have along it.

Proposition 11.5.5: Variance Along a Direction

Keep the data and the notation of theorem 11.5.3.

1. Let $w \in \mathbb{R}^n$ with $\|w\| = 1$. The m numbers $t_i = (d_i - \bar{d})^\top w$ have average 0, and their sample variance (definition 11.5.2) is

$$\frac{1}{m-1}\|Xw\|^2 = w^\top S w$$

called the *sample variance of the data along* w. For $w = v_j$ it equals $\sigma_j^2/(m-1)$.

2. The total sample variance is

$$\mathrm{tr} S = \frac{1}{m-1}\|X\|_F^2 = \frac{1}{m-1}\sum_{j=1}^{n} \sigma_j^2$$

3. If $w_1, \dots, w_n$ is any orthonormal basis of $\mathbb{R}^n$, then $\sum_{j=1}^n w_j^\top S w_j = \mathrm{tr} S$.

***Proof*:**

1. By definition 1.4.1 the i-th entry of Xw is the i-th row of X against w, which is $(d_i - \bar{d})^\top w = t_i$. Summing the entries, $\sum_{i=1}^m t_i = \mathbf{1}^\top X w = 0$ by (11.18), so the average of the t_i is 0 and their sample variance is

$$\frac{1}{m-1}\sum_{i=1}^{m}(t_i - 0)^2 = \frac{1}{m-1}\|Xw\|^2$$

by definition 5.2.1. Expanding the norm and inserting the definition of S,

$$\|Xw\|^2 = (Xw)^\top (Xw) = w^\top X^\top X w = (m-1)\, w^\top S w$$

which gives the stated equality. For $w = v_j$ the vector w is a unit vector, since the v_j are orthonormal, and $\|Xv_j\|^2 = \sigma_j^2$ by theorem 11.5.3(3).

2. By definition 2.5.7 the trace is the sum of the diagonal entries, so $\mathrm{tr} S = \sum_{j=1}^n S_{jj}$, and $S_{jj} = \frac{1}{m-1}\sum_{i=1}^m X_{ij}^2$ by definition 1.4.1. Summing over j,

$$\mathrm{tr} S = \frac{1}{m-1}\sum_{j=1}^{n}\sum_{i=1}^{m} X_{ij}^2 = \frac{1}{m-1}\|X\|_F^2$$

by definition 11.4.5, and $\|X\|_F^2 = \sum_{j=1}^n \sigma_j^2$ by proposition 11.4.6, the terms with $j > p$ contributing nothing because $\sigma_j = 0$ there, by the convention fixed in theorem 11.5.3.

3. Let $W \in M_n(\mathbb{R})$ have columns $w_1, \dots, w_n$. These are orthonormal, so $W^\top W = I_n$ and $WW^\top = I_n$ by proposition 7.1.5. By definition 1.4.1 the (j,j) entry of $W^\top S W$ is the j-th row of $W^\top$ against the j-th column of SW, that is $w_j^\top S w_j$. Hence, using definition 2.5.7

and then proposition 2.5.8 with the two factors $W^\top$ and SW,

$$\sum_{j=1}^{n} w_j^\top S w_j = \mathrm{tr}(W^\top SW) = \mathrm{tr}(SWW^\top) = \mathrm{tr}S$$

All three parts are established, completing the proof.

Part (2) calls $\mathrm{tr}S$ a total, and part (3) is what makes the word accurate: $\mathrm{tr}S$ is the sum of the sample variances of the n recorded quantities, and it is also the sum of the sample variances along the vectors of any other orthonormal basis, so it is attached to the data rather than to the coordinates chosen to record them. Part (1) then splits that total across the principal directions, the j-th of them taking the share

$$\frac{\sigma_j^2}{\sigma_1^2 + \cdots + \sigma_n^2}$$

For the four observations of example 11.5.4 the sample variances along v_1 and v_2 are $18/3 = 6$ and $2/3$, summing to $20/3$, which is $\mathrm{tr}S = (10 + 10)/3$; the first principal direction takes the share $18/20 = 9/10$ of the total.

Whether the observations lie close to a subspace of small dimension is not yet a precise question: it needs a measurement of "close" and a competition among subspaces to decide. Both are given by section 11.4. The distance from the table to a table of rank at most k is measured in the Frobenius norm, and corollary 11.4.11 names the matrix of rank at most k that minimizes that distance. The dimension k runs only to $p = \min(m, n)$, where the truncations of definition 11.4.8 stop, and that loses nothing: $Xv_j = 0$ for $j > p$ by theorem 11.5.3(2), so every centered observation has zero coordinate along each such v_j and already lies in $\mathrm{span}\{v_1, \ldots, v_p\}$.

Corollary 11.5.6: The Best Rank-k Summary

Keep the data and the notation of theorem 11.5.3, let k be an integer with $1 \leq k \leq p$, and put

$$W_k = \operatorname{span}\{v_1, \dots, v_k\}, \qquad P_k = \sum_{j=1}^{k} v_j v_j^\top \in M_n(\mathbb{R})$$

1. P_k is the matrix of the orthogonal projection of $\mathbb{R}^n$ onto W_k, and $P_k^\top = P_k$.
2. $XP_k = X_k$, the rank-k truncation of X (definition 11.4.8), and the i-th row of X_k is $\big(P_k(d_i - \bar{d})\big)^\top$.
3. Let $W \subseteq \mathbb{R}^n$ be a subspace with $\dim W \leq k$ and let $P_W \in M_n(\mathbb{R})$ be the matrix of the orthogonal projection onto W. Then

$$\sum_{i=1}^{m} \big\|(d_i - \bar{d}) - P_W(d_i - \bar{d})\big\|^2 \;\geq\; \sum_{j=k+1}^{n} \sigma_j^2$$

with equality when $W = W_k$.

4. $\displaystyle\sum_{i=1}^{m} \big\|P_k(d_i - \bar{d})\big\|^2 = \sum_{j=1}^{k} \sigma_j^2$, which is $m-1$ times the sum of the sample variances of the data along $v_1, \dots, v_k$.

***Proof*:**

1. Let $Q_k \in M_{n\times k}(\mathbb{R})$ have columns $v_1, \dots, v_k$, which are orthonormal and span W_k. Then $P_k = \sum_{j=1}^{k} v_j v_j^\top = Q_k Q_k^\top$ by definition 1.4.1, and theorem 6.1.5(2) identifies $Q_k Q_k^\top$ as the orthogonal projection matrix onto W_k. Symmetry is part of that identification, and also follows from $(v_j v_j^\top)^\top = v_j v_j^\top$ (proposition 2.3.5).

2. Since $k \leq p$, theorem 11.5.3(2) gives $Xv_j = \sigma_j u_j$ for every $j \leq k$, so

$$XP_k = \sum_{j=1}^{k} (Xv_j)v_j^\top = \sum_{j=1}^{k} \sigma_j u_j v_j^\top$$

which is the rank-k truncation X_k of definition 11.4.8. By definition 1.4.1 the i-th row of a product MN is the i-th row of M times N, so the i-th row of XP_k is $(d_i - \bar{d})^\top P_k$, and transposing with proposition 2.3.5 and part (1) turns this into $\big(P_k(d_i - \bar{d})\big)^\top$.

3. Choose an orthonormal basis $q_1, \dots, q_s$ of W (corollary 5.3.11), where $s = \dim W \leq k$. By theorem 6.1.5(2) again, now with the q_j in place of the v_j, the matrix $\sum_{j=1}^{s} q_j q_j^\top$ is the orthogonal projection onto W; that projection is unique, being determined by the decomposition of theorem 5.5.3, so $P_W = \sum_{j=1}^{s} q_j q_j^\top$ and $P_W^\top = P_W$.

The i-th row of $X - XP_W$ is $\big((I_n - P_W)(d_i - \bar{d})\big)^\top$, by the row computation of part (2). Since the square of the Frobenius norm of a matrix is the sum of the squares of all its entries (definition 11.4.5), and that sum may be taken row by row,

$$\|X - XP_W\|_F^2 = \sum_{i=1}^{m} \left\|(d_i - \bar{d}) - P_W(d_i - \bar{d})\right\|^2 \tag{11.20}$$

so the left side of the asserted inequality is $\|X - XP_W\|_F^2$.

Next, $\operatorname{rank}(XP_W) \le k$. Indeed P_W maps $\mathbb{R}^n$ into W by the first display of part (1), and onto W, since $P_W w = w$ for $w \in W$ by the uniqueness of the decomposition $w = w + 0$. The column space of XP_W is the set of vectors $XP_W y$ with $y \in \mathbb{R}^n$ (definition 2.3.1), which is therefore the image $X(W) = \{Xz \big| z \in W\}$. Let $T\colon W \to \mathbb{R}^m$ be the linear map sending z to Xz, whose image is $X(W)$. Applying theorem 2.4.10 to T gives $\dim W = \dim\ker T + \dim X(W)$, whence $\dim X(W) \le \dim W \le k$, and corollary 2.3.12 identifies the rank of a matrix with the dimension of its column space.

Write $r = \operatorname{rank} X$ and split into two cases. If $k \ge r$ then $\sigma_j = 0$ for every $j > k$ by corollary 11.2.2, so the right side of the asserted inequality is 0 and the inequality holds, the left side being a sum of squares. Equality at $W = W_k$ holds in this case because $\sum_{j=1}^{n} v_j v_j^\top = I_n$: the (a,b) entry of that sum is $\sum_j V_{aj}V_{bj}$, which is the (a,b) entry of $VV^\top$ by definition 1.4.1, and $VV^\top = I_n$ by proposition 7.1.5. Multiplying by X and distributing as in part (2),

$$X = X\sum_{j=1}^{n} v_j v_j^\top = \sum_{j=1}^{n}(Xv_j)v_j^\top = \sum_{j=1}^{k}(Xv_j)v_j^\top = XP_k$$

the third equality because $Xv_j = 0$ for every $j > k$: such a j has $\sigma_j = 0$, and $\|Xv_j\|^2 = \sigma_j^2$ by theorem 11.5.3(3). Hence $X - XP_{W_k} = X - XP_k = 0$, so the left side of the asserted inequality vanishes as well.

If instead $k < r$, then k is smaller than the rank of X and XP_W is a matrix of rank at most k, which are the hypotheses of corollary 11.4.11. That corollary applied to X gives

$$\|X - XP_W\|_F^2 \;\ge\; \|X - X_k\|_F^2 = \sum_{j=k+1}^{n} \sigma_j^2$$

Combining this with (11.20) is the asserted inequality, and taking $W = W_k$, for which $XP_{W_k} = XP_k = X_k$ by part (2), gives equality.

4. By part (1) the vectors $P_k(d_i - \bar{d})$ and $(d_i - \bar{d}) - P_k(d_i - \bar{d})$ are orthogonal, so theorem 5.3.2 gives

$$\|d_i - \bar{d}\|^2 = \|P_k(d_i - \bar{d})\|^2 + \left\|(d_i - \bar{d}) - P_k(d_i - \bar{d})\right\|^2$$

Summing over i and using (11.20) with $W = W_k$, for which $P_{W_k} = P_k$, together with the equality case of part (3) and with

$$\sum_{i=1}^{m}\|d_i - \bar{d}\|^2 = \|X\|_F^2 = \sum_{j=1}^{n}\sigma_j^2$$

the first equality being the row-by-row reading of the Frobenius norm used in (11.20), applied to X itself, and the second being proposition 11.5.5(2),

$$\sum_{i=1}^{m}\|P_k(d_i - \bar{d})\|^2 = \sum_{j=1}^{n}\sigma_j^2 - \sum_{j=k+1}^{n}\sigma_j^2 = \sum_{j=1}^{k}\sigma_j^2$$

> By proposition 11.5.5(1) the sample variance along v_j is $\sigma_j^2/(m-1)$, so this sum is $m-1$ times the sum of those variances for $j \le k$.
>
> All four parts are established, completing the proof.

Part (3) is the precise form of the claim that the data lie close to a subspace of dimension k. Among all subspaces of $\mathbb{R}^n$ of dimension at most k, the span of the first k principal directions makes the total squared distance from the centered observations as small as it can be, and that smallest value is $\sigma_{k+1}^2 + \cdots + \sigma_n^2$. Translating back by $\bar{d}$, the affine subspace $\bar{d} + W_k$ is the best fit of dimension k among those passing through the sample mean, and part (2) identifies the point of $\bar{d} + W_k$ nearest to d_i as $\bar{d} + P_k(d_i - \bar{d})$, which is $\bar{d}$ plus the i-th row of the truncation X_k: nearest, because theorem 6.1.1 identifies the orthogonal projection as the closest point of the subspace. Part (4) measures the same approximation from the other side. The k retained directions carry the fraction

$$\frac{\sigma_1^2 + \cdots + \sigma_k^2}{\sigma_1^2 + \cdots + \sigma_n^2}$$

of the total sample variance, and what is discarded is exactly the sum of squared distances in part (3).

For the four observations of example 11.5.4 and $k = 1$, the subspace W_1 is the line $\mathbb{R}\,(1,1)^\top$ and $\bar{d} + W_1$ is the line $x_2 = x_1 - 1$ through $(3,2)^\top$. The rank-one truncation is

$$X_1 = \sigma_1 u_1 v_1^\top = \frac{3}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \\ -1 & -1 \\ -1 & -1 \end{pmatrix}$$

so the four nearest points on that line are $\bar{d}$ plus its rows, namely $(9/2, 7/2)^\top$ twice and $(3/2, 1/2)^\top$ twice. The four displacements from the observations to those points are the rows of

$$X - X_1 = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \\ 1 & -1 \\ -1 & 1 \end{pmatrix}$$

each of length $1/\sqrt{2}$, so the total squared distance is $4 \cdot \frac{1}{2} = 2$, which is σ_2^2 as part (3) requires, and the single retained direction carries $18/20$ of the total sample variance.

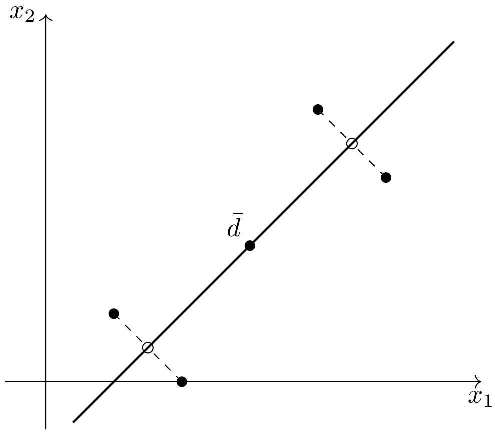


Figure 11.4: The four observations (filled), the sample mean $\bar{d}$, the line $\bar{d} + W_1$, and the two distinct points of the rank-one summary (hollow)

Exercise 11.5.1

1. Let the four observations be $(5,3)^\top$, $(3,5)^\top$, $(1,-1)^\top$ and $(-1,1)^\top$ in $\mathbb{R}^2$. Compute the sample mean, the centered data matrix X, the matrix $X^\top X$ and the sample covariance matrix S. Determine the singular values of X, the two principal directions, the sample variance along each, and the share of the total sample variance taken by the first. Then write down the line $\bar{d} + W_1$ of corollary 11.5.6 and verify by direct computation of the four distances that the total squared distance from the observations to it is σ_2^2.

2. Let the four observations be $(1,2,3)^\top$, $(2,1,3)^\top$, $(4,4,8)^\top$ and $(1,1,2)^\top$ in $\mathbb{R}^3$. Compute $\bar{d}$ and X, and exhibit a nonzero $w \in \mathbb{R}^3$ with $Xw = 0$. Deduce that $\sigma_3 = 0$, that $\mathrm{rank}X = 2$, and that all four observations lie in one affine plane; name that plane by an equation. Compute $\mathrm{tr}S$ and $\sigma_1^2 + \sigma_2^2$ without computing σ_1 and σ_2 separately.

3. Let D be a data matrix with sample mean $\bar{d}$, centered data matrix X and sample covariance matrix S.

    1. Show that replacing every observation d_i by $d_i + c$, for a fixed $c \in \mathbb{R}^n$, leaves X and S unchanged.

    2. Let $Q \in M_n(\mathbb{R})$ satisfy $Q^\top Q = I_n$ and replace every d_i by Qd_i. Show that the new centered data matrix is $XQ^\top$ and the new sample covariance matrix is $QSQ^\top$. Deduce that if $v_1, \ldots, v_n$ are principal directions of the original data then $Qv_1, \ldots, Qv_n$ are principal directions of the new data, with the same singular values and the same sample variances along them.

4. Let $C = \mathrm{diag}(c_1, \ldots, c_n)$ with every $c_j > 0$, and replace the data matrix D by DC, which rescales the j-th recorded quantity by c_j. Show that the centered data matrix becomes XC and the sample covariance matrix becomes CSC. Then take the four observations $(3,1)^\top$, $(1,2)^\top$, $(-1,1)^\top$, $(1,0)^\top$ in $\mathbb{R}^2$ and $C = \mathrm{diag}(1,3)$, and show that the first principal direction of the rescaled data is not the first principal direction of the original data.

5. Keep the notation of theorem 11.5.3 and let $1 \leq k \leq n-1$. Show that $\sigma_{k+1} = 0$ if and only if there is a subspace $W \subseteq \mathbb{R}^n$ with $\dim W \leq k$ such that $d_i - \bar{d} \in W$ for every i, and that W may then be taken to be $W_k = \operatorname{span}\{v_1, \ldots, v_k\}$.

6. Let $a \in \mathbb{R}^n$ have $\|a\| = 1$, let $c \in \mathbb{R}^n$, and let $t_1, \ldots, t_m$ be real numbers, not all zero, with $\sum_i t_i = 0$. Set $d_i = c + t_i a$ and let $t = (t_1, \ldots, t_m)^\top \in \mathbb{R}^m$. Show that $\bar{d} = c$, that $X = ta^\top$, that $\operatorname{rank} X = 1$, that $\sigma_1 = \|t\|$ and every other singular value is zero, and that $S = \frac{\|t\|^2}{m-1} aa^\top$. Show also that the sample variance of the data along any unit vector orthogonal to a is zero.

7. Let $X \in M_{m \times n}(\mathbb{R})$ be a centered data matrix and S its sample covariance matrix. Show that S and X have the same null space (definition 2.3.3), and deduce that $\operatorname{rank} S = \operatorname{rank} X$ and that the number of nonzero eigenvalues of S, counted with multiplicity, equals $\operatorname{rank} X$.

8. Let the four observations be $(2,1)^\top$, $(1,2)^\top$, $(0,1)^\top$ and $(1,0)^\top$ in $\mathbb{R}^2$, with centered data matrix X. Compute S and show that every unit vector of $\mathbb{R}^2$ is a principal direction of this data, and that the sample variance along every unit vector is the same. Compute the two singular values of X and say why this example is consistent with proposition 11.1.6.

12

Generalized Eigenspaces and Canonical Forms

Chapter 11 asked nothing of its matrix and got a diagonal form, at the cost of using one orthonormal basis at the input and a different one at the output. That freedom is unavailable to an operator. An operator maps a space to itself, the two ends are the same space, and a single basis has to serve both. The question of chapter 4 returns in the form it was left in, since the similarity class is what one basis can change and the singular values are blind to it.

Chapter 4 answered that question when the answer is a diagonal matrix, and gave the obstruction: the geometric multiplicities must sum to n, and for a defective matrix they do not. This chapter takes the obstruction as the subject. Over $\mathbb{C}$ the characteristic polynomial always splits, and what is achievable by similarity is then completely determined: a diagonal matrix when the multiplicities agree, and otherwise a matrix as close to diagonal as the defect permits, with the failure recorded in a pattern of ones just above the diagonal. Two matrices are similar exactly when their patterns match, which is the classification the chapter exists to prove.

Section 12.1 repairs the eigenspaces by enlarging them until they do fill the space, and splits $\mathbb{C}^n$ into the pieces they cut out. Section 12.2 analyses the single piece, where the operator is a scalar plus a nilpotent one, and assembles the Jordan form and its uniqueness. Section 12.3 descends to $\mathbb{R}$, using the conjugate-pair structure of chapter 9 to say what a real matrix can be brought to by a real change of basis. Section 12.4 extracts from the form a splitting into a diagonalizable part and a nilpotent part that commute, and observes that both are polynomials in the original matrix. Section 12.5 spends the form on a definition of $f(A)$, in particular of e^A and of square roots, and section 12.6 spends it on two equations in which the unknown is itself a matrix.

12.1 Generalized Eigenspaces and the Primary Decomposition over $\mathbb{C}$

Theorem 4.3.8 located the obstruction exactly: A is diagonalizable over k precisely when the geometric multiplicities of its eigenvalues sum to n, and example 4.2.8 exhibits a matrix for which they do not. The eigenspaces are then too small to fill k^n between them, and no choice of basis can repair that, since the eigenspaces do not depend on a basis.

What can be repaired is the definition. An eigenvector satisfies $(A - \lambda I)v = 0$, while the vectors this section collects satisfy $(A - \lambda I)^j v = 0$ for some j, which is a weaker condition and admits more vectors. Over $\mathbb{C}$ the enlarged spaces do fill $\mathbb{C}^n$, and the rest of the chapter works inside one of them at a time.

Two facts about polynomials are needed first. Neither is in the book, and both are used again in sections 12.2, 12.4 and 12.5.

Lemma 12.1.1: A Combination of Polynomials with No Common Root

Let $f_1, \ldots, f_m$ be nonzero polynomials over $\mathbb{C}$ with no common root: no $\mu \in \mathbb{C}$ satisfies $f_1(\mu) = \cdots = f_m(\mu) = 0$. Then there are polynomials $g_1, \ldots, g_m$ over $\mathbb{C}$ with

$$g_1 f_1 + \cdots + g_m f_m = 1$$

***Proof*:**
Let S be the set of all polynomials $g_1 f_1 + \cdots + g_m f_m$ as the g_i range over polynomials over $\mathbb{C}$. Then S is closed under addition and under multiplication by any polynomial, since $\sum_i g_i f_i + \sum_i g_i' f_i = \sum_i (g_i + g_i') f_i$ and $h \sum_i g_i f_i = \sum_i (h g_i) f_i$. It contains a nonzero element, namely f_1 itself, obtained by taking $g_1 = 1$ and the other $g_i = 0$.

Among the nonzero elements of S choose one, d, of least degree, which is possible because degrees are nonnegative integers. Fix i and divide with remainder: by lemma 4.4.5 there are q and r with $f_i = qd + r$ and either $r = 0$ or $\deg r < \deg d$. Now $r = f_i - qd$ lies in S, because $f_i \in S$ and $qd \in S$ and S is closed under the operations just named. If $r \neq 0$ then r is a nonzero element of S of degree less than $\deg d$, contradicting the choice of d. So $r = 0$ and d divides f_i, for every i.

Suppose d had a root $\mu \in \mathbb{C}$. Writing $f_i = q_i d$ gives $f_i(\mu) = q_i(\mu) d(\mu) = 0$ for every i, so μ would be a common root of $f_1, \ldots, f_m$, which the hypothesis forbids. So d has no root in $\mathbb{C}$, and by the fundamental theorem of algebra a nonconstant polynomial over $\mathbb{C}$ does have one. Hence d is a constant, and it is nonzero. Writing $d = \sum_i g_i f_i$ and dividing through by that constant gives the asserted identity.

The second fact separates out the highest power of $t - \lambda$ dividing a polynomial. Section 12.2 uses it to pin the size of the largest Jordan block, and section 12.5 to expand a polynomial in powers of $t - \lambda$.

Lemma 12.1.2: Factoring Out the Highest Power of a Root

Let p be a nonzero polynomial over k and let $\lambda \in k$. Then there are a unique integer $s \geq 0$ and a unique polynomial h over k with

$$p(t) = (t-\lambda)^s h(t) \qquad \text{and} \qquad h(\lambda) \neq 0$$

***Proof*:**
Existence, by induction on $\deg p$. If $p(\lambda) \neq 0$ take $s = 0$ and $h = p$. Otherwise $p(\lambda) = 0$, so corollary 4.4.6(1) gives $p = (t-\lambda)p_1$ with $\deg p_1 = \deg p - 1$, and $p_1 \neq 0$ since $p \neq 0$. By the inductive hypothesis $p_1 = (t-\lambda)^{s_1} h$ with $h(\lambda) \neq 0$, and then $p = (t-\lambda)^{s_1+1} h$. The induction is legitimate because the degree drops by one at each step and is nonnegative.

Uniqueness. Suppose $(t-\lambda)^s h = (t-\lambda)^{s'} h'$ with $h(\lambda) \neq 0$ and $h'(\lambda) \neq 0$, and suppose $s < s'$. A product of nonzero polynomials is nonzero, since degrees add (lemma 4.3.3), so from $(t-\lambda)^s\big(h - (t-\lambda)^{s'-s}h'\big) = 0$ and $(t-\lambda)^s \neq 0$ we get $h = (t-\lambda)^{s'-s}h'$. Evaluating at λ and using $s' - s \geq 1$ gives $h(\lambda) = 0$, a contradiction. So $s' \leq s$, and by symmetry $s = s'$. Cancelling $(t-\lambda)^s$ the same way gives $h = h'$.

Nilpotence is the other notion the section needs, and it is what the enlarged eigenspaces will exhibit. It is stated over $\mathbb{R}$ as well as $\mathbb{C}$ because section 12.3 uses it over $\mathbb{R}$.

Definition 12.1.3: Nilpotent Matrix

Let k be $\mathbb{R}$ or $\mathbb{C}$. A matrix $N \in M_n(k)$ is *nilpotent* if $N^m = 0$ for some integer $m \geq 1$. The least such m is the *index of nilpotency* of N. A linear map $T\colon W \to W$ on a finite-dimensional space is nilpotent if $T^m = 0$ for some $m \geq 1$.

Proposition 12.1.4: Nilpotent Exactly When the Characteristic Polynomial Is t^n

Let k be $\mathbb{R}$ or $\mathbb{C}$ and let $N \in M_n(k)$. Then N is nilpotent if and only if $\chi_N(t) = t^n$. When N is nilpotent its index of nilpotency is at most n.

***Proof*:**
Suppose first that $\chi_N(t) = t^n$. Then theorem 4.4.8 gives $N^n = \chi_N(N) = 0$, so N is nilpotent with index at most n. This also proves the final sentence, once the converse gives $\chi_N = t^n$.

Conversely, suppose $N^m = 0$. Read N as an element of $M_n(\mathbb{C})$, which is legitimate for $k = \mathbb{R}$ as well and does not change χ_N. Let $\mu \in \mathbb{C}$ be a root of χ_N. By theorem 4.1.10 there is $v \in \mathbb{C}^n$ with $v \neq 0$ and $Nv = \mu v$, whence $N^m v = \mu^m v$ by induction. The left side is 0 and $v \neq 0$, so $\mu^m = 0$ and therefore $\mu = 0$.

So 0 is the only root of χ_N in $\mathbb{C}$. Applying lemma 12.1.2 with $\lambda = 0$ writes $\chi_N(t) = t^s h(t)$ with $h(0) \neq 0$. If h were nonconstant it would have a root in $\mathbb{C}$ by the fundamental theorem of algebra, and that root would be a root of χ_N different from 0, which we have excluded. So h is a nonzero constant, and $\chi_N = ct^s$. Comparing degrees gives $s = n$, since χ_N is monic of degree n (proposition 4.1.8), and comparing leading coefficients gives $c = 1$.

Definition 12.1.5: Generalized Eigenspace

Let $A \in M_n(\mathbb{C})$ and $\lambda \in \mathbb{C}$. The *generalized eigenspace* of A for λ is

$$K_\lambda(A) = \operatorname{null}\big((A - \lambda I_n)^n\big)$$

the set of $v \in \mathbb{C}^n$ with $(A - \lambda I_n)^n v = 0$. Its nonzero elements are the *generalized eigenvectors* of A for λ.

The exponent n looks arbitrary, and the next proposition is what makes it not so: the kernels of the powers stop growing by step n at the latest, so raising the exponent further collects nothing new.

Proposition 12.1.6: The Chain of Kernels Stabilizes by Step n

Let $M \in M_n(k)$ and write $V_j = \operatorname{null}(M^j)$ for $j \geq 0$, so that $V_0 = \{0\}$.

1. $V_j \subseteq V_{j+1}$ for every $j \geq 0$.
2. If $V_j = V_{j+1}$ for some j, then $V_i = V_j$ for every $i \geq j$.
3. $V_j = V_n$ for every $j \geq n$.

***Proof*:**
For (1), if $M^j v = 0$ then $M^{j+1}v = M(M^j v) = 0$.

For (2), it is enough by induction to show that $V_j = V_{j+1}$ forces $V_{j+1} = V_{j+2}$. One inclusion is (1). For the other, let $v \in V_{j+2}$, so $M^{j+2}v = 0$. Then $M^{j+1}(Mv) = 0$, so $Mv \in V_{j+1} = V_j$, so $M^j(Mv) = 0$, which says $M^{j+1}v = 0$ and puts v in V_{j+1}.

For (3), each V_j is a subspace, being a null space (example 2.1.4), and there are two cases. If $V_j = V_{j+1}$ for some $j \leq n-1$, then (2) gives $V_i = V_j$ for every $i \geq j$, and in particular $V_i = V_n$ for every $i \geq n$.

Otherwise every one of the inclusions $V_0 \subseteq V_1 \subseteq \cdots \subseteq V_n$ is strict. A strict inclusion of subspaces raises the dimension (corollary 2.2.12), so $\dim V_0 < \dim V_1 < \cdots < \dim V_n$ is a strictly increasing chain of $n+1$ integers beginning at $\dim V_0 = 0$, so $\dim V_j \geq j$ for each j, and in particular $\dim V_n \geq n$. Since V_n is a subspace of k^n this forces $V_n = k^n$, and then $V_i = k^n = V_n$ for every $i \geq n$ because each V_i lies in k^n and contains V_n by (1).

Proposition 12.1.7: The Generalized Eigenspace Is Invariant, and the Shift Is Nilpotent

Let $A \in M_n(\mathbb{C})$ and $\lambda \in \mathbb{C}$. Then $K_\lambda(A)$ is a subspace of $\mathbb{C}^n$, it is invariant under A (definition 4.1.6), and the restriction of $A - \lambda I_n$ to $K_\lambda(A)$ is nilpotent of index at most n.

***Proof*:**
It is a null space, hence a subspace (example 2.1.4).

For invariance, note that A commutes with $(A - \lambda I_n)^n$: both are polynomials in A, and

$p(A)q(A) = (pq)(A) = (qp)(A) = q(A)p(A)$ by proposition 4.4.2. So for $v \in K_\lambda(A)$,

$$(A - \lambda I_n)^n (Av) = A\big((A - \lambda I_n)^n v\big) = A0 = 0$$

which puts Av in $K_\lambda(A)$.

The subspace is also invariant under $A - \lambda I_n$, since $(A - \lambda I_n)v = Av - \lambda v$ and both terms lie in $K_\lambda(A)$. Its restriction to $K_\lambda(A)$ satisfies $(A - \lambda I_n)^n v = 0$ for every v there, by the very definition of $K_\lambda(A)$, so that restriction is nilpotent with index at most n.

Theorem 12.1.8: The Primary Decomposition over $\mathbb{C}$

Let $A \in M_n(\mathbb{C})$ with $n \geq 1$ and let $\lambda_1, \dots, \lambda_r$ be its distinct eigenvalues, which by theorem 4.1.14 number at least one. Then

$$\mathbb{C}^n = K_{\lambda_1}(A) \oplus \cdots \oplus K_{\lambda_r}(A)$$

in the sense of definition 2.1.13.

***Proof*:**
Write $K_i = K_{\lambda_i}(A)$ and put

$$q_i(t) = \prod_{l \neq i} (t - \lambda_l)^n \qquad \text{and} \qquad Q(t) = \prod_{l=1}^{r} (t - \lambda_l)^n$$

with the empty product read as 1, which is the case $r = 1$.

Step 1: $Q(A) = 0$. By the fundamental theorem of algebra and corollary 4.4.6(1) applied repeatedly, the monic degree-n polynomial χ_A factors over $\mathbb{C}$ as $\chi_A(t) = \prod_l (t - \lambda_l)^{a_l}$ with $a_l \geq 1$ and $\sum_l a_l = n$, and its roots are exactly the eigenvalues (theorem 4.1.10). Each $a_l \leq n$, so $Q = \chi_A \cdot \prod_l (t - \lambda_l)^{n - a_l}$, and therefore $Q(A) = \big(\prod_l (A - \lambda_l I)^{n - a_l}\big)\chi_A(A) = 0$ by proposition 4.4.2 and theorem 4.4.8.

Step 2: projections. The polynomials $q_1, \dots, q_r$ are nonzero and have no common root: a common root would be some λ_j, since every root of q_1 is one of the λ_l with $l \neq 1$, and $q_j(\lambda_j) = \prod_{l \neq j} (\lambda_j - \lambda_l)^n \neq 0$ because the λ_l are distinct. (For $r = 1$ the single polynomial $q_1 = 1$ has no root at all.) So lemma 12.1.1 gives g_i with $\sum_i g_i q_i = 1$. Put $P_i = g_i(A) q_i(A)$. Evaluating the identity at A and using proposition 4.4.2,

$$P_1 + \cdots + P_r = I_n \tag{12.1}$$

Step 3: $\operatorname{im} P_i \subseteq K_i$. Since $(t - \lambda_i)^n q_i(t) = Q(t)$,

$$(A - \lambda_i I)^n P_i = g_i(A)\,(A - \lambda_i I)^n q_i(A) = g_i(A) Q(A) = 0$$

using Step 1 and proposition 4.4.2, the factors commuting because all are polynomials in A. So $P_i v \in K_i$ for every v.

Step 4: the sum is everything. For $v \in \mathbb{C}^n$, eq. (12.1) gives $v = P_1 v + \cdots + P_r v$, a sum of elements of $K_1, \dots, K_r$ by Step 3.

Step 5: the sum is direct. By proposition 2.1.14 it suffices to show that $w_1 + \cdots + w_r = 0$ with $w_i \in K_i$ forces every $w_i = 0$. First, if $i \neq j$ then q_j carries the factor $(t - \lambda_i)^n$, so $q_j(A) w_i = 0$.

The factors of $q_j(A)$ commute, and one of them is $(A - \lambda_i I)^n$, which kills w_i. Hence $P_j w_i = 0$ for $i \neq j$. Applying P_j to $\sum_i w_i = 0$ therefore leaves $P_j w_j = 0$. On the other hand eq. (12.1) applied to w_j gives $w_j = \sum_i P_i w_j = P_j w_j$, again because $P_i w_j = 0$ for $i \neq j$. So $w_j = P_j w_j = 0$, for each j, as we sought to show.

Each summand is invariant, so the decomposition converts at once into a statement about the matrix.

Corollary 12.1.9: The Block Diagonal Form of the Primary Decomposition

Let $A \in M_n(\mathbb{C})$ have distinct eigenvalues $\lambda_1, \dots, \lambda_r$ and put $d_i = \dim K_{\lambda_i}(A)$. Choosing a basis of each $K_{\lambda_i}(A)$ and concatenating them gives an invertible $P \in M_n(\mathbb{C})$ with

$$P^{-1}AP = \begin{pmatrix} B_1 & & \\ & \ddots & \\ & & B_r \end{pmatrix}, \qquad B_i = \lambda_i I_{d_i} + N_i$$

where each $N_i \in M_{d_i}(\mathbb{C})$ is nilpotent of index at most n.

***Proof*:**
Theorem 12.1.8 makes $\mathbb{C}^n$ the direct sum of the $K_{\lambda_i}(A)$, and proposition 12.1.7 makes each invariant under A. So proposition 4.2.6 applies: P is invertible and $P^{-1}AP$ is block diagonal with ith block the matrix B_i of $A|_{K_{\lambda_i}(A)}$ in the chosen basis.

Write $N_i = B_i - \lambda_i I_{d_i}$, so $B_i = \lambda_i I_{d_i} + N_i$ by construction. The matrix of $(A - \lambda_i I_n)|_{K_{\lambda_i}(A)}$ in that basis is $B_i - \lambda_i I_{d_i} = N_i$, and that restriction is nilpotent of index at most n by proposition 12.1.7. A map is zero exactly when its matrix in a basis is zero, so $N_i^n = 0$.

The dimensions are now forced, and they are the multiplicities Ch. 4 could not otherwise reach. Theorem 4.3.7 bounded the geometric multiplicity by the algebraic one and left the gap unexplained. The generalized eigenspace closes it exactly.

Proposition 12.1.10: The Generalized Eigenspace Has Dimension the Algebraic Multiplicity

Let $A \in M_n(\mathbb{C})$ and let λ be an eigenvalue of A with algebraic multiplicity a (definition 4.3.4). Then $\dim K_\lambda(A) = a$.

***Proof*:**
Keep the notation of corollary 12.1.9 and let λ_i be the eigenvalue in question, with a_i its algebraic multiplicity.

Step 1: the characteristic polynomial factors blockwise. Theorem 3.5.2(1) is a statement about two blocks, so apply it repeatedly: for $r \geq 2$, grouping $\operatorname{diag}(B_1, \dots, B_r)$ as the 2×2 block matrix with corner blocks B_1 and $\operatorname{diag}(B_2, \dots, B_r)$ and zero off-diagonal blocks gives $\chi_{P^{-1}AP} = \chi_{B_1} \cdot \chi_{\operatorname{diag}(B_2,\dots,B_r)}$, and induction on r yields $\chi_{P^{-1}AP} = \chi_{B_1}\chi_{B_2}\cdots\chi_{B_r}$. Similar matrices have the same characteristic polynomial (corollary 4.1.16), so $\chi_A = \prod_i \chi_{B_i}$.

Step 2: each block's characteristic polynomial. Fix i. Then

$$\chi_{B_i}(t) = \det\bigl(tI_{d_i} - \lambda_i I_{d_i} - N_i\bigr) = \det\bigl((t-\lambda_i)I_{d_i} - N_i\bigr) = \chi_{N_i}(t-\lambda_i)$$

and $\chi_{N_i}(t) = t^{d_i}$ by proposition 12.1.4, since N_i is nilpotent. So $\chi_{B_i}(t) = (t-\lambda_i)^{d_i}$ and

$$\chi_A(t) = \prod_{l=1}^{r}(t-\lambda_l)^{d_l}$$

Step 3: comparing with the multiplicities. By definition 4.3.4 and proposition 4.3.5, $\chi_A(t) = \prod_l (t-\lambda_l)^{a_l}$ as well. Apply lemma 12.1.2 to χ_A at $\lambda = \lambda_i$. The first factorization exhibits $\chi_A = (t-\lambda_i)^{d_i} h$ with $h = \prod_{l\neq i}(t-\lambda_l)^{d_l}$, and $h(\lambda_i) \neq 0$ because the λ_l are distinct. The second exhibits it as $(t-\lambda_i)^{a_i} h'$ with $h'(\lambda_i) \neq 0$ for the same reason. The uniqueness clause of lemma 12.1.2 gives $d_i = a_i$.

Example 12.1.11: A Primary Decomposition of a 4×4 Matrix

Let

$$A = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix} \in M_4(\mathbb{C})$$

The matrix is upper triangular, so its characteristic polynomial is the product of t minus the diagonal entries (proposition 4.1.12): $\chi_A(t) = (t-2)^3(t-5)$. The distinct eigenvalues are 2, with algebraic multiplicity 3, and 5, with algebraic multiplicity 1.

The ordinary eigenspace for 2 is too small. Indeed

$$A - 2I_4 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

has rank 2, so $\dim E_2(A) = 4 - 2 = 2$ by proposition 4.1.5, short of the algebraic multiplicity 3. By theorem 4.3.8 the matrix is therefore not diagonalizable, since $\dim E_2 + \dim E_5 = 2 + 1 = 3 < 4$.

Now compute the generalized eigenspaces. Squaring,

$$(A - 2I_4)^2 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 9 \end{pmatrix}$$

which has rank 1, so $\dim \operatorname{null}\bigl((A-2I_4)^2\bigr) = 3$. Higher powers change only the corner entry, from 9 to 27 and so on, so the chain has already stabilized, as proposition 12.1.6 promises. Hence

$$K_2(A) = \operatorname{span}\{e_1, e_2, e_3\}, \qquad K_5(A) = \operatorname{span}\{e_4\}$$

the second because $A - 5I_4$ is invertible on the first three coordinates and kills e_4. Their dimensions are 3 and 1, matching the algebraic multiplicities as proposition 12.1.10 requires, and together they span $\mathbb{C}^4$, which is theorem 12.1.8 for this matrix.

Taking $P = I_4$, already adapted to the decomposition, the block form of corollary 12.1.9 is

$$A = \begin{pmatrix} 2 & 1 & 0 & \\ 0 & 2 & 0 & \\ 0 & 0 & 2 & \\ & & & 5 \end{pmatrix} = \begin{pmatrix} 2I_3 + N_1 & 0 \\ 0 & 5I_1 + N_2 \end{pmatrix}, \qquad N_1 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad N_2 = 0$$

and $N_1^2 = 0$, so N_1 is nilpotent of index 2.

Everything above needed χ_A to factor into linear factors, which over $\mathbb{C}$ is automatic and over a general field is not: the rotation of example 4.1.13 has no real eigenvalue at all, so its generalized eigenspaces over $\mathbb{R}$ would be trivial and could not decompose $\mathbb{R}^2$. Section 12.3 treats the real case by descending from $\mathbb{C}$ rather than by working over $\mathbb{R}$ throughout.

Exercise 12.1.1

1. For $A = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix} \in M_2(\mathbb{C})$, compute $E_3(A)$ and $K_3(A)$, and verify that $\dim K_3(A)$ is the algebraic multiplicity of 3.

2. Let
$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 4 \end{pmatrix} \in M_3(\mathbb{C})$$
Find the distinct eigenvalues, compute each generalized eigenspace, and exhibit the block diagonal form of corollary 12.1.9.

3. Let $N \in M_n(k)$ be nilpotent of index m. Show that $\operatorname{null}(N) \subsetneq \operatorname{null}(N^2) \subsetneq \cdots \subsetneq \operatorname{null}(N^m)$, with every inclusion strict, and deduce that $m \leq n$ without invoking proposition 12.1.4.

4. Show that a nilpotent matrix is invertible only if $n = 0$, and that $I_n + N$ is invertible for every nilpotent $N \in M_n(k)$. For the second part, find an explicit inverse as a polynomial in N.

5. Let $A \in M_n(\mathbb{C})$ have a single eigenvalue λ. Show that $K_\lambda(A) = \mathbb{C}^n$ and that $A - \lambda I_n$ is nilpotent.

6. Let $f_1 = t^2 - 1$ and $f_2 = t^2 - 4$ over $\mathbb{C}$. Verify that they have no common root, and find explicit g_1, g_2 with $g_1 f_1 + g_2 f_2 = 1$, as lemma 12.1.1 guarantees.

12.2 Nilpotent Operators and the Jordan Form

Corollary 12.1.9 reduced a complex matrix to blocks $\lambda_i I + N_i$ with N_i nilpotent, so the whole question is now what a nilpotent operator looks like in a well-chosen basis. This section answers that, assembles the answer into a normal form for every complex matrix, and proves the form unique. The uniqueness is what makes it a classification: two matrices are similar exactly when their forms agree.

Four small facts are used along the way and are collected first.

Lemma 12.2.1: Rank Is a Similarity Invariant

Let $A \in M_n(k)$ and let $P \in M_n(k)$ be invertible. Then $\operatorname{rank}(P^{-1}AP) = \operatorname{rank} A$.

***Proof*:**
First $\operatorname{rank}(AP) = \operatorname{rank} A$. The map $x \mapsto Px$ carries $\operatorname{null}(AP)$ into $\operatorname{null} A$, since $APx = 0$ says $Px \in \operatorname{null} A$. It is injective because P is, and it is onto $\operatorname{null} A$, because $y \in \operatorname{null} A$ is the image of $P^{-1}y$, which lies in $\operatorname{null}(AP)$ as $AP(P^{-1}y) = Ay = 0$. So $\dim \operatorname{null}(AP) = \dim \operatorname{null} A$, and theorem 2.3.11 turns equal nullities into equal ranks.

Next $\operatorname{rank}(P^{-1}AP) = \operatorname{rank}(AP)$. Write $B = AP$. Then $\operatorname{null}(P^{-1}B) = \operatorname{null} B$, because $P^{-1}Bx = 0$ holds exactly when $Bx = 0$, the matrix P^{-1} being injective. Again theorem 2.3.11 gives equal ranks, and combining the two equalities proves the lemma.

Lemma 12.2.2: The Binomial Theorem for Commuting Matrices

Let $A, B \in M_n(k)$ with $AB = BA$. For integers $0 \le j \le m$ let $\binom{m}{j}$ be defined by $\binom{m}{0} = \binom{m}{m} = 1$ and $\binom{m}{j} = \binom{m-1}{j-1} + \binom{m-1}{j}$ for $0 < j < m$. Then

$$(A+B)^m = \sum_{j=0}^{m} \binom{m}{j} A^j B^{m-j} \qquad \text{for every } m \ge 0$$

***Proof*:**
Induction on m. At $m = 0$ both sides are I_n. Assume the identity for $m-1$, multiply it by $A+B$, and use $BA^j = A^jB$, which follows from $AB = BA$ by induction on j. This gives

$$(A+B)^m = \sum_{j=0}^{m-1} \binom{m-1}{j} A^{j+1} B^{m-1-j} + \sum_{j=0}^{m-1} \binom{m-1}{j} A^j B^{m-j}$$

Re-indexing the first sum by $j \mapsto j-1$ turns it into $\sum_{j=1}^{m} \binom{m-1}{j-1} A^j B^{m-j}$. Adding, the coefficient of A^jB^{m-j} is $\binom{m-1}{j-1} + \binom{m-1}{j} = \binom{m}{j}$ for $0 < j < m$, and is 1 at $j = 0$ and at $j = m$.

Lemma 12.2.3: A Sum or Difference of Commuting Nilpotents Is Nilpotent

Let $A, B \in M_n(k)$ be nilpotent with $AB = BA$. Then $A + B$ and $A - B$ are nilpotent.

***Proof*:**
Say $A^p = 0$ and $B^q = 0$. By lemma 12.2.2,

$$(A+B)^{p+q-1} = \sum_{j=0}^{p+q-1} \binom{p+q-1}{j} A^j B^{p+q-1-j}$$

Fix j. If $j \ge p$ then $A^j = A^pA^{j-p} = 0$. If $j \le p-1$ then $p+q-1-j \ge q$, so $B^{p+q-1-j} = 0$. Every term vanishes, so $(A+B)^{p+q-1} = 0$. For $A - B$, the matrix $-B$ is nilpotent, since

$(-B)^q = (-1)^q B^q = 0$, and commutes with A, so the case just proved applies.

Lemma 12.2.4: Divisors with No Common Root Multiply

Let f, g, p be polynomials over $\mathbb{C}$ with f and g nonzero and having no common root. If f divides p and g divides p, then fg divides p.

***Proof*:**
By lemma 12.1.1 there are a, b with $af + bg = 1$. Write $p = fp_1$ and $p = gp_2$. Then

$$p = p(af + bg) = a(gp_2)f + b(fp_1)g = fg(ap_2 + bp_1)$$

so fg divides p.

Definition 12.2.5: Jordan Chain

Let N be a nilpotent operator on a finite-dimensional space W over k, and let $v \in W$ satisfy $N^m v = 0$ and $N^{m-1}v \neq 0$. The list

$$N^{m-1}v,\ N^{m-2}v,\ \ldots,\ Nv,\ v$$

is the *Jordan chain* of length m generated by v. Its first entry $N^{m-1}v$ is the *bottom* of the chain and v is its *top*. The bottom lies in $\ker N$, since $N(N^{m-1}v) = N^m v = 0$.

Lemma 12.2.6: A Jordan Chain Is Linearly Independent

In the notation of definition 12.2.5, the list $N^{m-1}v, \ldots, Nv, v$ is linearly independent.

***Proof*:**
A vanishing combination of the list is $\sum_{j=0}^{m-1} c_j N^j v = 0$ for scalars c_j, the list being the same m vectors in another order. Suppose some $c_j \neq 0$ and let i be the least index with $c_i \neq 0$, so that

$$\sum_{j=i}^{m-1} c_j N^j v = 0$$

Apply N^{m-1-i}. The term with $j = i$ becomes $c_i N^{m-1}v$. A term with $j > i$ becomes $c_j N^{m-1-i+j}v$, and $j \geq i+1$ makes the exponent $m - 1 - i + j \geq m$, so that term is 0 because $N^m v = 0$. What remains is $c_i N^{m-1}v = 0$, and $N^{m-1}v \neq 0$ by hypothesis, so $c_i = 0$. That contradicts the choice of i, so every c_j is zero.

Definition 12.2.7: Jordan Block

For $m \geq 1$ let $N_m \in M_m(k)$ have 1 in every superdiagonal entry and 0 elsewhere, so that $N_1 = (0)$ and

$$N_2 = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad N_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

The *Jordan block* of size m for $\lambda \in k$ is $J_m(\lambda) = \lambda I_m + N_m$, and a *Jordan matrix* is a block diagonal matrix each of whose diagonal blocks is a Jordan block.

The matrix N_m sends e_1 to 0 and e_j to e_{j-1} for $j \geq 2$, so $e_1, \ldots, e_m$ is exactly the Jordan chain of length m generated by e_m. Iterating, N_m^j sends e_i to e_{i-j} when $i > j$ and to 0 otherwise, so the nonzero images are the independent vectors $e_1, \ldots, e_{m-j}$ and

$$\operatorname{rank}\bigl(N_m^j\bigr) = \max(m - j,\ 0) \tag{12.2}$$

In particular $N_m^m = 0$ while $N_m^{m-1} \neq 0$: the index of nilpotency of N_m is exactly m.

The point of a chain is that it converts into a block. If a basis of W is a disjoint union of Jordan chains, each written bottom to top, then N carries each basis vector to the one before it in its own chain, and the first of each chain to 0. That is precisely the action of N_m on $e_1, \ldots, e_m$, so the matrix of N in such a basis is block diagonal with blocks $N_{m_1}, \ldots, N_{m_p}$, the m_i being the chain lengths. So the theorem to prove is that such a basis exists.

Theorem 12.2.8: A Nilpotent Operator Has a Basis of Jordan Chains

Let k be $\mathbb{R}$ or $\mathbb{C}$, let W be a finite-dimensional vector space over k, and let $N\colon W \to W$ be nilpotent. Then W has a basis that is a disjoint union of Jordan chains for N. Equivalently, in a suitable basis the matrix of N is block diagonal with every block of the form N_m.

***Proof*:**

Induction on $\dim W$. If $\dim W = 0$ the empty basis serves. If $N = 0$, any basis serves: each $v \neq 0$ satisfies $Nv = 0$ and $N^0 v = v \neq 0$, so it is a chain of length 1.

Suppose $N \neq 0$ and put $U = \operatorname{im} N$. Then $U \neq \{0\}$ because $N \neq 0$. Also $U \neq W$: otherwise N would be surjective, hence injective by theorem 2.4.10 (applicable to an abstract finite-dimensional W by box 2.2.16), hence bijective, and then $N^m = 0$ would give $W = N^{-m}(0) = \{0\}$, contradicting $N \neq 0$. So $0 < \dim U < \dim W$. Moreover $N(U) \subseteq U$, so N restricts to a nilpotent operator on U, and the inductive hypothesis applies to it.

Step 1: a basis of U from chains. By induction U has a basis that is a disjoint union of Jordan chains for $N|_U$, say with tops $u_1, \ldots, u_p$ and lengths $m_1, \ldots, m_p$. So the vectors $N^s u_i$ with $1 \leq i \leq p$ and $0 \leq s \leq m_i - 1$ form a basis of U, and $\dim U = m_1 + \cdots + m_p$.

Step 2: lift each chain by one. Each u_i lies in $U = \operatorname{im} N$, so choose $w_i \in W$ with $Nw_i = u_i$. Then $N^{m_i} w_i = N^{m_i - 1} u_i \neq 0$ and $N^{m_i+1} w_i = N^{m_i} u_i = 0$, so w_i generates a Jordan chain of length $m_i + 1$.

Step 3: fill out the kernel. Put $b_i = N^{m_i - 1} u_i$, the bottom of the ith chain. Each b_i lies in $\ker N$, and $b_1, \ldots, b_p$ are among the basis vectors of U from Step 1, hence independent. Extend them to a basis $b_1, \ldots, b_p, z_1, \ldots, z_q$ of $\ker N$, which is possible by theorem 2.2.9.

Step 4: the assembled list is a basis of W. Let

$$\mathcal{B} = \{\, N^s w_i : 1 \le i \le p,\ 0 \le s \le m_i \,\} \cup \{\, z_1, \dots, z_q \,\}$$

Its size is $\sum_i (m_i + 1) + q = \dim U + p + q$. On the other hand theorem 2.4.10, read for an abstract space via box 2.2.16, gives $\dim W = \dim \ker N + \dim \operatorname{im} N = (p+q) + \dim U$, since Step 3 exhibits a basis of $\ker N$ with $p+q$ entries. So $\mathcal{B}$ has exactly $\dim W$ entries, and by corollary 2.2.13 it is enough to prove it independent.

Suppose $\sum_{i,s} c_{i,s} N^s w_i + \sum_j d_j z_j = 0$. Apply N. Each z_j dies, and $N^{s+1} w_i = N^s u_i$, so

$$\sum_i \sum_{s=0}^{m_i} c_{i,s} N^s u_i = 0$$

The terms with $s = m_i$ vanish because $N^{m_i} u_i = 0$, leaving a vanishing combination of the Step-1 basis of U. Hence $c_{i,s} = 0$ whenever $0 \le s \le m_i - 1$.

The original relation therefore collapses to $\sum_i c_{i,m_i} N^{m_i} w_i + \sum_j d_j z_j = 0$, and $N^{m_i} w_i = N^{m_i - 1} u_i = b_i$, so this reads $\sum_i c_{i,m_i} b_i + \sum_j d_j z_j = 0$. The b_i and z_j together form a basis of $\ker N$, hence are independent, so all c_{i,m_i} and all d_j vanish. Every coefficient is zero, and $\mathcal{B}$ is a basis.

Finally $\mathcal{B}$ is a disjoint union of chains: the vectors $N^s w_i$ with $0 \le s \le m_i$ are the chain generated by w_i, and each z_j is a chain of length 1, being a nonzero element of $\ker N$. The matrix statement follows from the paragraph preceding the theorem.

Theorem 12.2.9: Every Complex Matrix Has a Jordan Form

Let $A \in M_n(\mathbb{C})$ with $n \ge 1$. Then there is an invertible $P \in M_n(\mathbb{C})$ with $P^{-1}AP$ a Jordan matrix, the diagonal entries of its blocks being eigenvalues of A.

***Proof*:**

By corollary 12.1.9 there is an invertible P_1 with $P_1^{-1} A P_1$ block diagonal, the ith block being $B_i = \lambda_i I_{d_i} + N_i$ with N_i nilpotent and λ_i an eigenvalue of A. Apply theorem 12.2.8 to each N_i: there is an invertible $Q_i \in M_{d_i}(\mathbb{C})$ with $Q_i^{-1} N_i Q_i$ block diagonal, its blocks being matrices N_m. Since $Q_i^{-1}(\lambda_i I_{d_i}) Q_i = \lambda_i I_{d_i}$, we get

$$Q_i^{-1} B_i Q_i = \lambda_i I_{d_i} + Q_i^{-1} N_i Q_i$$

which is block diagonal with blocks $\lambda_i I_m + N_m = J_m(\lambda_i)$.

Let $Q = \operatorname{diag}(Q_1, \dots, Q_r)$, which is invertible with inverse $\operatorname{diag}(Q_1^{-1}, \dots, Q_r^{-1})$ by proposition 3.5.1. Then $Q^{-1}(P_1^{-1} A P_1) Q$ is block diagonal with Jordan blocks, again by proposition 3.5.1, so $P = P_1 Q$ works.

The blocks are forced, not merely available. The count of blocks of each size is computed from ranks, and ranks do not change under similarity.

Proposition 12.2.10: The Block Counts Are Determined by Ranks

Let $J \in M_n(\mathbb{C})$ be a Jordan matrix, let $\lambda \in \mathbb{C}$, and for $j \geq 0$ put $r_j = \operatorname{rank}\big((J - \lambda I_n)^j\big)$. Then for every $j \geq 1$,

$$r_{j-1} - r_j = \#\{\text{blocks of } J \text{ for } \lambda \text{ of size} \geq j\}$$

***Proof*:**
First, ranks add over a block diagonal matrix. If $D = \operatorname{diag}(C_1, \ldots, C_s)$ then $Dx = 0$ holds exactly when each block annihilates the corresponding piece of x, so $\operatorname{null} D$ is the direct sum of the $\operatorname{null} C_t$ embedded in their coordinate blocks, whence $\dim \operatorname{null} D = \sum_t \dim \operatorname{null} C_t$ and theorem 2.3.11 gives $\operatorname{rank} D = \sum_t \operatorname{rank} C_t$.

Now $(J - \lambda I_n)^j$ is block diagonal, its block at a Jordan block $J_m(\mu)$ being $\big((\mu - \lambda)I_m + N_m\big)^j$. There are two cases. If $\mu \neq \lambda$ the matrix $(\mu - \lambda)I_m + N_m$ is upper triangular with every diagonal entry $\mu - \lambda \neq 0$, so it is invertible by proposition 1.5.2(2), and so is its jth power. That block contributes m to r_j for every j, and so contributes 0 to the difference $r_{j-1} - r_j$. If $\mu = \lambda$ the block is N_m^j, contributing $\max(m - j, 0)$ by eq. (12.2).

So $r_{j-1} - r_j$ is the sum, over the blocks for λ only, of $\max(m - j + 1, 0) - \max(m - j, 0)$. For a block of size $m \geq j$ that difference is $(m - j + 1) - (m - j) = 1$. For $m < j$ both terms are 0 and the difference is 0. The sum therefore counts the blocks for λ of size at least j.

Theorem 12.2.11: The Jordan Form Is Unique up to the Order of the Blocks

Let $A \in M_n(\mathbb{C})$ and suppose J and J' are Jordan matrices with $J = P^{-1}AP$ and $J' = R^{-1}AR$ for invertible P, R. Then J and J' have the same blocks with the same multiplicities, and so differ only in the order in which the blocks are listed.

***Proof*:**
Fix $\lambda \in \mathbb{C}$ and $j \geq 1$. The matrix J is similar to A, so $J - \lambda I_n = P^{-1}(A - \lambda I_n)P$ and hence $(J - \lambda I_n)^j = P^{-1}(A - \lambda I_n)^j P$. By lemma 12.2.1 its rank equals $\operatorname{rank}\big((A - \lambda I_n)^j\big)$, a number depending only on A. The same holds for J'. So J and J' have the same numbers r_j, and proposition 12.2.10 then gives, for every λ and every $j \geq 1$, the same count of blocks for λ of size at least j.

The number of blocks for λ of size exactly j is the count for size at least j minus the count for size at least $j + 1$, so those agree too. A Jordan matrix is determined by the multiset of pairs (eigenvalue, block size), so J and J' agree up to the order of the blocks.

Corollary 12.2.12: Similar Exactly When the Jordan Forms Agree

Let $A, B \in M_n(\mathbb{C})$. Then A and B are similar if and only if they have the same Jordan form up to the order of the blocks.

***Proof*:**
If A and B have a common Jordan form J, say $J = P^{-1}AP$ and $J = R^{-1}BR$, then $A = PJP^{-1} = PR^{-1}BRP^{-1} = (RP^{-1})^{-1}B(RP^{-1})$, so A and B are similar.

Conversely suppose $B = S^{-1}AS$ and let $J = P^{-1}AP$ be a Jordan form of A, which exists by theorem 12.2.9. Then $J = P^{-1}SBS^{-1}P = (S^{-1}P)^{-1}B(S^{-1}P)$, so J is also a Jordan form of B, and theorem 12.2.11 makes any Jordan form of B equal to J up to the order of the blocks.

The minimal polynomial, which chapter 4 could compute only case by case, is now read off the block sizes.

Corollary 12.2.13: The Minimal Polynomial from the Block Sizes

Let $A \in M_n(\mathbb{C})$ have distinct eigenvalues $\lambda_1, \dots, \lambda_r$, and for each i let m_i be the largest size of a Jordan block for λ_i in a Jordan form of A. Then

$$\mu_A(t) = (t - \lambda_1)^{m_1} \cdots (t - \lambda_r)^{m_r}$$

***Proof*:**
Let $J = P^{-1}AP$ be a Jordan form. By proposition 4.4.13(3) the matrices A and J have the same minimal polynomial, so it suffices to compute μ_J.

Step 1: when a polynomial annihilates one block. Let p be a nonzero polynomial and consider $J_m(\lambda) = \lambda I_m + N_m$. Write $p = (t-\lambda)^s h$ with $h(\lambda) \neq 0$, which is lemma 12.1.2. Then $p(J_m(\lambda)) = N_m^s\, h(J_m(\lambda))$ by proposition 4.4.2. Now $h(J_m(\lambda))$ is upper triangular with every diagonal entry $h(\lambda)$: indeed $J_m(\lambda)$ is upper triangular with diagonal entries λ, and proposition 1.5.2(1) makes any power of it upper triangular with diagonal entries λ^j, so any polynomial in it is upper triangular with diagonal entries the corresponding polynomial in λ. Since $h(\lambda) \neq 0$, proposition 1.5.2(2) makes $h(J_m(\lambda))$ invertible. Hence $p(J_m(\lambda)) = 0$ if and only if $N_m^s = 0$, that is, if and only if $s \geq m$, which says exactly that $(t - \lambda)^m$ divides p.

Step 2: all blocks at once. A polynomial p satisfies $p(J) = 0$ exactly when p annihilates every diagonal block, since $p(J)$ is block diagonal with blocks $p(J_m(\lambda))$. By Step 1 this says $(t - \lambda_i)^{m_i}$ divides p for every i, the largest block size being the binding condition at λ_i.

Step 3: the product. The polynomials $(t - \lambda_i)^{m_i}$ for distinct i have no common root, so lemma 12.2.4, applied repeatedly, shows that $p(J) = 0$ forces $q = \prod_i (t - \lambda_i)^{m_i}$ to divide p. Conversely $q(J) = 0$ by Step 2, since $(t - \lambda_i)^{m_i}$ divides q for each i. So q is a monic annihilating polynomial dividing every annihilating polynomial, and hence is of least degree among them. By the uniqueness clause of theorem 4.4.4, $\mu_J = q$.

Example 12.2.14: Two Matrices with One Characteristic Polynomial and Two Jordan Forms

Let

$$A = \begin{pmatrix} 5 & 1 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix}, \qquad B = \begin{pmatrix} 5 & 1 & 0 \\ 0 & 5 & 1 \\ 0 & 0 & 5 \end{pmatrix}$$

Both are upper triangular with every diagonal entry 5, so $\chi_A = \chi_B = (t-5)^3$ by proposition 4.1.12, and 5 is the only eigenvalue of each. The characteristic polynomial therefore cannot separate them.

The ranks do.

Both matrices are already Jordan matrices: A is $\operatorname{diag}(J_2(5), J_1(5))$ and B is $J_3(5)$. Compute with $\lambda = 5$, writing $r_j = \operatorname{rank}\big((\cdot - 5I_3)^j\big)$:

$$A - 5I_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad B - 5I_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

For A the ranks are $r_0 = 3$, $r_1 = 1$, $r_2 = 0$. For B they are $r_0 = 3$, $r_1 = 2$, $r_2 = 1$, $r_3 = 0$.

Apply proposition 12.2.10. For A the number of blocks of size at least 1 is $r_0 - r_1 = 2$, and of size at least 2 is $r_1 - r_2 = 1$, so A has two blocks, one of size 1 and one of size 2. For B the counts are $r_0 - r_1 = 1$ at every stage up to 3, so B has a single block, of size 3. By corollary 12.2.12 the two matrices are not similar.

Corollary 12.2.13 reads the minimal polynomials off the largest block: $\mu_A = (t-5)^2$ and $\mu_B = (t-5)^3$. Neither equals $t - 5$, so by theorem 4.4.15 neither matrix is diagonalizable, which the block sizes already said.

The same normal form is reached by a different route in *Core Algebra* [9], from the structure theorem for modules over a principal ideal domain.

Exercise 12.2.1

1. Compute the Jordan form of $\begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$, giving the block sizes from the ranks of the powers of $A - 2I_3$ and $A - 3I_3$ as in proposition 12.2.10.
2. Let $N \in M_4(\mathbb{C})$ be nilpotent with $\operatorname{rank} N = 2$ and $\operatorname{rank} N^2 = 1$. Determine the sizes of the Jordan blocks of N, and write the Jordan matrix.
3. Show that two nilpotent matrices in $M_n(\mathbb{C})$ are similar if and only if $\operatorname{rank} N^j = \operatorname{rank} M^j$ for every $j \geq 1$.
4. Let $A \in M_n(\mathbb{C})$. Show that A is diagonalizable if and only if every Jordan block of A has size 1, and deduce theorem 4.4.15 from corollary 12.2.13.
5. Exhibit two matrices in $M_4(\mathbb{C})$ with the same characteristic polynomial *and* the same minimal polynomial that are not similar. (Consider block sizes summing to 4 with the same largest part.)
6. Let $A \in M_n(\mathbb{C})$ have a single eigenvalue λ and satisfy $\operatorname{rank}(A - \lambda I_n) = n - 1$. Show that the Jordan form of A is the single block $J_n(\lambda)$.

12.3 The Real Jordan Form

Everything so far has been over $\mathbb{C}$, and for a reason: theorem 12.2.9 needs the characteristic polynomial to split, which over $\mathbb{R}$ it need not do. The quarter turn of example 4.1.13 has characteristic polynomial $t^2 + 1$ and no real eigenvalue at all, so its generalized eigenspaces over $\mathbb{R}$ are trivial and cannot decompose $\mathbb{R}^2$. A real matrix still has a normal form under real similarity, and this section finds it by working over $\mathbb{C}$ and descending, which is the method chapter 9 was built for.

Throughout, $A \in M_n(\mathbb{R})$, read also as an element of $M_n(\mathbb{C})$, and $\sigma \colon \mathbb{C}^n \to \mathbb{C}^n$ is entrywise conjugation, $\sigma(z) = \overline{z}$. By proposition 9.2.8(1) this is a conjugation on $\mathbb{C}^n$ in the sense of definition 9.2.1, and its fixed set is exactly $\mathbb{R}^n$. By part (2) of the same proposition, A being real is exactly the statement $\overline{Az} = A\overline{z}$ for every z, so σ commutes with A.

The first observation is what makes the descent possible, and both cases below use it, so it comes before the case split.

Lemma 12.3.1: Conjugation Interchanges Conjugate Generalized Eigenspaces

Let $A \in M_n(\mathbb{R})$ and $\lambda \in \mathbb{C}$. Then

$$\sigma\big(K_\lambda(A)\big) = K_{\overline{\lambda}}(A)$$

In particular $K_\lambda(A)$ is carried to itself when λ is real, and $\dim_{\mathbb{C}} K_\lambda(A) = \dim_{\mathbb{C}} K_{\overline{\lambda}}(A)$.

***Proof*:**
Let $v \in K_\lambda(A)$, so $(A - \lambda I_n)^n v = 0$. Conjugating entrywise and using proposition 5.5.11 together with $\overline{A} = A$,

$$0 = \overline{(A - \lambda I_n)^n v} = \big(\overline{A - \lambda I_n}\big)^n \overline{v} = (A - \overline{\lambda} I_n)^n \sigma(v)$$

so $\sigma(v) \in K_{\overline{\lambda}}(A)$. This shows $\sigma(K_\lambda) \subseteq K_{\overline{\lambda}}$. Applying the same with $\overline{\lambda}$ in place of λ gives $\sigma(K_{\overline{\lambda}}) \subseteq K_\lambda$, and applying σ to that inclusion, together with $\sigma \circ \sigma = \mathrm{id}$, gives $K_{\overline{\lambda}} \subseteq \sigma(K_\lambda)$. The two inclusions give equality.

If λ is real then $\overline{\lambda} = \lambda$ and the equality reads $\sigma(K_\lambda) = K_\lambda$. The dimension statement is proposition 9.2.3.

Definition 12.3.2: Real Jordan Block

For $a, b \in \mathbb{R}$ with $b \neq 0$ write

$$C(a,b) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix} \in M_2(\mathbb{R})$$

The *real Jordan block* of size $2m$ for the pair $a \pm bi$ is the $2m \times 2m$ real matrix

$$J^{\mathbb{R}}_{2m}(a,b) = \begin{pmatrix} C(a,b) & I_2 & & \\ & C(a,b) & \ddots & \\ & & \ddots & I_2 \\ & & & C(a,b) \end{pmatrix}$$

with m copies of $C(a,b)$ on the diagonal and I_2 in each superdiagonal block. For $m = 1$ it is $C(a,b)$ itself. A *real Jordan matrix* is a block diagonal real matrix whose blocks are ordinary Jordan blocks $J_m(\lambda)$ with $\lambda \in \mathbb{R}$ and real Jordan blocks $J^{\mathbb{R}}_{2m}(a,b)$.

The orientation of $C(a,b)$ is a choice, and the book has already made it once, so it is worth saying which. Proposition 9.4.6(2) records that for $w = x+iy$ with $Aw = (a+bi)w$,

$$Ax = ax - by \qquad \text{and} \qquad Ay = bx + ay$$

so in the ordered basis (x, y) the matrix is $\left(\begin{smallmatrix} a & b \\ -b & a \end{smallmatrix}\right)$, the transpose of $C(a,b)$. Lemma 7.5.2 in chapter 7 instead takes the second basis vector to be a negative multiple of y, and reaches $C(a,b)$. This section follows chapter 7: the ordered basis is $(x, -y)$, and then

$$Ax = a\,x + b\,(-y), \qquad A(-y) = -bx - ay = (-b)\,x + a\,(-y)$$

whose coefficient columns are $(a,b)^\top$ and $(-b,a)^\top$, giving $C(a,b)$. A reader comparing with proposition 9.4.6 should expect the transpose, and the difference is exactly the sign on the second basis vector.

Lemma 12.3.3: A Real Basis from a Complex Jordan Chain

Let $A \in M_n(\mathbb{R})$, let $\lambda = a+bi$ with $b \neq 0$, and let $w_0, w_1, \ldots, w_{m-1}$ be vectors of $K_\lambda(A)$ with

$$(A - \lambda I_n)w_0 = 0 \qquad \text{and} \qquad (A-\lambda I_n)w_j = w_{j-1} \ \ (1 \leq j \leq m-1)$$

that is, a Jordan chain for $(A - \lambda I_n)$ restricted to $K_\lambda(A)$, written bottom to top. Write $w_j = x_j + iy_j$ with $x_j, y_j \in \mathbb{R}^n$. Then

1. the $2m$ real vectors $x_0, -y_0, x_1, -y_1, \ldots, x_{m-1}, -y_{m-1}$ are linearly independent over $\mathbb{R}$, and
2. in that order they span a real subspace invariant under A, on which the matrix of A is $J^{\mathbb{R}}_{2m}(a,b)$.

***Proof*:**
For (2), rewrite the chain relations. From $Aw_j = \lambda w_j + w_{j-1}$, separating real and imaginary parts of $\lambda w_j + w_{j-1} = (ax_j - by_j + x_{j-1}) + i(bx_j + ay_j + y_{j-1})$,

$$Ax_j = ax_j - by_j + x_{j-1}, \qquad Ay_j = bx_j + ay_j + y_{j-1}$$

with the convention $x_{-1} = y_{-1} = 0$, which is the case $j = 0$. Hence

$$Ax_j = a\,x_j + b\,(-y_j) + x_{j-1}, \qquad A(-y_j) = (-b)\,x_j + a\,(-y_j) + (-y_{j-1})$$

Reading the coefficients of the ordered list $x_0, -y_0, \ldots, x_{m-1}, -y_{m-1}$, the pair $(x_j, -y_j)$ contributes the diagonal block $C(a,b)$ and, for $j \geq 1$, the identity I_2 into the preceding pair. That is $J^{\mathbb{R}}_{2m}(a,b)$. The displayed identities also show the span is carried into itself by A.

For (1), first note that $\sigma(w_j) = \overline{w_j}$ lies in $K_{\overline{\lambda}}(A)$ by lemma 12.3.1, and that $K_\lambda(A) \cap K_{\overline{\lambda}}(A) = \{0\}$ because $\lambda \neq \overline{\lambda}$ and theorem 12.1.8 makes the generalized eigenspaces of distinct eigenvalues independent.

The list $w_0, \ldots, w_{m-1}$ is linearly independent over $\mathbb{C}$, being a Jordan chain (lemma 12.2.6), and so is $\overline{w_0}, \ldots, \overline{w_{m-1}}$, by proposition 9.2.3. Since the two chains lie in subspaces meeting only in 0, the combined list $w_0, \ldots, w_{m-1}, \overline{w_0}, \ldots, \overline{w_{m-1}}$ is independent over $\mathbb{C}$ and spans a complex subspace S with $\dim_{\mathbb{C}} S = 2m$.

Now $x_j = \frac{1}{2}(w_j + \overline{w_j})$ and $-y_j = \frac{i}{2}(w_j - \overline{w_j})$, so all $2m$ real vectors lie in S. Conversely $w_j = x_j - i(-y_j)$ and $\overline{w_j} = x_j + i(-y_j)$, so they span S over $\mathbb{C}$. A spanning list of $2m$ vectors in a space of complex dimension $2m$ is a basis (corollary 2.2.13), hence independent over $\mathbb{C}$, and independence over $\mathbb{C}$ of real vectors implies independence over $\mathbb{R}$, a real vanishing combination being in particular a complex one.

Theorem 12.3.4: The Real Jordan Form

Let $A \in M_n(\mathbb{R})$ with $n \geq 1$. Then there is an invertible $P \in M_n(\mathbb{R})$ such that $P^{-1}AP$ is a real Jordan matrix: block diagonal with blocks $J_m(\lambda)$ for the real eigenvalues λ of A and blocks $J^{\mathbb{R}}_{2m}(a,b)$ for the non-real eigenvalues $a \pm bi$.

***Proof*:**
Step 1: a σ-stable decomposition. Let $\lambda_1, \ldots, \lambda_r$ be the distinct eigenvalues of A in $\mathbb{C}$. Theorem 12.1.8 gives $\mathbb{C}^n = K_{\lambda_1} \oplus \cdots \oplus K_{\lambda_r}$. Group the summands: each real λ_i alone, and each non-real λ_i together with $K_{\overline{\lambda_i}}$, which is another summand because $\overline{\lambda_i}$ is again an eigenvalue by lemma 12.3.1 and proposition 9.4.6(1). Write the grouping as $\mathbb{C}^n = S_1 \oplus \cdots \oplus S_q$. Each S_t is carried to itself by σ, by lemma 12.3.1, and each is carried to itself by A, by proposition 12.1.7.

Step 2: the decomposition descends to $\mathbb{R}^n$. Put $T_t = S_t \cap \mathbb{R}^n$. If $v \in \mathbb{R}^n$, write $v = s_1 + \cdots + s_q$ with $s_t \in S_t$. Applying σ gives $v = \sigma(v) = \sigma(s_1) + \cdots + \sigma(s_q)$ with $\sigma(s_t) \in S_t$, and the decomposition of a vector in a direct sum is unique (definition 2.1.13), so $\sigma(s_t) = s_t$, that is $s_t \in T_t$. Hence $\mathbb{R}^n = T_1 + \cdots + T_q$, and the sum is direct because it is direct already in $\mathbb{C}^n$. Each T_t is invariant under A, since S_t is and A is real.

Step 3: a real eigenvalue. Suppose $S_t = K_\lambda$ with $\lambda \in \mathbb{R}$. By lemma 12.3.1 the restriction $\sigma|_{S_t}$ is a conjugation on S_t, so its fixed set T_t is a real form of S_t by theorem 9.2.6(1), and $\dim_{\mathbb{R}} T_t = \dim_{\mathbb{C}} S_t$ by corollary 9.2.7(1). The operator $A - \lambda I_n$ maps T_t into itself, being real, and is nilpotent on S_t by proposition 12.1.7, hence nilpotent on the subset T_t. Now theorem 12.2.8 applies *over* $\mathbb{R}$, which is why that theorem was stated for $k \in \{\mathbb{R}, \mathbb{C}\}$, and gives a basis of T_t that is a disjoint union of Jordan chains for $(A - \lambda I_n)|_{T_t}$. In that basis the matrix of $A - \lambda I_n$ is block diagonal with blocks N_m, so the matrix of A is block diagonal with blocks $\lambda I_m + N_m = J_m(\lambda)$.

Step 4: a conjugate pair. Suppose $S_t = K_\lambda \oplus K_{\overline{\lambda}}$ with $\lambda = a + bi$, $b \neq 0$. Apply theorem 12.2.8 over $\mathbb{C}$ to $(A - \lambda I_n)$ on K_λ, obtaining a basis of K_λ that is a disjoint union of Jordan chains. For each chain, of length m, lemma 12.3.3 produces $2m$ real vectors spanning an A-invariant real subspace on which the matrix of A is $J^{\mathbb{R}}_{2m}(a,b)$.

Collect the real vectors from all the chains of K_λ. Their number is $2\dim_{\mathbb{C}} K_\lambda = \dim_{\mathbb{C}} S_t$, using lemma 12.3.1 for $\dim_{\mathbb{C}} K_{\overline{\lambda}} = \dim_{\mathbb{C}} K_\lambda$. They lie in T_t, which has $\dim_{\mathbb{R}} T_t = \dim_{\mathbb{C}} S_t$ by theorem 9.2.6(1) and corollary 9.2.7(1) applied to $\sigma|_{S_t}$. They are independent over $\mathbb{R}$: a vanishing combination restricted to one chain's vectors is handled by lemma 12.3.3(1), and combinations across chains reduce to the independence over $\mathbb{C}$ of the union of the chains and their conjugates, which holds because the chains form a basis of K_λ and conjugation carries that basis to a basis of $K_{\overline{\lambda}}$ (proposition 9.2.3), the two summands meeting only in 0. Having the right number and being independent, they form a basis of T_t (corollary 2.2.13).

Step 5: assemble. Concatenating the bases of $T_1, \ldots, T_q$ gives a basis of $\mathbb{R}^n$ by Step 2, and P is the real matrix whose columns are those vectors, invertible by proposition 2.3.13. Each T_t is A-invariant, so proposition 4.2.6 makes $P^{-1}AP$ block diagonal with the blocks computed in

Steps 3 and 4.

Corollary 12.3.5: The Real Jordan Form Is Unique up to the Order of the Blocks

Two real Jordan matrices that are similar over $\mathbb{R}$ have the same blocks with the same multiplicities. Consequently the real Jordan form of $A \in M_n(\mathbb{R})$ is determined by A up to the order of its blocks.

***Proof*:**
Let R be a real Jordan matrix. Read over $\mathbb{C}$, each block $J_m(\lambda)$ with λ real is already a complex Jordan block. For a block $J^{\mathbb{R}}_{2m}(a, b)$, put $\lambda = a + bi$ and note that $\chi_{C(a,b)}(t) = (t - a)^2 + b^2 = (t - \lambda)(t - \overline{\lambda})$ has two distinct roots, so corollary 4.2.4 gives an invertible $S \in M_2(\mathbb{C})$ with $S^{-1}C(a, b)S = \operatorname{diag}(\lambda, \overline{\lambda})$.

Let $T = \operatorname{diag}(S, \ldots, S)$ with m copies. Conjugating blockwise by T (proposition 3.5.1) replaces each diagonal block $C(a, b)$ by $\operatorname{diag}(\lambda, \overline{\lambda})$ and each superdiagonal block I_2 by $S^{-1}I_2S = I_2$. The result is the $2m \times 2m$ matrix carrying $\lambda, \overline{\lambda}$ alternately down the diagonal and 1 in the entries two places above the diagonal. Permuting the basis to list the odd positions first and the even positions second, a similarity since a permutation matrix is invertible, separates it into $\operatorname{diag}\big(J_m(\lambda), J_m(\overline{\lambda})\big)$.

So the complex Jordan form of R is determined by the block data of R, and conversely the block data of R is recovered from the complex Jordan form: the blocks for real eigenvalues are the real blocks, and the blocks for a non-real λ pair off with those for $\overline{\lambda}$, each pair $J_m(\lambda), J_m(\overline{\lambda})$ coming from one block $J^{\mathbb{R}}_{2m}(a, b)$.

Now if two real Jordan matrices are similar over $\mathbb{R}$ they are similar over $\mathbb{C}$, so theorem 12.2.11 makes their complex Jordan forms agree up to order, and the recovery just described makes their real block data agree up to order.

Example 12.3.6: A Real 4×4 with a Repeated Conjugate Pair
Let

$$A = \begin{pmatrix} 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \in M_4(\mathbb{R})$$

Read as two 2×2 blocks, $A = \left(\begin{smallmatrix} C & I_2 \\ 0 & C \end{smallmatrix}\right)$ with $C = C(0, 1) = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. So A is already in the form $J^{\mathbb{R}}_4(0, 1)$ of definition 12.3.2.

Its characteristic polynomial is $(t^2 + 1)^2$ by theorem 3.5.2(1), since $\chi_C(t) = t^2 + 1$. The complex eigenvalues are i and $-i$, each of algebraic multiplicity 2, and there is no real eigenvalue: the real quarter turn of example 4.1.13 sits inside A twice.

Over $\mathbb{C}$ the form is $\operatorname{diag}\big(J_2(i), J_2(-i)\big)$, as corollary 12.3.5 predicts for a single real block $J^{\mathbb{R}}_4(0, 1)$. One checks the block count directly: $\operatorname{rank}(A - iI_4) = 3$ and $\operatorname{rank}(A - iI_4)^2 = 2$, so at $\lambda = i$ the counts of proposition 12.2.10 are $4 - 3 = 1$ block of size at least 1 and $3 - 2 = 1$ of size at least 2, giving one block of size exactly 2.

The contrast with $\operatorname{diag}\big(C(0, 1), C(0, 1)\big)$ is the point: that matrix has the same characteristic polynomial $(t^2 + 1)^2$ and is diagonalizable over $\mathbb{C}$, while A is not, so the two are not similar even over $\mathbb{C}$, let alone over $\mathbb{R}$.

Exercise 12.3.1

1. Find the real Jordan form of the rotation $\begin{pmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{pmatrix}$ for $\sin\theta \neq 0$, and identify a and b.
2. Let $A \in M_3(\mathbb{R})$ have characteristic polynomial $(t-2)(t^2+4)$. Determine the real Jordan form of A completely, and explain why no further information about A is needed.
3. Show that a real matrix with no real eigenvalue has even size.
4. Let $w = x + iy$ satisfy $Aw = \lambda w$ with A real, $\lambda = a + bi$ and $b \neq 0$. Show that x and y are nonzero and that neither is a real multiple of the other.
5. Verify directly that $C(a,b)$ has characteristic polynomial $(t-a)^2 + b^2$, and deduce that $C(a,b)$ is similar over $\mathbb{C}$ to $\operatorname{diag}(a+bi,\ a-bi)$.
6. Let $A \in M_n(\mathbb{R})$ be such that $A^2 = -I_n$. Show that n is even and that the real Jordan form of A is a block diagonal matrix with $n/2$ copies of $C(0,1)$.

12.4 The Jordan-Chevalley Decomposition

A Jordan matrix is the sum of two matrices of very different character: the diagonal, carrying the eigenvalues, and the pattern of ones above it, which is nilpotent. The two commute. This section shows that the splitting is a property of the original matrix rather than of the basis that exhibited it, that it is unique, and that both pieces are polynomials in the matrix. The last of these is the substantive part, and it is what makes the decomposition useful: a polynomial in A commutes with everything A commutes with, so the two pieces inherit every symmetry A has.

Four lemmas are needed, and the first is used by all the others.

Lemma 12.4.1: A Polynomial in A Commutes with Whatever Commutes with A

Let $A, B \in M_n(k)$ with $AB = BA$, and let p be a polynomial over k. Then $Bp(A) = p(A)B$.

***Proof*:**
First $BA^j = A^jB$ for every $j \geq 0$, by induction: it is the hypothesis at $j = 1$ and trivial at $j = 0$, and $BA^{j+1} = (BA^j)A = A^jBA = A^jAB = A^{j+1}B$. Writing $p(t) = \sum_j c_j t^j$ and using proposition 1.4.3(2) to move B through the sum and the scalars, $Bp(A) = \sum_j c_j BA^j = \sum_j c_j A^j B = p(A)B$.

The next lemma is the one that lets a diagonalizable matrix be recognised without computing its minimal polynomial. It is stated for an annihilating polynomial rather than the minimal one on purpose: the book has no unique factorization, so knowing that μ_A *divides* a product of distinct linear factors would not by itself say that μ_A is one.

Lemma 12.4.2: A Split Squarefree Annihilator Forces Diagonalizability

Let $A \in M_n(\mathbb{C})$ and suppose $p(A) = 0$ for some

$$p(t) = (t - \mu_1)(t - \mu_2) \cdots (t - \mu_s)$$

with $\mu_1, \ldots, \mu_s \in \mathbb{C}$ pairwise distinct. Then A is diagonalizable.

***Proof*:**
For each i put $f_i(t) = \prod_{l \neq i}(t - \mu_l)$, the empty product being 1. These are nonzero and have no common root: any root of f_1 is some μ_l with $l \neq 1$, while $f_l(\mu_l) = \prod_{j \neq l}(\mu_l - \mu_j) \neq 0$ because the μ_j are distinct. So lemma 12.1.1 gives $g_1, \ldots, g_s$ with $\sum_i g_i f_i = 1$, and evaluating at A with proposition 4.4.2,

$$\sum_{i=1}^{s} g_i(A) f_i(A) = I_n$$

Fix i and let $v \in \mathbb{C}^n$. Since $(t - \mu_i) f_i(t) = p(t)$,

$$(A - \mu_i I_n)\, g_i(A) f_i(A) v = g_i(A)\, p(A)\, v = 0$$

so $g_i(A) f_i(A) v$ lies in the eigenspace $E_{\mu_i}(A)$, all the factors commuting because all are polynomials in A.

Therefore every $v = \sum_i g_i(A) f_i(A) v$ is a sum of eigenvectors of A, and the eigenvectors of A span $\mathbb{C}^n$. By theorem 2.2.9(2) a spanning list contains a basis, so $\mathbb{C}^n$ has a basis of eigenvectors of A, and theorem 4.2.2 makes A diagonalizable.

Lemma 12.4.3: A Matrix That Is Both Diagonalizable and Nilpotent Is Zero

Let $A \in M_n(k)$ be diagonalizable and nilpotent. Then $A = 0$.

***Proof*:**
Write $A = PDP^{-1}$ with $D = \operatorname{diag}(d_1, \ldots, d_n)$. If $A^m = 0$ then $D^m = P^{-1} A^m P = 0$, and $D^m = \operatorname{diag}(d_1^m, \ldots, d_n^m)$, so every $d_j^m = 0$ and hence every $d_j = 0$. Thus $D = 0$ and $A = P0P^{-1} = 0$.

Lemma 12.4.4: Commuting Diagonalizable Matrices Are Simultaneously Diagonalizable

Let k be $\mathbb{R}$ or $\mathbb{C}$.

1. If $D \in M_n(k)$ is diagonalizable with distinct eigenvalues $\mu_1, \ldots, \mu_s$, then $k^n = E_{\mu_1}(D) \oplus \cdots \oplus E_{\mu_s}(D)$.
2. If $D \in M_n(\mathbb{C})$ is diagonalizable and $W \subseteq \mathbb{C}^n$ is a nonzero subspace invariant under D, then the restriction $D|_W$ is diagonalizable, in the sense that W has a basis of eigenvectors of D.
3. If $D, \tilde{D} \in M_n(\mathbb{C})$ are both diagonalizable and $D\tilde{D} = \tilde{D}D$, then there is an invertible $S \in M_n(\mathbb{C})$ with $S^{-1}DS$ and $S^{-1}\tilde{D}S$ both diagonal. In particular $D - \tilde{D}$ is diagonalizable.

***Proof*:**
For (1), proposition 4.2.5(2) says that D is diagonalizable exactly when $\sum_i \dim E_{\mu_i}(D) = n$, and that the concatenation of bases of the eigenspaces is then a basis of k^n. Spanning is immediate from that, and directness follows from proposition 2.1.14: if $\sum_i w_i = 0$ with $w_i \in E_{\mu_i}(D)$ and some $w_i \neq 0$, the nonzero ones would be eigenvectors for distinct eigenvalues summing to zero, contradicting theorem 4.2.3.

For (2), let $\mathcal{B}$ be a basis of W and let $M = [D|_W]_{\mathcal{B}\mathcal{B}}$ be the matrix of the restriction, which is defined because W is D-invariant. By theorem 4.4.15 the minimal polynomial μ_D is a product of distinct linear factors. Now $\mu_D(M) = 0$: the assignment $T \mapsto [T]_{\mathcal{B}\mathcal{B}}$ carries composites to products by theorem 2.4.9 and sums and scalar multiples to sums and scalar multiples, so $[\mu_D(D|_W)]_{\mathcal{B}\mathcal{B}} = \mu_D(M)$, while $\mu_D(D|_W)$ is the restriction of $\mu_D(D) = 0$ to W and so is the zero map. Lemma 12.4.2 then makes M diagonalizable, and a basis of $\mathbb{C}^{\dim W}$ of eigenvectors of M pulls back through the coordinate map (proposition 2.2.5) to a basis of W of eigenvectors of $D|_W$, which are eigenvectors of D.

For (3), decompose $\mathbb{C}^n = \bigoplus_i E_{\mu_i}(D)$ by (1). Each $E_{\mu_i}(D)$ is invariant under $\tilde{D}$, by lemma 7.6.1(2) in the matrix form its closing sentence records. By (2) each $E_{\mu_i}(D)$ has a basis of eigenvectors of $\tilde{D}$. Every vector of $E_{\mu_i}(D)$ is an eigenvector of D for μ_i or zero, so those basis vectors are eigenvectors of both matrices.

Concatenating over i gives a basis of $\mathbb{C}^n$, by (1), consisting of common eigenvectors. Let S have those vectors as columns. It is invertible by proposition 2.3.13, and theorem 4.2.2 makes both $S^{-1}DS$ and $S^{-1}\tilde{D}S$ diagonal. Their difference $S^{-1}(D - \tilde{D})S$ is then diagonal, so $D - \tilde{D}$ is diagonalizable.

The decomposition itself now follows, and the polynomial statement comes first because uniqueness depends on it.

Proposition 12.4.5: The Two Parts Are Polynomials in the Matrix

Let $A \in M_n(\mathbb{C})$ with $n \geq 1$. There is a polynomial q over $\mathbb{C}$ with $q(0) = 0$ such that, writing $D = q(A)$ and $N = A - D$,

1. D is diagonalizable and N is nilpotent,
2. $DN = ND$, and
3. both D and N are polynomials in A with zero constant term.

***Proof*:**
Let $\lambda_1, \ldots, \lambda_r$ be the distinct eigenvalues of A and $K_i = K_{\lambda_i}(A)$, so that $\mathbb{C}^n = K_1 \oplus \cdots \oplus K_r$ by theorem 12.1.8. Put $q_i(t) = \prod_{l \neq i}(t - \lambda_l)^n$, the empty product being 1. As in the proof of theorem 12.1.8 these have no common root, so lemma 12.1.1 gives g_i with $\sum_i g_i q_i = 1$. Set

$$q_0 = \sum_{i=1}^{r} \lambda_i \, g_i q_i$$

Step 1: $q_0(A)$ acts as λ_j on K_j. Let $v \in K_j$. For $i \neq j$ the polynomial q_i carries the factor $(t - \lambda_j)^n$, and $(A - \lambda_j I_n)^n v = 0$, so $q_i(A)v = 0$, the factors commuting as polynomials in A. Evaluating $\sum_i g_i q_i = 1$ at A and applying it to v leaves $g_j(A)q_j(A)v = v$. Applying $q_0(A)$ to v likewise leaves the single term $\lambda_j g_j(A) q_j(A) v = \lambda_j v$.

Step 2: the constant term. If some $\lambda_i = 0$, say $\lambda_1 = 0$, then the $i = 1$ term of q_0 vanishes and every remaining q_i carries the factor $(t - \lambda_1)^n = t^n$, so t divides q_0 and $q_0(0) = 0$, so take $q = q_0$. Otherwise no λ_i is zero, so $\chi_A(0) = \prod_i (-\lambda_i)^{a_i} \neq 0$, and

$$q = q_0 - \frac{q_0(0)}{\chi_A(0)} \chi_A$$

satisfies $q(0) = 0$ and $q(A) = q_0(A)$, because $\chi_A(A) = 0$ by theorem 4.4.8.

Step 3: the two parts. Put $D = q(A)$. By Step 1, D acts as λ_j on K_j, so every nonzero vector of K_j is an eigenvector of D. Concatenating bases of the K_j gives a basis of $\mathbb{C}^n$ of eigenvectors of D (theorem 12.1.8), and theorem 4.2.2 makes D diagonalizable.

Put $N = A - D$. On K_j it acts as $A - \lambda_j I_n$, which is nilpotent there of index at most n by proposition 12.1.7. Each K_j is invariant under A and under D, hence under N, so N^n kills every K_j and therefore kills $\mathbb{C}^n$: N is nilpotent.

Finally $D = q(A)$ and $N = A - q(A) = \tilde{q}(A)$ with $\tilde{q}(t) = t - q(t)$, and $\tilde{q}(0) = 0 - q(0) = 0$. Both being polynomials in A, they commute by proposition 4.4.2.

Theorem 12.4.6: The Jordan-Chevalley Decomposition

Let $A \in M_n(\mathbb{C})$ with $n \geq 1$. There is exactly one pair (D, N) of matrices in $M_n(\mathbb{C})$ with

$$A = D + N, \qquad D \text{ diagonalizable}, \qquad N \text{ nilpotent}, \qquad DN = ND$$

and for that pair, D and N are polynomials in A with zero constant term.

***Proof*:**
Existence, and the polynomial statement, are proposition 12.4.5.

For uniqueness, suppose $A = D + N = \tilde{D} + \tilde{N}$ with both pairs satisfying the four conditions. Then $\tilde{D}$ and $\tilde{N}$ each commute with $A = \tilde{D} + \tilde{N}$, since $\tilde{D}\tilde{N} = \tilde{N}\tilde{D}$. By proposition 12.4.5 the matrices D and N are polynomials in A, so lemma 12.4.1 makes each of them commute with $\tilde{D}$ and with $\tilde{N}$.

Now $D - \tilde{D} = \tilde{N} - N$. The left side is a difference of two commuting diagonalizable matrices, hence diagonalizable by lemma 12.4.4(3). The right side is a difference of two commuting nilpotent matrices, hence nilpotent by lemma 12.2.3. Being equal, the common value is both diagonalizable and nilpotent, so it is 0 by lemma 12.4.3. Hence $D = \tilde{D}$ and $N = \tilde{N}$.

Corollary 12.4.7: The Two Parts Commute with the Commutant of A

Let $A \in M_n(\mathbb{C})$ with Jordan-Chevalley decomposition $A = D + N$, and let $B \in M_n(\mathbb{C})$ satisfy $AB = BA$. Then $DB = BD$ and $NB = BN$.

***Proof*:**
By theorem 12.4.6 the matrices D and N are polynomials in A, so lemma 12.4.1 applies to each.

This does not follow from the defining conditions on their own. Those say only that D and N commute with each other, which would leave open the possibility that some B commuting with A shuffles the two parts. That it cannot is a consequence of their being polynomials in A, and is the reason proposition 12.4.5 was proved before the theorem rather than deduced from it.

Definition 12.4.8: Unipotent Matrix

A matrix $U \in M_n(k)$ is *unipotent* if $U - I_n$ is nilpotent (definition 12.1.3).

Proposition 12.4.9: The Multiplicative Jordan Decomposition

Let $A \in M_n(\mathbb{C})$ be invertible. Then there is exactly one pair (D, U) with $A = DU$, D diagonalizable, U unipotent and $DU = UD$. Both are polynomials in A, and D is the diagonalizable part of the additive decomposition.

***Proof*:**
Let $A = D + N$ be the decomposition of theorem 12.4.6. The matrix D is invertible: it acts as λ_j on K_j by the construction in proposition 12.4.5, and no λ_j is zero, since A invertible means 0 is not an eigenvalue of A (corollary 3.2.8 and theorem 4.1.10). A diagonalizable matrix with no zero eigenvalue is invertible, its diagonal form having nonzero diagonal entries.

Put $U = D^{-1}A$. Then $DU = A$, and $U = D^{-1}(D+N) = I_n + D^{-1}N$. Now D^{-1} commutes with N: it commutes with D, and D commutes with N, so $D^{-1}N = D^{-1}NDD^{-1} = D^{-1}DND^{-1} = ND^{-1}$. Hence $(D^{-1}N)^n = D^{-n}N^n = 0$, so $D^{-1}N$ is nilpotent and U is unipotent. The same commuting gives $DU = UD$.

For uniqueness, suppose $A = \tilde{D}\tilde{U}$ with $\tilde{D}$ diagonalizable, $\tilde{U}$ unipotent and $\tilde{D}\tilde{U} = \tilde{U}\tilde{D}$. Put $\tilde{N} = \tilde{D}(\tilde{U} - I_n) = A - \tilde{D}$. Then $\tilde{N}$ is nilpotent, being a product of the commuting matrices $\tilde{D}$ and $\tilde{U} - I_n$ with the second nilpotent, so $(\tilde{N})^n = (\tilde{D})^n(\tilde{U} - I_n)^n = 0$, and $\tilde{N}$ commutes with $\tilde{D}$. So $A = \tilde{D} + \tilde{N}$ satisfies the conditions of theorem 12.4.6, whence $\tilde{D} = D$ and then $\tilde{U} = (\tilde{D})^{-1}A = D^{-1}A = U$.

Both D and U are polynomials in A: this is theorem 12.4.6 for D, and $U = D^{-1}A$ with D^{-1} a polynomial in D by corollary 4.4.10, hence a polynomial in A.

Example 12.4.10: A Jordan-Chevalley Decomposition Computed
Let
$$A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{pmatrix} \in M_3(\mathbb{C})$$
which is upper triangular, so $\chi_A(t) = (t-3)^2(t-5)$ by proposition 4.1.12, and the distinct eigenvalues are 3 and 5. Reading the block structure, $K_3(A) = \operatorname{span}\{e_1, e_2\}$ and $K_5(A) = \operatorname{span}\{e_3\}$.

The parts are visible: $D = \operatorname{diag}(3,3,5)$ acts as 3 on K_3 and as 5 on K_5, and
$$N = A - D = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$
with $N^2 = 0$, and $DN = ND$ because D is scalar on each block. So $A = D+N$ is the decomposition of theorem 12.4.6.

To see D as a polynomial in A, follow proposition 12.4.5 in the simpler form available here: a polynomial q with $q \equiv 3$ on K_3 and $q \equiv 5$ on K_5 will do, and since $(A - 3I_3)^2$ kills K_3 while $A - 5I_3$ kills K_5, it is enough to interpolate. Take
$$q(t) = 3 + \tfrac{1}{2}(t-3)^2$$
Then $q(A) = 3I_3 + \frac{1}{2}(A - 3I_3)^2$, and $(A - 3I_3)^2 = \operatorname{diag}(0,0,4)$, so $q(A) = 3I_3 + \operatorname{diag}(0,0,2) = \operatorname{diag}(3,3,5) = D$. Its constant term is $q(0) = 3 + \frac{9}{2} \neq 0$, which proposition 12.4.5 would correct by subtracting a multiple of χ_A, and the correction changes nothing about $q(A)$.

Multiplicatively, A is invertible and
$$U = D^{-1}A = \begin{pmatrix} 1 & \frac{1}{3} & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$
with $U - I_3$ nilpotent, so $A = DU$ is the decomposition of proposition 12.4.9.

The same decomposition organizes the structure theory of algebraic groups and Lie algebras, where [18] develops it in its own terms.

Exercise 12.4.1

1. Compute the Jordan-Chevalley decomposition of $\begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ and of $\begin{pmatrix} 2 & 0 \\ 0 & 7 \end{pmatrix}$.
2. Show that A is diagonalizable if and only if the nilpotent part of its Jordan-Chevalley decomposition is 0, and nilpotent if and only if the diagonalizable part is 0.
3. Let $A = D + N$ be the Jordan-Chevalley decomposition. Show that D and A have the same characteristic polynomial, and that D and A have the same eigenvalues with the same algebraic multiplicities.
4. Give two commuting matrices in $M_2(\mathbb{C})$, one diagonalizable and one nilpotent, whose sum is $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, and confirm that the pair is the one theorem 12.4.6 predicts.
5. Show that the hypothesis $DN = ND$ cannot be dropped from theorem 12.4.6: exhibit a matrix written in two different ways as (diagonalizable) + (nilpotent) with the two summands not commuting.
6. Let $A \in M_n(\mathbb{C})$ be invertible with multiplicative decomposition $A = DU$. Show that A and D have the same determinant.

12.5 Matrix Functions, e^A, and Square Roots

A polynomial can be applied to a matrix because a matrix can be multiplied by itself and added to itself, and nothing else is required. Other functions have no such reading: there is no operation on matrices corresponding to sin, and e^A means nothing until it is given a meaning. The Jordan form gives one. On a single block the answer is forced by a finite computation, because the nilpotent part vanishes after finitely many steps, and the whole construction stays inside polynomial algebra. No limit is taken anywhere in this section.

Only one fact about the exponential is used, and it is the seventh of the results this book borrows: there is a function $z \mapsto e^z$ on $\mathbb{C}$ with $e^0 = 1$, with $e^{z+w} = e^z e^w$, and with $|e^{a+bi}| = e^a$. The section uses the first two of those and nothing further. In particular e^z is never expanded as a series here.

Lemma 12.5.1: Expansion of a Polynomial About a Point

Let p be a polynomial over k of degree $d \geq 0$ and let $\lambda \in k$. Then there are unique scalars $c_0, \ldots, c_d$ with

$$p(t) = \sum_{j=0}^{d} c_j (t - \lambda)^j$$

***Proof*:**
Existence, by induction on d. If $d = 0$ then p is a constant c_0. Otherwise divide by $t - \lambda$ using lemma 4.4.5: $p = (t - \lambda)q + r$ with $r = 0$ or $\deg r < 1$, so r is a constant, which we call c_0. Comparing degrees, $\deg q = d - 1$, so by induction $q = \sum_{j=0}^{d-1} c_{j+1}(t - \lambda)^j$ for suitable scalars, and multiplying by $t - \lambda$ and adding c_0 gives the displayed form.

Uniqueness. Suppose $\sum_j c_j(t - \lambda)^j = \sum_j \tilde{c}_j(t - \lambda)^j$. Setting $t = \lambda$ gives $c_0 = \tilde{c}_0$. Subtracting and dividing by $t - \lambda$, which is legitimate because degrees add (lemma 4.3.3) so a product with $t - \lambda$ vanishes only if the other factor does, reduces to the same statement one degree lower, and induction finishes.

The expansion is exactly what a polynomial does to a Jordan block.

Lemma 12.5.2: A Polynomial on a Jordan Block

Let p be a polynomial over $\mathbb{C}$, let $\lambda \in \mathbb{C}$, and let $J_m(\lambda) = \lambda I_m + N_m$. Writing $p(t) = \sum_j c_j(t - \lambda)^j$ as in lemma 12.5.1,

$$p\big(J_m(\lambda)\big) = \sum_{j=0}^{m-1} c_j N_m^j$$

In particular $p(J_m(\lambda))$ is upper triangular with every diagonal entry $c_0 = p(\lambda)$, and it depends on p only through $c_0, \ldots, c_{m-1}$.

***Proof*:**
Evaluation at a matrix respects sums and products (proposition 4.4.2), and $J_m(\lambda) - \lambda I_m = N_m$, so $p(J_m(\lambda)) = \sum_j c_j N_m^j$. The terms with $j \geq m$ vanish because $N_m^m = 0$ (definition 12.2.7). Each N_m^j with $j \geq 1$ is strictly upper triangular and $N_m^0 = I_m$, so the sum is upper triangular with diagonal entries c_0, and $c_0 = p(\lambda)$ by setting $t = \lambda$ in the expansion.

Definition 12.5.3: A Function of a Matrix

Let $A \in M_n(\mathbb{C})$ with distinct eigenvalues $\lambda_1, \ldots, \lambda_r$, and let m_i be the largest size of a Jordan block of A for λ_i, as in corollary 12.2.13. *Data for a function at A* is a family of complex numbers

$$c_{i,j} \qquad 1 \le i \le r, \quad 0 \le j \le m_i - 1$$

Given such data, fix a Jordan form $J = P^{-1}AP$ (theorem 12.2.9) and define $f(J)$ blockwise: on a block $J_m(\lambda_i)$, which has $m \le m_i$, put

$$f\big(J_m(\lambda_i)\big) = \sum_{j=0}^{m-1} c_{i,j} N_m^j$$

and let $f(J)$ be the block diagonal matrix assembled from these. Finally

$$f(A) = Pf(J)P^{-1}$$

When f is a function of a complex variable with enough derivatives at each λ_i, the data are taken to be $c_{i,j} = f^{(j)}(\lambda_i)/j!$. The definition uses only the numbers, never the function.

That definition names a Jordan form and a matrix P, neither of which is unique, so it is not yet a definition of anything. The repair is to identify $f(A)$ with a polynomial in A, an expression in which no choice appears.

Proposition 12.5.4: A Function of a Matrix Is a Polynomial in It

In the notation of definition 12.5.3, there is a polynomial p over $\mathbb{C}$ whose expansion about λ_i has $c_{i,j}$ as its coefficient of $(t - \lambda_i)^j$ for every i and every $0 \le j \le m_i - 1$. For any such p,

$$f(A) = p(A)$$

***Proof*:**
Existence of p. Put $u_i(t) = (t - \lambda_i)^{m_i}$ and $v_i(t) = \prod_{l \ne i} u_l(t)$. The v_i are nonzero and have no common root, since $v_i(\lambda_i) \ne 0$ and every root of v_i is some λ_l with $l \ne i$. So lemma 12.1.1 gives g_i with $\sum_i g_i v_i = 1$. Let $h_i(t) = \sum_{j=0}^{m_i - 1} c_{i,j}(t - \lambda_i)^j$ and put

$$p = \sum_{i=1}^{r} h_i\, g_i\, v_i$$

Fix i. For $l \ne i$ the product $g_l v_l$ carries the factor $u_i = (t - \lambda_i)^{m_i}$, so modulo u_i the sum reduces to $h_i g_i v_i$, and $g_i v_i = 1 - \sum_{l \ne i} g_l v_l \equiv 1$ modulo u_i for the same reason. Hence $p \equiv h_i$ modulo $(t - \lambda_i)^{m_i}$, which says exactly that p and h_i have the same coefficients of $(t - \lambda_i)^j$ for $j \le m_i - 1$: the difference is divisible by $(t - \lambda_i)^{m_i}$, so its expansion about λ_i (lemma 12.5.1) starts at degree m_i.

$f(A) = p(A)$. Let $J = P^{-1}AP$ be the Jordan form used in definition 12.5.3. On a block $J_m(\lambda_i)$ with $m \le m_i$, lemma 12.5.2 gives $p(J_m(\lambda_i)) = \sum_{j<m} c_{i,j} N_m^j$, since the first m expansion coefficients of p about λ_i are $c_{i,0}, \ldots, c_{i,m-1}$ and $m \le m_i$. That is precisely $f(J_m(\lambda_i))$. Evaluating a polynomial at a block diagonal matrix acts blockwise (proposition 3.5.1), so $p(J) = f(J)$, and

then

$$p(A) = p(PJP^{-1}) = Pp(J)P^{-1} = Pf(J)P^{-1} = f(A)$$

the second equality being proposition 4.4.13(1).

Corollary 12.5.5: The Definition Does Not Depend on the Jordan Form Chosen

In definition 12.5.3 the matrix $f(A)$ is the same for every choice of Jordan form J and of invertible P with $J = P^{-1}AP$.

***Proof*:**
Proposition 12.5.4 produces a polynomial p, depending only on the data $c_{i,j}$ and the numbers λ_i, m_i, and shows that the matrix built from any particular J and P equals $p(A)$. The expression $p(A)$ names neither J nor P, so all such choices give the same matrix.

Two different polynomials with the prescribed coefficients also give the same matrix: if p and $\tilde{p}$ both have them, then $p - \tilde{p}$ is divisible by $(t - \lambda_i)^{m_i}$ for each i, hence by their product, which is μ_A by corollary 12.2.13, the factors for distinct i having no common root so that lemma 12.2.4 applied repeatedly combines them. Therefore $(p - \tilde{p})(A) = 0$.

Definition 12.5.6: The Matrix Exponential

Let $A \in M_n(\mathbb{C})$ with distinct eigenvalues $\lambda_1, \dots, \lambda_r$ and largest block sizes m_i. The *matrix exponential* e^A is $f(A)$ in the sense of definition 12.5.3 for the data

$$c_{i,j} = \frac{e^{\lambda_i}}{j!} \qquad 0 \le j \le m_i - 1$$

the numbers e^{λ_i} being the seventh borrowed result. Concretely, on a Jordan block,

$$e^{J_m(\lambda)} = e^{\lambda} \sum_{j=0}^{m-1} \frac{1}{j!} N_m^j$$

a finite sum. Taking $n = 1$ and $A = (z)$ recovers the scalar e^z, so the notation is consistent.

Two special cases are worth recording, because everything later reduces to them.

Proposition 12.5.7: The Exponential of a Diagonalizable or Nilpotent Matrix

Let $A \in M_n(\mathbb{C})$.

1. If $A = UDU^{-1}$ with $D = \mathrm{diag}(d_1, \dots, d_n)$, then $e^A = U\mathrm{diag}(e^{d_1}, \dots, e^{d_n})U^{-1}$.
2. If A is nilpotent, then $e^A = \sum_{j=0}^{n-1} \frac{1}{j!} A^j$, a finite sum.

***Proof*:**
For (1), a diagonal matrix is a Jordan matrix all of whose blocks have size 1, so D itself serves as the Jordan form. On a block of size 1 for d, definition 12.5.6 gives the single entry e^d, so $e^D = \mathrm{diag}(e^{d_1}, \ldots, e^{d_n})$ and $e^A = Ue^DU^{-1}$ by definition 12.5.3.

For (2), the only eigenvalue of A is 0 (proposition 12.1.4), so the data are $c_{0,j} = e^0/j! = 1/j!$, and proposition 12.5.4 identifies e^A with $p(A)$ for any polynomial p whose expansion about 0 begins $\sum_{j<m} t^j/j!$, where $m \le n$ is the largest block size. The polynomial $\sum_{j=0}^{n-1} t^j/j!$ is such a p, and evaluating it at A gives the stated sum.

Proposition 12.5.8: The Exponential of a Sum of Commuting Matrices

Let $A, B \in M_n(\mathbb{C})$ with $AB = BA$. Then

$$e^{A+B} = e^A e^B$$

***Proof*:**
Write $A = D + N$ and $B = D' + N'$ for the Jordan-Chevalley decompositions (theorem 12.4.6). All four matrices commute with one another: D and N are polynomials in A and D', N' are polynomials in B, so lemma 12.4.1 applies twice, first to move B past D and N, then to move each of D, N past D' and N'.

Step 1: two commuting diagonalizable matrices. By lemma 12.4.4(3) there is an invertible S with $S^{-1}DS = \Lambda$ and $S^{-1}D'S = \Lambda'$ both diagonal. Then $S^{-1}(D + D')S = \Lambda + \Lambda'$ is diagonal, and by proposition 12.5.7(1) all three exponentials are computed entrywise on the diagonals. For each diagonal position the identity reads $e^{d+d'} = e^d e^{d'}$, which is the borrowed property of the scalar exponential. Hence $e^{D+D'} = e^D e^{D'}$.

Step 2: two commuting nilpotent matrices. Here $N + N'$ is nilpotent by lemma 12.2.3, so proposition 12.5.7(2) writes all three exponentials as finite sums. Using lemma 12.2.2,

$$e^{N+N'} = \sum_{j\ge0} \frac{(N+N')^j}{j!} = \sum_{j\ge0} \frac{1}{j!} \sum_{a+b=j} \binom{j}{a} N^a N'^b = \sum_{a\ge0}\sum_{b\ge0} \frac{N^a N'^b}{a!\,b!} = e^N e^{N'}$$

every sum having finitely many nonzero terms because N and N' are nilpotent, so no question of convergence arises.

Step 3: a diagonalizable and a commuting nilpotent matrix. Let $X = Y + Z$ with Y diagonalizable, Z nilpotent and $YZ = ZY$, so that Y and Z are the Jordan-Chevalley parts of X by uniqueness in theorem 12.4.6. As established in proposition 12.4.5, Y acts as λ_i on the generalized eigenspace K_i of X, and $Z = X - Y$ acts there as $X - \lambda_i I_n$. Fix i and let $v \in K_i$. By definition 12.5.6 and proposition 12.5.4, e^X acts on K_i as $e^{\lambda_i} \sum_j Z^j/j!$, while e^Y acts there as the scalar e^{λ_i} by proposition 12.5.7(1) and e^Z as $\sum_j Z^j/j!$ by proposition 12.5.7(2). So $e^X v = e^Y e^Z v$ for every v in every K_i, and these span $\mathbb{C}^n$ by theorem 12.1.8. Hence $e^X = e^Y e^Z$.

Step 4. The Jordan-Chevalley parts of $A + B$ are $D + D'$ and $N + N'$: the first is diagonalizable by lemma 12.4.4(3), the second nilpotent by lemma 12.2.3, they commute, and their sum is $A + B$, so uniqueness in theorem 12.4.6 identifies them. Applying Step 3 three times, then Steps 1 and 2,

$$e^{A+B} = e^{D+D'} e^{N+N'} = e^D e^{D'} e^N e^{N'}$$

Every exponential here is a polynomial in one of D, D', N, N' (proposition 12.5.4), and those four commute, so lemma 12.4.1 lets the middle two factors be exchanged:

$$e^{A+B} = \left(e^D e^N\right)\left(e^{D'} e^{N'}\right) = e^A e^B$$

the last step by Step 3 applied to $A = D + N$ and to $B = D' + N'$.

Example 12.5.9: The Identity Fails Without Commuting

Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Then $AB = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ while $BA = \left(\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right)$, so A and B do not commute.

The matrix A is diagonal, so $e^A = \operatorname{diag}(e^1, e^0) = \left(\begin{smallmatrix} e & 0 \\ 0 & 1 \end{smallmatrix}\right)$ by proposition 12.5.7(1). The matrix B is nilpotent with $B^2 = 0$, so $e^B = I_2 + B = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ by proposition 12.5.7(2). Hence

$$e^A e^B = \begin{pmatrix} e & e \\ 0 & 1 \end{pmatrix}$$

Now $A + B = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 0 \end{smallmatrix}\right)$ has characteristic polynomial $t(t-1)$, with the two distinct roots 0 and 1, so it is diagonalizable by corollary 4.2.4. Eigenvectors are $(1, 0)^\top$ for 1 and $(1, -1)^\top$ for 0, so with $U = \left(\begin{smallmatrix} 1 & 1 \\ 0 & -1 \end{smallmatrix}\right)$, which satisfies $U^{-1} = U$,

$$e^{A+B} = U \begin{pmatrix} e & 0 \\ 0 & 1 \end{pmatrix} U^{-1} = \begin{pmatrix} e & e-1 \\ 0 & 1 \end{pmatrix}$$

The two results differ in the upper right entry, by $e - (e-1) = 1$. Note that establishing the difference needs no numerical information about e whatever: it is enough that $e \neq e - 1$.

Proposition 12.5.10: The Determinant of an Exponential

Let $A \in M_n(\mathbb{C})$. Then $\det\left(e^A\right) = e^{\operatorname{tr} A}$. In particular e^A is invertible, with inverse e^{-A}.

***Proof*:**

Let $J = P^{-1}AP$ be a Jordan form. Then $e^A = Pe^J P^{-1}$, so $\det e^A = \det e^J$ by theorem 3.2.10. The matrix e^J is block diagonal, and on a block $J_m(\lambda)$ it is upper triangular with every diagonal entry e^λ, by lemma 12.5.2 together with proposition 12.5.4. So e^J is upper triangular with diagonal entries $e^{\lambda_1}, \ldots, e^{\lambda_n}$, listing the eigenvalues of A with multiplicity, and proposition 3.2.6 gives

$$\det e^A = e^{\lambda_1} e^{\lambda_2} \cdots e^{\lambda_n} = e^{\lambda_1 + \cdots + \lambda_n} = e^{\operatorname{tr} A}$$

the middle step by repeated use of $e^{z+w} = e^z e^w$ and the last by proposition 4.1.15.

Since A and $-A$ commute, proposition 12.5.8 gives $e^A e^{-A} = e^0 = I_n$, the last equality because the zero matrix is diagonal with every entry 0 and $e^0 = 1$. So e^A is invertible with inverse e^{-A}.

Proposition 12.5.11: Every Invertible Complex Matrix Has a Square Root

Let $A \in M_n(\mathbb{C})$ be invertible. Then there is $S \in M_n(\mathbb{C})$ with $S^2 = A$. The square root is not unique: for $n \geq 1$ both I_n and $-I_n$ square to I_n.

***Proof*:**
Step 1: $I_m + M$ with M nilpotent. Suppose $M^m = 0$ and look for $S = I_m + \sum_{j=1}^{m-1} c_j M^j$. Squaring and collecting powers of M, using that M commutes with itself so no ordering question arises,

$$S^2 = I_m + 2c_1 M + \sum_{j=2}^{m-1} \Big(2c_j + \sum_{a+b=j,\ a,b\geq 1} c_a c_b\Big) M^j$$

modulo $M^m = 0$. Setting this equal to $I_m + M$ requires $2c_1 = 1$ and, for $2 \leq j \leq m-1$, $2c_j = -\sum_{a+b=j,\ a,b\geq 1} c_a c_b$. The right side of each equation involves only $c_1, \ldots, c_{j-1}$, so the coefficients are determined one after another, and the process stops at $j = m-1$ because $M^m = 0$. With those c_j, $S^2 = I_m + M$.

Step 2: a Jordan block. Let $J_m(\lambda)$ be a block of A. Since A is invertible, 0 is not an eigenvalue (corollary 3.2.8 with theorem 4.1.10), so $\lambda \neq 0$. By the fundamental theorem of algebra the polynomial $t^2 - \lambda$ has a root μ, so $\mu^2 = \lambda$ and $\mu \neq 0$. Then

$$J_m(\lambda) = \lambda I_m + N_m = \lambda\Big(I_m + \tfrac{1}{\lambda} N_m\Big)$$

and $\frac{1}{\lambda}N_m$ is nilpotent, so Step 1 produces S_0 with $S_0^2 = I_m + \frac{1}{\lambda}N_m$. Then $(\mu S_0)^2 = \lambda S_0^2 = J_m(\lambda)$.

Step 3: assemble. Let $J = P^{-1}AP$ be a Jordan form and let S_J be the block diagonal matrix whose blocks are the square roots from Step 2. Then $S_J^2 = J$ by proposition 3.5.1, and $S = PS_JP^{-1}$ satisfies $S^2 = PS_J^2P^{-1} = PJP^{-1} = A$.

For non-uniqueness, I_n and $-I_n$ both square to I_n and differ whenever $n \geq 1$.

Invertibility is not a technical convenience. Without it the conclusion is false.

Example 12.5.12: A Matrix with No Square Root

Let $N = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right) \in M_2(\mathbb{C})$ and suppose $S^2 = N$ for some $S \in M_2(\mathbb{C})$. Then $S^4 = N^2 = 0$, so S is nilpotent, and proposition 12.1.4 gives $\chi_S(t) = t^2$. By theorem 4.4.8, $S^2 = \chi_S(S) = 0$. But $S^2 = N \neq 0$, a contradiction. So N has no square root.

The obstruction is exactly the failure of invertibility: N is a single Jordan block for the eigenvalue 0, and Step 2 of proposition 12.5.11 divided by λ.

For a positive semidefinite matrix the situation is different again, and the book has already settled it. Existence follows from corollary 7.3.3: such a P is $U\mathrm{diag}(\lambda_1, \ldots, \lambda_n)U^*$ with every $\lambda_i \geq 0$ by proposition 11.3.3, so $U\mathrm{diag}(\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n})U^*$ is a positive semidefinite square root. Proposition 11.3.6(1) is the statement that such a root is unique among positive semidefinite matrices. Neither claim follows from proposition 12.5.11, which produces some square root and says nothing about definiteness.

Example 12.5.13: The Exponential of a Matrix That Is Not Diagonalizable

Let $A = \left(\begin{smallmatrix} 3 & 1 \\ 0 & 3 \end{smallmatrix}\right)$, which is $J_2(3)$ and so is already in Jordan form. It is not diagonalizable, since it has the single eigenvalue 3 and is not $3I_2$ (theorem 4.2.2).

By definition 12.5.6 with $m = 2$, the data are $c_{1,0} = e^3$ and $c_{1,1} = e^3$, so

$$e^A = e^3\big(I_2 + N_2\big) = \begin{pmatrix} e^3 & e^3 \\ 0 & e^3 \end{pmatrix}$$

As a check, $\det e^A = e^6$ and $\operatorname{tr} A = 6$, agreeing with proposition 12.5.10.

A square root is available by proposition 12.5.11, since A is invertible. Following Step 1 with $M = \frac{1}{3}N_2$ and $m = 2$, only $c_1 = \frac{1}{2}$ is needed, so $S_0 = I_2 + \frac{1}{6}N_2$, and $\mu = \sqrt{3}$ gives

$$S = \sqrt{3}\Big(I_2 + \tfrac{1}{6}N_2\Big) = \begin{pmatrix} \sqrt{3} & \frac{\sqrt{3}}{6} \\ 0 & \sqrt{3} \end{pmatrix}, \qquad S^2 = \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}$$

Two remarks close the section, both pointing outside it. First, the function $f(A)$ has an equivalent description as a contour integral of the resolvent, which is the form that survives into infinite dimensions where no Jordan form is available. The operator-algebra versions of a functional calculus live with the spectral theory of [16] and [4]. Second, [8] and [18] introduce e^A instead as the sum of the series $\sum_{k \geq 0} A^k/k!$, and prove there that it converges and agrees with the matrix defined here. This book takes the finite definition because it needs no notion of convergence.

Exercise 12.5.1

1. Compute e^A for $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and for $A = \begin{pmatrix} 2 & 0 \\ 0 & -1 \end{pmatrix}$.
2. Let $A = J_3(\lambda)$. Write out e^A explicitly as a 3×3 matrix.
3. Show that e^A is diagonalizable whenever A is, and give an example of a non-diagonalizable A for which e^A is also not diagonalizable.
4. Show that $\det e^A > 0$ for every $A \in M_n(\mathbb{R})$, and deduce that no real A satisfies $e^A = \begin{pmatrix} -1 & 0 \\ 0 & 2 \end{pmatrix}$.
5. Find all square roots of I_2 in $M_2(\mathbb{C})$ that are diagonal, and conclude that proposition 12.5.11 cannot be strengthened to give uniqueness.
6. Let A be nilpotent with $A^3 = 0$. Following Step 1 of proposition 12.5.11, compute the coefficients c_1 and c_2 for a square root of $I + A$, and verify the answer by squaring.

12.6 The Sylvester and Lyapunov Equations

Every equation in this book so far has had a column for its unknown. This section takes the unknown to be a matrix, which changes nothing in principle, since the matrices of a fixed shape form a vector space and a matrix equation that is linear in the unknown is a linear map on that space. What it changes in practice is that the coefficients of the linear map are themselves built from matrices, and the spectrum of the map turns out to be computed from the spectra of those. The chapter's normal forms are what make that computation possible.

Three small facts come first.

Lemma 12.6.1: The Space of $m \times n$ Matrices

Let k be $\mathbb{R}$ or $\mathbb{C}$. Then $M_{m\times n}(k)$ is a vector space over k of dimension mn, a basis being the *matrix units* E_{ij}, where E_{ij} has 1 in position (i,j) and 0 elsewhere.

***Proof*:**
The eight conditions of definition 2.2.14 hold entrywise, addition and scalar multiplication of matrices being defined entrywise. Any $X = (x_{ij})$ satisfies $X = \sum_{i,j} x_{ij}E_{ij}$, so the E_{ij} span. If $\sum_{i,j} c_{ij}E_{ij} = 0$ then reading off the (i,j) entry gives $c_{ij} = 0$, so they are independent. There are mn of them.

Results of chapter 2 stated for subspaces of k^n apply here: $M_{m\times n}(k)$ is not literally such a subspace, but box 2.2.16 records that those results use nothing beyond the eight conditions and the existence of a basis. That licence is used below without further comment for theorem 2.4.10 and proposition 2.4.7.

Lemma 12.6.2: The Determinant of a Conjugate

Let $B \in M_n(\mathbb{C})$. Then $\det \overline{B} = \overline{\det B}$.

***Proof*:**
Induction on n. For $n = 1$ the claim is $\overline{b_{11}} = \overline{b_{11}}$. For $n \geq 2$, expand along the first row with theorem 3.3.4. Deleting a row and a column commutes with conjugating every entry, so the $(1,j)$ minor of $\overline{B}$ is the entrywise conjugate of the $(1,j)$ minor of B. Hence

$$\det \overline{B} = \sum_j (-1)^{1+j}\overline{b_{1j}}\det\big(\overline{B_{1j}}\big) = \sum_j (-1)^{1+j}\overline{b_{1j}}\;\overline{\det B_{1j}} = \overline{\sum_j (-1)^{1+j} b_{1j}\det B_{1j}}$$

the middle step by induction and the last because conjugation preserves sums and products and fixes the integers $(-1)^{1+j}$. The final expression is $\overline{\det B}$.

Lemma 12.6.3: The Eigenvalues of the Adjoint

Let $A \in M_n(\mathbb{C})$. Then $\chi_{A^*}(t) = \overline{\chi_A(\bar{t}\,)}$ for every $t \in \mathbb{C}$, and μ is an eigenvalue of A^* if and only if $\overline{\mu}$ is an eigenvalue of A.

***Proof*:**
First, $\det(M^*) = \overline{\det M}$ for every $M \in M_n(\mathbb{C})$: indeed $M^* = \overline{M}^{\top}$ by definition 5.5.12, so $\det(M^*) = \det(\overline{M})$ by theorem 3.2.11, and that is $\overline{\det M}$ by lemma 12.6.2.

Next, $(\bar{t}I_n - A)^* = tI_n - A^*$, by proposition 5.5.13 and because the conjugate transpose of $\bar{t}I_n$ is tI_n. Therefore

$$\chi_{A^*}(t) = \det(tI_n - A^*) = \det\big((\bar{t}I_n - A)^*\big) = \overline{\det(\bar{t}I_n - A)} = \overline{\chi_A(\bar{t}\,)}$$

Finally $\chi_{A^*}(\mu) = 0$ holds exactly when $\overline{\chi_A(\overline{\mu})} = 0$, that is when $\chi_A(\overline{\mu}) = 0$, and theorem 4.1.10 identifies the roots of a characteristic polynomial with the eigenvalues.

Definition 12.6.4: The Sylvester Operator

Let $A \in M_m(k)$ and $B \in M_n(k)$. The *Sylvester operator* of the pair is the map

$$S\colon M_{m\times n}(k) \to M_{m\times n}(k), \qquad S(X) = AX - XB$$

It is linear, since $A(X+X') - (X+X')B = S(X) + S(X')$ and $S(cX) = cS(X)$. The *Sylvester equation* for a given $C \in M_{m\times n}(k)$ is $S(X) = C$, that is $AX - XB = C$.

The spectrum of S is where the chapter's earlier work is spent. Since the book attaches eigenvalues to matrices rather than to operators (definition 4.1.2), the statement is made about the matrix of S in the basis of matrix units.

Proposition 12.6.5: The Eigenvalues of the Sylvester Operator

Let $A \in M_m(\mathbb{C})$ and $B \in M_n(\mathbb{C})$ have eigenvalues $\lambda_1, \dots, \lambda_m$ and $\mu_1, \dots, \mu_n$ respectively, each listed as often as its multiplicity as a root of the characteristic polynomial. Let $[S]$ be the matrix of the Sylvester operator in the basis of matrix units. Then

$$\chi_{[S]}(t) = \prod_{i=1}^{m}\prod_{j=1}^{n}\big(t - (\lambda_i - \mu_j)\big)$$

so the eigenvalues of $[S]$ are exactly the differences $\lambda_i - \mu_j$, with multiplicity.

***Proof*:**
Step 1: triangularize. By theorem 7.2.1 there are unitary $U \in M_m(\mathbb{C})$ and $V \in M_n(\mathbb{C})$ with $T = U^*AU$ and $R = V^*BV$ upper triangular. The diagonal entries of T are $\lambda_1, \dots, \lambda_m$ and those of R are $\mu_1, \dots, \mu_n$, by corollary 7.2.2. Define $\Phi\colon M_{m\times n}(\mathbb{C}) \to M_{m\times n}(\mathbb{C})$ by $\Phi(X) = U^*XV$, which is linear and bijective, its inverse being $Y \mapsto UYV^*$. Then

$$\Phi(S(X)) = U^*(AX - XB)V = TU^*XV - U^*XVR = \Psi(\Phi(X))$$

where $\Psi(Y) = TY - YR$. So $\Phi \circ S = \Psi \circ \Phi$, and $[S]$ and $[\Psi]$ are similar (theorem 2.5.4), hence have the same characteristic polynomial (corollary 4.1.16).

Step 2: Ψ is triangular in a suitable order. Compute on a matrix unit. Since $TE_{ij} = \sum_k T_{ki}E_{kj}$ and T is upper triangular, so that $T_{ki} = 0$ for $k > i$,

$$TE_{ij} = \lambda_i E_{ij} + \sum_{k<i} T_{ki}E_{kj}$$

and since $E_{ij}R = \sum_l R_{jl}E_{il}$ with $R_{jl} = 0$ for $l < j$,

$$E_{ij}R = \mu_j E_{ij} + \sum_{l>j} R_{jl}E_{il}$$

Subtracting,

$$\Psi(E_{ij}) = (\lambda_i - \mu_j)E_{ij} + \sum_{k<i} T_{ki}E_{kj} - \sum_{l>j} R_{jl}E_{il}$$

Order the matrix units by decreasing i, breaking ties by increasing j. Every unit occurring in the two sums comes strictly later in that order: a term E_{kj} has $k < i$, so a later position by the first key, and a term E_{il} has the same i and $l > j$, so a later position by the second. Hence the matrix of Ψ in this ordered basis has its only nonzero entries on and below the diagonal, and its diagonal entries are the $\lambda_i - \mu_j$.

Step 3. A lower triangular matrix has characteristic polynomial the product of t minus its diagonal entries (proposition 4.1.12), which is the displayed product, and by Step 1 that is also $\chi_{[S]}$.

Theorem 12.6.6: When the Sylvester Equation Is Uniquely Solvable

Let $A \in M_m(\mathbb{C})$ and $B \in M_n(\mathbb{C})$. The equation $AX - XB = C$ has exactly one solution X for every $C \in M_{m\times n}(\mathbb{C})$ if and only if A and B have no eigenvalue in common.

***Proof*:**
The equation has exactly one solution for every C precisely when the linear map S of definition 12.6.4 is a bijection. Since S maps $M_{m\times n}(\mathbb{C})$ to itself and that space is finite-dimensional (lemma 12.6.1), theorem 2.4.10 makes S bijective exactly when it is injective, and proposition 2.4.7 makes it injective exactly when $\ker S = \{0\}$, that is when 0 is not an eigenvalue of $[S]$.

By proposition 12.6.5 the eigenvalues of $[S]$ are the differences $\lambda_i - \mu_j$, so 0 is one of them exactly when some $\lambda_i = \mu_j$, which is exactly when A and B share an eigenvalue.

When the spectra do meet, solvability depends on C, and the criterion is a statement about similarity of block matrices rather than about C directly.

Theorem 12.6.7: Roth's Removal Rule

Let $A \in M_m(\mathbb{C})$, $B \in M_n(\mathbb{C})$ and $C \in M_{m\times n}(\mathbb{C})$, and put

$$M = \begin{pmatrix} A & C \\ 0 & B \end{pmatrix}, \qquad M_0 = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}$$

Then $AX - XB = C$ has a solution if and only if M and M_0 are similar.

***Proof*:**
Solvable implies similar. If $AX - XB = C$, put $P = \left(\begin{smallmatrix} I_m & -X \\ 0 & I_n \end{smallmatrix}\right)$, which is invertible with inverse $\left(\begin{smallmatrix} I_m & X \\ 0 & I_n \end{smallmatrix}\right)$, as proposition 3.5.1 confirms. Then

$$P^{-1}M_0P = \begin{pmatrix} I_m & X \\ 0 & I_n \end{pmatrix}\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix}\begin{pmatrix} I_m & -X \\ 0 & I_n \end{pmatrix} = \begin{pmatrix} A & -AX + XB \\ 0 & B \end{pmatrix}$$

and $-AX + XB = -C$. Replacing X by $-X$ throughout gives M itself, so M and M_0 are similar.

Similar implies solvable. This is the argument of Flanders and Wimmer. Work in $M_{m+n}(\mathbb{C})$ and define two linear maps on it,

$$T_1(Z) = MZ - ZM_0, \qquad T_2(Z) = M_0 Z - ZM_0$$

Step 1: the kernels have equal dimension. By hypothesis there is an invertible P with $MP = PM_0$, so $M_0 P^{-1} = P^{-1}M$. If $Z \in \ker T_1$, that is $MZ = ZM_0$, then $W = P^{-1}Z$ satisfies $M_0 W = P^{-1}MZ = P^{-1}ZM_0 = WM_0$, so $W \in \ker T_2$. The map $Z \mapsto P^{-1}Z$ is linear and bijective on $M_{m+n}(\mathbb{C})$, and the same computation run with P in place of P^{-1} carries $\ker T_2$ back into $\ker T_1$. So it restricts to a bijection between the kernels and $\dim\ker T_1 = \dim\ker T_2$.

Step 2: both kernels map onto the same target. Write an element of $M_{m+n}(\mathbb{C})$ in blocks as $Z = \left(\begin{smallmatrix} R & U \\ V & W \end{smallmatrix}\right)$ with $R \in M_m(\mathbb{C})$, $U \in M_{m\times n}(\mathbb{C})$, $V \in M_{n\times m}(\mathbb{C})$ and $W \in M_n(\mathbb{C})$. Multiplying out with proposition 3.5.1, $Z \in \ker T_2$ says

$$AR = RA, \qquad AU = UB, \qquad BV = VA, \qquad BW = WB$$

while $Z \in \ker T_1$ says

$$AR + CV = RA, \qquad AU + CW = UB, \qquad BV = VA, \qquad BW = WB$$

The last two conditions are the same in both lists. Let $\mathcal{S}$ be the space of pairs (V, W) satisfying them, a subspace of $M_{n\times m}(\mathbb{C}) \times M_n(\mathbb{C})$, and let $\varphi_i \colon \ker T_i \to \mathcal{S}$ take Z to its bottom block row (V, W). Both maps are linear.

Step 3: the kernels of φ_1 and φ_2 coincide. Setting $V = 0$ and $W = 0$ in the two lists leaves $AR = RA$ and $AU = UB$ in each case, the terms CV and CW having vanished. So

$$\ker\varphi_1 = \ker\varphi_2 = \{\, (R, U) : AR = RA,\ AU = UB \,\}$$

identifying a matrix with zero bottom row with its top row.

Step 4: φ_2 is onto, hence so is φ_1. Given $(V, W) \in \mathcal{S}$, the matrix $\left(\begin{smallmatrix} 0 & 0 \\ V & W \end{smallmatrix}\right)$ lies in $\ker T_2$, by the first list, and φ_2 sends it to (V, W). So $\operatorname{im}\varphi_2 = \mathcal{S}$. Now apply theorem 2.4.10 to each φ_i:

$$\dim\ker T_i = \dim\ker\varphi_i + \dim\operatorname{im}\varphi_i$$

The left sides agree by Step 1 and the first right-hand terms agree by Step 3, so $\dim\operatorname{im}\varphi_1 = \dim\operatorname{im}\varphi_2 = \dim\mathcal{S}$. A subspace of $\mathcal{S}$ of the full dimension is $\mathcal{S}$ (corollary 2.2.12), so φ_1 is onto as well.

Step 5: read off the solution. The pair $(0, I_n)$ lies in $\mathcal{S}$, since $B0 = 0A$ and $BI_n = I_nB$. By Step 4 there is $Z = \left(\begin{smallmatrix} R & U \\ 0 & I_n \end{smallmatrix}\right) \in \ker T_1$. Its second condition in the $\ker T_1$ list reads $AU + CI_n = UB$, that is

$$A(-U) - (-U)B = C$$

so $X = -U$ solves the equation.

Corollary 12.6.8: Removing the Corner Block

Let $A \in M_m(\mathbb{C})$ and $B \in M_n(\mathbb{C})$ have no eigenvalue in common. Then for every $C \in M_{m\times n}(\mathbb{C})$ the matrices $\left(\begin{smallmatrix} A & C \\ 0 & B \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} A & 0 \\ 0 & B \end{smallmatrix}\right)$ are similar.

***Proof*:**
Theorem 12.6.6 makes $AX - XB = C$ solvable for every C, and theorem 12.6.7 converts a solution into the similarity.

The Lyapunov equation is the Sylvester equation with B taken to be $-A^*$, and it is where an order on the scalars enters: the statement is about positive definiteness, which needs $\mathbb{R}$ or $\mathbb{C}$ and has no analogue over a general field.

Theorem 12.6.9: The Lyapunov Equation

Let $A \in M_n(\mathbb{C})$ and let $Q \in M_n(\mathbb{C})$ be positive definite (definition 11.3.2). Consider

$$A^*X + XA = -Q \tag{12.3}$$

1. If every eigenvalue of A has strictly negative real part, then eq. (12.3) has exactly one solution X, and that X is Hermitian.
2. Conversely, if some positive definite X satisfies eq. (12.3), then every eigenvalue of A has strictly negative real part.

***Proof*:**
Read eq. (12.3) as a Sylvester equation by taking the first coefficient to be A^* and the second to be $-A$: with those choices definition 12.6.4 gives $S(X) = A^*X - X(-A) = A^*X + XA$, so eq. (12.3) is $S(X) = -Q$.

For (1), let $\lambda_1, \ldots, \lambda_n$ be the eigenvalues of A. Those of A^* are the $\overline{\lambda_i}$ by lemma 12.6.3, and those of $-A$ are the $-\lambda_j$, since $\chi_{-A}(t) = (-1)^n \chi_A(-t)$. The two lists are disjoint: a common value would mean $\overline{\lambda_i} = -\lambda_j$ for some i, j, that is $\overline{\lambda_i} + \lambda_j = 0$, whereas

$$\operatorname{Re}\big(\overline{\lambda_i} + \lambda_j\big) = \operatorname{Re}\lambda_i + \operatorname{Re}\lambda_j < 0$$

So theorem 12.6.6 gives exactly one solution X.

That X is Hermitian: taking conjugate transposes in eq. (12.3) and using proposition 5.5.13,

$$(A^*X + XA)^* = A^*X^* + X^*A$$

while $(-Q)^* = -Q$ because a positive definite matrix is Hermitian by definition 11.3.2. So X^* satisfies eq. (12.3) too, and uniqueness forces $X^* = X$.

For (2), let λ be an eigenvalue of A with $Av = \lambda v$ and $v \neq 0$. Multiplying eq. (12.3) on the left by v^* and on the right by v,

$$v^*A^*Xv + v^*XAv = (Av)^*Xv + v^*X(Av) = \overline{\lambda}\, v^*Xv + \lambda\, v^*Xv = 2\operatorname{Re}(\lambda)\, v^*Xv$$

and the right side of the equation gives $-v^*Qv$. Both v^*Xv and v^*Qv are strictly positive, X and Q being positive definite and $v \neq 0$, so $2\operatorname{Re}(\lambda) = -v^*Qv/v^*Xv < 0$.

One clause is missing, and it is worth saying plainly which. Part (1) produces a unique Hermitian X, and part (2) needs a *positive definite* X to run. That the X of part (1) is in fact positive definite is true and is not proved here: the standard argument constructs it as an integral over the solutions of a differential equation, and this book has no integration among the results it borrows. The gap costs less than it looks. Part (2) already turns the equation into a usable stability test, because exhibiting one positive definite solution for one positive definite Q certifies the whole spectrum.

Corollary 12.6.10: The Quadratic Form Attached to a Solution

Let X be Hermitian and satisfy $A^*X + XA = -Q$ with Q positive definite. Then for every $x \neq 0$,

$$2\,\mathrm{Re}\big(x^*XAx\big) = -\,x^*Qx < 0$$

***Proof*:**
Put $z = x^*XAx$, a scalar. Its conjugate is $\overline{z} = z^* = x^*A^*X^*x = x^*A^*Xx$, using proposition 5.5.13 and $X^* = X$. Hence

$$x^*(A^*X + XA)x = \overline{z} + z = 2\,\mathrm{Re}(z)$$

and the left side is $-x^*Qx$, which is strictly negative for $x \neq 0$ because Q is positive definite.

The quantity x^*Xx therefore decreases along the solutions of the linear differential equation $\dot{x} = Ax$, which is where the equation gets its name. That reading belongs to the theory of stability rather than to this book, and [8] develops it in the more general setting of Lyapunov functions for nonlinear systems.

Example 12.6.11: A Sylvester Equation and a Lyapunov Equation
Sylvester. Take

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}, \qquad B = \begin{pmatrix} 3 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

so $m = 2$, $n = 1$. The eigenvalues are $1, 2$ for A and 3 for B, disjoint, so theorem 12.6.6 promises exactly one solution. Writing $X = (x_1, x_2)^\top$, the equation $AX - XB = C$ reads $x_1 - 3x_1 = 1$ and $2x_2 - 3x_2 = 1$, so $X = (-\frac{1}{2},\ -1)^\top$. By corollary 12.6.8 the 3×3 matrix with corner C is similar to the block diagonal one, and theorem 12.6.7 exhibits the similarity explicitly through this X.

Lyapunov. Take $A = \mathrm{diag}(-1, -2)$, whose eigenvalues -1 and -2 have negative real parts, and $Q = I_2$. Then $A^* = A$ and eq. (12.3) reads $AX + XA = -I_2$. Writing $X = (x_{ij})$, the (i,j) entry of $AX + XA$ is $(a_i + a_j)x_{ij}$ with $a_1 = -1$, $a_2 = -2$, so

$$x_{11} = \tfrac{1}{2}, \qquad x_{12} = x_{21} = 0, \qquad x_{22} = \tfrac{1}{4}$$

The solution is unique, as part (1) requires, and Hermitian. It also happens to be positive definite, its eigenvalues $\frac{1}{2}$ and $\frac{1}{4}$ being positive (proposition 11.3.3). That is the clause the theorem leaves unproved, visible here in a case where it can be checked by hand.

Exercise 12.6.1

1. Solve $AX - XB = C$ for $A = \left(\begin{smallmatrix} 2 & 0 \\ 0 & 5 \end{smallmatrix}\right)$, $B = \left(\begin{smallmatrix} 1 \end{smallmatrix}\right)$ and $C = \left(\begin{smallmatrix} 1 \\ 2 \end{smallmatrix}\right)$, and say why theorem 12.6.6 guarantees the answer is the only one.
2. Let $A = B = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$. Show that $AX - XB = C$ is not solvable for $C = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$, and deduce from theorem 12.6.7 that the corresponding 4×4 block matrices are not similar.
3. Using proposition 12.6.5, compute the eigenvalues of the Sylvester operator for $A = \operatorname{diag}(1, 2)$ and $B = \operatorname{diag}(0, 1)$, and confirm that 0 occurs exactly once.
4. Solve the Lyapunov equation $A^*X + XA = -I_2$ for $A = \left(\begin{smallmatrix} -1 & 1 \\ 0 & -1 \end{smallmatrix}\right)$, and check that the solution is Hermitian and positive definite.
5. Let $A = \operatorname{diag}(i, -i)$. Show that $A^*X + XA = -Q$ fails to have a unique solution, and identify the coincidence of eigenvalues that theorem 12.6.6 is detecting.

Index

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